

General Information

Prof. Antonella Ferrara

<https://scholar.google.com/citations?user=r5JuMskAAAAJ&hl=en>

Course Teaching Material:

KIRO UNIPV <https://elearning.unipv.it/>

- [504462 - PROCESS CONTROL 2025-26 - PROF.SSA FERRARA ANTONELLA](#)

Lecture Time-table:

<http://www-3.unipv.it/ingserv/orario2526/1sem/insegnamenti/ProCont.html>

Exams:

<https://studentionline.unipv.it/esse3/Home.do>

<https://kirotesting.unipv.it/>

Introduction

- The course and its objectives
- Kiro UNIPV
- Teaching material
- The preparatory course
- The exam
- Office hours
- The Intelligent Robotics Lab
- Theses and research
- The role of Ph.D. Students and PostDocs

Introduction

- Program of the course:

Advanced SISO control schemes:

Pre-filters and parallel compensators, two degrees of freedom control schemes, compensation of measurable disturbances, systems with delays and Smith Predictor, Padé approximation, decoupling in the frequency domain, control of open loop unstable systems.

Advanced MIMO control schemes:

Decoupling based control schemes, decentralized control, relative gain array.

PID controllers:

Features and properties. Rules for the empirical calibration. Wind-up and anti wind-up schemes.

Digital control:

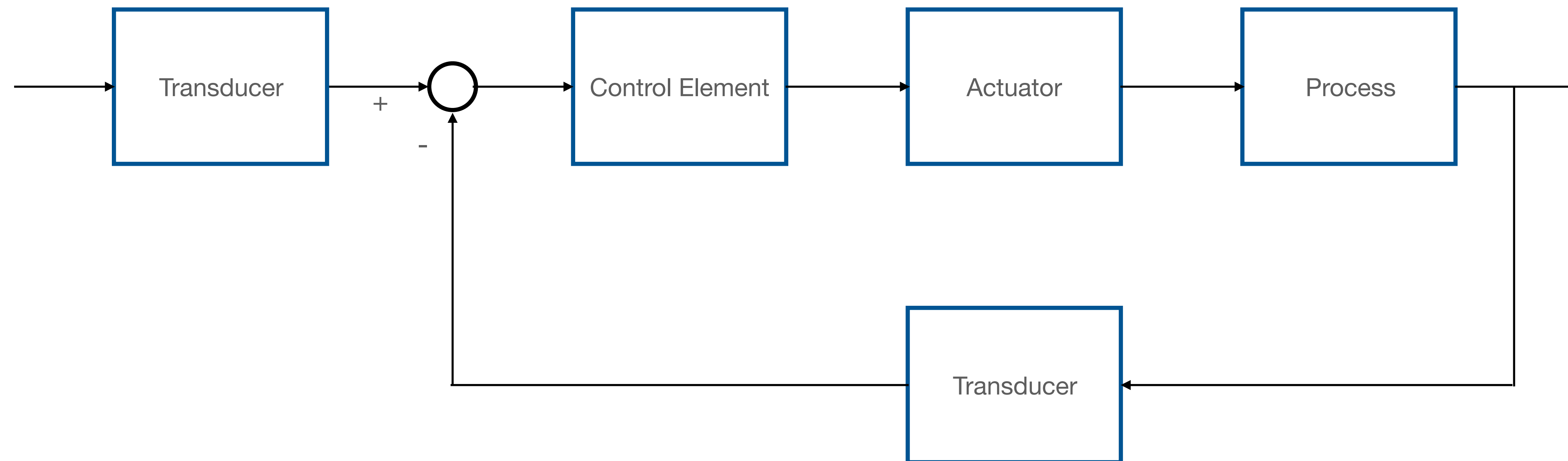
Discrete-time systems. The concept of equilibrium for discrete-time systems. Stability of linear time-invariant discrete-time systems. Jury test. Digital control schemes. Zeta transform and its properties. Transfer functions in the z domain. Sampling and aliasing. Choice of the sampling time. Zero-order-Hold. Discretization of continuous-time controllers. Bilinear transformation, Euler and Tustin methods.

Introduction

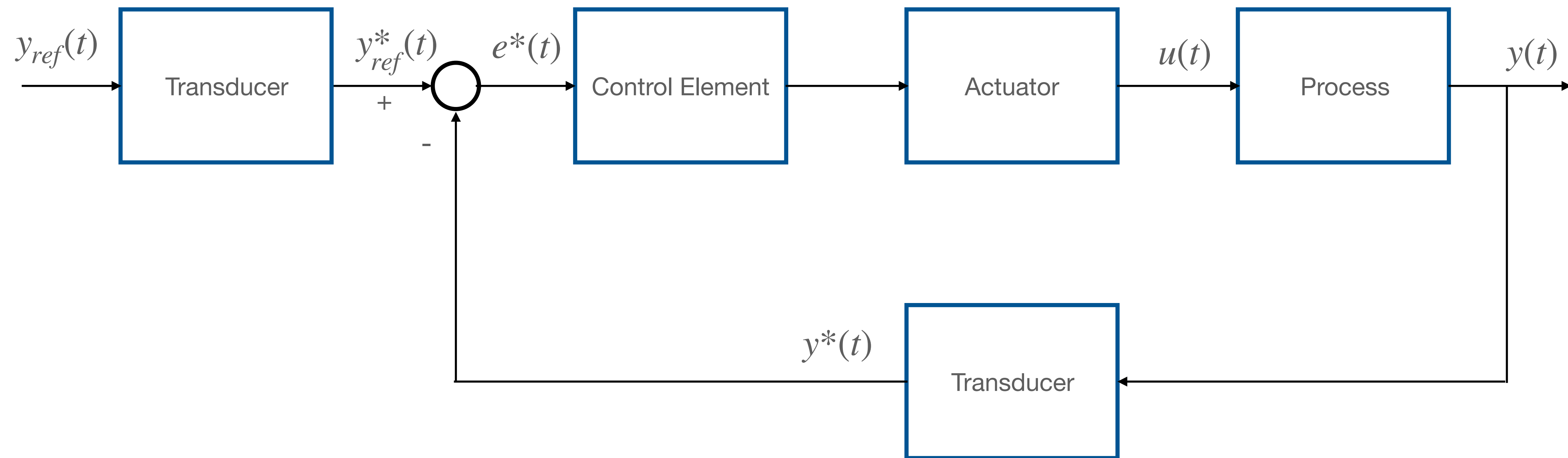
- Some of the figures in these slides, kindly provided by McGraw-Hill, are those of the Textbook:



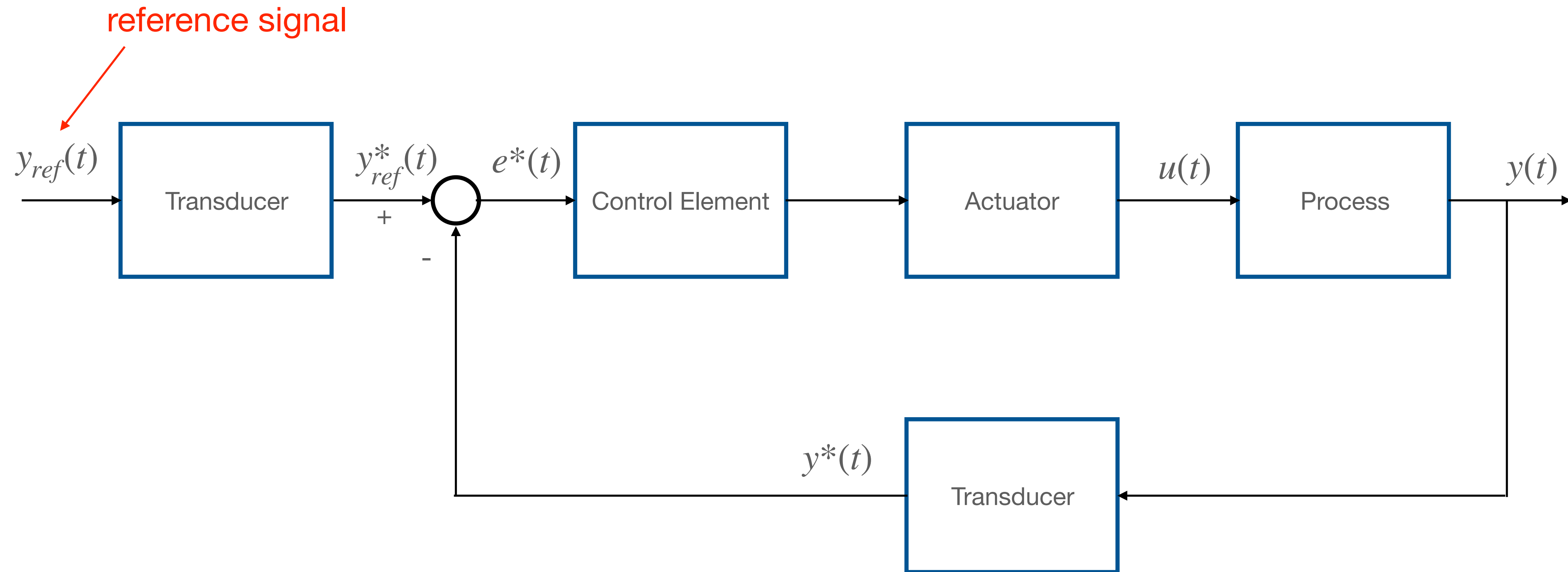
Basic Control Scheme



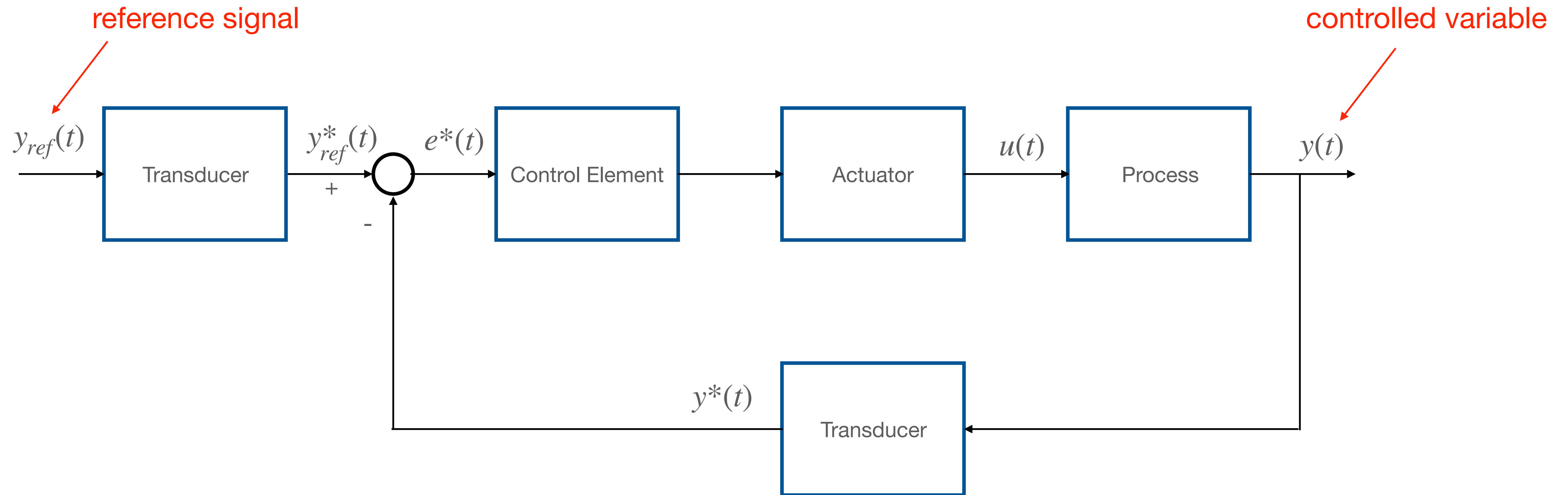
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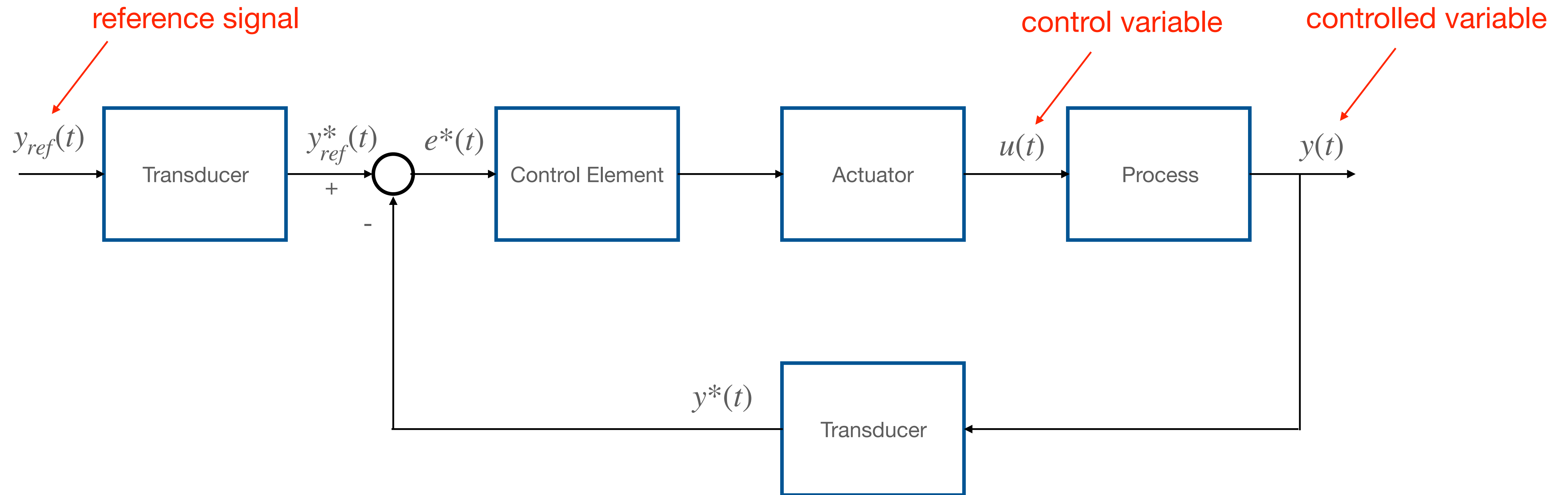
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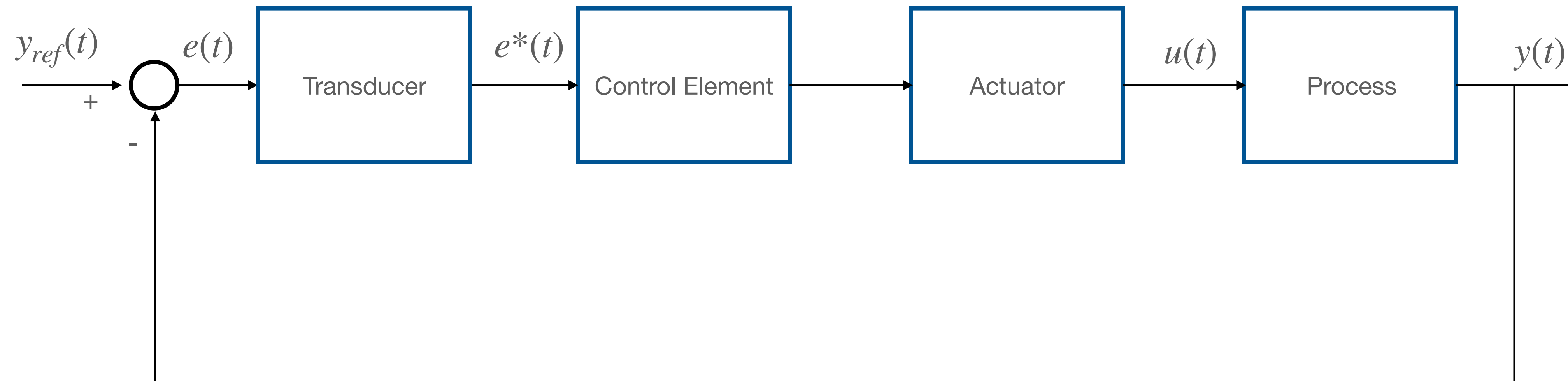
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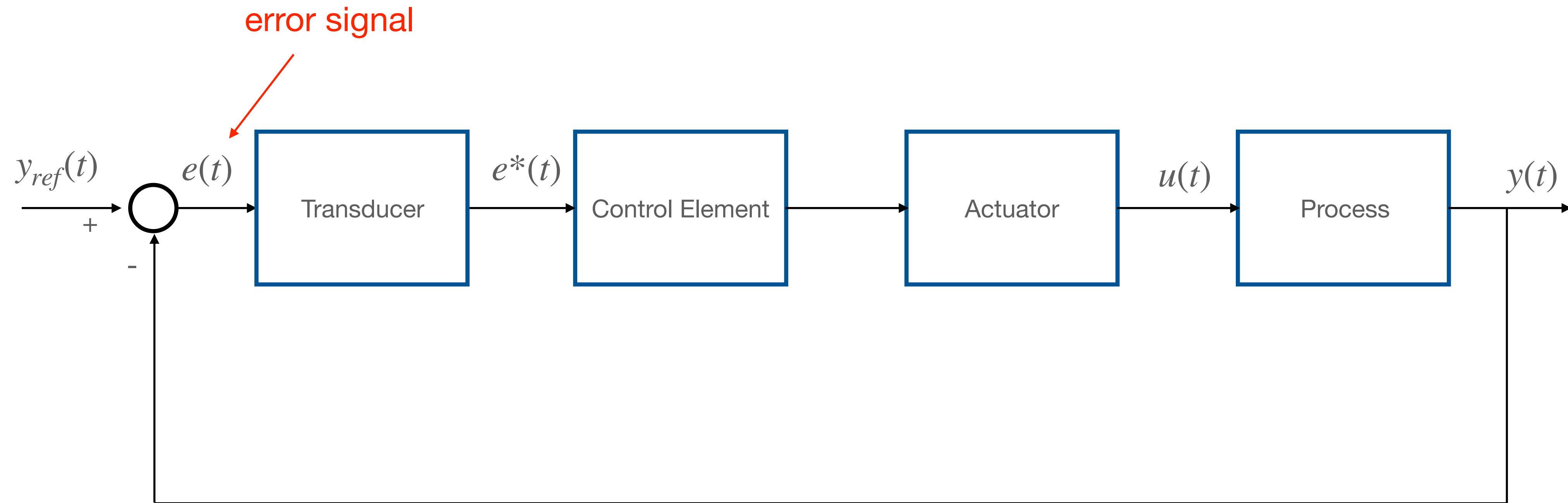
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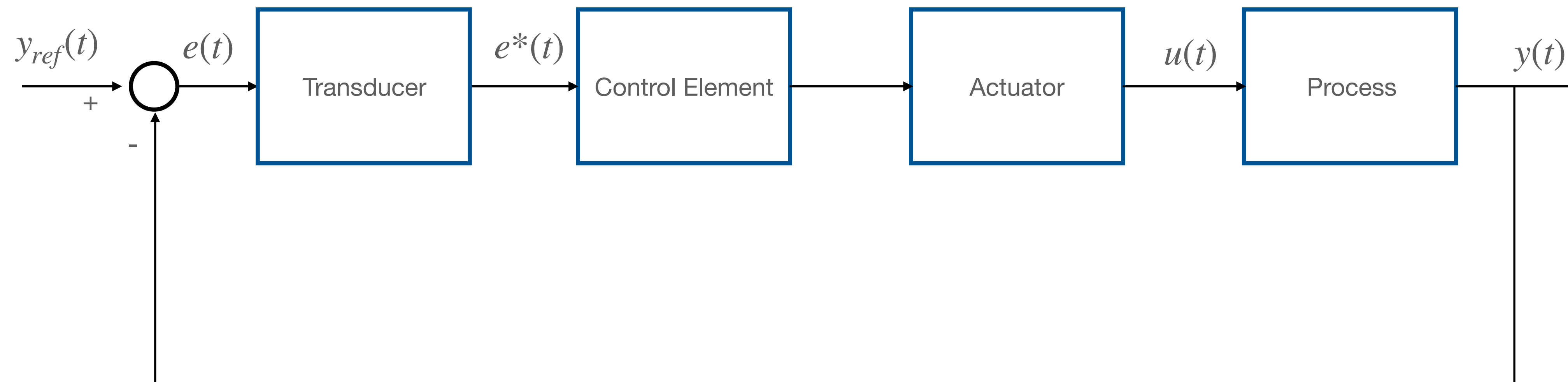
Basic Control Scheme



Basic Control Scheme

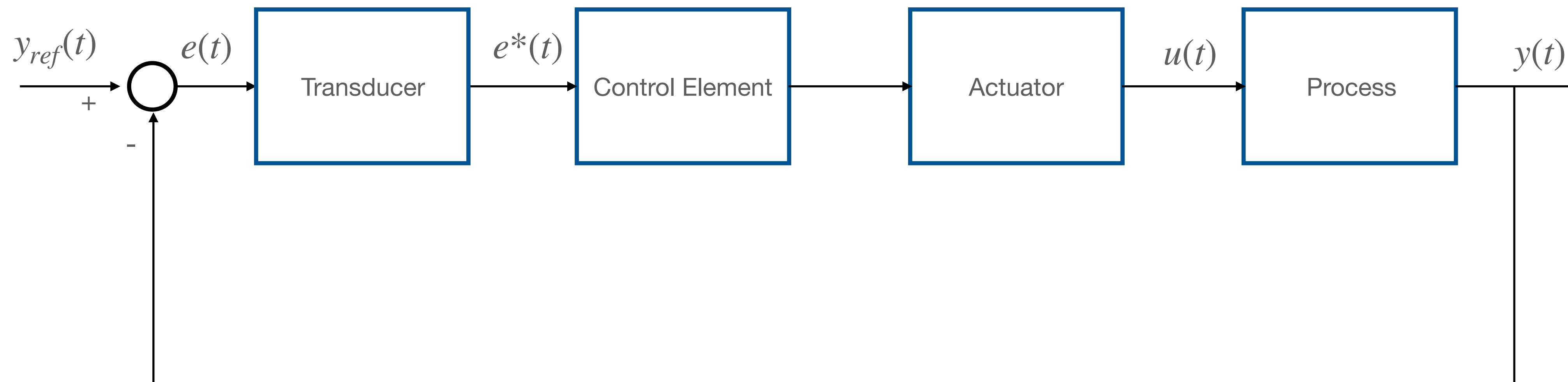


Basic Control Scheme



Assumption: Linear Time-Invariant (LTI) Dynamic Systems

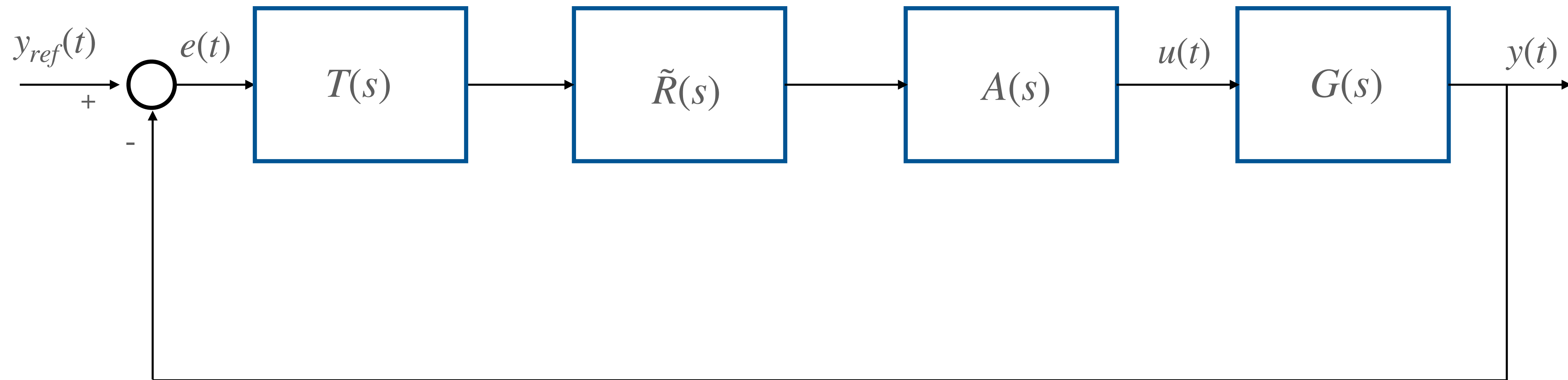
Basic Control Scheme



Assumption: Linear Time-Invariant (LTI) Dynamic Systems

They can be modelled via Transfer Functions

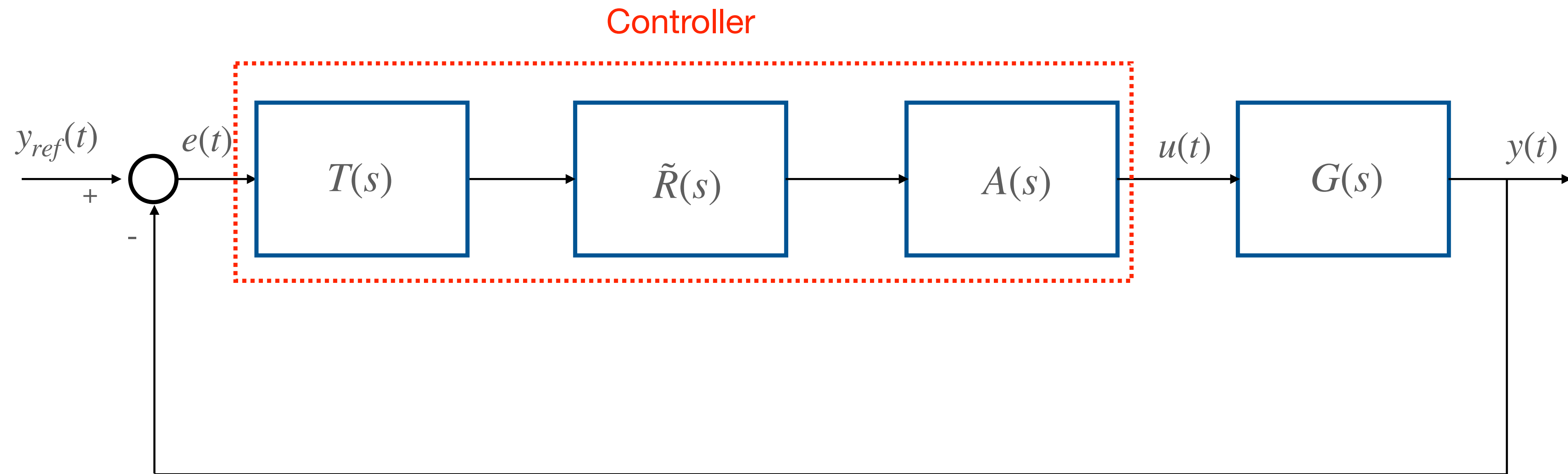
Basic Control Scheme



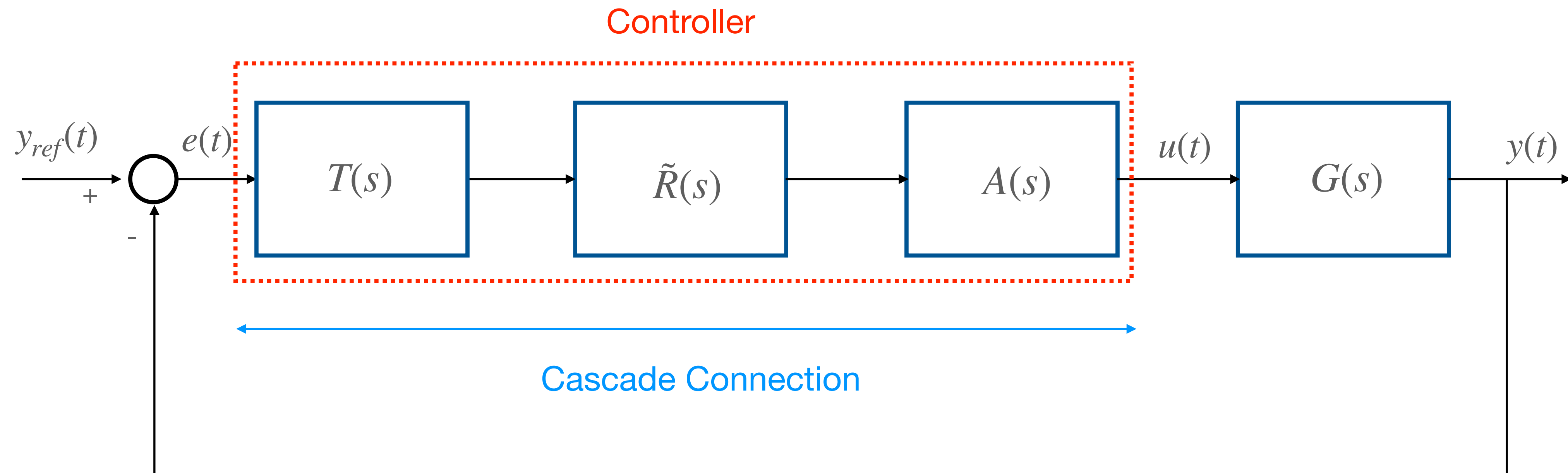
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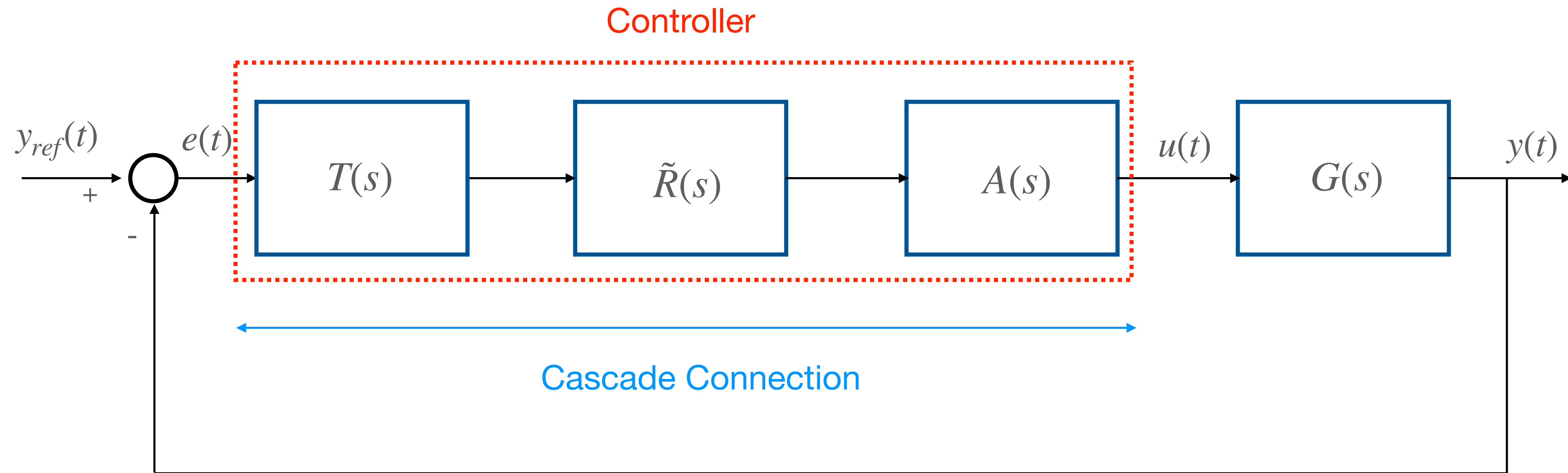
Basic Control Scheme



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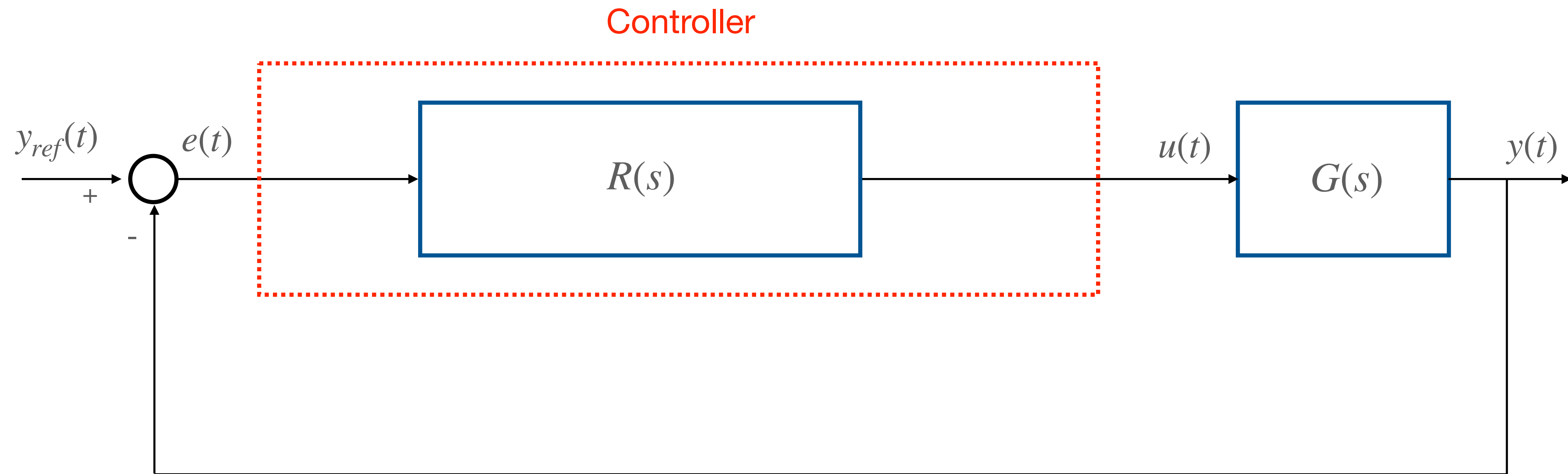


Basic Control Scheme



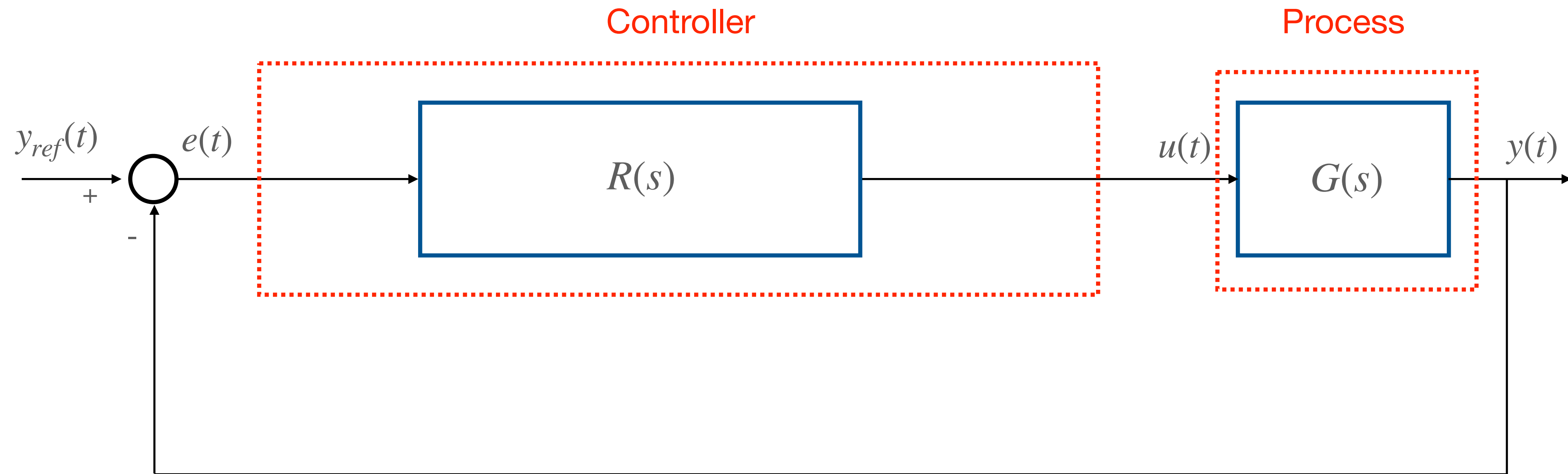
Block Algebra Rule: $T(s) \cdot \tilde{R}(s) \cdot A(s) = R(s)$

Basic Control Scheme

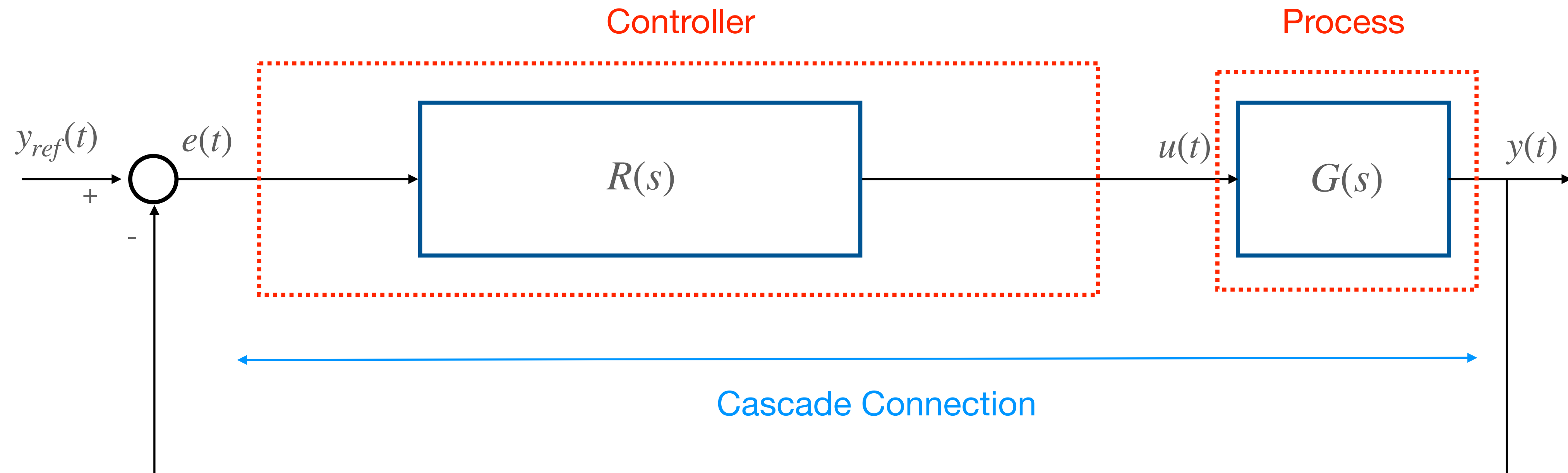


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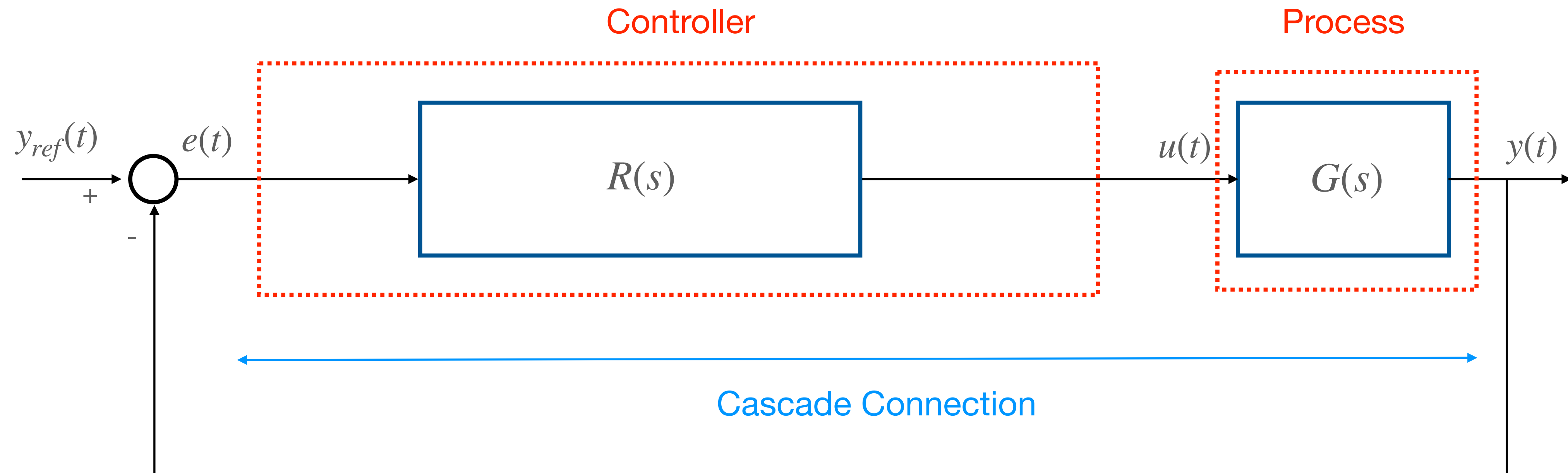
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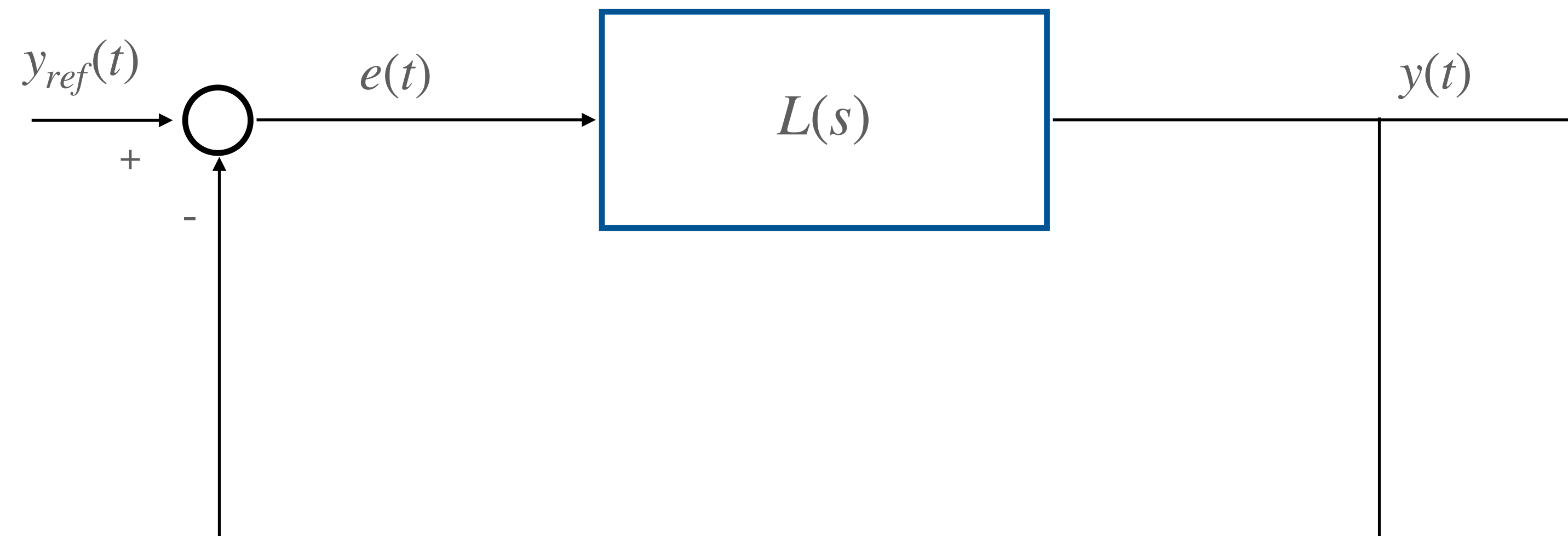


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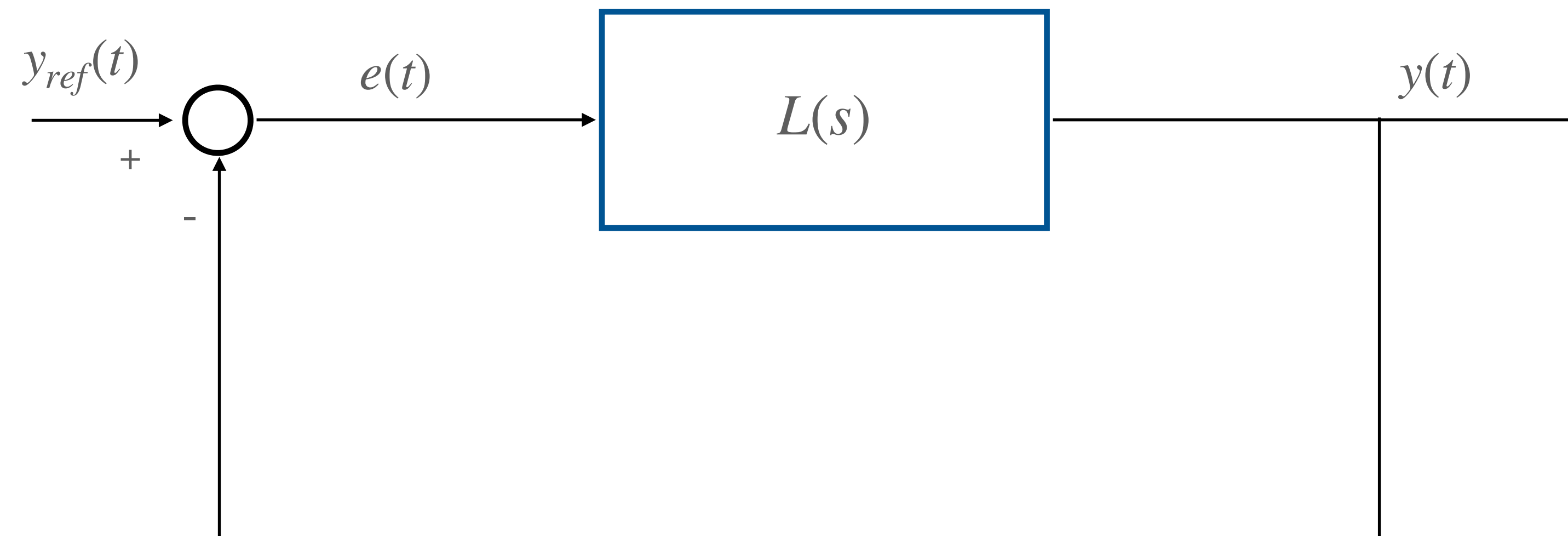
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Basic Control Scheme



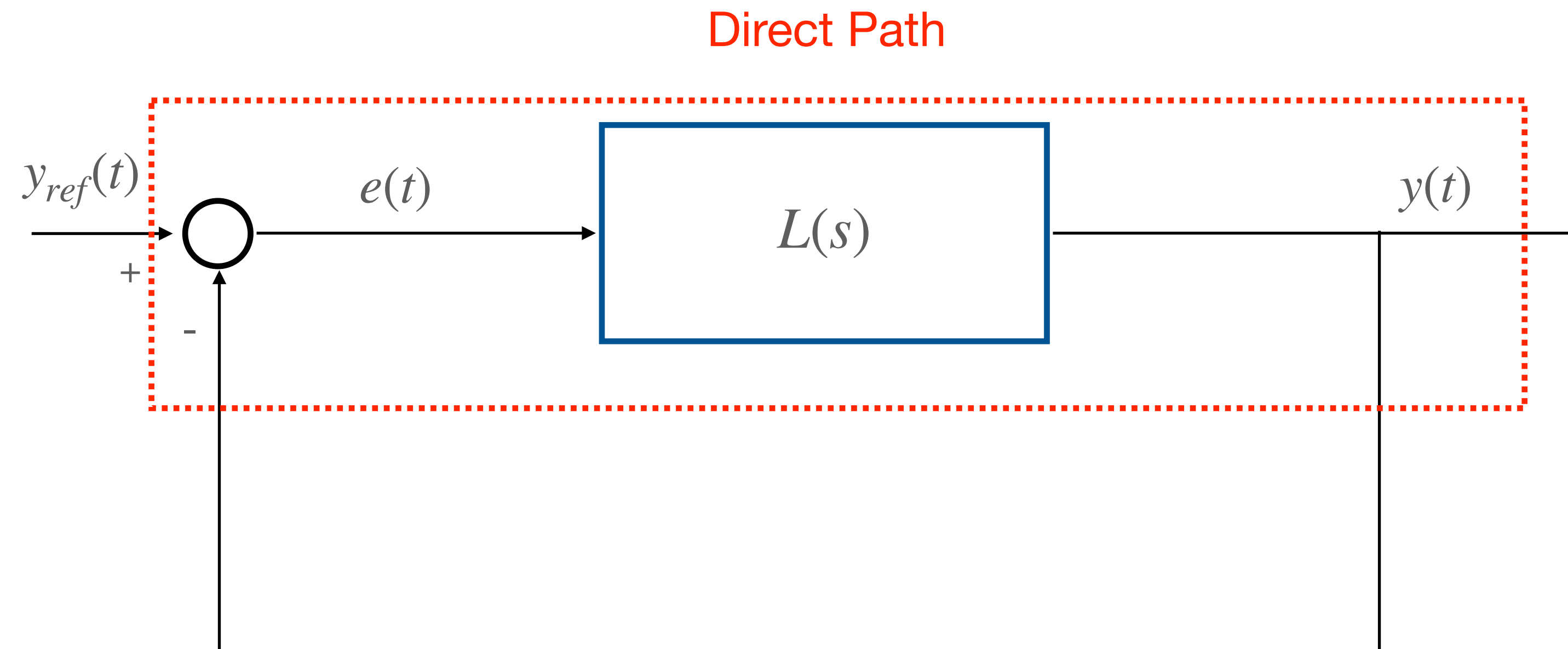
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Unitary Feedback Control Scheme



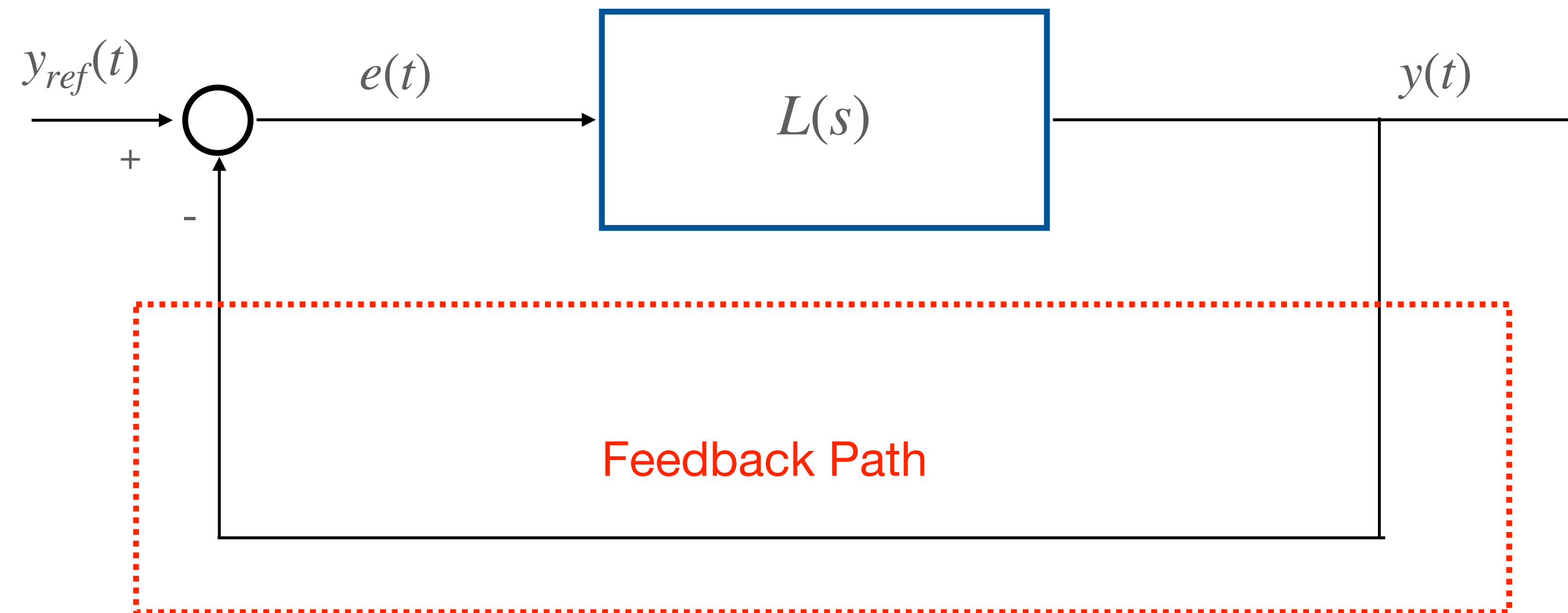
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Unitary Feedback Control Scheme



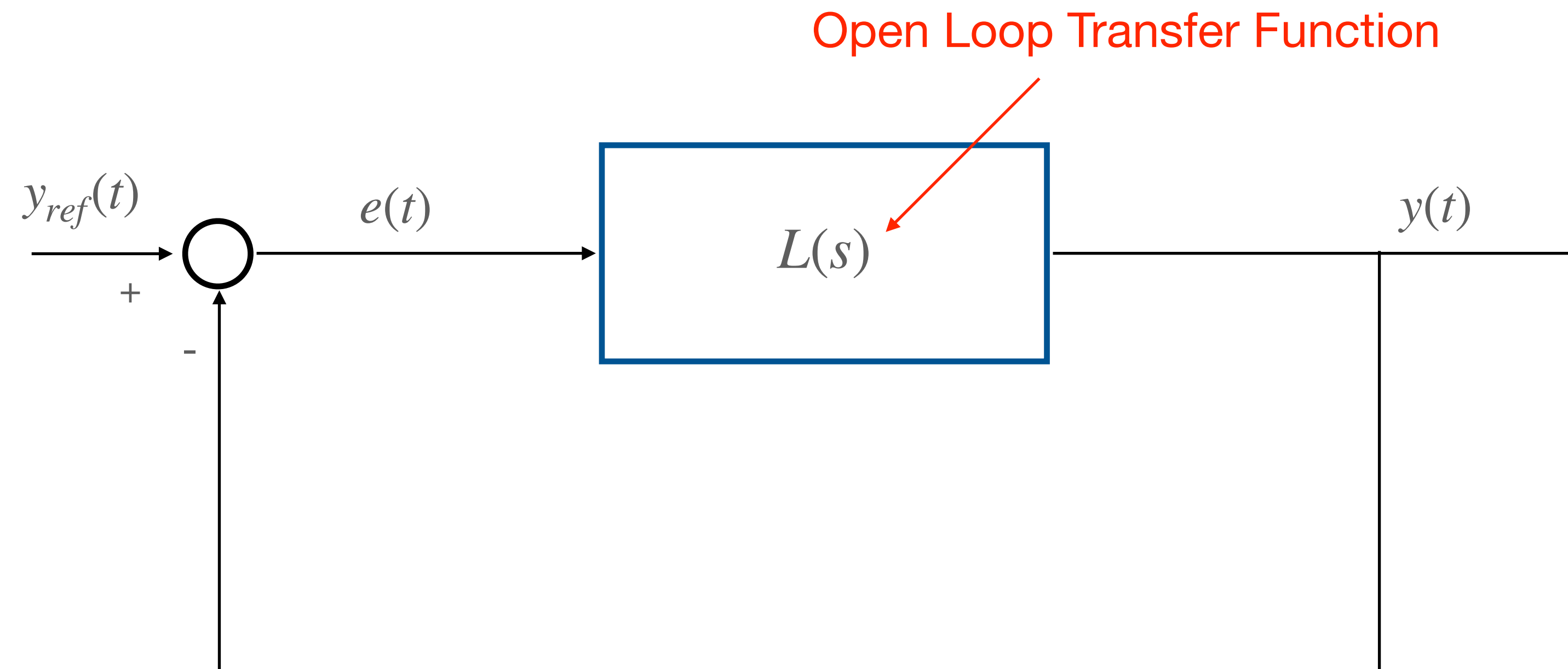
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Unitary Feedback Control Scheme



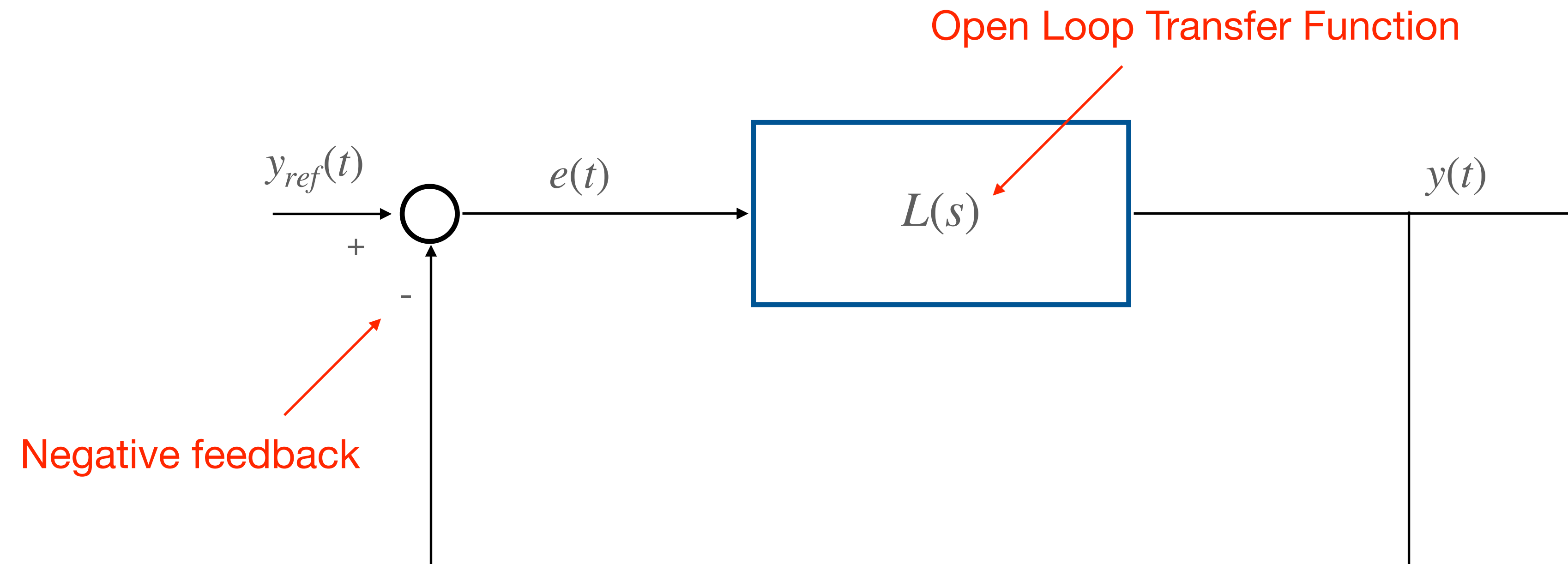
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Unitary Feedback Control Scheme



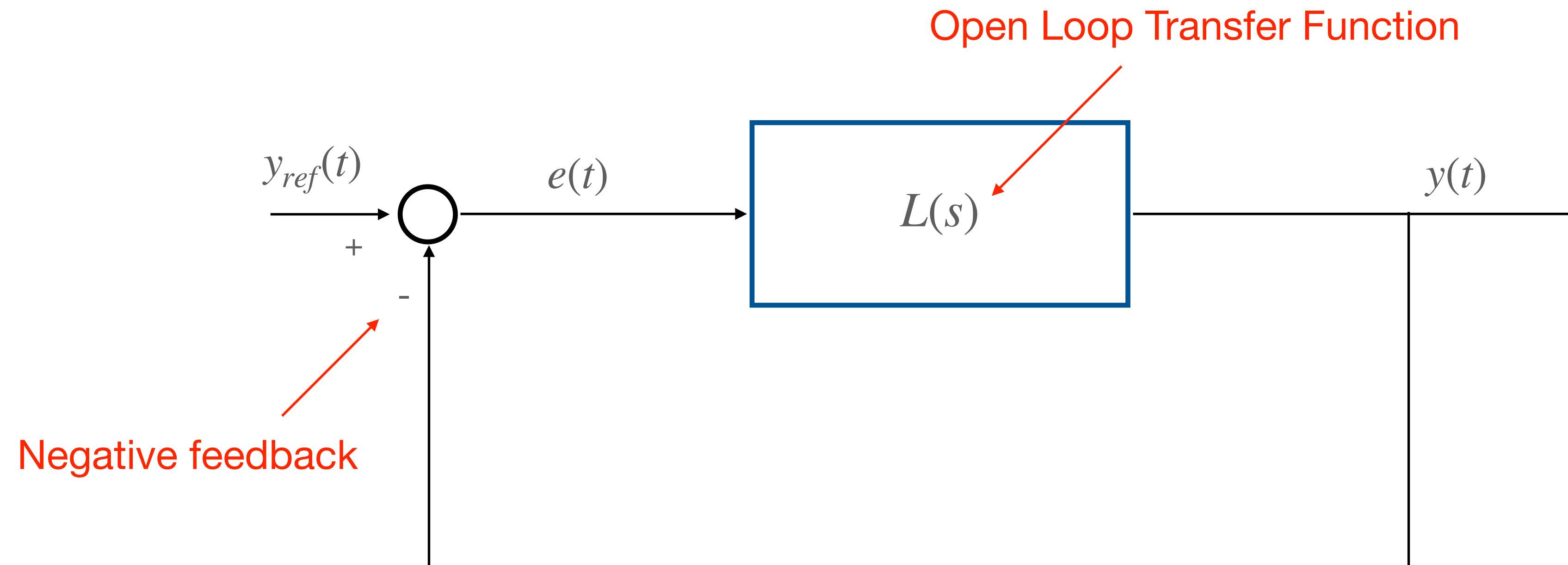
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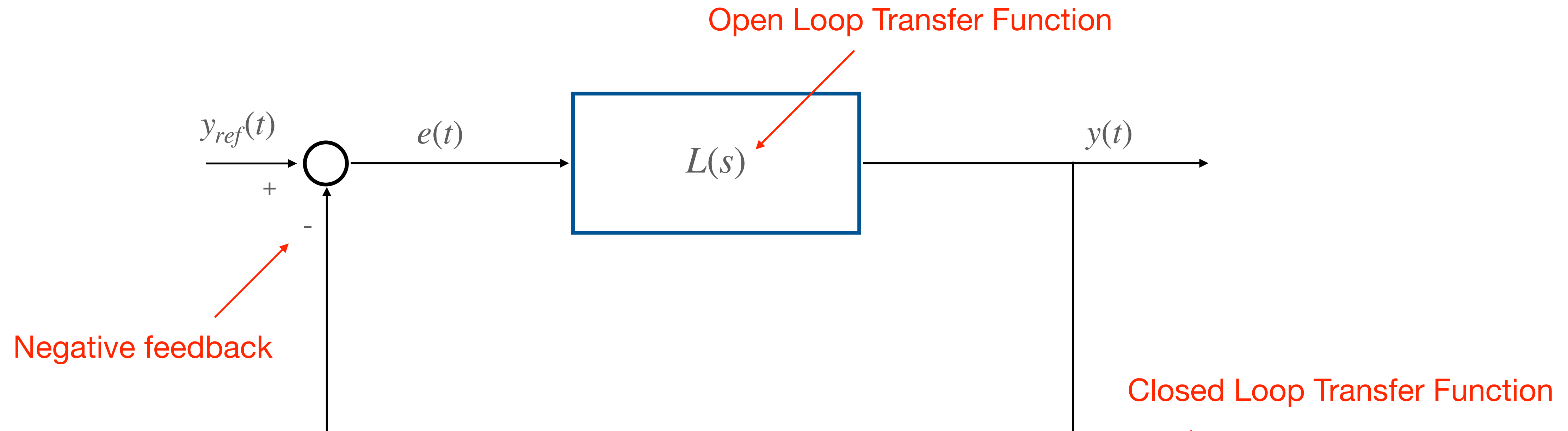
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Unitary Feedback Control Scheme



Block Algebra Rule:
$$F(s) = \frac{R(s) \cdot G(s)}{1 + R(s) \cdot G(s)} = \frac{L(s)}{1 + L(s)}$$

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Limitations of the Basic Unitary Feedback Control Scheme

- Limited bandwidth in case of measurement disturbances
- Limited bandwidth in case of model uncertainties
- Limited bandwidth in case of delays
- No “control moderation” in case of reference signal characterized by sharp transitions in the waveform (abrupt transitions and near-vertical edges in the waveform)
- Possible difficulties in enforcing steady-state precision with step reference signal via the classical approach

Limitations of the Basic Unitary Feedback Control Scheme

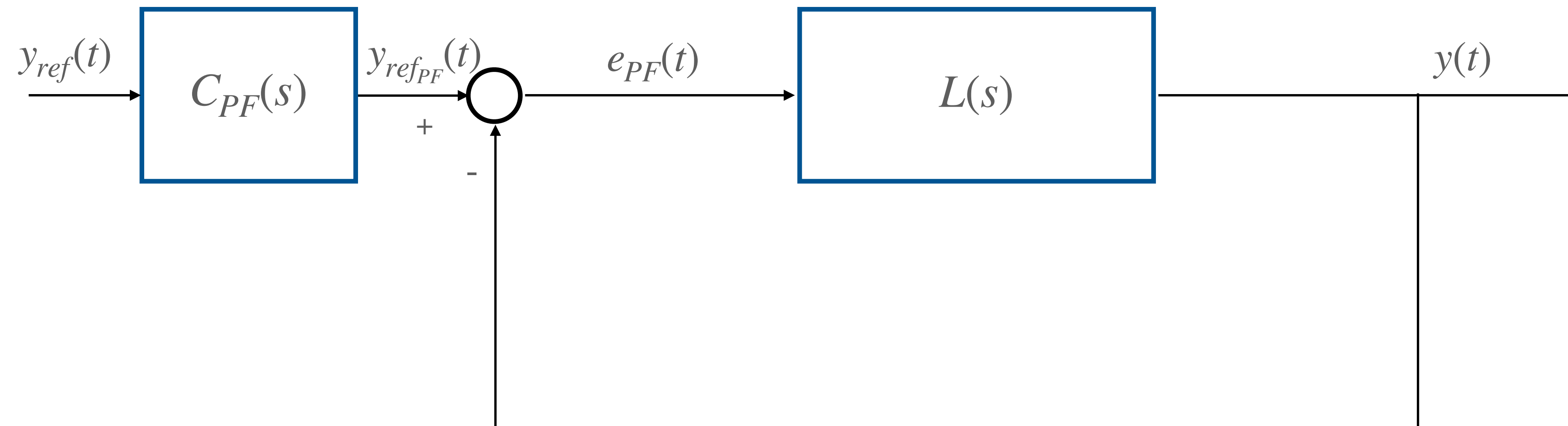
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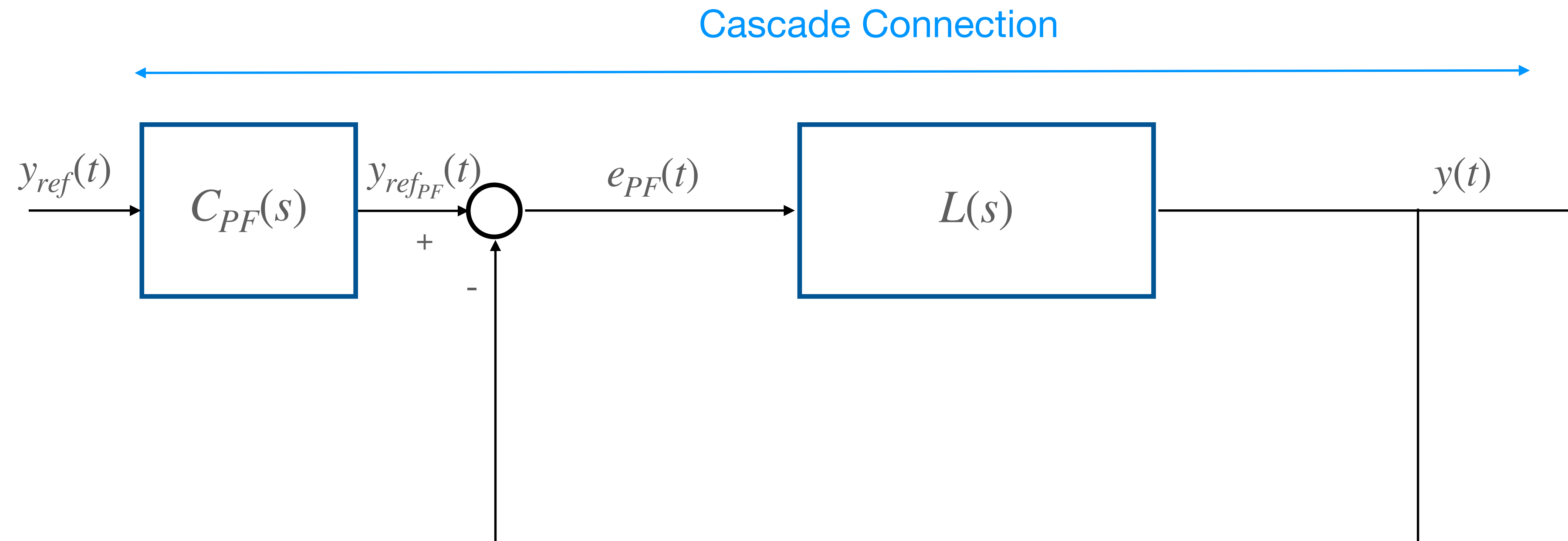
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Possible solution: add a Pre-Filter to the basic unitary feedback control scheme

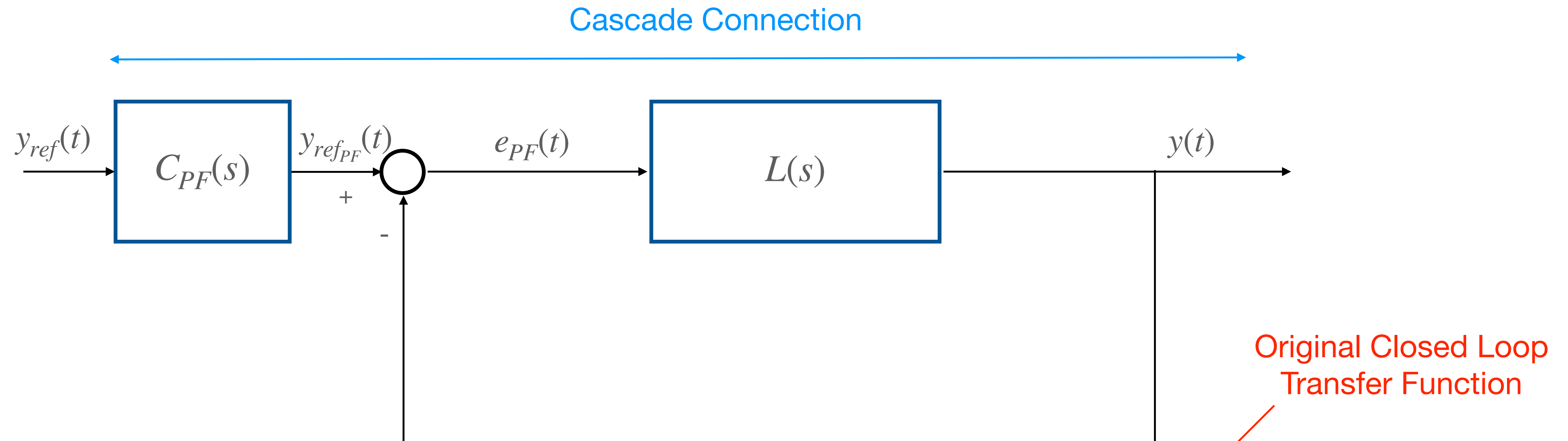
Pre-filter Based Control Scheme



Pre-filter Based Control Scheme

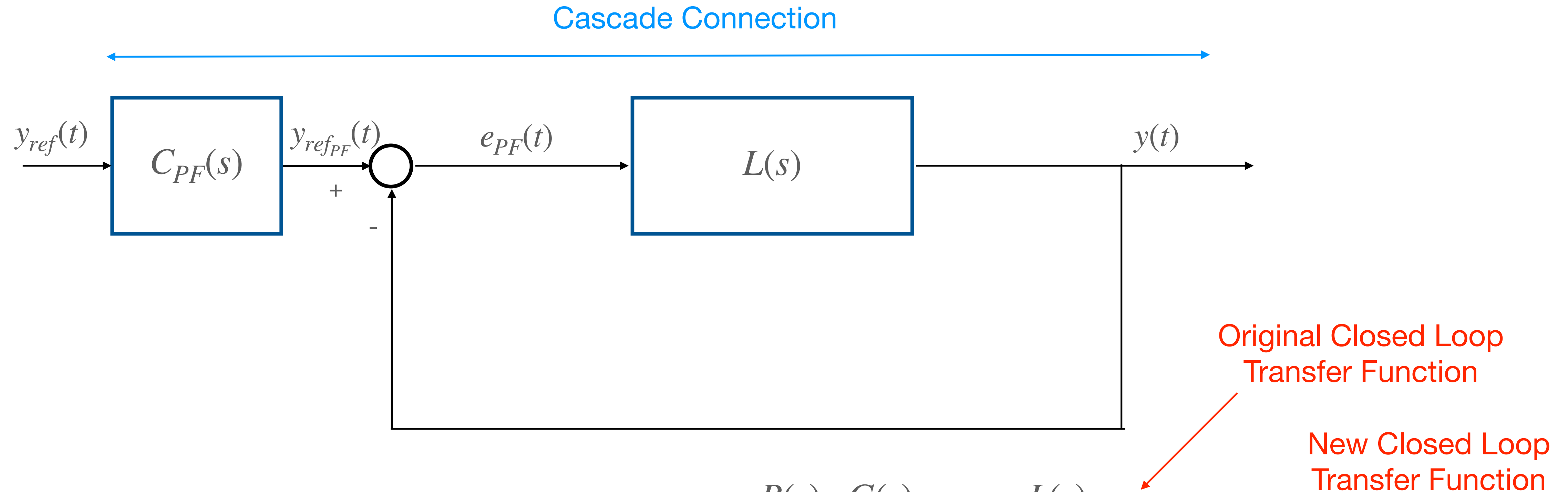


Pre-filter Based Control Scheme



Block Algebra Rule:
$$F(s) = \frac{R(s) \cdot G(s)}{1 + R(s) \cdot G(s)} = \frac{L(s)}{1 + L(s)}$$

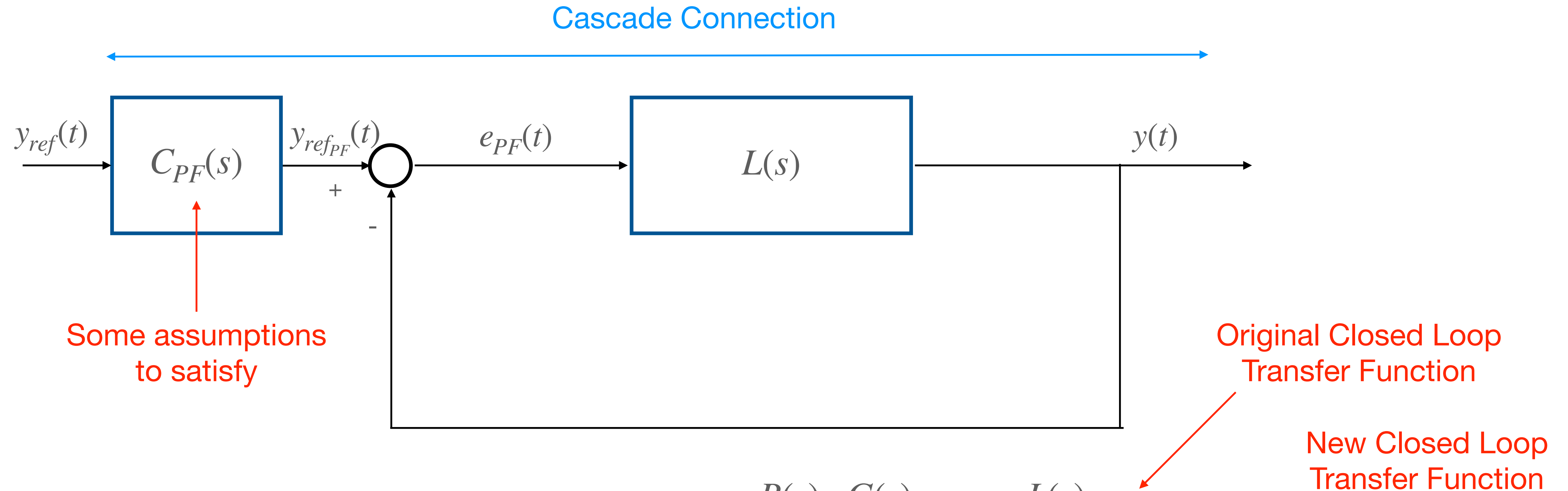
Pre-filter Based Control Scheme



Block Algebra Rule: $F(s) = \frac{R(s) \cdot G(s)}{1 + R(s) \cdot G(s)} = \frac{L(s)}{1 + L(s)}$

$$F_{PF}(s) = C_{PF}(s) \cdot F(s) = C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)}$$

Pre-filter Based Control Scheme

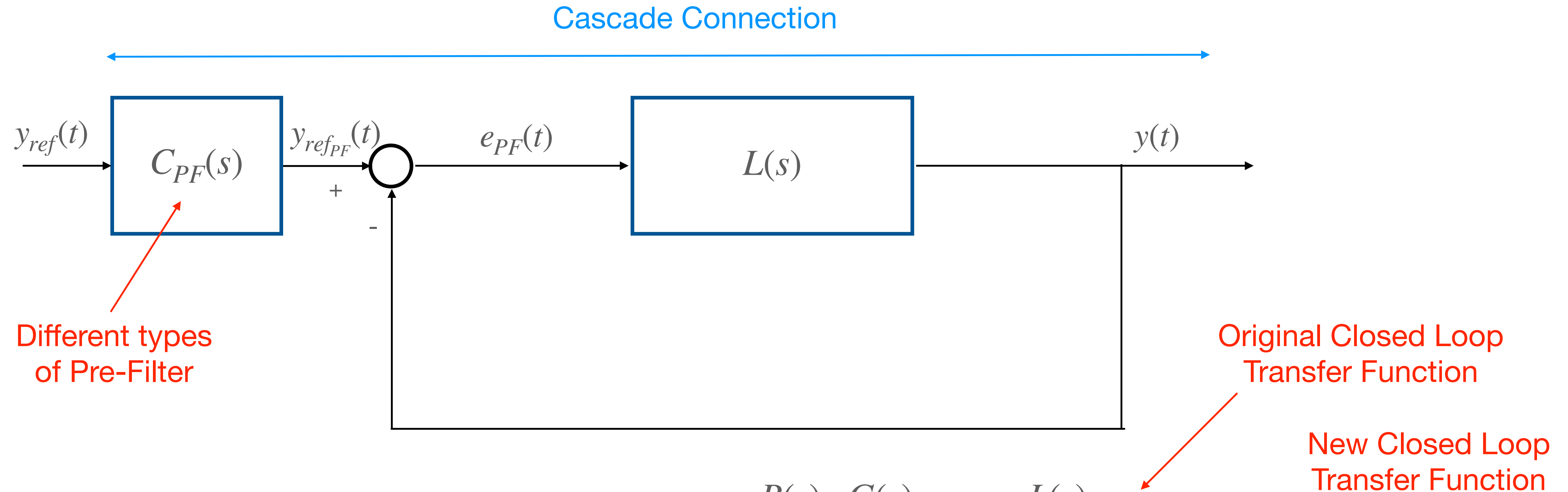


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Pre-filter Based Control Scheme

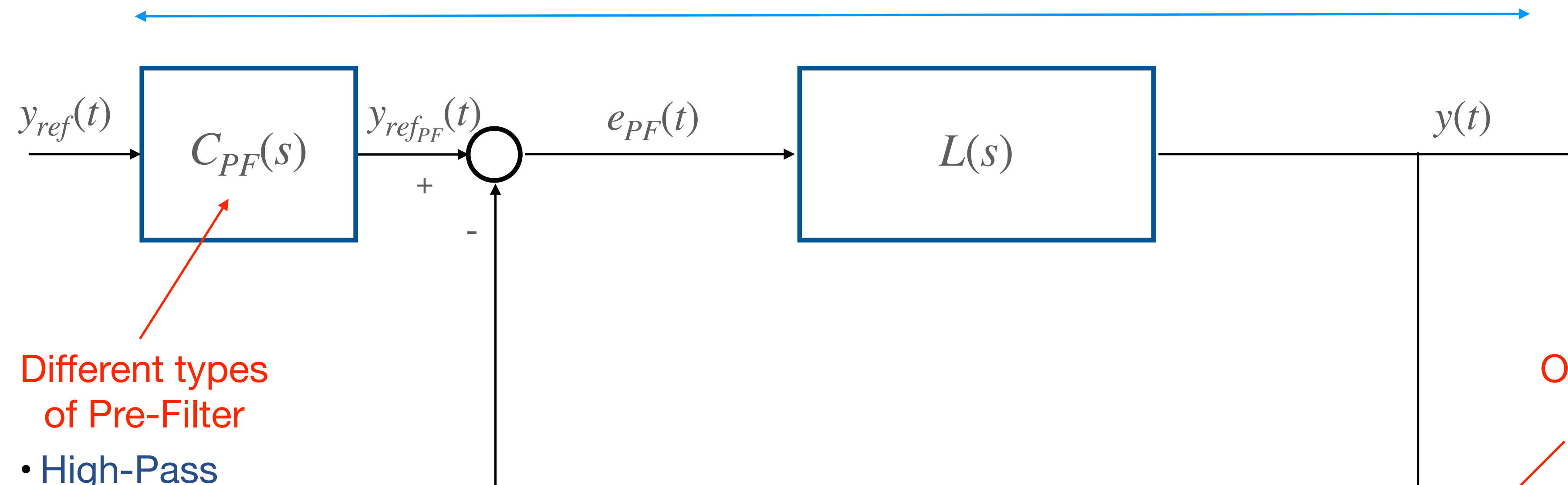


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Pre-filter Based Control Scheme

Cascade Connection



Different types
of Pre-Filter

- High-Pass
- Low-Pass
- For steady-state compensation

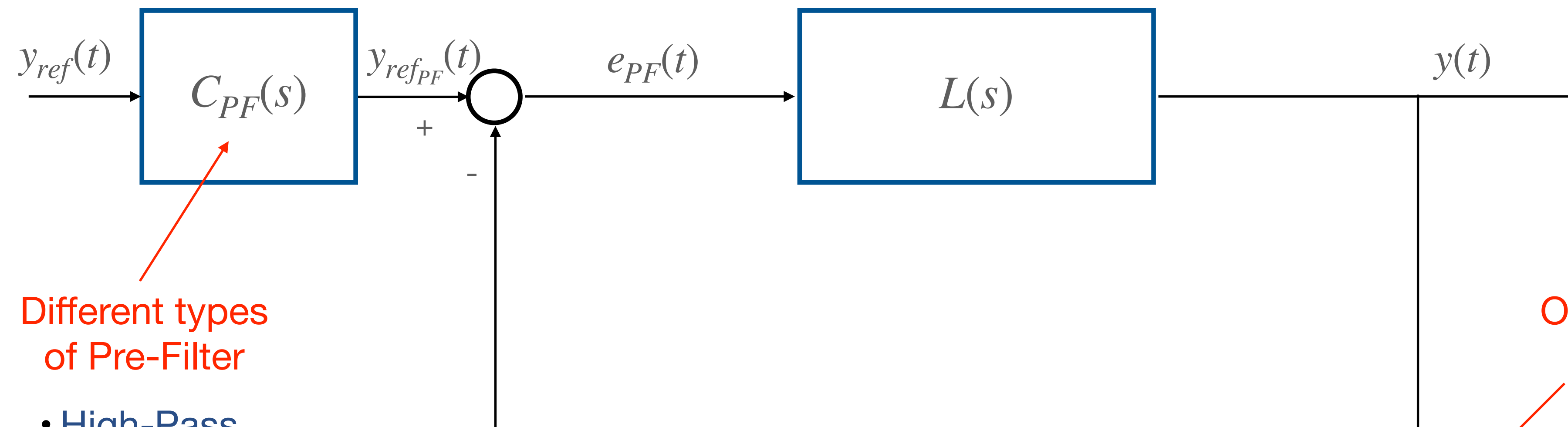
Original Closed Loop
Transfer Function

New Closed Loop
Transfer Function

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Pre-filter Based Control Scheme: Case 1



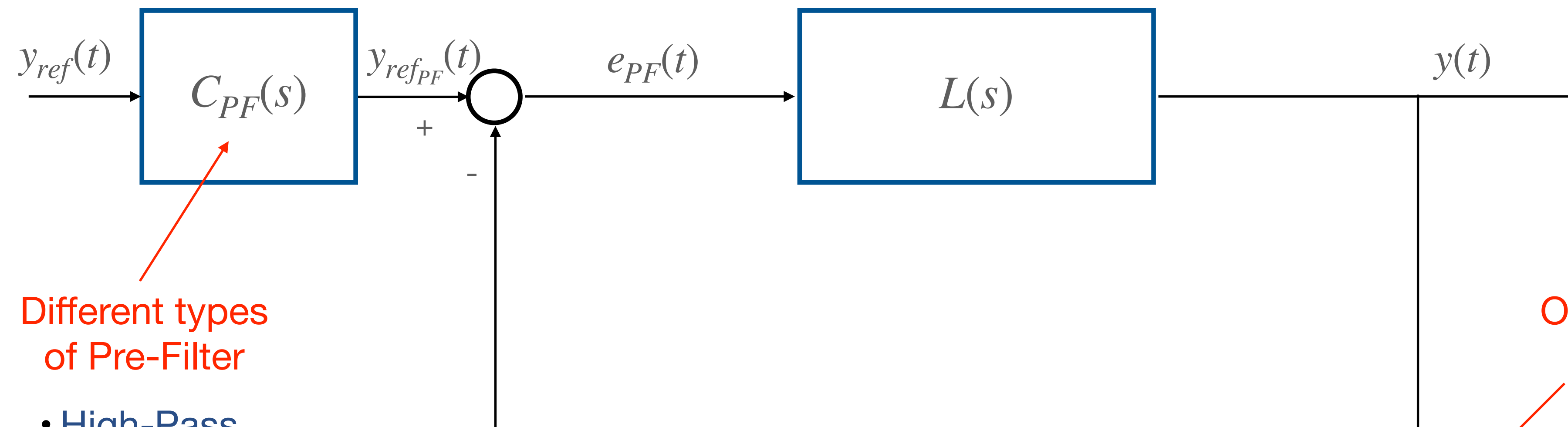
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Original Closed Loop Transfer Function

New Closed Loop Transfer Function

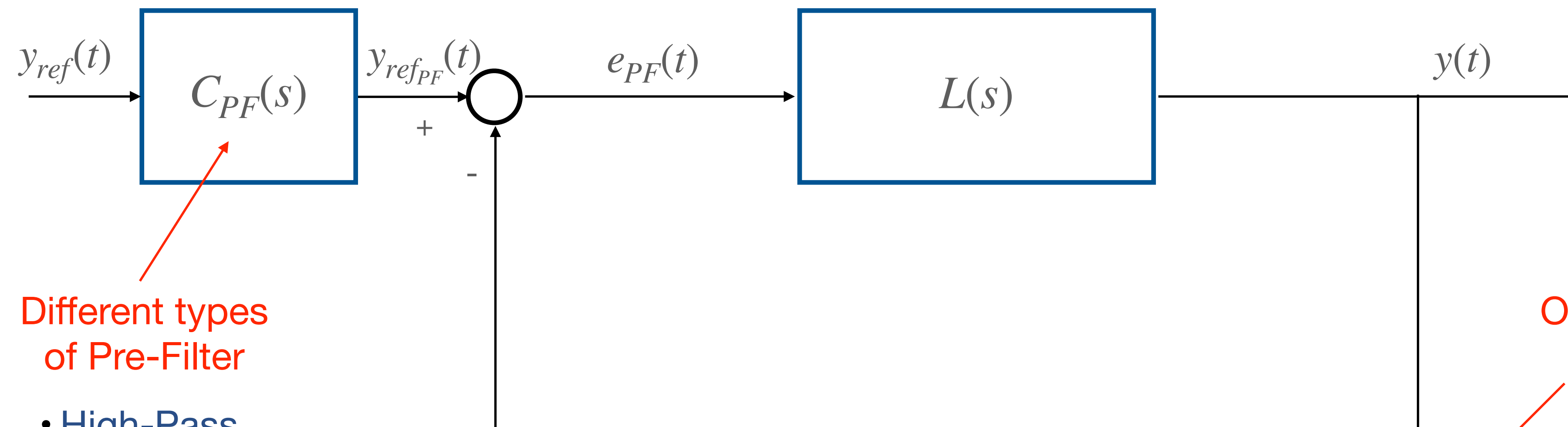
Pre-filter Based Control Scheme: Case 1



Block Algebra Rule: $F(s) = \frac{R(s) \cdot G(s)}{1 + R(s) \cdot G(s)} = \frac{L(s)}{1 + L(s)}$

$$F_{PF}(s) = C_{PF}(s) \cdot F(s) = C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = \frac{\mu(1 + \tau' s)}{(1 + \tau s)} \cdot \frac{L(s)}{1 + L(s)}$$

Pre-filter Based Control Scheme: Case 1



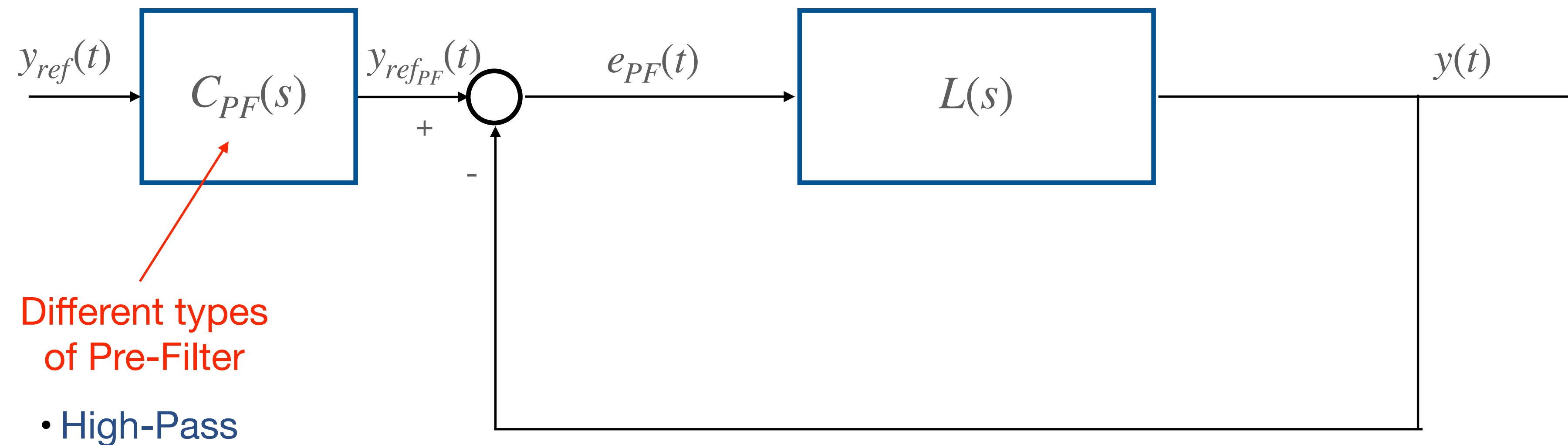
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The zero precedes the pole on the ω axis

Pre-filter Based Control Scheme: Case 1



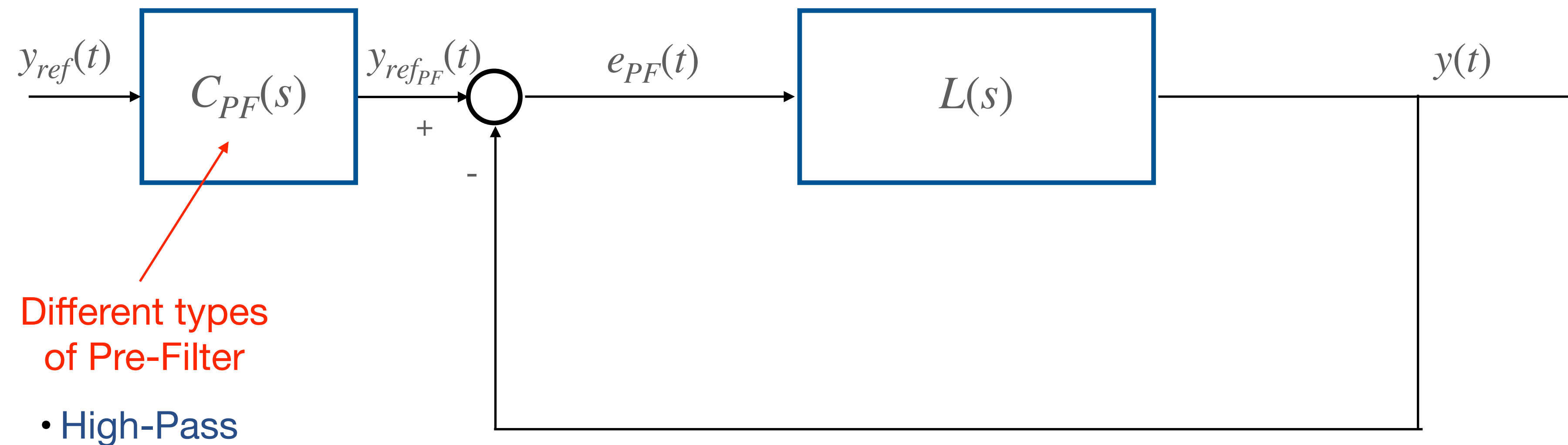
Assumptions:

- $C_{PF}(s)$ As. Stable
- Proper
- Unitary gain

The zero precedes the pole on the ω axis

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Pre-filter Based Control Scheme: Case 1



Assumptions:

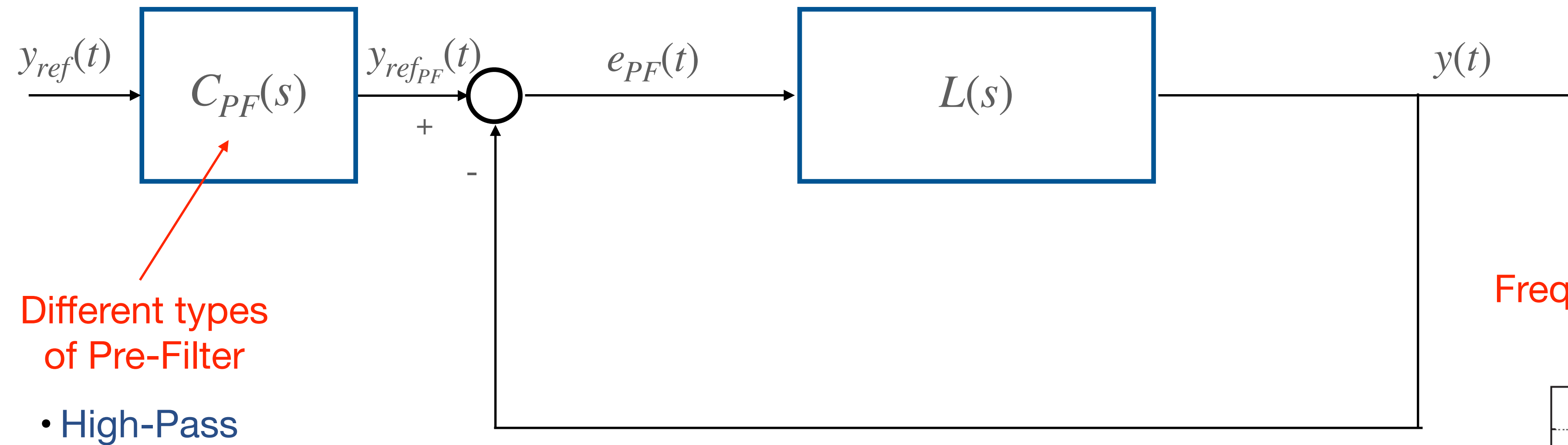
- $C_{PF}(s)$ As. Stable
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$$\lim_{s \rightarrow 0} \frac{\mu(1 + \tau' s)}{1 + \tau s} = \mu = 1$$

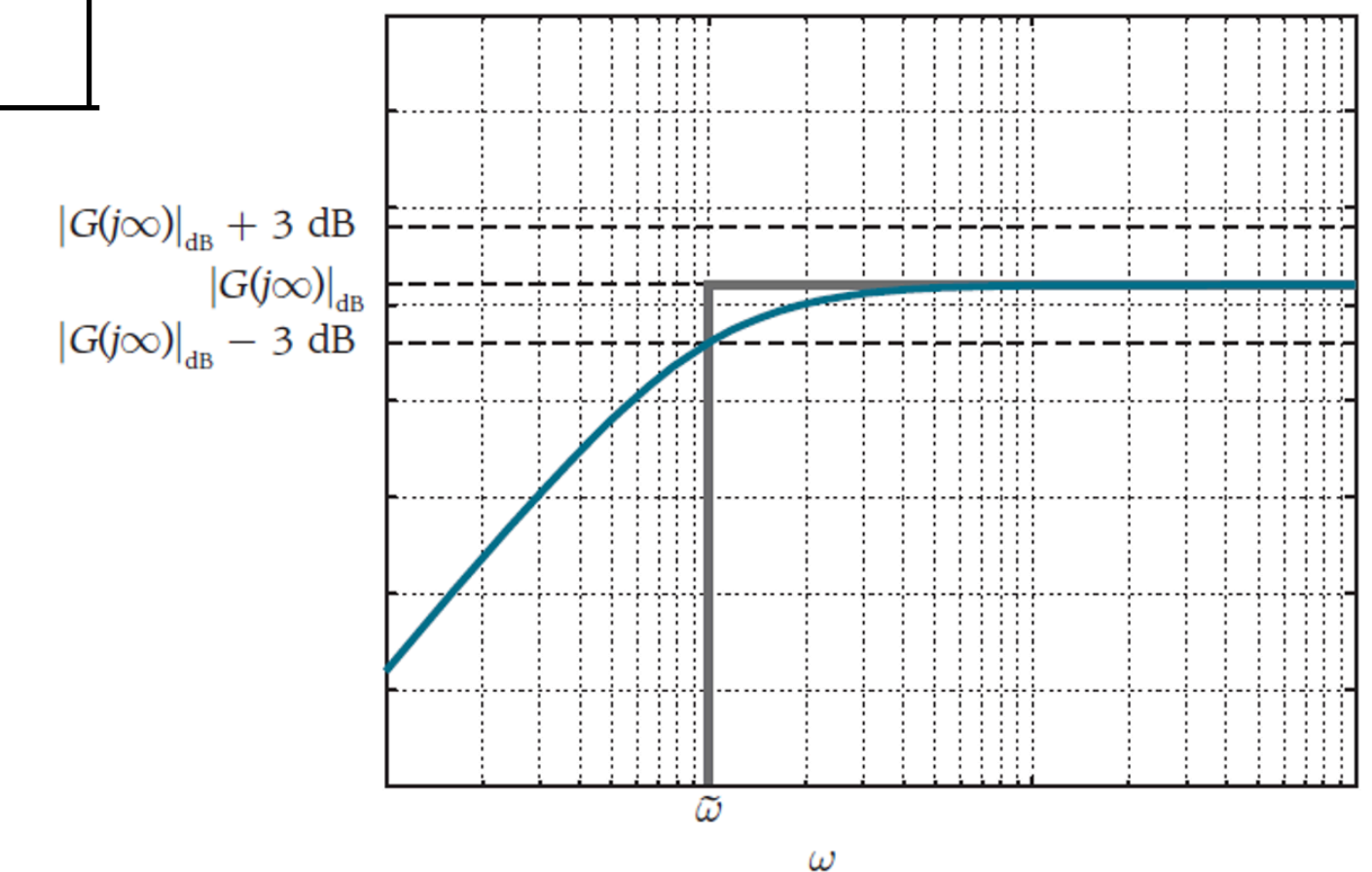
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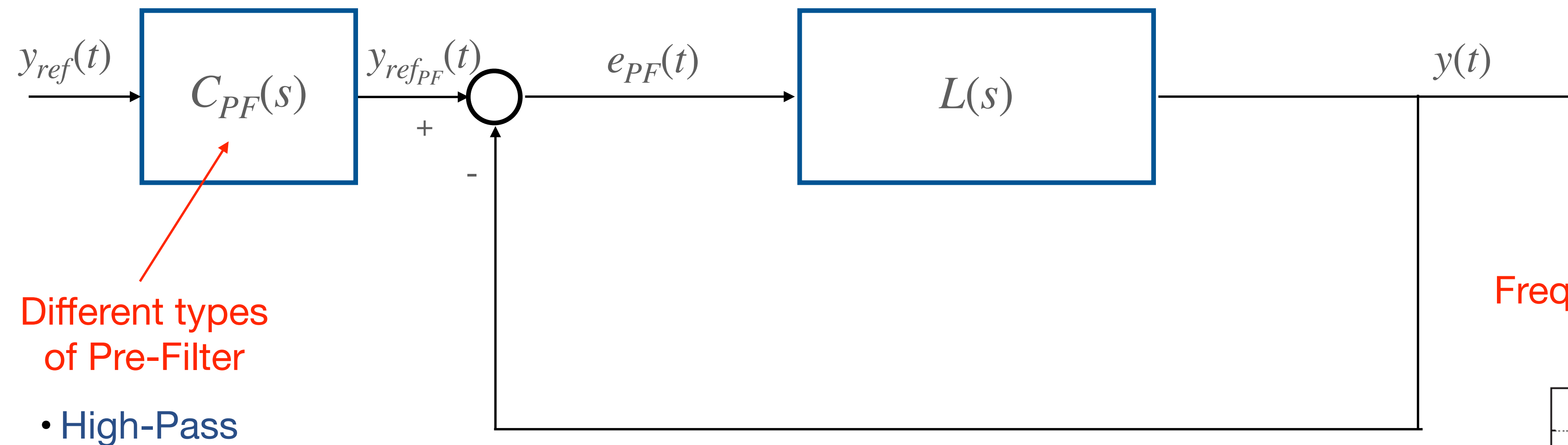
Pre-filter Based Control Scheme: Case 1



Frequency response of HP filter (magnitude)



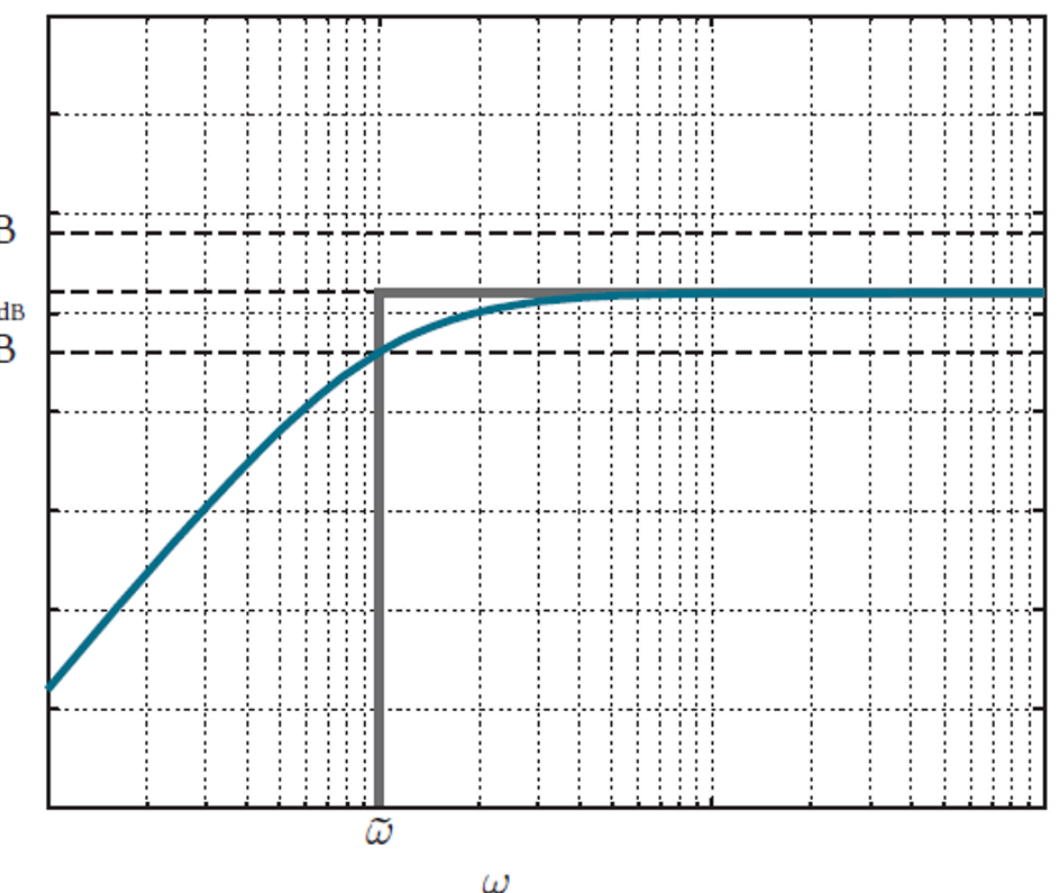
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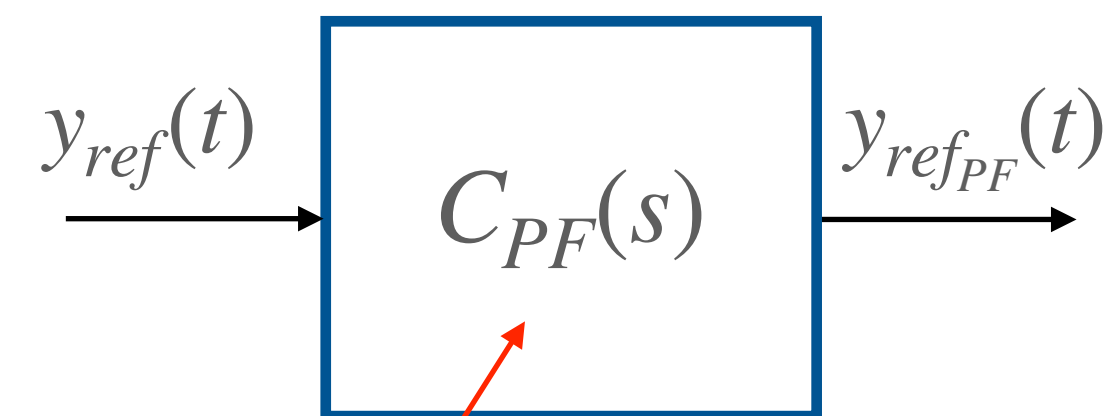
Frequency response of HP filter (magnitude)

An HP Pre-filter can be used to to enlarge the closed-loop bandwidth $\omega_{BW_{CL}}$ a posteriori

$|G(j\infty)|_{dB} + 3 \text{ dB}$
 $|G(j\infty)|_{dB}$
 $|G(j\infty)|_{dB} - 3 \text{ dB}$



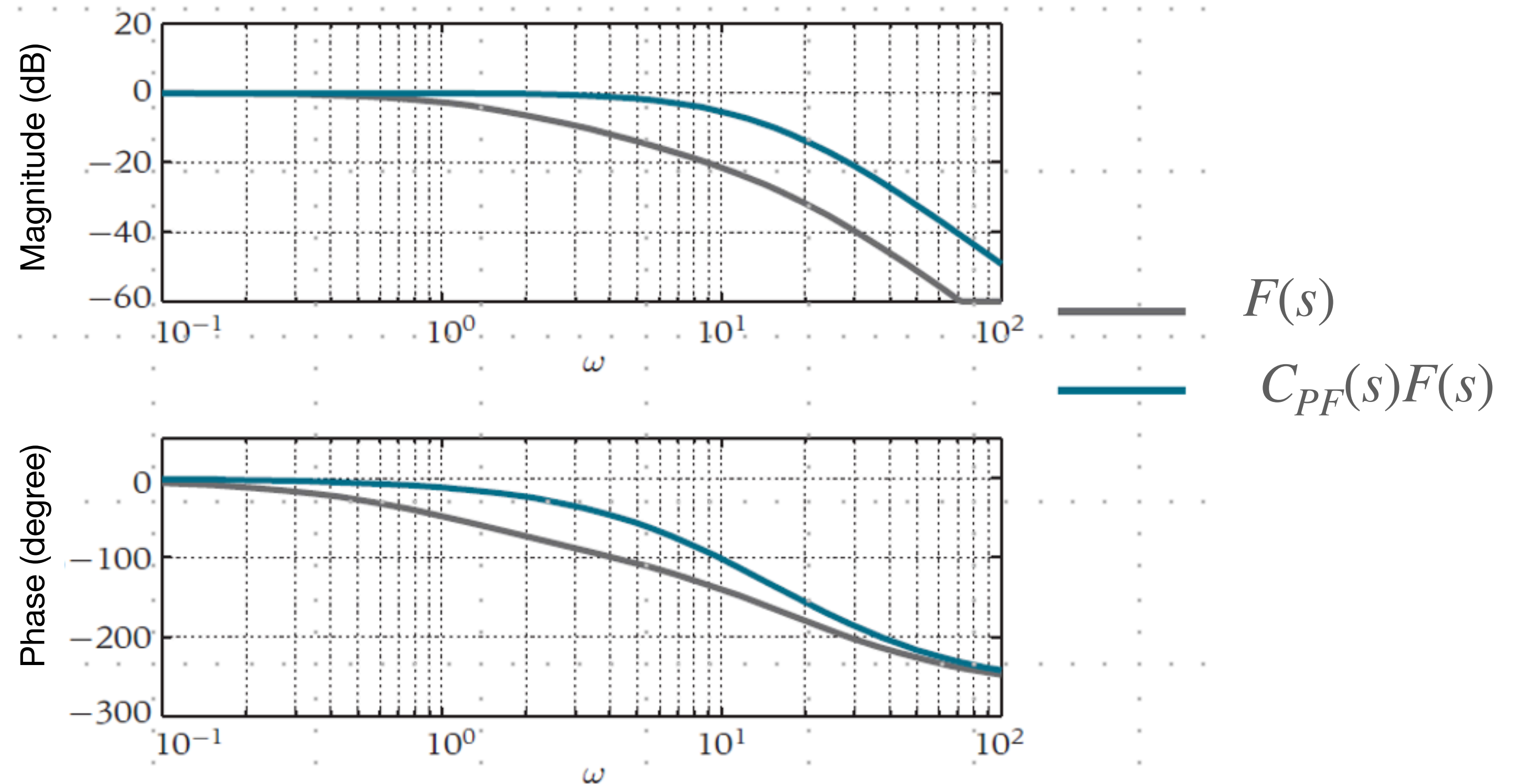
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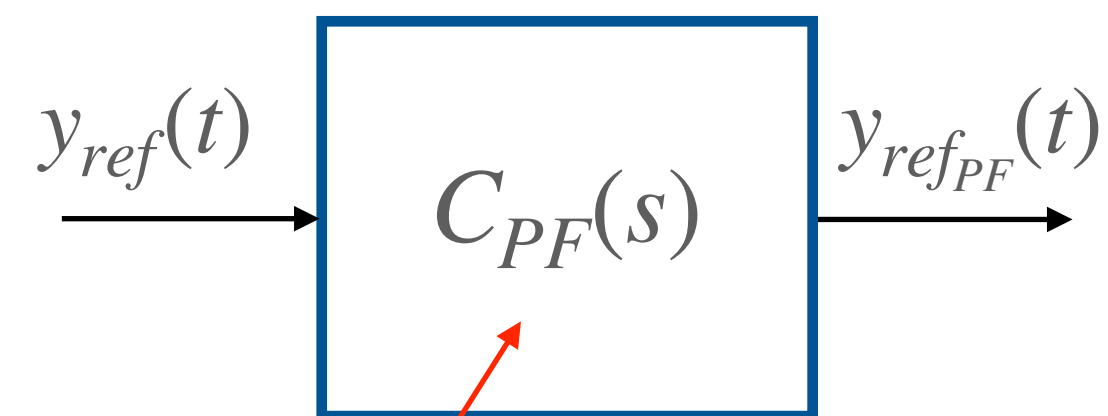
Different types
of Pre-Filter

- High-Pass

Frequency domain



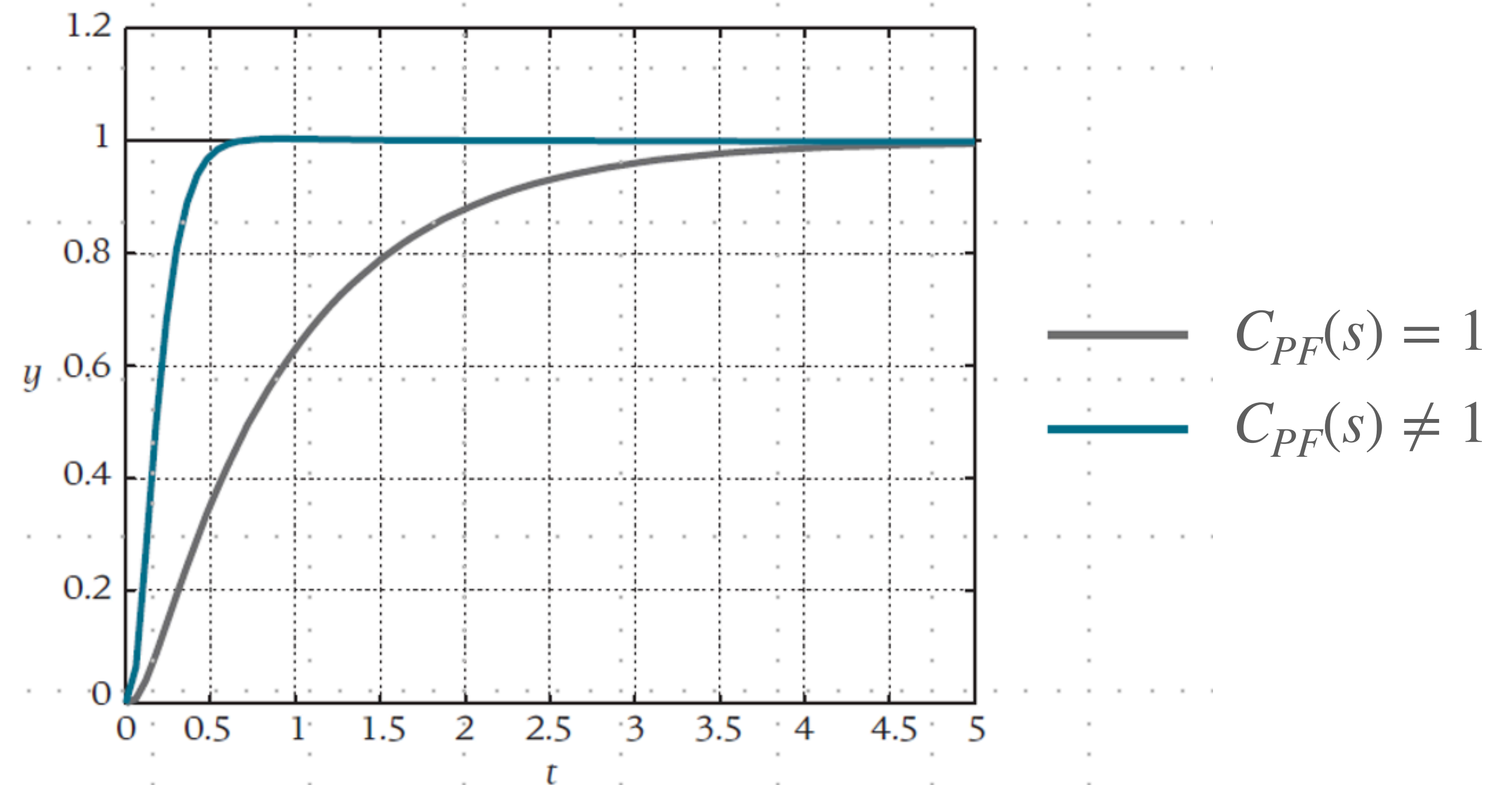
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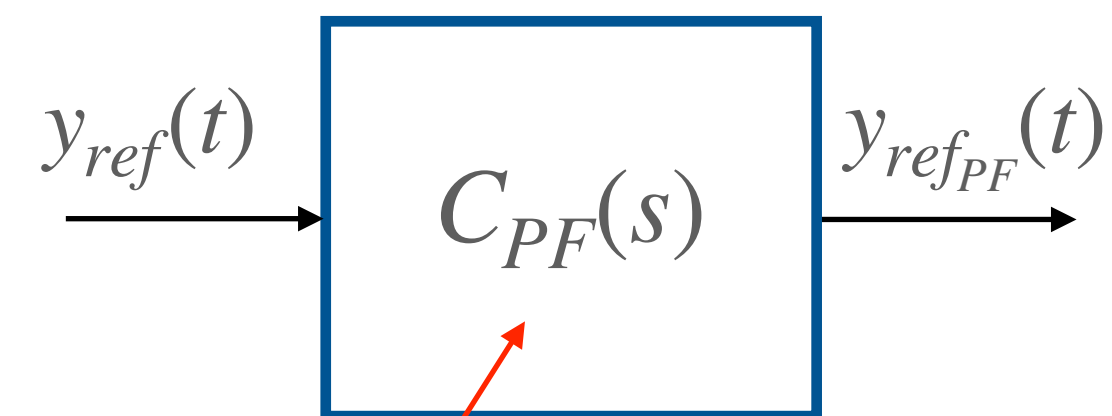
Different types
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- High-Pass

Time domain



Pre-filter Based Control Scheme: Case 1

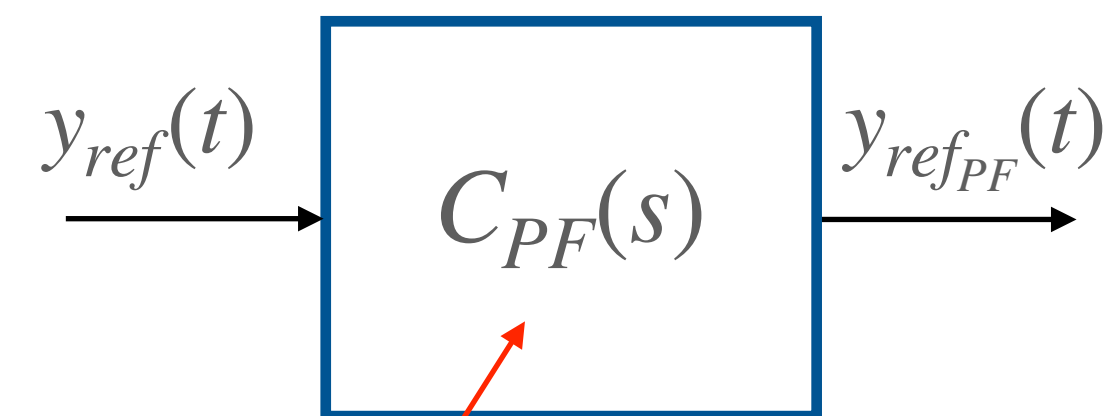


Different types
of Pre-Filter

- High-Pass

- Limited bandwidth in case of measurement disturbances

Pre-filter Based Control Scheme: Case 1

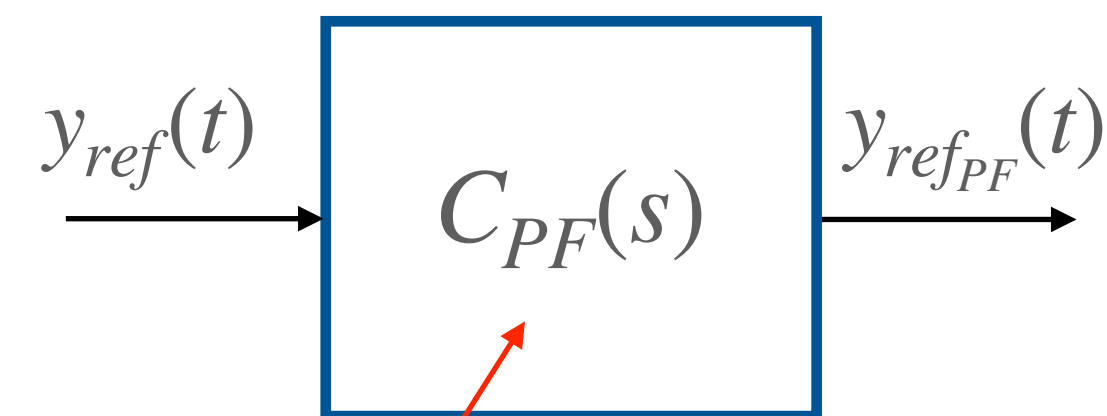


Different types
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- High-Pass

- Limited bandwidth in case of measurement disturbances
- Limited bandwidth in case of model uncertainties
- Limited bandwidth in case of delays

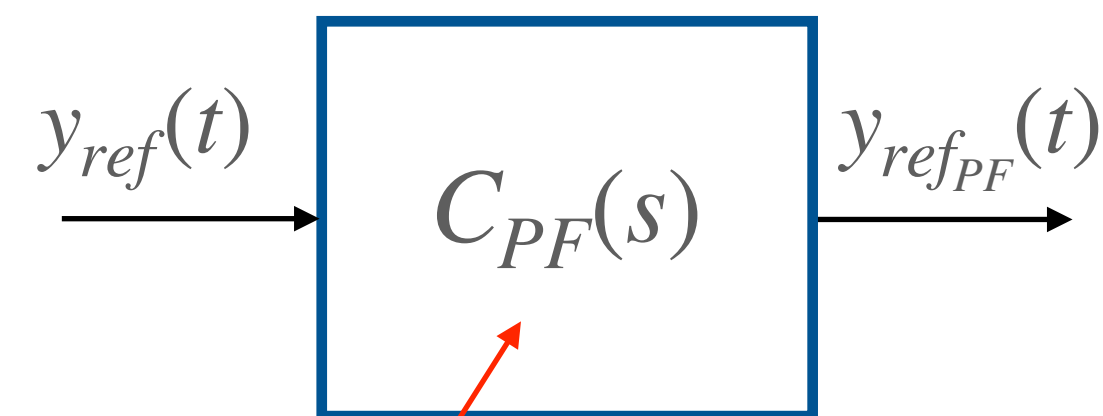
Pre-filter Based Control Scheme: Case 1



Different types
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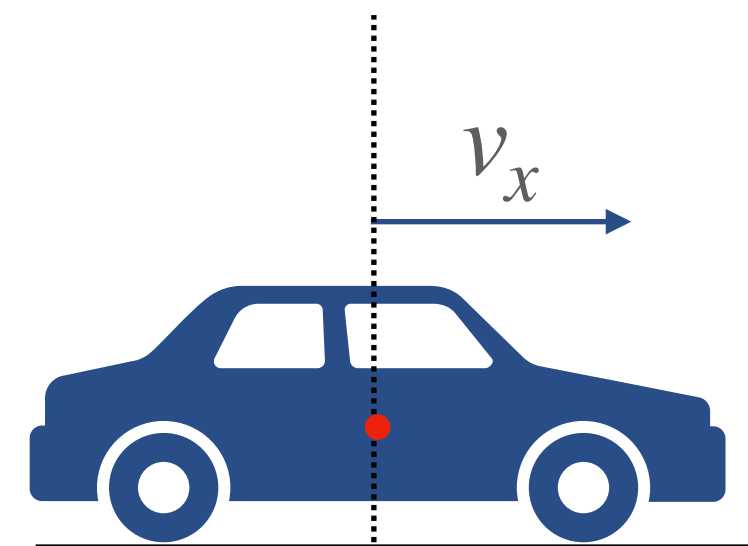
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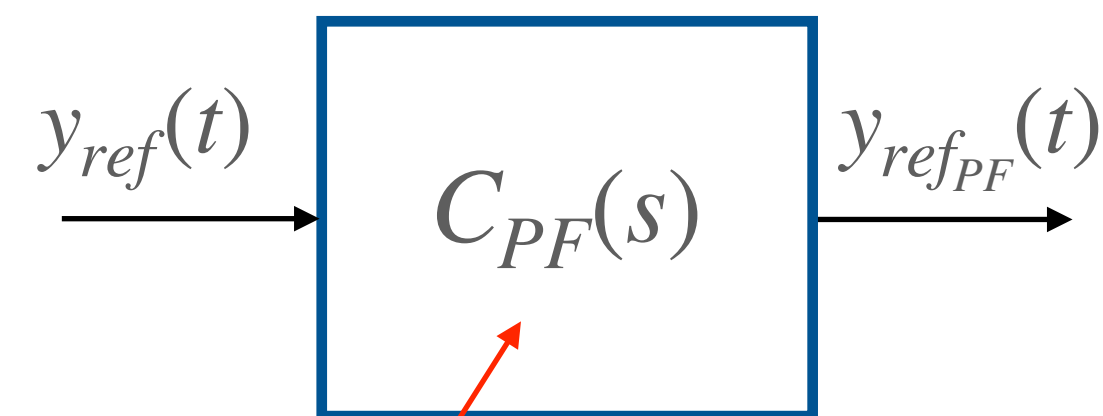
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Example:



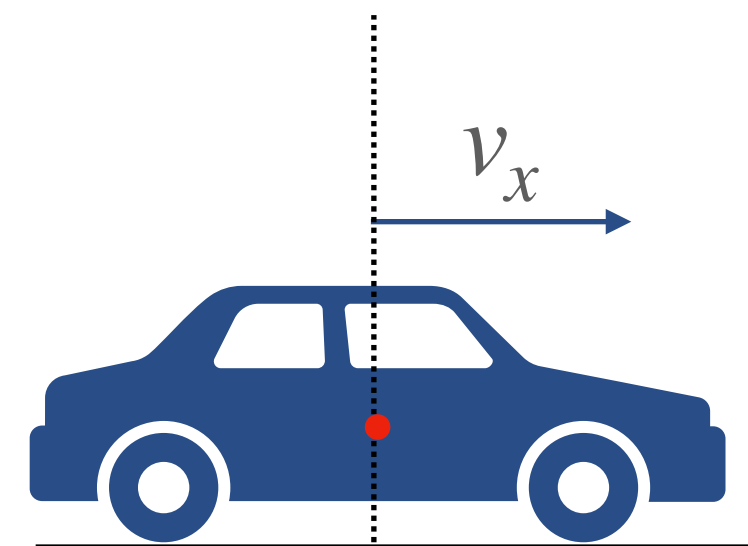
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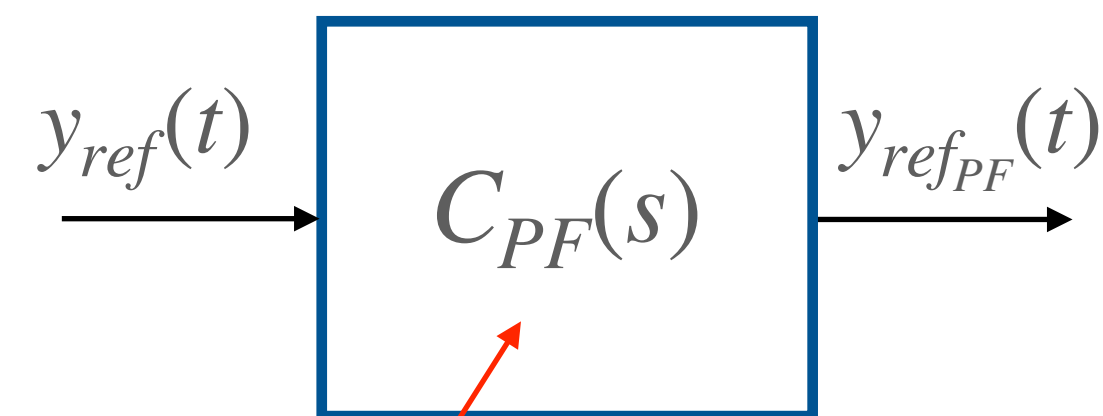
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$$\lambda := \frac{\omega r - v_x}{\max(\omega r, v_x)}$$

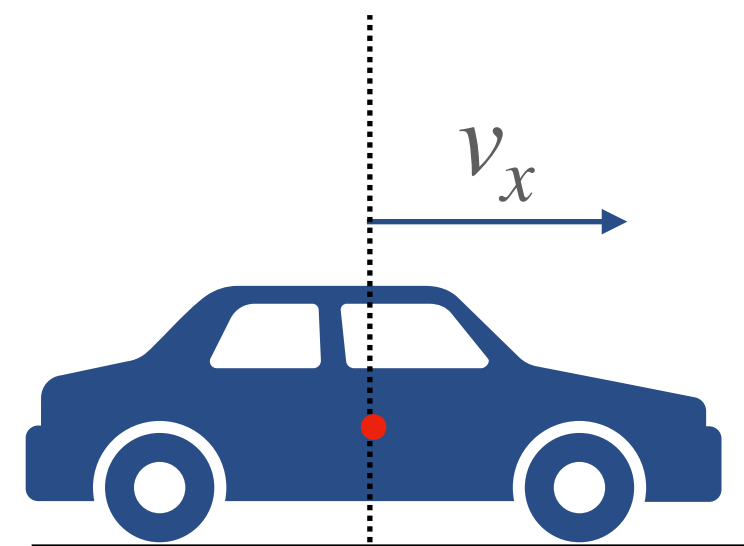
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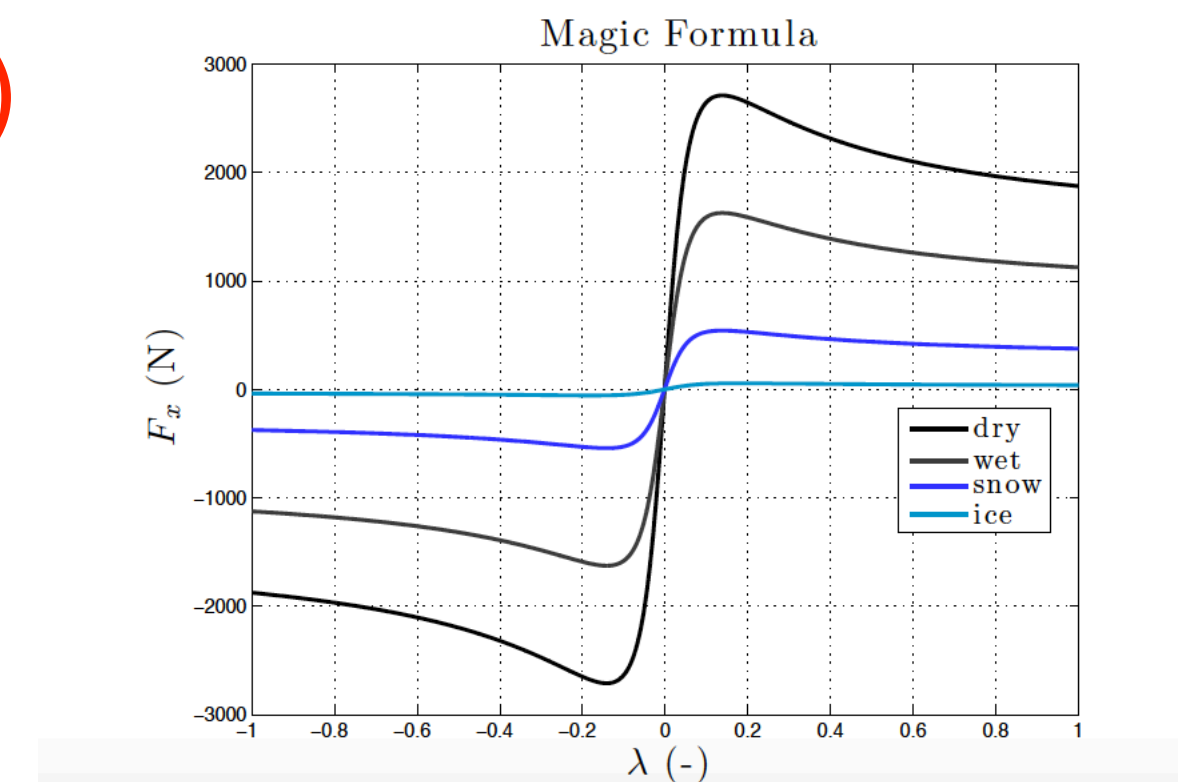
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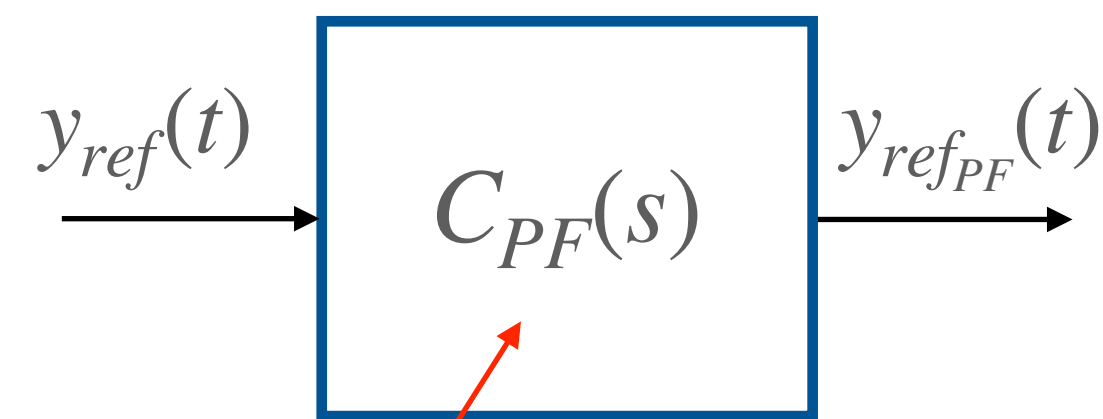
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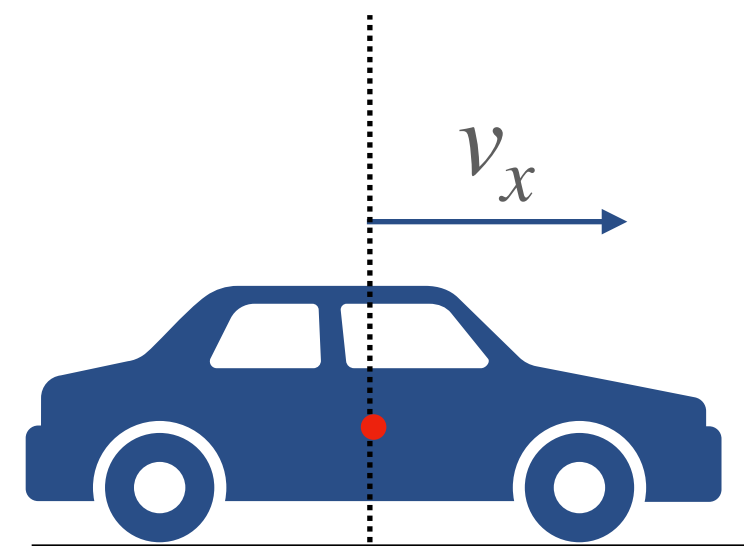
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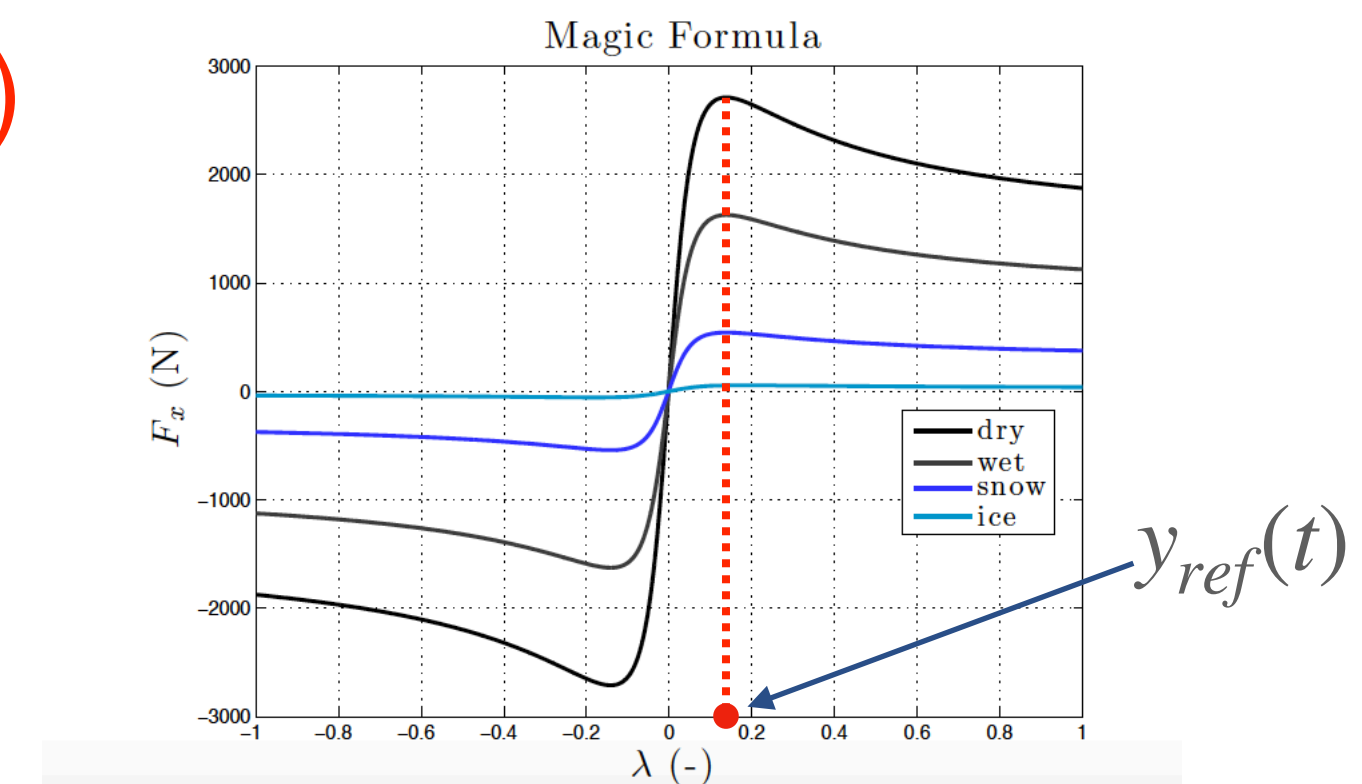
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- No “control moderation” in case of reference signal characterized by sharp transitions in the waveform (abrupt transitions and near-vertical edges in the waveform)

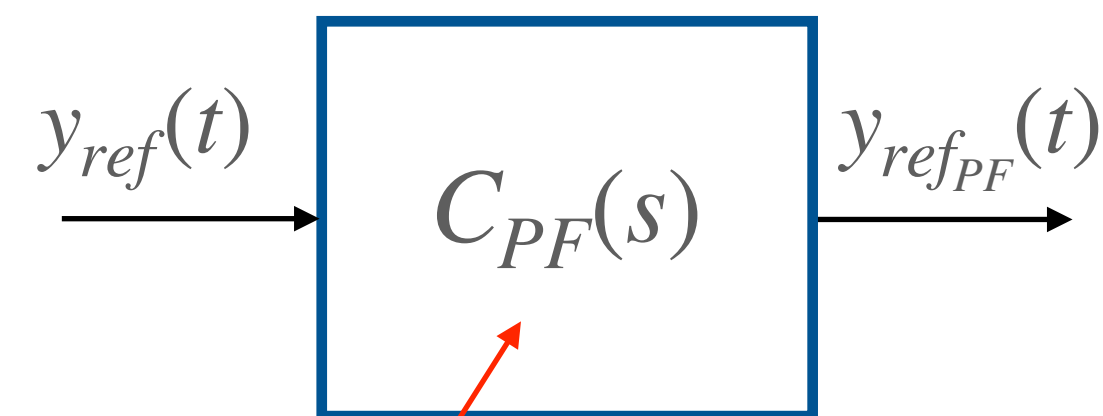
Example:



$$\lambda := \frac{\omega r - v_x}{\max(\omega r, v_x)}$$



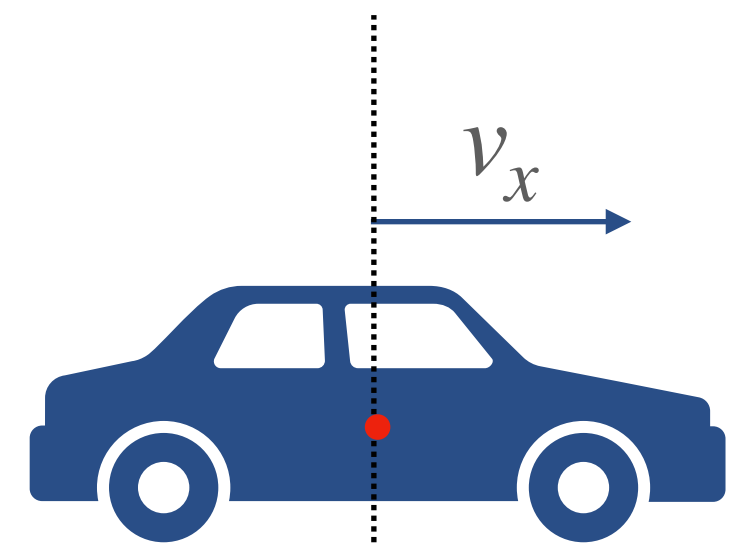
Pre-filter Based Control Scheme: Case 1



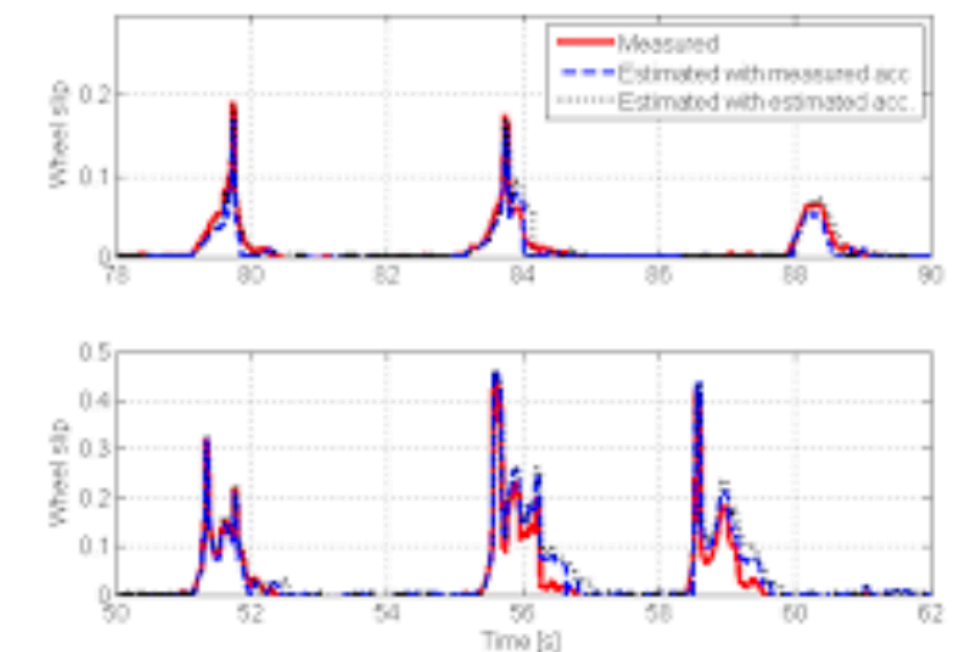
Different types
of Pre-Filter

- Limited bandwidth in case of measurement disturbances
- Limited bandwidth in case of model uncertainties
- Limited bandwidth in case of delays
- No “control moderation” in case of reference signal characterized by sharp transitions in the waveform (abrupt transitions and near-vertical edges in the waveform)

Example:

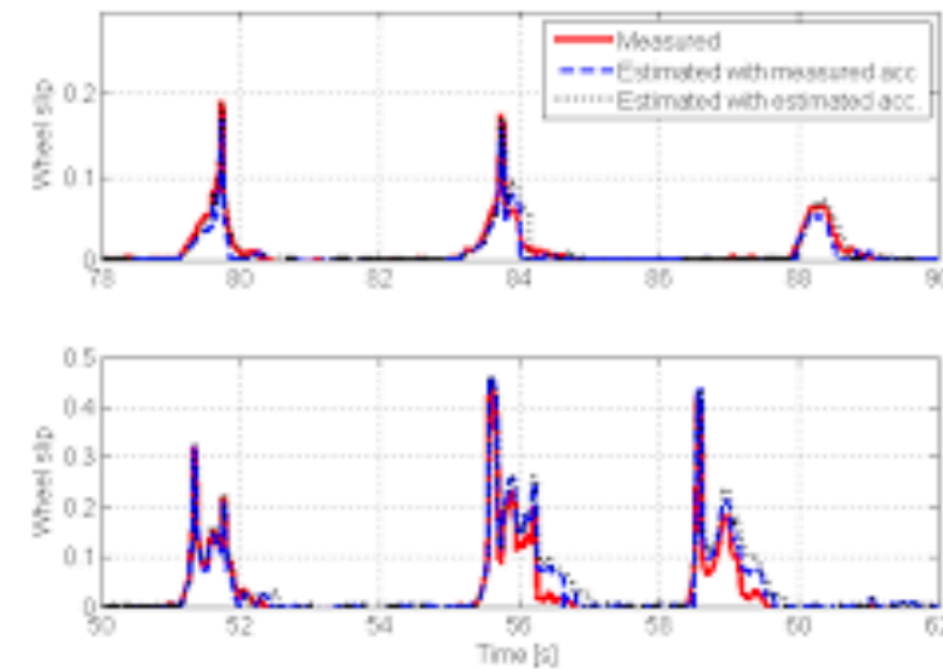
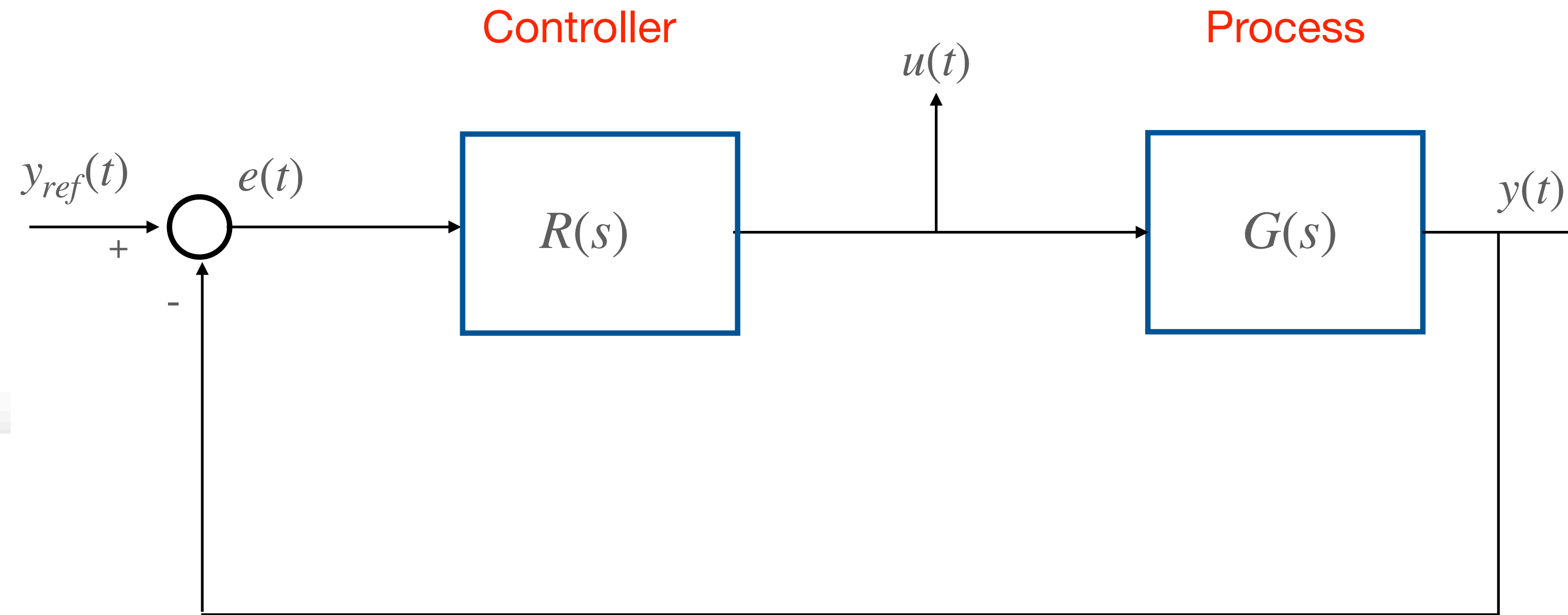


$$\lambda := \frac{\omega r - v_x}{\max(\omega r, v_x)}$$

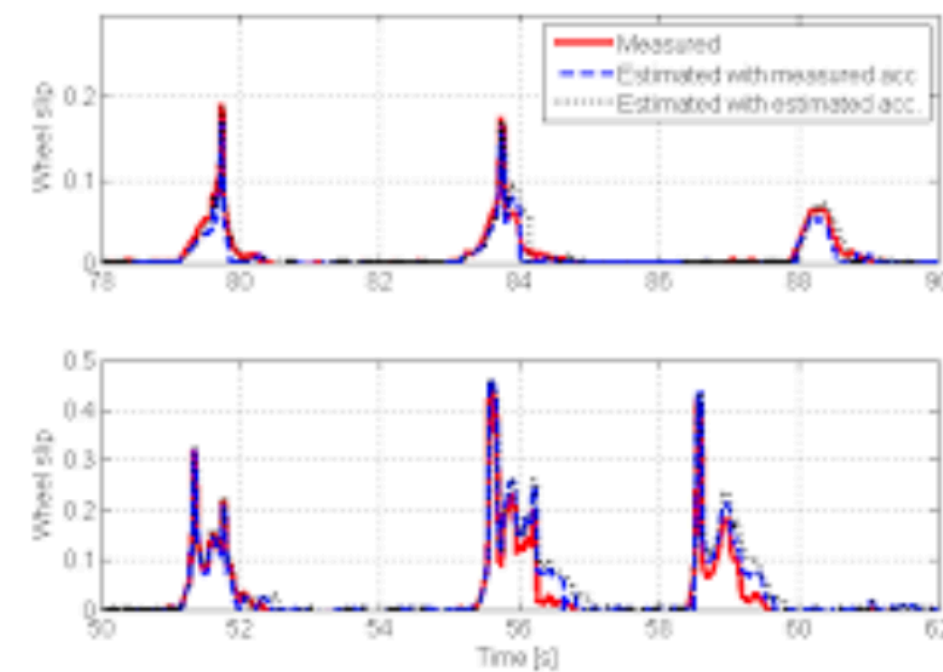


WHEEL SLIP

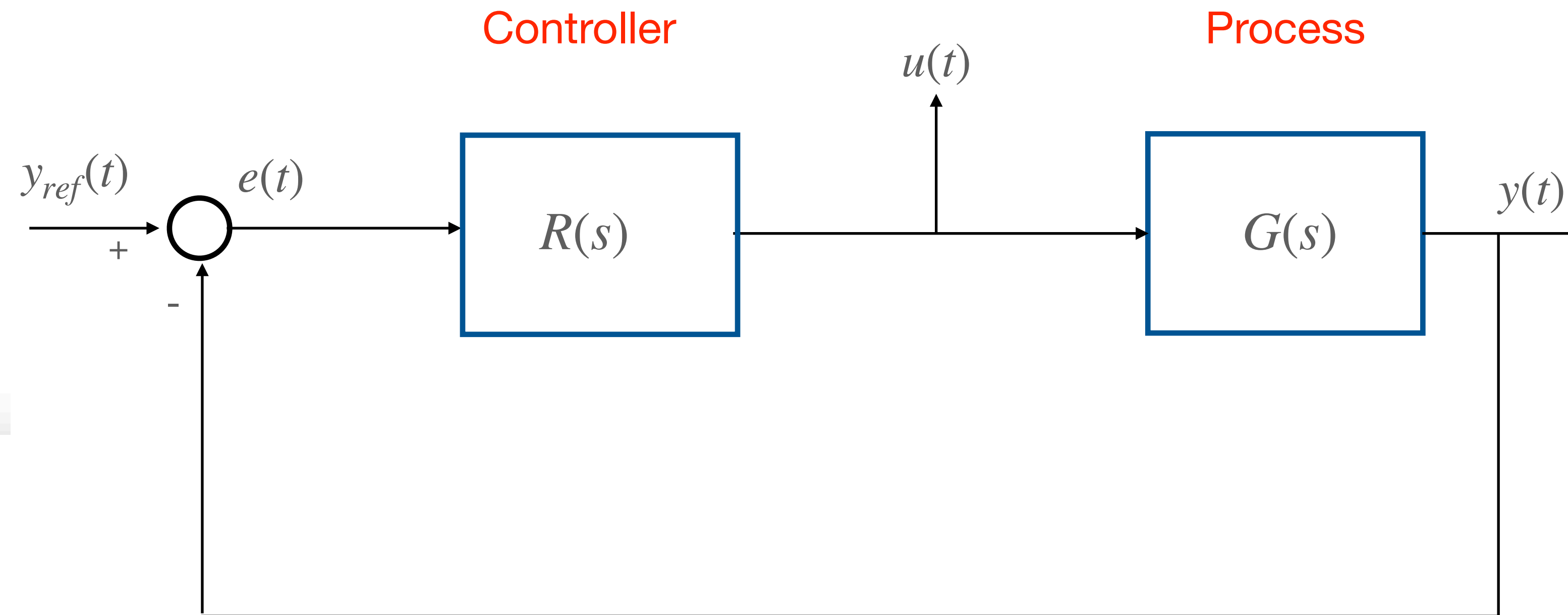
Basic Control Scheme

**WHEEL SLIP**

Basic Control Scheme

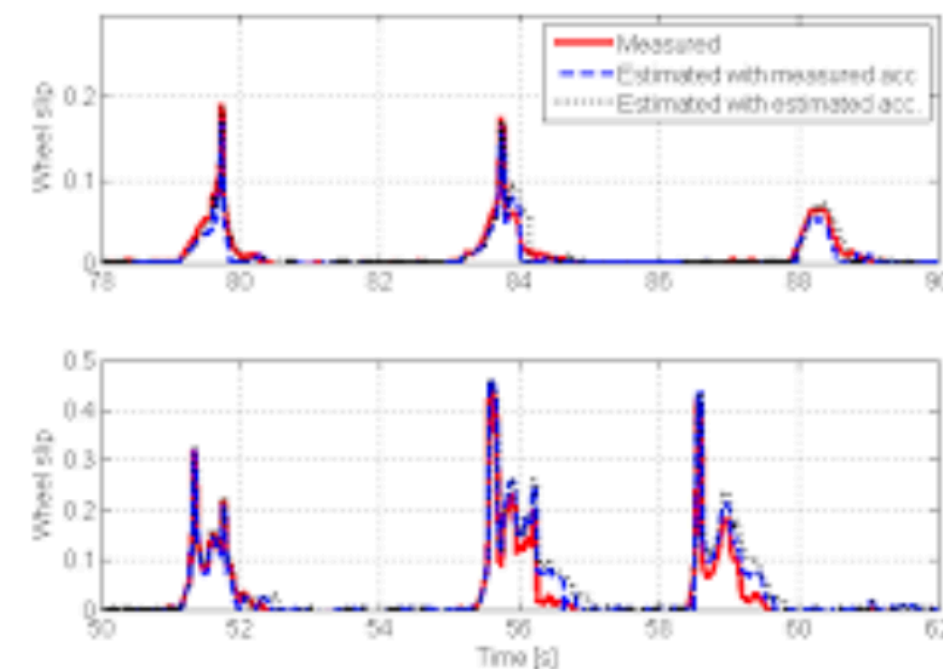


WHEEL SLIP

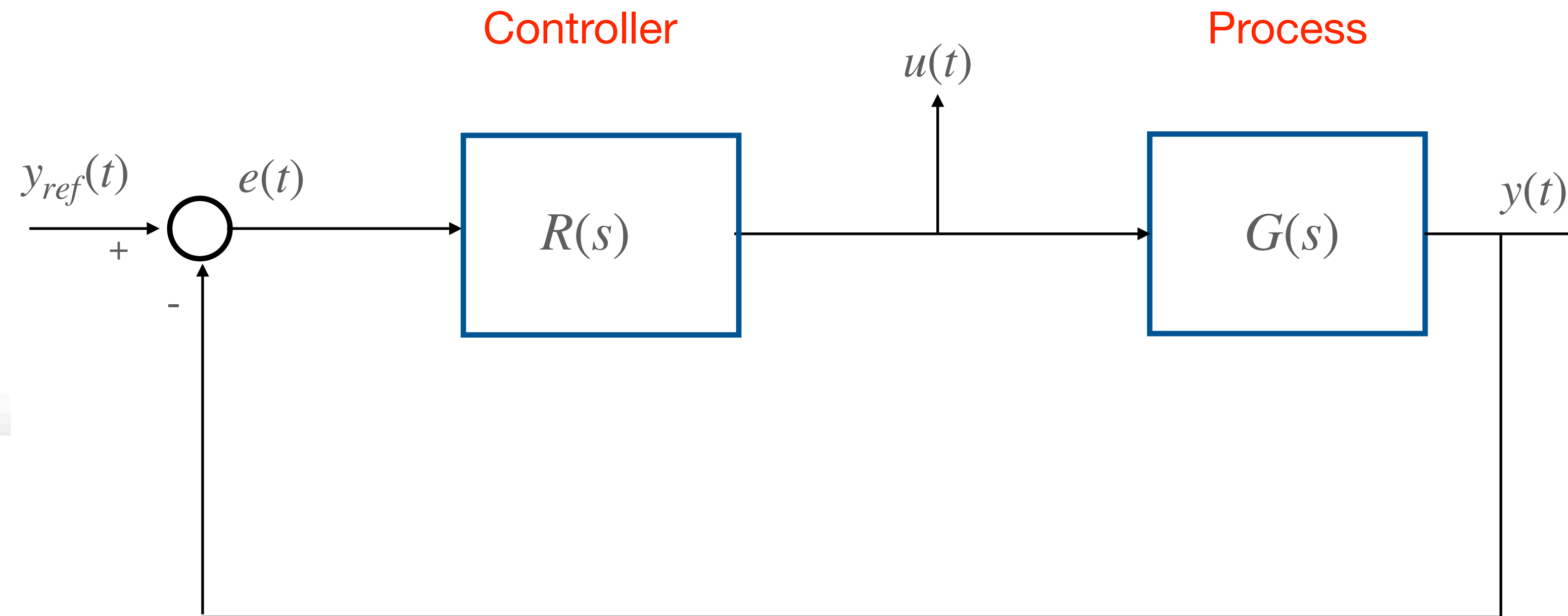


Block Algebra Rule: $y_{ref} \rightarrow u : Q(s) = \frac{R(s)}{1 + R(s)G(s)}$

Basic Control Scheme

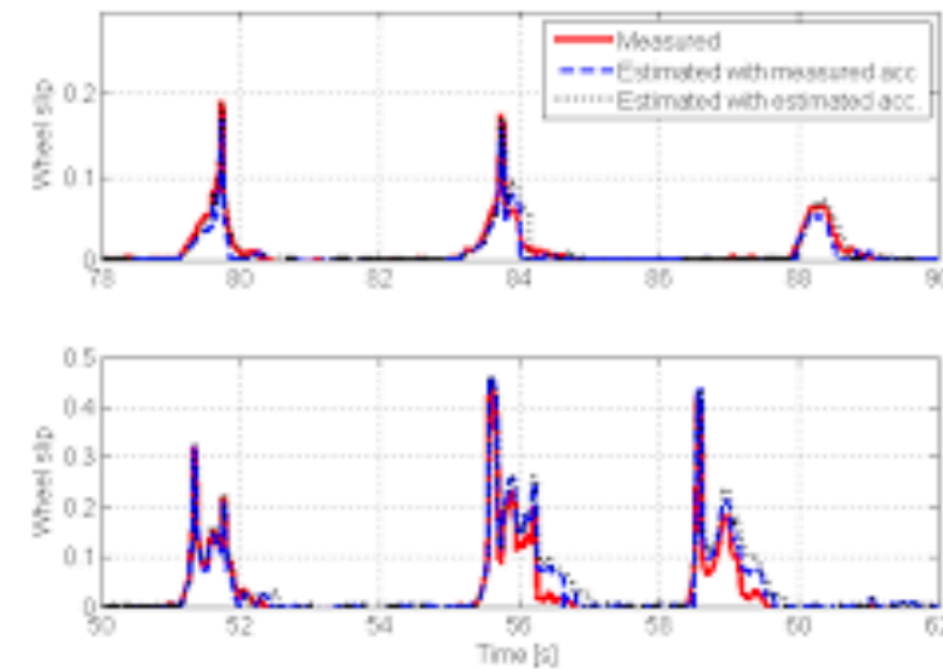


WHEEL SLIP

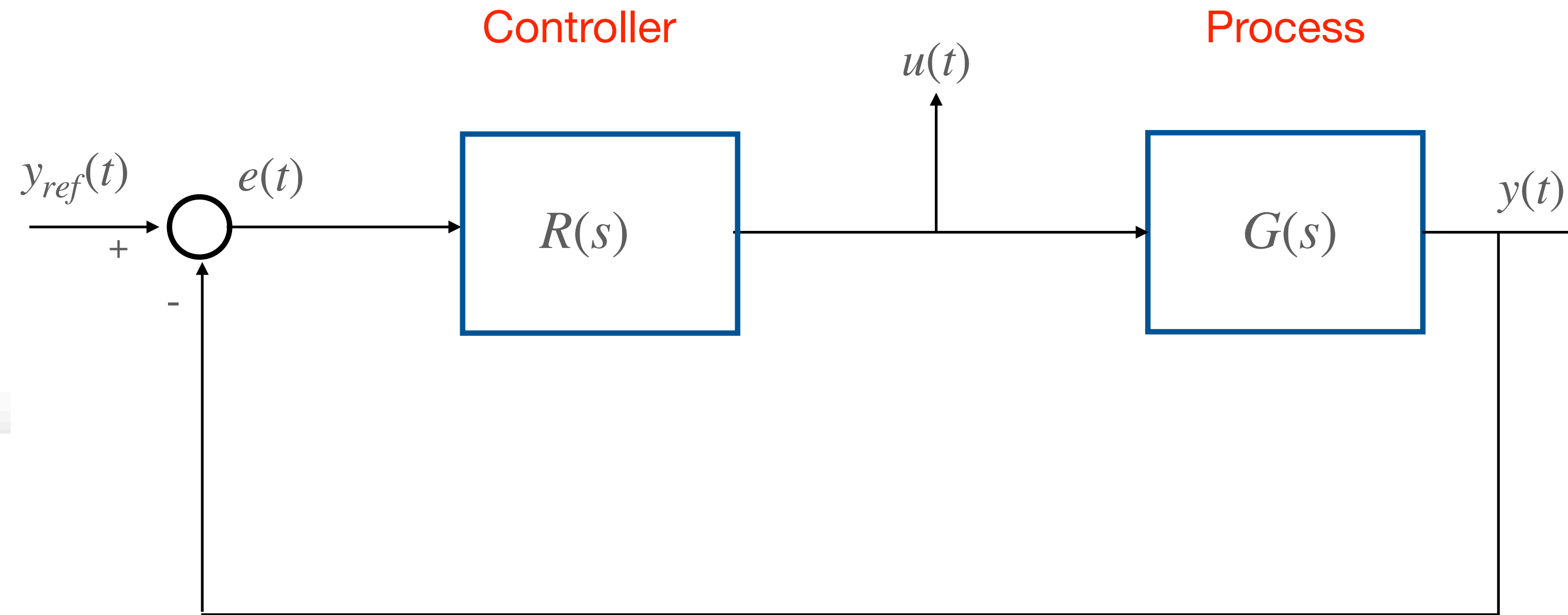


Block Algebra Rule: $y_{ref} \rightarrow u : Q(s) = \frac{R(s)}{1 + R(s)G(s)}$ Control Sensitivity Function

Basic Control Scheme



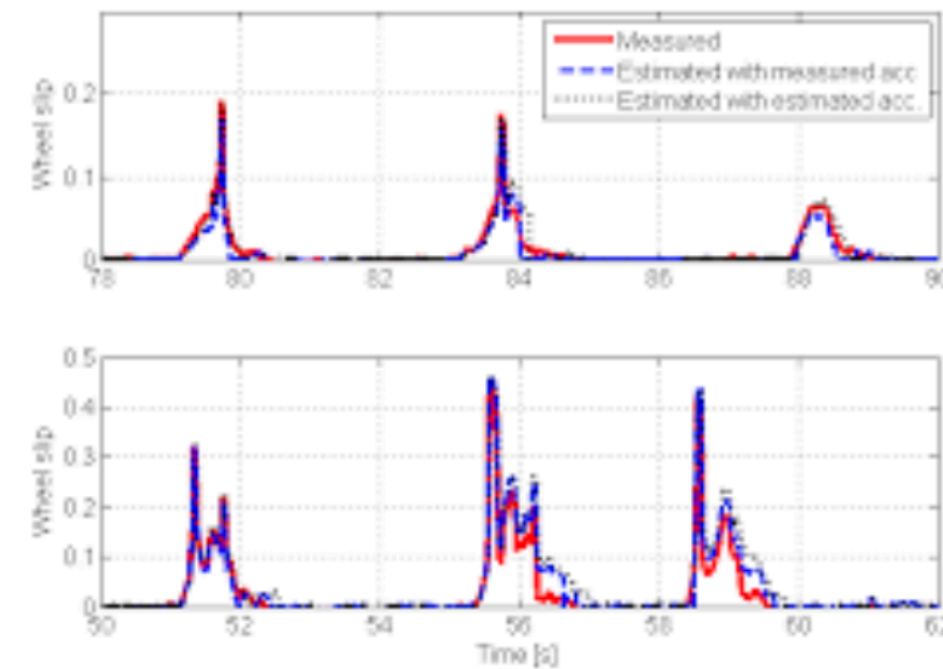
WHEEL SLIP



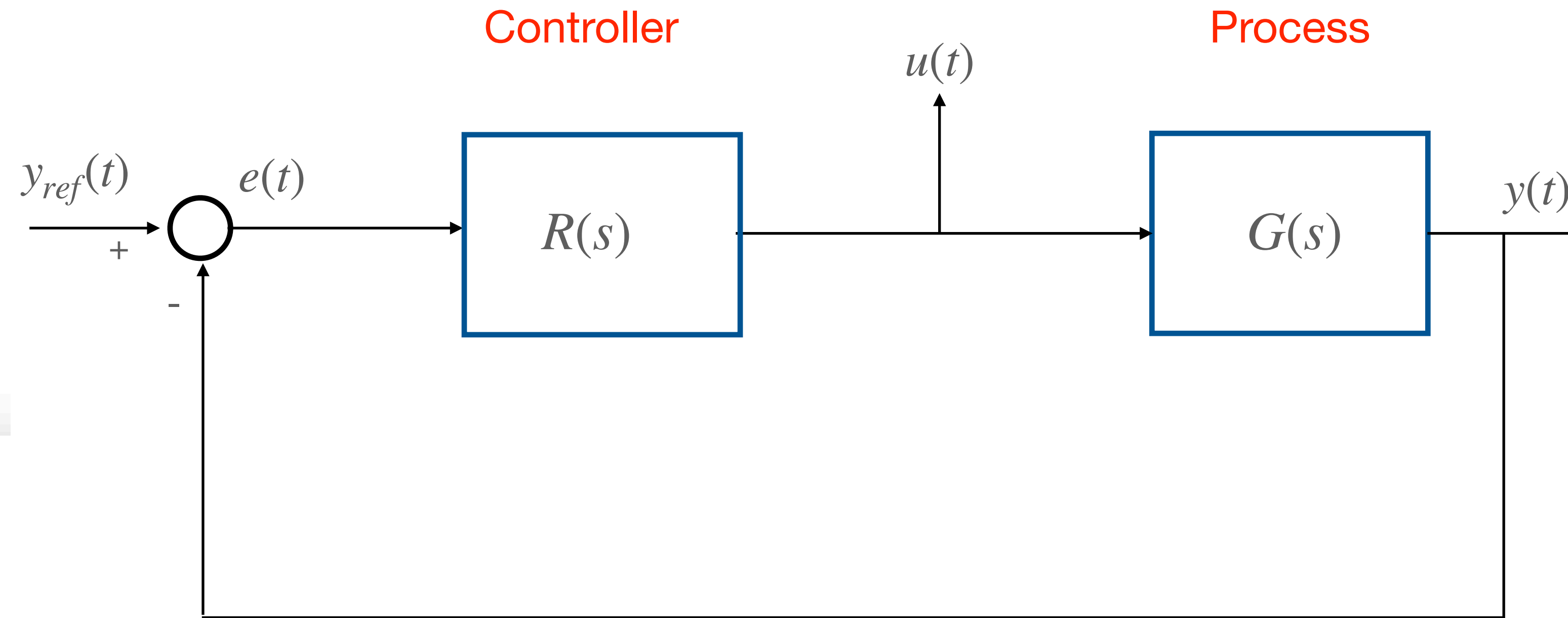
Block Algebra Rule: $y_{ref} \rightarrow u : Q(s) = \frac{R(s)}{1 + R(s)G(s)}$ Control Sensitivity Function

Recommendation: $|Q(j\omega)|$ sufficiently low $\forall \omega \in \mathbb{R}_{+ \cup \{0\}}$

Basic Control Scheme



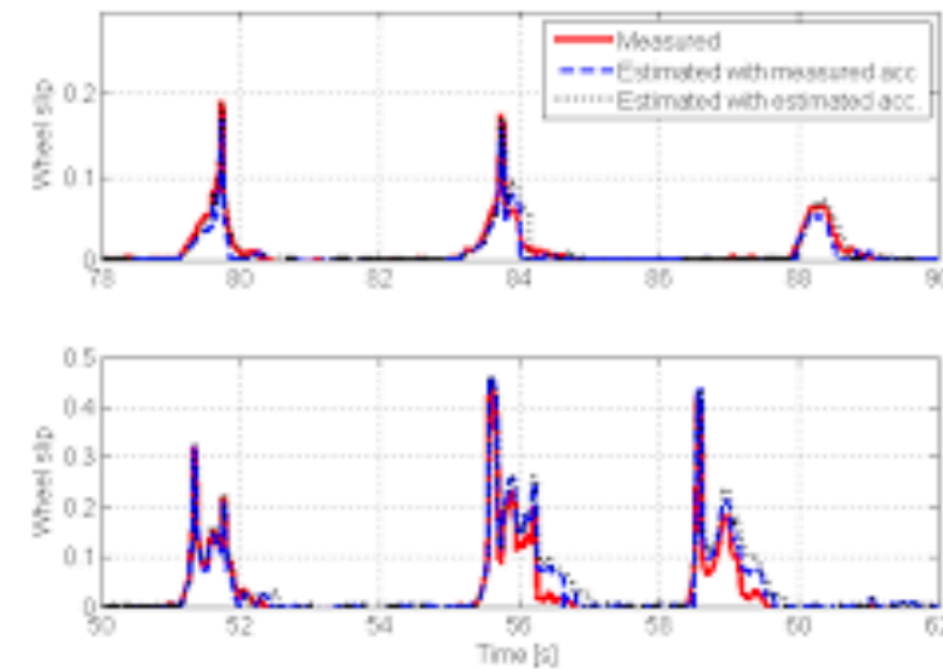
WHEEL SLIP



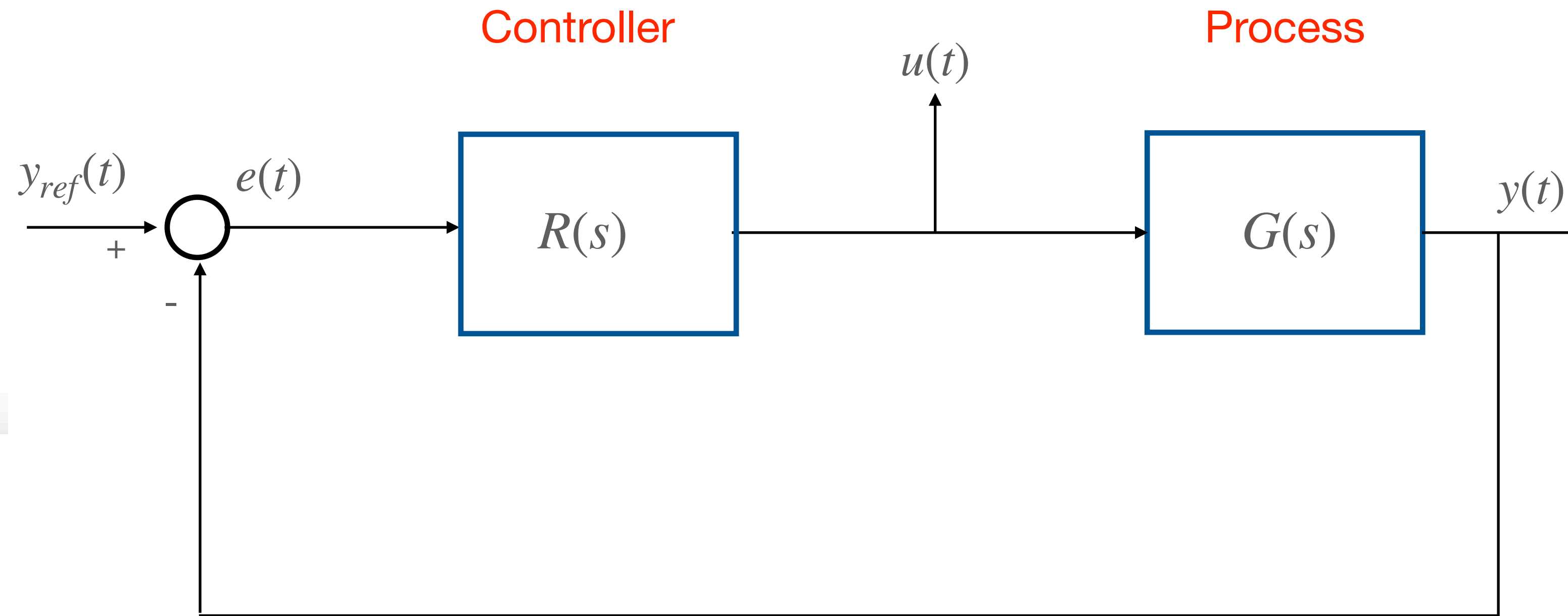
Block Algebra Rule: $y_{ref} \rightarrow u : Q(s) = \frac{R(s)}{1 + R(s)G(s)}$ Control Sensitivity Function

Recommendation: $|Q(j\omega)|$ sufficiently low $\forall \omega \in \mathbb{R}_{+ \cup \{0\}}$ $\rightarrow$ LF: $|Q(j\omega)| \rightarrow \left| \frac{1}{G(j\omega)} \right|$

Basic Control Scheme



WHEEL SLIP

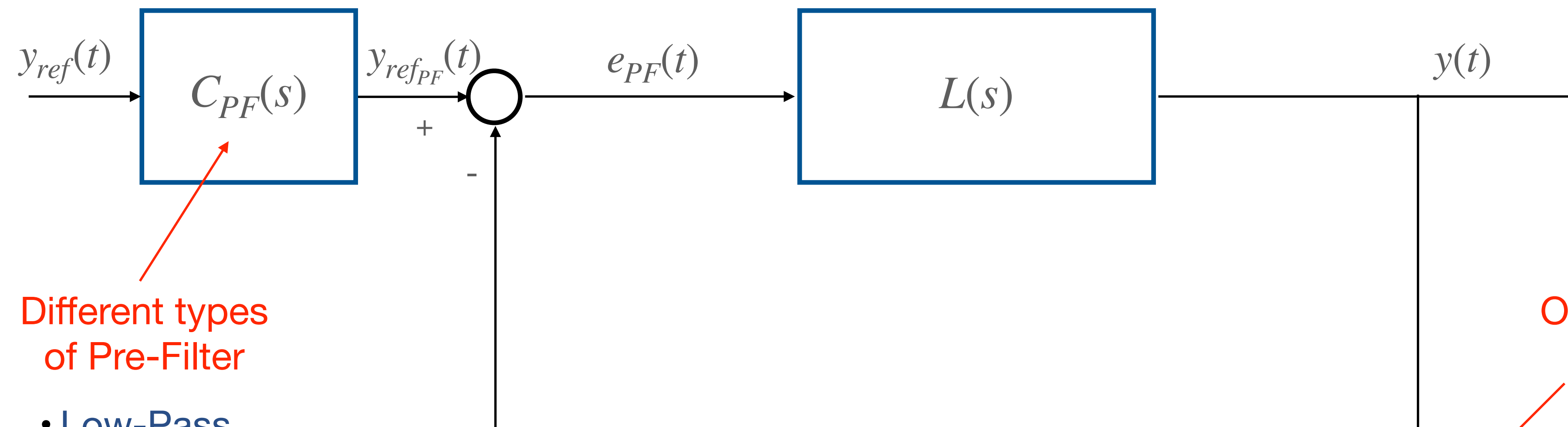


Block Algebra Rule: $y_{ref} \rightarrow u : Q(s) = \frac{R(s)}{1 + R(s)G(s)}$ **Control Sensitivity Function**

Recommendation: $|Q(j\omega)|$ sufficiently low $\forall \omega \in \mathbb{R}_{+ \cup \{0\}}$

- LF: $|Q(j\omega)| \rightarrow \left| \frac{1}{G(j\omega)} \right|$
- HF: $|Q(j\omega)| \rightarrow |R(j\omega)|$

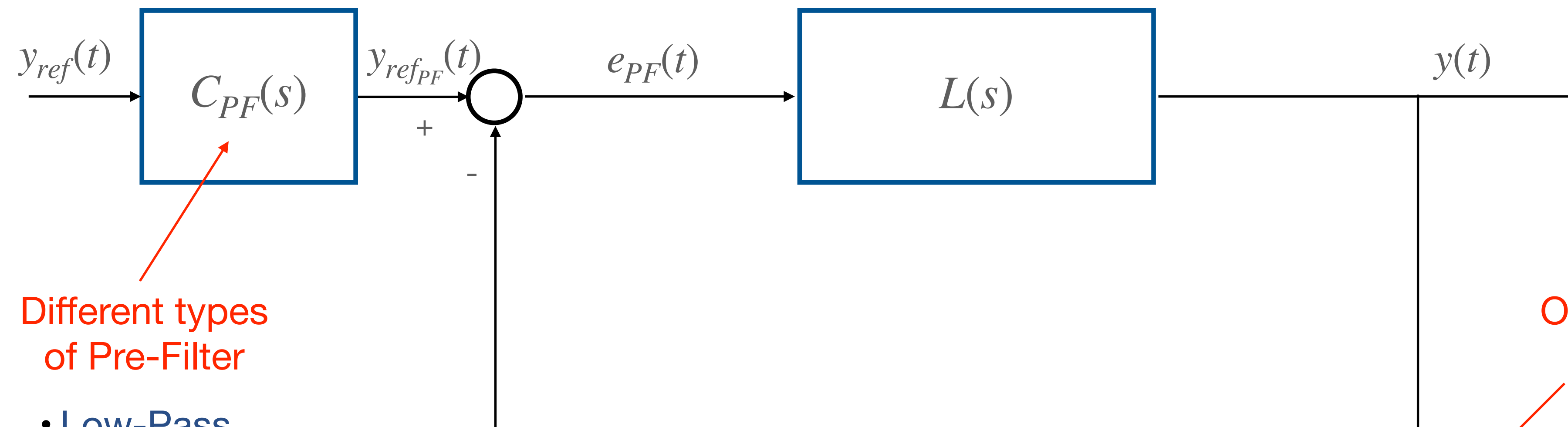
Pre-filter Based Control Scheme: Case 2



Block Algebra Rule: $F(s) = \frac{R(s) \cdot G(s)}{1 + R(s) \cdot G(s)} = \frac{L(s)}{1 + L(s)}$

$$F_{PF}(s) = C_{PF}(s) \cdot F(s) = C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = \frac{\mu(1 + \tau's)}{(1 + \tau s)} \cdot \frac{L(s)}{1 + L(s)}$$

Pre-filter Based Control Scheme: Case 2



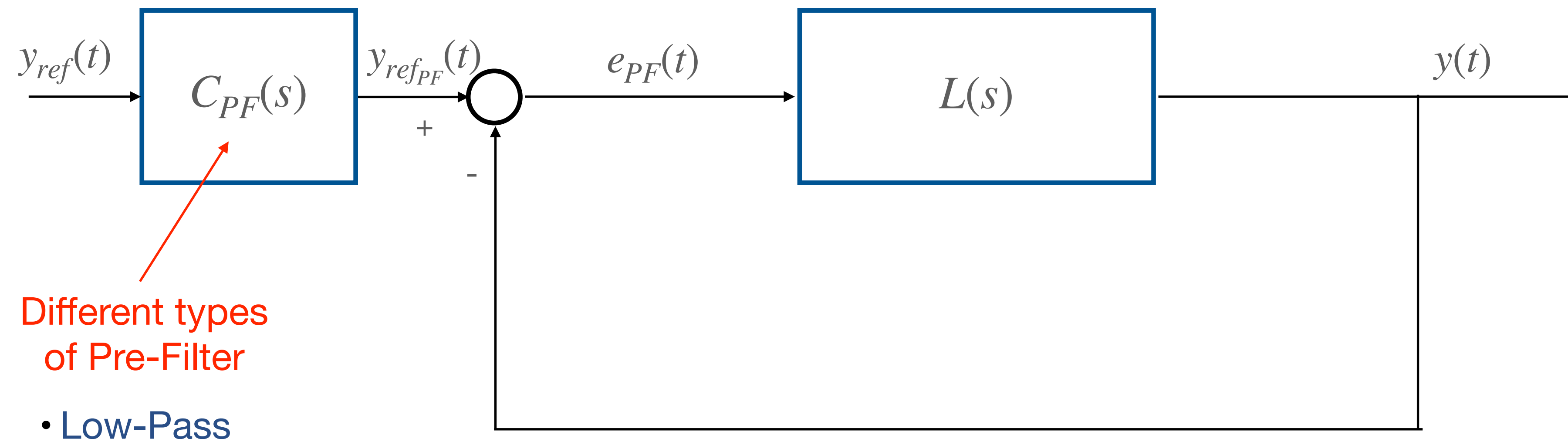
Block Algebra Rule:

$$F(s) = \frac{R(s) \cdot G(s)}{1 + R(s) \cdot G(s)} = \frac{L(s)}{1 + L(s)}$$

$$F_{PF}(s) = C_{PF}(s) \cdot F(s) = C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = \frac{\mu(1 + \tau's)}{(1 + \tau s)} \cdot \frac{L(s)}{1 + L(s)}$$

The pole precedes the zero on the ω axis

Pre-filter Based Control Scheme: Case 2



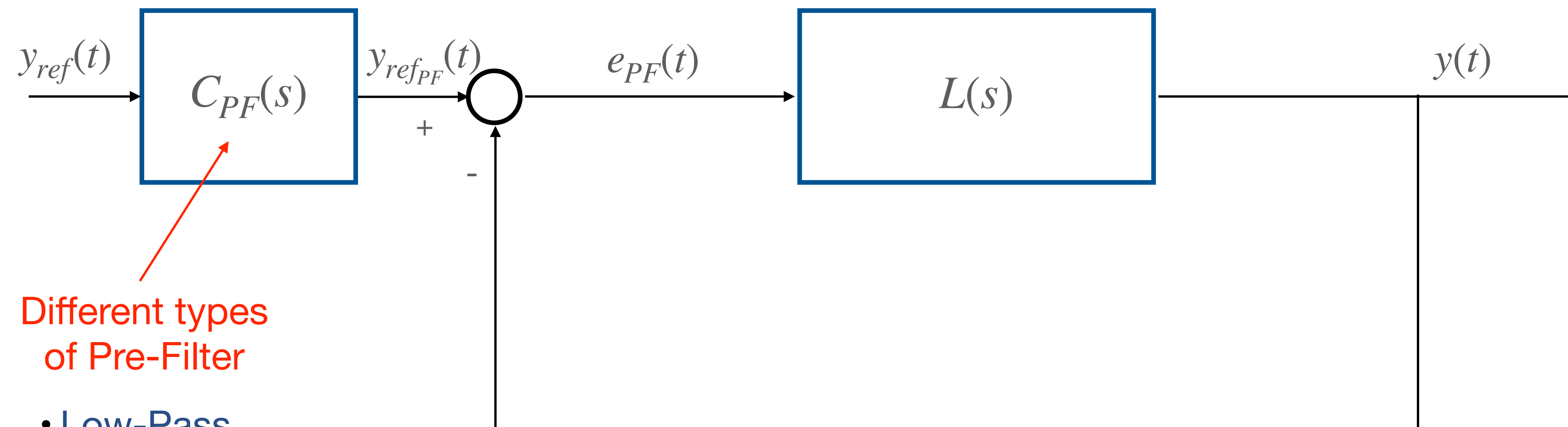
Assumptions:

- $C_{PF}(s)$ As. Stable
- Proper
- Unitary gain

The pole precedes the zero on the ω axis

$$F_{PF}(s) = C_{PF}(s) \cdot F(s) = C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = \frac{\mu(1 + \tau' s)}{(1 + \tau s)} \cdot \frac{L(s)}{1 + L(s)}$$

Pre-filter Based Control Scheme: Case 2



Assumptions:

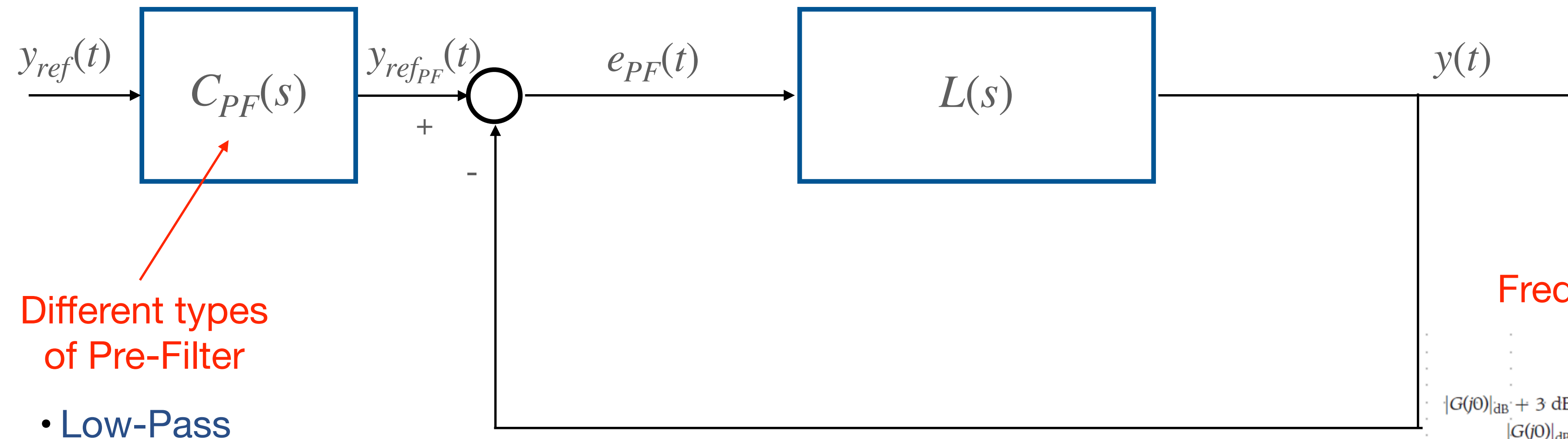
- $C_{PF}(s)$ As. Stable
- Proper
- Unitary gain

$$\lim_{s \rightarrow 0} \frac{\mu(1 + \tau' s)}{1 + \tau s} = \mu = 1$$

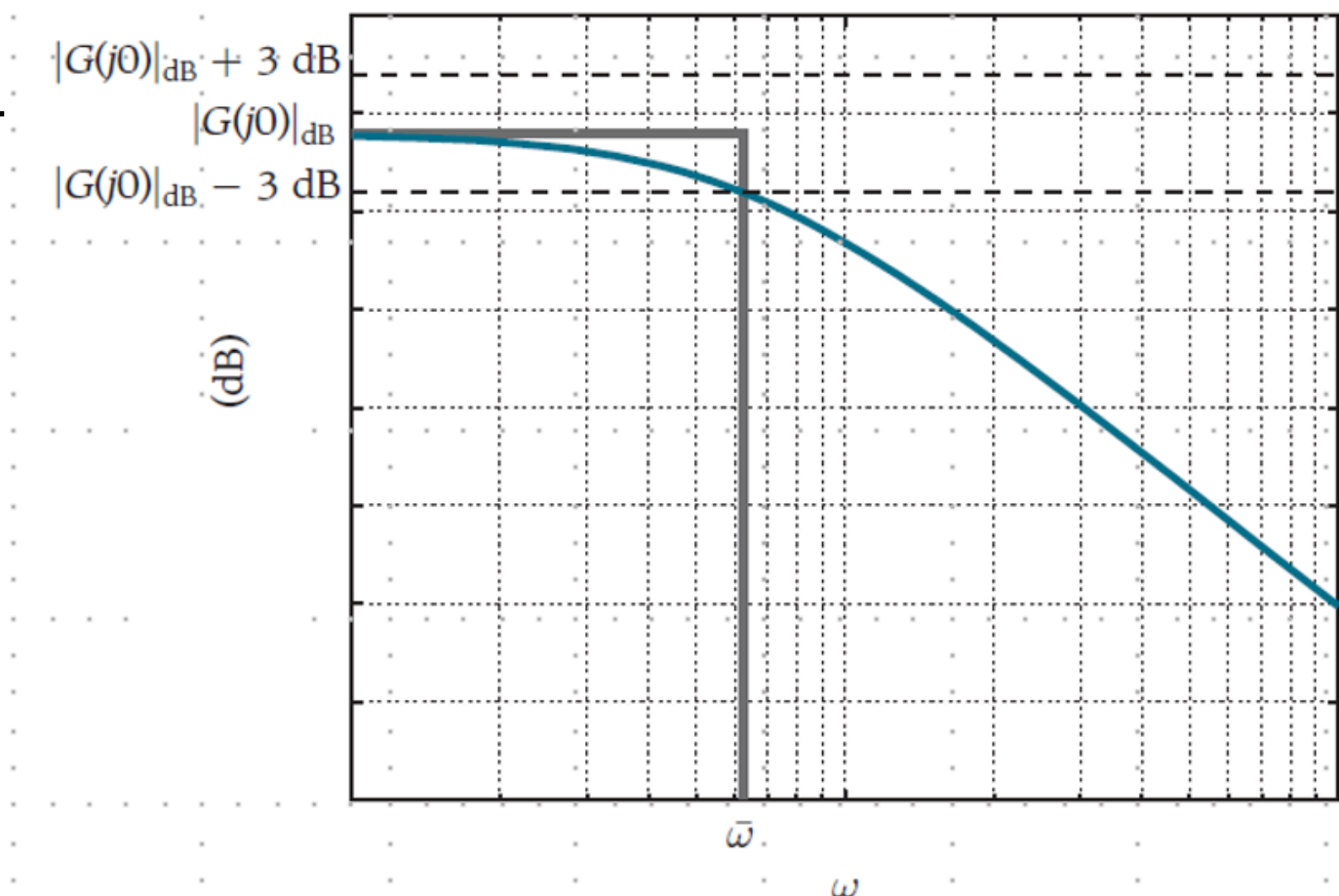
The pole precedes the zero on the ω axis

$$F_{PF}(s) = C_{PF}(s) \cdot F(s) = C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = \frac{\mu(1 + \tau' s)}{(1 + \tau s)} \cdot \frac{L(s)}{1 + L(s)}$$

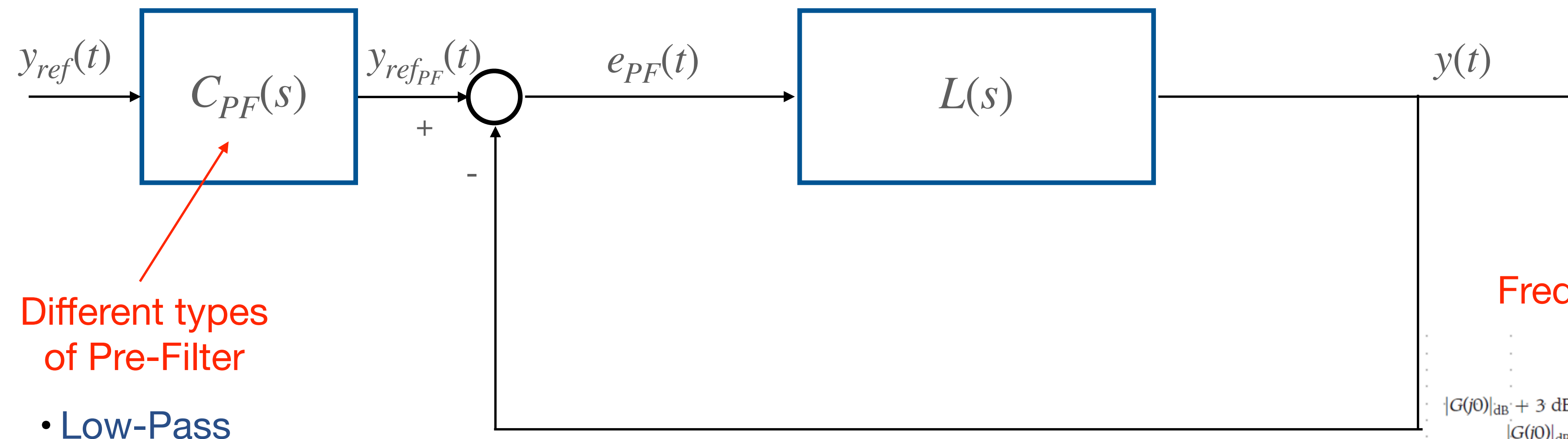
Pre-filter Based Control Scheme: Case 2



Frequency response of LP filter (magnitude)

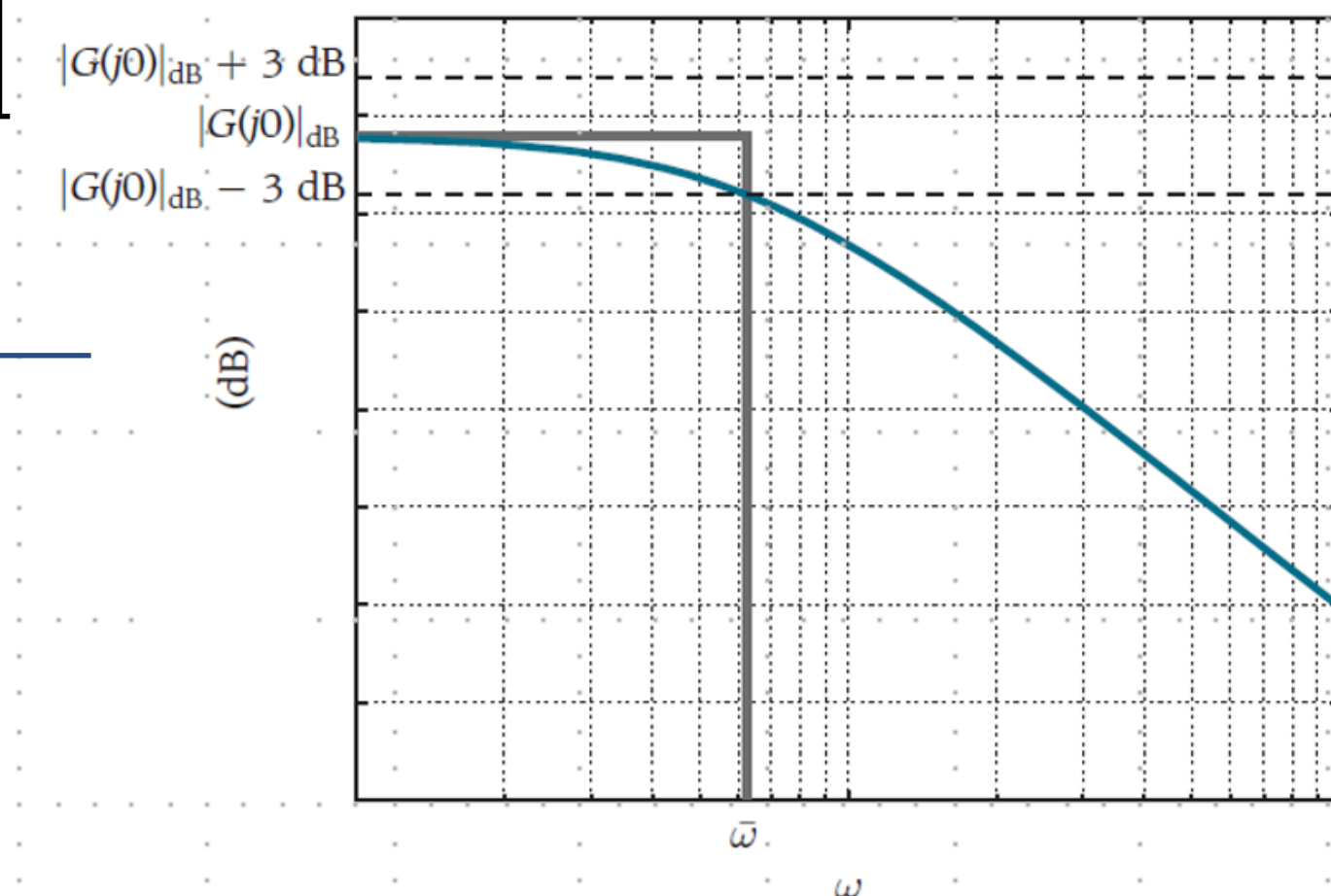


Pre-filter Based Control Scheme: Case 2

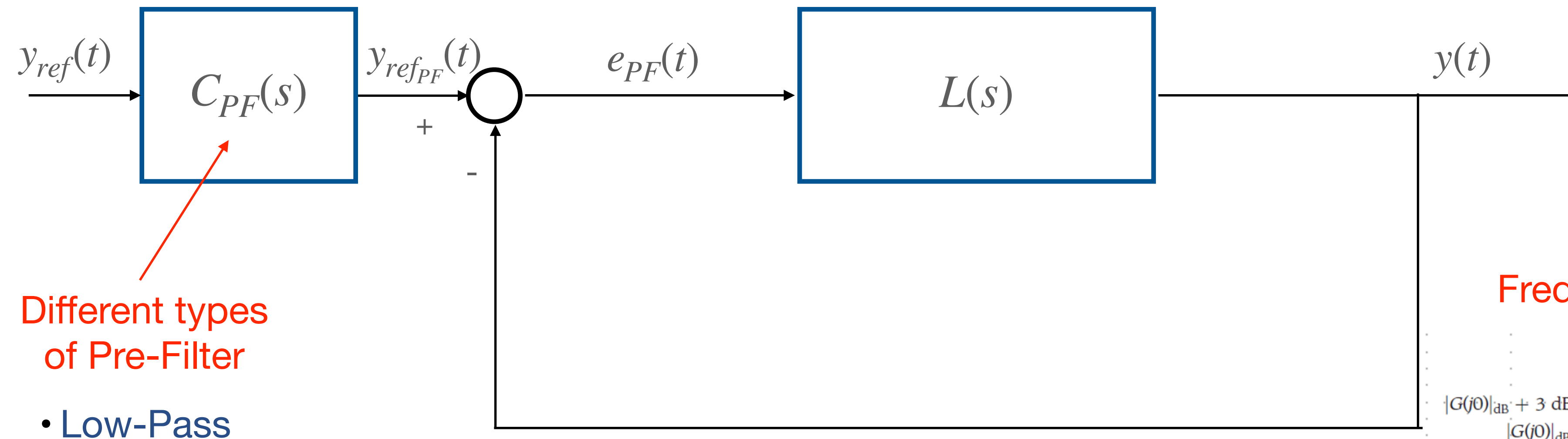


A LP Pre-filter is used to filter the reference signal to alleviate control peaking phenomena

Frequency response of LP filter (magnitude)



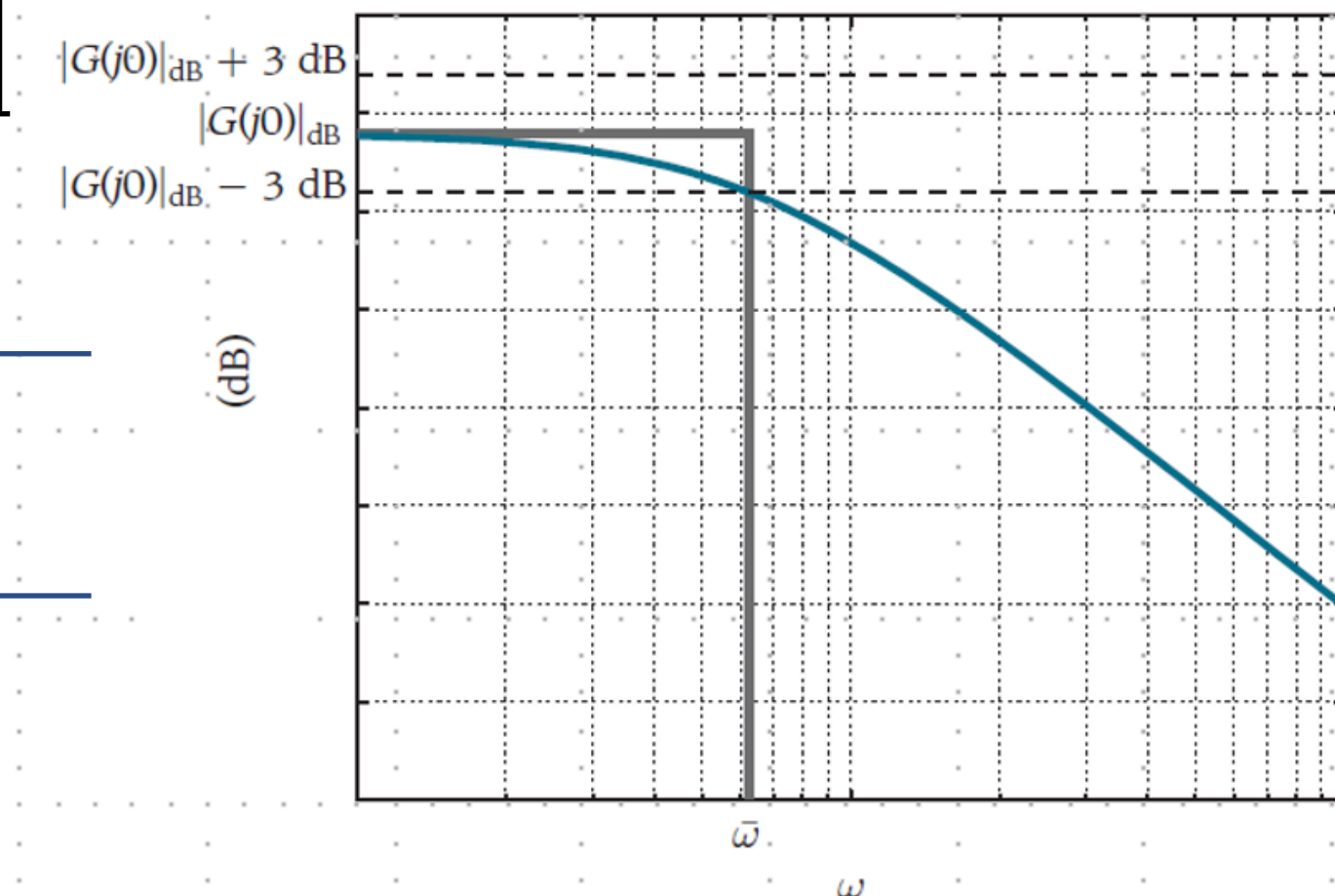
Pre-filter Based Control Scheme: Case 2



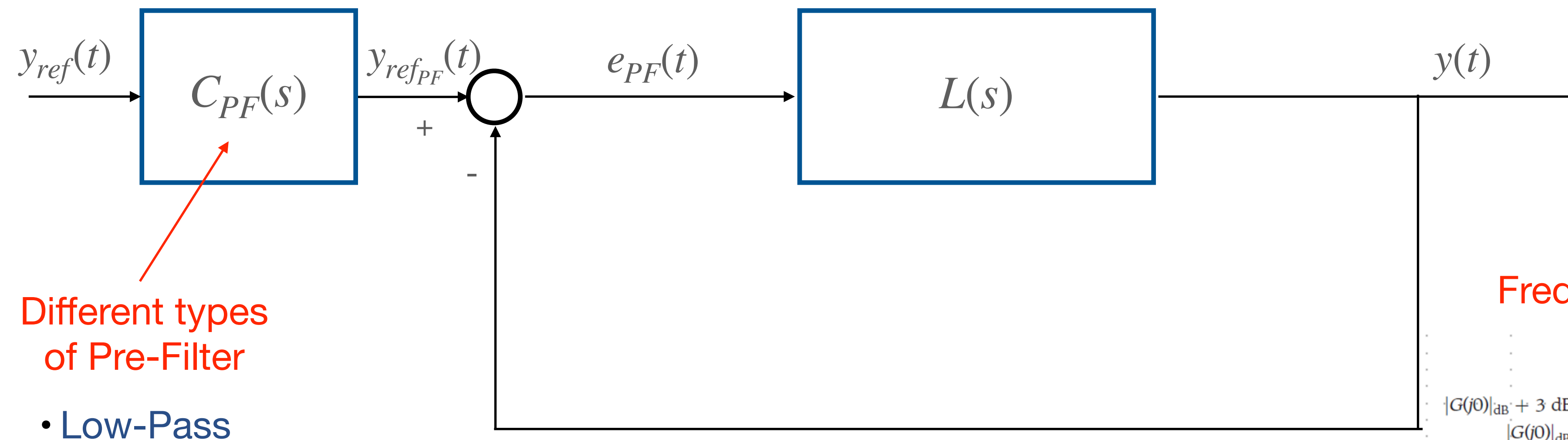
A LP Pre-filter is used to filter the reference signal to alleviate control peaking phenomena

Care must be taken not to reduce the closed-loop bandwidth $\omega_{BW_{CL}}$

Frequency response of LP filter (magnitude)



Pre-filter Based Control Scheme: Case 2

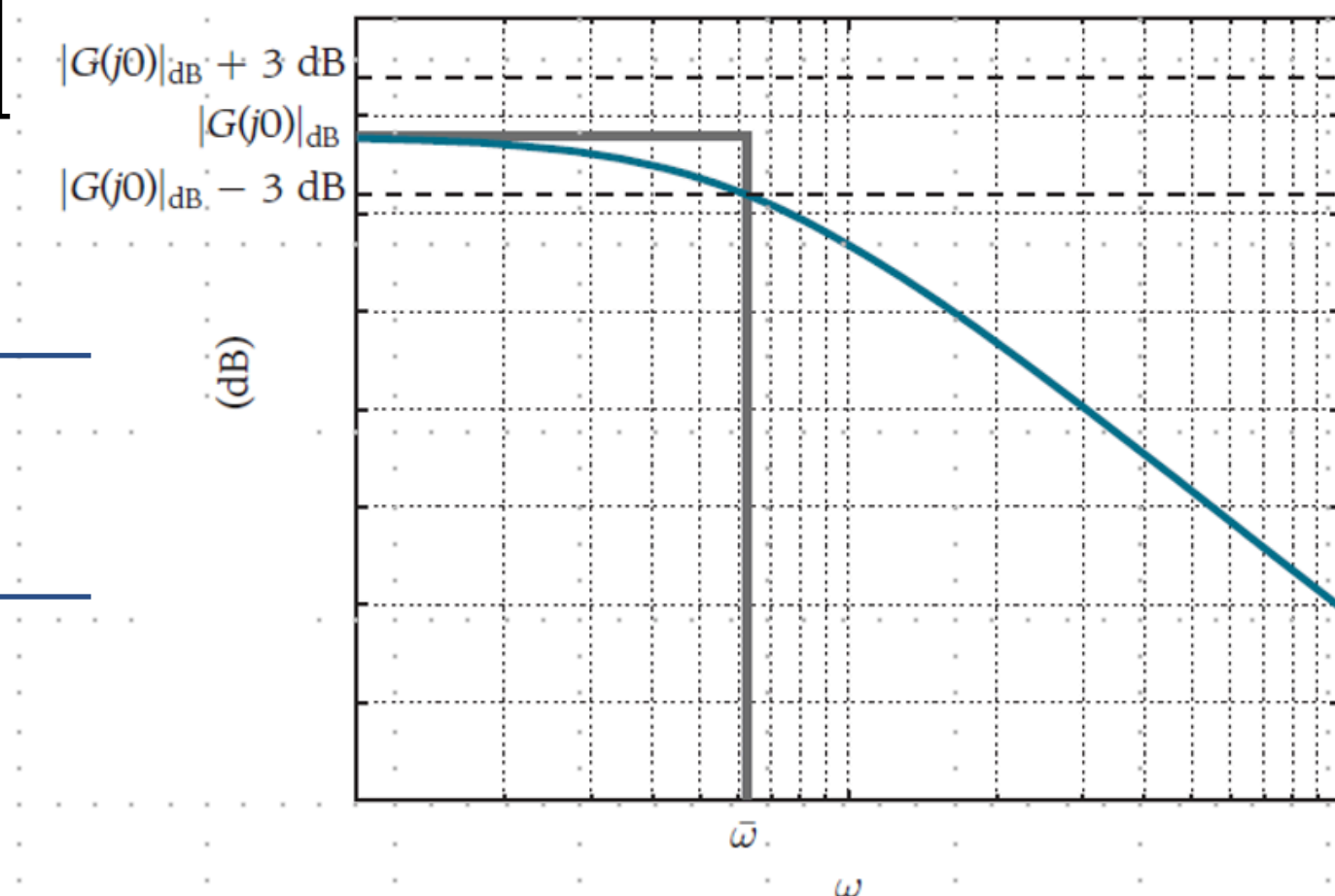


A LP Pre-filter is used to filter the reference signal to alleviate control peaking phenomena

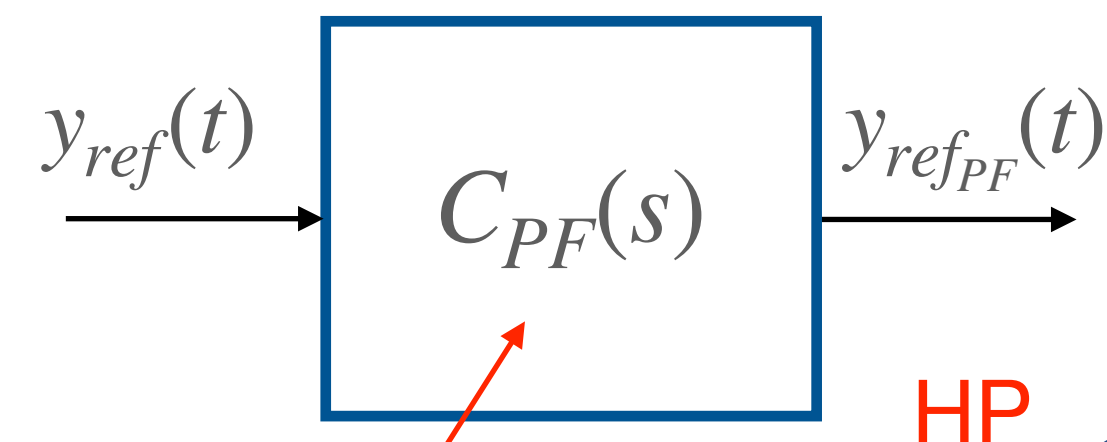
$$\omega_{BW_{PF}} \geq \omega_{BW_{CL}}$$

Care must be taken not to reduce the closed-loop bandwidth $\omega_{BW_{CL}}$

Frequency response of LP filter (magnitude)

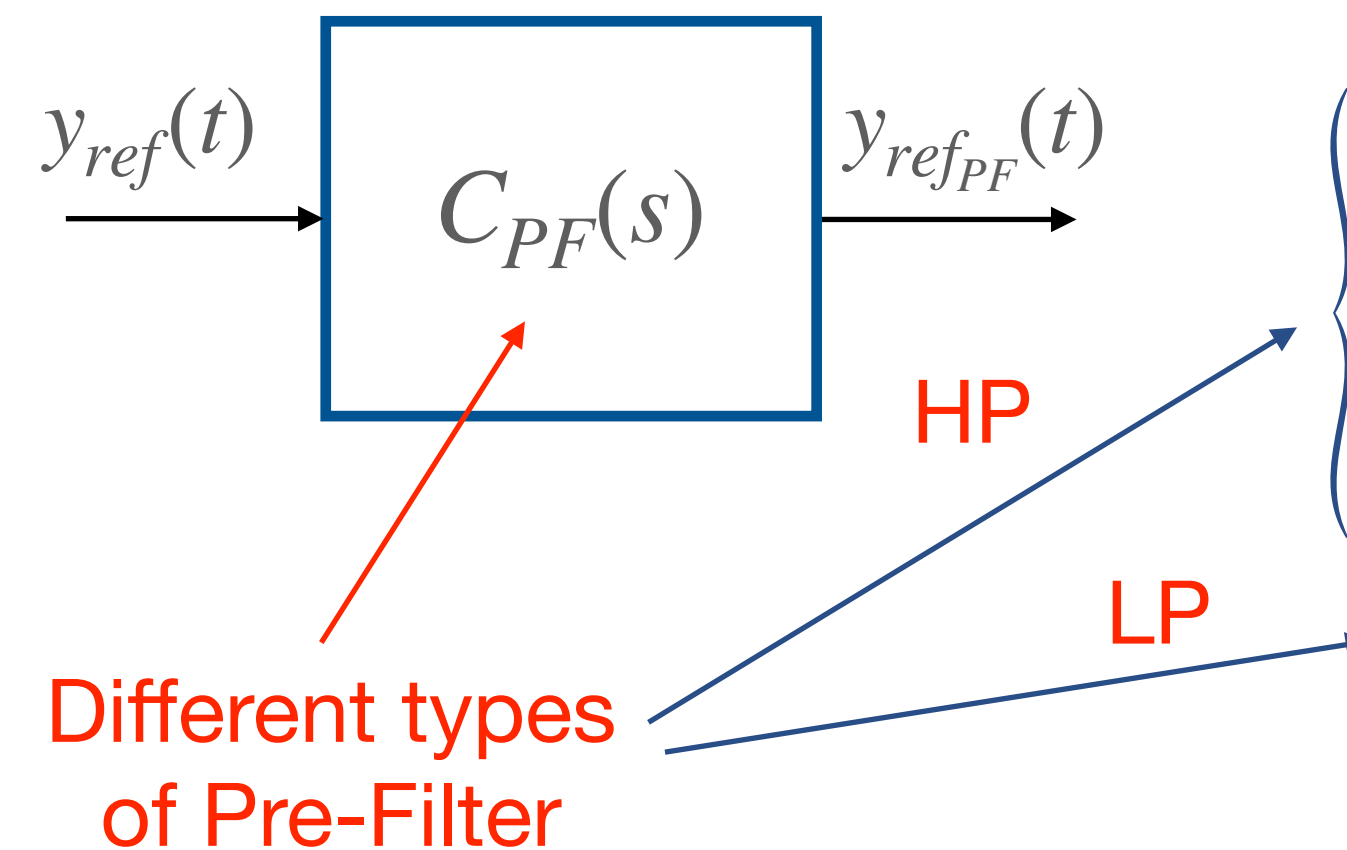


Pre-filter Based Control Scheme: Case 3



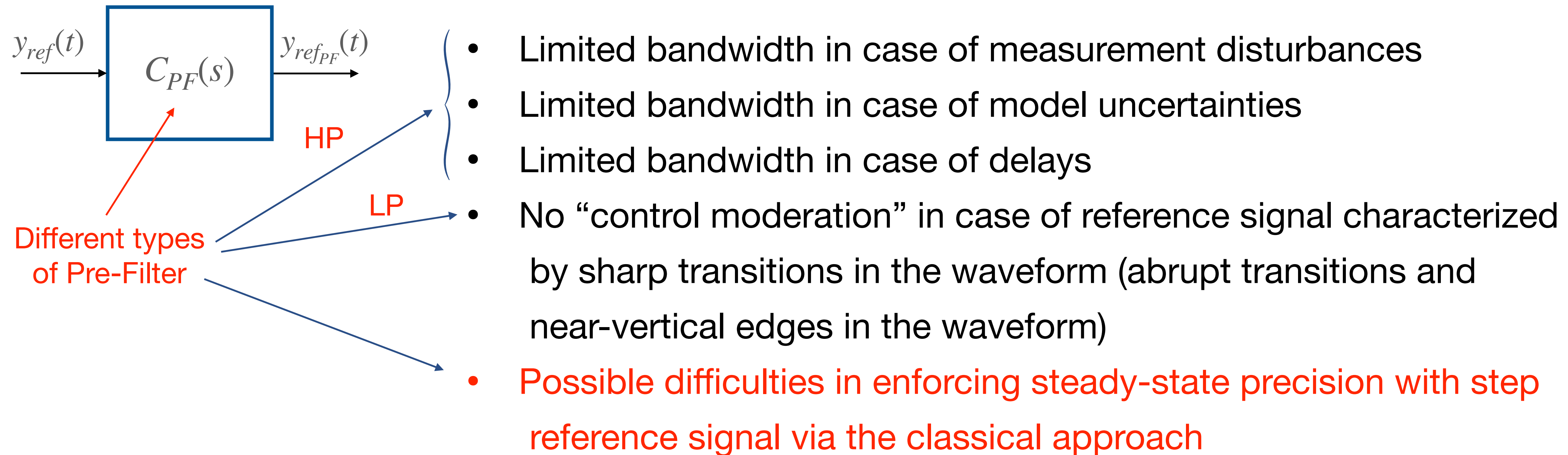
- Limited bandwidth in case of measurement disturbances
- Limited bandwidth in case of model uncertainties
- Limited bandwidth in case of delays
- No “control moderation” in case of reference signal characterized by sharp transitions in the waveform (abrupt transitions and near-vertical edges in the waveform)

Pre-filter Based Control Scheme: Case 3

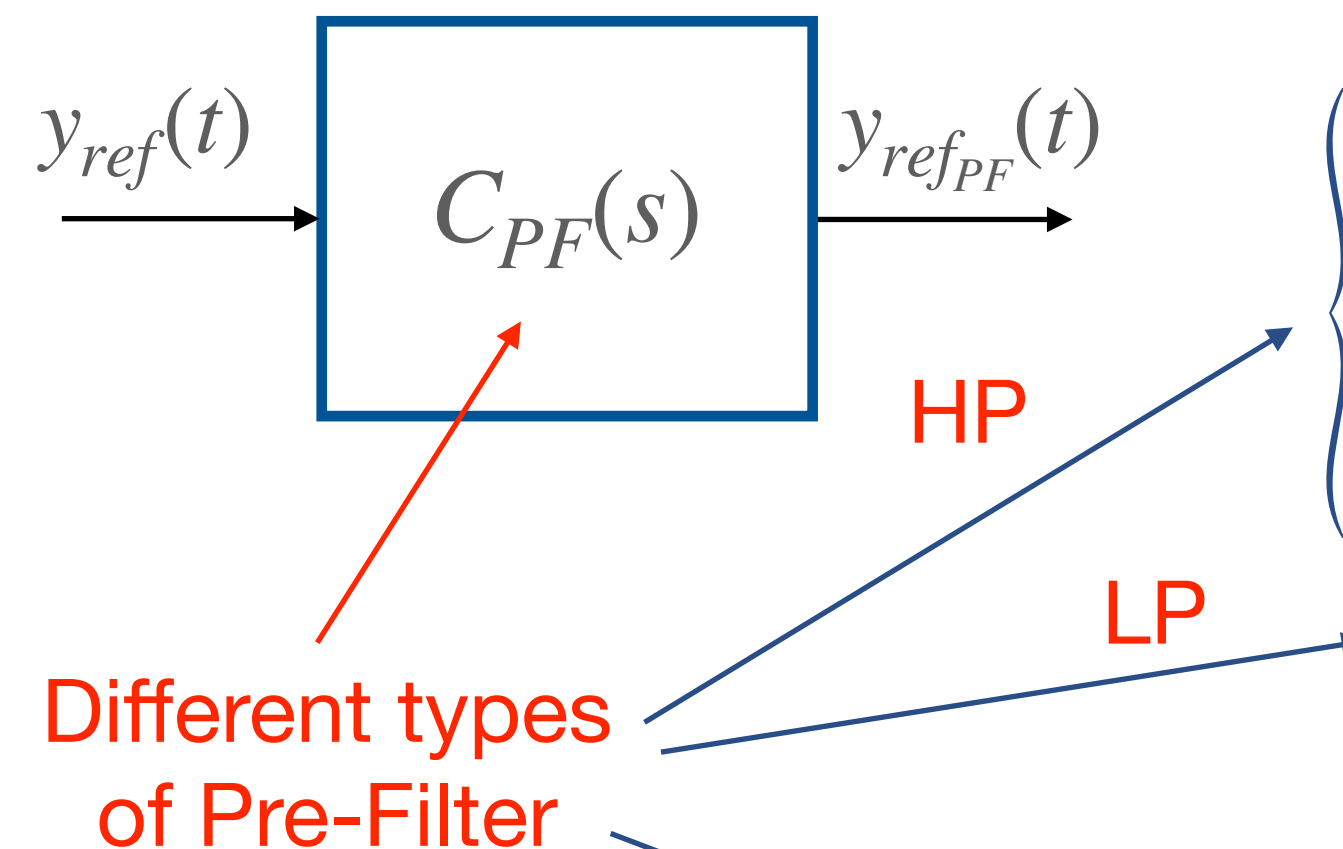


- Limited bandwidth in case of measurement disturbances
- Limited bandwidth in case of model uncertainties
- Limited bandwidth in case of delays
- No “control moderation” in case of reference signal characterized by sharp transitions in the waveform (abrupt transitions and near-vertical edges in the waveform)

Pre-filter Based Control Scheme: Case 3

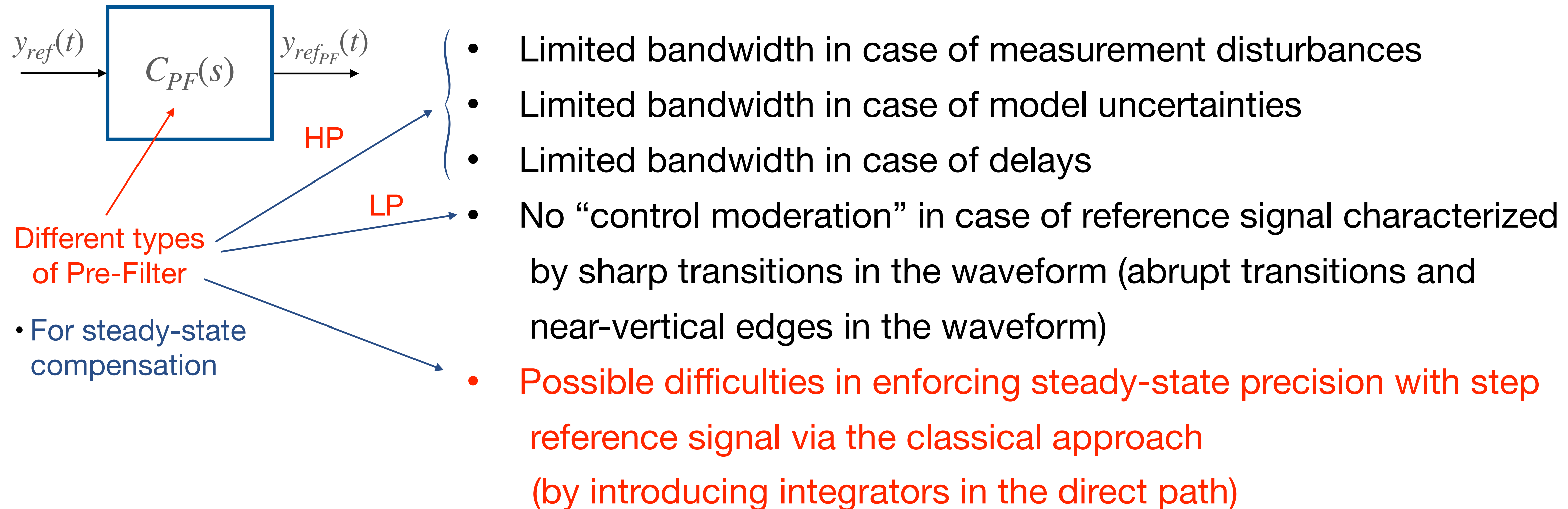


Pre-filter Based Control Scheme: Case 3

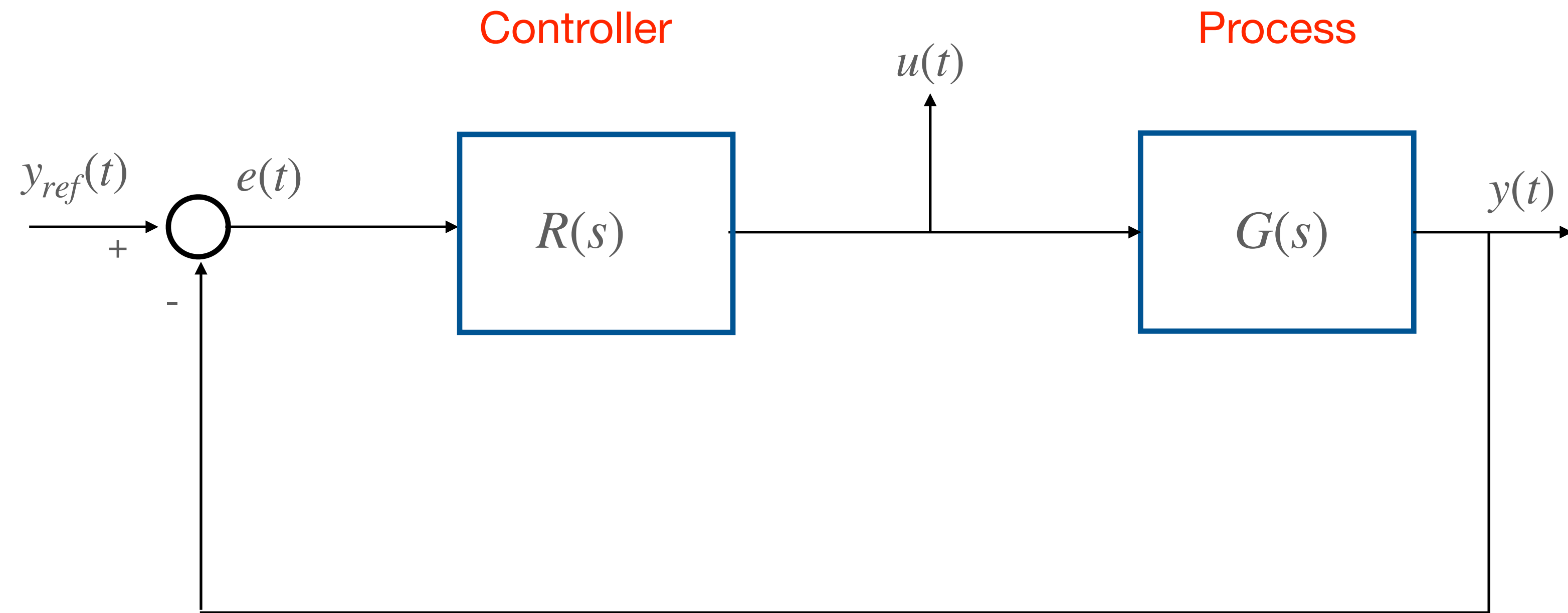


- Limited bandwidth in case of measurement disturbances
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- Limited bandwidth in case of delays
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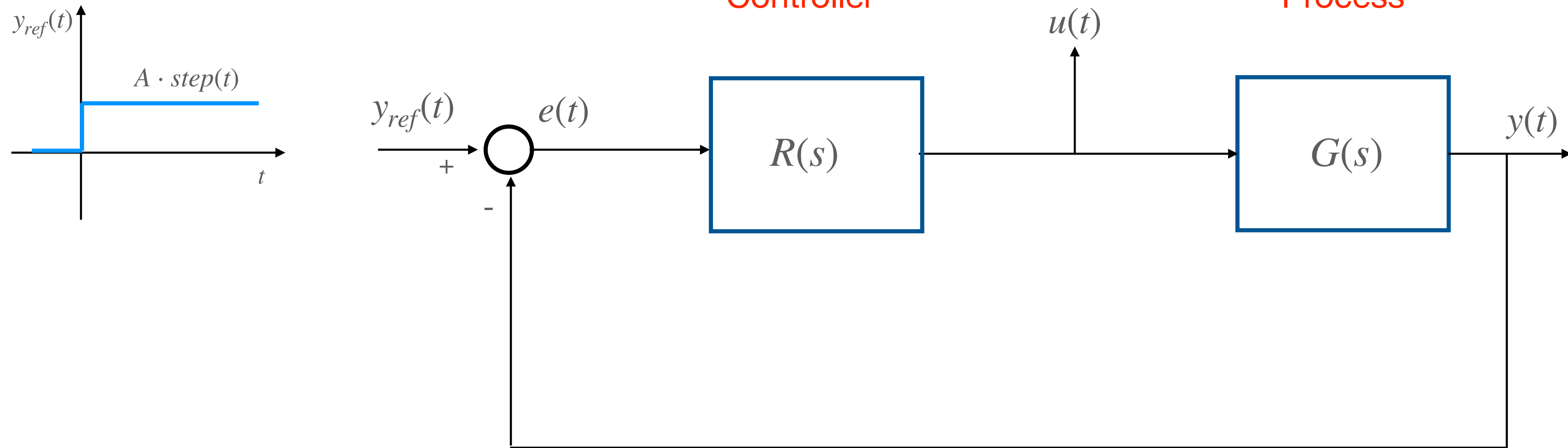
Pre-filter Based Control Scheme: Case 3



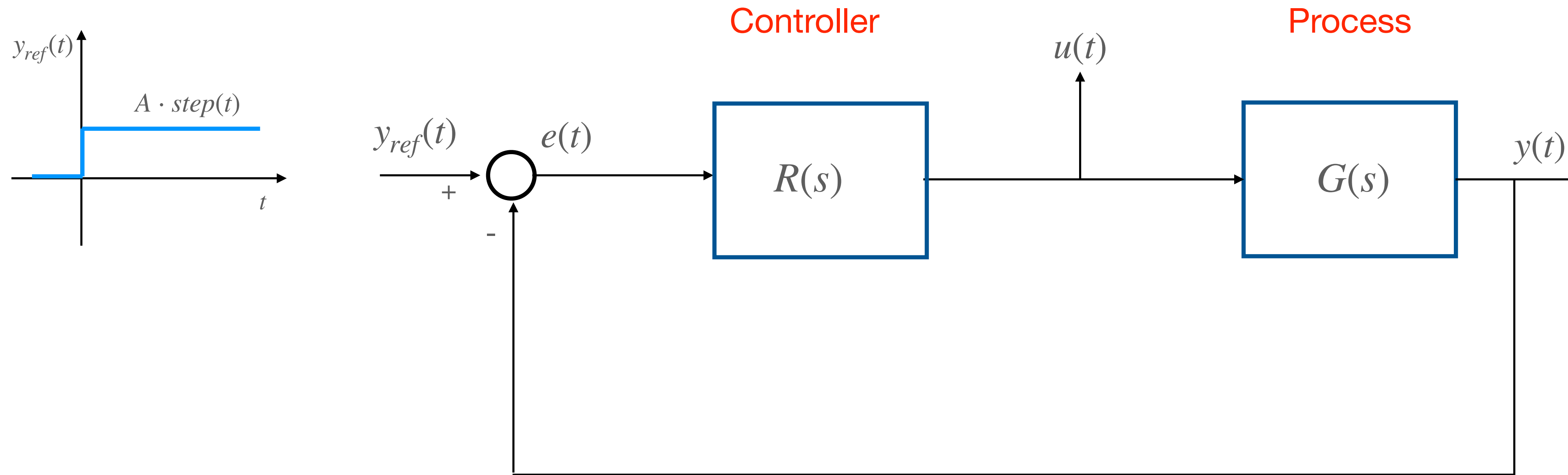
Basic Control Scheme



Basic Control Scheme



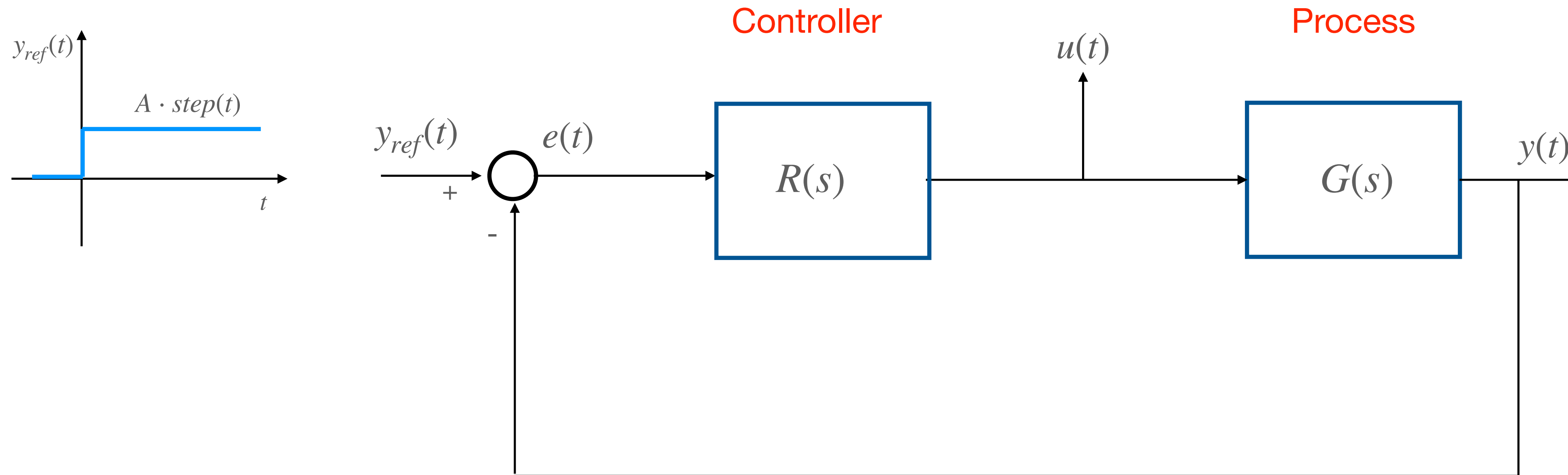
Basic Control Scheme



Block Algebra Rule: $y_{ref} \rightarrow y : F(s) = \frac{R(s)G(s)}{1 + R(s)G(s)} = \frac{L(s)}{1 + L(s)}$

$$y \approx y_{ref} : F(0) = 1$$

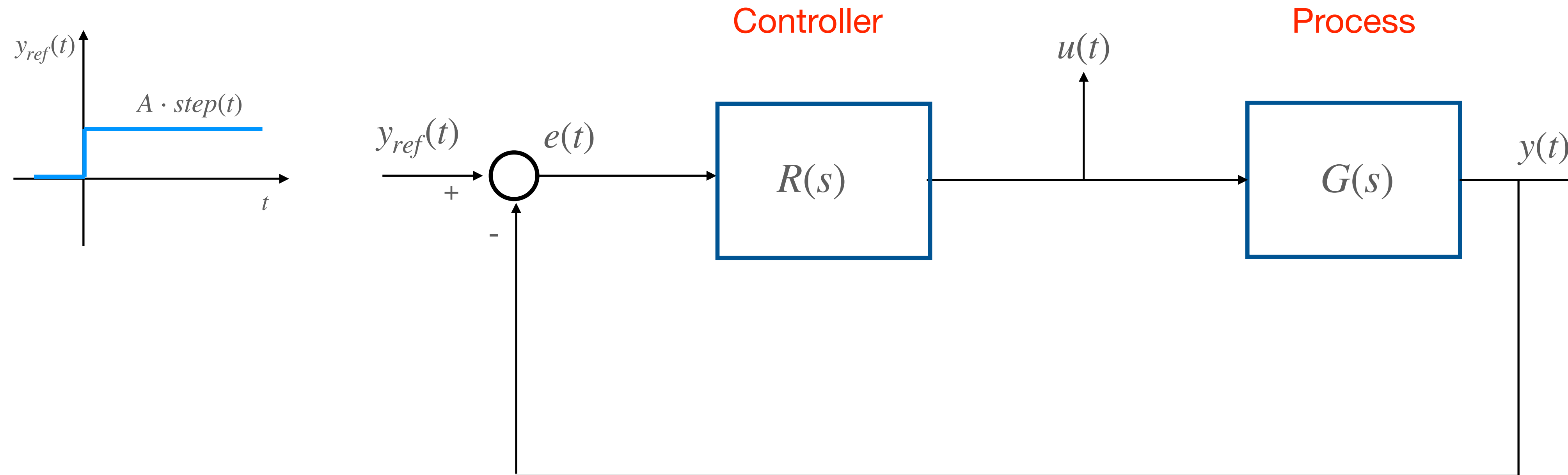
Basic Control Scheme



Block Algebra Rule: $y_{ref} \rightarrow y : F(s) = \frac{R(s)G(s)}{1 + R(s)G(s)} = \frac{L(s)}{1 + L(s)}$

$y \approx y_{ref} : F(0) = 1 \longrightarrow$ Impossible if $L(s)$ is not at least a TYPE 1 transfer function

Basic Control Scheme: Example

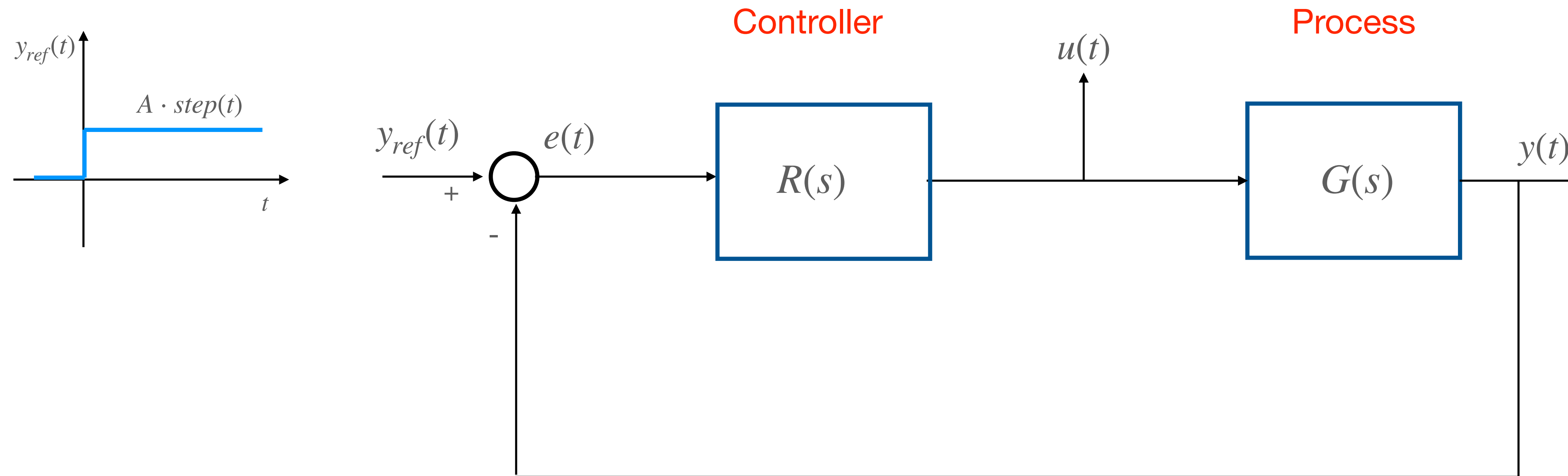


$$L(s) = \frac{100}{s + 10} \rightarrow F(s) = \frac{\frac{100}{s + 10}}{1 + \frac{100}{s + 10}} = \frac{100}{s + 110}$$

$$y_{ref} \rightarrow y : F(s) = \frac{R(s)G(s)}{1 + R(s)G(s)} = \frac{L(s)}{1 + L(s)}$$

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Basic Control Scheme: Example



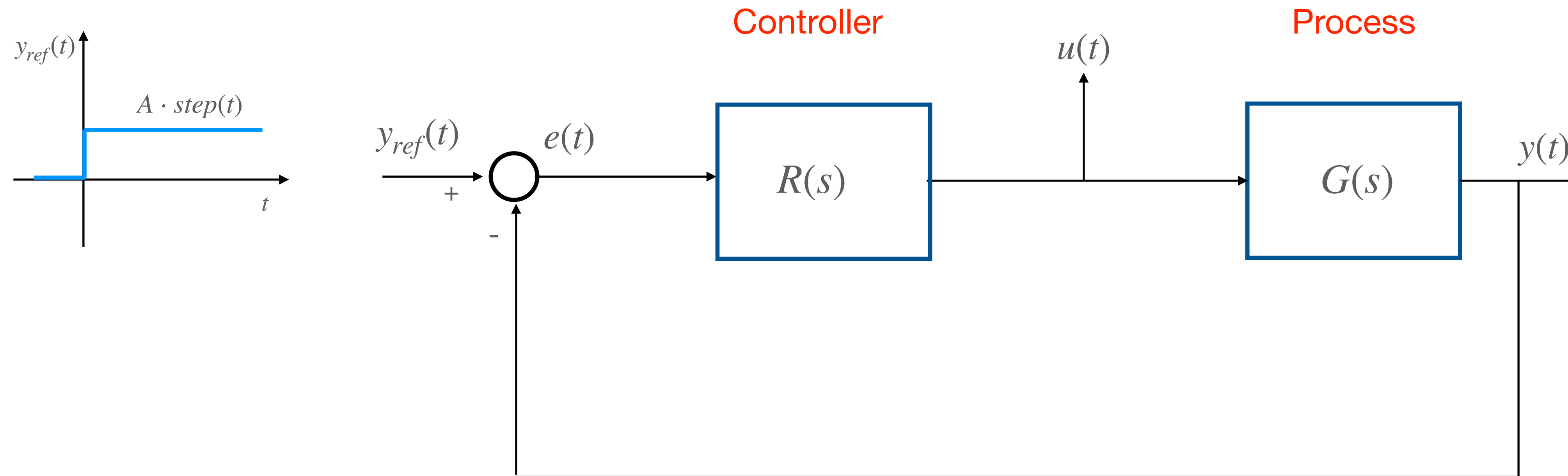
$$L(s) = \frac{100}{s + 10} \rightarrow F(s) = \frac{\frac{100}{s + 10}}{1 + \frac{100}{s + 10}} = \frac{100}{s + 110}$$

$$\lim_{s \rightarrow 0} F(s) = \frac{100}{110} = \frac{10}{11} \neq 1$$

$$y_{ref} \rightarrow y : F(s) = \frac{R(s)G(s)}{1 + R(s)G(s)} = \frac{L(s)}{1 + L(s)}$$

$y \approx y_{ref} : F(0) = 1 \longrightarrow$ Impossible if $L(s)$ is not at least a TYPE 1 transfer function

Basic Control Scheme: Example

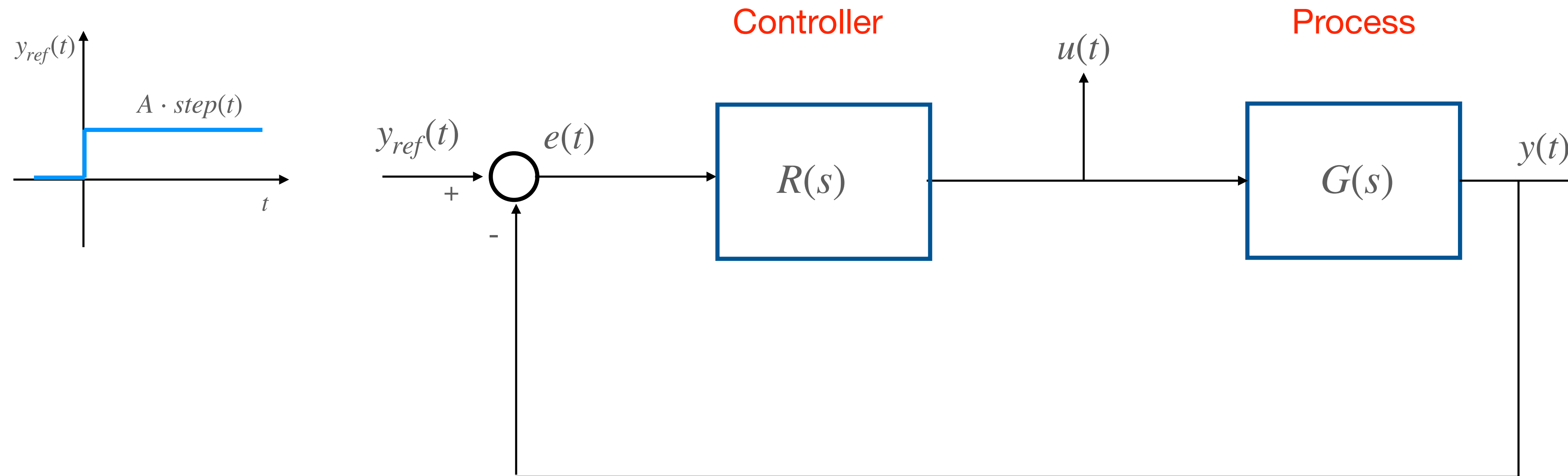


$$L(s) = \frac{100}{s(s+10)} \rightarrow F(s) = \frac{\frac{100}{s(s+10)}}{1 + \frac{100}{s(s+10)}} = \frac{100}{s(s+10) + 100}$$

$$y_{ref} \rightarrow y : F(s) = \frac{R(s)G(s)}{1 + R(s)G(s)} = \frac{L(s)}{1 + L(s)}$$

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Basic Control Scheme: Example



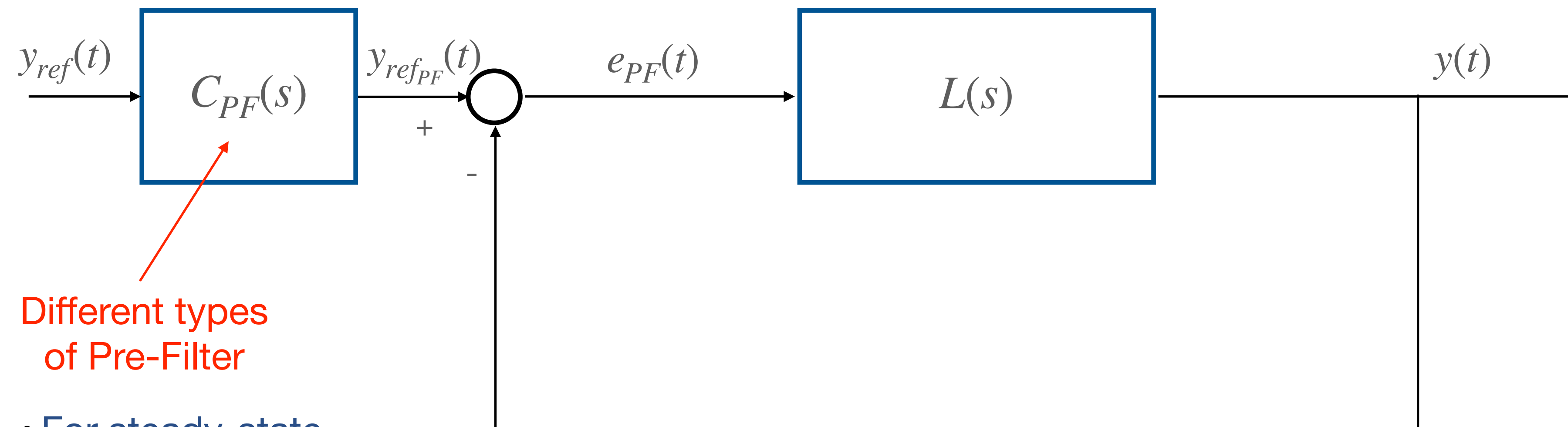
$$L(s) = \frac{100}{s(s+10)} \rightarrow F(s) = \frac{\frac{100}{s(s+10)}}{1 + \frac{100}{s(s+10)}} = \frac{100}{s(s+10) + 100}$$

$$\lim_{s \rightarrow 0} F(s) = \frac{100}{100} = 1$$

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$y \approx y_{ref} : F(0) = 1 \longrightarrow$ Impossible if $L(s)$ is not at least a TYPE 1 transfer function

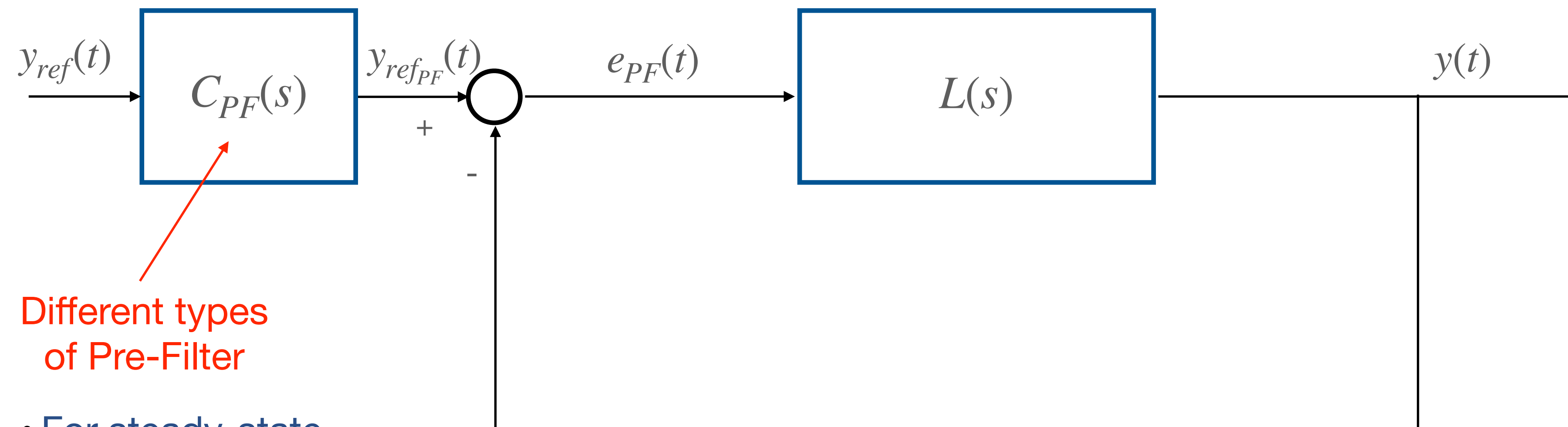
Pre-filter Based Control Scheme: Case 3



Different types
of Pre-Filter

- For steady-state compensation

Pre-filter Based Control Scheme: Case 3



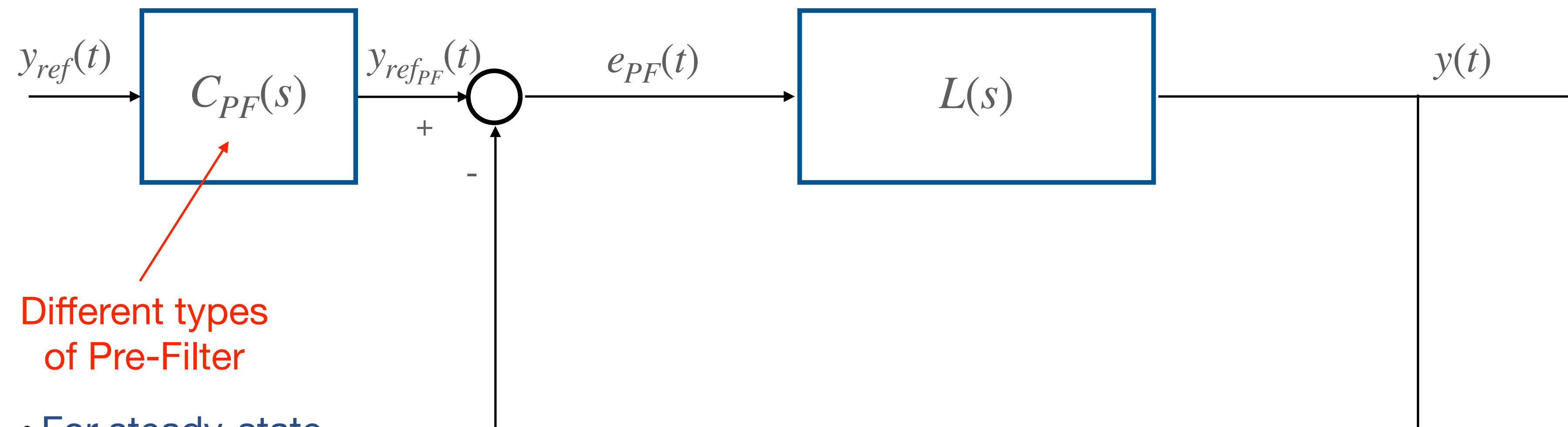
Different types
of Pre-Filter

- For steady-state compensation

Assumptions:

- $C_{PF}(s)$ As. Stable
- Proper
- Unitary gain

Pre-filter Based Control Scheme: Case 3



Different types
of Pre-Filter

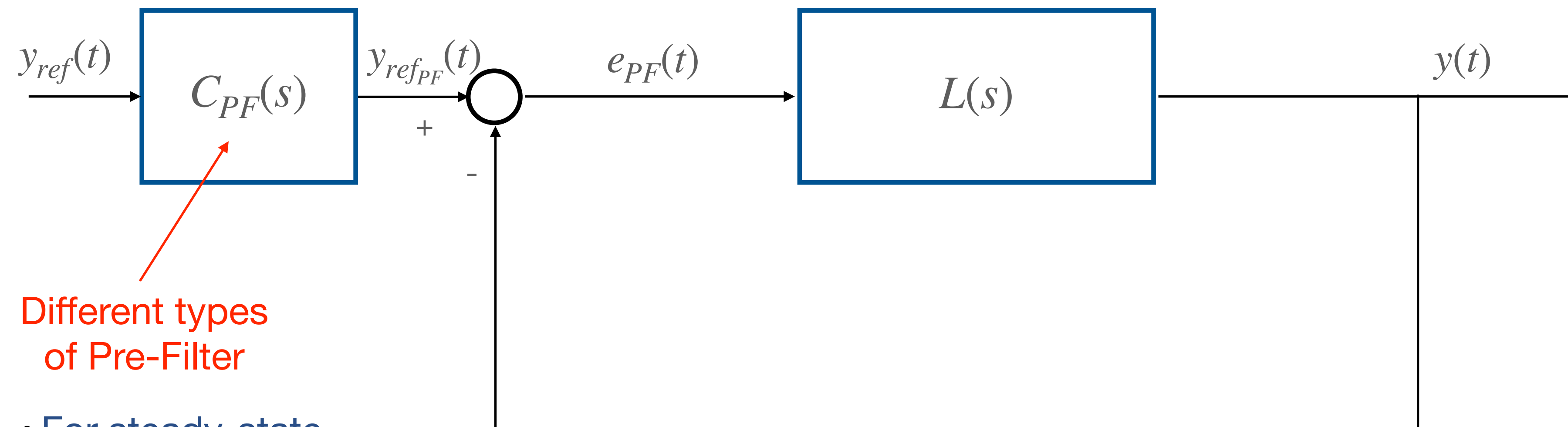
- For steady-state compensation

Assumptions:

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- Proper
- Unitary gain

Block Algebra Rule: $y_{ref} \rightarrow y : C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = C_{PF}(s) \cdot F(s)$

Pre-filter Based Control Scheme: Case 3



Different types
of Pre-Filter

- For steady-state compensation

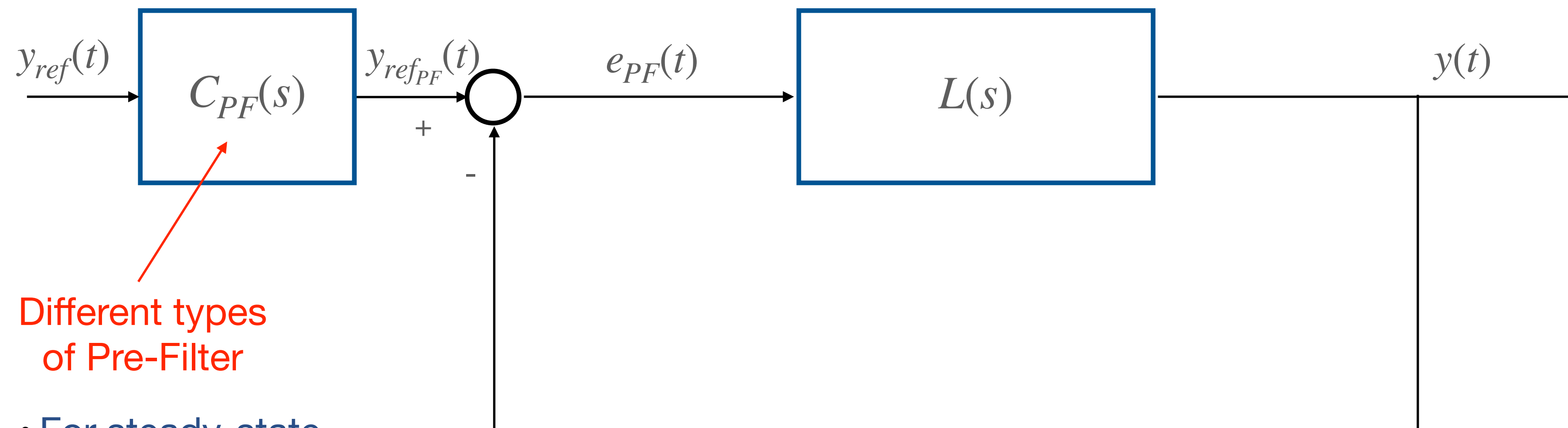
Assumptions:

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Block Algebra Rule: $y_{ref} \rightarrow y : C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = C_{PF}(s) \cdot F(s)$

$$y \approx y_{ref} : C_{PF}(0) \cdot F(0) = 1$$

Pre-filter Based Control Scheme: Case 3



Assumptions:

- $C_{PF}(s)$ As. Stable
- Proper
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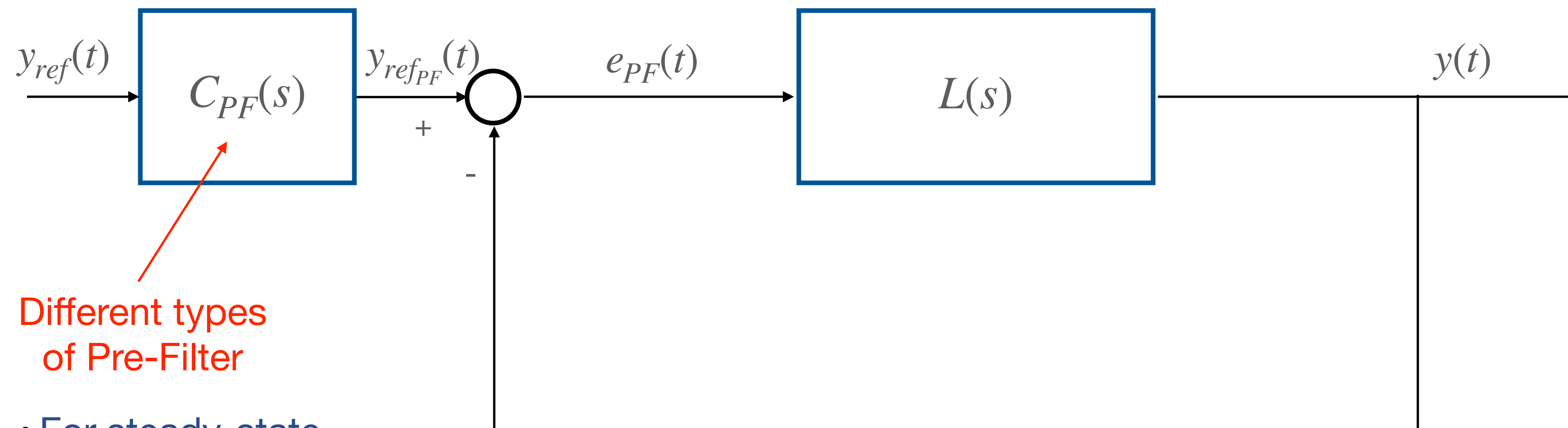
Different types
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- For steady-state compensation

Block Algebra Rule: $y_{ref} \rightarrow y : C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = C_{PF}(s) \cdot F(s)$

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Pre-filter Based Control Scheme: Case 3



Different types
of Pre-Filter

- For steady-state compensation

Assumptions:

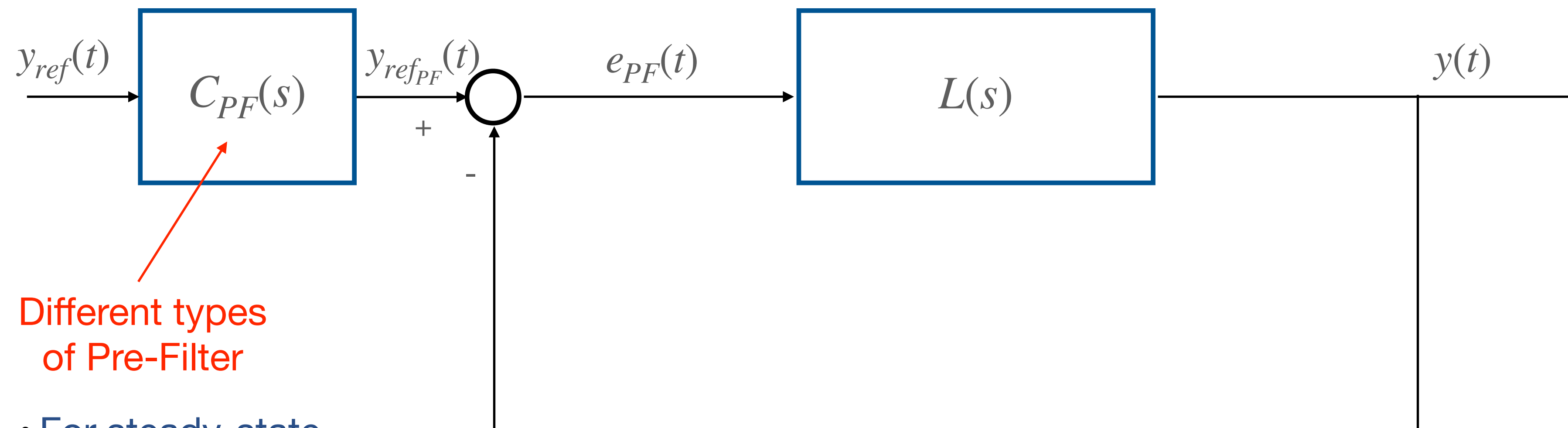
- $C_{PF}(s)$ As. Stable
- Proper
- Unitary gain

The filter could be
just a constant gain

Block Algebra Rule: $y_{ref} \rightarrow y : C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = C_{PF}(s) \cdot F(s)$

$$y \approx y_{ref} : C_{PF}(0) \cdot F(0) = 1 \longrightarrow C_{PF}(0) = \frac{1}{F(0)}$$

Pre-filter Based Control Scheme: Case 3



Assumptions:

- $C_{PF}(s)$ As. Stable
- Proper
- Unitary gain

Warning:
Use only when
strictly necessary!

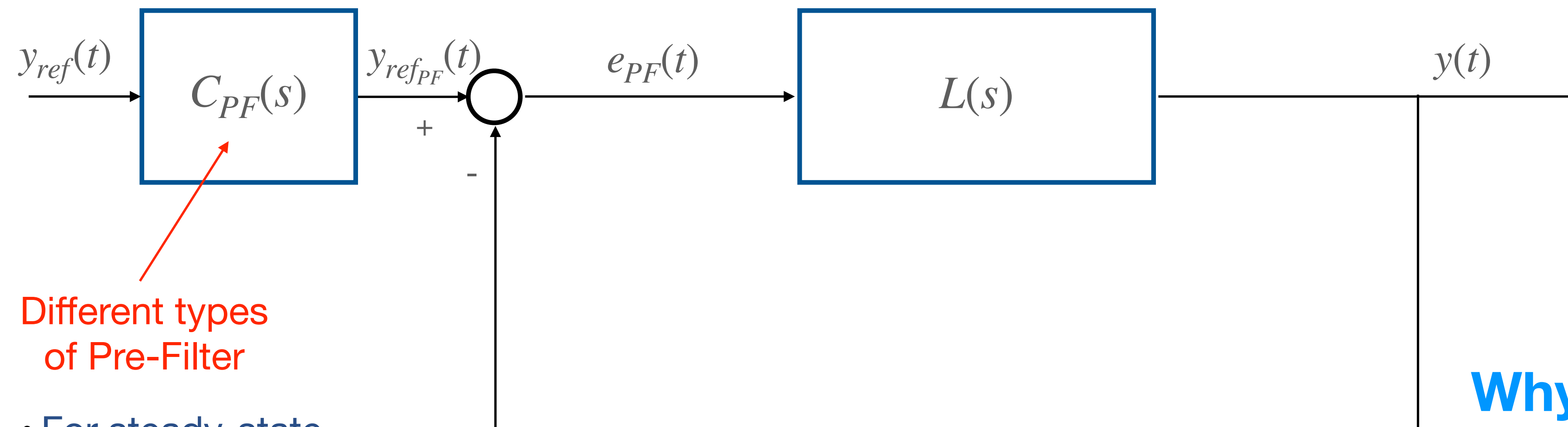
Different types
of Pre-Filter

- For steady-state compensation

Block Algebra Rule: $y_{ref} \rightarrow y : C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = C_{PF}(s) \cdot F(s)$

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Pre-filter Based Control Scheme: Case 3



Different types
of Pre-Filter

- For steady-state compensation

Assumptions:

- $C_{PF}(s)$ As. Stable
- Proper
- Unitary gain

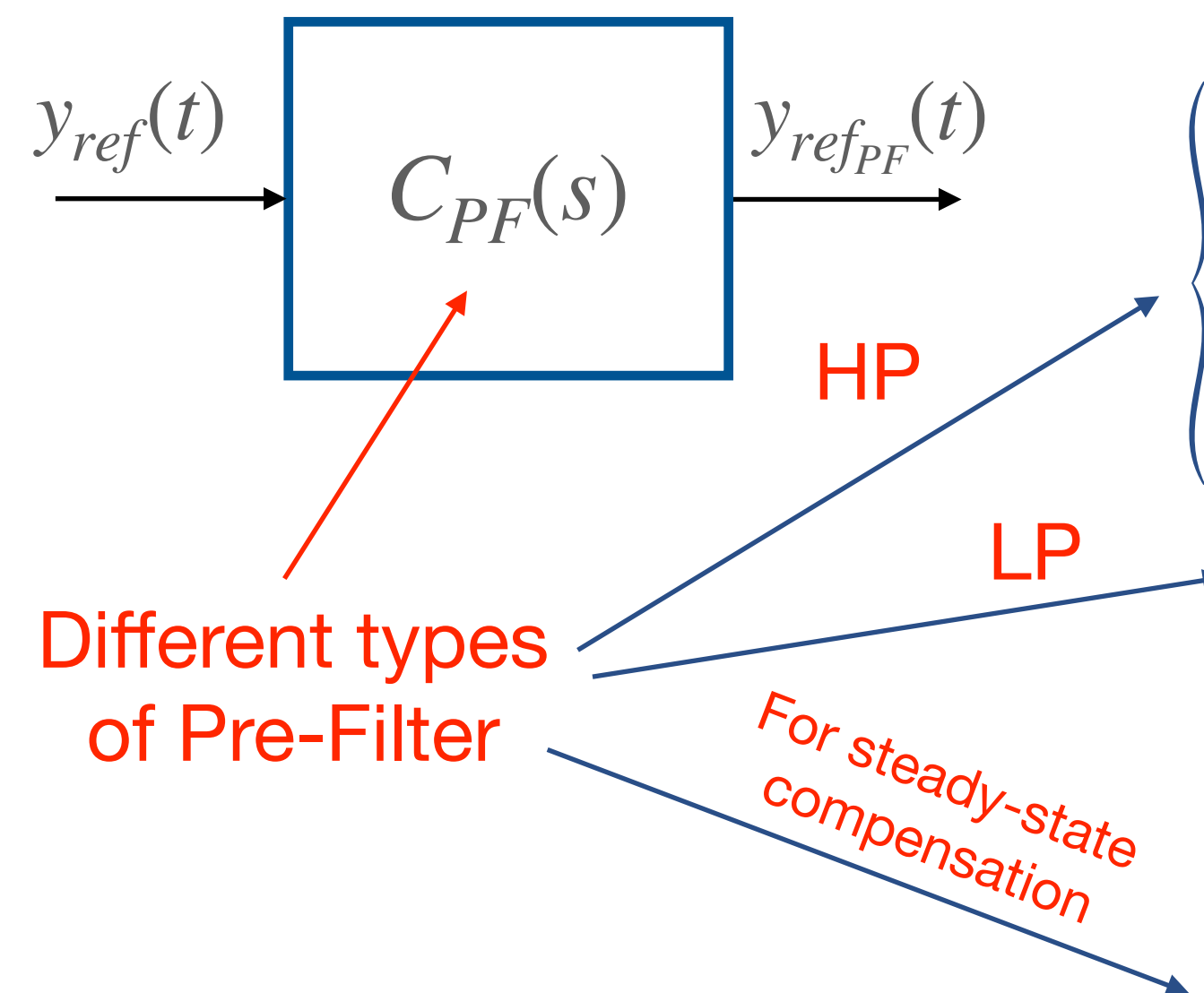
Why?

Warning:
Use only when
strictly necessary!

Block Algebra Rule: $y_{ref} \rightarrow y : C_{PF}(s) \cdot \frac{L(s)}{1 + L(s)} = C_{PF}(s) \cdot F(s)$

$$y \approx y_{ref} : C_{PF}(0) \cdot F(0) = 1 \longrightarrow C_{PF}(0) = \frac{1}{F(0)}$$

Pre-filter Based Control Scheme: Summary

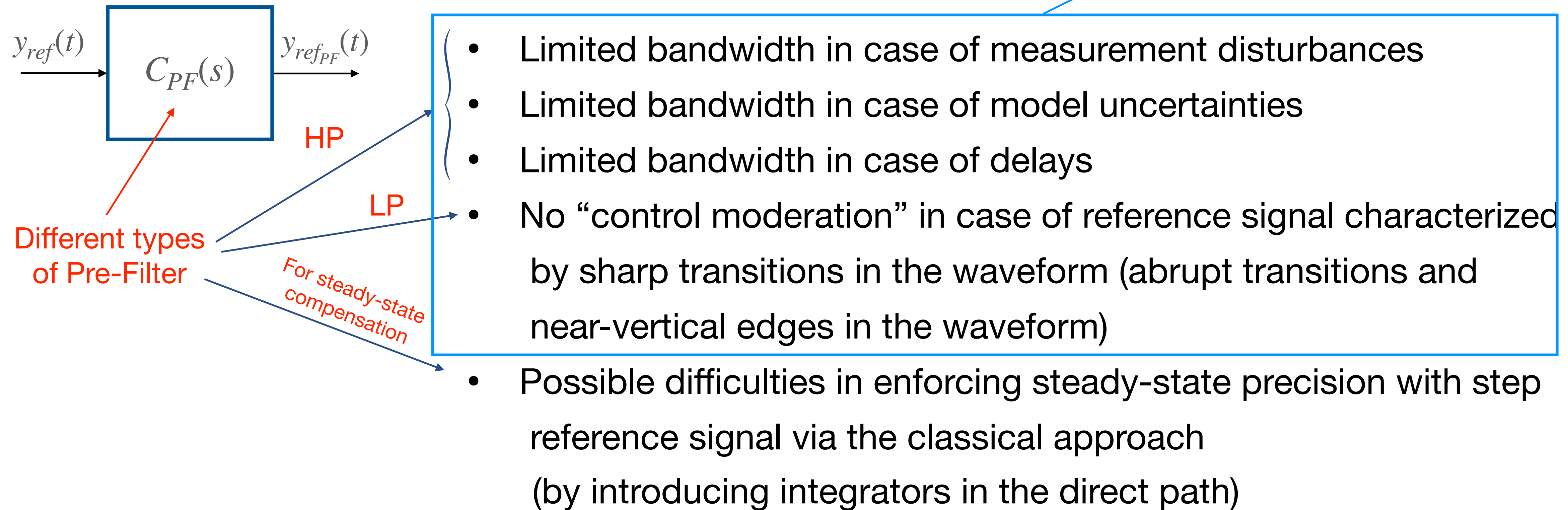


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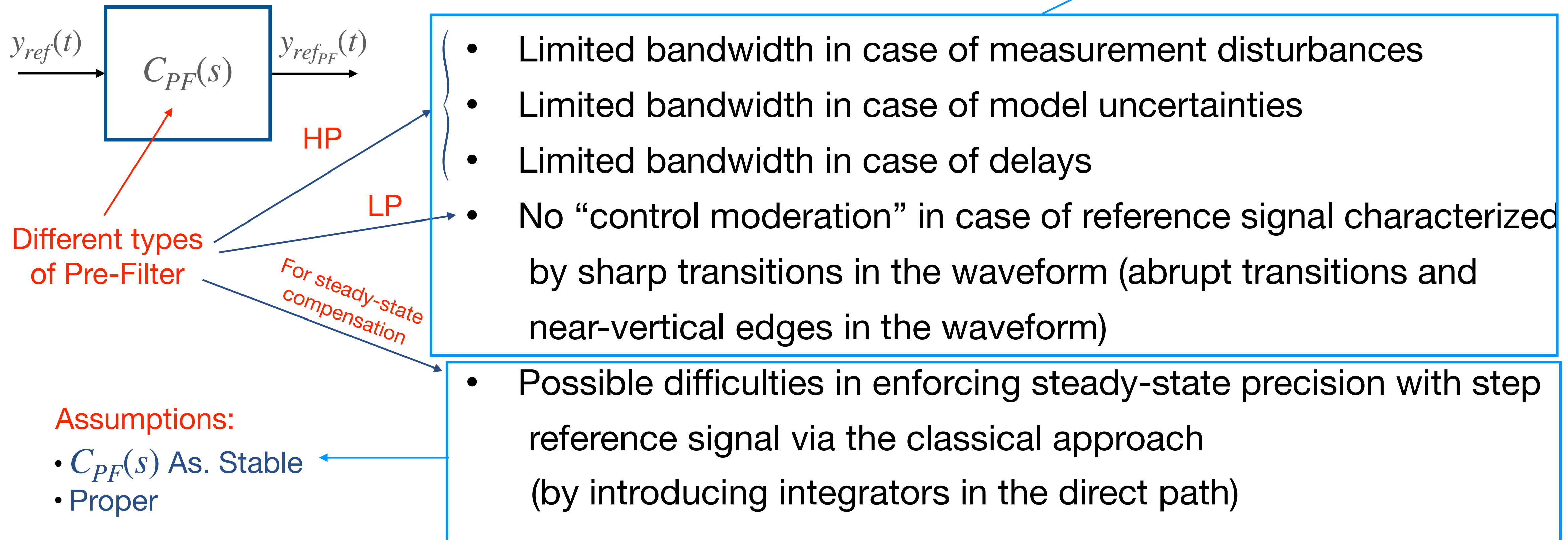
Pre-filter Based Control Scheme: Summary

Assumptions:

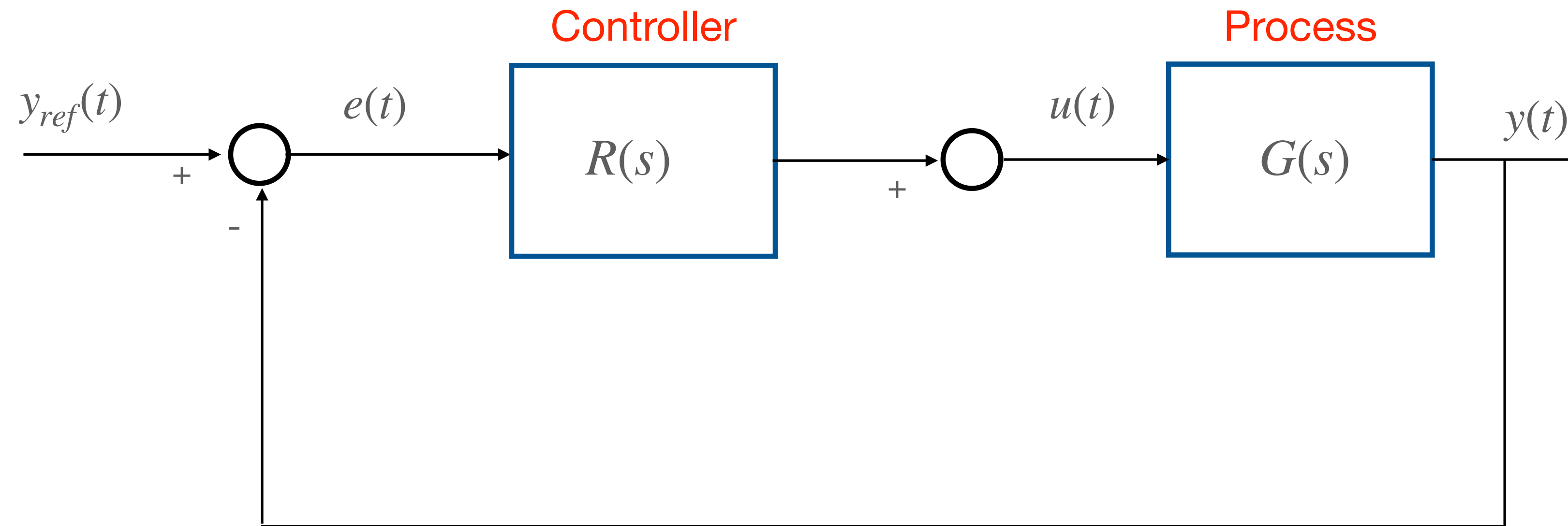
- $C_{PF}(s)$ As. Stable
- Proper
- Unitary gain



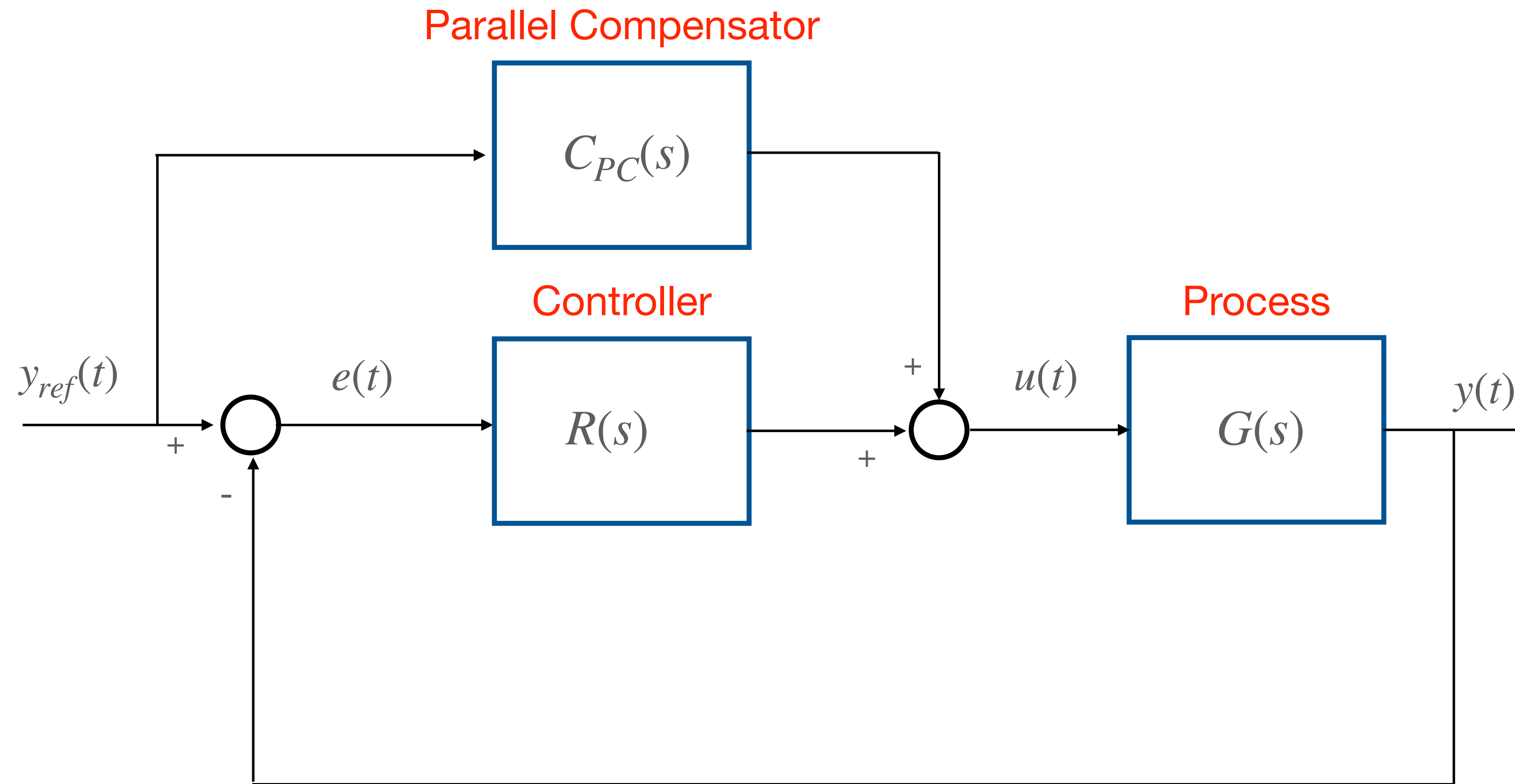
Pre-filter Based Control Scheme: Summary



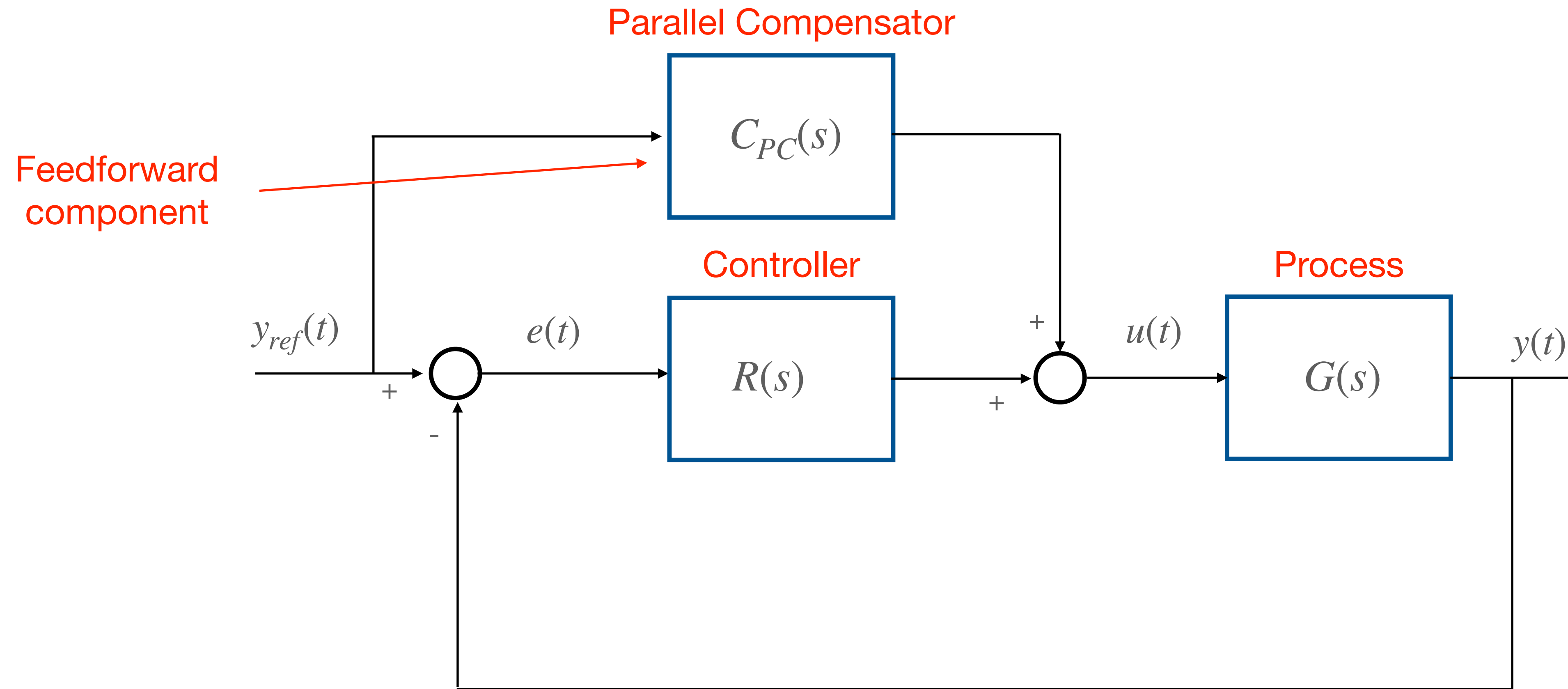
Parallel-compensator Based Control Scheme



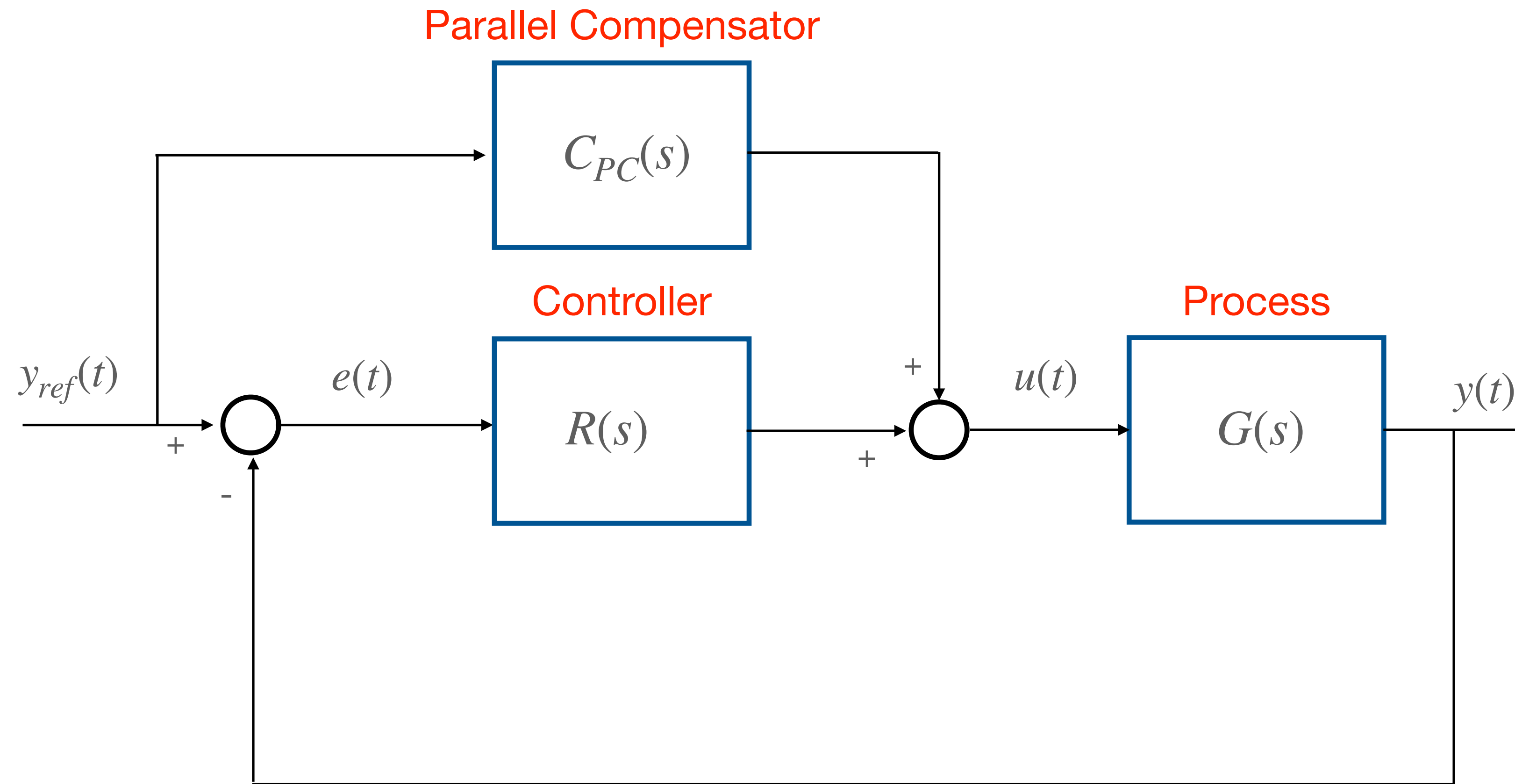
Parallel-compensator Based Control Scheme



Parallel-compensator Based Control Scheme

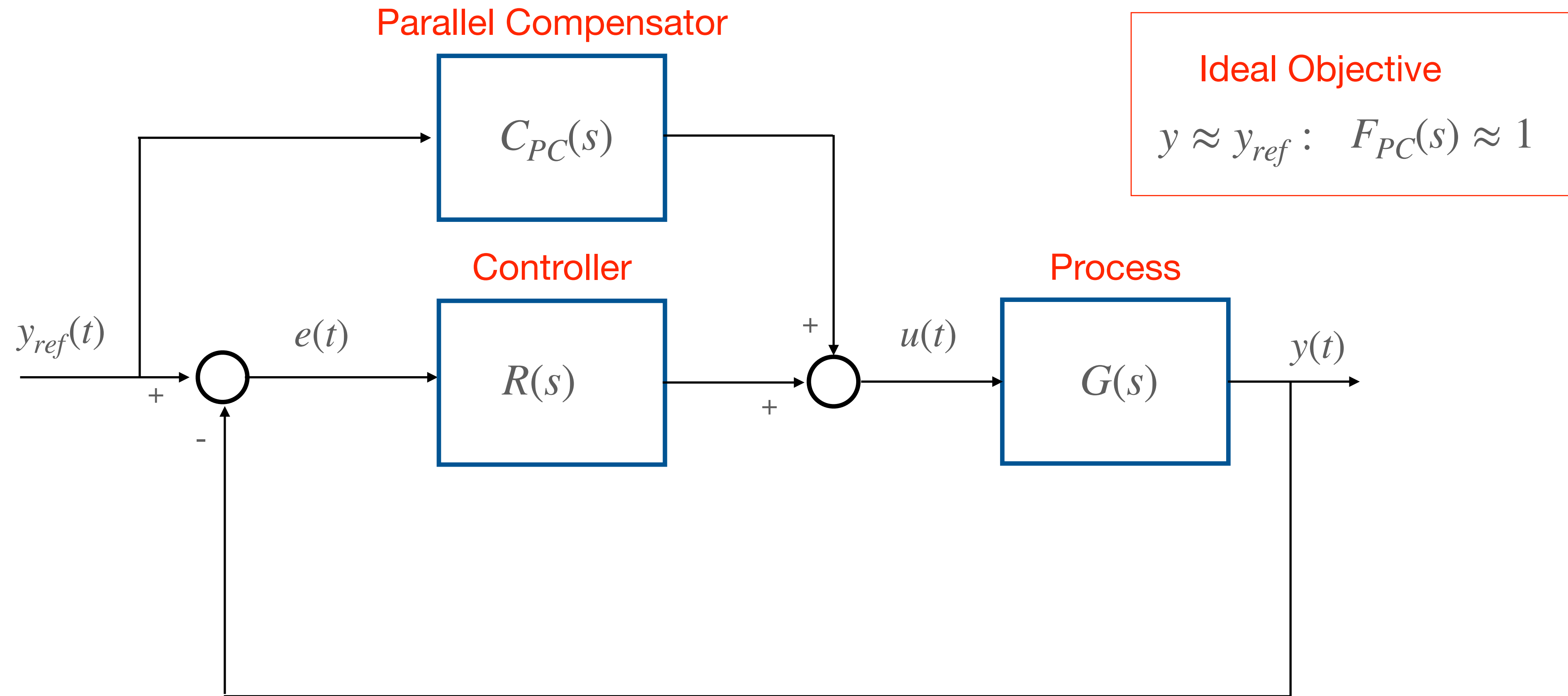


Parallel-compensator Based Control Scheme



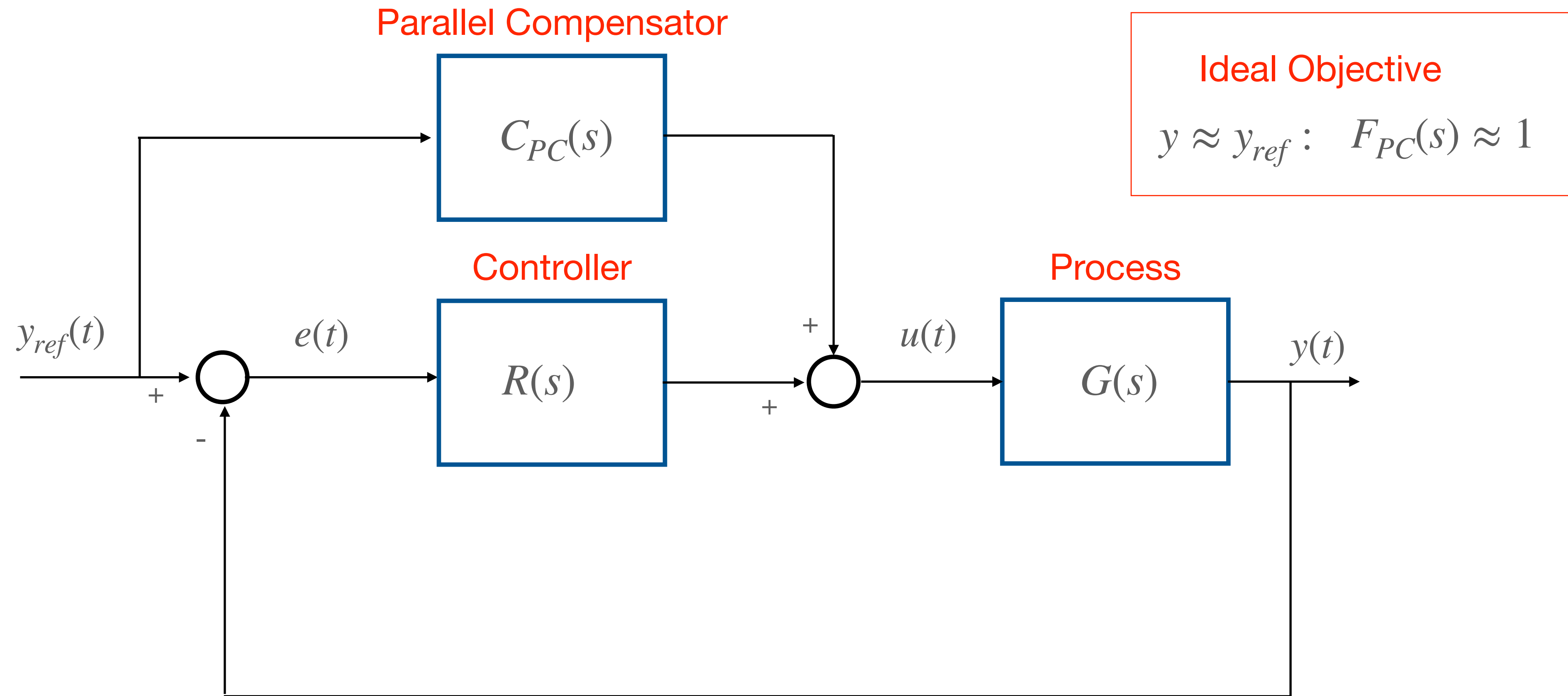
$$y_{ref} \rightarrow y : F_{PC}(s) = \frac{C_{PC}(s)G(s) + R(s)G(s)}{1 + R(s)G(s)}$$

Parallel-compensator Based Control Scheme



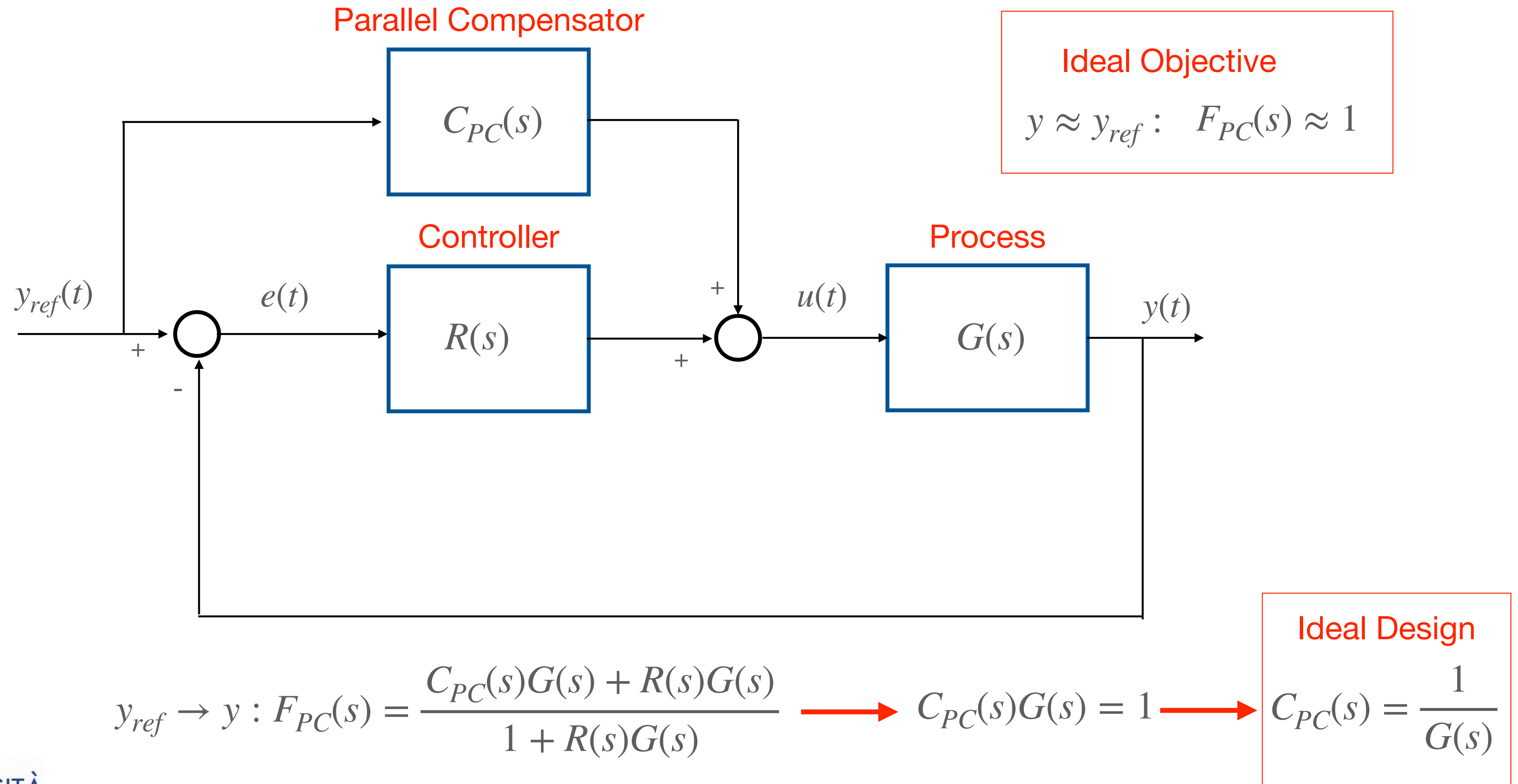
$$y_{ref} \rightarrow y : F_{PC}(s) = \frac{C_{PC}(s)G(s) + R(s)G(s)}{1 + R(s)G(s)}$$

Parallel-compensator Based Control Scheme

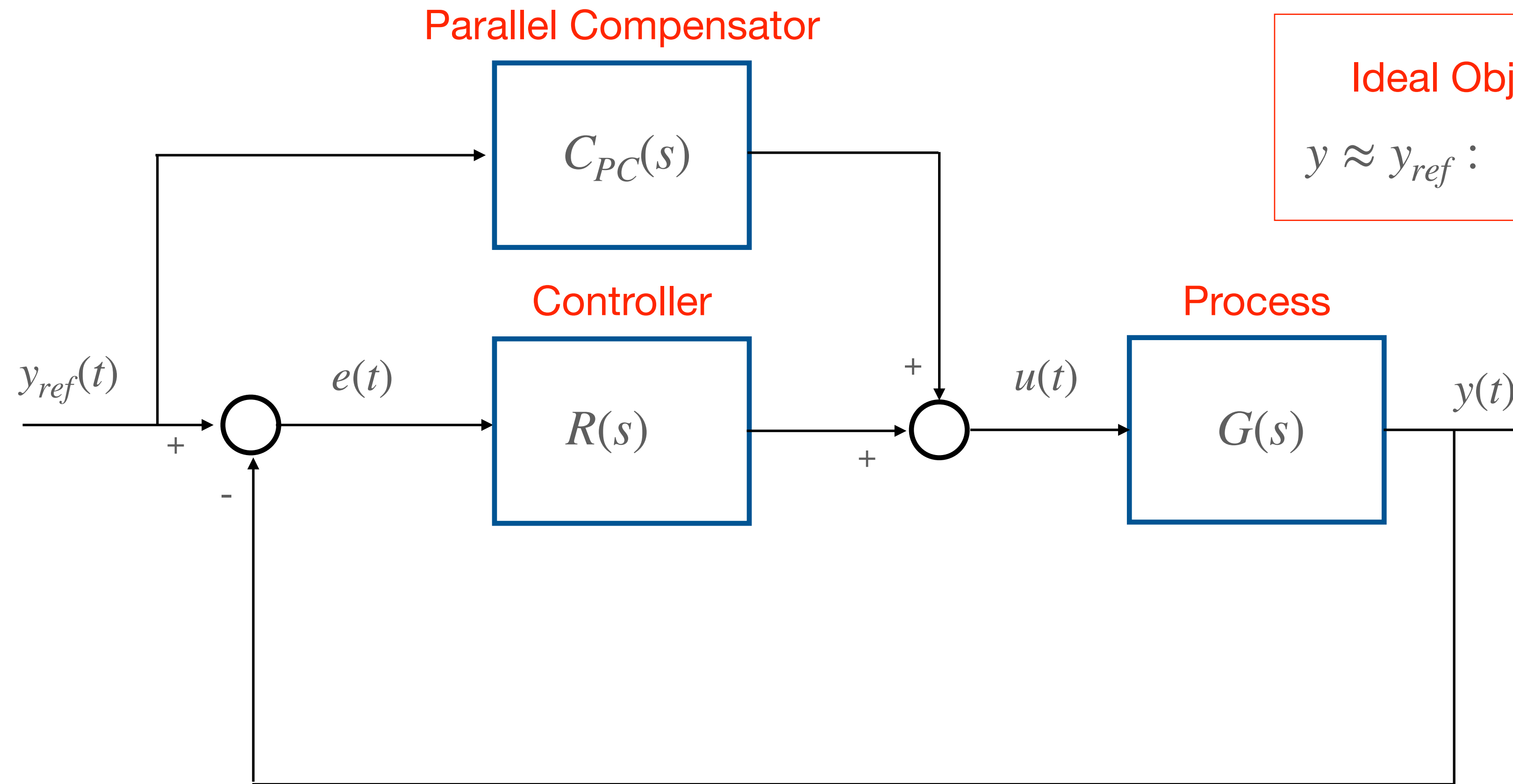


$$y_{ref} \rightarrow y : F_{PC}(s) = \frac{C_{PC}(s)G(s) + R(s)G(s)}{1 + R(s)G(s)} \longrightarrow C_{PC}(s)G(s) = 1$$

Parallel-compensator Based Control Scheme



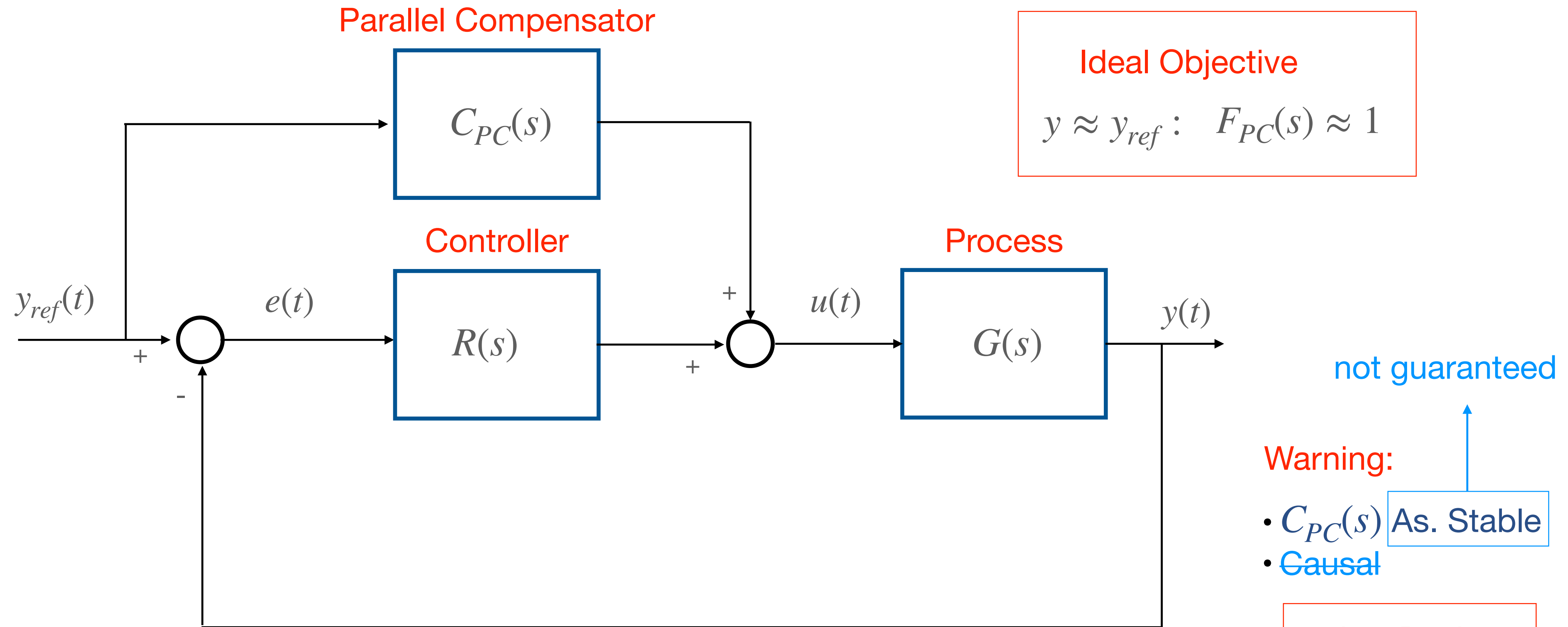
Parallel-compensator Based Control Scheme



$$y_{ref} \rightarrow y : F_{PC}(s) = \frac{C_{PC}(s)G(s) + R(s)G(s)}{1 + R(s)G(s)} \longrightarrow C_{PC}(s)G(s) = 1 \longrightarrow C_{PC}(s) = \frac{1}{G(s)}$$

Ideal Design

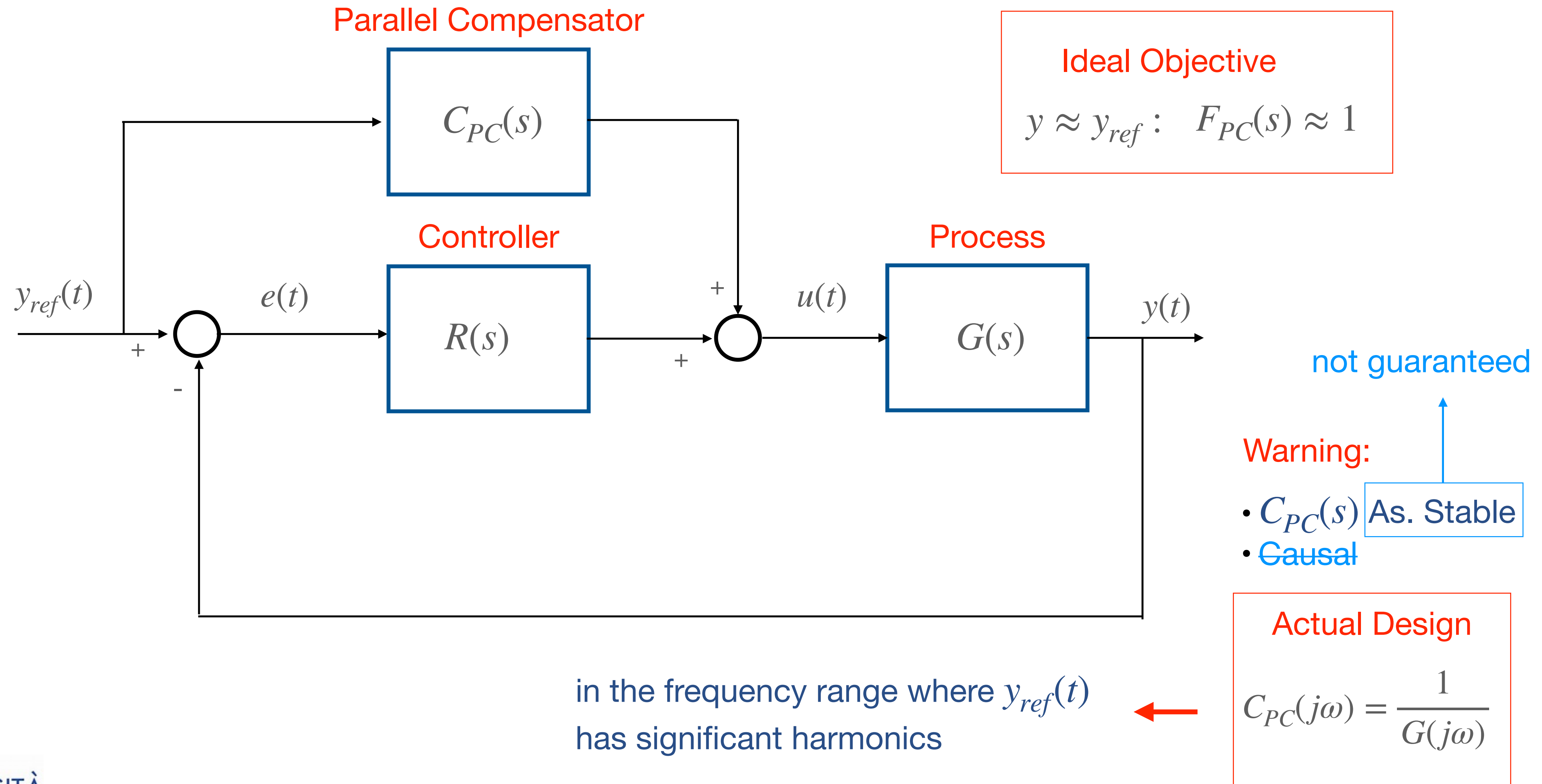
Parallel-compensator Based Control Scheme



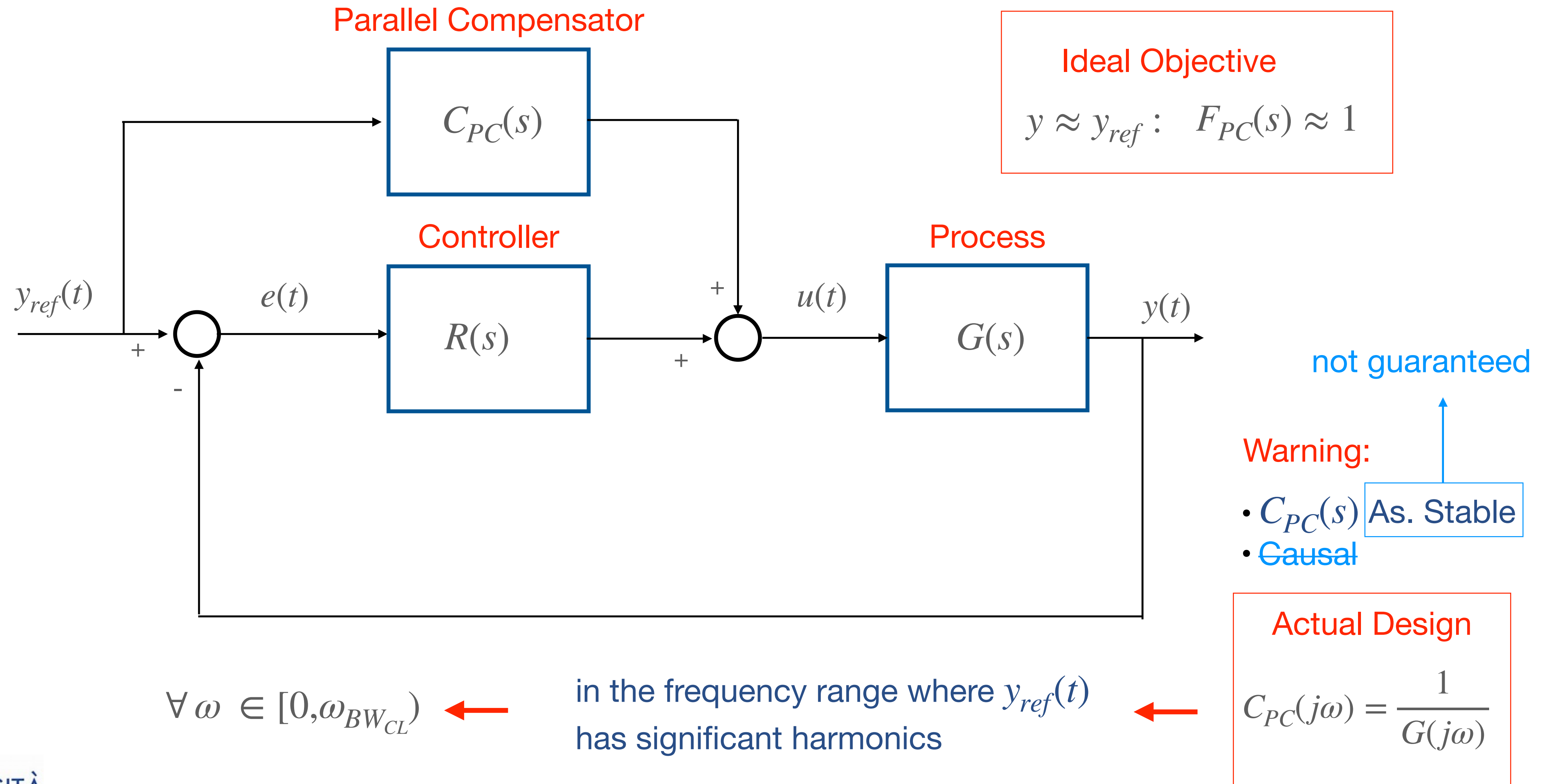
$$y_{ref} \rightarrow y : F_{PC}(s) = \frac{C_{PC}(s)G(s) + R(s)G(s)}{1 + R(s)G(s)} \longrightarrow C_{PC}(s)G(s) = 1 \longrightarrow C_{PC}(s) = \frac{1}{G(s)}$$

Ideal Design

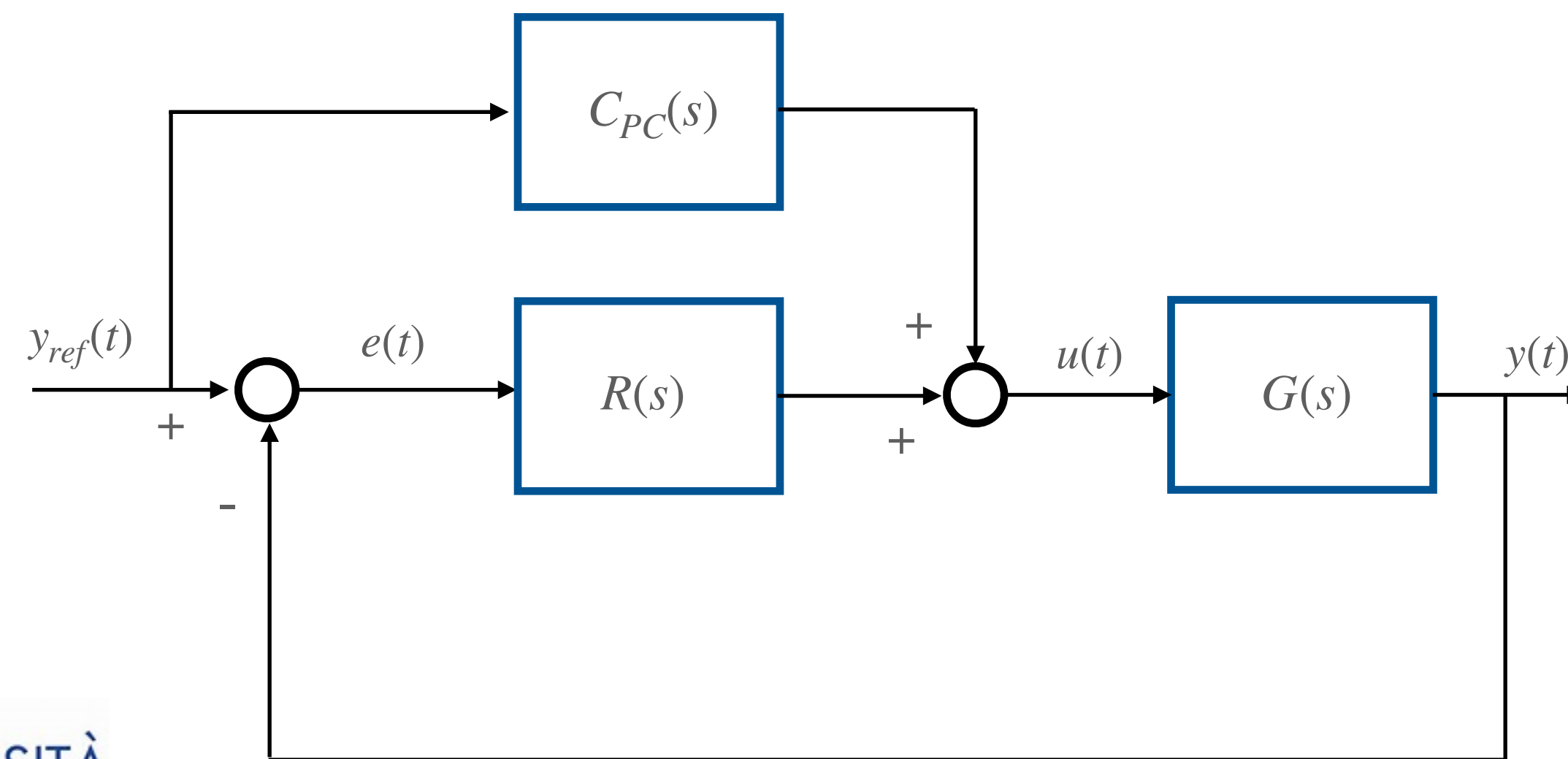
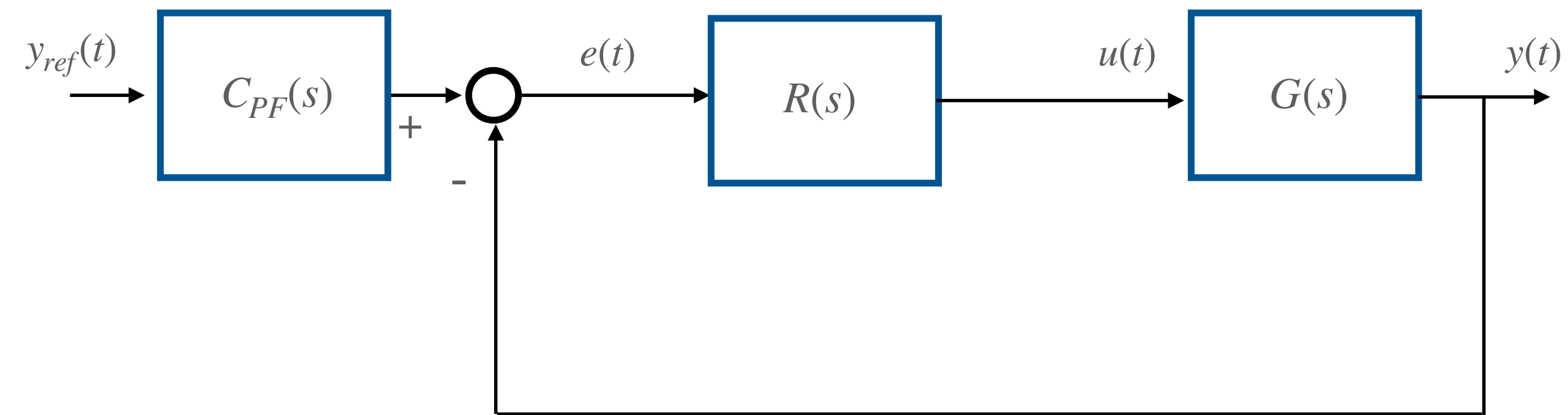
Parallel-compensator Based Control Scheme



Parallel-compensator Based Control Scheme

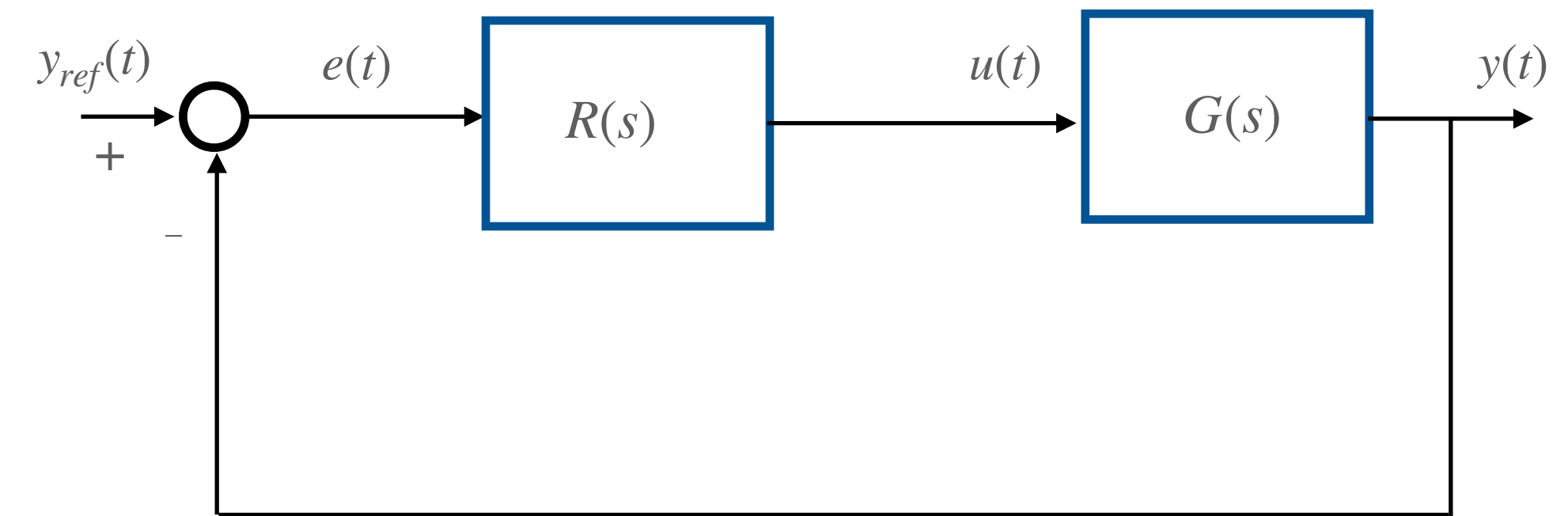
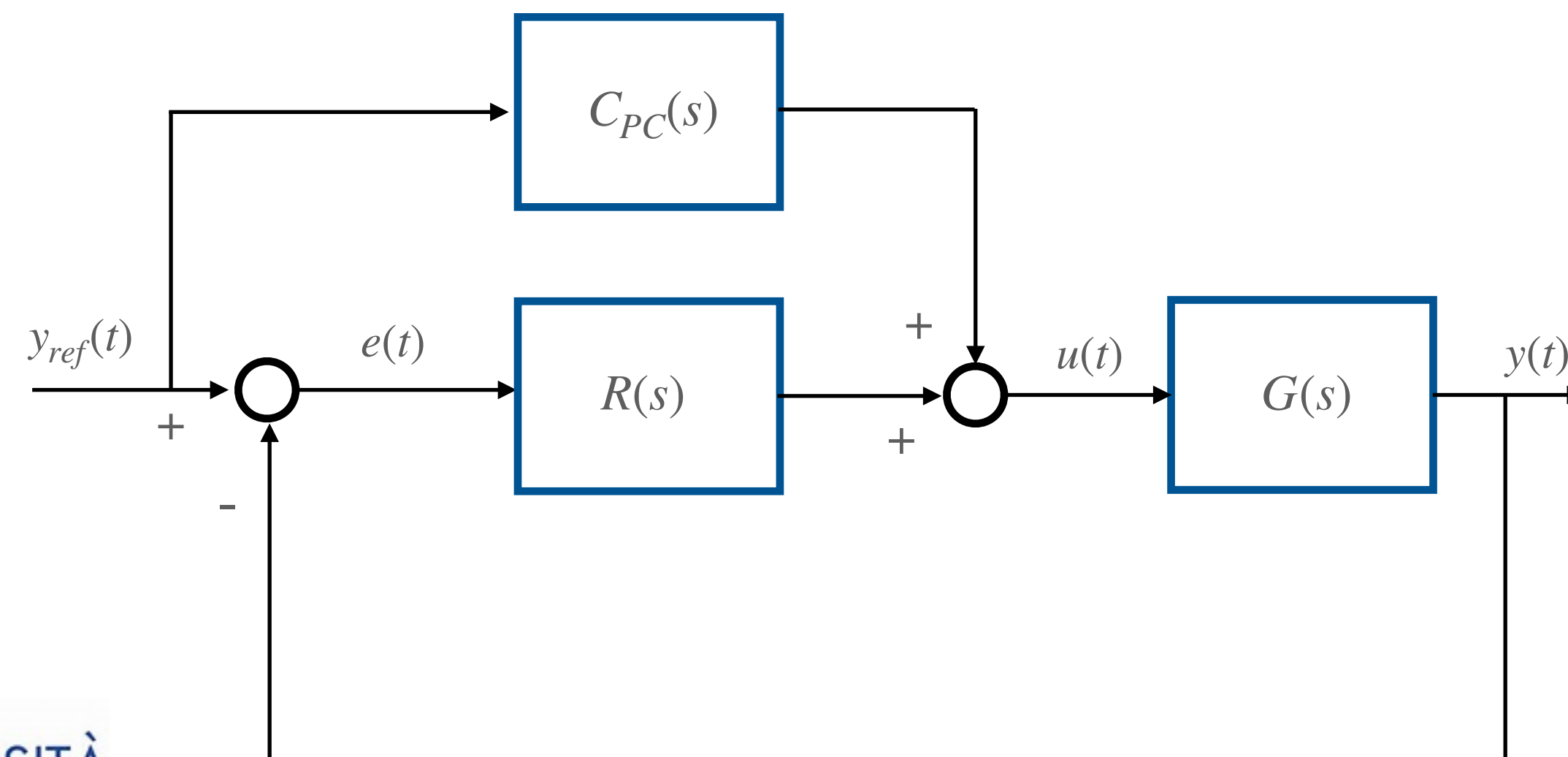
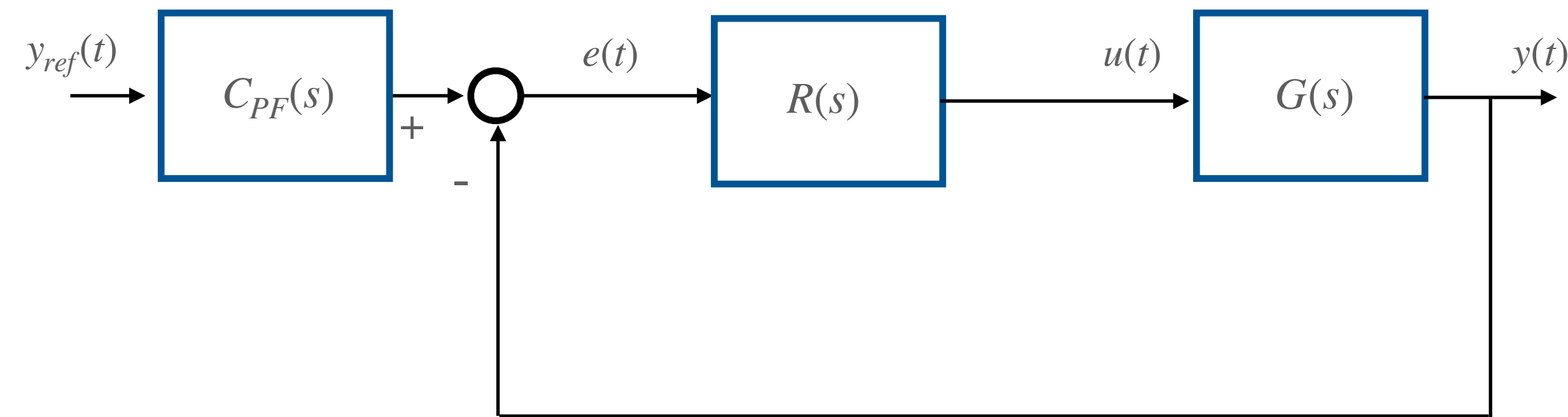


Prefilter-based and Parallel-compensator Based Control Schemes: 2DoF Control Schemes

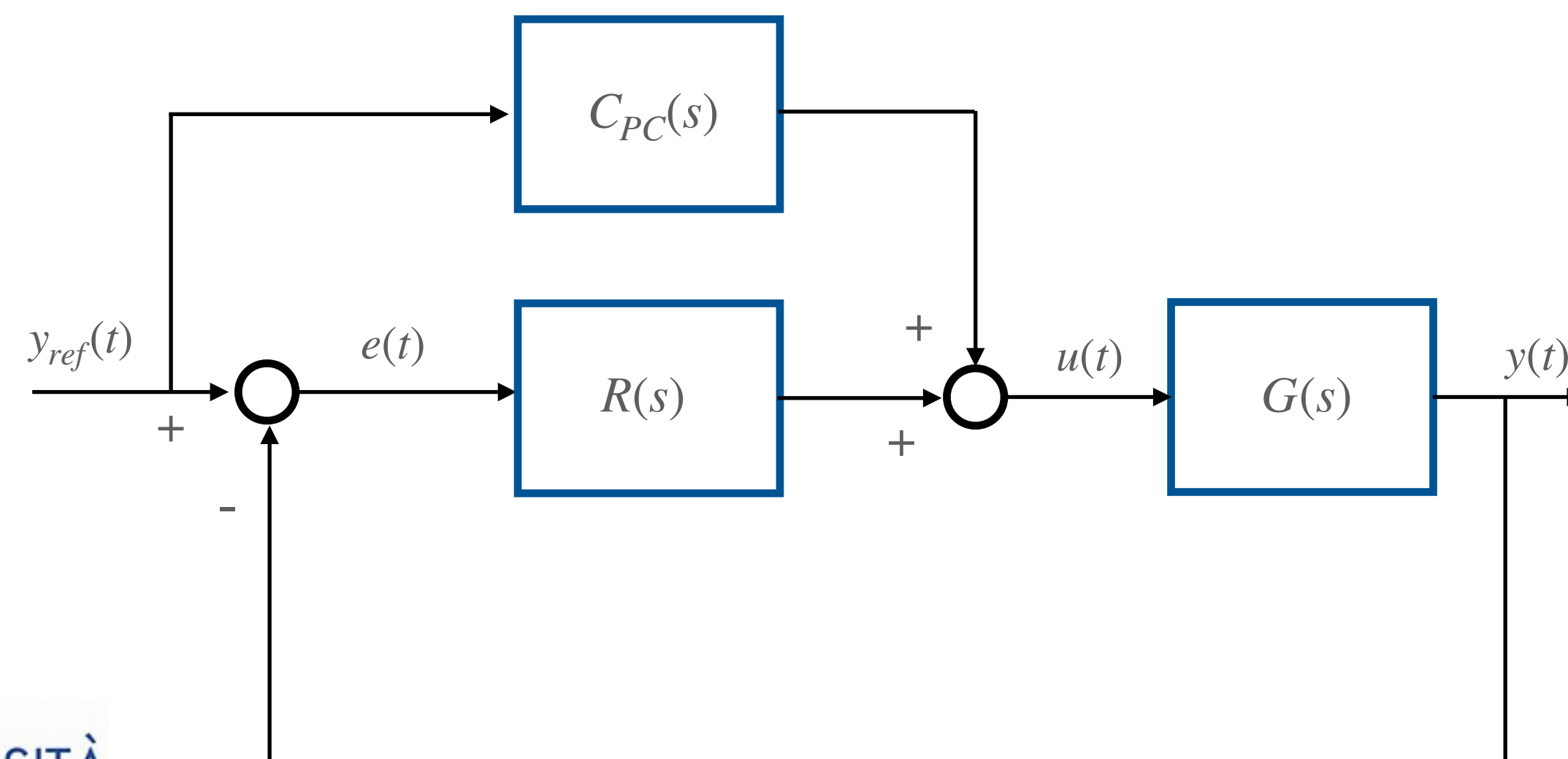
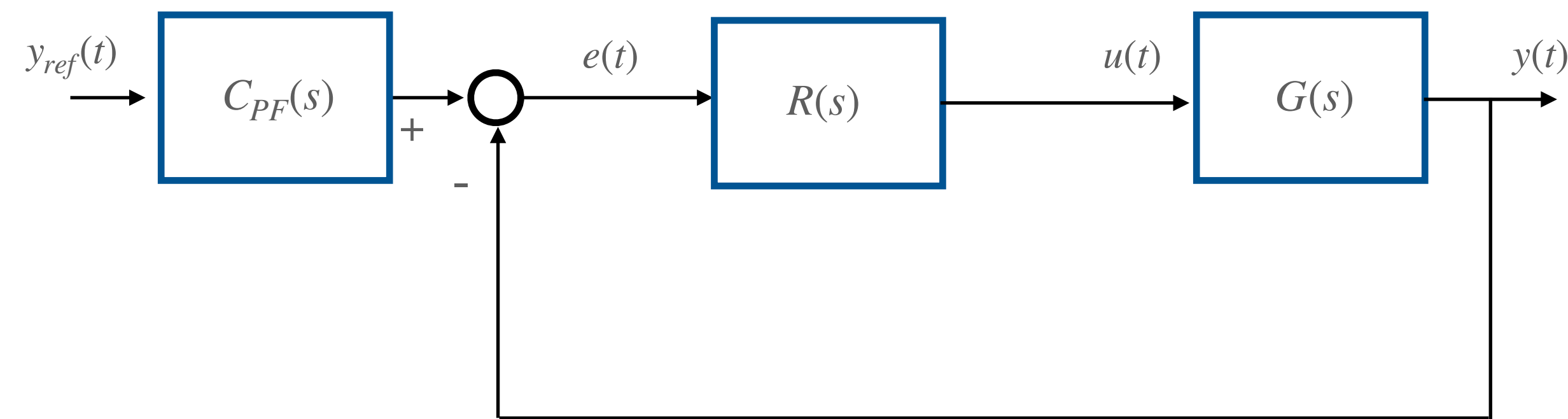


Prefilter-based and Parallel-compensator Based Control Schemes: 2DoF Control Schemes

1DoF scheme



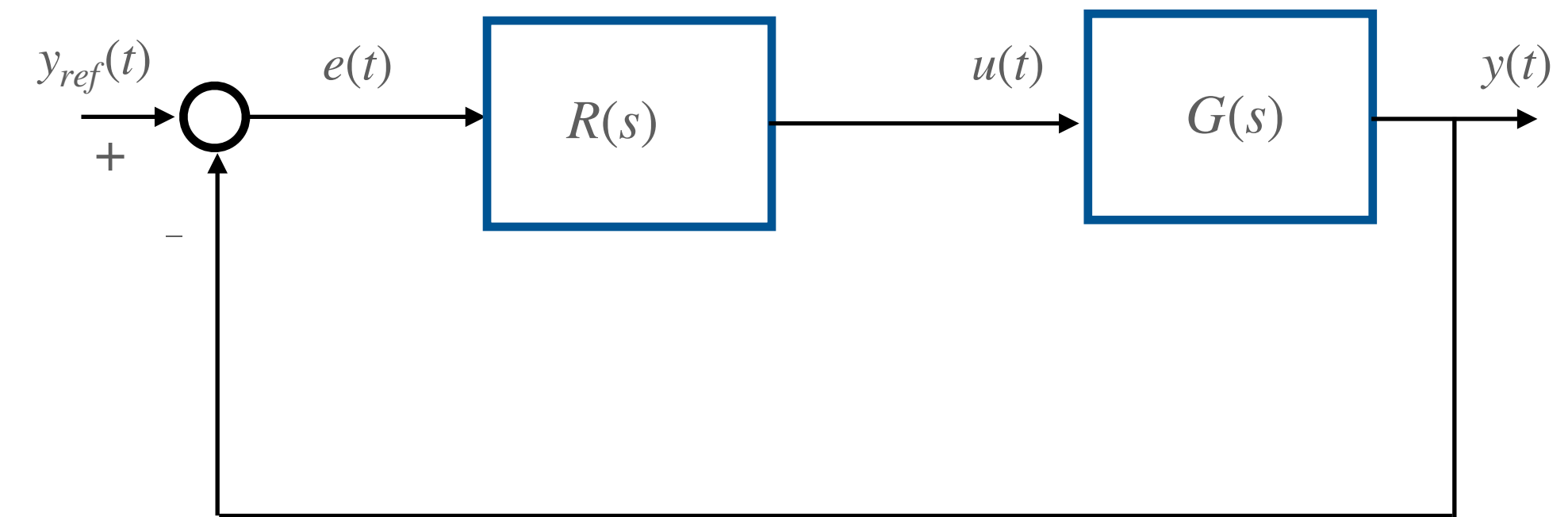
Prefilter-based and Parallel-compensator Based Control Schemes: 2DoF Control Schemes



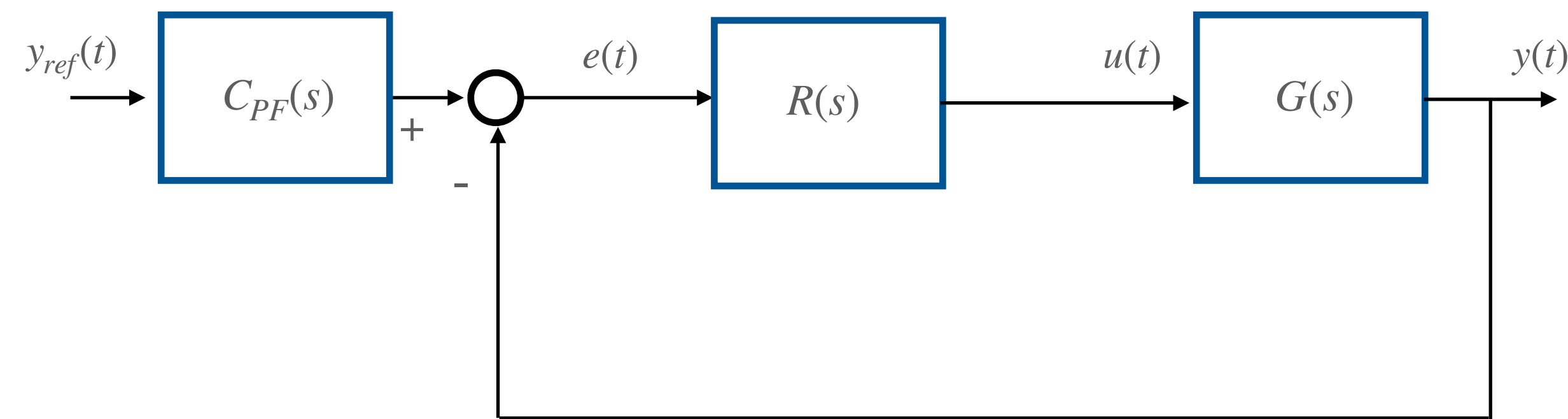
1DoF scheme

$$y_{ref} \rightarrow u : \frac{R(s)}{1 + R(s)Gs}$$

$$y \rightarrow u : - \frac{R(s)}{1 + R(s)Gs}$$



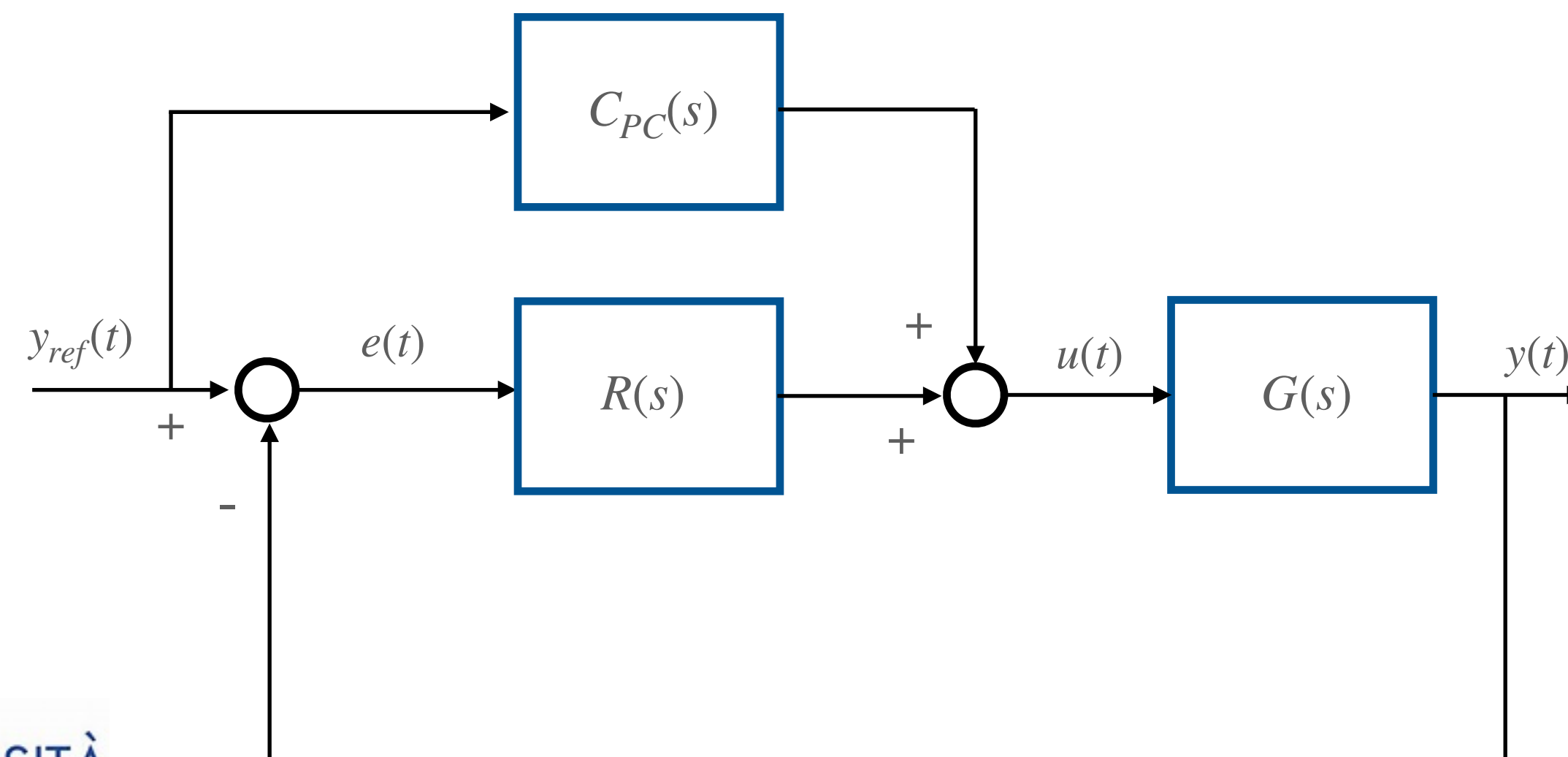
Prefilter-based and Parallel-compensator Based Control Schemes: 2DoF Control Schemes



2DoF scheme

$$y_{ref} \rightarrow u : \frac{C_{PF}(s)R(s)}{1 + R(s)Gs}$$

$$y \rightarrow u : - \frac{R(s)}{1 + R(s)Gs}$$

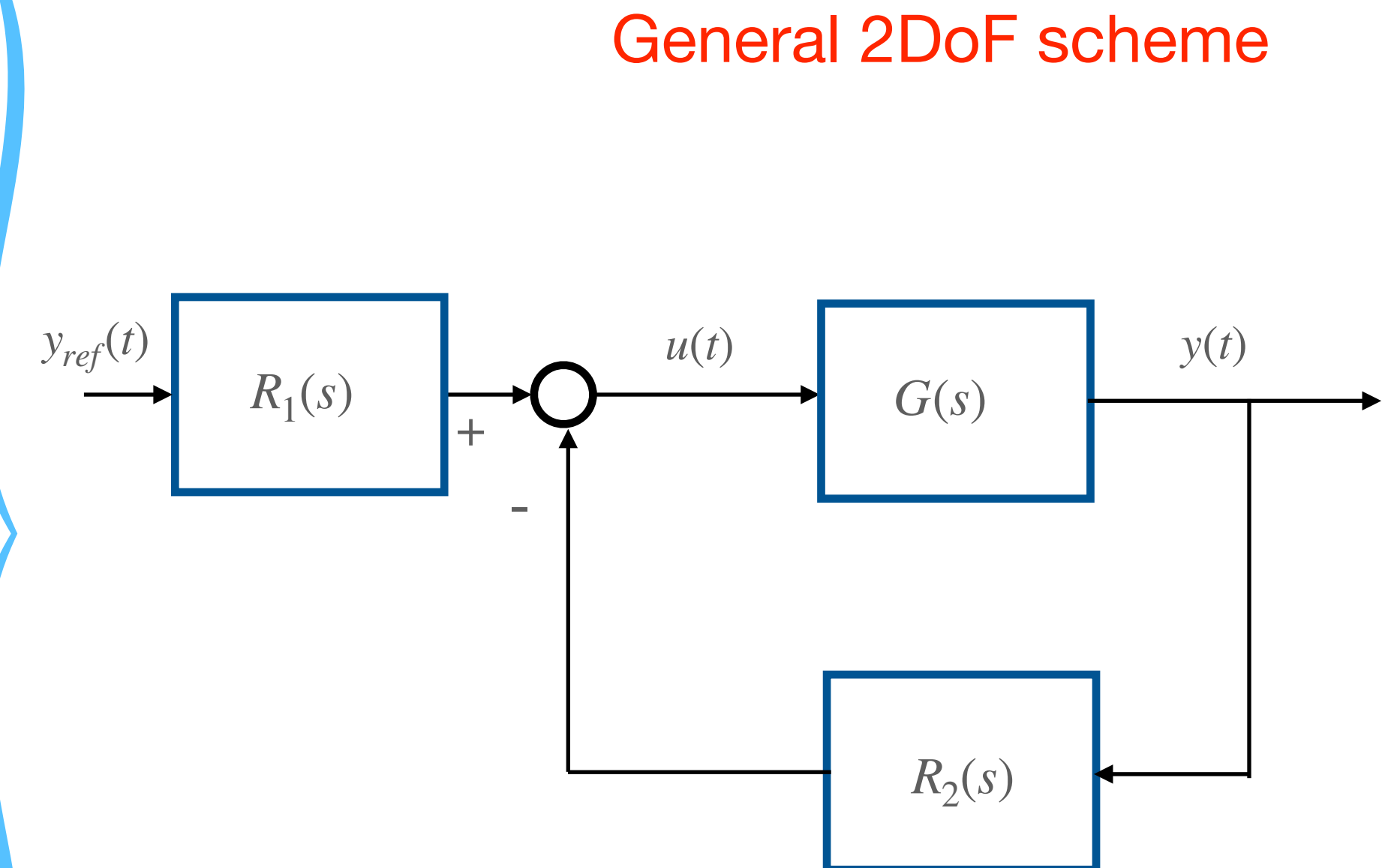
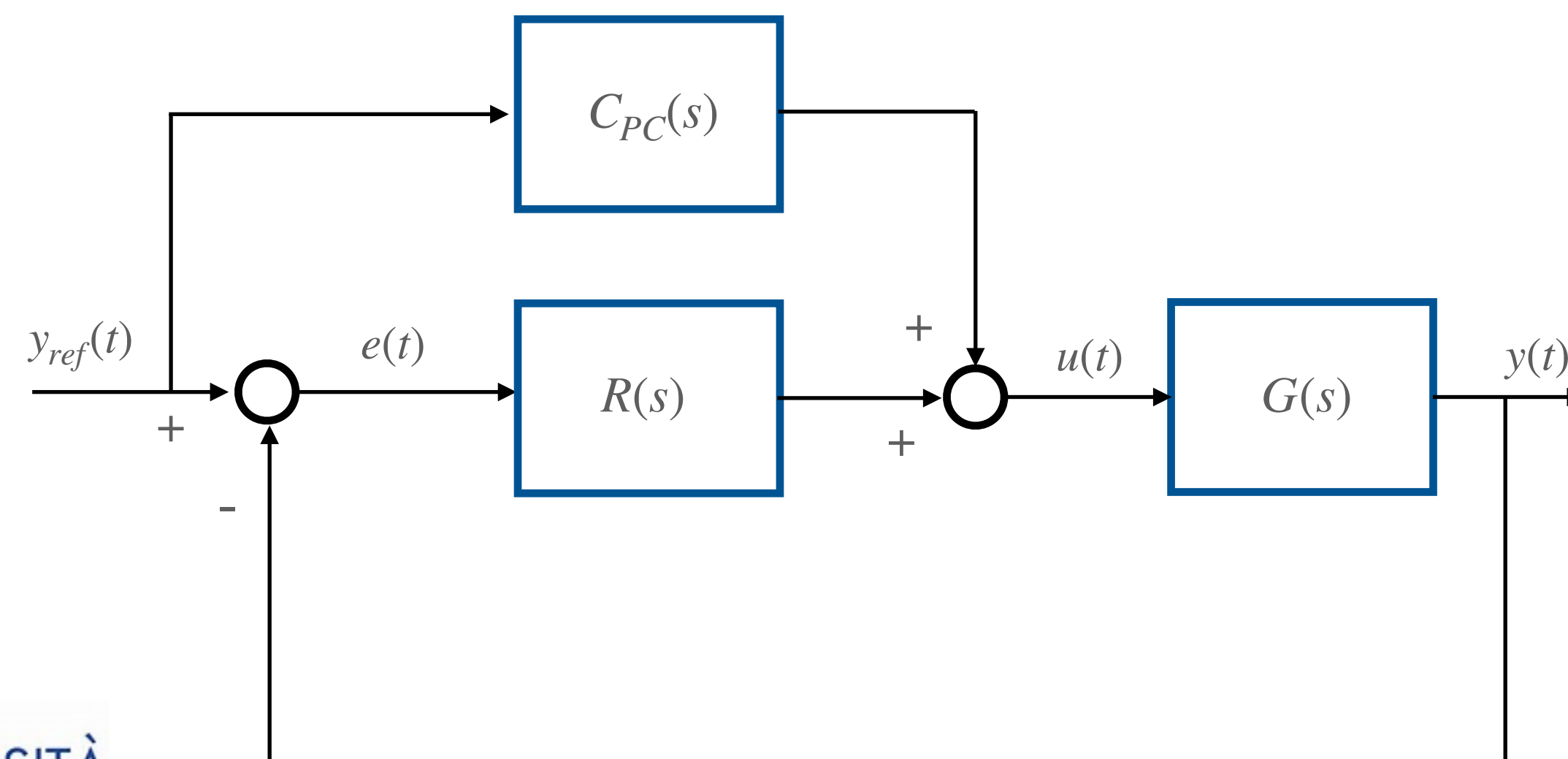
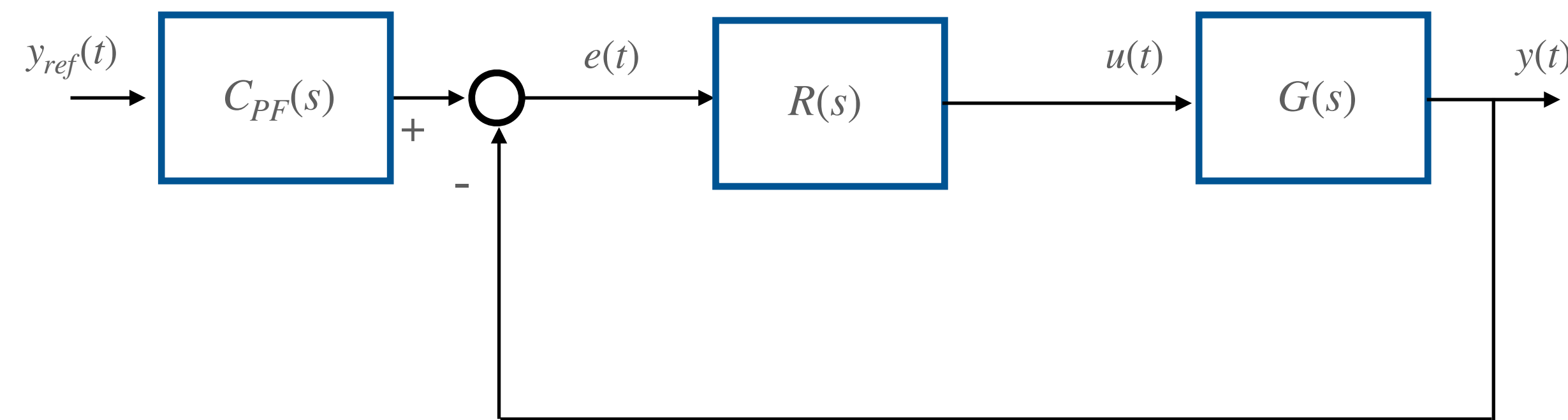


2DoF scheme

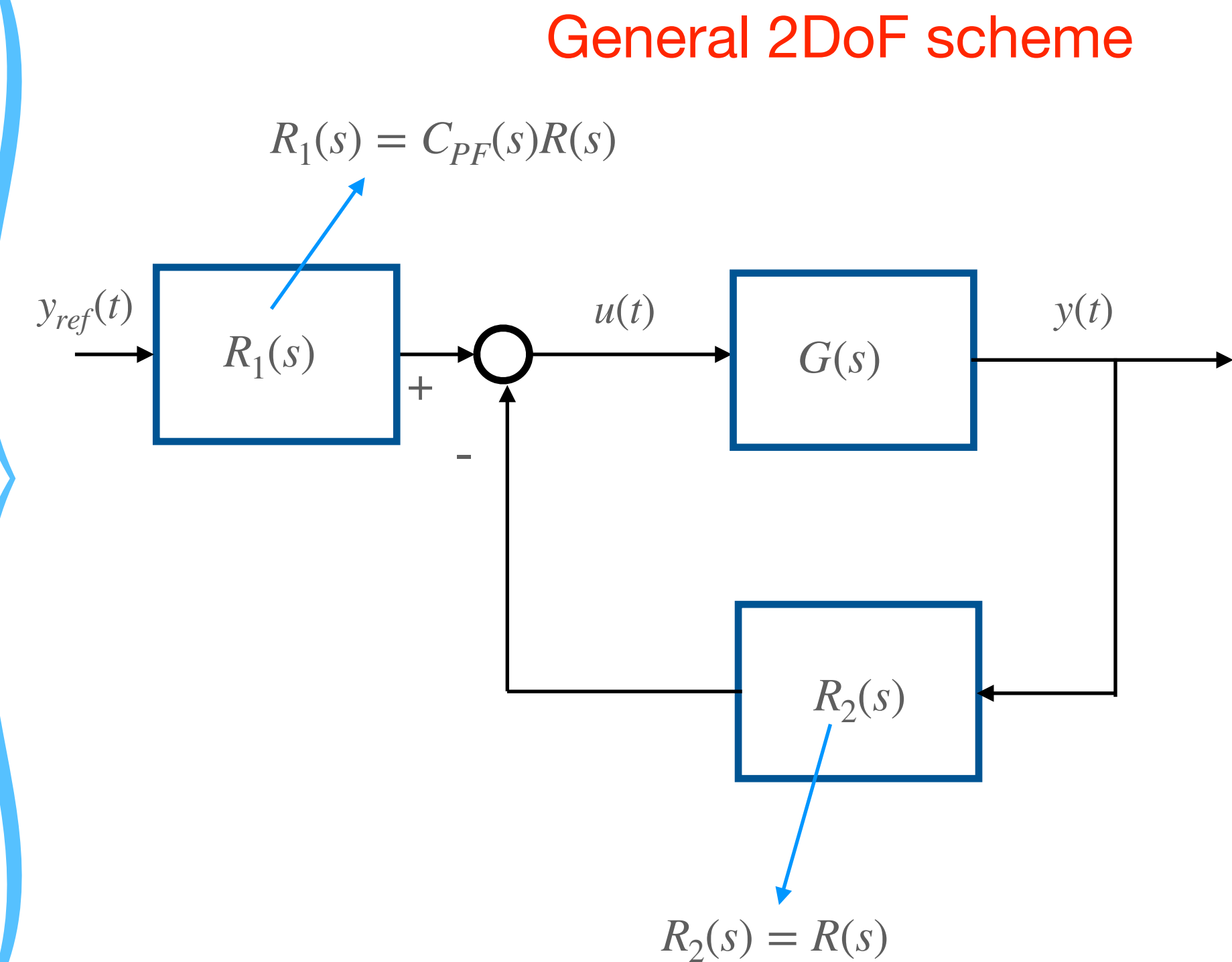
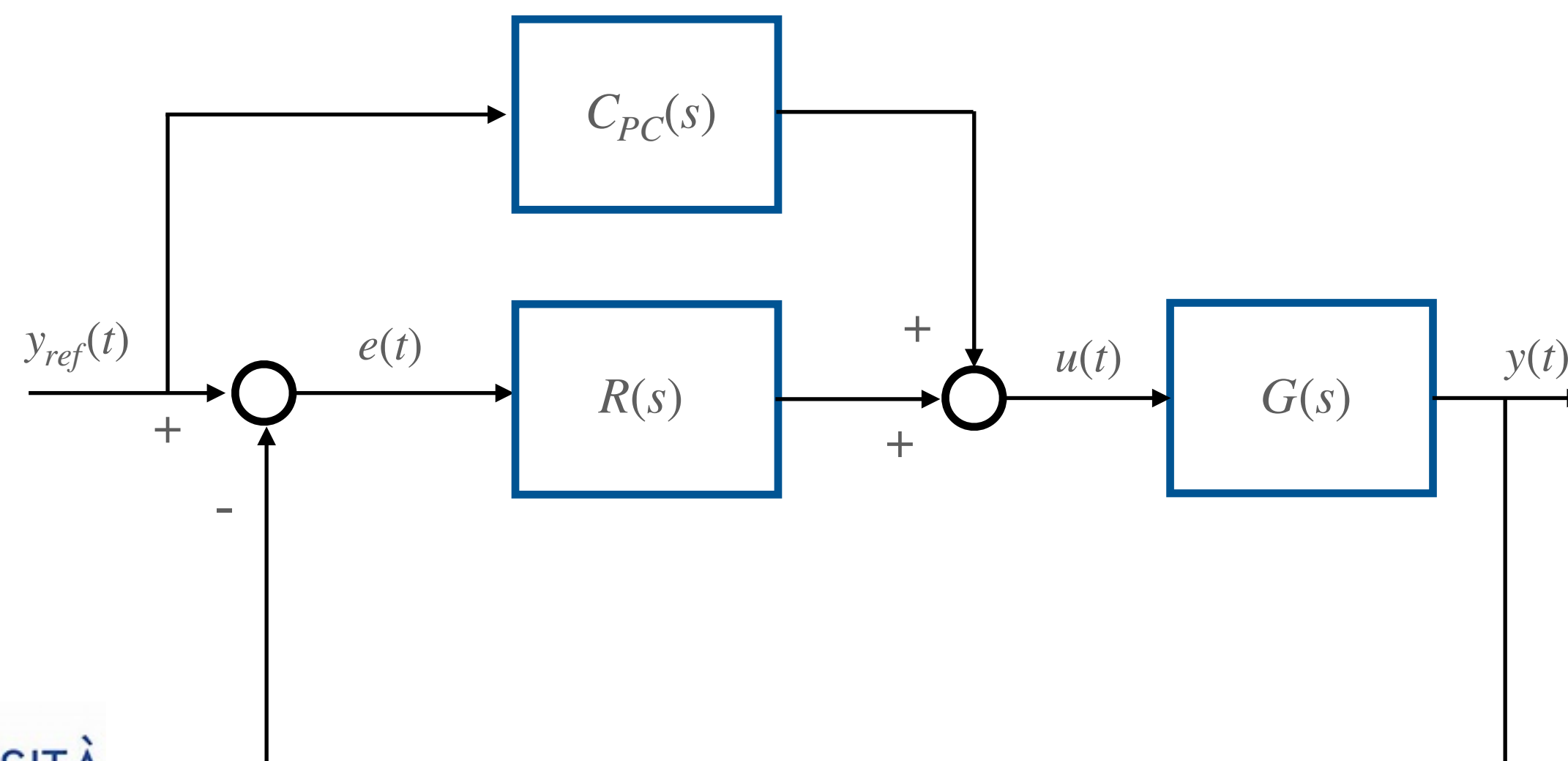
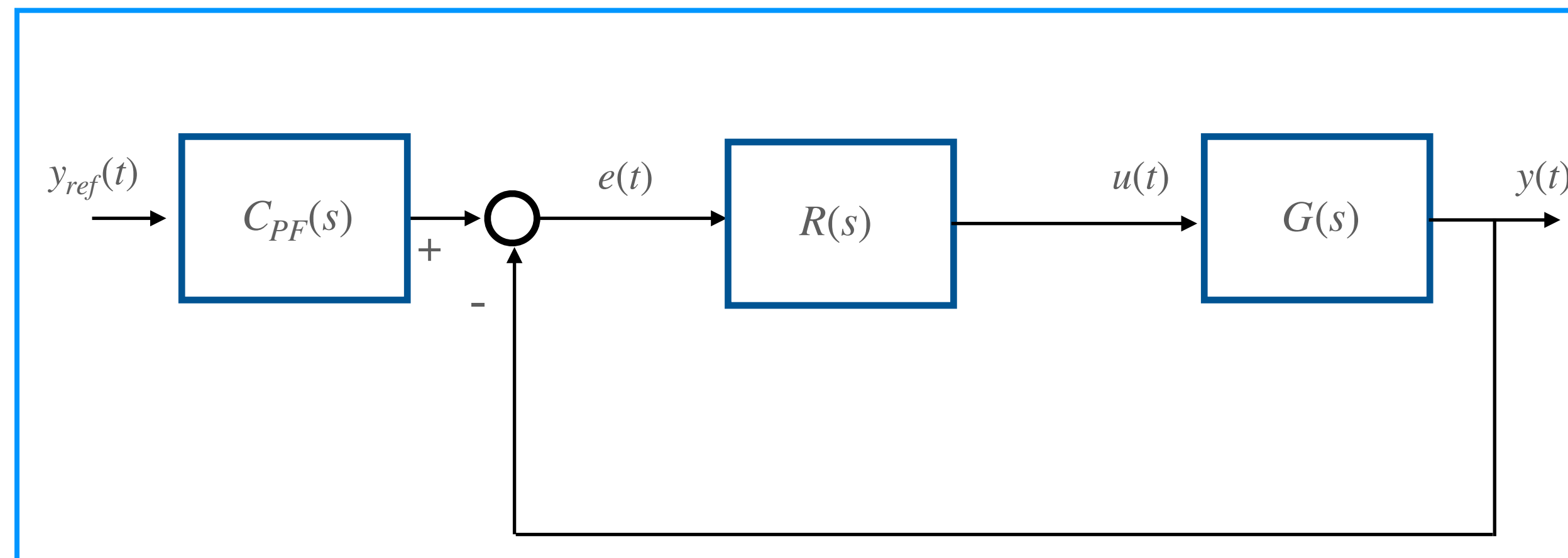
$$y_{ref} \rightarrow u : \frac{C_{PC}(s) + R(s)}{1 + R(s)Gs}$$

$$y \rightarrow u : - \frac{R(s)}{1 + R(s)Gs}$$

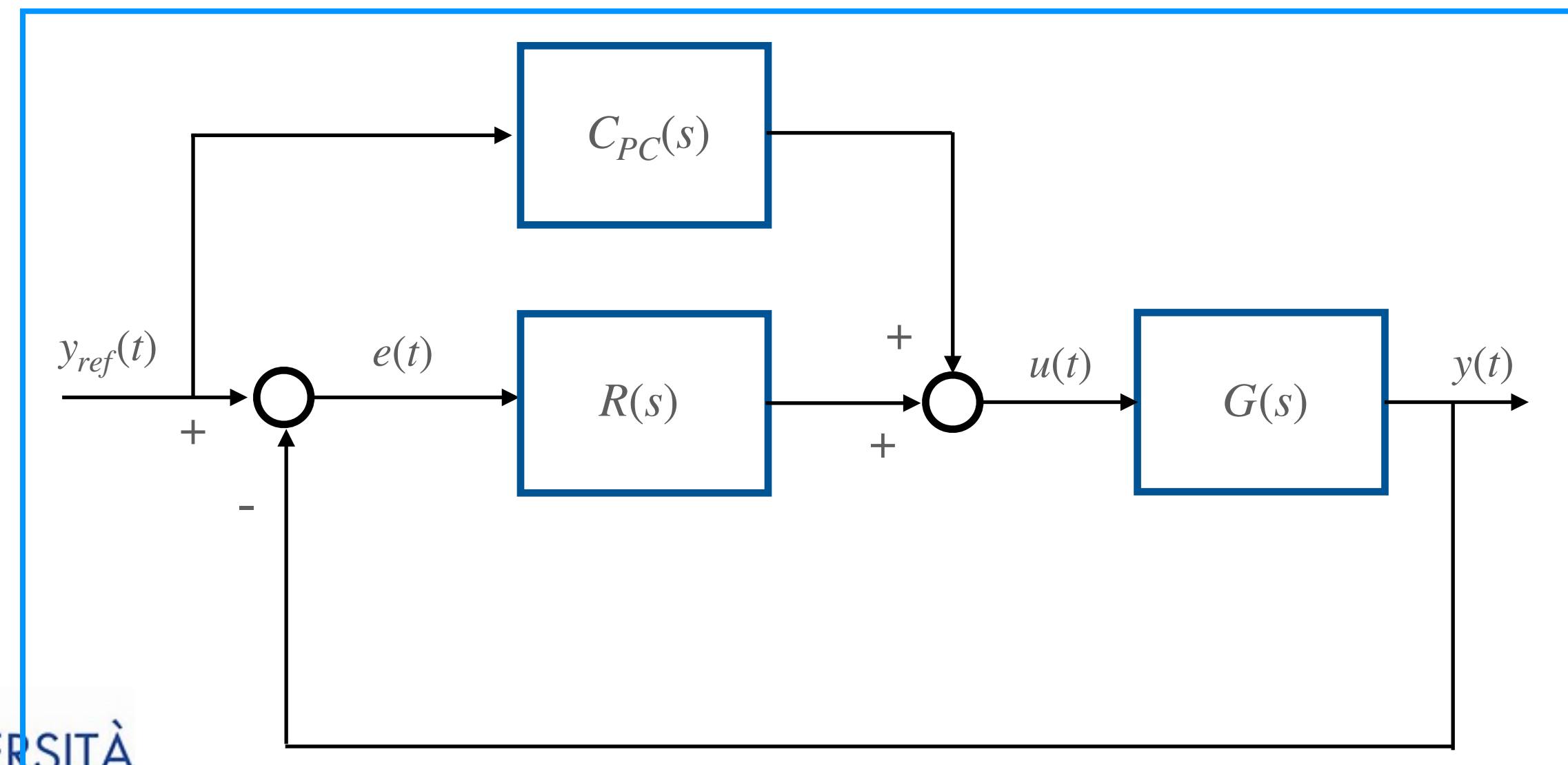
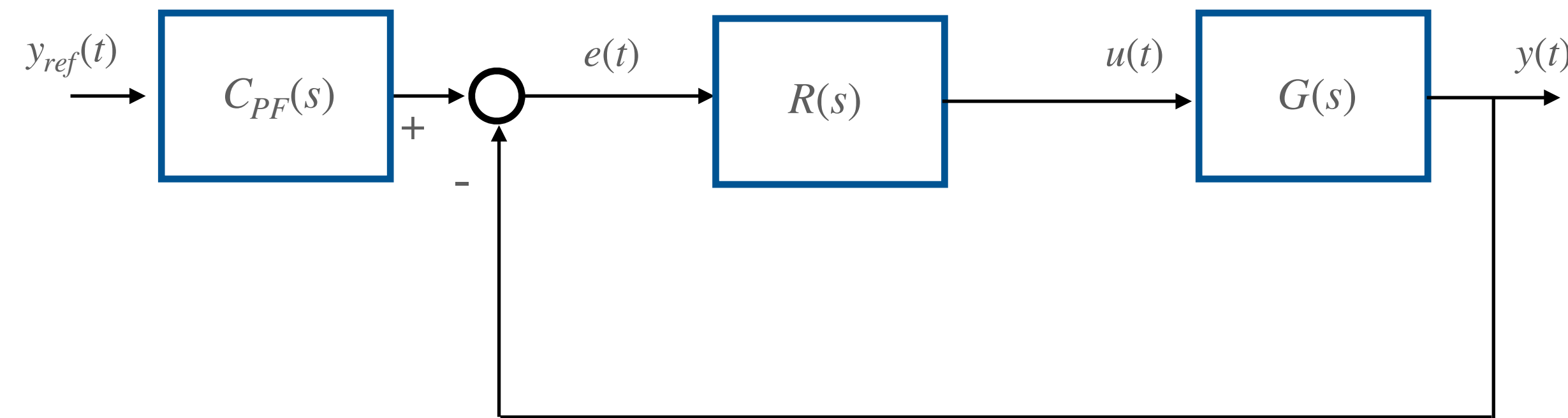
Prefilter-based and Parallel-compensator Based Control Schemes: 2DoF Control Schemes



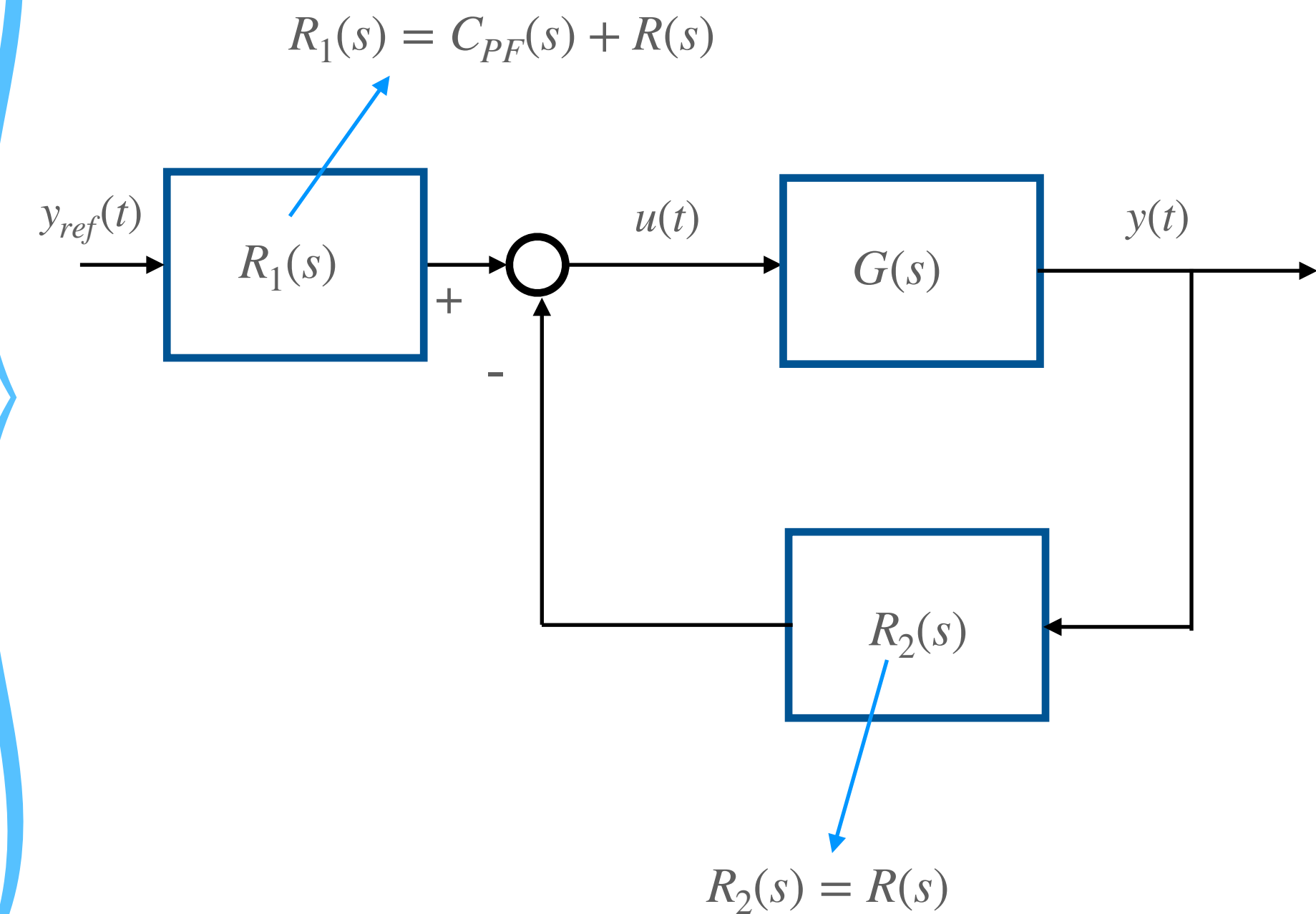
Prefilter-based and Parallel-compensator Based Control Schemes: 2DoF Control Schemes



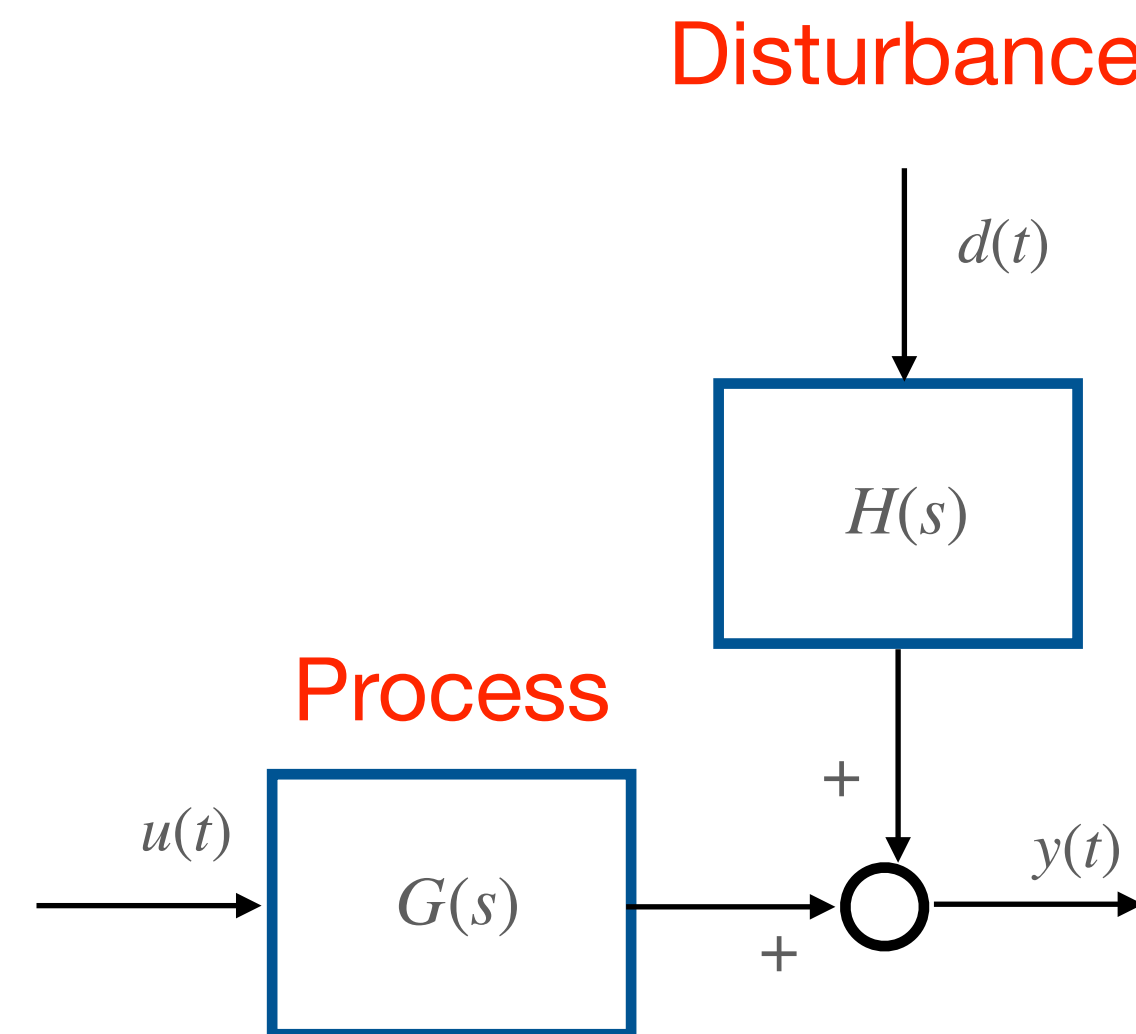
Prefilter-based and Parallel-compensator Based Control Schemes: 2DoF Control Schemes



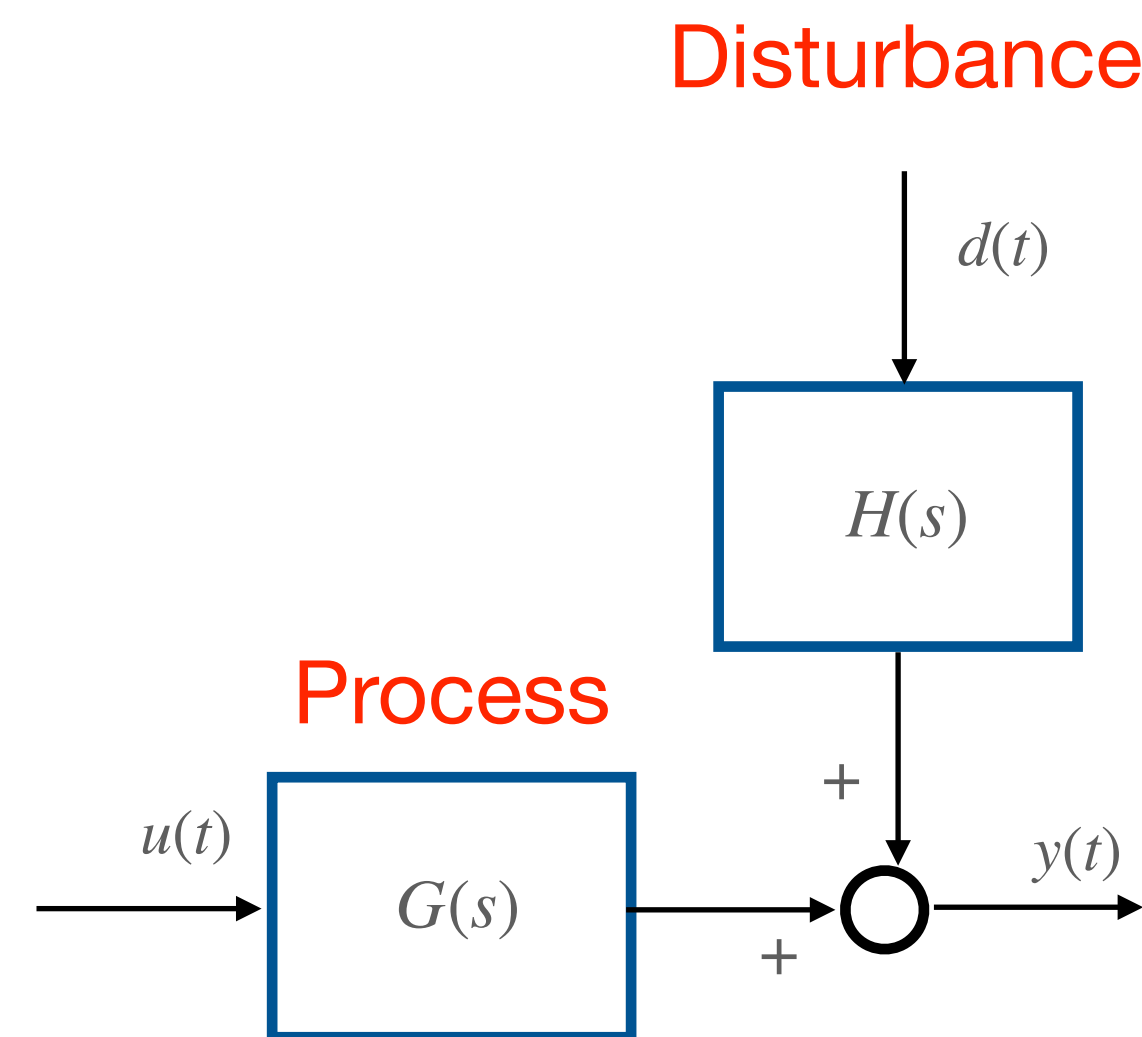
General 2DoF scheme



Control Scheme with Measurable Disturbance Compensation



Control Scheme with Measurable Disturbance Compensation

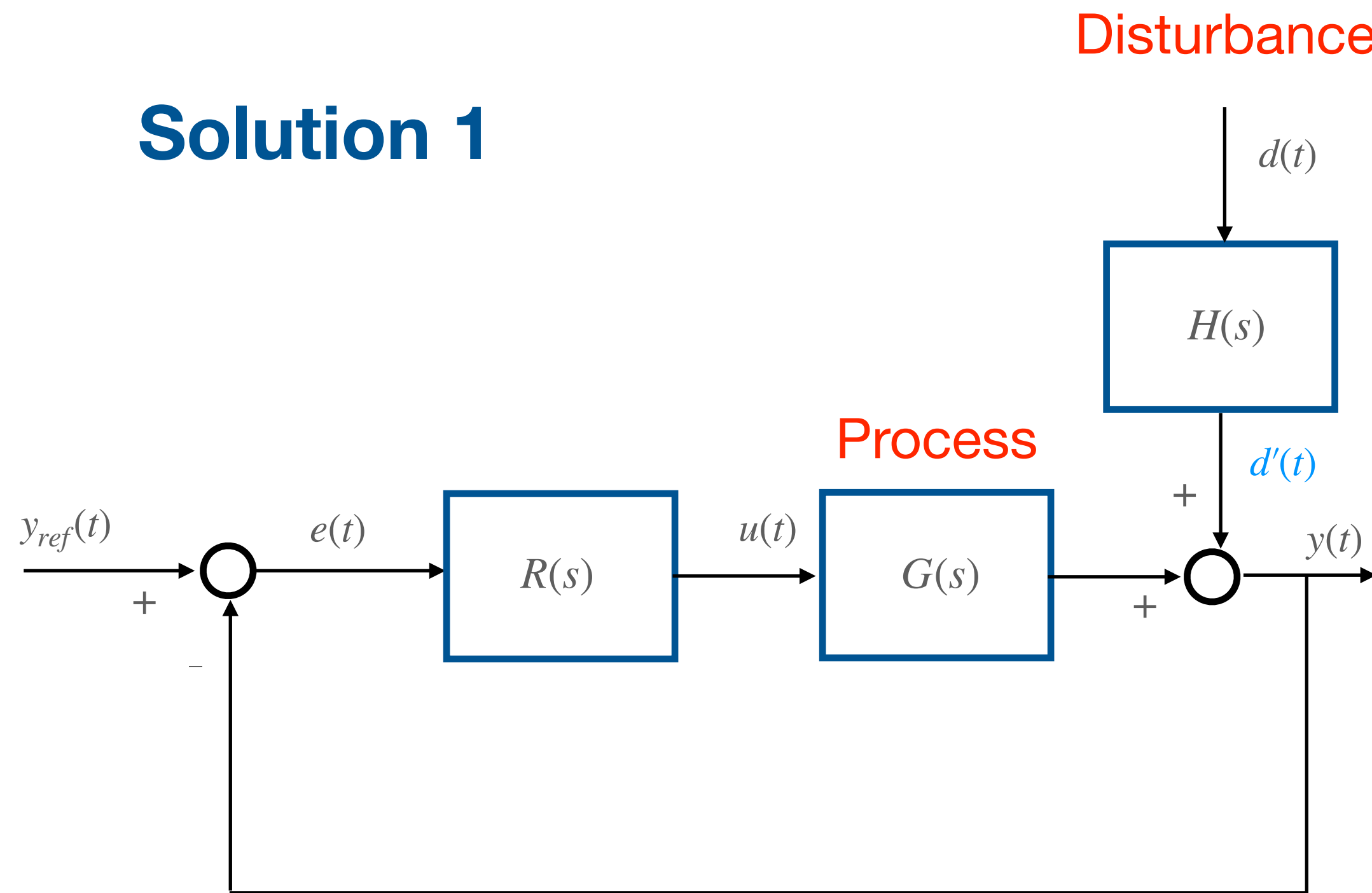


Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

Control Scheme with Measurable Disturbance Compensation

Solution 1

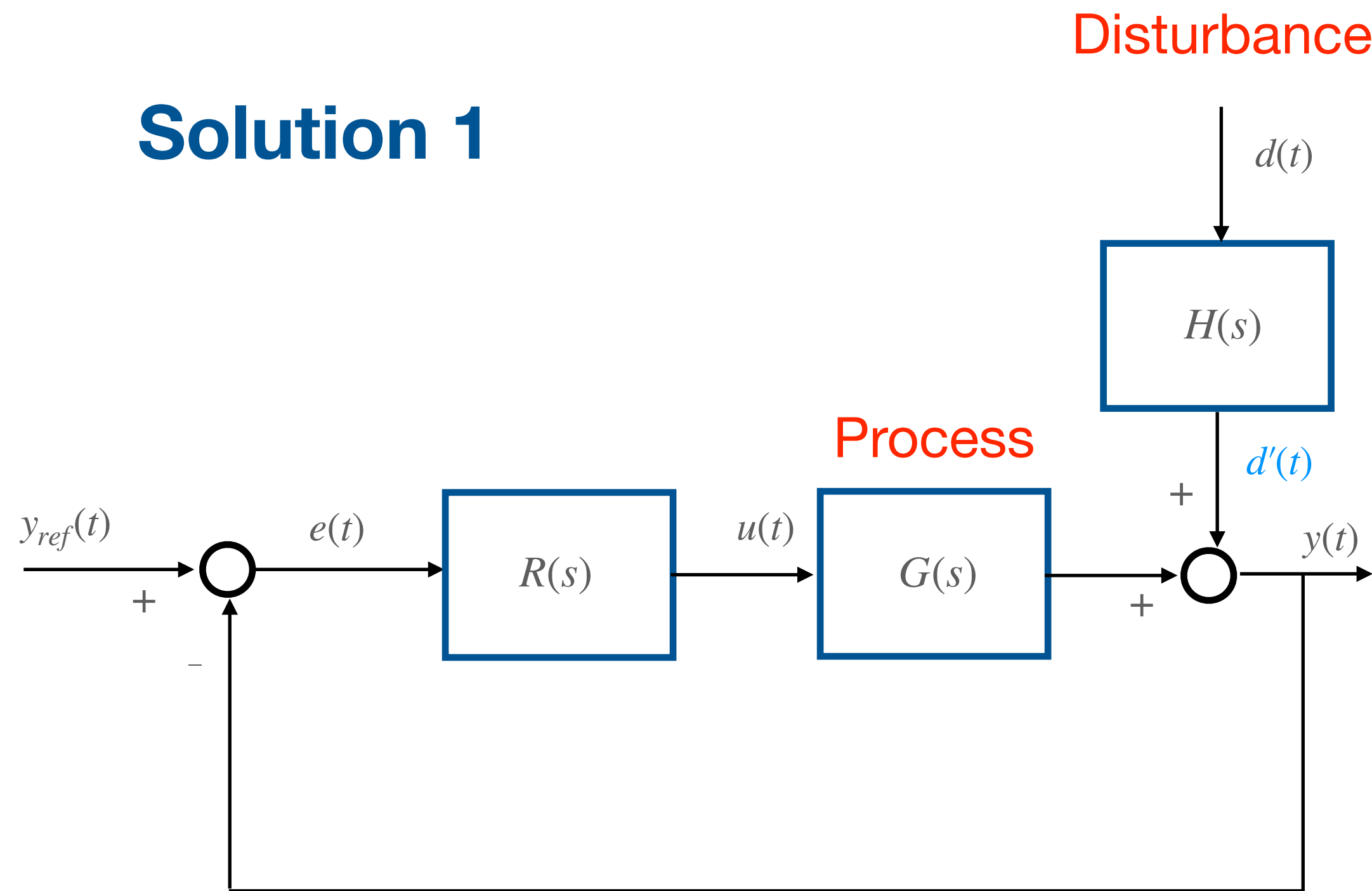


Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

Control Scheme with Measurable Disturbance Compensation

Solution 1



Assumptions:

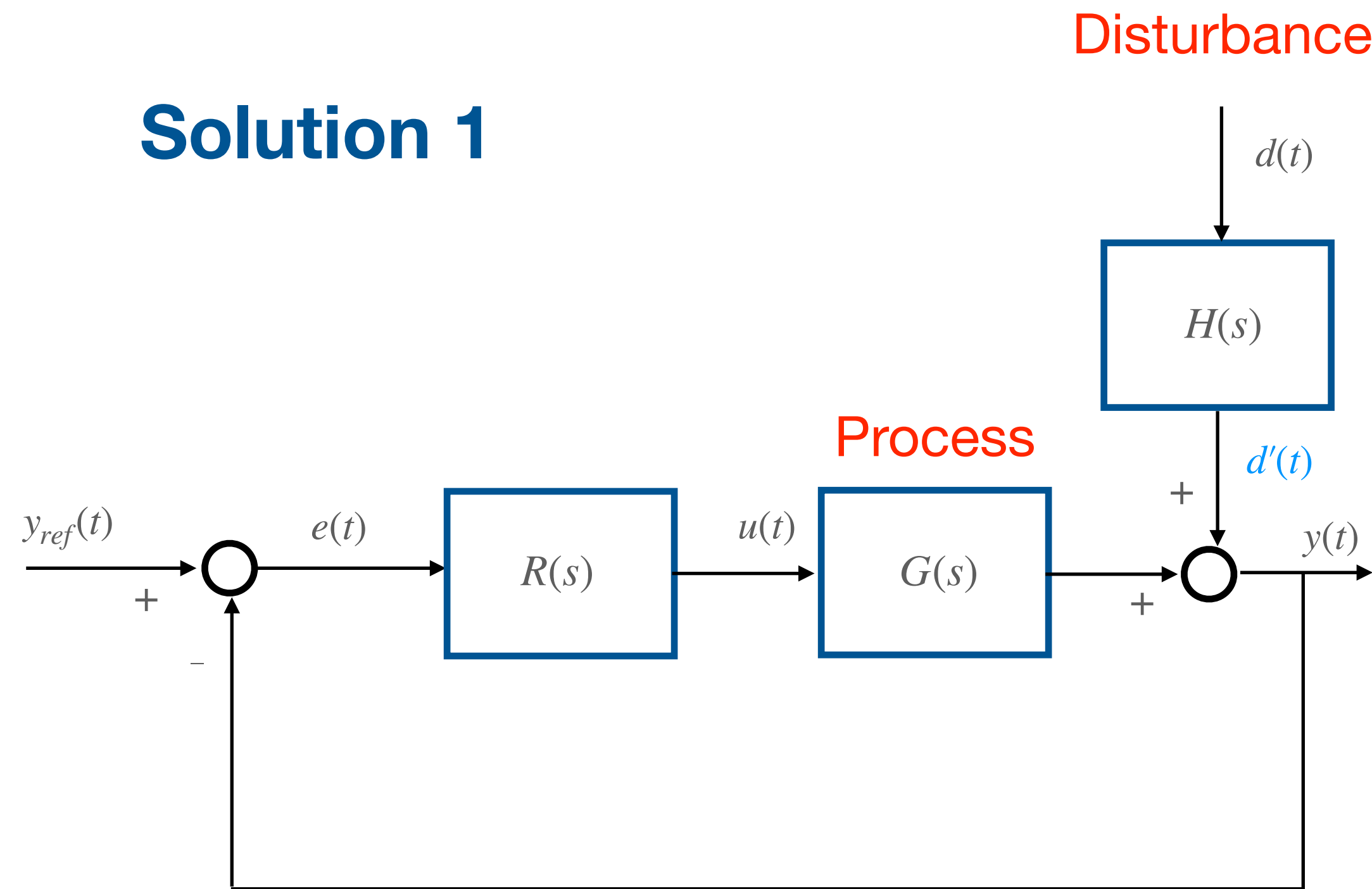
- $d(t)$ measurable
- $H(s)$ known (possibly 1)

$d' \rightarrow y : S(s) = \frac{1}{1 + R(s)G(s)}$

Sensitivity Function

Control Scheme with Measurable Disturbance Compensation

Solution 1



Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

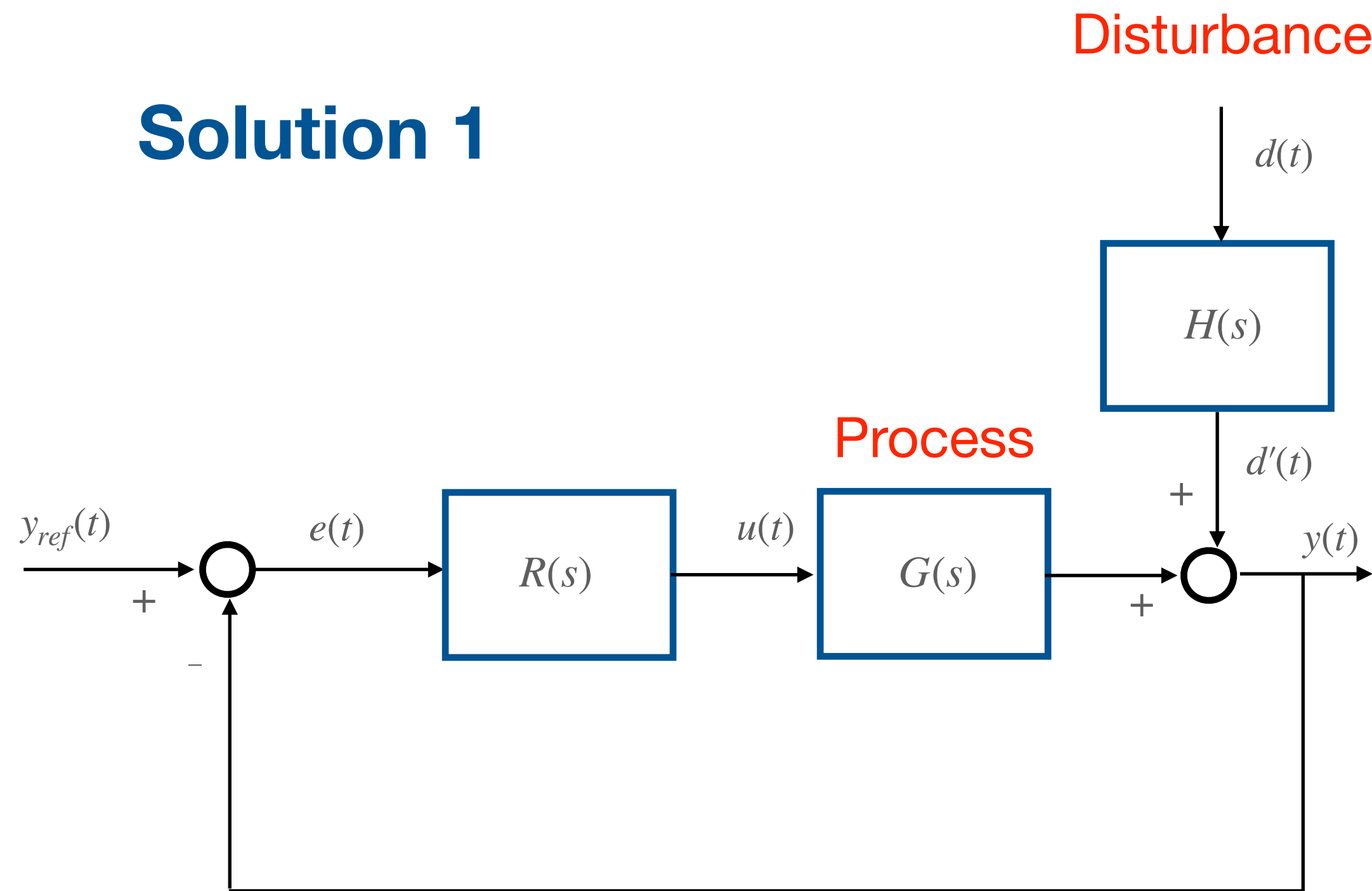
Sensitivity Function

$$d' \rightarrow y : S(s) = \frac{1}{1 + R(s)G(s)}$$

Recommendation: $|S(j\omega)|$ sufficiently low $\forall \omega$
in the frequency range where $d'(t)$
has significant harmonics

Control Scheme with Measurable Disturbance Compensation

Solution 1



Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

Sensitivity Function

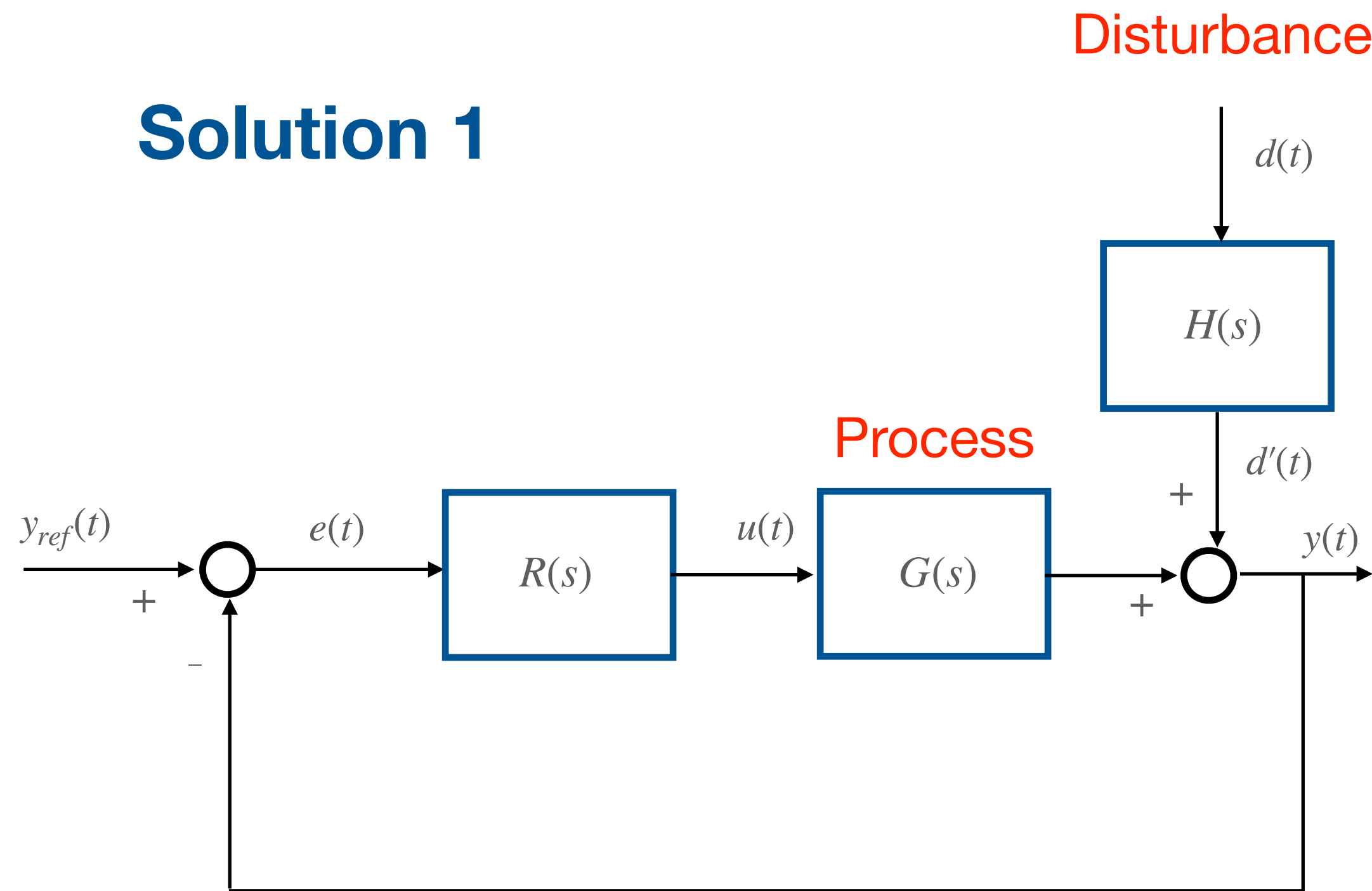
$$d' \rightarrow y : S(s) = \frac{1}{1 + R(s)G(s)}$$

Recommendation: $|S(j\omega)|$ sufficiently low $\forall \omega$ in the frequency range where $d'(t)$ has significant harmonics

$$\omega_{d'} \in [0, \omega_{BW_{CL}})$$

Control Scheme with Measurable Disturbance Compensation

Solution 1



Disturbance

Process

Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

Sensitivity Function

$$d' \rightarrow y : S(s) = \frac{1}{1 + R(s)G(s)}$$

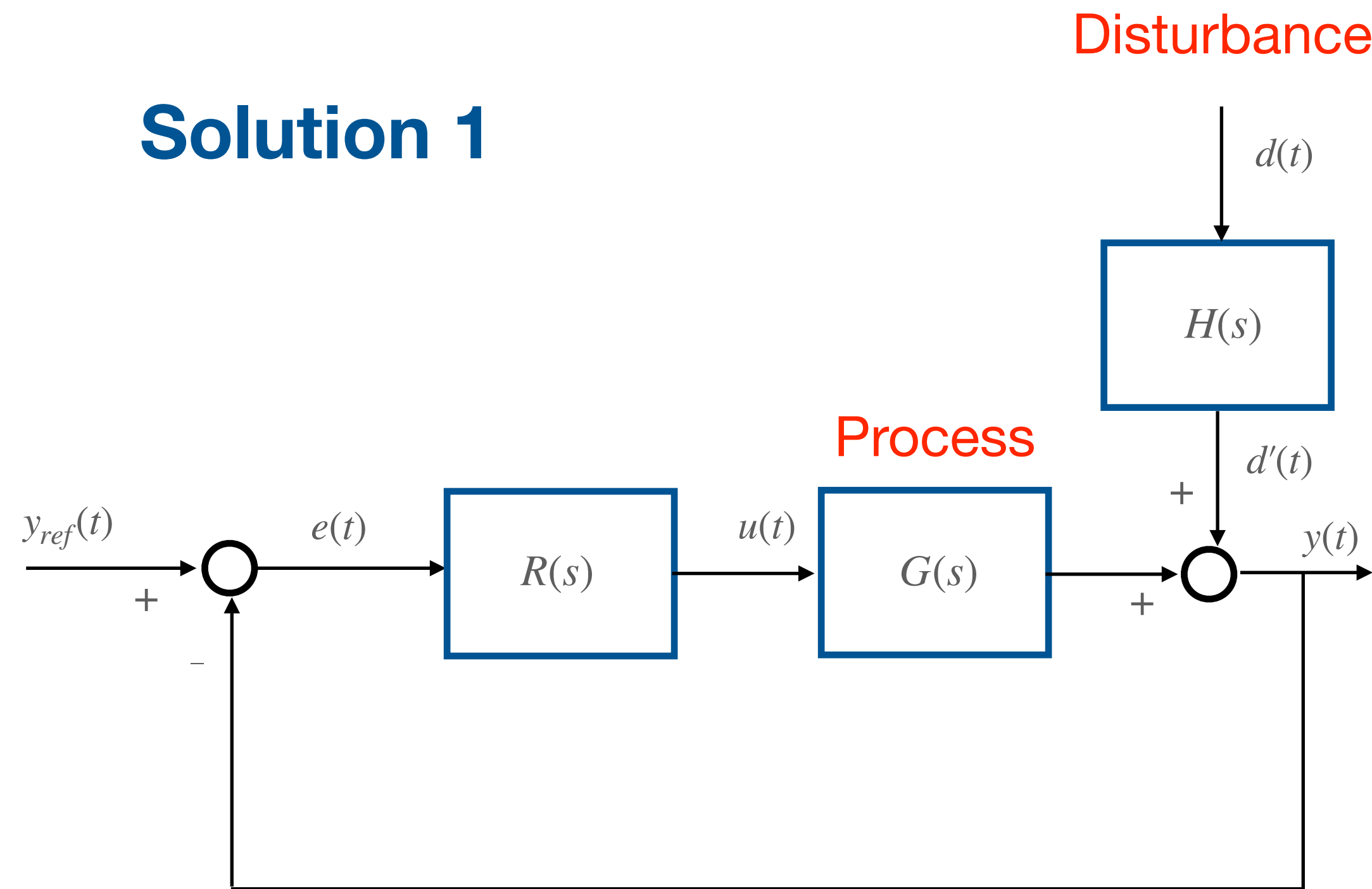
Recommendation: $|S(j\omega)|$ sufficiently low $\forall \omega$ in the frequency range where $d'(t)$ has significant harmonics

$$\omega_{d'} \in [0, \omega_{BW_{CL}})$$

$$|S(j\omega)| \begin{cases} \text{LF: } |S(j\omega)| \rightarrow \left| \frac{1}{L(j\omega)} \right| \\ \text{HF: } |S(j\omega)| \rightarrow 1 \end{cases}$$

Control Scheme with Measurable Disturbance Compensation

Solution 1



Disturbance

Process

Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

Sensitivity Function

$$d' \rightarrow y : S(s) = \frac{1}{1 + R(s)G(s)}$$

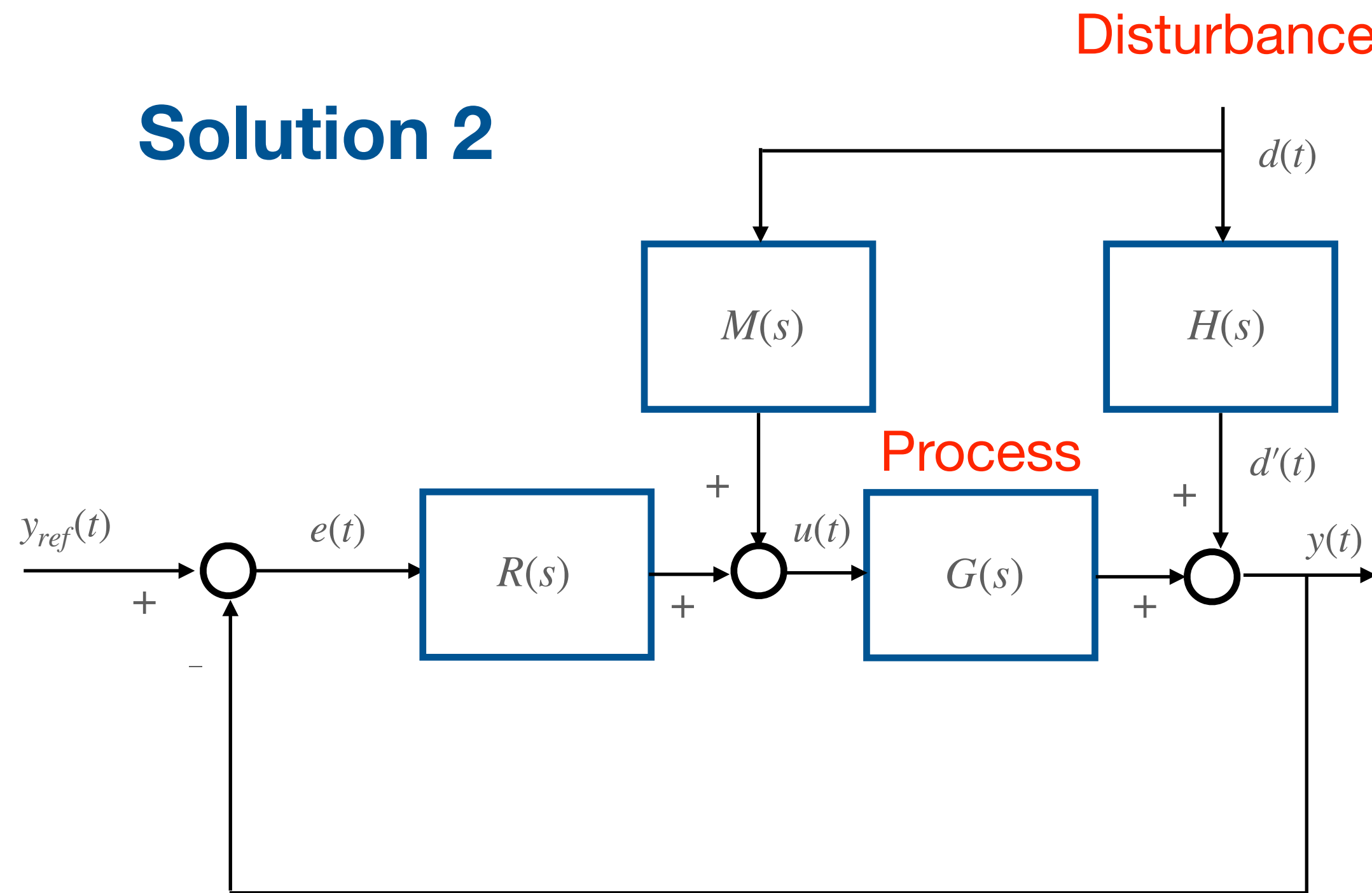
Recommendation: $|S(j\omega)|$ sufficiently low $\forall \omega$ in the frequency range where $d'(t)$ has significant harmonics

$$|S(j\omega)| \begin{cases} \text{LF: } |S(j\omega)| \rightarrow \left| \frac{1}{L(j\omega)} \right| \\ \text{HF: } |S(j\omega)| \rightarrow 1 \end{cases}$$

Recommendation: $|L(j\omega)|$ sufficiently high $\forall \omega \in [0, \omega_{BW_{CL}})$

Control Scheme with Measurable Disturbance Compensation

Solution 2

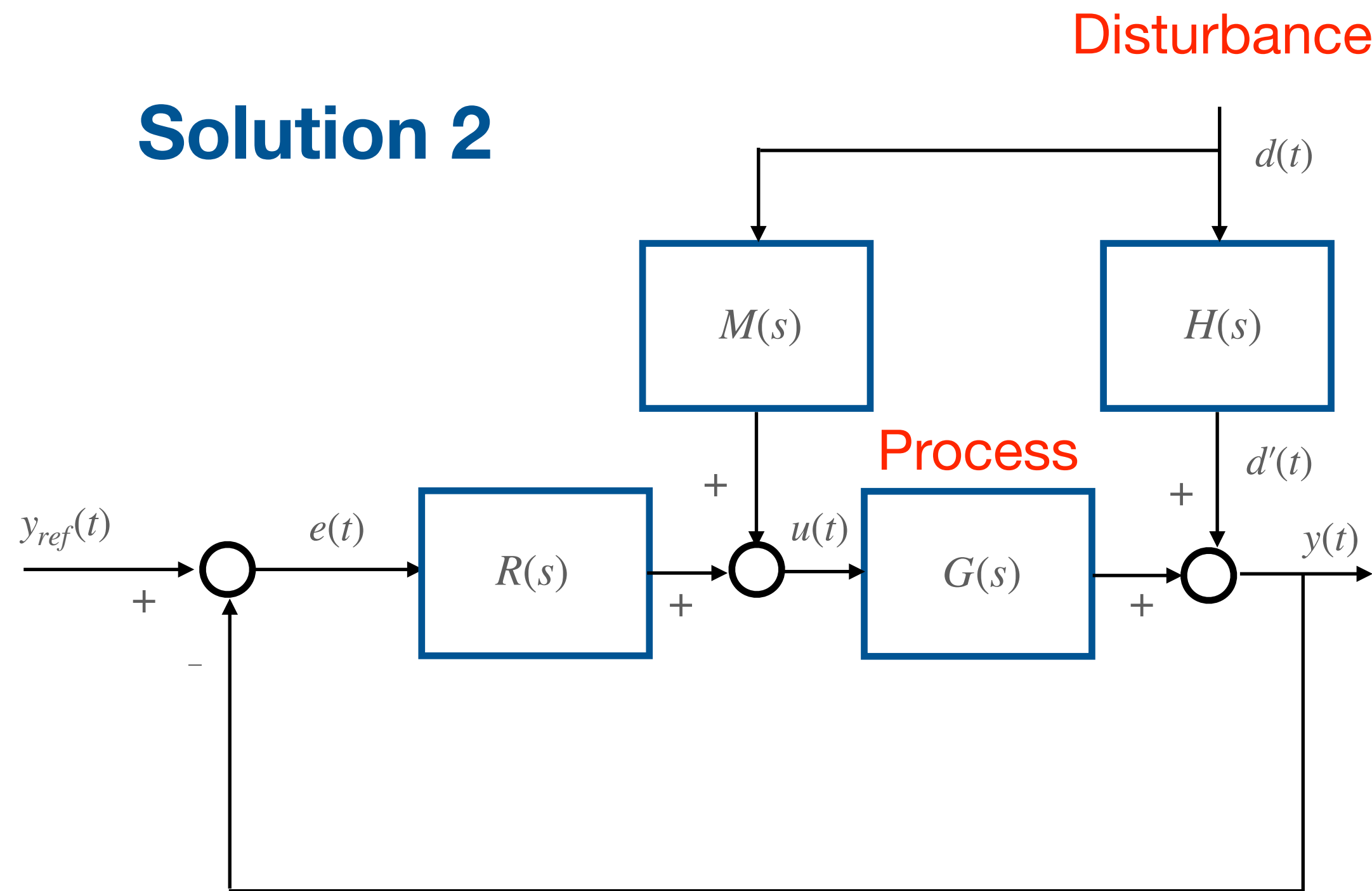


Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

Control Scheme with Measurable Disturbance Compensation

Solution 2



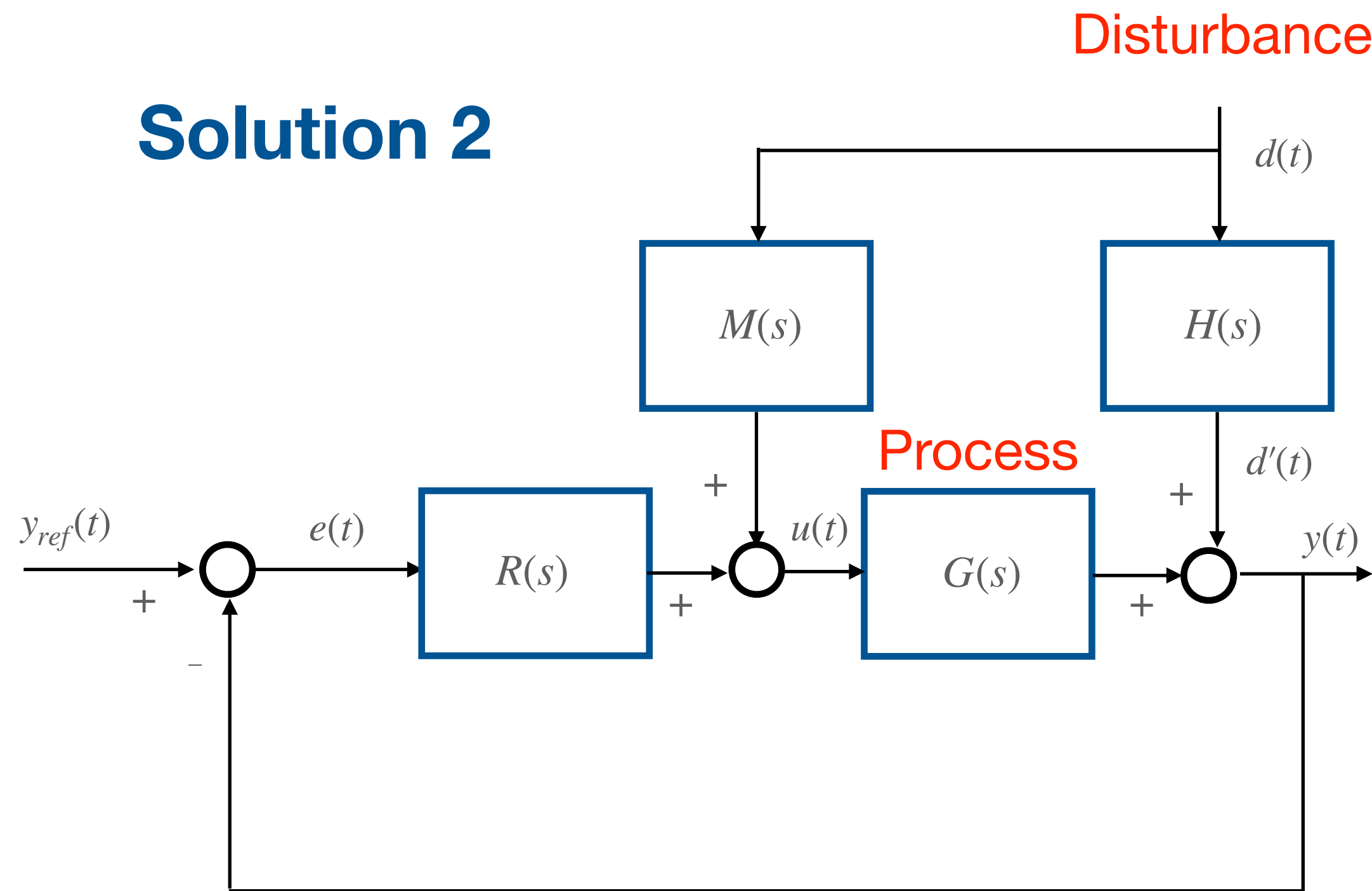
Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

$$d \rightarrow y : \frac{H(s) + M(s)G(s)}{1 + R(s)G(s)}$$

Control Scheme with Measurable Disturbance Compensation

Solution 2



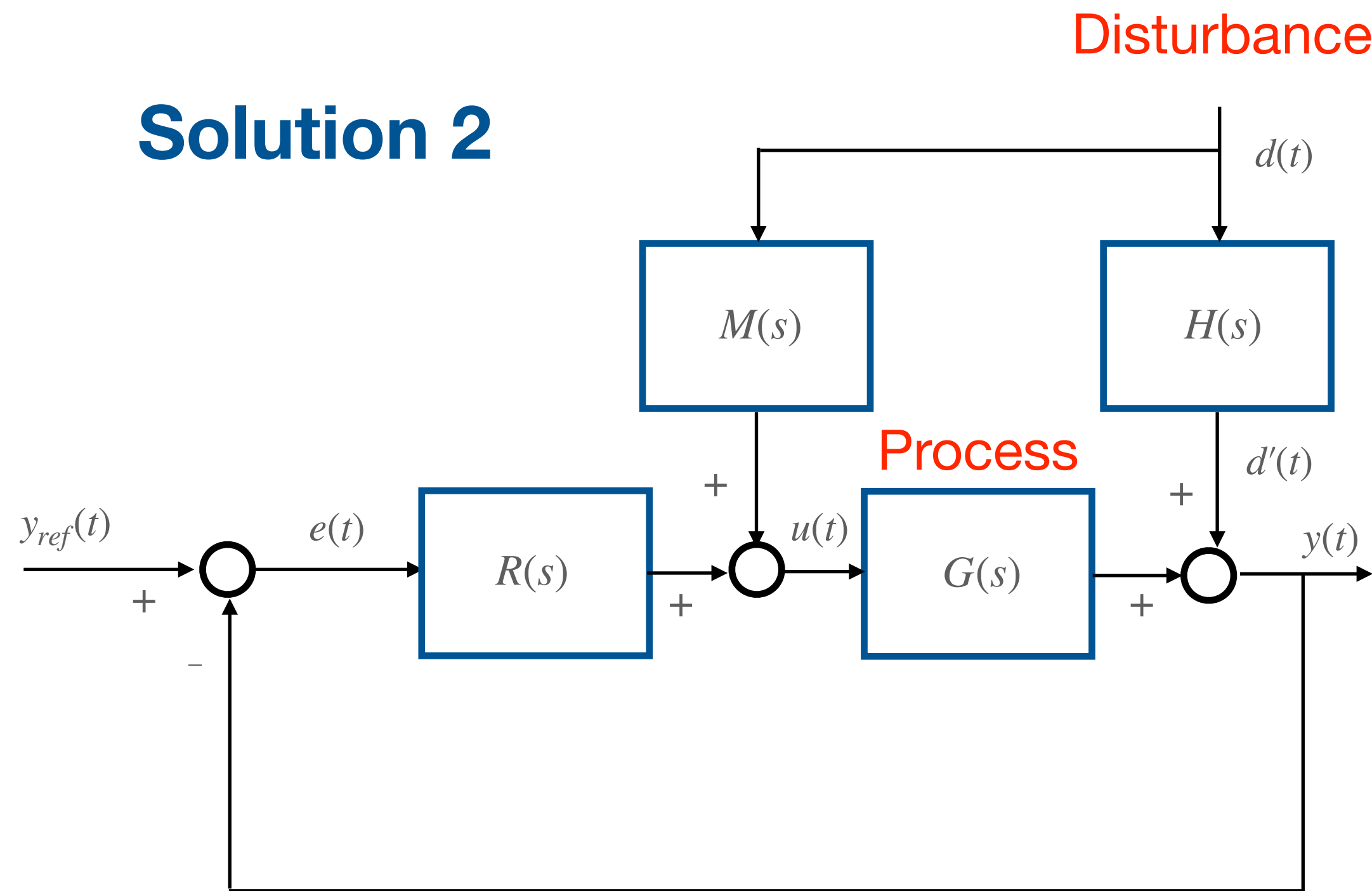
Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

$$d \rightarrow y : \frac{H(s) + M(s)G(s)}{1 + R(s)G(s)} \approx 0$$

Control Scheme with Measurable Disturbance Compensation

Solution 2



Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

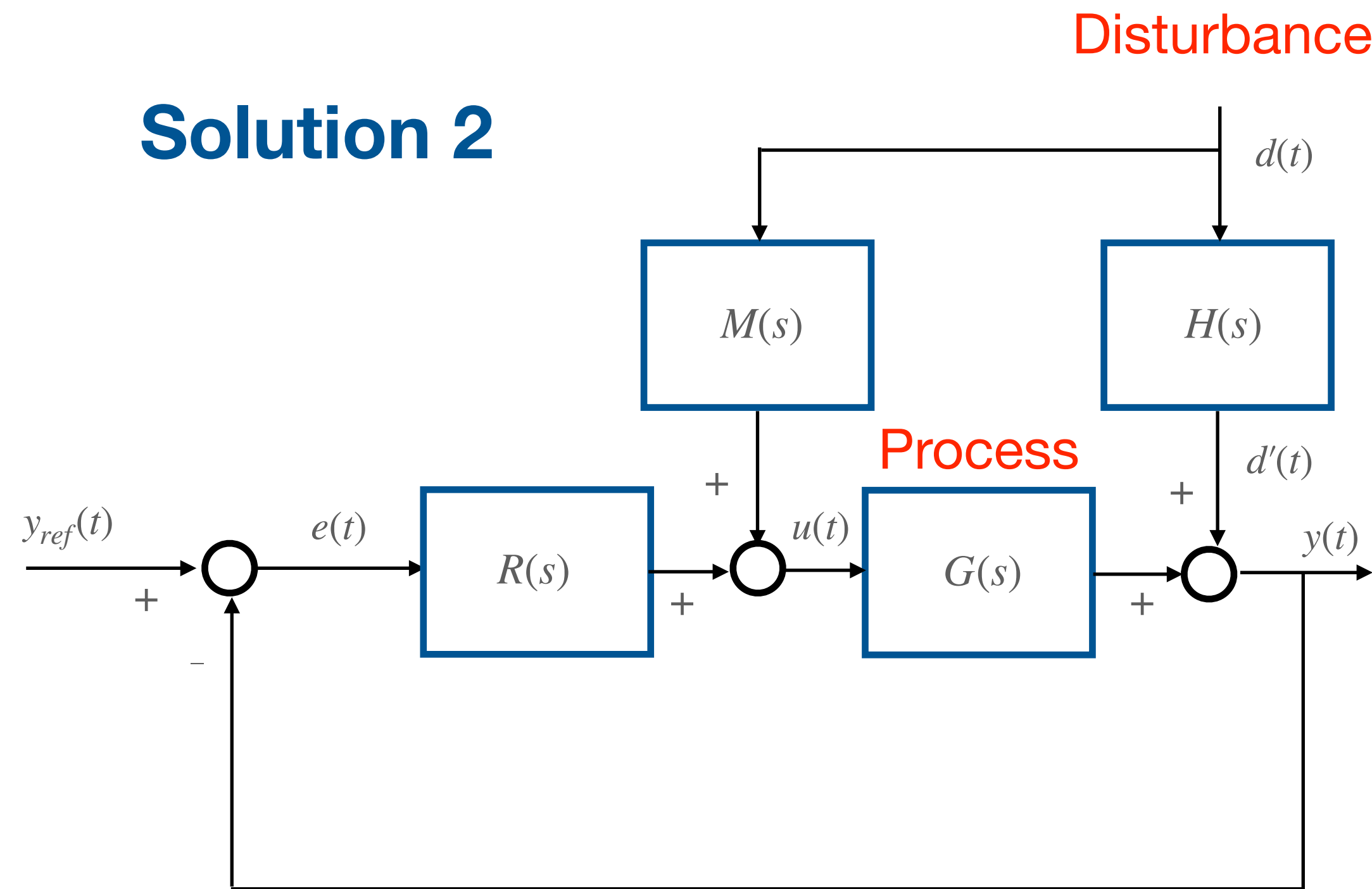
$$d \rightarrow y : \frac{H(s) + M(s)G(s)}{1 + R(s)G(s)} \approx 0$$



$$M(s) = -\frac{H(s)}{G(s)}$$

Control Scheme with Measurable Disturbance Compensation

Solution 2



Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

$$d \rightarrow y : \frac{H(s) + M(s)G(s)}{1 + R(s)G(s)} \approx 0$$



Ideal Design

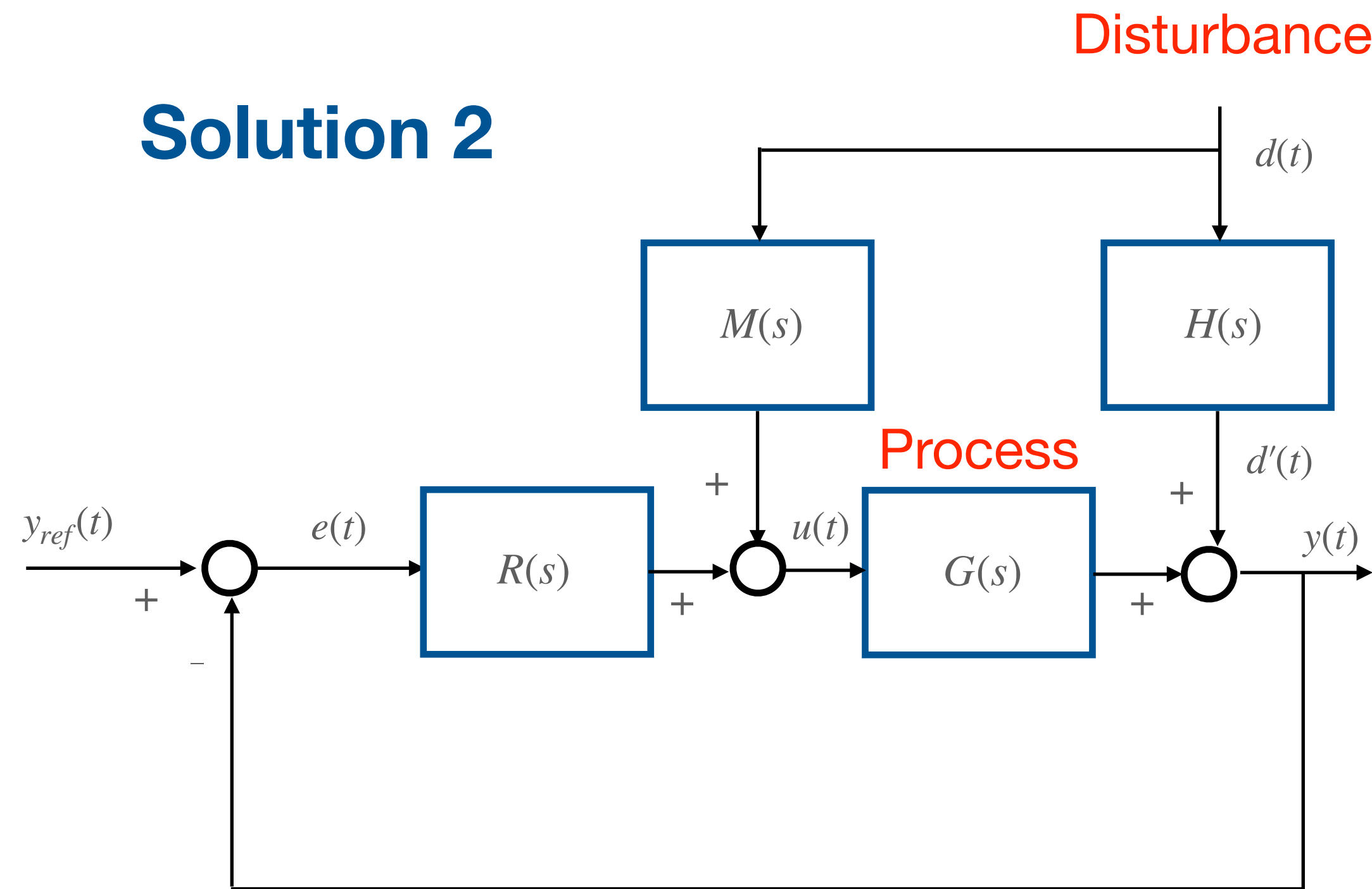
$$M(s) = -\frac{H(s)}{G(s)}$$

Warning:

- $M(s)$ As. Stable
- Causal

Control Scheme with Measurable Disturbance Compensation

Solution 2



Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

$$d \rightarrow y : \frac{H(s) + M(s)G(s)}{1 + R(s)G(s)} \approx 0$$

Ideal Design

$$M(s) = -\frac{H(s)}{G(s)}$$

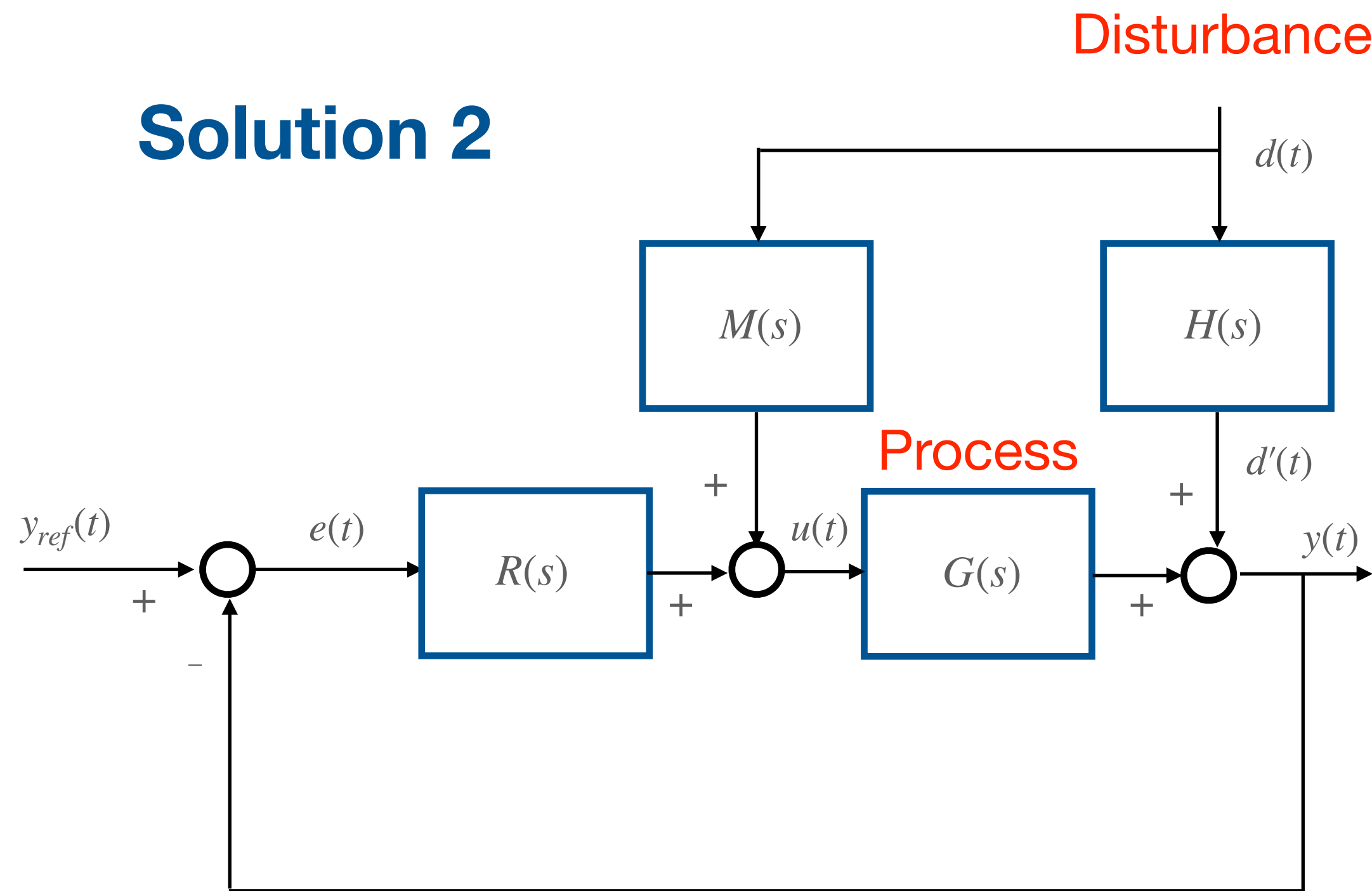
not guaranteed

Warning:

- $M(s)$ As. Stable
- Causal

Control Scheme with Measurable Disturbance Compensation

Solution 2



Disturbance

Process

Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

$$d \rightarrow y : \frac{H(s) + M(s)G(s)}{1 + R(s)G(s)} \approx 0$$

Actual Design

$$M(j\omega) = -\frac{H(j\omega)}{G(j\omega)}$$

in the frequency range where $d(t)$ has significant harmonics

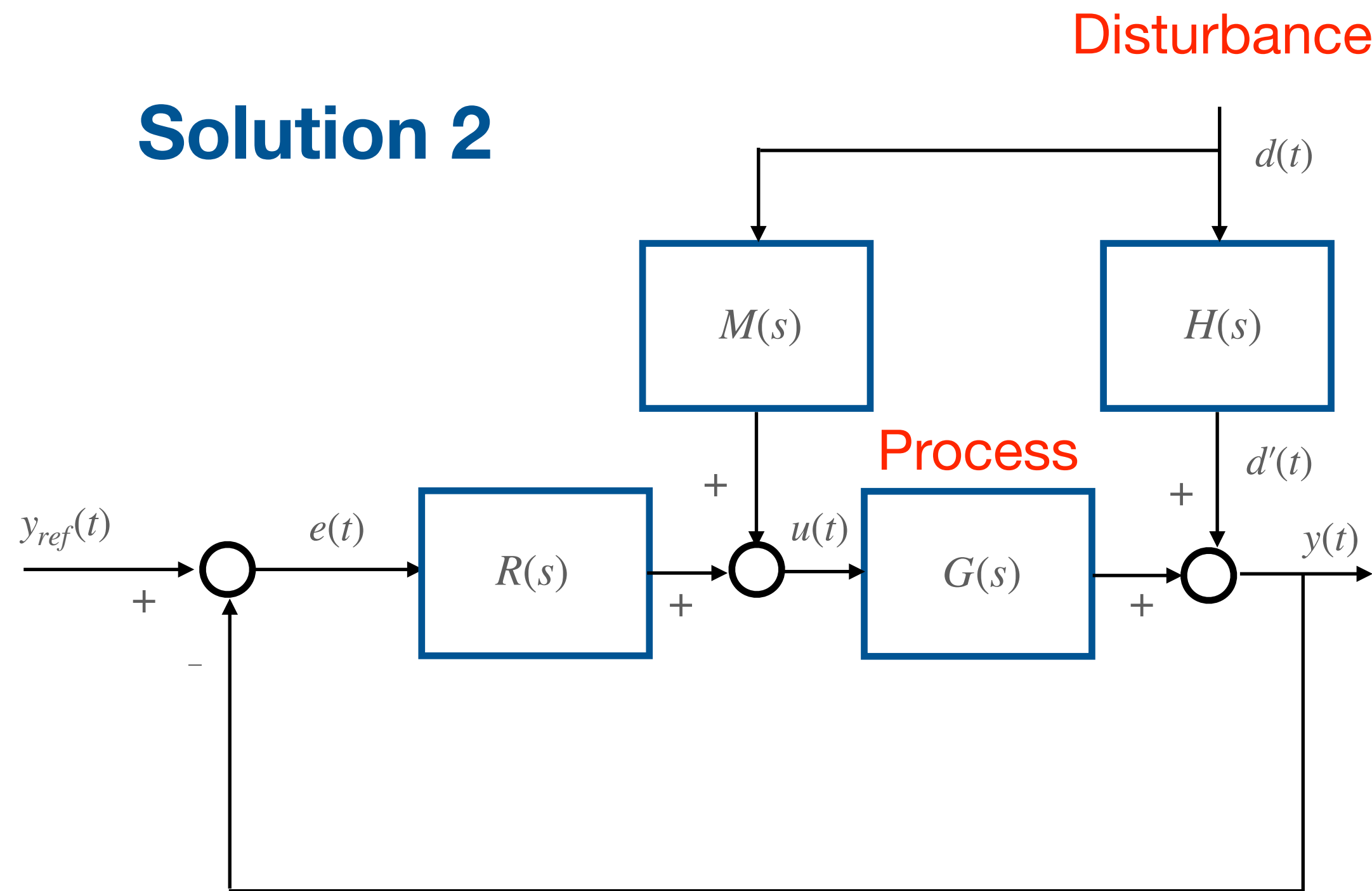
not guaranteed

Warning:

- $M(s)$ As. Stable
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Control Scheme with Measurable Disturbance Compensation

Solution 2



Disturbance

Process

Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

$$d \rightarrow y : \frac{H(s) + M(s)G(s)}{1 + R(s)G(s)} \approx 0$$

Actual Design

$$M(j\omega) = -\frac{H(j\omega)}{G(j\omega)}$$

not guaranteed

Warning:

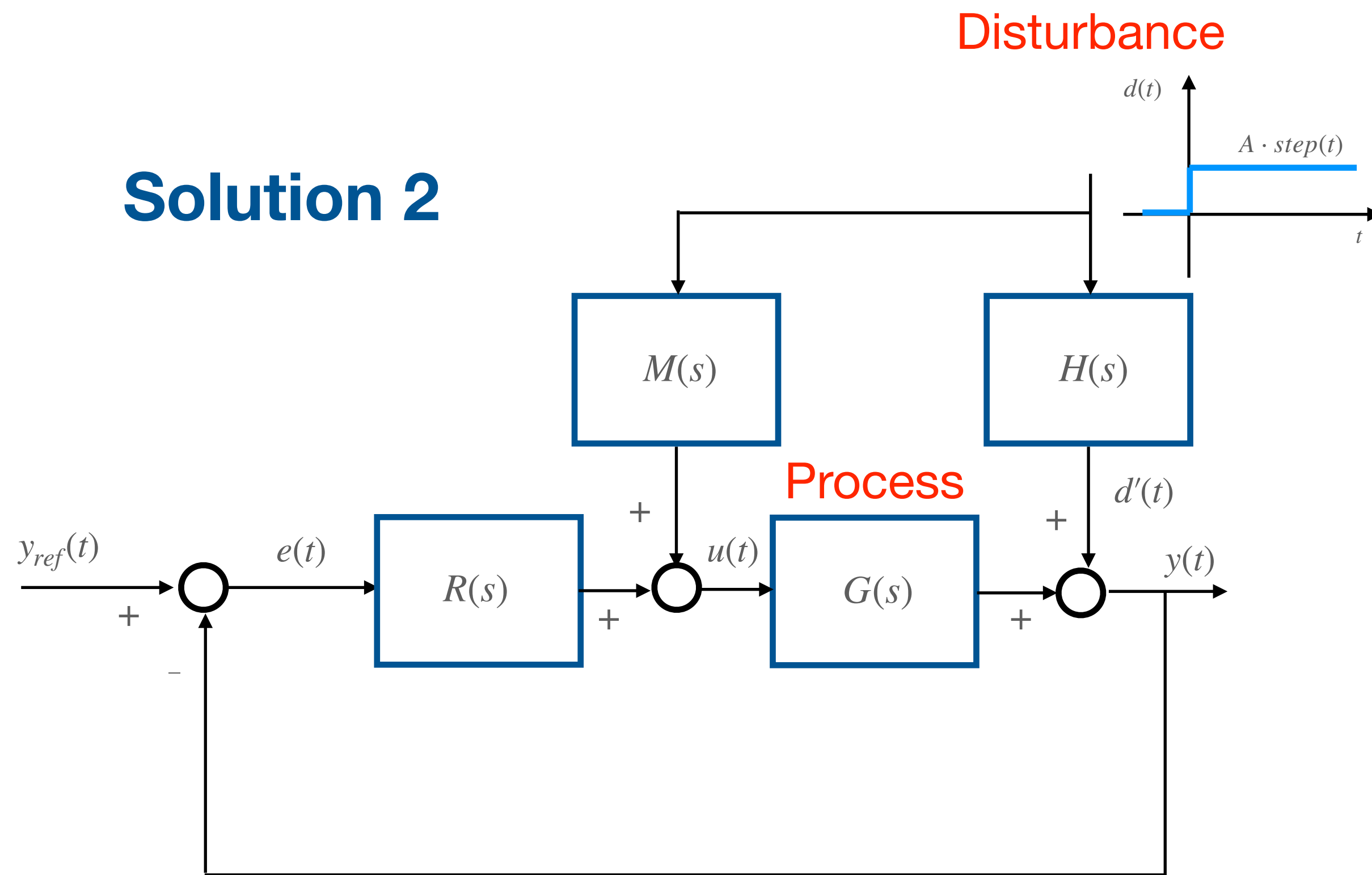
- $M(s)$ As. Stable
- Causal

Example 1:

$$d(t) = \sin(\hat{\omega} t) \longrightarrow M(s) \text{ s.t. } M(j\omega) \Big|_{\omega=\hat{\omega}} = -\frac{H(j\hat{\omega})}{G(j\hat{\omega})}$$

Control Scheme with Measurable Disturbance Compensation

Solution 2



Assumptions:

- $d(t)$ measurable
- $H(s)$ known (possibly 1)

$$d \rightarrow y : \frac{H(s) + M(s)G(s)}{1 + R(s)G(s)} \approx 0$$

Actual Design

$$M(j\omega) = -\frac{H(j\omega)}{G(j\omega)}$$

not guaranteed

Warning:

- $M(s)$ As. Stable
- Causal

Example 2:

static case

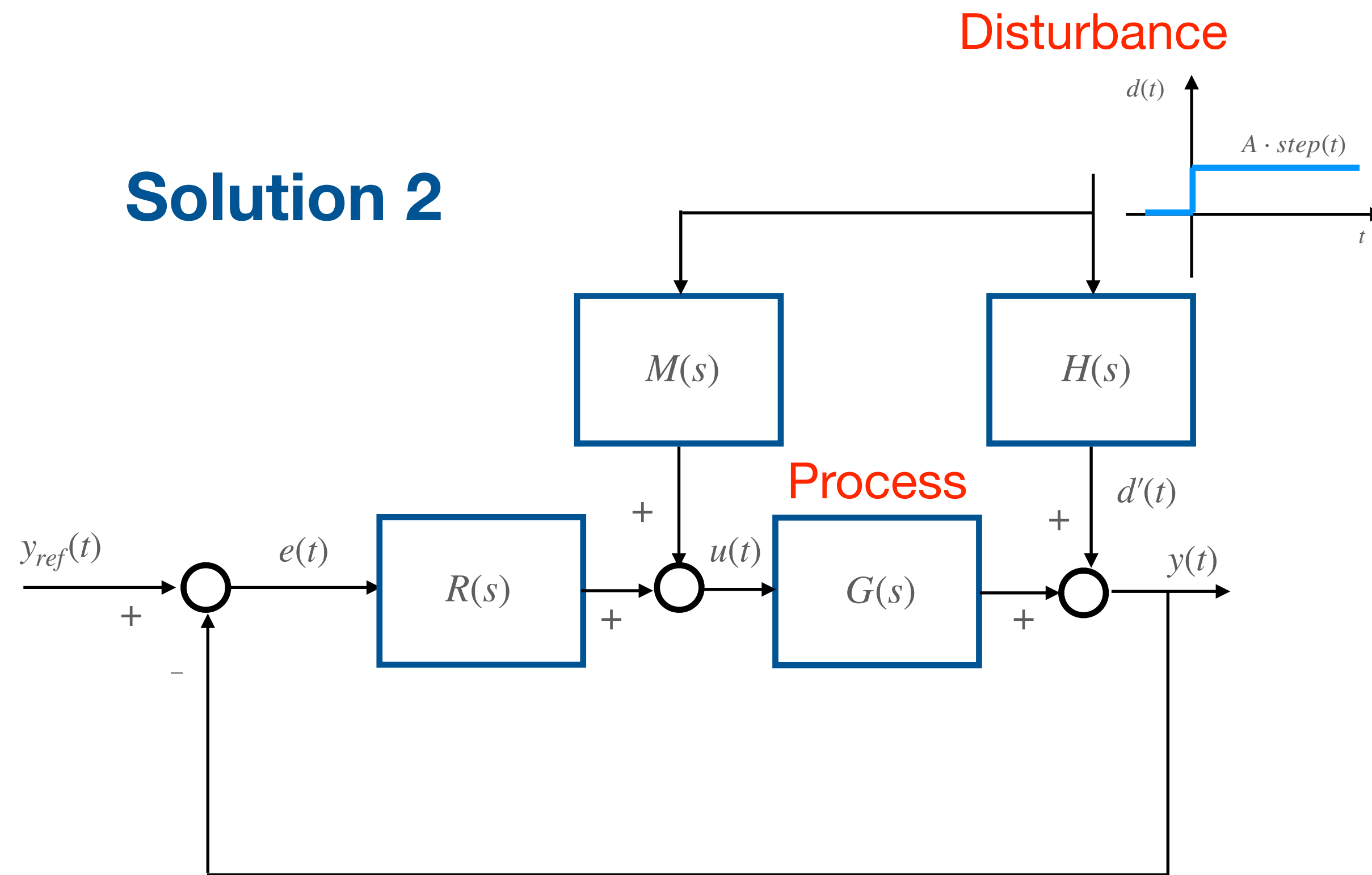
$$\hat{\omega} = 0$$

 $M(s)$ s.t.

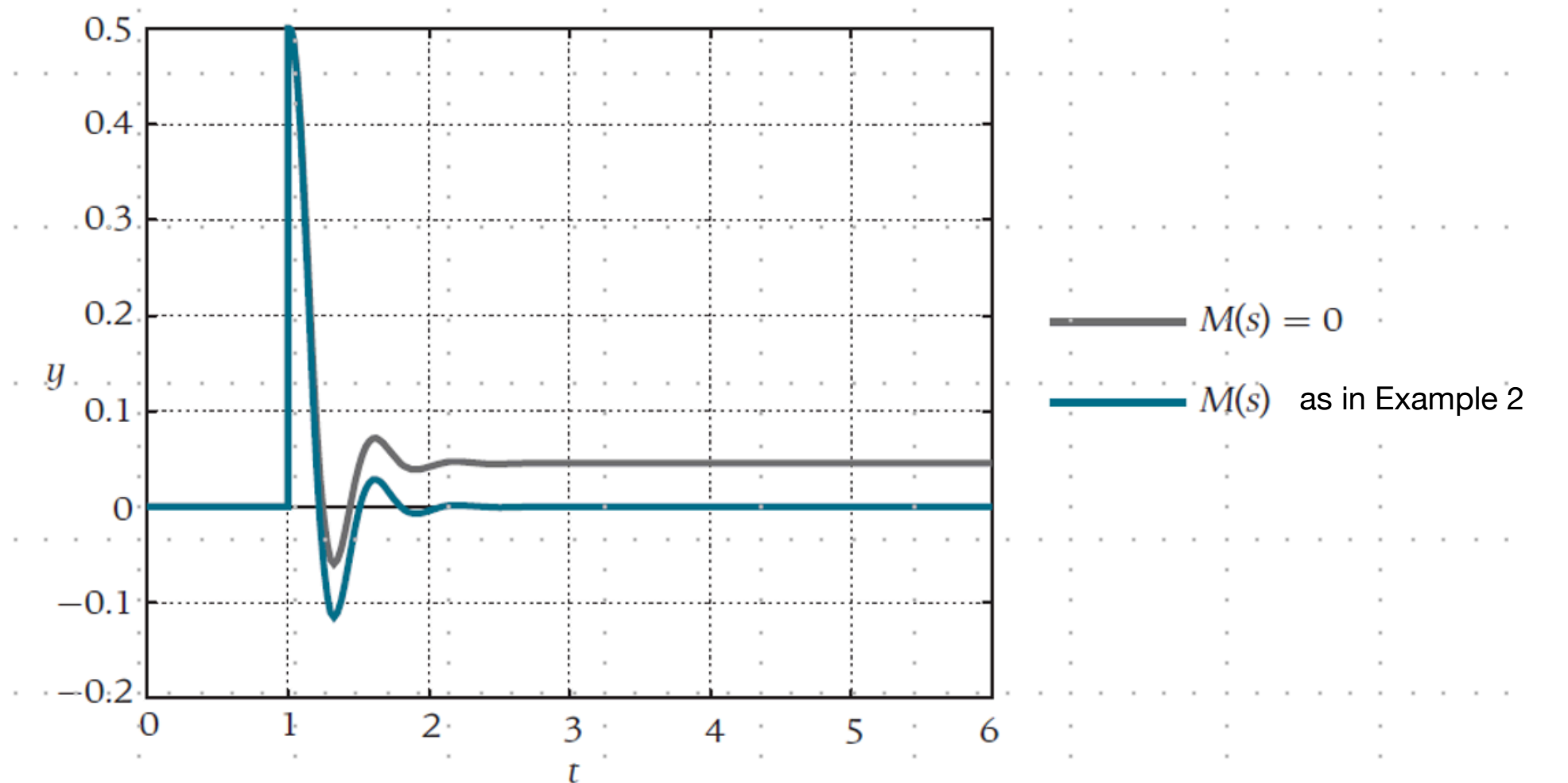
$$M(j\omega) \Big|_{\omega=0} = -\frac{H(0)}{G(0)}$$

Control Scheme with Measurable Disturbance Compensation

Solution 2



Response to a step disturbance



Example 2:

static case

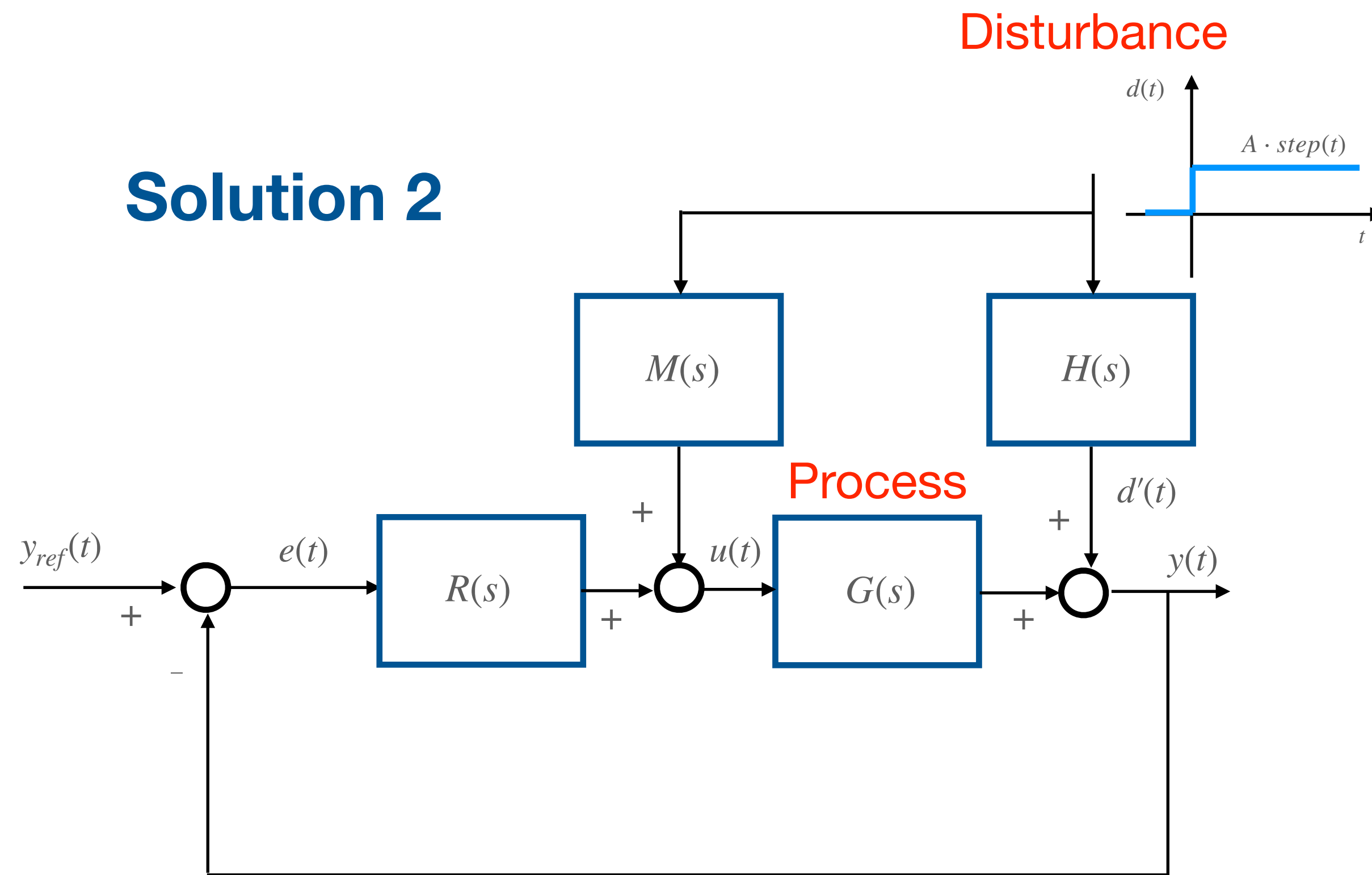
$$\hat{\omega} = 0$$

$M(s)$ s.t.

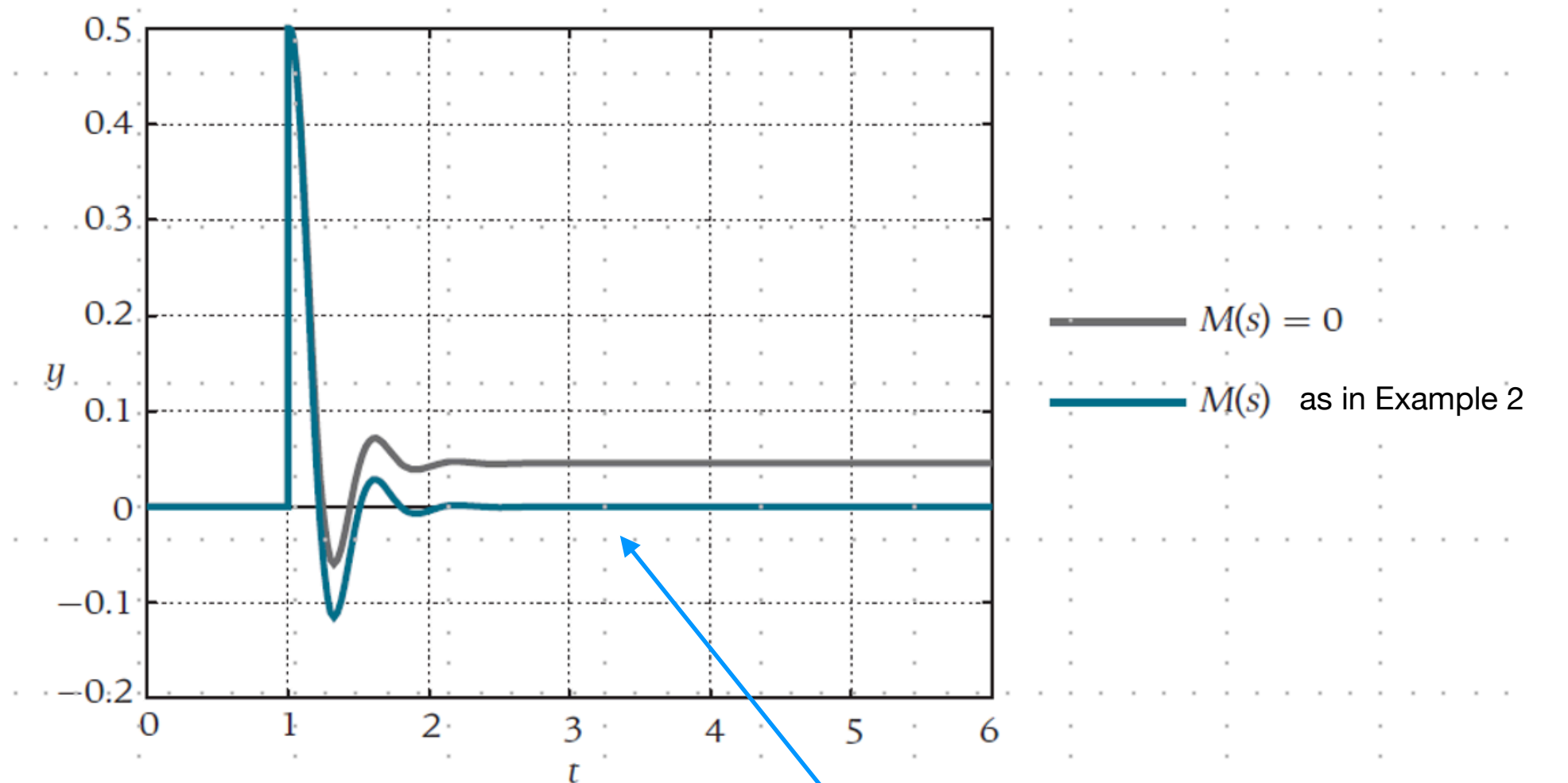
$$M(j\omega) \Big|_{\omega=0} = -\frac{H(0)}{G(0)}$$

Control Scheme with Measurable Disturbance Compensation

Solution 2



Response to a step disturbance



Example 2:

static case

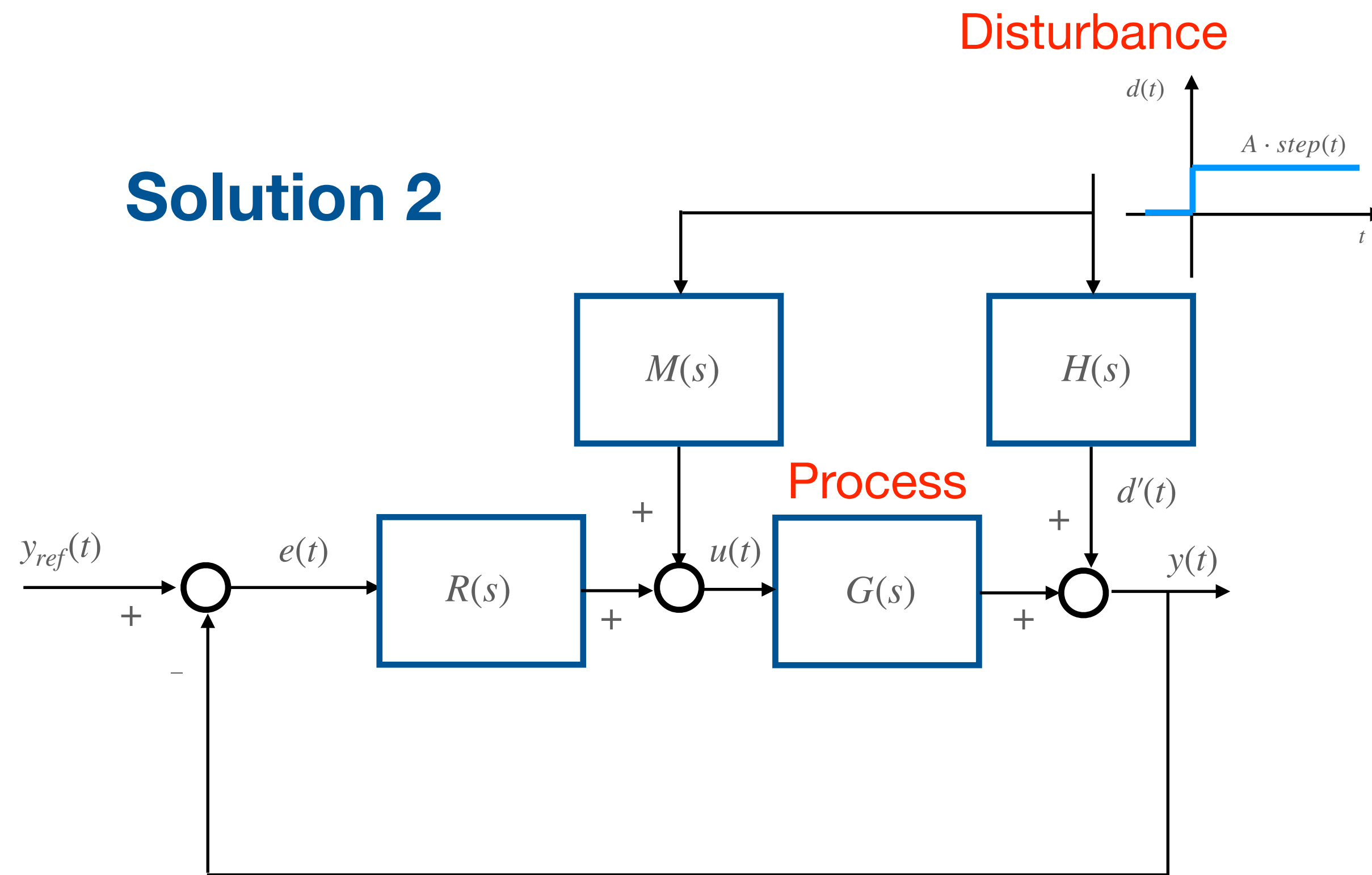
$$\hat{\omega} = 0$$

$M(s)$ s.t.

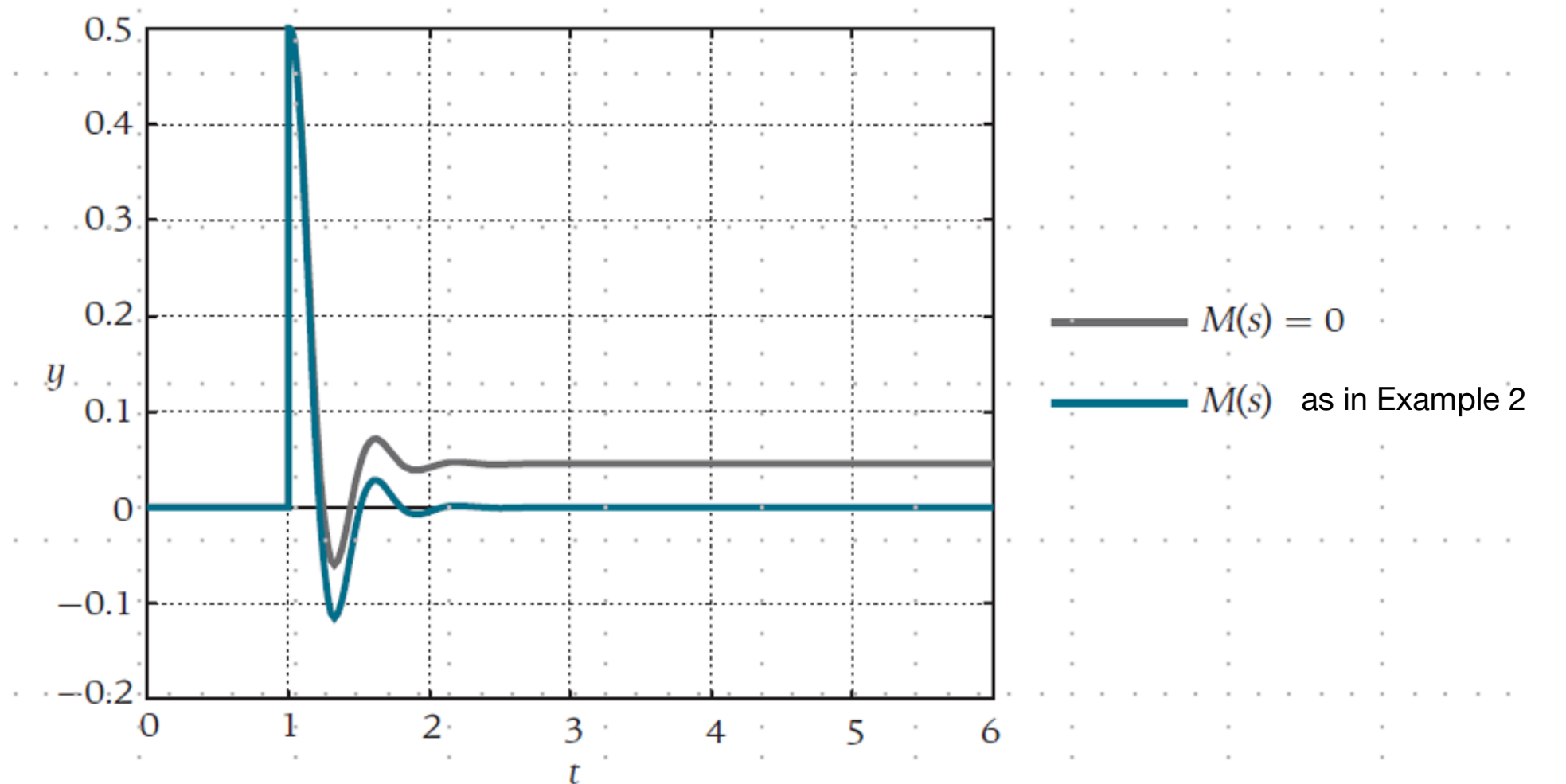
$$M(j\omega) \Big|_{\omega=0} = -\frac{H(0)}{G(0)}$$

Control Scheme with Measurable Disturbance Compensation

Solution 2



Response to a step disturbance



Example 2:

static case

$$\hat{\omega} = 0$$

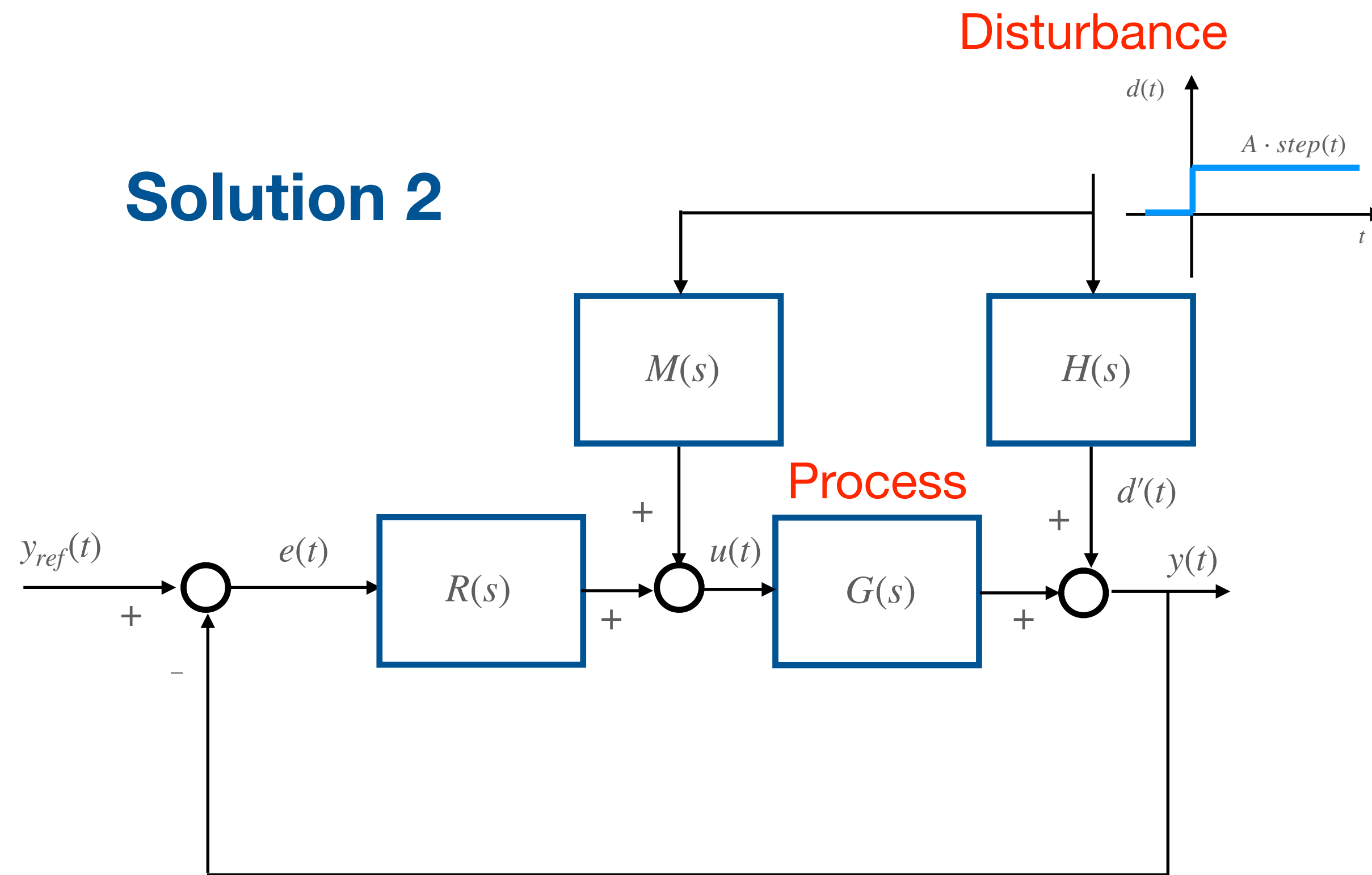
$M(s)$ s.t.

$$M(j\omega) \Big|_{\omega=0} = -\frac{H(0)}{G(0)}$$

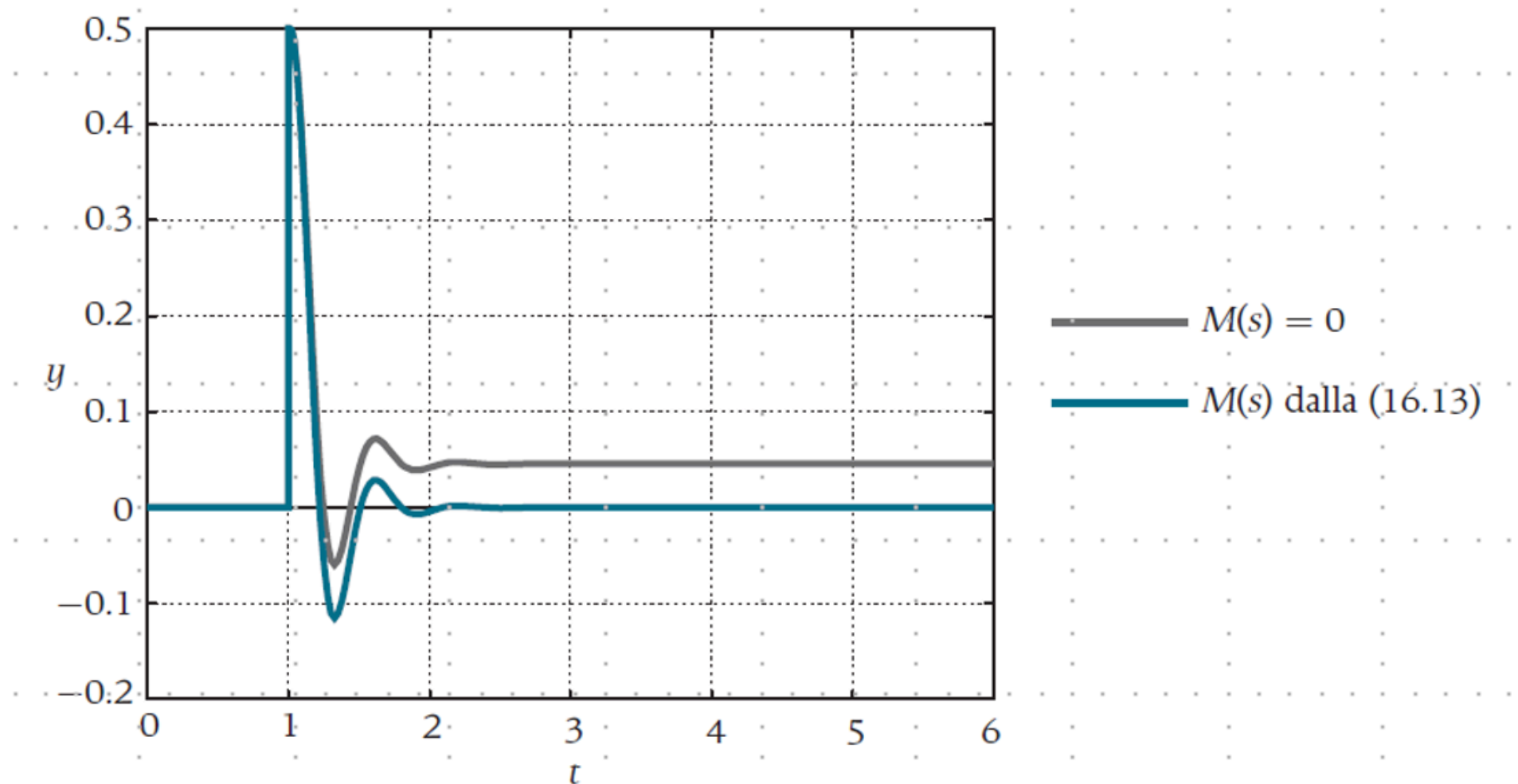
Is this static compensation robust?

Control Scheme with Measurable Disturbance Compensation

Solution 2



Response to a step disturbance



Example 2:

static case

$$\hat{\omega} = 0$$

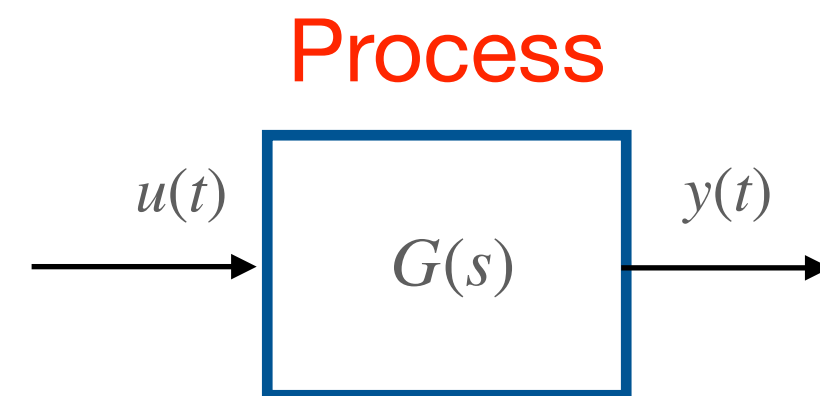
$M(s)$ s.t.

$$M(j\omega) \Big|_{\omega=0} = -\frac{H(0)}{G(0)}$$

Is this static compensation robust?

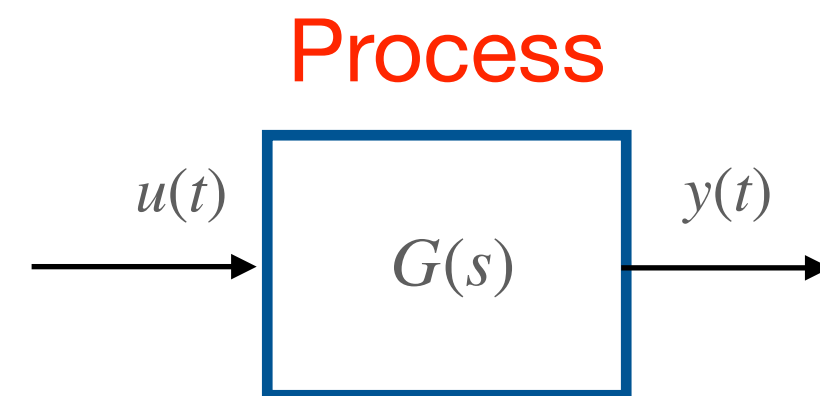
Is there a better alternative?

Control of LTI Systems with Delays



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

Control of LTI Systems with Delays

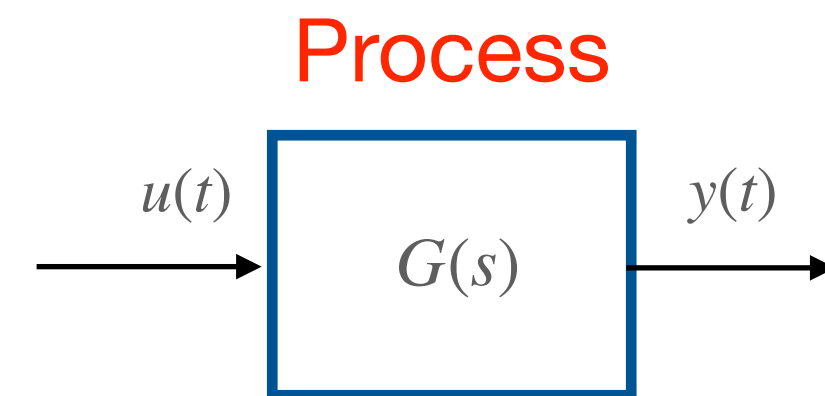


$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$



Image credit to: <https://www.gramconveyor.com/how-to-install-conveyor-belt/>

Control of LTI Systems with Delays



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

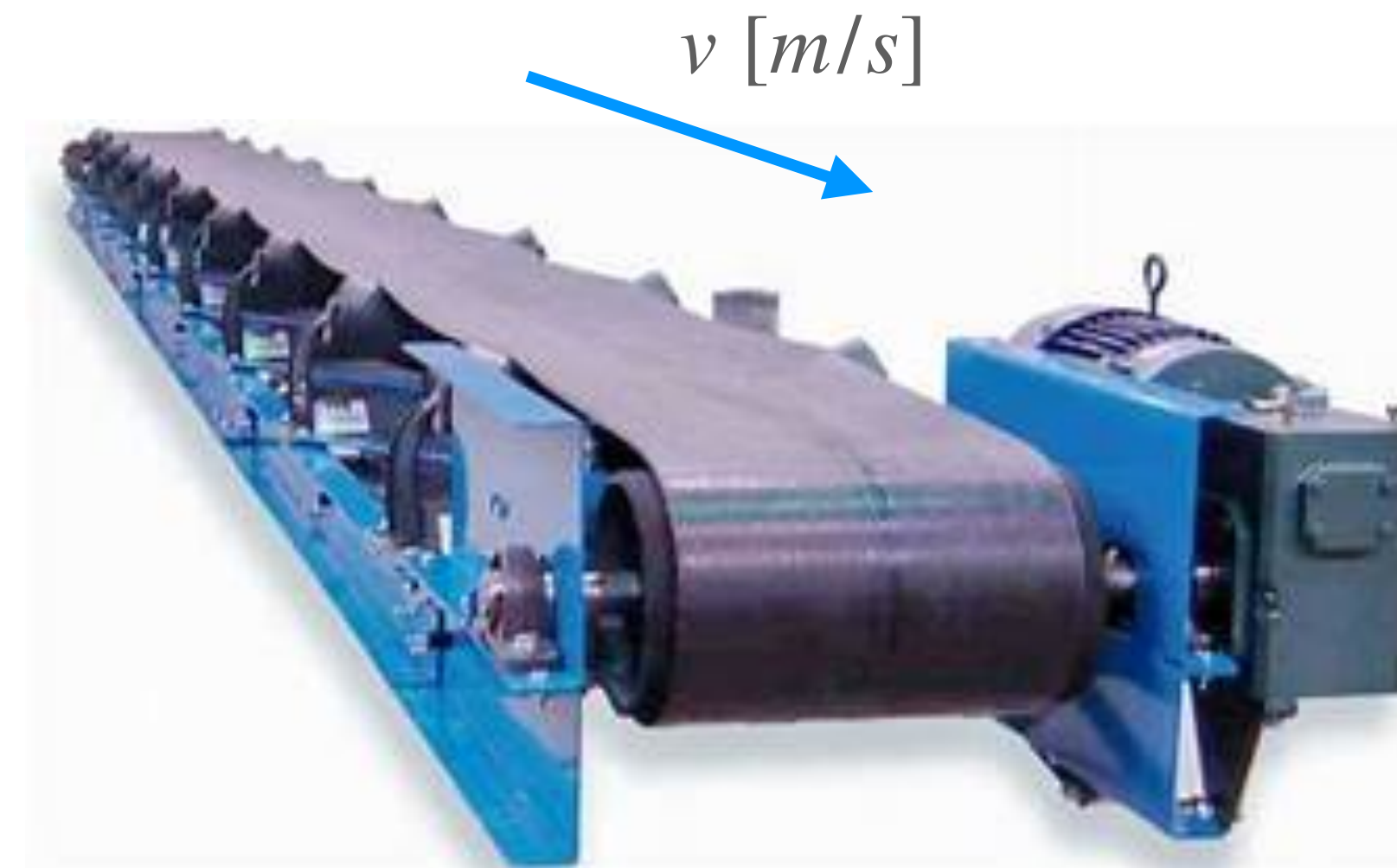
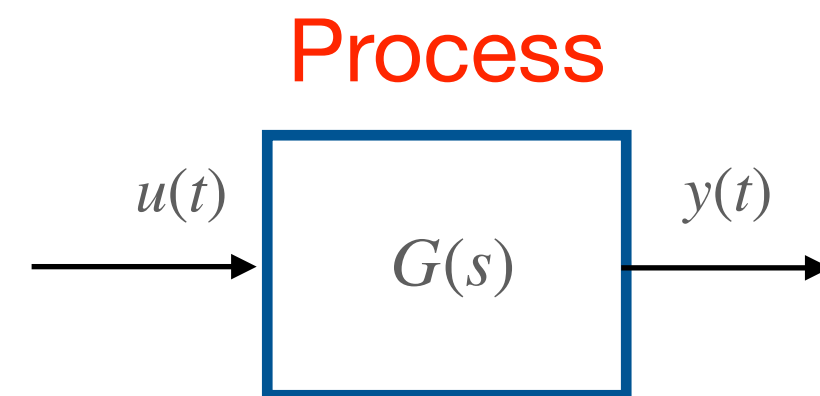


Image credit to: <https://www.gramconveyor.com/how-to-install-conveyor-belt/>

Control of LTI Systems with Delays



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

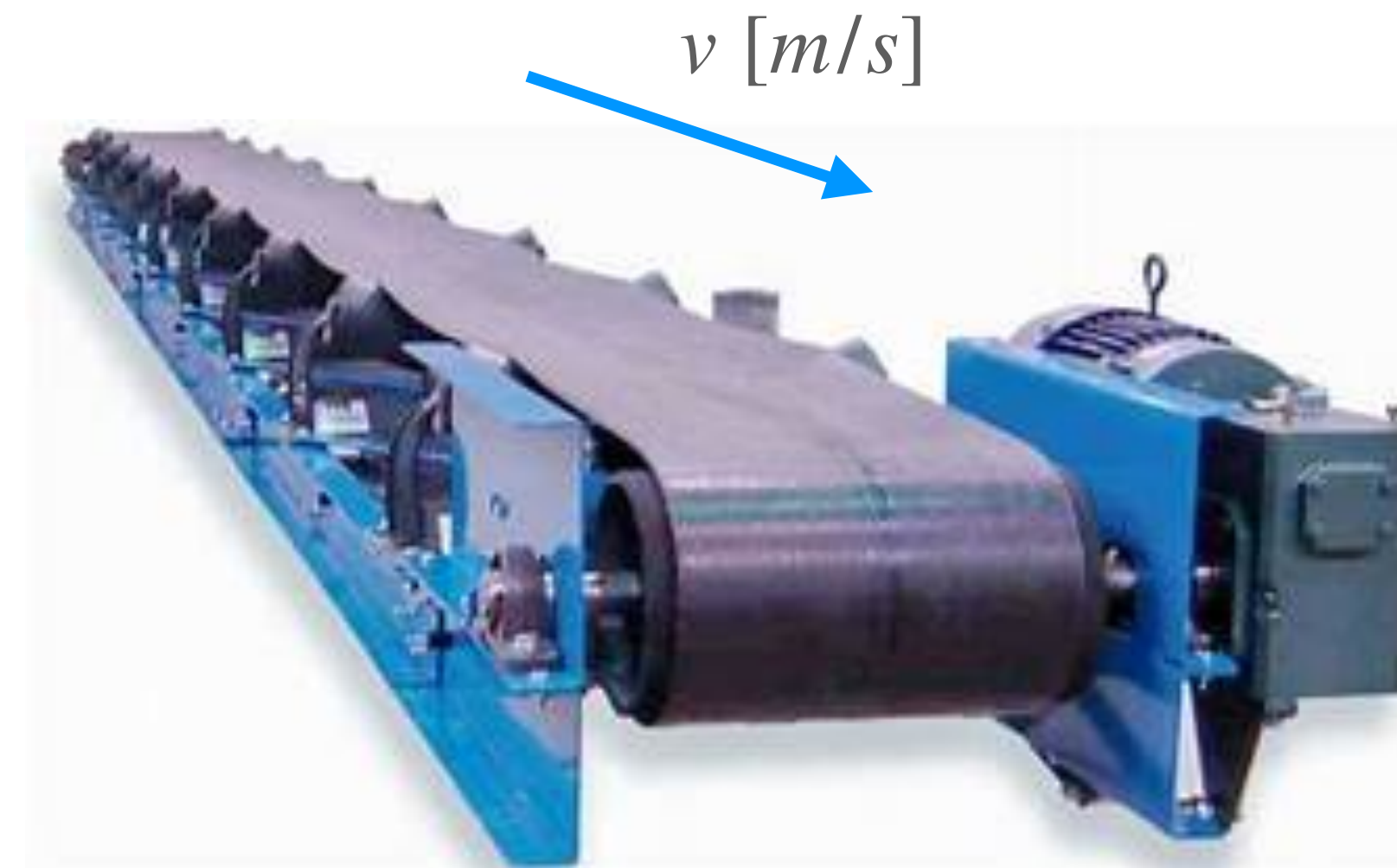
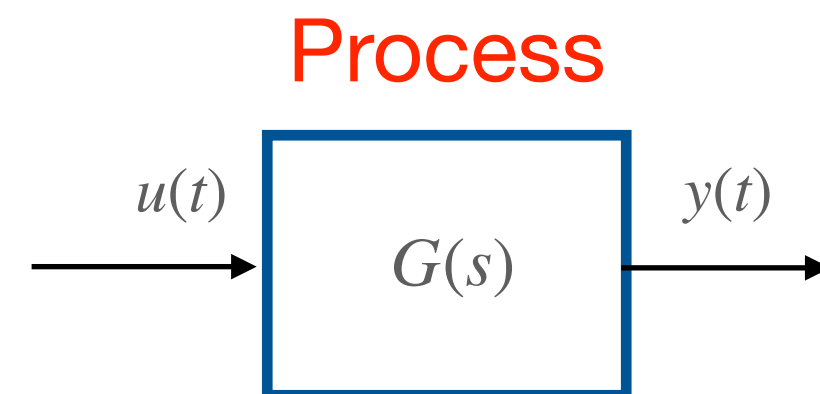


Image credit to: <https://www.gramconveyor.com/how-to-install-conveyor-belt/>

$$q_{OUT}(t) = q_{IN}(t - \tau) \quad \tau = \frac{L}{v}$$

Control of LTI Systems with Delays



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

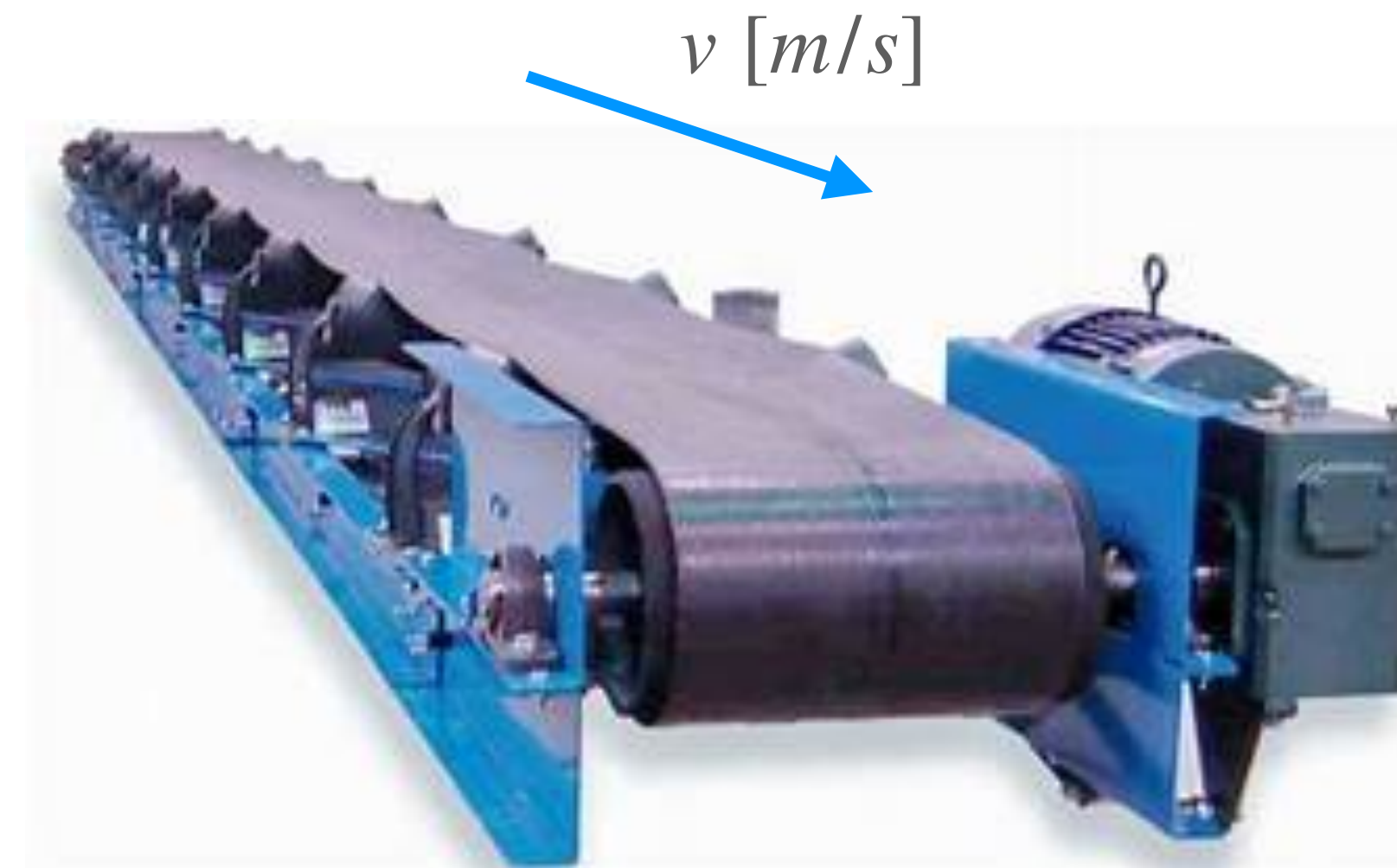
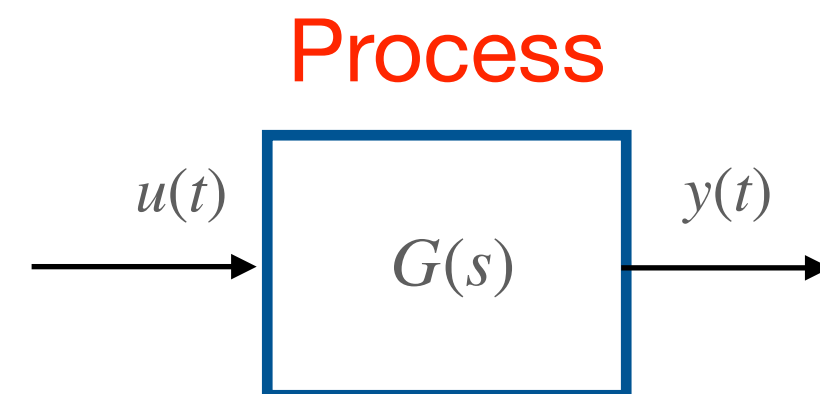


Image credit to: <https://www.gramconveyor.com/how-to-install-conveyor-belt/>

$$\mathcal{L} \begin{cases} q_{OUT}(t) = q_{IN}(t - \tau) \\ Q_{OUT}(s) = e^{-\tau s} Q_{IN}(s) \end{cases} \quad \tau = \frac{L}{v}$$

Control of LTI Systems with Delays



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

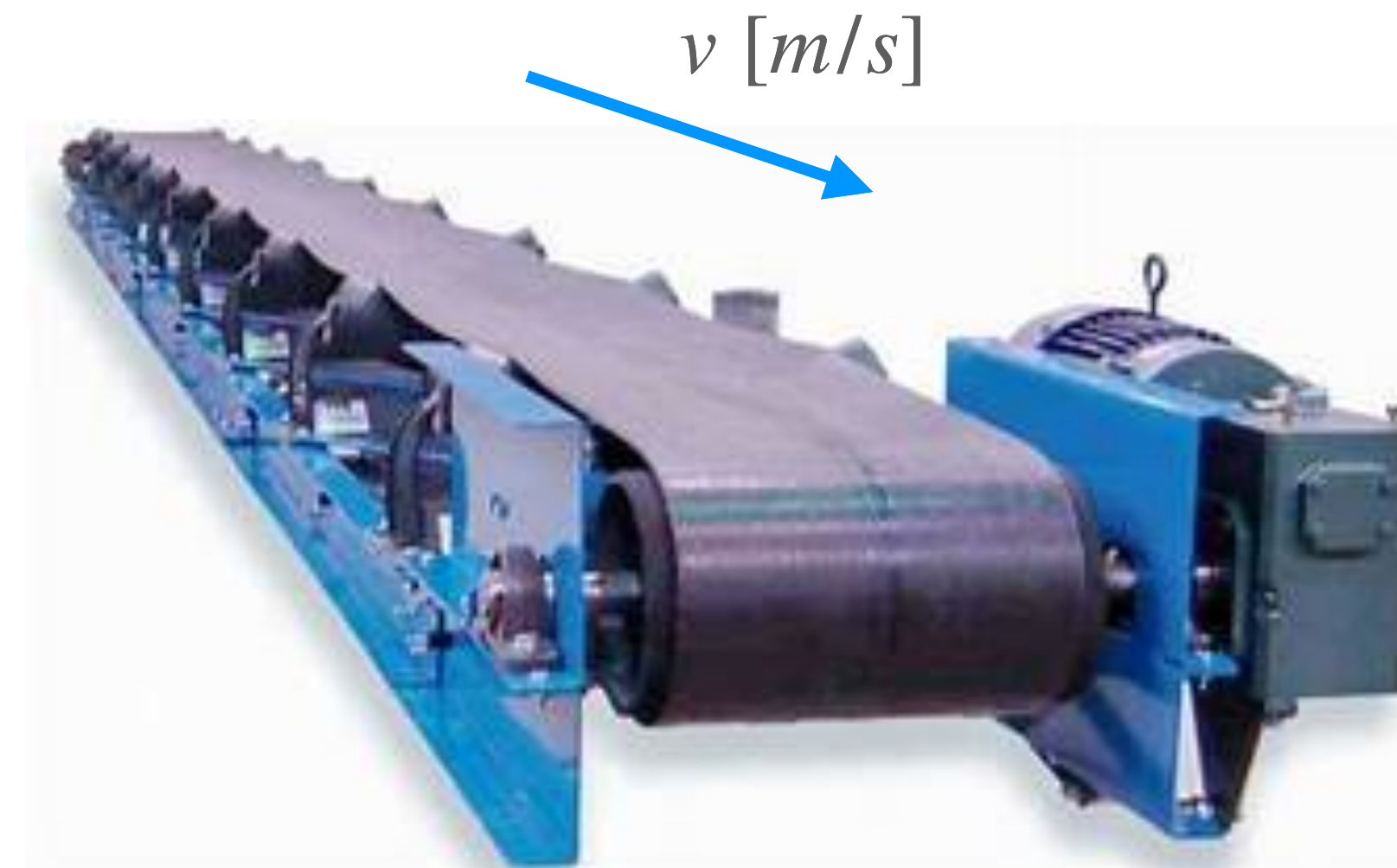
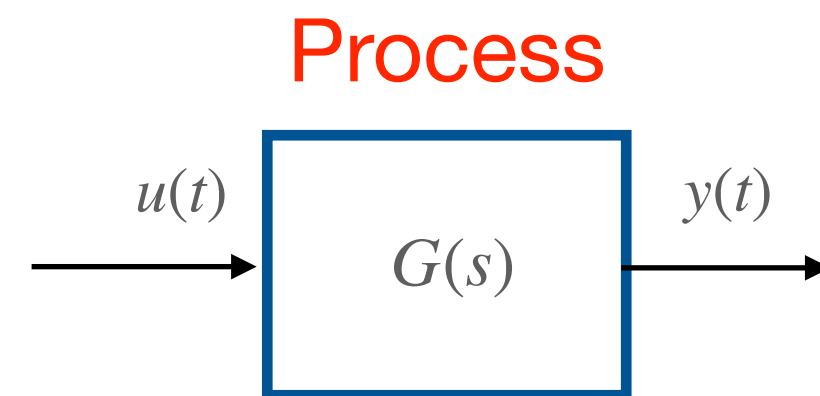


Image credit to: <https://www.gramconveyor.com/how-to-install-conveyor-belt/>

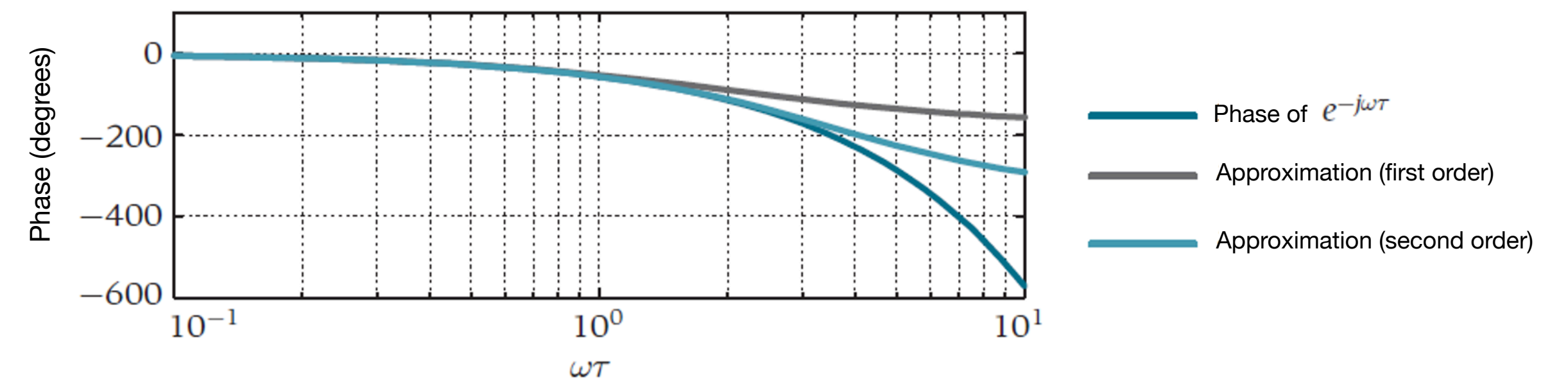
$$G(s) = e^{-\tau s} \quad \leftarrow \quad \mathcal{L} \quad \begin{aligned} q_{OUT}(t) &= q_{IN}(t - \tau) \\ Q_{OUT}(s) &= e^{-\tau s} Q_{IN}(s) \end{aligned} \quad \tau = \frac{L}{v}$$

Control of LTI Systems with Delays

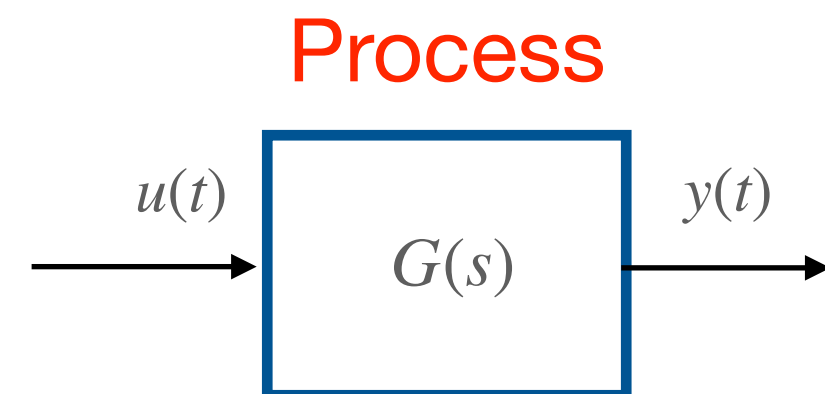


$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

Phase Lag due to the delay term

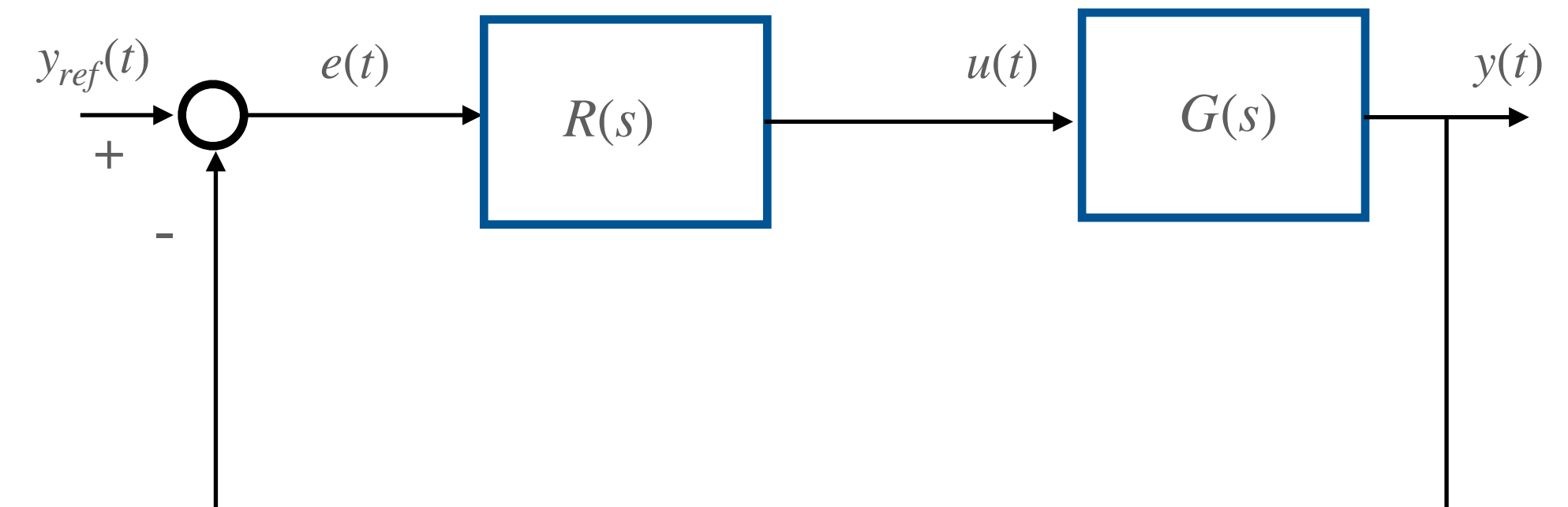
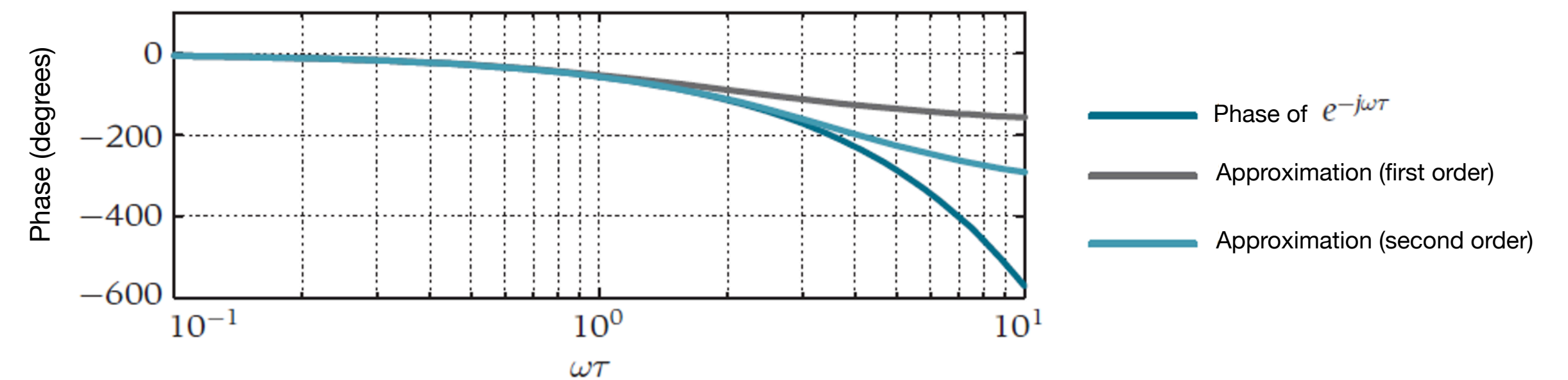


Control of LTI Systems with Delays

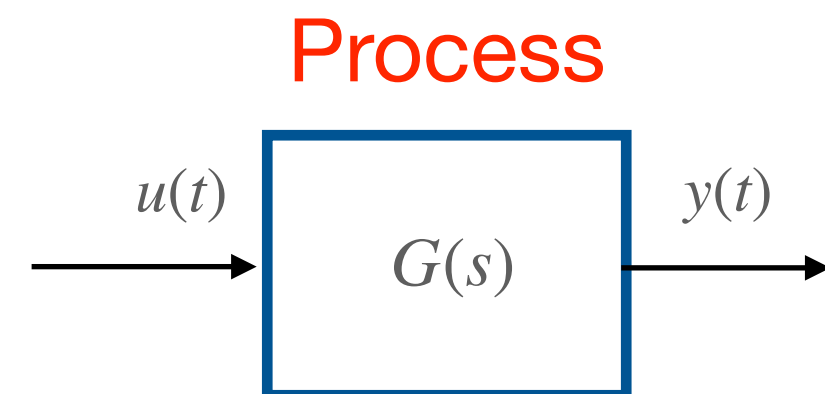


$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

Phase Lag due to the delay term

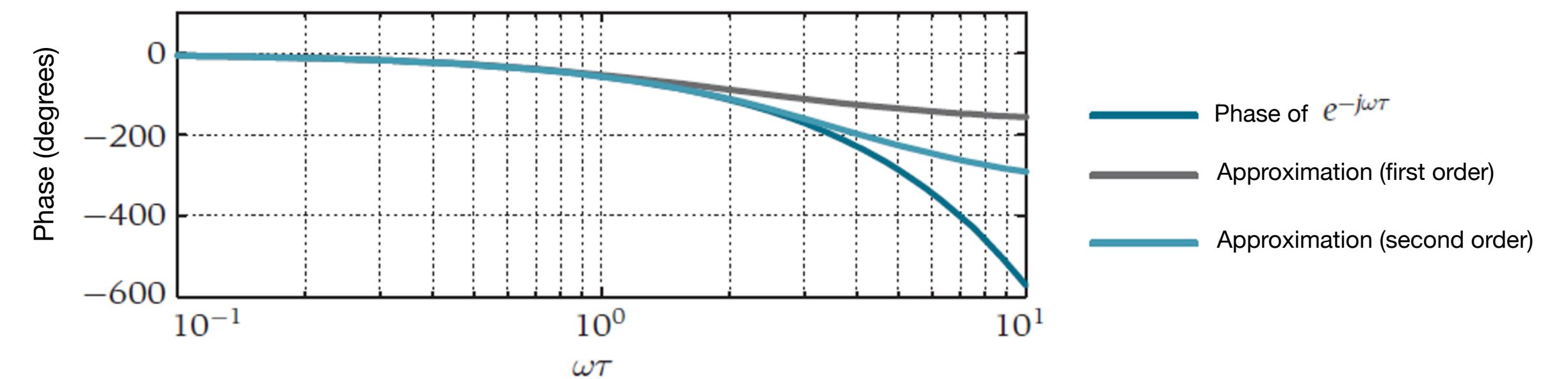


Control of LTI Systems with Delays

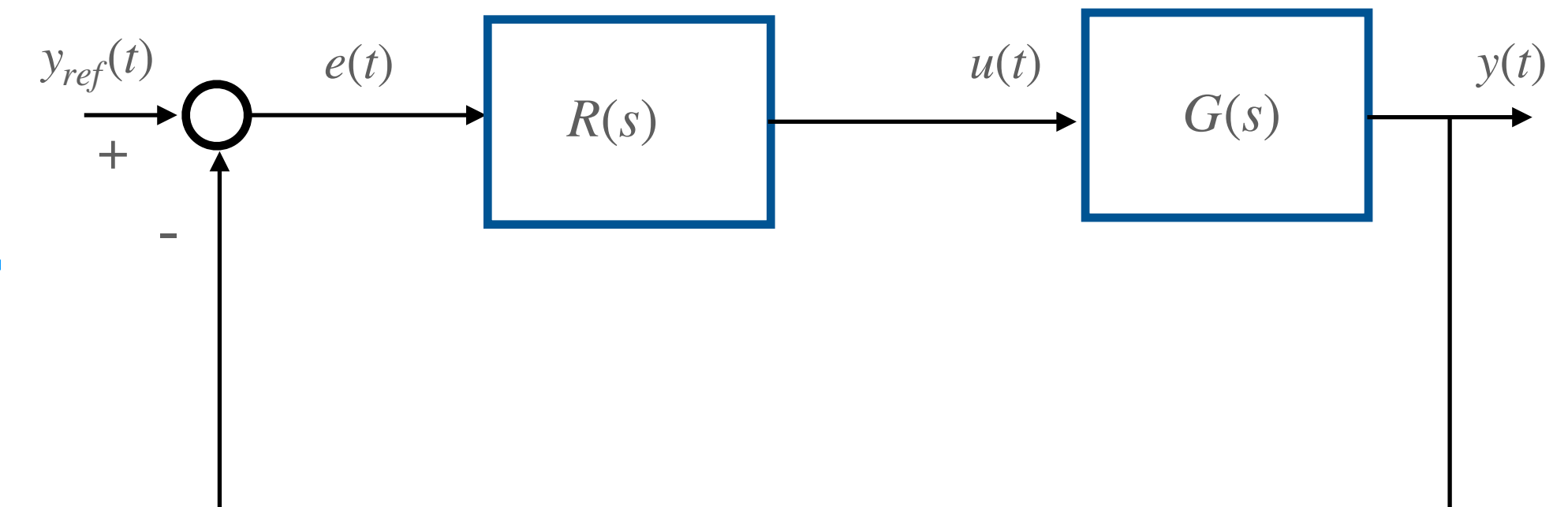


$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

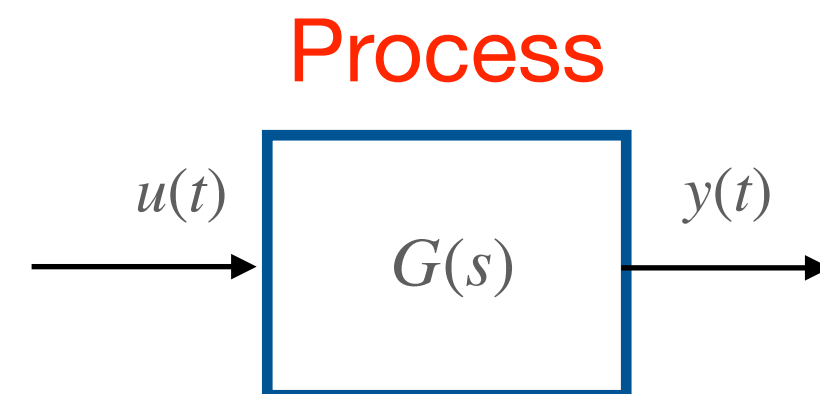
Phase Lag due to the delay term



Recommendation: $\omega_{BW_{CL}}$ sufficiently narrow



Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

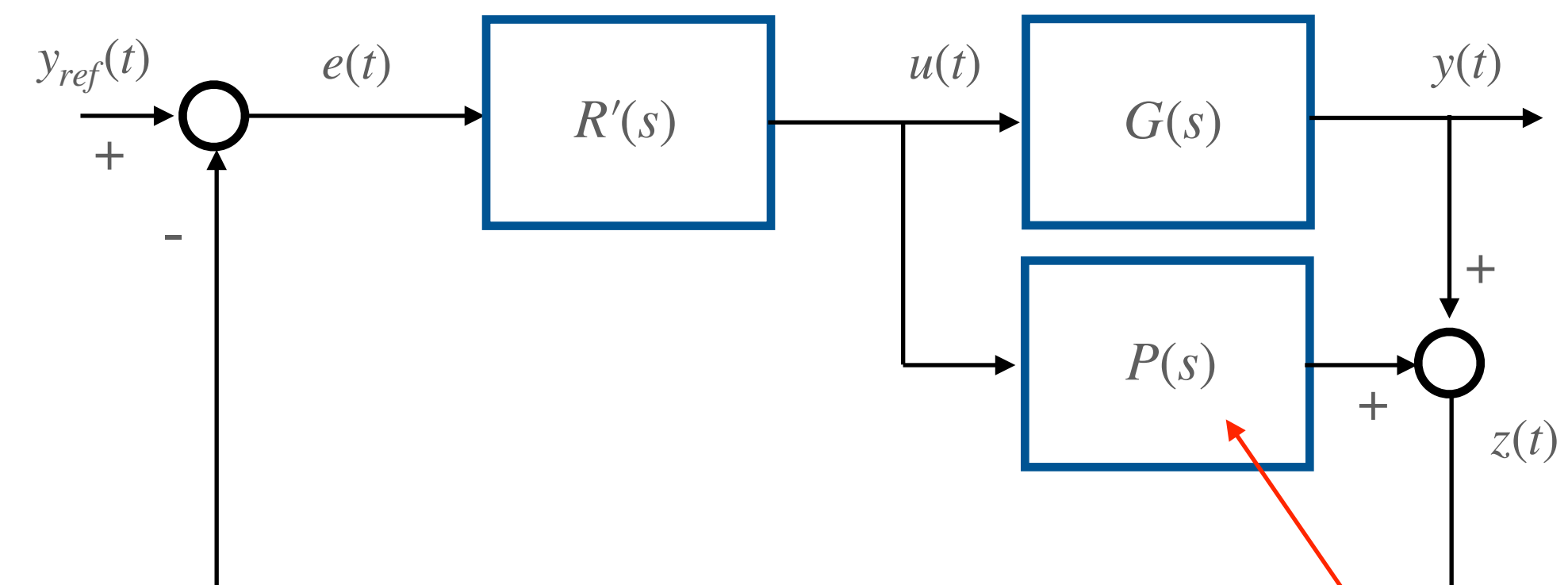
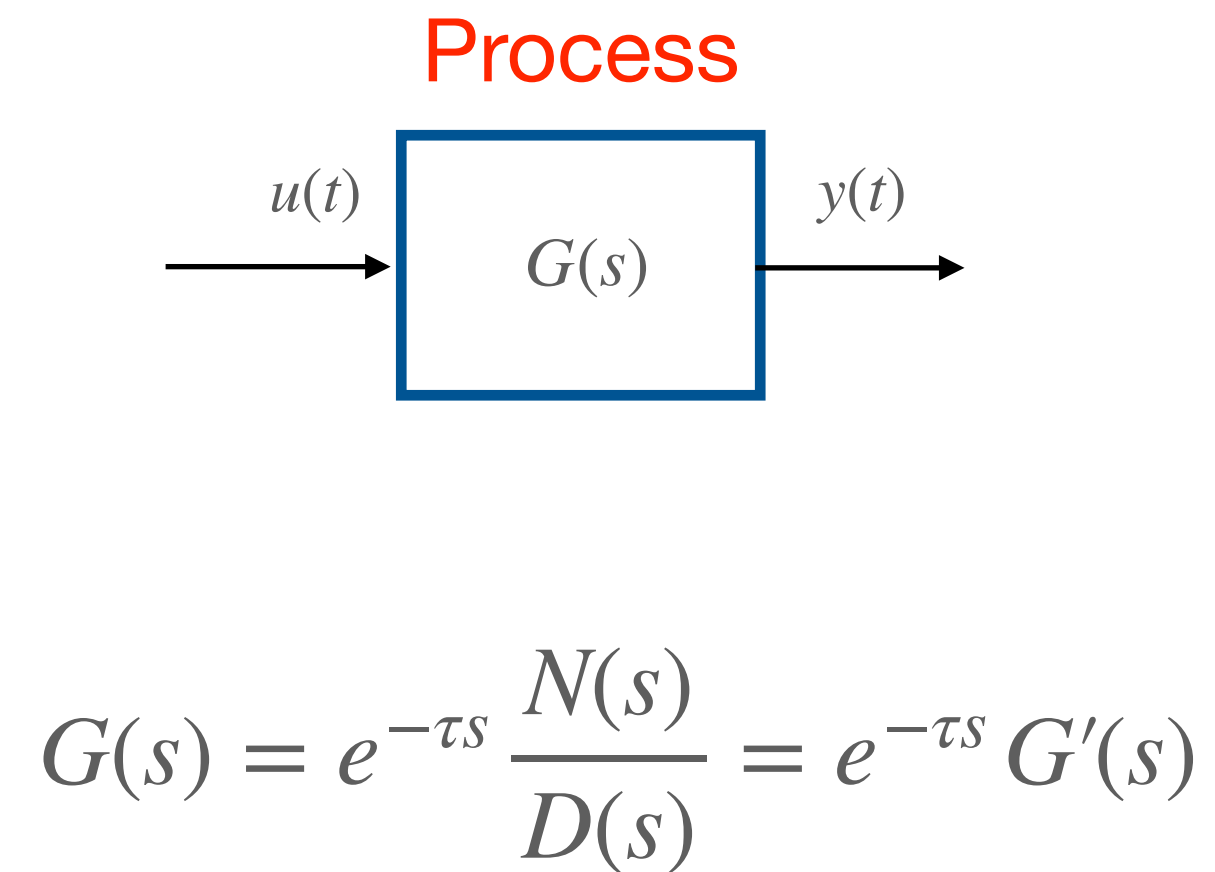


$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

Assumptions:

- $G'(s)$ As. Stable
- τ known

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

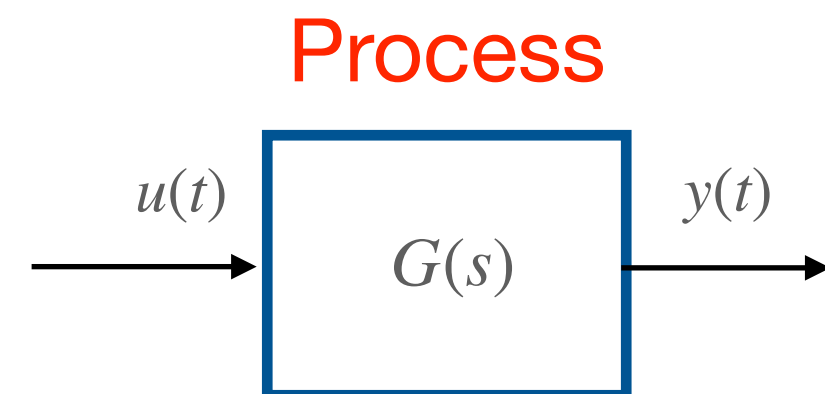


Smith Predictor

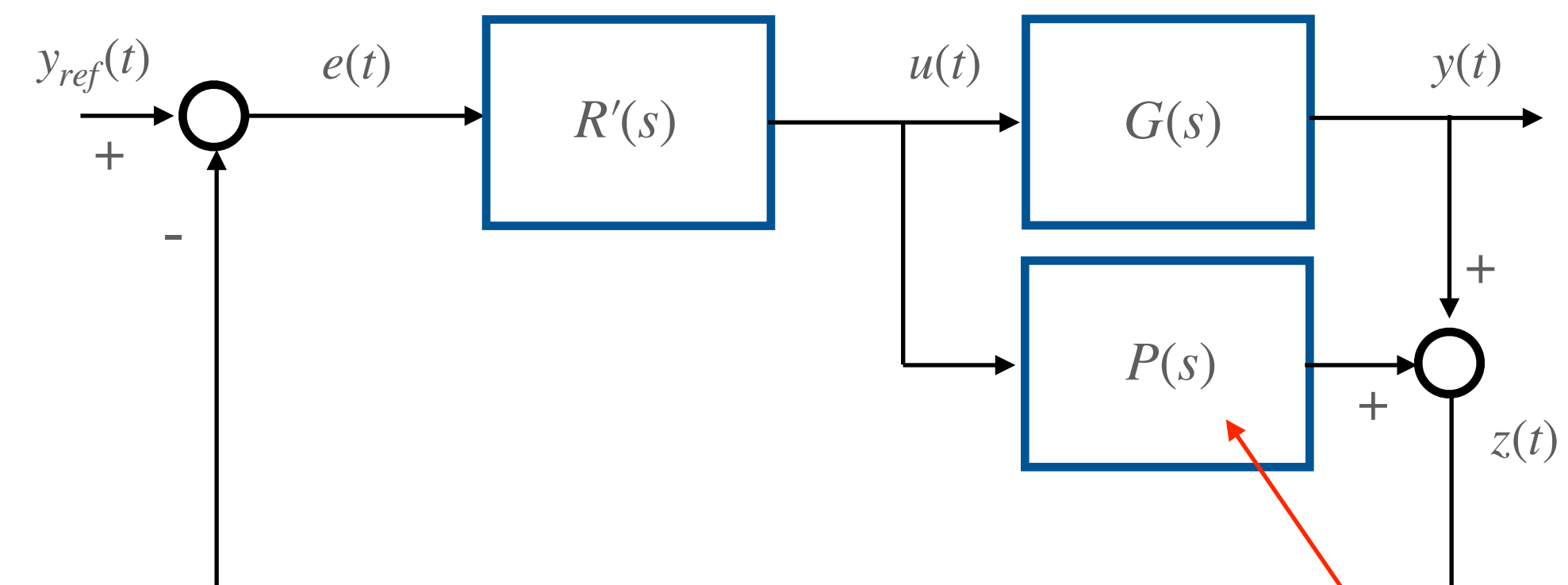
Assumptions:

- $G'(s)$ As. Stable
- τ known

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$



Smith Predictor

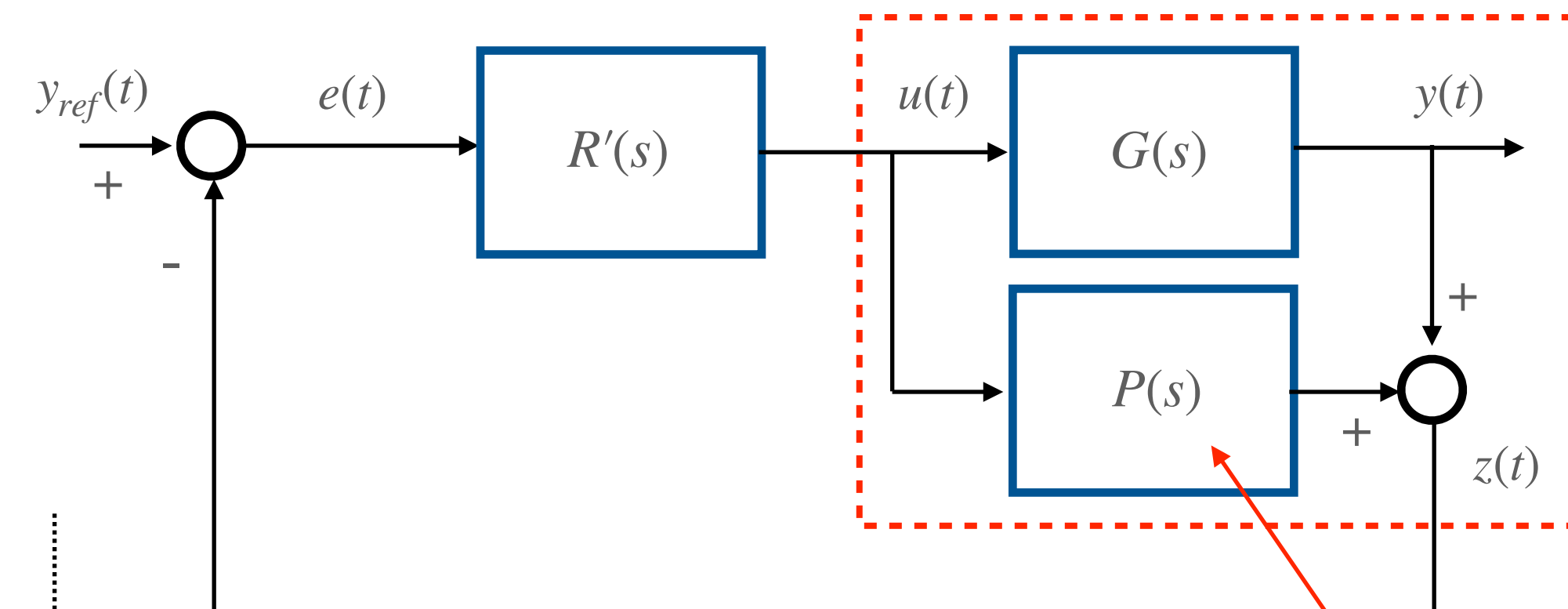
Assumptions:

- $G'(s)$ As. Stable
- τ known

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

$$\begin{aligned}
 u \rightarrow z : \quad G(s) + P(s) &= e^{-\tau s} G'(s) + (1 - e^{-\tau s}) G'(s) \\
 &= e^{-\tau s} G'(s) + G'(s) - e^{-\tau s} G'(s) \\
 &= G'(s)
 \end{aligned}$$

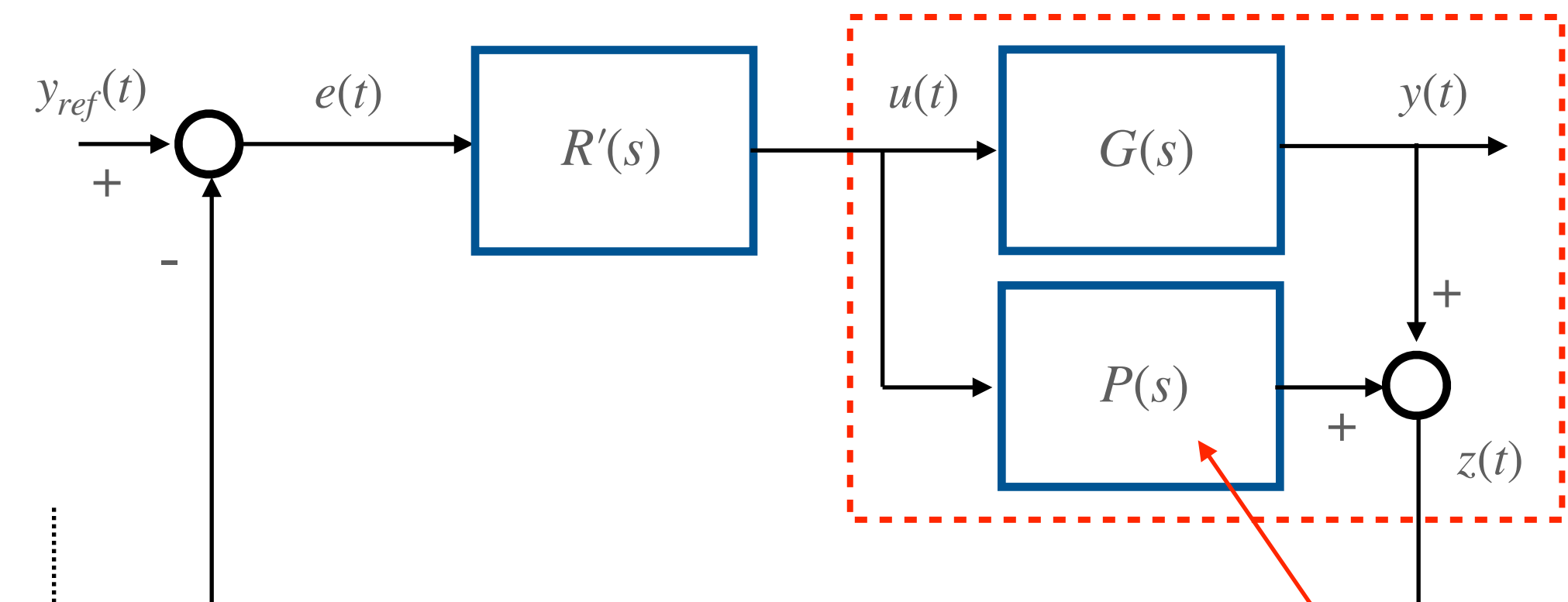


Smith Predictor

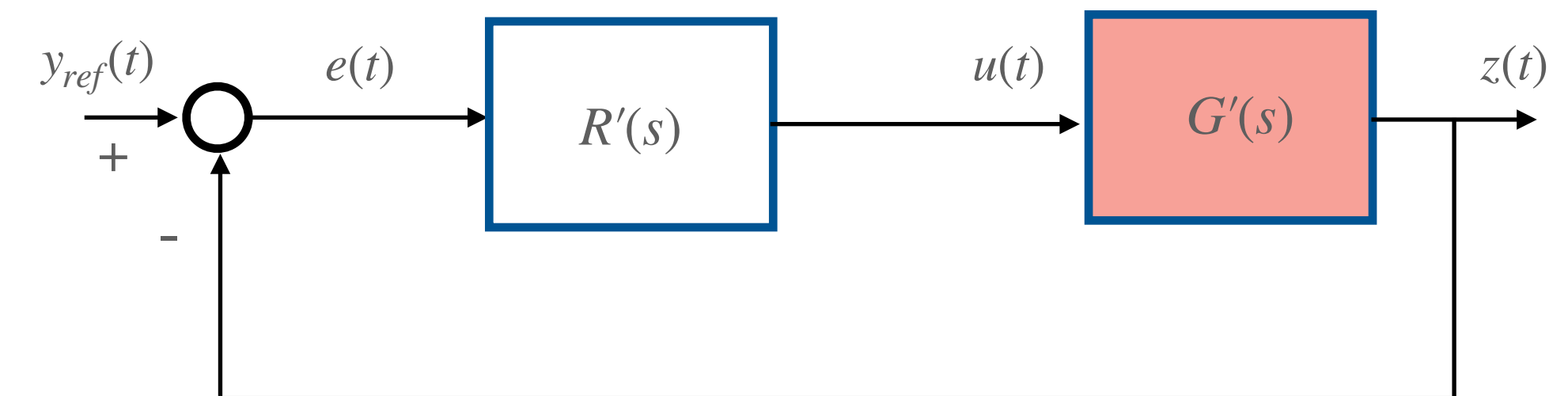
$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

$$\begin{aligned}
 u \rightarrow z : \quad G(s) + P(s) &= e^{-\tau s} G'(s) + (1 - e^{-\tau s}) G'(s) \\
 &= e^{-\tau s} G'(s) + G'(s) - e^{-\tau s} G'(s) \\
 &= G'(s)
 \end{aligned}$$



Smith Predictor



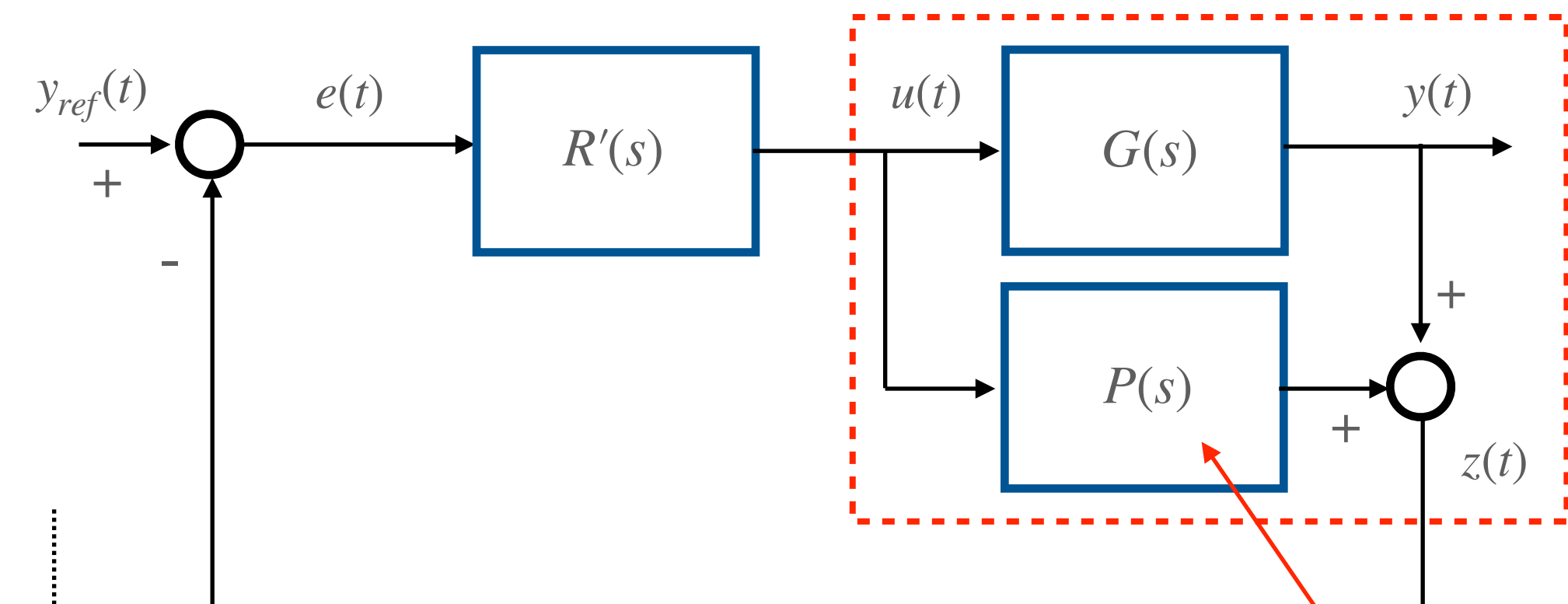
Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

$$\begin{aligned}
 u \rightarrow z : \quad G(s) + P(s) &= e^{-\tau s} G'(s) + (1 - e^{-\tau s}) G'(s) \\
 &= e^{-\tau s} G'(s) + G'(s) - e^{-\tau s} G'(s) \\
 &= G'(s)
 \end{aligned}$$



Warning:

$$z(t) \neq y(t)$$

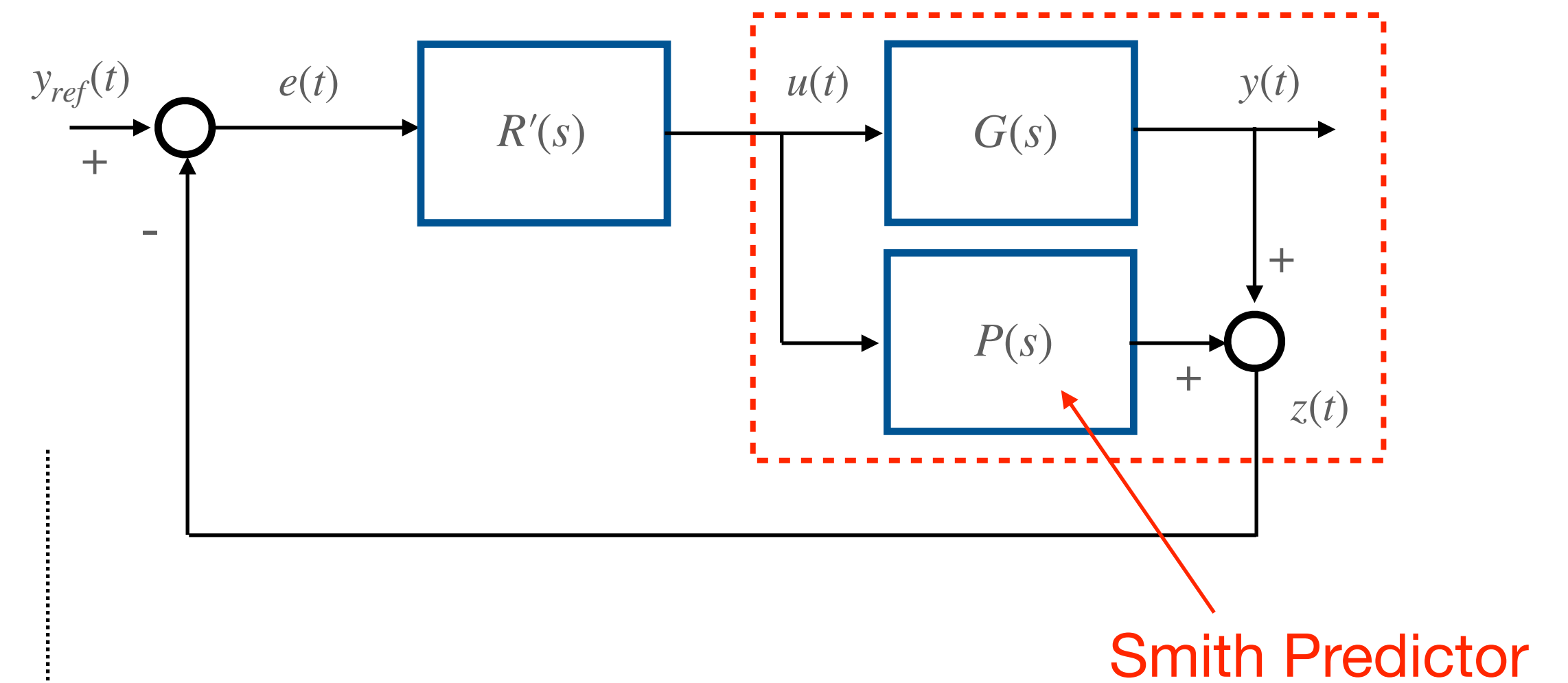


Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

$$z(t) \neq y(t) \rightarrow z(t) ?$$



$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

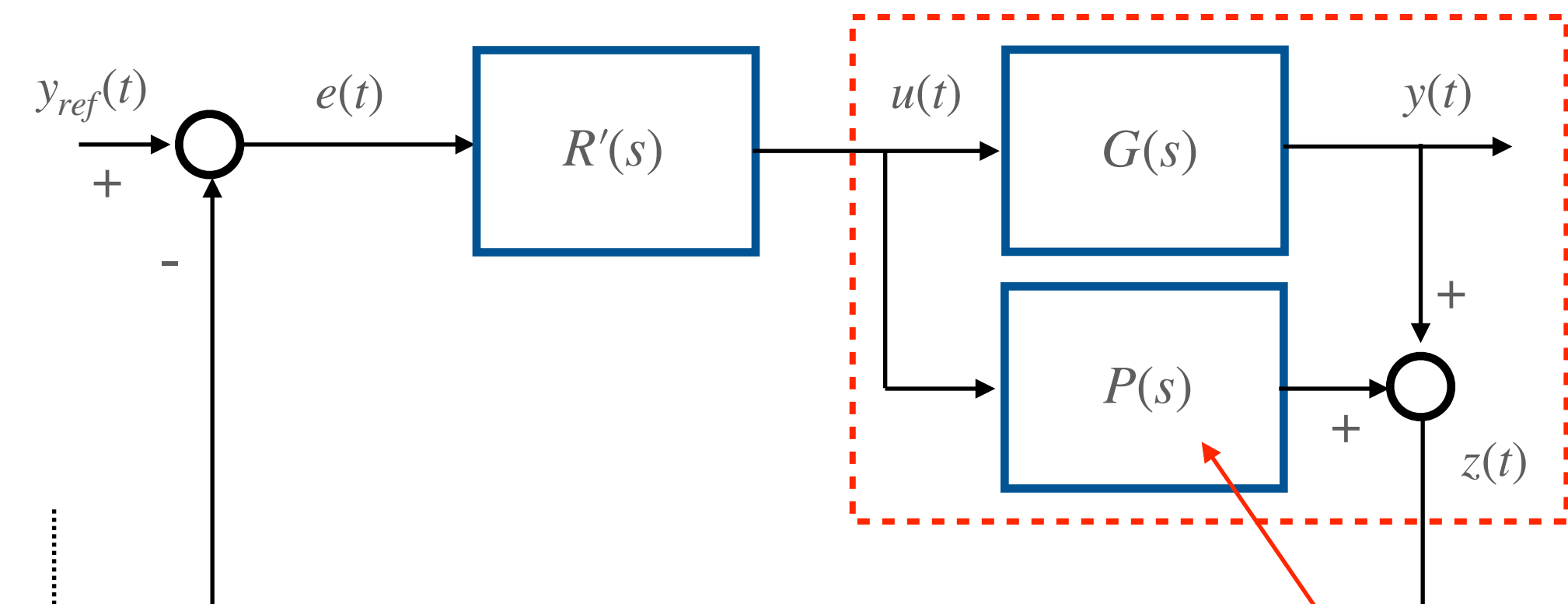
Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

$$z(t) \neq y(t) \rightarrow z(t) ?$$

In open loop:

$$u \rightarrow z : G'(s)$$

$$u \rightarrow y : e^{-\tau s} G'(s)$$



Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

$$z(t) \neq y(t) \rightarrow z(t) ?$$

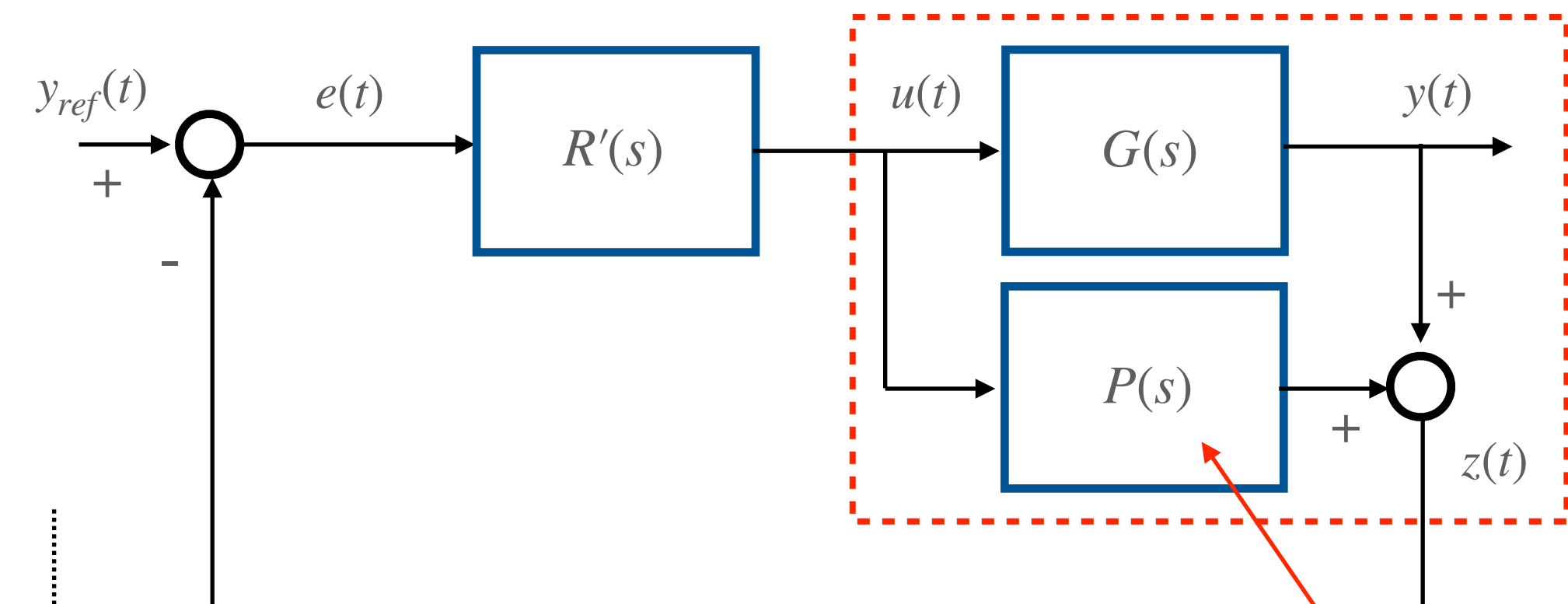
In open loop:

$$u \rightarrow z : G'(s)$$

$$u \rightarrow y : e^{-\tau s} G'(s)$$

$$\mathcal{L} \rightarrow Z(s) = G'(s) U(s)$$

$$Y(s) = e^{-\tau s} G'(s) U(s)$$



Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

$$z(t) \neq y(t) \rightarrow z(t) ?$$

In open loop:

$$u \rightarrow z : G'(s)$$

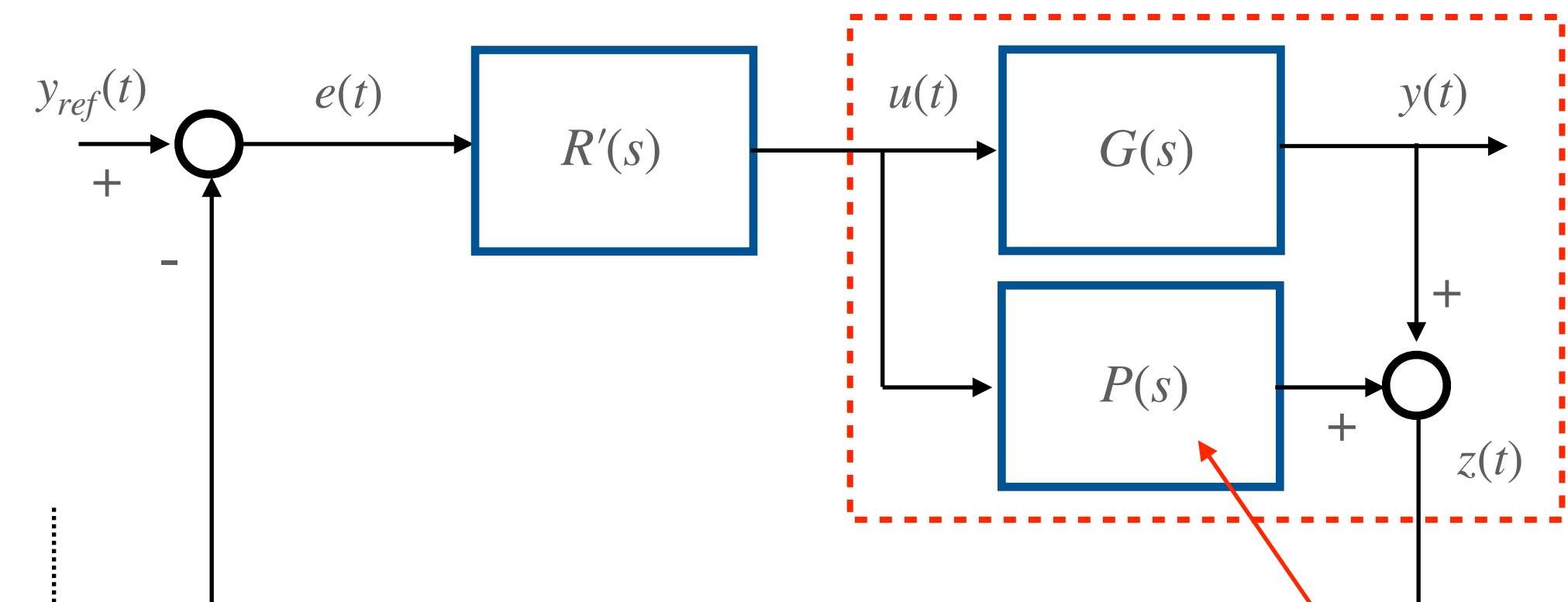
$$u \rightarrow y : e^{-\tau s} G'(s)$$

$$Z(s) = G'(s) U(s)$$

$$Y(s) = e^{-\tau s} G'(s) U(s)$$

$$Z(s) = e^{\tau s} Y(s)$$

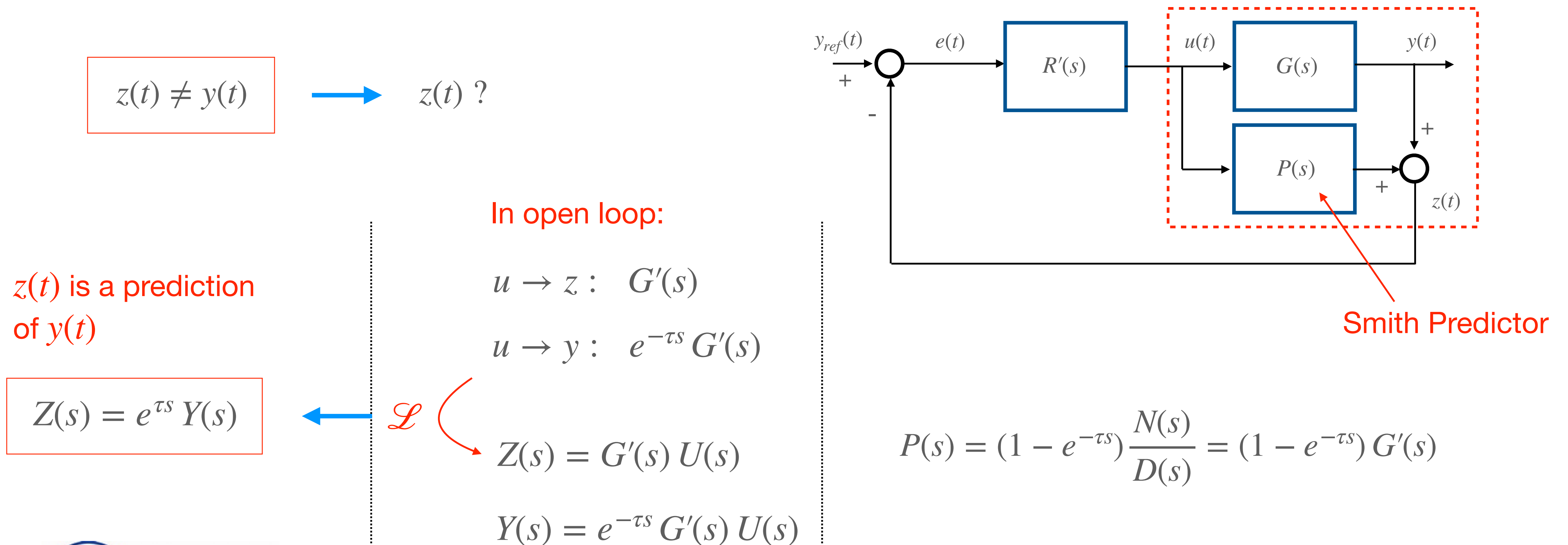
$\mathcal{L}$



Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

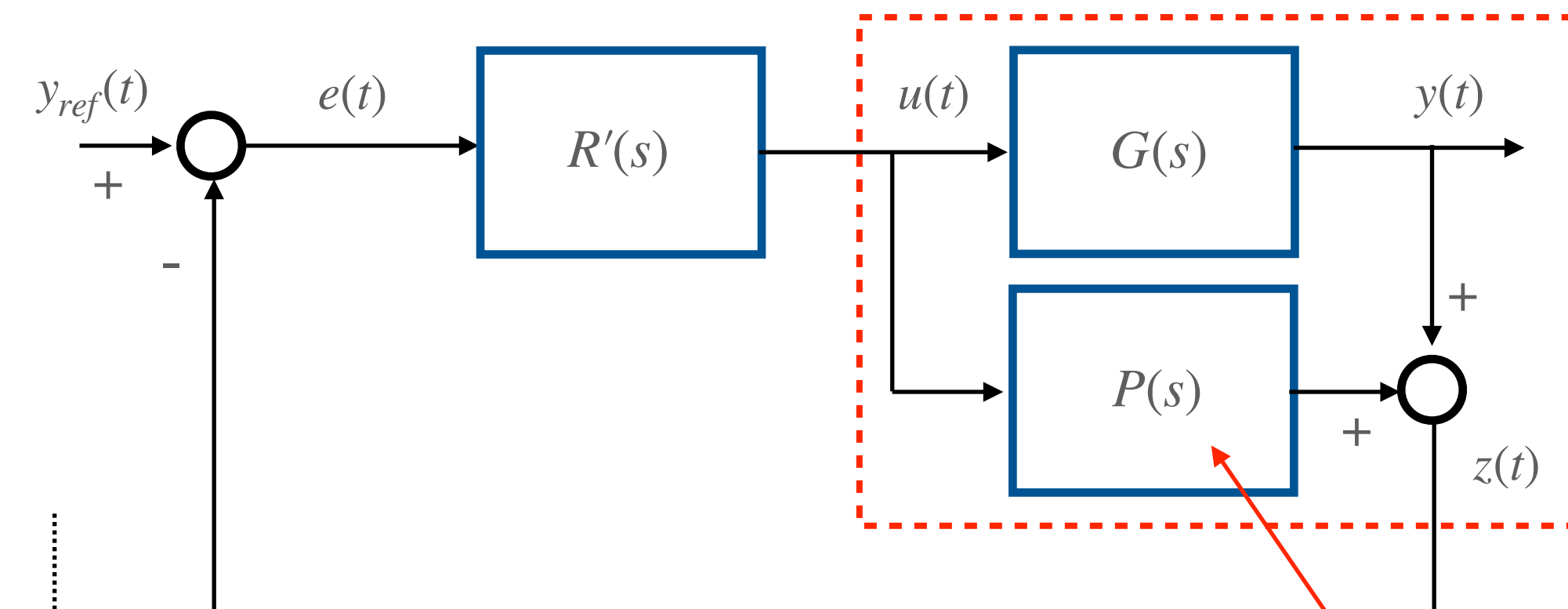


Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

Observation 1:

In the static case

$$P(0) = (1 - e^{-\tau_0}) G'(0) = 0$$



Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

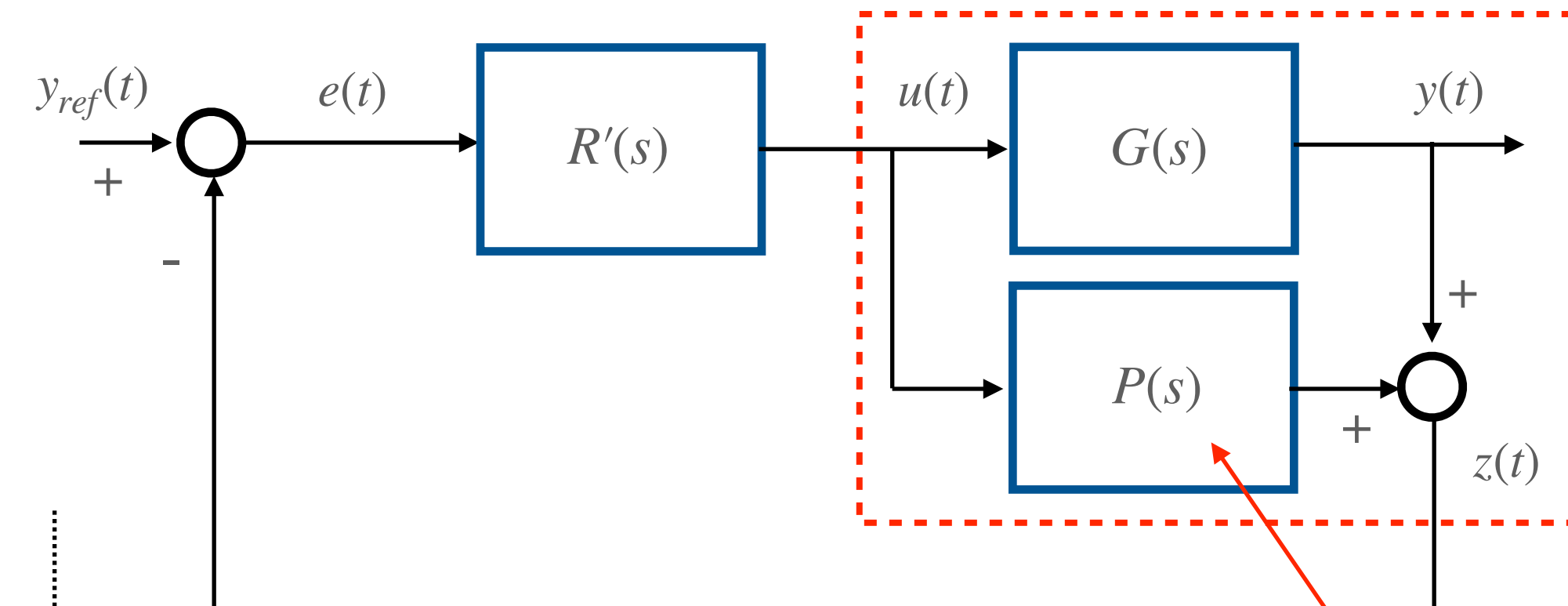
Observation 1:

In the static case

$$P(0) = (1 - e^{-\tau_0}) G'(0) = 0$$

Then, if the closed loop system is As. Stable
and $y_{ref}(t) = A \text{ step}(t)$:

$$z(t) \rightarrow y(t) \quad \text{for} \quad t \rightarrow \infty$$

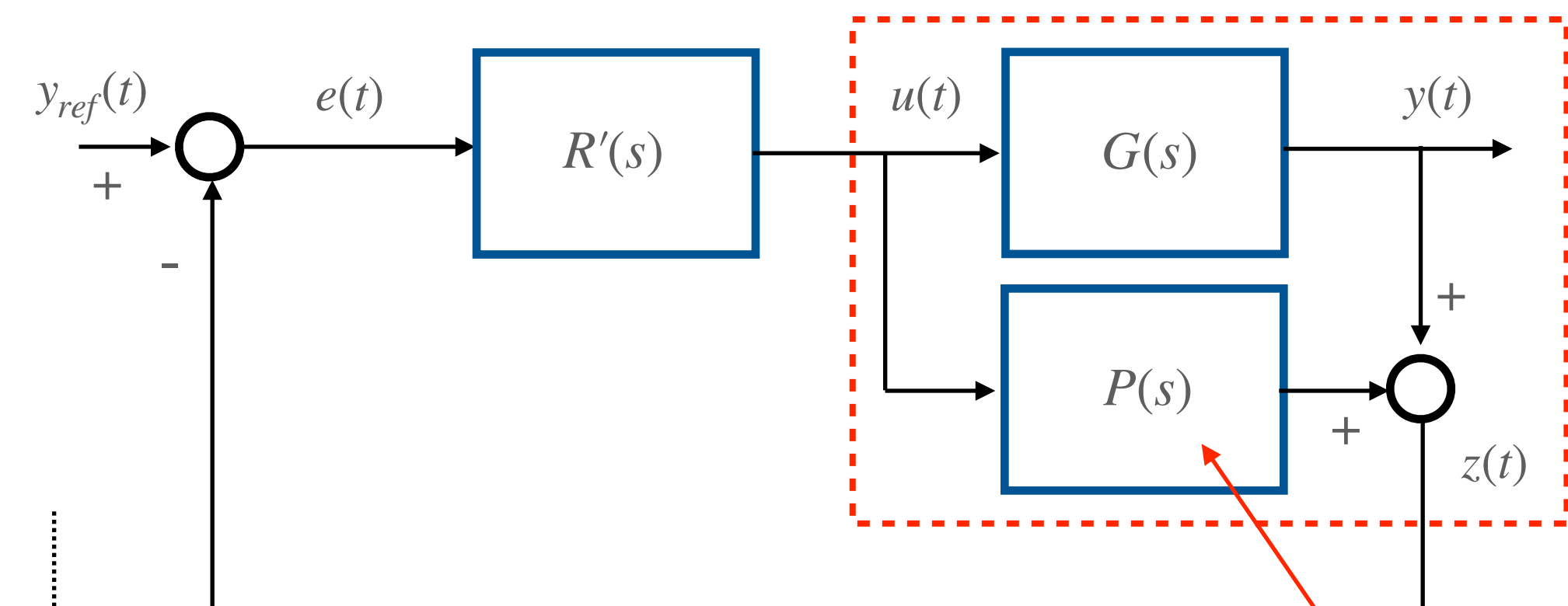
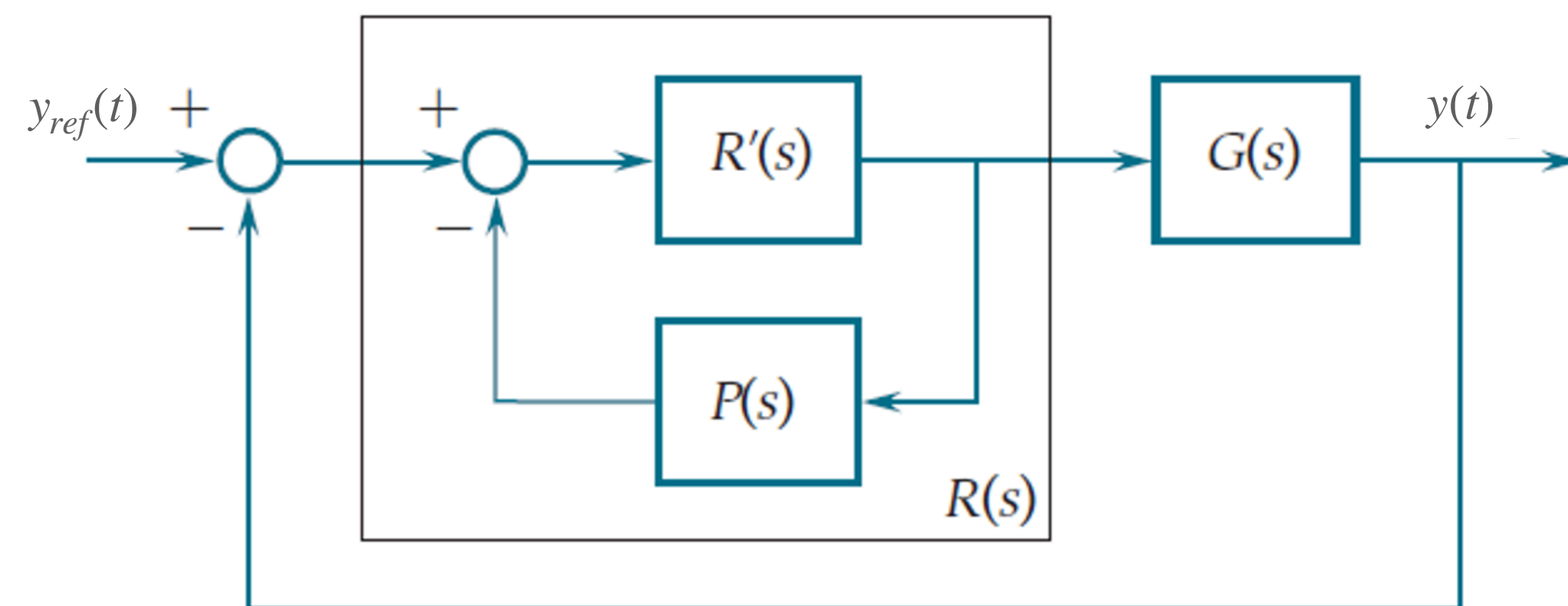


Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

Observation 2: alternative formulation

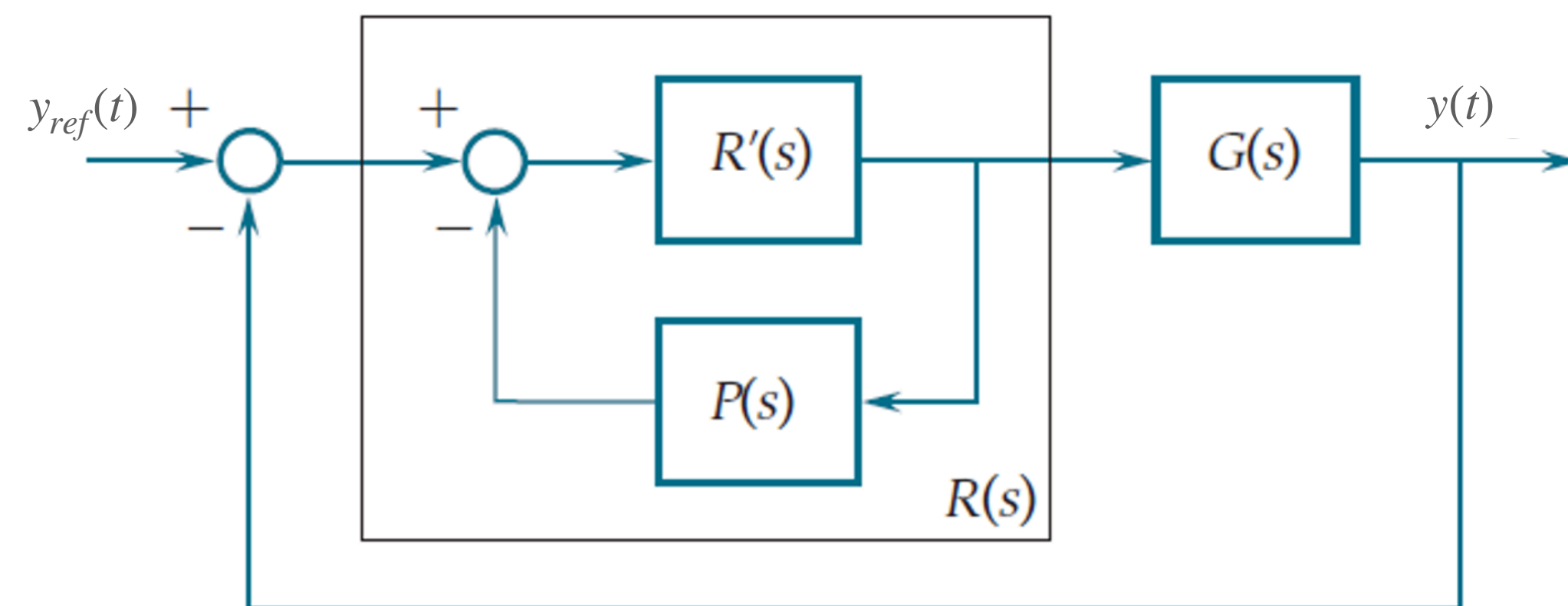


Smith Predictor

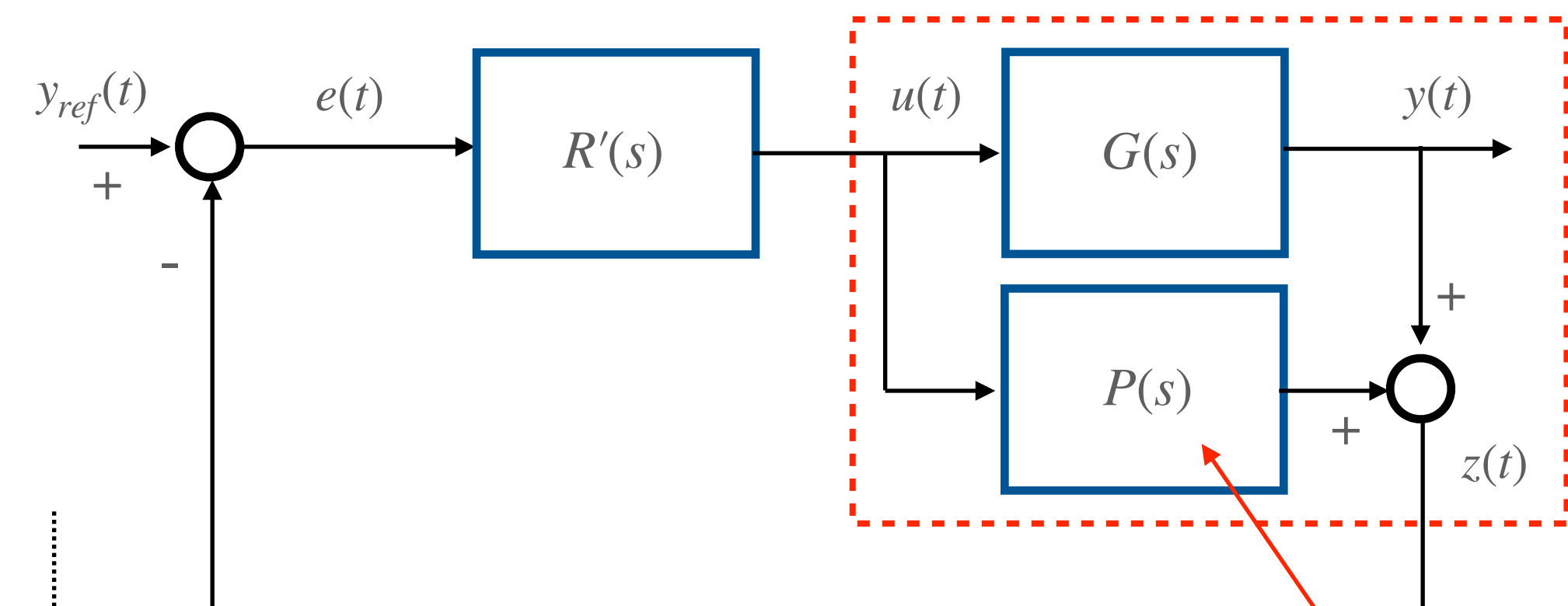
$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

Observation 2: alternative formulation



$$R(s) = \frac{R'(s)}{1 + R'(s)P(s)}$$

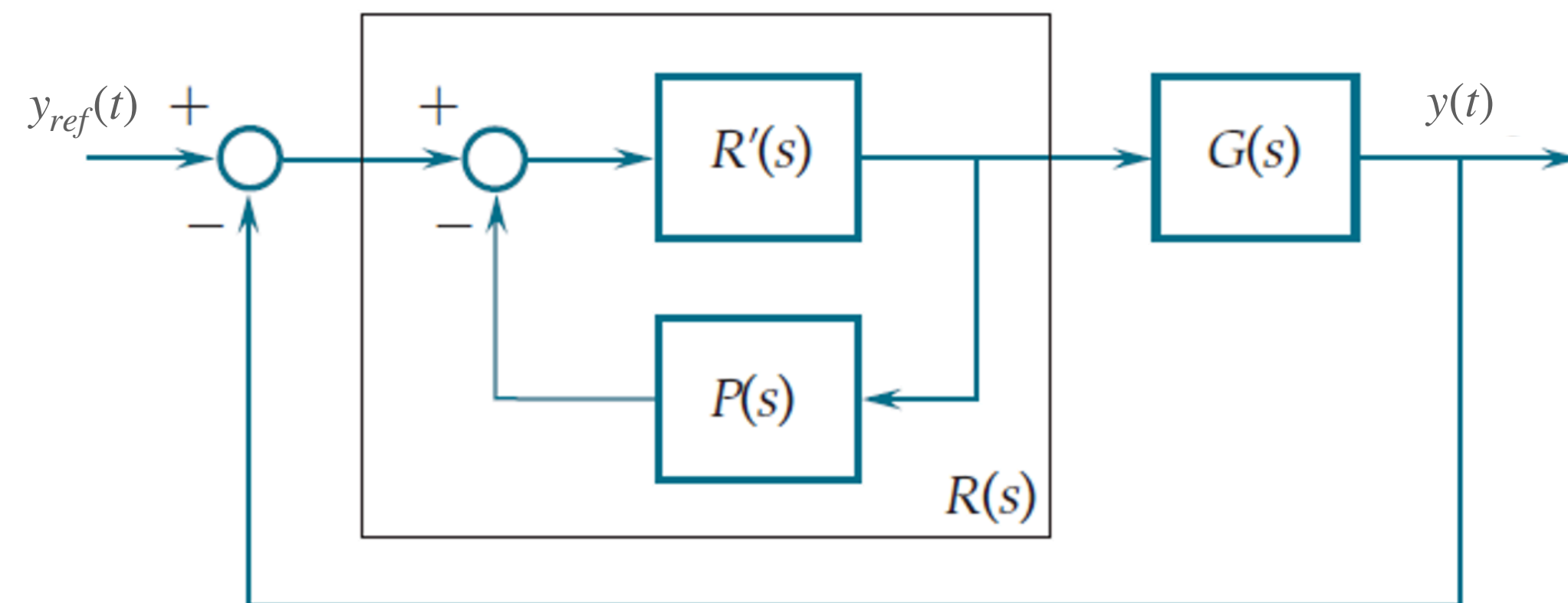


Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

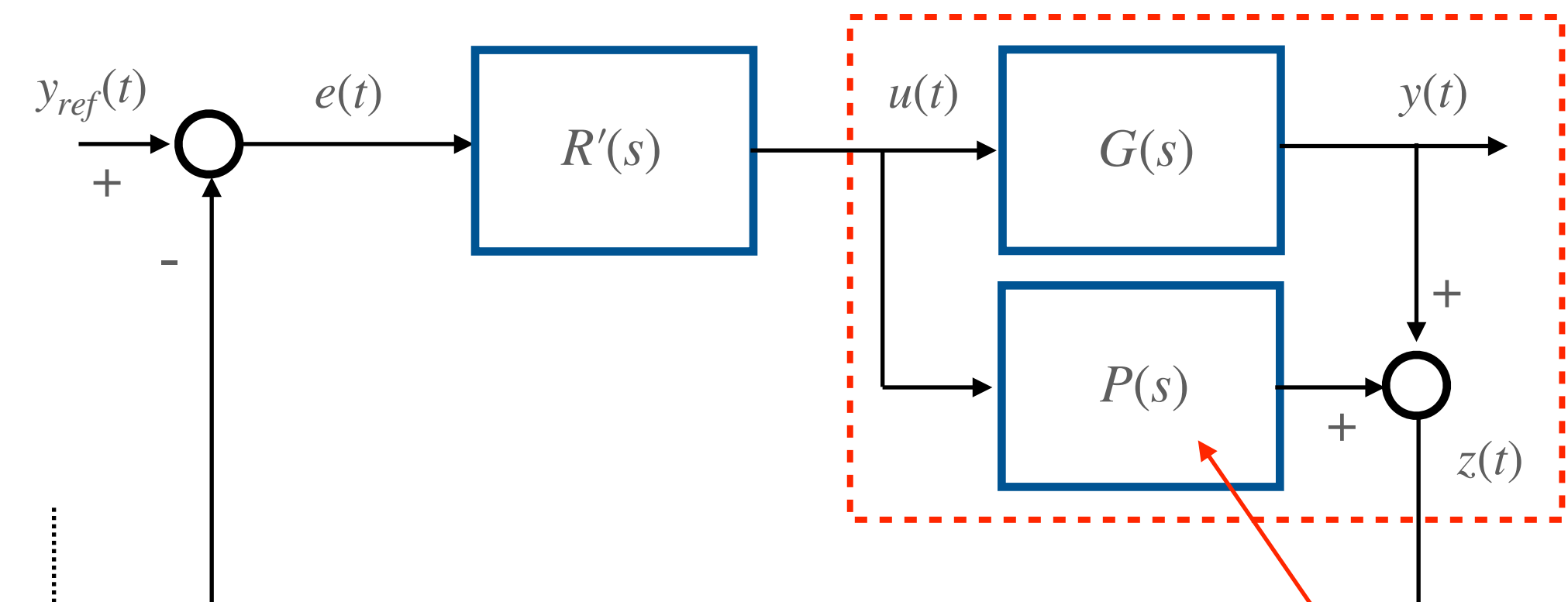
Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

Observation 2: alternative formulation



$$R(s) = \frac{R'(s)}{1 + R'(s)P(s)}$$

$$= \frac{N_{R'}(s)/D_{R'}(s)}{1 + (N_{R'}(s)/D_{R'}(s))(1 - e^{-\tau s})(N(s)/D(s))}$$

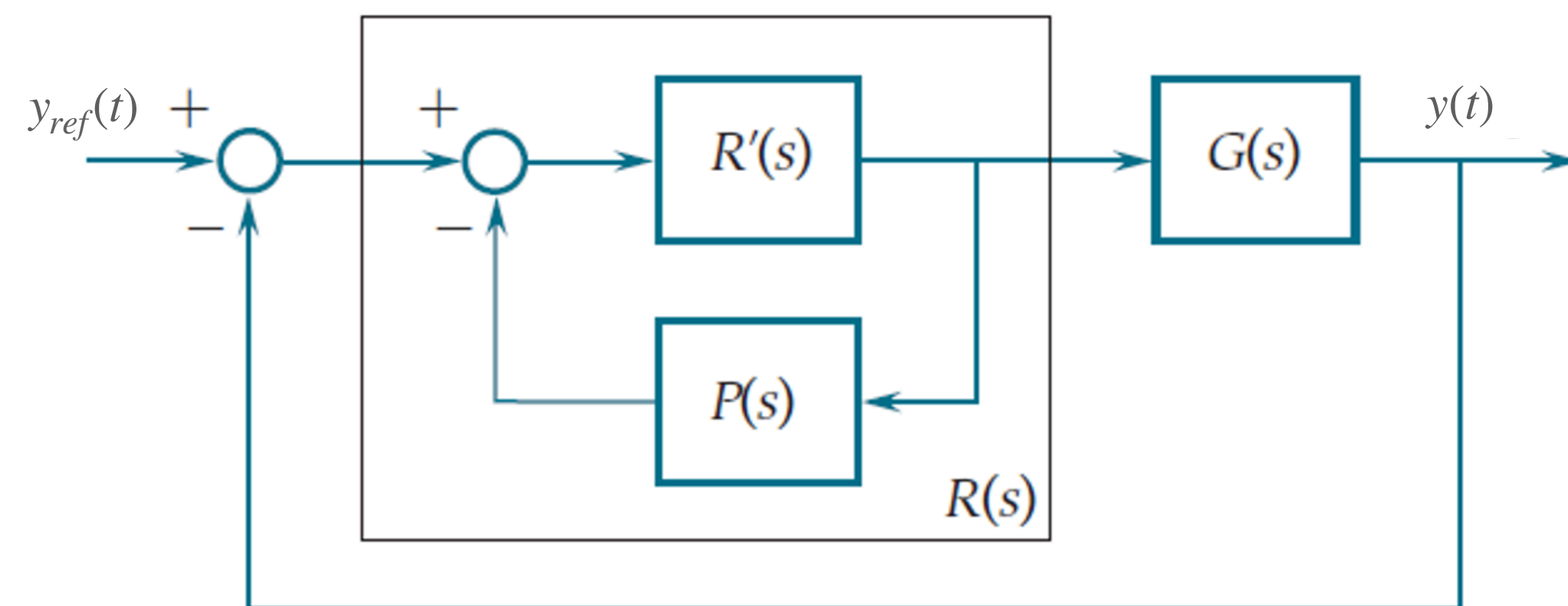


Smith Predictor

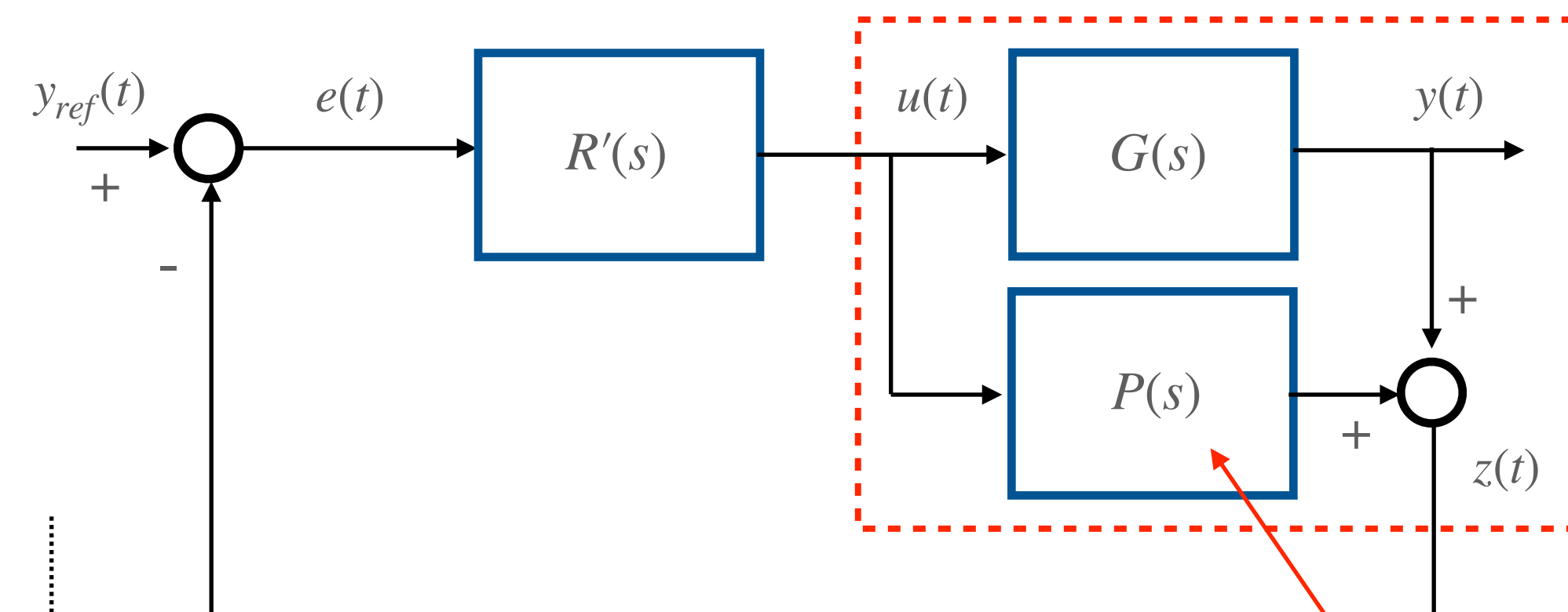
$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

Observation 2: alternative formulation



$$\begin{aligned} R(s) &= \frac{R'(s)}{1 + R'(s)P(s)} \\ &= \frac{N_{R'}(s)/D_{R'}(s)}{1 + (N_{R'}(s)/D_{R'}(s))(1 - e^{-\tau s})(N(s)/D(s))} \\ &= \frac{N_{R'}(s)D(s)}{D_{R'}(s)D(s) + N_{R'}(s)N(s)(1 - e^{-\tau s})} \end{aligned}$$

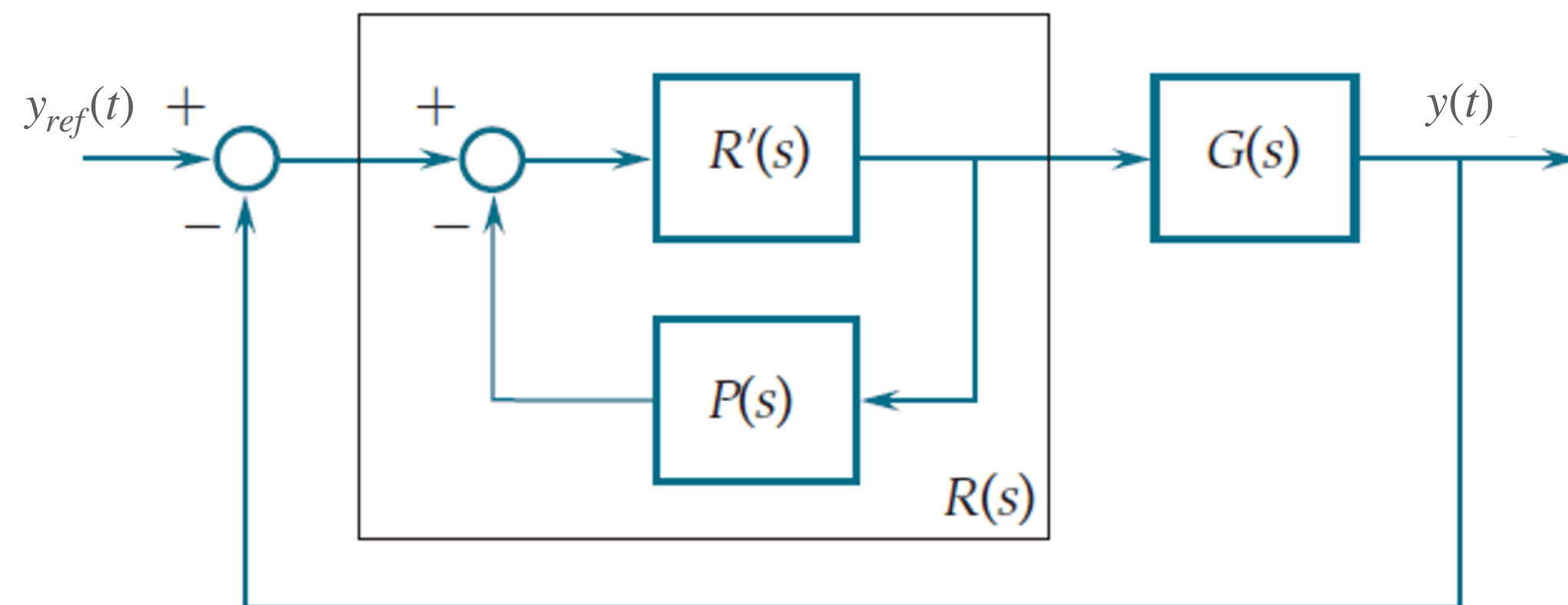


Smith Predictor

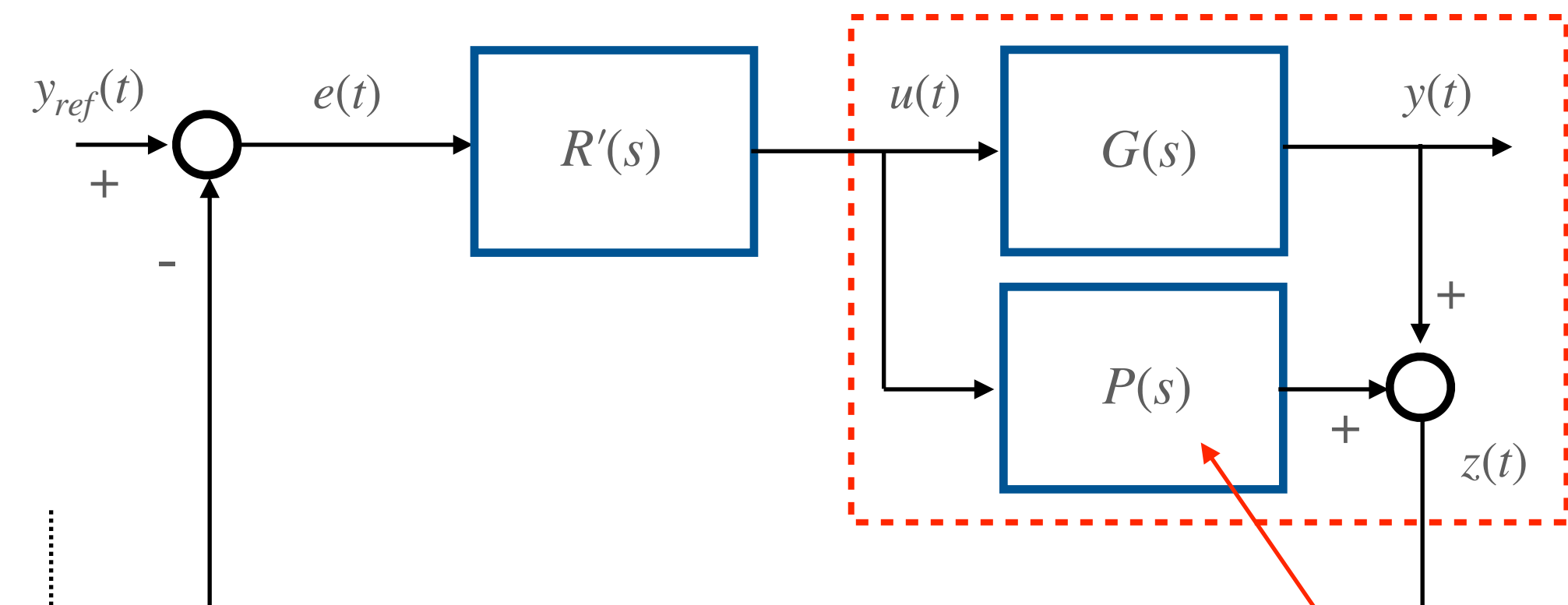
$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

Observation 2: alternative formulation



$$\begin{aligned}
 R(s) &= \frac{R'(s)}{1 + R'(s)P(s)} \\
 &= \frac{N_{R'}(s)/D_{R'}(s)}{1 + (N_{R'}(s)/D_{R'}(s))(1 - e^{-\tau s})(N(s)/D(s))} \\
 &= \frac{N_{R'}(s) \boxed{D(s)}}{D_{R'}(s)D(s) + N_{R'}(s)N(s)(1 - e^{-\tau s})}
 \end{aligned}$$



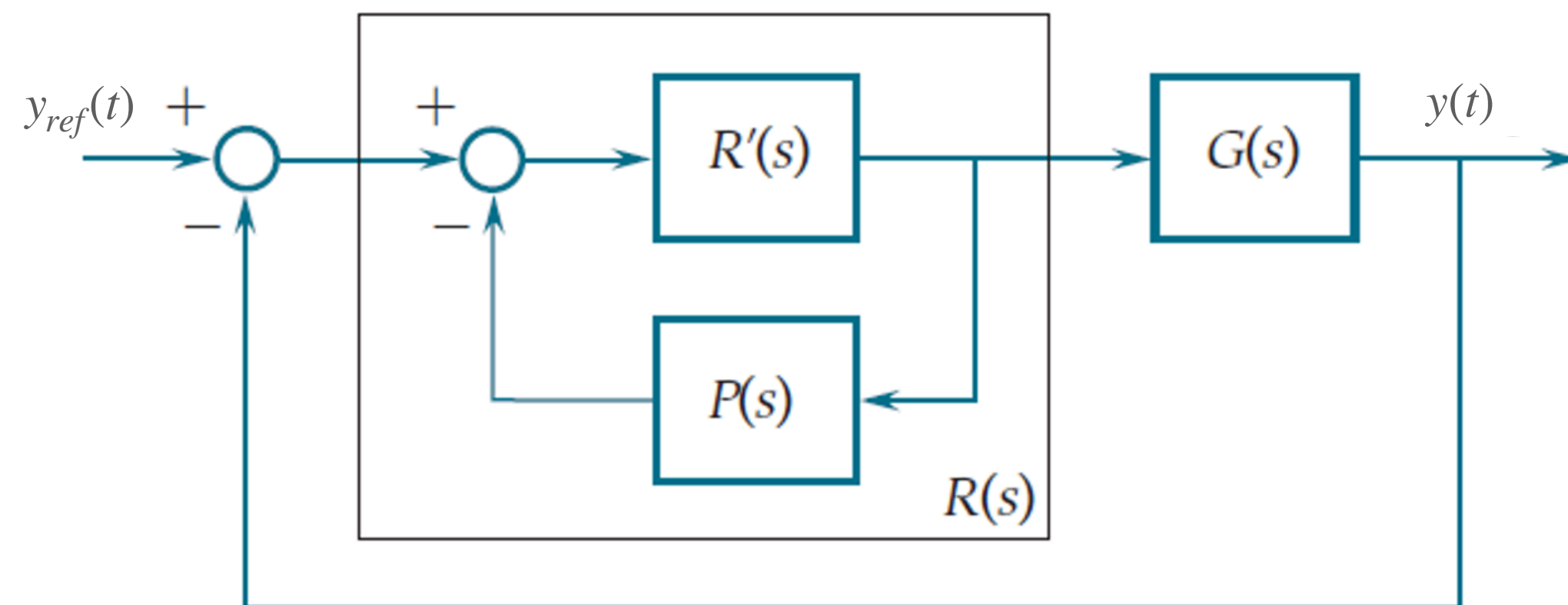
Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

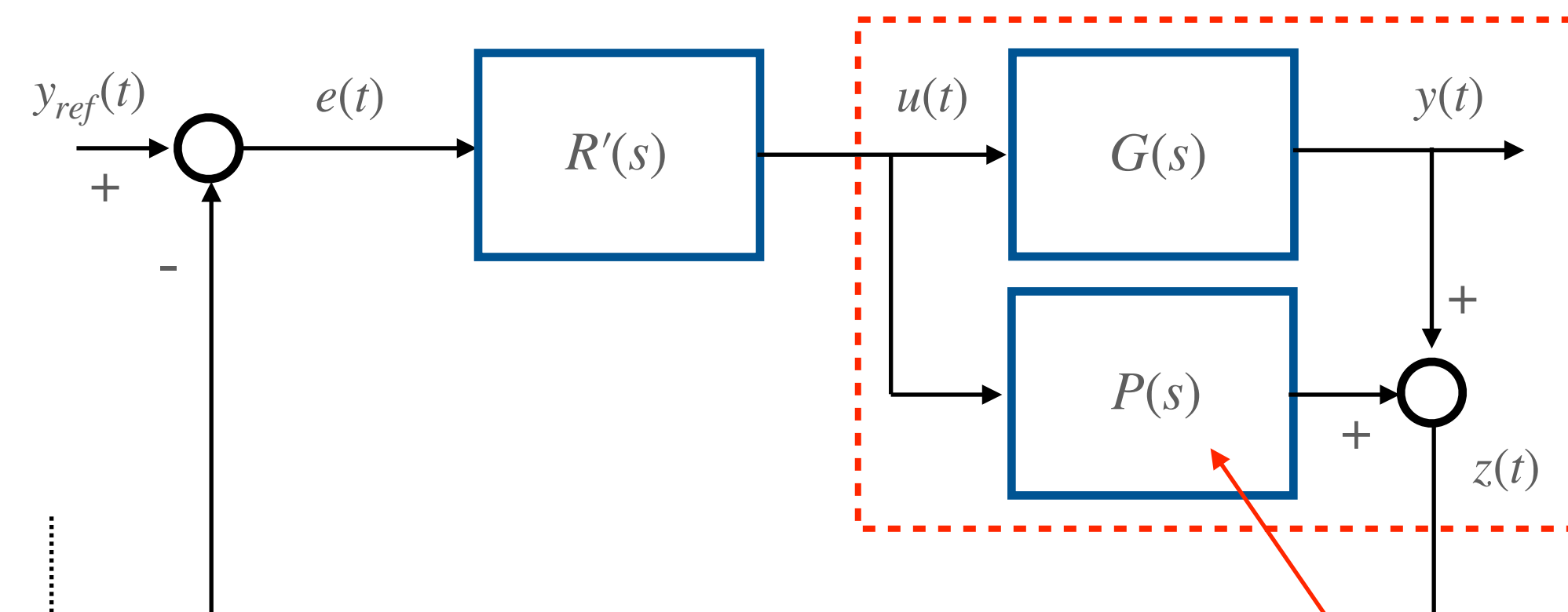
synthesis by cancellation

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

Observation 3:



$$\begin{aligned}
 R(s) &= \frac{R'(s)}{1 + R'(s)P(s)} \\
 &= \frac{N_{R'}(s)/D_{R'}(s)}{1 + (N_{R'}(s)/D_{R'}(s))(1 - e^{-\tau s})(N(s)/D(s))} \\
 &= \frac{N_{R'}(s)D(s)}{D_{R'}(s)D(s) + N_{R'}(s)N(s)(1 - e^{-\tau s})}
 \end{aligned}$$



Smith Predictor

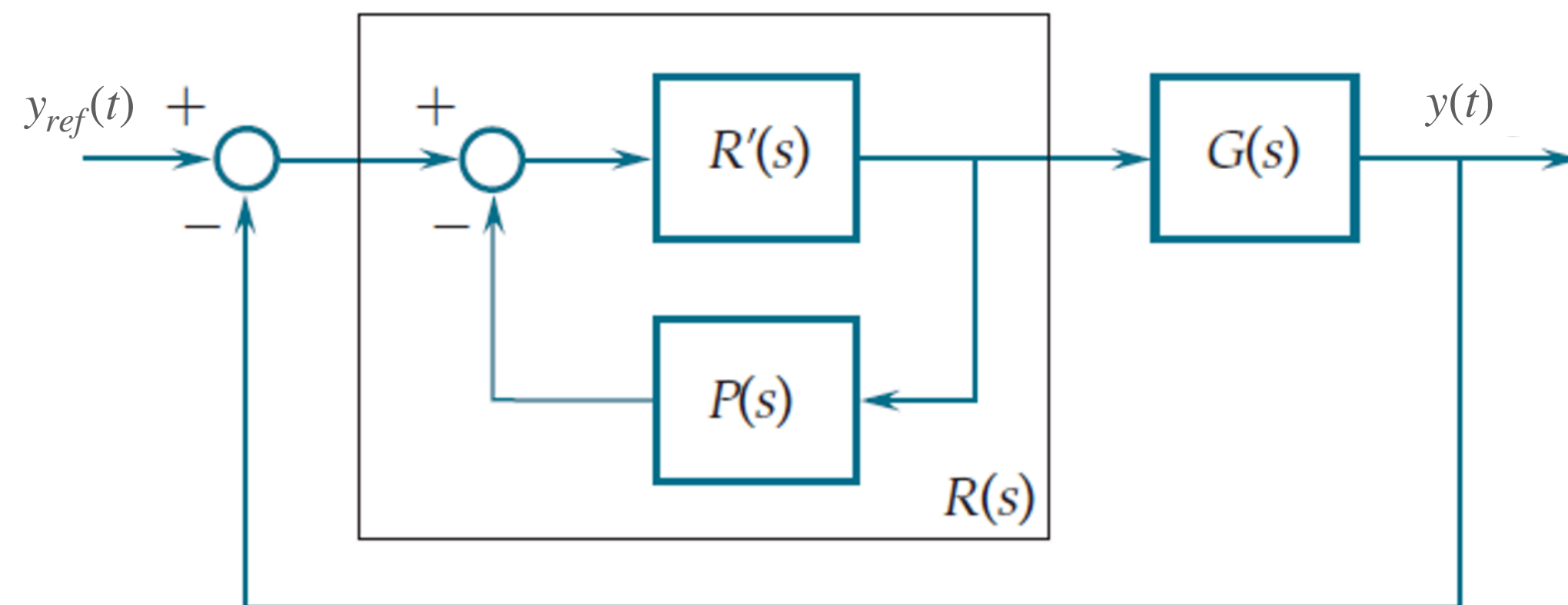
$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

$R(s)$ is a controller with a non-rational transfer function

Control of LTI Systems with Delays: **Smith Predictor** Based Control Scheme

Summary:

Alternative scheme



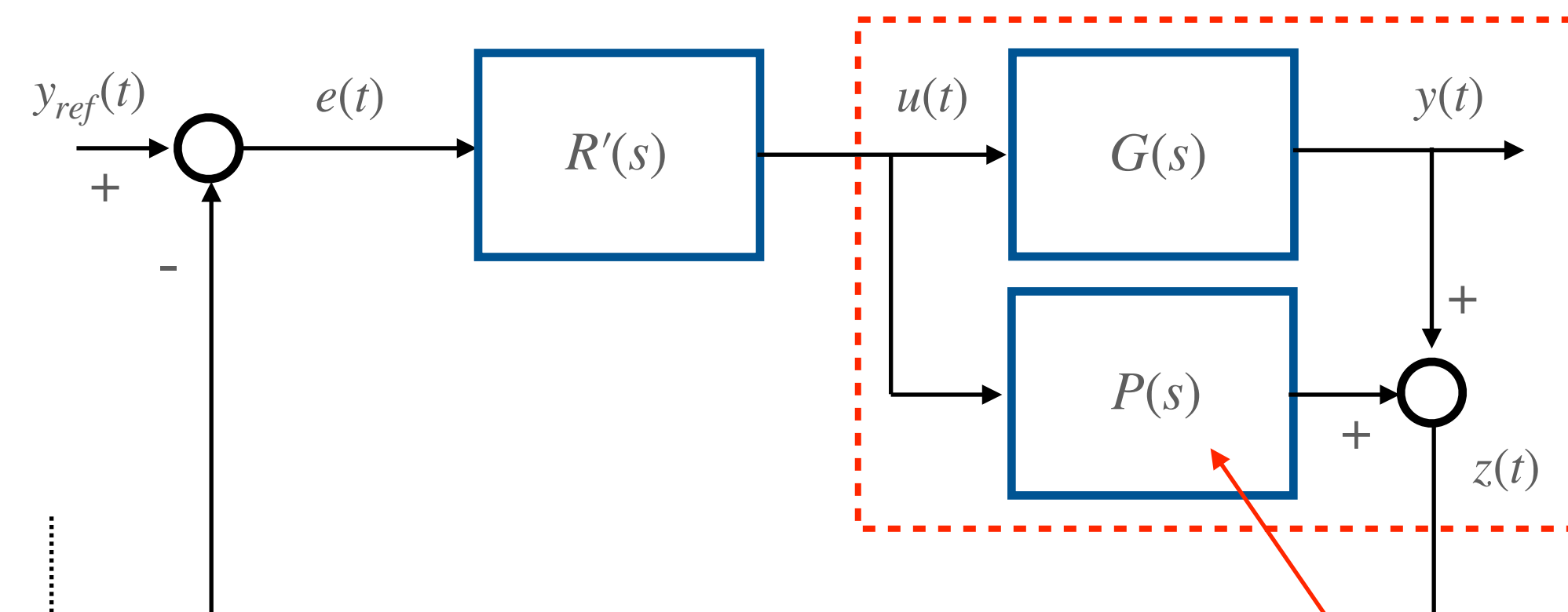
$z(t)$ is a prediction of $y(t)$

$$Z(s) = e^{\tau s} Y(s)$$

$y(t)$ is $z(t)$ delayed of τ seconds

$$Y(s) = e^{-\tau s} Z(s)$$

Classical scheme

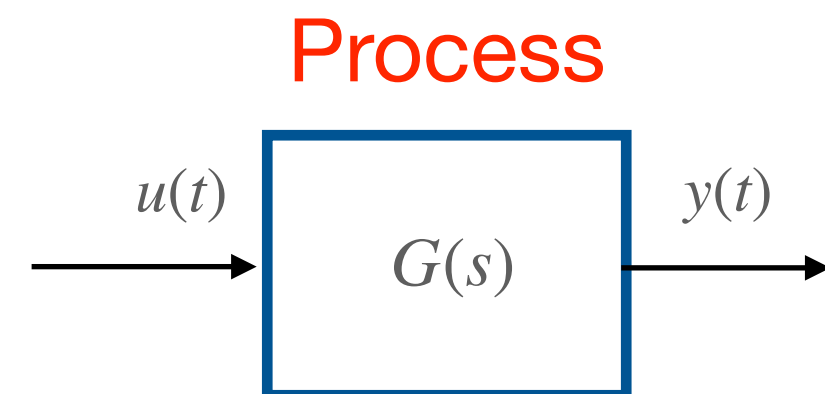


Smith Predictor

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

the delay has not disappeared!

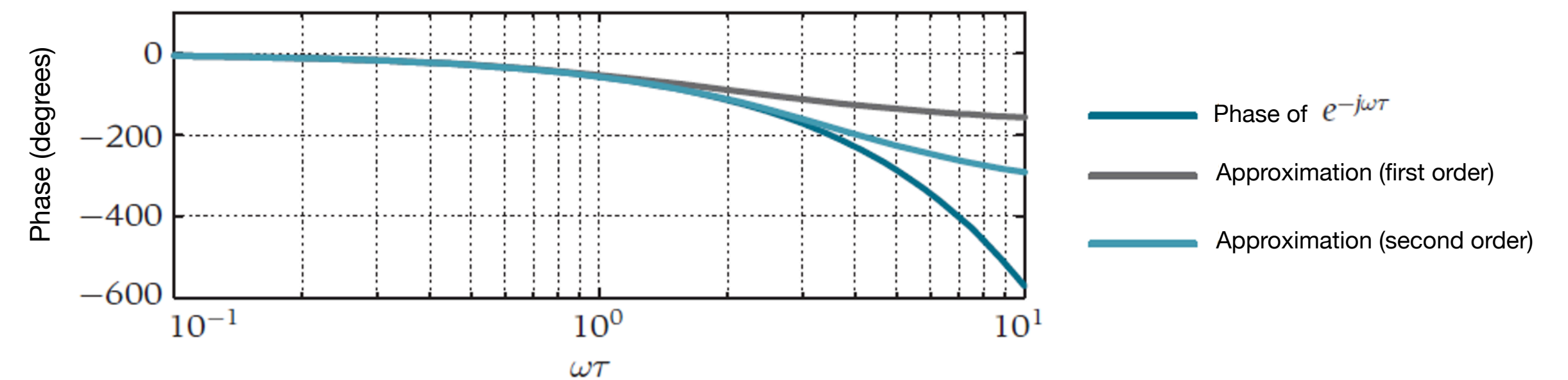
Control of LTI Systems with Delays: Padé Approximation



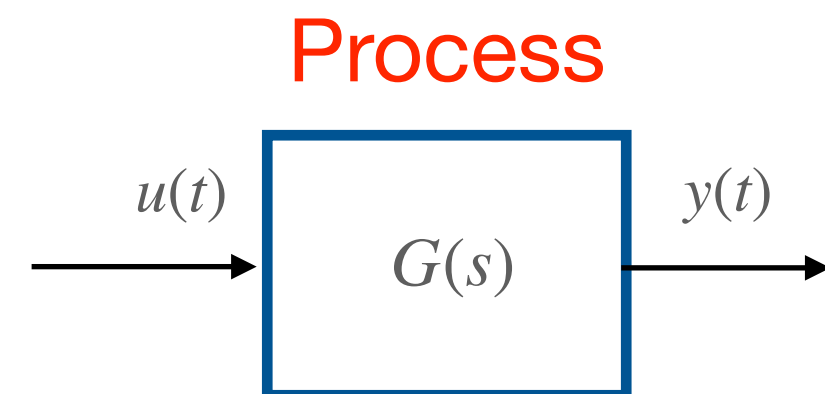
$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

replace with an approximation
based on a rational function of s

Phase Lag due to the delay term

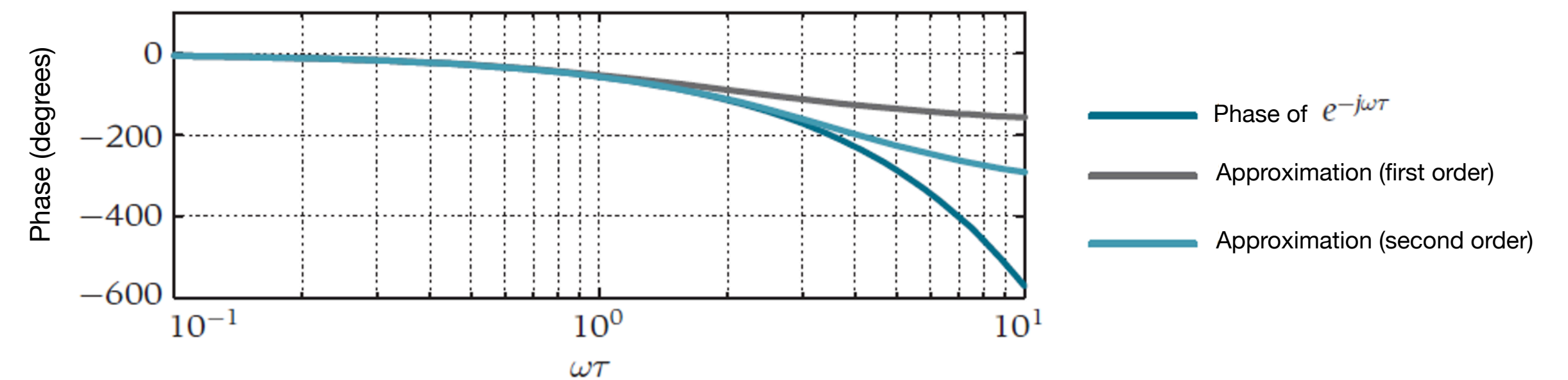


Control of LTI Systems with Delays: Padé Approximation



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

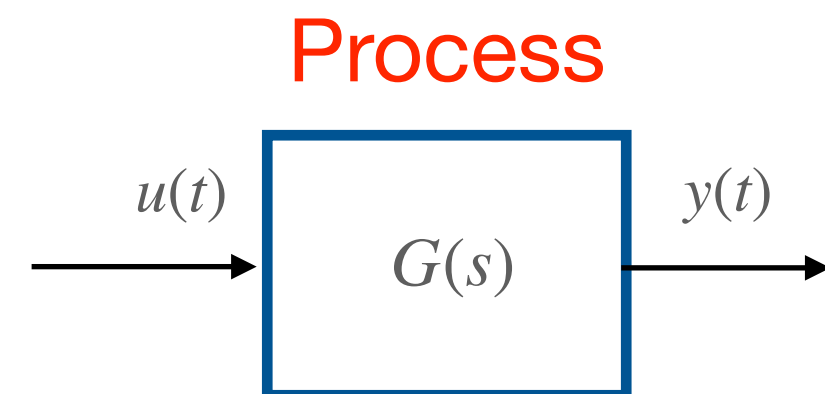
Phase Lag due to the delay term



Function to approximate: $e^{-\tau s}$

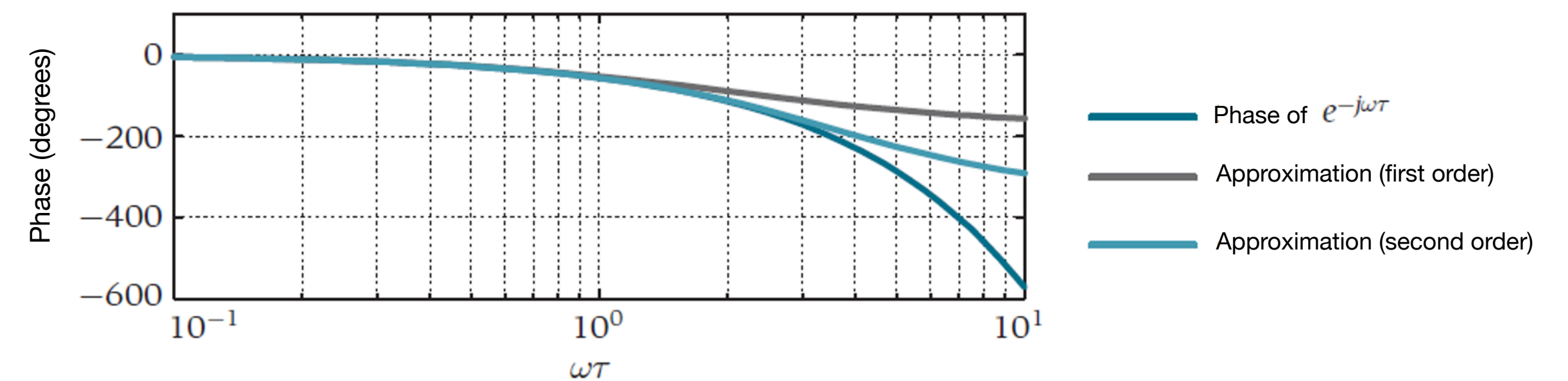
First order approximation: $G_I(s) = \mu \frac{1 + as}{1 + bs}$

Control of LTI Systems with Delays: Padé Approximation



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

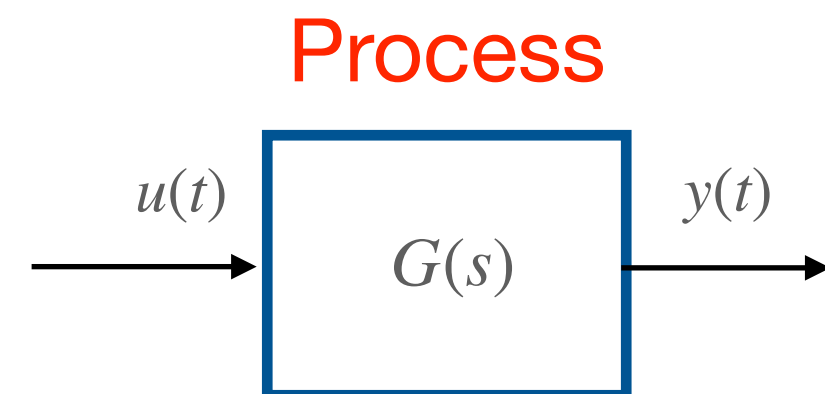
Phase Lag due to the delay term



Function to approximate: $e^{-\tau s}$

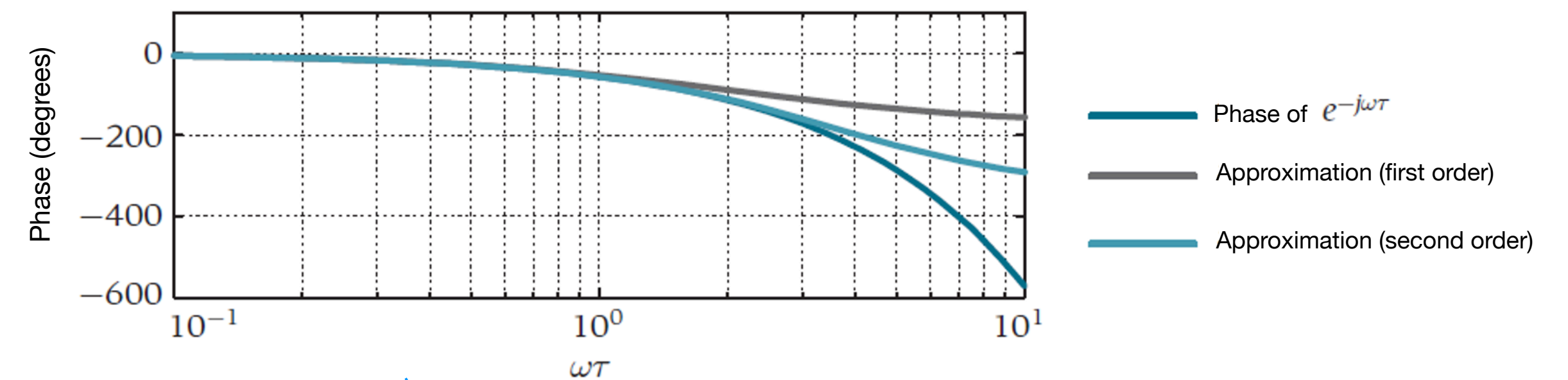
First order approximation: $G_I(s) = \mu \frac{1 + as}{1 + bs}$ → How to select the parameters?

Control of LTI Systems with Delays: Padé Approximation



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

Phase Lag due to the delay term

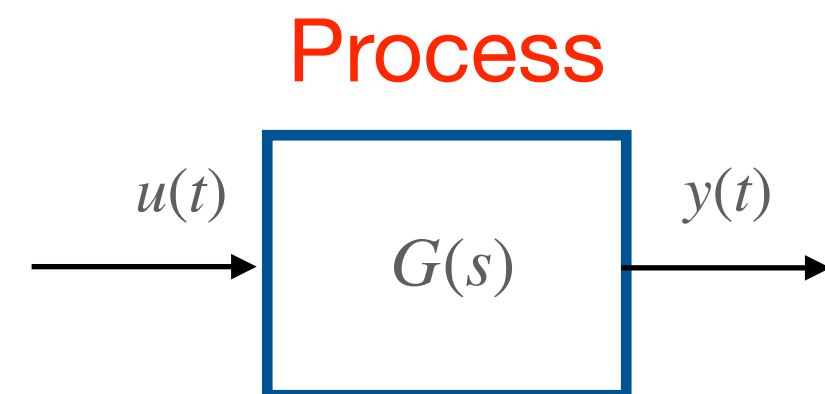


Function to approximate: $e^{-\tau s}$

First order approximation: $G_I(s) = \mu \frac{1 + as}{1 + bs}$

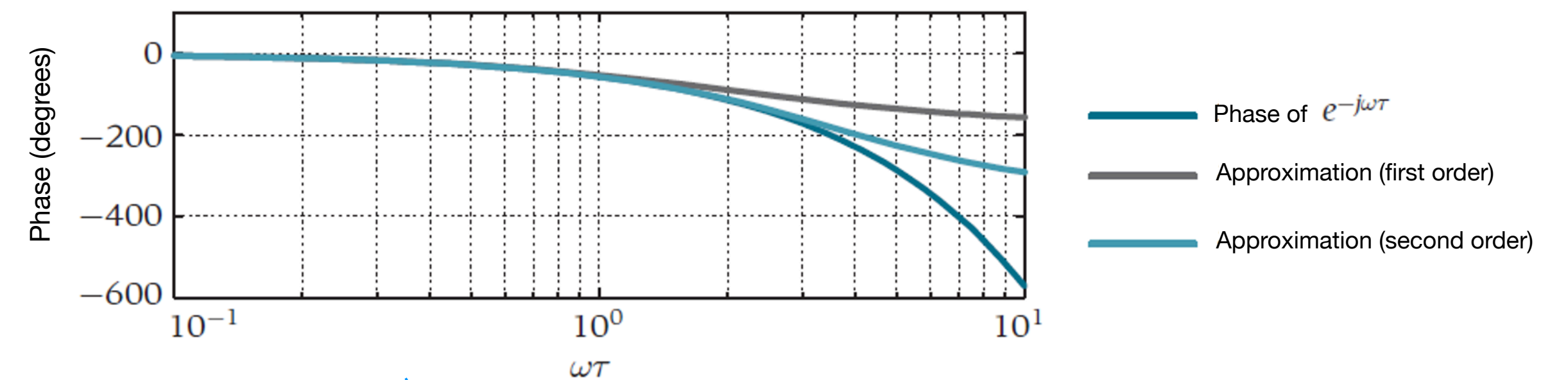
Use MacLaurin series expansion

Control of LTI Systems with Delays: Padé Approximation



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

Phase Lag due to the delay term



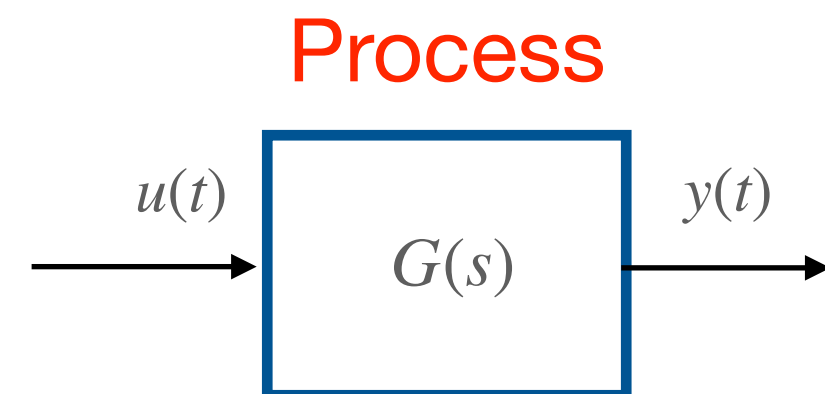
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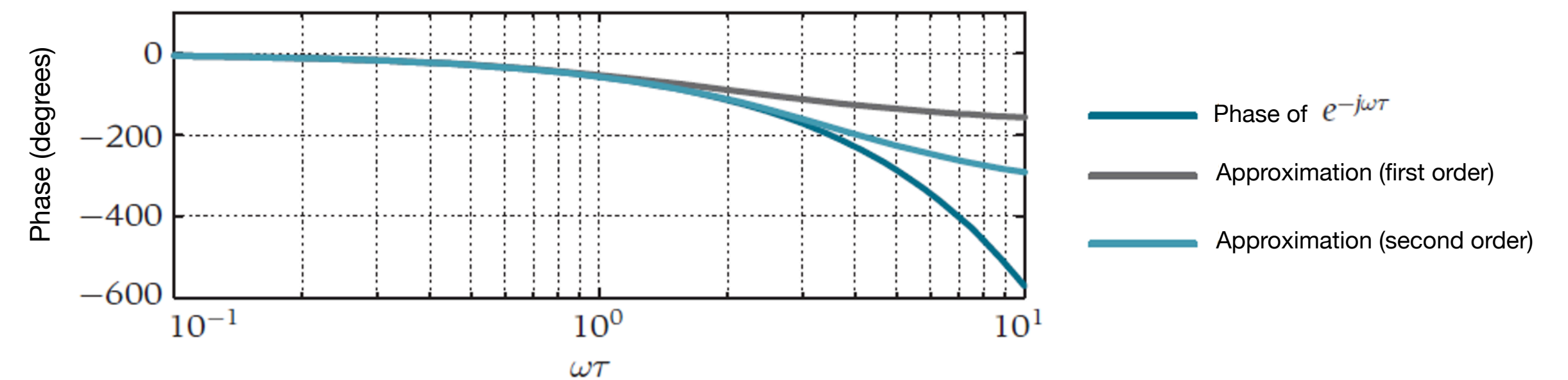
$$f(s) = f(0) + \frac{df}{ds}(0)s + \frac{1}{2!} \frac{d^2f}{ds^2}(0)s^2 + \frac{1}{3!} \frac{d^3f}{ds^3}(0)s^3 + \mathcal{O}(s^4)$$

Control of LTI Systems with Delays: Padé Approximation



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

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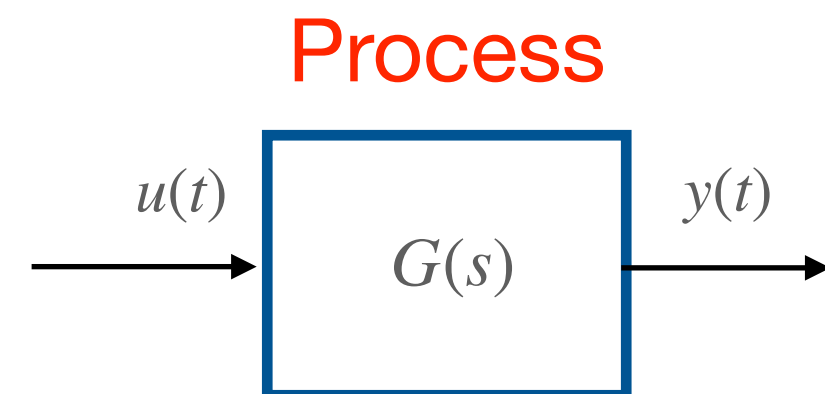


Function to approximate: $e^{-\tau s}$

McLaurin approximation: $e^{-\tau s} \approx 1 - \tau s + \frac{\tau^2 s^2}{2} + \dots$

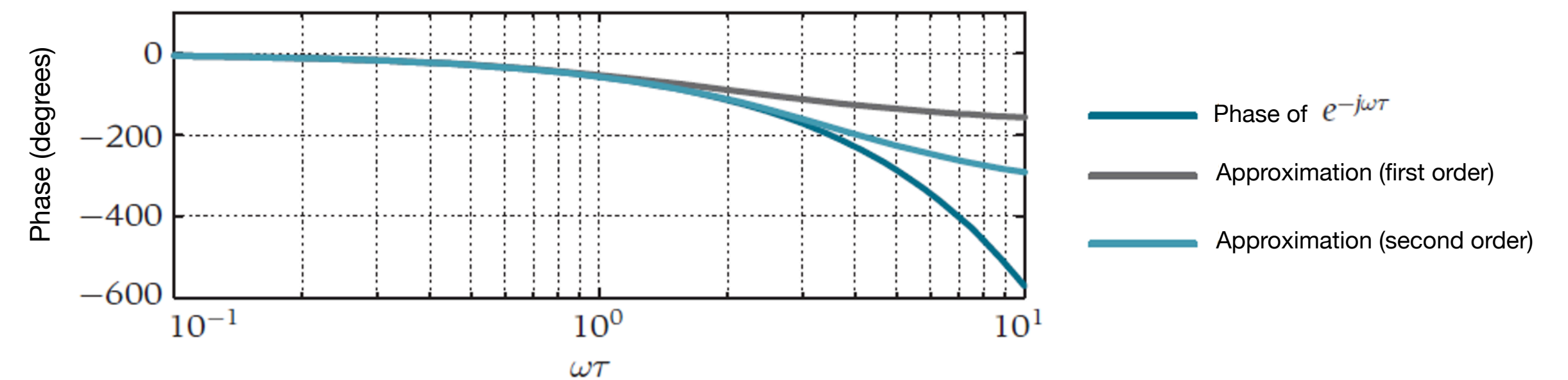
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Control of LTI Systems with Delays: Padé Approximation



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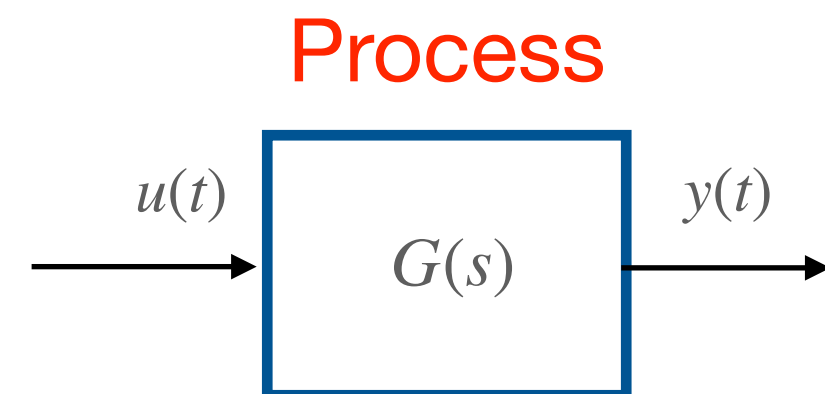
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McLaurin approximation: $G_I(s) = \mu \frac{1 + as}{1 + bs} \approx \mu + \mu(a - b)s - \mu b(a - b)s^2$

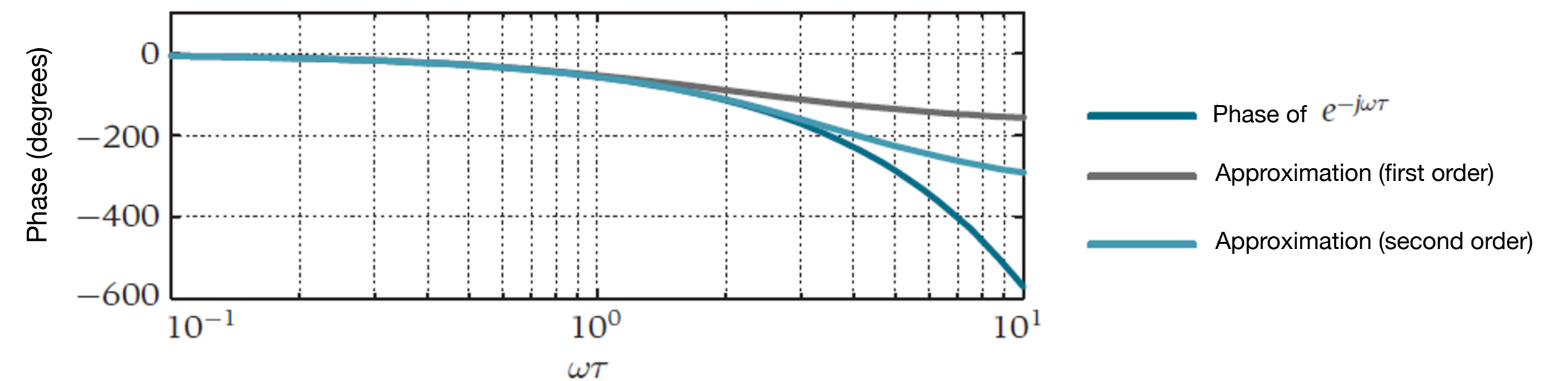
Control of LTI Systems with Delays: Padé Approximation



$$f(s) = f(0) + \frac{df}{ds}(0)s + \frac{1}{2!} \frac{d^2f}{ds^2}(0)s^2 + \frac{1}{3!} \frac{d^3f}{ds^3}(0)s^3 + \mathcal{O}(s^4)$$

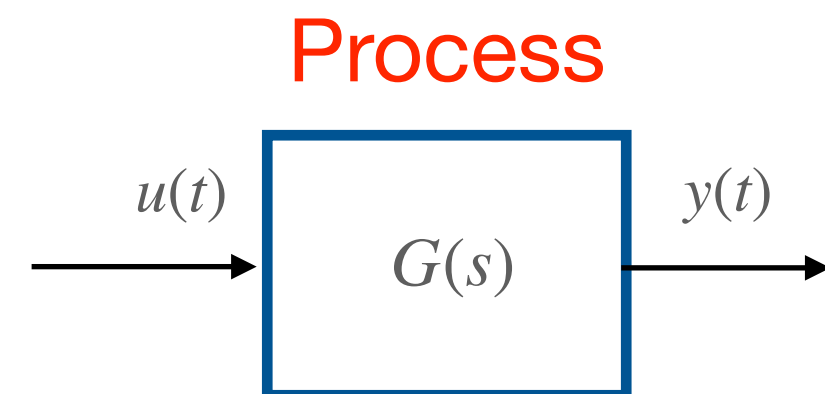
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Phase Lag due to the delay term



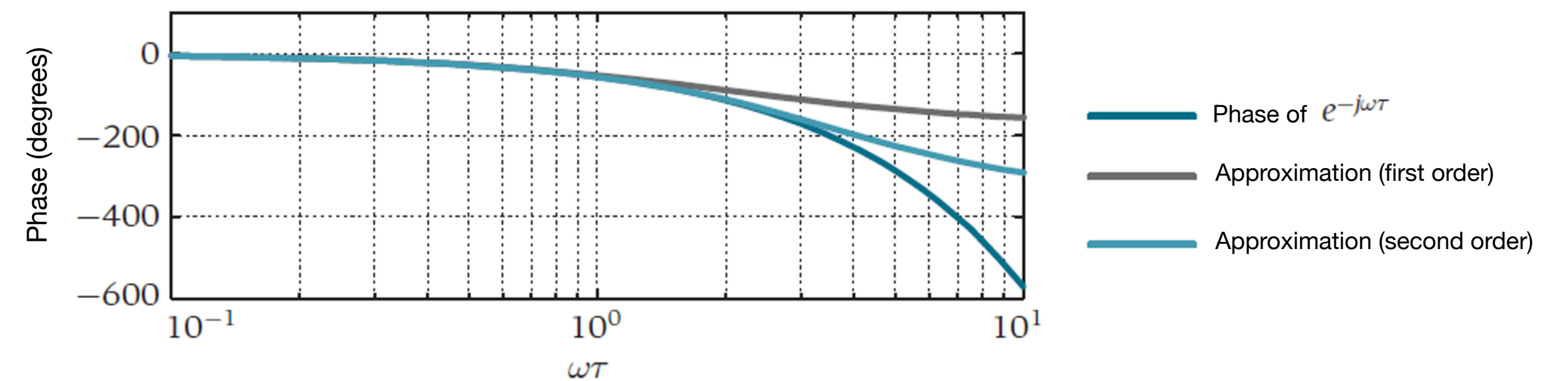
$\mu = 1$ $\mu(a - b) = -\tau$ $-\mu b(a - b) = \frac{\tau^2}{2}$	{	Function to approximate: McLaurin approximation: First order approximation: McLaurin approximation:	$e^{-\tau s}$ $e^{-\tau s} \approx 1 - \tau s + \frac{\tau^2 s^2}{2} + \dots$ $G_I(s) = \mu \frac{1 + as}{1 + bs}$ $G_I(s) = \mu \frac{1 + as}{1 + bs} \approx \mu + \mu(a - b)s - \mu b(a - b)s^2$
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Control of LTI Systems with Delays: Padé Approximation



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

Phase Lag due to the delay term



$$a = -\frac{\tau}{2}$$

$$b = \frac{\tau}{2}$$



$$\mu = 1$$

$$\mu(a - b) = -\frac{\tau}{2}$$

$$-\mu b(a - b) = \frac{\tau^2}{2}$$

Function to approximate:

$$e^{-\tau s}$$

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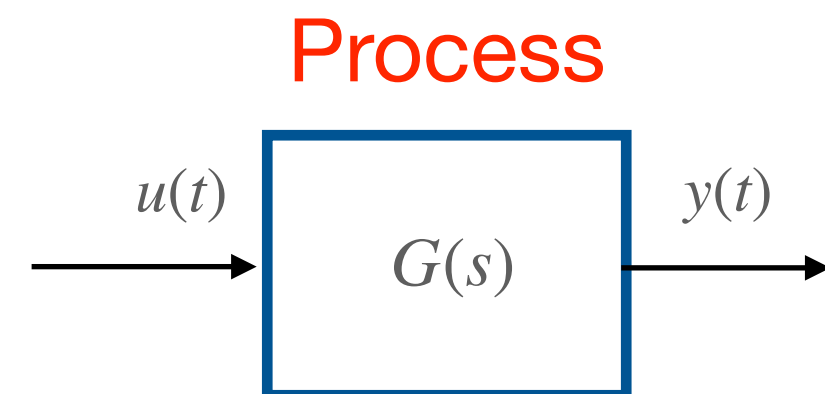
First order approximation:

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McLaurin approximation:

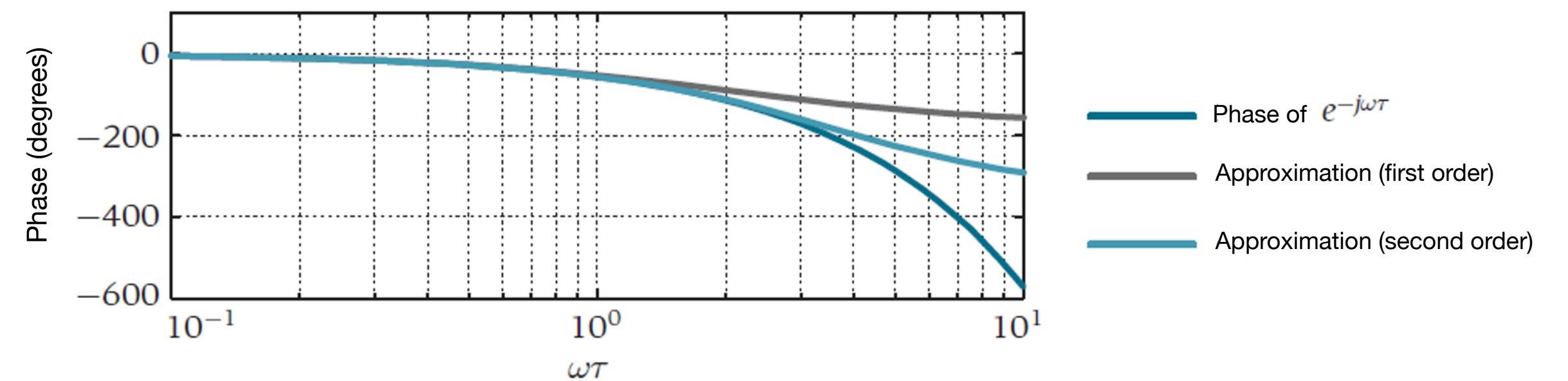
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Control of LTI Systems with Delays: Padé Approximation



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

Phase Lag due to the delay term



$$\begin{aligned} a &= -\frac{\tau}{2} \\ b &= \frac{\tau}{2} \end{aligned}$$



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Function to approximate:

McLaurin approximation:

First order approximation:

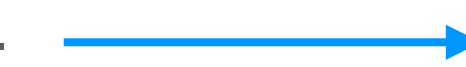
McLaurin approximation:

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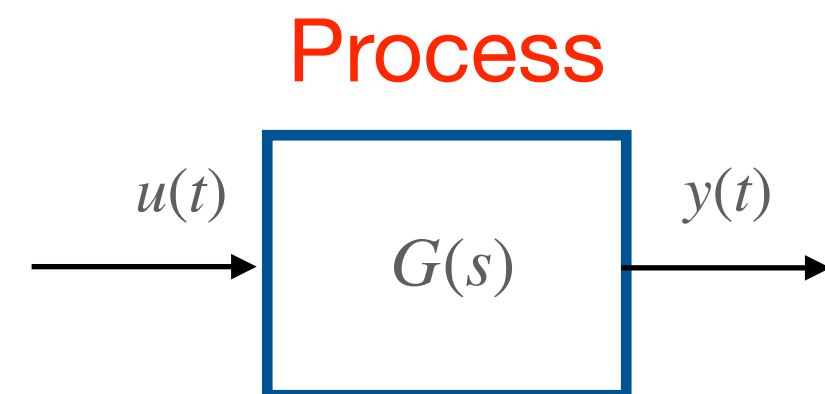
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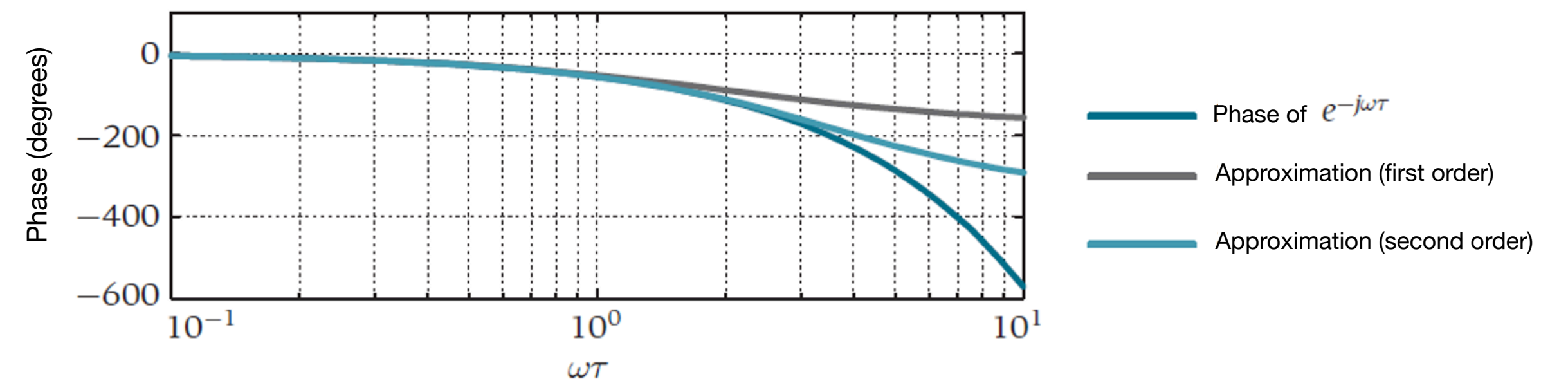
$$\boxed{G_I(s) = \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s}}$$

Control of LTI Systems with Delays: Padé Approximation



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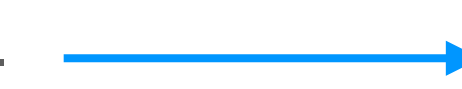
McLaurin approximation:

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$$G_I(s) = \mu \frac{1 + as}{1 + bs}$$

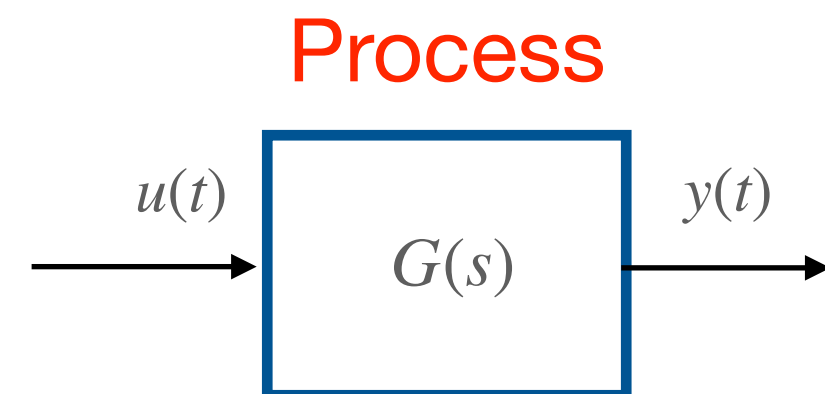
$$G_I(s) = \mu \frac{1 + as}{1 + bs} \approx \mu + \mu(a - b)s - \mu b(a - b)s^2$$



$$G_I(s) = \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s}$$

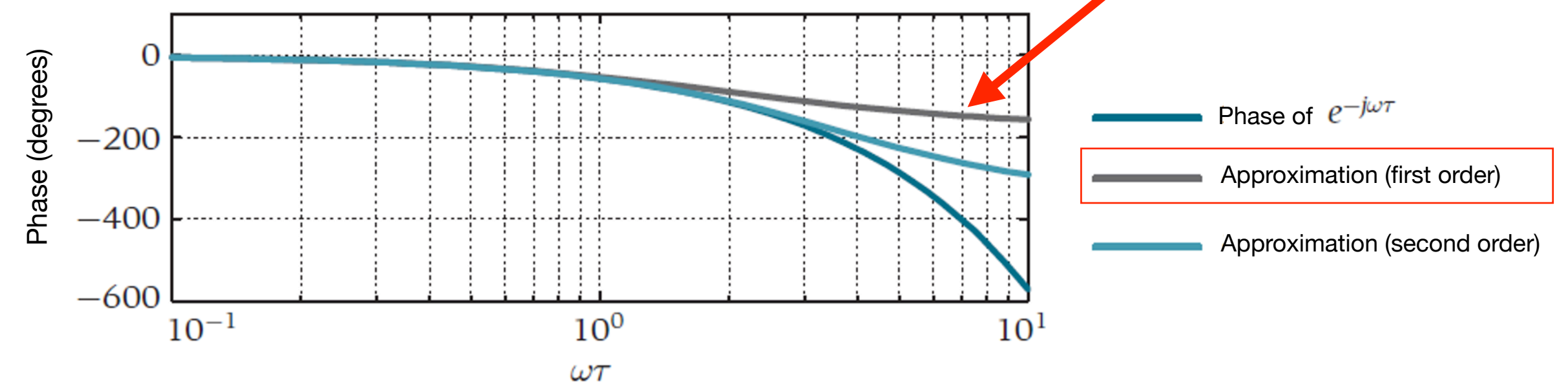
Magnitude
and phase ?

Control of LTI Systems with Delays: Padé Approximation



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

Phase Lag due to the delay term



$$\boxed{\begin{aligned} a &= -\frac{\tau}{2} \\ b &= \frac{\tau}{2} \end{aligned}}$$



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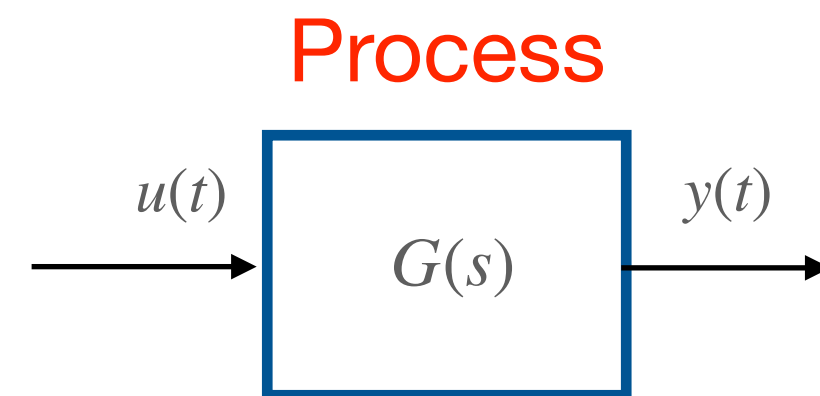
Magnitude and phase ?

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Control of LTI Systems with Delays

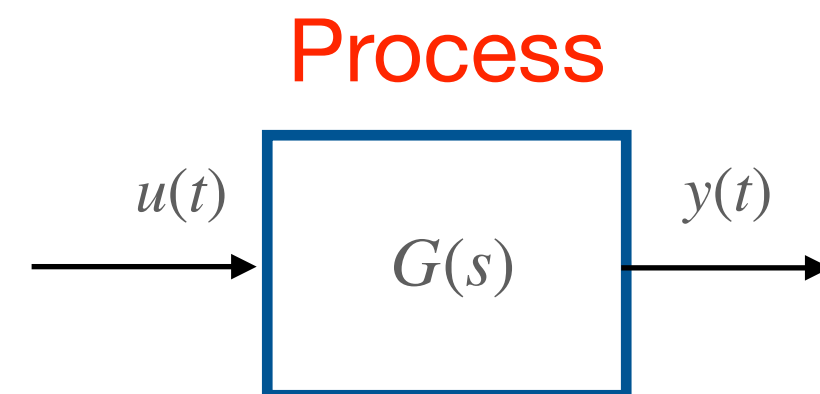
Summary:



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

Control of LTI Systems with Delays

Summary:



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

Modelling alternatives:

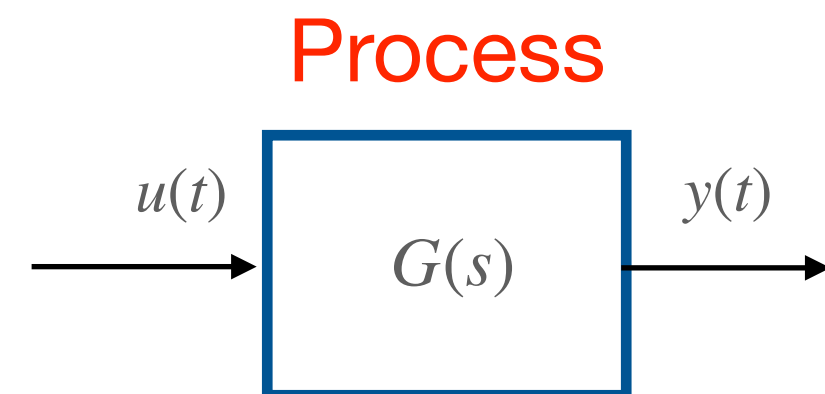
$$e^{-\tau s} \quad G_I(s) = \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s}$$

$$G_{II}(s) = \dots$$

$$G_{III}(s) = \dots$$

Control of LTI Systems with Delays

Summary:



$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = \boxed{e^{-\tau s}} G'(s)$$

$e^{-\tau s}$

$G_I(s) = \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s}$
 $G_{II}(s) = \dots$
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Modelling alternatives:

Design alternatives for R(s):

$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} = e^{-\tau s} G'(s)$$

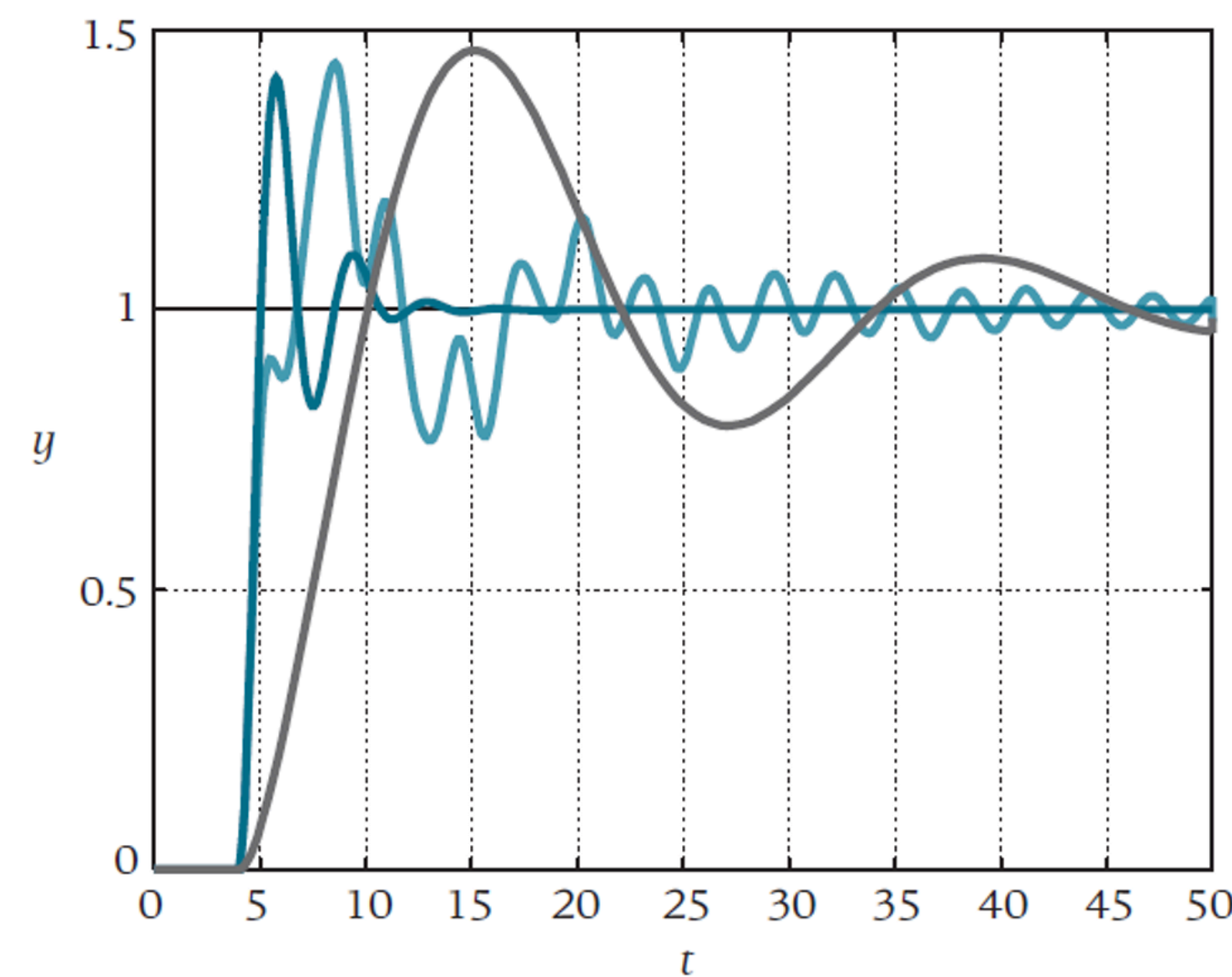
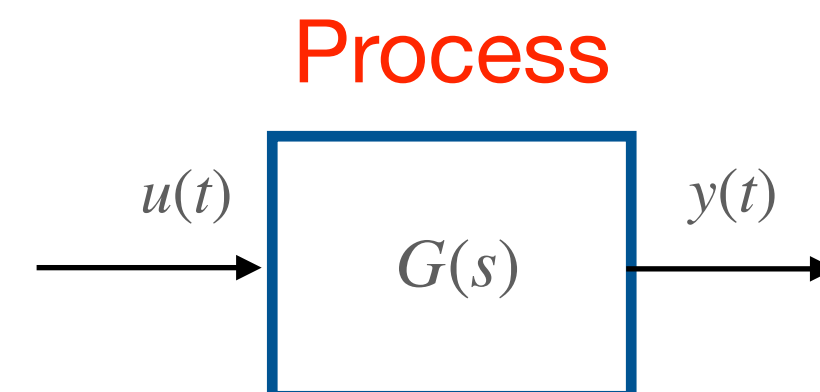
$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} = (1 - e^{-\tau s}) G'(s)$$

$$G(s) = e^{-\tau s} \frac{N(s)}{D(s)} \approx \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s} G'(s)$$

$$P(s) = (1 - e^{-\tau s}) \frac{N(s)}{D(s)} \approx \left(1 - \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s} \right) G'(s)$$

Control of LTI Systems with Delays

Summary:



Design alternatives for R(s):

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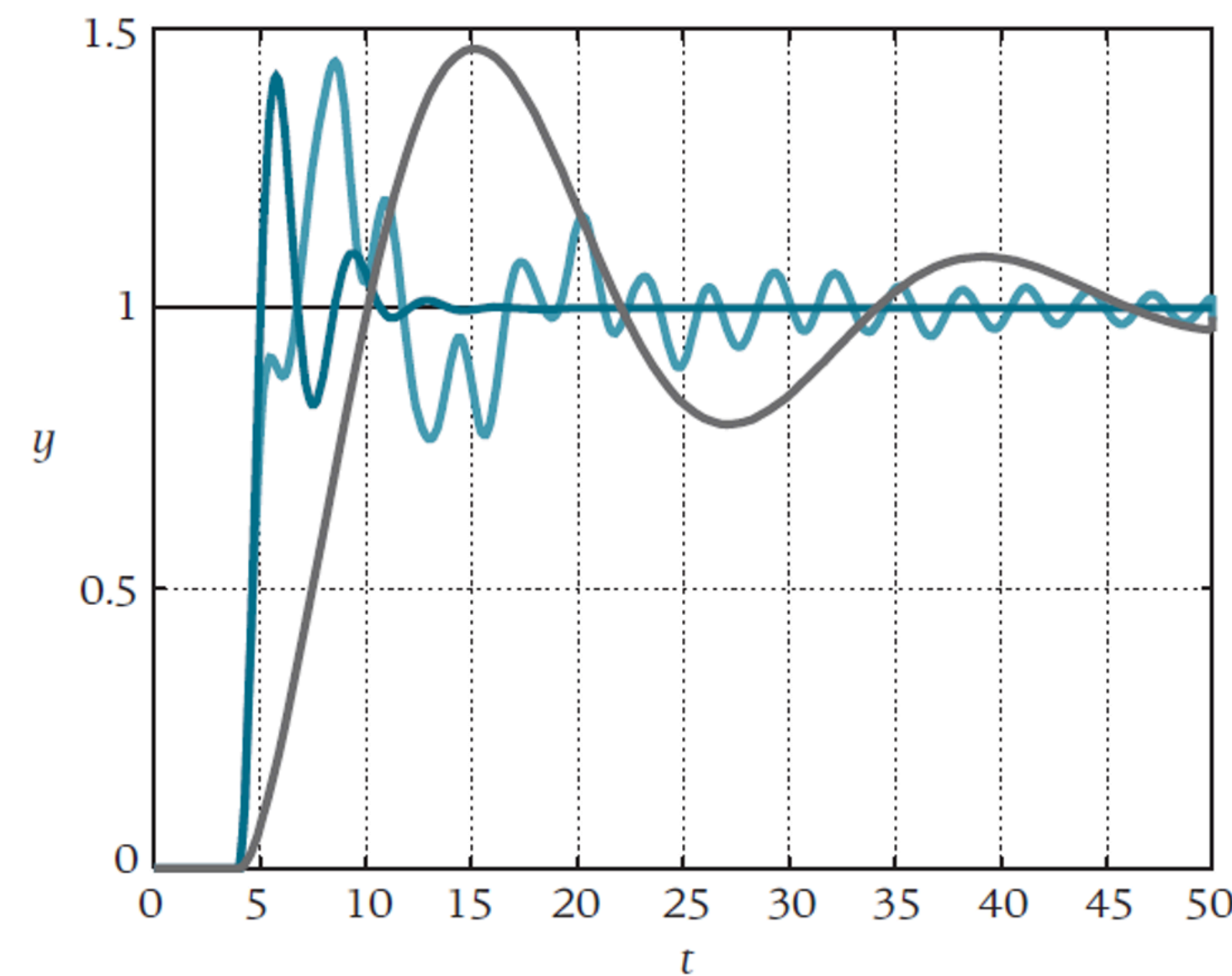
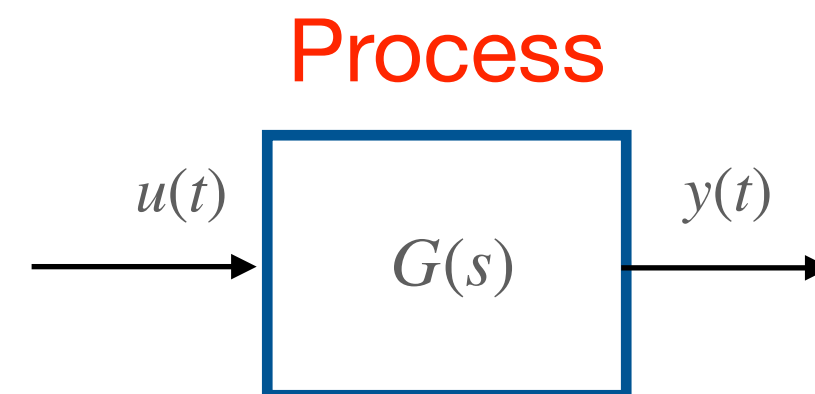
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Control of LTI Systems with Delays

Summary:



Design alternatives for R(s):

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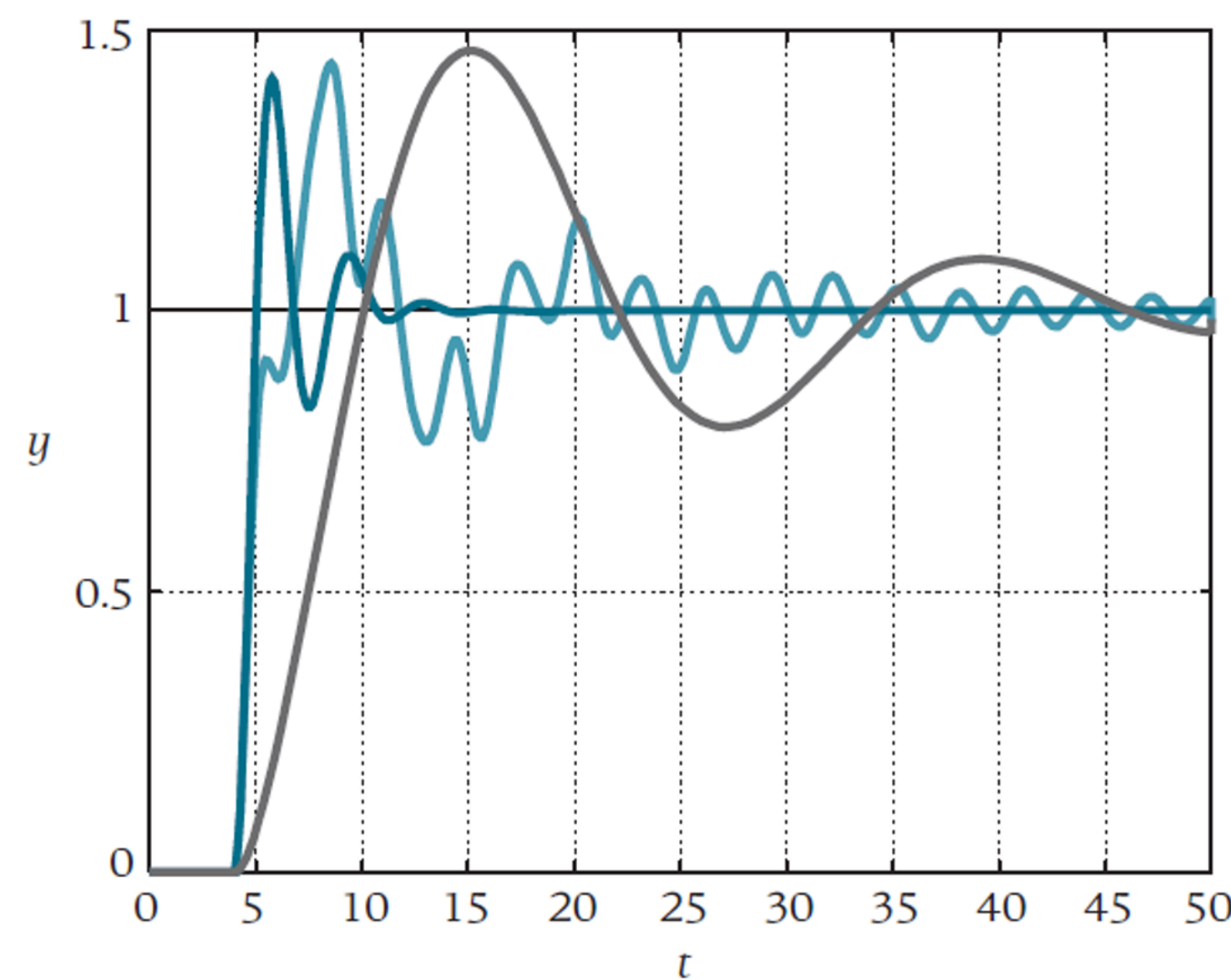
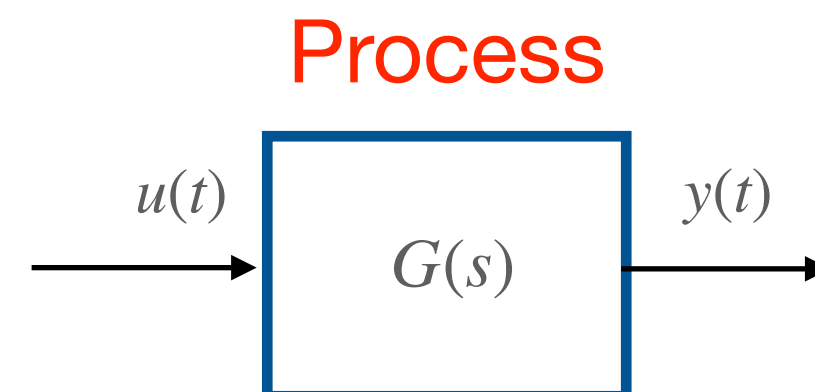
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Control of LTI Systems with Delays

Summary:



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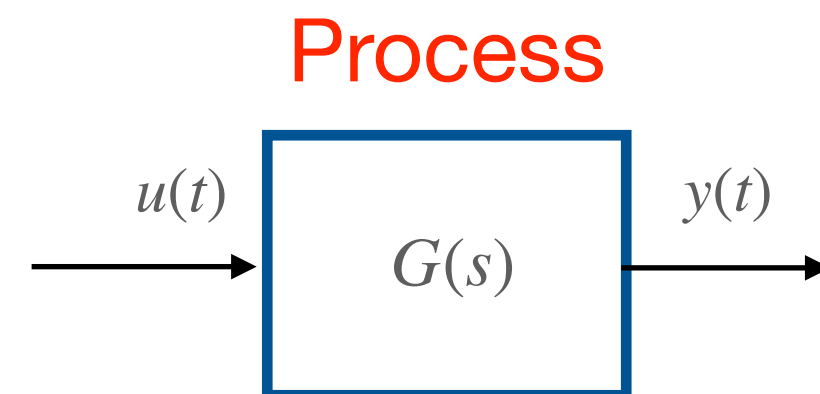
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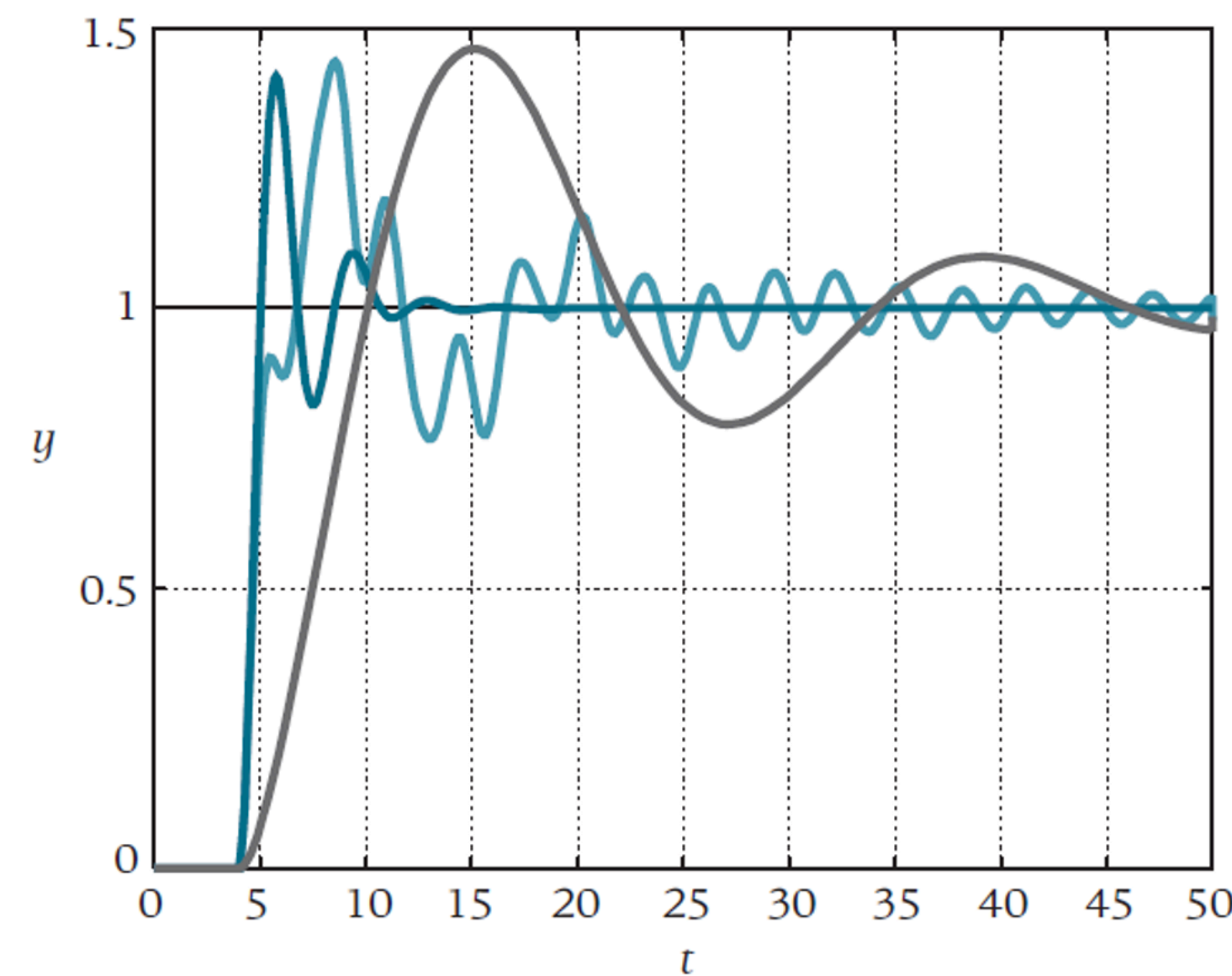
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Control of LTI Systems with Delays

Summary:



Exercise: Use the
UFCS and design
 $R(s)$ relying on

Design alternatives for $R(s)$:

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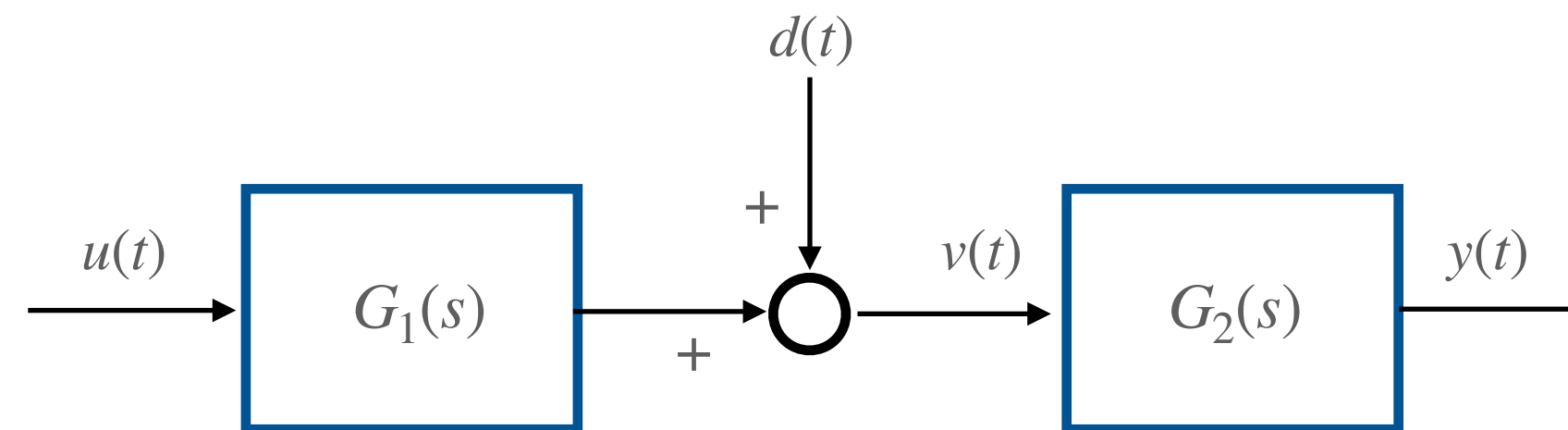
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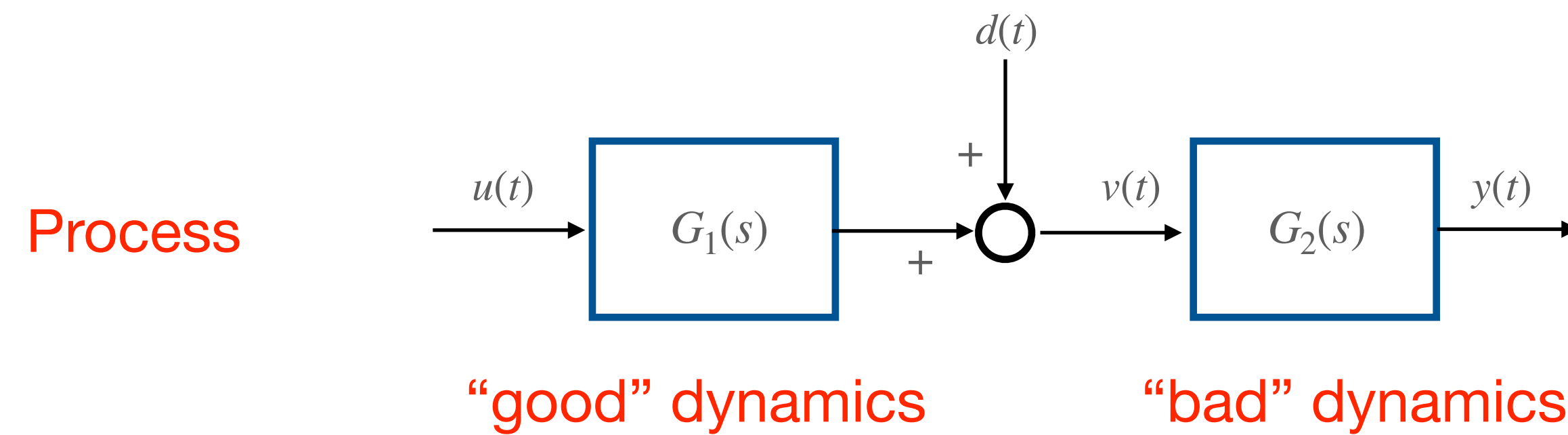
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Control Scheme with Decoupling in the Frequency Domain

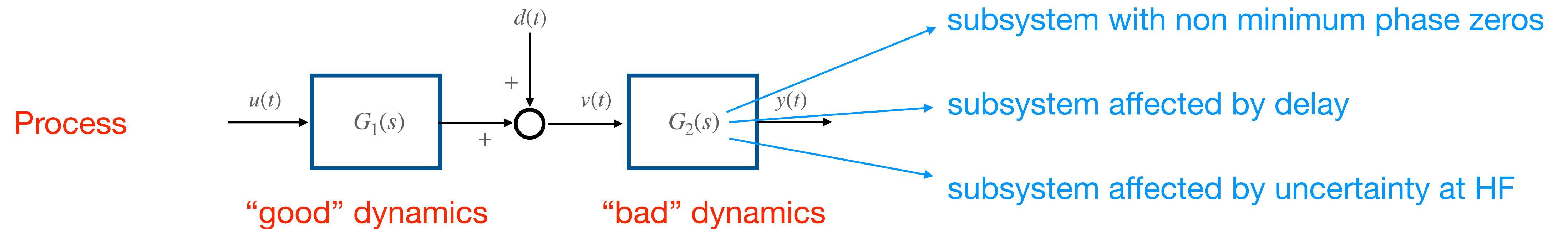
Process



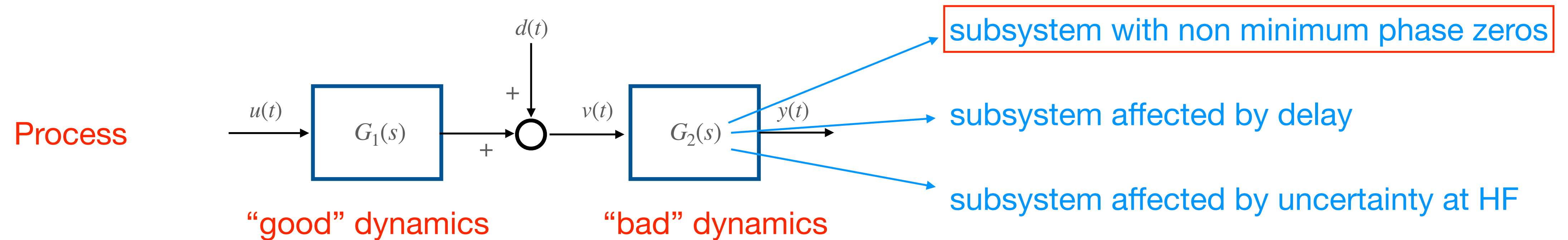
Control Scheme with Decoupling in the Frequency Domain



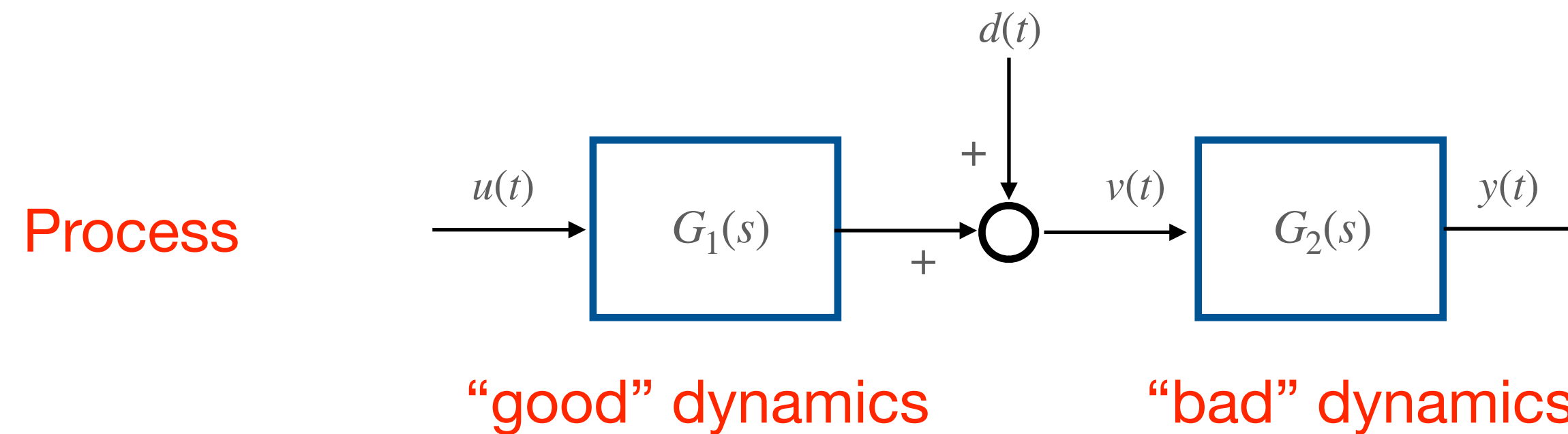
Control Scheme with Decoupling in the Frequency Domain



Control Scheme with Decoupling in the Frequency Domain



Control Scheme with Decoupling in the Frequency Domain

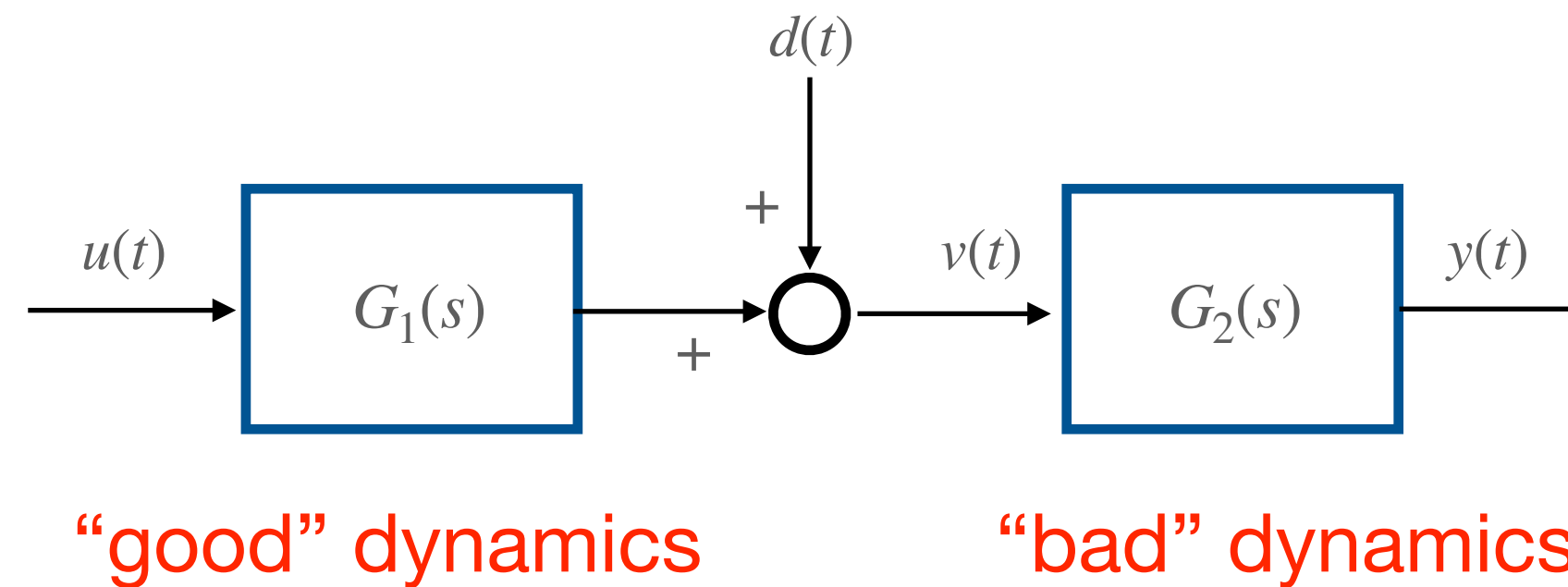


Assumptions:

- The performances that can be obtained by designing a control system for $G_1(s)$ alone are better than those that can be obtained for the entire system $G_1(s)G_2(s)$
- v is accessible (measurable)

Control Scheme with Decoupling in the Frequency Domain

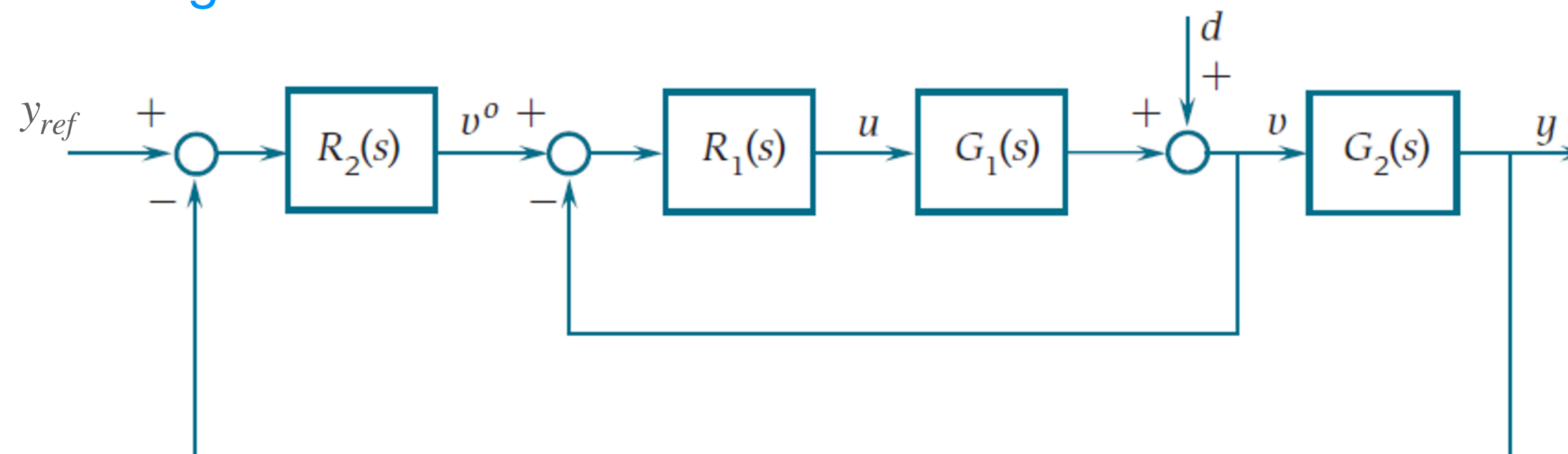
Process



Assumptions:

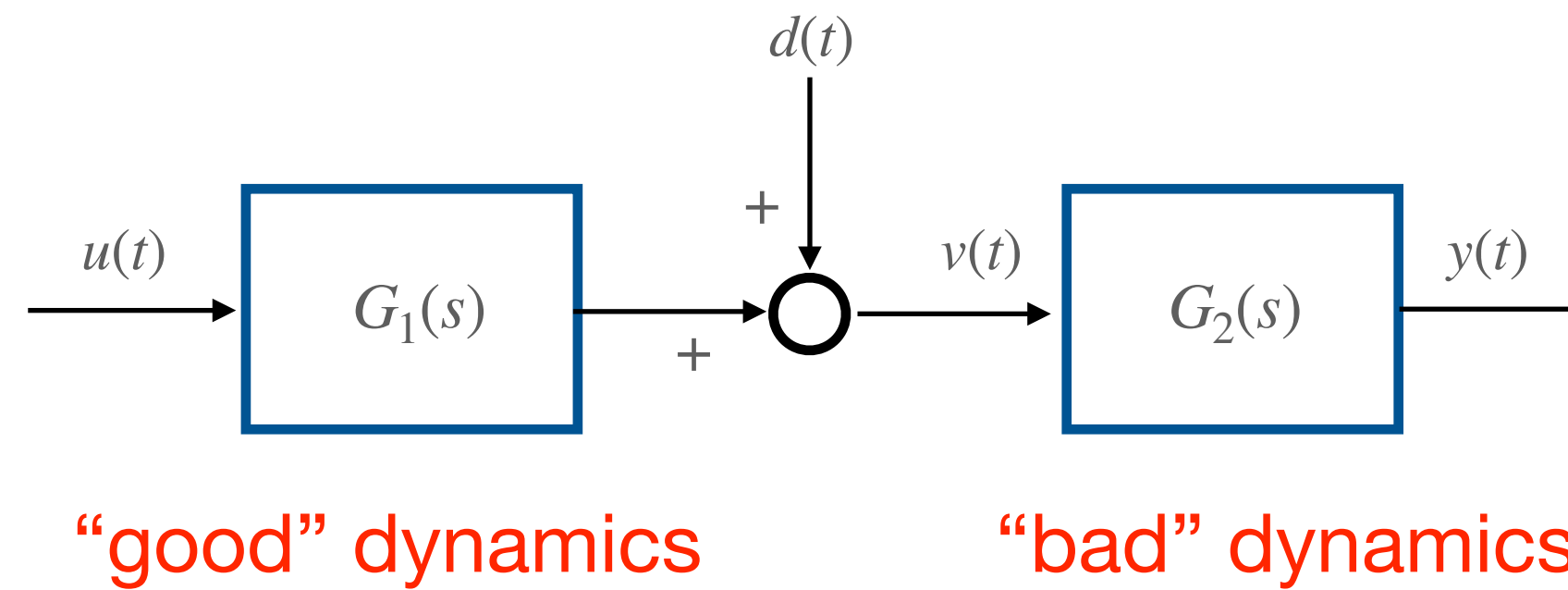
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Control scheme design

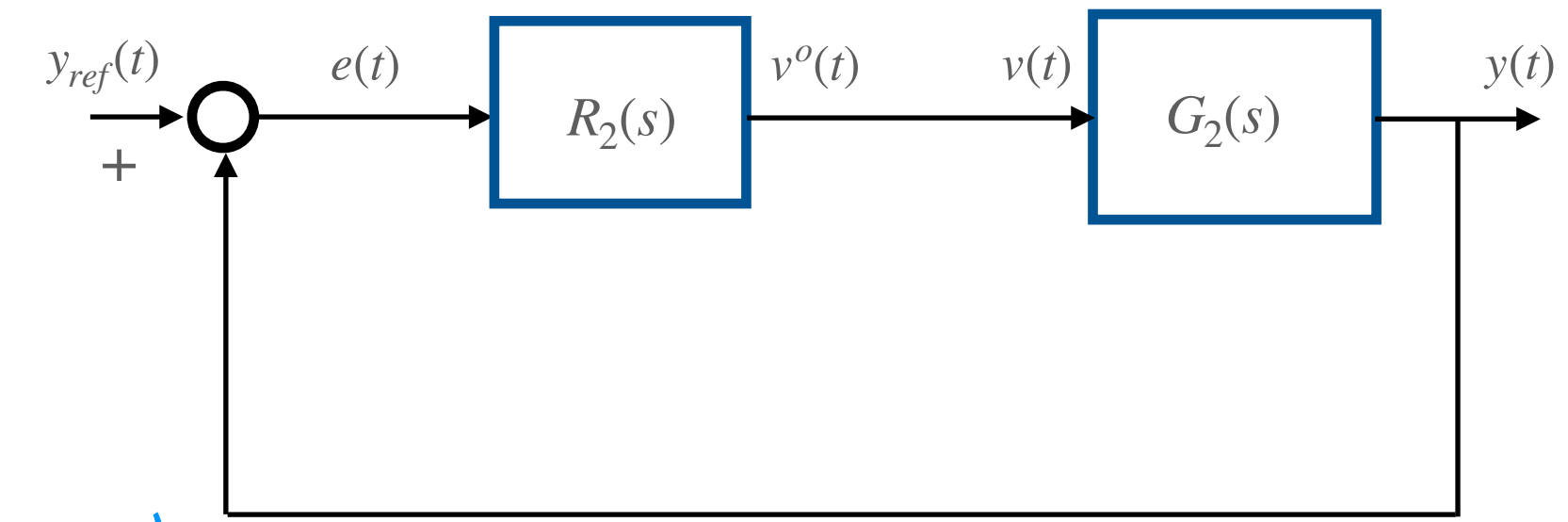
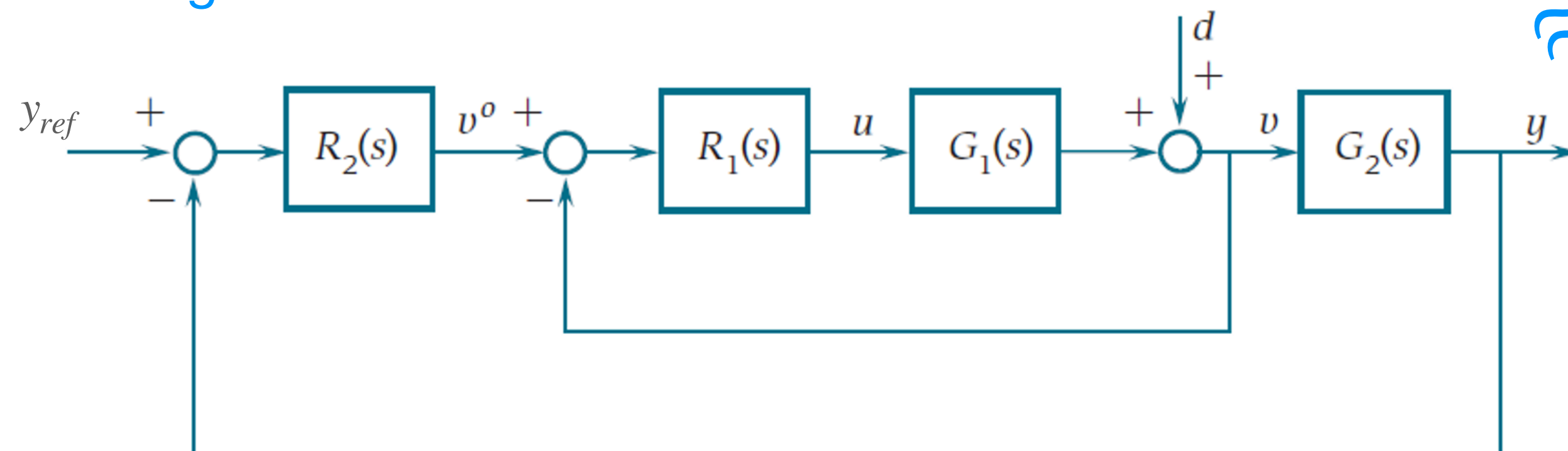


Control Scheme with Decoupling in the Frequency Domain

Process

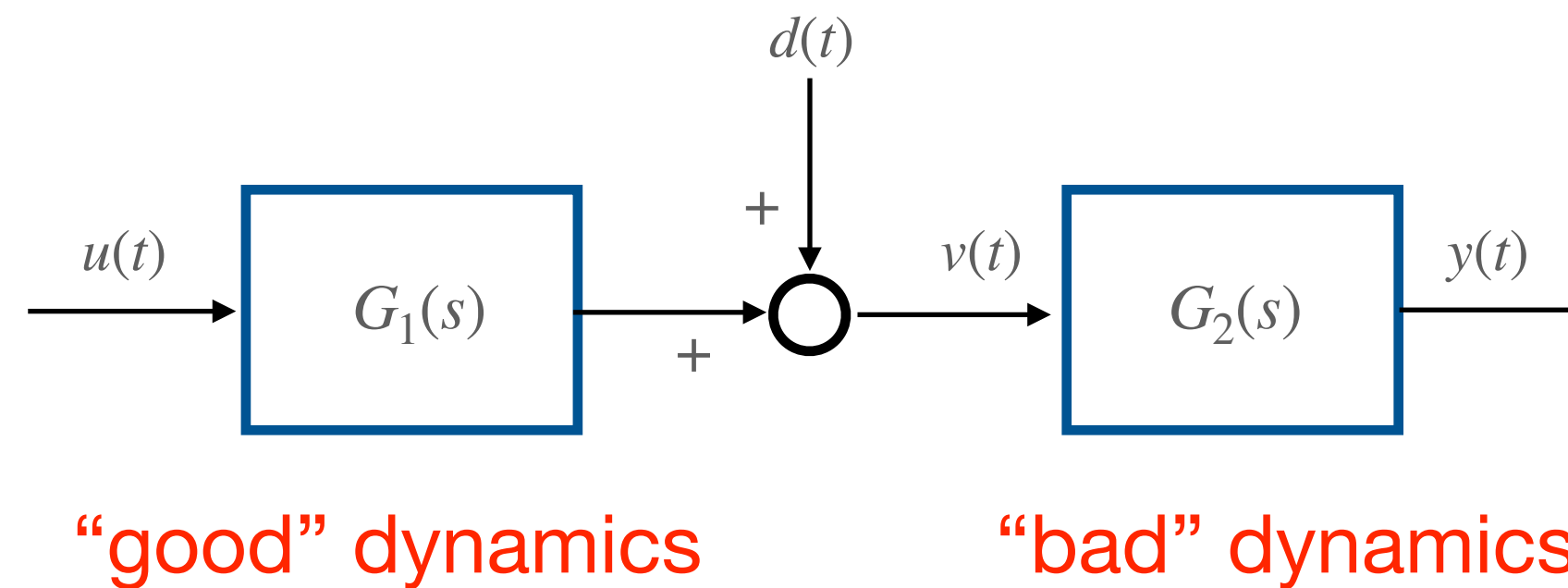


Control scheme design

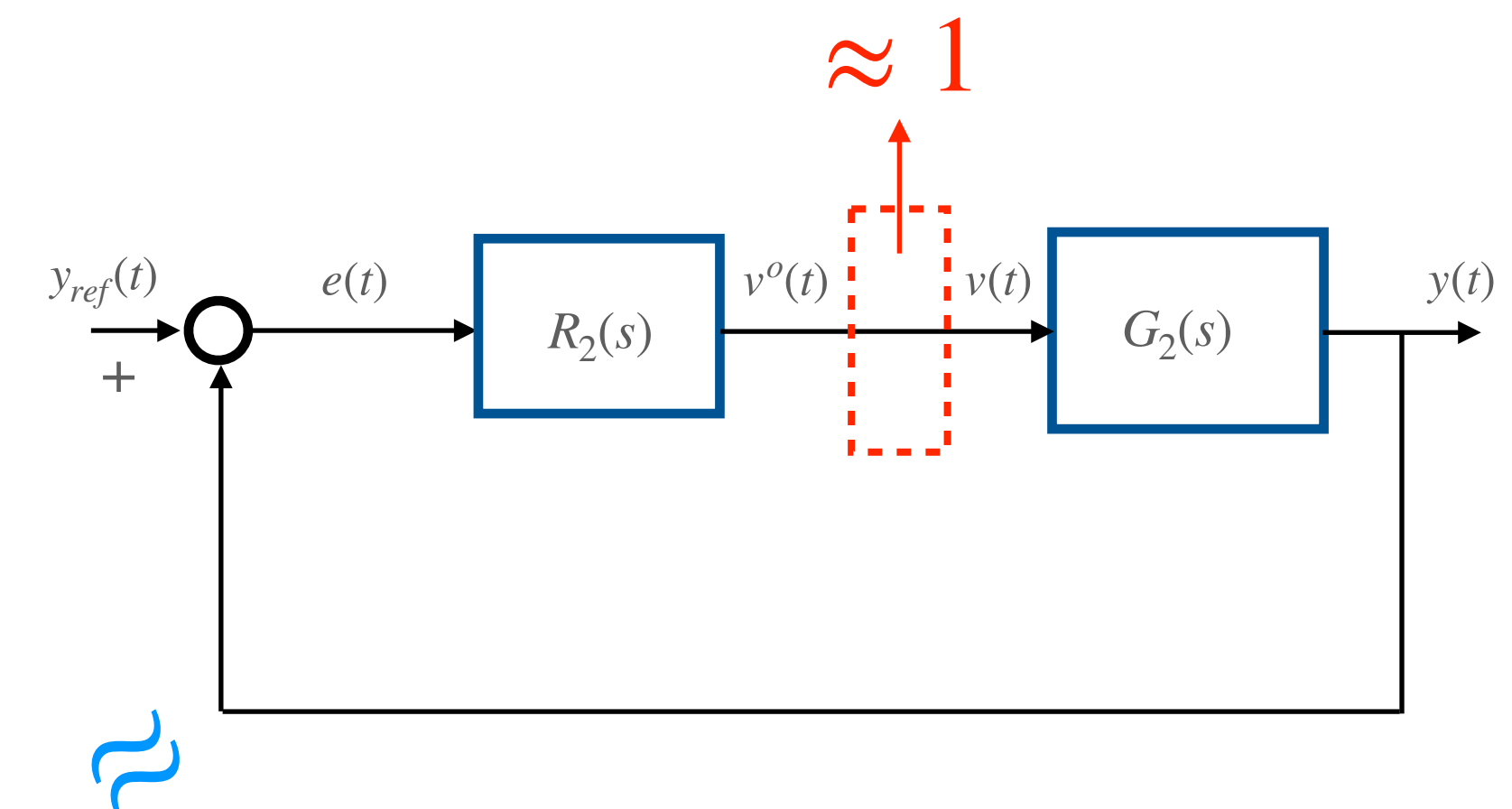
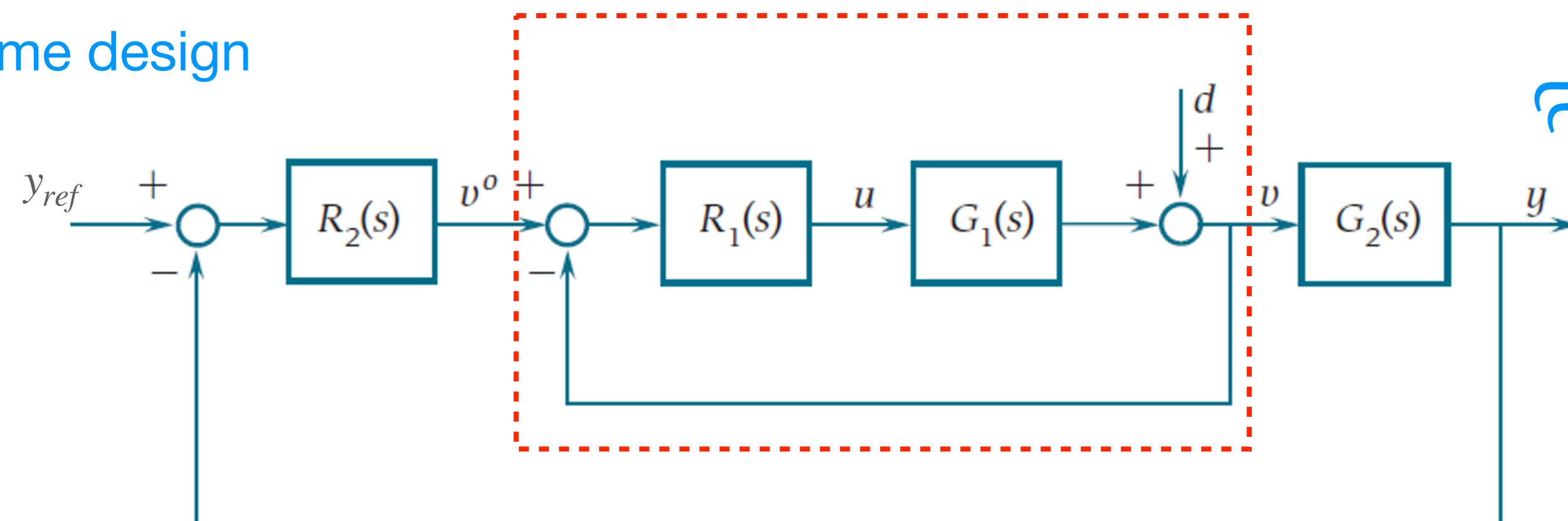


Control Scheme with Decoupling in the Frequency Domain

Process

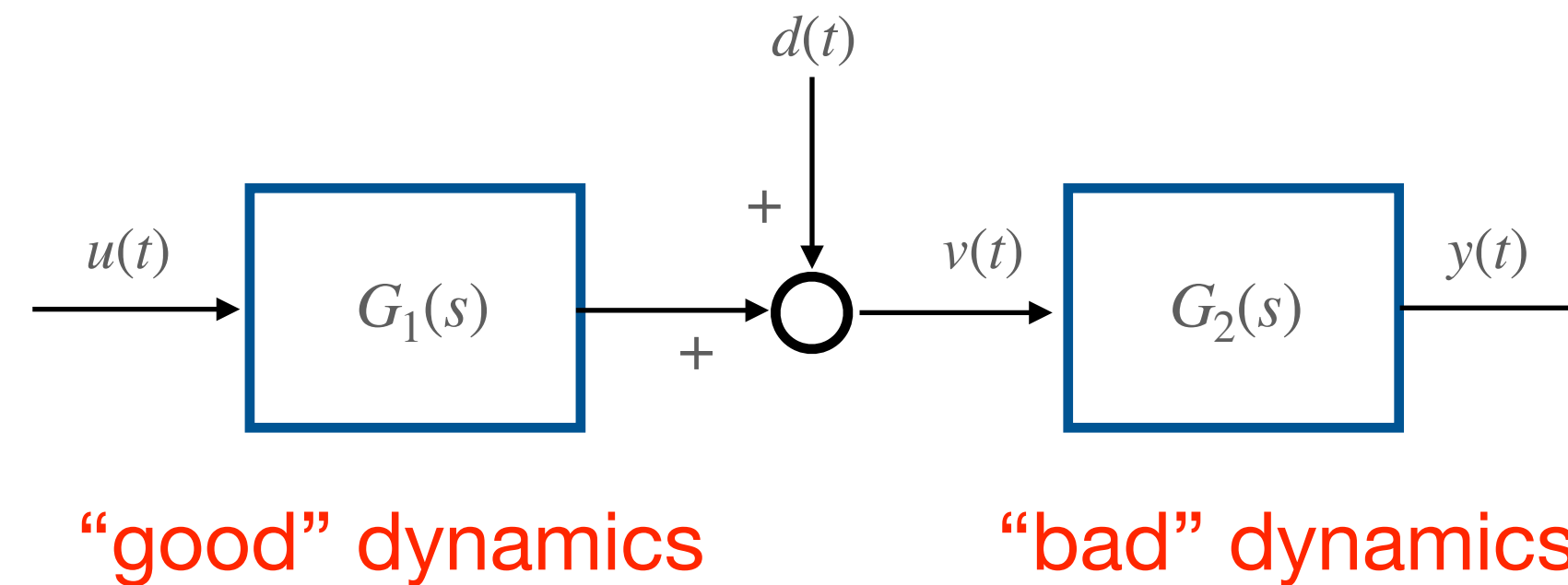


Control scheme design

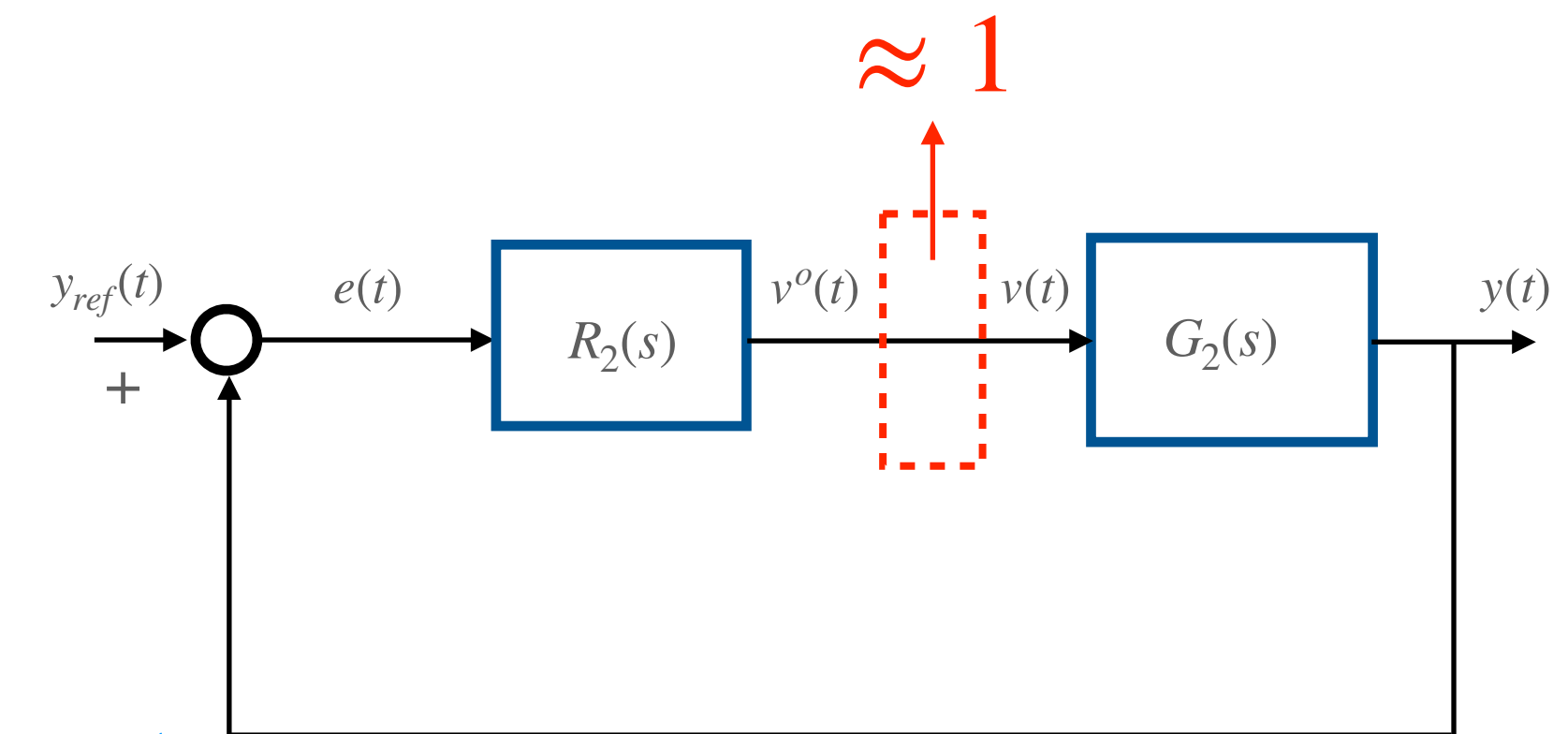
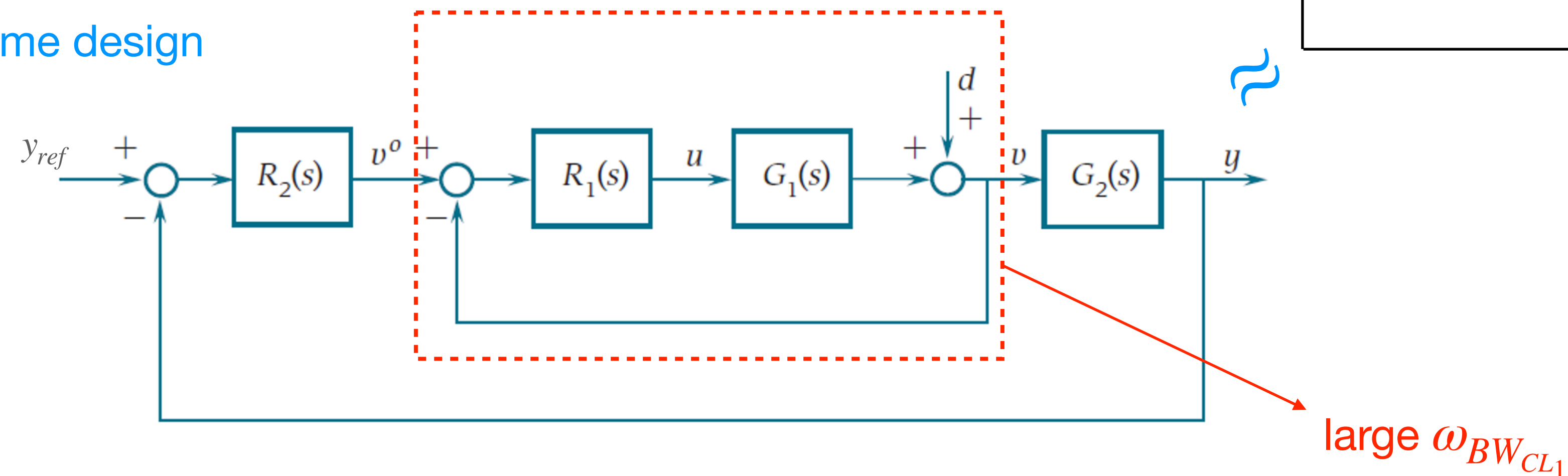


Control Scheme with Decoupling in the Frequency Domain

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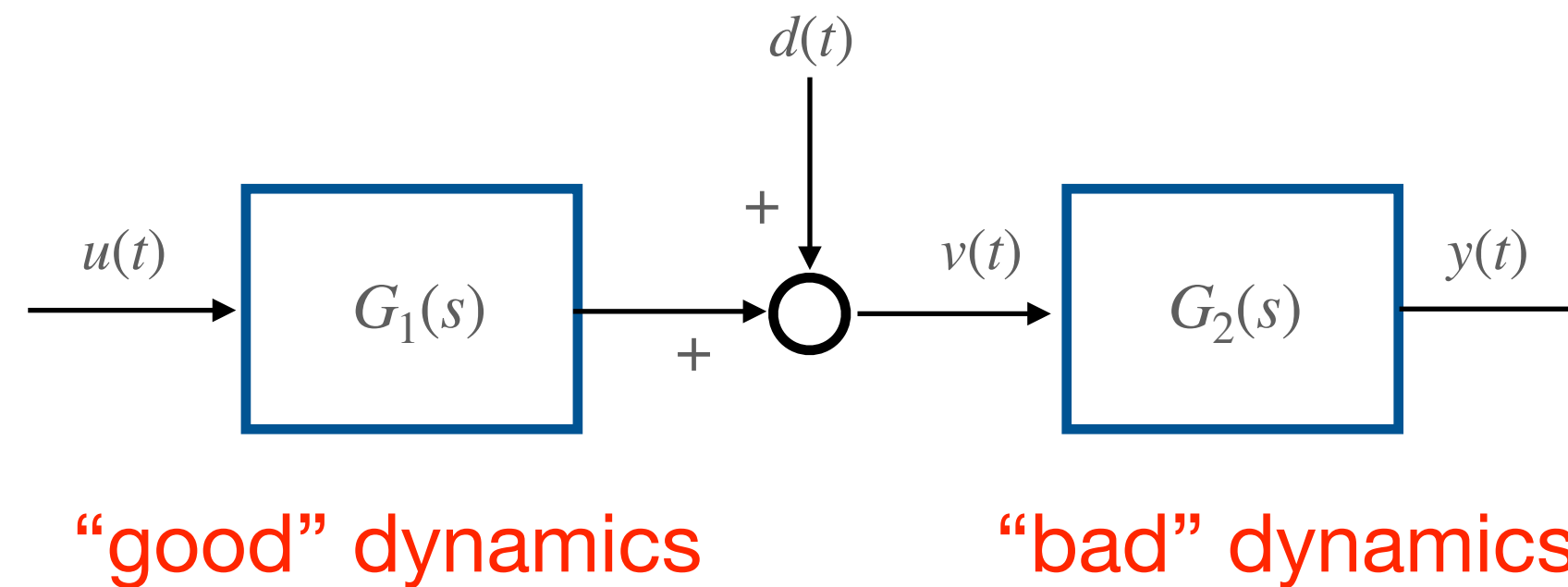


Control scheme design

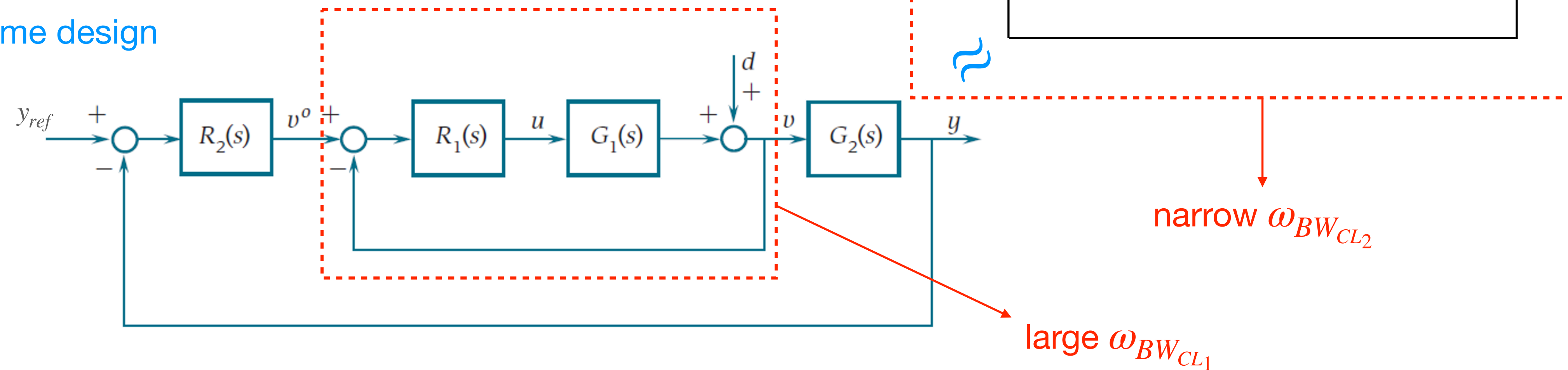


Control Scheme with Decoupling in the Frequency Domain

Process

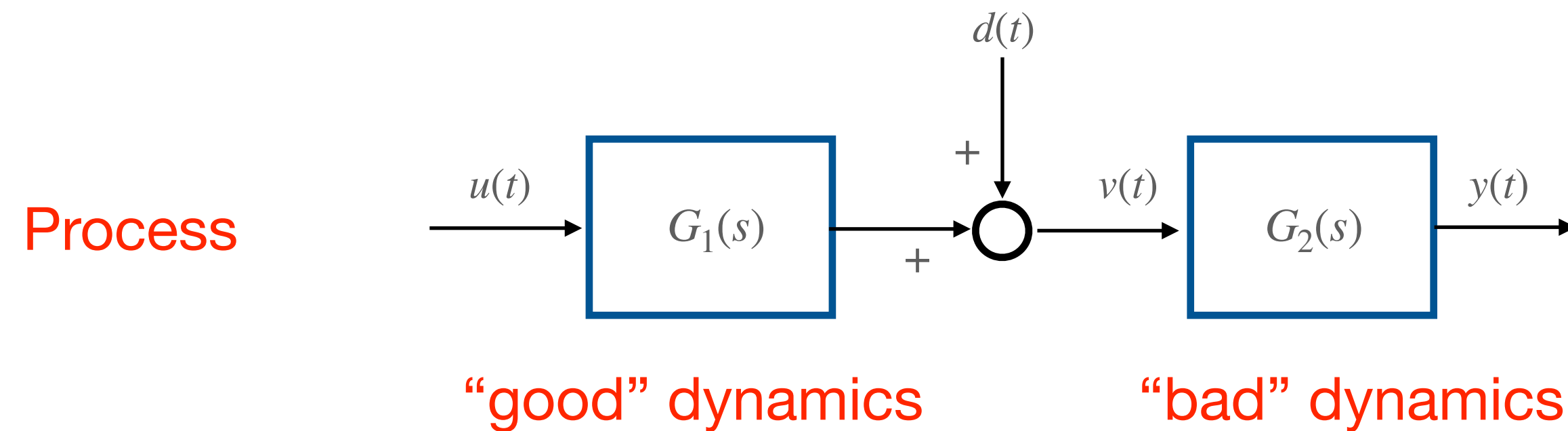


Control scheme design



Control Scheme with Decoupling in the Frequency Domain

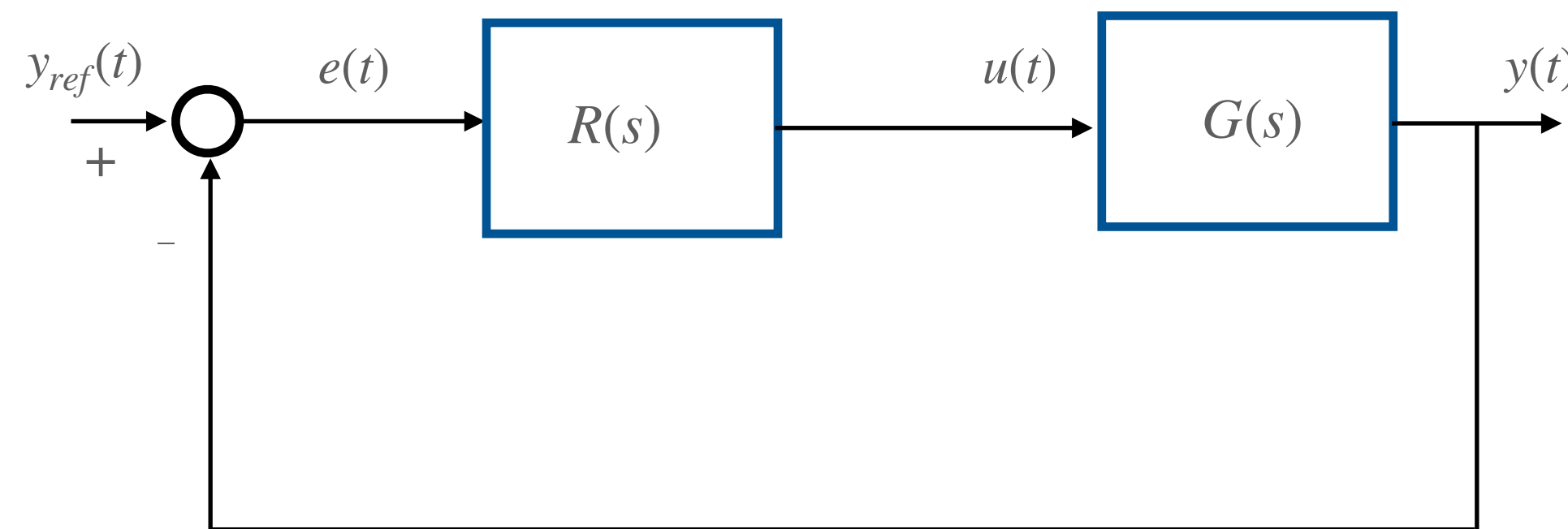
Example (from the textbook):



$$G_1(s) = \frac{1}{1 + 0.005s}$$

$$G_2(s) = \frac{e^{-4s}}{(1 + s)^2}$$

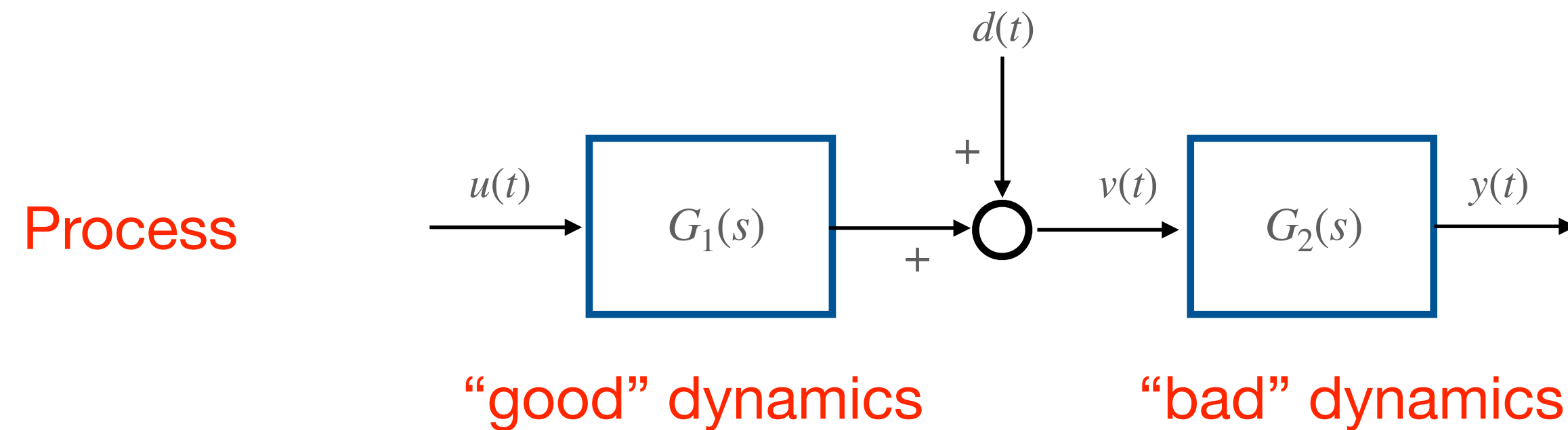
Control scheme design: Solution 1



$$R(s) = \frac{0.2(s + 1)}{s}$$

Control Scheme with Decoupling in the Frequency Domain

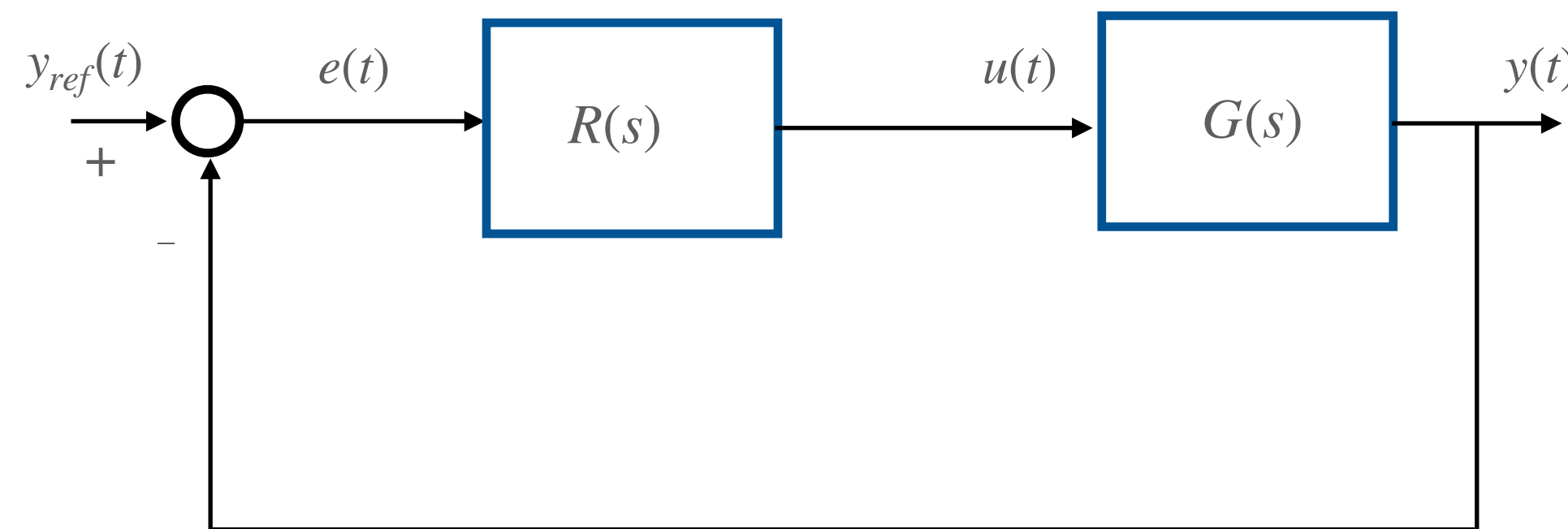
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$$G_2(s) = \frac{e^{-4s}}{(1 + s)^2}$$

Control scheme design: Solution 1

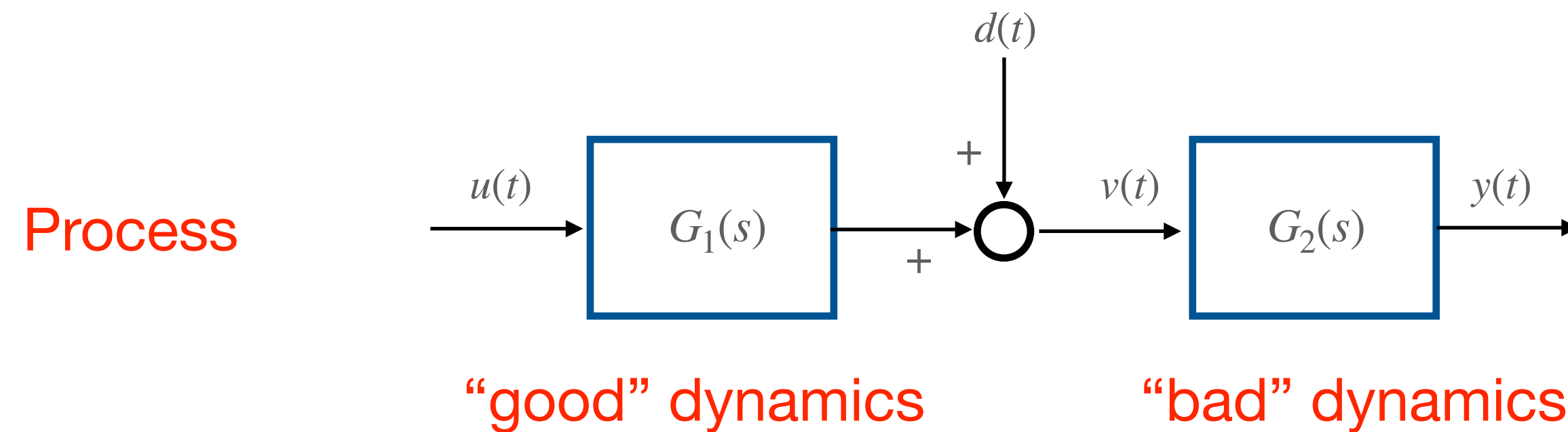


$$R(s) = \frac{0.2(s + 1)}{s}$$

→ Compute the closed loop bandwidth

Control Scheme with Decoupling in the Frequency Domain

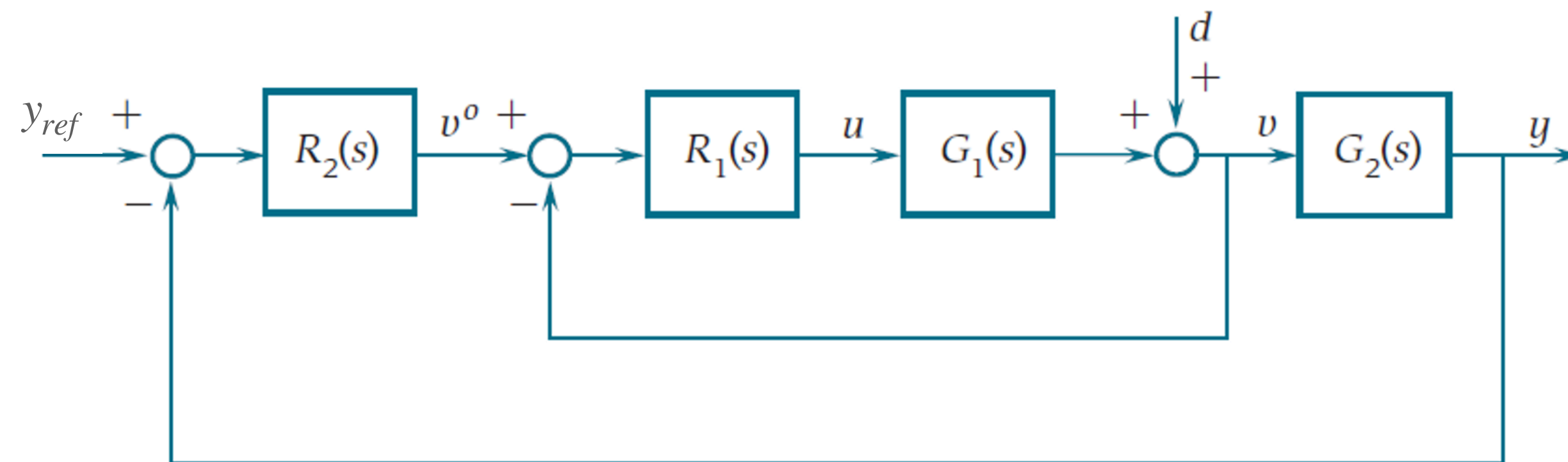
Example (from the textbook):



$$G_1(s) = \frac{1}{1 + 0.005s}$$

$$G_2(s) = \frac{e^{-4s}}{(1 + s)^2}$$

Control scheme design: Solution 2

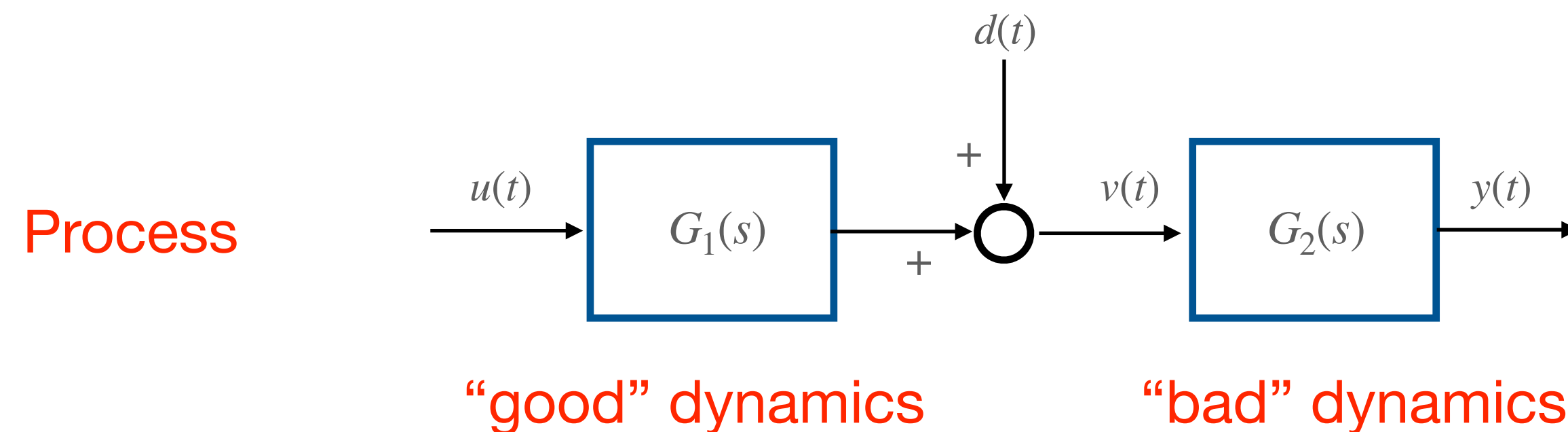


$$R_1(s) = 10$$

$$R_2(s) = \frac{0.2(s + 1)}{s}$$

Control Scheme with Decoupling in the Frequency Domain

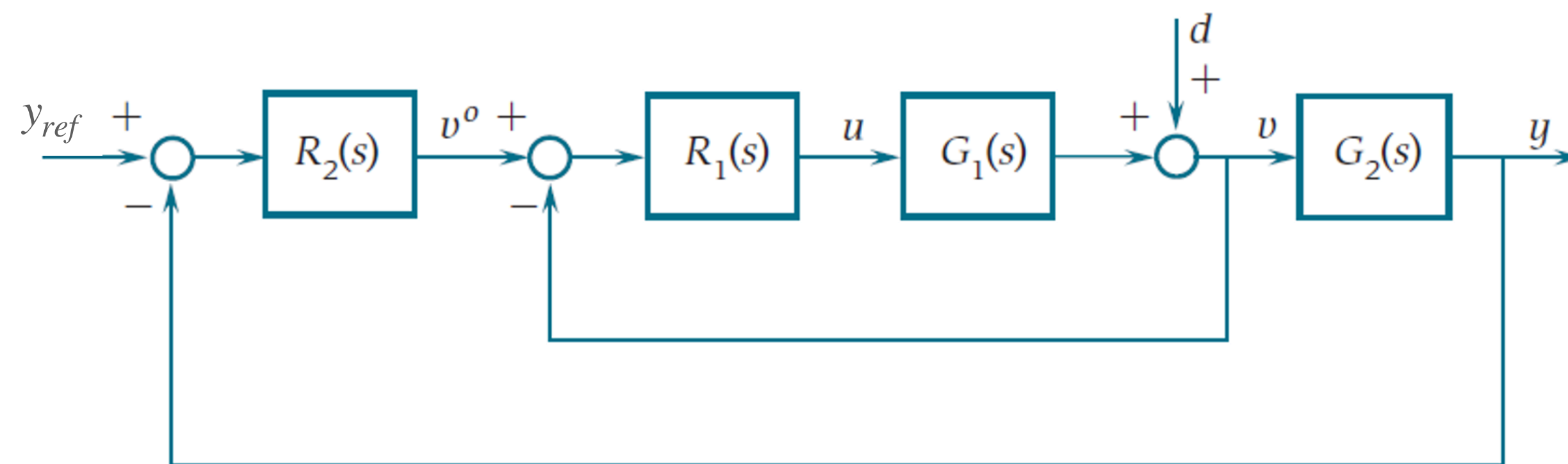
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Control scheme design: Solution 2



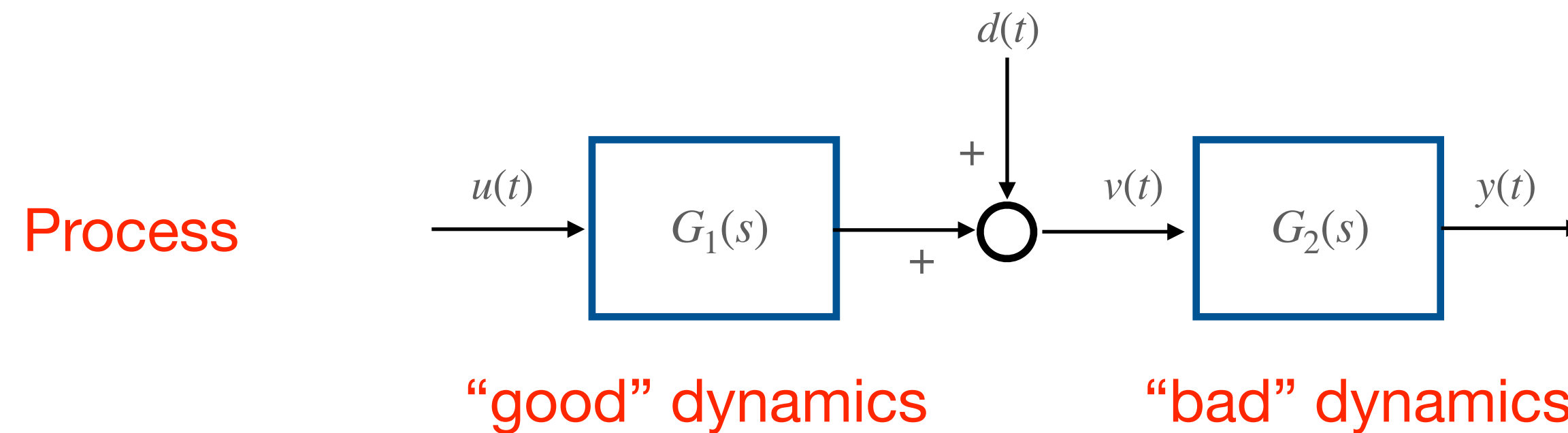
$$R_1(s) = 10$$

$$R_2(s) = \frac{0.2(s + 1)}{s}$$

Compute $\omega_{BW_{CL1}}$ and $\omega_{BW_{CL2}}$

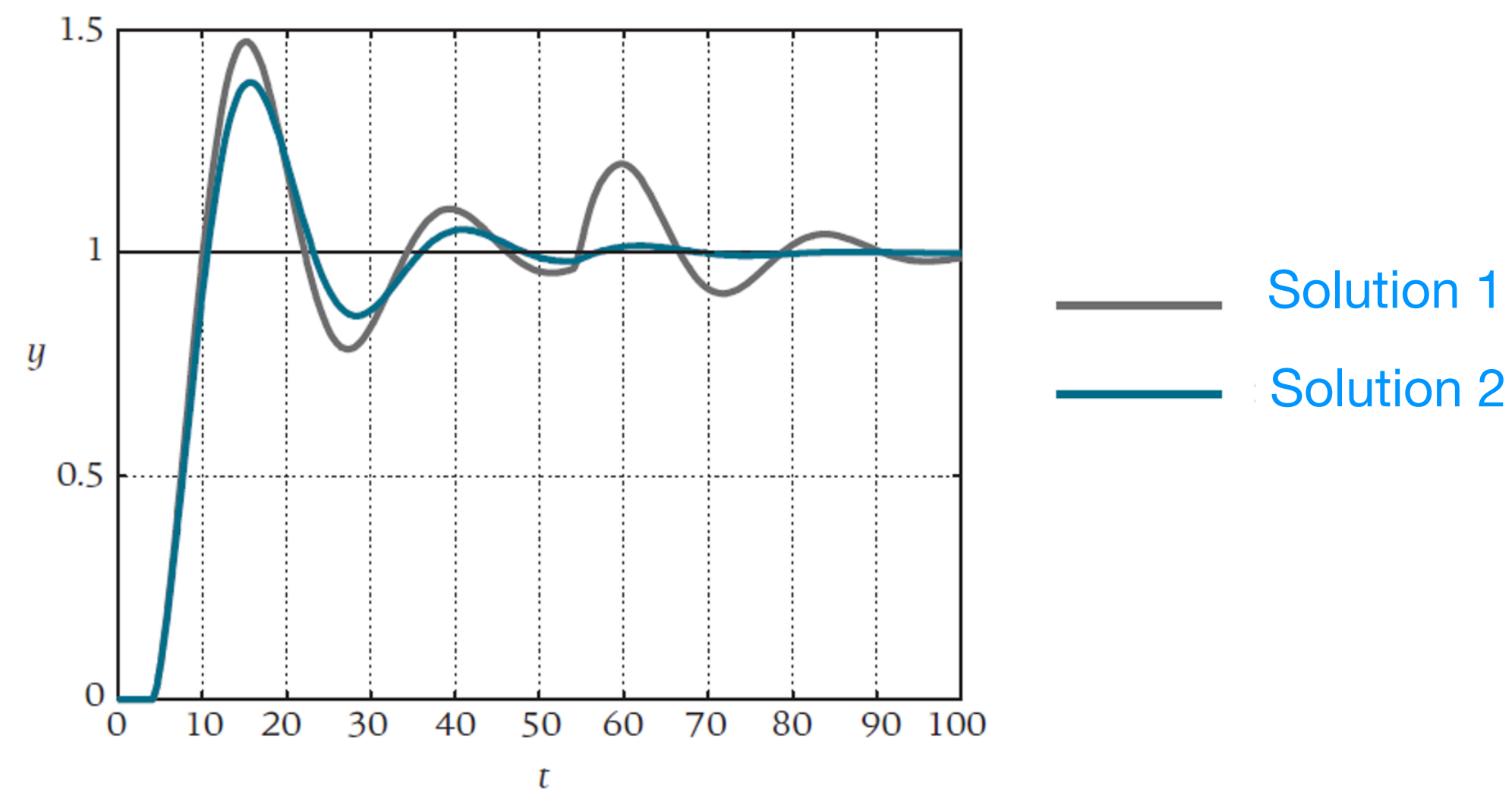
Control Scheme with Decoupling in the Frequency Domain

Example (from the textbook):



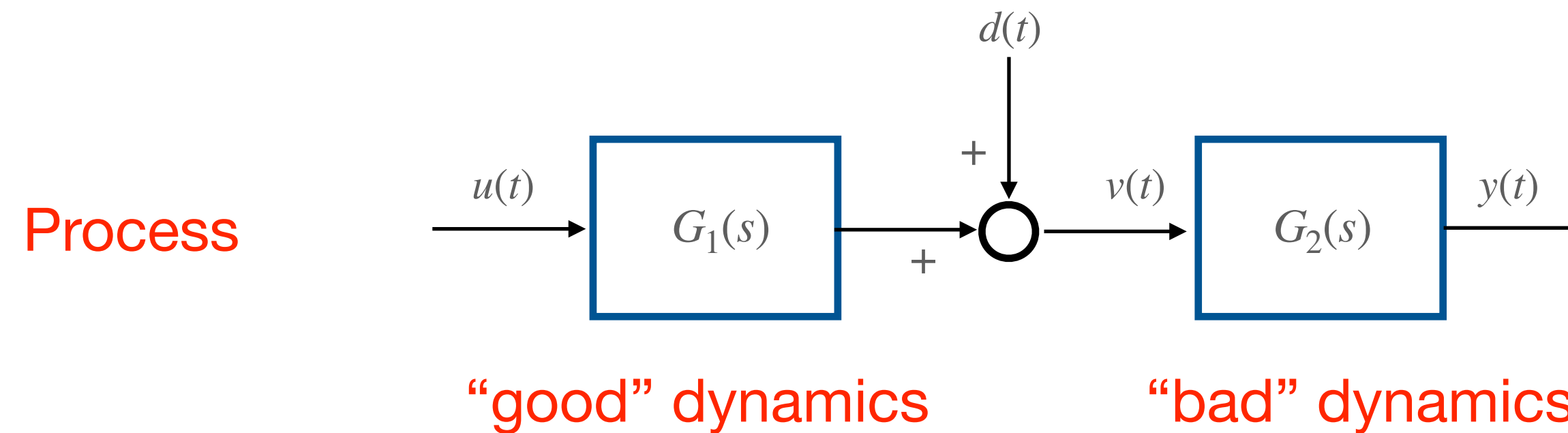
$$G_1(s) = \frac{1}{1 + 0.005s}$$

$$G_2(s) = \frac{e^{-4s}}{(1 + s)^2}$$



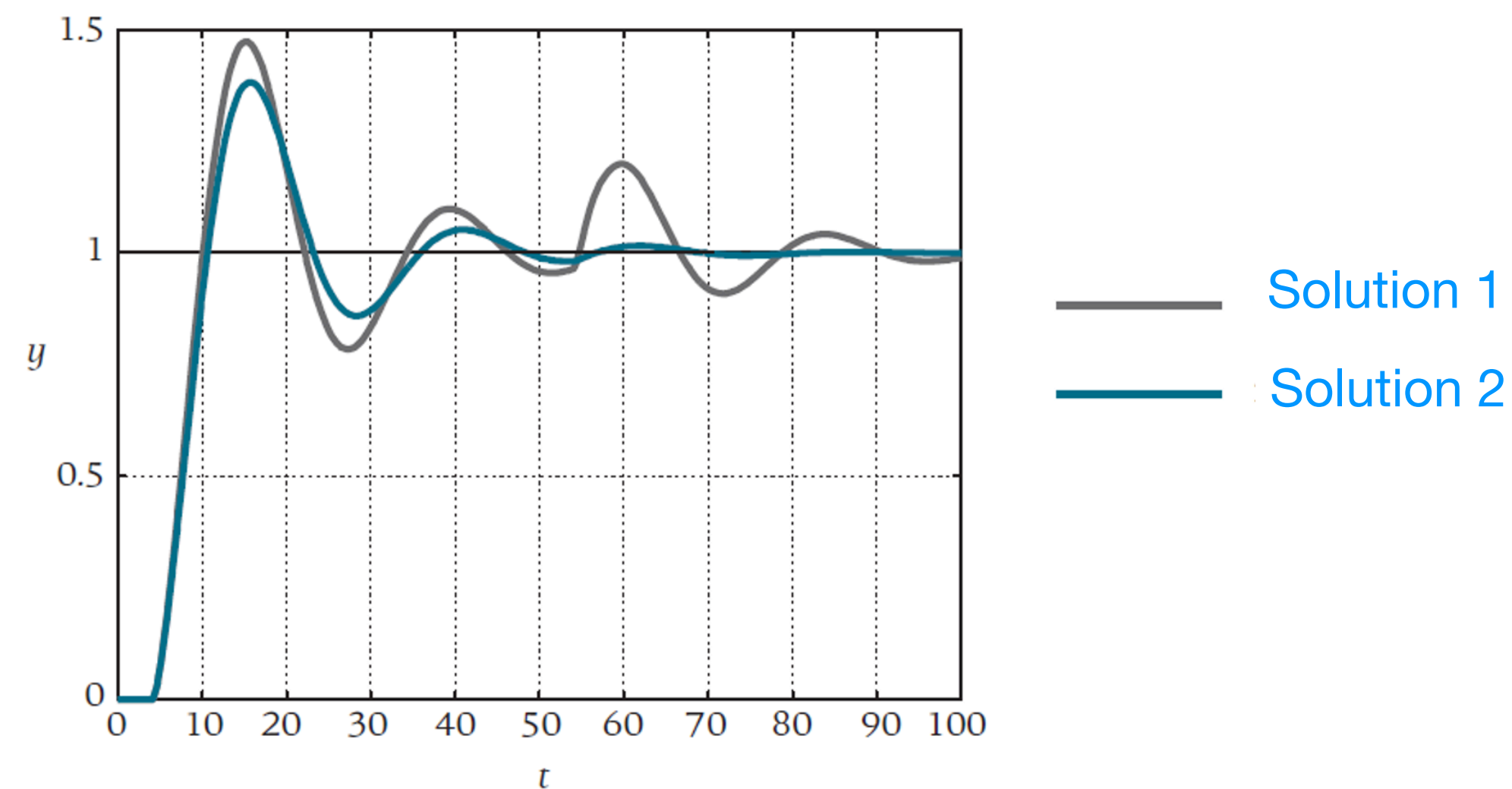
Control Scheme with Decoupling in the Frequency Domain

Example (from the textbook):



$$G_1(s) = \frac{1}{1 + 0.005s}$$

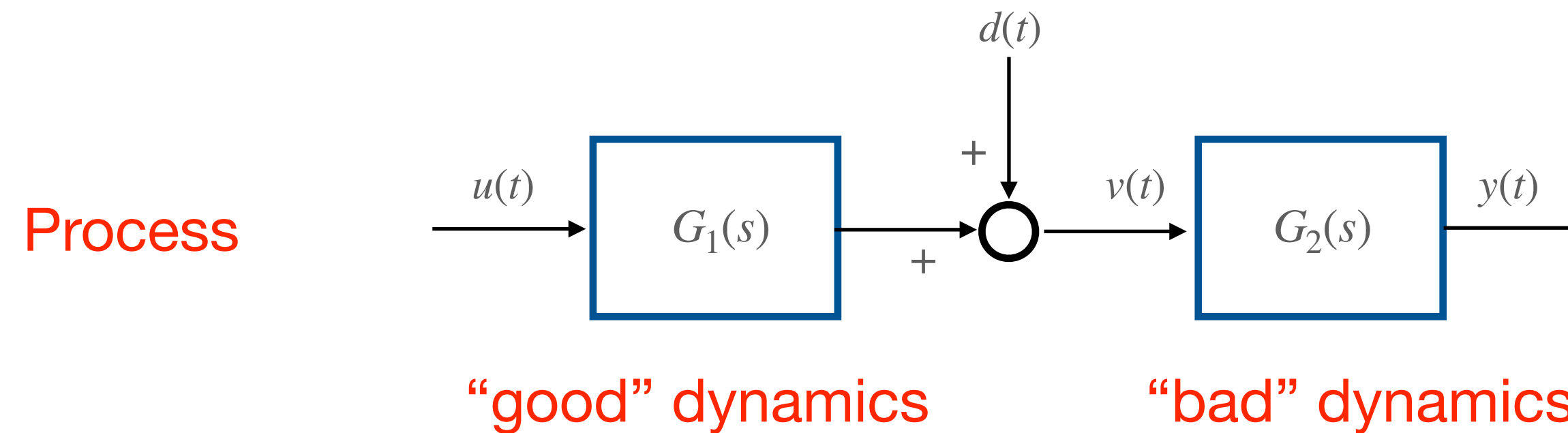
$$G_2(s) = \frac{e^{-4s}}{(1 + s)^2}$$



Exercise: formulate a similar example when $G_2(s)$ is rational with a real zero in $+1$

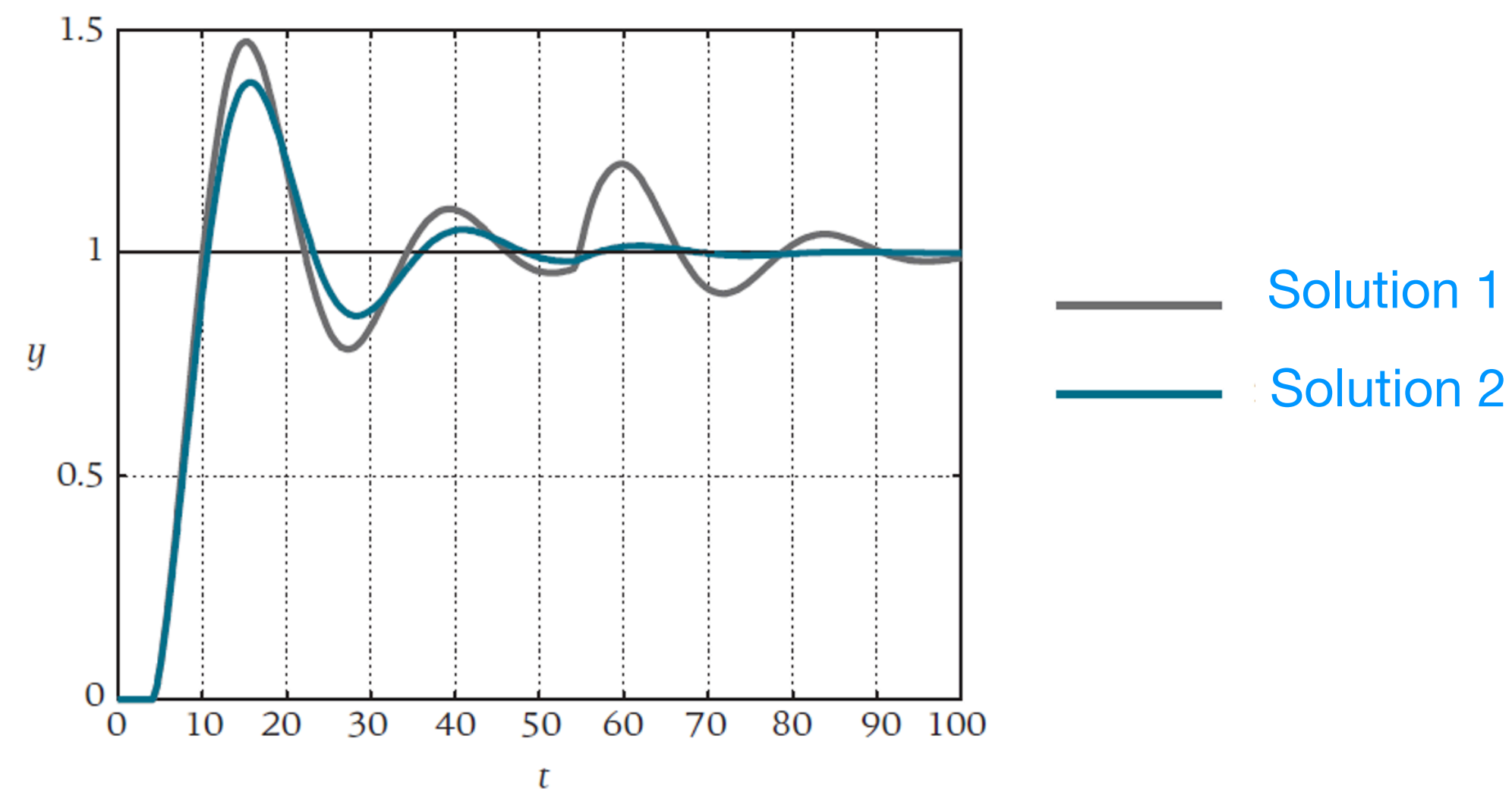
Control Scheme with Decoupling in the Frequency Domain

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$$G_1(s) = \frac{1}{1 + 0.005s}$$

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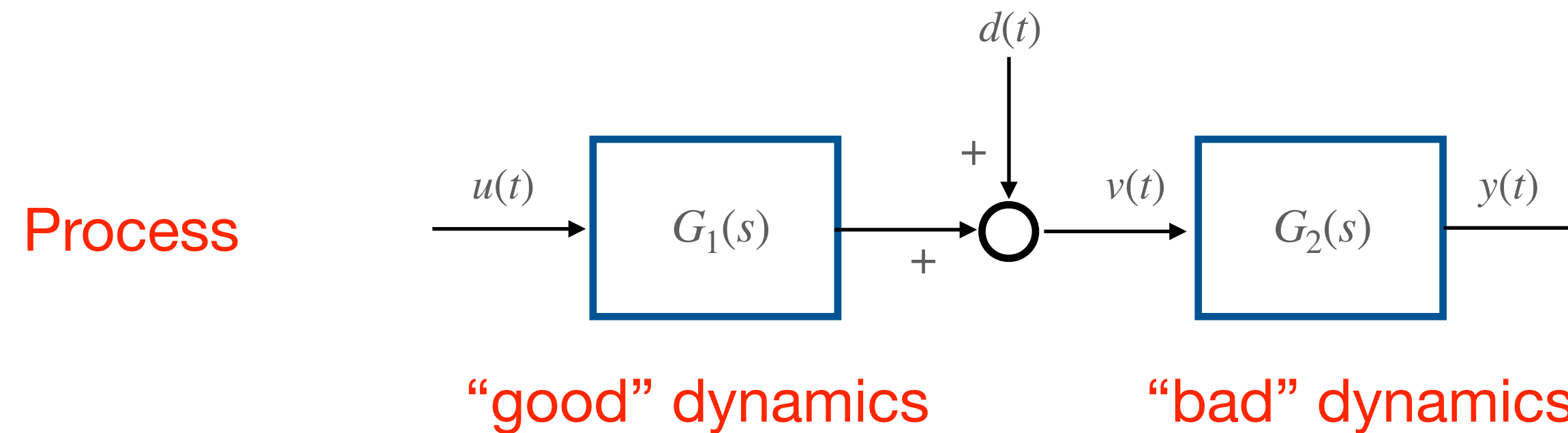


Exercise: formulate a similar example when $G_2(s)$ is rational with a real zero in $+1$

Can you use the unitary feedback control scheme and design a single controller $R(s)$? Try for different $d(t)$

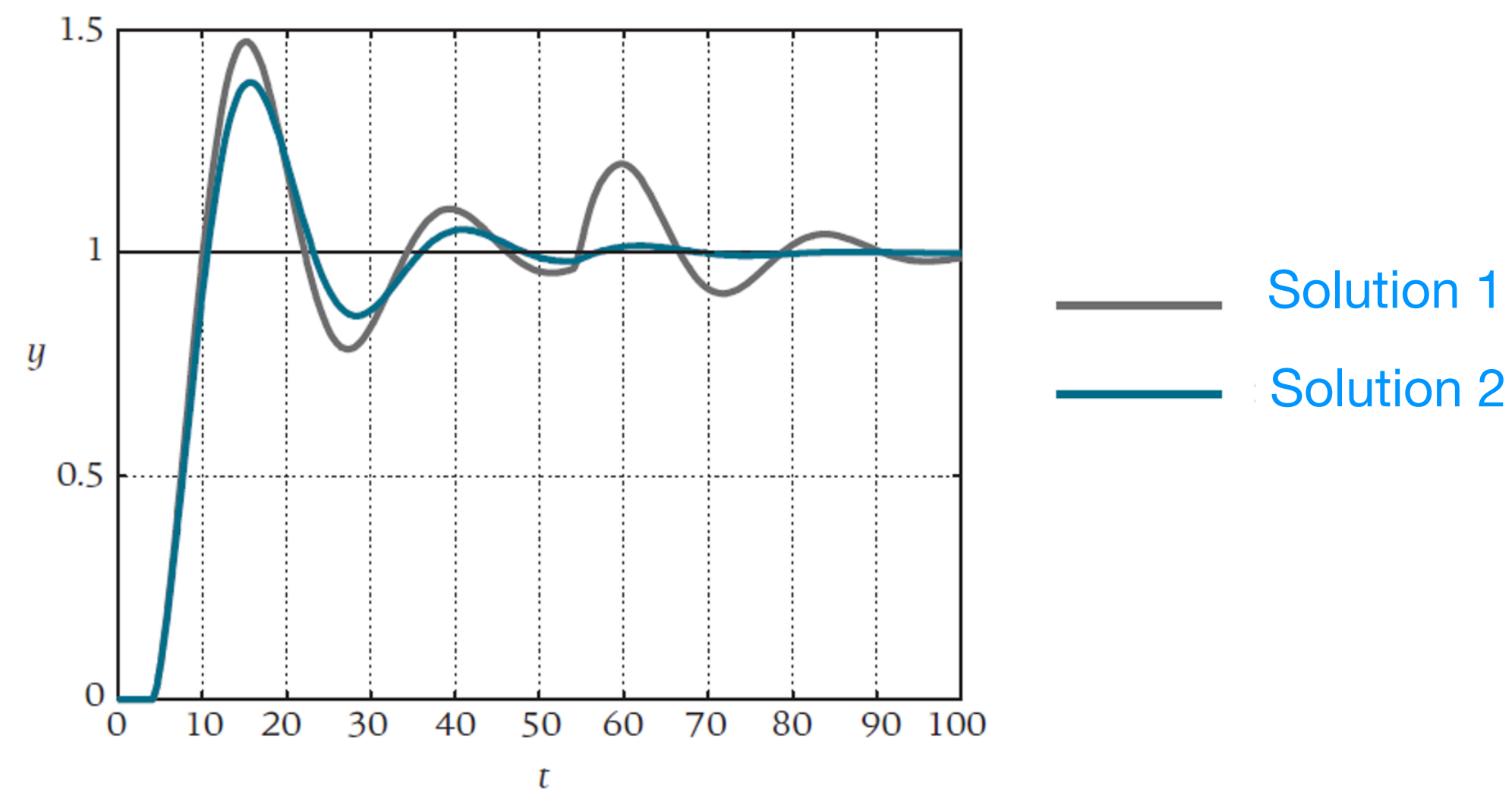
Control Scheme with Decoupling in the Frequency Domain

Example (from the textbook):



$$G_1(s) = \frac{1}{1 + 0.005s}$$

$$G_2(s) = \frac{e^{-4s}}{(1 + s)^2}$$



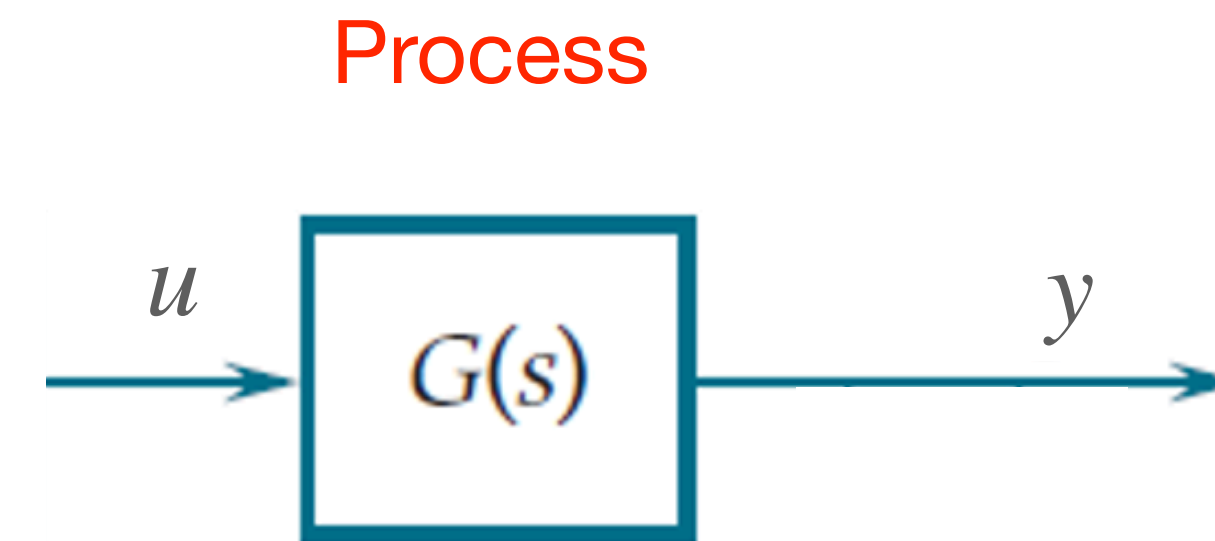
Exercise: formulate a similar example when $G_2(s)$ is rational with a real zero in $+1$

Use decoupling in the freq. dom. Find $R_1(s)$ and $R_2(s)$ and simulate the step response in MATLAB

Control of Open-loop Unstable Systems

Assumption:

- $G(s)$ open-loop unstable



Control of Open-loop Unstable Systems

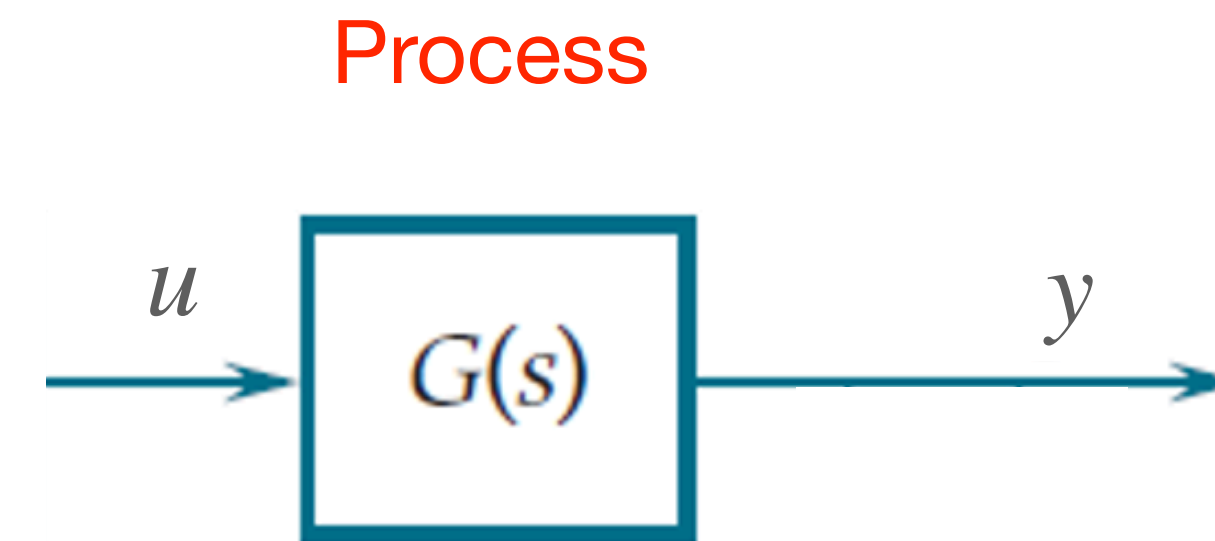
Assumption:

- $G(s)$ open-loop unstable



Bode Criterion: It requires that

- $L(s)$ has no poles with positive real part
- the Bode plot of the magnitude of $L(j\omega)$ crosses the 0 dB axis only once



Control of Open-loop Unstable Systems

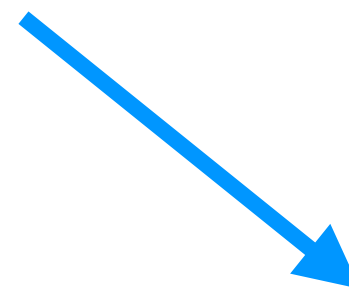
Assumption:

- $G(s)$ open-loop unstable

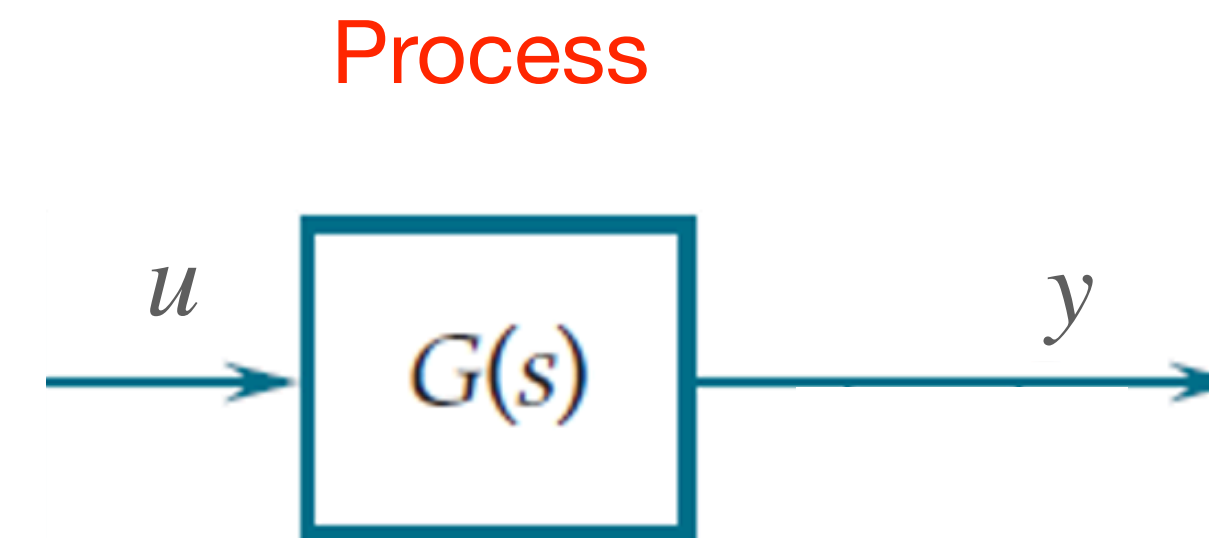


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It cannot be applied in this case!



Control of Open-loop Unstable Systems

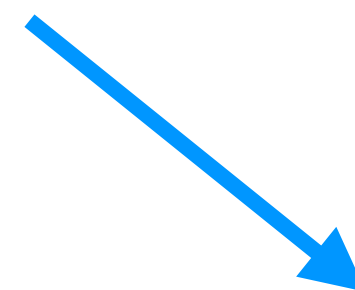
Assumption:

- $G(s)$ open-loop unstable

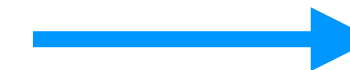


Bode Criterion: It requires that

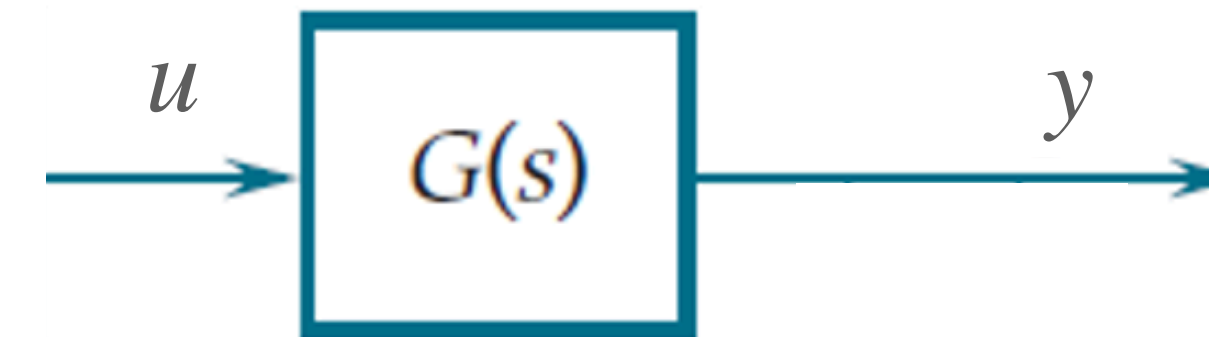
- $L(s)$ has no poles with positive real part
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It cannot be applied in this case!



Process



Alternatives:

- Nyquist Criterion
- Poles Assignment
- Root Locus

Control of Open-loop Unstable Systems

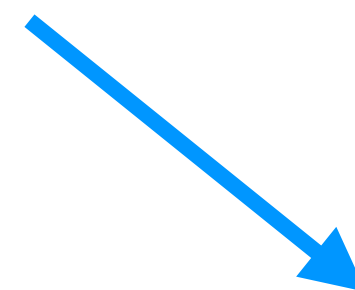
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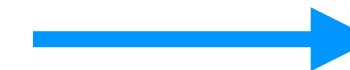


Bode Criterion: It requires that

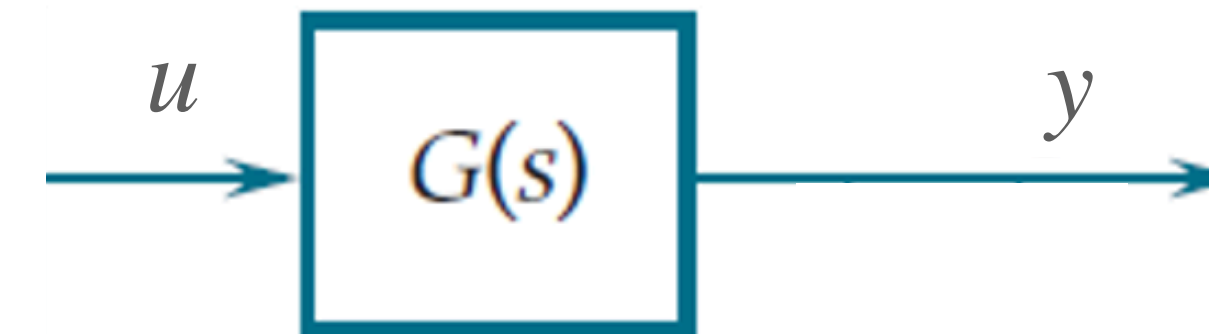
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It cannot be applied in this case!



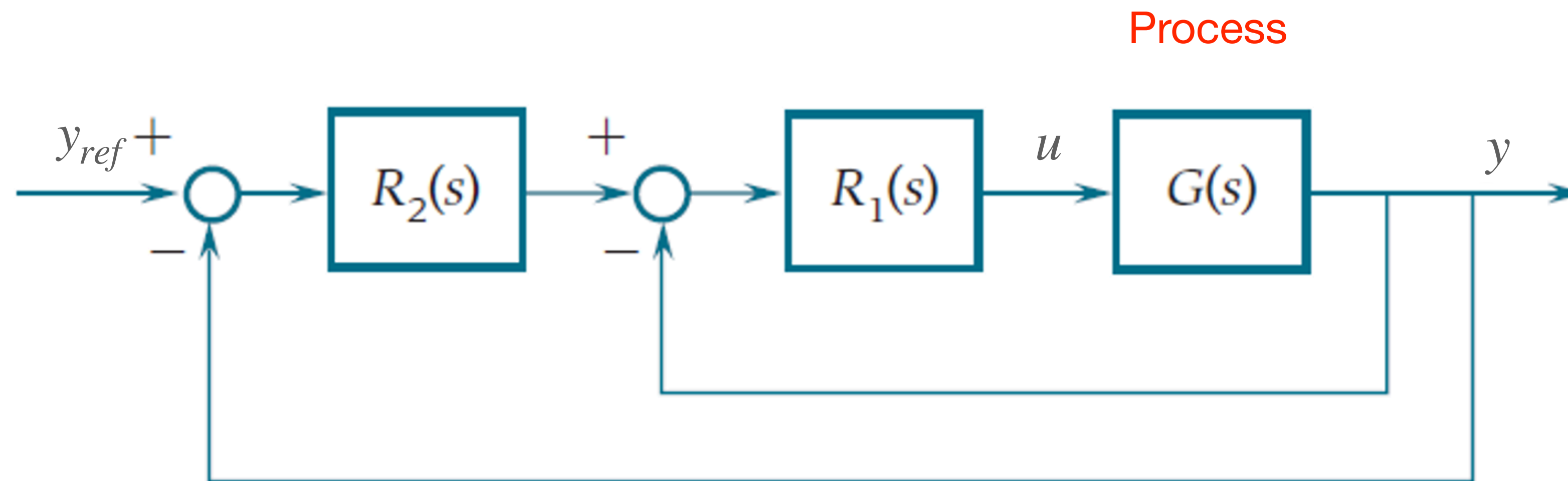
Process



Alternatives:

- Nyquist Criterion
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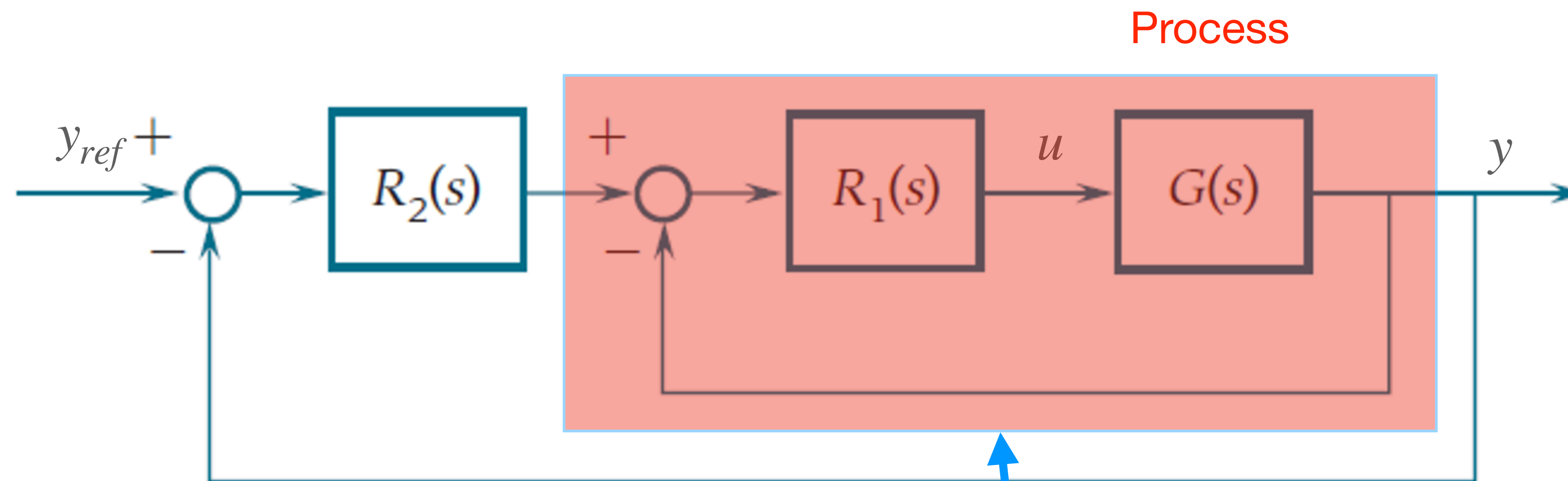
Control of Open-loop Unstable Systems



Alternatives:

- Nyquist Criterion
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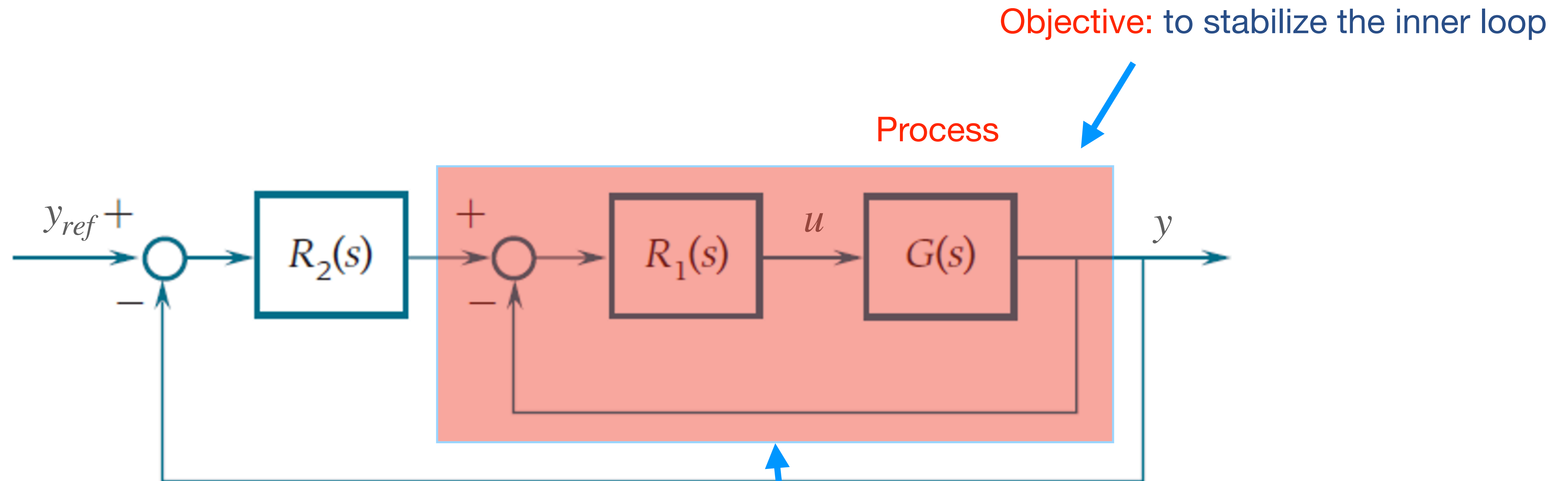
Control of Open-loop Unstable Systems



Alternatives:

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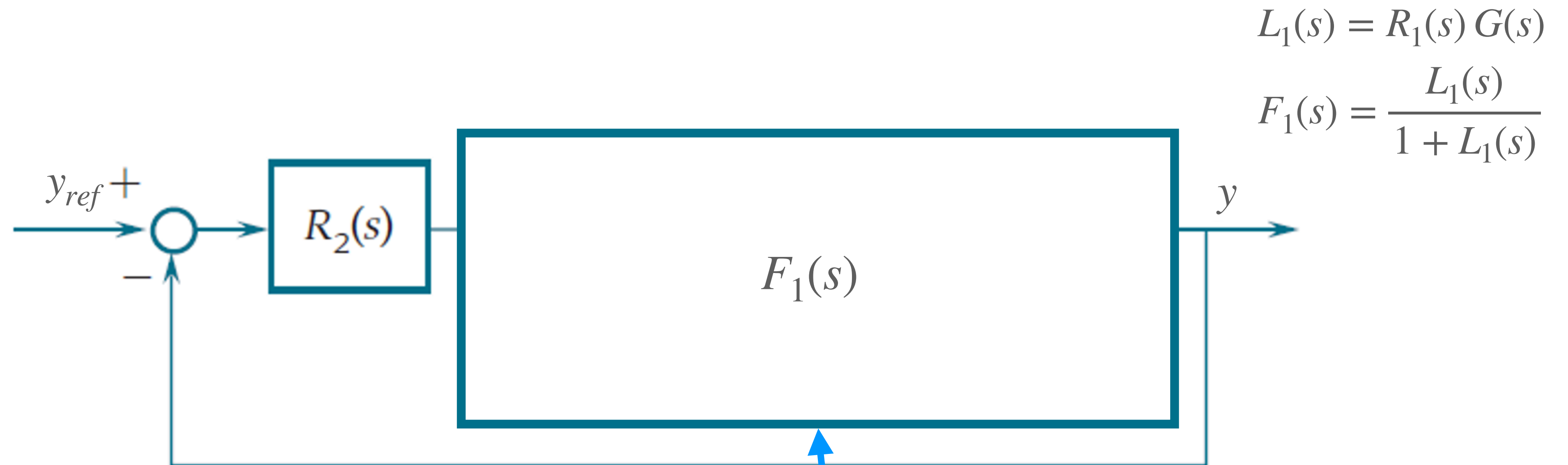
Control of Open-loop Unstable Systems



Alternatives:

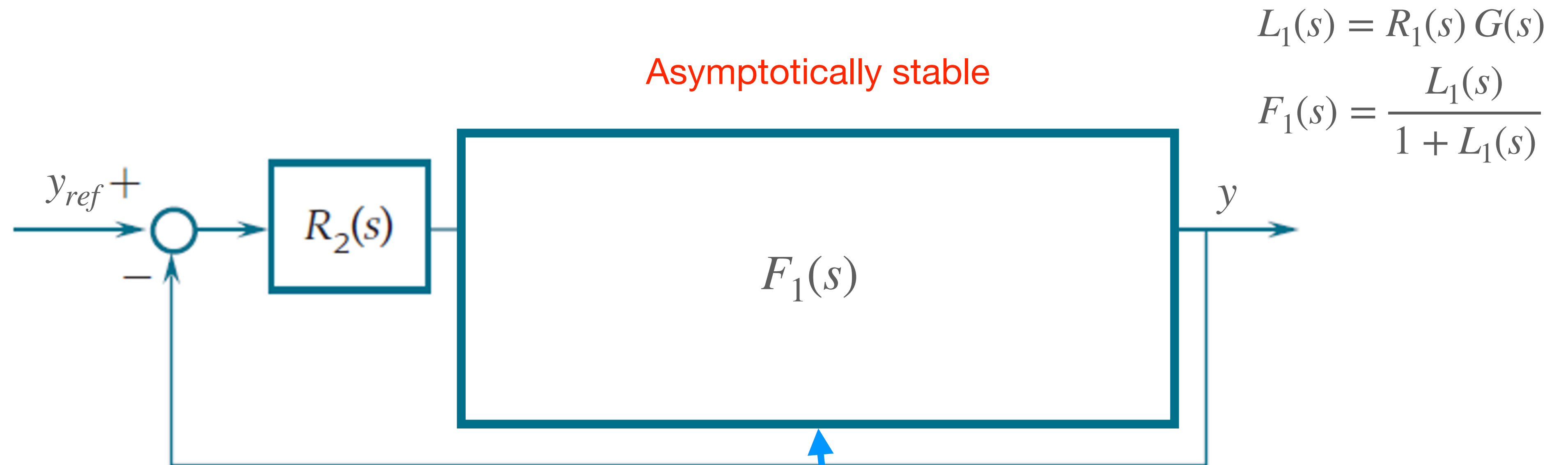
- Nyquist Criterion
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Control of Open-loop Unstable Systems

**Alternatives:**

- Nyquist Criterion
- Poles Assignment
- Root Locus

Control of Open-loop Unstable Systems



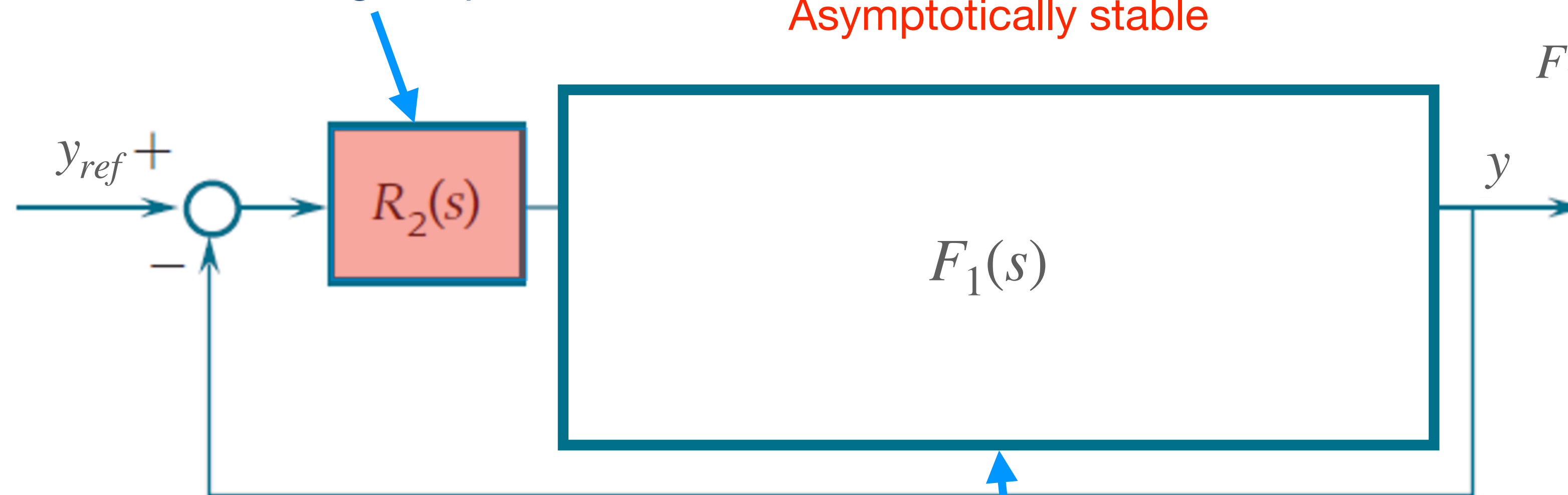
Alternatives:

- Nyquist Criterion
- Poles Assignment
- Root Locus

Control of Open-loop Unstable Systems

Objective: to stabilize the outer loop & satisfy the design requirements

Asymptotically stable



$$L_1(s) = R_1(s) G(s)$$

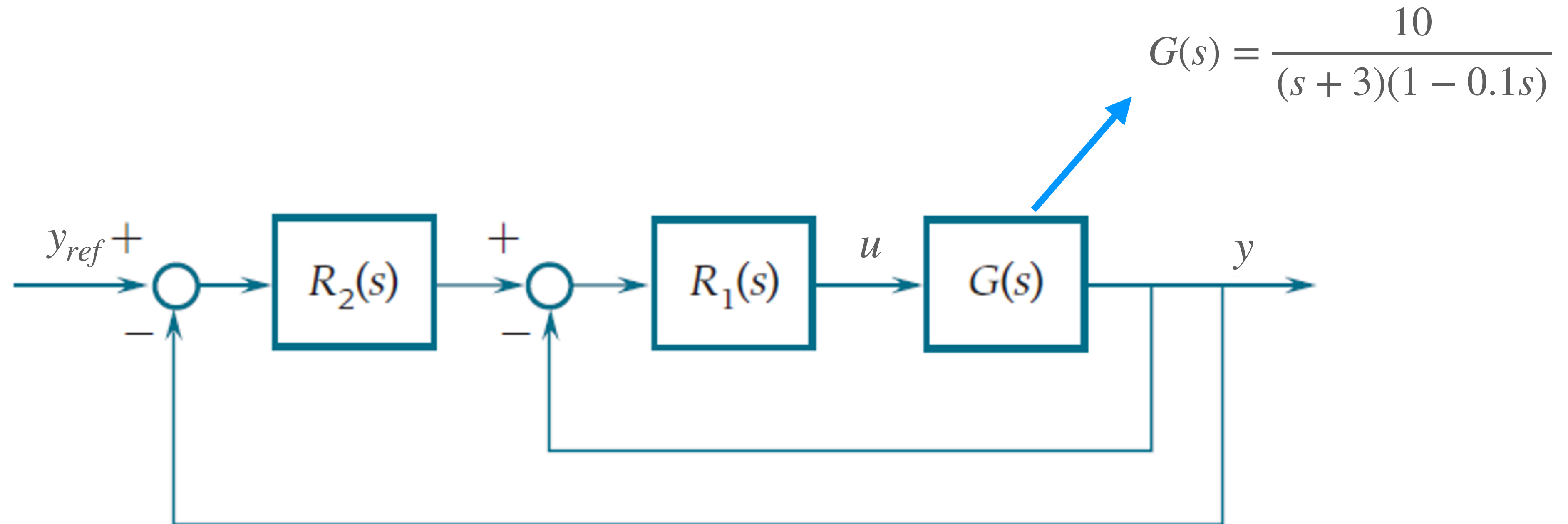
$$F_1(s) = \frac{L_1(s)}{1 + L_1(s)}$$

Alternatives:

- Nyquist Criterion
- Poles Assignment
- Root Locus

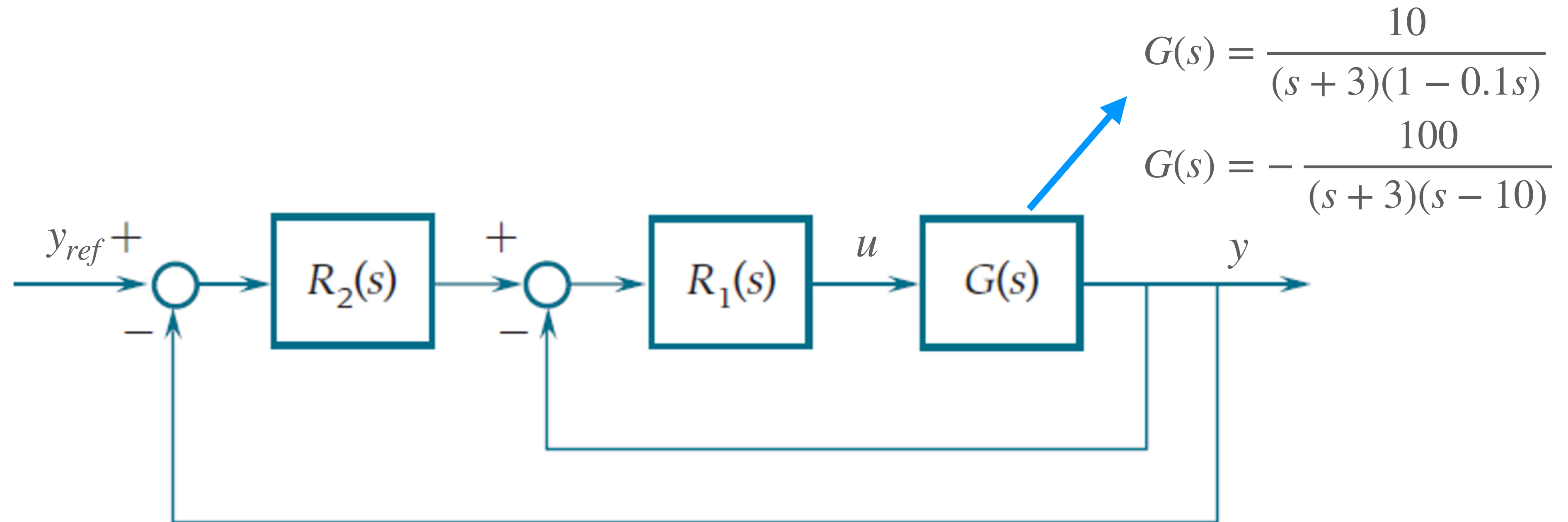
Control of Open-loop Unstable Systems

Example:



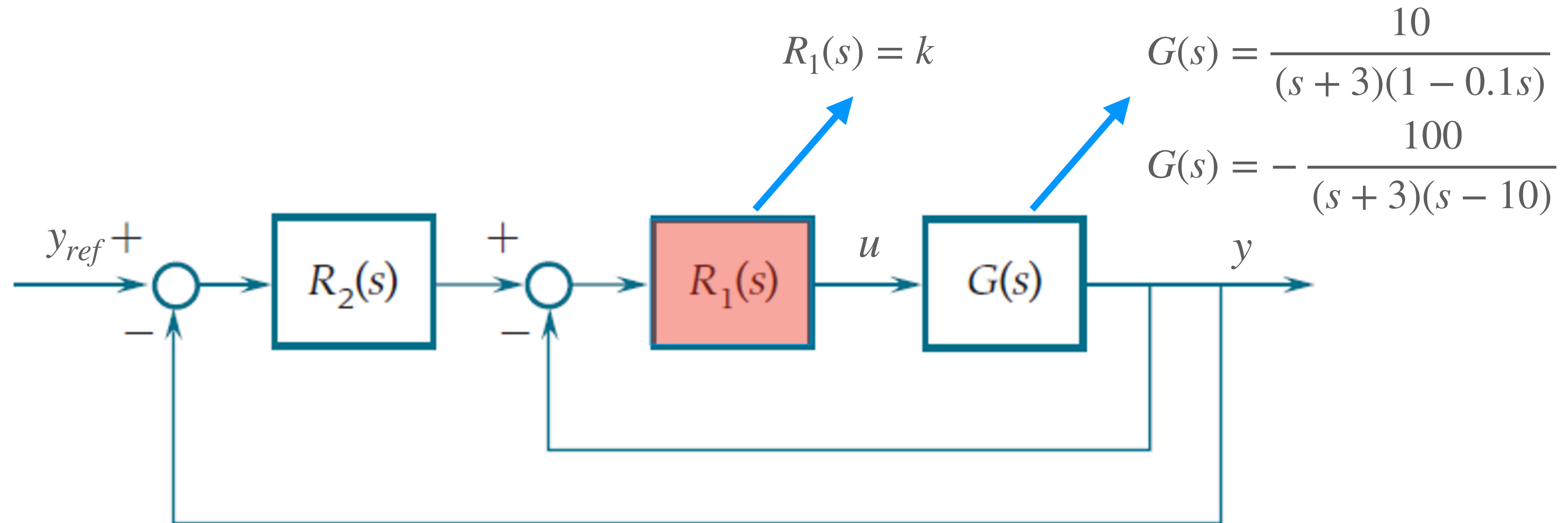
Control of Open-loop Unstable Systems

Example:



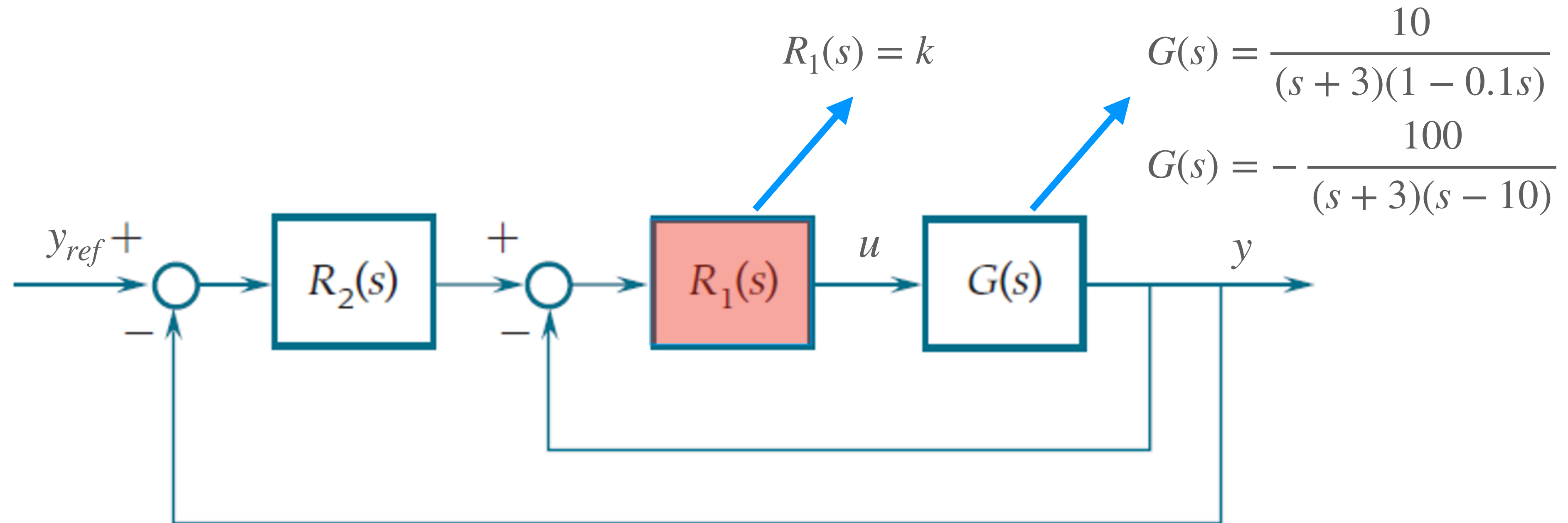
Control of Open-loop Unstable Systems

Example:



Control of Open-loop Unstable Systems

Example:

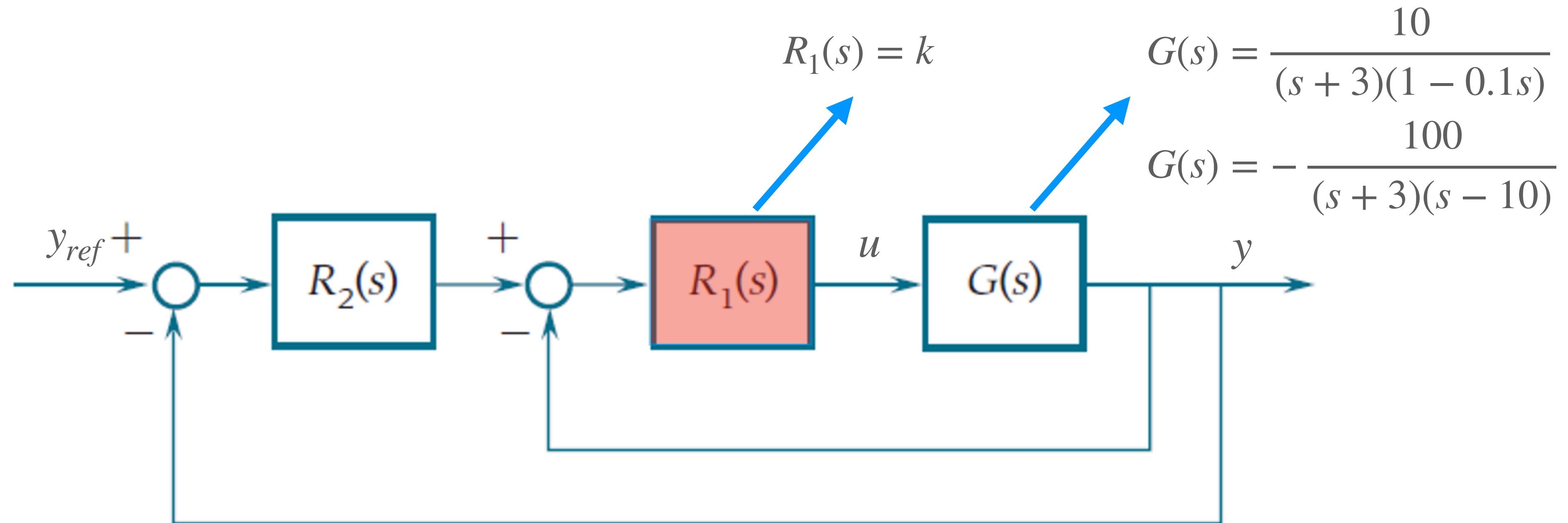


$$L_1(s) = R_1(s) G(s) = -\frac{100k}{(s + 3)(s - 10)}$$

$$F_1(s) = \frac{L_1(s)}{1 + L_1(s)}$$

Control of Open-loop Unstable Systems

Example:



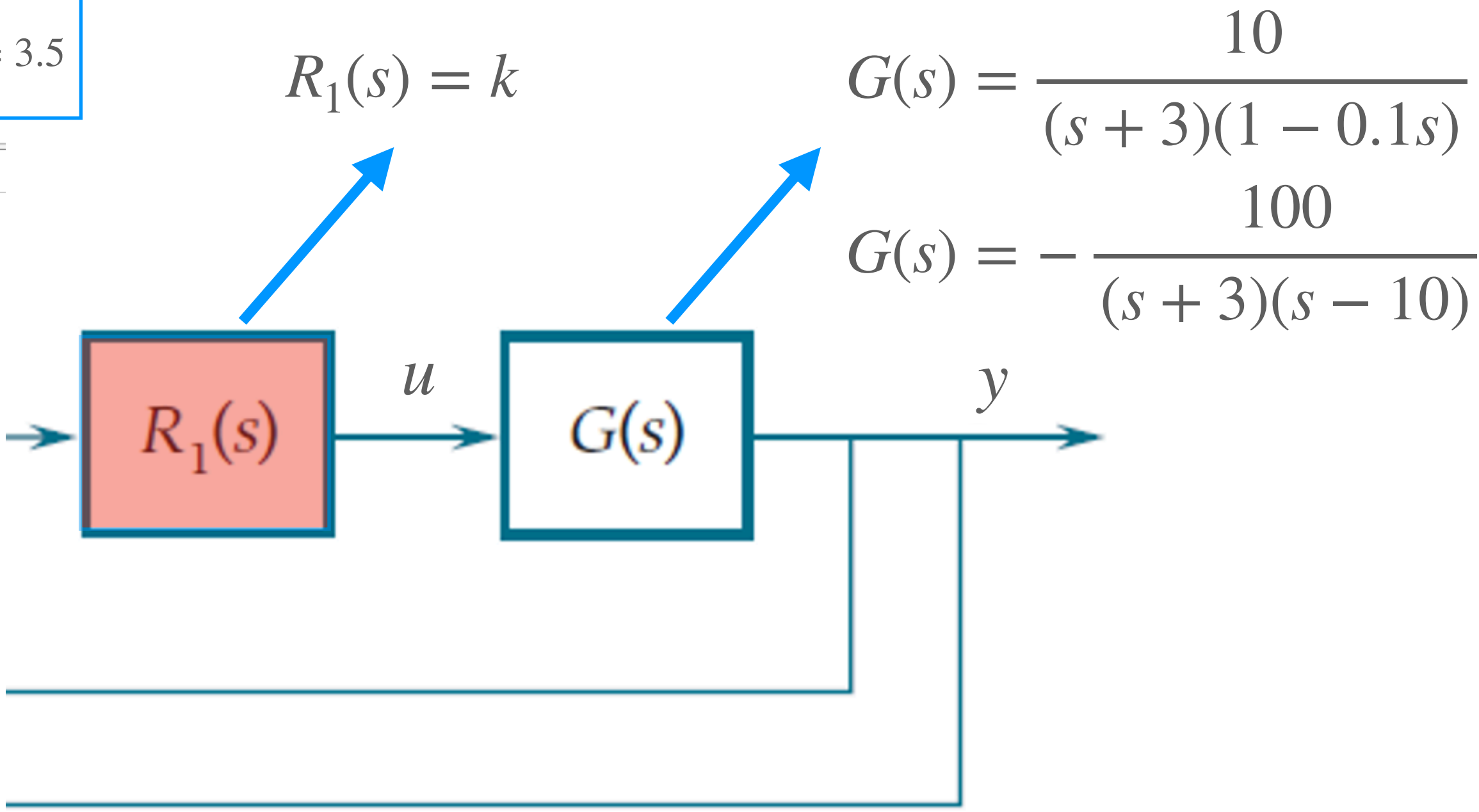
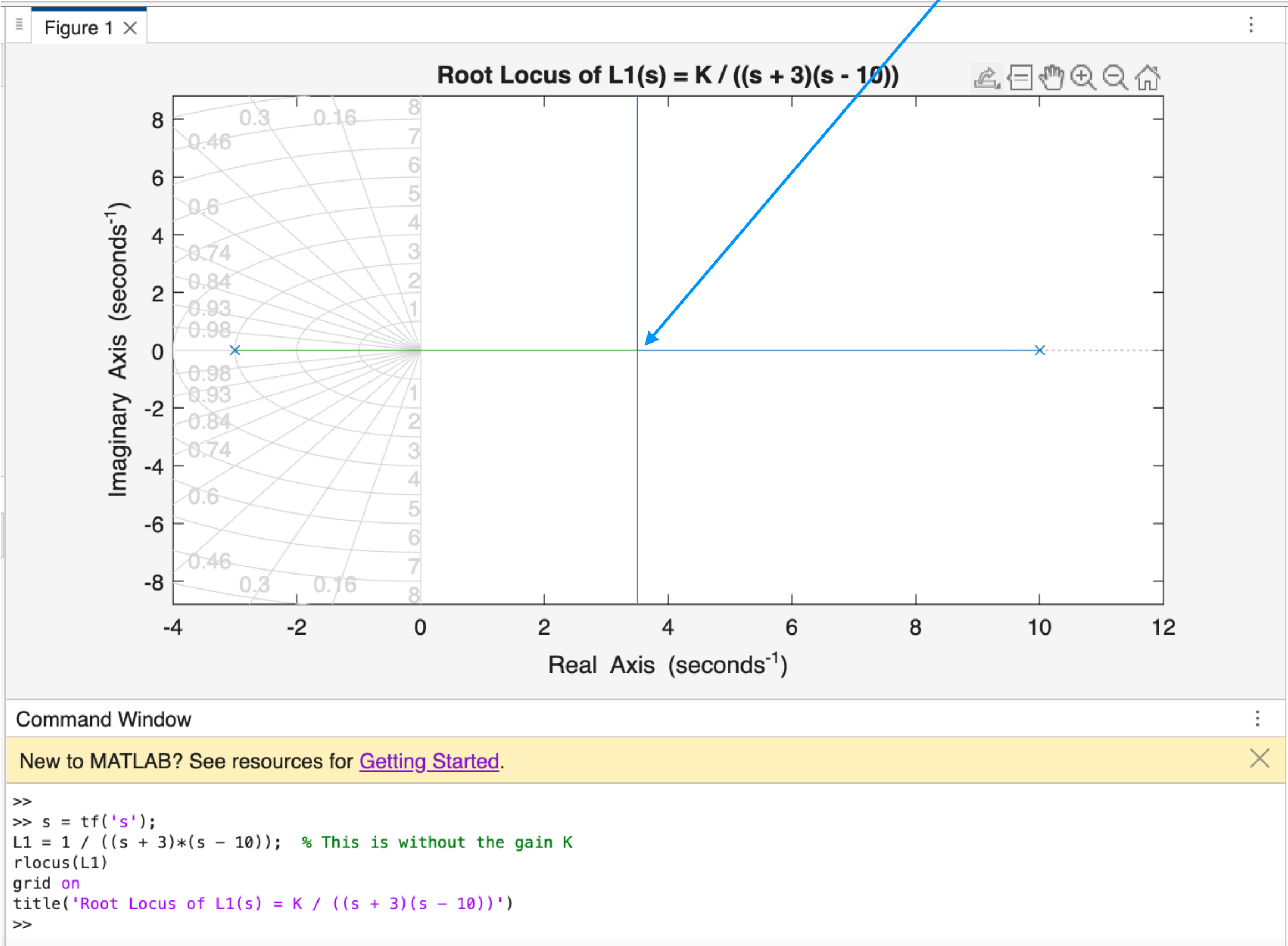
$$L_1(s) = R_1(s) G(s) = -\frac{100k}{(s+3)(s-10)} = \frac{K}{(s+3)(s-10)}$$

$$F_1(s) = \frac{L_1(s)}{1 + L_1(s)}$$

Control of Open-loop Unstable Systems

Example:

$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m}$$
$$\sigma_a = \frac{-3 + 10}{2 - 0} = \frac{7}{2} = 3.5$$



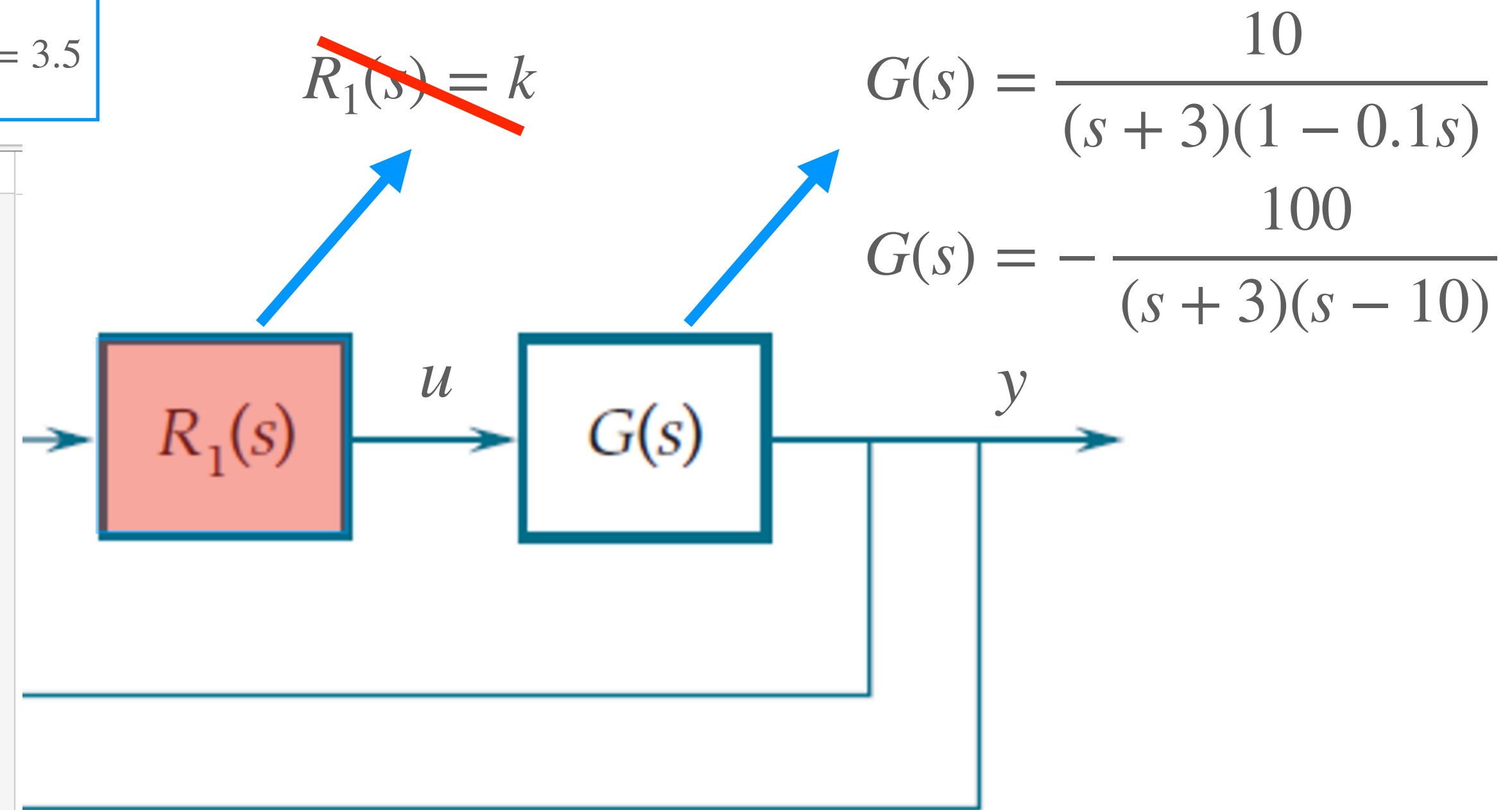
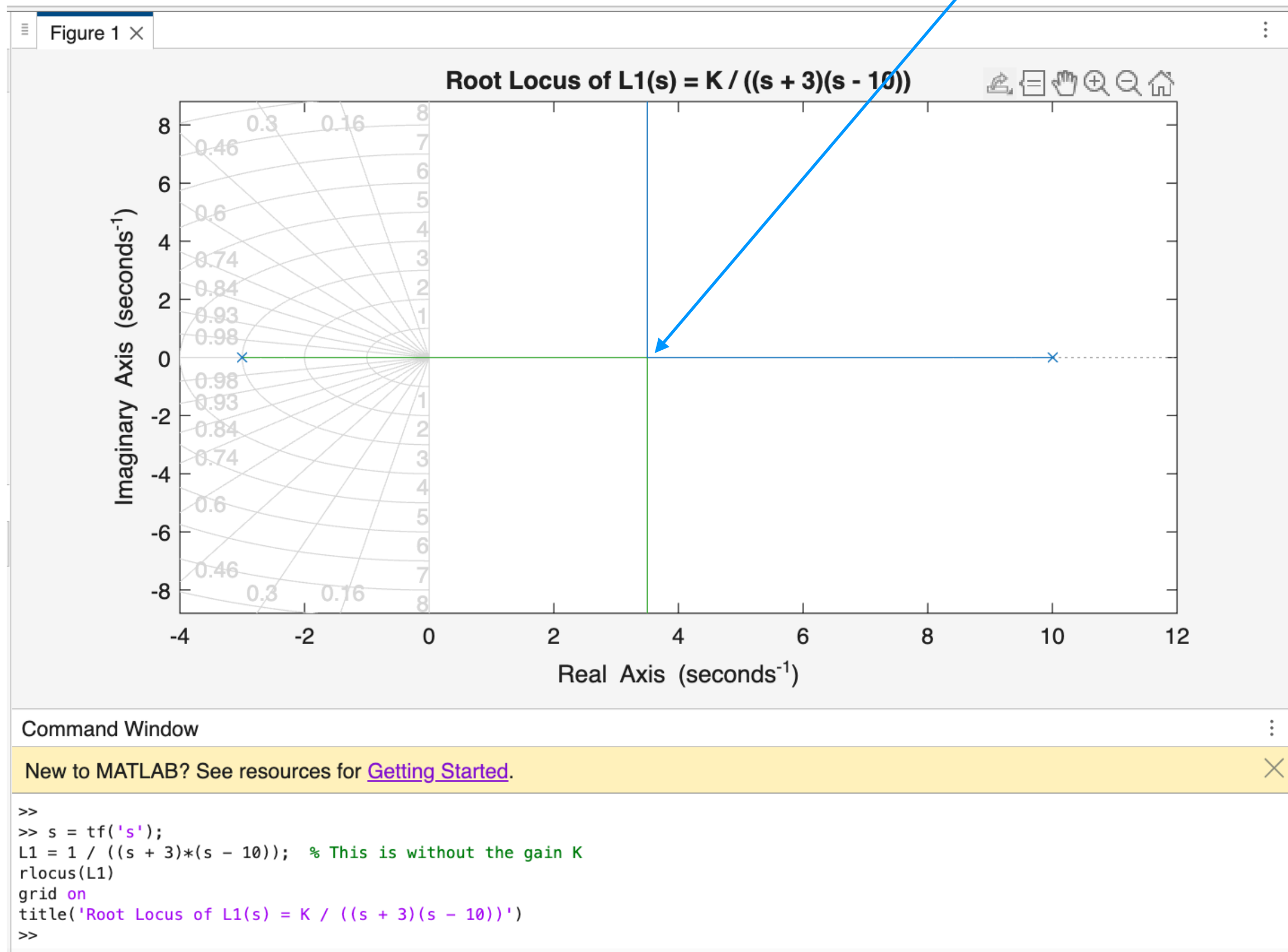
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Control of Open-loop Unstable Systems

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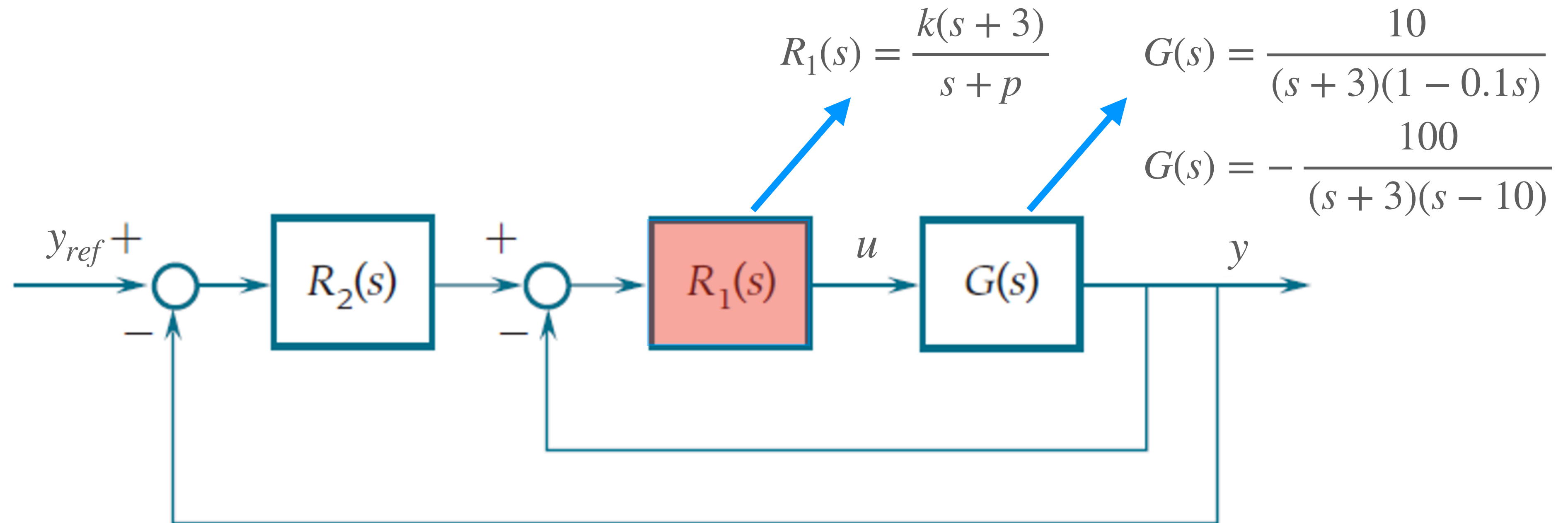


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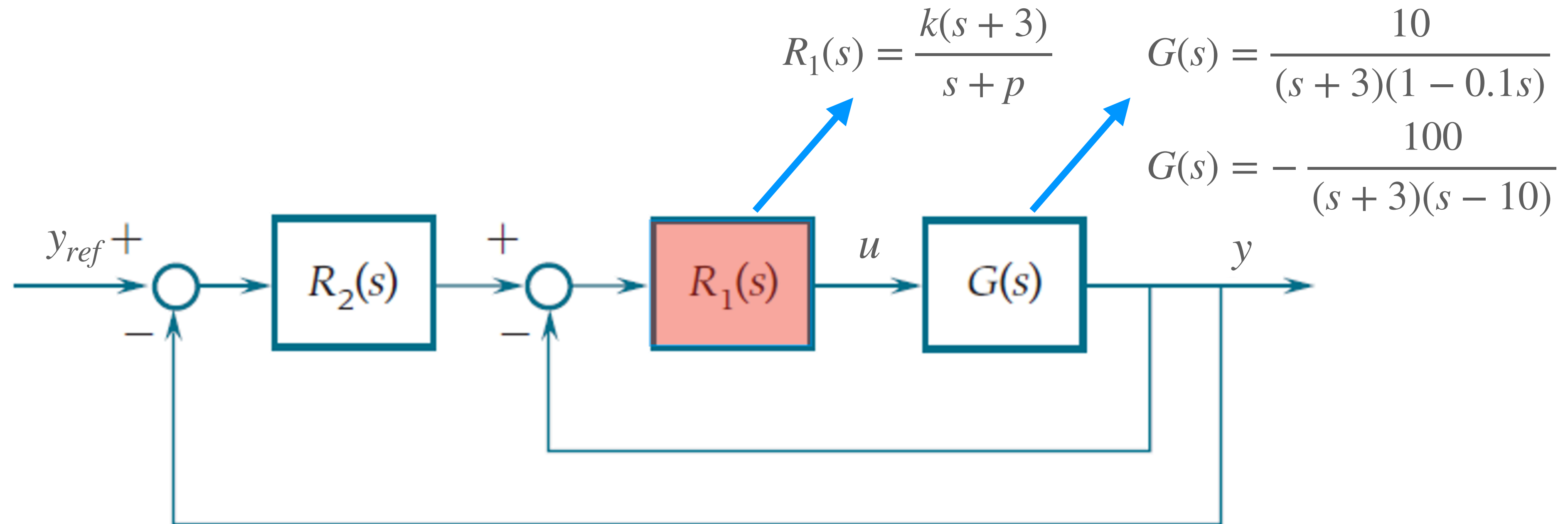
Control of Open-loop Unstable Systems

Example:



Control of Open-loop Unstable Systems

Example:

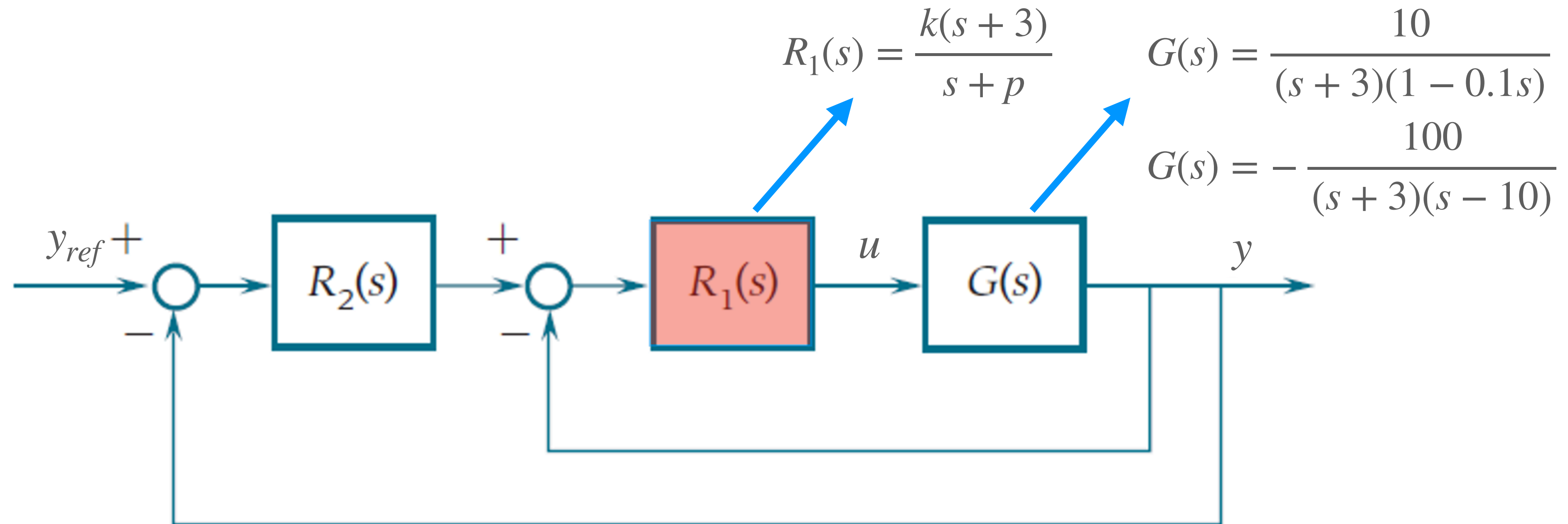


$$L_1(s) = R_1(s) G(s) = -\frac{100k}{(s+p)(s-10)} = \frac{K}{(s+p)(s-10)}$$

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Control of Open-loop Unstable Systems

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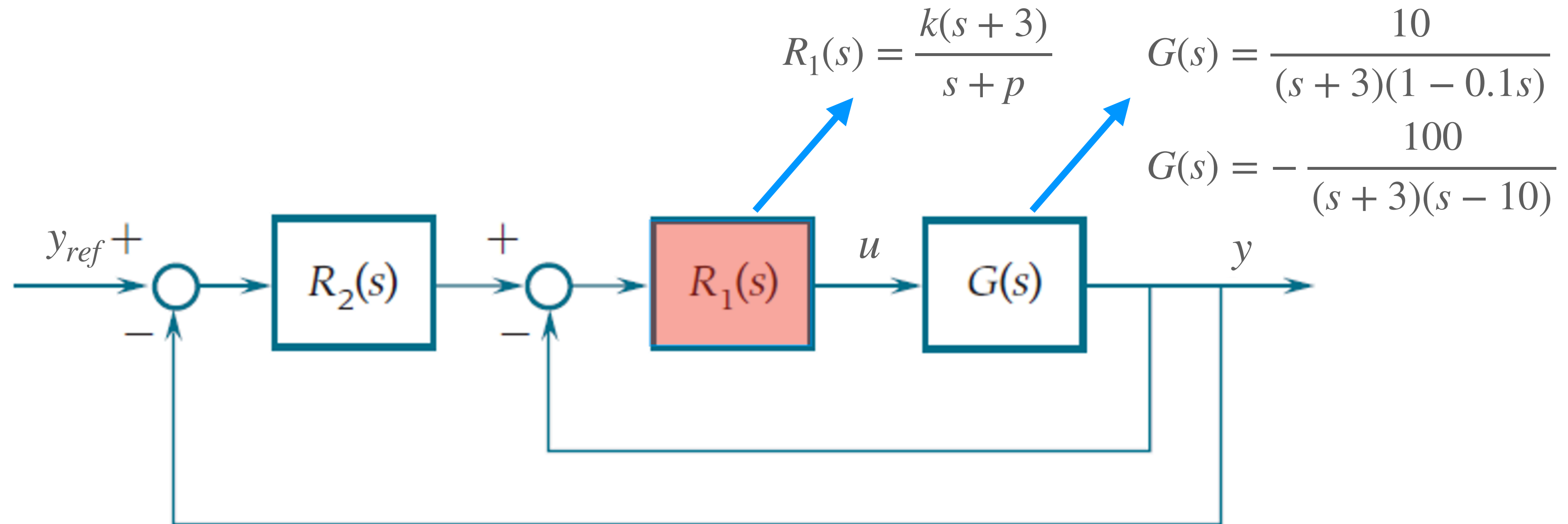
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?

Control of Open-loop Unstable Systems

Example:



$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m} \quad \sigma_a = \frac{(-p) + 10}{2 - 0} = \frac{10 - p}{2} =$$

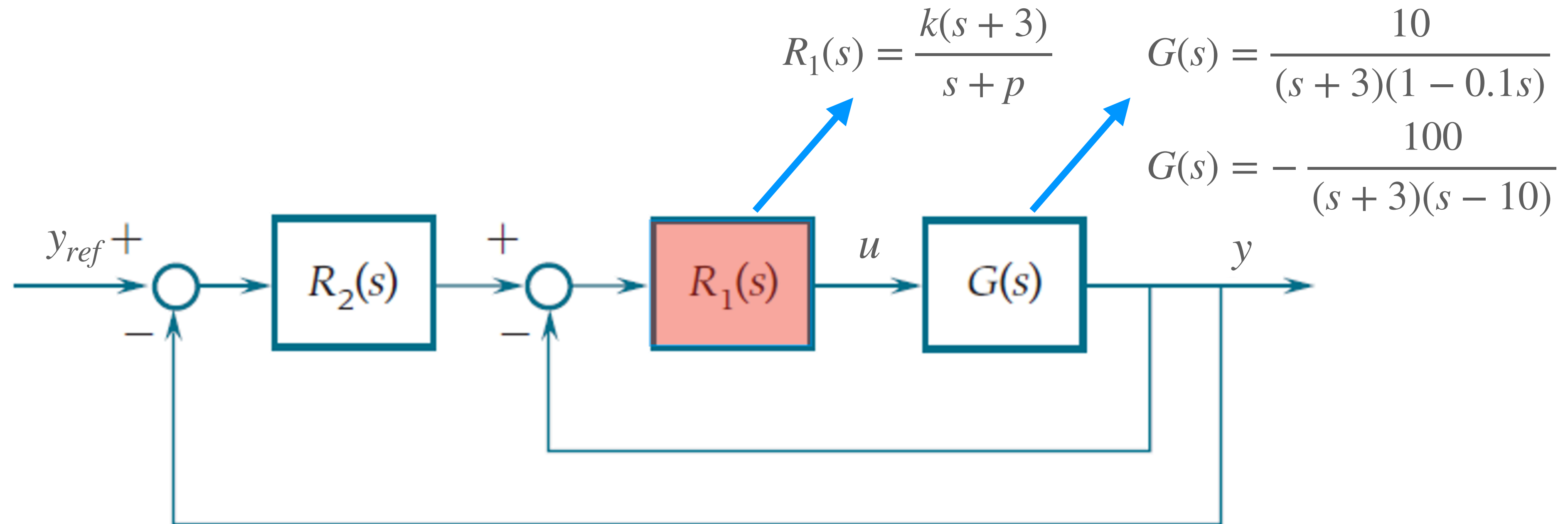
$$L_1(s) = R_1(s) G(s) = -\frac{100k}{(s+p)(s-10)} = \frac{K}{(s+p)(s-10)}$$

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?

Control of Open-loop Unstable Systems

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$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m} \quad \sigma_a = \frac{(-p) + 10}{2 - 0} = \frac{10 - p}{2} = -10$$

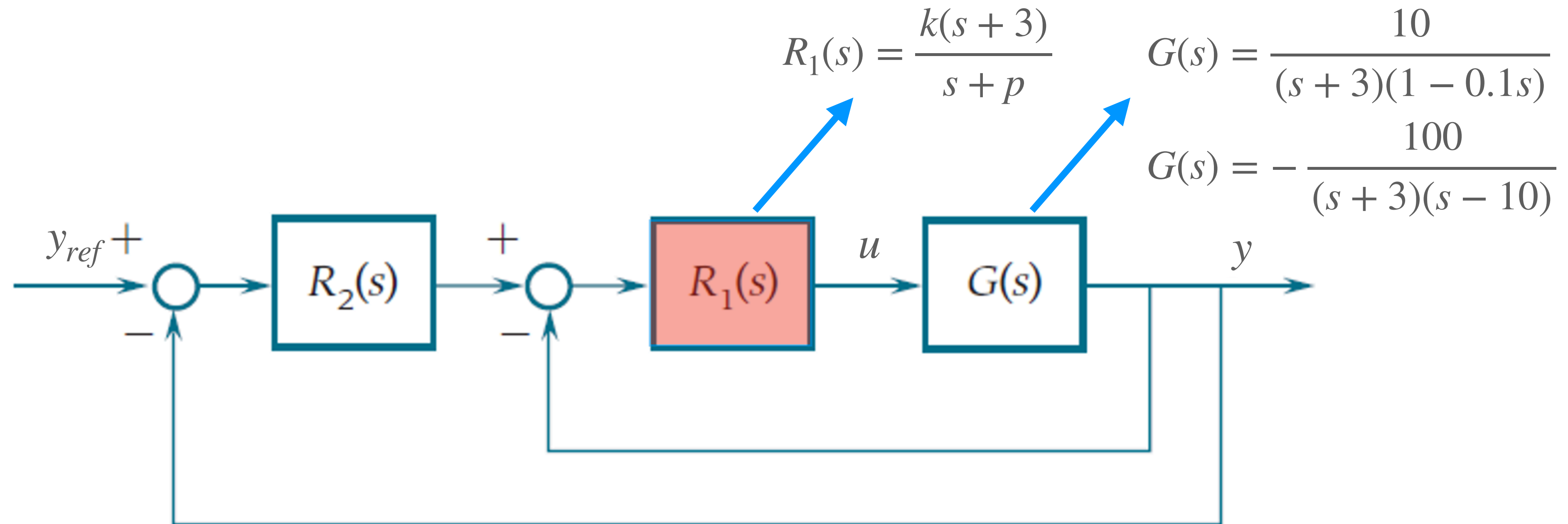
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Control of Open-loop Unstable Systems

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$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m} \quad \sigma_a = \frac{(-p) + 10}{2 - 0} = \frac{10 - p}{2} = -10$$

$$p = 30$$

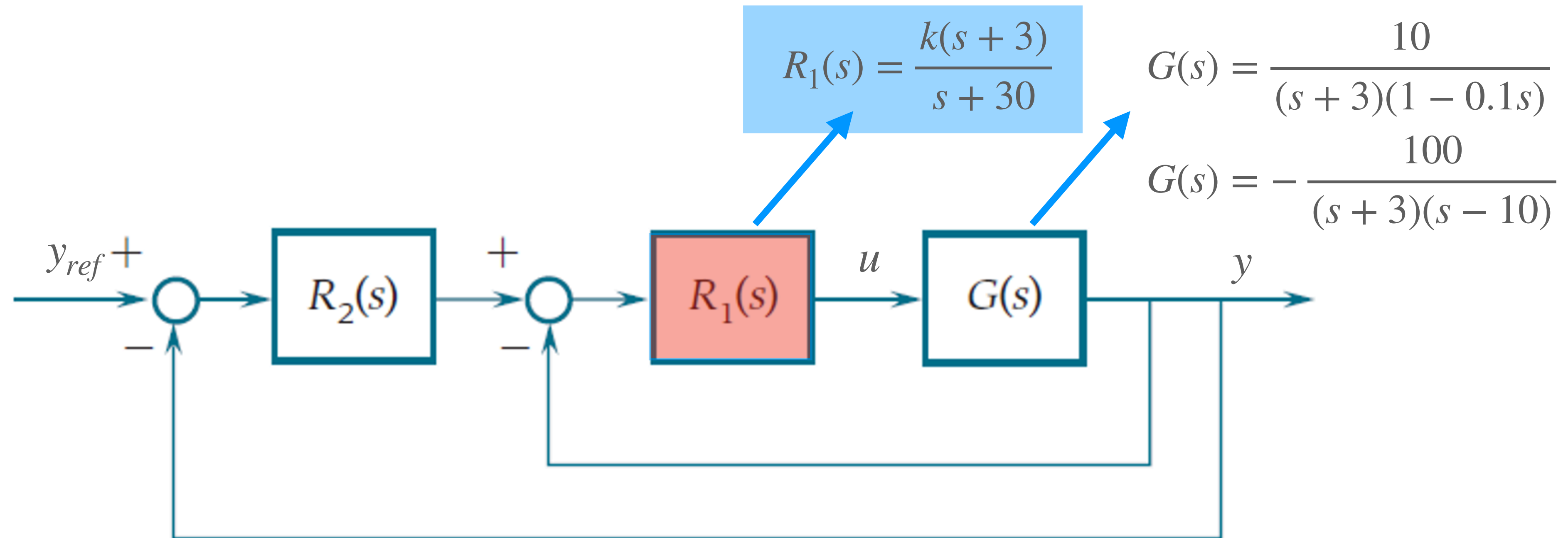
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Control of Open-loop Unstable Systems

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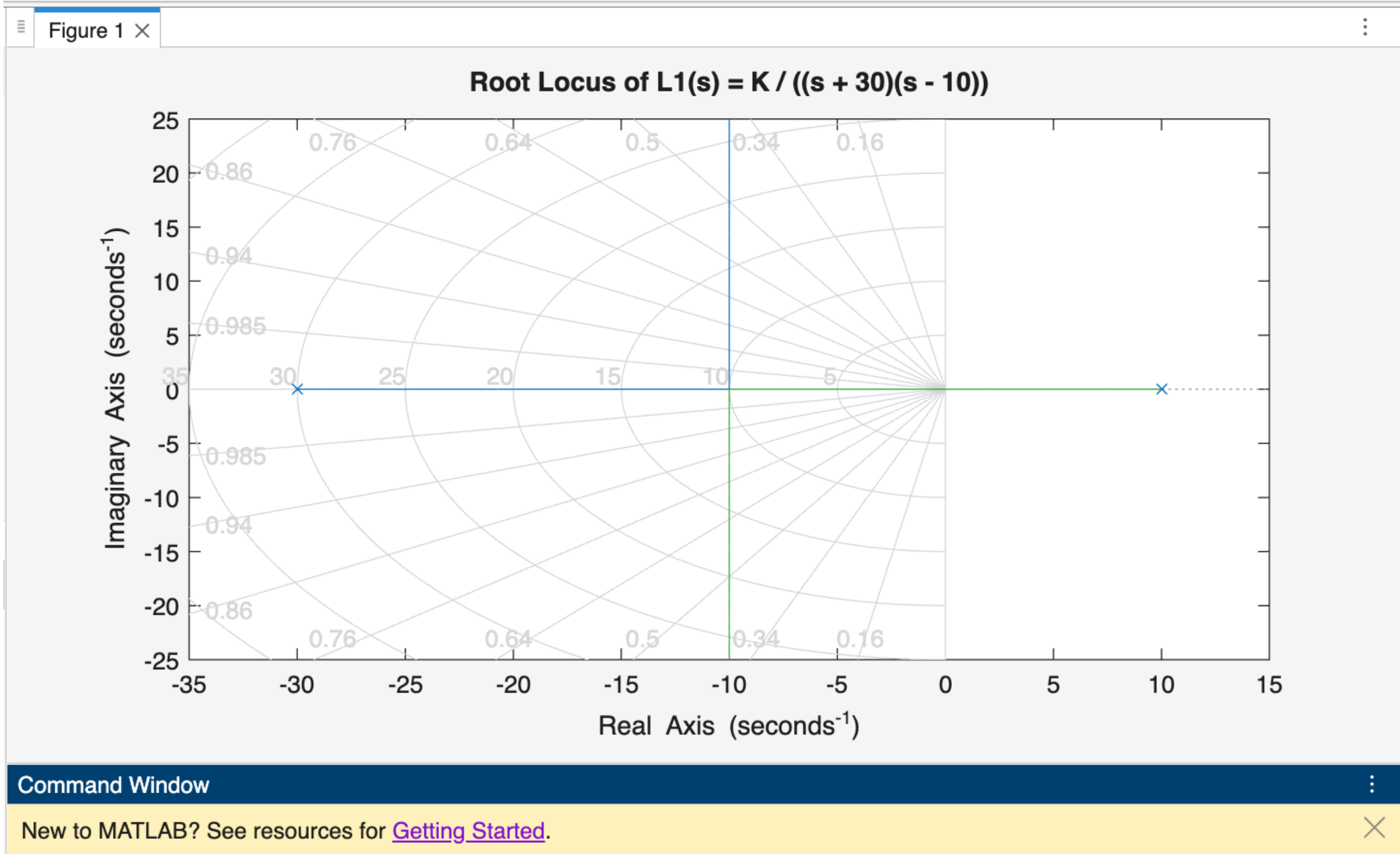
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?

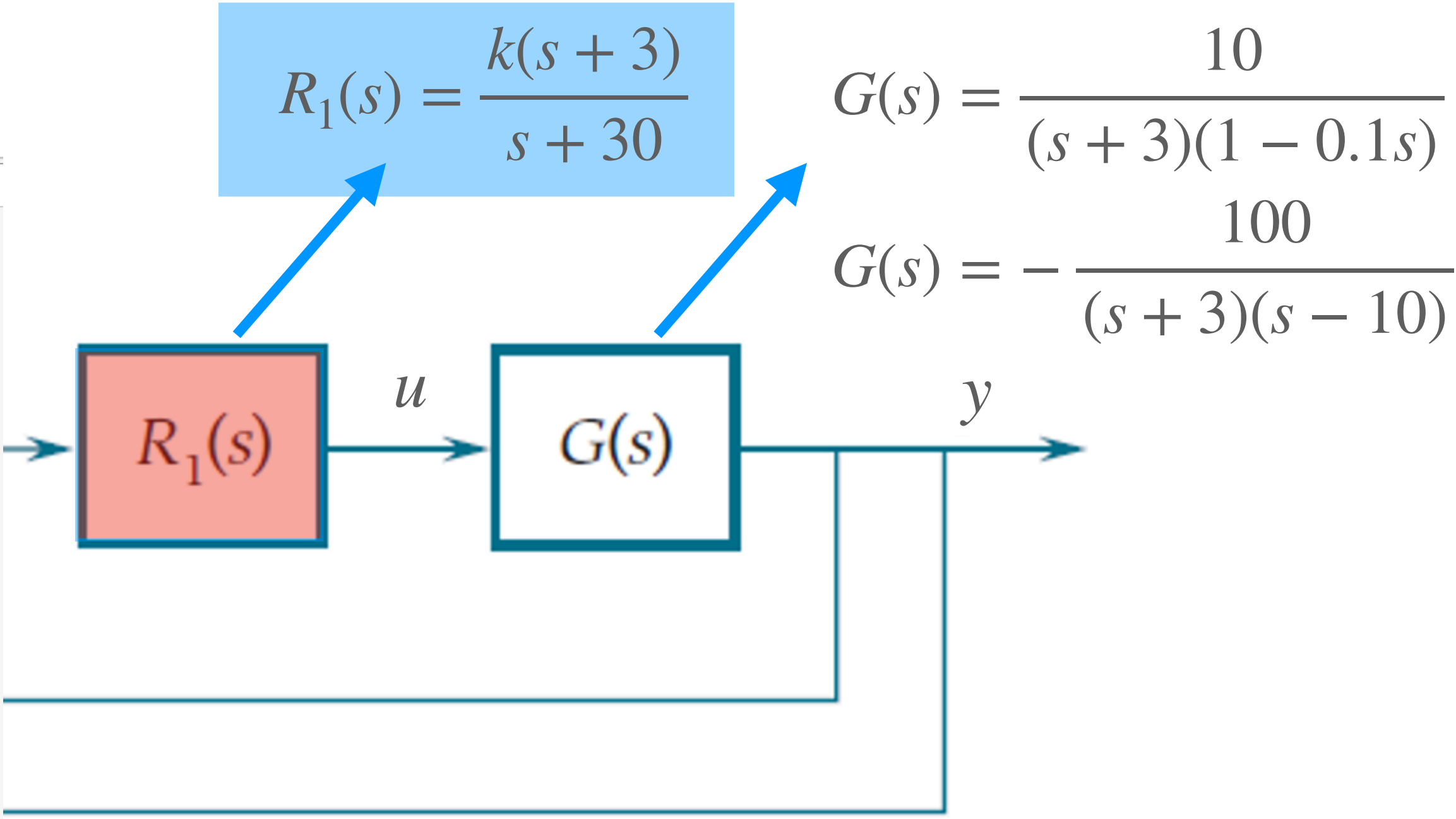
$$F_1(s) = \frac{L_1(s)}{1 + L_1(s)}$$

Control of Open-loop Unstable Systems

Example:



```
>> s = tf('s');  
L1 = 1 / ((s + 30)*(s - 10)); % This is without the gain K  
rlocus(L1)  
grid on  
title('Root Locus of L1(s) = K / ((s + 30)(s - 10))')  
>> |
```



$$L_1(s) = R_1(s) G(s) = -\frac{100k}{(s + p)(s - 10)} = \frac{K}{(s + p)(s - 10)}$$

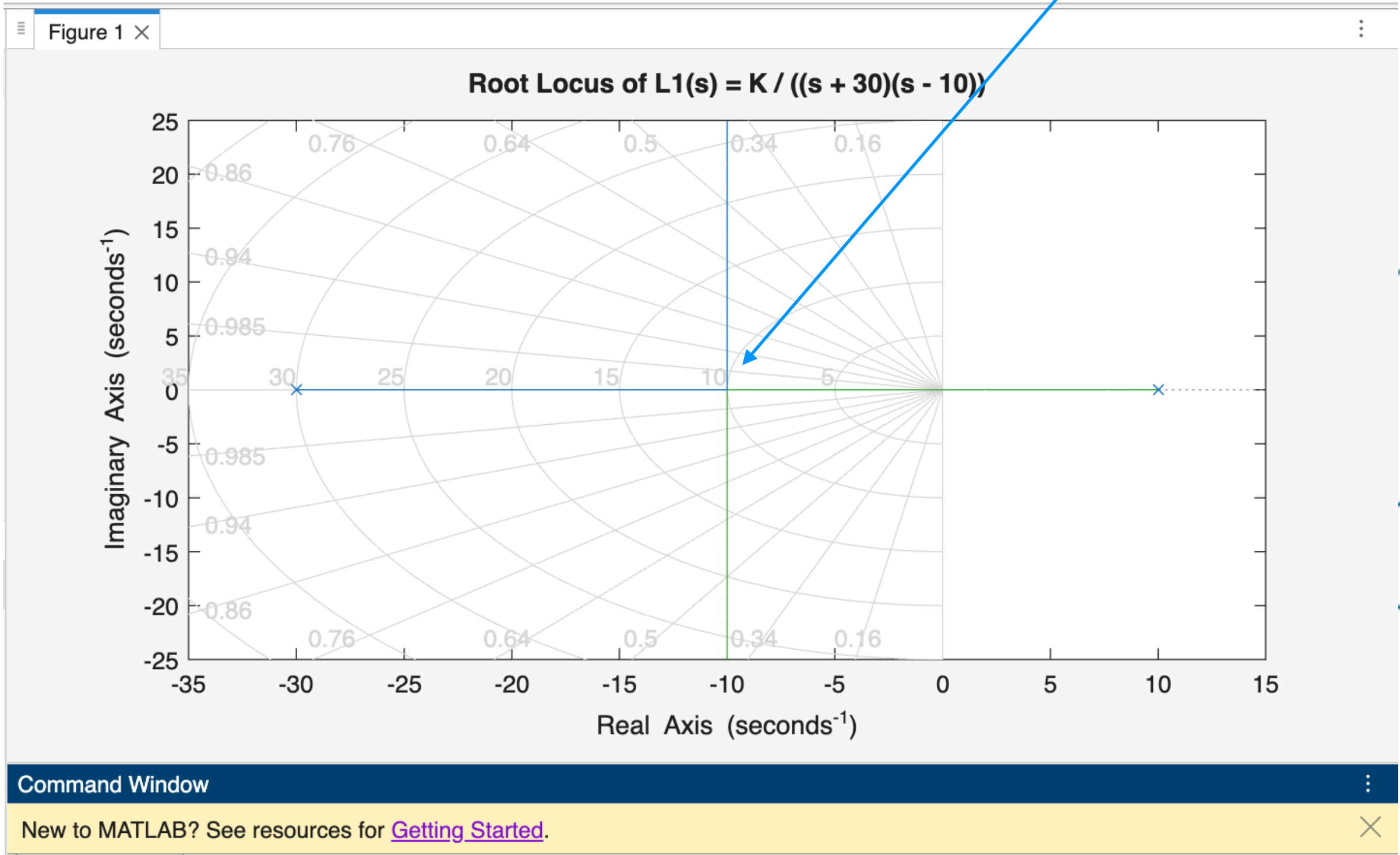
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?

Control of Open-loop Unstable Systems

Example:

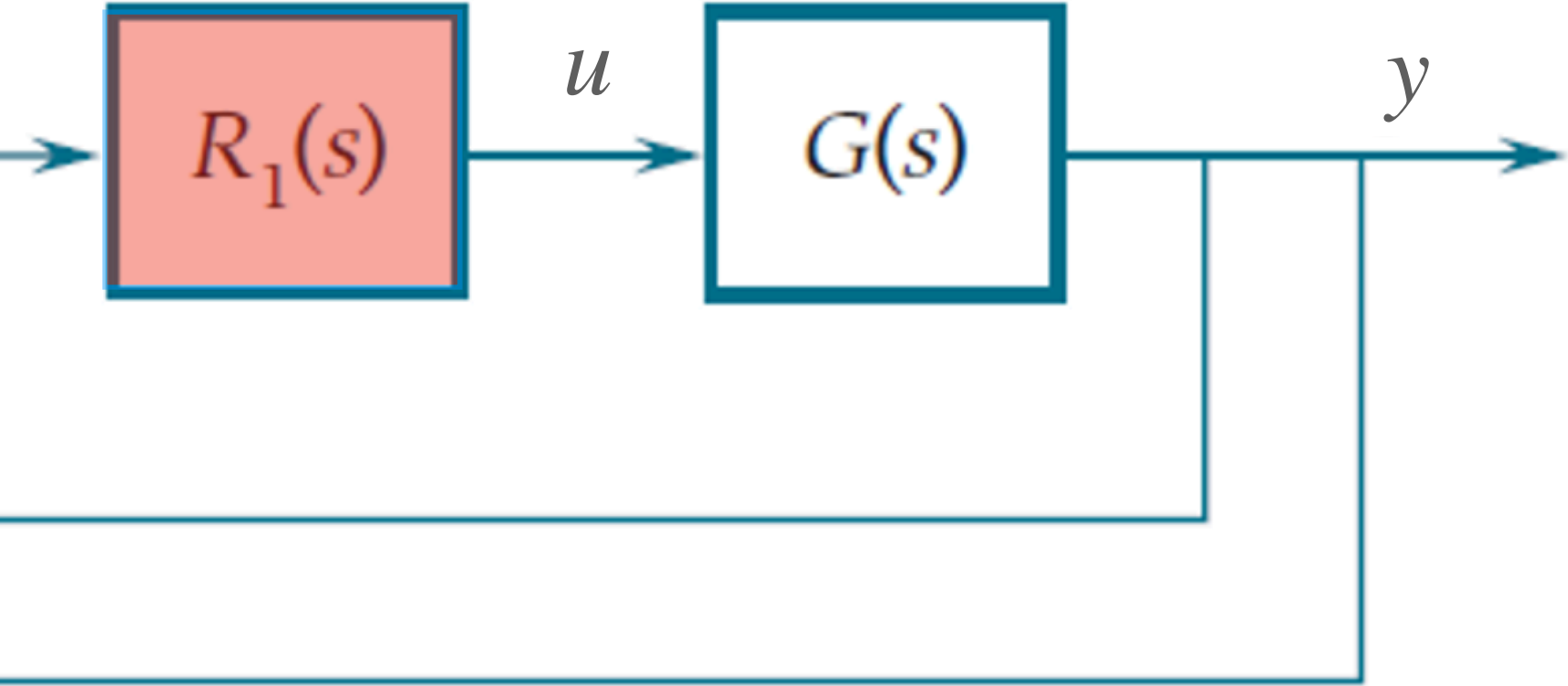
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$$R_1(s) = \frac{k(s + 3)}{s + 30}$$

$$G(s) = \frac{10}{(s + 3)(1 - 0.1s)}$$
$$G(s) = -\frac{100}{(s + 3)(s - 10)}$$



$$L_1(s) = R_1(s) G(s) = -\frac{100k}{(s + p)(s - 10)} = \frac{K}{(s + p)(s - 10)}$$

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?

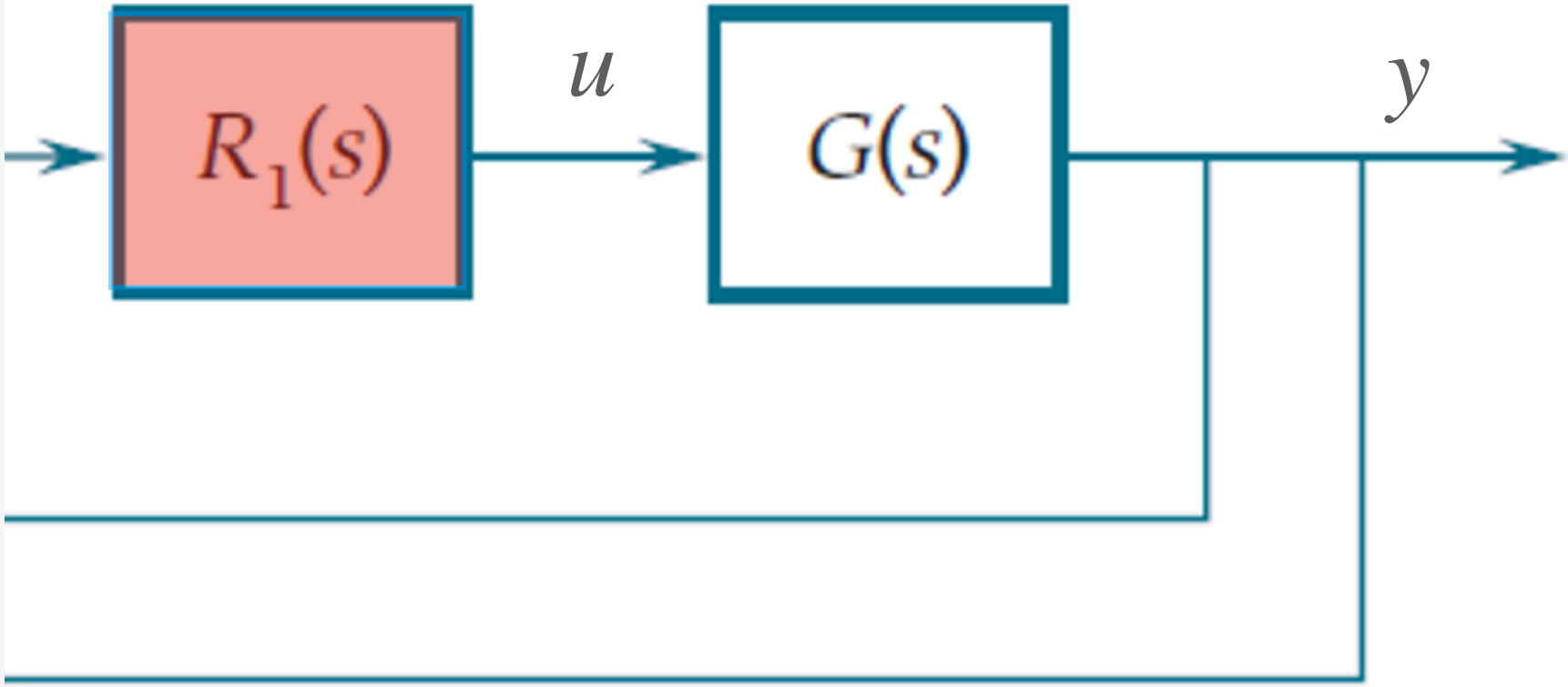
Control of Open-loop Unstable Systems

Example:

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$$R_1(s) = \frac{k(s + 3)}{s + 30}$$

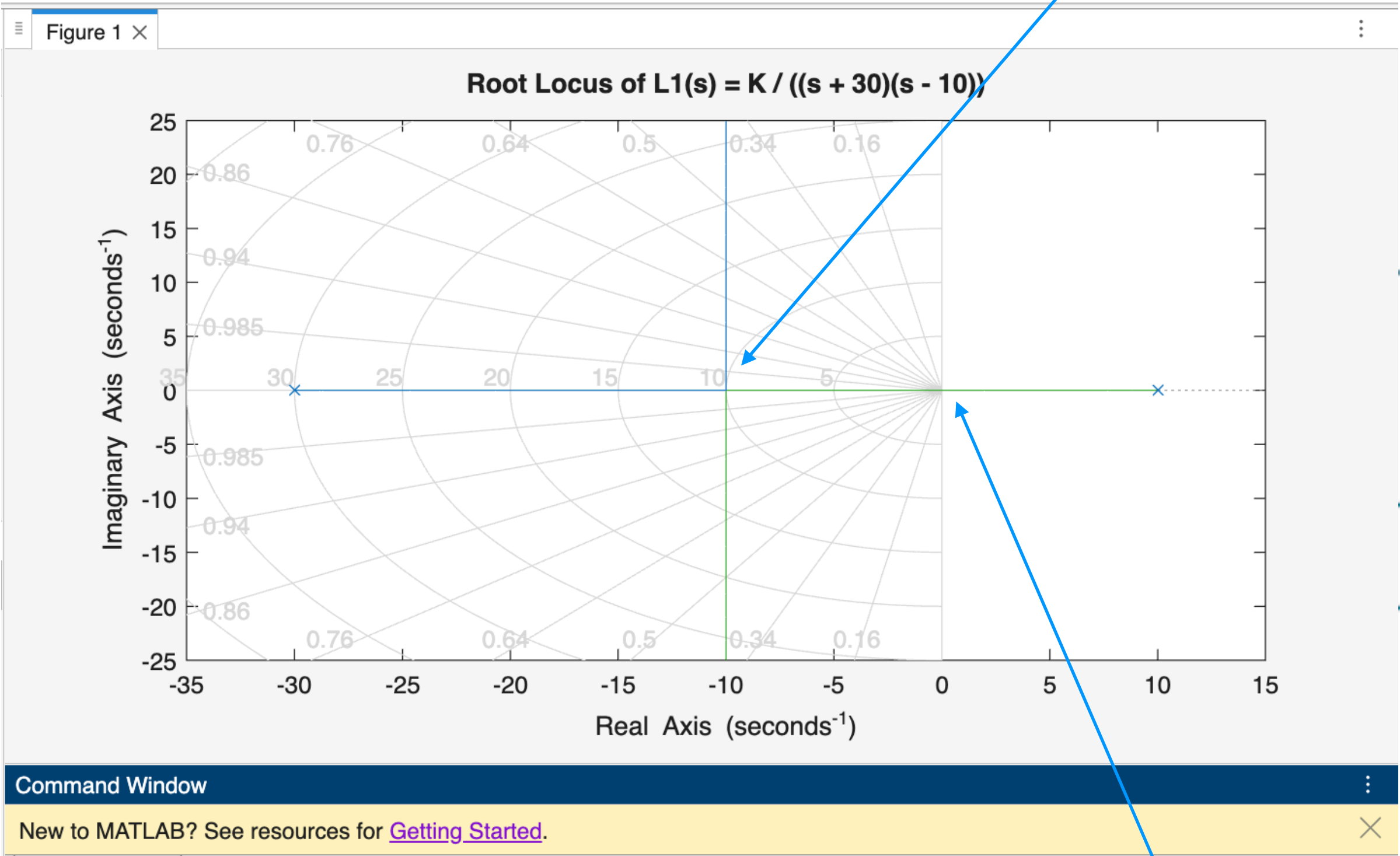
$$G(s) = \frac{10}{(s + 3)(1 - 0.1s)}$$
$$G(s) = -\frac{100}{(s + 3)(s - 10)}$$



$$L_1(s) = R_1(s) G(s) = -\frac{100k}{(s + p)(s - 10)} = \frac{K}{(s + \textcolor{blue}{p})(s - 10)}$$
$$F_1(s) = \frac{L_1(s)}{1 + L_1(s)}$$

?

$K^* ?$



Command Window

New to MATLAB? See resources for [Getting Started.](#)

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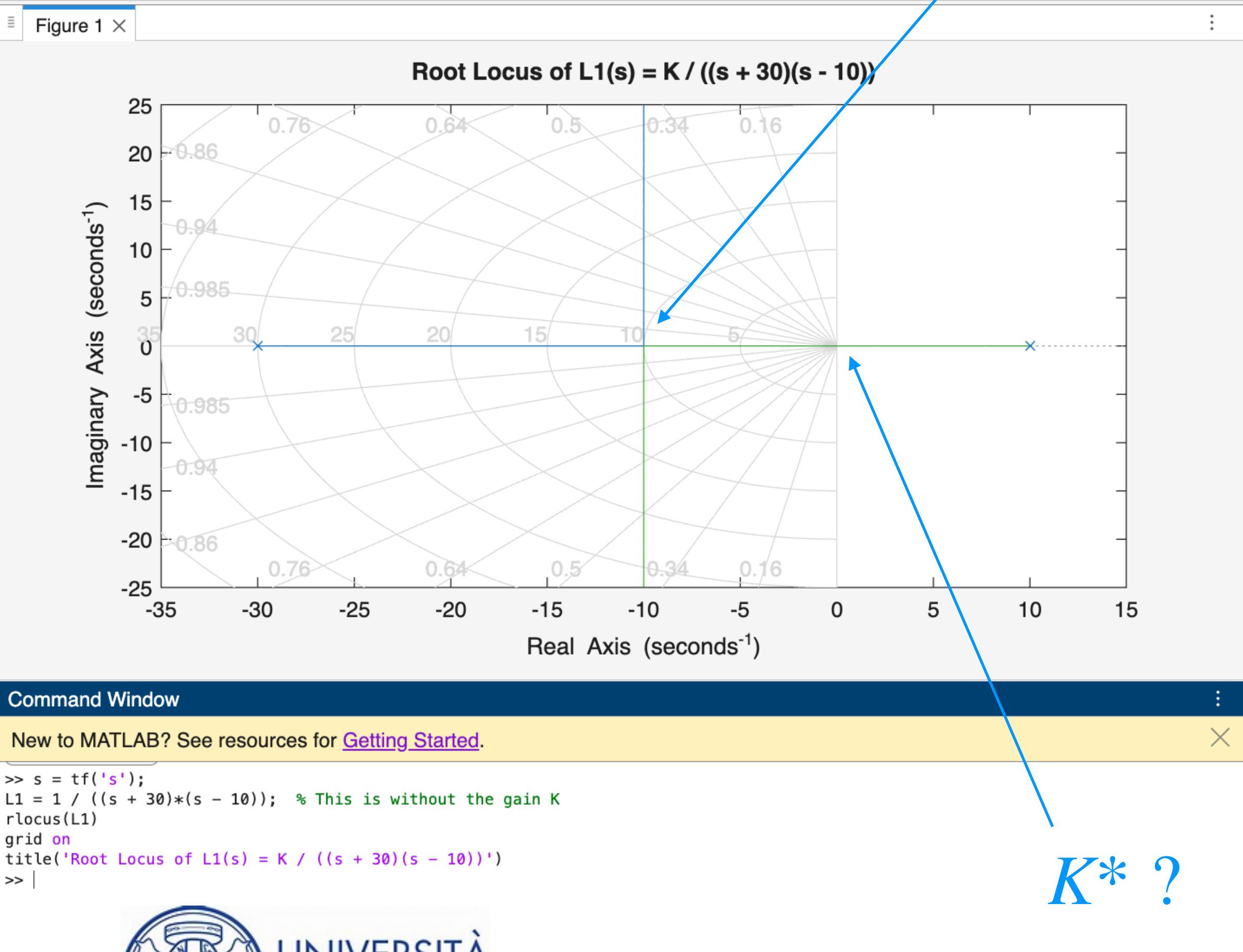
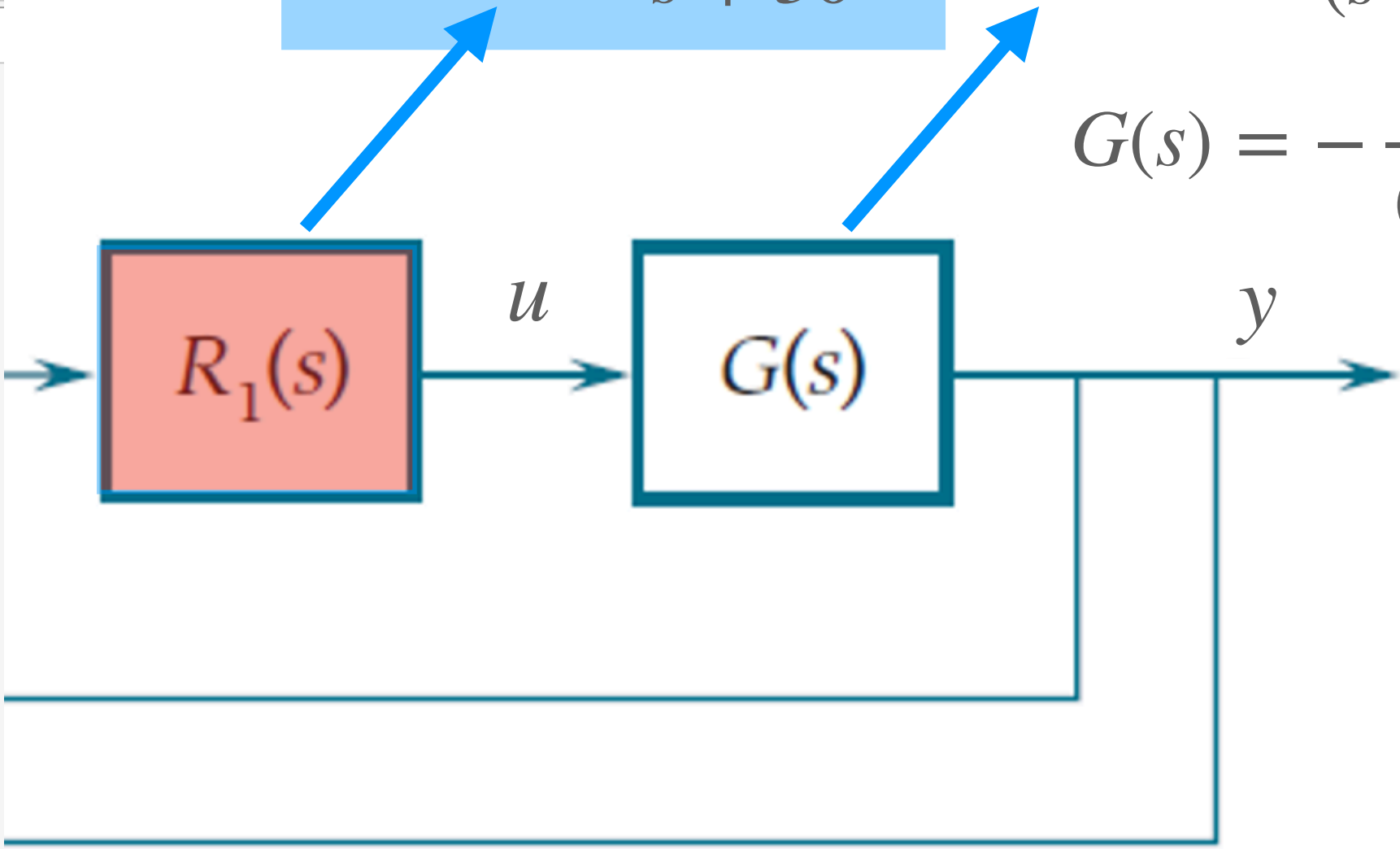

Control of Open-loop Unstable Systems

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$$G(s) = \frac{10}{(s + 3)(1 - 0.1s)}$$
$$G(s) = -\frac{100}{(s + 3)(s - 10)}$$



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>> |
```

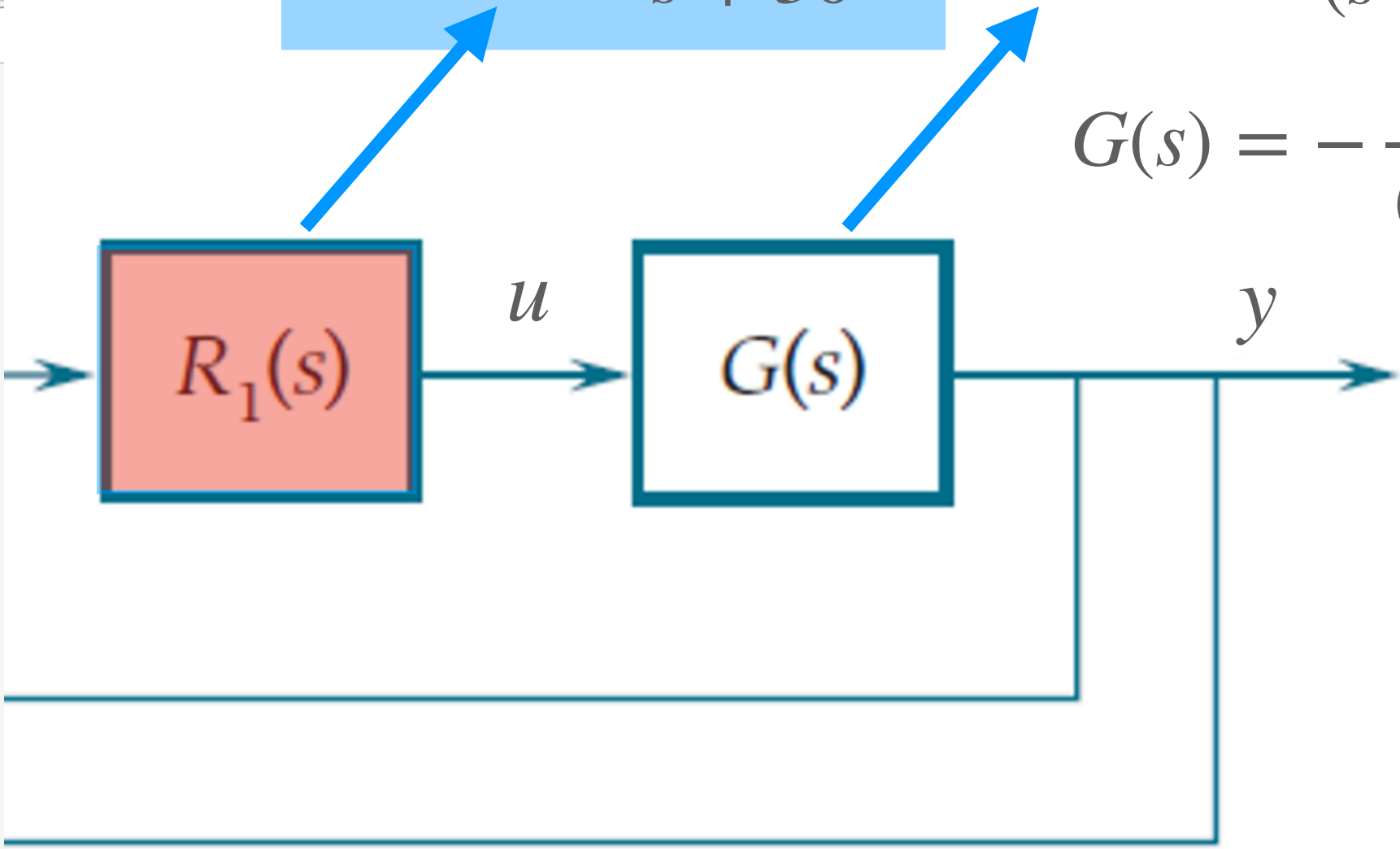
Control of Open-loop Unstable Systems

Example:

$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m} \quad \sigma_a = \frac{-30 + 10}{2 - 0} = \frac{-20}{2} = -10$$

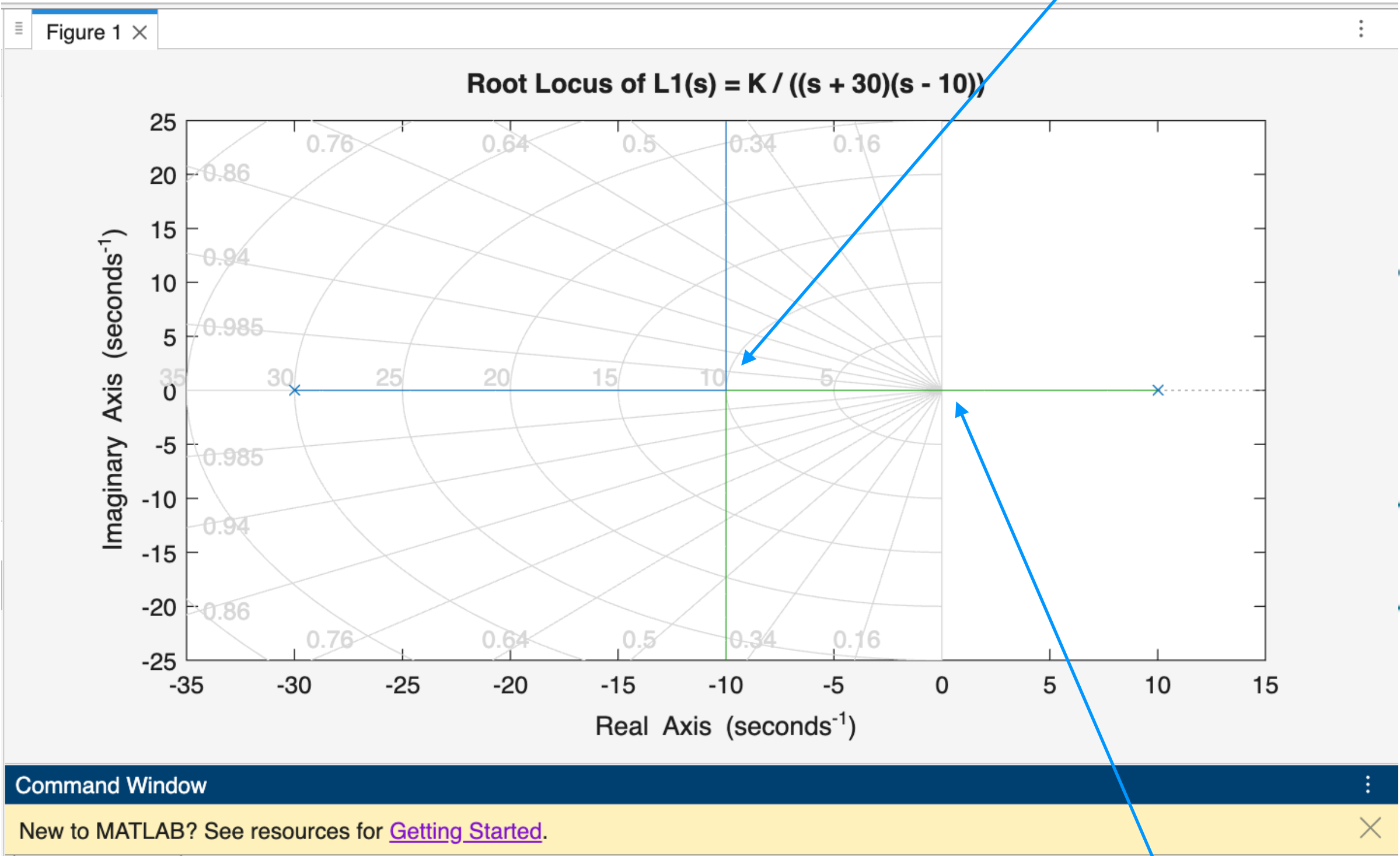
$$R_1(s) = \frac{k(s + 3)}{s + 30}$$

$$G(s) = \frac{10}{(s + 3)(1 - 0.1s)}$$
$$G(s) = -\frac{100}{(s + 3)(s - 10)}$$



$$|K|_{s=s^*} = \left| \frac{D(s^*)}{N(s^*)} \right|$$

$K^* ?$



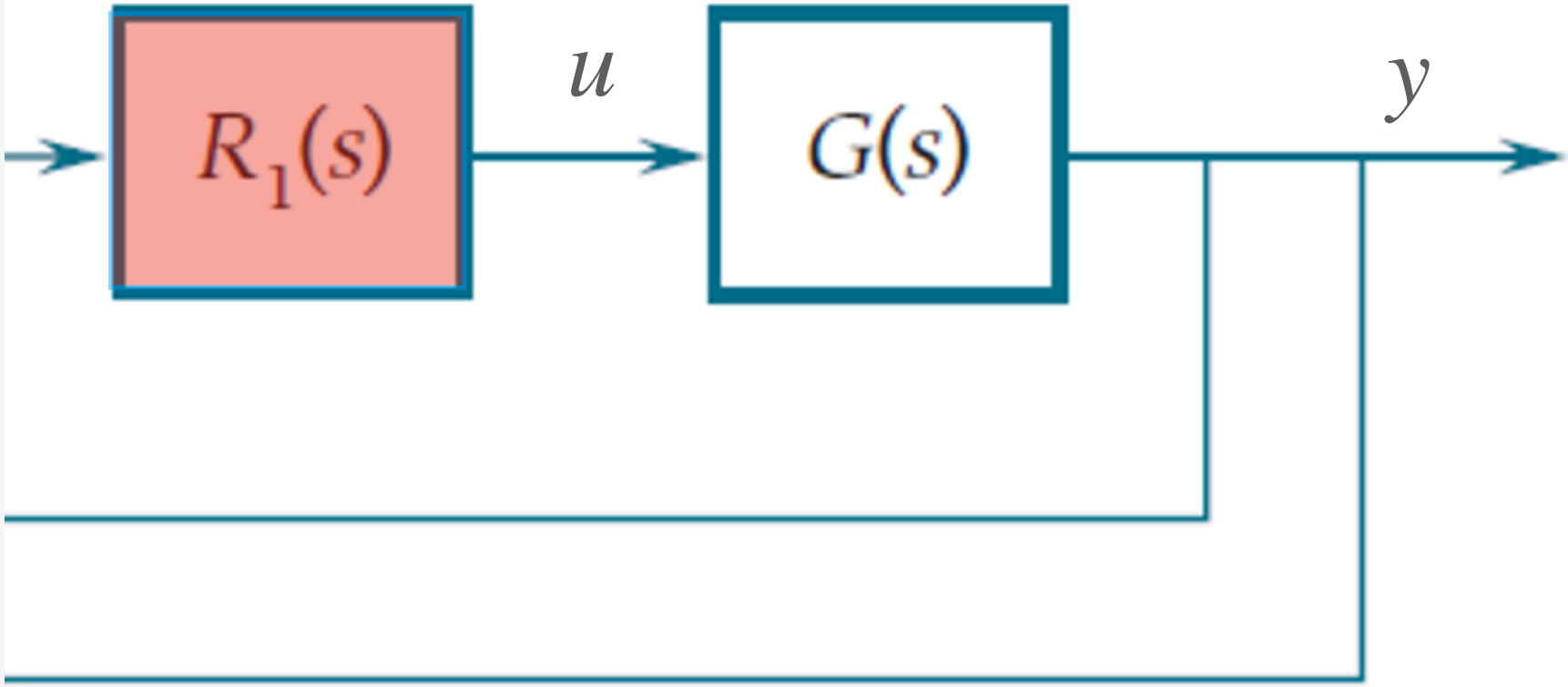
Control of Open-loop Unstable Systems

Example:

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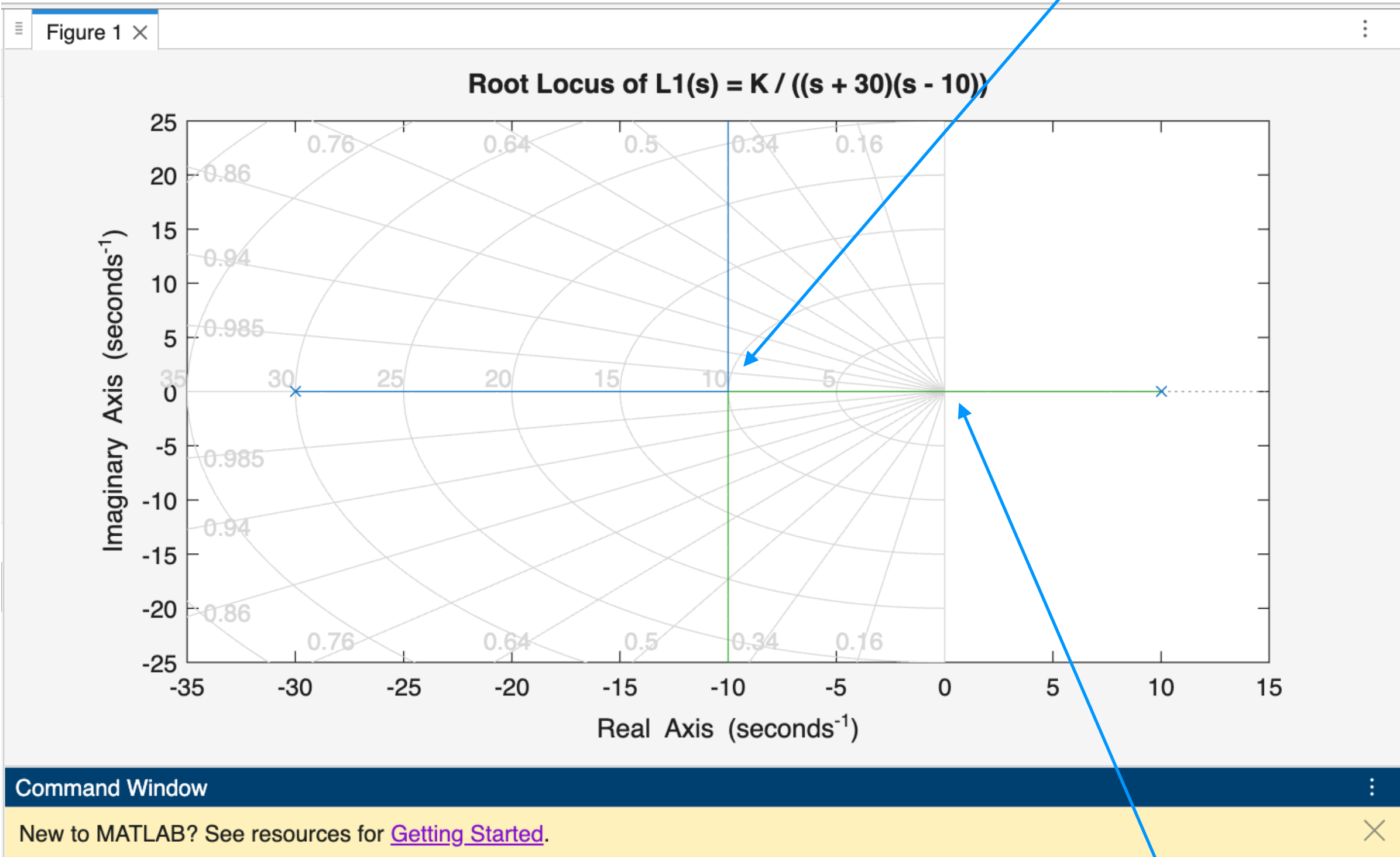
$$R_1(s) = \frac{k(s + 3)}{s + 30}$$

$$G(s) = \frac{10}{(s + 3)(1 - 0.1s)}$$
$$G(s) = -\frac{100}{(s + 3)(s - 10)}$$



$$|K|_{s=s^*} = \left| \frac{D(s^*)}{N(s^*)} \right| = \left| \frac{(s^* + 30)(s^* - 10)}{1} \right|$$

$K^* ?$



Command Window

New to MATLAB? See resources for [Getting Started](#).

```
>> s = tf('s');
L1 = 1 / ((s + 30)*(s - 10)); % This is without the gain K
rlocus(L1)
grid on
title('Root Locus of L1(s) = K / ((s + 30)(s - 10))')
>> |
```

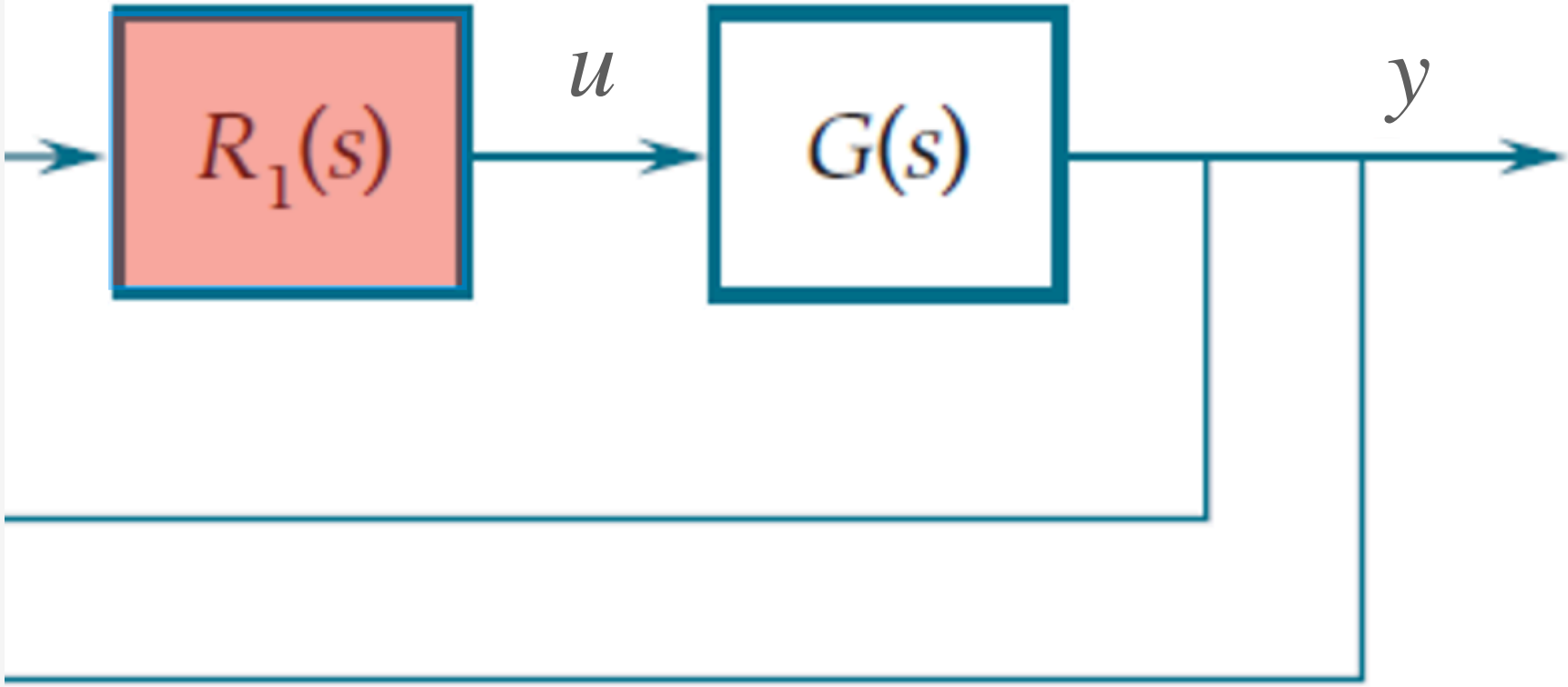

Control of Open-loop Unstable Systems

Example:

$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m}$$
$$\sigma_a = \frac{-30 + 10}{2 - 0} = \frac{-20}{2} = -10$$

$$R_1(s) = \frac{k(s + 3)}{s + 30}$$

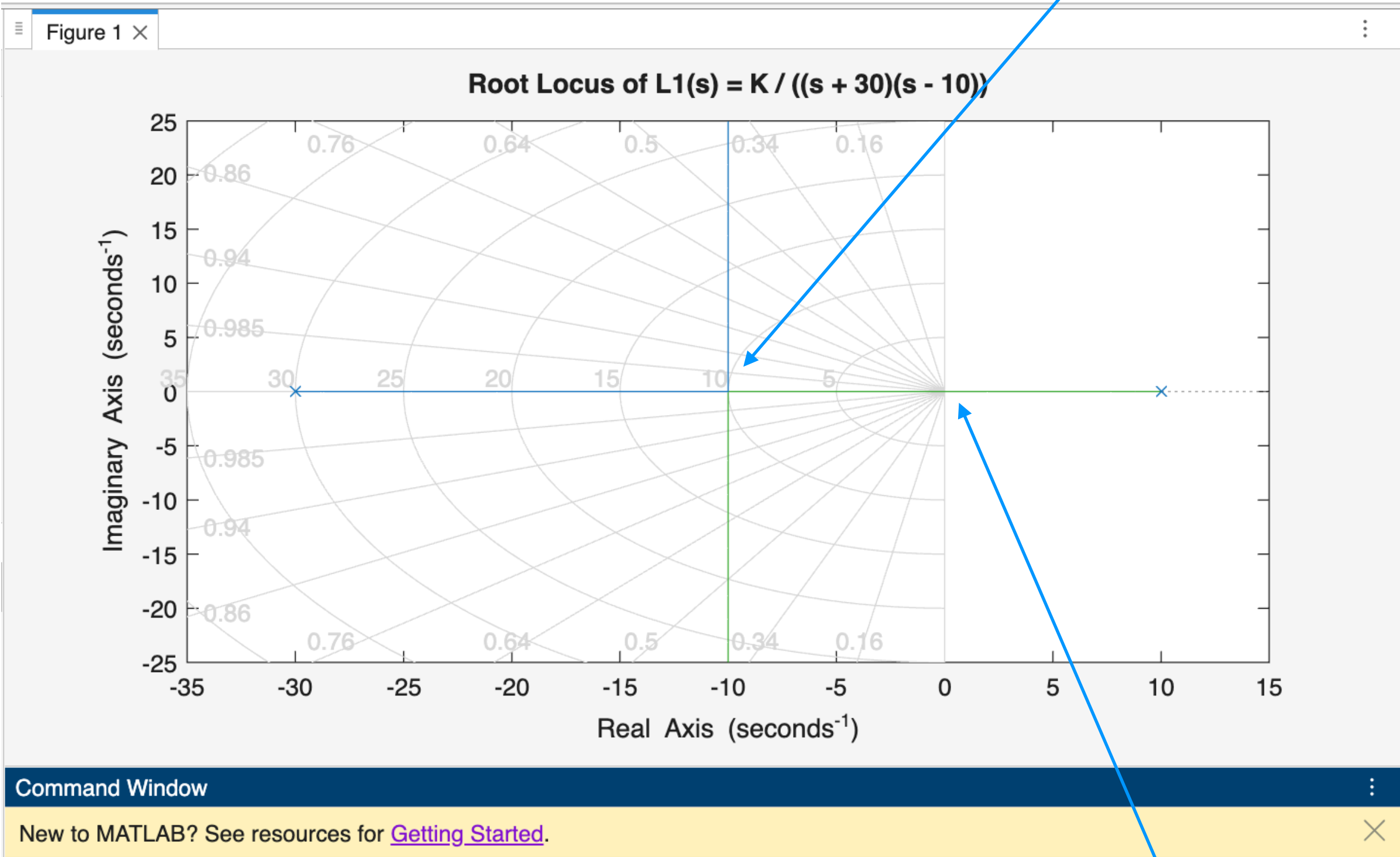
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$$|K|_{s=s^*} = \left| \frac{D(s^*)}{N(s^*)} \right| = \left| \frac{(s^* + 30)(s^* - 10)}{1} \right|$$

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$K^* ? \longrightarrow$



Command Window

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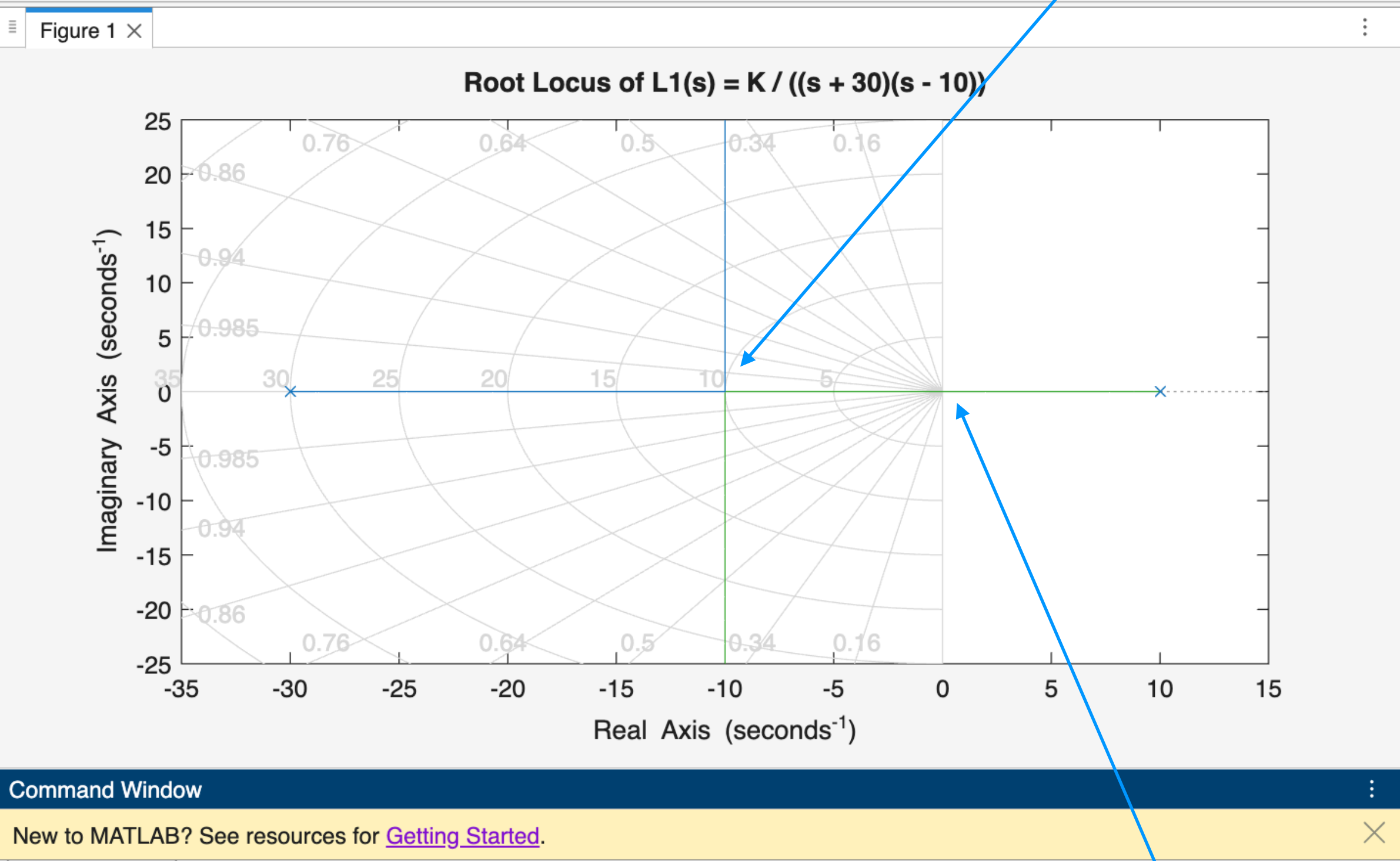
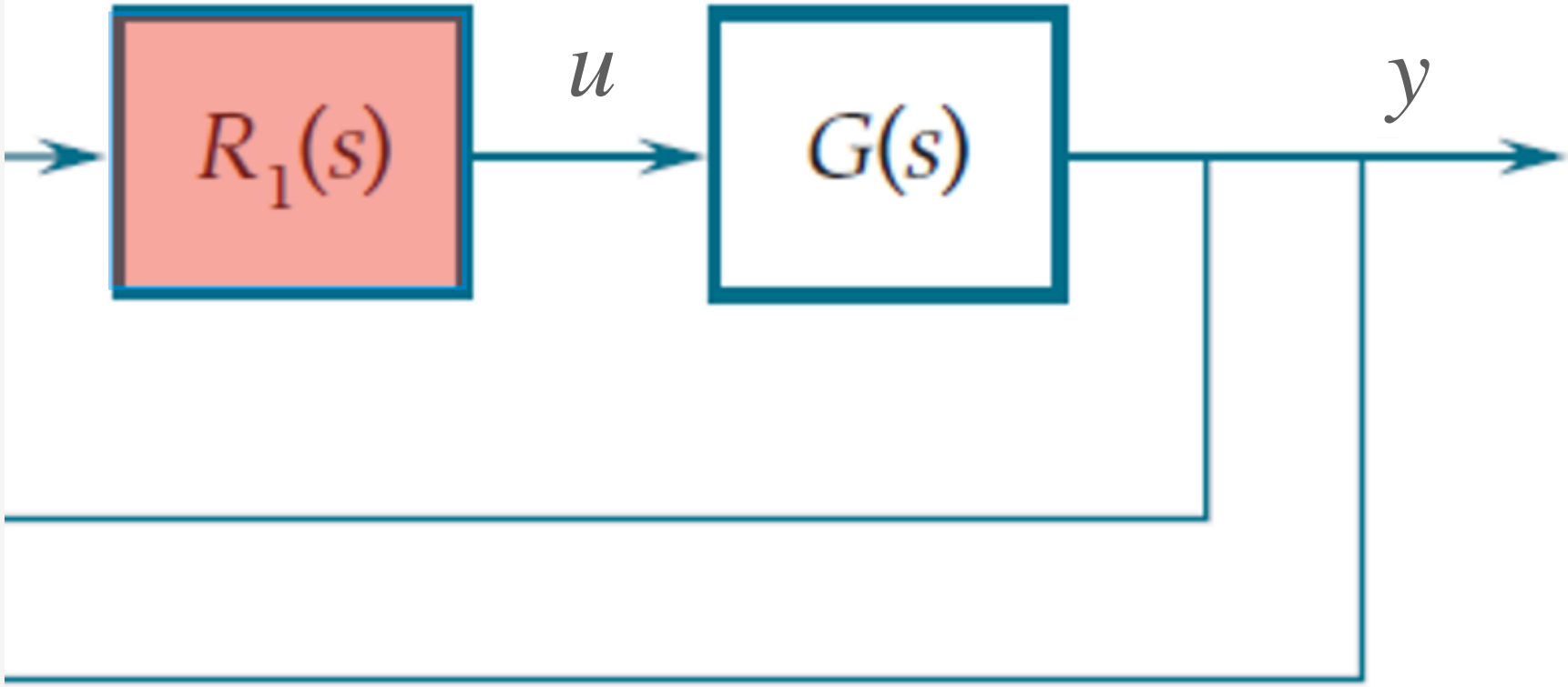
Control of Open-loop Unstable Systems

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Command Window

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$$|K|_{s=s^*} = \left| \frac{D(s^*)}{N(s^*)} \right| = \left| \frac{(s^* + 30)(s^* - 10)}{1} \right|$$

$K^* ?$



$$|K|_{s=0} = \left| \frac{D(0)}{N(0)} \right| = \left| \frac{(0 + 30)(0 - 10)}{1} \right| = 300$$

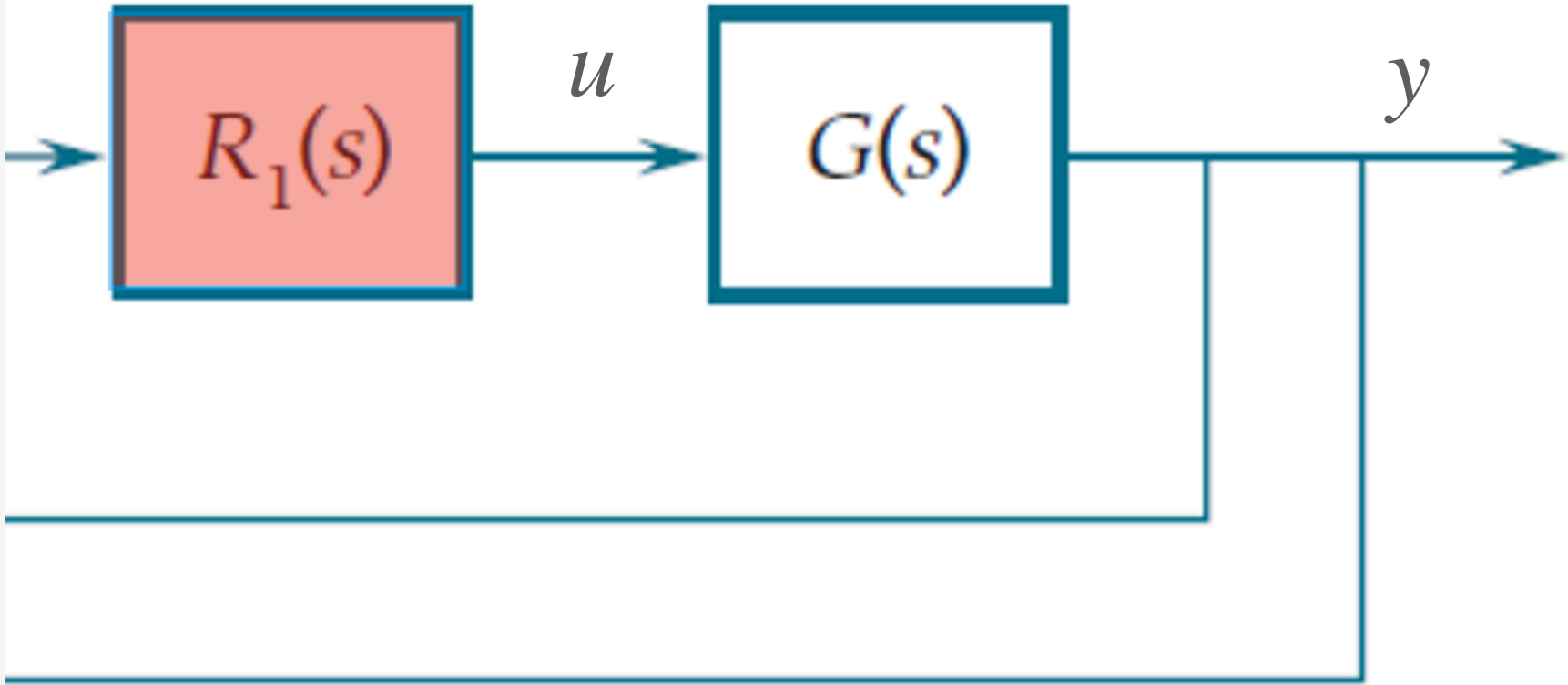
Control of Open-loop Unstable Systems

Example:

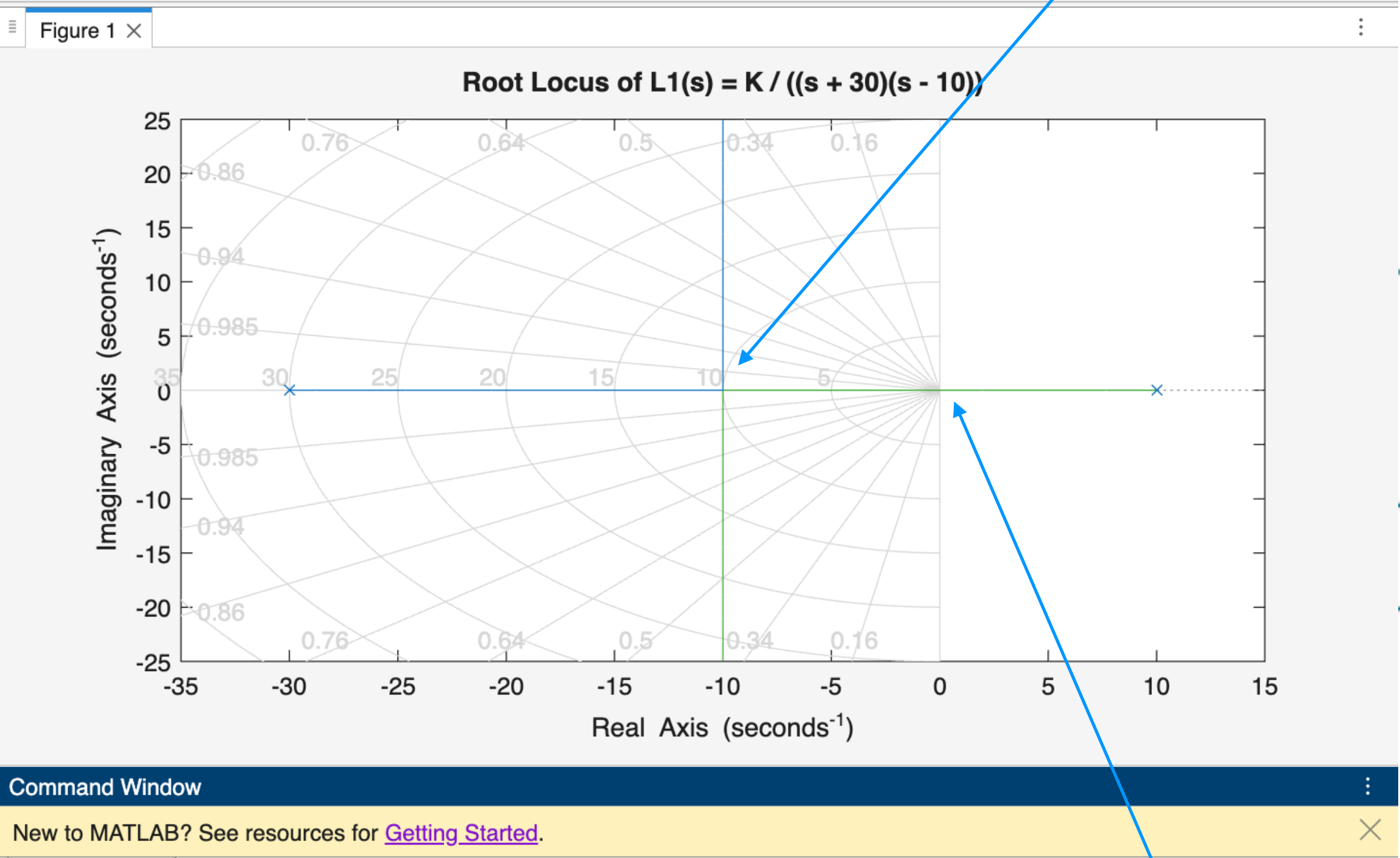
$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m}$$
$$\sigma_a = \frac{-30 + 10}{2 - 0} = \frac{-20}{2} = -10$$

$$R_1(s) = \frac{k(s + 3)}{s + 30}$$

$$G(s) = \frac{10}{(s + 3)(1 - 0.1s)}$$
$$G(s) = -\frac{100}{(s + 3)(s - 10)}$$



- To conclude for $R_1(s)$:
- Choose K
 - Find k considering that $-100k = K$



Command Window

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$K^* ?$ →

Control of Open-loop Unstable Systems

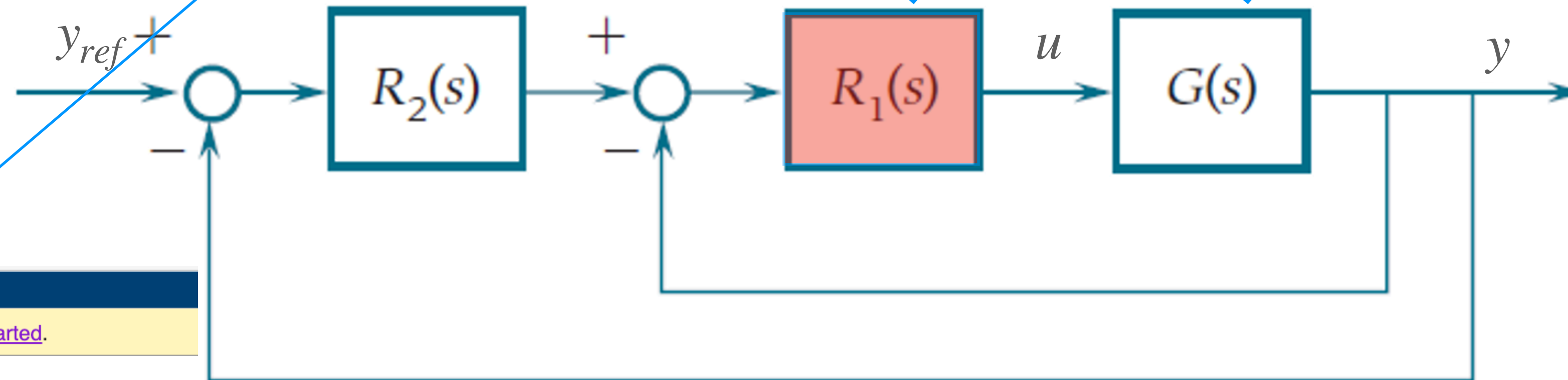
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$$G(s) = -\frac{100}{(s + 3)(s - 10)}$$



My choice:

```
Command Window
New to MATLAB? See resources for Getting Started.
>> desired_pole = -10;
K_at_sigma_a = rlocfind(L1, desired_pole);
K_at_sigma_a
K_at_sigma_a =
400
```

To conclude for $R_1(s)$:

- Choose K
- Find k considering that $-100k = K$

 $K^* ?$

$$|K|_{s=s^*} = \left| \frac{D(s^*)}{N(s^*)} \right| = \left| \frac{(s^* + 30)(s^* - 10)}{1} \right|$$

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Control of Open-loop Unstable Systems

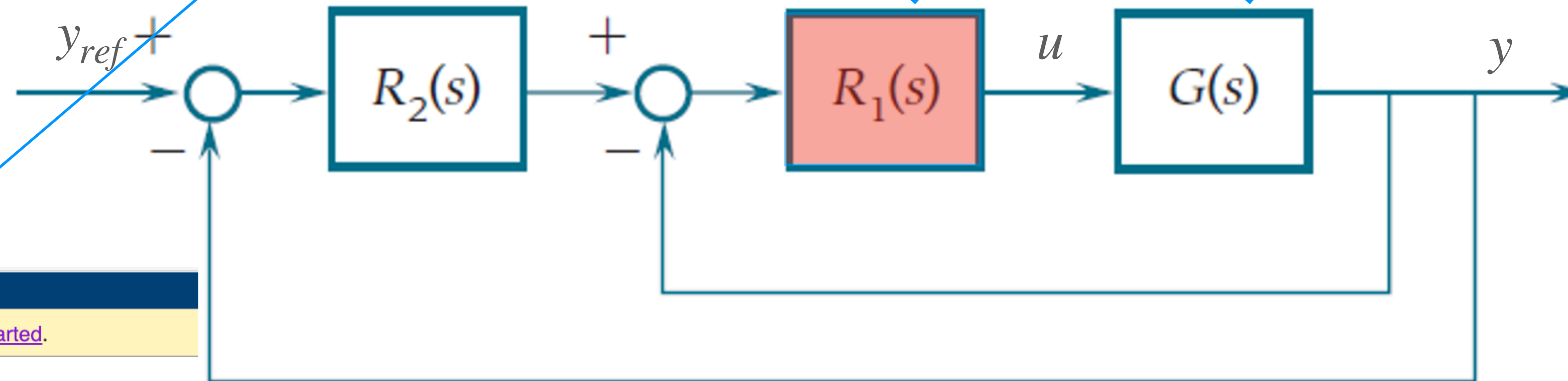
Example:

$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m} \quad \sigma_a = \frac{-30 + 10}{2 - 0} = \frac{-20}{2} = -10$$

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K_at_sigma_a =
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To conclude for $R_1(s)$:

- Choose K
- Find k considering that $-100k = K$

$$-100k = K$$

$$k = -\frac{K}{100}$$

$K^* ?$



$$|K|_{s=s^*} = \left| \frac{D(s^*)}{N(s^*)} \right| = \left| \frac{(s^* + 30)(s^* - 10)}{1} \right|$$

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Control of Open-loop Unstable Systems

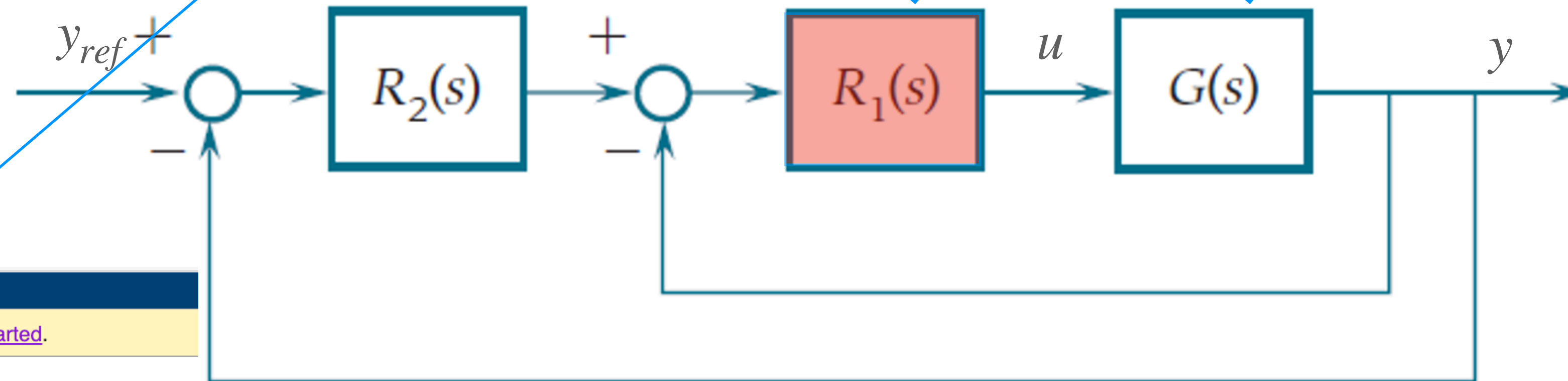
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```

To conclude for $R_1(s)$:

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$$-100k = K$$

$$k = -\frac{K}{100}$$

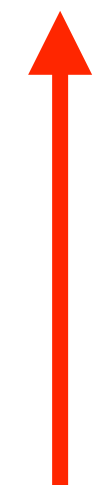
$$R_1(s) = -\frac{4(s + 3)}{s + 30}$$

$K^* ?$



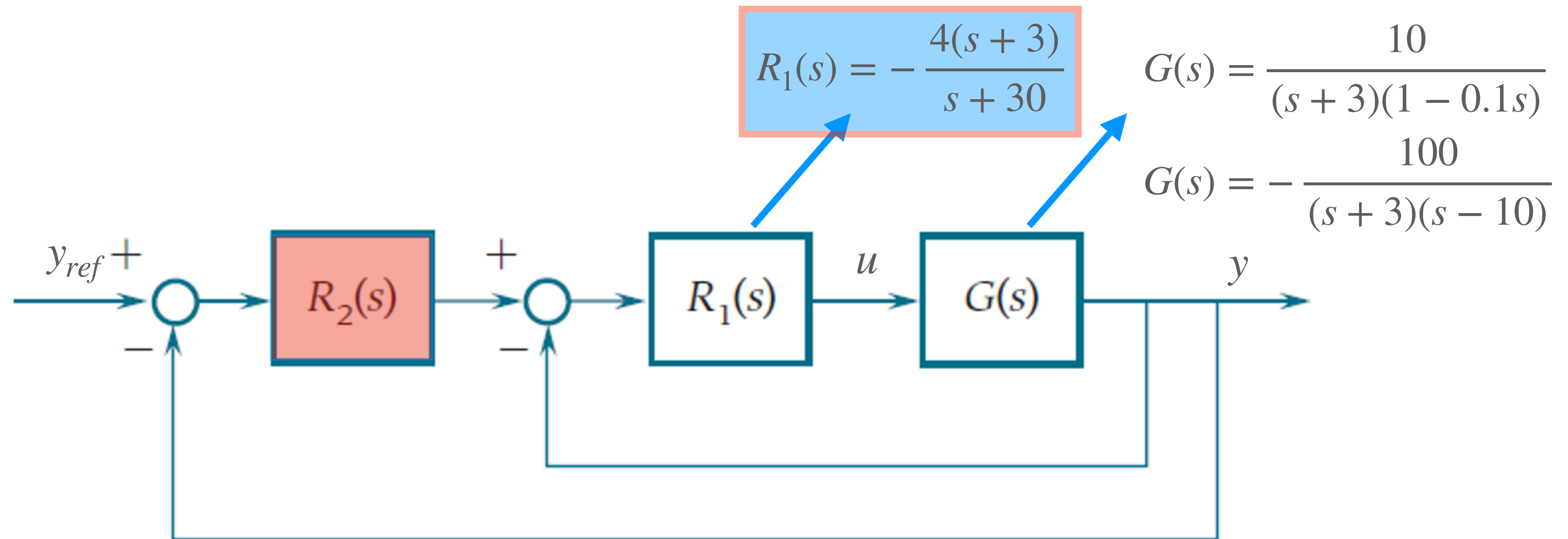
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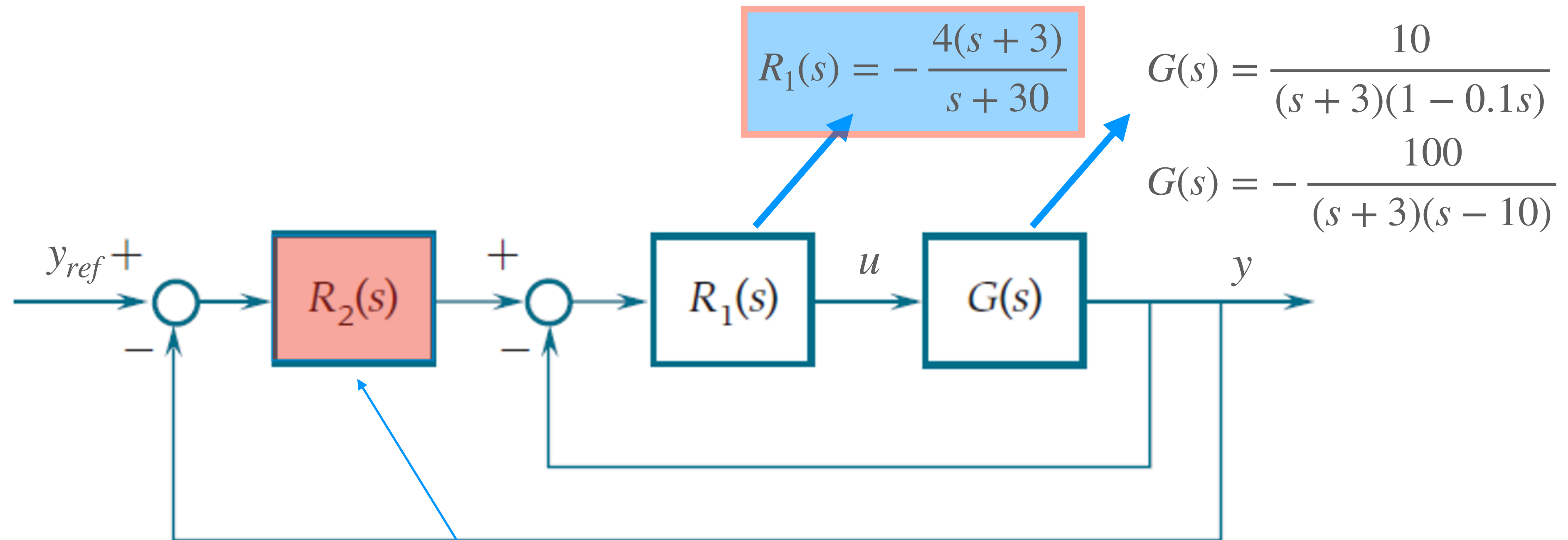
Control of Open-loop Unstable Systems

Example:



Control of Open-loop Unstable Systems

Example:



To conclude for $R_2(s)$:

- Compute $F_1(s)$
- Design $R_2(s)$ based on $F_1(s)$ using standard loop-shaping design guided by the Bode criterion

General Information

Prof. Antonella Ferrara

<https://scholar.google.com/citations?user=r5JuMskAAAAJ&hl=en>

Course Teaching Material:

KIRO UNIPV <https://elearning.unipv.it/>

- [504462 - PROCESS CONTROL 2025-26 - PROF.SSA FERRARA ANTONELLA](#)

Lecture Time-table:

<http://www-3.unipv.it/ingserv/orario2526/1sem/insegnamenti/ProCont.html>

Exams:

<https://studentionline.unipv.it/esse3/Home.do>

<https://kirotesting.unipv.it/>

Introduction

- Program of the course:

Advanced SISO control schemes:

Pre-filters and parallel compensators, two degrees of freedom control schemes, compensation of measurable disturbances, systems with delays and Smith Predictor, Padé approximation, decoupling in the frequency domain, control of open loop unstable systems.

Advanced MIMO control schemes:

Decoupling based control schemes, decentralized control, relative gain array.

PID controllers:

Features and properties. Rules for the empirical calibration. Wind-up and anti wind-up schemes.

Digital control:

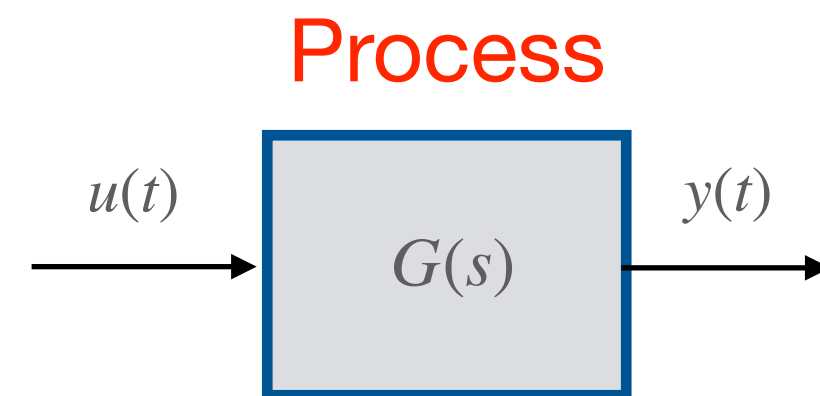
Discrete-time systems. The concept of equilibrium for discrete-time systems. Stability of linear time-invariant discrete-time systems. Jury test. Digital control schemes. Zeta transform and its properties. Transfer functions in the z domain. Sampling and aliasing. Choice of the sampling time. Zero-order-Hold. Discretization of continuous-time controllers. Bilinear transformation, Euler and Tustin methods.

Introduction

- Some of the figures in these slides, kindly provided by McGraw-Hill, are those of the Textbook:



Control of MIMO Systems

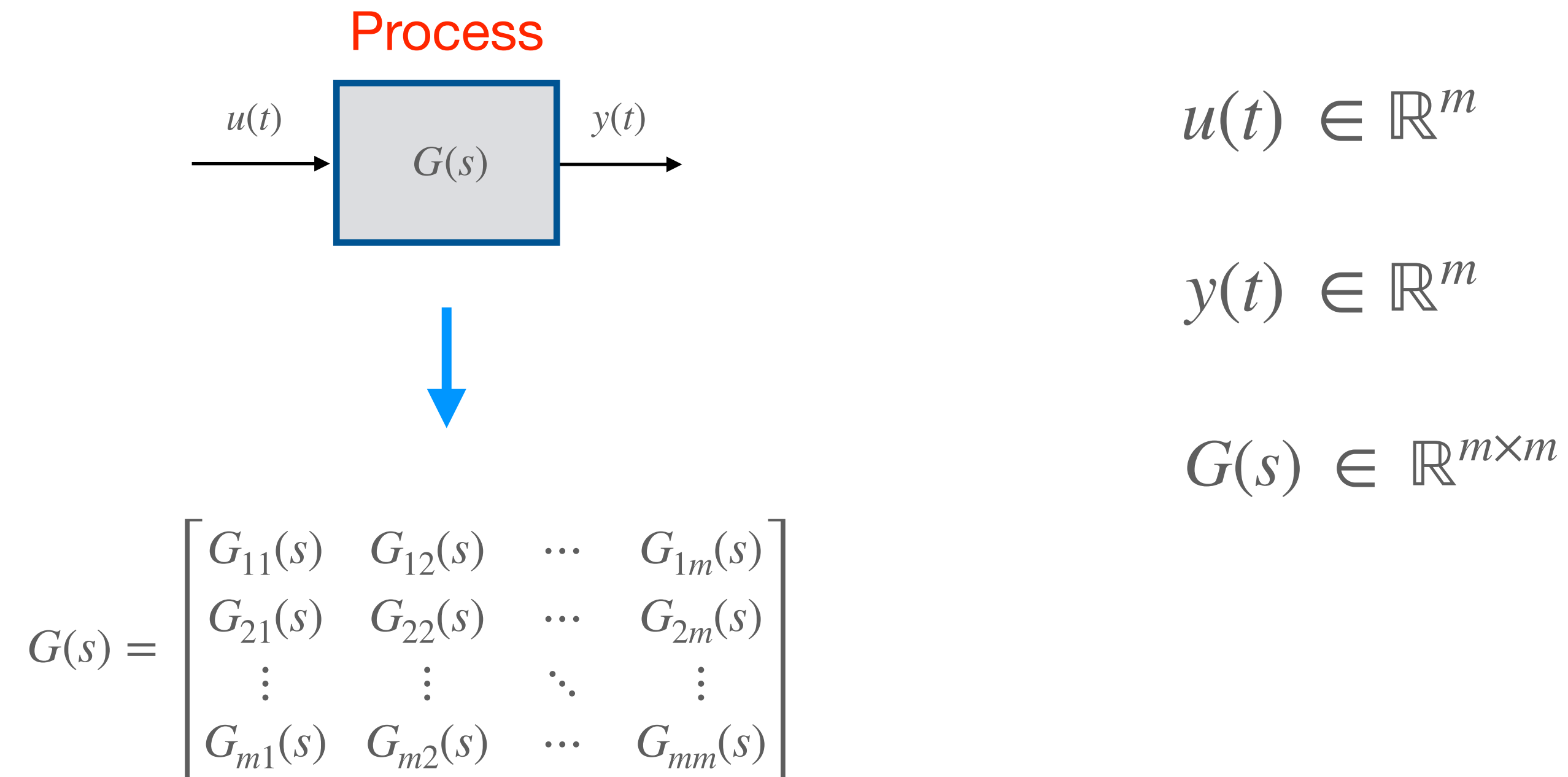


$$u(t) \in \mathbb{R}^m$$

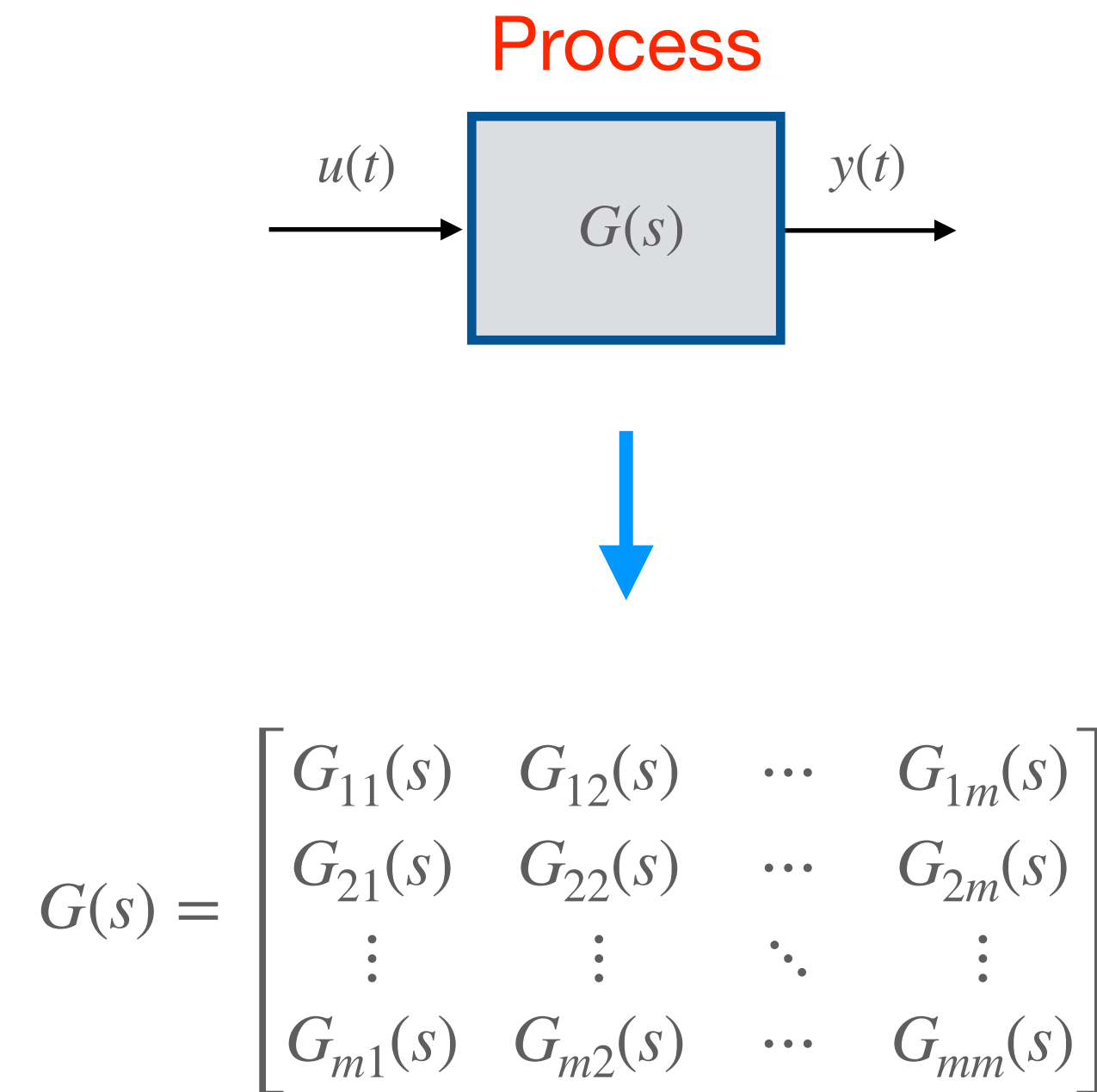
$$y(t) \in \mathbb{R}^m$$

$$G(s) \in \mathbb{R}^{m \times m}$$

Control of MIMO Systems



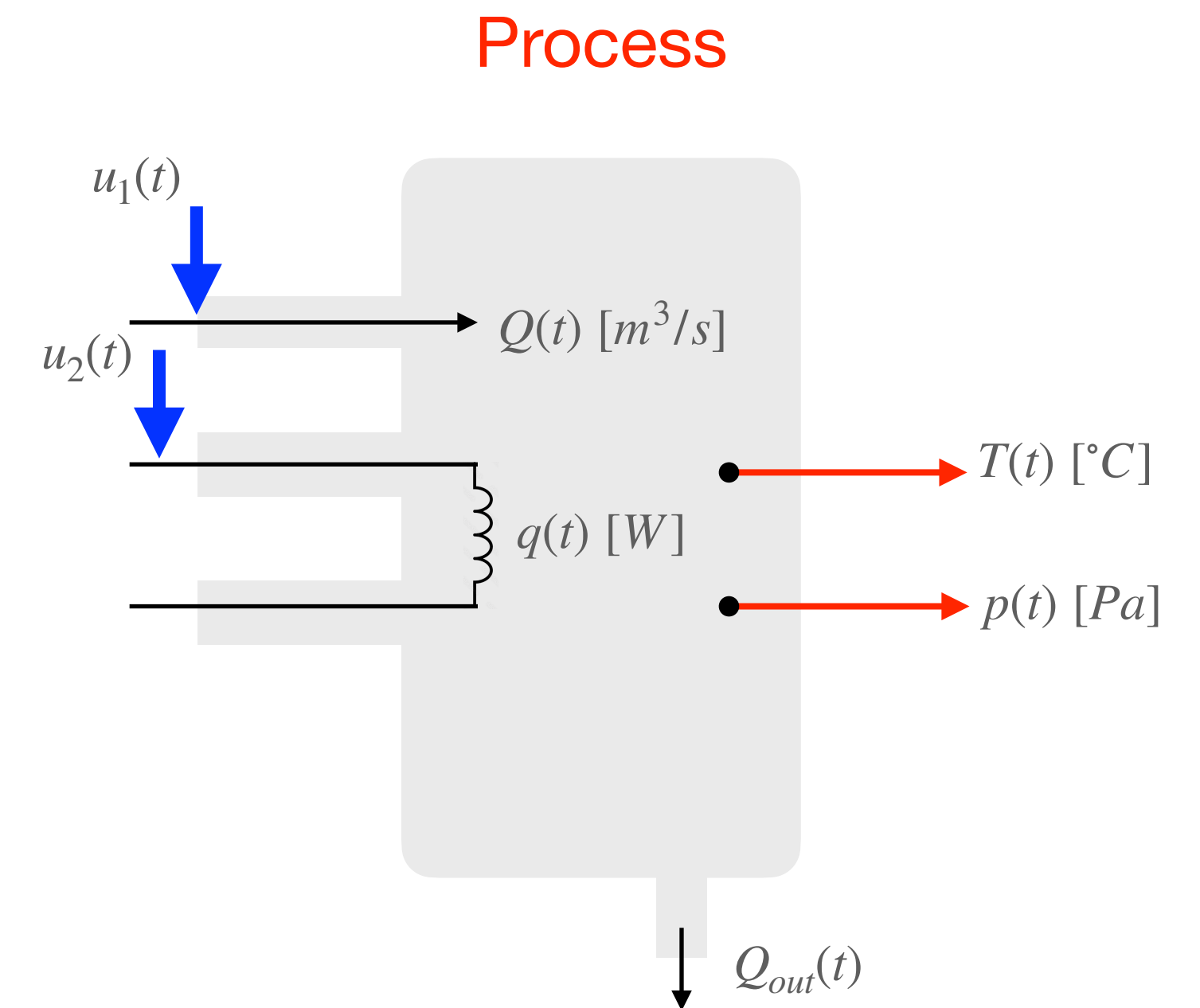
Control of MIMO Systems



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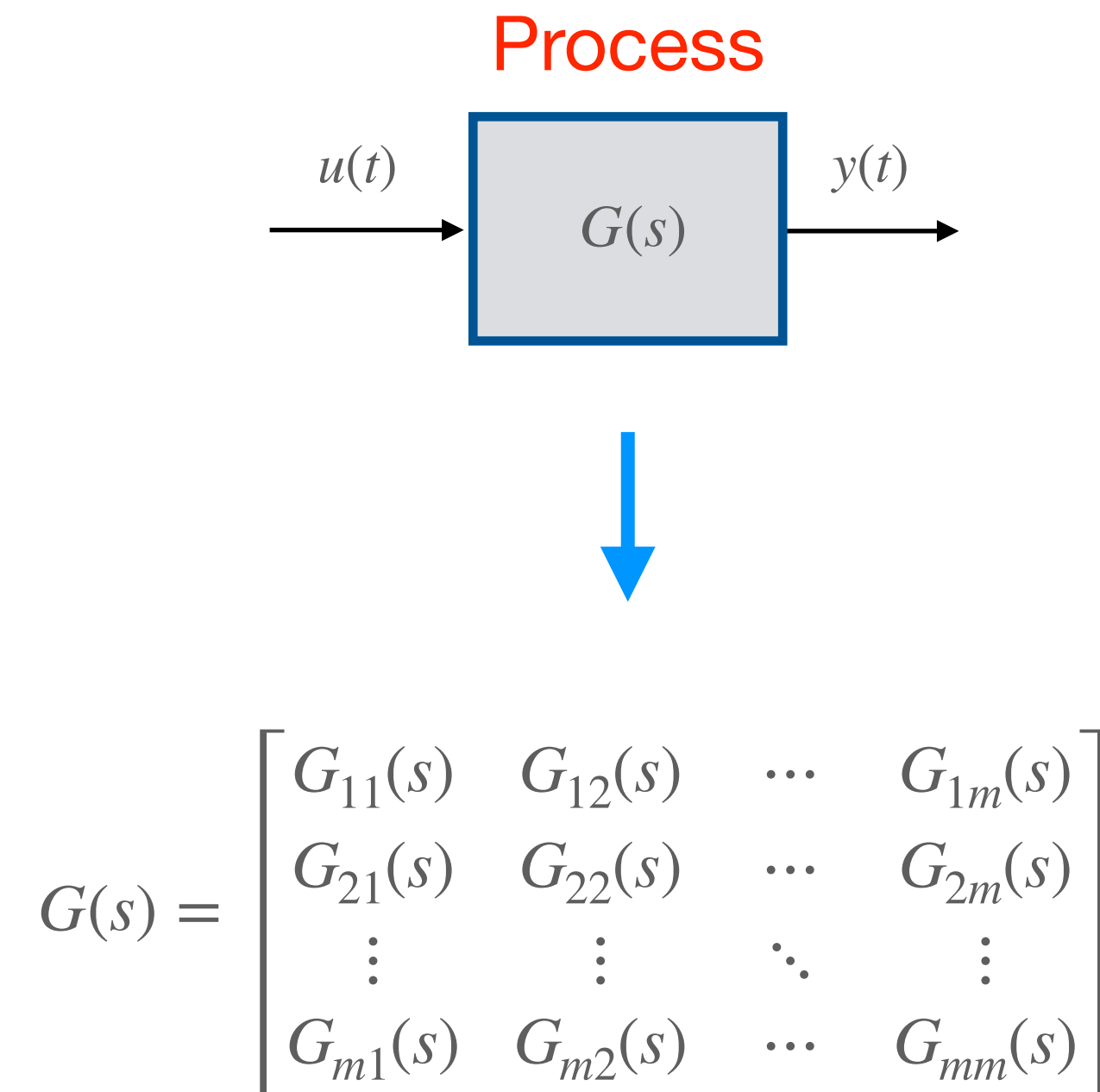
$$y(t) \in \mathbb{R}^m$$

$$G(s) \in \mathbb{R}^{m \times m}$$



Thermal fluid heating system

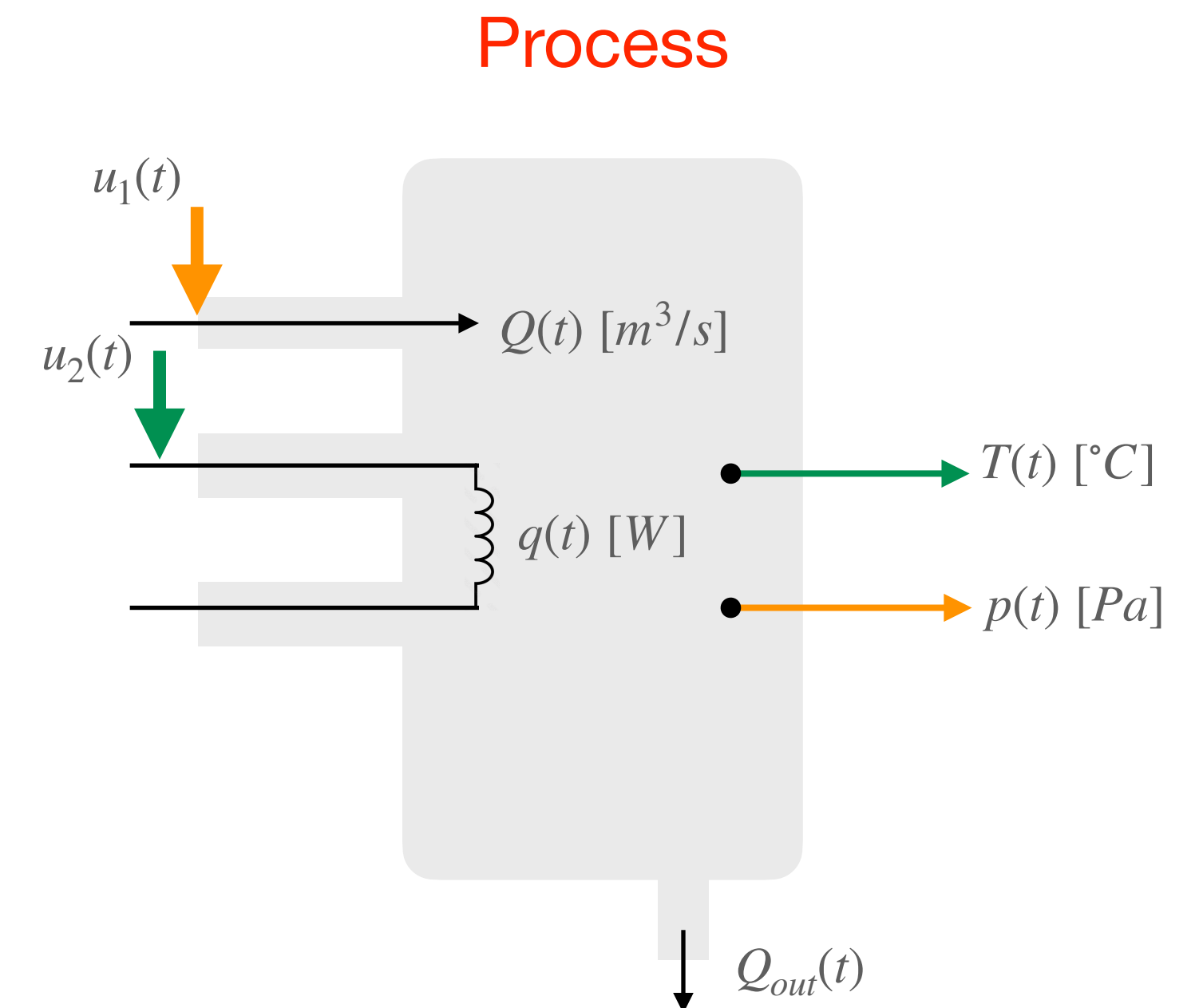
Control of MIMO Systems



$$u(t) \in \mathbb{R}^m$$

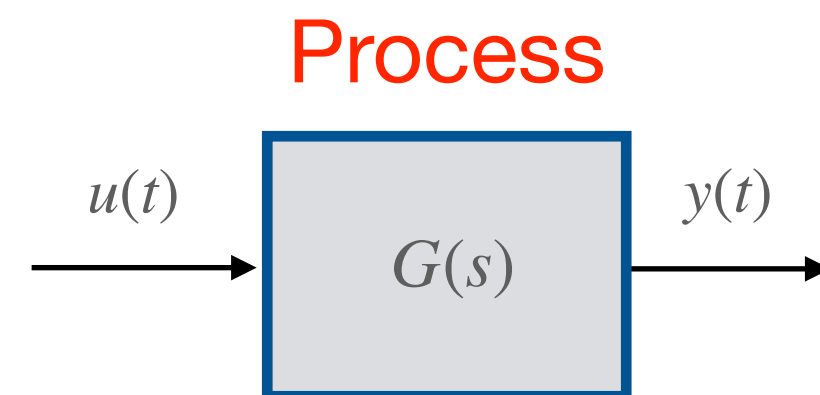
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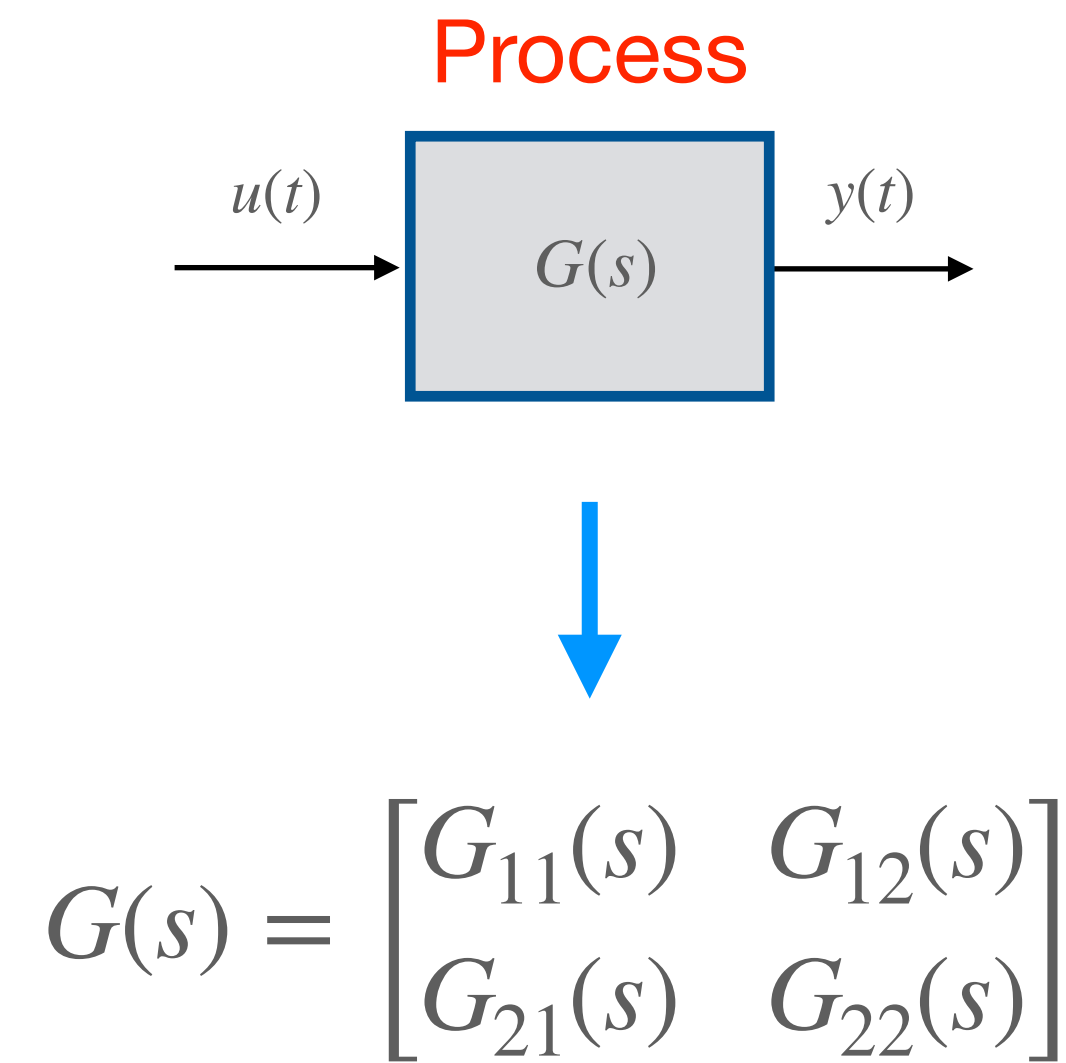
Decoupling Based Control Schemes



Assumptions:

- $G(s) \in \mathbb{R}^{2 \times 2}$

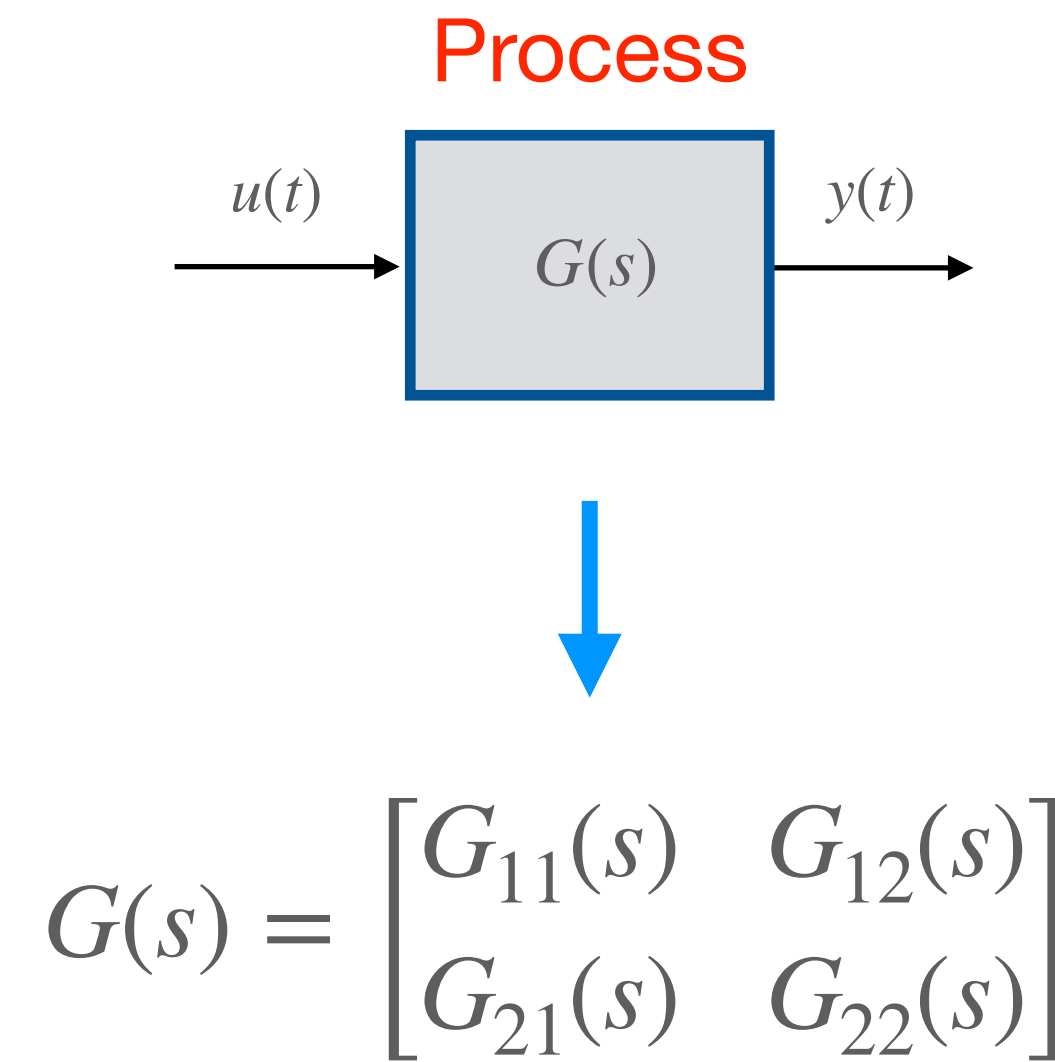
Decoupling Based Control Schemes



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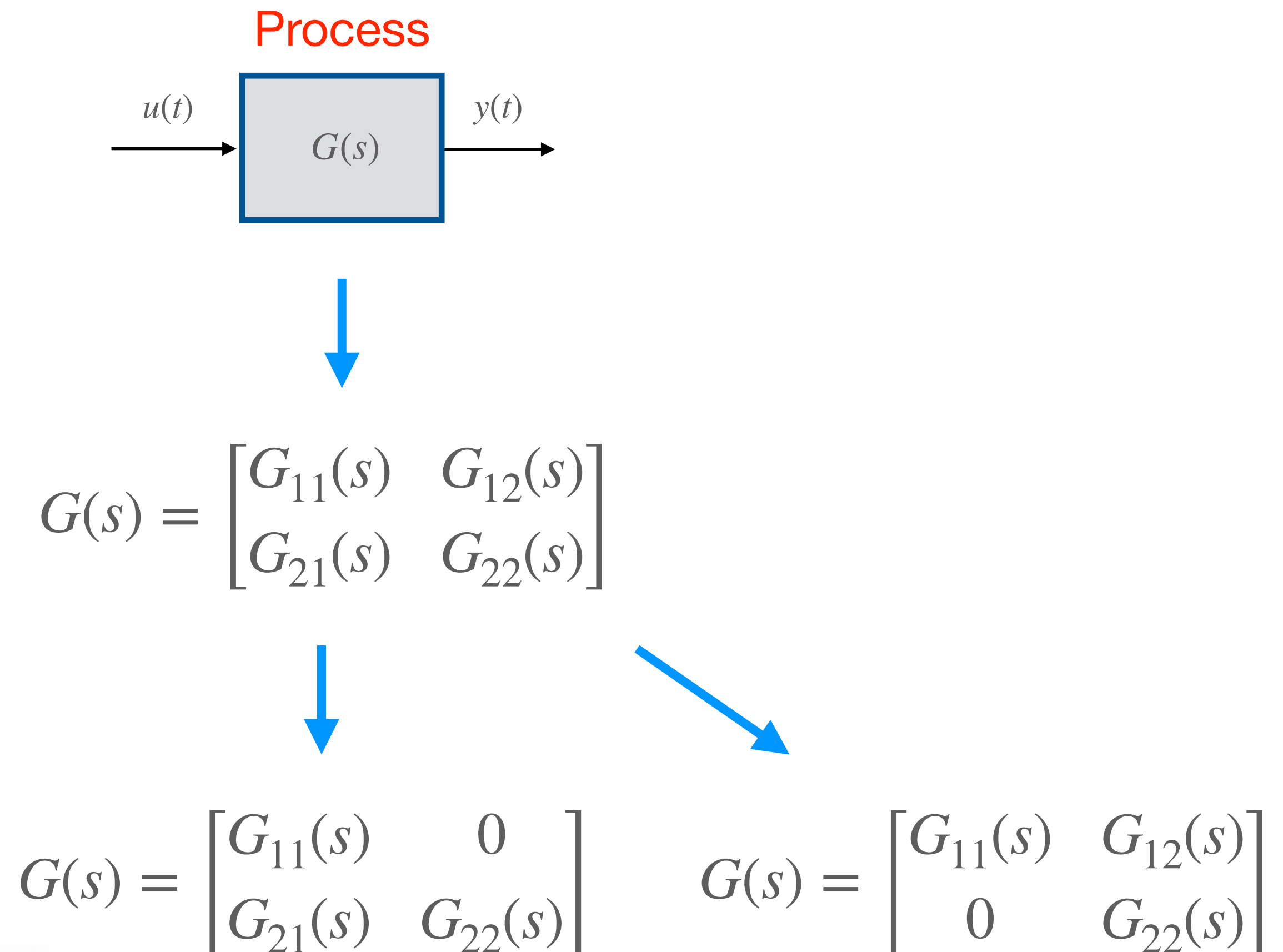
Decoupling Based Control Schemes



Assumptions:

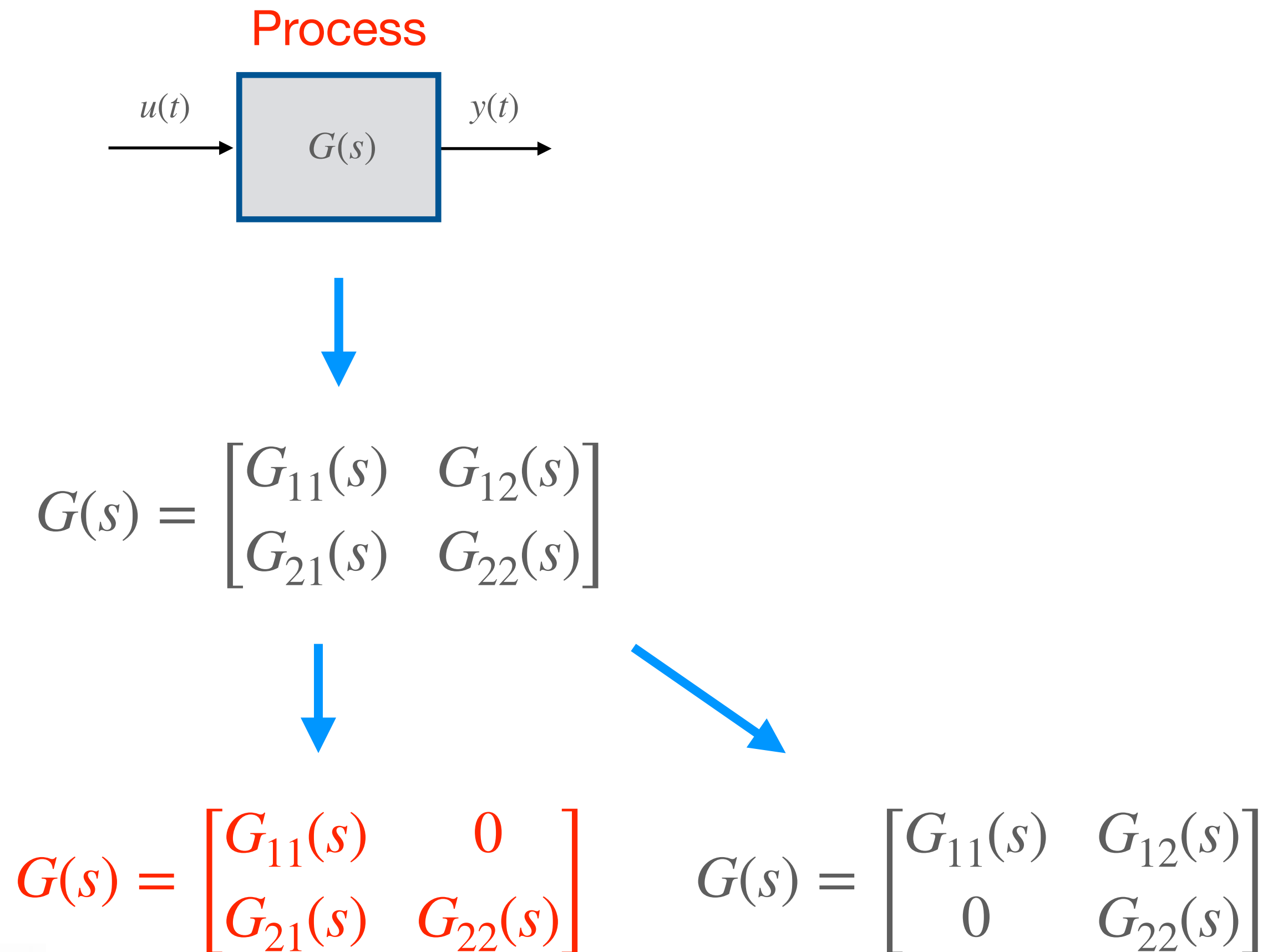
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Decoupling Based Control Schemes

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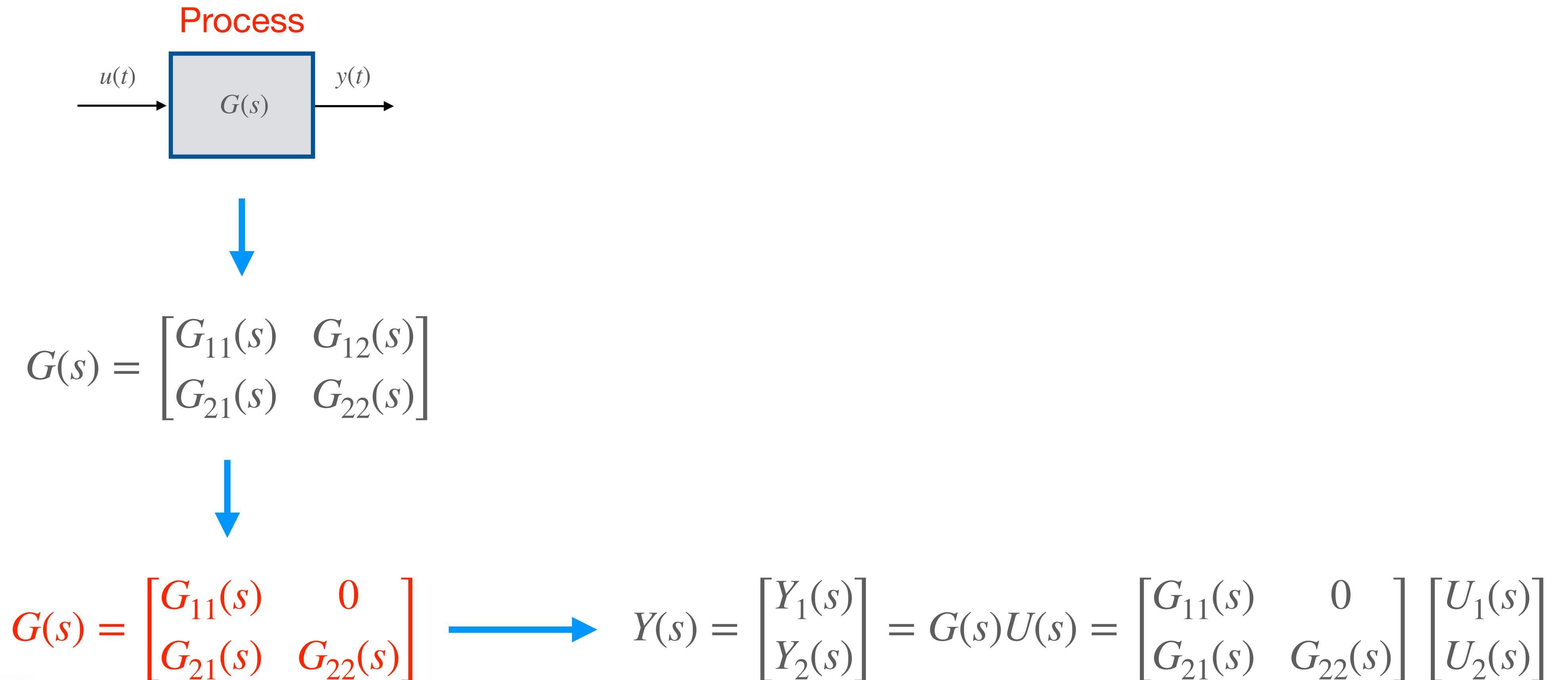
Decoupling Based Control Schemes



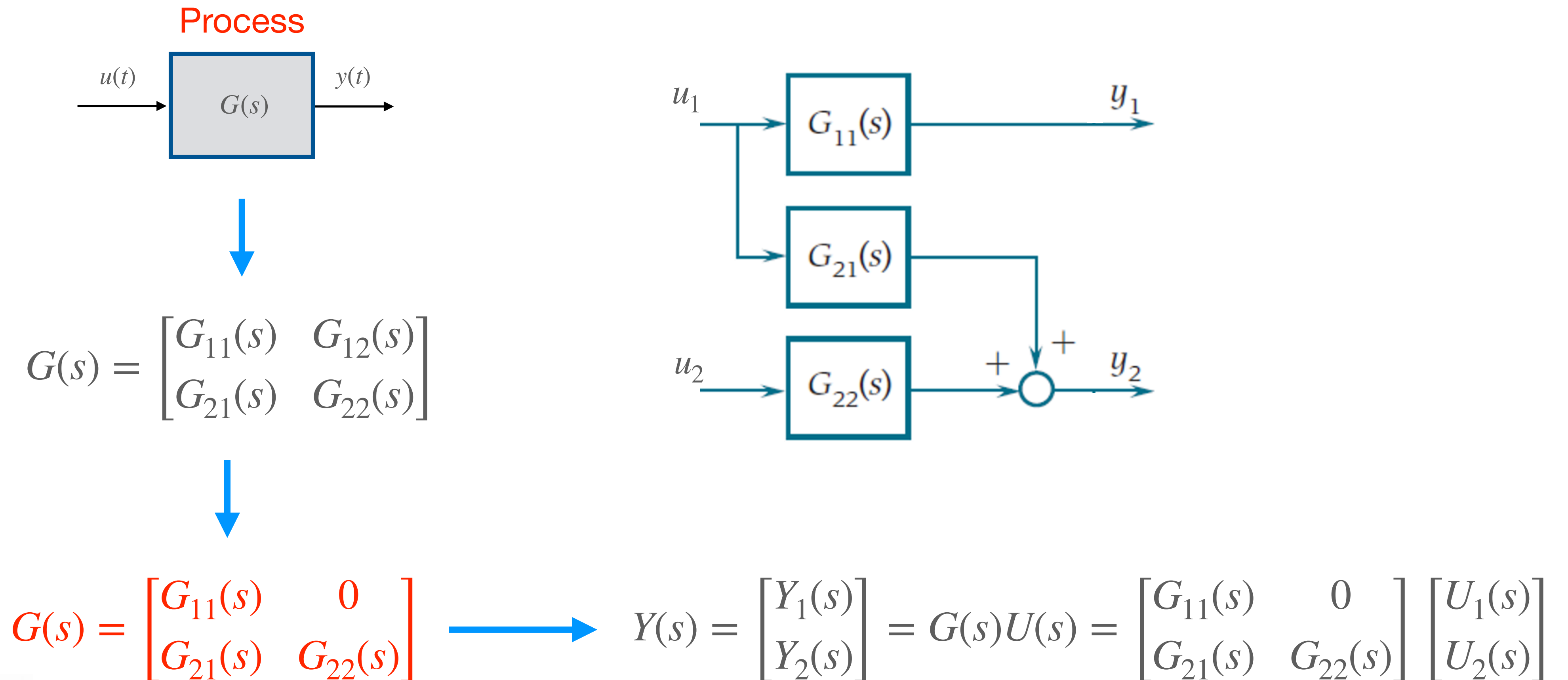
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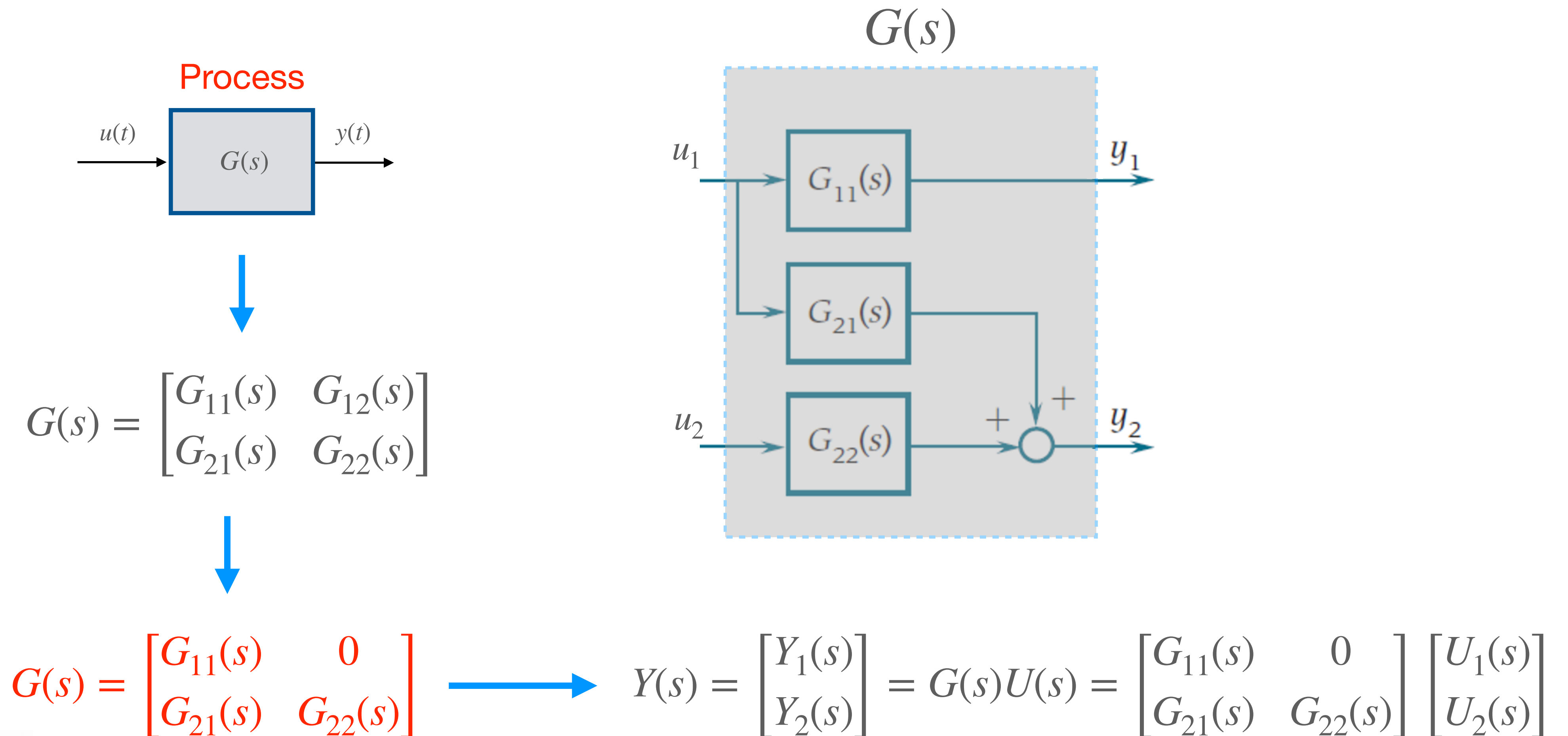
Decoupling Based Control Schemes



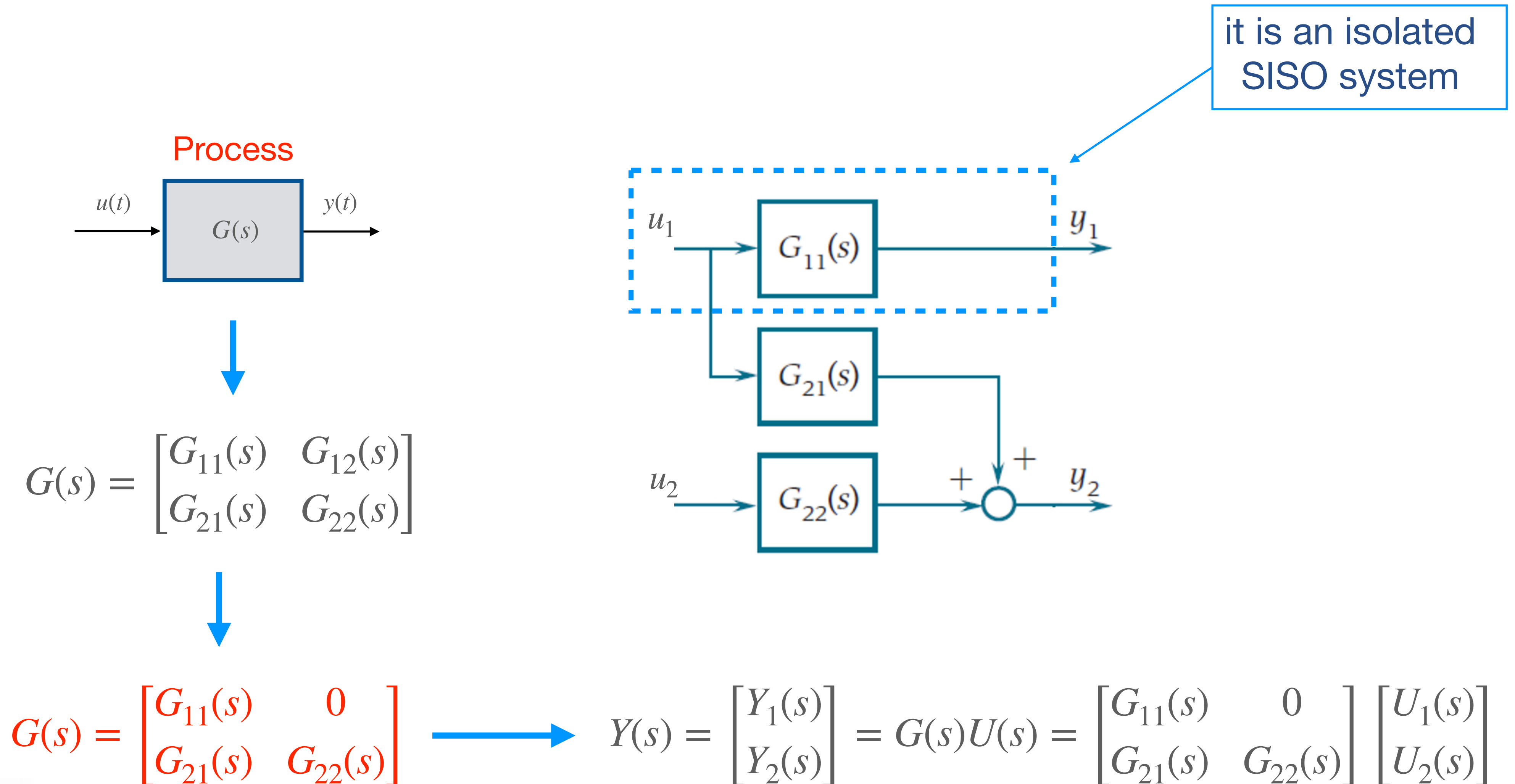
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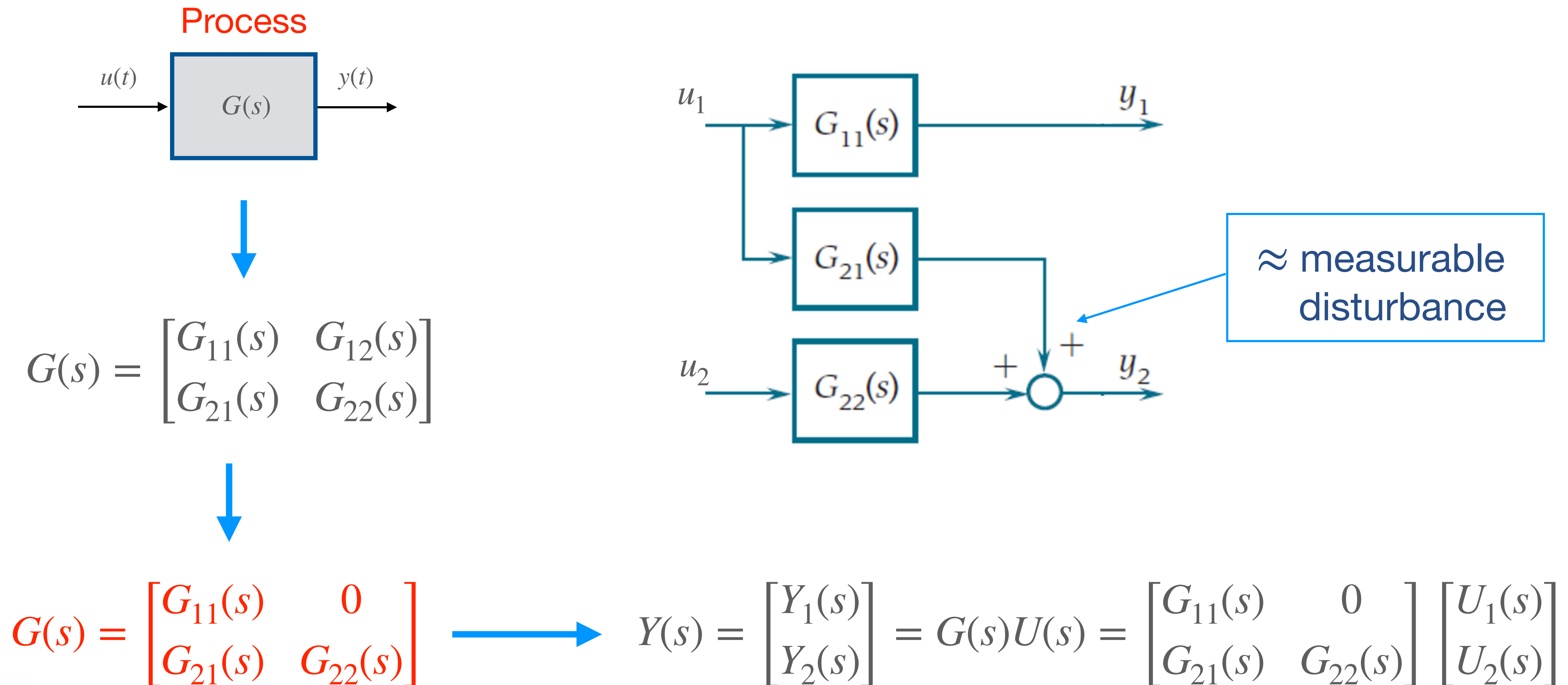
Decoupling Based Control Schemes



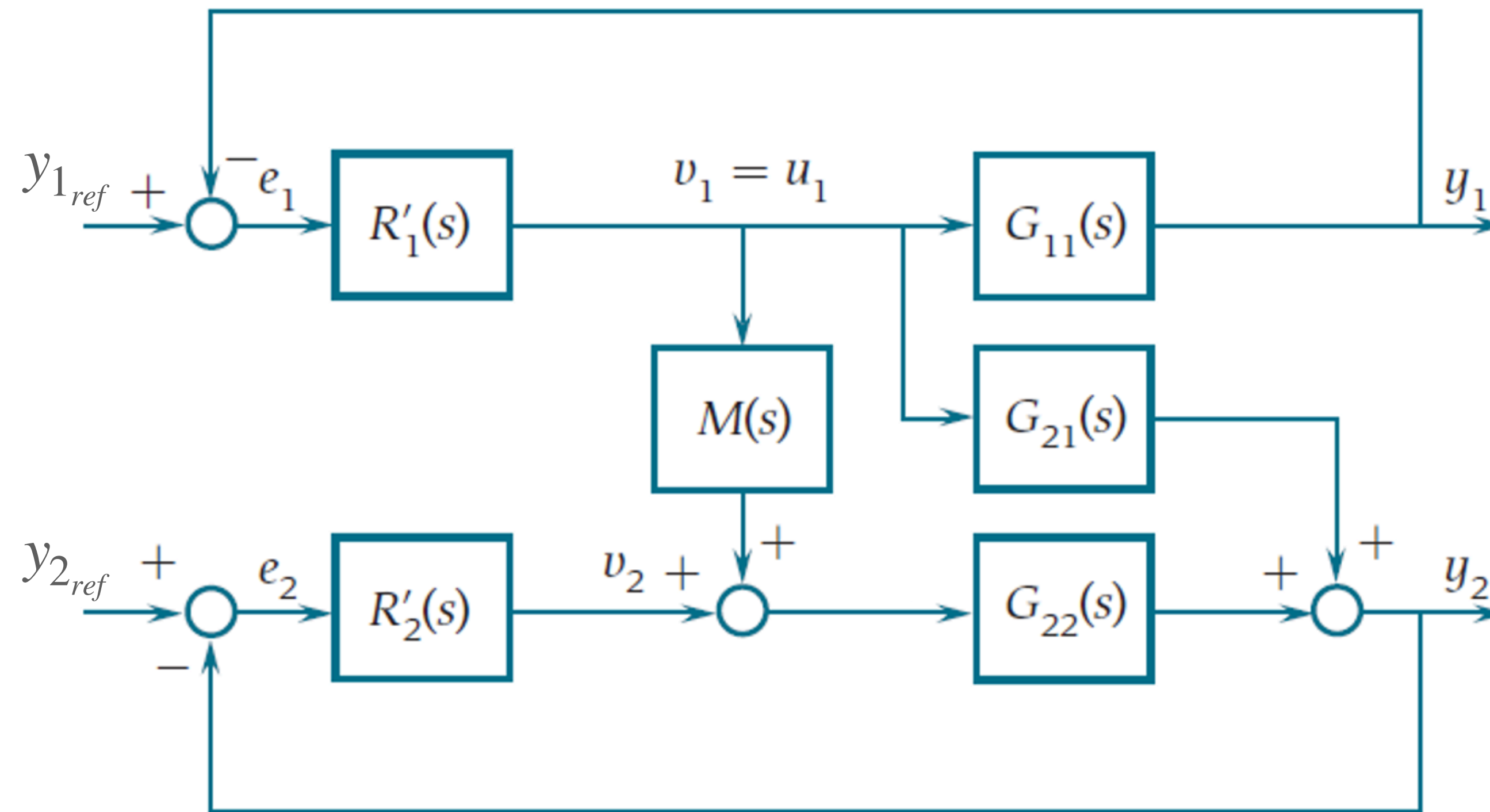
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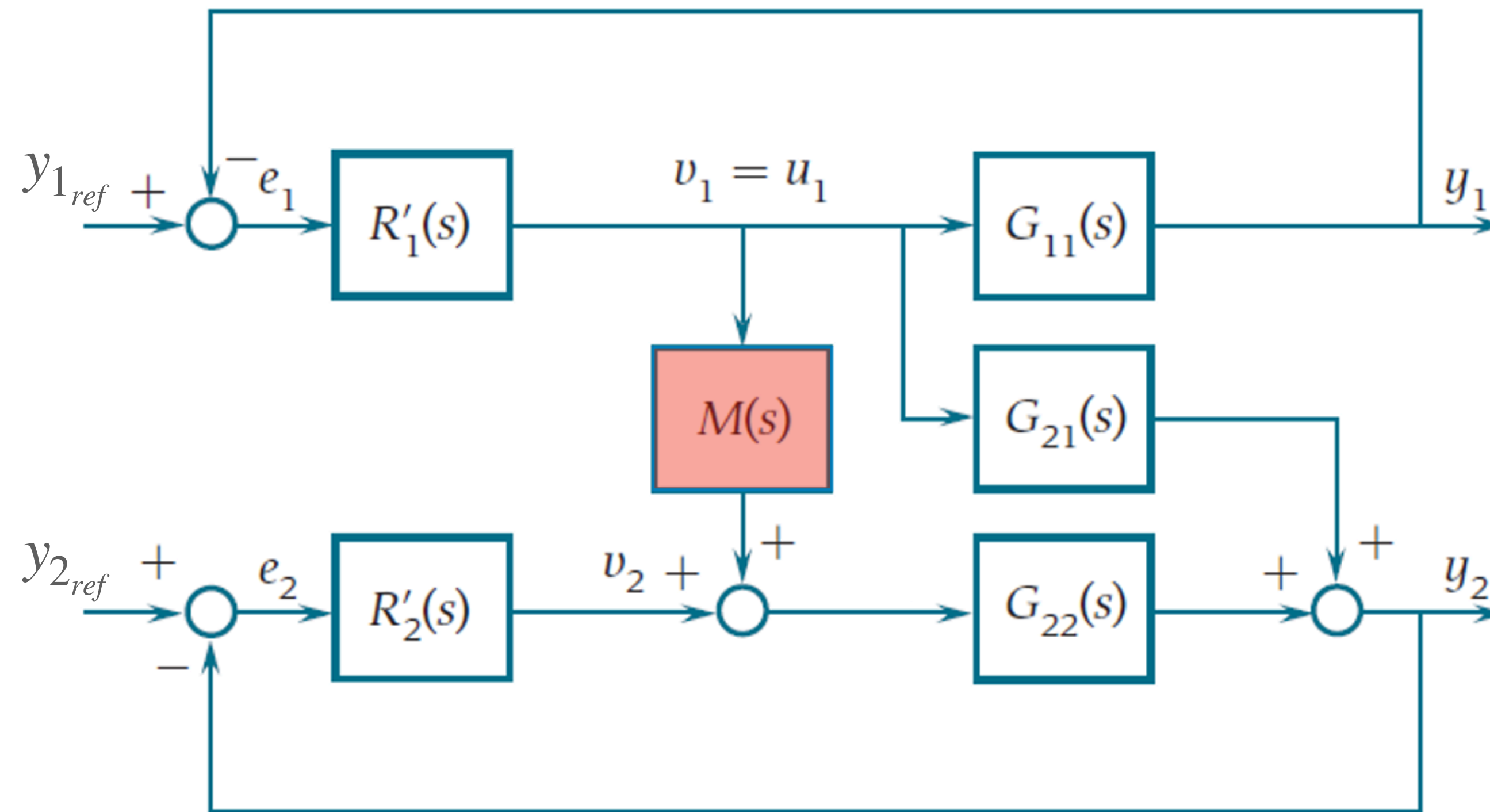
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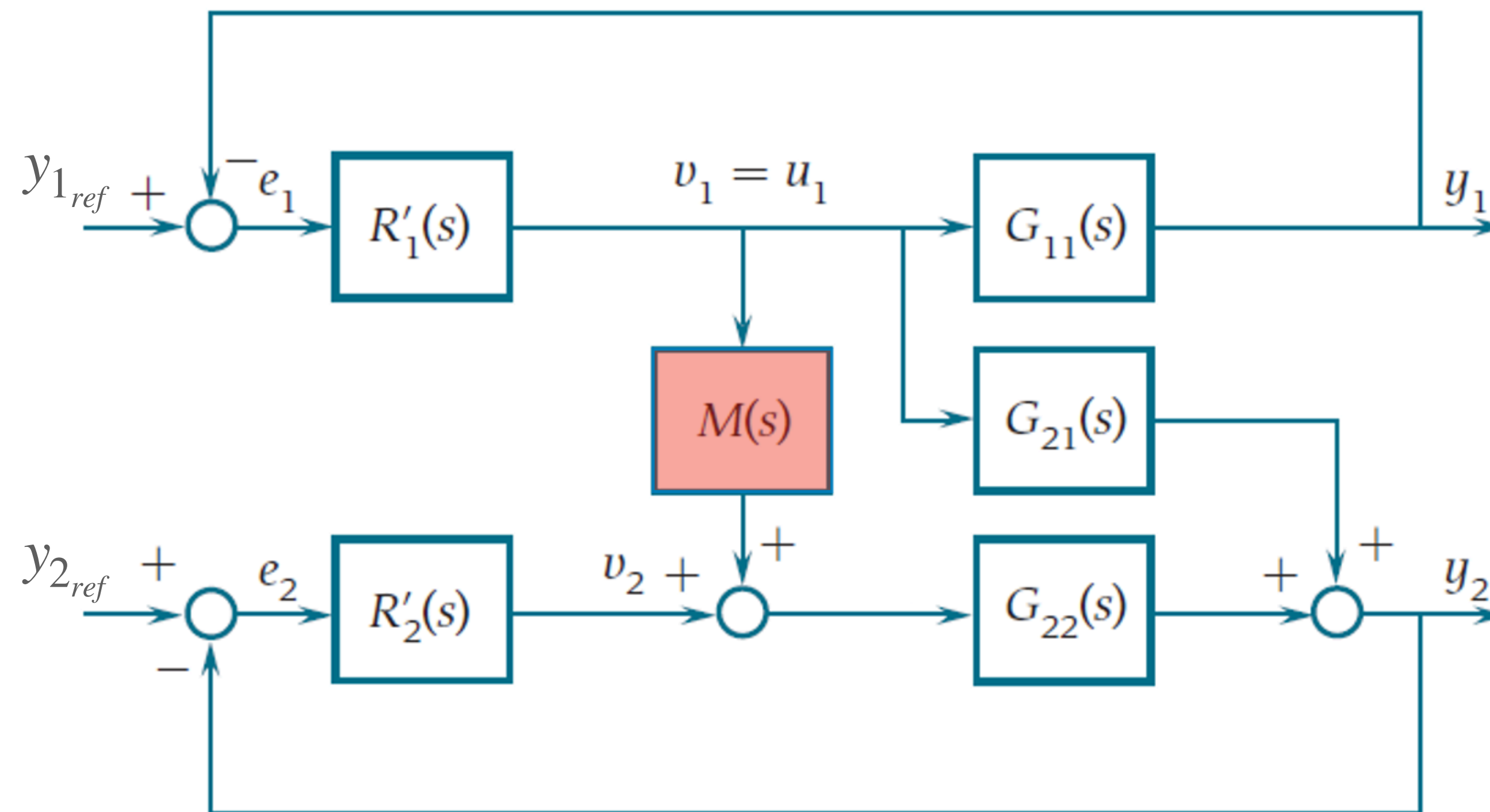
Decoupling Based Control Schemes



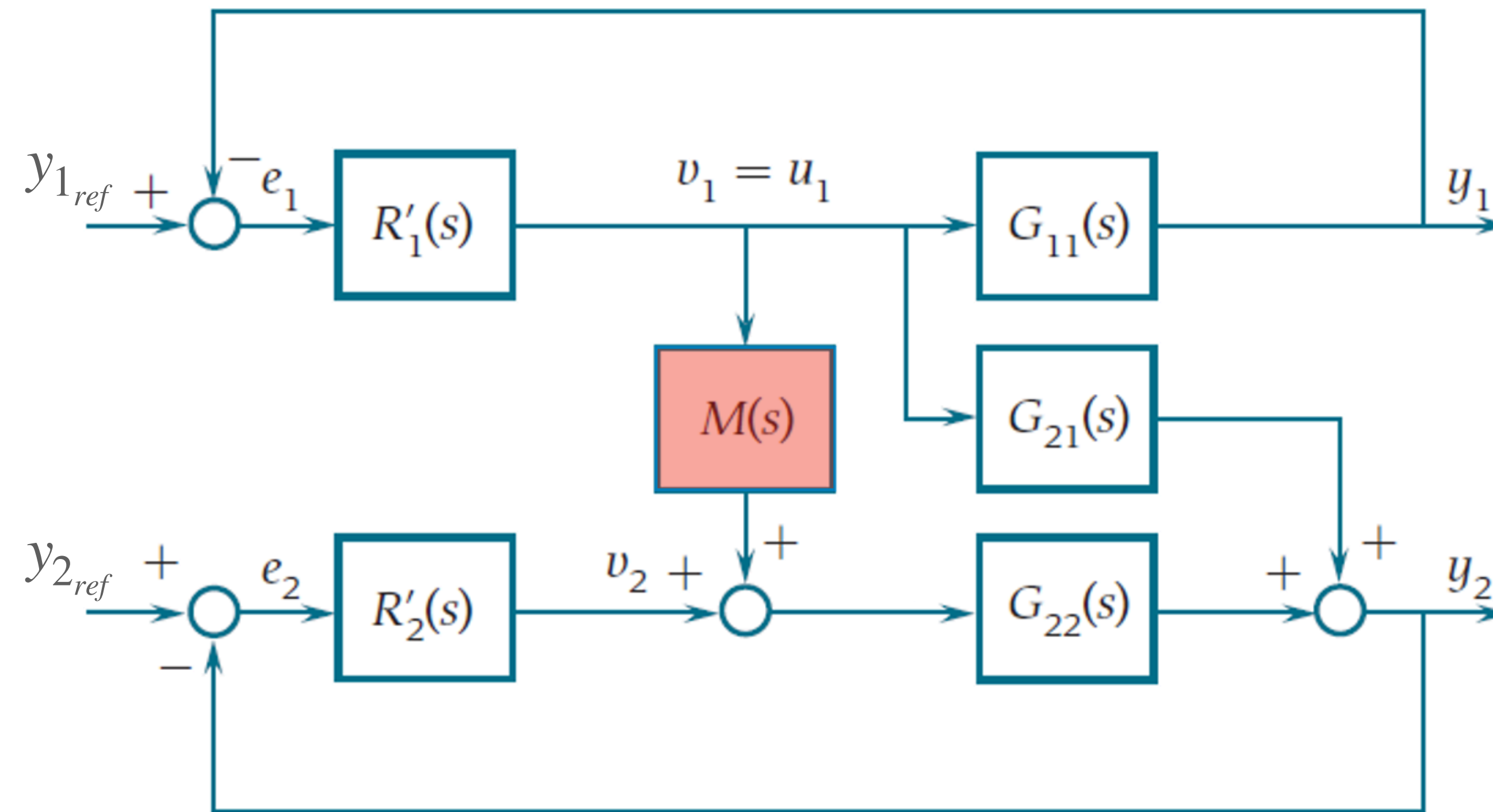
Decoupling Based Control Schemes



Decoupling Based Control Schemes



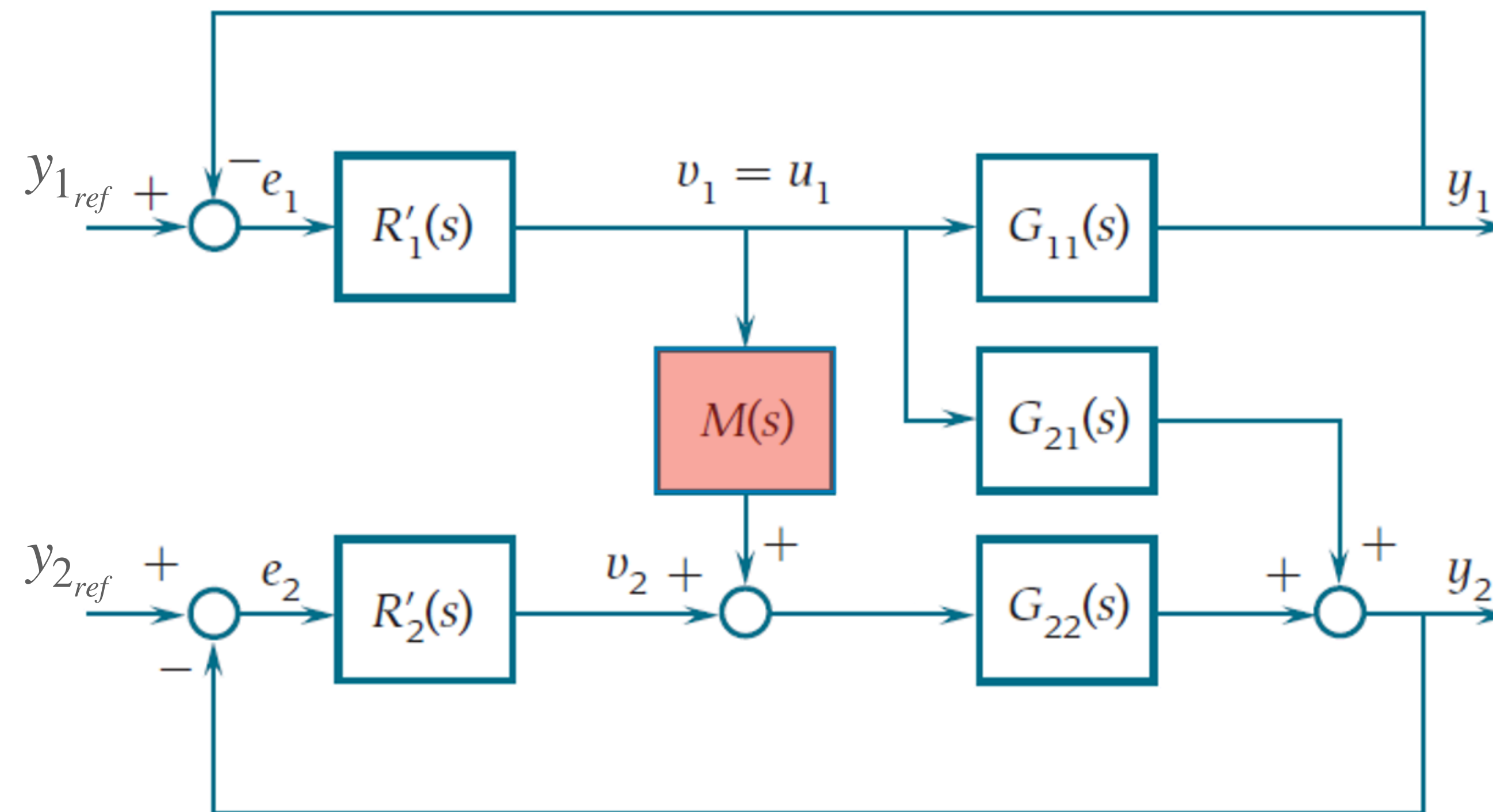
Decoupling Based Control Schemes



$$M(s)G_{22}(s) + G_{21}(s) = 0$$

$$M(s) = -\frac{G_{21}(s)}{G_{22}(s)}$$

Decoupling Based Control Schemes

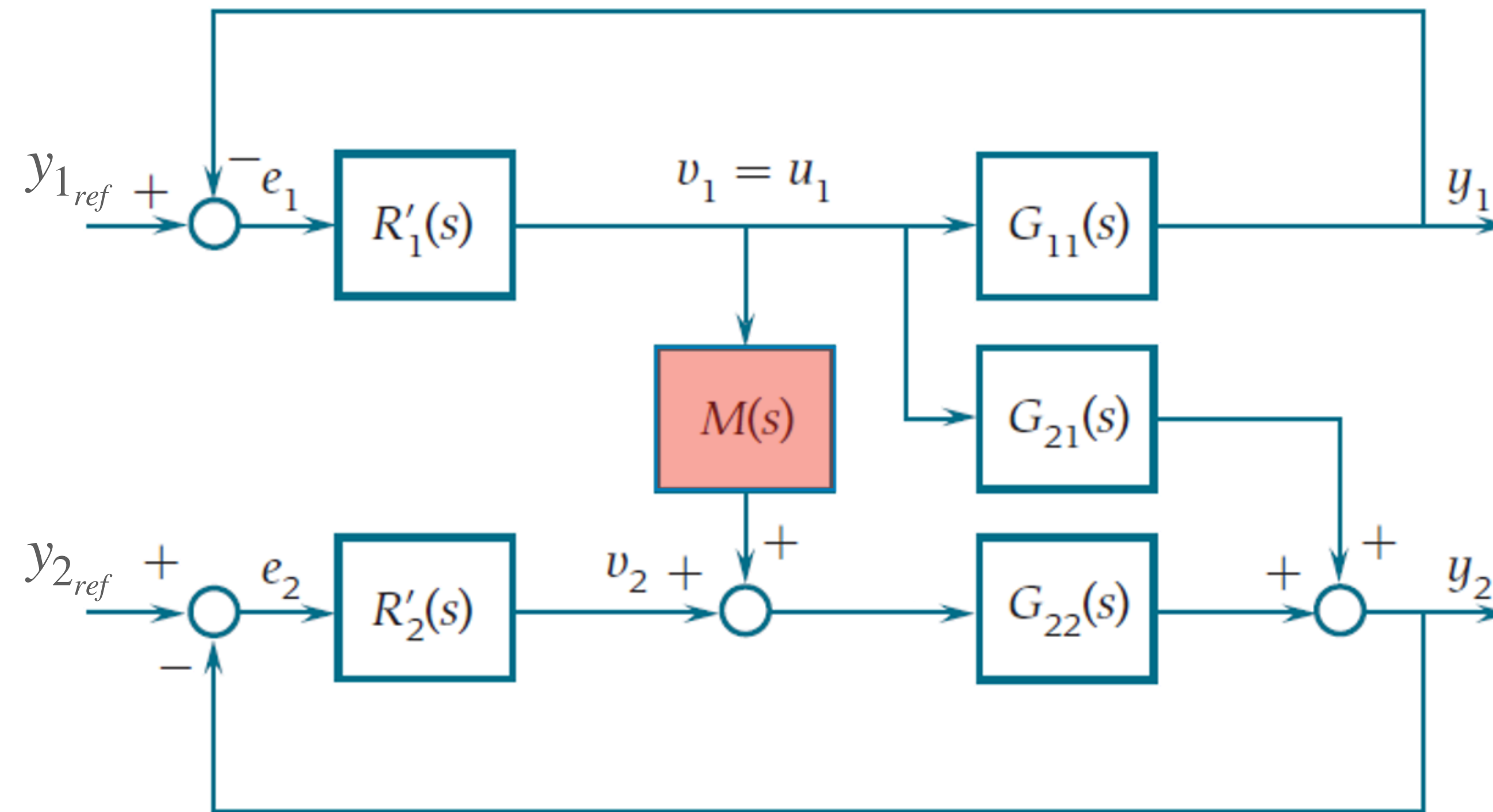


$$M(s)G_{22}(s) + G_{21}(s) = 0$$

$$M(s) = -\frac{G_{21}(s)}{G_{22}(s)}$$

or an approximation of it
in the frequency domain
if the system is not causal

Decoupling Based Control Schemes



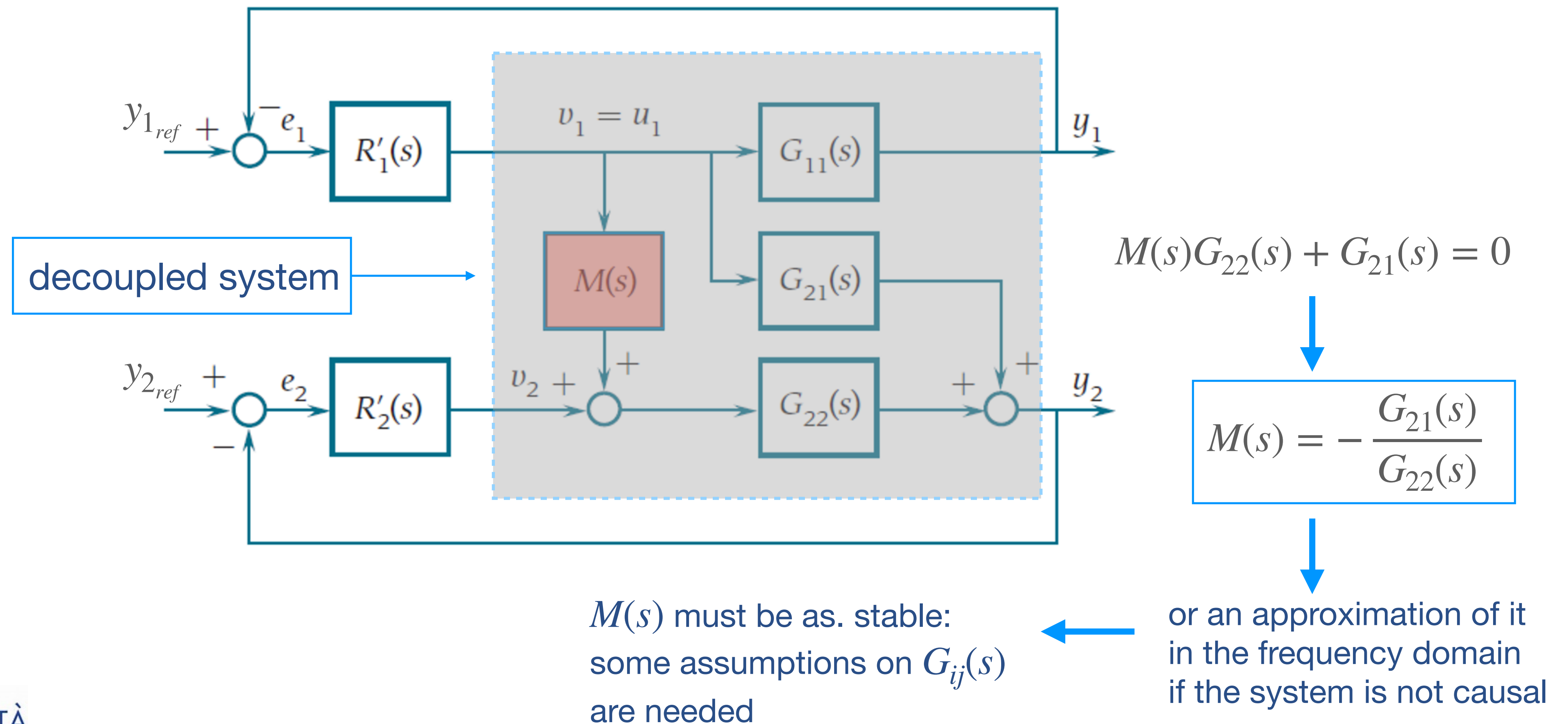
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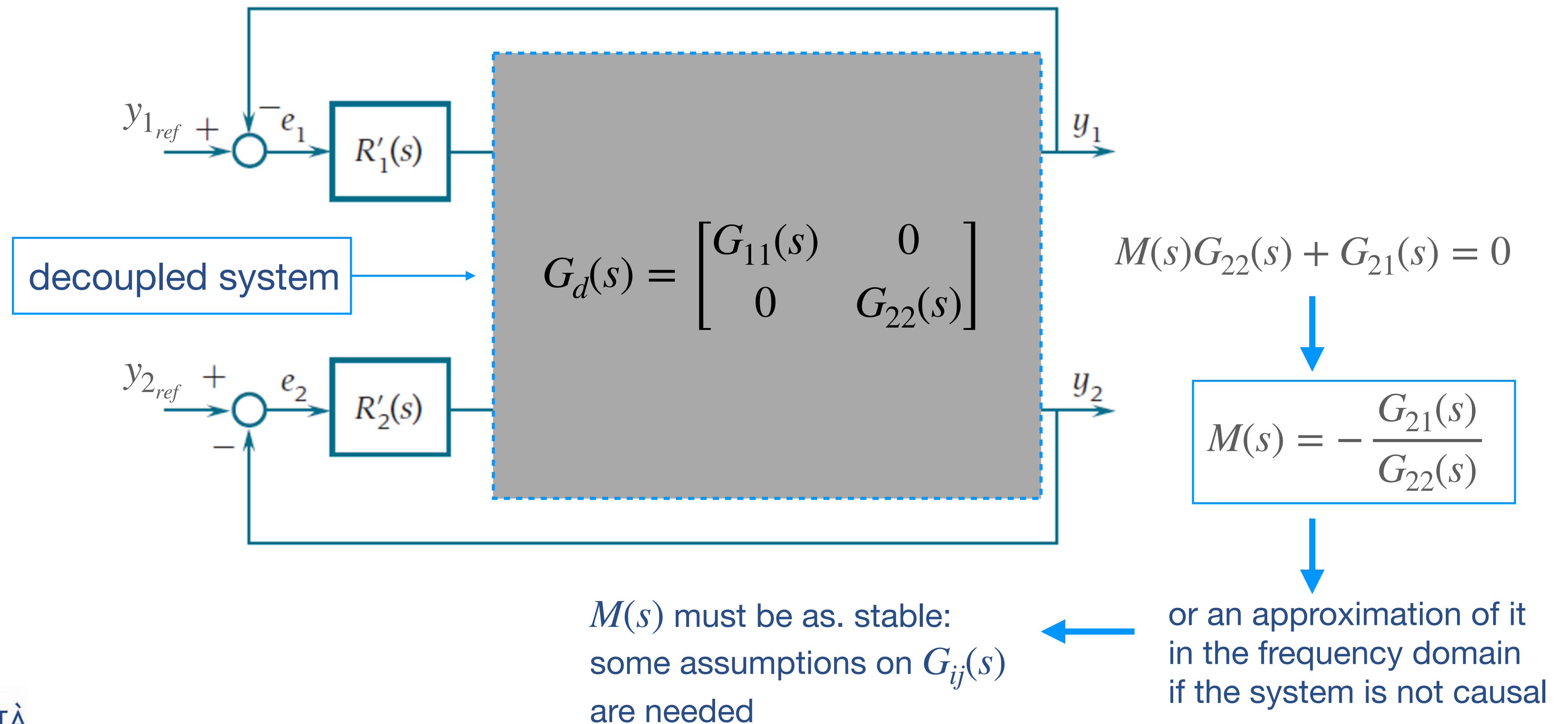
$M(s)$ must be as. stable:
some assumptions on $G_{ij}(s)$
are needed

or an approximation of it
in the frequency domain
if the system is not causal

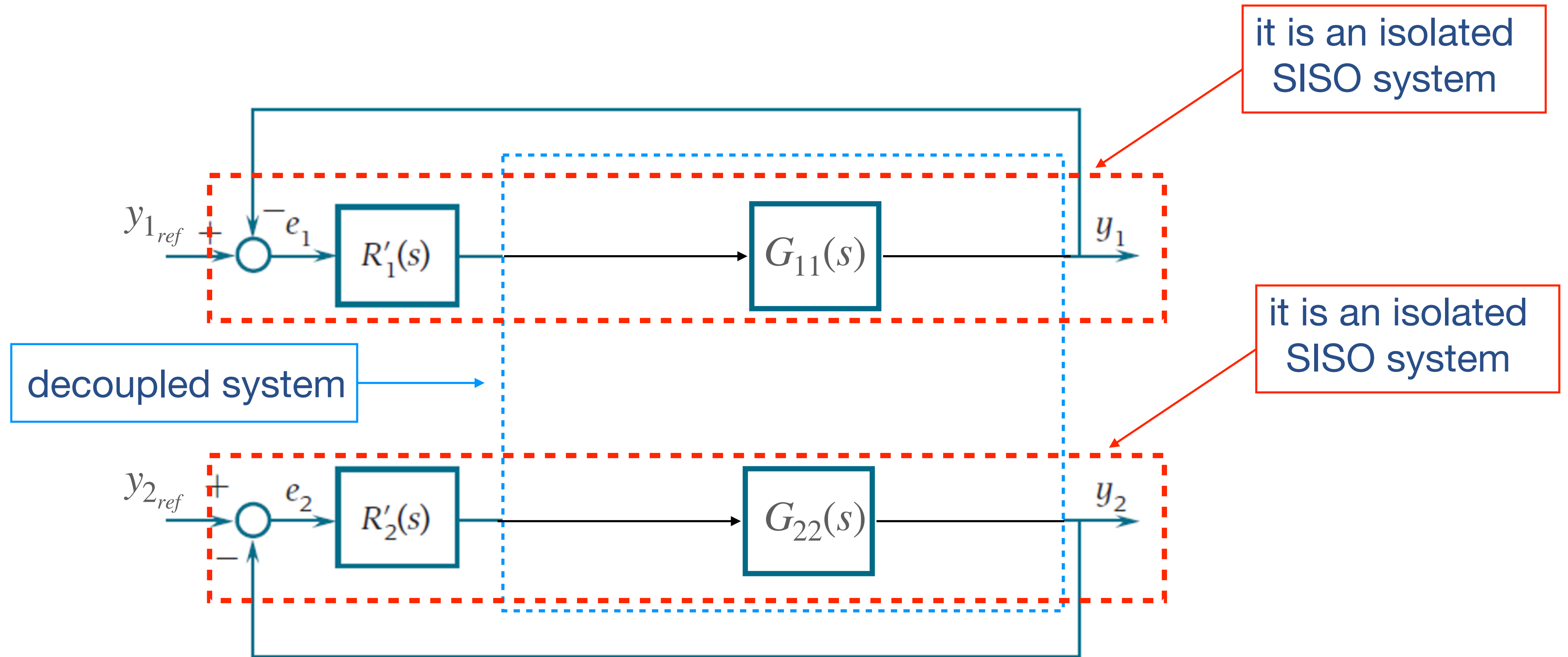
Decoupling Based Control Schemes



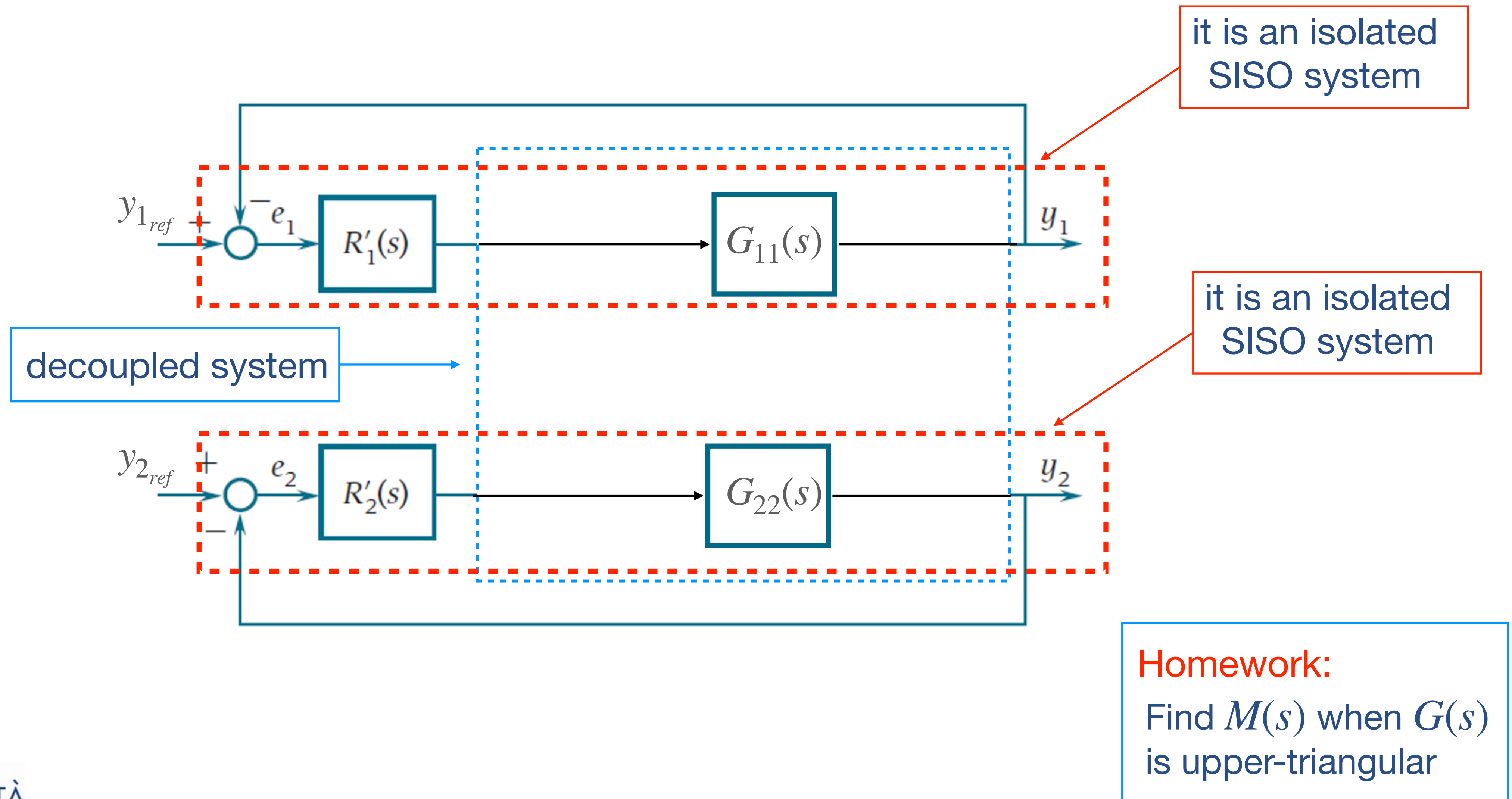
Decoupling Based Control Schemes



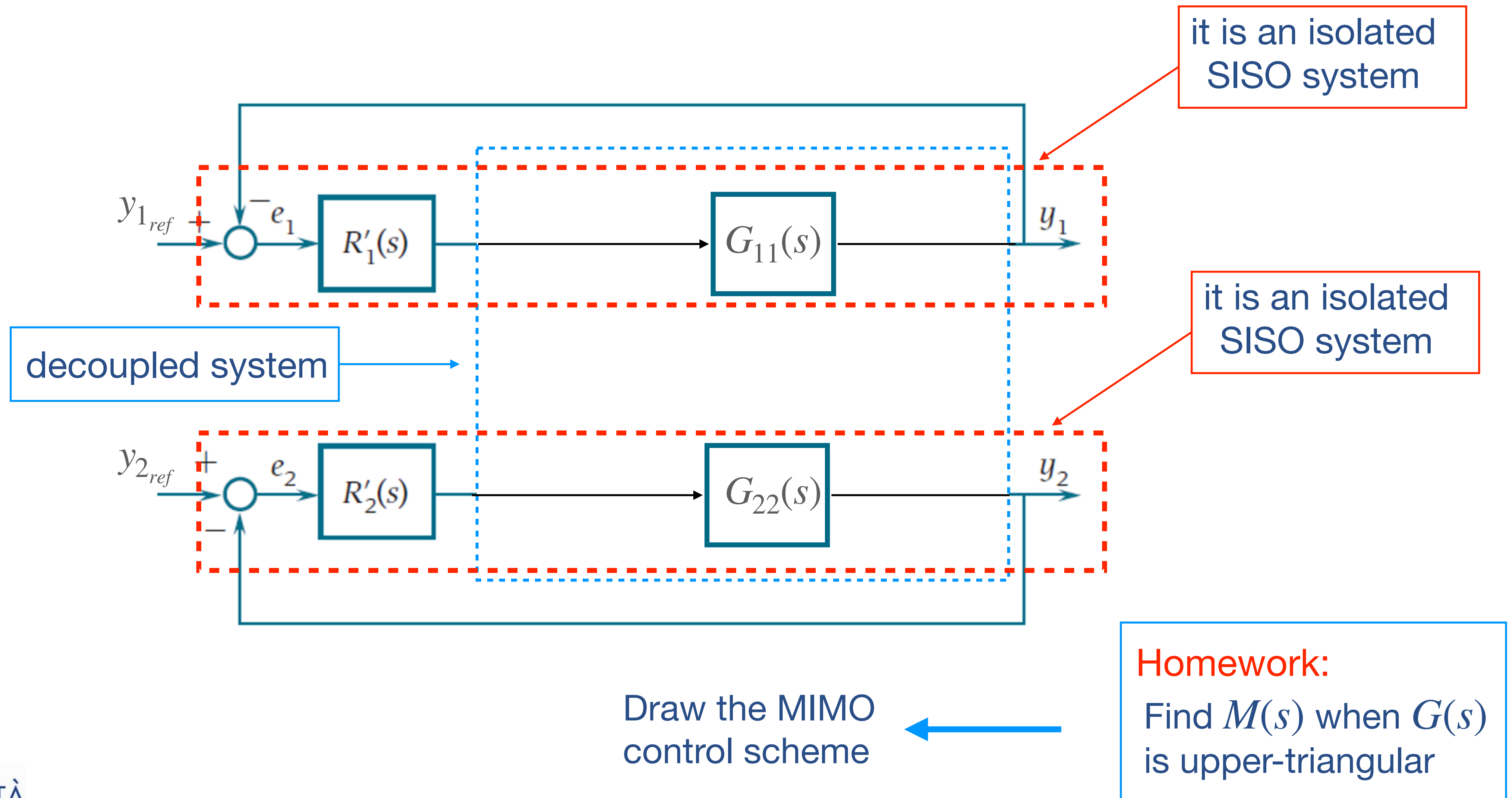
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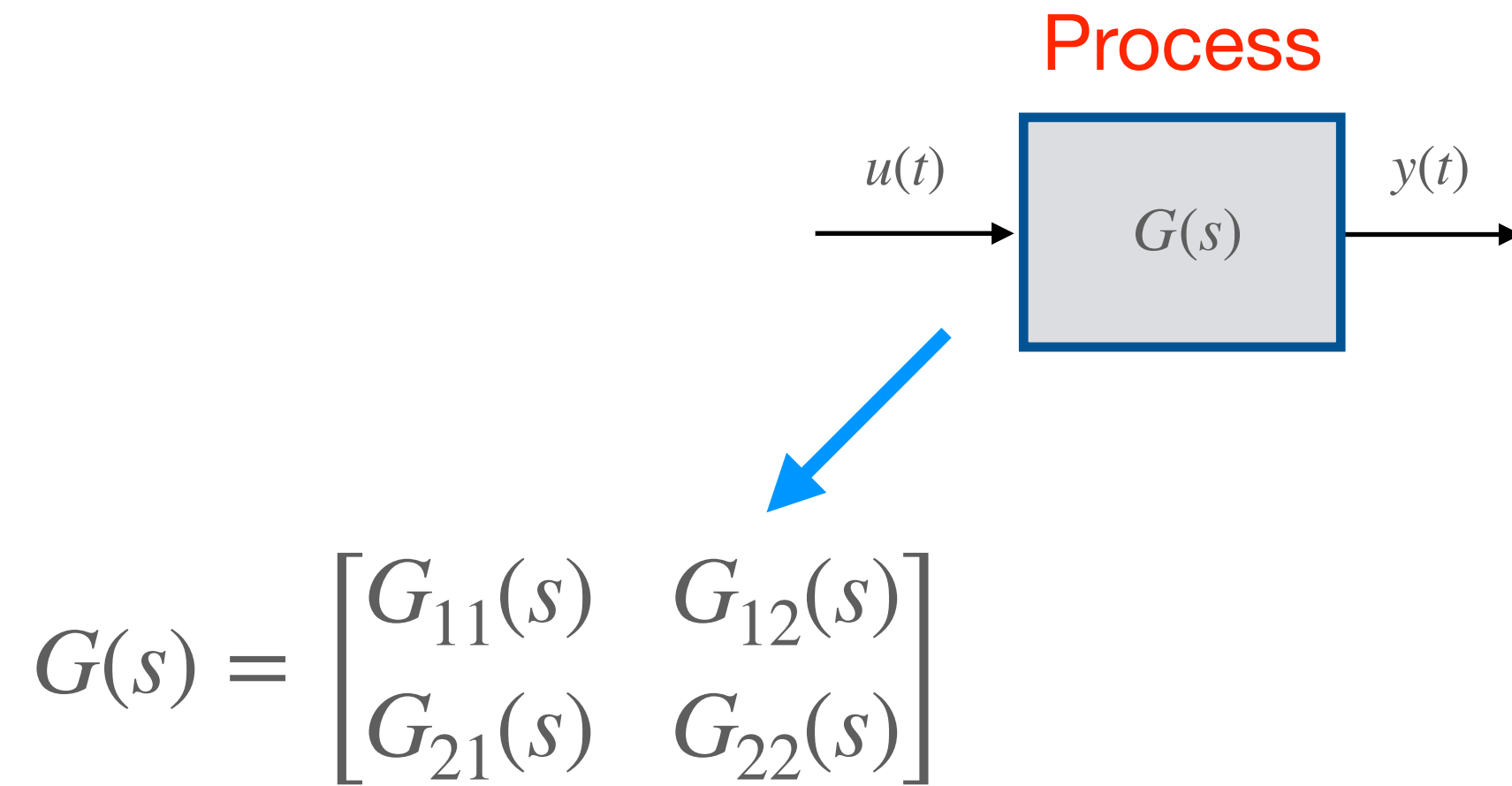
Decoupling Based Control Schemes



Decoupling Based Control Schemes



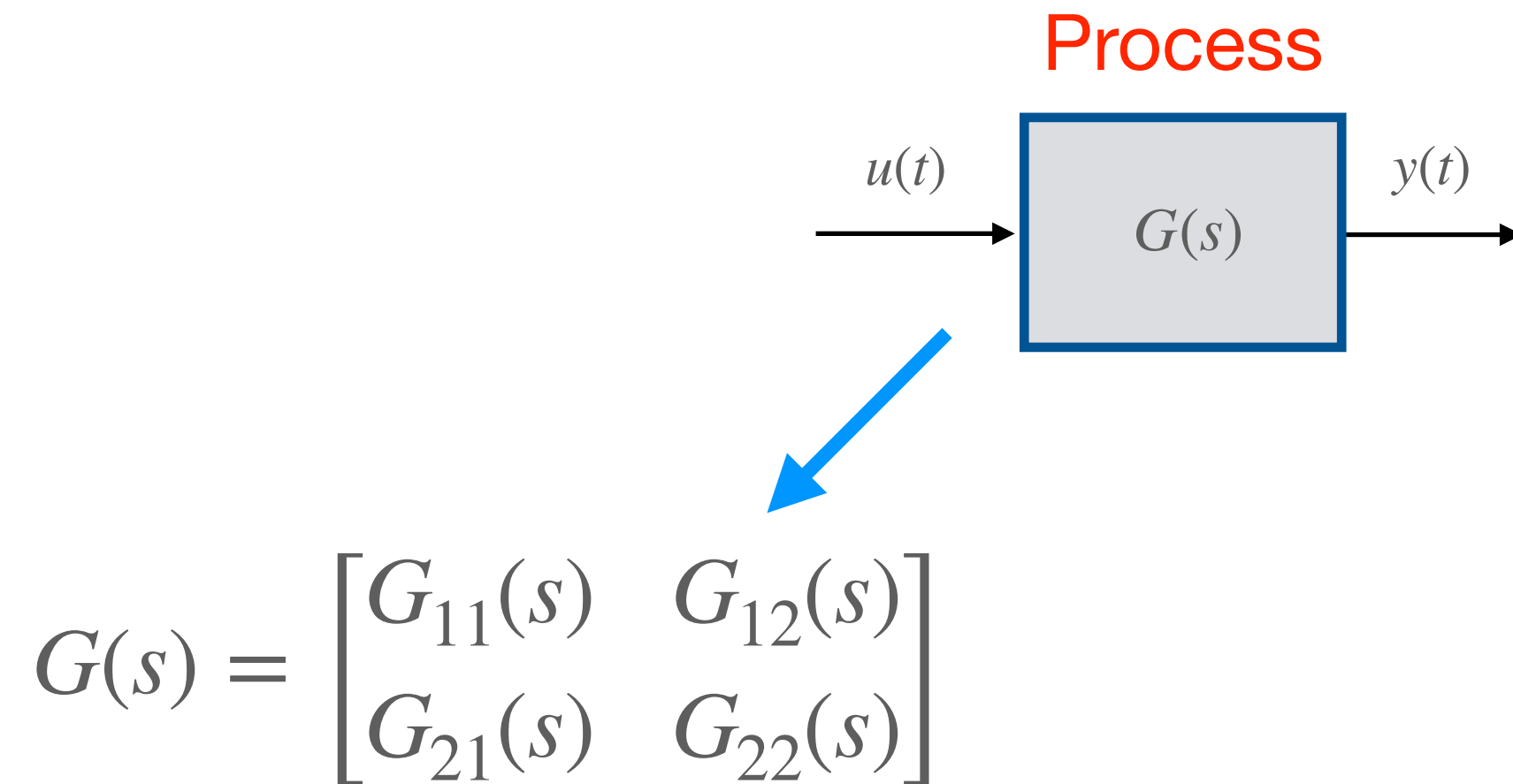
Decoupling Based Control Schemes



Assumptions:

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- $G(s)$ full matrix

Decoupling Based Control Schemes



Assumptions:

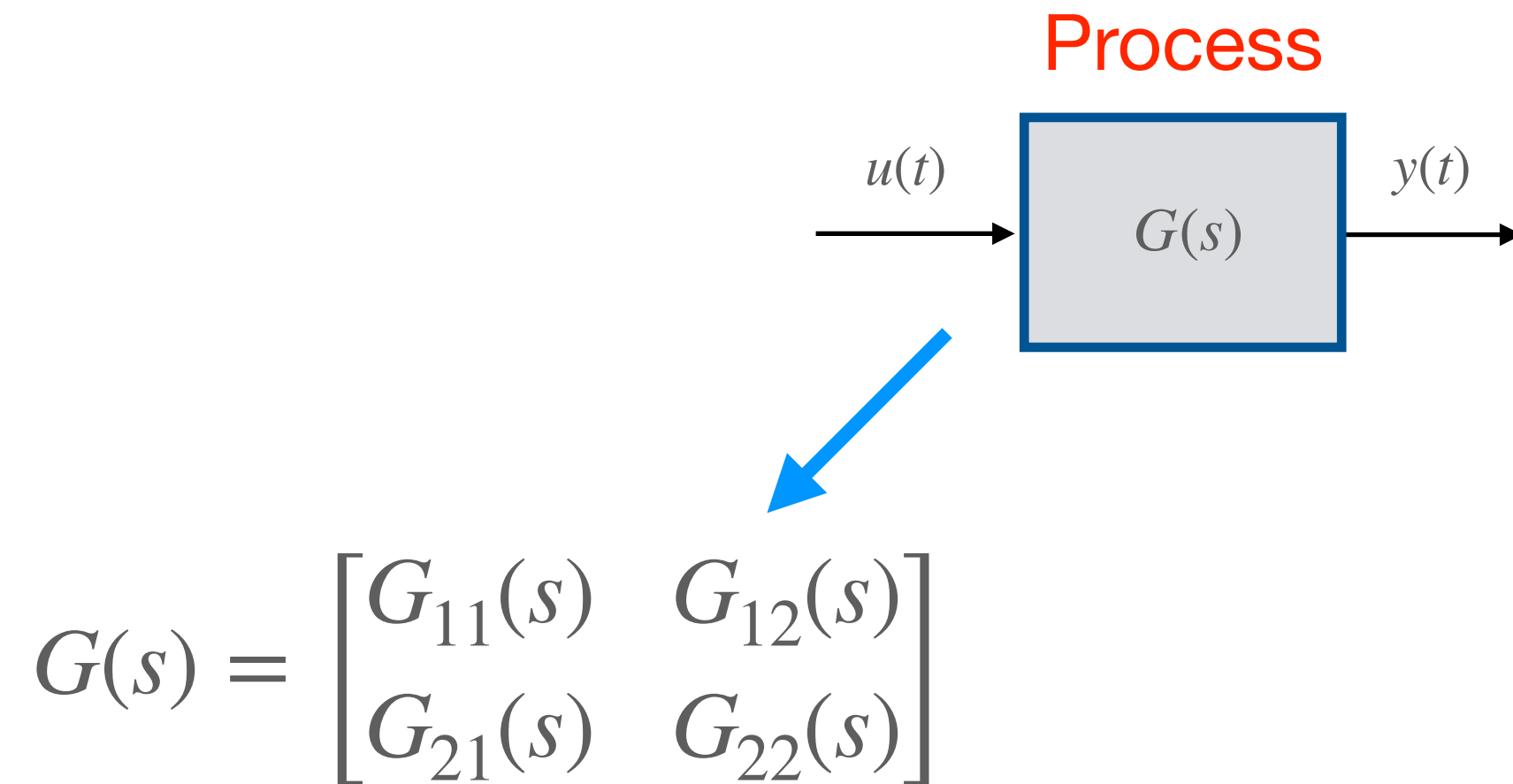
- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix



Sufficient conditions:

- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$

Decoupling Based Control Schemes



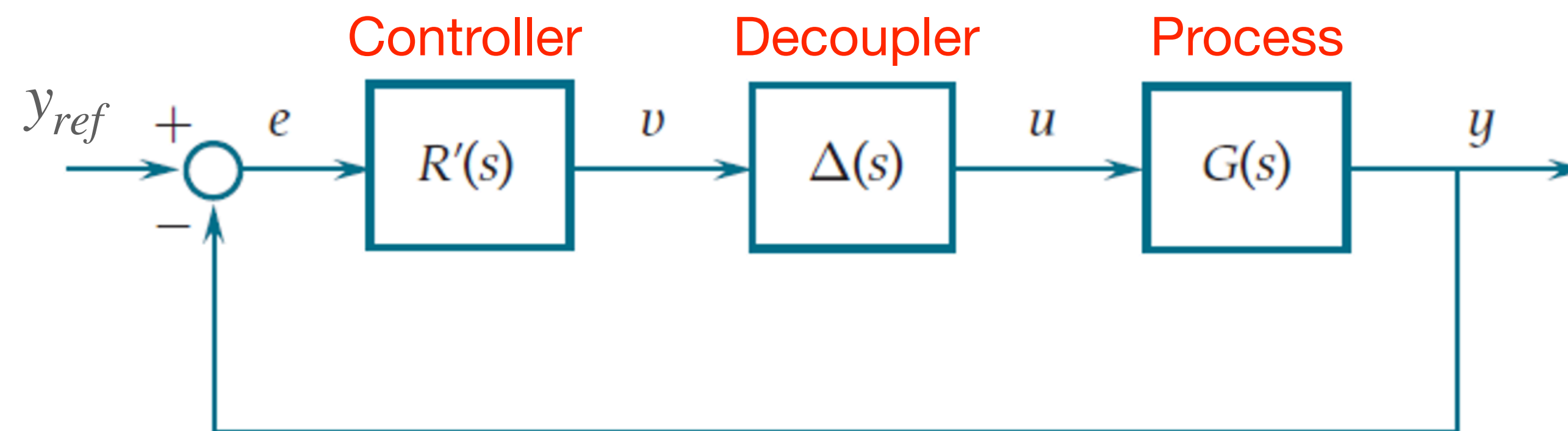
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- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix

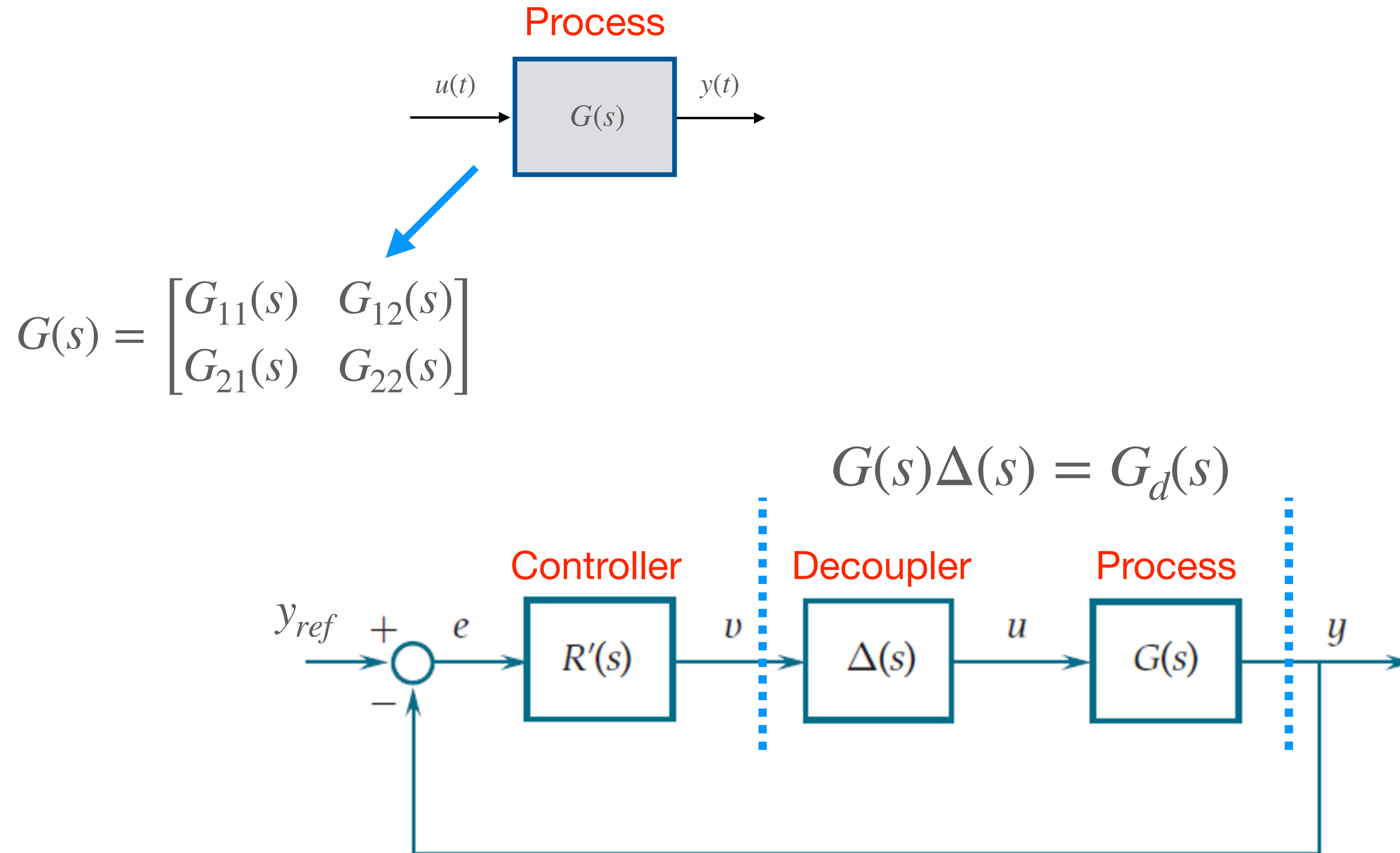


Sufficient conditions:

- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$



Decoupling Based Control Schemes



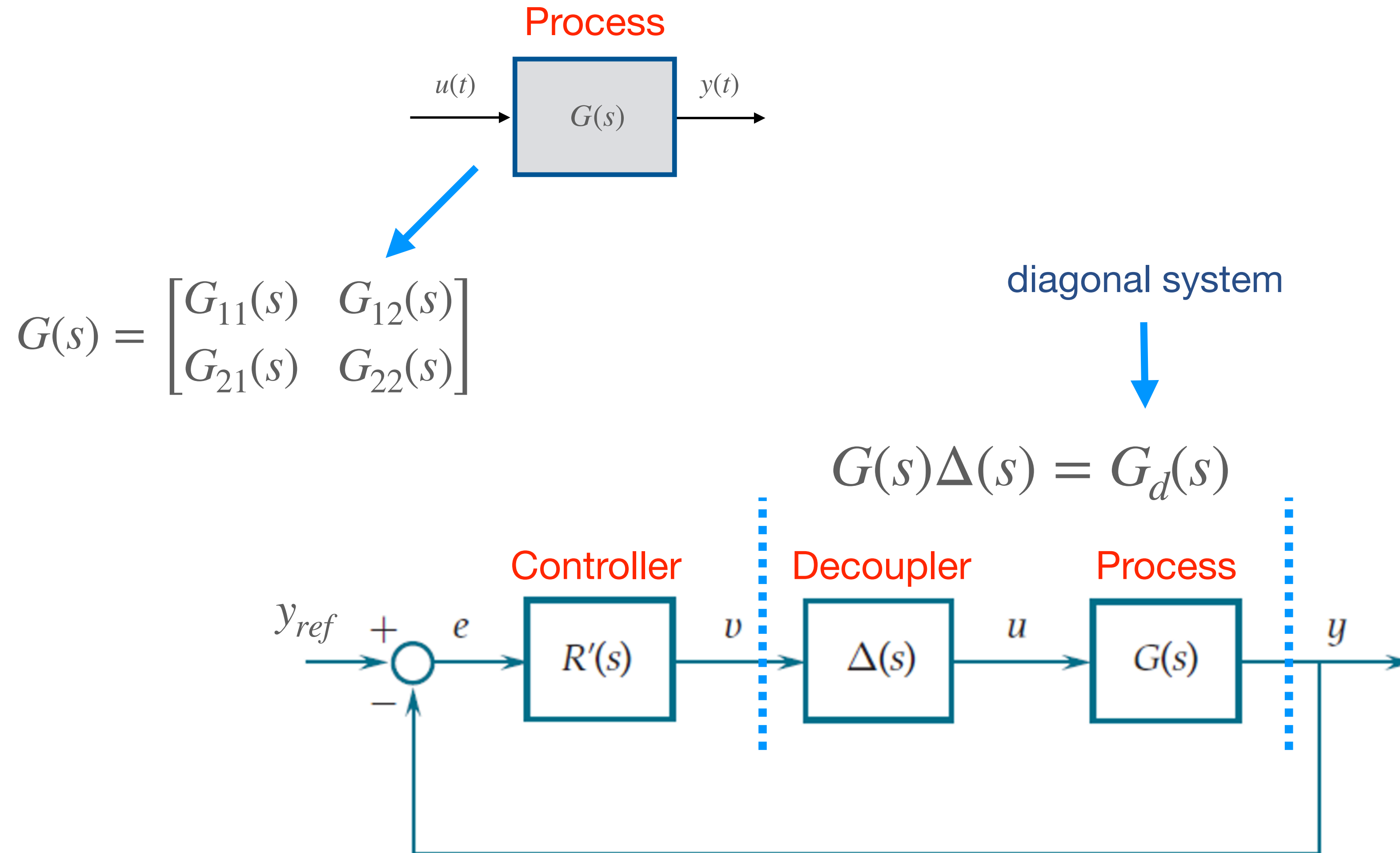
Assumptions:

- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix

Sufficient conditions:

- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$

Decoupling Based Control Schemes



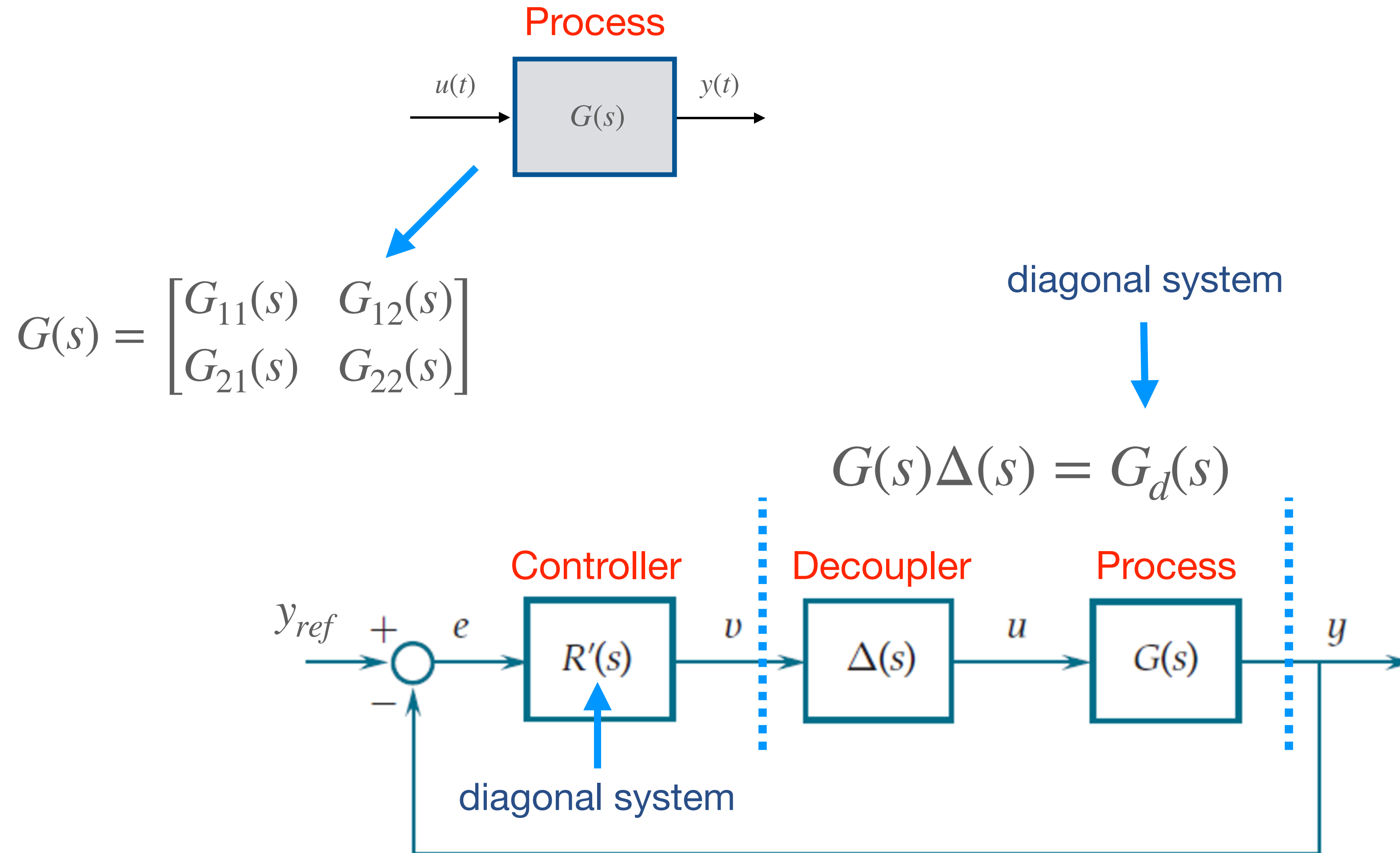
Assumptions:

- $G(s) \in \mathbb{R}^{2 \times 2}$
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Sufficient conditions:

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Decoupling Based Control Schemes



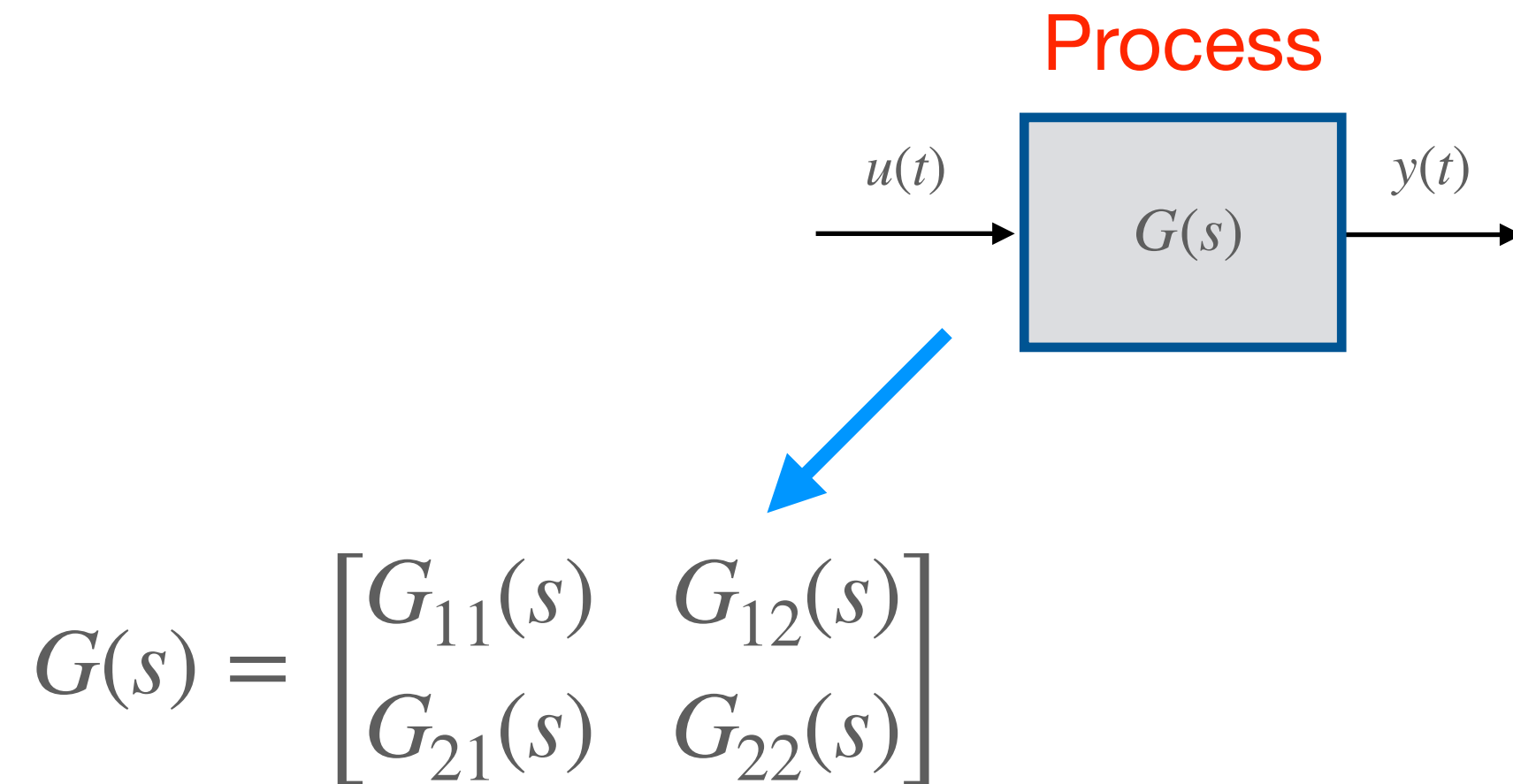
Assumptions:

- $G(s) \in \mathbb{R}^{2 \times 2}$
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Sufficient conditions:

- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$

Decoupling Based Control Schemes



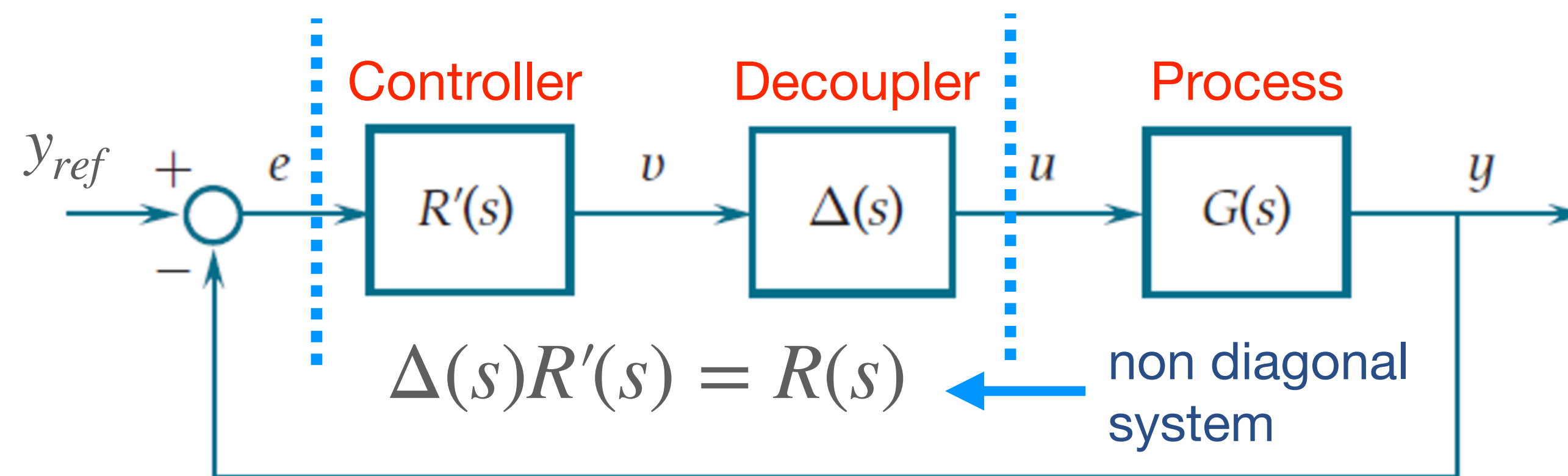
Assumptions:

- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix

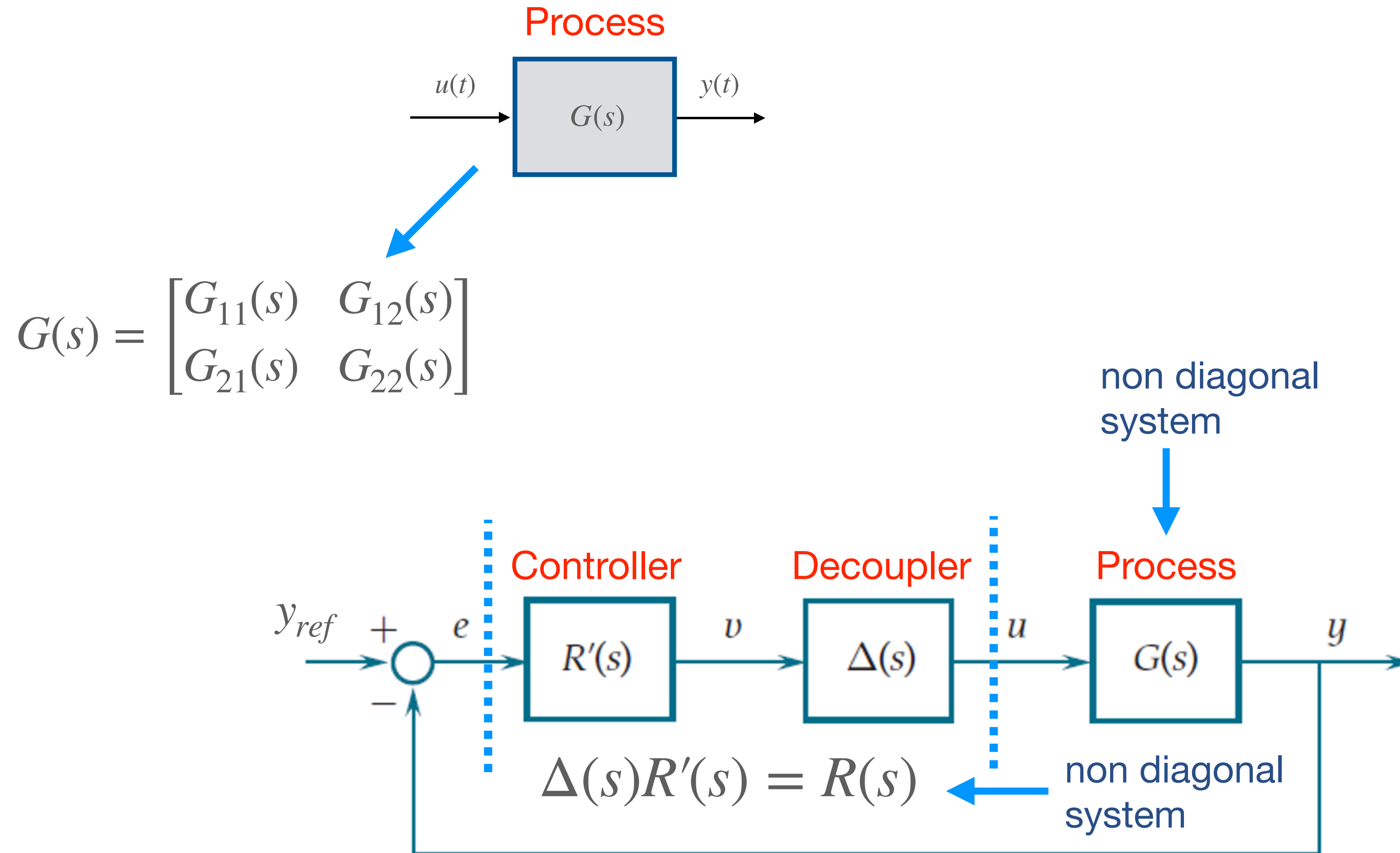


Sufficient conditions:

- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$



Decoupling Based Control Schemes



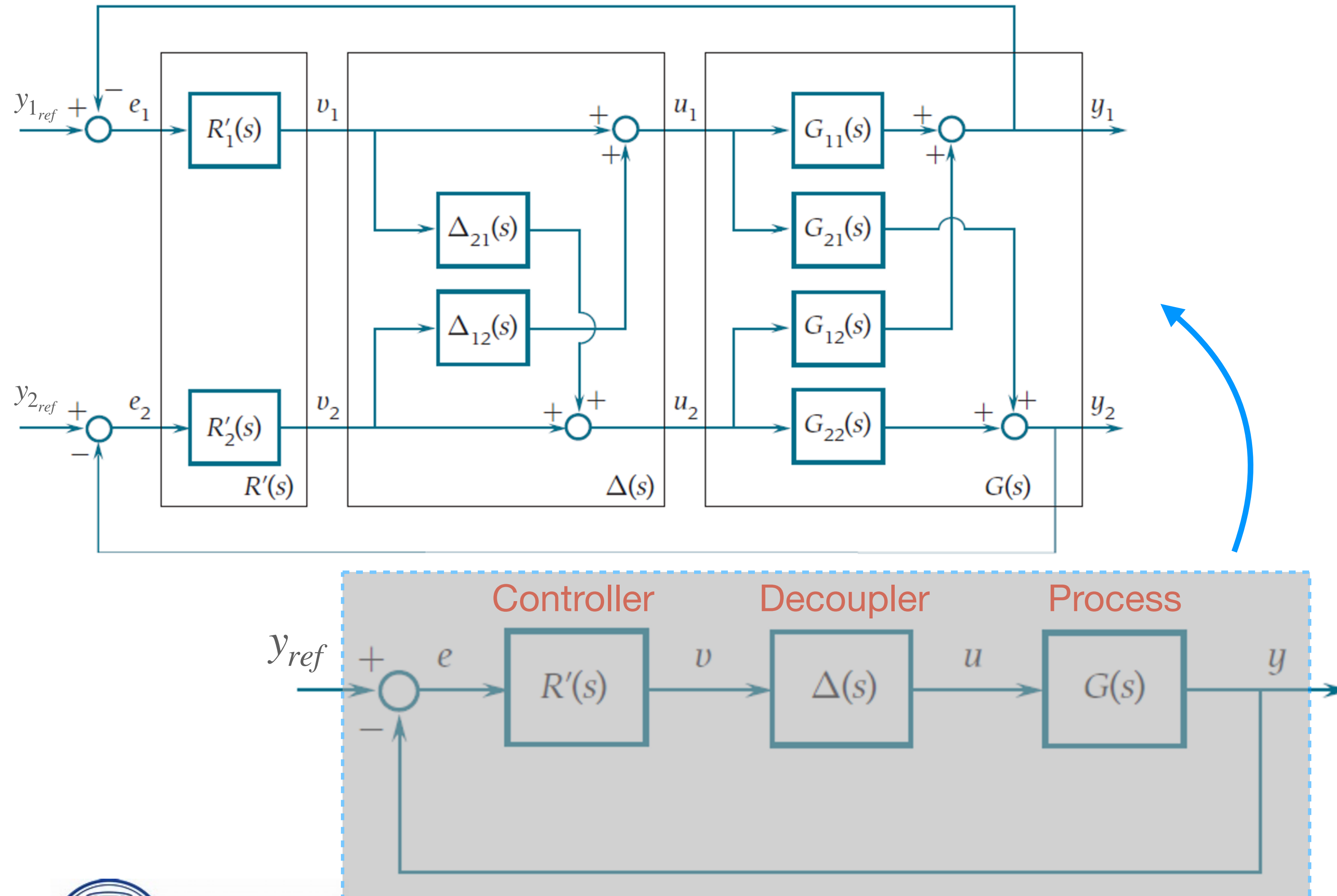
Assumptions:

- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix

Sufficient conditions:

- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$

Decoupling Based Control Schemes: **Forward Decoupling**



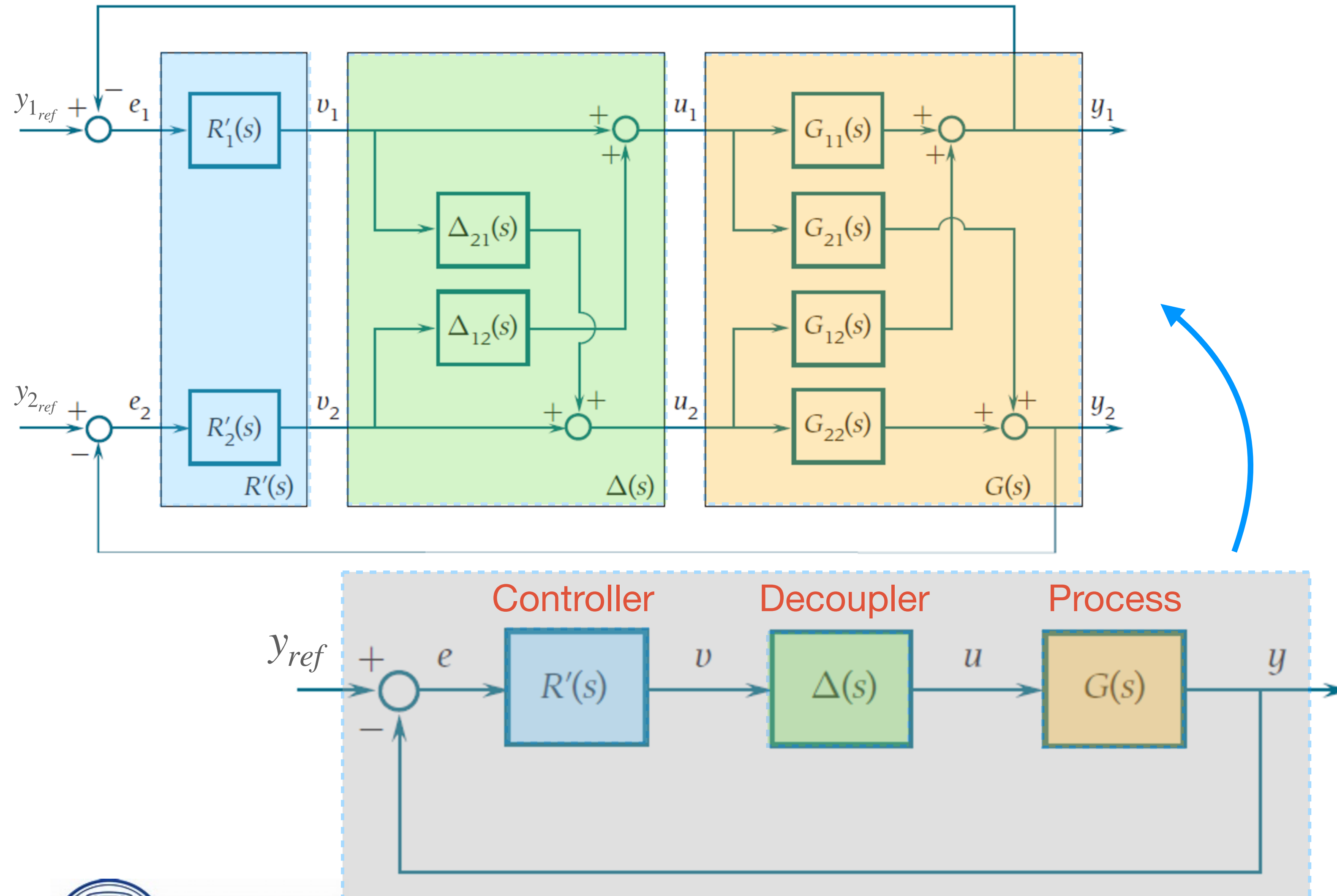
Assumptions:

- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix



Sufficient conditions:

- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \forall s$
s.t. $\text{Re}(s) \geq 0$

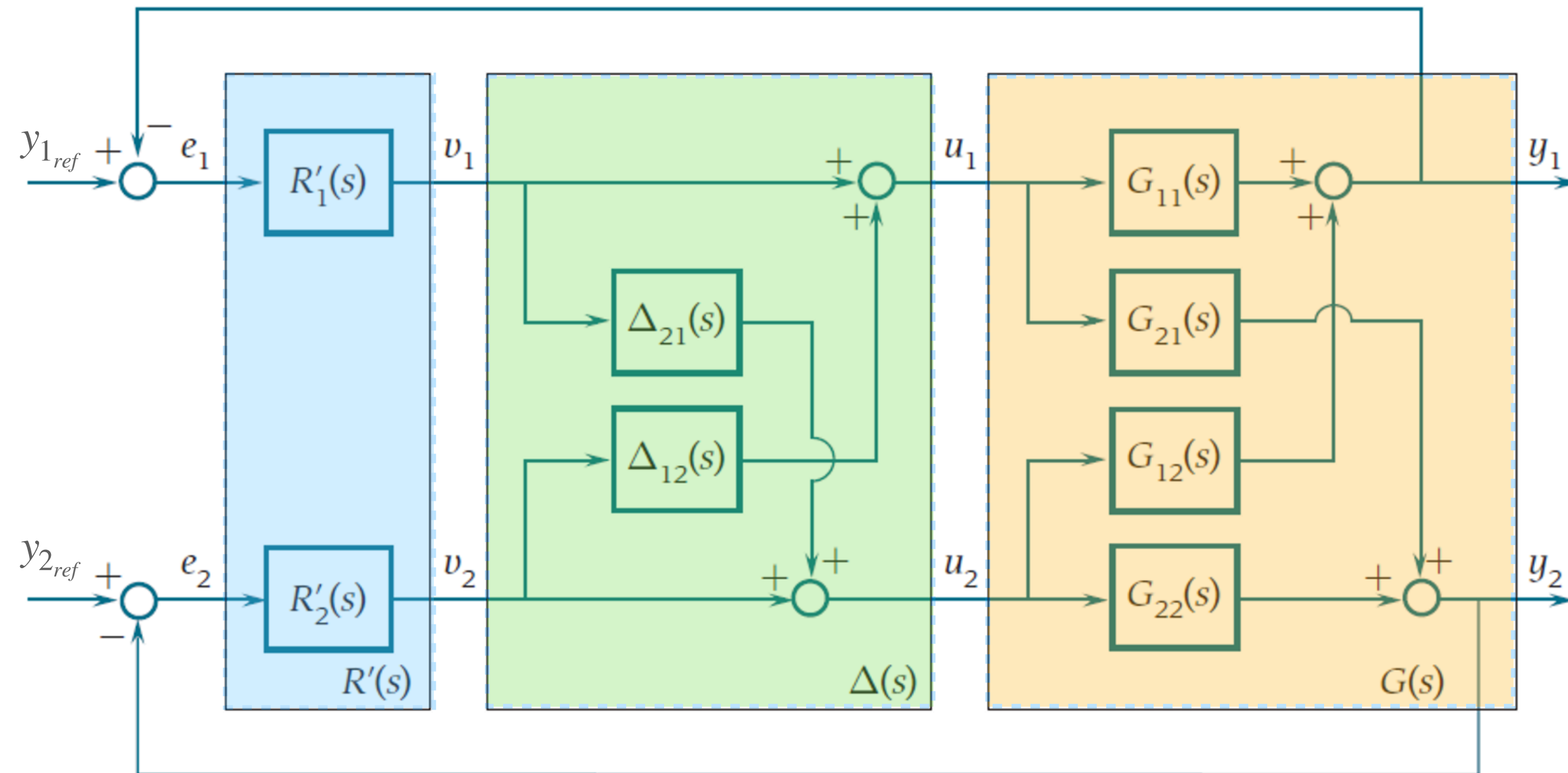
Decoupling Based Control Schemes: **Forward Decoupling****Assumptions:**

- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix

**Sufficient conditions:**

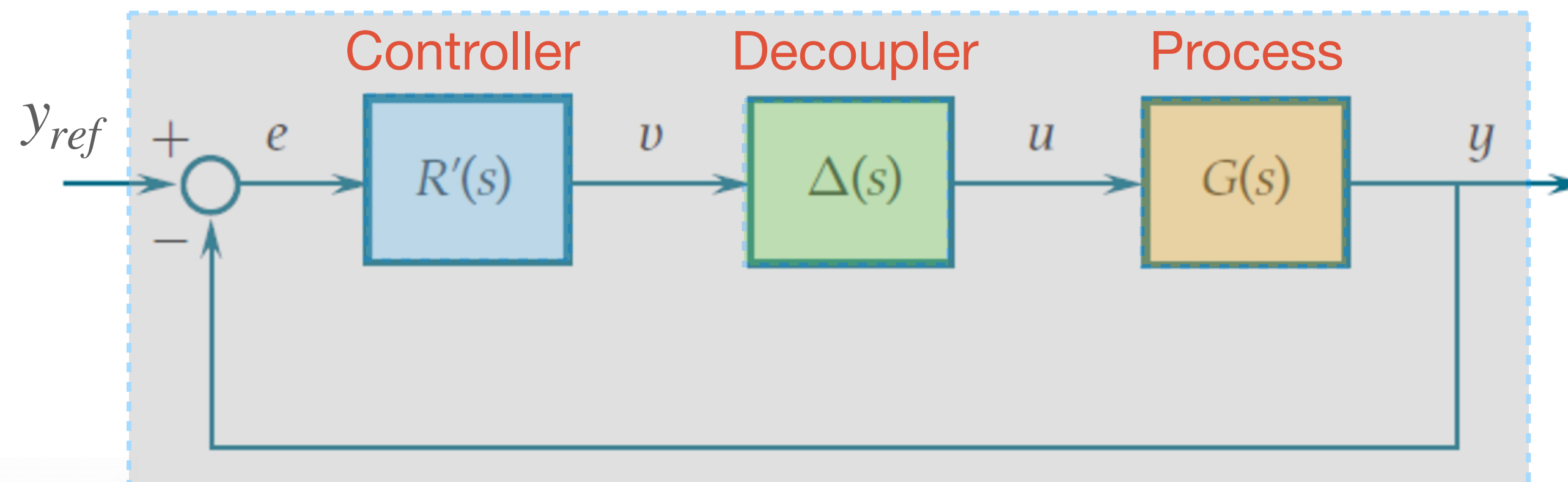
- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$

Decoupling Based Control Schemes: **Forward Decoupling**

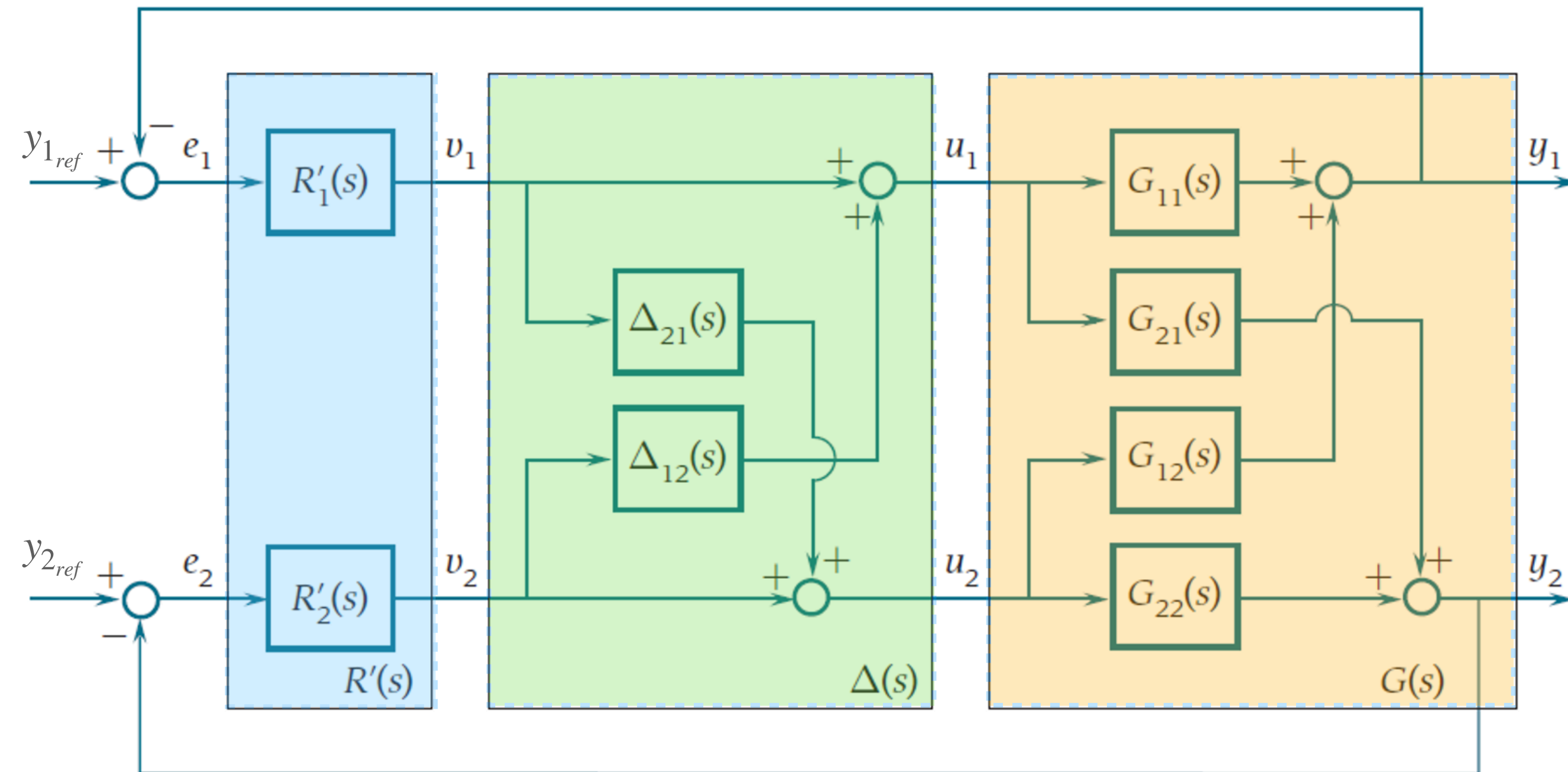


$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{d_{11}}(s) & 0 \\ 0 & G_{d_{22}}(s) \end{bmatrix}$$



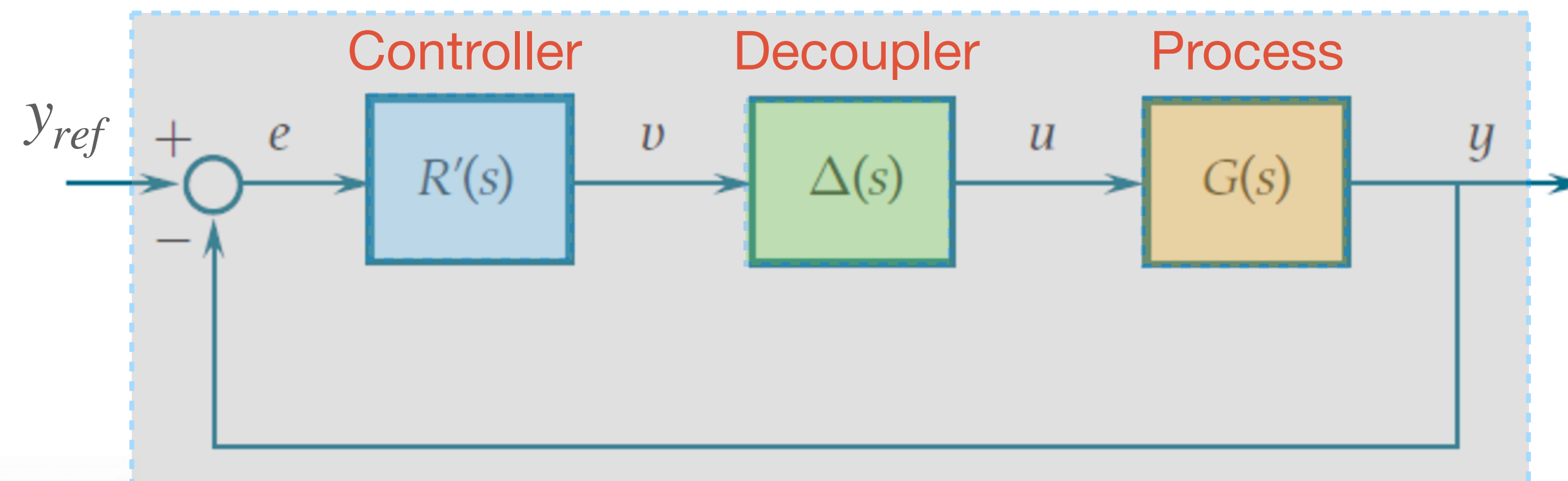
Decoupling Based Control Schemes: **Forward Decoupling**



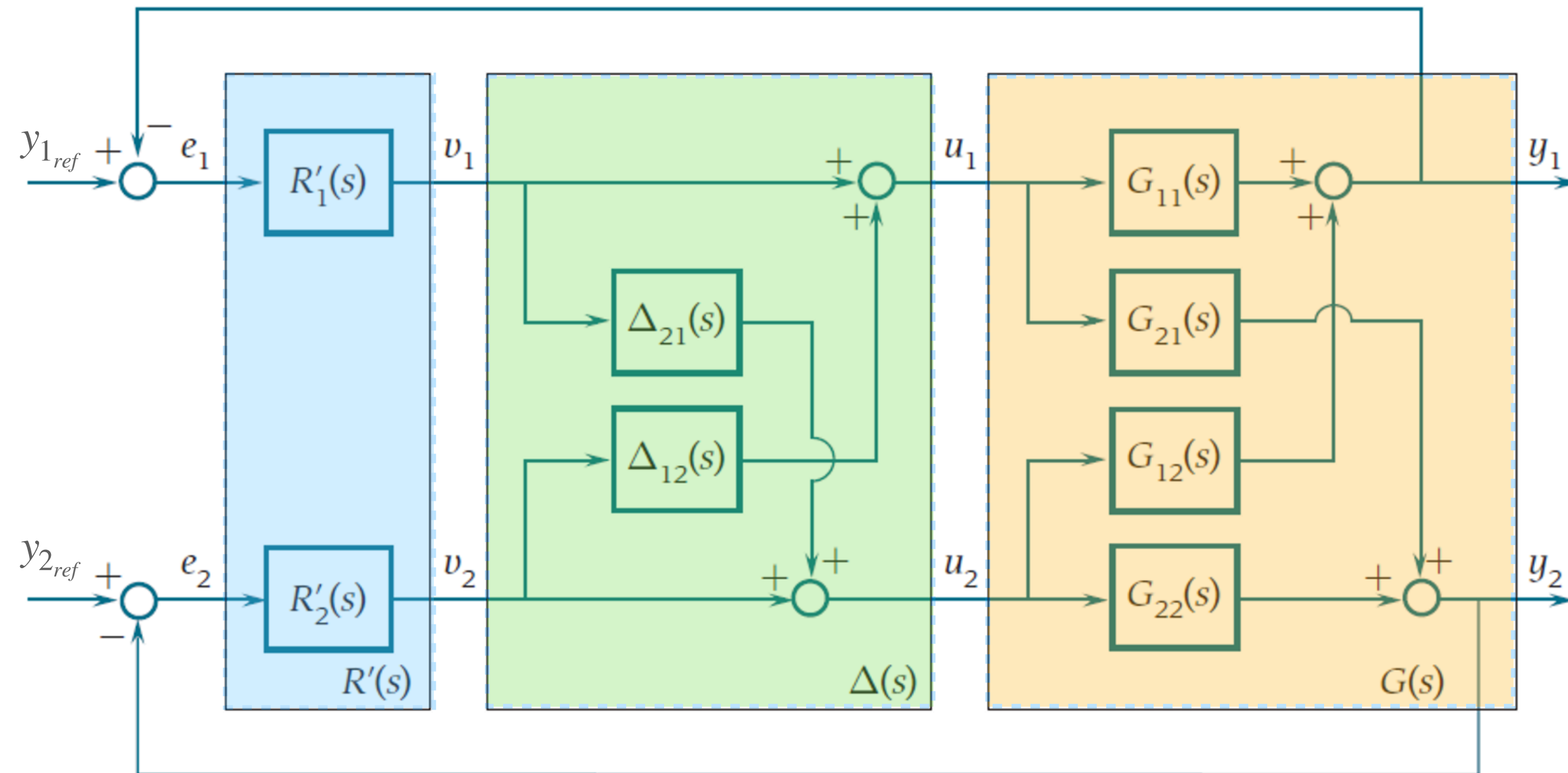
$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

unknowns



Decoupling Based Control Schemes: **Forward Decoupling**



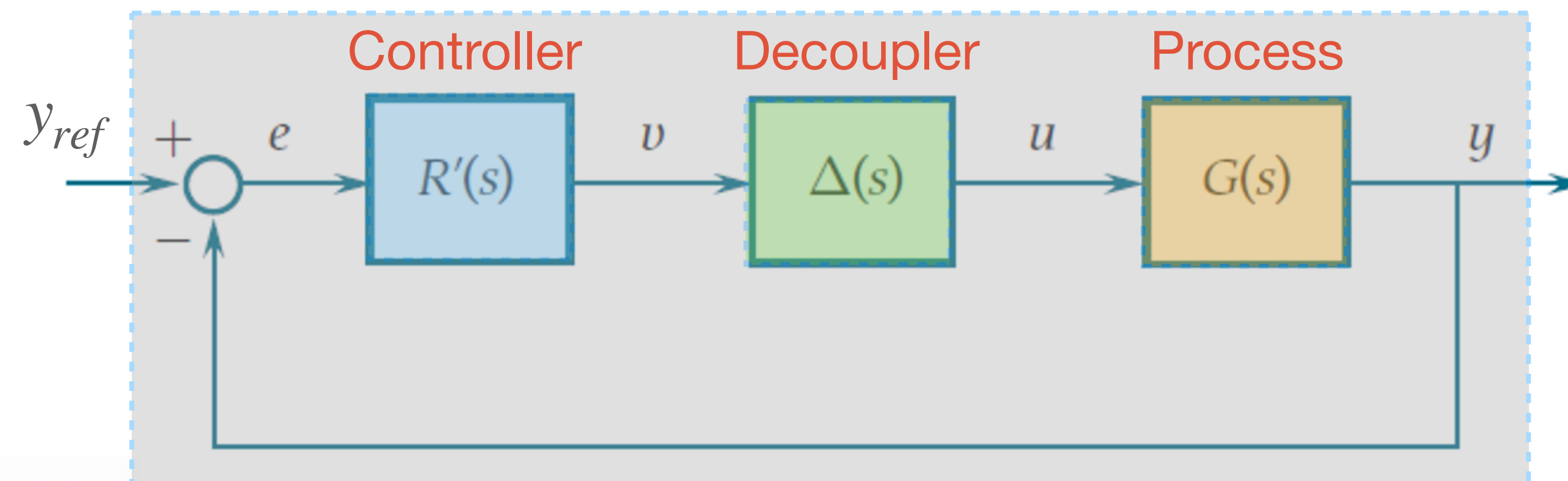
$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

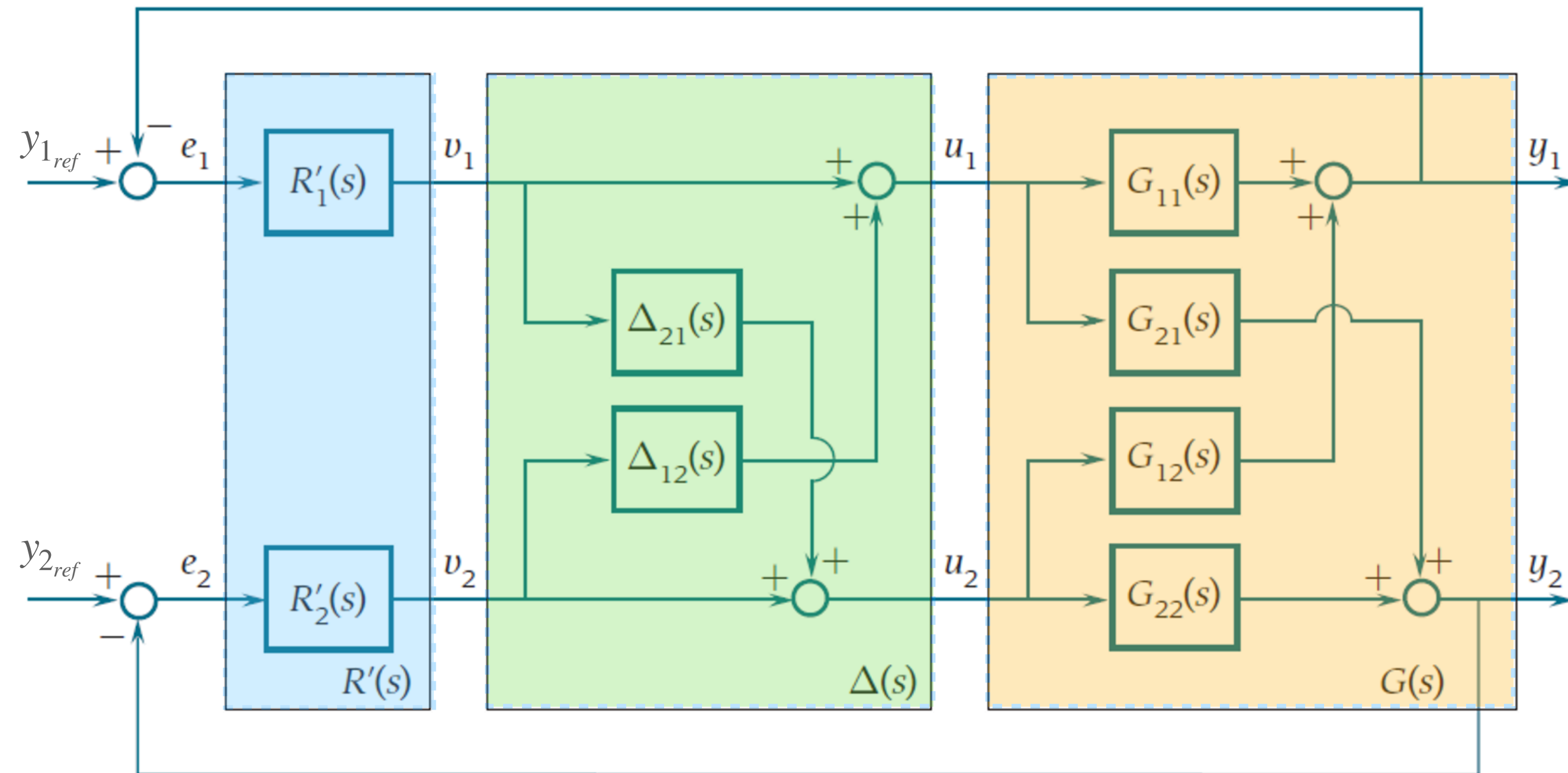
unknowns



$$\Delta_{11}(s) = \Delta_{22}(s) = 1$$



Decoupling Based Control Schemes: **Forward Decoupling**



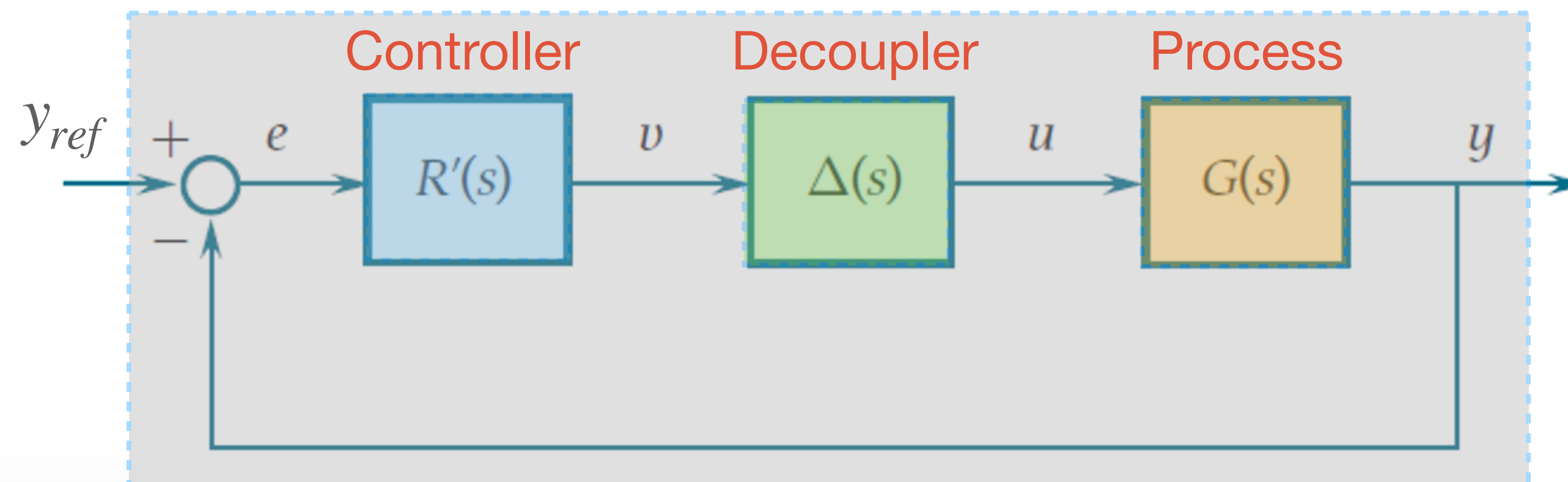
$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} 1 & \Delta_{12}(s) \\ \Delta_{21}(s) & 1 \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

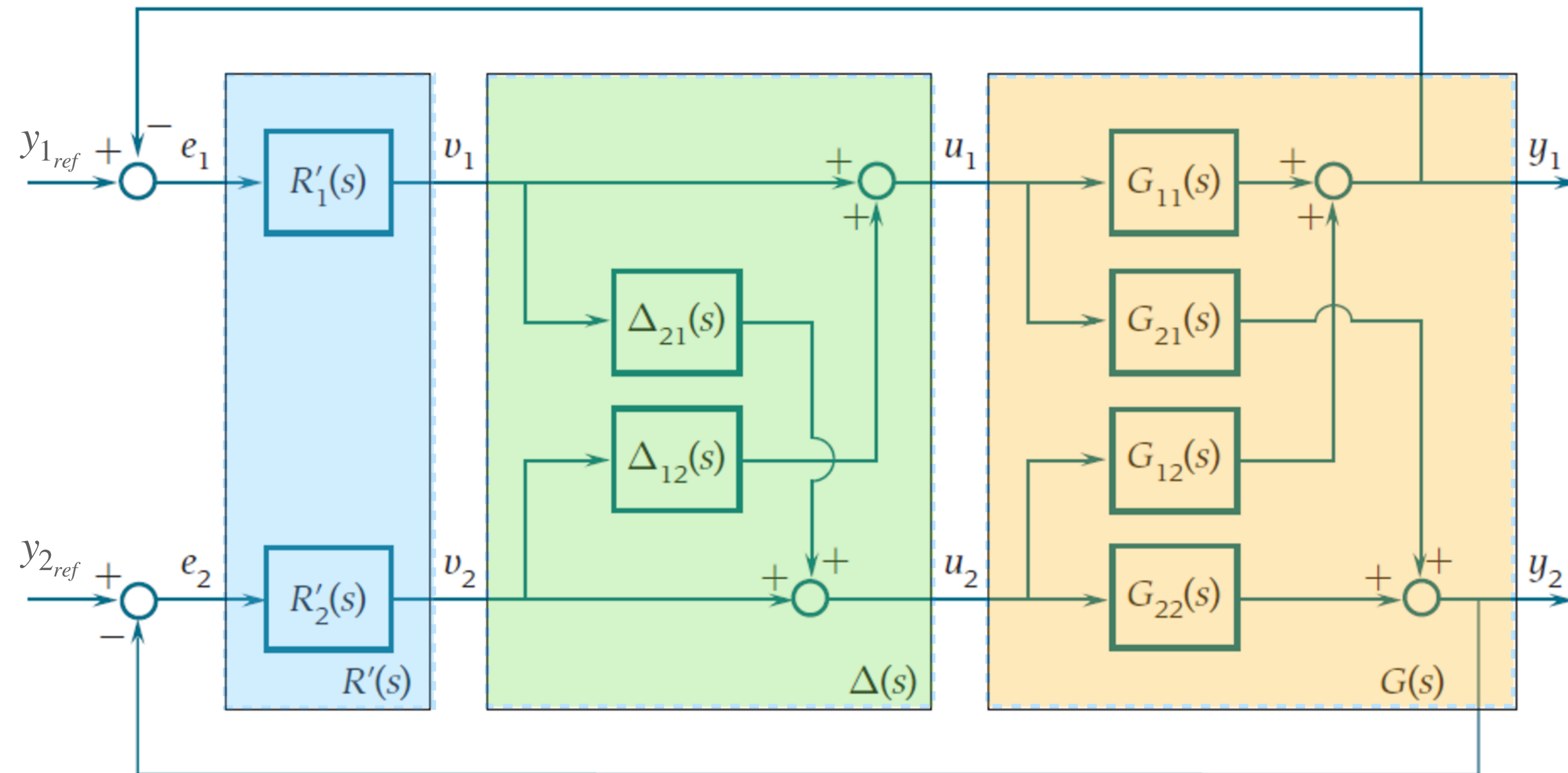
unknowns



$$\Delta_{11}(s) = \Delta_{22}(s) = 1$$



Decoupling Based Control Schemes: **Forward Decoupling**



$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} 1 & \Delta_{12}(s) \\ \Delta_{21}(s) & 1 \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

unknowns

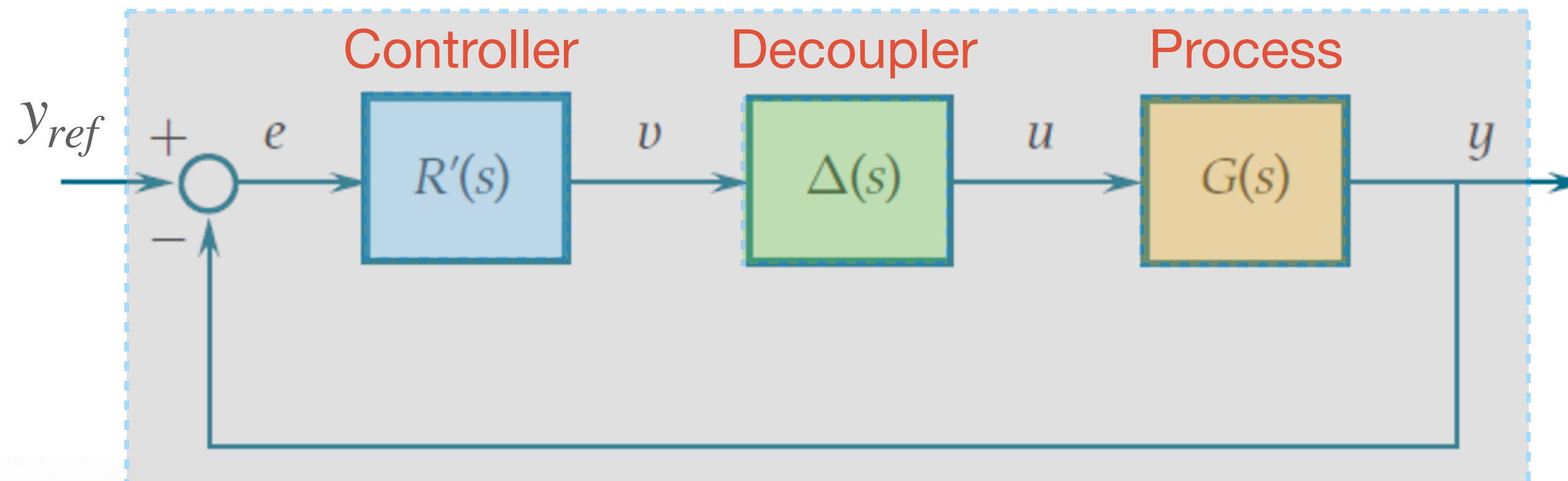


$$G_{11}(s) + G_{12}(s)\Delta_{21}(s) = G_{d11}(s)$$

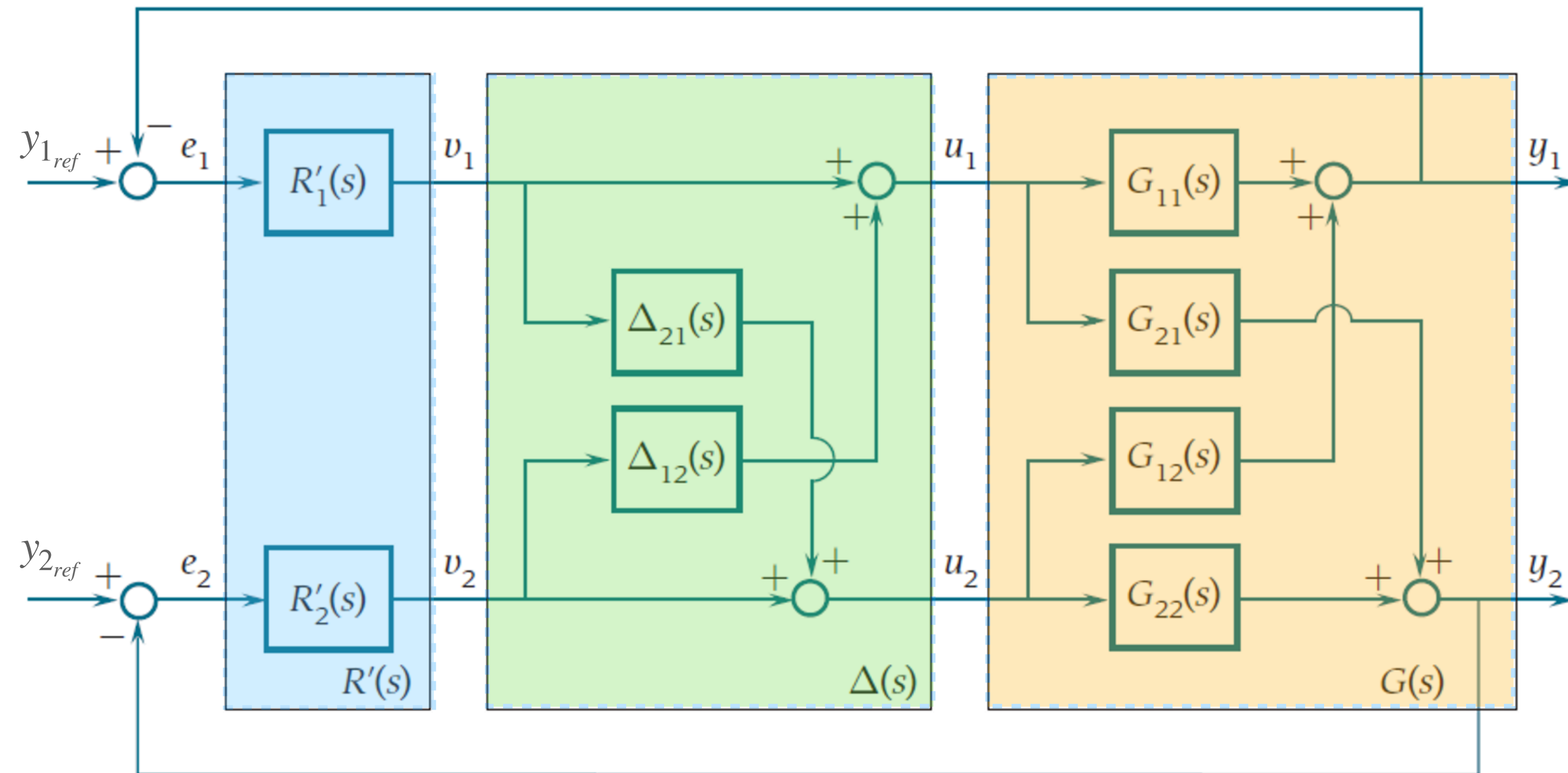
$$G_{11}(s)\Delta_{12}(s) + G_{12}(s) = 0$$

$$G_{21}(s) + G_{22}(s)\Delta_{21}(s) = 0$$

$$G_{21}(s)\Delta_{12}(s) + G_{22}(s) = G_{d22}(s)$$



Decoupling Based Control Schemes: **Forward Decoupling**



$$G(s)\Delta(s) = G_d(s)$$

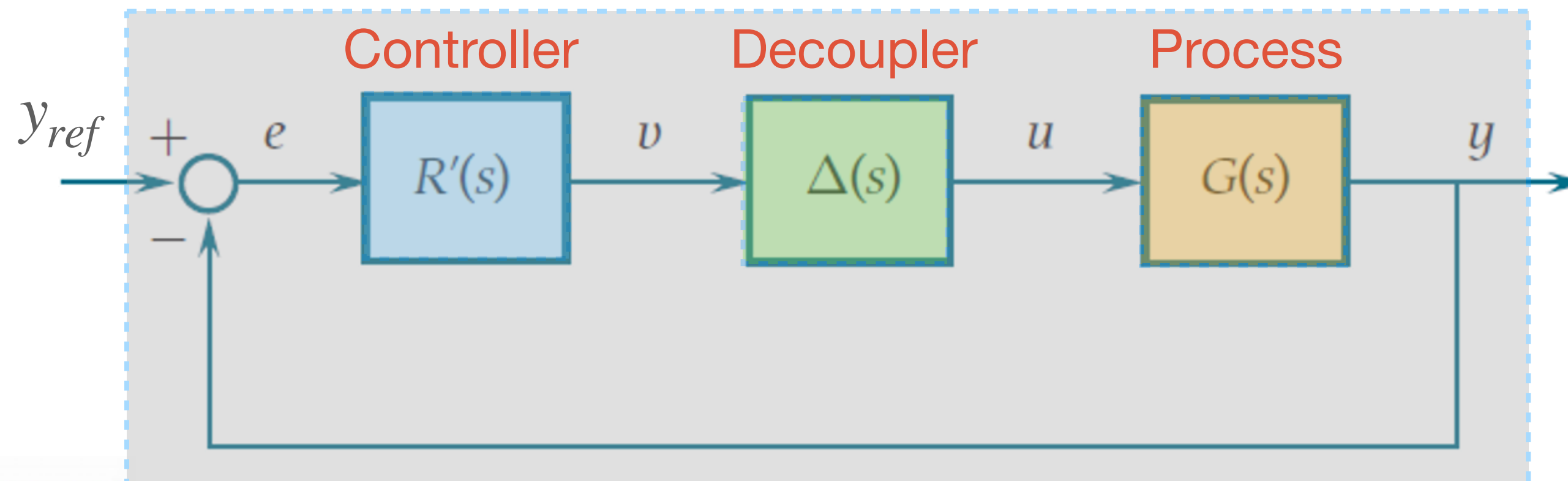
$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} 1 & \Delta_{12}(s) \\ \Delta_{21}(s) & 1 \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

unknowns

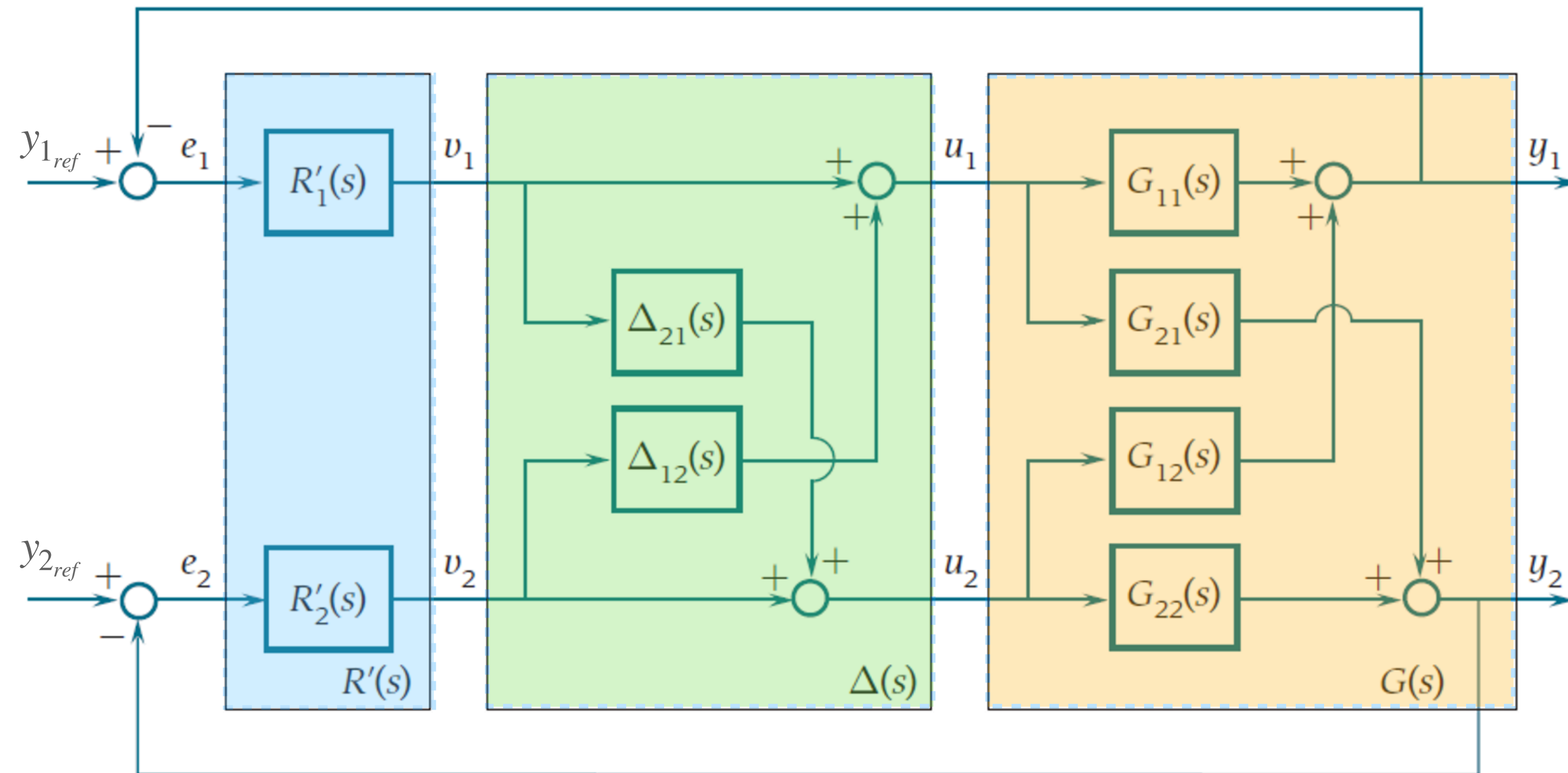


$$G_{11}(s) + G_{12}(s)\Delta_{21}(s) = G_{d11}(s)$$

$$\begin{aligned} G_{11}(s)\Delta_{12}(s) + G_{12}(s) &= 0 \\ G_{21}(s) + G_{22}(s)\Delta_{21}(s) &= 0 \\ G_{21}(s)\Delta_{12}(s) + G_{22}(s) &= G_{d22}(s) \end{aligned} \rightarrow \begin{aligned} &\Delta_{12}(s) \\ &\Delta_{21}(s) \end{aligned}$$



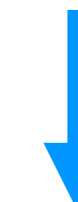
Decoupling Based Control Schemes: **Forward Decoupling**



$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} 1 & \Delta_{12}(s) \\ \Delta_{21}(s) & 1 \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

unknowns



$$G_{11}(s) + G_{12}(s)\Delta_{21}(s) = G_{d11}(s)$$

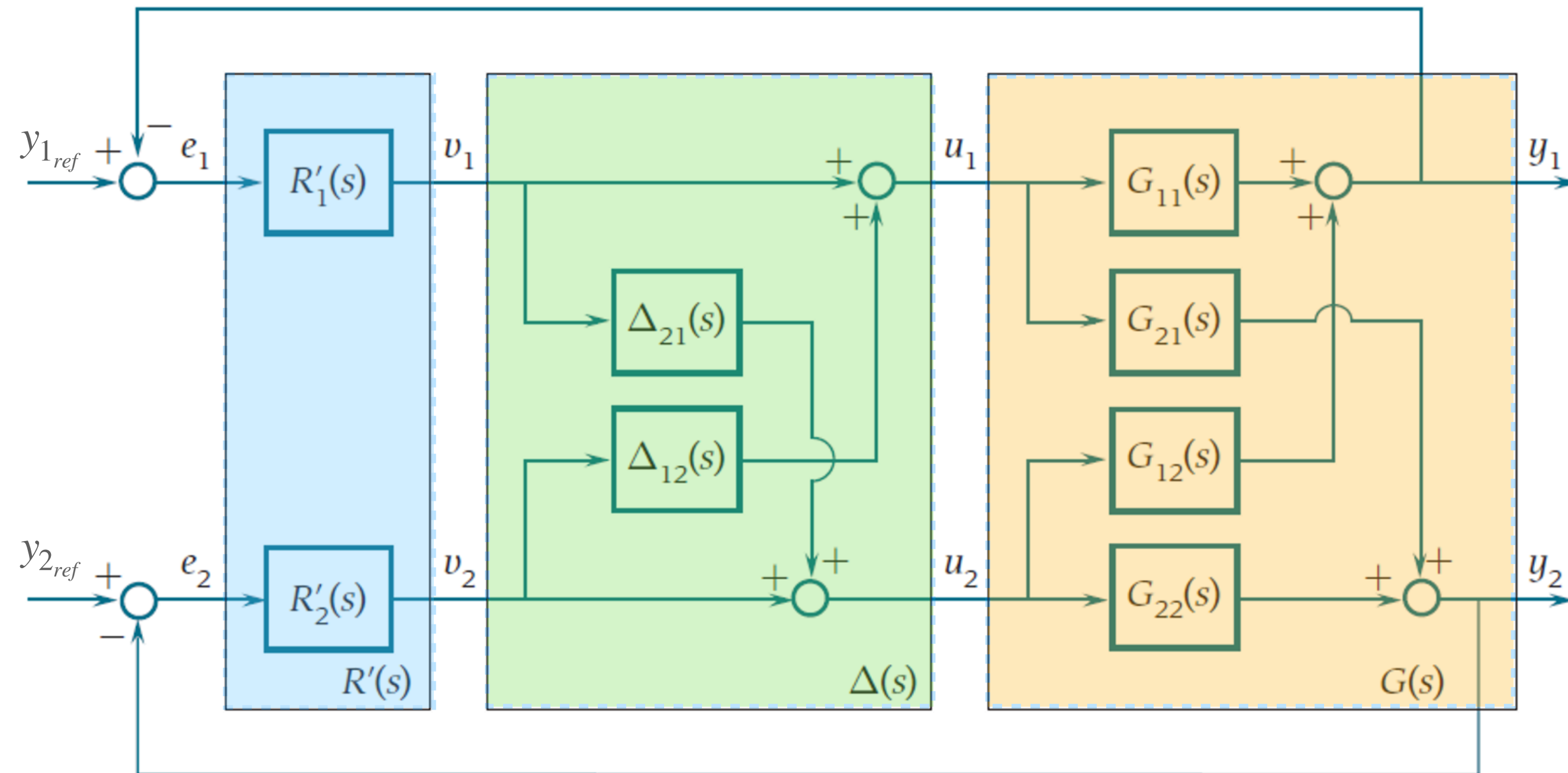
$$\begin{aligned} G_{11}(s)\Delta_{12}(s) + G_{12}(s) &= 0 \\ G_{21}(s) + G_{22}(s)\Delta_{21}(s) &= 0 \\ G_{21}(s)\Delta_{12}(s) + G_{22}(s) &= G_{d22}(s) \end{aligned}$$

$$\Delta_{12}(s)$$

$$\Delta_{21}(s)$$

they must be causal!

Decoupling Based Control Schemes: **Forward Decoupling**



$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} 1 & \Delta_{12}(s) \\ \Delta_{21}(s) & 1 \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

unknowns



$$G_{11}(s) + G_{12}(s)\Delta_{21}(s) = G_{d11}(s)$$

$$G_{11}(s)\Delta_{12}(s) + G_{12}(s) = 0$$

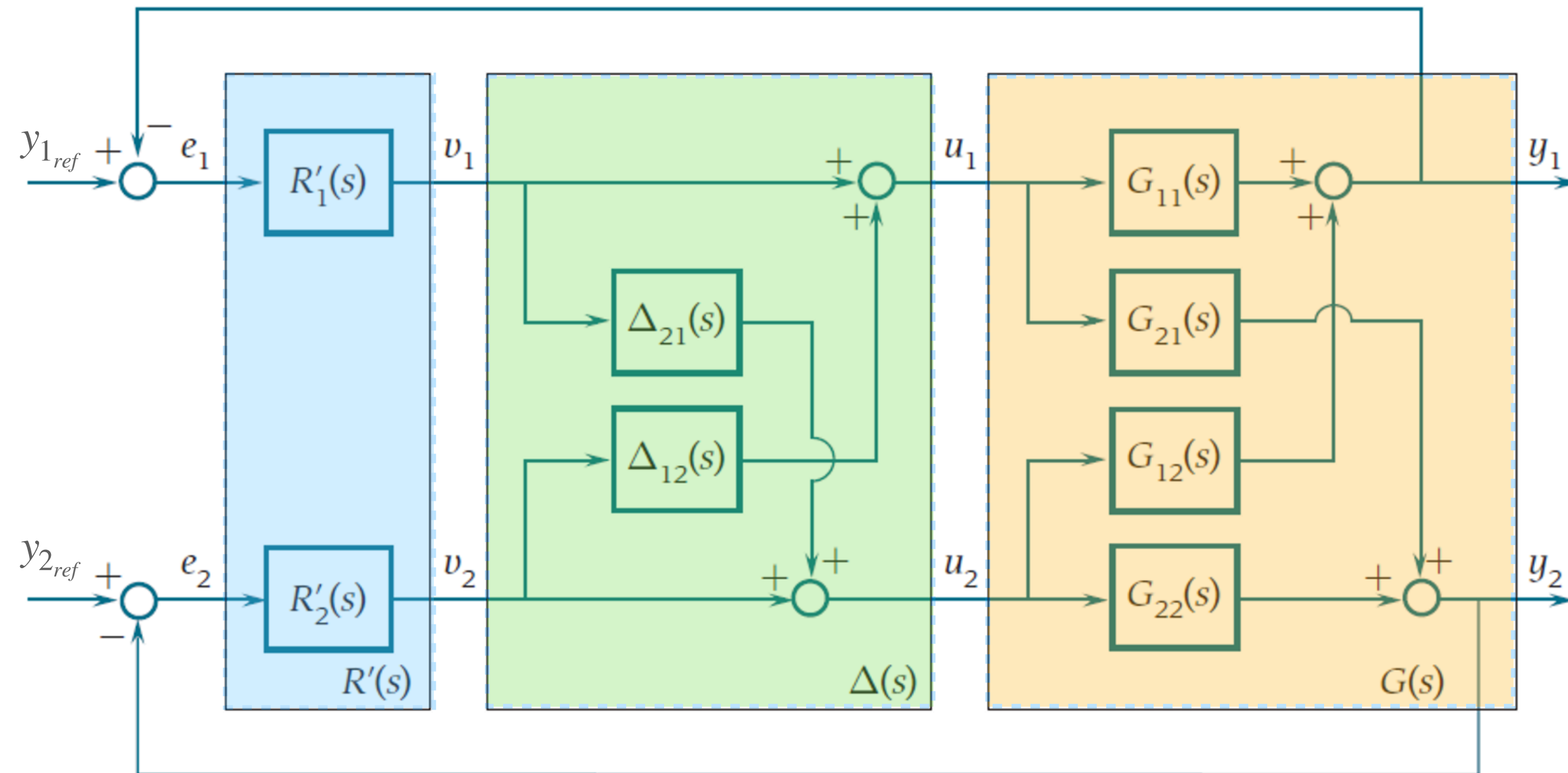
$$G_{21}(s) + G_{22}(s)\Delta_{21}(s) = 0$$

$$G_{21}(s)\Delta_{12}(s) + G_{22}(s) = G_{d22}(s)$$

$\Delta_{12}(s)$

$\Delta_{21}(s)$

Decoupling Based Control Schemes: **Forward Decoupling**



$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} 1 & \Delta_{12}(s) \\ \Delta_{21}(s) & 1 \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

unknowns



$$G_{11}(s) + G_{12}(s)\Delta_{21}(s) = G_{d11}(s)$$

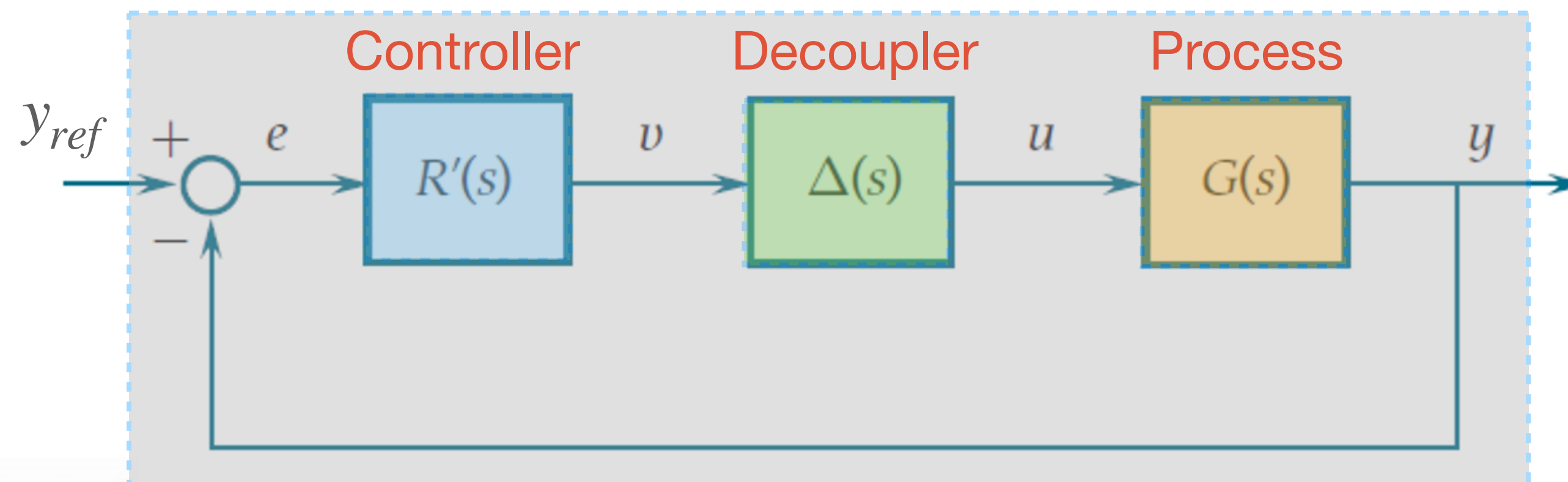
$$G_{11}(s)\Delta_{12}(s) + G_{12}(s) = 0$$

$$G_{21}(s) + G_{22}(s)\Delta_{21}(s) = 0$$

$$G_{21}(s)\Delta_{12}(s) + G_{22}(s) = G_{d22}(s)$$

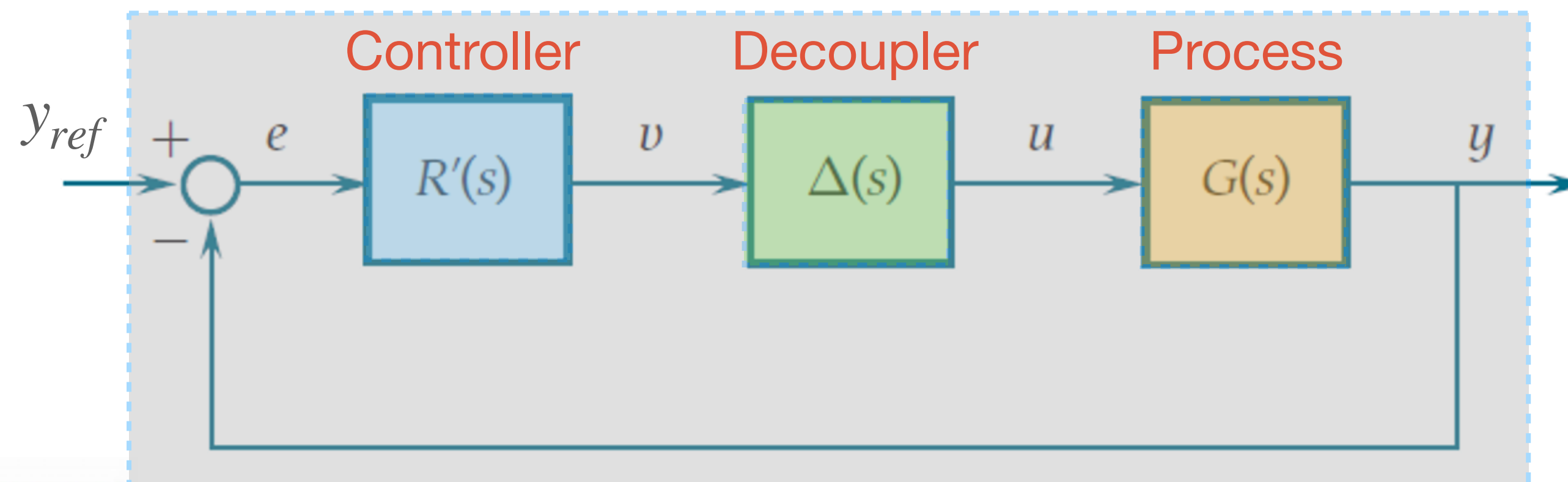
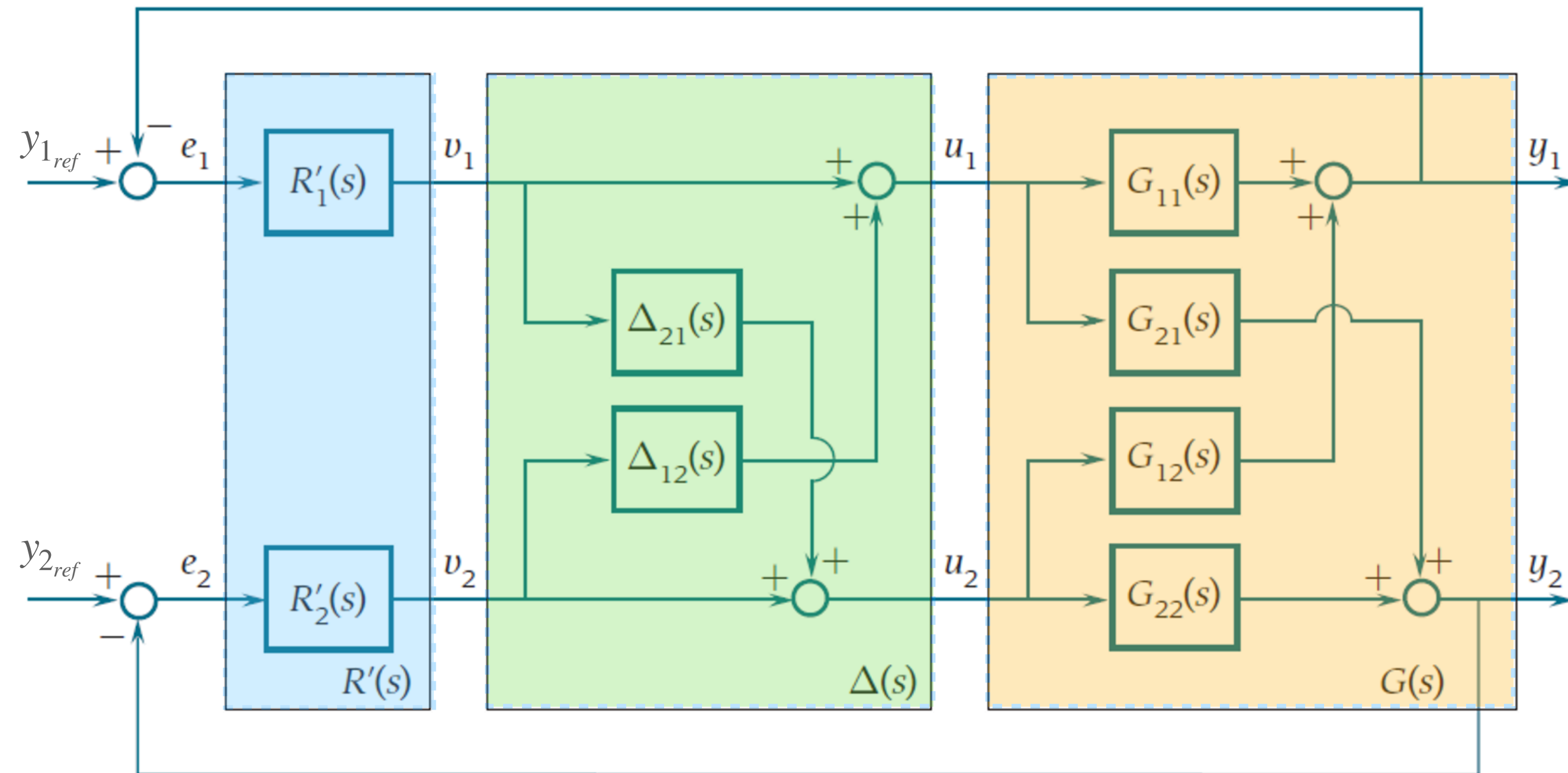
$\Delta_{12}(s)$

$\Delta_{21}(s)$



design: $R'(s) = \begin{bmatrix} R'_1(s) & 0 \\ 0 & R'_2(s) \end{bmatrix}$

Decoupling Based Control Schemes: **Forward Decoupling**



$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} 1 & \Delta_{12}(s) \\ \Delta_{21}(s) & 1 \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

unknowns

$$G_{11}(s) + G_{12}(s)\Delta_{21}(s) = G_{d11}(s)$$

$$G_{11}(s)\Delta_{12}(s) + G_{12}(s) = 0$$

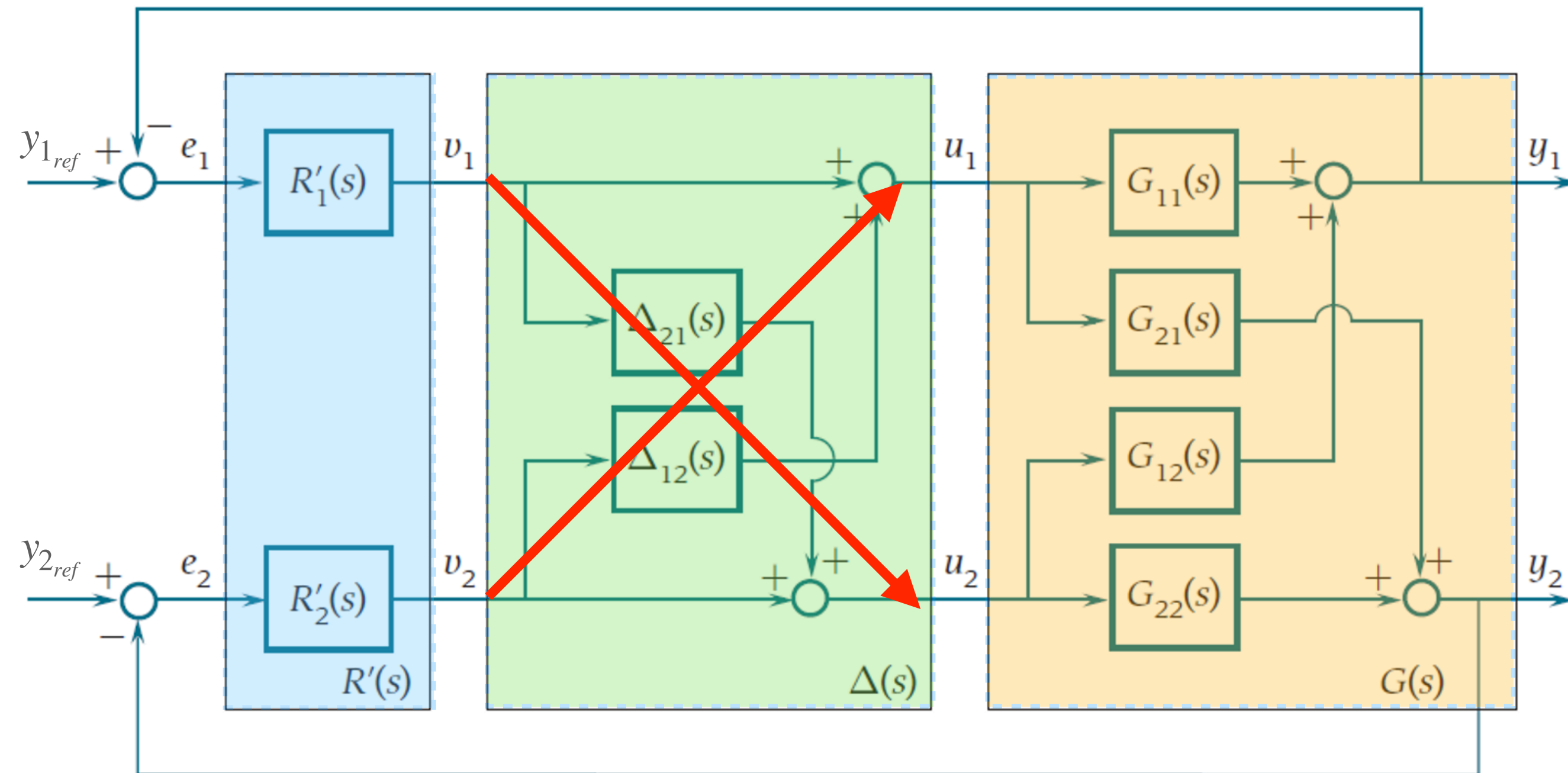
$$G_{21}(s) + G_{22}(s)\Delta_{21}(s) = 0$$

$$G_{21}(s)\Delta_{12}(s) + G_{22}(s) = G_{d22}(s)$$

$$G_{d11}(s)$$

design: $R'(s) = \begin{bmatrix} R'_1(s) & 0 \\ 0 & R'_2(s) \end{bmatrix}$

Decoupling Based Control Schemes: **Forward Decoupling**



$$G(s)\Delta(s) = G_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} 1 & \Delta_{12}(s) \\ \Delta_{21}(s) & 1 \end{bmatrix} = \begin{bmatrix} G_{d11}(s) & 0 \\ 0 & G_{d22}(s) \end{bmatrix}$$

unknowns

$$G_{11}(s) + G_{12}(s)\Delta_{21}(s) = G_{d11}(s)$$

$$G_{11}(s)\Delta_{12}(s) + G_{12}(s) = 0$$

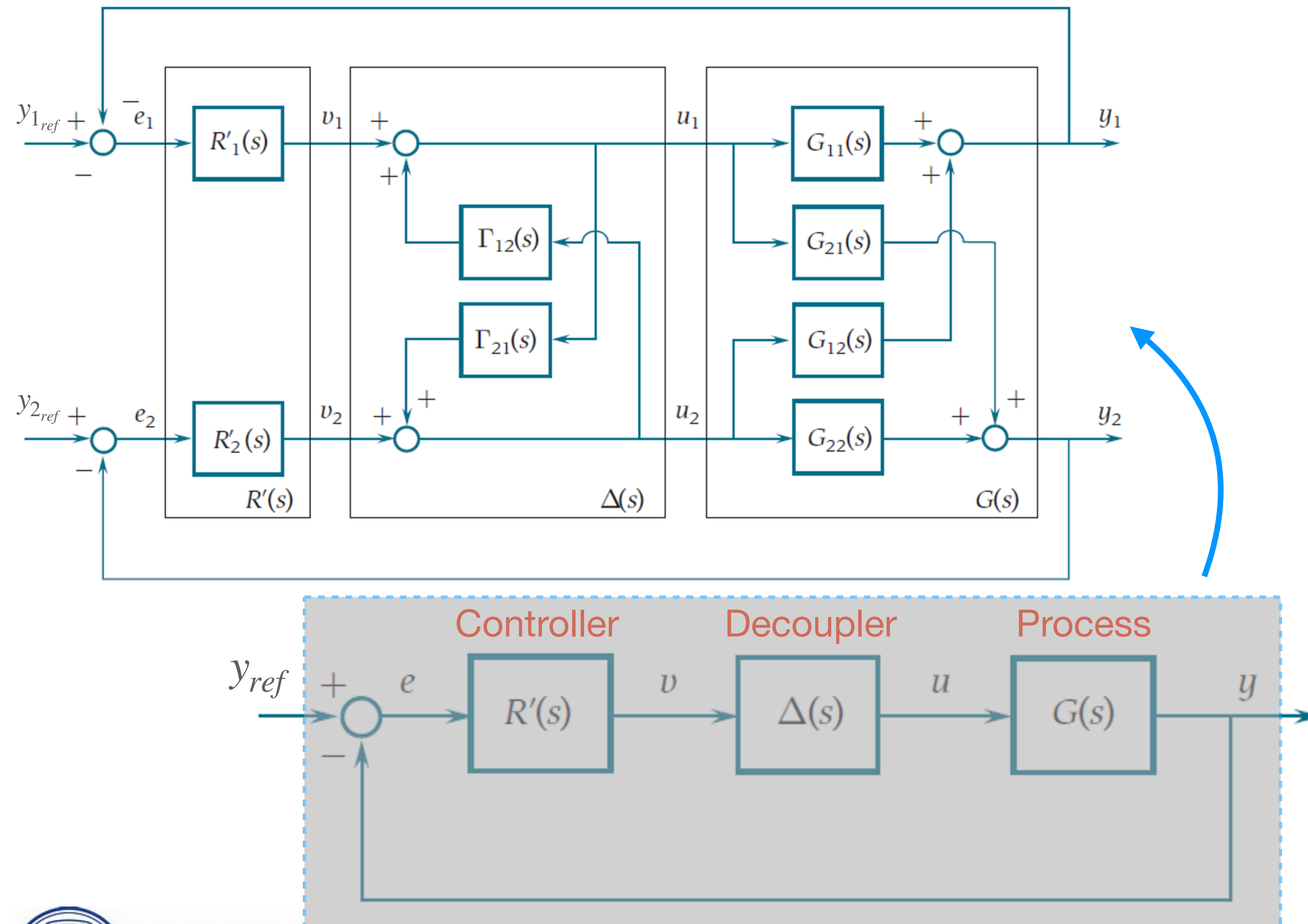
$$G_{21}(s) + G_{22}(s)\Delta_{21}(s) = 0$$

$$G_{21}(s)\Delta_{12}(s) + G_{22}(s) = G_{d22}(s)$$

$$G_{d11}(s)$$

design: $R'(s) = \begin{bmatrix} R'_1(s) & 0 \\ 0 & R'_2(s) \end{bmatrix}$

Decoupling Based Control Schemes: **Backward Decoupling**



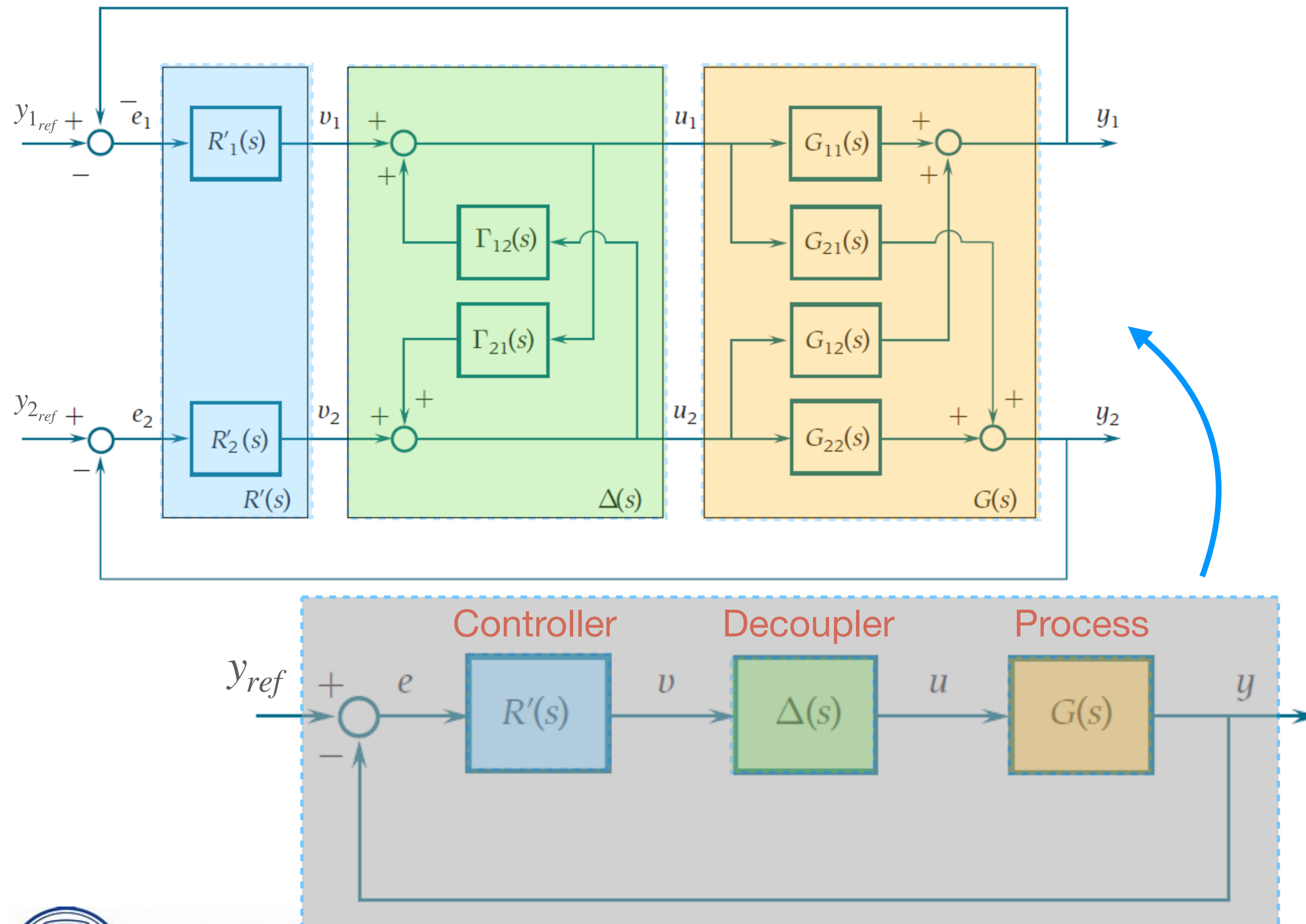
Assumptions:

- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix



Sufficient conditions:

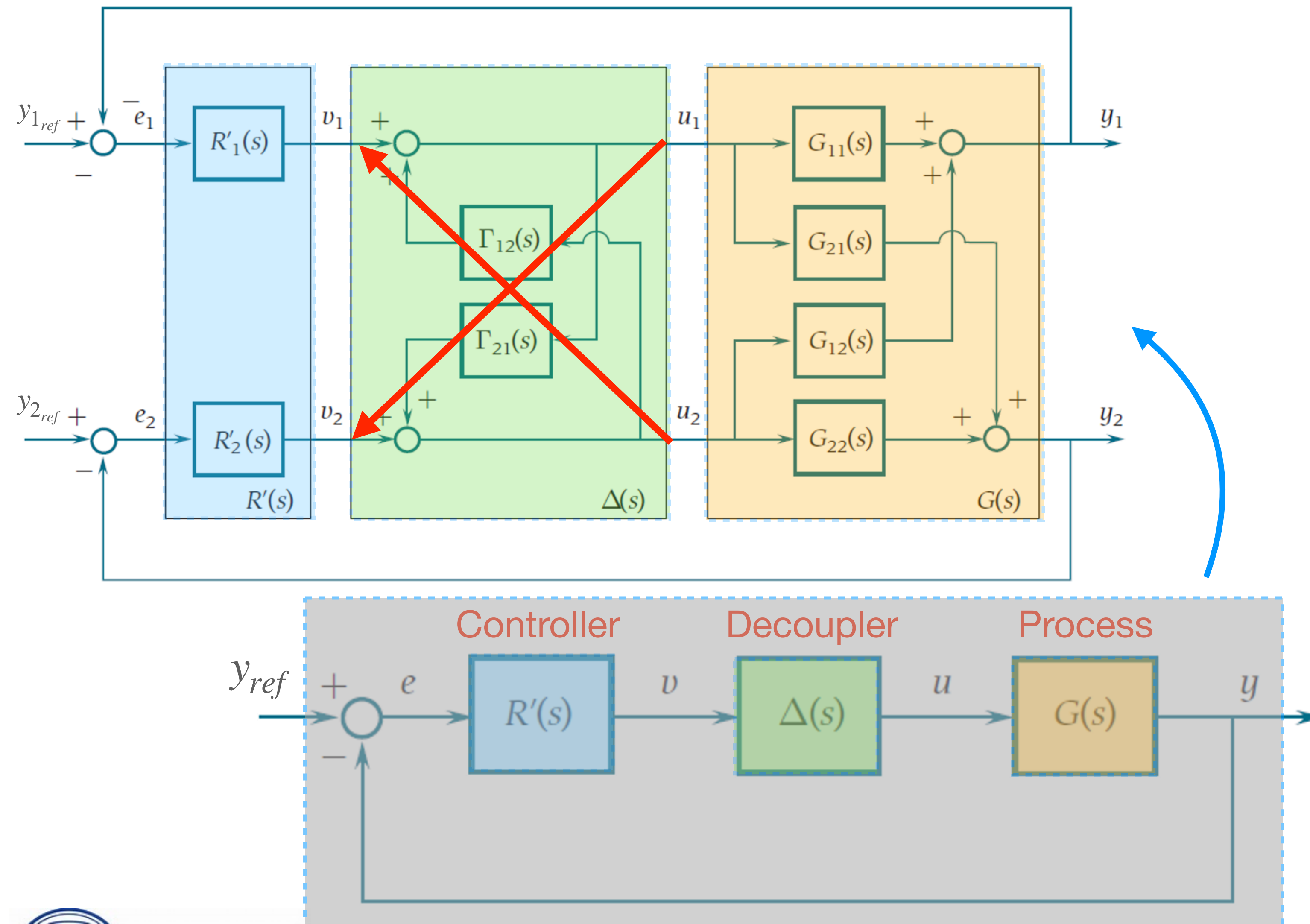
- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$

Decoupling Based Control Schemes: **Backward Decoupling****Assumptions:**

- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix

**Sufficient conditions:**

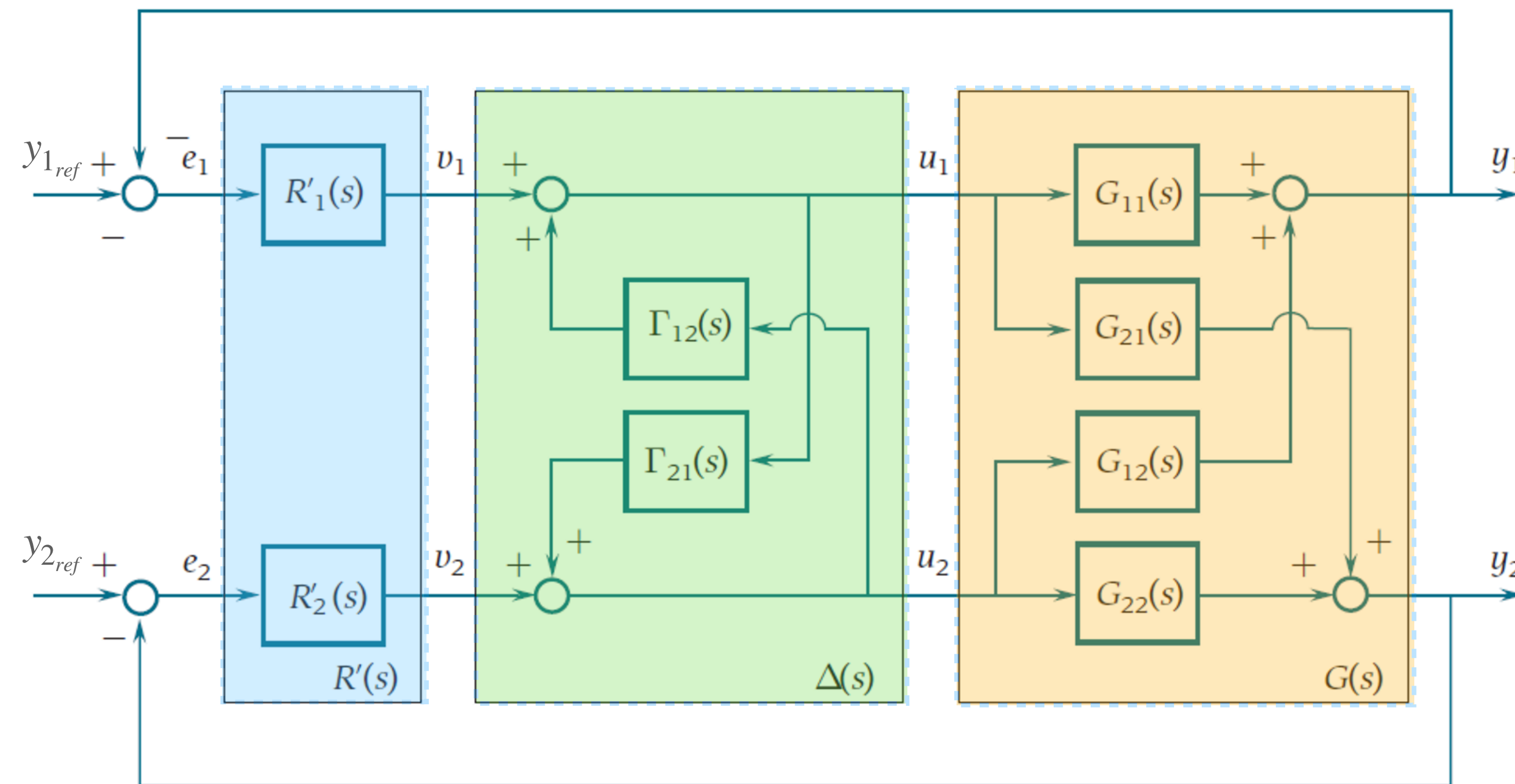
- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \quad \forall s$
s.t. $\text{Re}(s) \geq 0$

Decoupling Based Control Schemes: **Backward Decoupling****Assumptions:**

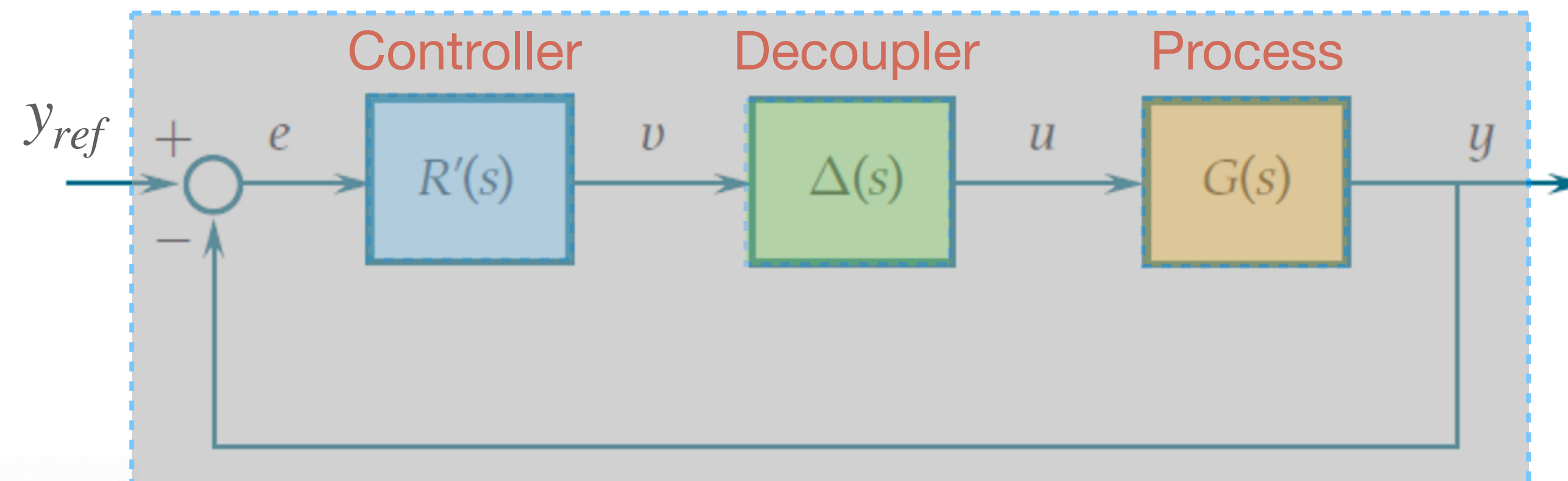
- $G(s) \in \mathbb{R}^{2 \times 2}$
- $G(s)$ full matrix

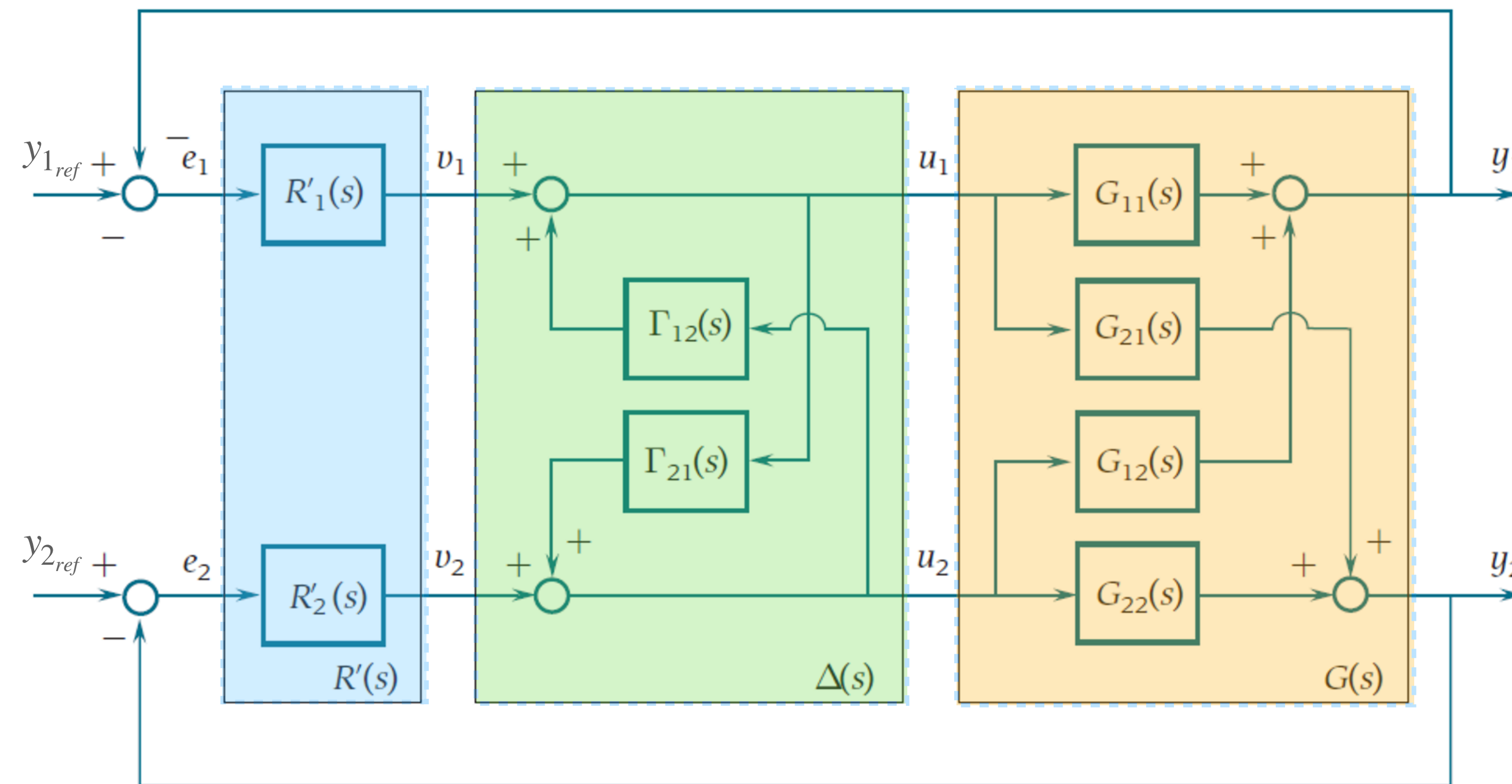
**Sufficient conditions:**

- $G(s)$ rational function
- $G(s)$ As. Stable
- $\det G(s) \neq 0 \forall s$
s.t. $\text{Re}(s) \geq 0$

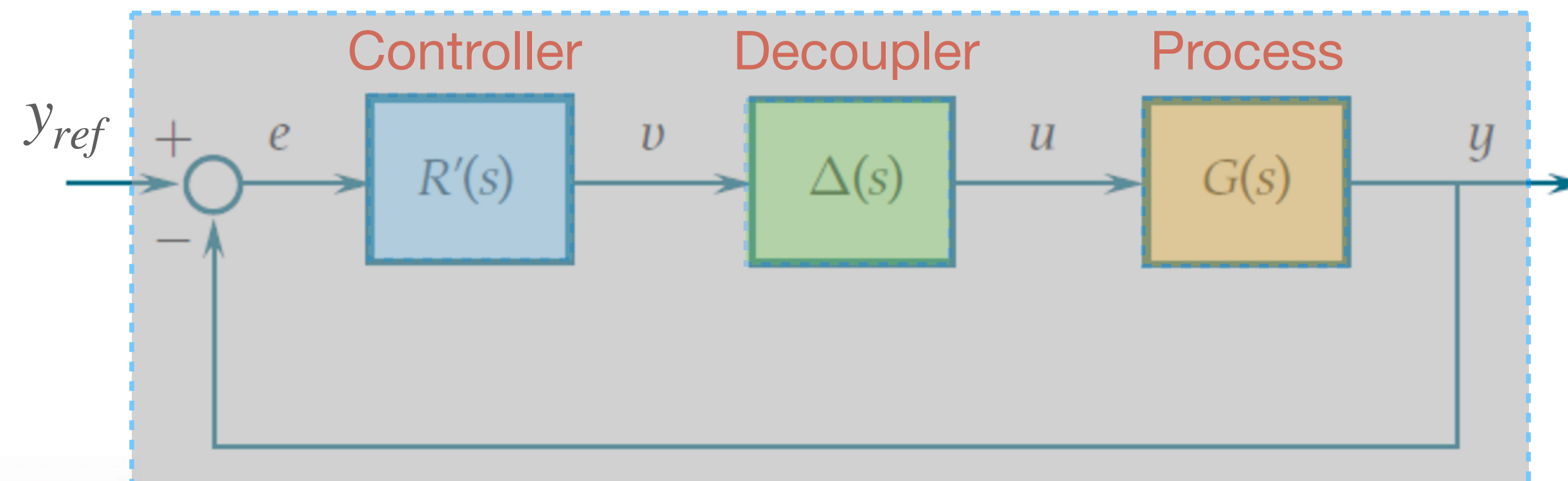
Decoupling Based Control Schemes: **Backward Decoupling**

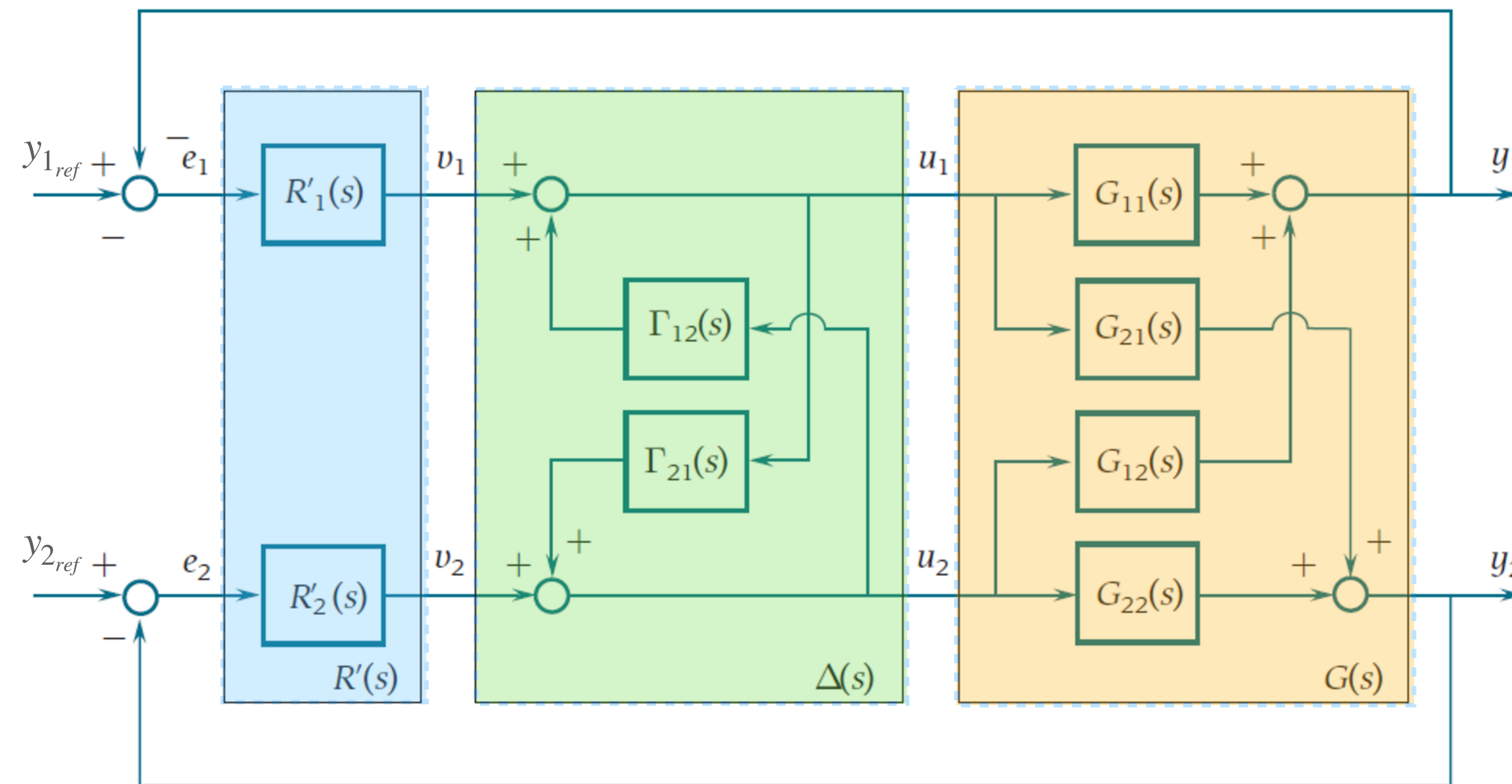
$$G(s)\Delta(s) = \tilde{G}_d(s)$$



Decoupling Based Control Schemes: **Backward Decoupling**

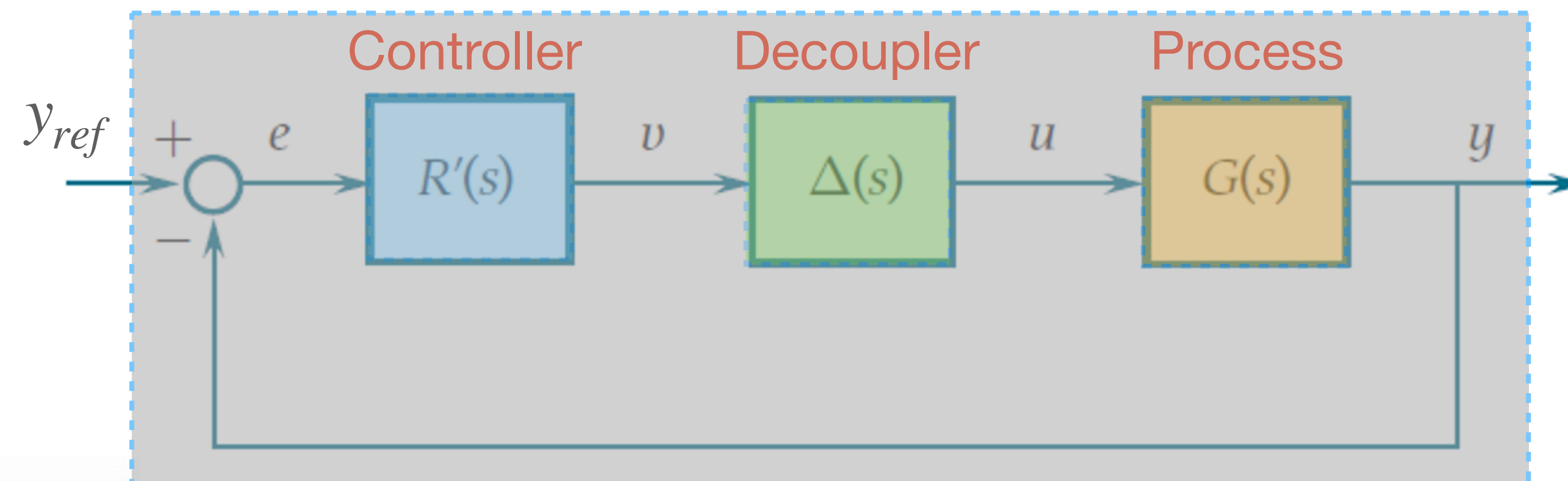
$$G(s)\Delta(s) = \tilde{G}_d(s)$$



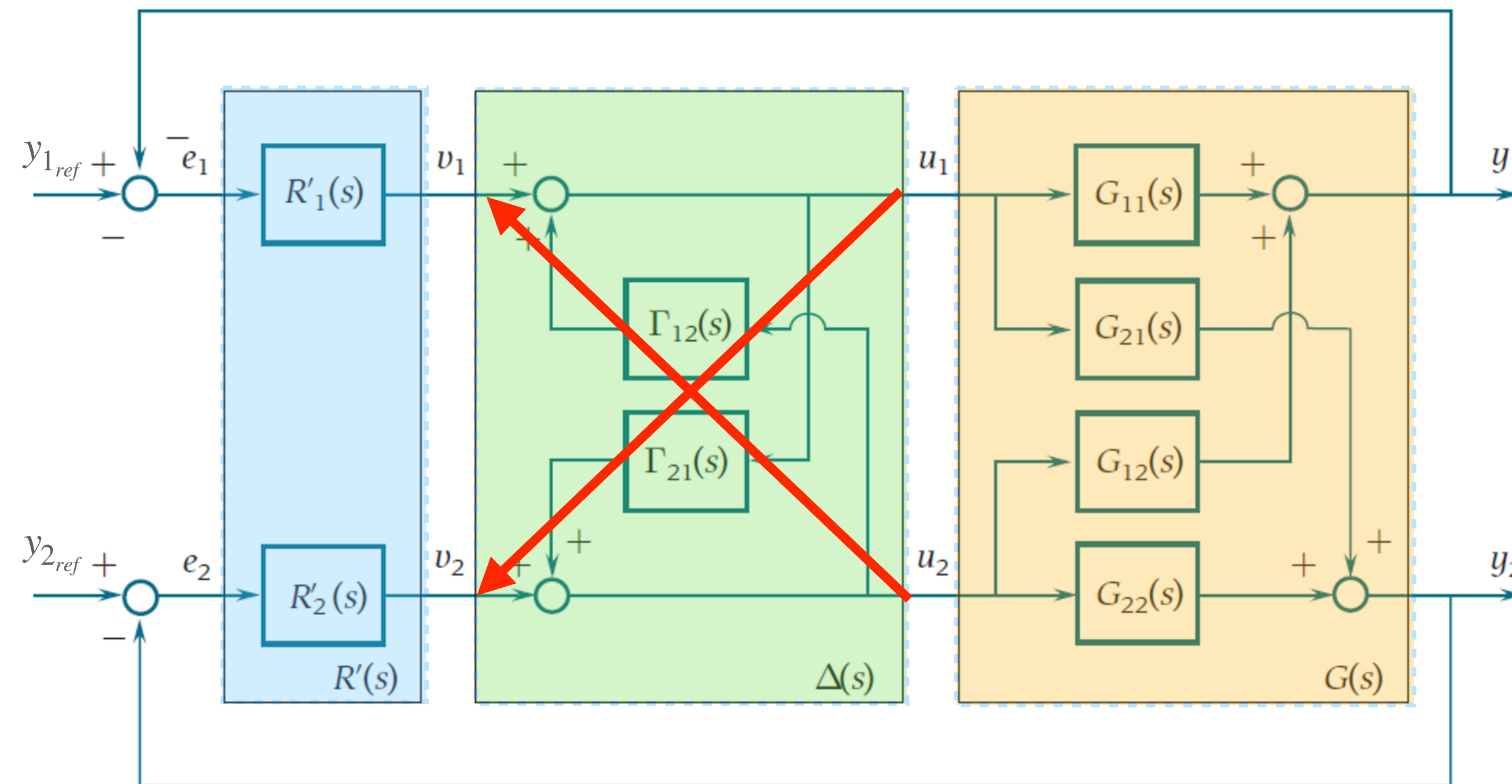
Decoupling Based Control Schemes: **Backward Decoupling**

$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$



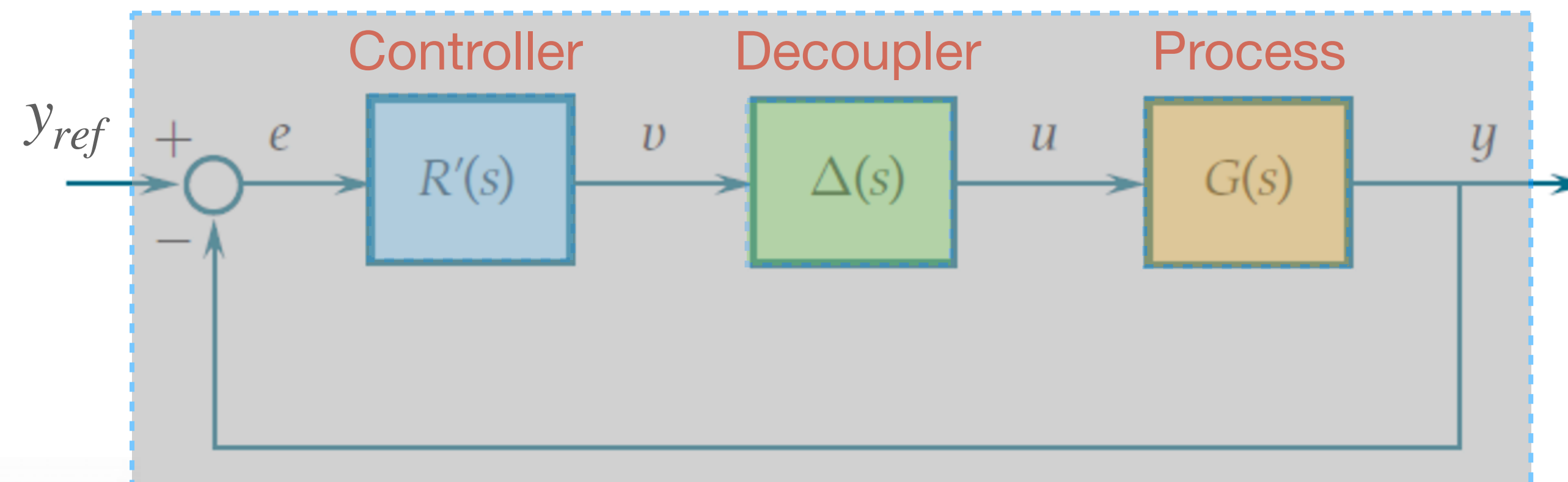
Decoupling Based Control Schemes: **Backward Decoupling**



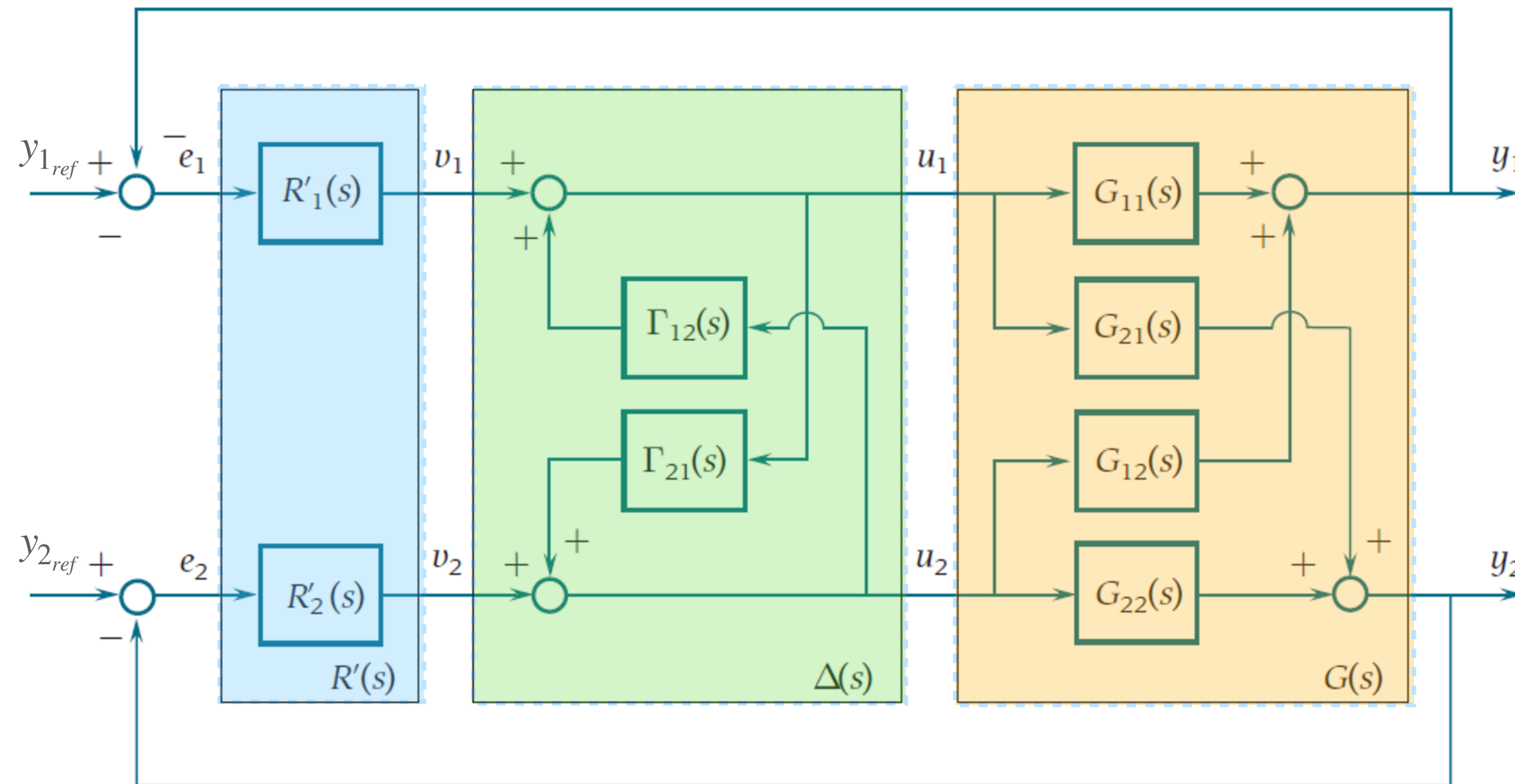
$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

$$U(s) = \Gamma(s)U(s) + V(s)$$



Decoupling Based Control Schemes: **Backward Decoupling**

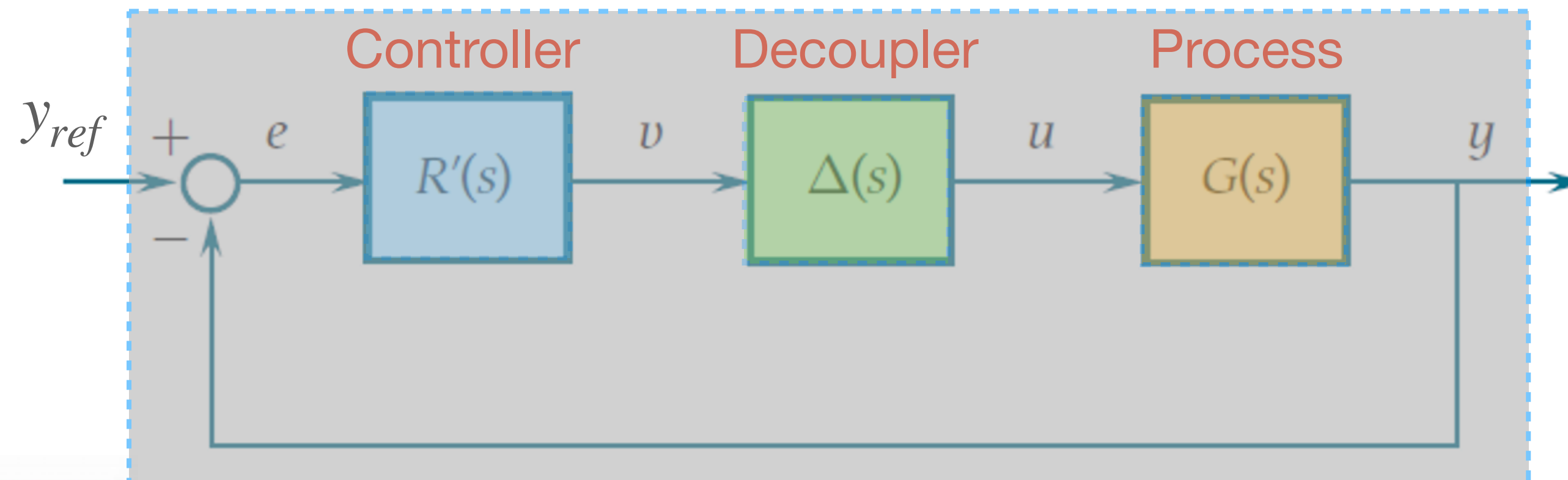


$$G(s)\Delta(s) = \tilde{G}_d(s)$$

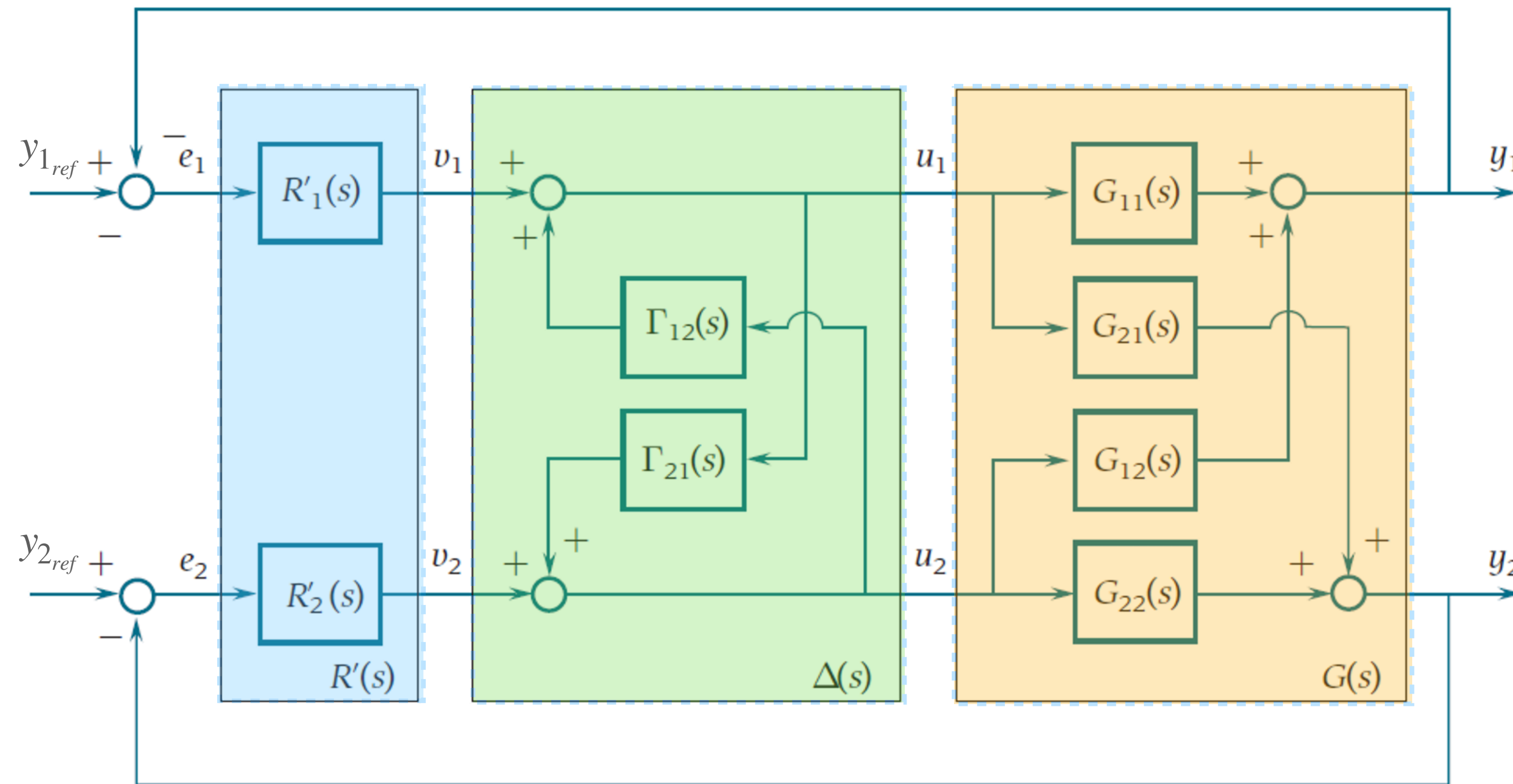
$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

$$U(s) = \Gamma(s)U(s) + V(s)$$

$$U(s) - \Gamma(s)U(s) = V(s)$$



Decoupling Based Control Schemes: **Backward Decoupling**



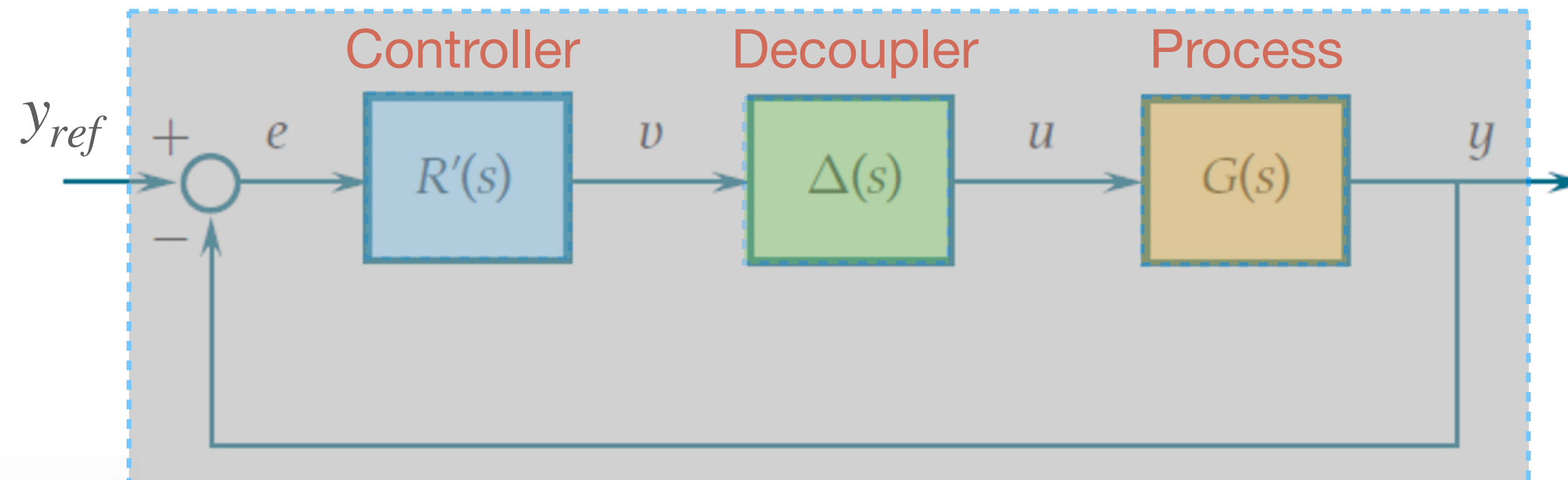
$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

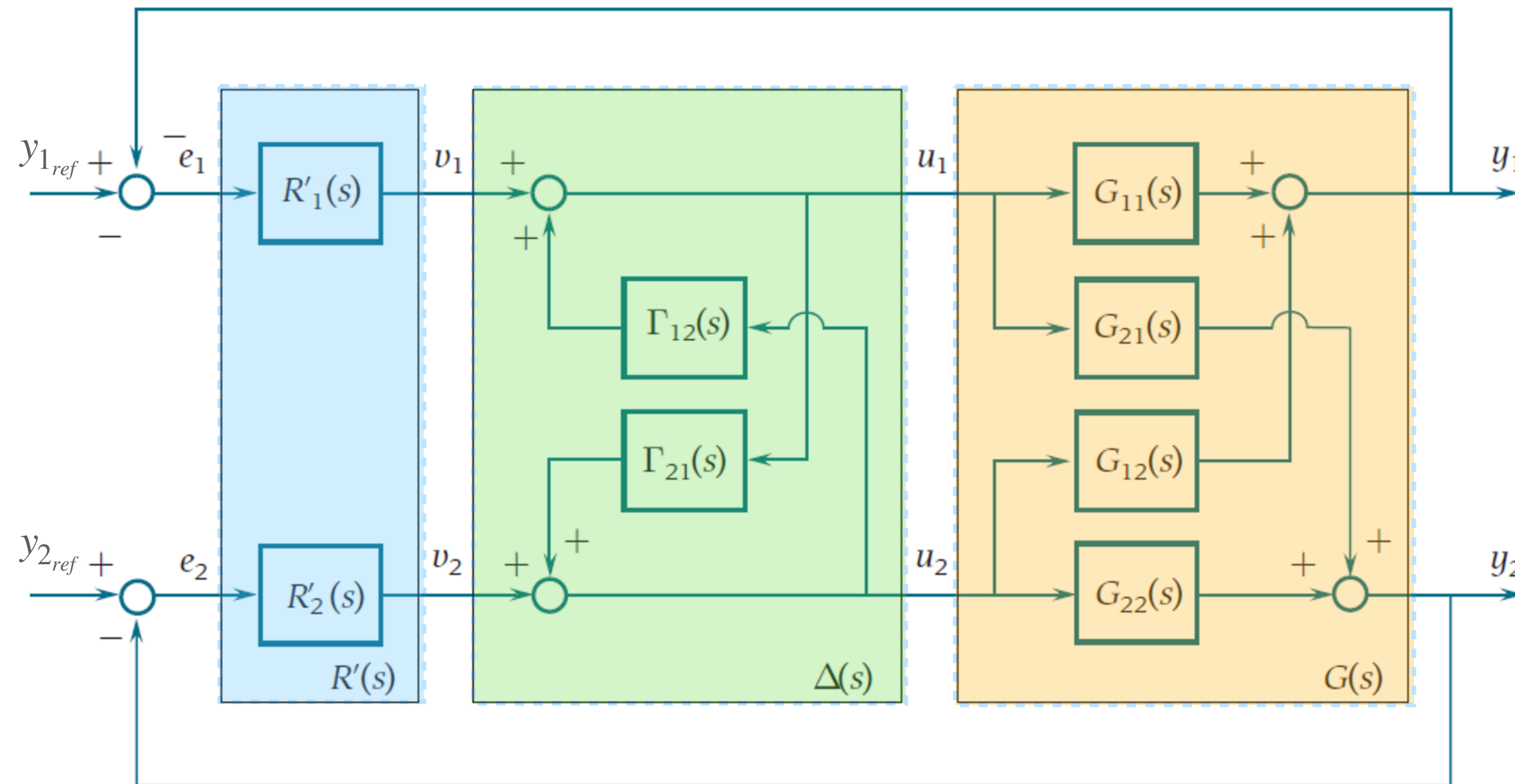
$$U(s) = \Gamma(s)U(s) + V(s)$$

$$U(s) - \Gamma(s)U(s) = V(s)$$

$$[I - \Gamma(s)]U(s) = V(s)$$



Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

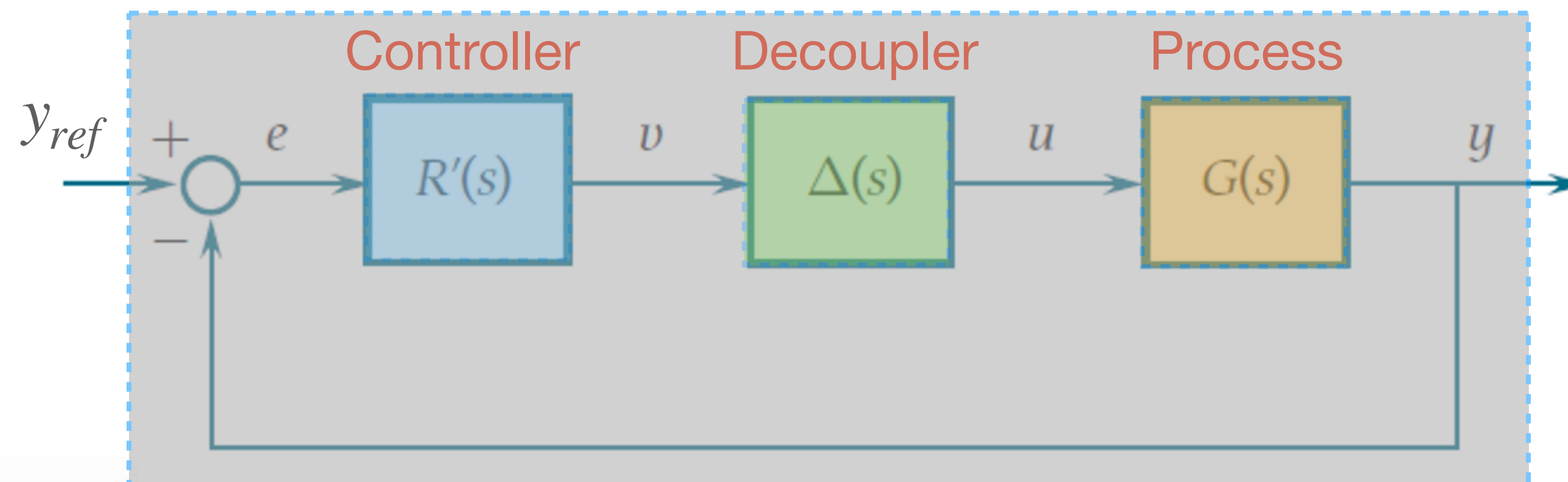
$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

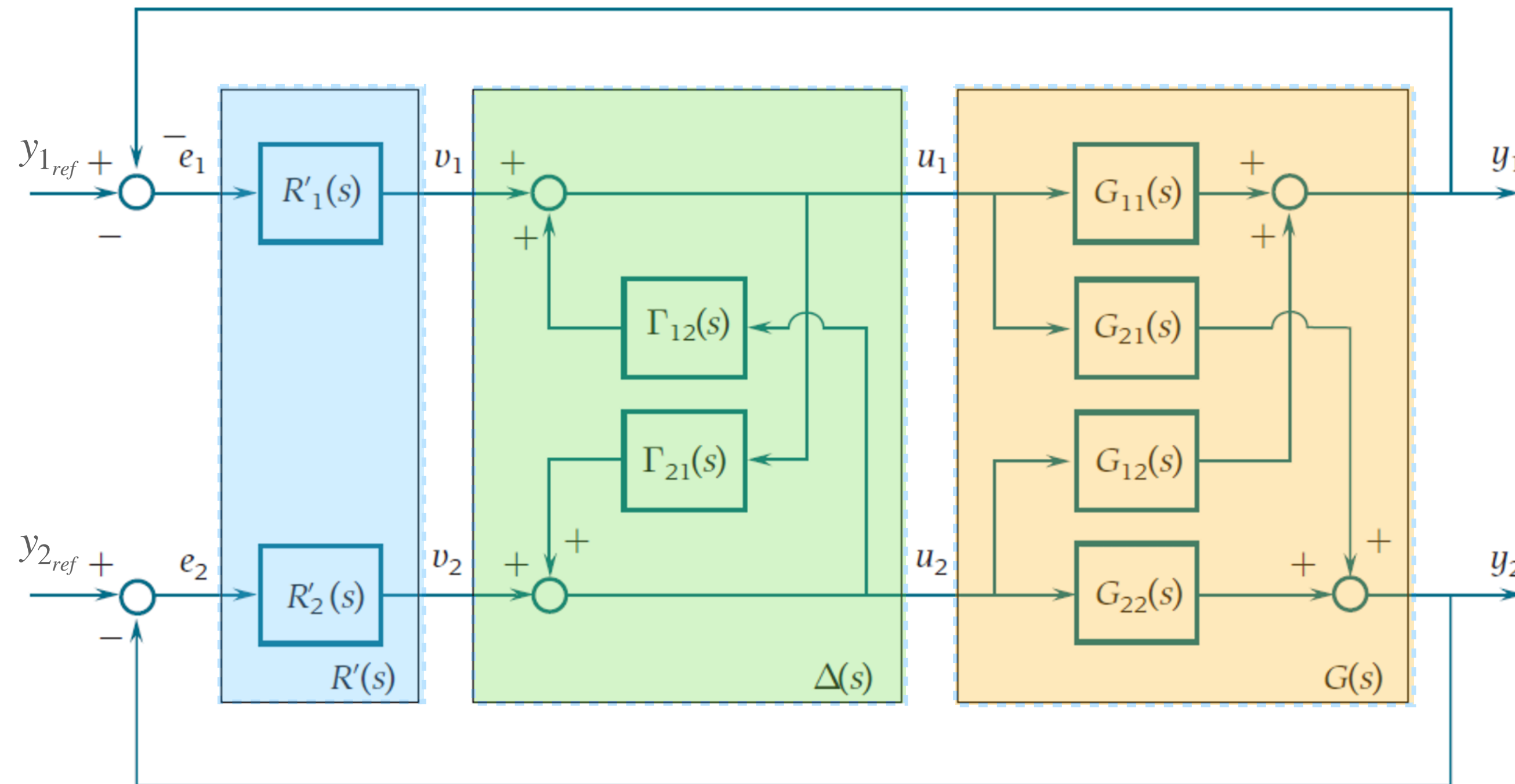
$$U(s) = \Gamma(s)U(s) + V(s)$$

$$U(s) - \Gamma(s)U(s) = V(s)$$

$$[I - \Gamma(s)]U(s) = V(s)$$

$$U(s) = [I - \Gamma(s)]^{-1}V(s)$$



Decoupling Based Control Schemes: **Backward Decoupling**

$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

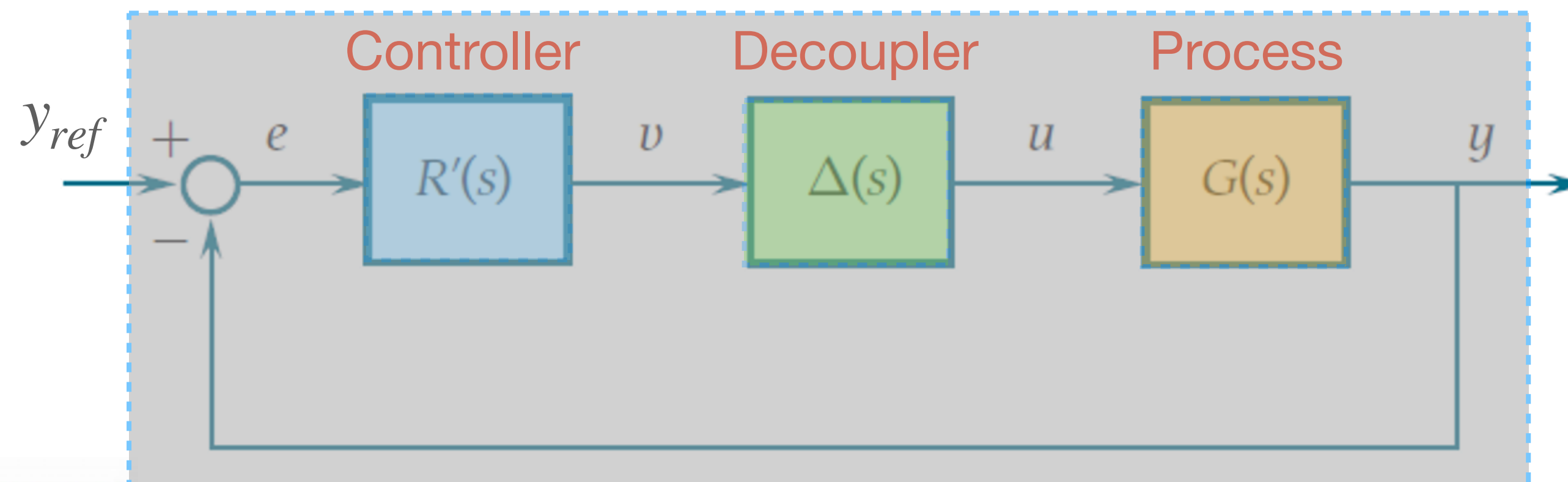
$$U(s) = \Gamma(s)U(s) + V(s)$$

$$U(s) - \Gamma(s)U(s) = V(s)$$

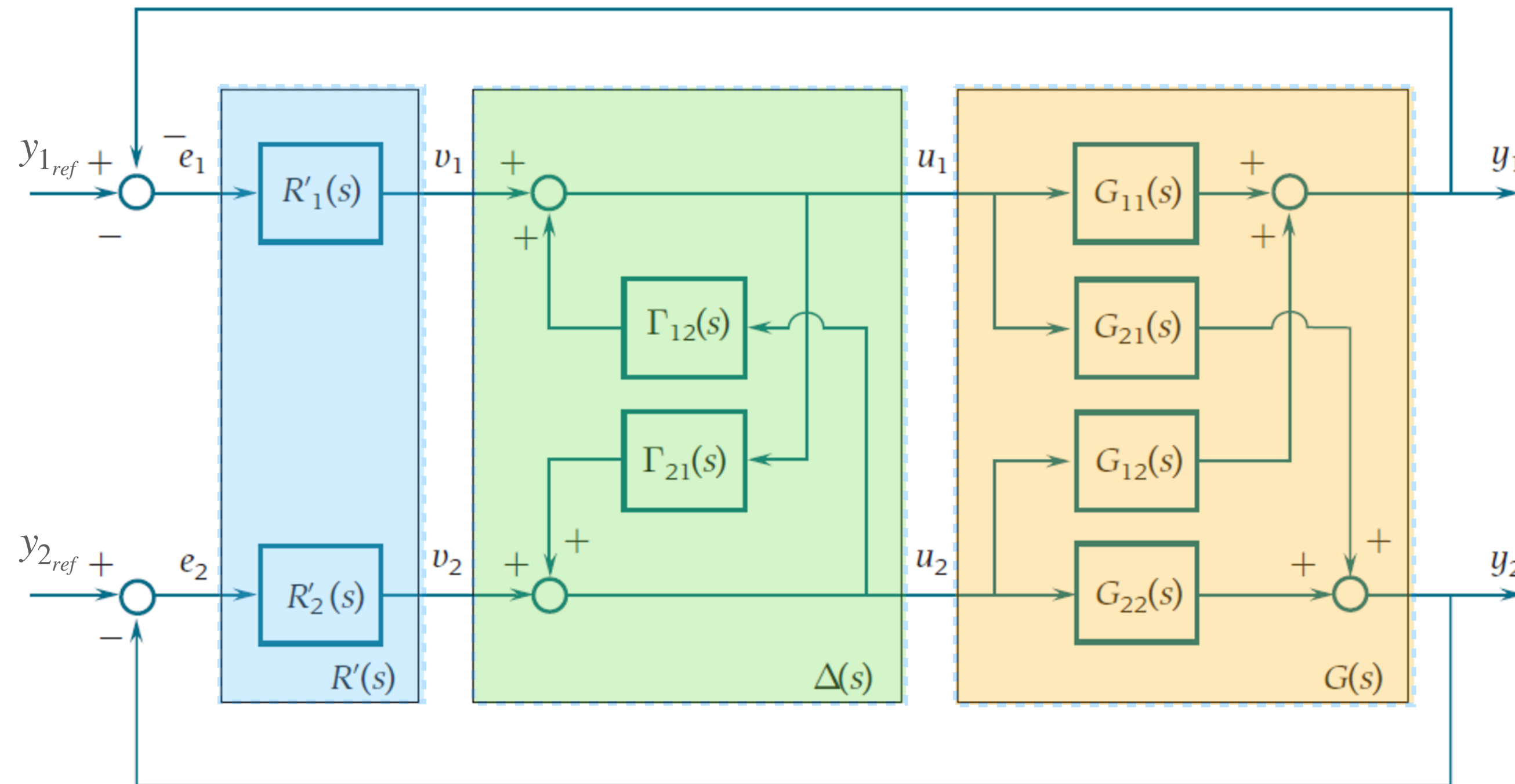
$$[I - \Gamma(s)]U(s) = V(s)$$

$$U(s) = [I - \Gamma(s)]^{-1}V(s)$$

?



Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

$$U(s) = \Gamma(s)U(s) + V(s)$$

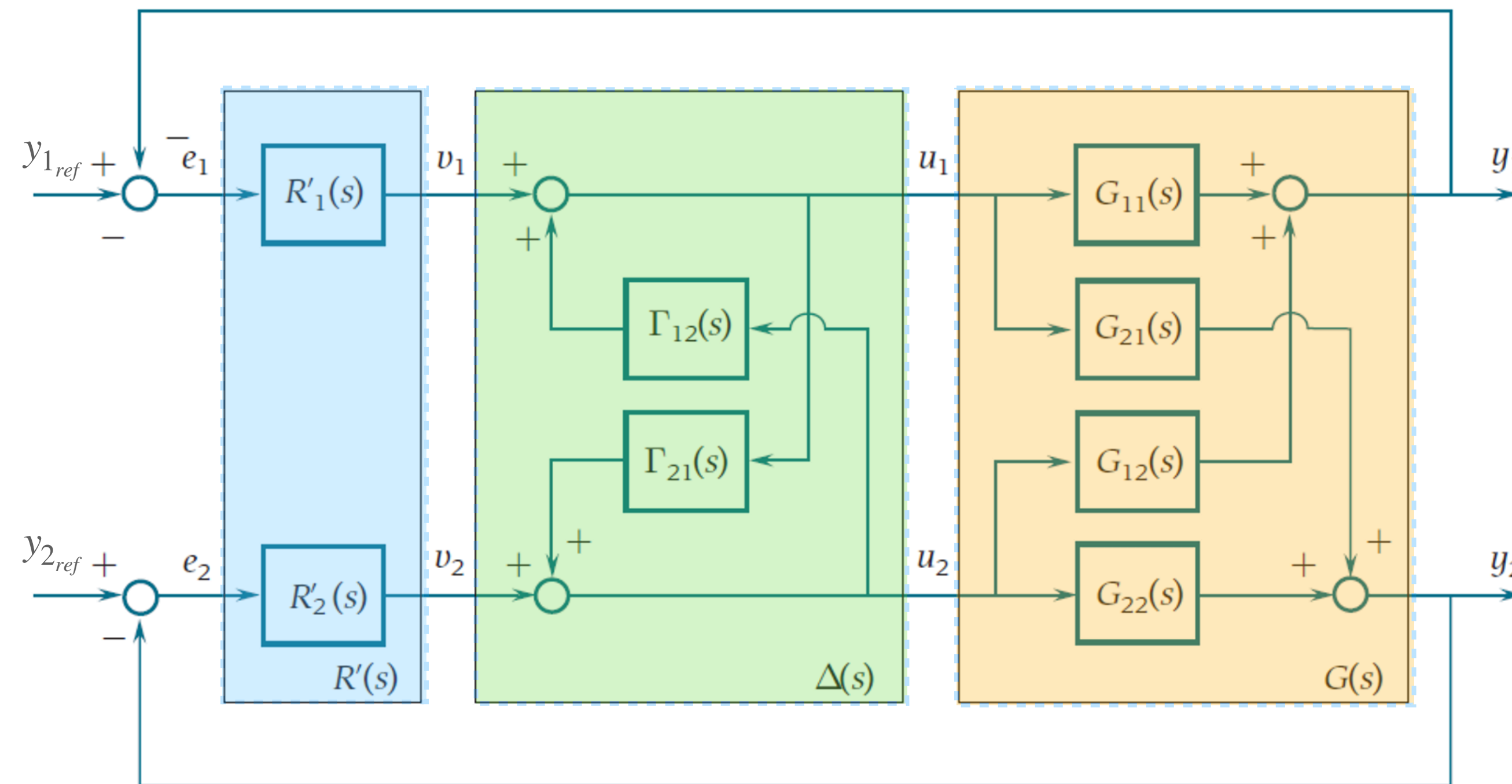
$$U(s) - \Gamma(s)U(s) = V(s)$$

$$[I - \Gamma(s)]U(s) = V(s)$$

$$U(s) = [I - \Gamma(s)]^{-1}V(s)$$

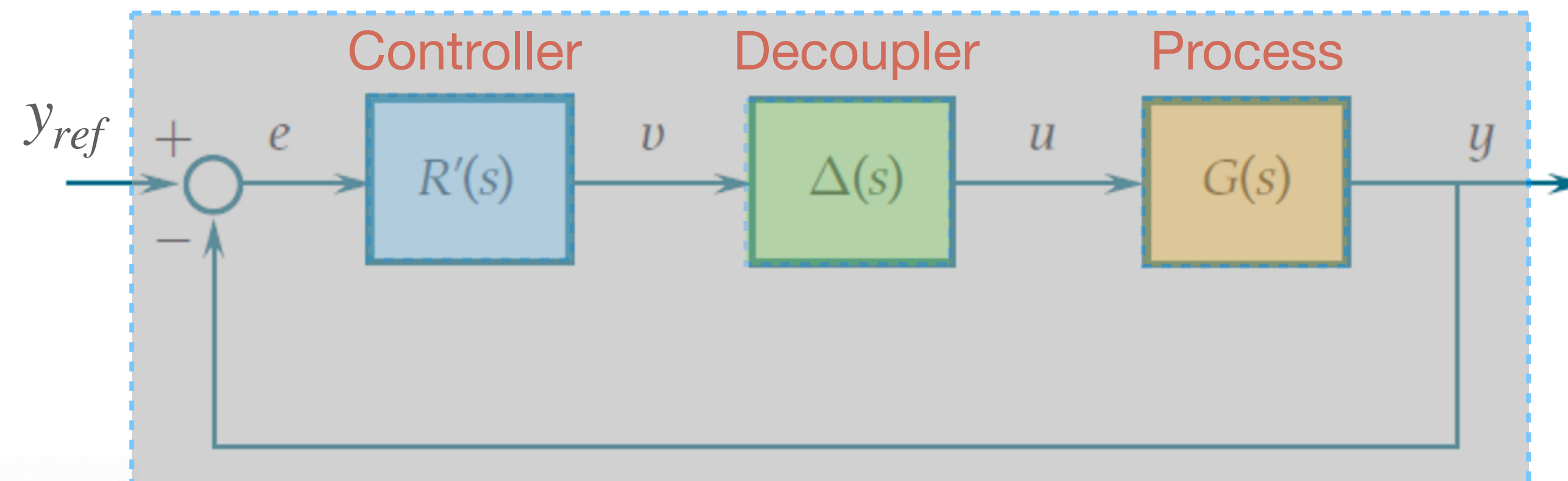
$$\Delta(s)$$

$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

Decoupling Based Control Schemes: **Backward Decoupling**

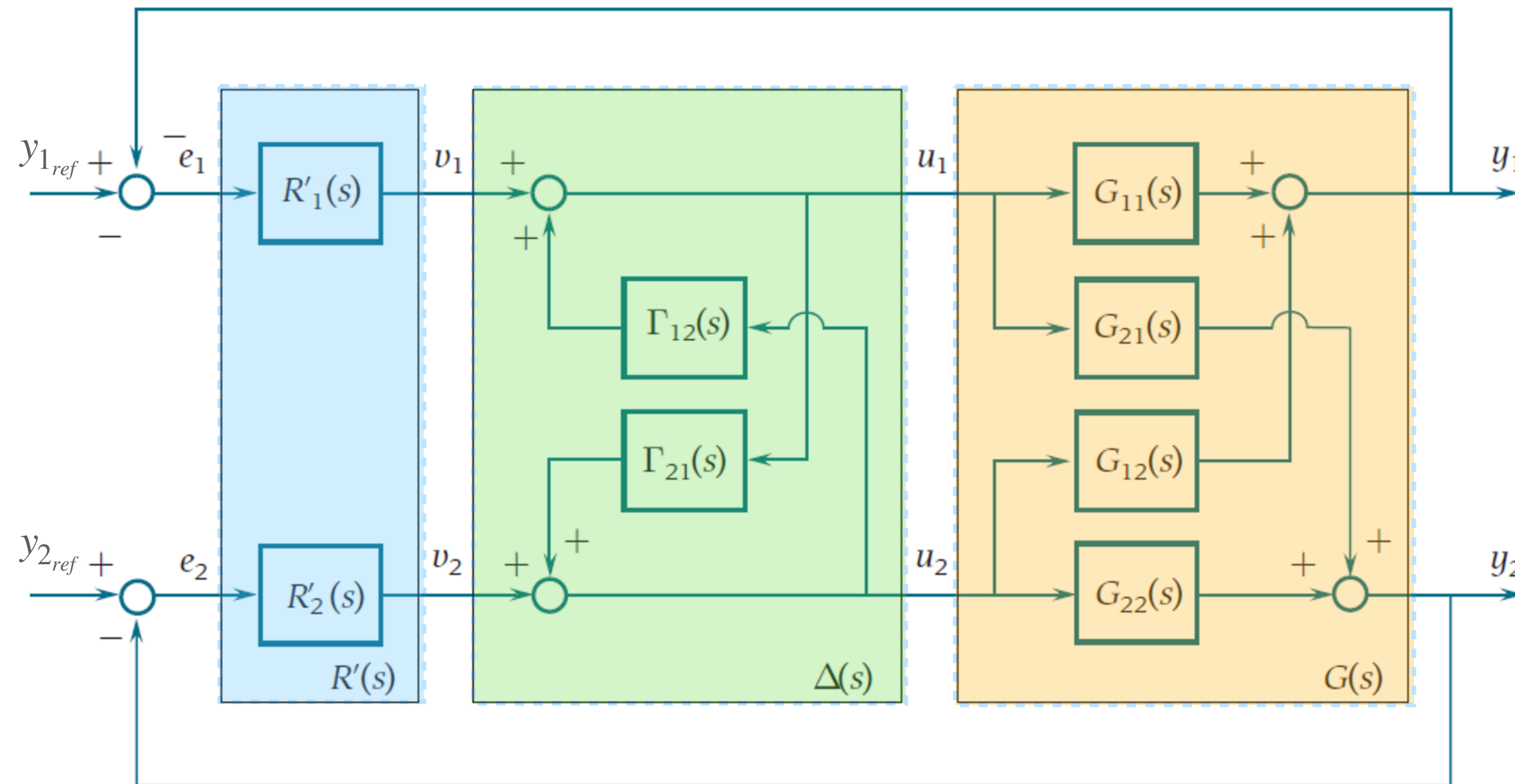
$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$



$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

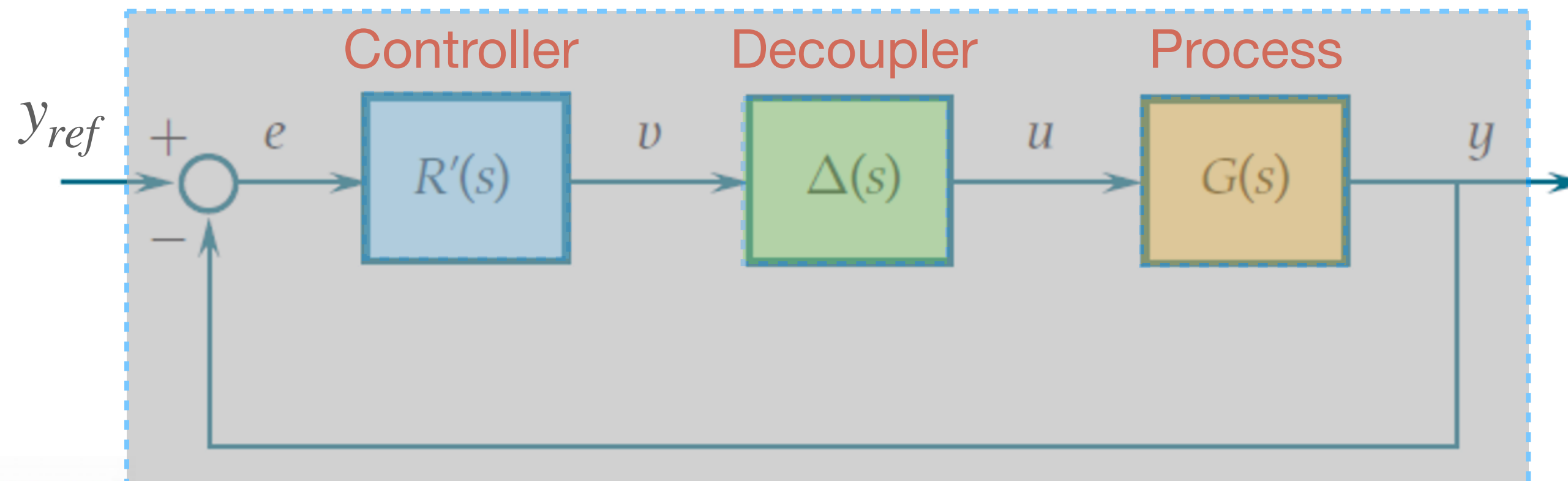
Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

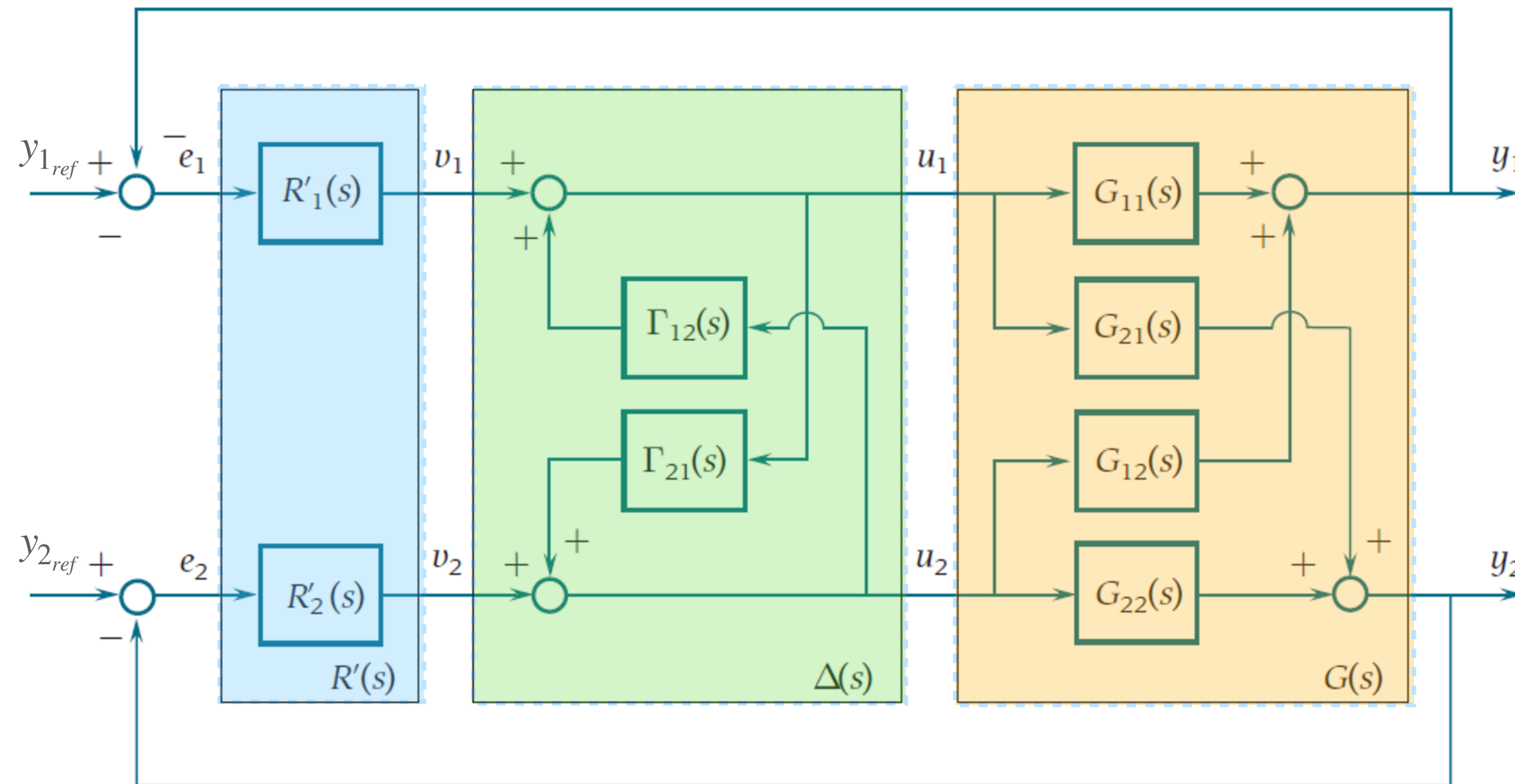
$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

$$G(s)[I - \Gamma(s)]^{-1} = \tilde{G}_d(s)$$



$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

Decoupling Based Control Schemes: **Backward Decoupling**

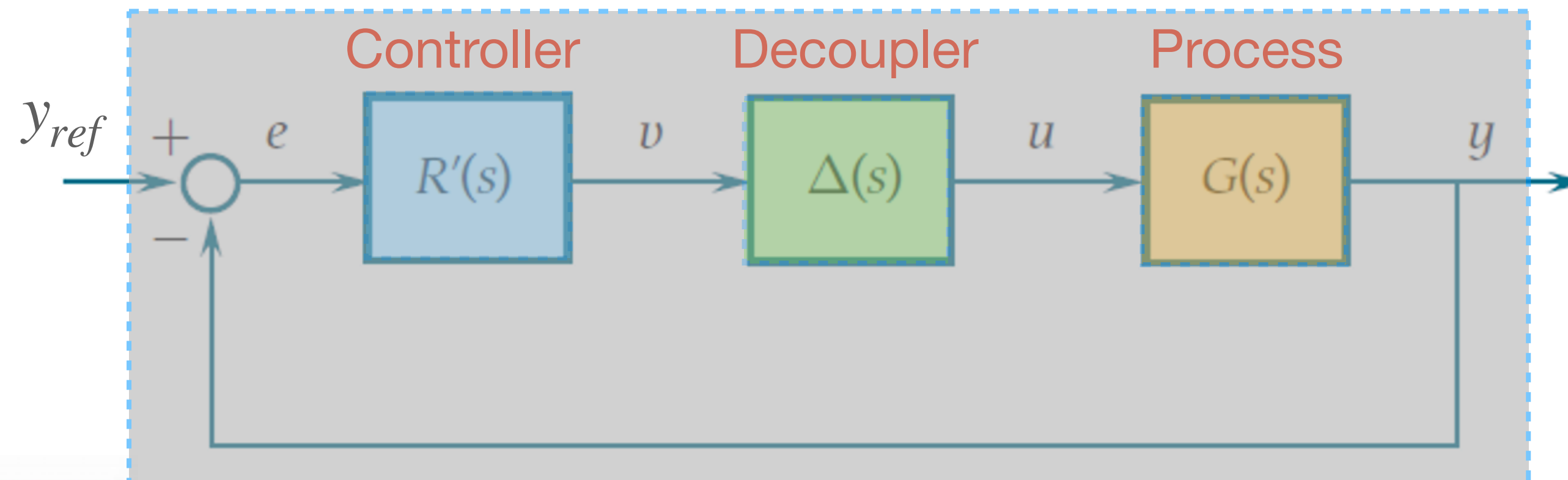


$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

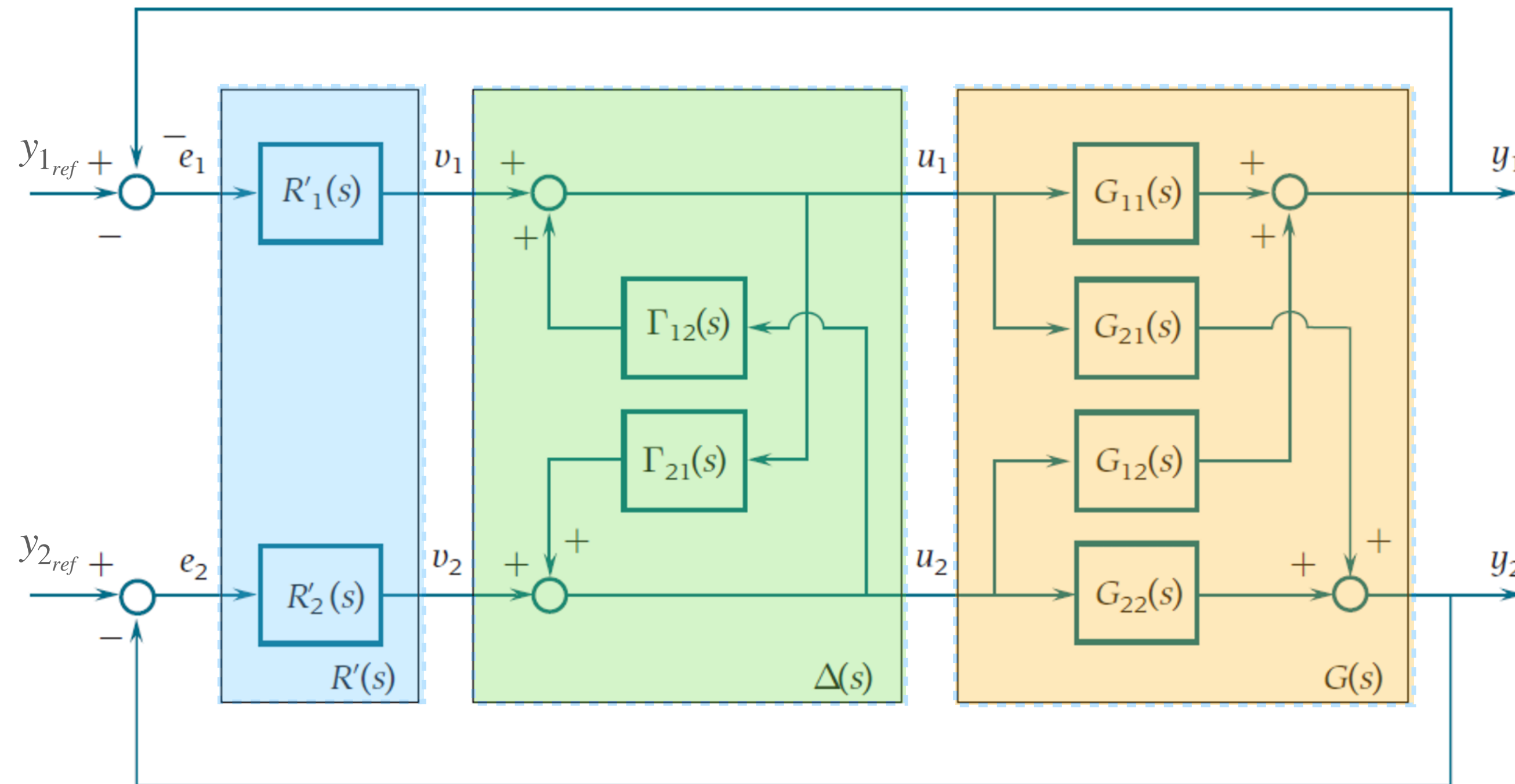
$$G(s)[I - \Gamma(s)]^{-1} = \tilde{G}_d(s)$$

$$G(s) = \tilde{G}_d(s)[I - \Gamma(s)]$$



$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

Decoupling Based Control Schemes: **Backward Decoupling**



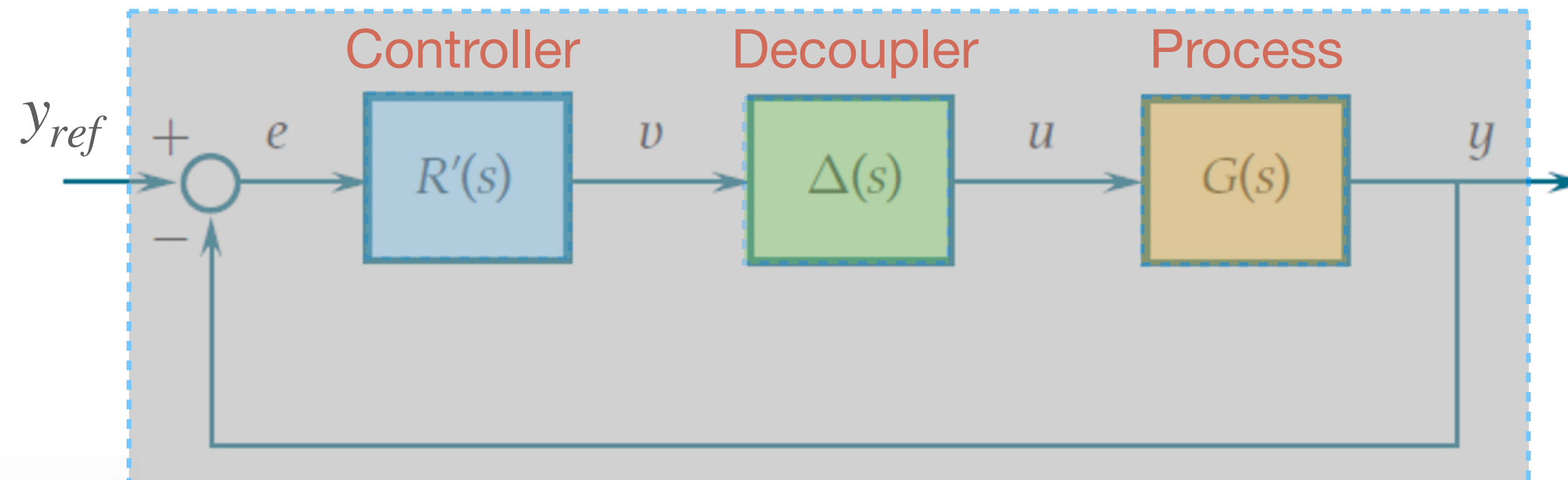
$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

$$G(s)[I - \Gamma(s)]^{-1} = \tilde{G}_d(s)$$

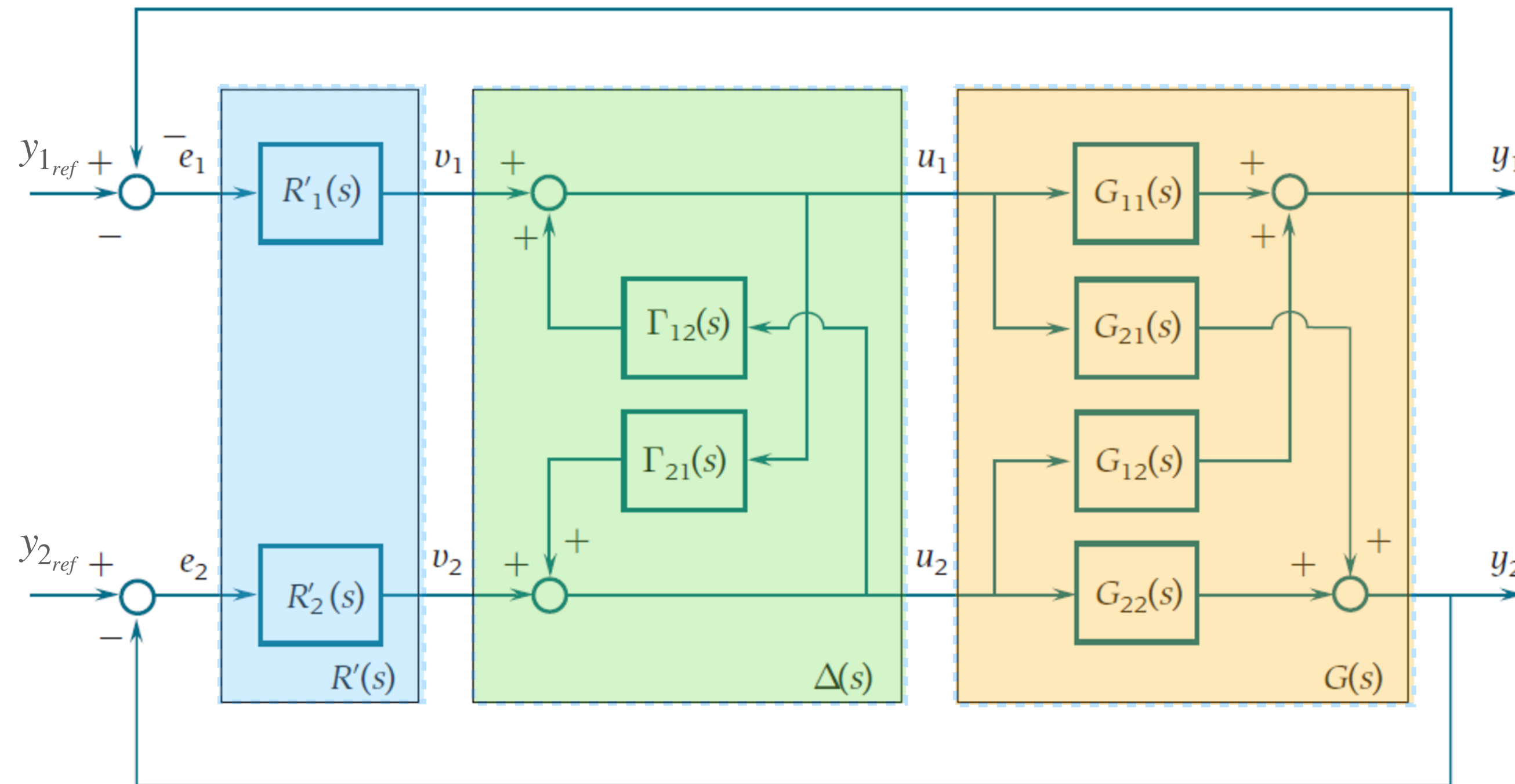
$$G(s) = \tilde{G}_d(s)[I - \Gamma(s)]$$

$$G(s) = \tilde{G}_d(s) - \tilde{G}_d(s)\Gamma(s)$$



$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

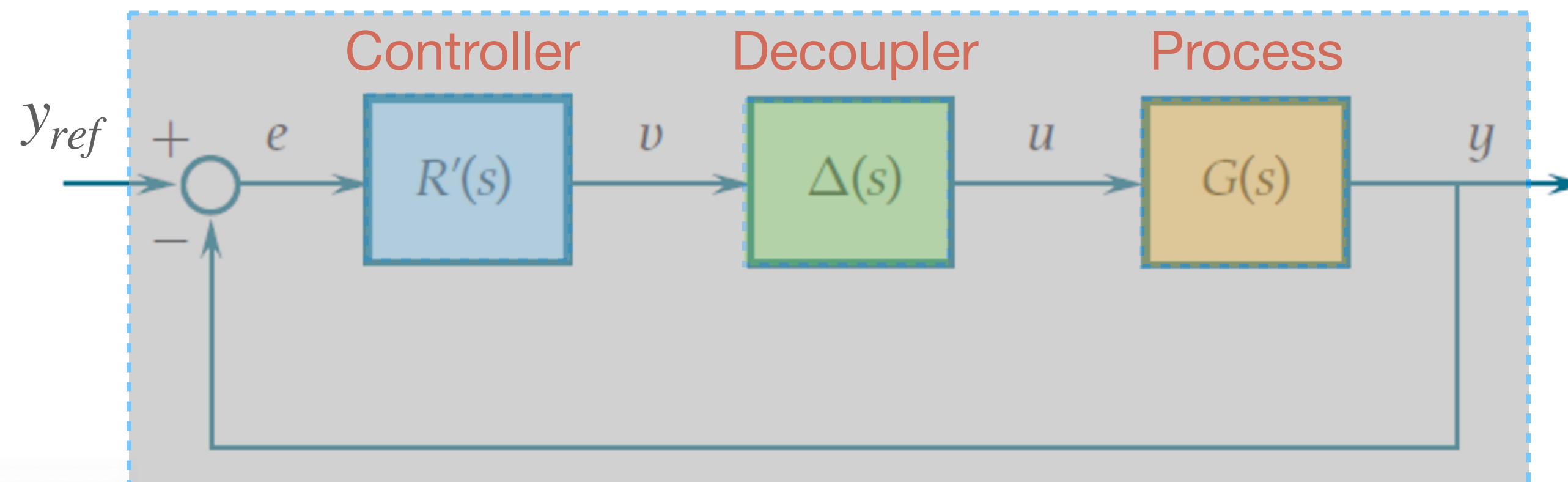
$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

$$G(s)[I - \Gamma(s)]^{-1} = \tilde{G}_d(s)$$

$$G(s) = \tilde{G}_d(s)[I - \Gamma(s)]$$

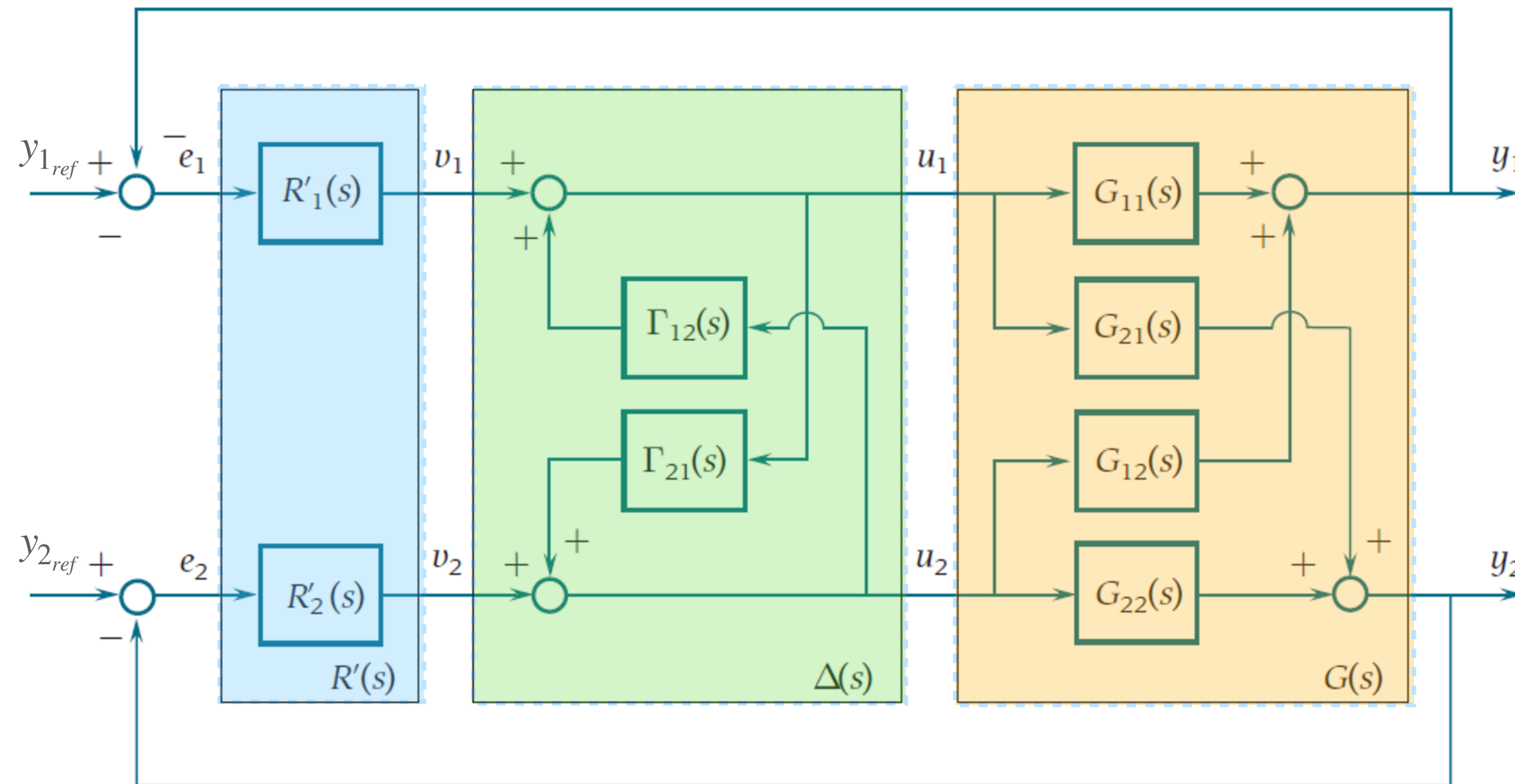
$$G(s) = \tilde{G}_d(s) - \tilde{G}_d(s)\Gamma(s)$$

$$\tilde{G}_d(s)\Gamma(s) = \tilde{G}_d(s) - G(s)$$



$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

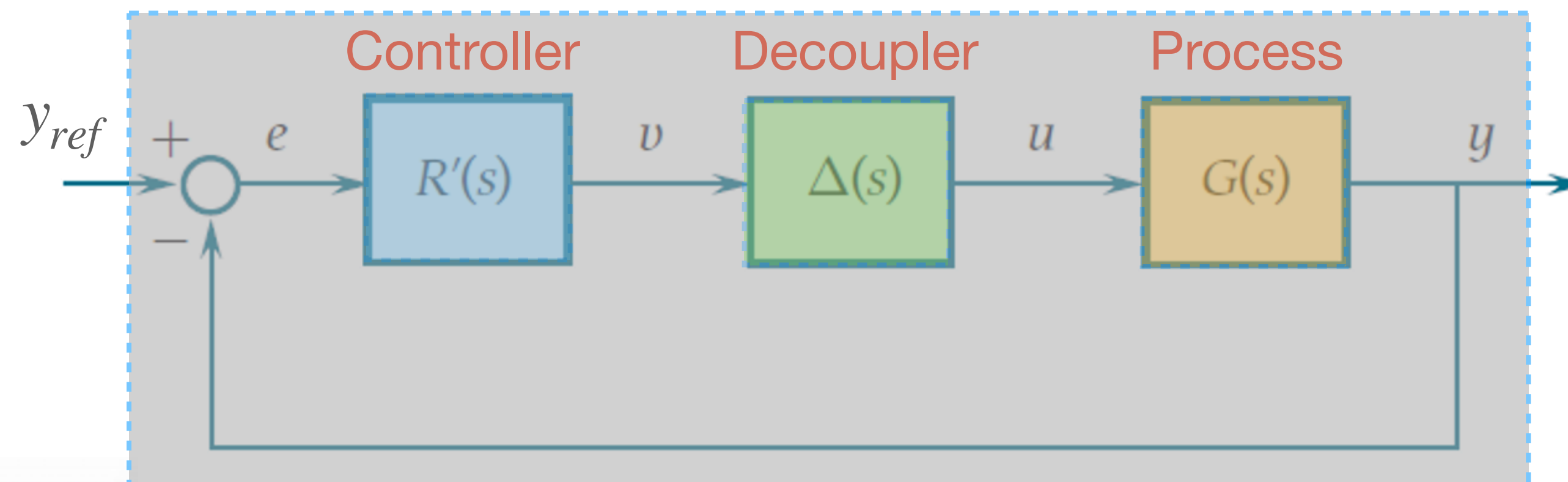
$$G(s)[I - \Gamma(s)]^{-1} = \tilde{G}_d(s)$$

$$G(s) = \tilde{G}_d(s)[I - \Gamma(s)]$$

$$G(s) = \tilde{G}_d(s) - \tilde{G}_d(s)\Gamma(s)$$

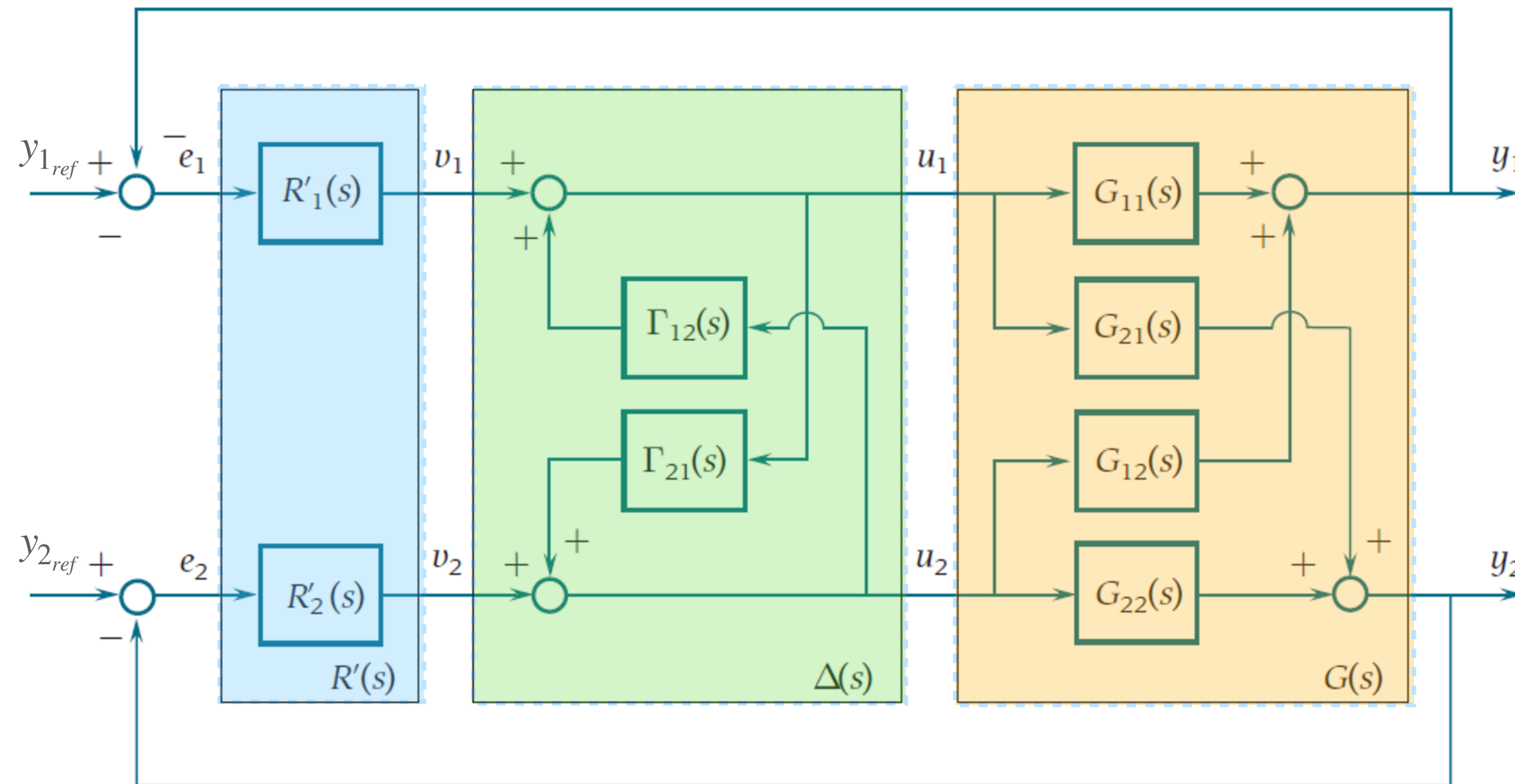
$$\tilde{G}_d(s)\Gamma(s) = \tilde{G}_d(s) - G(s)$$

$$\Gamma(s) = \tilde{G}_d^{-1}(s)[\tilde{G}_d(s) - G(s)]$$



$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

$$G(s)[I - \Gamma(s)]^{-1} = \tilde{G}_d(s)$$

$$G(s) = \tilde{G}_d(s)[I - \Gamma(s)]$$

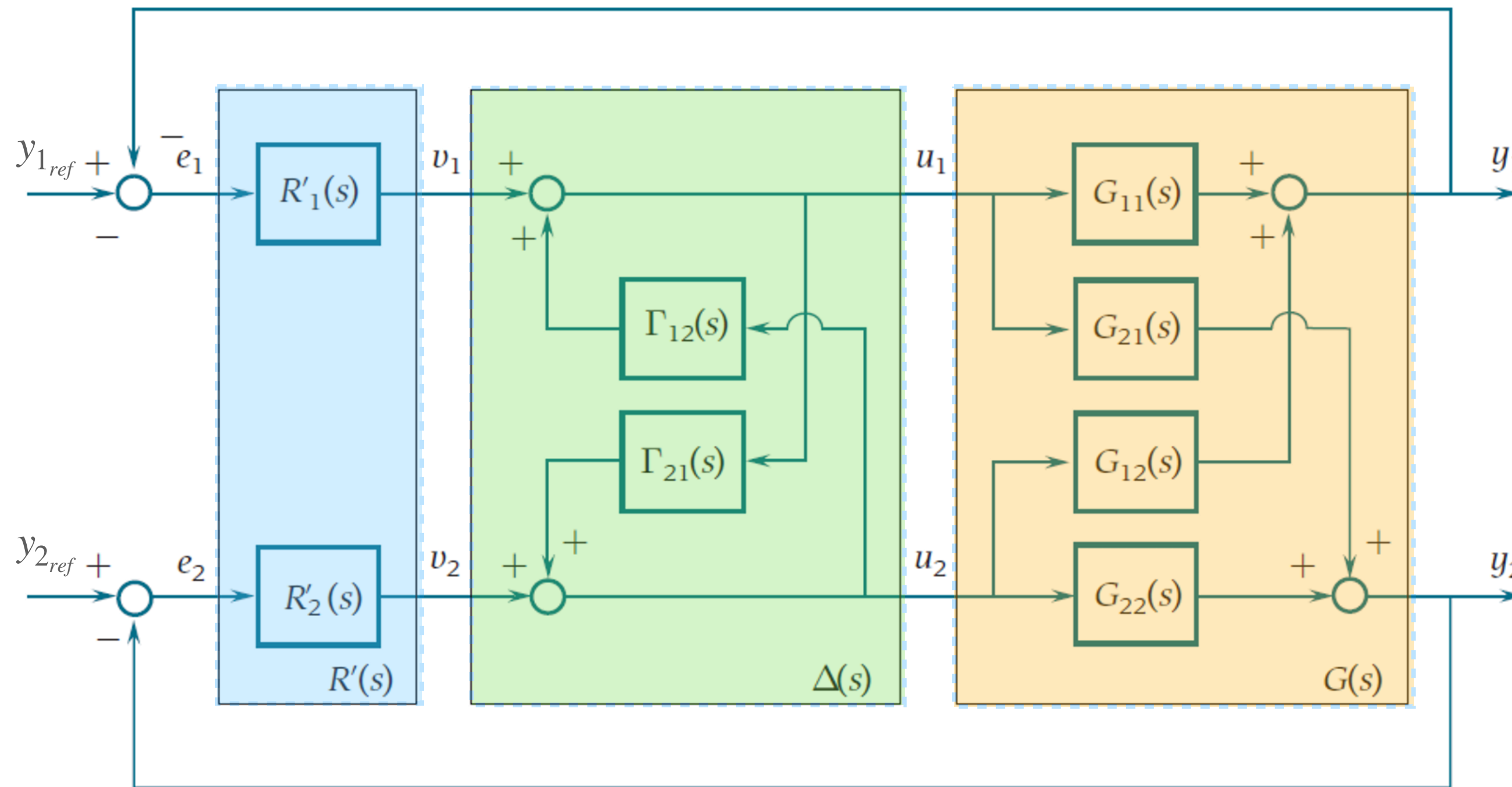
$$G(s) = \tilde{G}_d(s) - \tilde{G}_d(s)\Gamma(s)$$

$$\tilde{G}_d(s)\Gamma(s) = \tilde{G}_d(s) - G(s)$$

$$\Gamma(s) = \tilde{G}_d^{-1}(s)[\tilde{G}_d(s) - G(s)]$$

$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

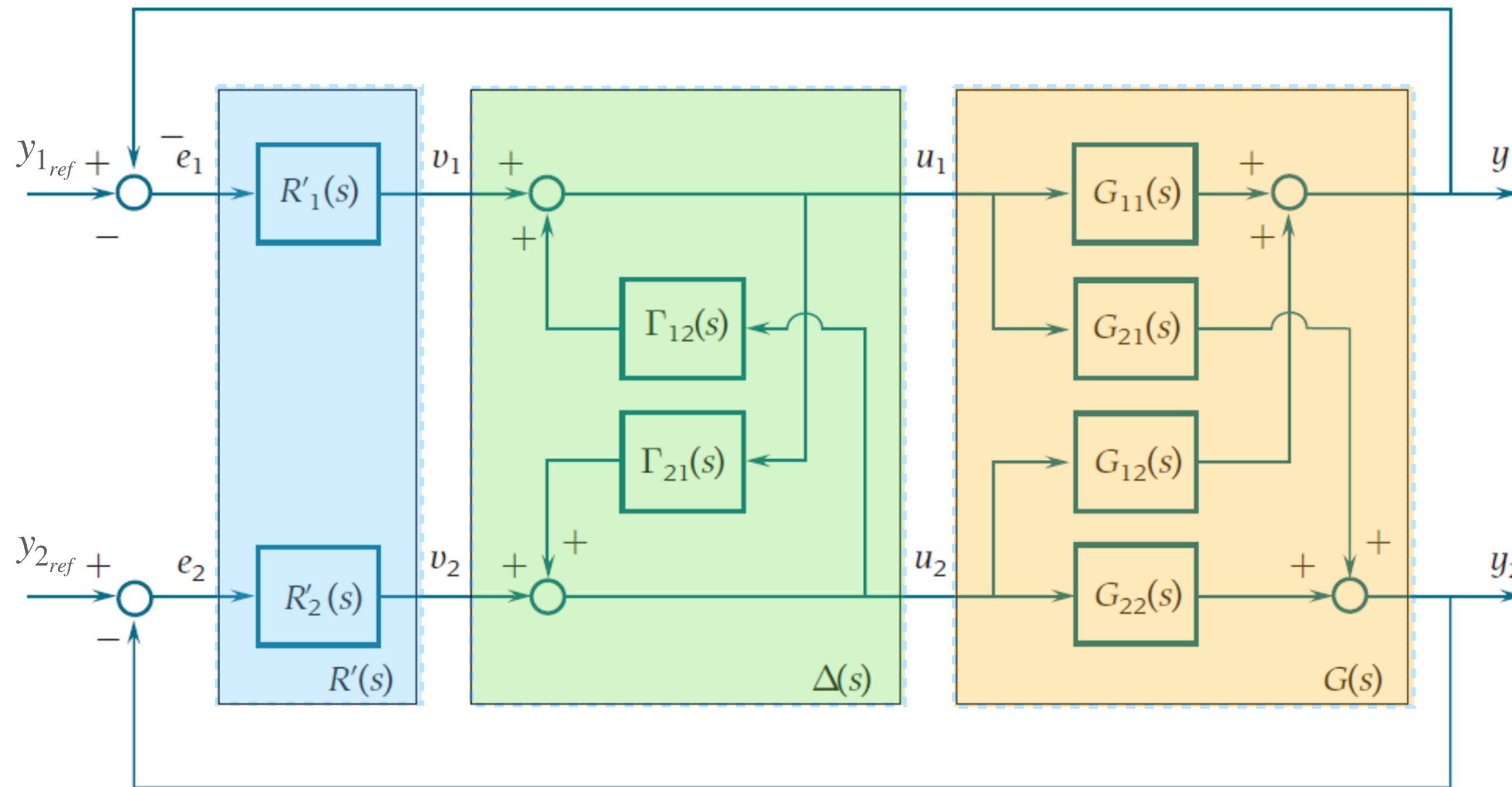
$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

In the 2×2 case:

$$\Gamma(s) = \tilde{G}_d^{-1}(s)[\tilde{G}_d(s) - G(s)]$$

$$\Gamma(s) = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}^{-1} \left\{ \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix} - \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \right\}$$

Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

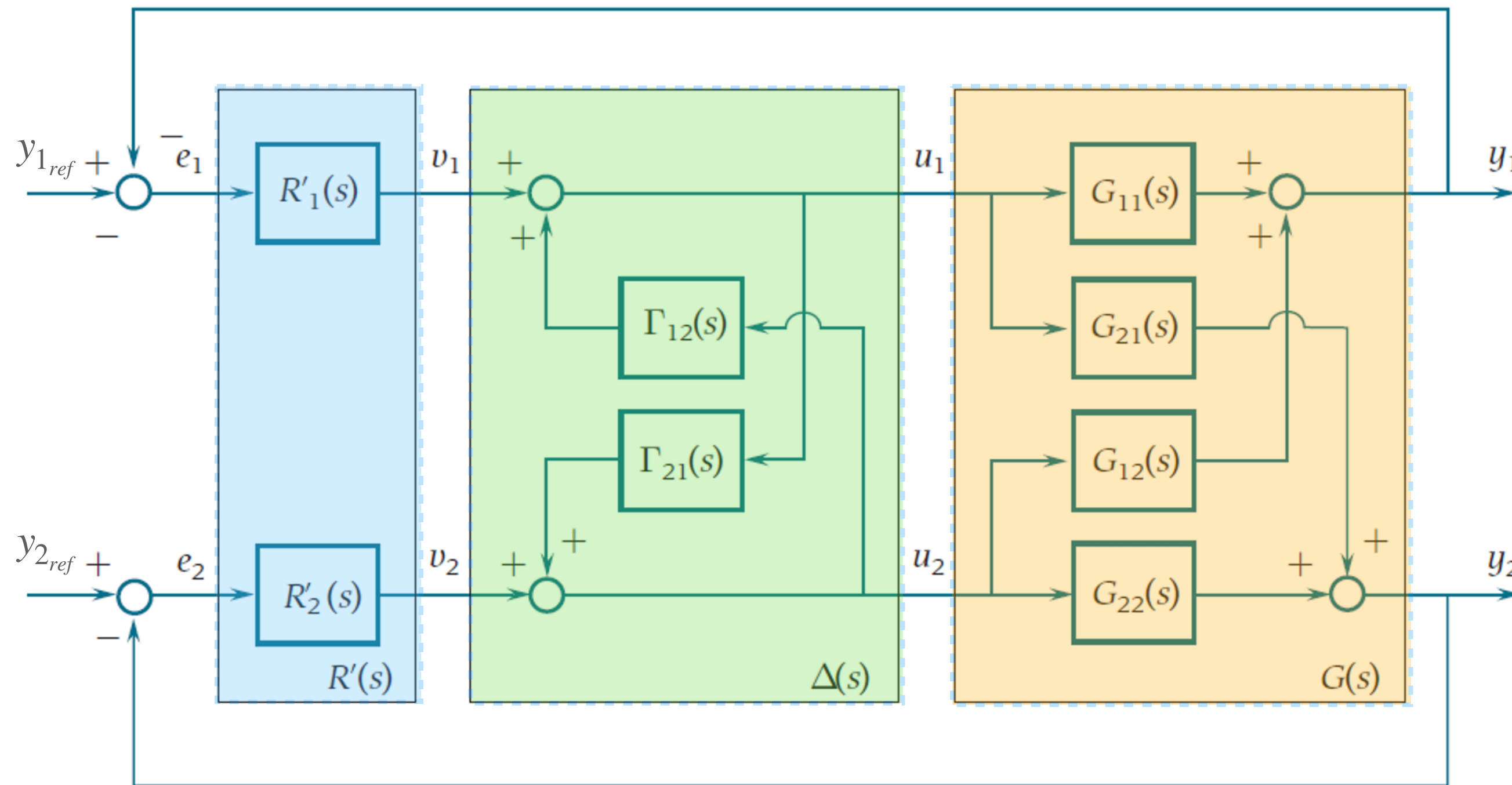
In the 2×2 case:

$$\Gamma(s) = \tilde{G}_d^{-1}(s)[\tilde{G}_d(s) - G(s)]$$

$$\Gamma(s) = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}^{-1} \left\{ \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix} - \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \right\}$$

$$\Gamma(s) = \begin{bmatrix} \frac{1}{G_{11}(s)} & 0 \\ 0 & \frac{1}{G_{22}(s)} \end{bmatrix} \begin{bmatrix} 0 & -G_{12}(s) \\ -G_{21}(s) & 0 \end{bmatrix}$$

Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

In the 2×2 case:

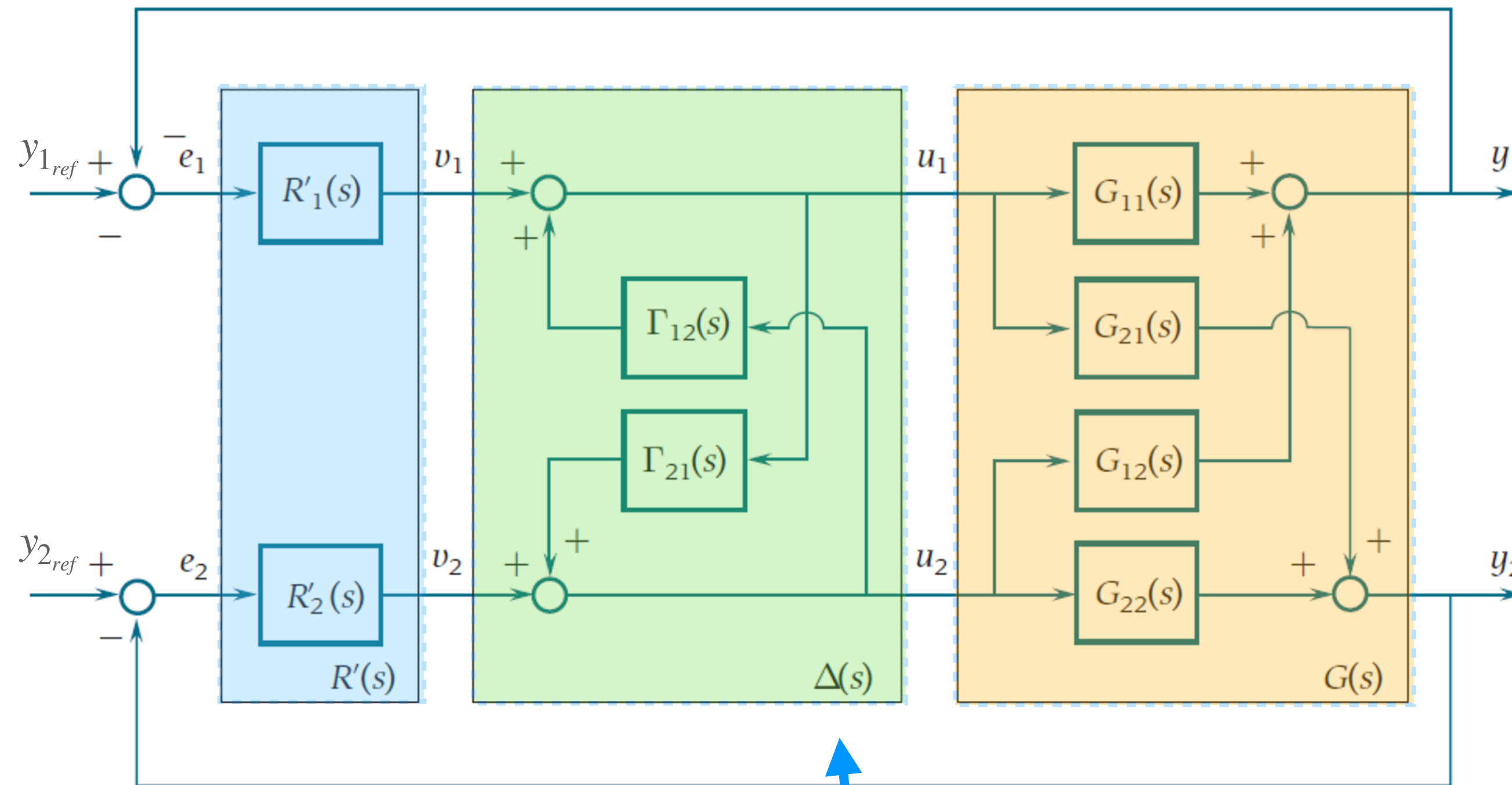
$$\Gamma(s) = \tilde{G}_d^{-1}(s)[\tilde{G}_d(s) - G(s)]$$

$$\Gamma(s) = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}^{-1} \left\{ \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix} - \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \right\}$$

$$\Gamma(s) = \begin{bmatrix} \frac{1}{G_{11}(s)} & 0 \\ 0 & \frac{1}{G_{22}(s)} \end{bmatrix} \begin{bmatrix} 0 & -G_{12}(s) \\ -G_{21}(s) & 0 \end{bmatrix}$$

$$\Gamma(s) = \begin{bmatrix} 0 & -\frac{G_{12}(s)}{G_{11}(s)} \\ -\frac{G_{21}(s)}{G_{22}(s)} & 0 \end{bmatrix}$$

Decoupling Based Control Schemes: **Backward Decoupling**



$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

In the 2×2 case:

$$\Gamma(s) = \tilde{G}_d^{-1}(s)[\tilde{G}_d(s) - G(s)]$$

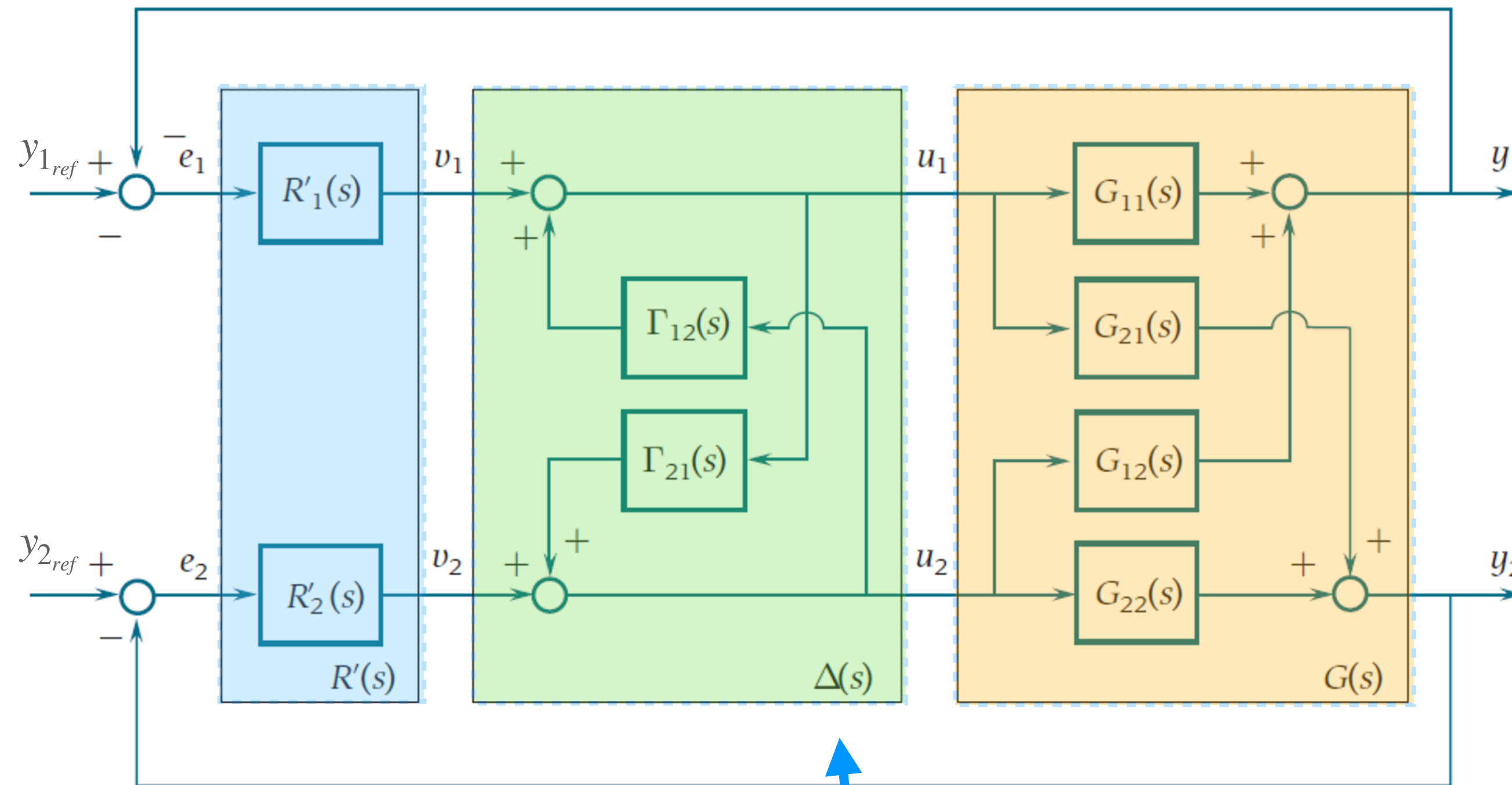
$$\Gamma(s) = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}^{-1} \left\{ \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix} - \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \right\}$$

$$\Gamma(s) = \begin{bmatrix} \frac{1}{G_{11}(s)} & 0 \\ 0 & \frac{1}{G_{22}(s)} \end{bmatrix} \begin{bmatrix} 0 & -G_{12}(s) \\ -G_{21}(s) & 0 \end{bmatrix}$$

$$\Gamma(s) = \begin{bmatrix} 0 & -\frac{G_{12}(s)}{G_{11}(s)} \\ -\frac{G_{21}(s)}{G_{22}(s)} & 0 \end{bmatrix}$$

$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

Decoupling Based Control Schemes: **Backward Decoupling**



$\Gamma(s)$ must be causal:
Approximations in the frequency domain of its component $\Gamma_{ij}(s)$ must be adopted if this is not true

$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

$$G(s)\Delta(s) = \tilde{G}_d(s)$$

$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

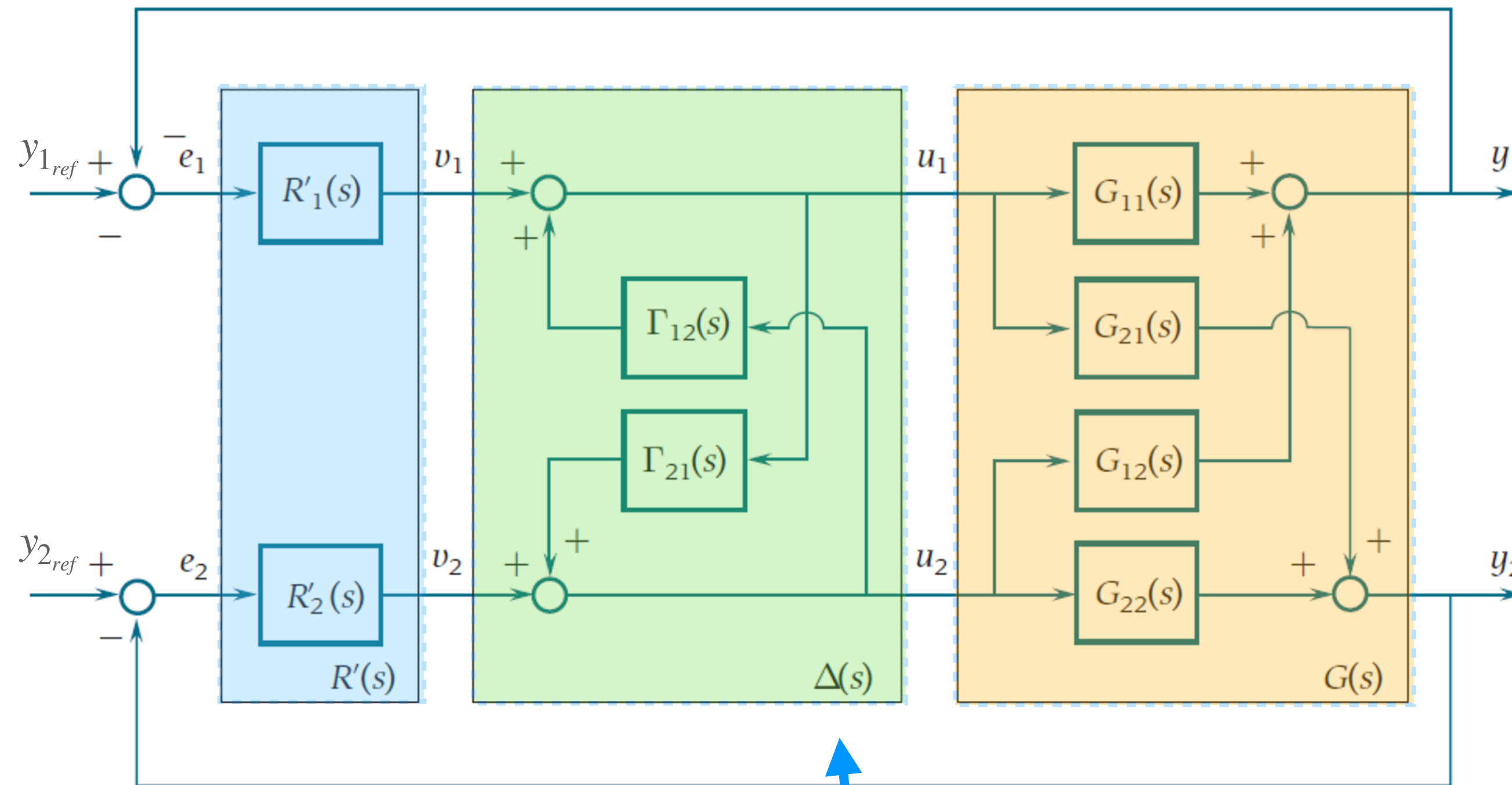
In the 2×2 case:

$$\Gamma(s) = \tilde{G}_d^{-1}(s)[\tilde{G}_d(s) - G(s)]$$

$$\Gamma(s) = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}^{-1} \left\{ \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix} - \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \right\}$$

$$\Gamma(s) = \begin{bmatrix} \frac{1}{G_{11}(s)} & 0 \\ 0 & \frac{1}{G_{22}(s)} \end{bmatrix} \begin{bmatrix} 0 & -G_{12}(s) \\ -G_{21}(s) & 0 \end{bmatrix}$$

$$\Gamma(s) = \begin{bmatrix} 0 & -\frac{G_{12}(s)}{G_{11}(s)} \\ -\frac{G_{21}(s)}{G_{22}(s)} & 0 \end{bmatrix}$$

Decoupling Based Control Schemes: **Backward Decoupling**

$\Gamma(s)$ must be causal:
 Approximations in the frequency domain of its component $\Gamma_{ij}(s)$ must be adopted if this is not true

$$\Delta(s) = [I - \Gamma(s)]^{-1}$$

$$\Gamma(s) = \begin{bmatrix} 0 & -\frac{G_{12}(s)}{G_{11}(s)} \\ -\frac{G_{21}(s)}{G_{22}(s)} & 0 \end{bmatrix}$$

$$G(s)\Delta(s) = \tilde{G}_d(s)$$

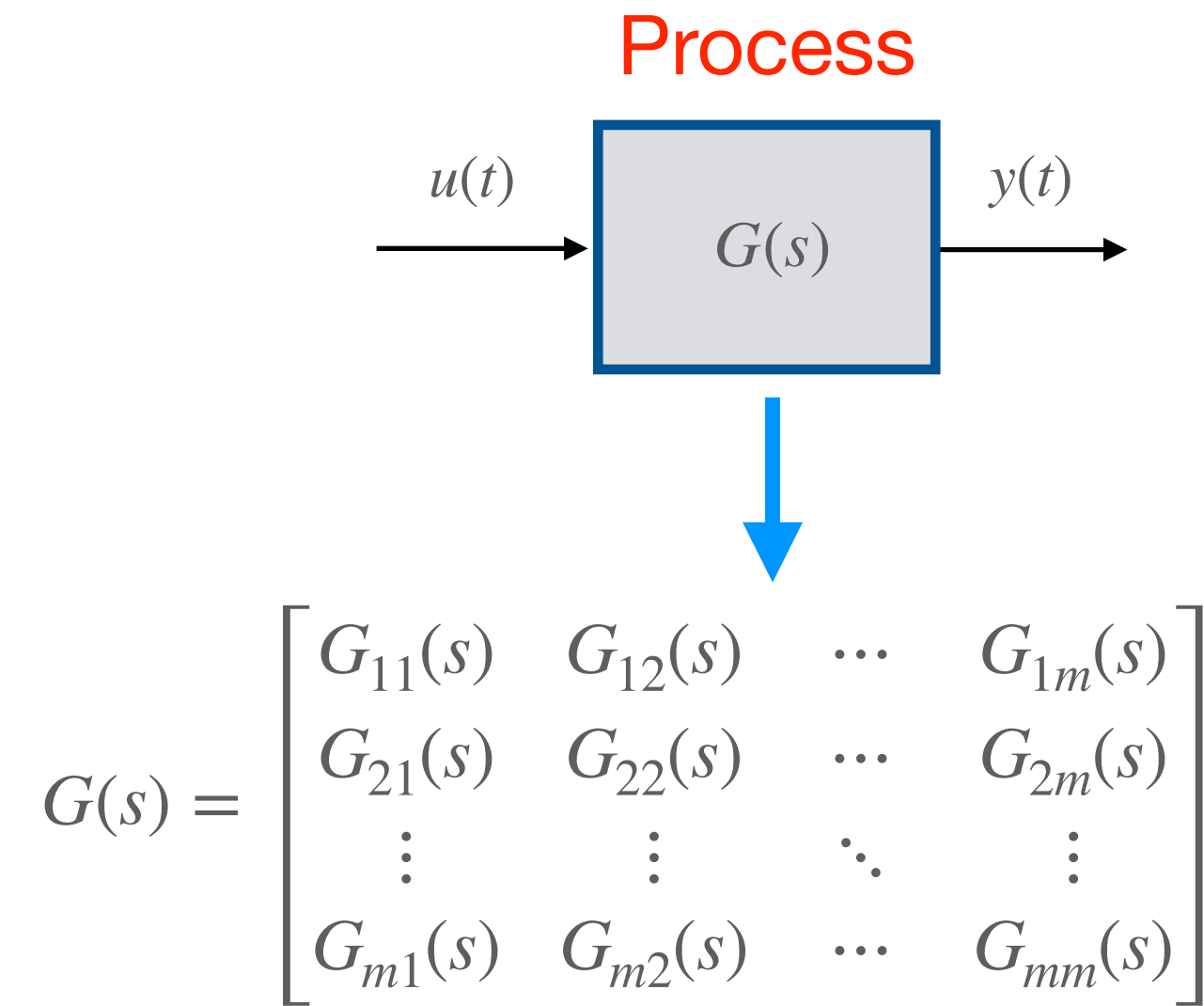
$$\begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} \Delta_{11}(s) & \Delta_{12}(s) \\ \Delta_{21}(s) & \Delta_{22}(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & 0 \\ 0 & G_{22}(s) \end{bmatrix}$$

To conclude:

design: $R'(s) = \begin{bmatrix} R'_1(s) & 0 \\ 0 & R'_2(s) \end{bmatrix}$

Arrows indicate connections from $R'_1(s)$ to $G_{11}(s)$ and from $R'_2(s)$ to $G_{22}(s)$.

Centralized vs. Decentralized MIMO Control Schemes



$$u(t) \in \mathbb{R}^m$$

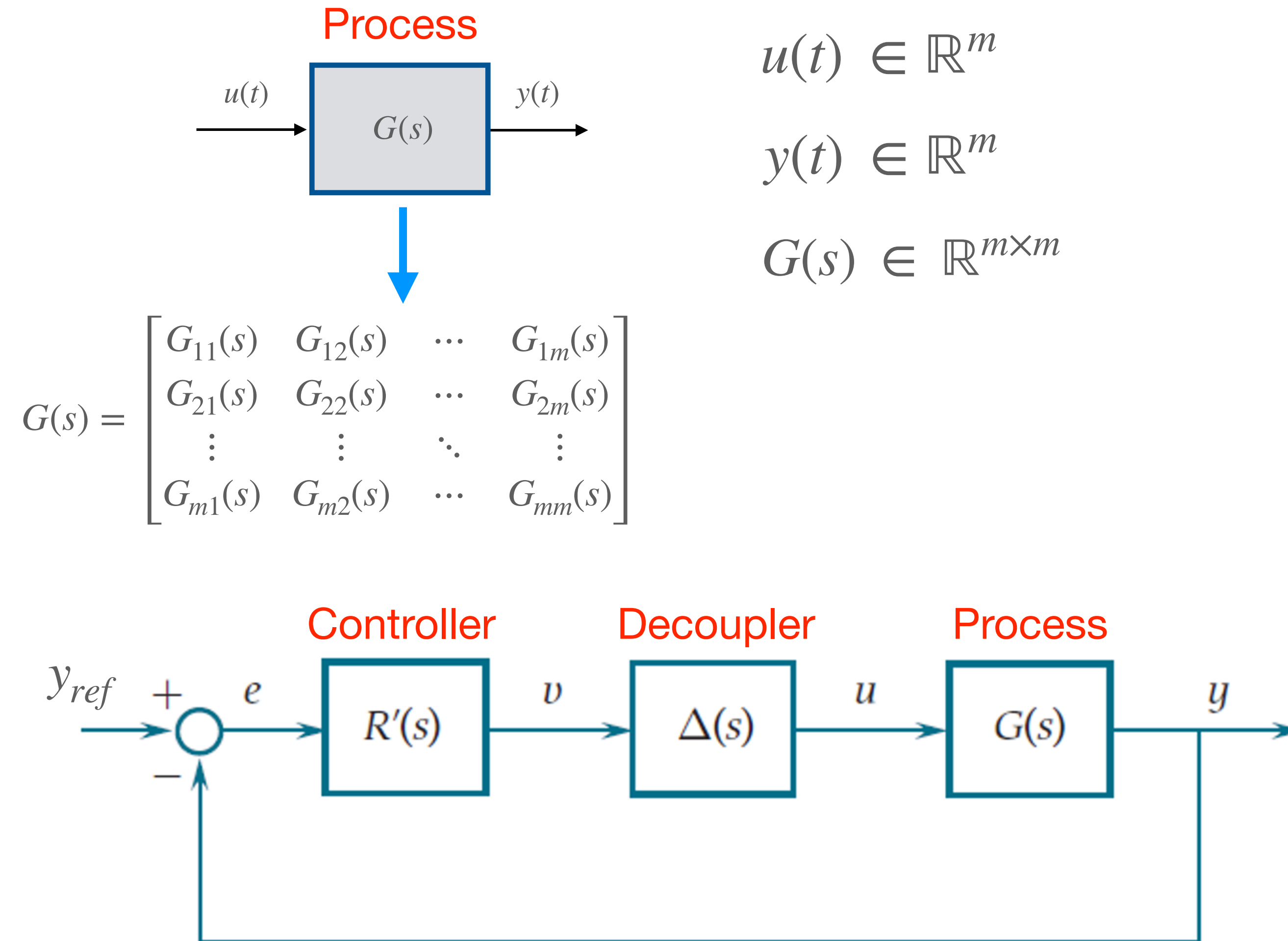
$$y(t) \in \mathbb{R}^m$$

$$G(s) \in \mathbb{R}^{m \times m}$$

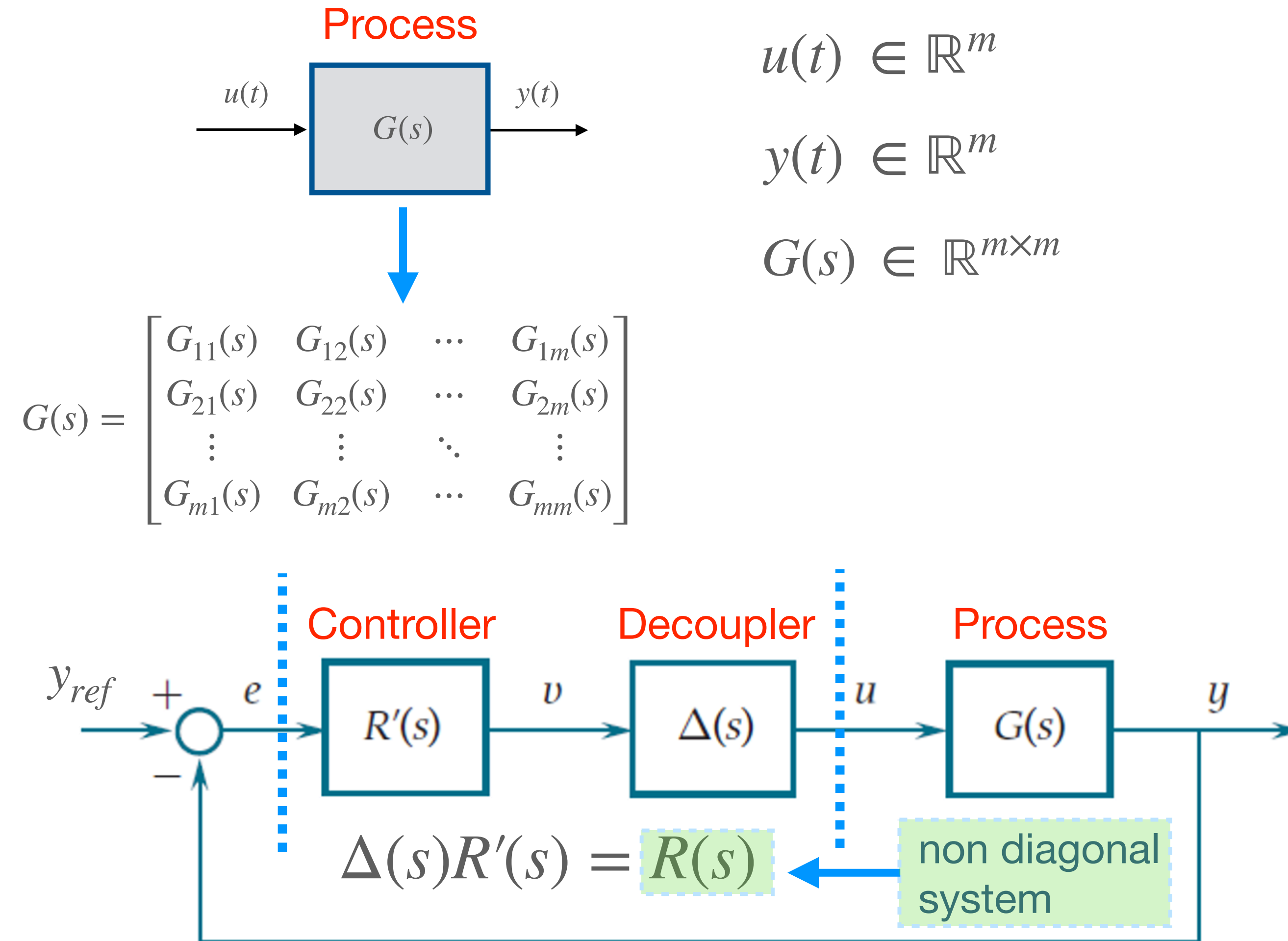
Assumptions:

- $G(s) \in \mathbb{R}^{m \times m}$
- $G(s)$ full matrix

Centralized vs. Decentralized MIMO Control Schemes



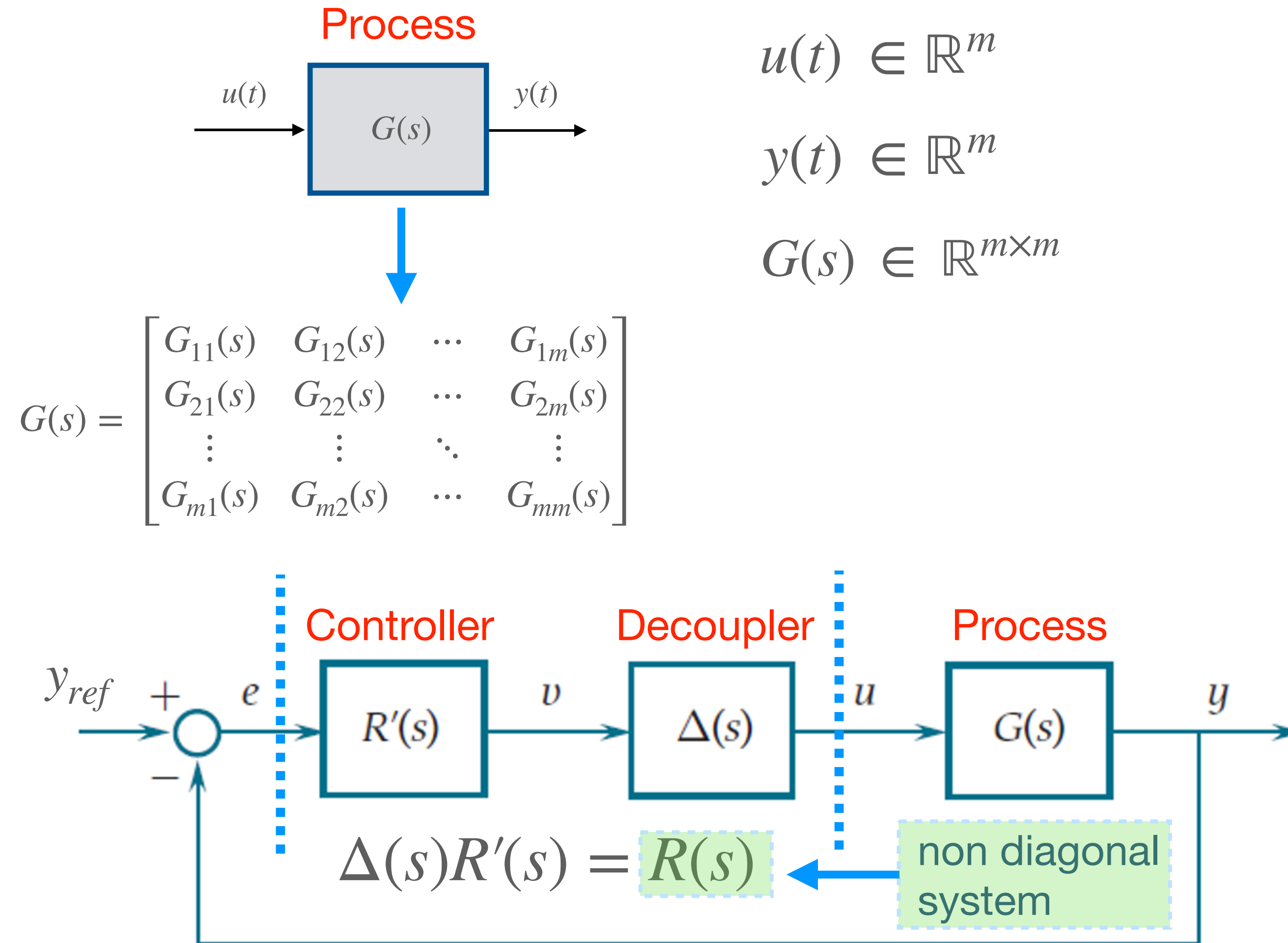
Centralized vs. Decentralized MIMO Control Schemes



Assumptions:

- $G(s) \in \mathbb{R}^{m \times m}$
- $G(s)$ full matrix

Centralized vs. Decentralized MIMO Control Schemes

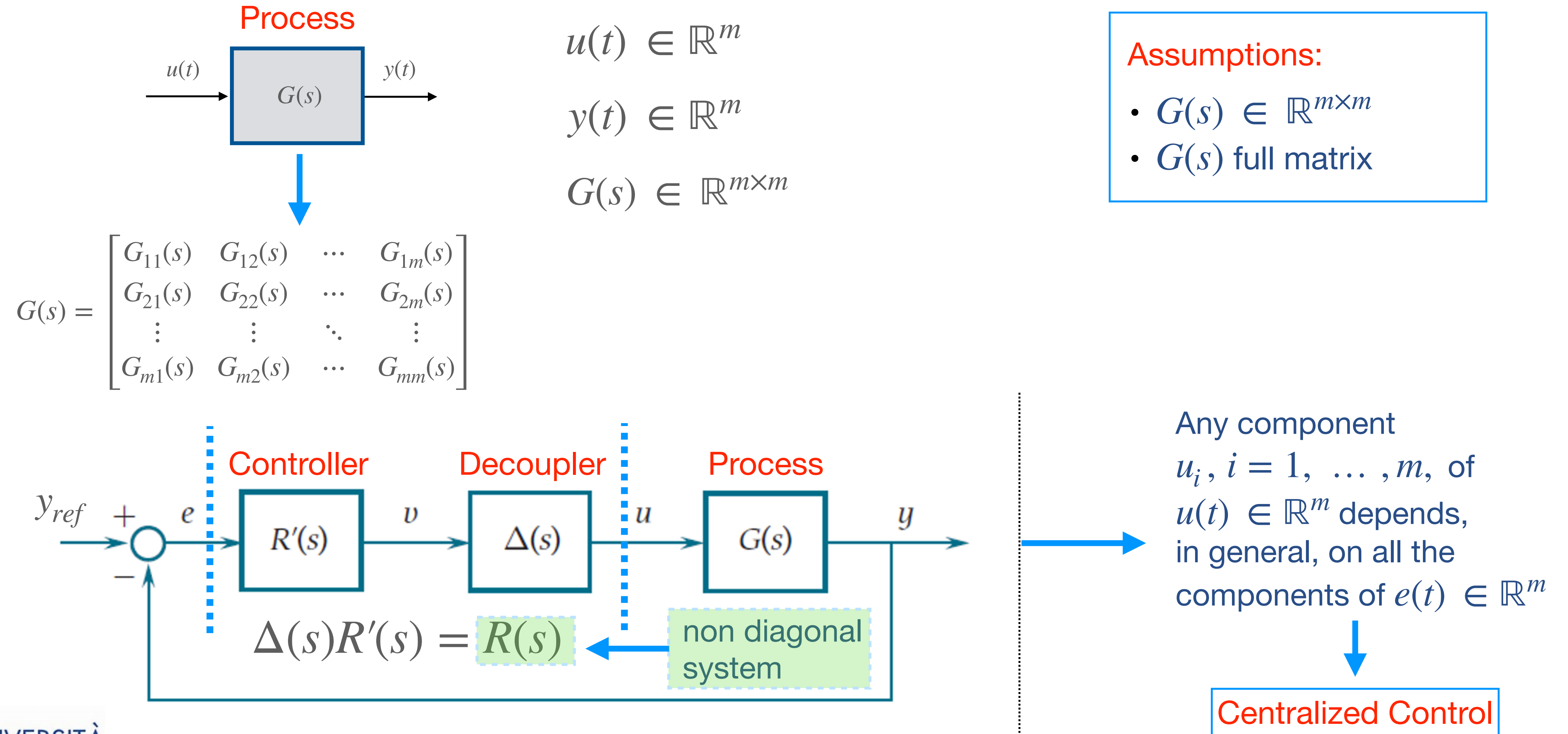


Assumptions:

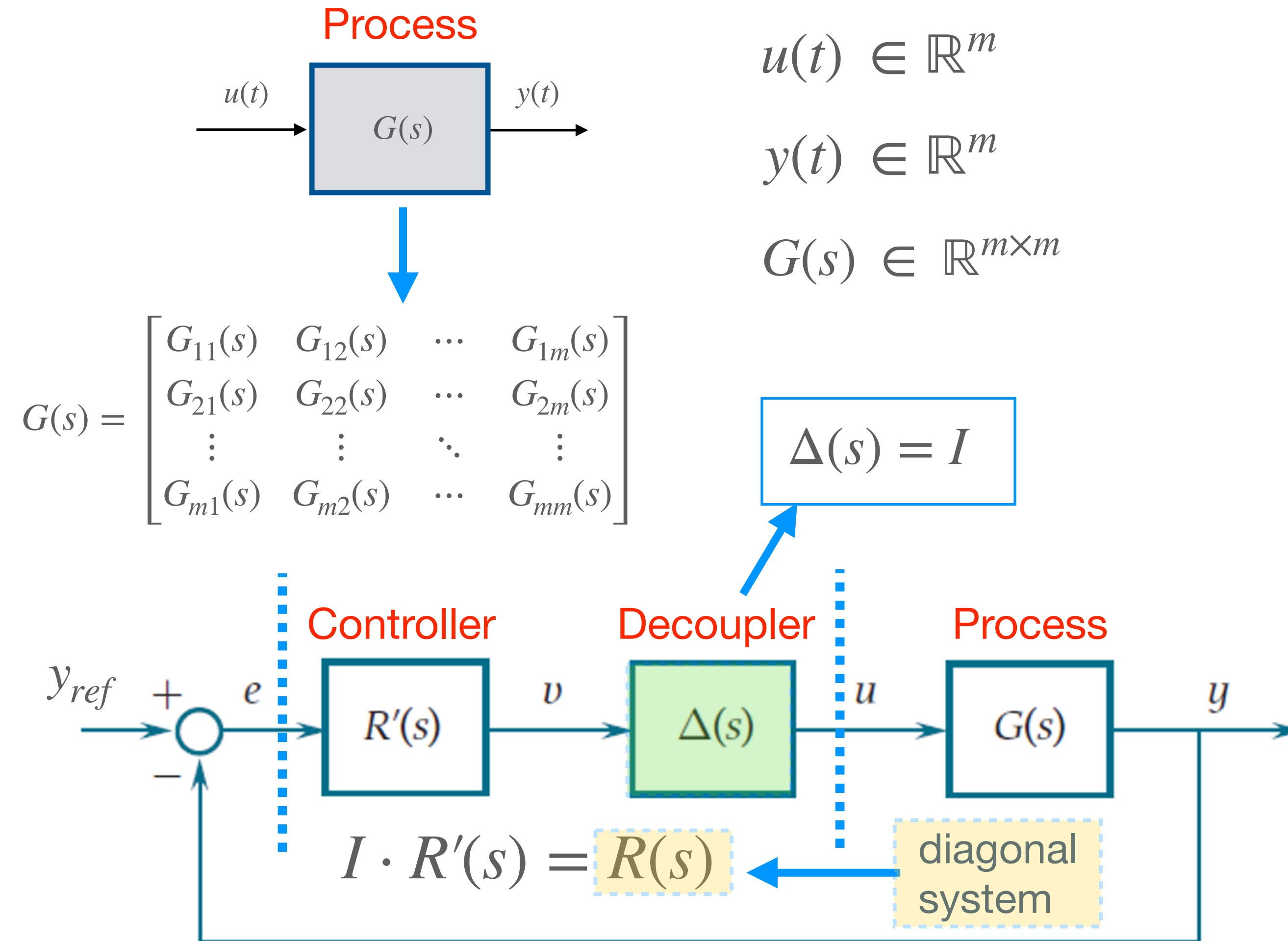
- $G(s) \in \mathbb{R}^{m \times m}$
- $G(s)$ full matrix

Any component $u_i, i = 1, \dots, m$, of $u(t) \in \mathbb{R}^m$ depends, in general, on all the components of $e(t) \in \mathbb{R}^m$

Centralized vs. Decentralized MIMO Control Schemes

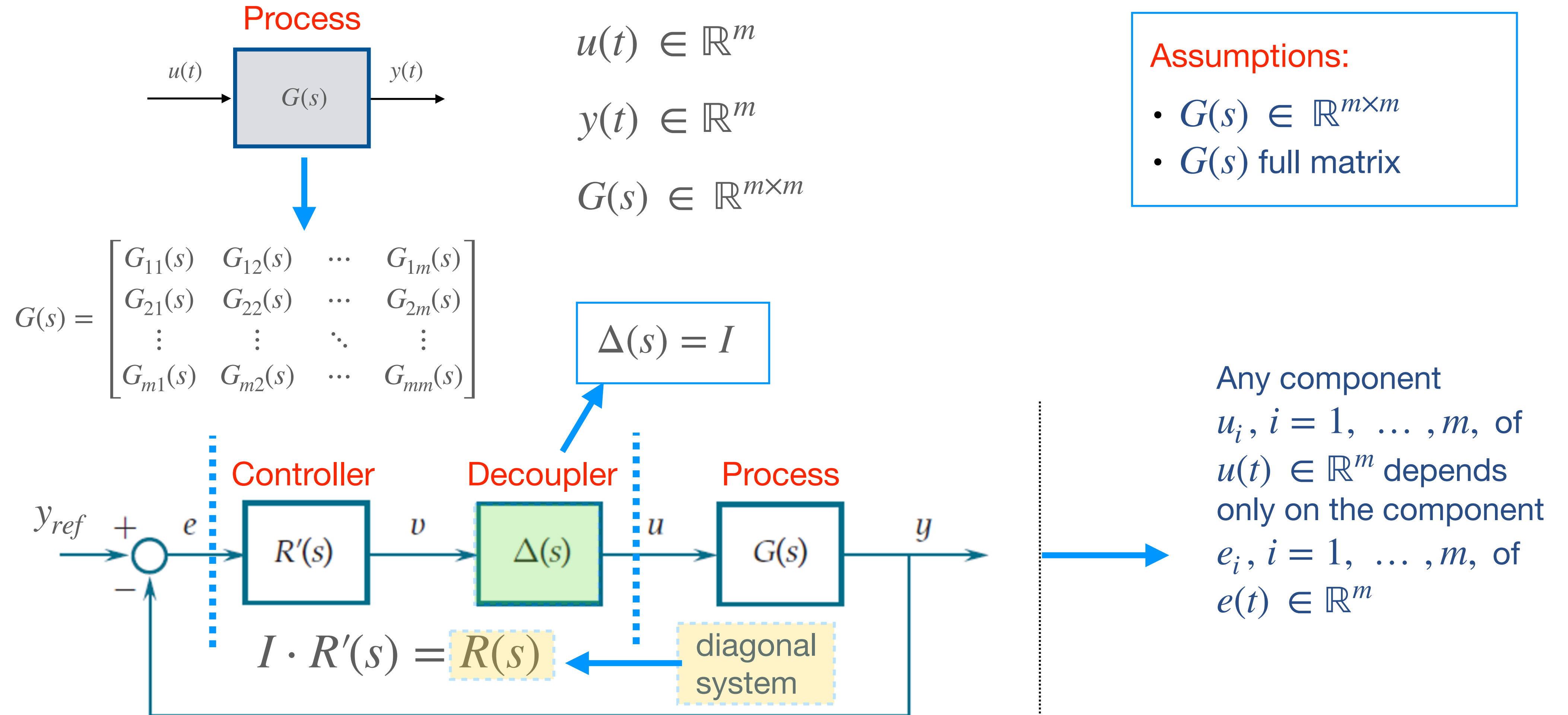


Centralized vs. Decentralized MIMO Control Schemes

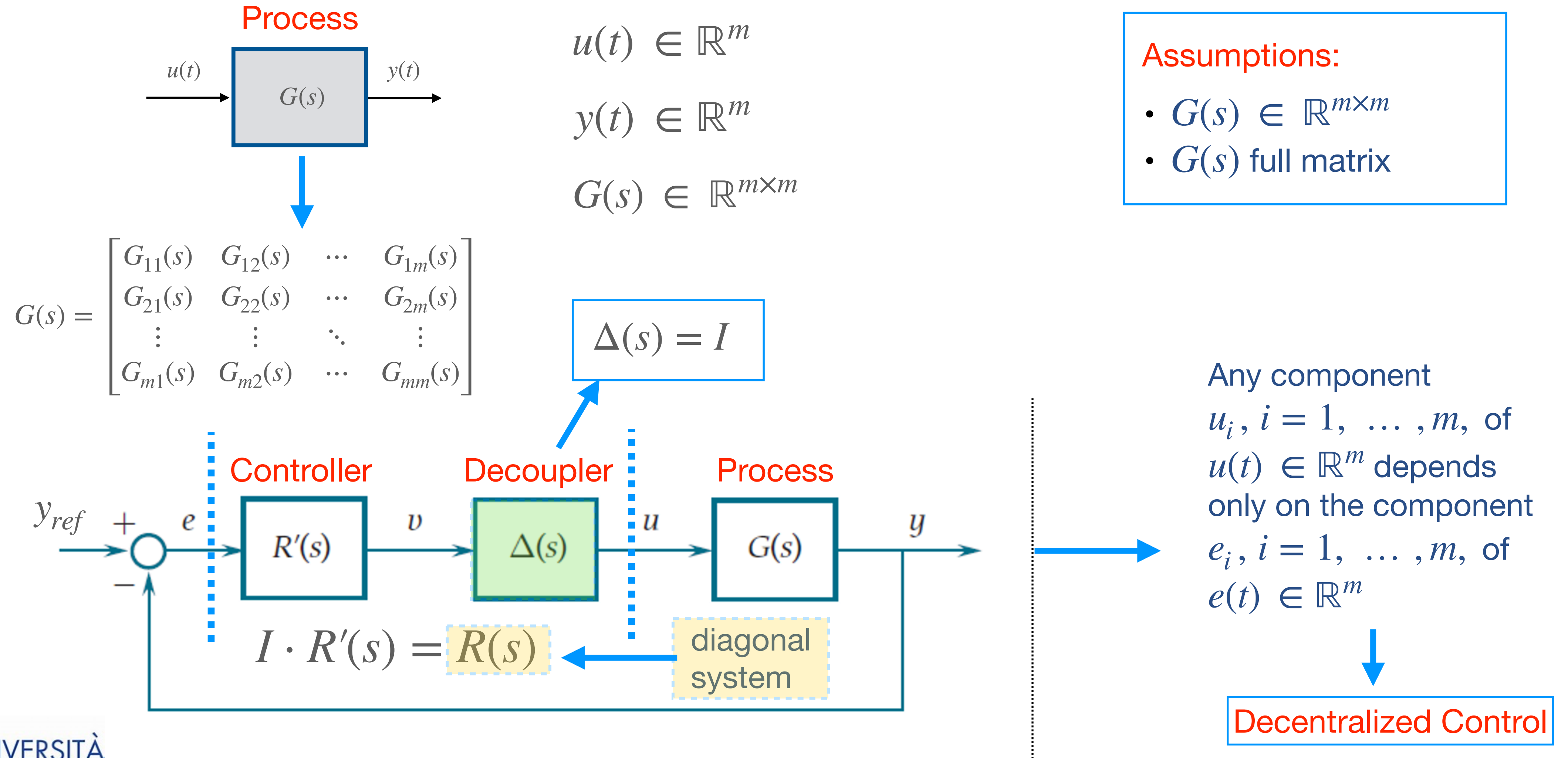
**Assumptions:**

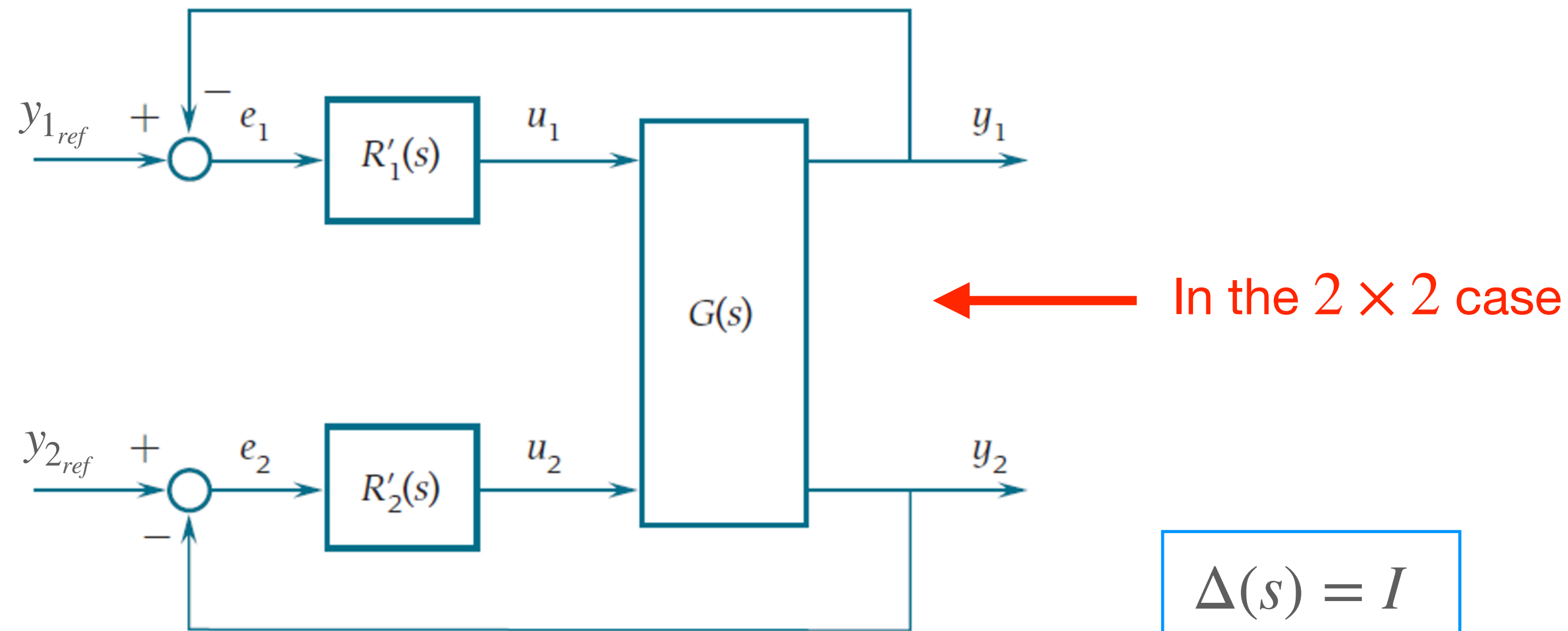
- $G(s) \in \mathbb{R}^{m \times m}$
- $G(s)$ full matrix

Centralized vs. Decentralized MIMO Control Schemes

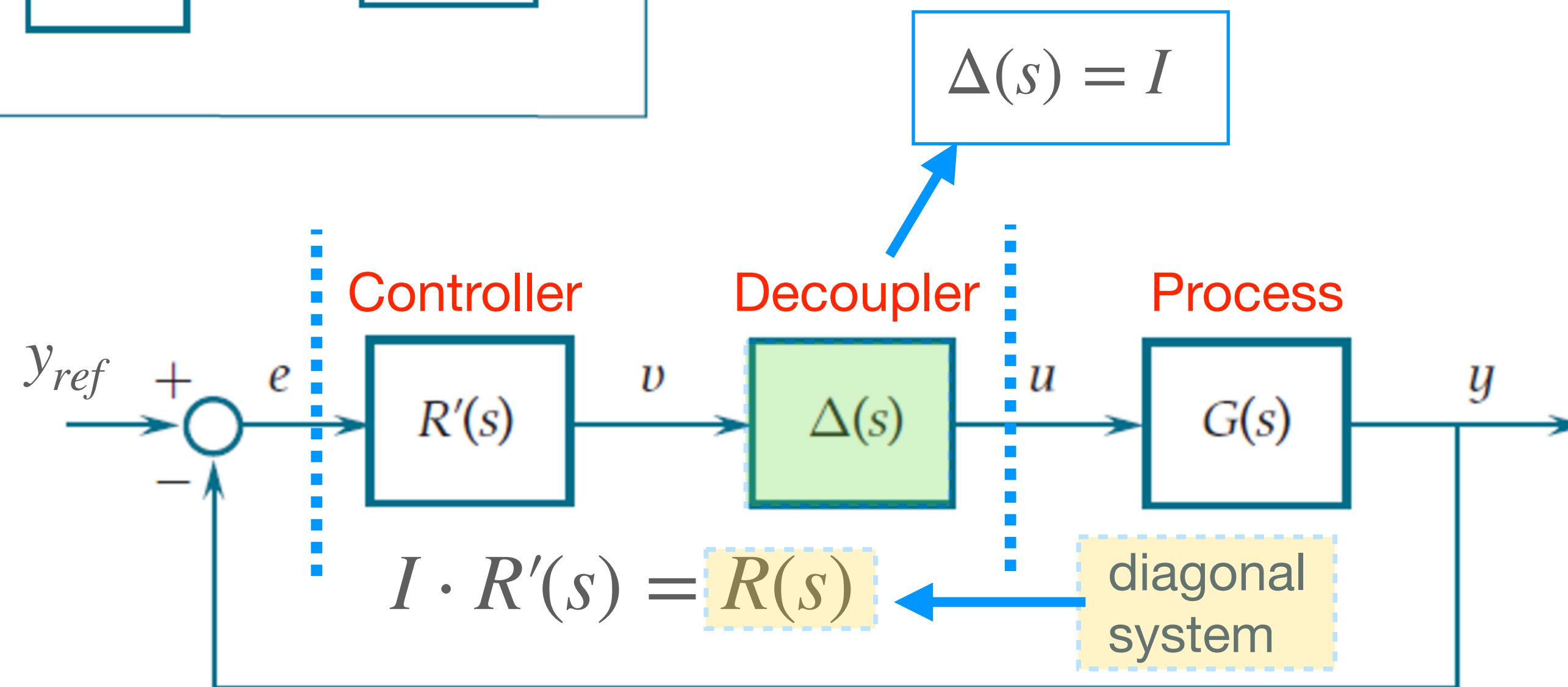


Centralized vs. Decentralized MIMO Control Schemes



Centralized vs. **Decentralized** MIMO Control Schemes**Assumptions:**

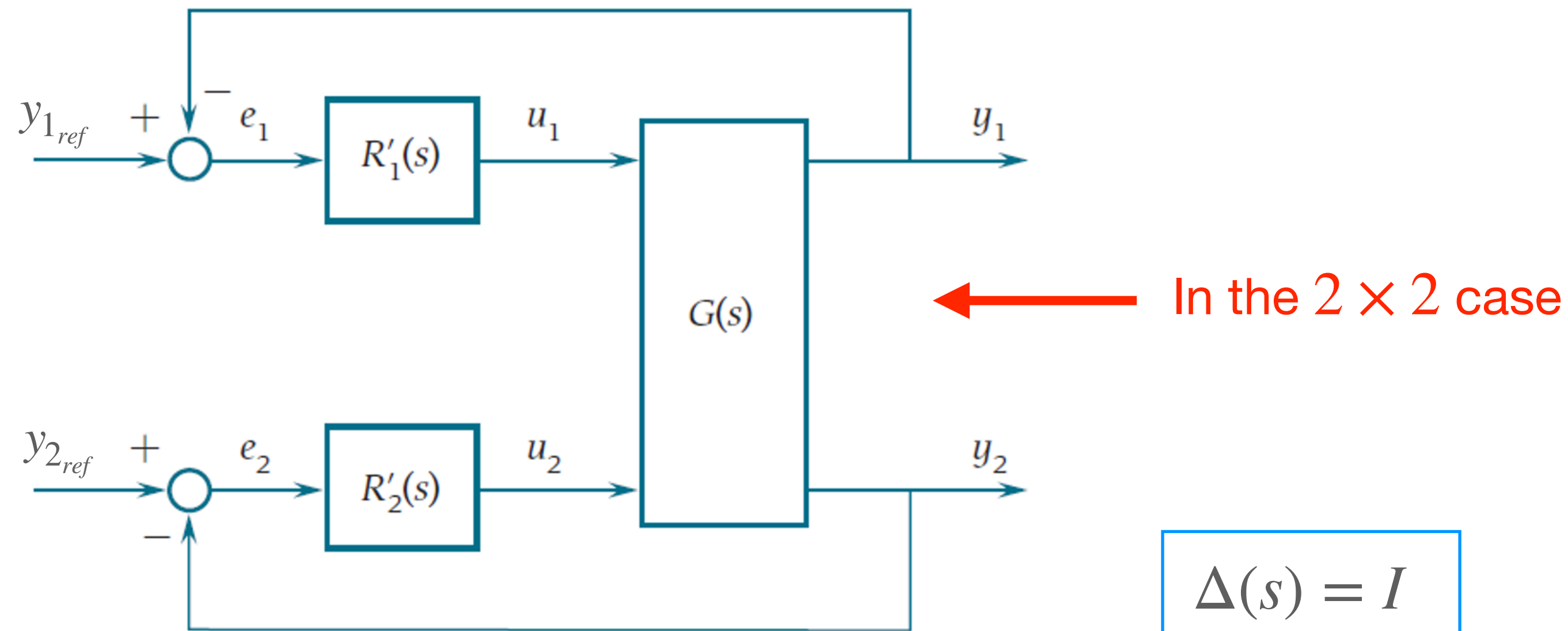
- $G(s) \in \mathbb{R}^{m \times m}$
- $G(s)$ full matrix



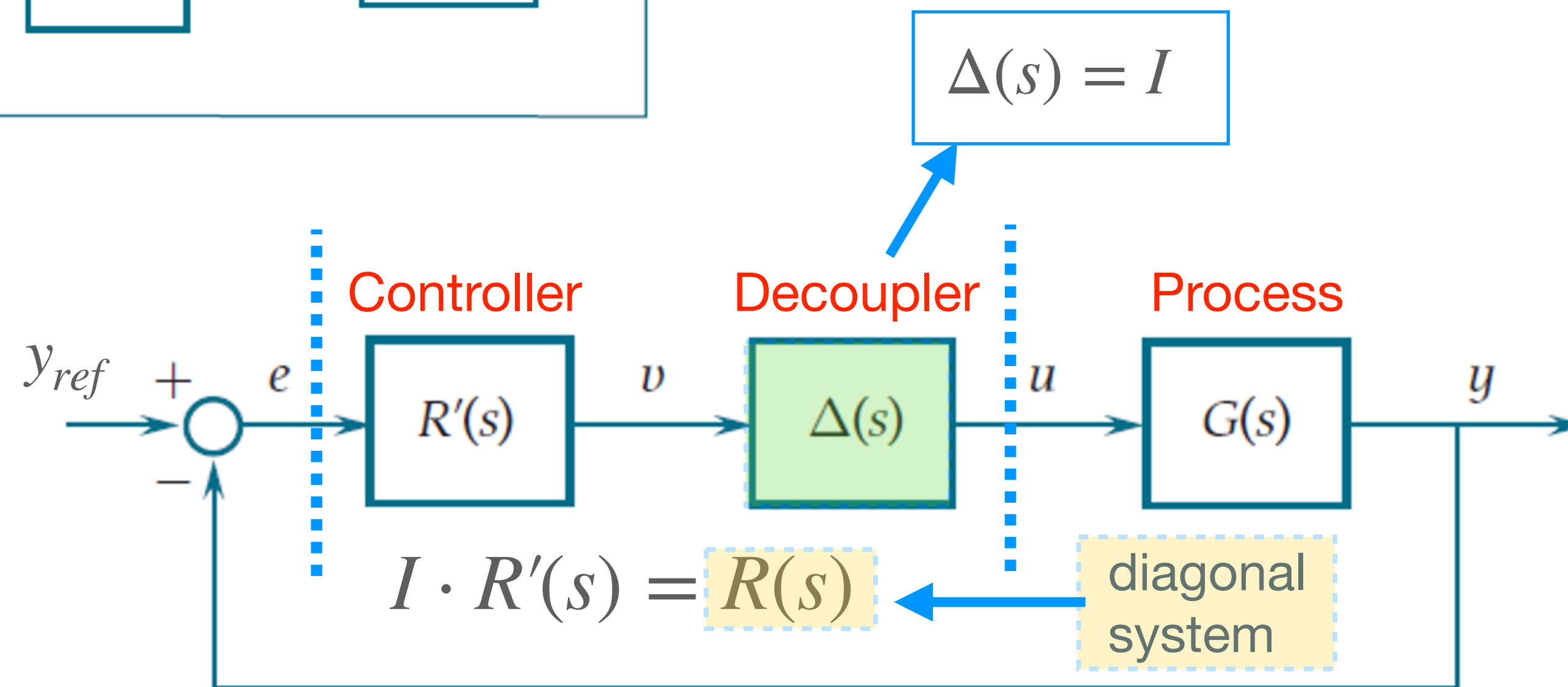
Any component $u_i, i = 1, \dots, m$, of $u(t) \in \mathbb{R}^m$ depends only on the component $e_i, i = 1, \dots, m$, of $e(t) \in \mathbb{R}^m$

Decentralized Control

Decentralized MIMO Control Schemes: Heuristic Method

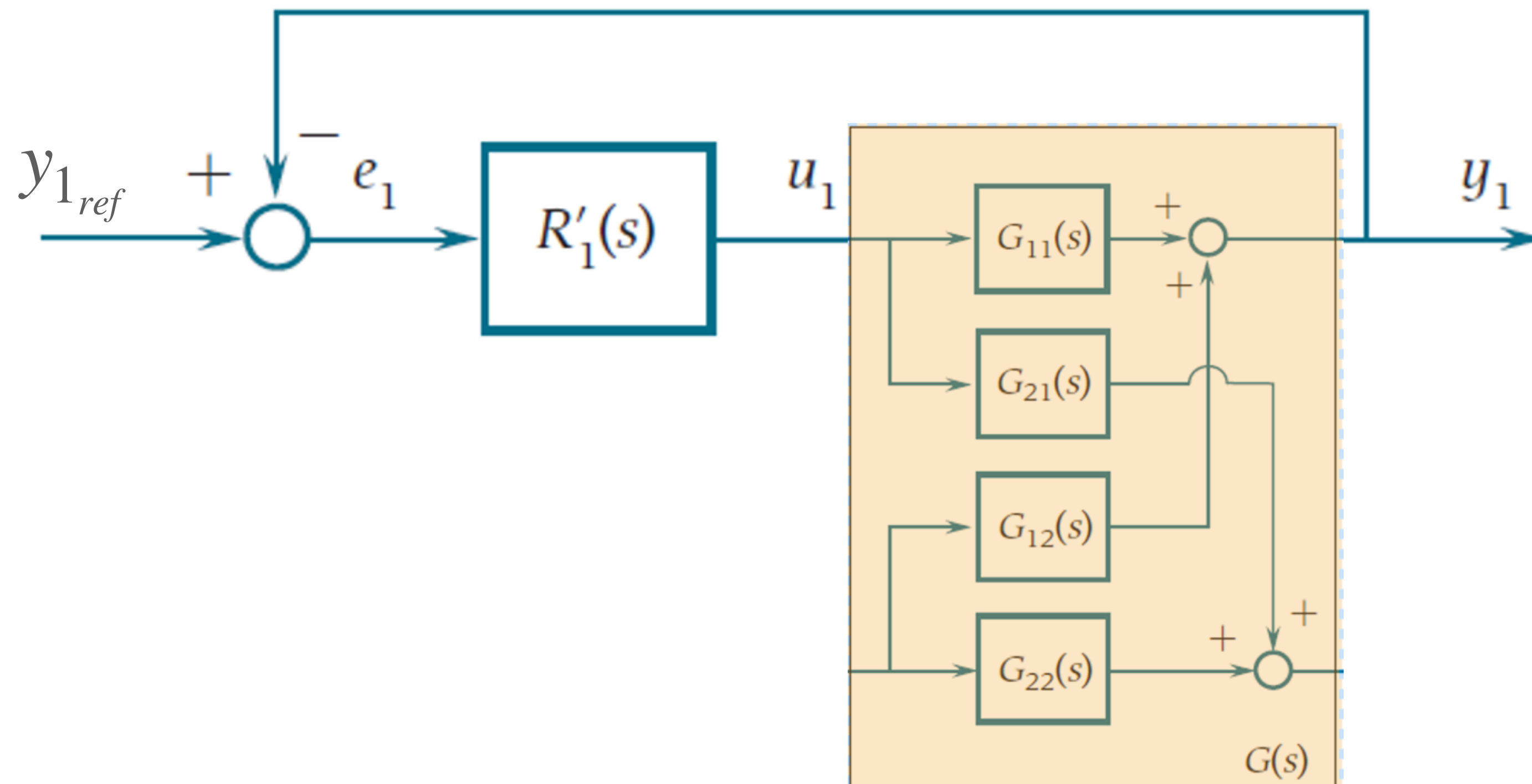


Design $R'_j(s)$ taking into account that the controllers from $R'_1(s)$ to $R'_{j-1}(s)$ have already been inserted into the control system



Decentralized MIMO Control Schemes: Heuristic Method

In the 2×2 case



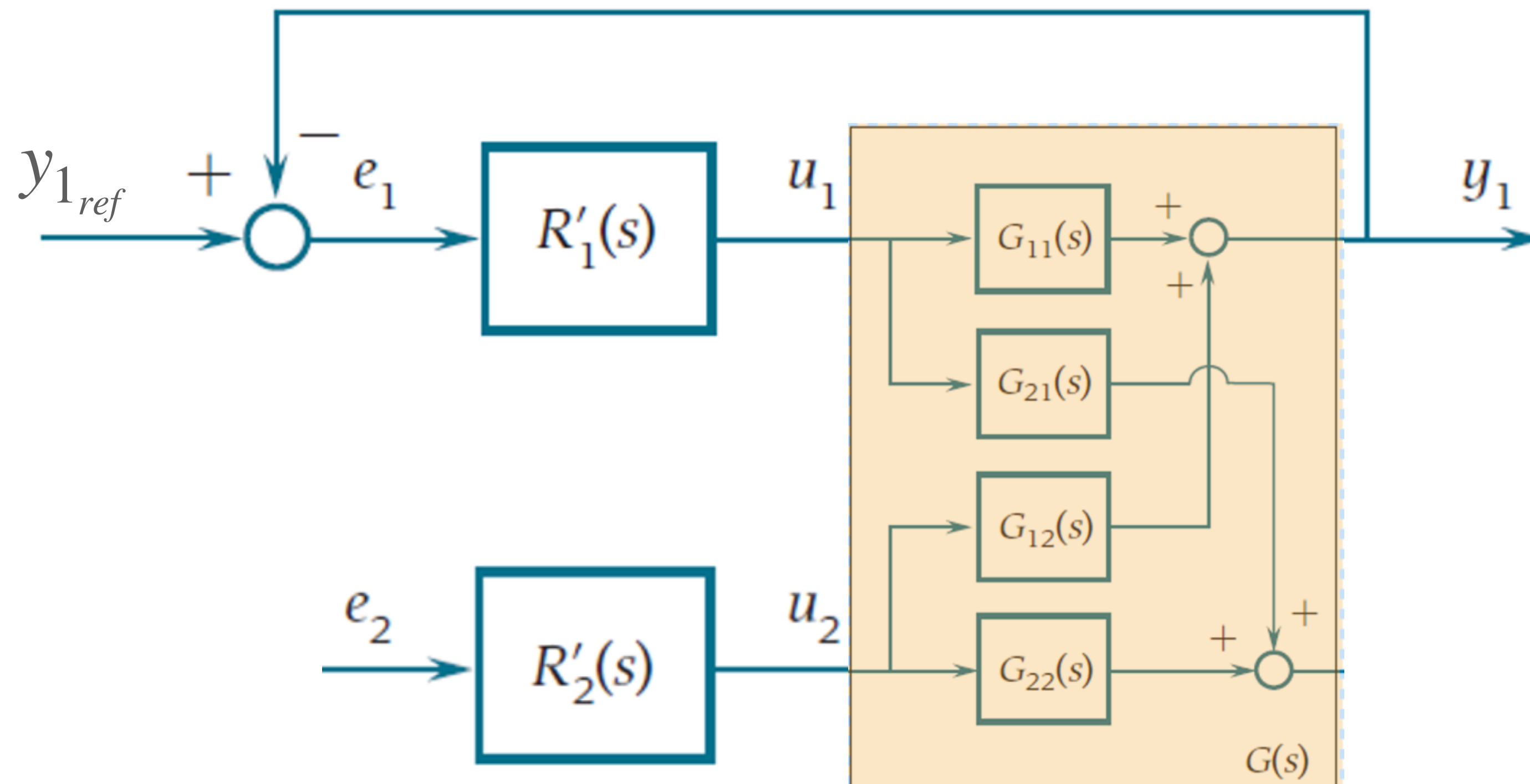
Design $R'_j(s)$ taking into account that the controllers from $R'_1(s)$ to $R'_{j-1}(s)$ have already been inserted into the control system

Step 1:

Design $R'_1(s)$ based on $G_{11}(s)$

Decentralized MIMO Control Schemes: Heuristic Method

In the 2×2 case



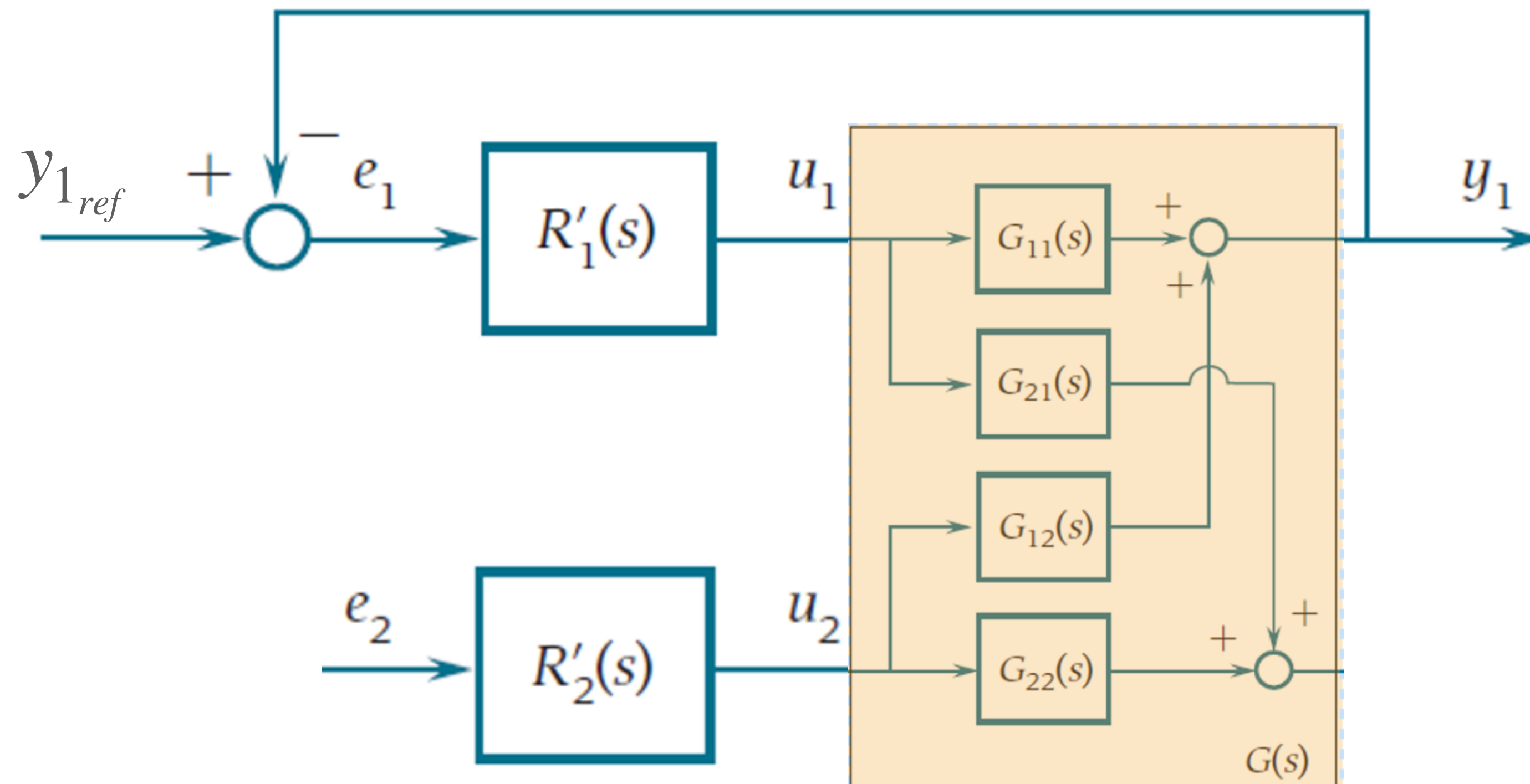
Design $R'_j(s)$ taking into account that the controllers from $R'_1(s)$ to $R'_{j-1}(s)$ have already been inserted into the control system

Step 1:

Design $R'_1(s)$ based on $G_{11}(s)$

Step 2:

Design $R'_2(s)$ based on $G'_{22}(s)$

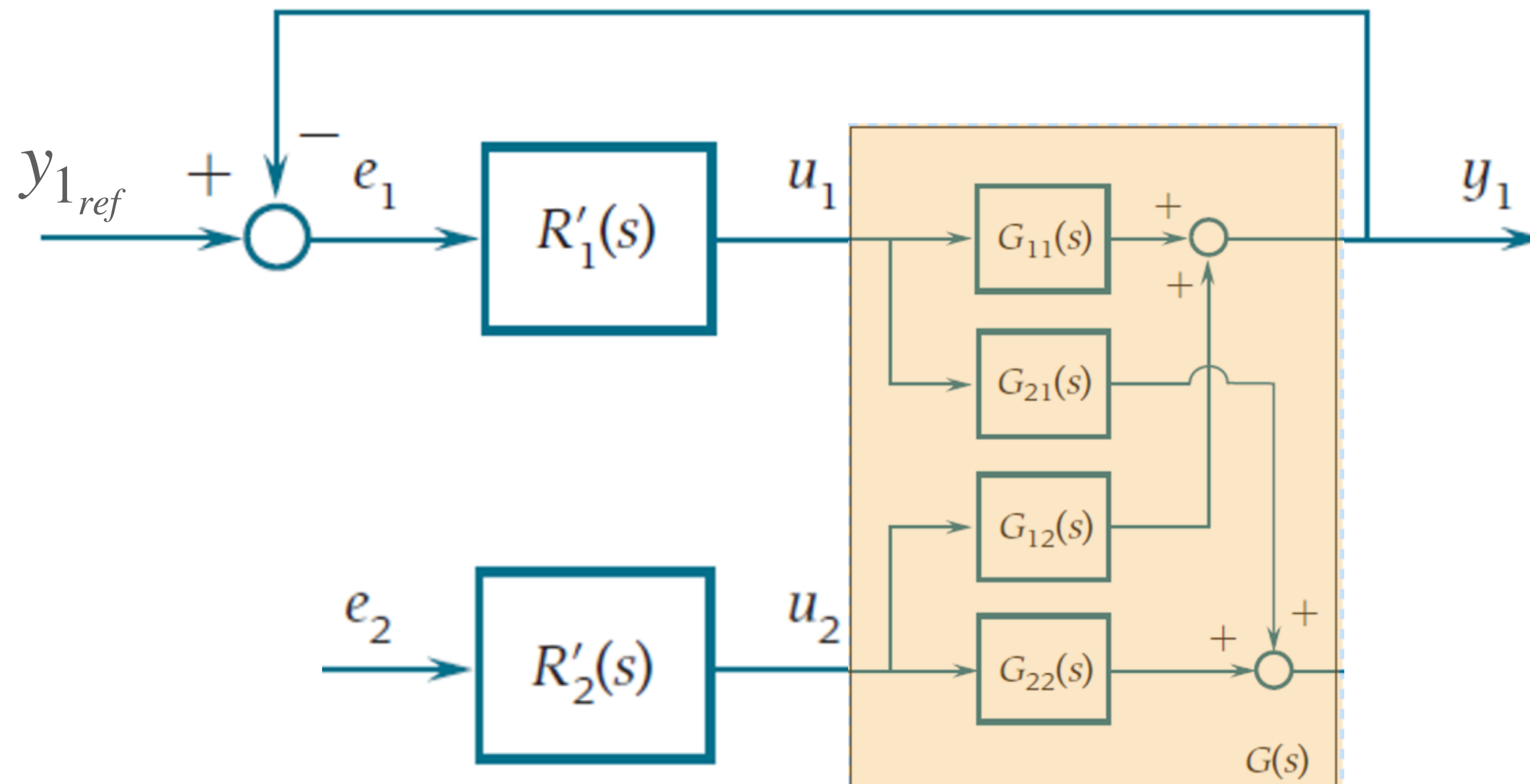
Decentralized MIMO Control Schemes: Heuristic MethodIn the 2×2 case

Design $R'_j(s)$ taking into account that the controllers from $R'_1(s)$ to $R'_{j-1}(s)$ have already been inserted into the control system

Step 1:Design $R'_1(s)$ based on $G_{11}(s)$ **Step 2:**Design $R'_2(s)$ based on $G'_{22}(s)$?

Decentralized MIMO Control Schemes: Heuristic Method

In the 2×2 case



Design $R'_j(s)$ taking into account that the controllers from $R'_1(s)$ to $R'_{j-1}(s)$ have already been inserted into the control system

Step 1:

Design $R'_1(s)$ based on $G_{11}(s)$

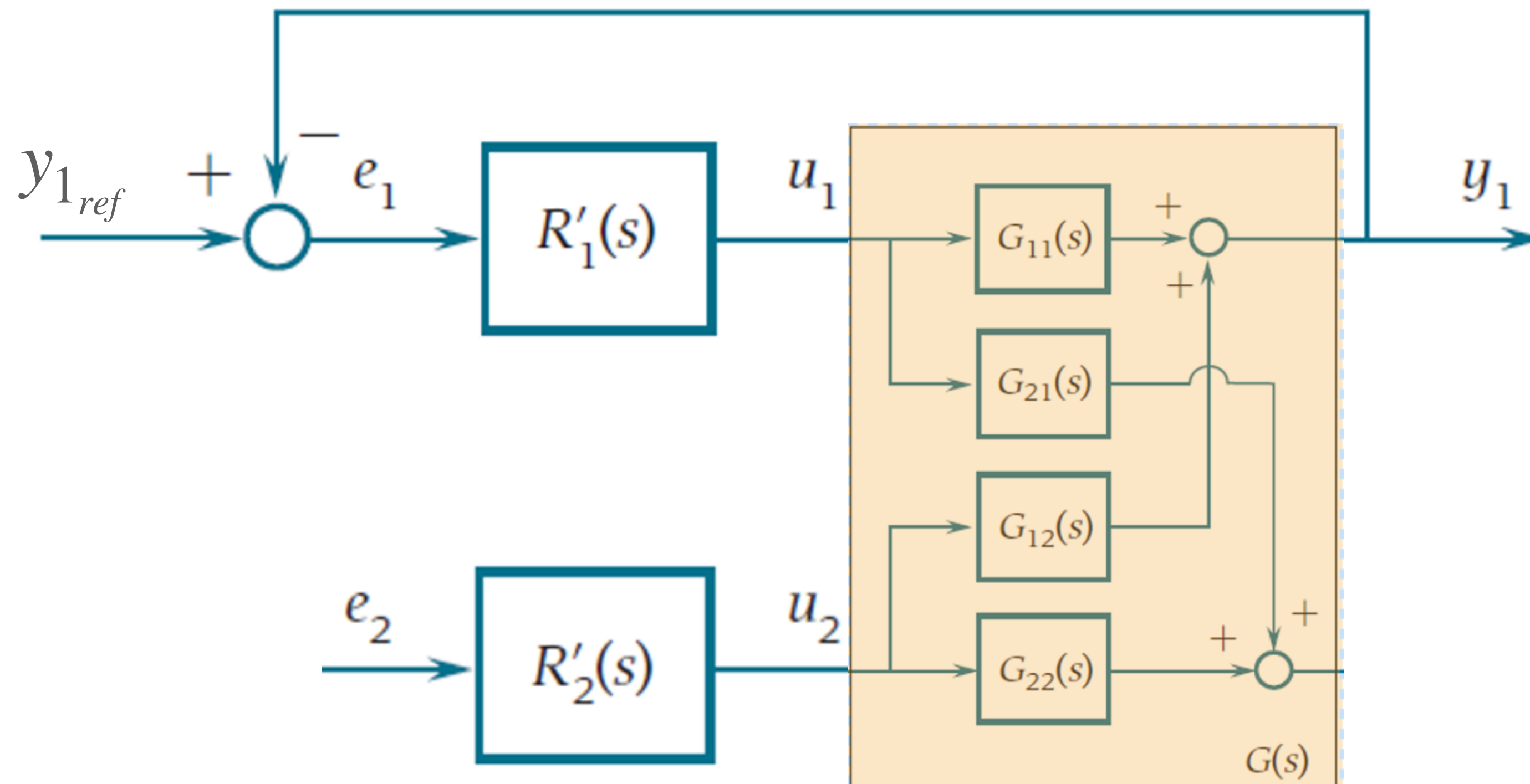
Step 2:

Design $R'_2(s)$ based on $G'_{22}(s)$?

$$G'_{22}(s) = G_{22}(s) - \frac{G_{12}(s)R'_1(s)G_{21}(s)}{1 + R'_1(s)G_{11}(s)}$$

Decentralized MIMO Control Schemes: Heuristic Method

In the 2×2 case



Design $R'_j(s)$ taking into account that the controllers from $R'_1(s)$ to $R'_{j-1}(s)$ have already been inserted into the control system

Step 1:

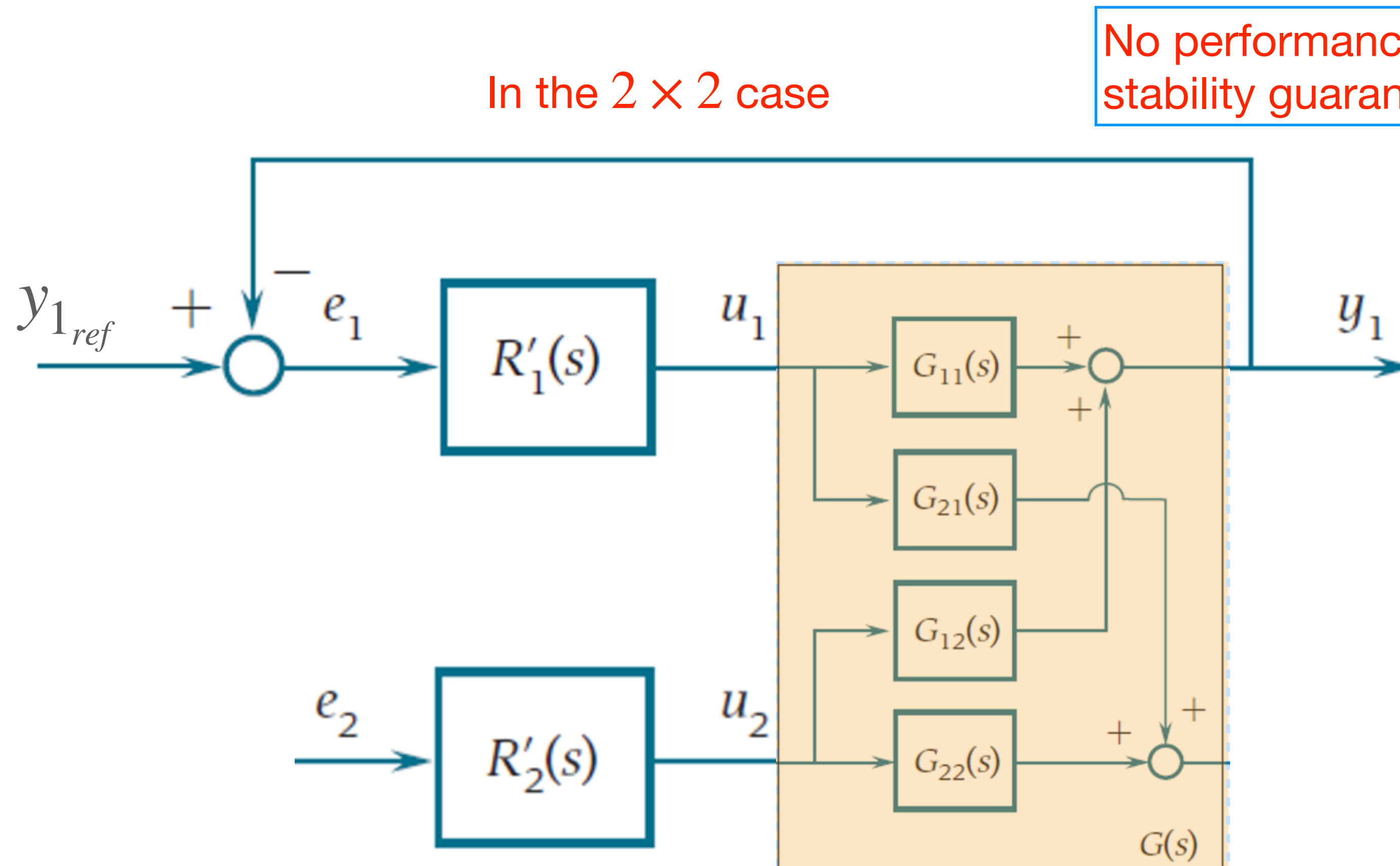
Design $R'_1(s)$ based on $G_{11}(s)$

Step 2:

Design $R'_2(s)$ based on $G'_{22}(s)$

In general ($m \times m$ case):

Design $R'_j(s)$ based on $G'_{jj}(s)$

Decentralized MIMO Control Schemes: Heuristic Method

Design $R'_j(s)$ taking into account that the controllers from $R'_1(s)$ to $R'_{j-1}(s)$ have already been inserted into the control system

Step 1:

Design $R'_1(s)$ based on $G_{11}(s)$

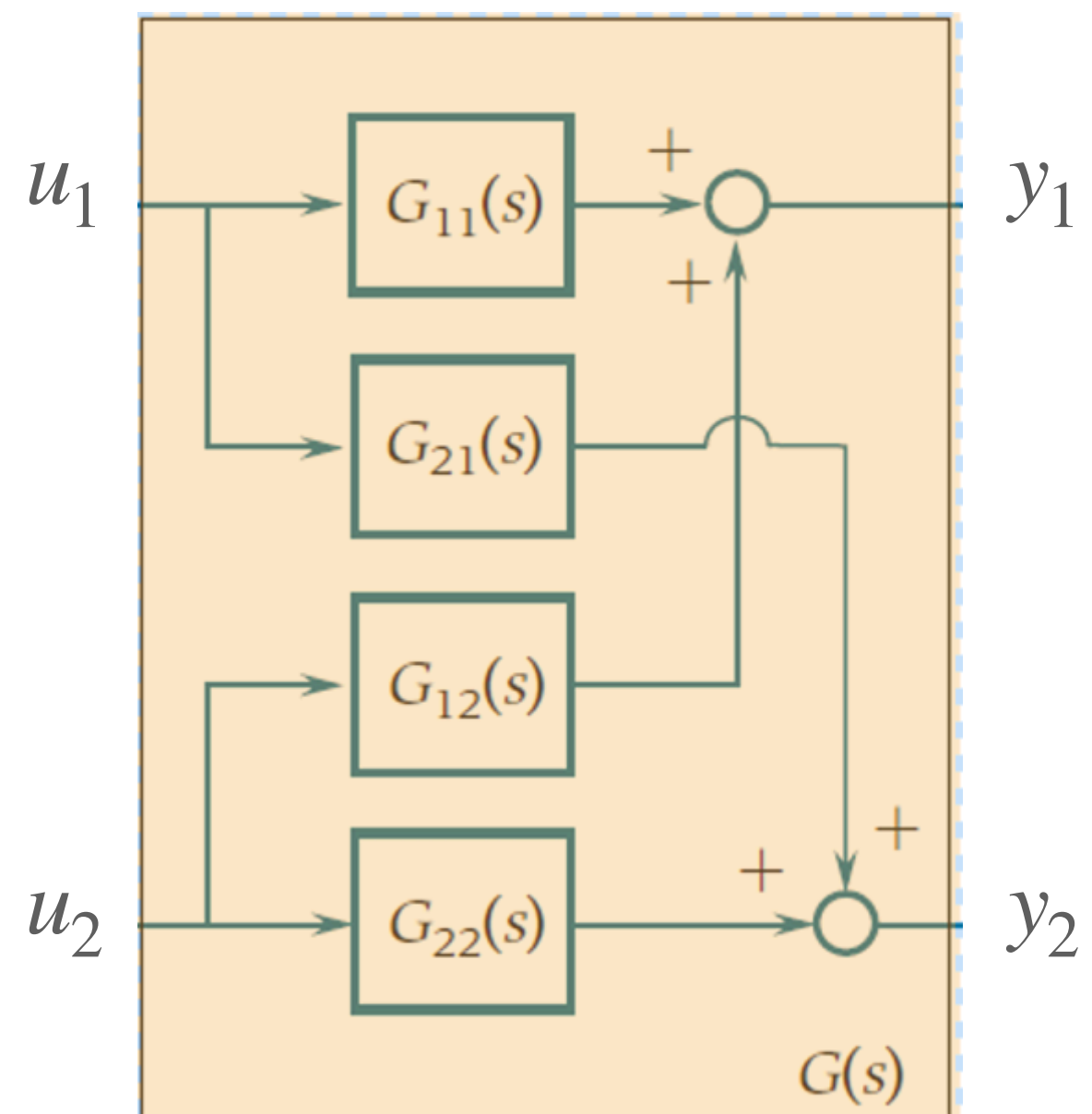
Step 2:

Design $R'_2(s)$ based on $G'_{22}(s)$

In general ($m \times m$ case):

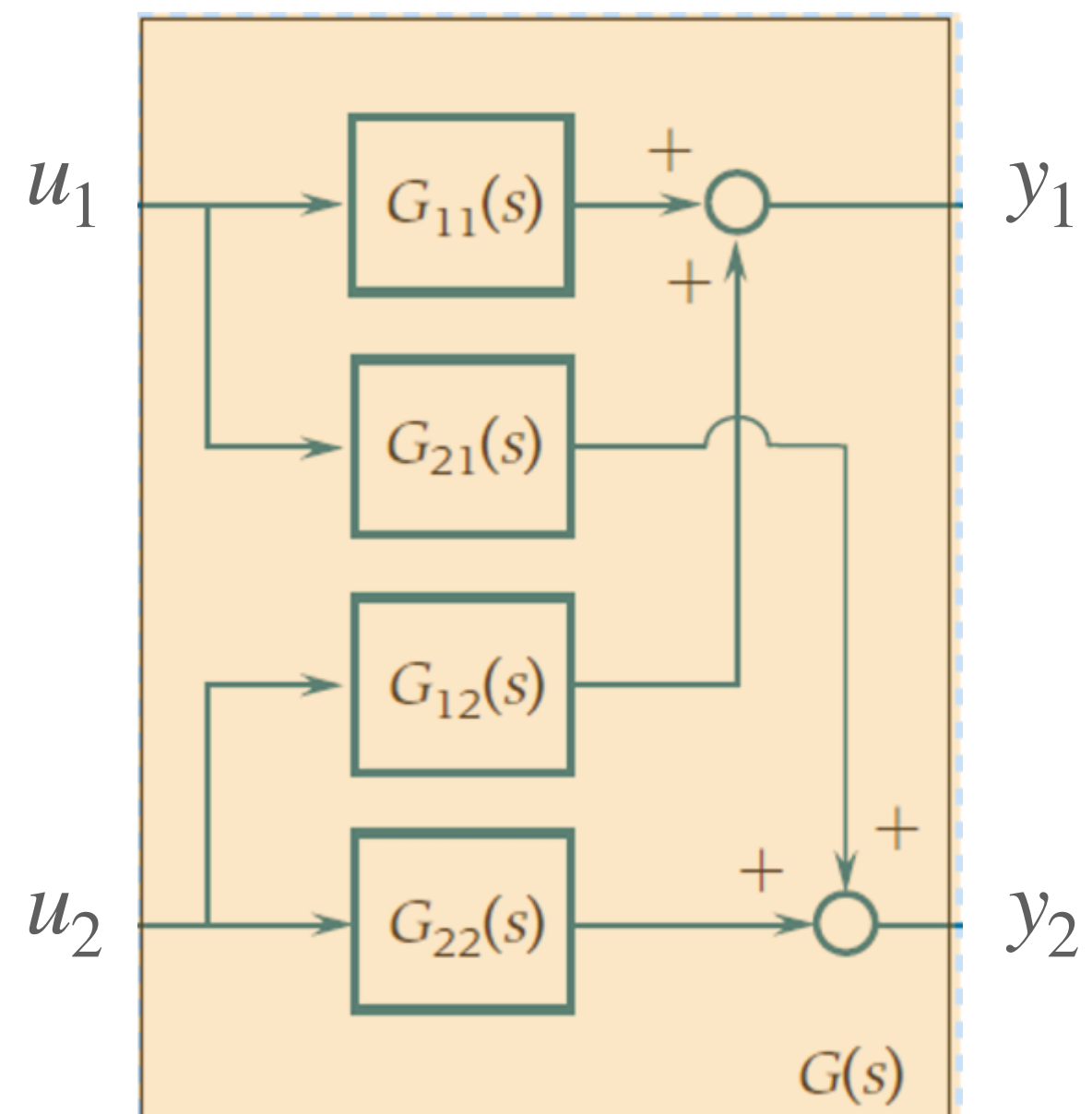
Design $R'_j(s)$ based on $G'_{jj}(s)$

Relative Gain Array



Aim: Design a decentralized control scheme

Relative Gain Array



Aim: Design a decentralized control scheme

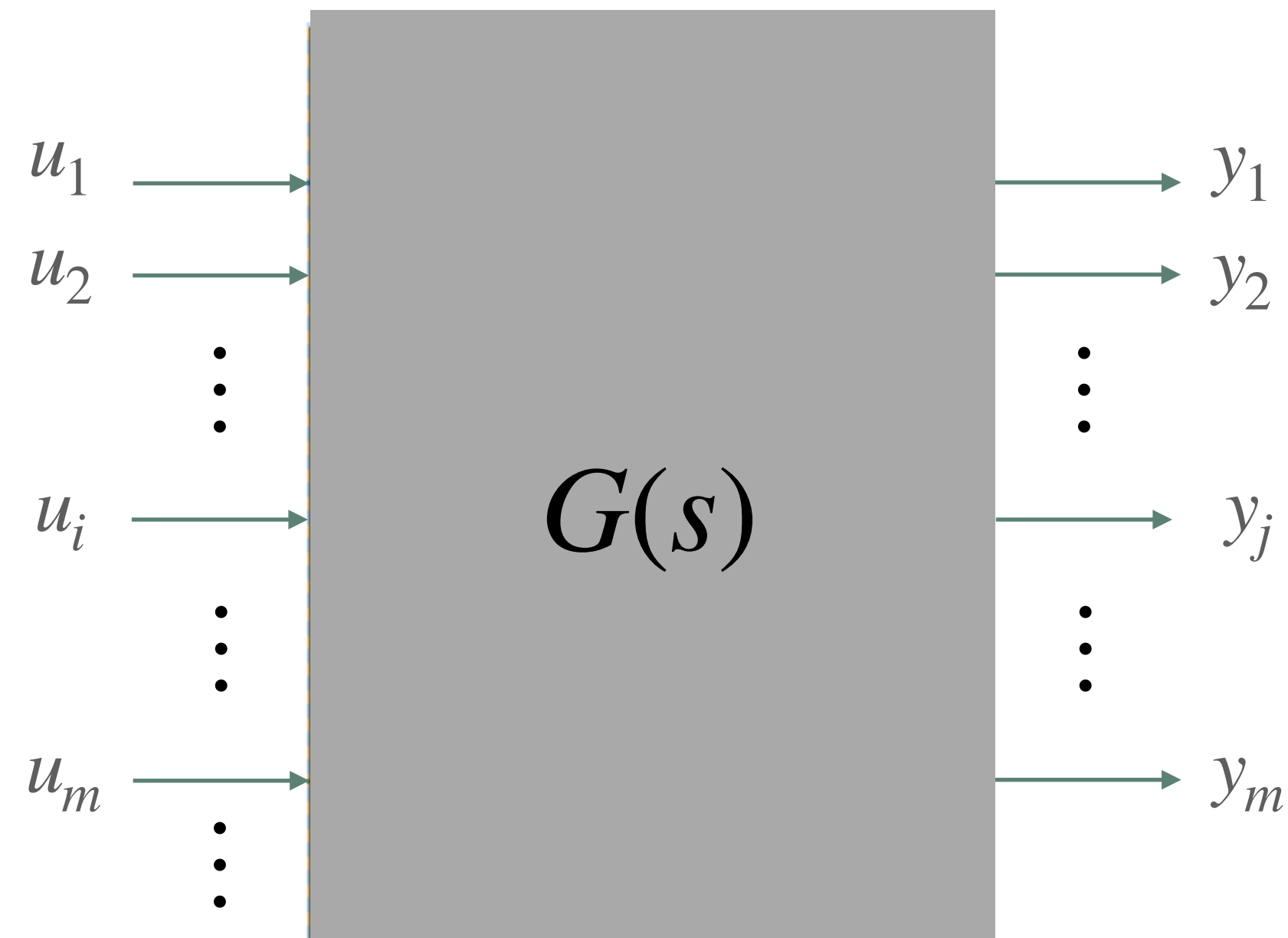
Question: Which pairings are the best?

$$\{u_1 \rightarrow y_1\} \cup \{u_2 \rightarrow y_2\}$$

$$\{u_1 \rightarrow y_2\} \cup \{u_2 \rightarrow y_1\}$$

?

Relative Gain Array



Aim: Design a decentralized control scheme

Question: Which pairings are the best?

$$\{u_1 \rightarrow y_1\} \cup \{u_2 \rightarrow y_2\}$$

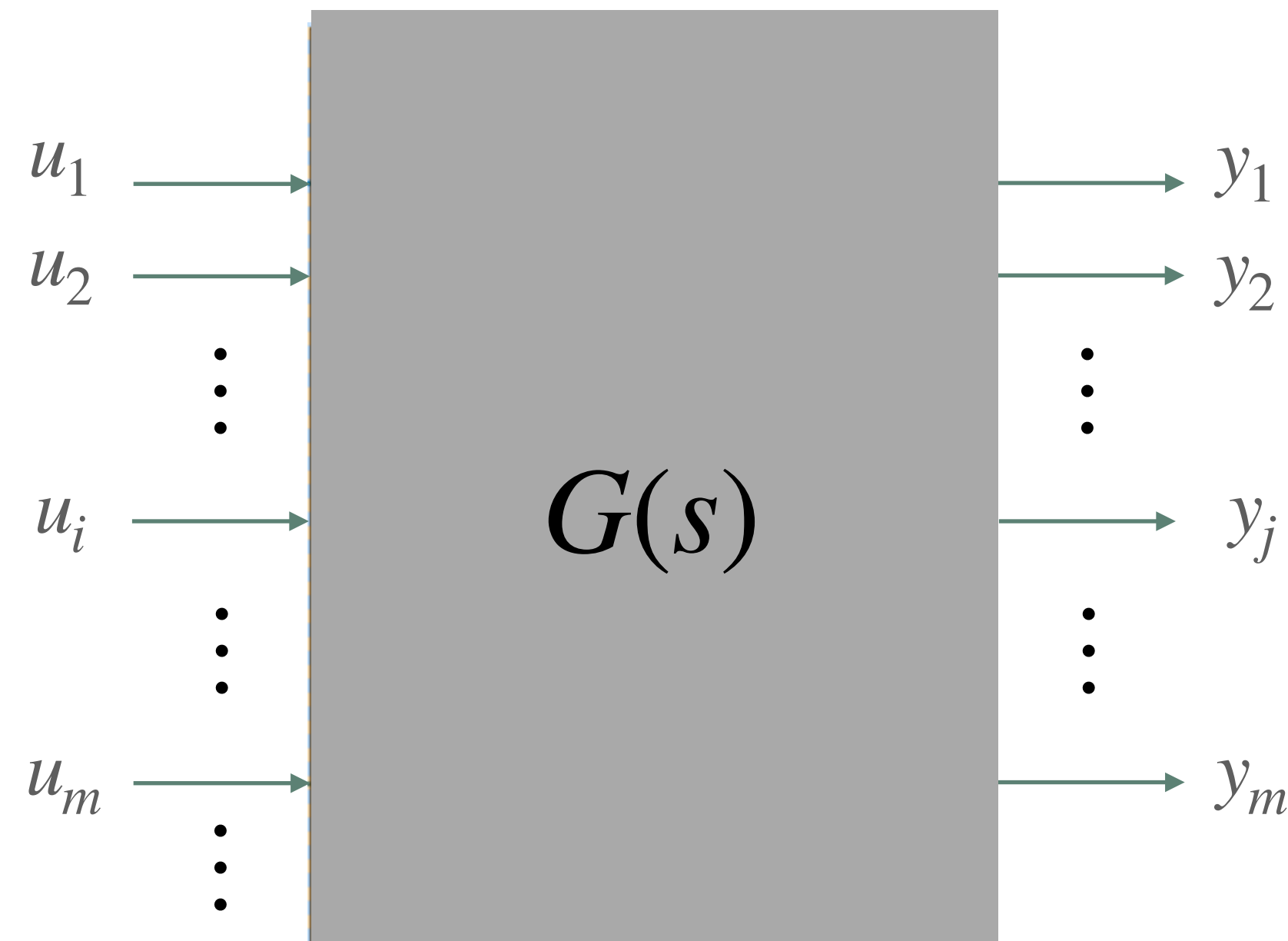
$$\{u_1 \rightarrow y_2\} \cup \{u_2 \rightarrow y_1\}$$

In the $m \times m$ case: Which pairings are the best?

$$\{u_i \rightarrow y_j\}, \quad i, j = 1, \dots, m \quad ?$$

Relative Gain Array

Method to determine the best I/O pairings



Aim: Design a decentralized control scheme

Question: Which pairings are the best?

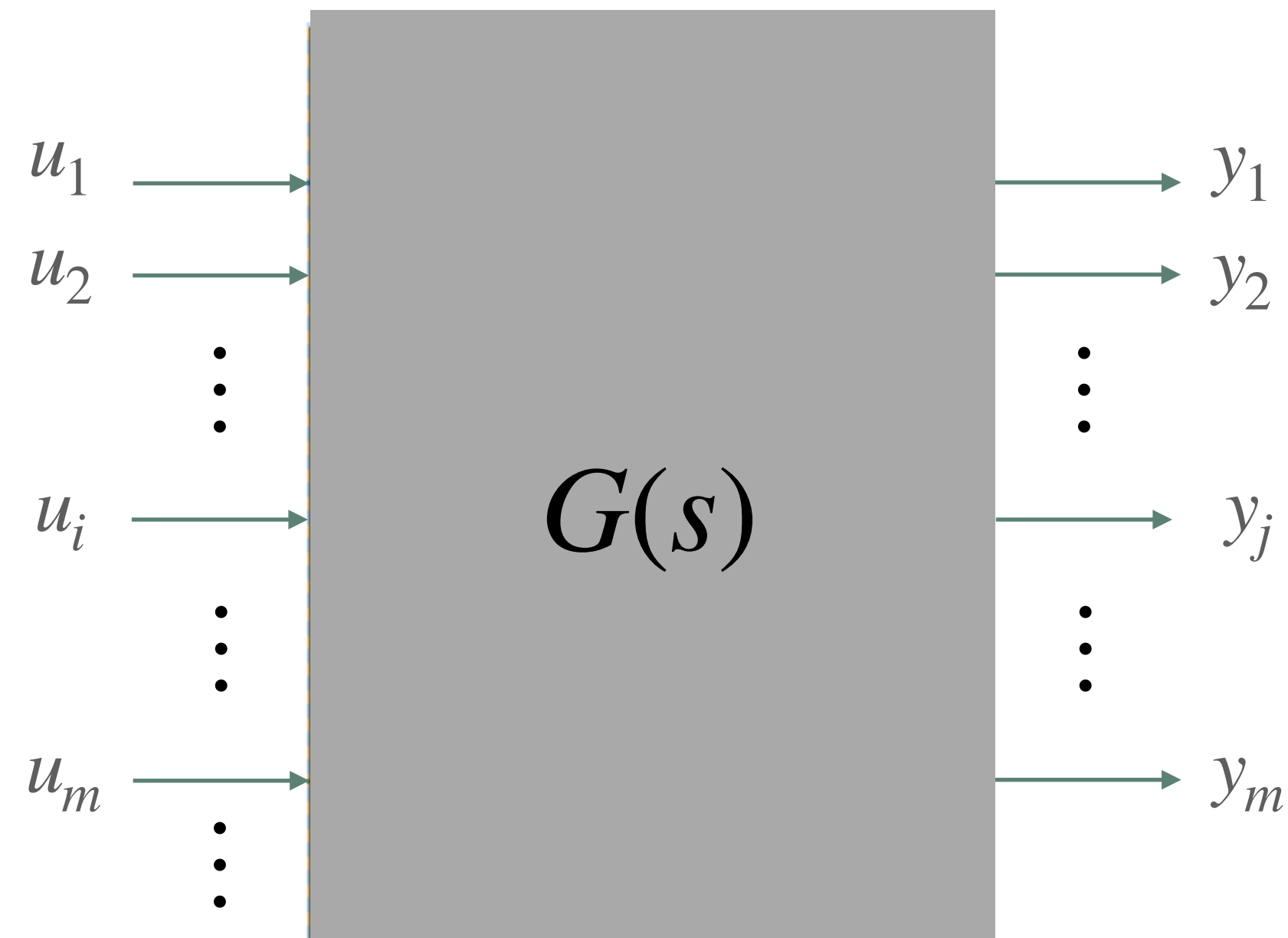
$$\{u_1 \rightarrow y_1\} \cup \{u_2 \rightarrow y_2\}$$

$$\{u_1 \rightarrow y_2\} \cup \{u_2 \rightarrow y_1\}$$

In the $m \times m$ case: Which pairings are the best?

$$\{u_i \rightarrow y_j\}, \quad i, j = 1, \dots, m \quad ?$$

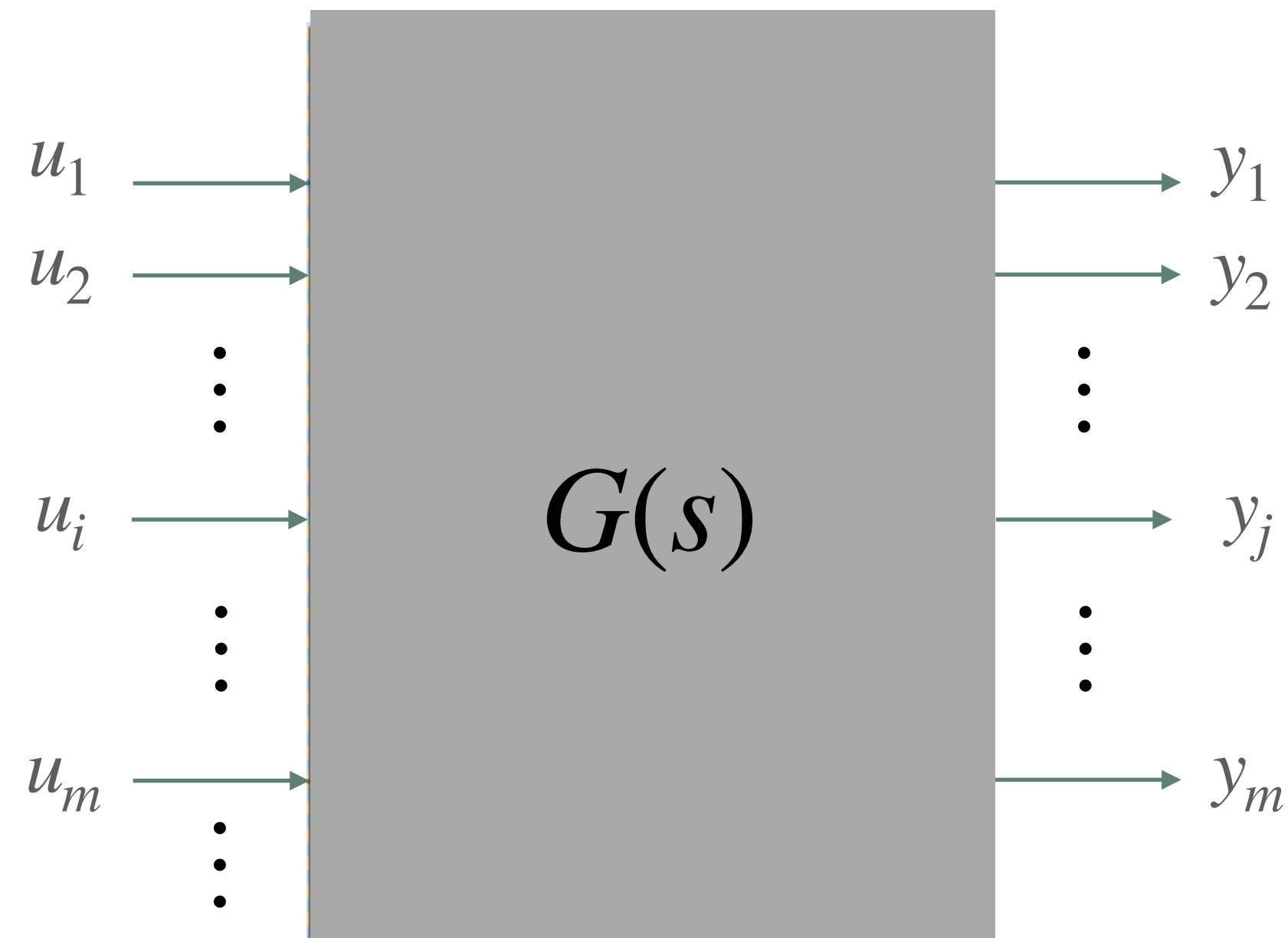
Relative Gain Array



Assumptions:

- Static case ($\omega = 0$)
- $G(s)$ As. Stable
- $\det G(0) \neq 0$
- System at the equilibrium at given constant values of the control variables

Relative Gain Array



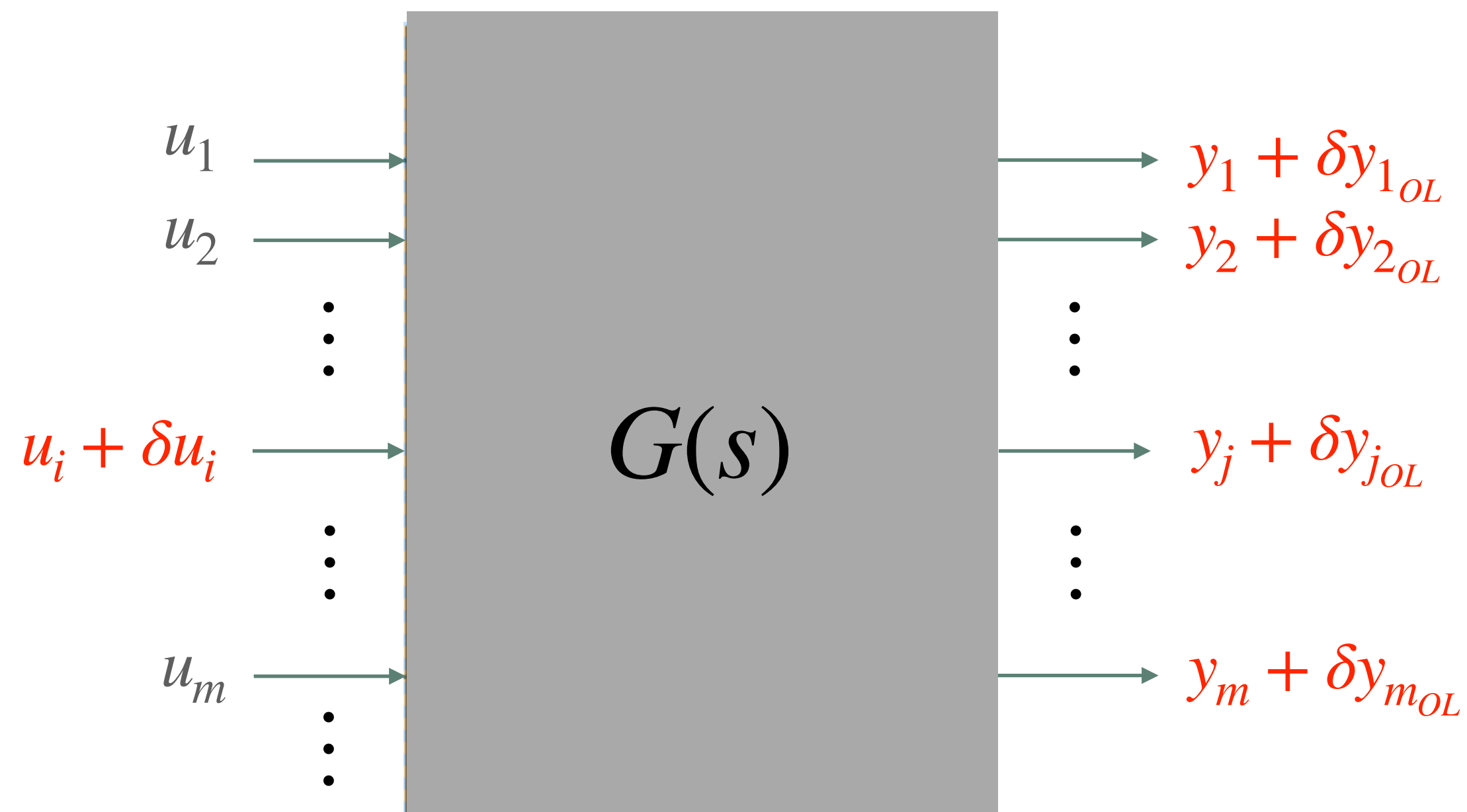
Assumptions:

- Static case ($\omega = 0$)
- $G(s)$ As. Stable
- $\det G(0) \neq 0$
- System at the equilibrium at given constant values of the control variables

Two tests are performed:

- 1) Open loop test
- 2) Closed loop test

Relative Gain Array



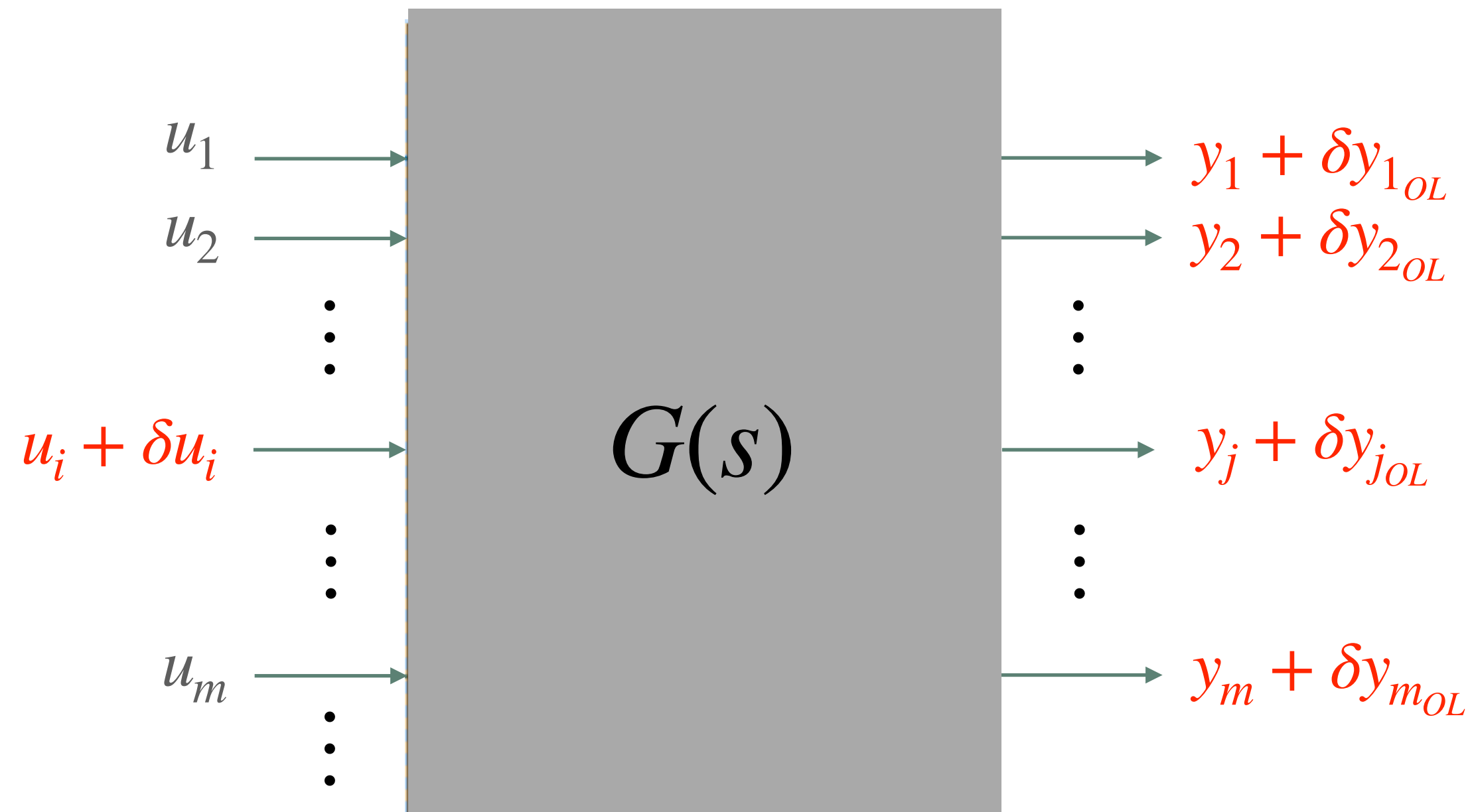
Assumptions:

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Relative Gain Array



Assumptions:

- Static case ($\omega = 0$)
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Two tests are performed:

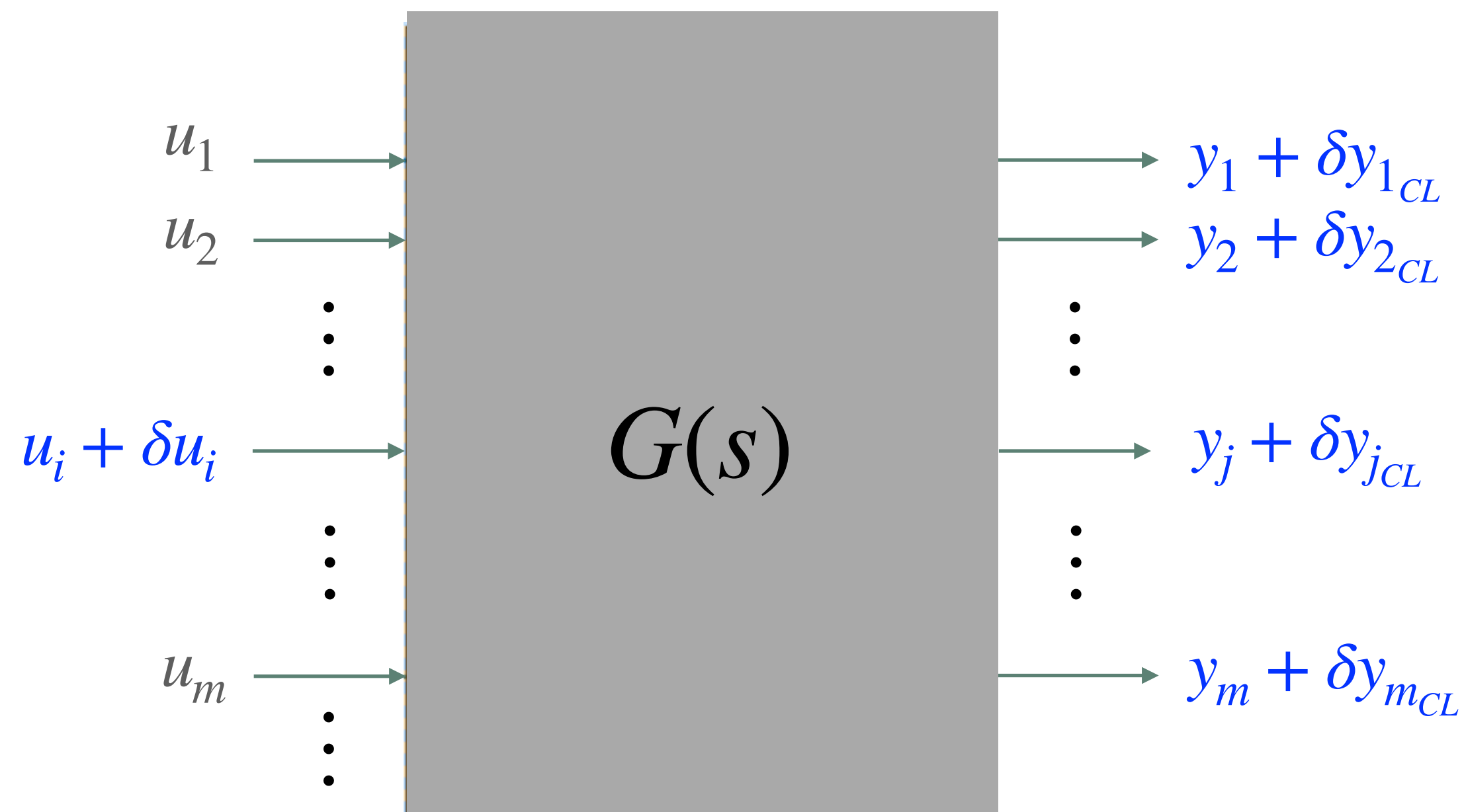
- 1) Open loop test
- 2) Closed loop test

$$\frac{\delta y_{j_{OL}}}{\delta u_i} = G_{ji}(0) = g_{ji}$$

$$\forall \quad i, j = 1, \dots, m$$

open loop gain

Relative Gain Array



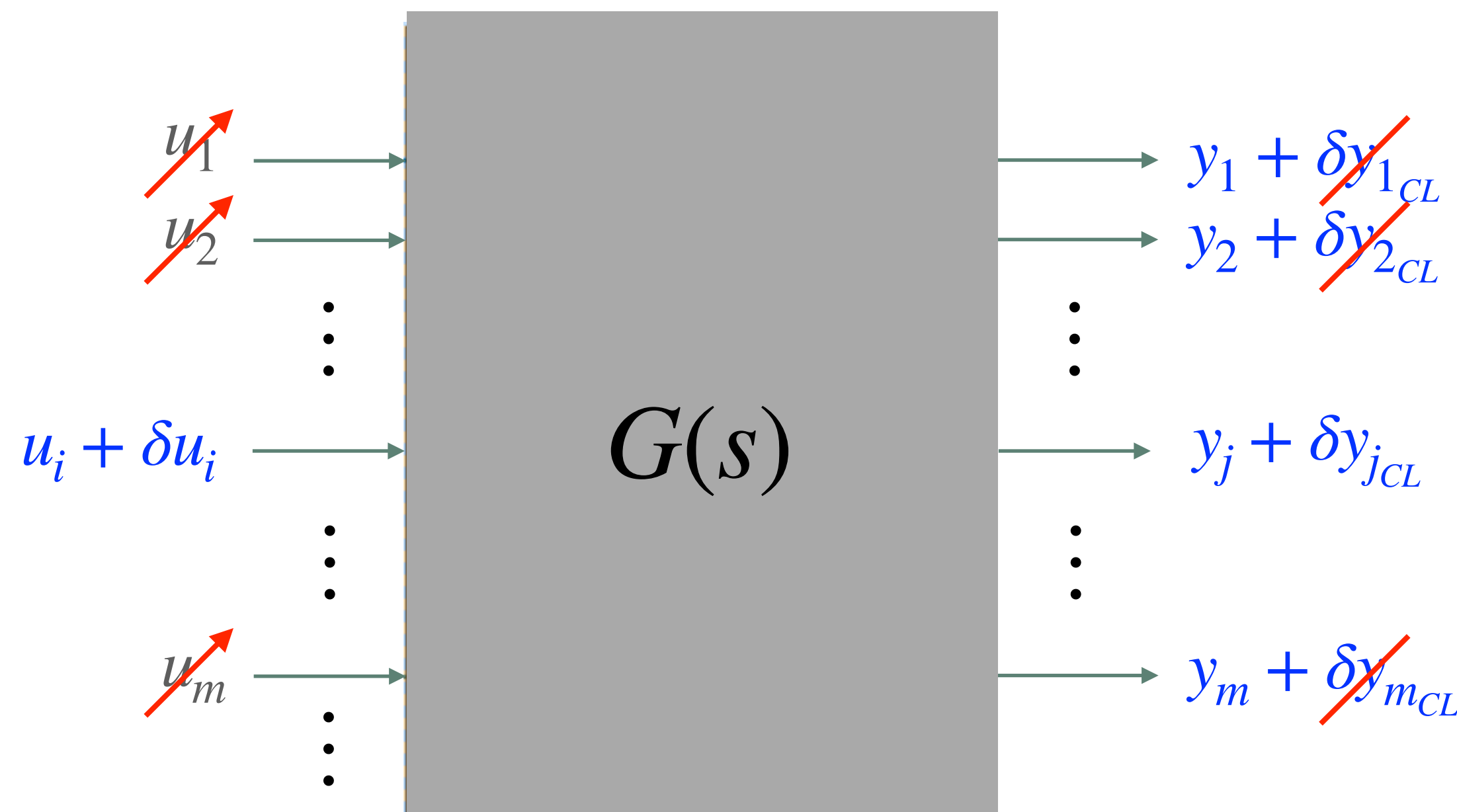
Assumptions:

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Relative Gain Array



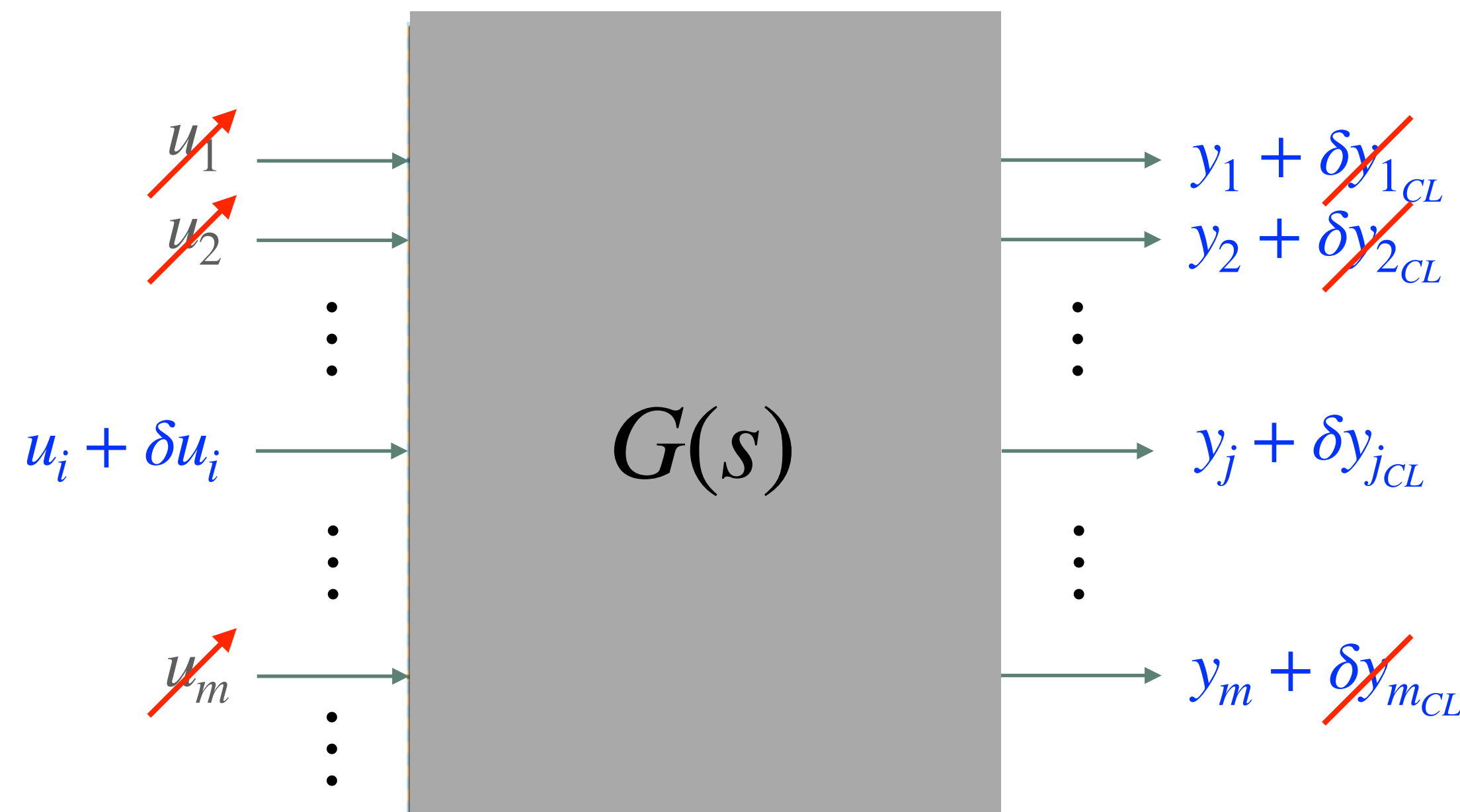
Assumptions:

- Static case ($\omega = 0$)
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Relative Gain Array



Assumptions:

- Static case ($\omega = 0$)
- $G(s)$ As. Stable
- $\det G(0) \neq 0$
- System at the equilibrium at given constant values of the control variables

Two tests are performed:

- 1) Open loop test
- 2) Closed loop test

closed loop gain

$$\frac{\delta y_{jCL}}{\delta u_i} = h_{ji}$$

$\forall \quad i, j = 1, \dots, m$

Relative Gain Array

$$\forall \quad i, j = 1, \dots, m$$

$$\frac{\delta y_{jOL}}{\delta u_i} = G_{ji}(0) = g_{ji}$$

$$\frac{\delta y_{jCL}}{\delta u_i} = h_{ji}$$


$$\lambda_{ji} = \frac{g_{ji}}{h_{ji}}$$

Assumptions:

- Static case ($\omega = 0$)
- $G(s)$ As. Stable
- $\det G(0) \neq 0$
- System at the equilibrium at given constant values of the control variables

Two tests are performed:

- 1) Open loop test
- 2) Closed loop test

Relative Gain Array

$$\forall \quad i, j = 1, \dots, m$$

$$\frac{\delta y_{jOL}}{\delta u_i} = G_{ji}(0) = g_{ji}$$

$$\frac{\delta y_{jCL}}{\delta u_i} = h_{ji}$$

$$\lambda_{ji} = \frac{g_{ji}}{h_{ji}}$$

$$\Lambda = \begin{bmatrix} \lambda_{11}(s) & \lambda_{12}(s) & \cdots & \lambda_{1m}(s) \\ \lambda_{21}(s) & \lambda_{22}(s) & \cdots & \lambda_{2m}(s) \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{m1}(s) & \lambda_{m2}(s) & \cdots & \lambda_{mm}(s) \end{bmatrix}$$

Assumptions:

- Static case ($\omega = 0$)
- $G(s)$ As. Stable
- $\det G(0) \neq 0$
- System at the equilibrium at given constant values of the control variables

Two tests are performed:

- 1) Open loop test
- 2) Closed loop test

Relative Gain Array

$$\forall \quad i, j = 1, \dots, m$$

$$\frac{\delta y_{jOL}}{\delta u_i} = G_{ji}(0) = g_{ji}$$

$$\frac{\delta y_{jCL}}{\delta u_i} = h_{ji}$$

$$\lambda_{ji} = \frac{g_{ji}}{h_{ji}}$$

$$\Lambda = \begin{bmatrix} \lambda_{11}(s) & \lambda_{12}(s) & \dots & \lambda_{1m}(s) \\ \lambda_{21}(s) & \lambda_{22}(s) & \dots & \lambda_{2m}(s) \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{m1}(s) & \lambda_{m2}(s) & \dots & \lambda_{mm}(s) \end{bmatrix}$$

Assumptions:

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- $\det G(0) \neq 0$
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Two tests are performed:

- 1) Open loop test
- 2) Closed loop test

Relative Gain Array (RGA)

Relative Gain Array

$$\forall \quad i, j = 1, \dots, m$$

$$\frac{\delta y_{jOL}}{\delta u_i} = G_{ji}(0) = g_{ji}$$

$$\frac{\delta y_{jCL}}{\delta u_i} = h_{ji}$$

$$\lambda_{ji} = \frac{g_{ji}}{h_{ji}}$$

$$\Lambda = \begin{bmatrix} \lambda_{11}(s) & \lambda_{12}(s) & \dots & \lambda_{1m}(s) \\ \lambda_{21}(s) & \lambda_{22}(s) & \dots & \lambda_{2m}(s) \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{m1}(s) & \lambda_{m2}(s) & \dots & \lambda_{mm}(s) \end{bmatrix}$$

Assumptions:

- Static case ($\omega = 0$)
- $G(s)$ As. Stable
- $\det G(0) \neq 0$
- System at the equilibrium at given constant values of the control variables

Two tests are performed:

- 1) Open loop test
- 2) Closed loop test

Relative Gain Array (RGA)

The best pairings correspond to $\lambda_{ji} > 0$ and close to 1

Relative Gain Array

$$\forall \quad i, j = 1, \dots, m$$

$$\frac{\delta y_{jOL}}{\delta u_i} = G_{ji}(0) = g_{ji}$$

$$\frac{\delta y_{jCL}}{\delta u_i} = h_{ji}$$

$$\lambda_{ji} = \frac{g_{ji}}{h_{ji}}$$

$$\Lambda = \begin{bmatrix} \lambda_{11}(s) & \lambda_{12}(s) & \cdots & \lambda_{1m}(s) \\ \lambda_{21}(s) & \lambda_{22}(s) & \cdots & \lambda_{2m}(s) \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{m1}(s) & \lambda_{m2}(s) & \cdots & \lambda_{mm}(s) \end{bmatrix}$$

Alternative approach:

- Compute

$$\Lambda = G(0) \circ [G(0)^{-1}]^T$$

Relative Gain Array (RGA)

Relative Gain Array

$$\forall \quad i, j = 1, \dots, m$$

$$\frac{\delta y_{jOL}}{\delta u_i} = G_{ji}(0) = g_{ji} \quad \frac{\delta y_{jCL}}{\delta u_i} = h_{ji}$$

$$\lambda_{ji} = \frac{g_{ji}}{h_{ji}}$$

$$\Lambda = \begin{bmatrix} \lambda_{11}(s) & \lambda_{12}(s) & \dots & \lambda_{1m}(s) \\ \lambda_{21}(s) & \lambda_{22}(s) & \dots & \lambda_{2m}(s) \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{m1}(s) & \lambda_{m2}(s) & \dots & \lambda_{mm}(s) \end{bmatrix}$$

Alternative approach:

- Compute

$$\Lambda = G(0) \circ [G(0)^{-1}]^T$$

Properties of the RGA:

- $\sum_{i=1}^m \lambda_{ji} = 1, \quad \text{for } j = 1, \dots, m$
- $\sum_{j=1}^m \lambda_{ji} = 1, \quad \text{for } i = 1, \dots, m$
- elements independent of the measurement units used to model $G(s)$
- Λ diagonal if $G(s)$ is triangular or diagonal

Relative Gain Array (RGA)

Relative Gain Array

Example 1:

Given

$$G(s) = \begin{bmatrix} \frac{5}{10s+1} & \frac{1}{5s+1} & \frac{0.5}{8s+1} \\ \frac{2}{7s+1} & \frac{4}{9s+1} & \frac{1}{6s+1} \\ \frac{1}{4s+1} & \frac{0.5}{3s+1} & \frac{3}{5s+1} \end{bmatrix}$$

compute the RGA Λ and determine the best input-output pairings

Relative Gain Array

Example 1:

Given

$$G(s) = \begin{bmatrix} \frac{5}{10s+1} & \frac{1}{5s+1} & \frac{0.5}{8s+1} \\ \frac{2}{7s+1} & \frac{4}{9s+1} & \frac{1}{6s+1} \\ \frac{1}{4s+1} & \frac{0.5}{3s+1} & \frac{3}{5s+1} \end{bmatrix}$$

compute the RGA Λ and determine the best input-output pairings

By using the formula:

$$\Lambda = G(0) \circ [G(0)^{-1}]^T$$

```

/MATLAB Drive/RGA_example.m
1  %% RGA Example 3x3 System with Automatic Coupling Selection
2  clear; clc;
3
4  %% Define Laplace variable
5  s = tf('s');
6
7  %% Define the 3x3 transfer function matrix G(s)
8  G = [ 5/(10*s+1)  1/(5*s+1)  0.5/(8*s+1);
9        2/(7*s+1)  4/(9*s+1)  1/(6*s+1);
10       1/(4*s+1)  0.5/(3*s+1)  3/(5*s+1) ];
11
12  %% Compute DC gain
13  G0 = dcgain(G);
14
15  %% Compute RGA at steady state
16  RGA = G0 .* transpose(inv(G0));
17
18  %% Display RGA numerically
19  disp('RGA matrix at steady state:');
20  disp(RGA);
21

```

Relative Gain Array

Example 1:

Given

$$G(s) = \begin{bmatrix} \frac{5}{10s+1} & \frac{1}{5s+1} & \frac{0.5}{8s+1} \\ \frac{2}{7s+1} & \frac{4}{9s+1} & \frac{1}{6s+1} \\ \frac{1}{4s+1} & \frac{0.5}{3s+1} & \frac{3}{5s+1} \end{bmatrix}$$

compute the RGA Λ and determine the best input-output pairings

By using the formula:

$$\Lambda = G(0) \circ [G(0)^{-1}]^T$$

```

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10       1/(4*s+1)  0.5/(3*s+1)  3/(5*s+1) ];
11
12 %% Compute DC gain
13 G0 = dcgain(G);
14
15 %% Compute RGA at steady state
16 RGA = G0 .* transpose(inv(G0));
17
18 %% Display RGA numerically
19 disp('RGA matrix at steady state:');
20 disp(RGA);
21

```

Command Window

New to MATLAB? See resources for [Getting Started](#).

RGA matrix at steady state:

```

1.1275  -0.0980  -0.0294
-0.1078  1.1373  -0.0294
-0.0196  -0.0392  1.0588

```

Suggested input-output pairings based on RGA:

```

Output y3 ↔ Input u3 (RGA = 1.059)
Output y1 ↔ Input u1 (RGA = 1.127)
Output y2 ↔ Input u2 (RGA = 1.137)

```

Best pairings:

$\{u_1 \rightarrow y_1\} \cup \{u_2 \rightarrow y_2\} \cup \{u_3 \rightarrow y_3\}$

Relative Gain Array

Example 2:

Given

$$G(s) = \begin{bmatrix} \frac{2}{1+s} & \frac{1}{1+s} \\ \frac{-1}{1+s} & \frac{2}{1+s} \end{bmatrix}$$

compute the RGA Λ and determine the best input-output pairings

Relative Gain Array

Example 2:

Given

$$G(s) = \begin{bmatrix} \frac{2}{1+s} & \frac{1}{1+s} \\ \frac{-1}{1+s} & \frac{2}{1+s} \end{bmatrix}$$

compute the RGA Λ and determine the best input-output pairings

By using the formula:

$$\Lambda = G(0) \circ [G(0)^{-1}]^T$$

/MATLAB Drive/RGA_example_2.m

```
1 %% RGA Example 2x2 System with Automatic Coupling Selection
2 clear; clc;
3
4 %% Define Laplace variable
5 s = tf('s');
6
7 %% Define 2x2 transfer function matrix G(s) with given static gains
8 G = [ 2/(1+s)  1/(1+s);
9       -1/(1+s) 2/(1+s) ];
10
11 %% Compute DC gain
12 G0 = dcgain(G);
13
14 disp('DC gain G(0):');
15 disp(G0);
16
17 %% Compute RGA
18 RGA = G0 .* transpose(inv(G0));
19 disp('RGA matrix:');
20 disp(RGA);
```


Relative Gain Array

Example 2:

Given

$$G(s) = \begin{bmatrix} \frac{2}{1+s} & \frac{1}{1+s} \\ \frac{-1}{1+s} & \frac{2}{1+s} \end{bmatrix}$$

compute the RGA Λ and determine the best input-output pairings

Best pairings:

$$\{u_1 \rightarrow y_1\} \cup \{u_2 \rightarrow y_2\}$$

By using the formula:

$$\Lambda = G(0) \circ [G(0)^{-1}]^T$$

/MATLAB Drive/RGA_example_2.m

```

1  %% RGA Example 2x2 System with Automatic Coupling Selection
2  clear; clc;
3
4  %% Define Laplace variable
5  s = tf('s');
6
7  %% Define 2x2 transfer function matrix G(s) with given static gains
8  G = [ 2/(1+s)    1/(1+s);
9        -1/(1+s)   2/(1+s) ];
10
11 %% Compute DC gain
12 G0 = dcgain(G);
13
14 disp('DC gain G(0):');
15 disp(G0);
16
17 %% Compute RGA
18 RGA = G0 .* transpose(inv(G0));
19 disp('RGA matrix:');
20 disp(RGA);

```

Command Window

New to MATLAB? See resources for [Getting Started](#).

RGA matrix:
 0.8000 0.2000
 0.2000 0.8000

Suggested input-output pairings:
 Output y1 ↔ Input u1 (RGA = 0.800)
 Output y2 ↔ Input u2 (RGA = 0.800)



General Information

Prof. Antonella Ferrara

<https://scholar.google.com/citations?user=r5JuMskAAAAJ&hl=en>

Course Teaching Material:

KIRO UNIPV <https://elearning.unipv.it/>

- [504462 - PROCESS CONTROL 2025-26 - PROF.SSA FERRARA ANTONELLA](#)

Lecture Time-table:

<http://www-3.unipv.it/ingserv/orario2526/1sem/insegnamenti/ProCont.html>

Exams:

<https://studentionline.unipv.it/esse3/Home.do>

<https://kirotesting.unipv.it/>

Introduction

- Program of the course:

Advanced SISO control schemes:

Pre-filters and parallel compensators, two degrees of freedom control schemes, compensation of measurable disturbances, systems with delays and Smith Predictor, Padé approximation, decoupling in the frequency domain, control of open loop unstable systems.

Advanced MIMO control schemes:

Decoupling based control schemes, decentralized control, relative gain array.

PID controllers:

Features and properties. Rules for the empirical calibration. Wind-up and anti wind-up schemes.

Digital control:

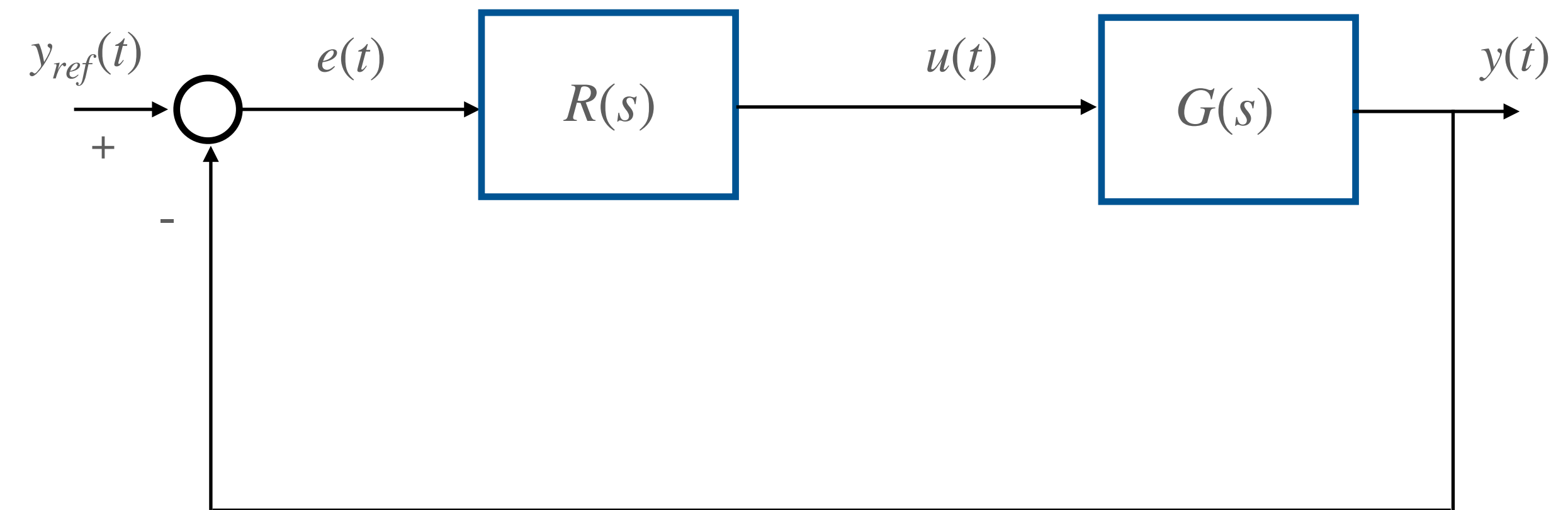
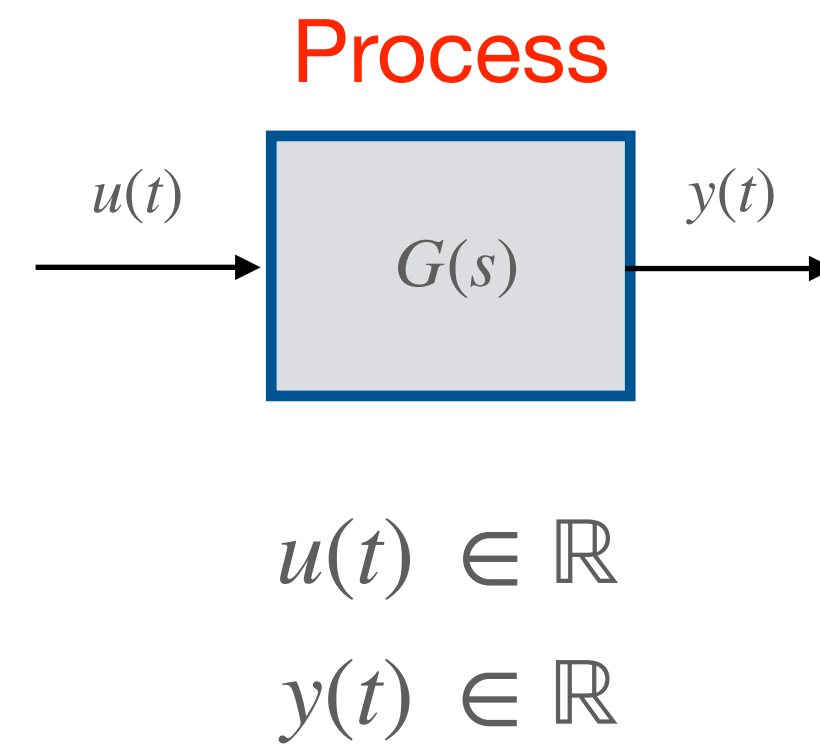
Discrete-time systems. The concept of equilibrium for discrete-time systems. Stability of linear time-invariant discrete-time systems. Jury test. Digital control schemes. Zeta transform and its properties. Transfer functions in the z domain. Sampling and aliasing. Choice of the sampling time. Zero-order-Hold. Discretization of continuous-time controllers. Bilinear transformation, Euler and Tustin methods.

Introduction

- Some of the figures in these slides, kindly provided by McGraw-Hill, are those of the Textbook:

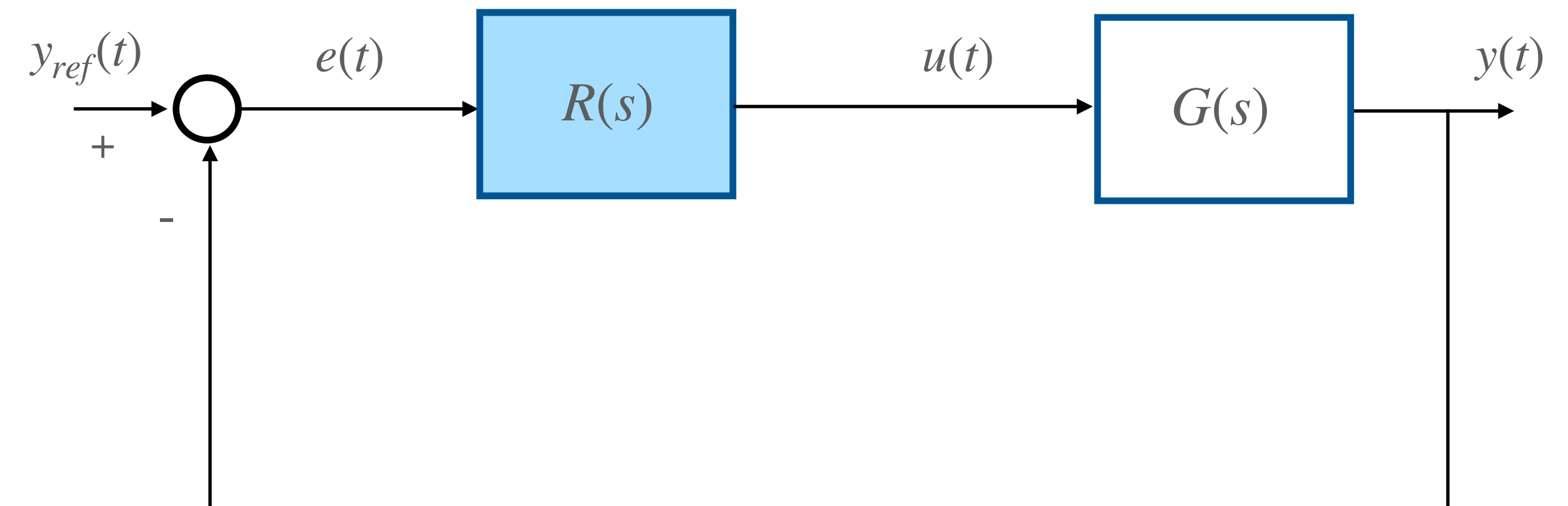
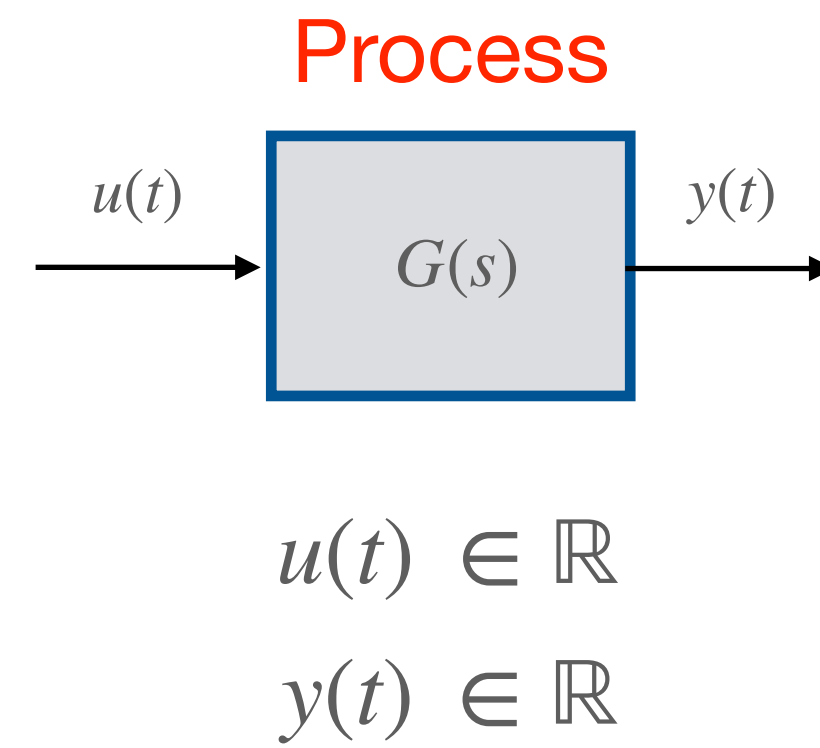


Design of PID Controllers

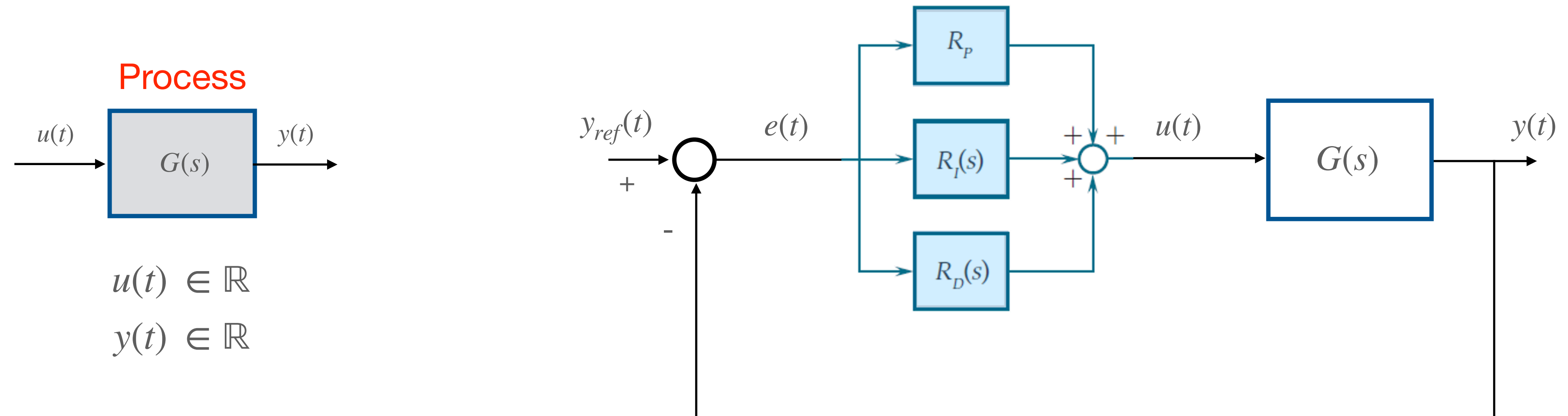


Unitary Feedback Control Scheme

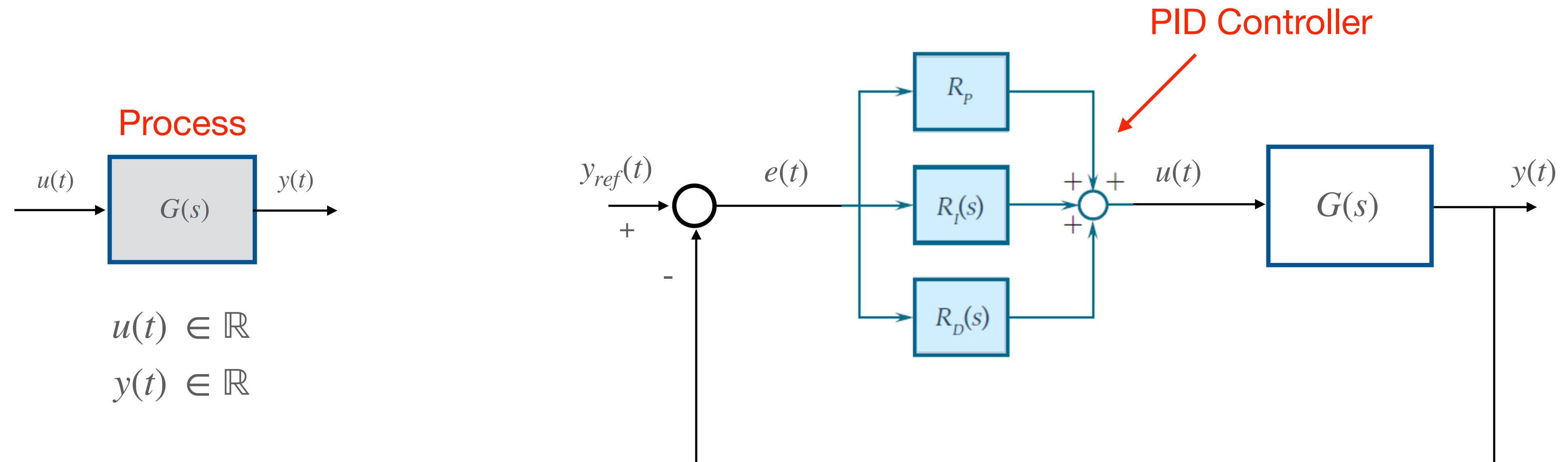
Design of PID Controllers



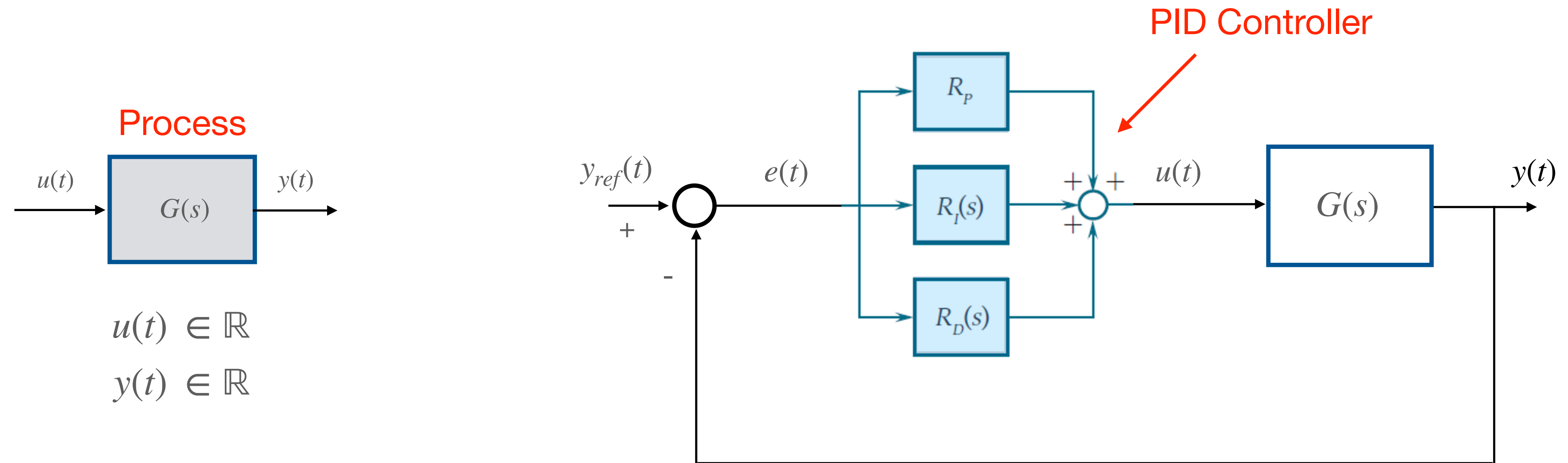
Design of PID Controllers



Design of PID Controllers

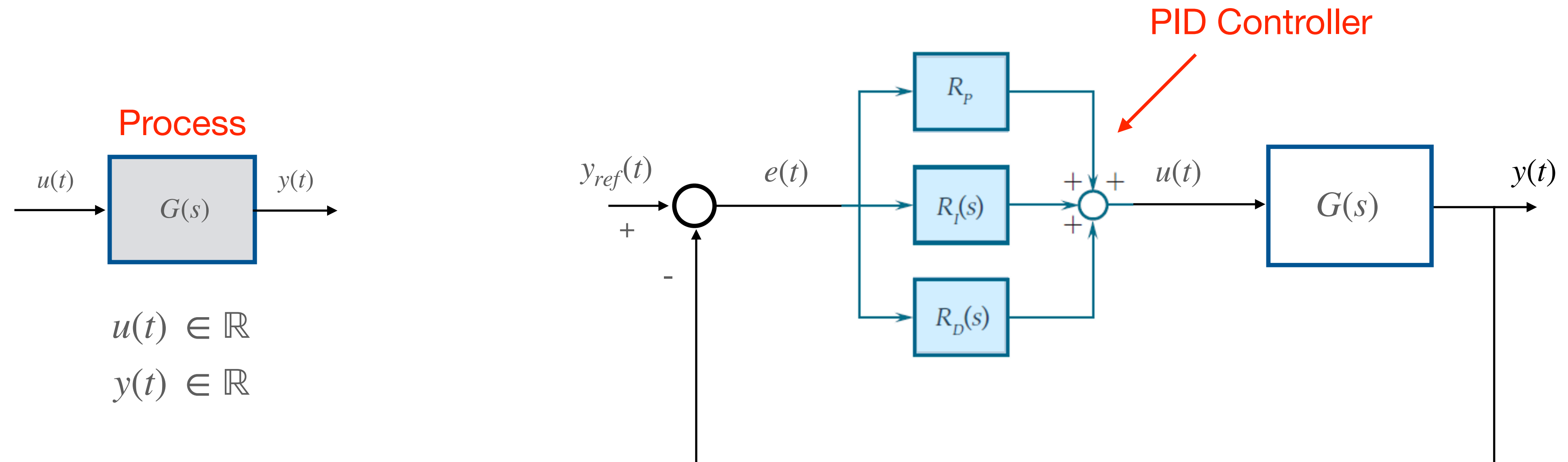


Design of PID Controllers



Ideal PID Controller: $u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$

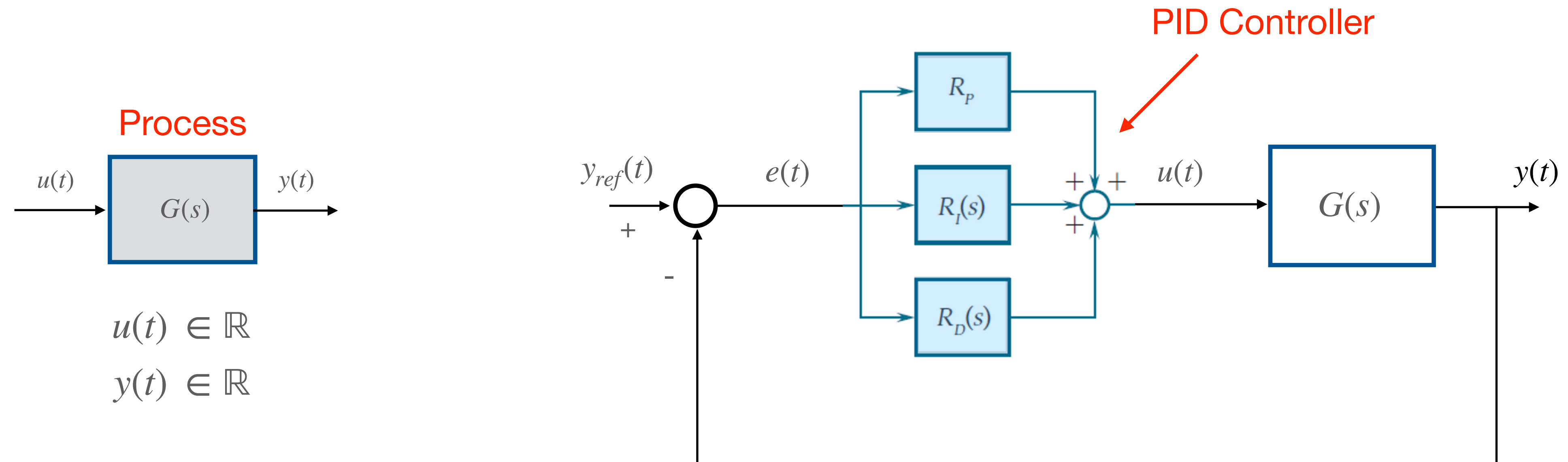
Design of PID Controllers



Ideal PID Controller: $u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$

$$\mathcal{L}\{u(t)\} = U(s) = \mathcal{L}\left\{K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}\right\} = \left(K_P + \frac{K_I}{s} + K_D s\right) E(s)$$

Design of PID Controllers

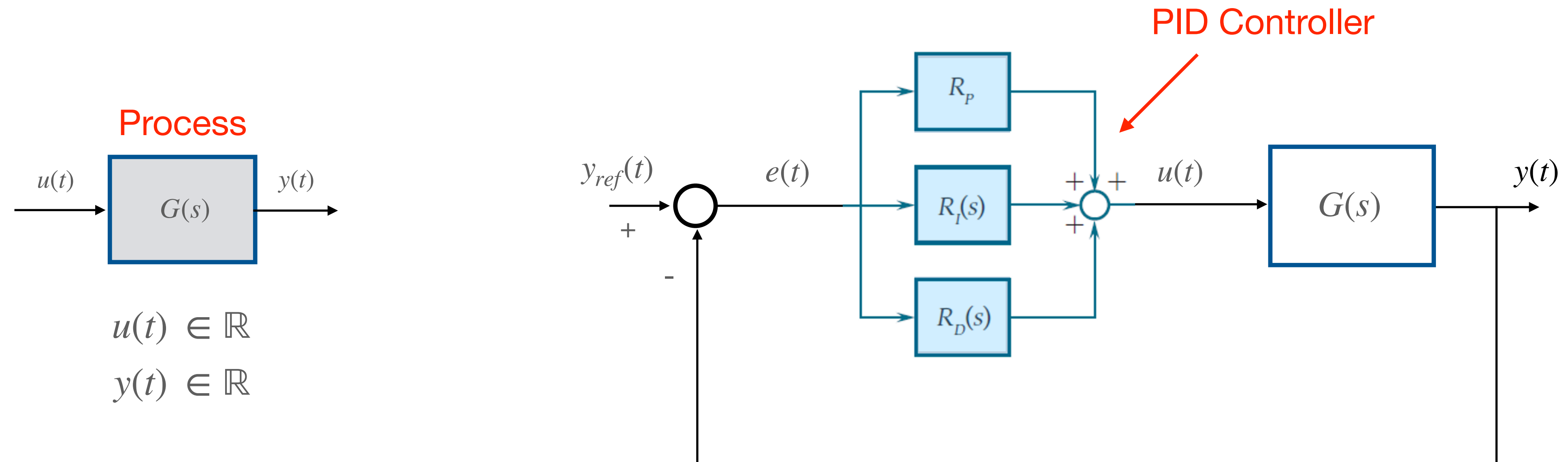


$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = \frac{K_D s^2 + K_P s + K_I}{s}$$

Ideal PID Controller: $u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$

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Design of PID Controllers



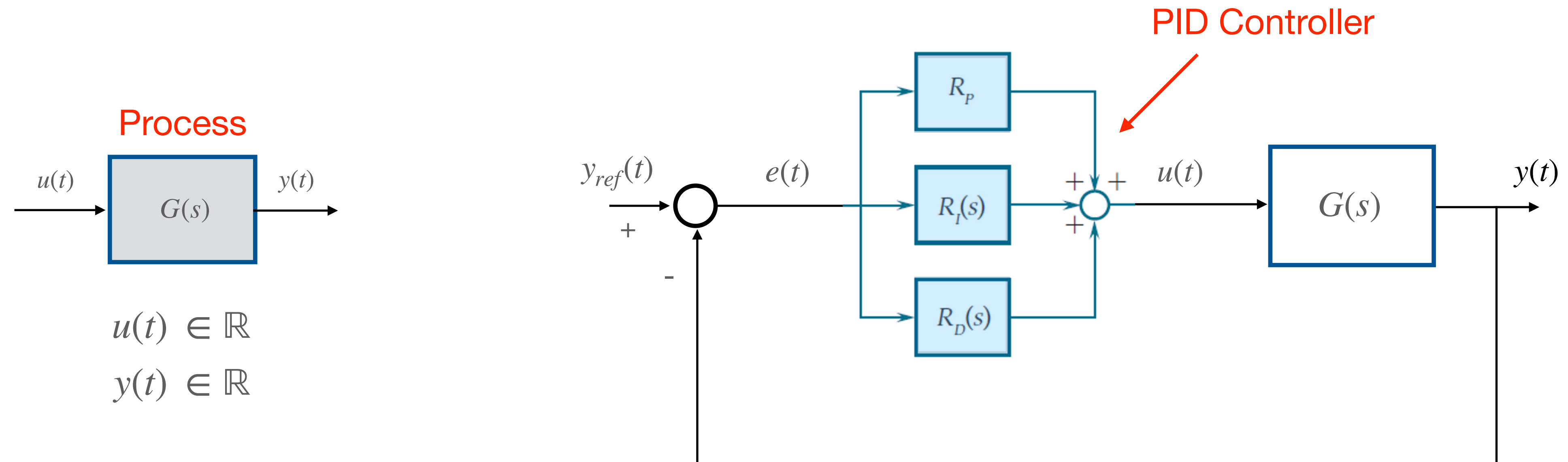
It is not causal

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = \frac{K_D s^2 + K_P s + K_I}{s}$$

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Design of PID Controllers



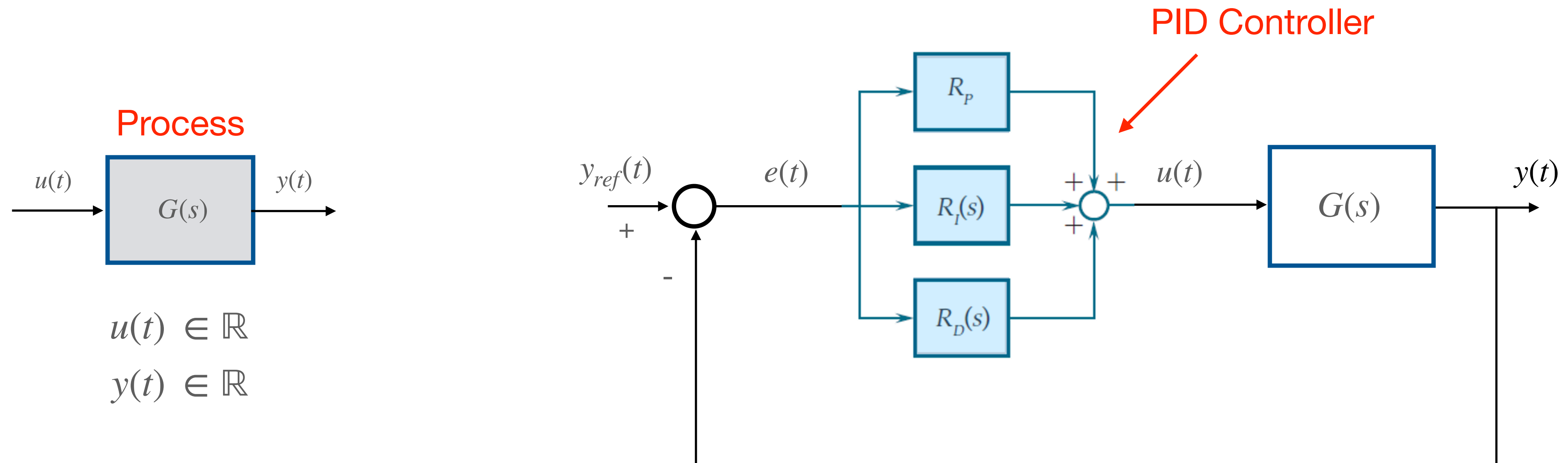
Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Ideal PID Controller: $u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$

$$\mathcal{L}\{u(t)\} = U(s) = \mathcal{L}\left\{ K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt} \right\} = \left(K_P + \frac{K_I}{s} + K_D s \right) E(s)$$

Design of PID Controllers



Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

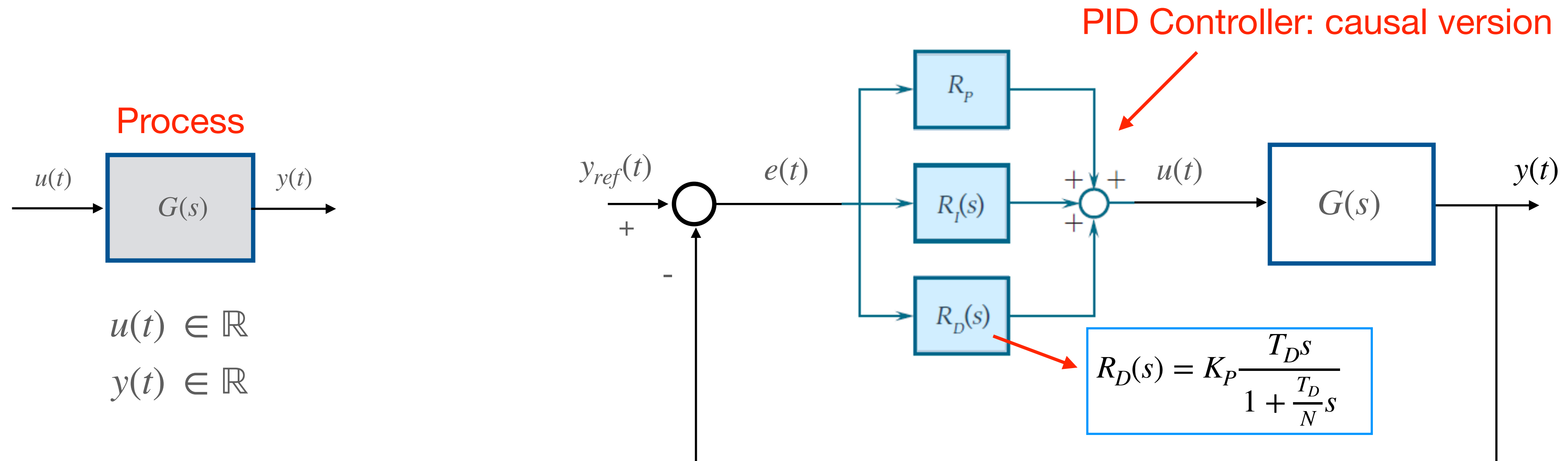
To make it
causal

$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D s}{1 + \frac{T_D}{N} s} \right)$$

Ideal PID Controller: $u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$

$$\mathcal{L}\{u(t)\} = U(s) = \mathcal{L}\left\{ K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt} \right\} = \left(K_P + \frac{K_I}{s} + K_D s \right) E(s)$$

Design of PID Controllers



Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

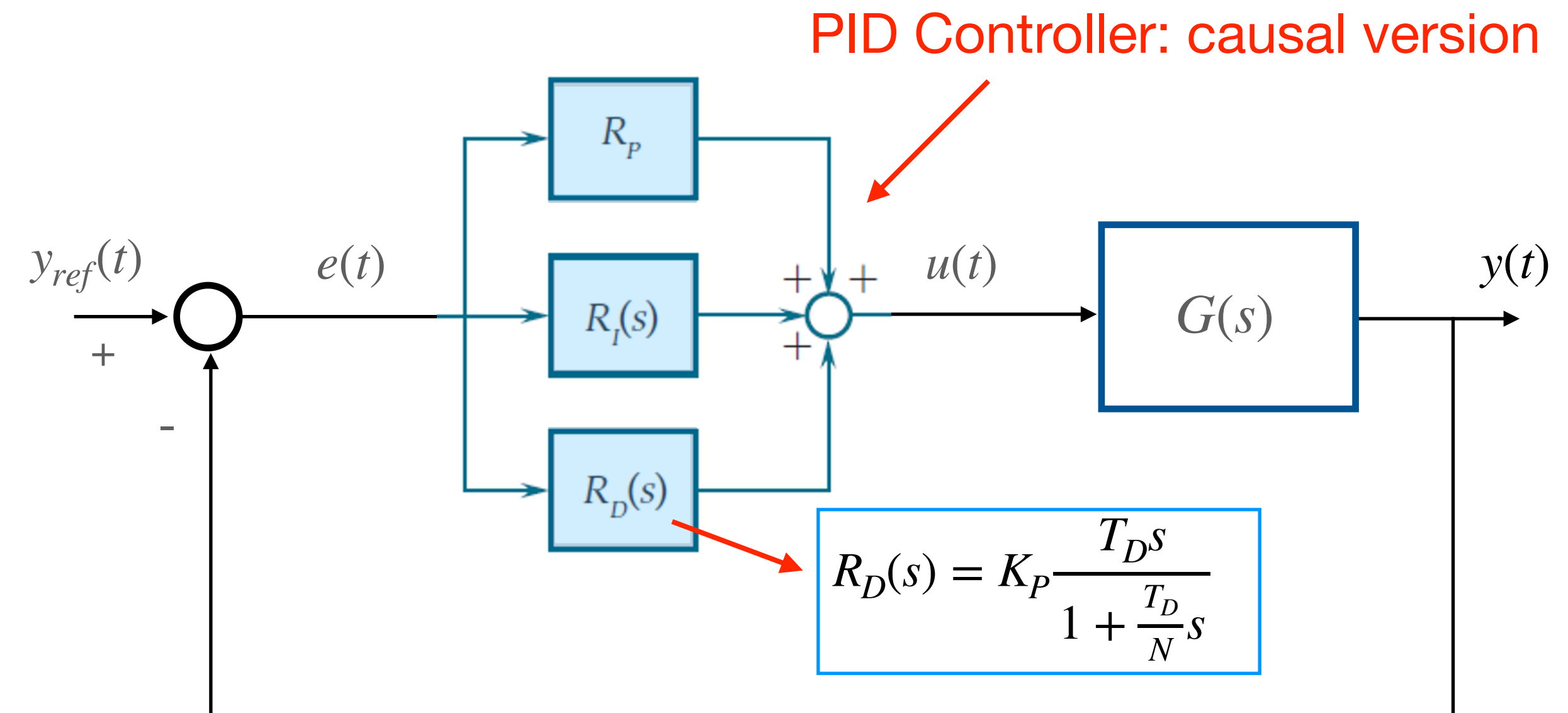
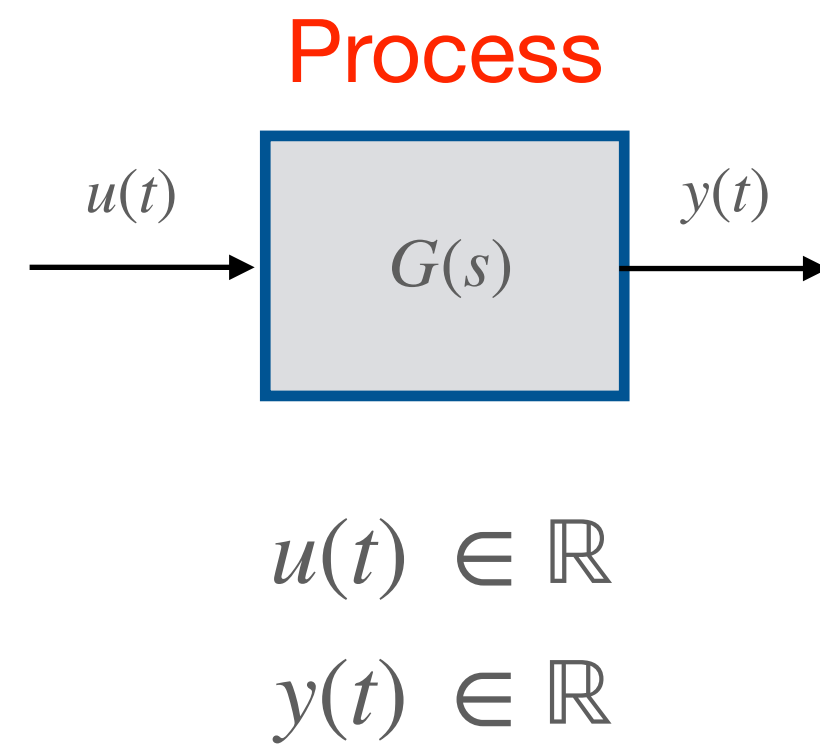
To make it
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$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D s}{1 + \frac{T_D}{N} s} \right)$$

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Design of PID Controllers

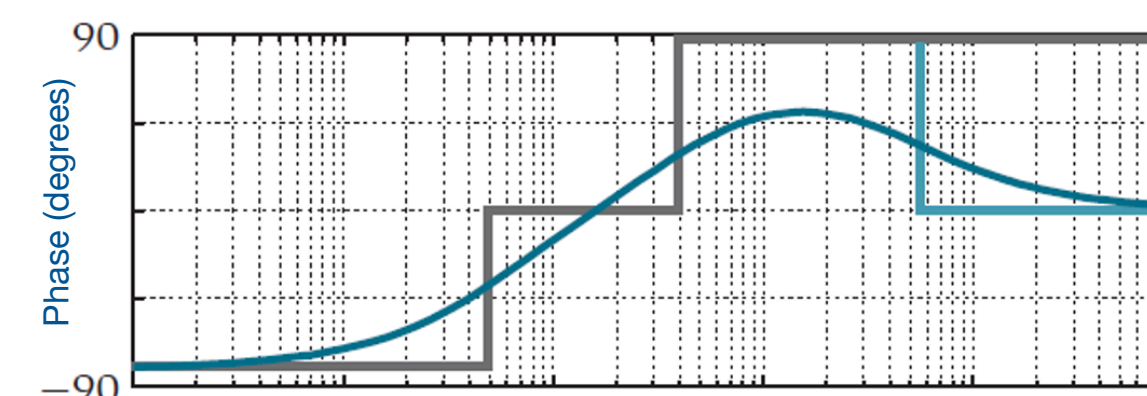
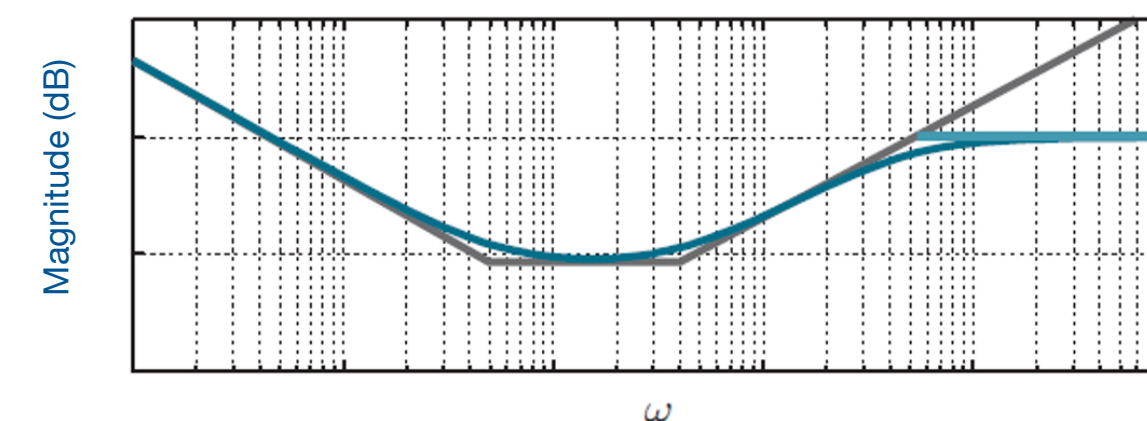


Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

It is causal

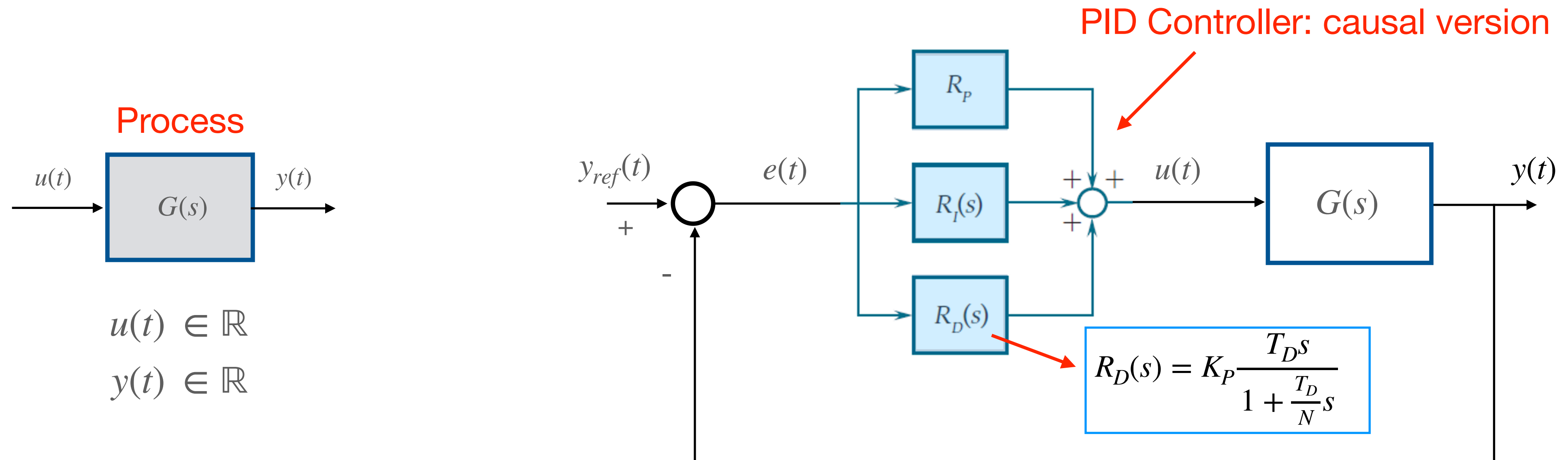
$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D s}{1 + \frac{T_D}{N} s} \right)$$



Effect of the new pole on the Bode plots

— Actual plots
— PID Controller: ideal version (asympt. plot)
— PID Controller: causal version (asympt. plot)

Design of PID Controllers



Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Effect of the new pole on the PID zeros:

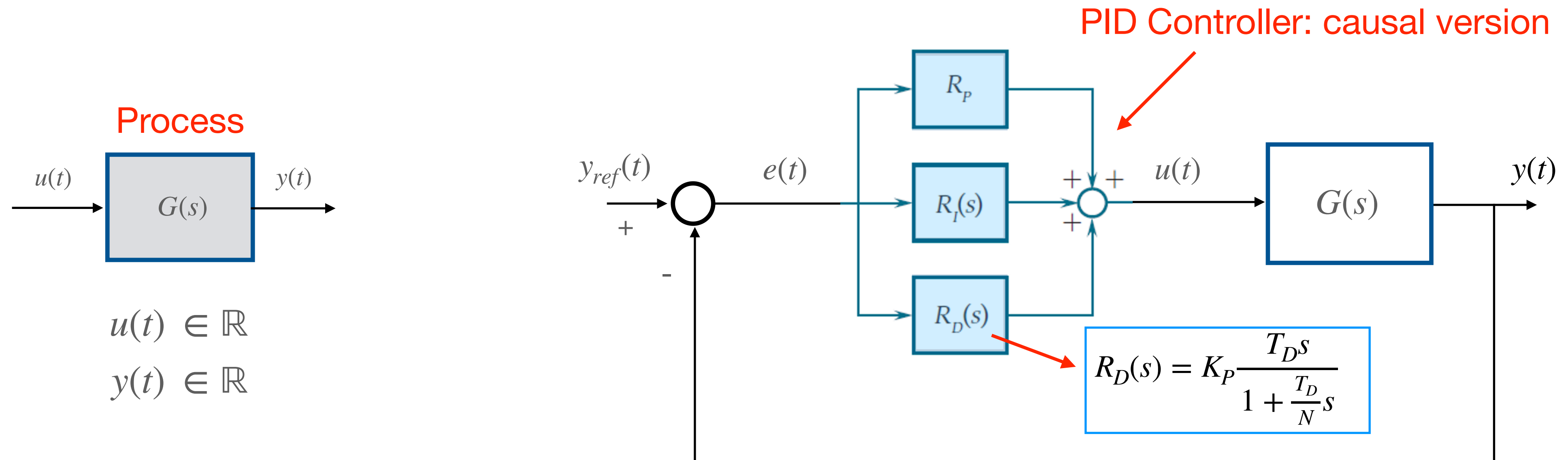
It is causal

$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D s}{1 + \frac{T_D}{N} s} \right)$$



$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D N s}{T_D s + N} \right) = \frac{K_P \left[T_D s^2 + \left(1 + \frac{T_D}{T_I} \right) s + \frac{1}{T_I} N \right]}{s(T_D s + N)}$$

Design of PID Controllers



Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Effect of the new pole on the PID zeros:

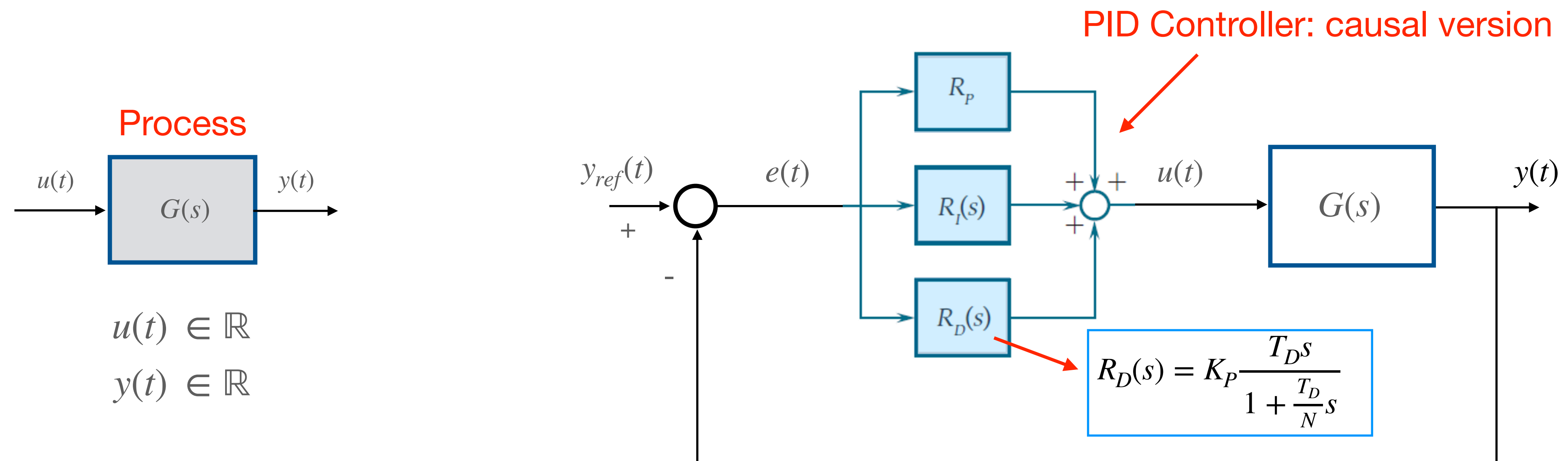
It is causal

$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D s}{1 + \frac{T_D}{N} s} \right)$$



$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D N s}{T_D s + N} \right) = \frac{K_P \left[T_D s^2 + \left(1 + \frac{T_D}{T_I} \right) s + \frac{1}{T_I} N \right]}{s(T_D s + N)}$$

Design of PID Controllers



Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Effect of the new pole on the PID zeros:

It is causal

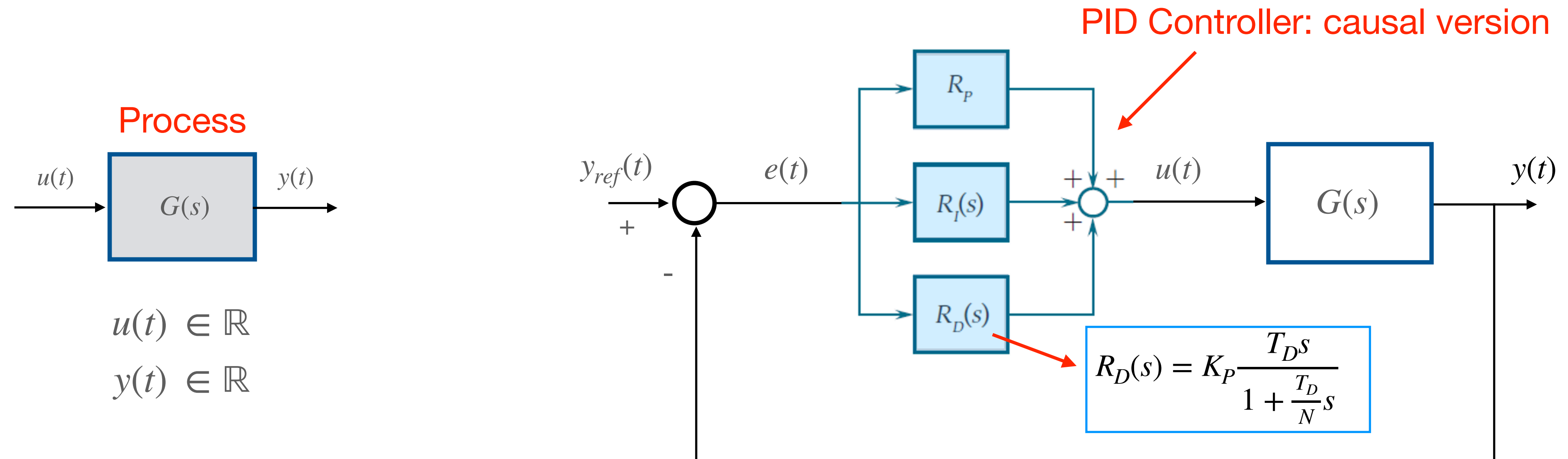
$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D s}{1 + \frac{T_D}{N} s} \right)$$



$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D N s}{T_D s + N} \right) = \frac{K_P \left[T_D s^2 + \left(1 + \frac{T_D}{T_I} \right) s + \frac{1}{T_I} N \right]}{s(T_D s + N)}$$

+∞ the zeros are practically the same as before

Design of PID Controllers



Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Effect of the new pole on the PID zeros:

when $T_I = 4T_D$ the zeros coincide in $s = -\frac{1}{2T_D}$

It is causal

$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D s}{1 + \frac{T_D}{N} s} \right)$$

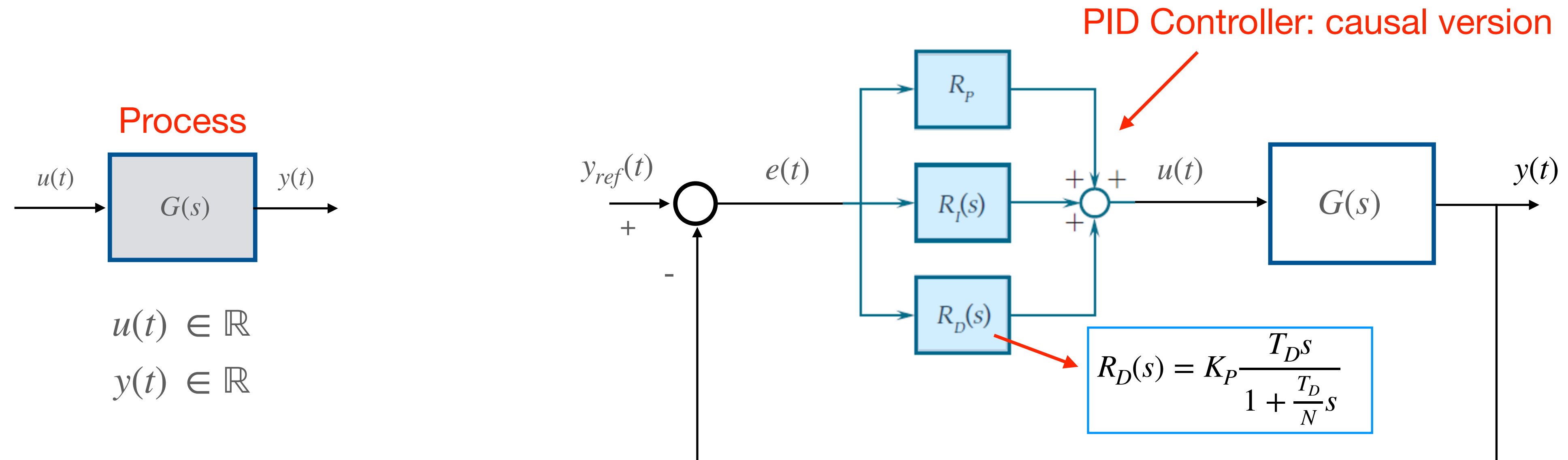


$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D N s}{T_D s + N} \right)$$

$$= \frac{K_P \left[T_D s^2 + \left(1 + \frac{T_D}{T_I} \right) s + \frac{1}{T_I} N \right]}{s(T_D s + N)}$$

$+\infty$ the zeros are practically the same as before

Design of PID Controllers



Alternative representation:

$$R_{PID_{id}}(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Effect of the new pole on the PID zeros:

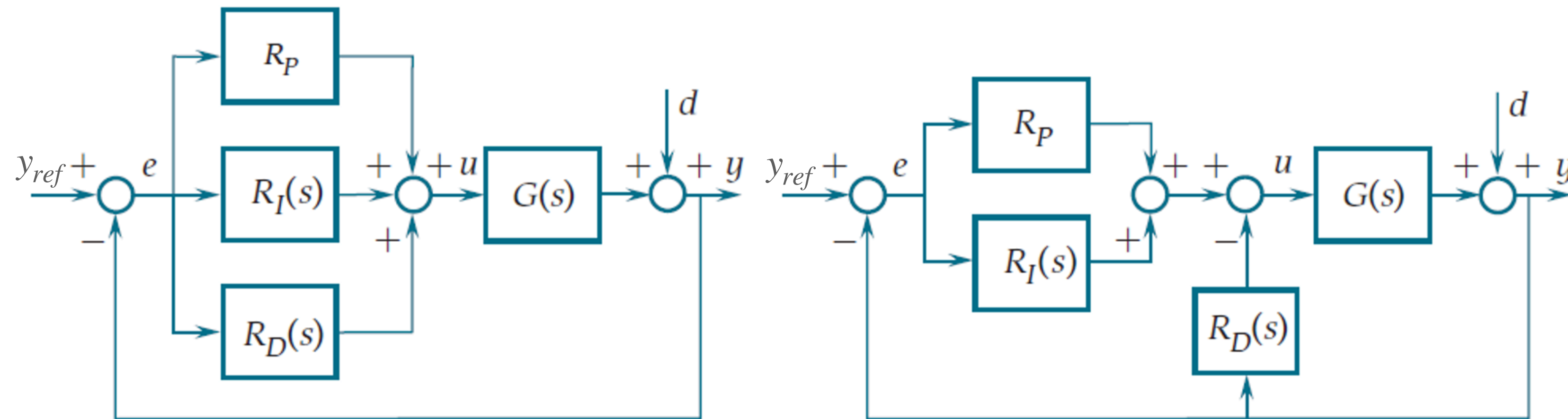
choice made to simplify PID parameter tuning
when $T_I = 4T_D$ the zeros coincide in $s = -\frac{1}{2T_D}$

It is causal

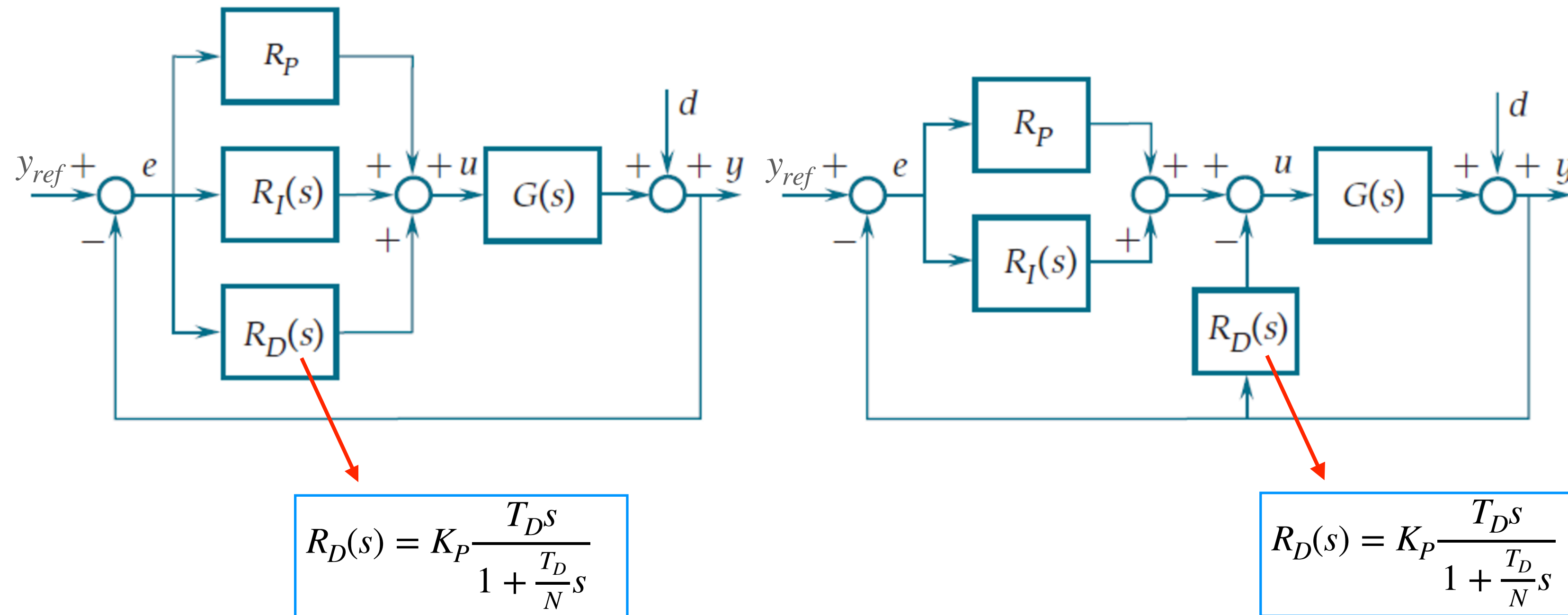
$$R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D s}{1 + \frac{T_D}{N} s} \right) \longrightarrow R_{PID}(s) = K_P \left(1 + \frac{1}{T_I s} + \frac{T_D N s}{T_D s + N} \right) = \frac{K_P \left[T_D s^2 + \left(1 + \frac{T_D}{T_I} \right) s + \frac{1}{T_I} N \right]}{s(T_D s + N)}$$

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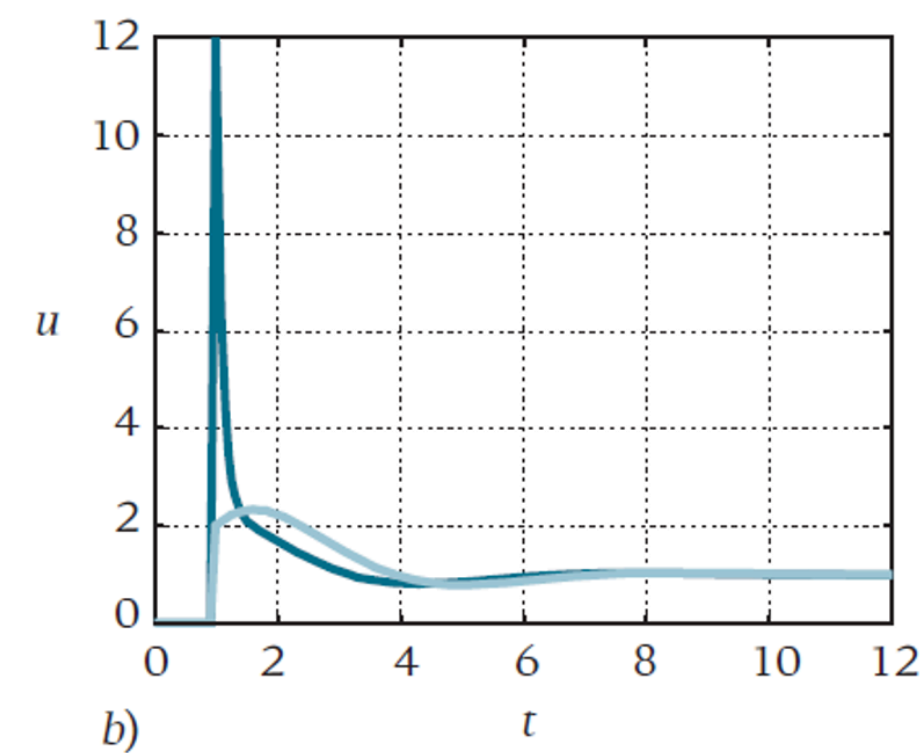
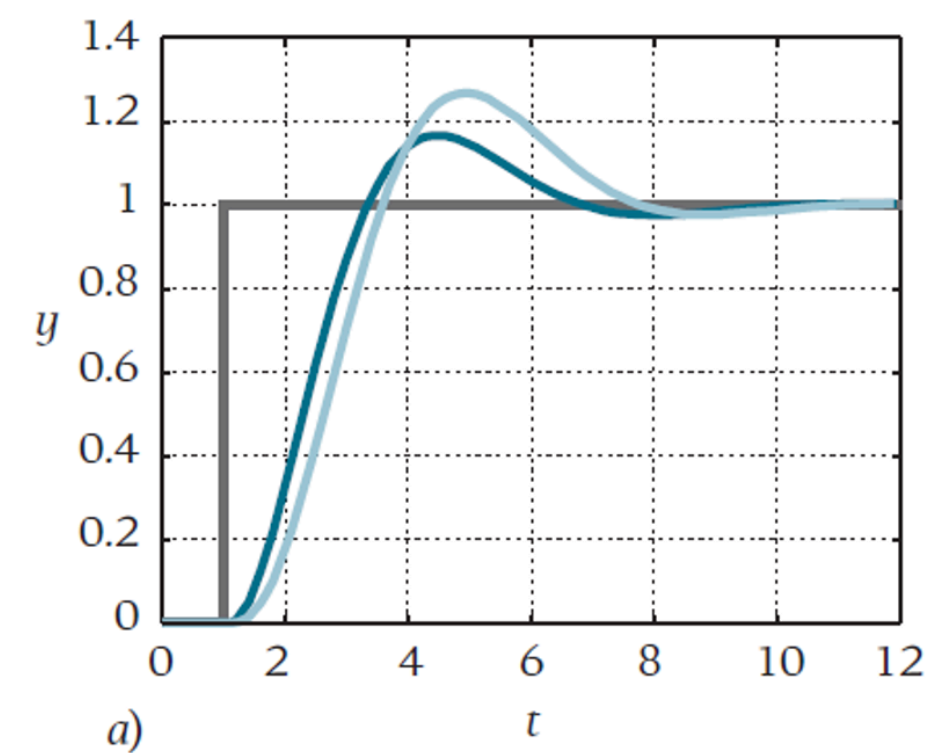
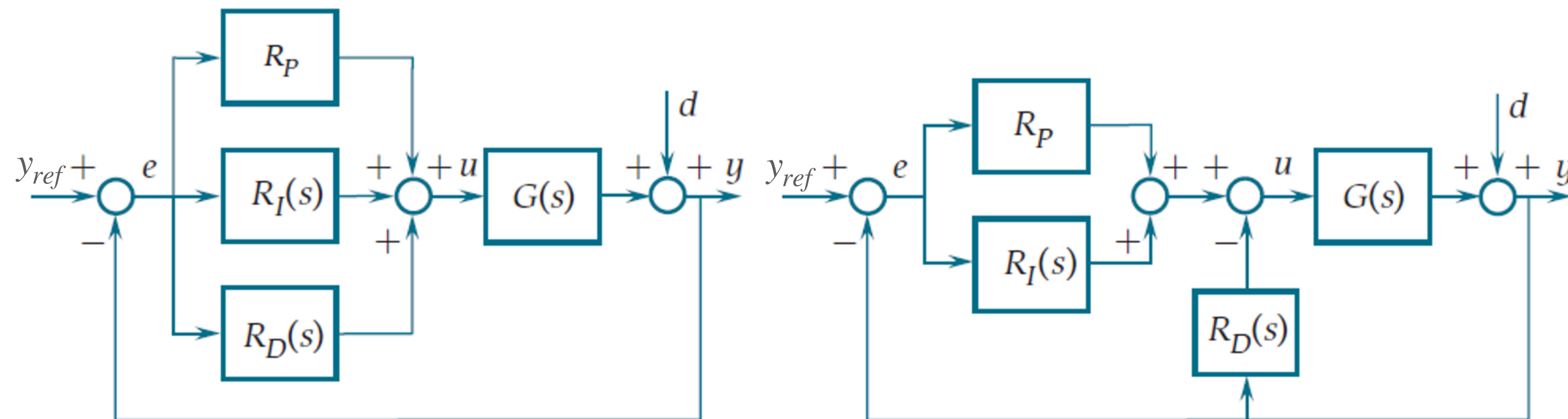
Practical Implementation of PID Controllers



Practical Implementation of PID Controllers

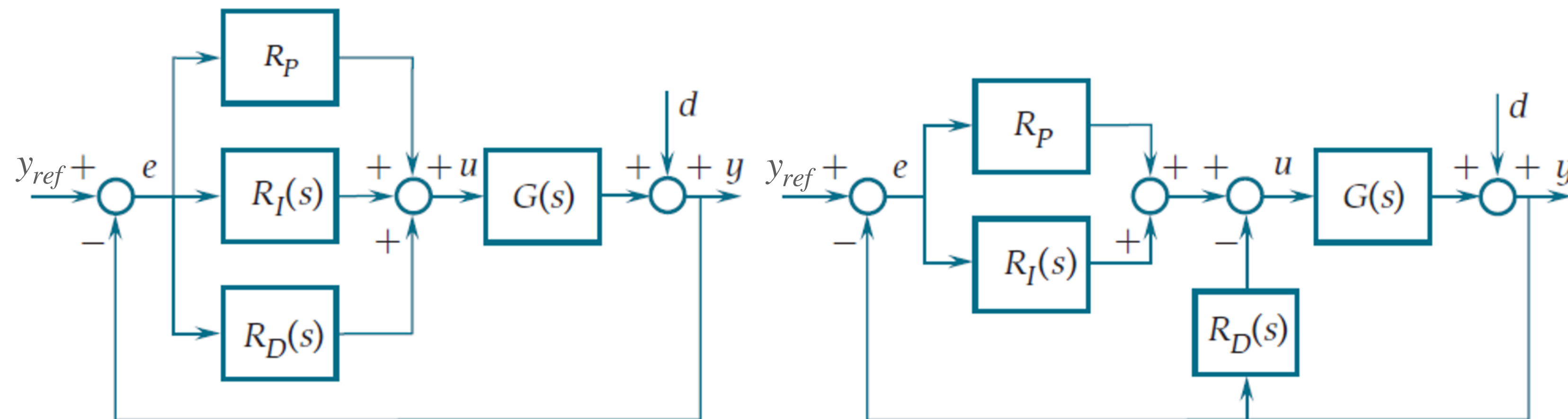


Practical Implementation of PID Controllers

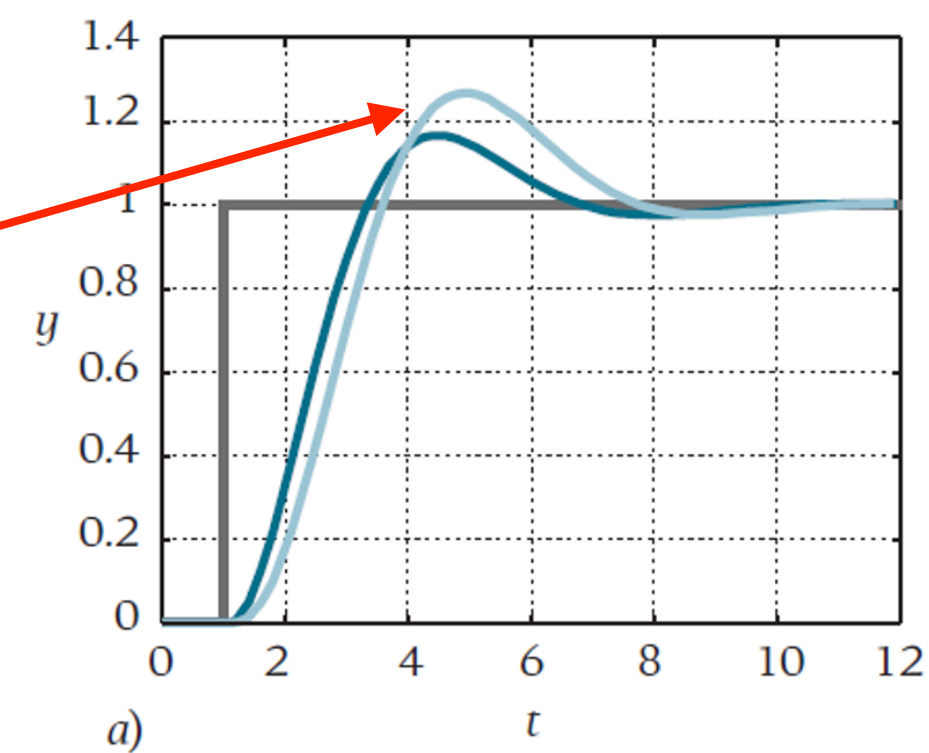


— Error differentiation — Output differentiation

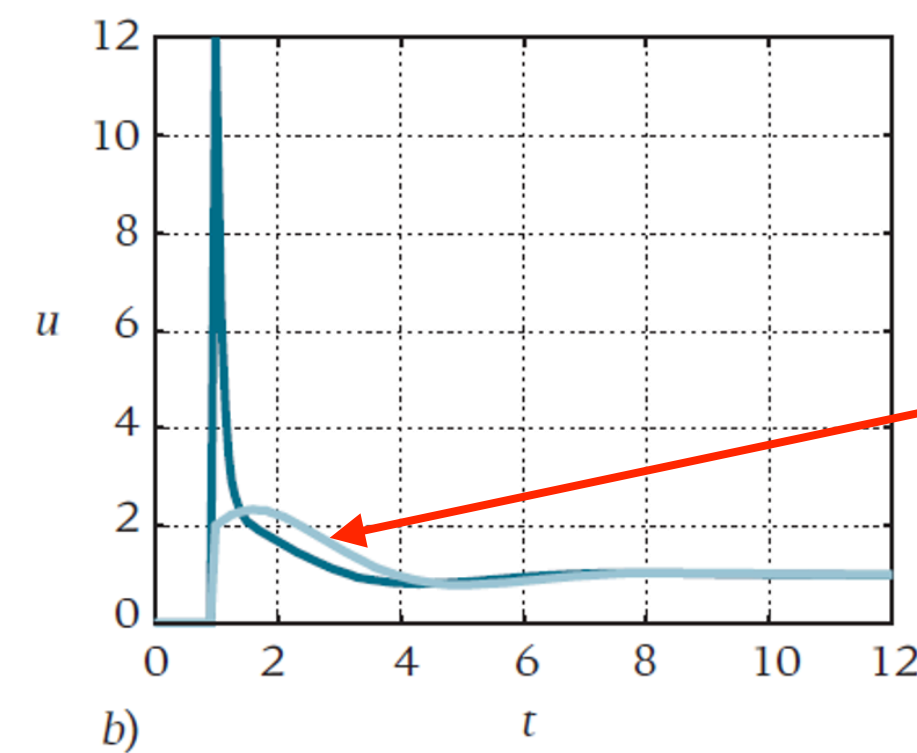
Practical Implementation of PID Controllers



Lower quality in
the transient phase



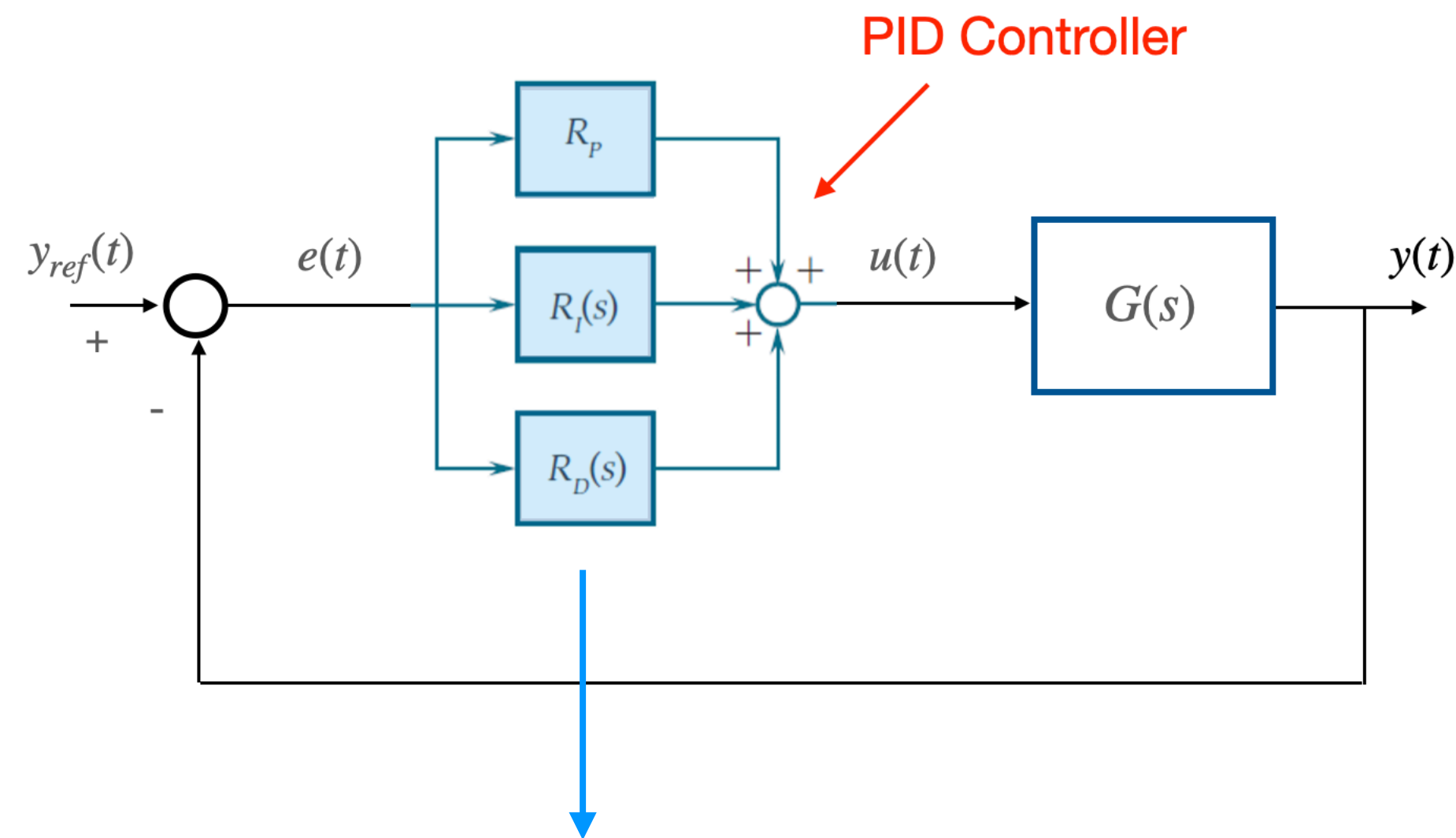
Error
differentiation



Less peaking

Output
differentiation

Practical Implementation of PID Controllers: **Tuning Rules**

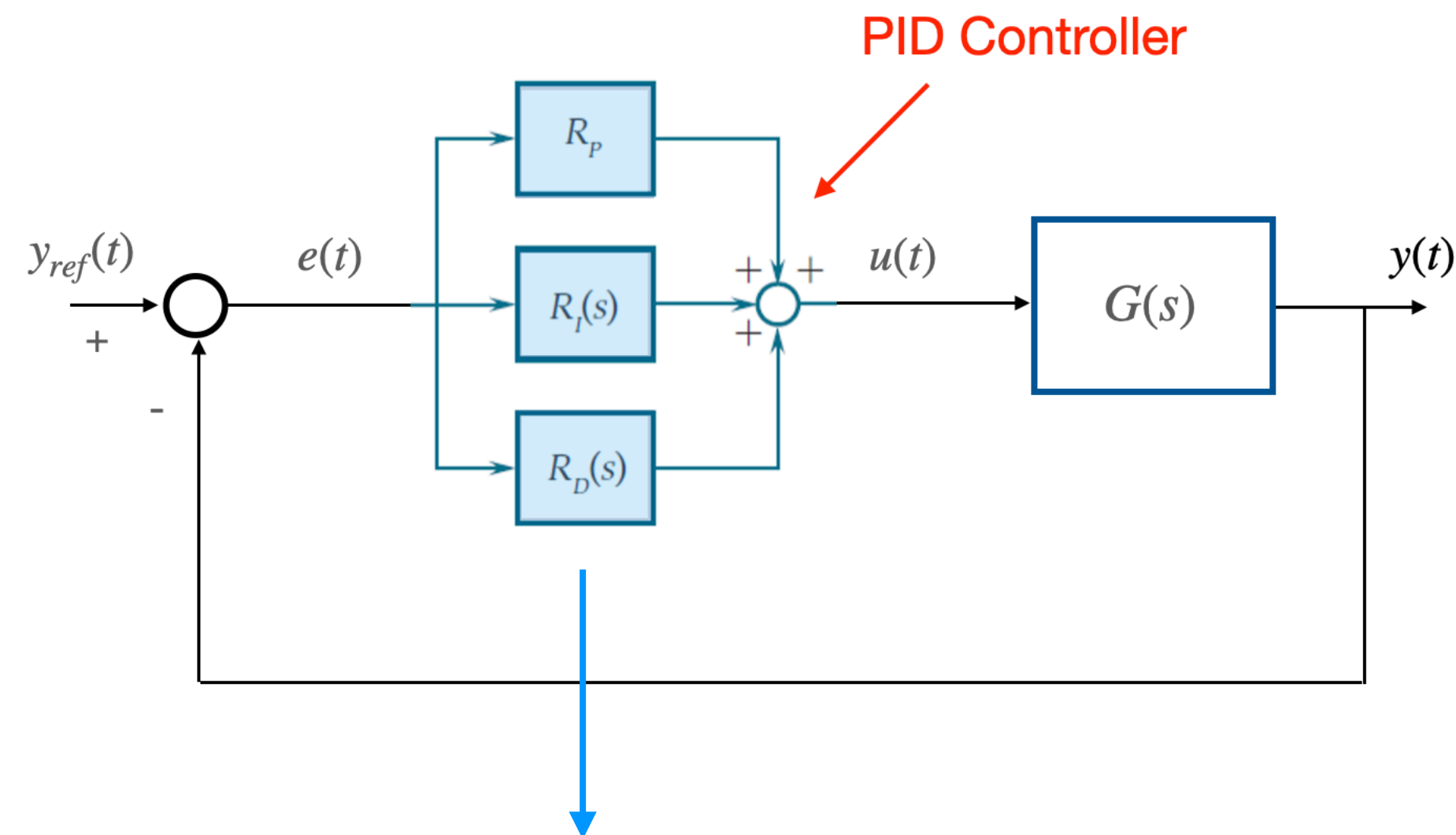


Closed-loop Ziegler-Nichols Method

Testing phase:

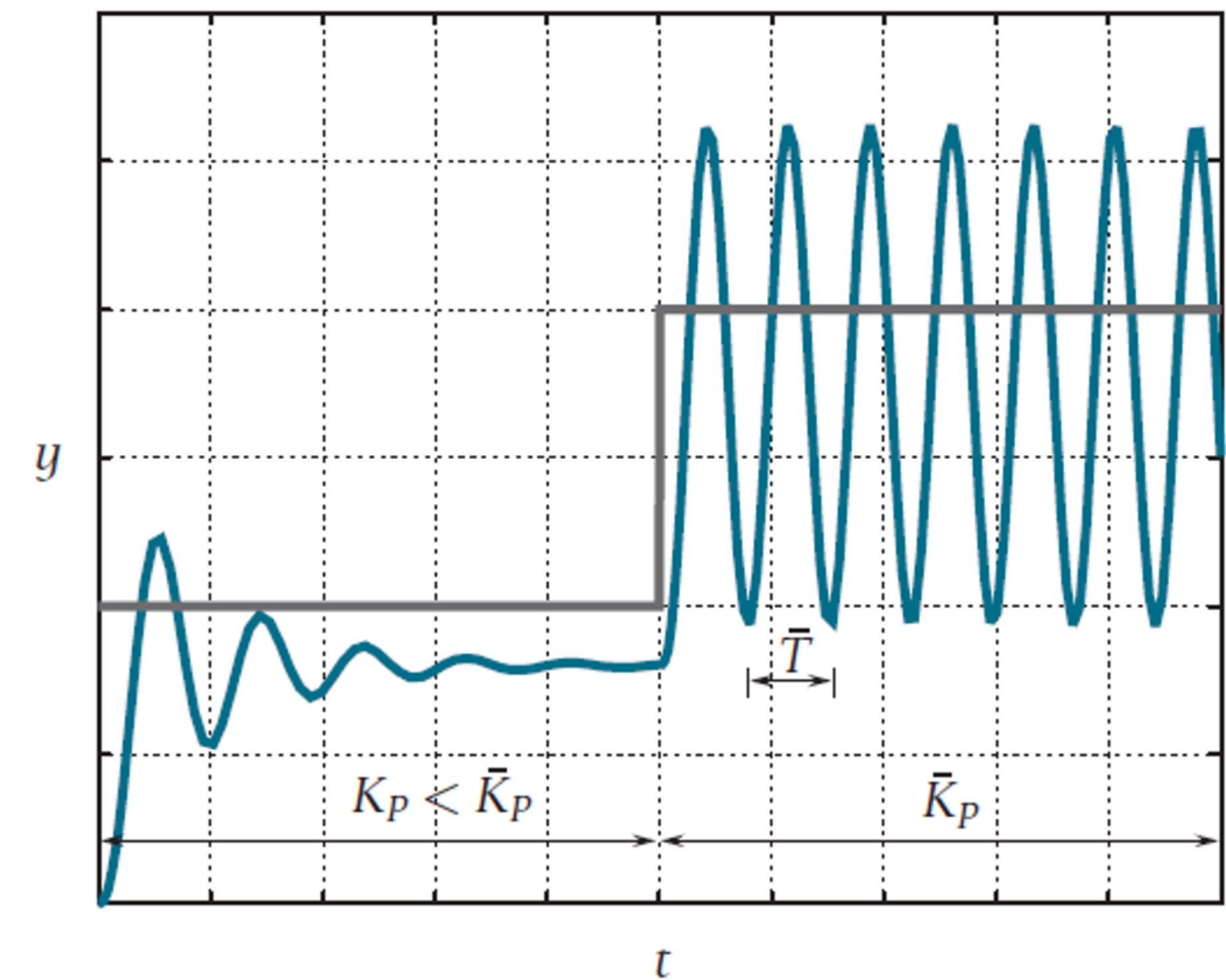
$$R_P(s) = K_P, \quad R_I(s) = 0, \quad R_D(s) = 0$$

Practical Implementation of PID Controllers: **Tuning Rules**



$$R_P(s) = K_P, \quad R_I(s) = 0, \quad R_D(s) = 0$$

Closed-loop Ziegler-Nichols Method

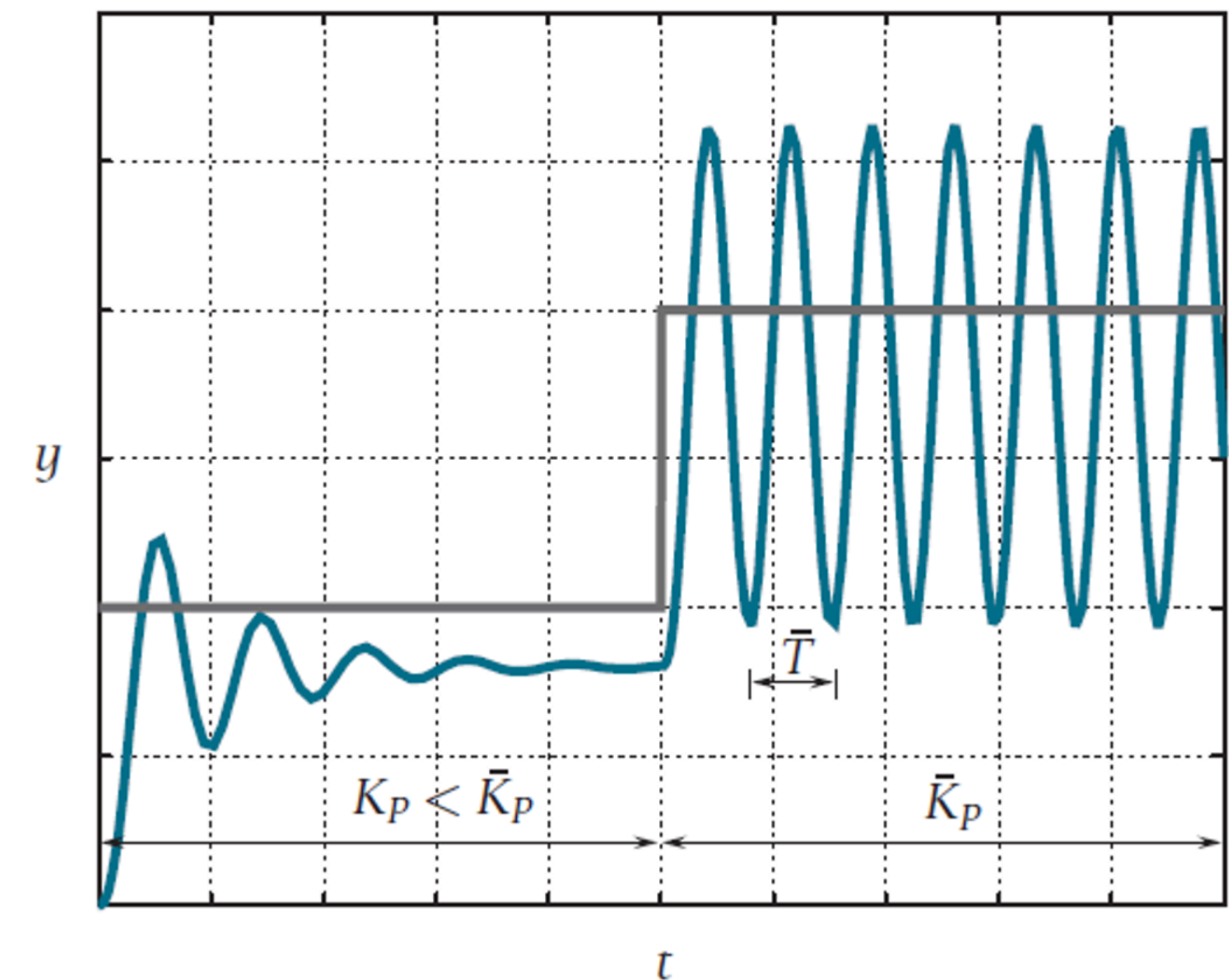


Practical Implementation of PID Controllers: **Tuning Rules**

Closed-loop Ziegler-Nichols Method

	K_P	T_I	T_D
P	$0.5\bar{K}_P$		
PI	$0.45\bar{K}_P$	$0.8\bar{T}$	
PID	$0.6\bar{K}_P$	$0.5\bar{T}$	$0.125\bar{T}$

$$R_{PID_{id}}(s) = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$



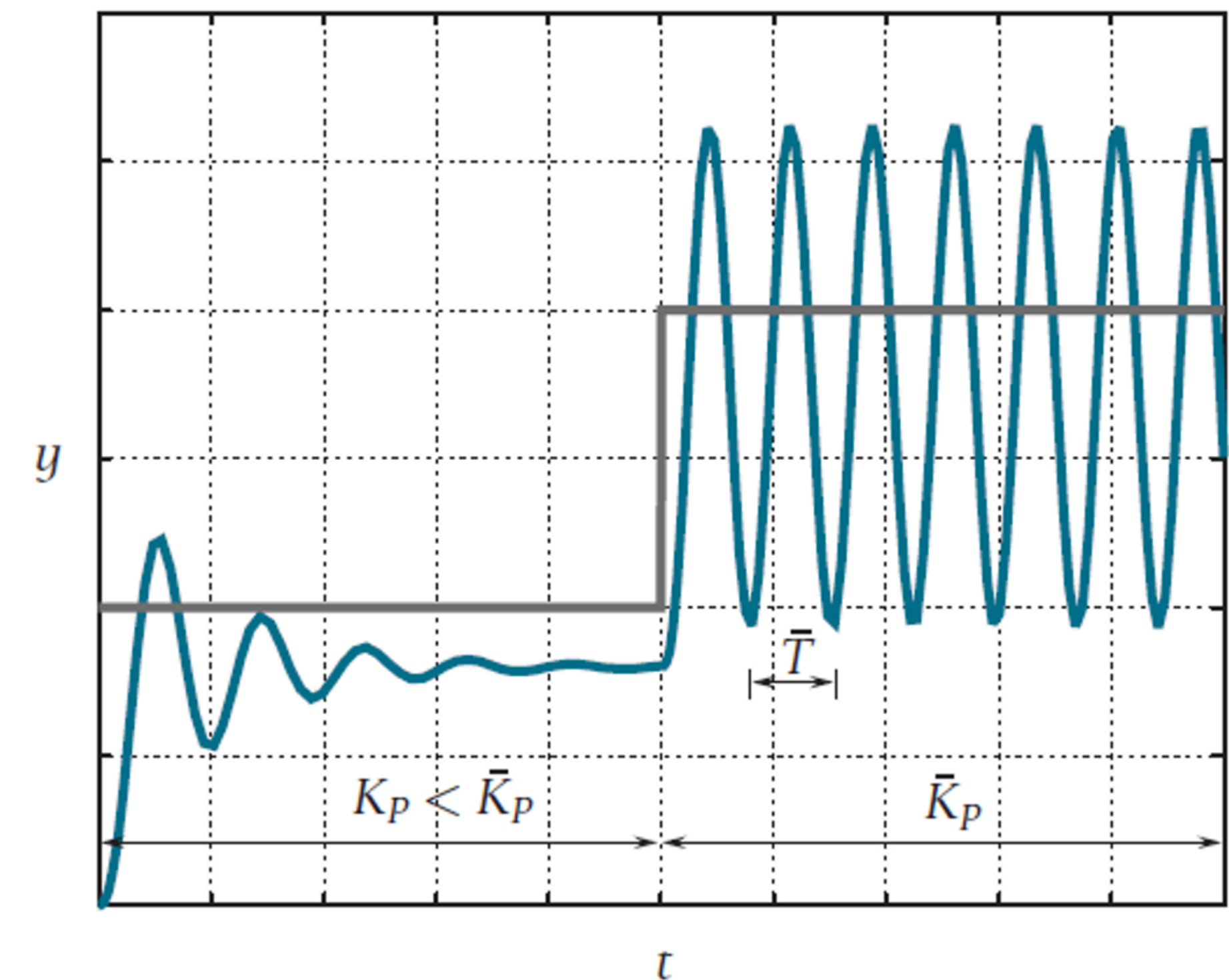
Practical Implementation of PID Controllers: **Tuning Rules**

Closed-loop Ziegler-Nichols Method

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Note that $T_I = 4T_D \rightarrow$ the PID zeros coincide in $s = -\frac{1}{2T_D}$



Practical Implementation of PID Controllers: **Tuning Rules**

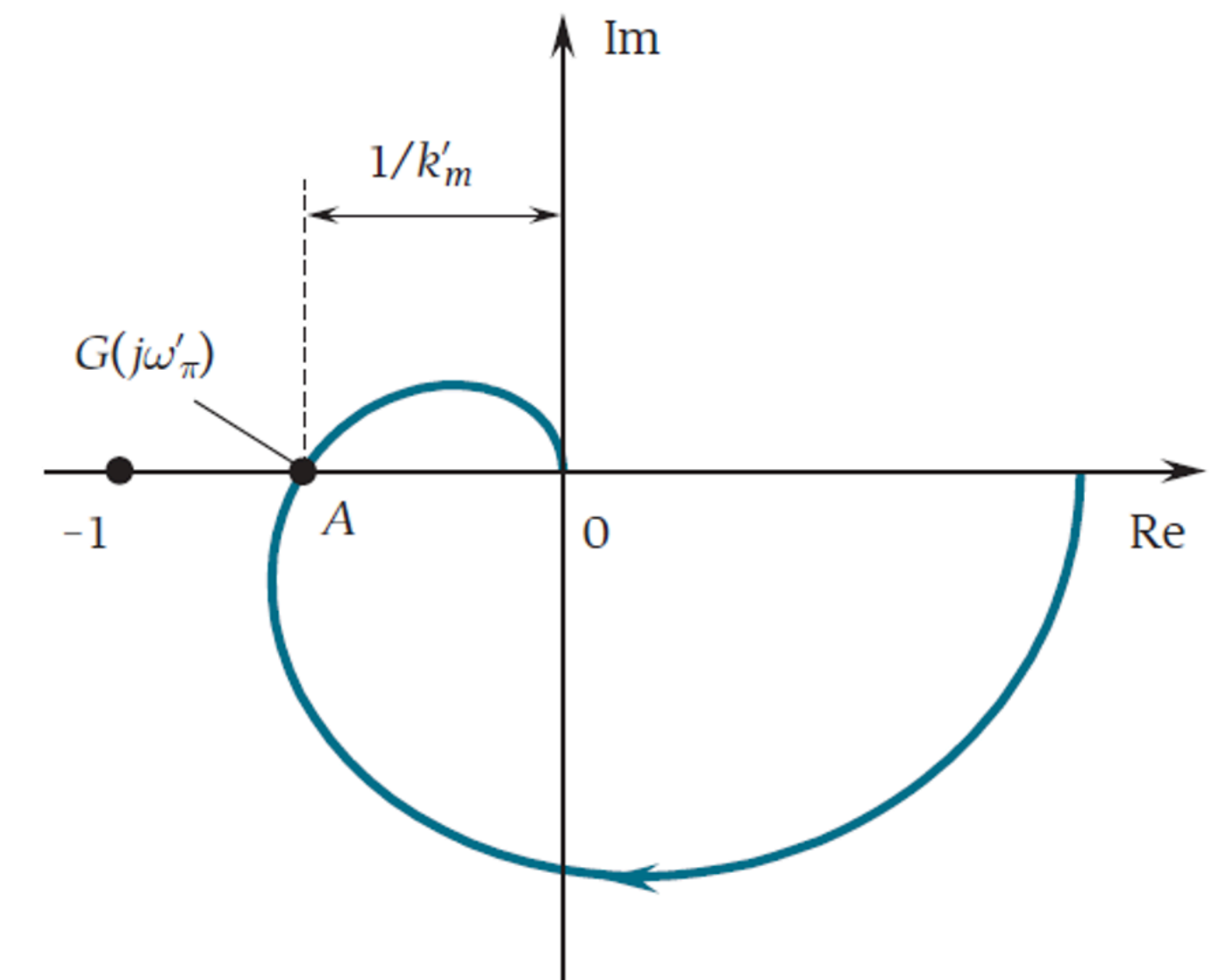
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Note that $T_I = 4T_D \rightarrow$ the PID zeros coincide in $s = -\frac{1}{2T_D}$

Closed-loop Ziegler-Nichols Method: Interpretation



$\bar{K}_p = k'_m \rightarrow$ Gain Margin of $G(s)$

$\bar{T} = \frac{\pi}{\omega'_\pi} \rightarrow$ where ω'_π is the pulse corresponding to point A

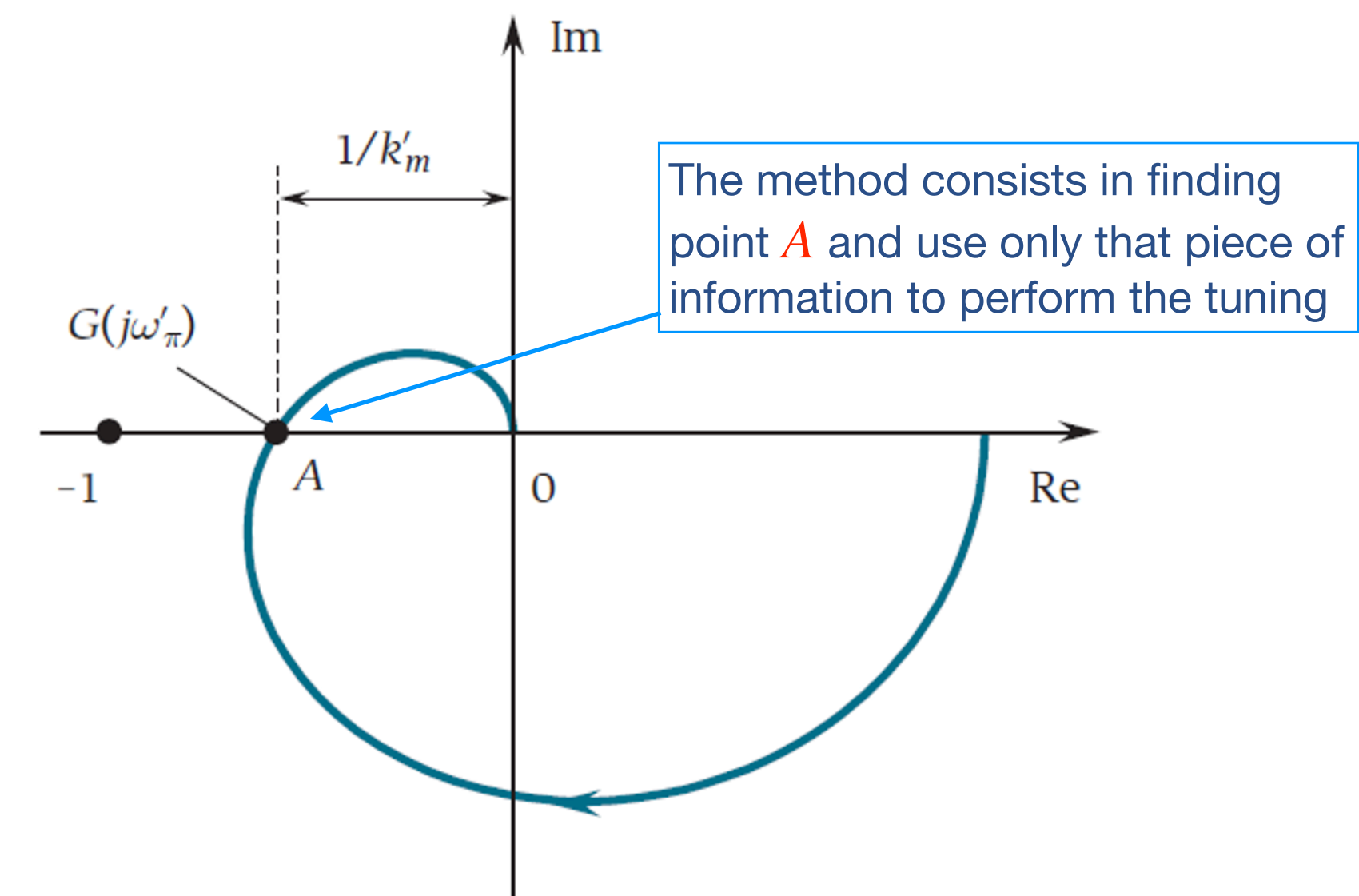
Practical Implementation of PID Controllers: **Tuning Rules**

	K_P	T_I	T_D
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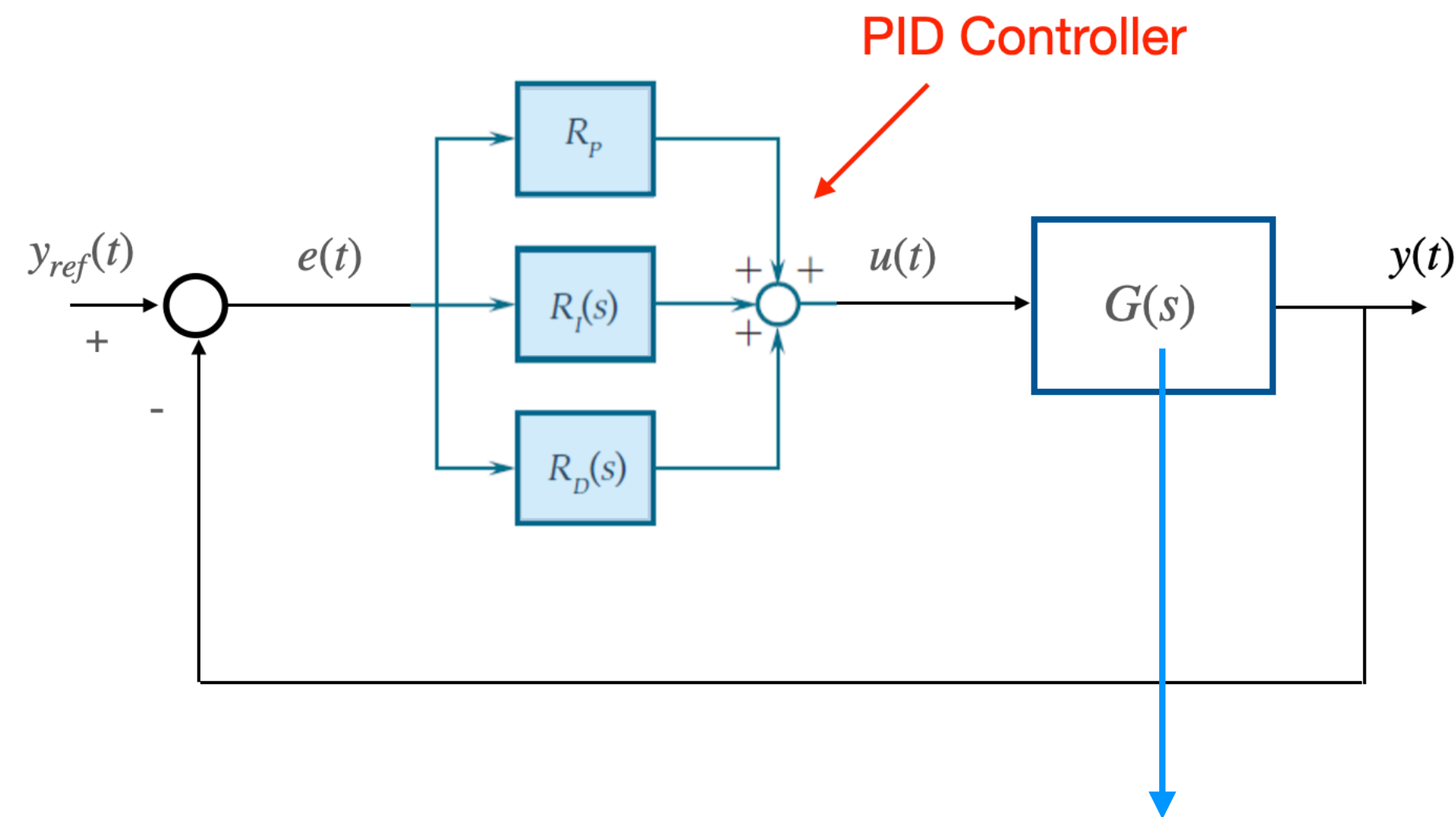
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Practical Implementation of PID Controllers: **Tuning Rules**

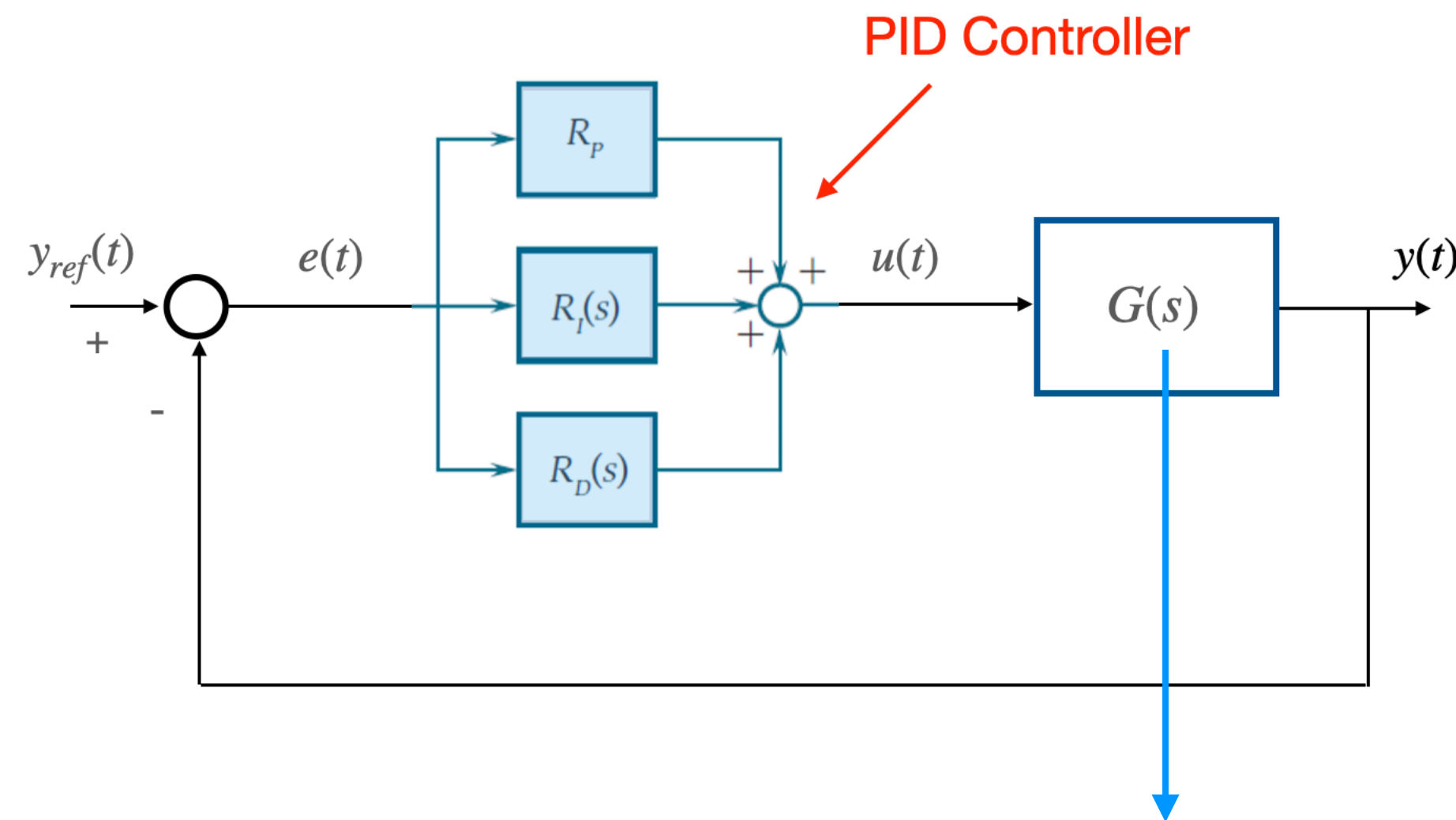


Open-loop Methods

Approximate Model of the Process:

$$G_a(s) = \frac{\mu e^{-\tau s}}{1 + Ts}$$

Practical Implementation of PID Controllers: **Tuning Rules**



Open-loop Methods

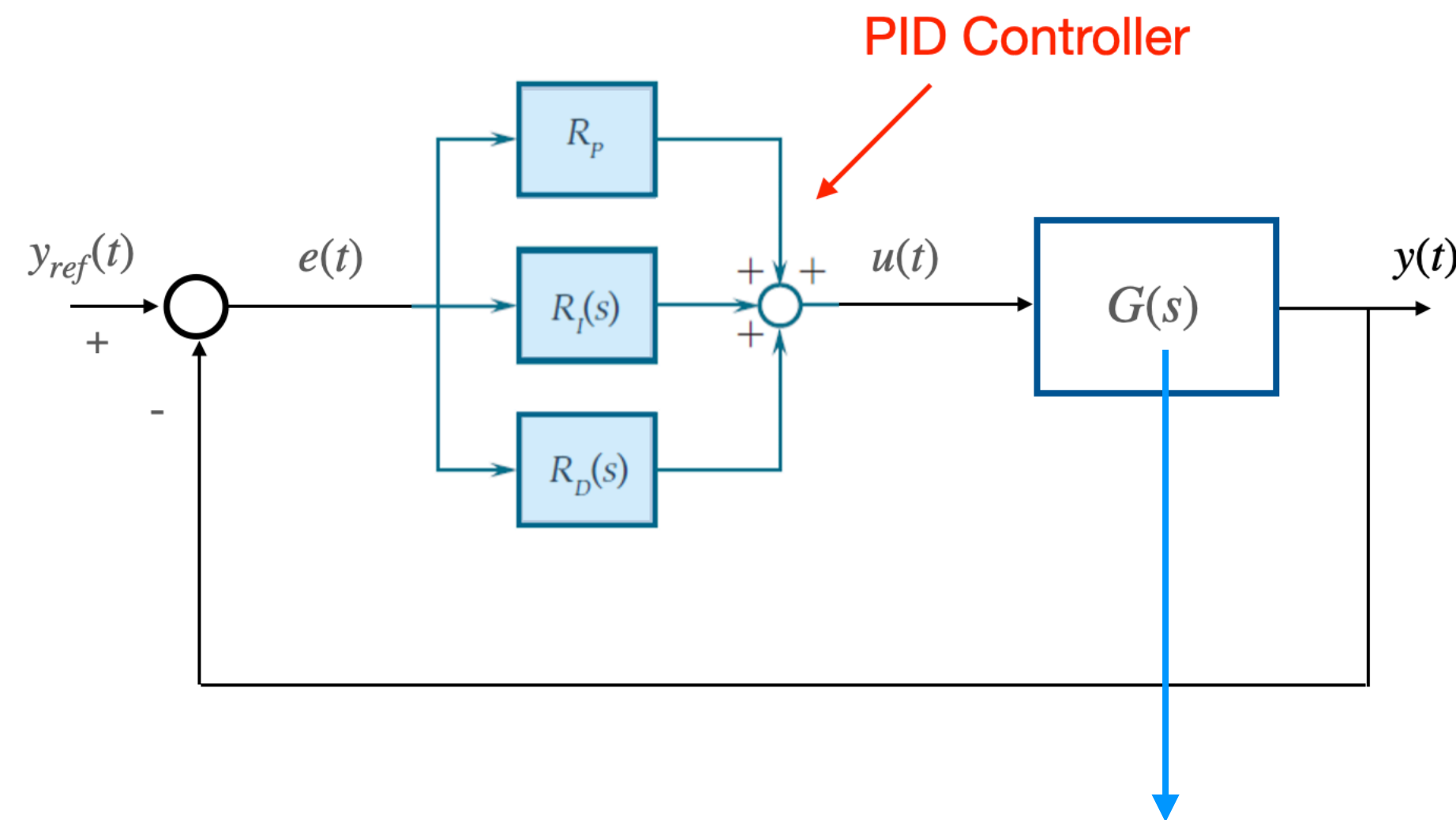
Approximate Model of the Process:

$$G_a(s) = \frac{\mu e^{-\tau s}}{1 + Ts}$$

equivalent delay

equivalent time-constant

Practical Implementation of PID Controllers: **Tuning Rules**



Approximate Model of the Process:

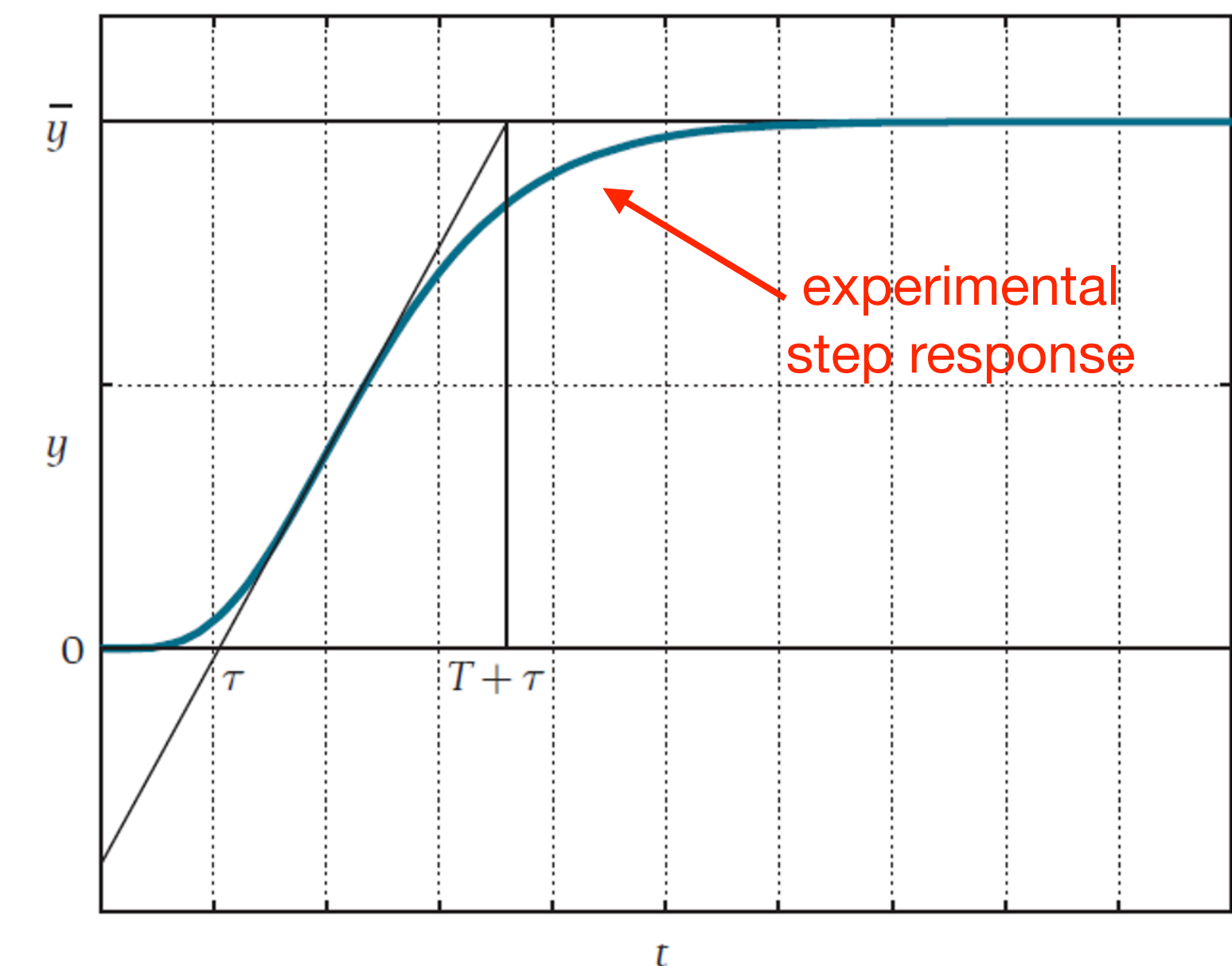
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equivalent delay

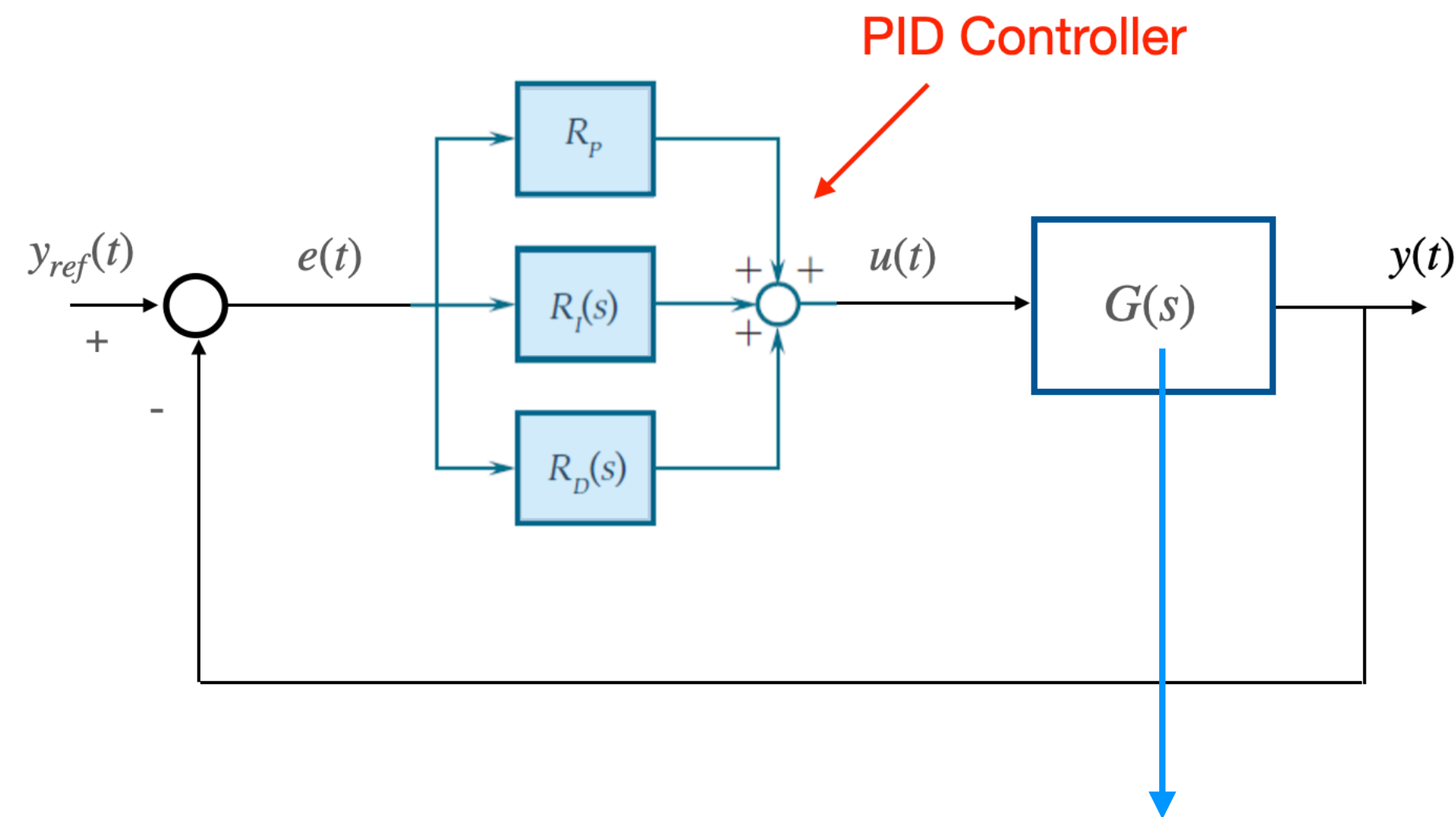
equivalent time-constant

Open-loop Methods

Parameters τ , T determination via the Tangent Method



Practical Implementation of PID Controllers: **Tuning Rules**



Approximate Model of the Process:

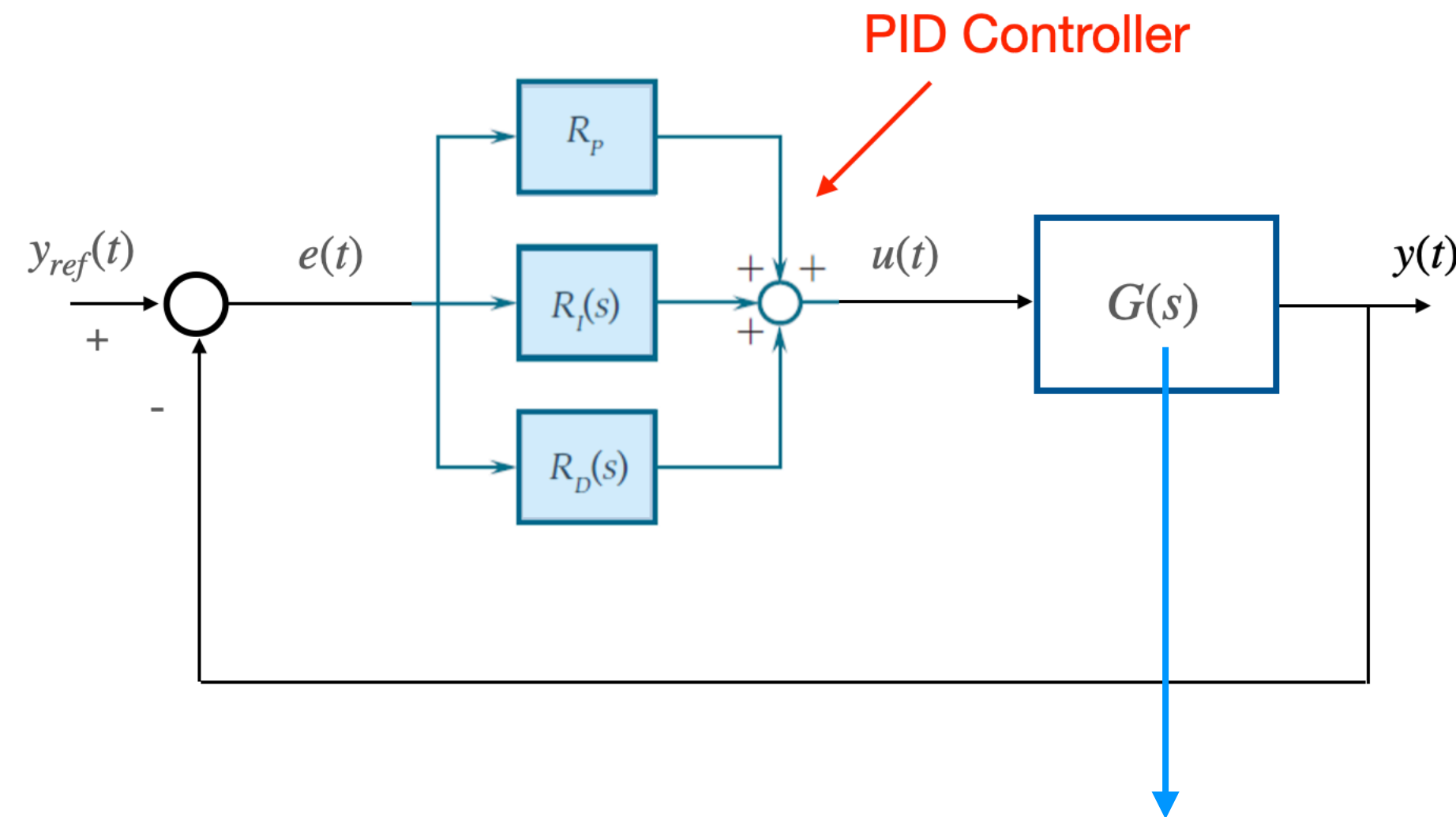
$$G_a(s) = \frac{\mu e^{-\tau s}}{1 + Ts}$$

Open-loop Ziegler-Nichols Method

	K_P	T_I	T_D
P	$\frac{T}{\mu\tau}$		
PI	$\frac{0.9T}{\mu\tau}$	3τ	
PID	$\frac{1.2T}{\mu\tau}$	2τ	0.5τ

$$R_{PID_{id}}(s) = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Practical Implementation of PID Controllers: **Tuning Rules**



Approximate Model of the Process:

$$G_a(s) = \frac{\mu e^{-\tau s}}{1 + Ts}$$

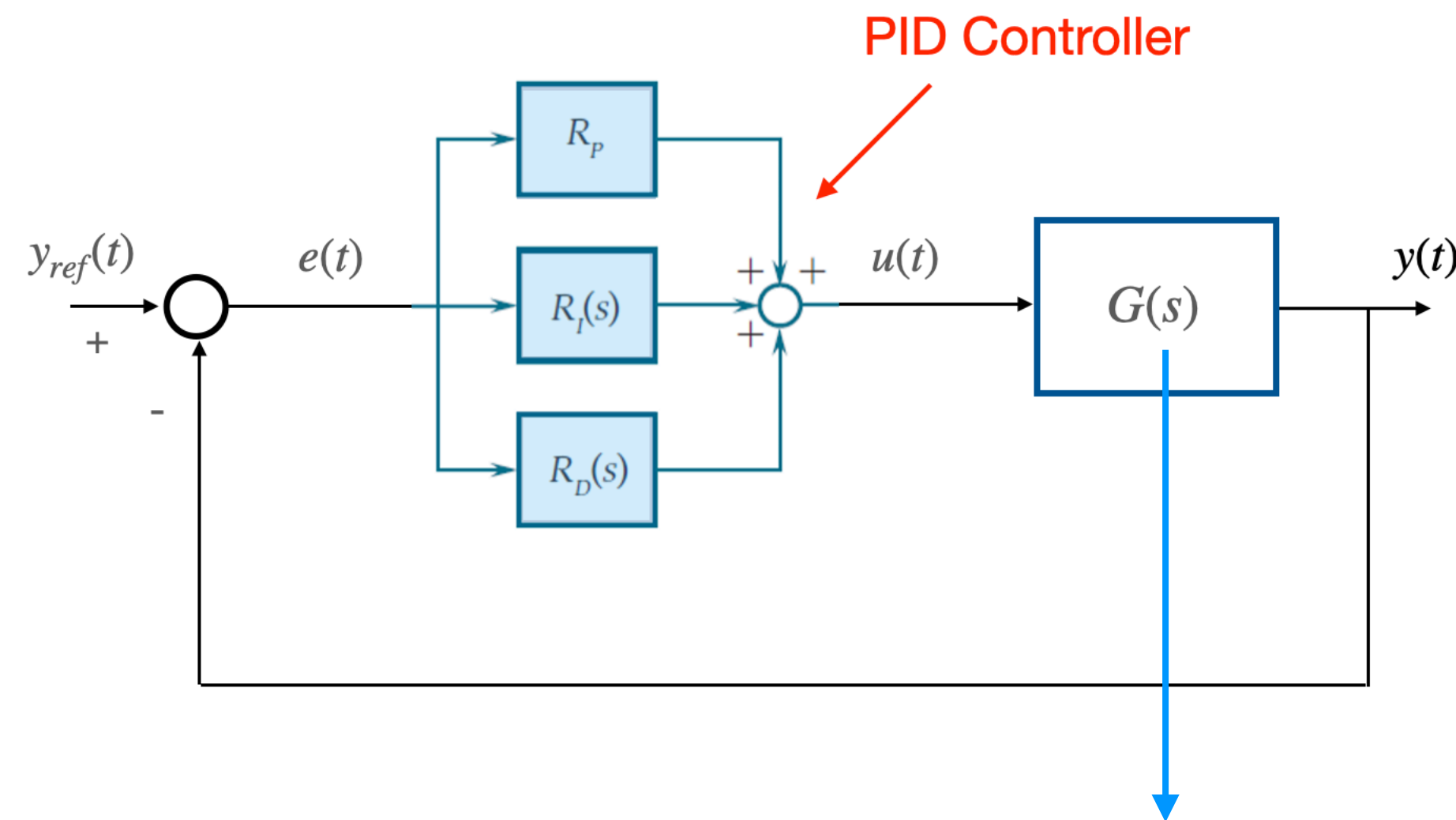
Open-loop Ziegler-Nichols Method

	K_P	T_I	T_D
P	$\frac{T}{\mu\tau}$		
PI	$\frac{0.9T}{\mu\tau}$	3τ	
PID	$\frac{1.2T}{\mu\tau}$	2τ	0.5τ

$$R_{PID_{id}}(s) = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Note that $T_I = 4T_D \rightarrow$ the PID zeros coincide in $s = -\frac{1}{2T_D} = -\frac{1}{\tau}$

Practical Implementation of PID Controllers: **Tuning Rules**



Open-loop Cohen-Coon Method

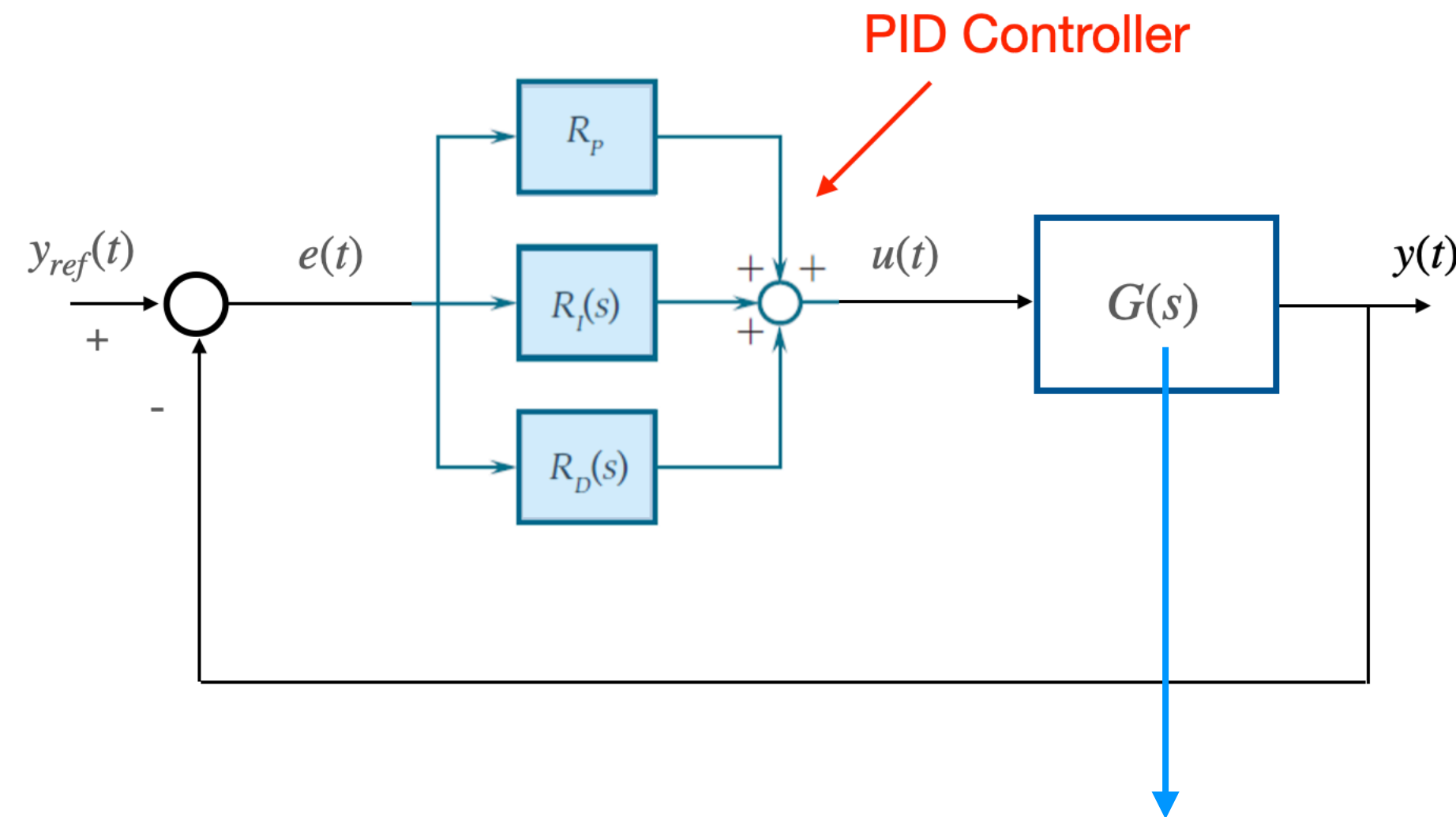
	K_P	T_I	T_D
P	$\frac{3T + \tau}{3\mu\tau}$		
PI	$\frac{10.8T + \tau}{12\mu\tau}$	$\tau \frac{30T + 3\tau}{9T + 20\tau}$	
PID	$\frac{16T + 3\tau}{12\mu\tau}$	$\tau \frac{32T + 6\tau}{12\tau}$	$\frac{4T\tau}{11T + 2\tau}$

Approximate Model of the Process:

$$G_a(s) = \frac{\mu e^{-\tau s}}{1 + Ts}$$

$$R_{PIDid}(s) = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Practical Implementation of PID Controllers: **Tuning Rules**



Approximate Model of the Process:

$$G_a(s) = \frac{\mu e^{-\tau s}}{1 + Ts}$$

Open-loop Internal Model Control (IMC) Method

	K_P	T_I	T_D
PI	$\frac{T}{\mu(\tau + T_f)}$	T	
PID	$\frac{T + 0.5\tau}{\mu(T_f + 0.5\tau)}$	$T + 0.5\tau$	$\frac{0.5\tau T}{T + 0.5\tau}$

$$R_{PIDid}(s) = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

additional parameter: if T_f increases, then $\omega_{BW_{CL}}$ decreases and the phase and gain margins increase

Practical Implementation of PID Controllers: **Tuning Rules**

Comparison:

	φ_m	k_m	ω_c
Cohen - Coon	31°	7.9	0.74
IMC ($T_f = 0.4$)	45°	10.8	0.6
IMC ($T_f = 0.8$)	53°	12.8	0.5
IMC ($T_f = 1.2$)	59°	14.4	0.42

For system: $G(s) = \frac{1}{(1+s)^3}$ approximated as: $G_a(s) = \frac{e^{-0.8s}}{1+3.7s}$

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Note that: The IMC method produces more moderate control systems w.r.t. those obtained via the Cohen-Coon Method

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For system: $G(s) = \frac{1}{(1 + s)^3}$ approximated as: $G_a(s) = \frac{e^{-0.8s}}{1 + 3.7s}$

Exercise:
 Determine the Control Sensitivity Functions for a PI tuned using Cohen-Coon or the IMC in the table for the considered $G(s)$

Open-loop Internal Model Control (IMC) Method

	K_P	T_I	T_D
PI	$\frac{T}{\mu(\tau + T_f)}$	T	
PID	$\frac{T + 0.5\tau}{\mu(T_f + 0.5\tau)}$	$T + 0.5\tau$	$\frac{0.5\tau T}{T + 0.5\tau}$

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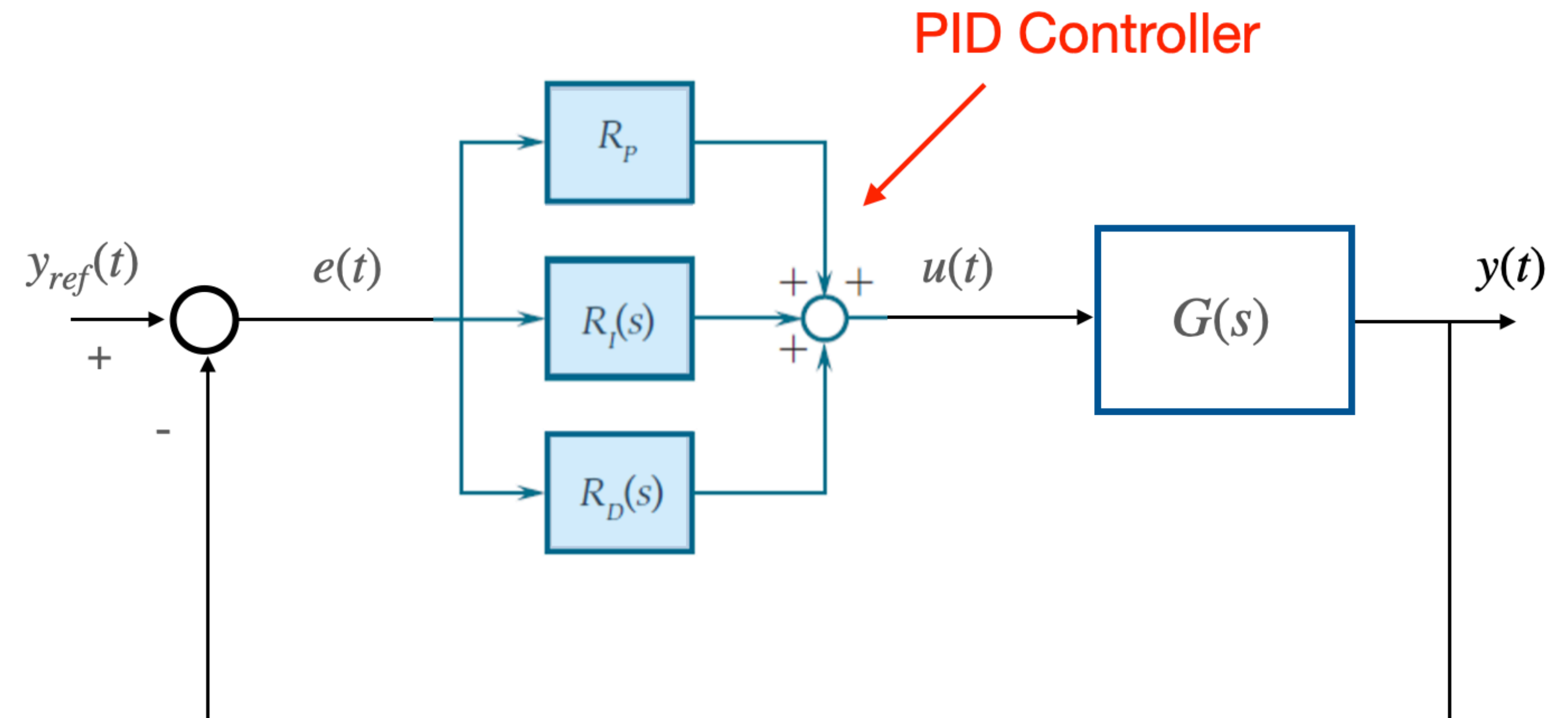
Practical Implementation of PID Controllers: **Example**

Process model:

$$G(s) = \frac{100}{(s + 10)(s + 30)(s + 5)}$$

```
/MATLAB Drive/PID_example1.m  
1 %% System definition  
2 s = tf('s');  
3 G = 100 / ((s + 10)*(s + 30)*(s+5));  
4
```

Control scheme:



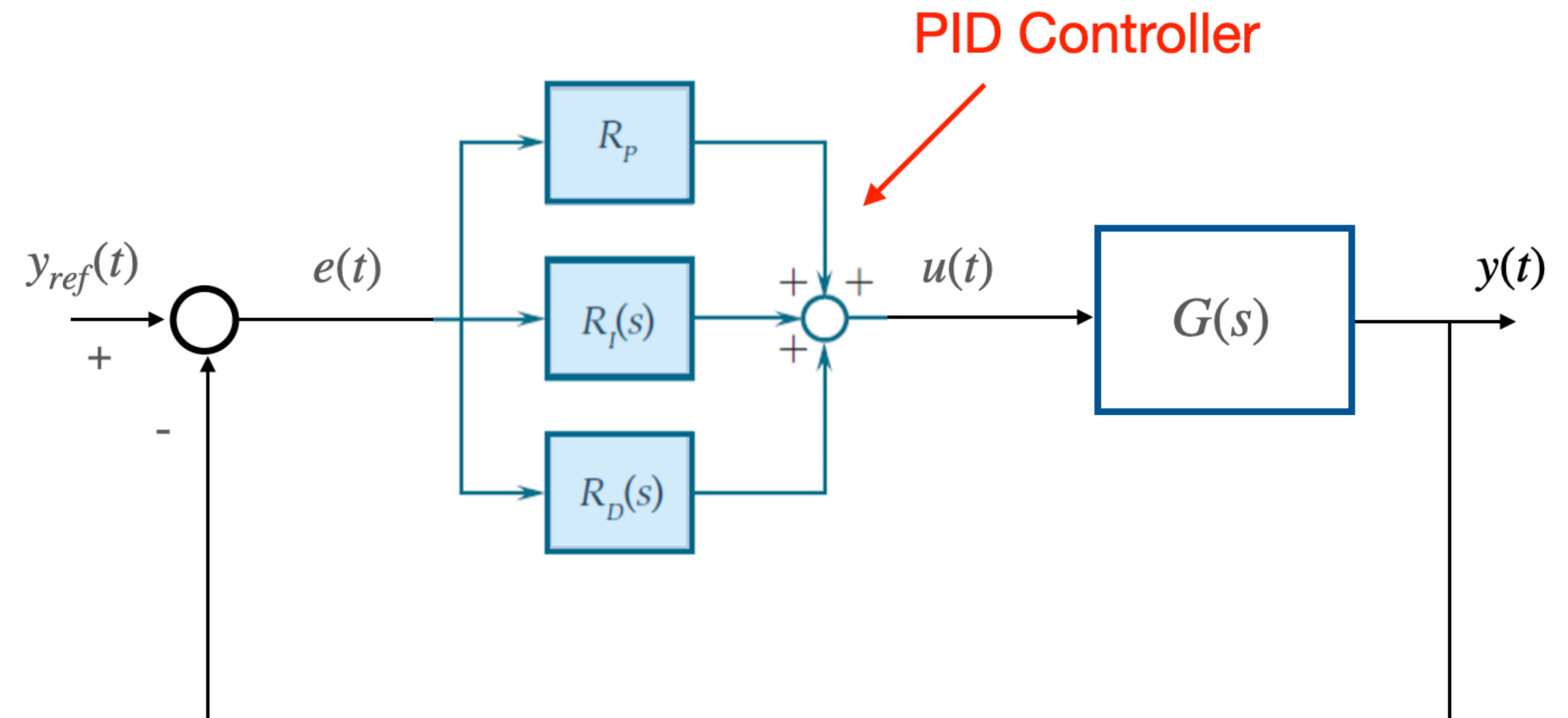
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Control scheme:



Objective: Compare the performance of the closed loop system with PID designed via CL Ziegler-Nichols rules and PID designed via OL Ziegler-Nichols rules

Practical Implementation of PID Controllers: **Example**

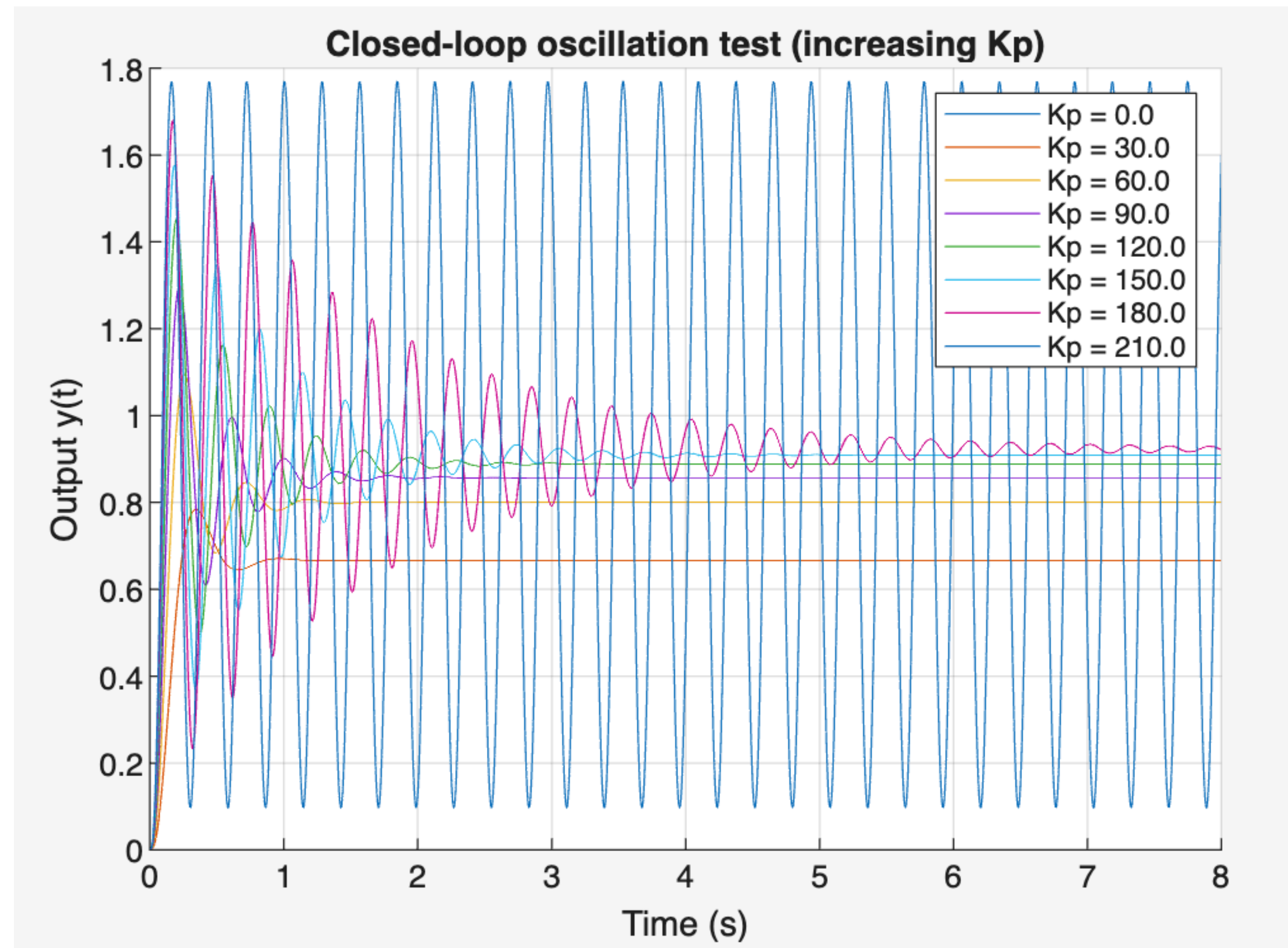
Process model:

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/MATLAB Drive/PID_example1.m

```
1 %% System definition
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4
```

Simulation for increasing K_P :



Practical Implementation of PID Controllers: Example

Process model:

$$G(s) = \frac{100}{(s + 10)(s + 30)(s + 5)}$$

/MATLAB Drive/PID_example1.m

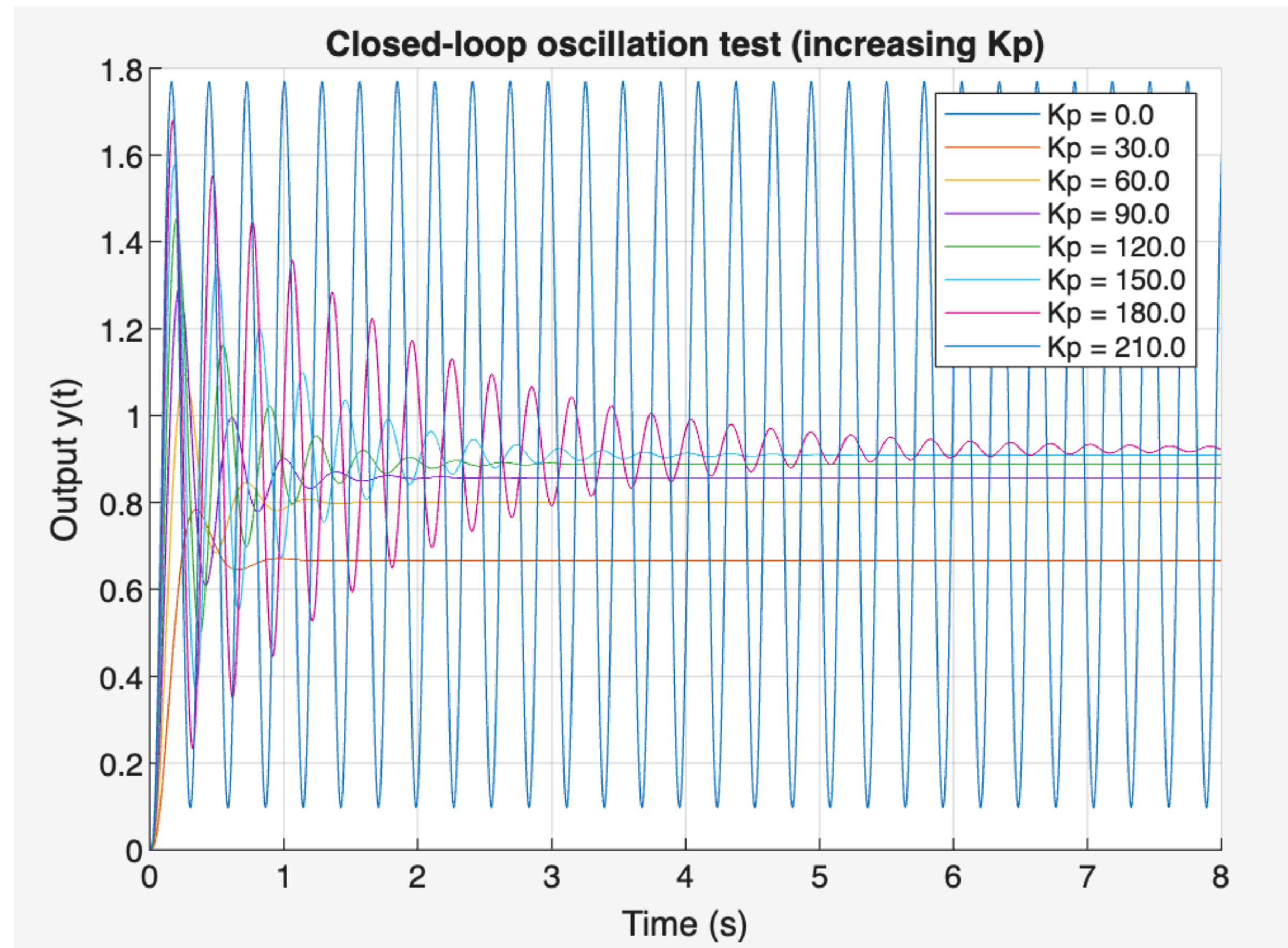
```
1 %% System definition
2 s = tf('s');
3 G = 100 / ((s + 10)*(s + 30)*(s+5));
4
```

PID calibration:

```
Estimated critical gain Kcr = 210.00
Estimated oscillation period Tcr = 0.281 s

Closed-loop ZN PID (with derivative filter, tau_f=1.000000e-02):
Kp = 126.000, Ti = 0.141, Td = 0.035
```

Simulation for increasing K_p :



Practical Implementation of PID Controllers: **Example**

Process model:

$$G(s) = \frac{100}{(s + 10)(s + 30)(s + 5)}$$

Approximate Model of the Process:

$$G_a(s) = \frac{\mu e^{-\tau s}}{1 + Ts}$$

$$\mu = 0.067 \quad \tau = 0.053 \quad T = 0.175$$

Practical Implementation of PID Controllers: Example

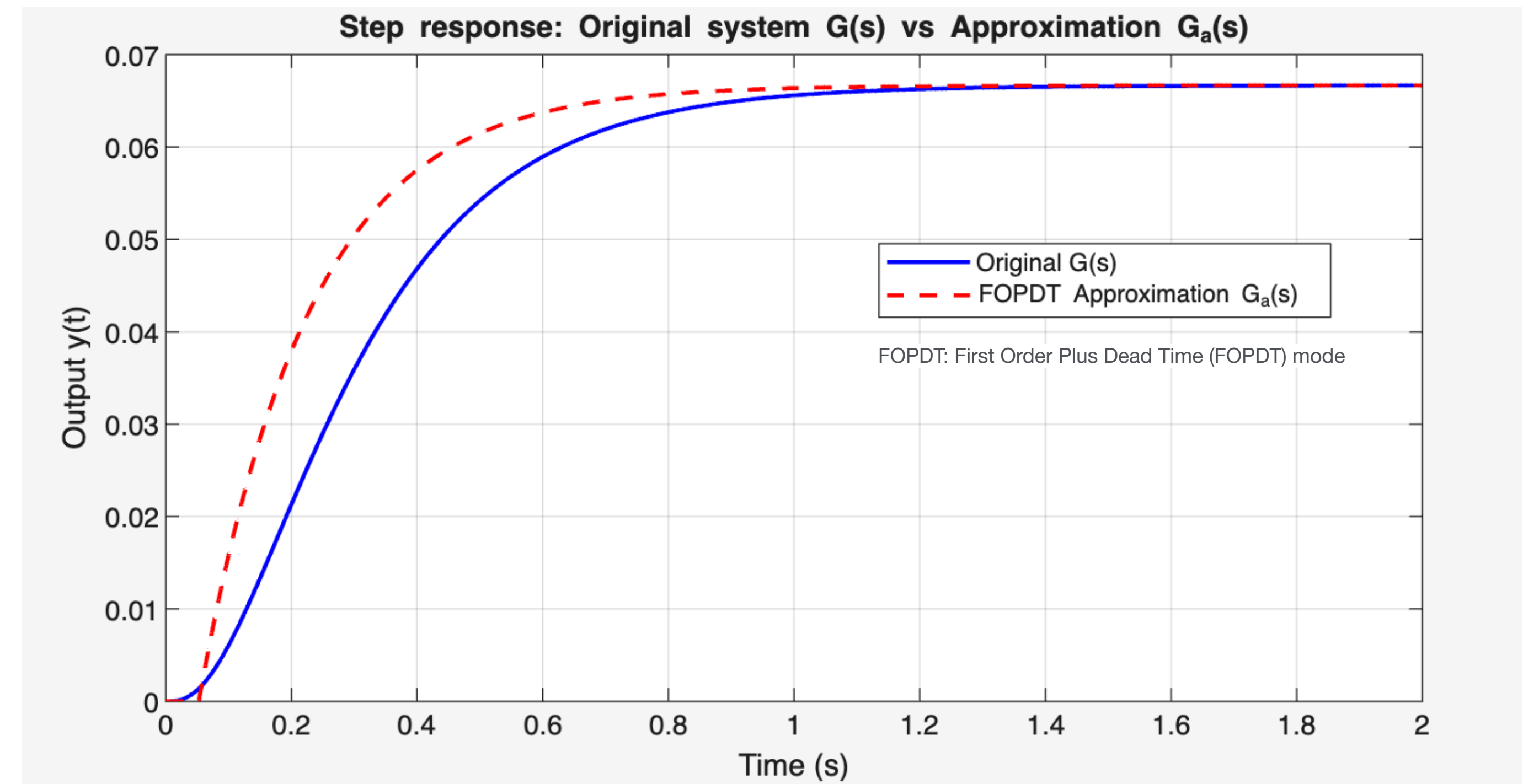
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Approximate Model of the Process:

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$$\mu = 0.067 \quad \tau = 0.053 \quad T = 0.175$$



```

76 %% === COMPARE STEP RESPONSE: REAL G(s) vs APPROXIMATED G_a(s) ===
77 t_sim = 0:0.001:2;
78 [yG, tG] = step(G, t_sim);
79 [yGa, tGa] = step(G_a, t_sim);
80
81 figure;
82 plot(tG, yG, 'b', 'LineWidth', 1.5); hold on;
83 plot(tGa, yGa, 'r--', 'LineWidth', 1.5);
84 xlabel('Time (s)');
85 ylabel('Output y(t)');
86 title('Step response: Original system G(s) vs Approximation G_a(s)');
87 legend('Original G(s)', 'FOPDT Approximation G_a(s)');
88 grid on;
89

```

Practical Implementation of PID Controllers: **Example**

Process model:

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Approximate Model of the Process:

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$$\mu = 0.067 \quad \tau = 0.053 \quad T = 0.175$$

Open-loop Ziegler-Nichols Method

	K_P	T_I	T_D
P	$\frac{T}{\mu\tau}$		
PI	$\frac{0.9T}{\mu\tau}$	3τ	
PID	$\frac{1.2T}{\mu\tau}$	2τ	0.5τ

Open-loop ZN PID:

$$K_p = 60.000, T_i = 0.105, T_d = 0.026$$

Practical Implementation of PID Controllers: Example

Example:

/MATLAB Drive/PID_example1.m

```
1 %% System definition
2 s = tf('s');
3 G = 100 / ((s + 10)*(s + 30)*(s+5));
4
```

Estimated critical gain $K_{cr} = 210.00$
Estimated oscillation period $T_{cr} = 0.281$ s

Closed-loop ZN PID (with derivative filter, $\tau_f=1.000000e-02$):
 $K_p = 126.000$, $T_i = 0.141$, $T_d = 0.035$

Approximate FOPDT model parameters:
 $\mu = 0.067$, $\tau = 0.053$, $T = 0.175$

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 $K_p = 60.000$, $T_i = 0.105$, $T_d = 0.026$



Practical Implementation of PID Controllers: Example

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/MATLAB Drive/PID_example1.m

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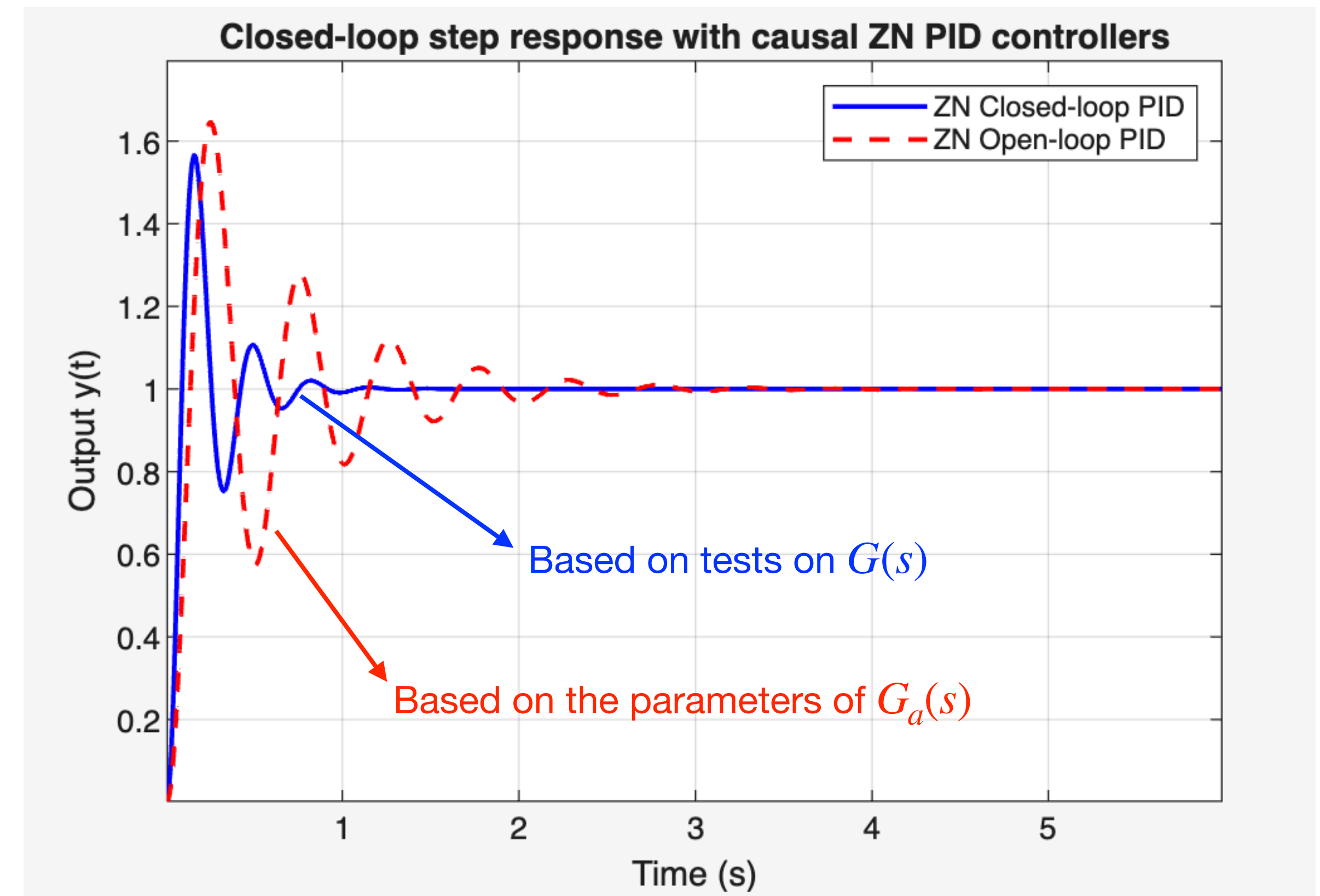
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Practical Implementation of PID Controllers: Example

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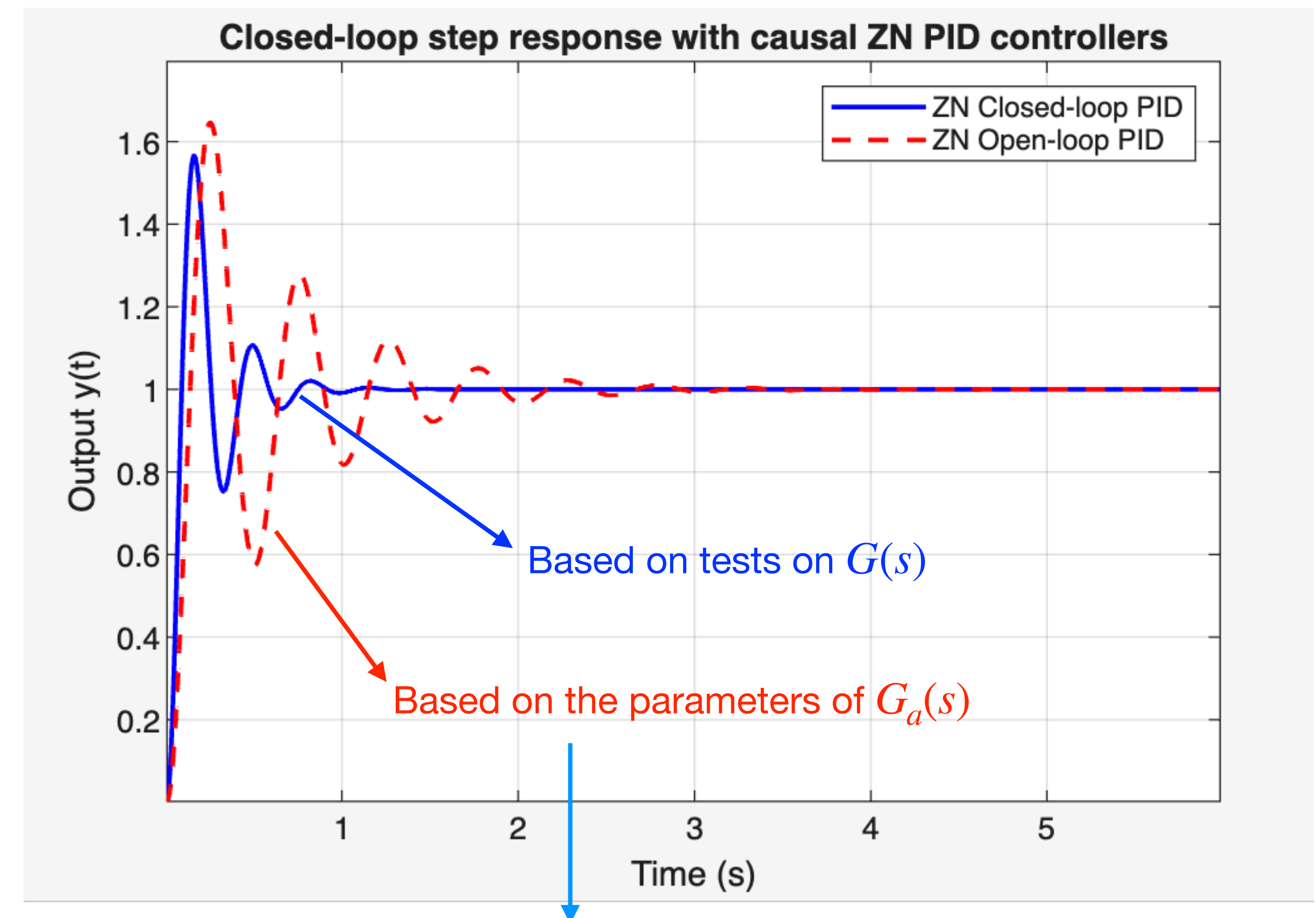
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in the example we are approximating a 3rd order system with a first order system with delay!

Practical Implementation of PID Controllers: **Example**

Process model:

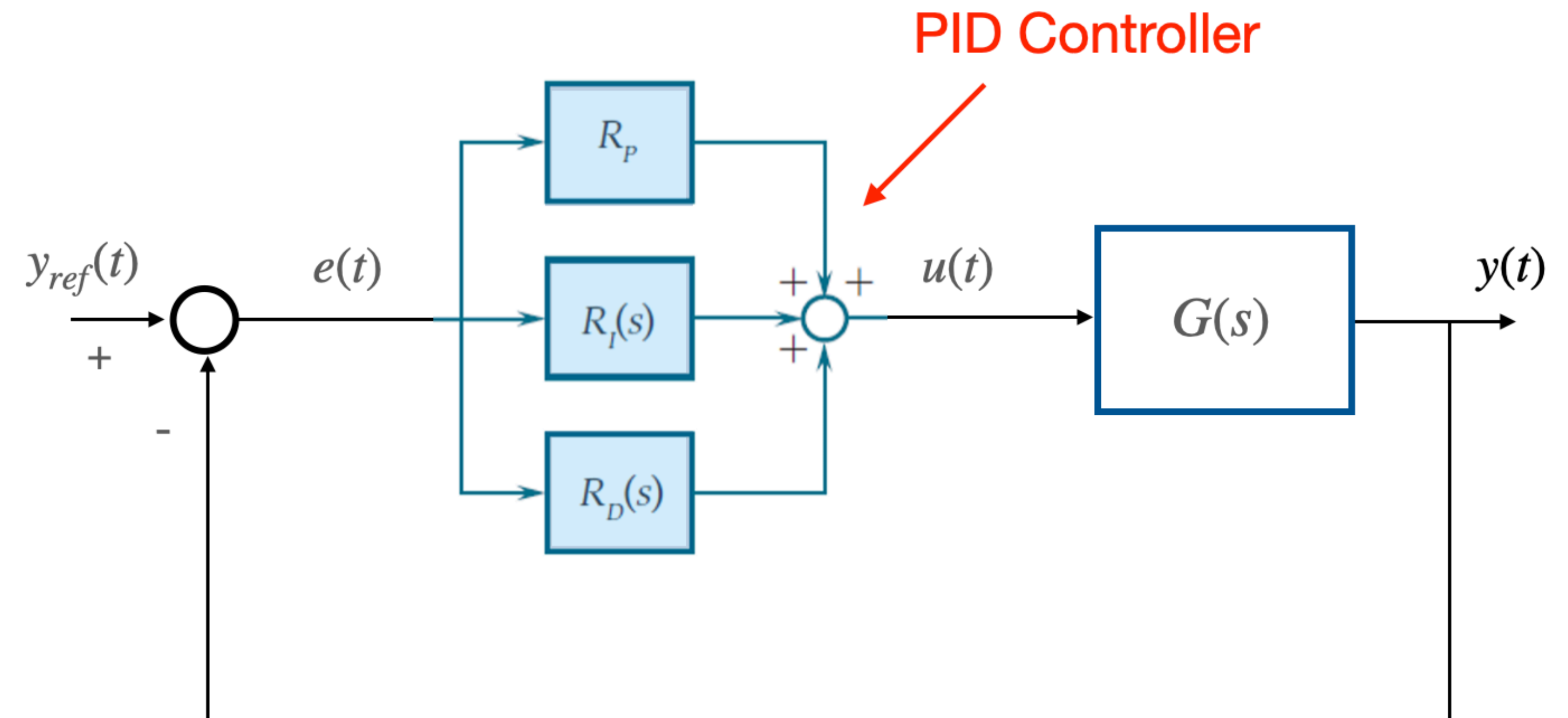
$$G(s) = \frac{100}{(s + 10)(s + 30)(s + 5)}$$

Approximate Model of the Process:

$$G_a(s) = \frac{\mu e^{-\tau s}}{1 + Ts}$$

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Control scheme:



Practical Implementation of PID Controllers: Example

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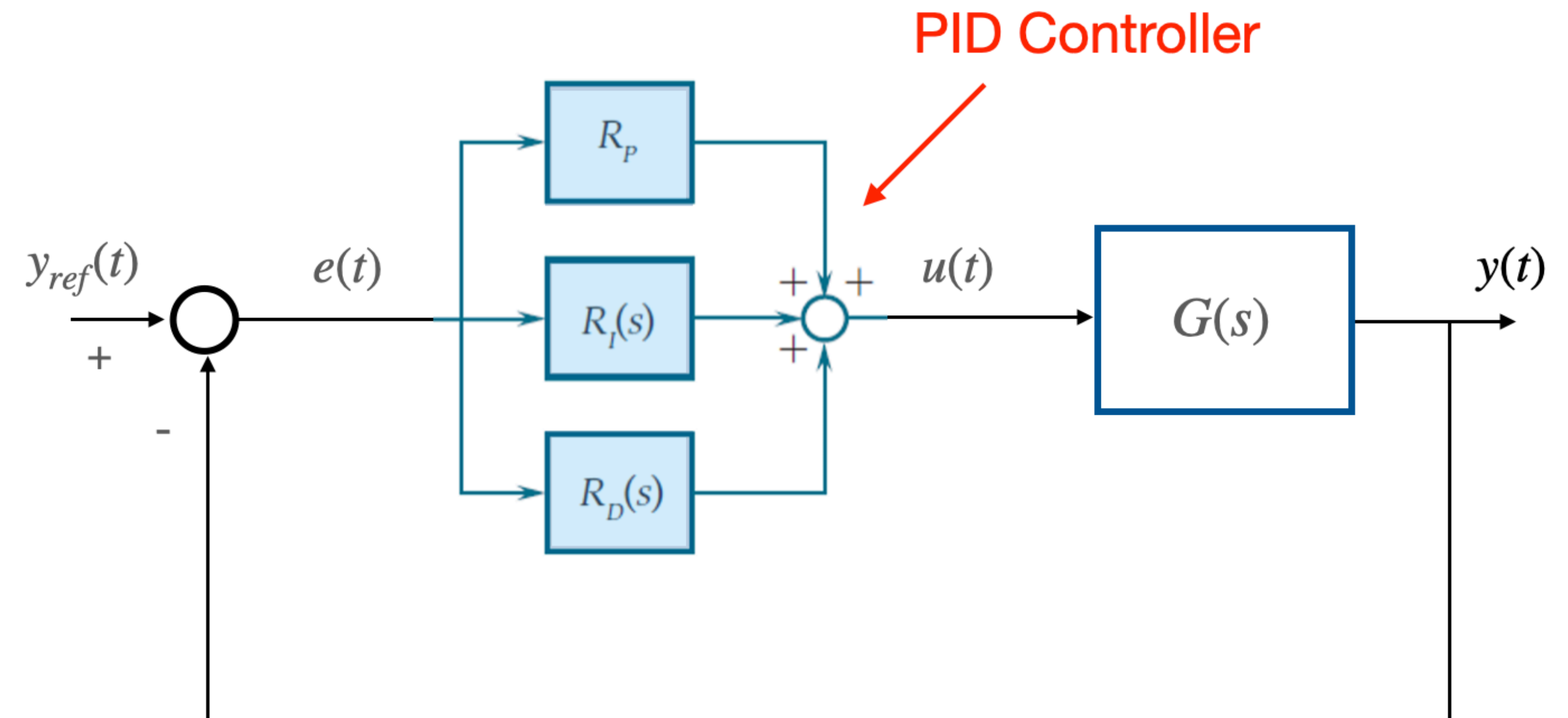
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Objective: Compare the performance of the closed loop system with PID designed via Cohen-Coon rules and PID designed via the IMC method

Practical Implementation of PID Controllers: Example

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Cohen-Coon PID (with derivative filter, tau_f=0.01):

Kp = 23.4620, Ki = 169.5603, Kd = 0.4321

IMC PID (T_f = 0.4, tau_f=0.01):

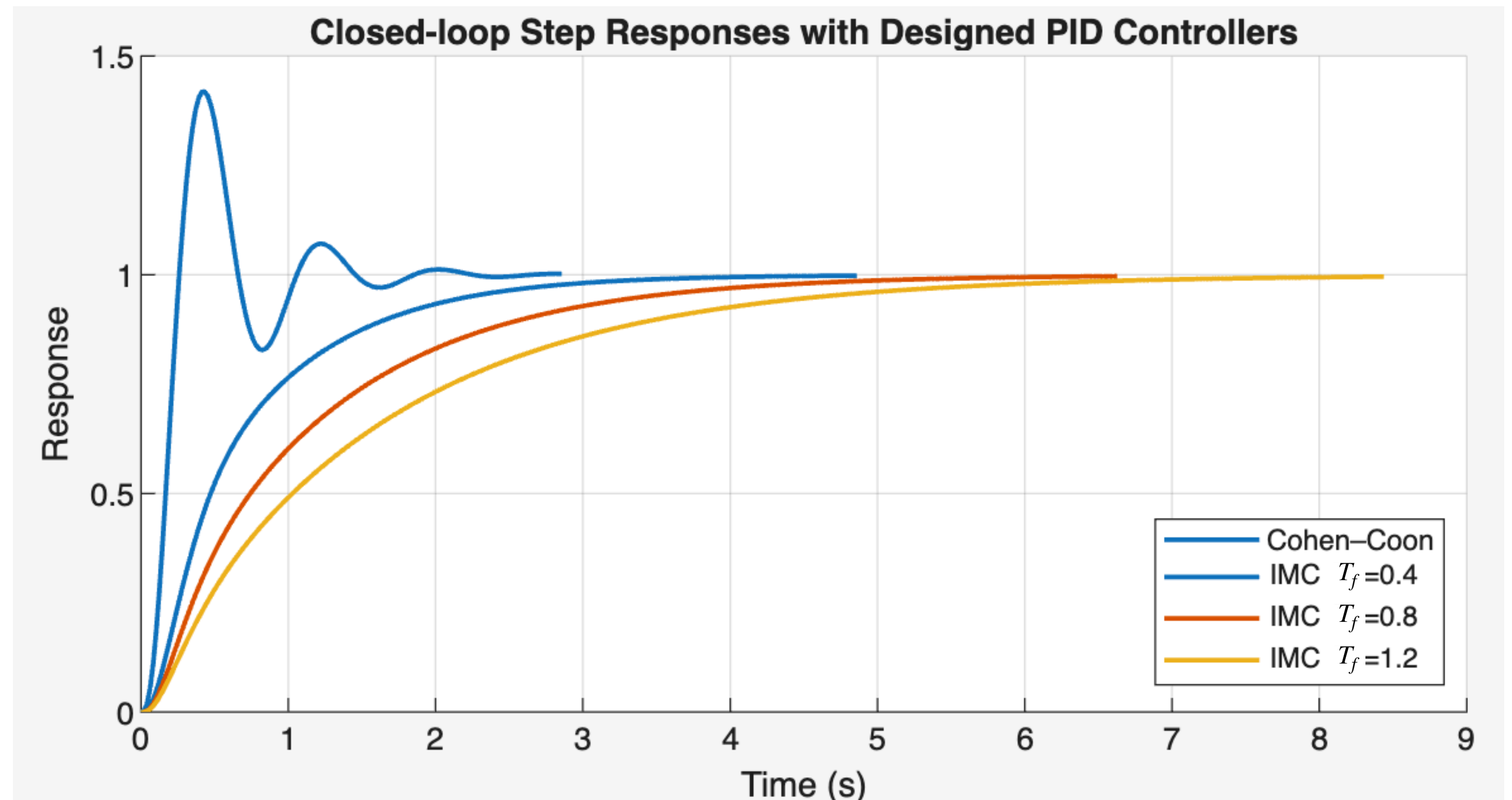
Kp = 6.8182, Ki = 20.4545, Kd = 0.6136

IMC PID (T_f = 0.8, tau_f=0.01):

Kp = 4.4118, Ki = 13.2353, Kd = 0.3971

IMC PID (T_f = 1.2, tau_f=0.01):

Kp = 3.2609, Ki = 9.7826, Kd = 0.2935



Practical Implementation of PID Controllers: Example

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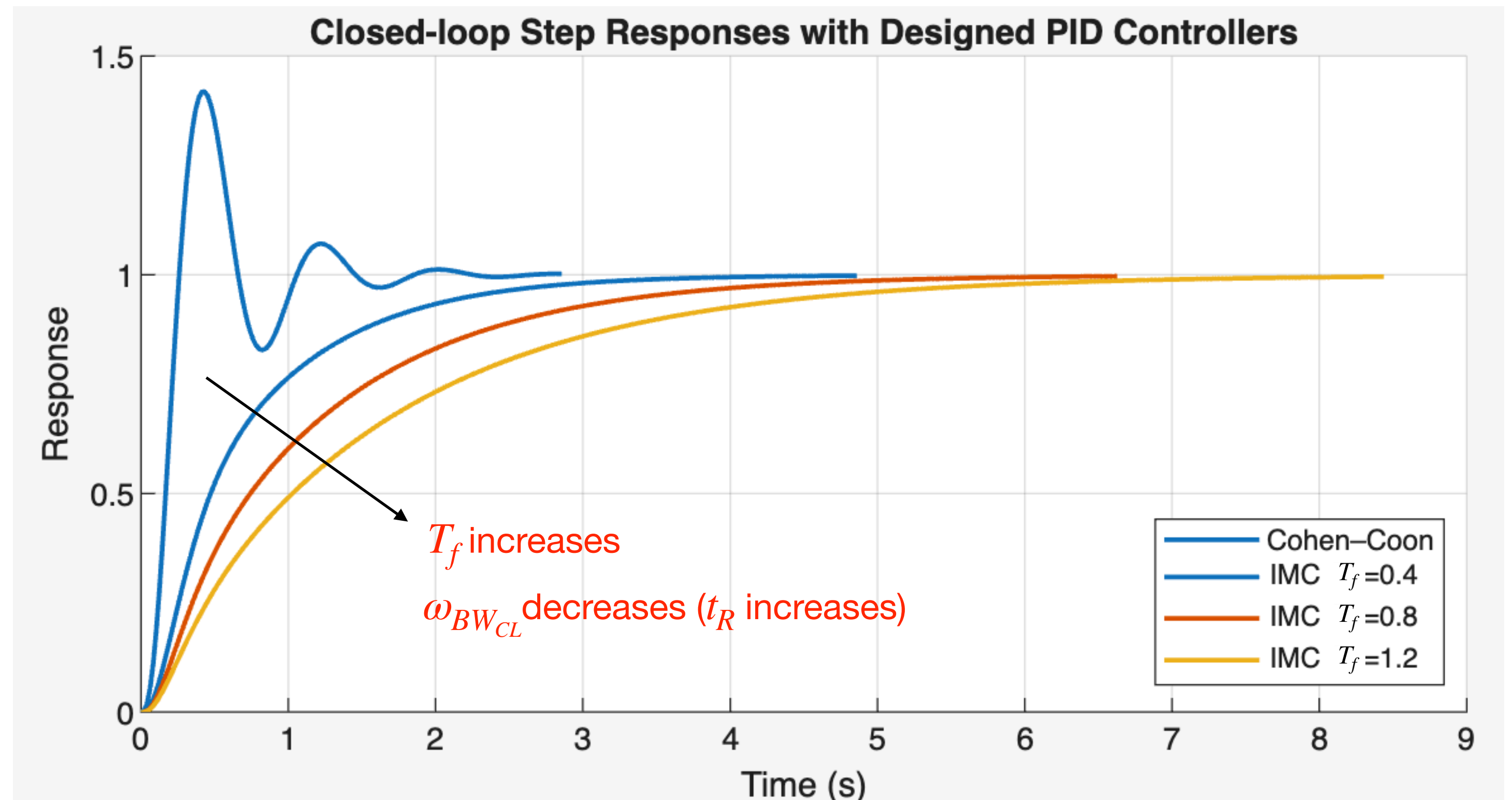
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additional parameter: if T_f increases, then $\omega_{BW_{CL}}$ decreases and the phase and gain margins increase

Practical Implementation of PID Controllers: **Example**

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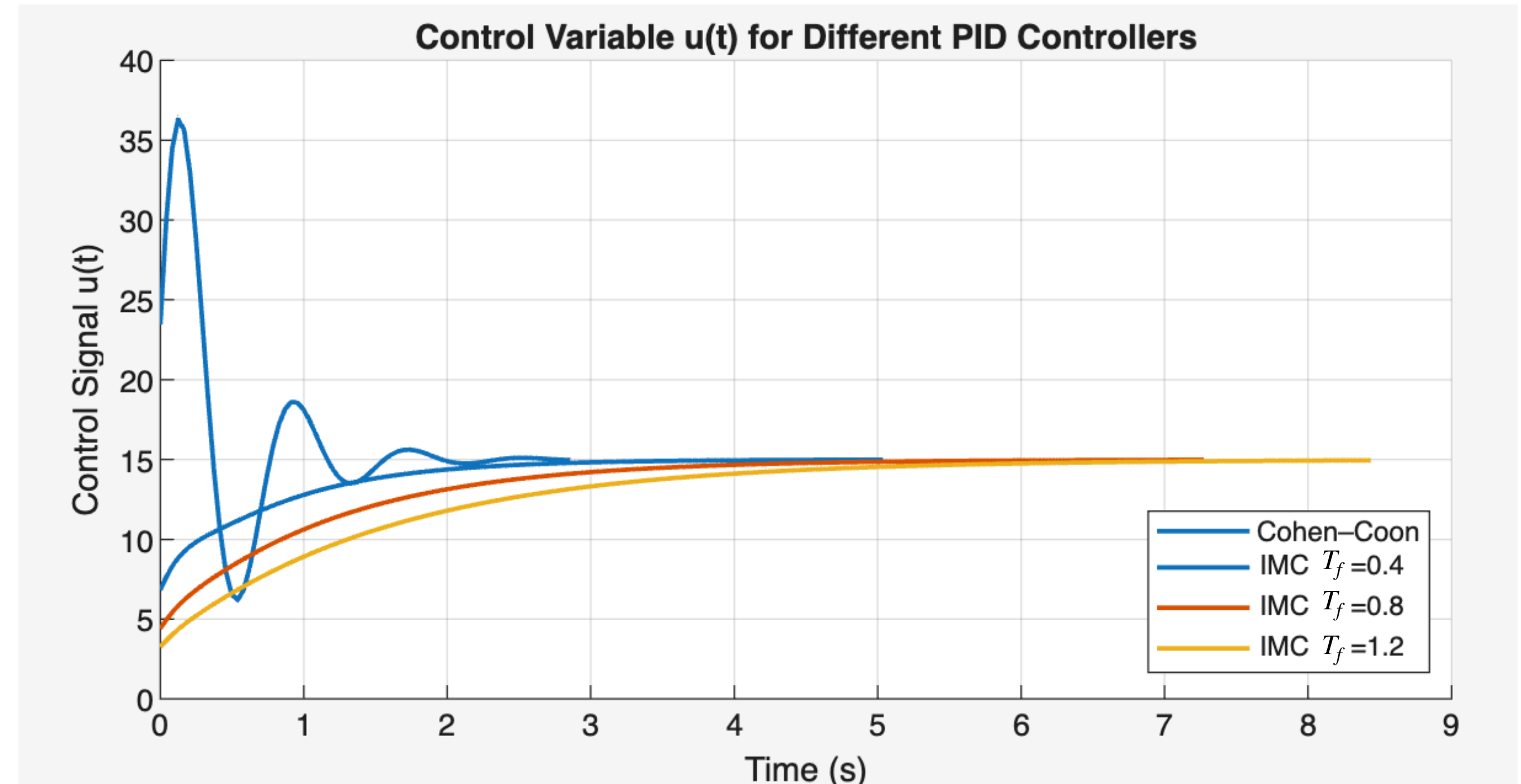
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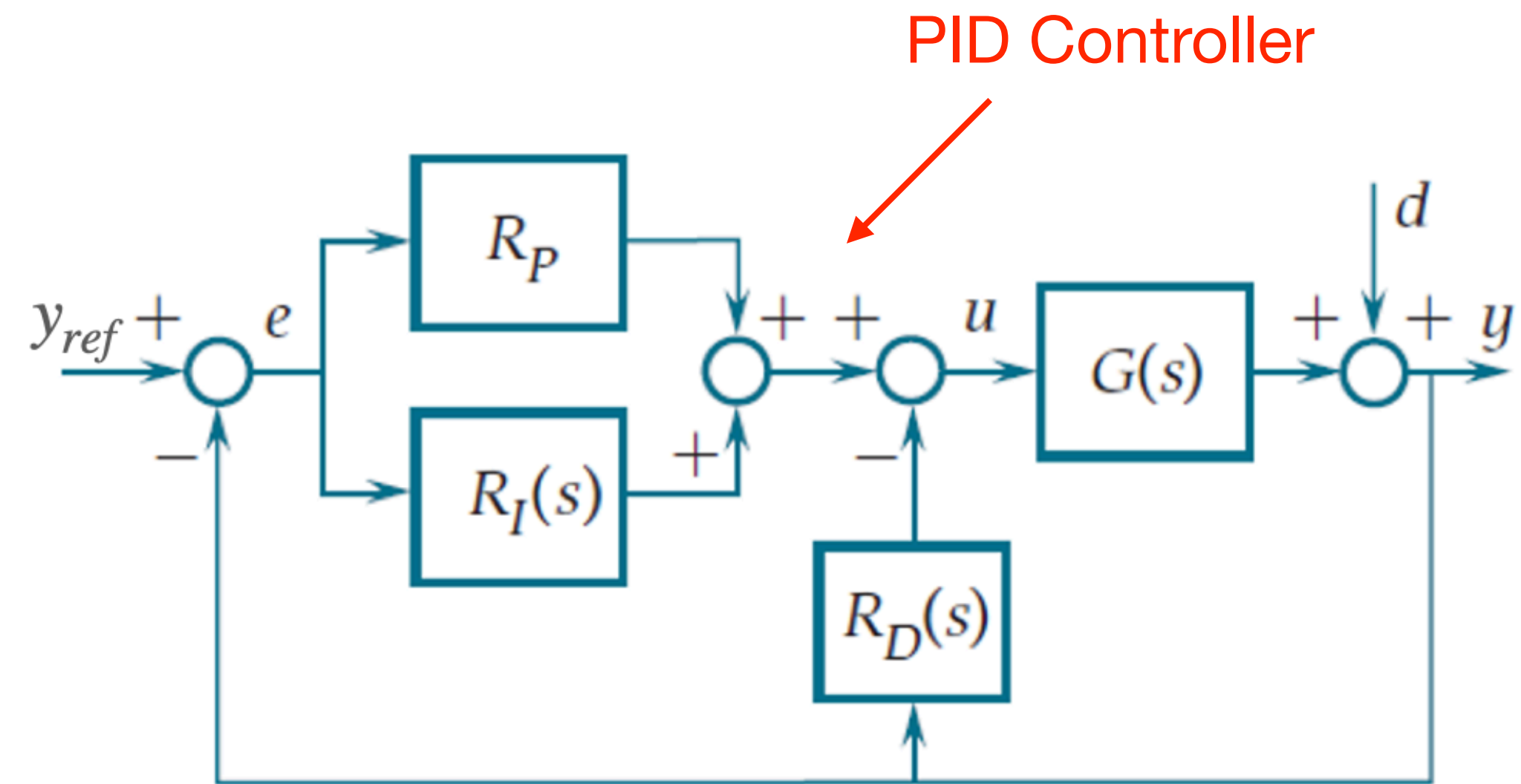
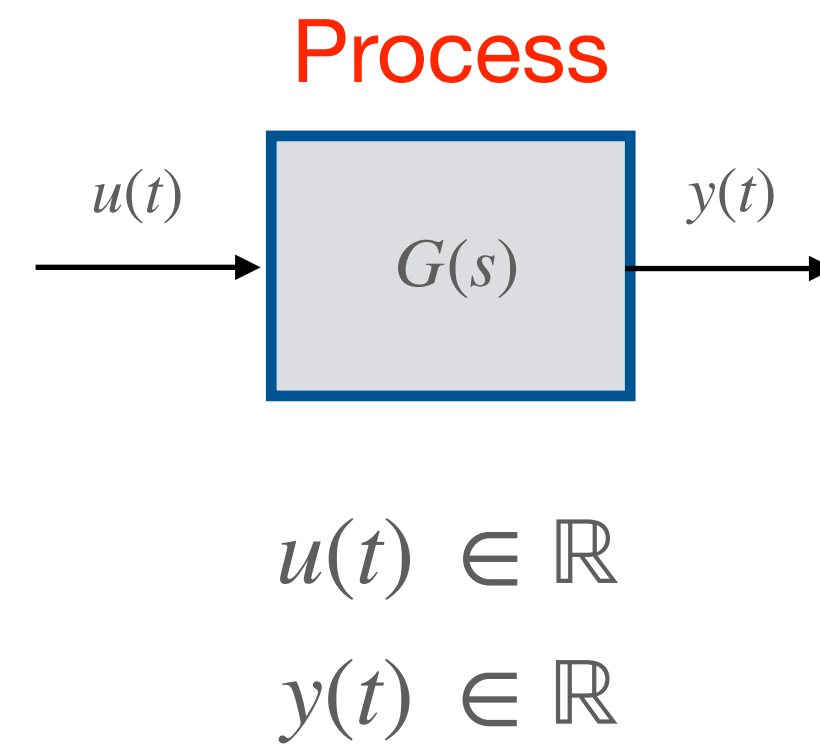
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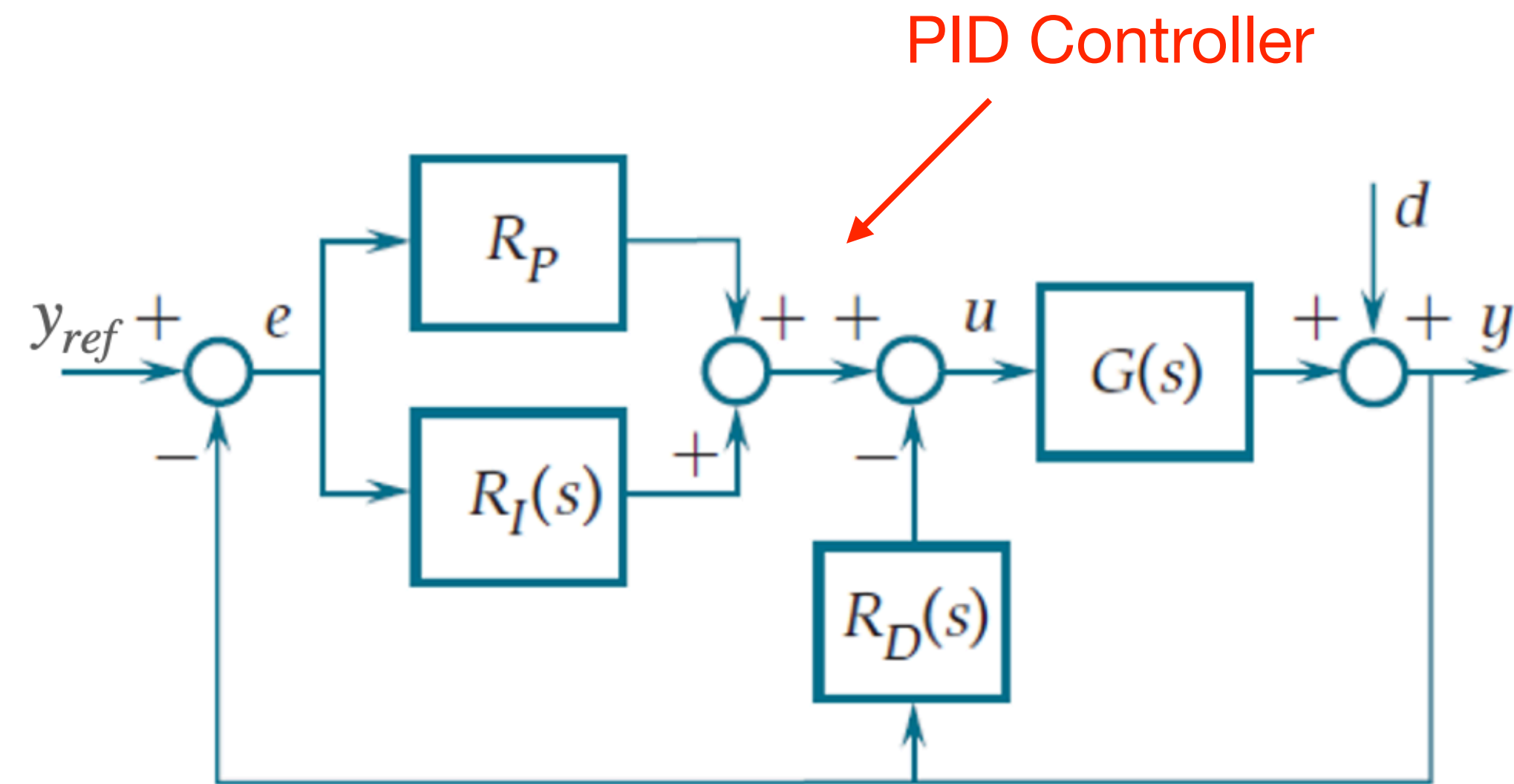
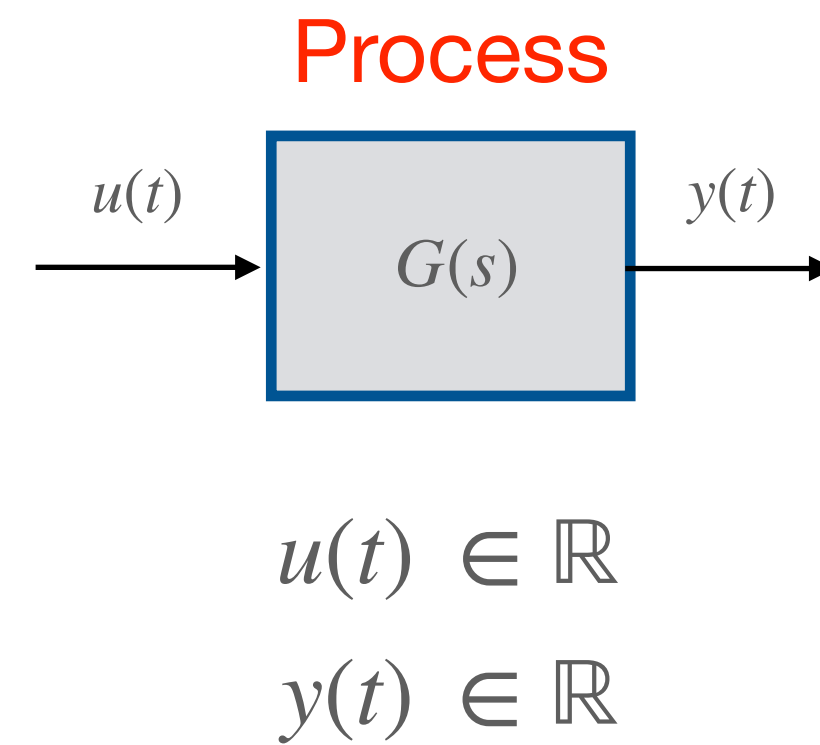


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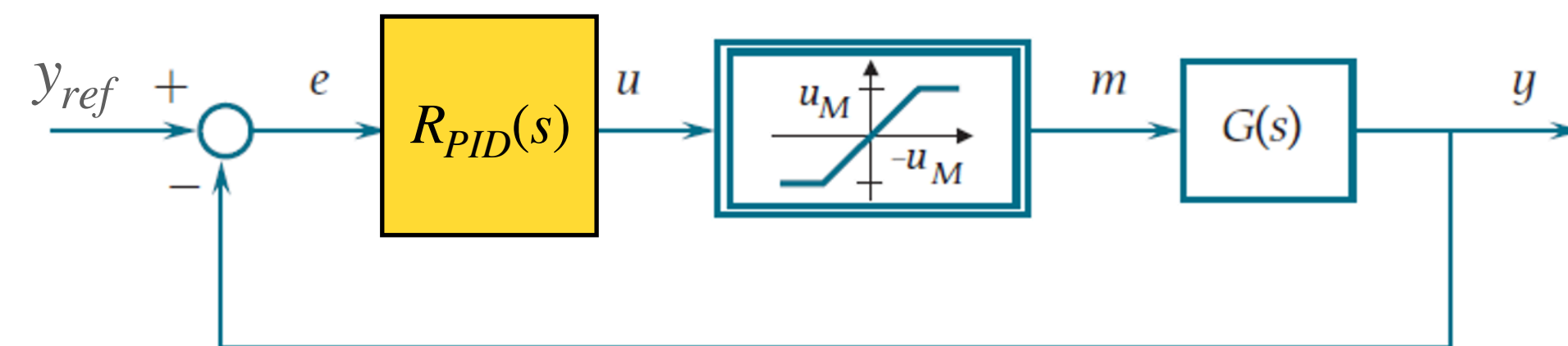
PID Controllers: Wind-up Effect



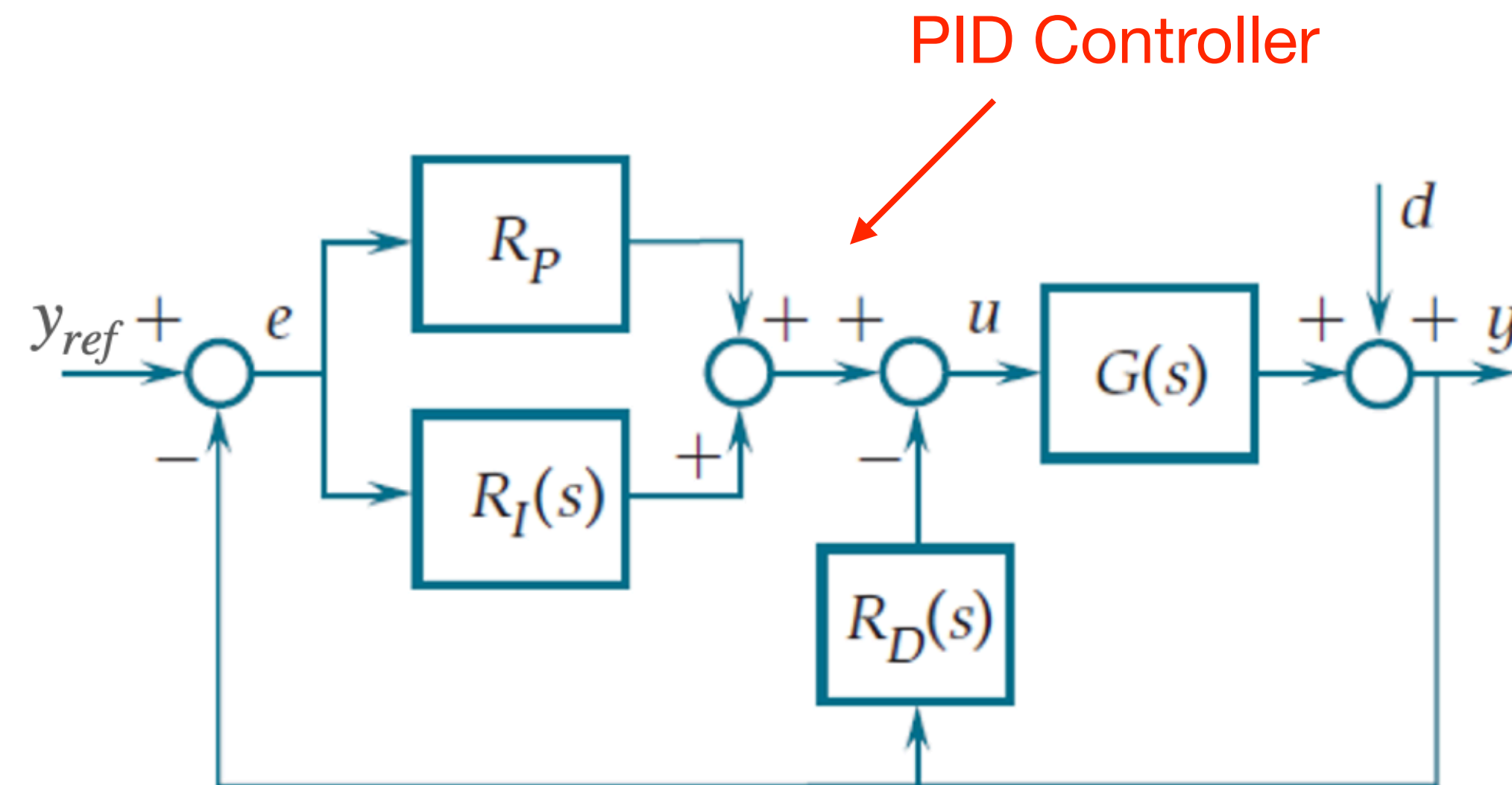
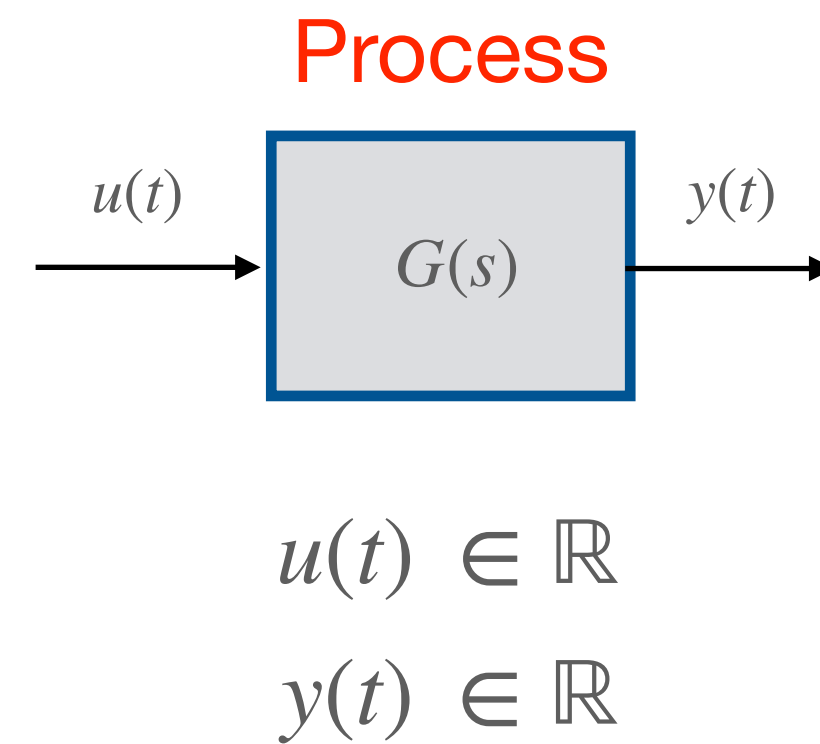
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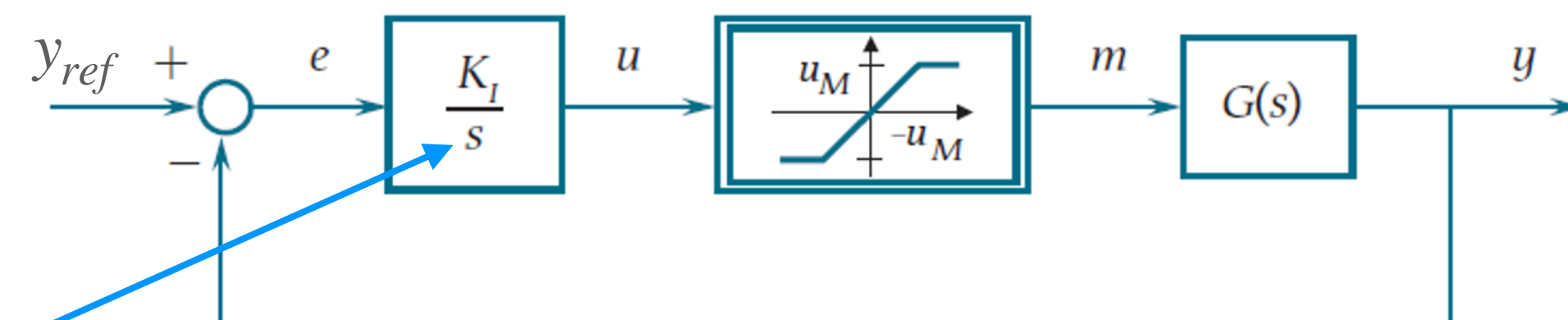
Realistic situation



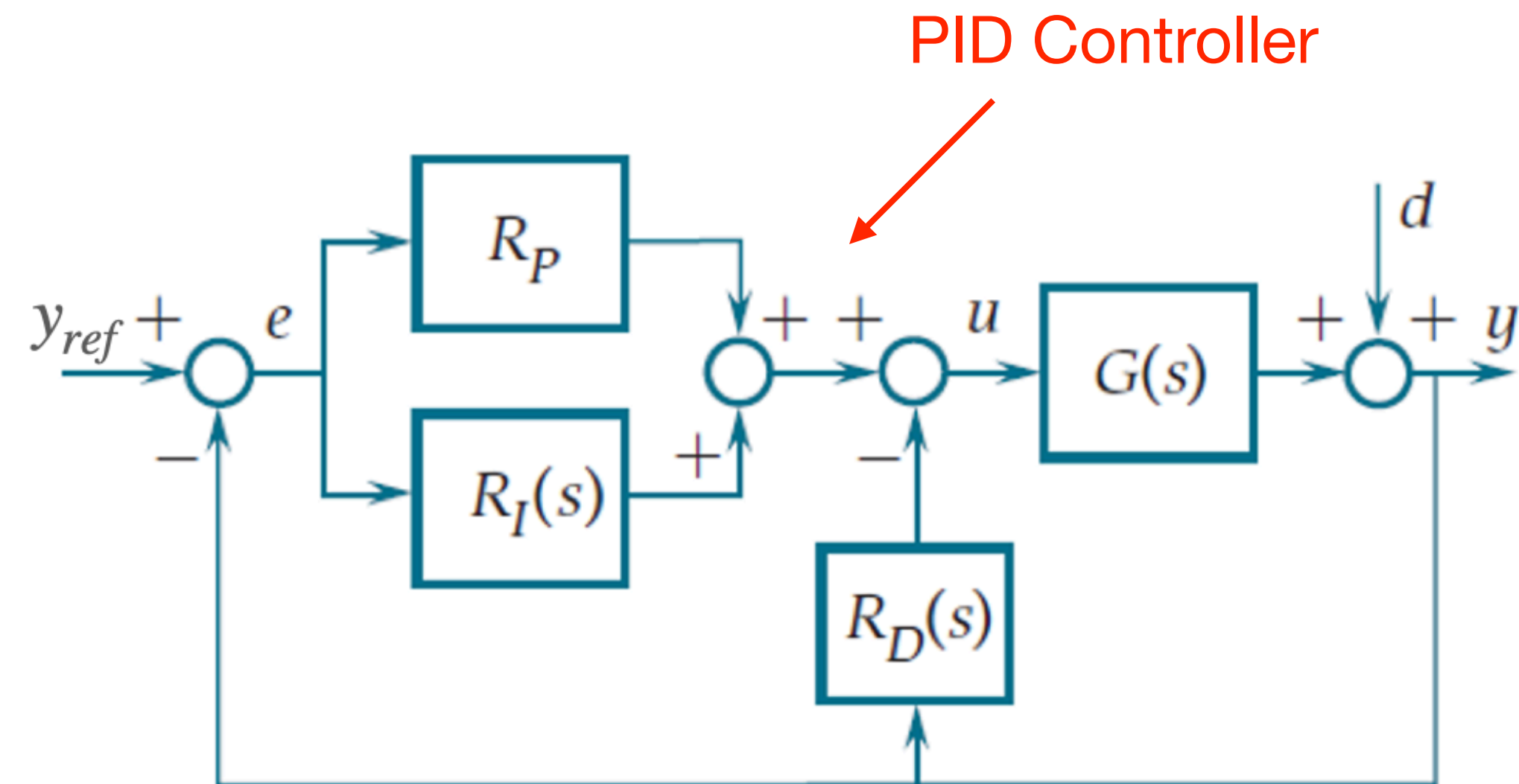
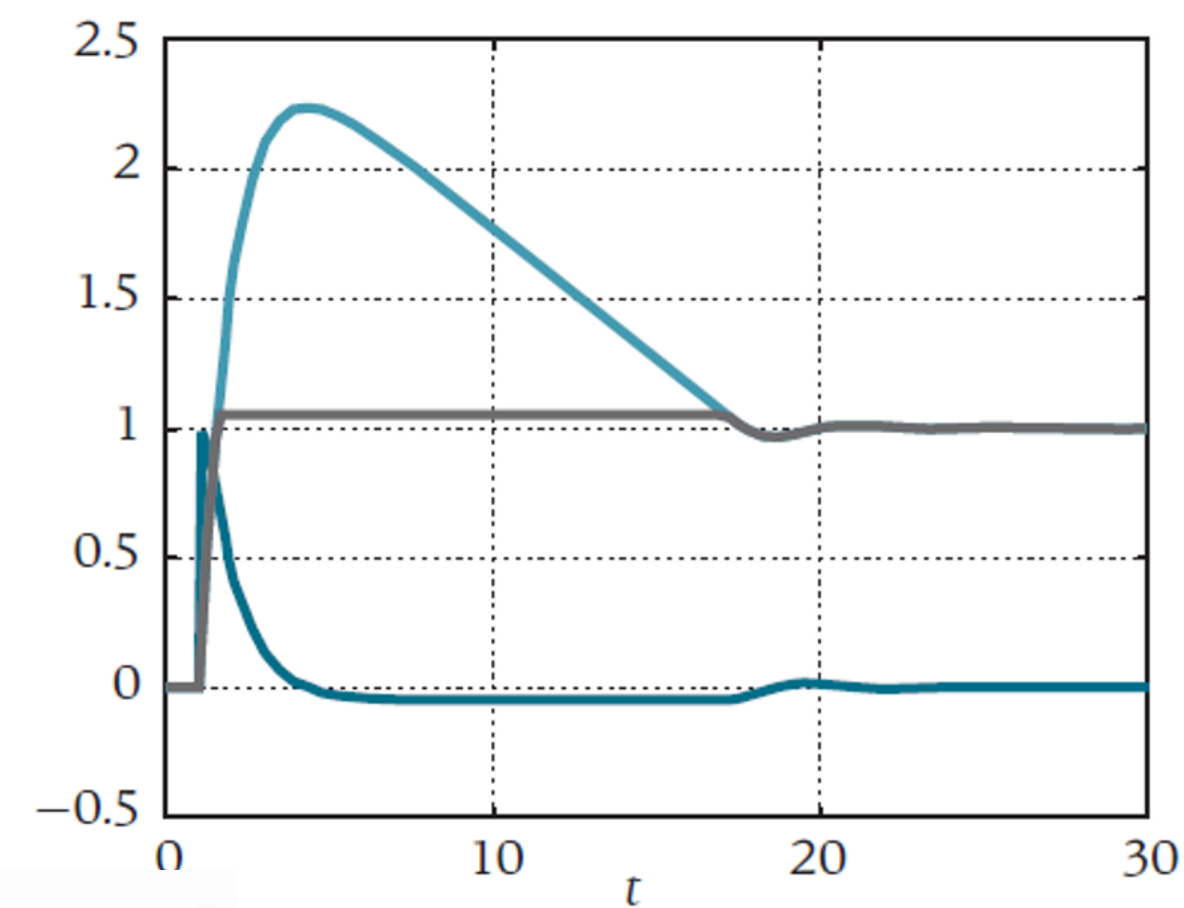
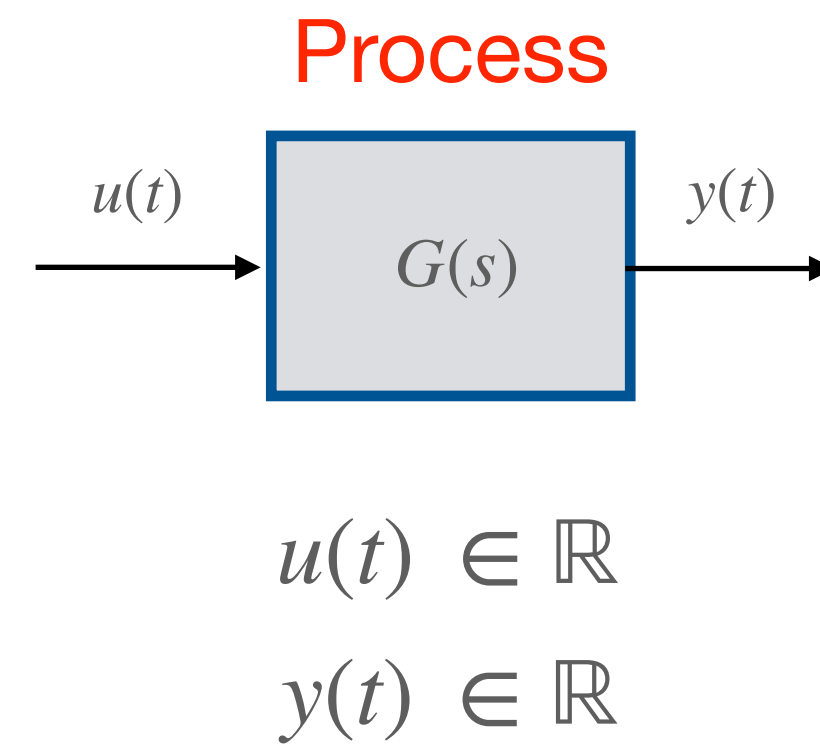
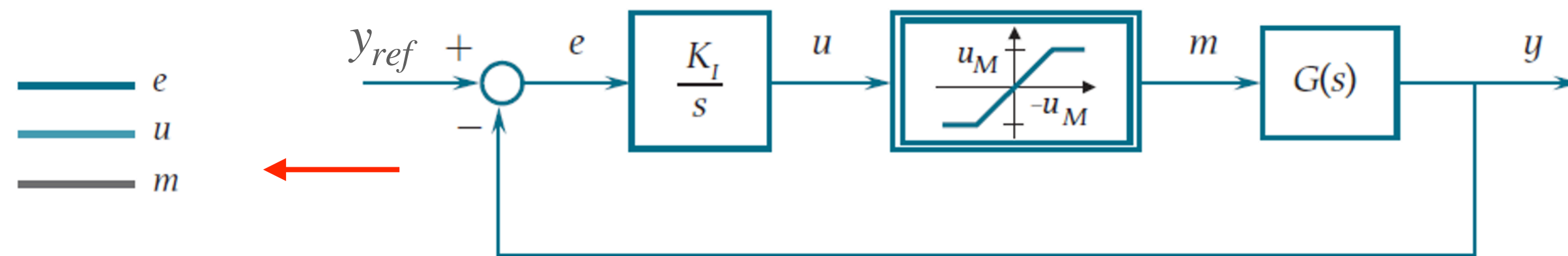
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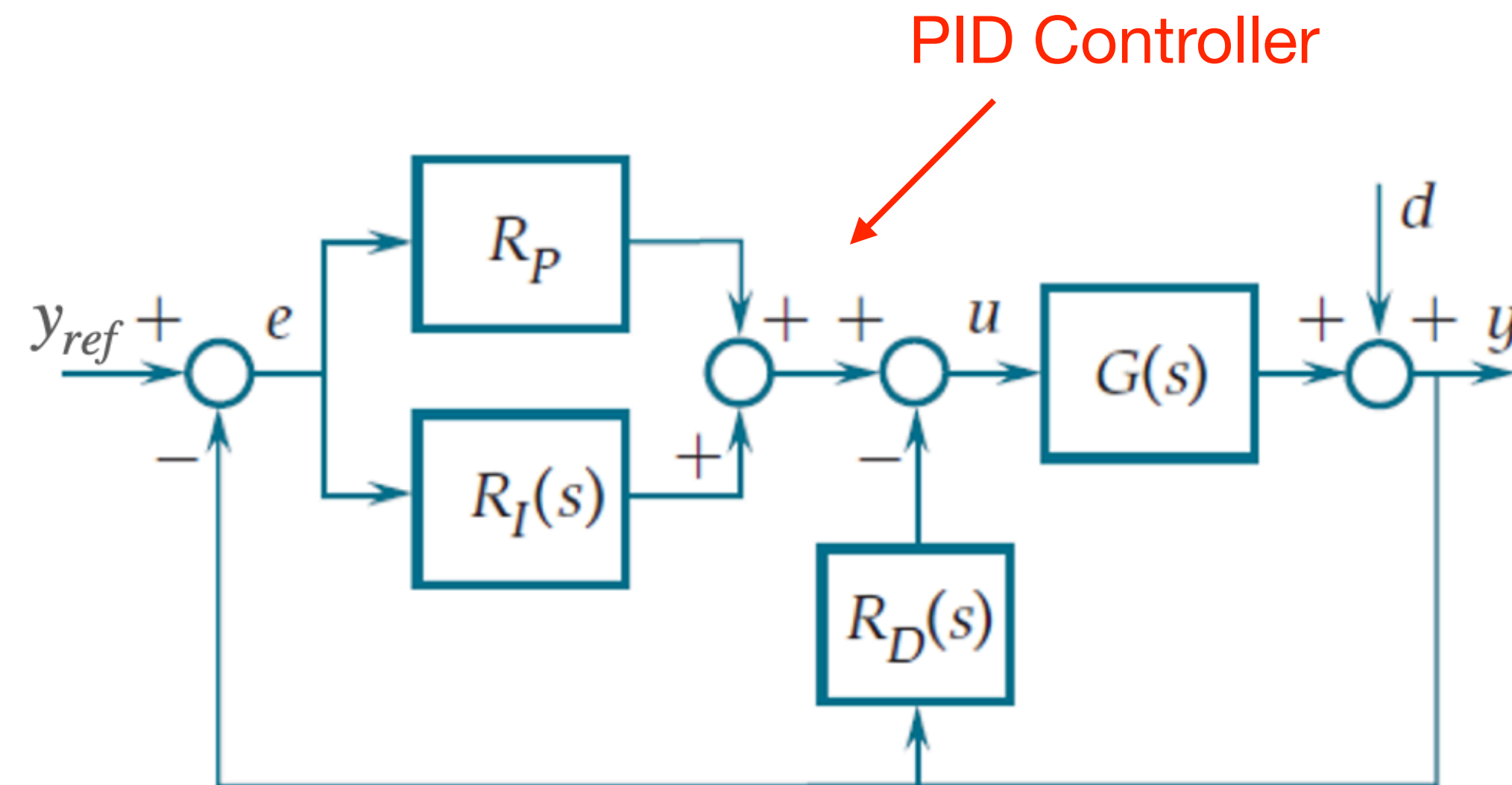
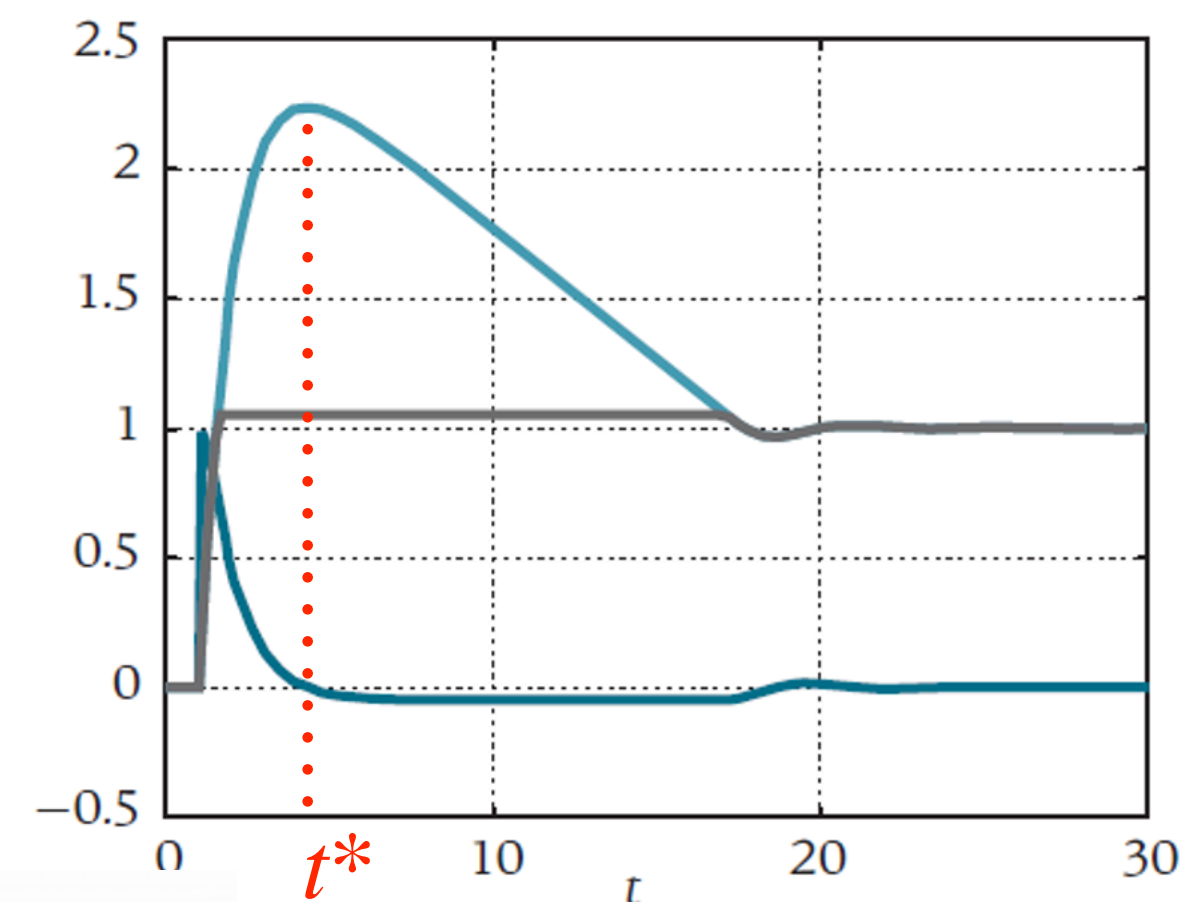
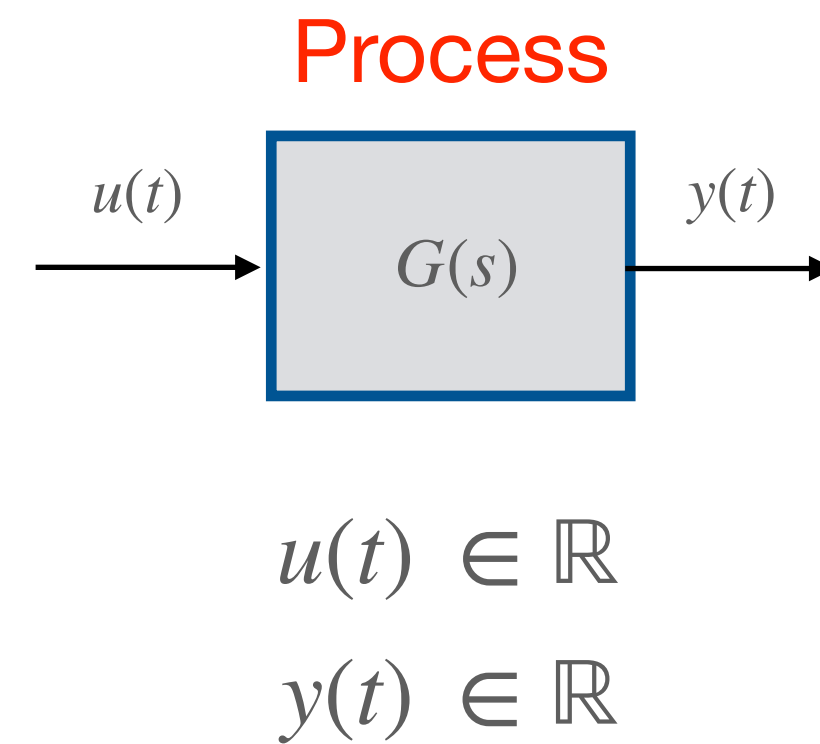
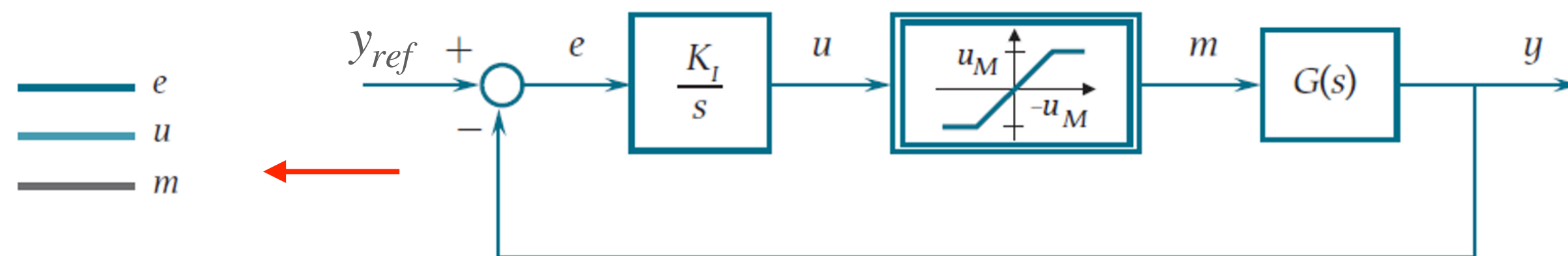


Realistic situation

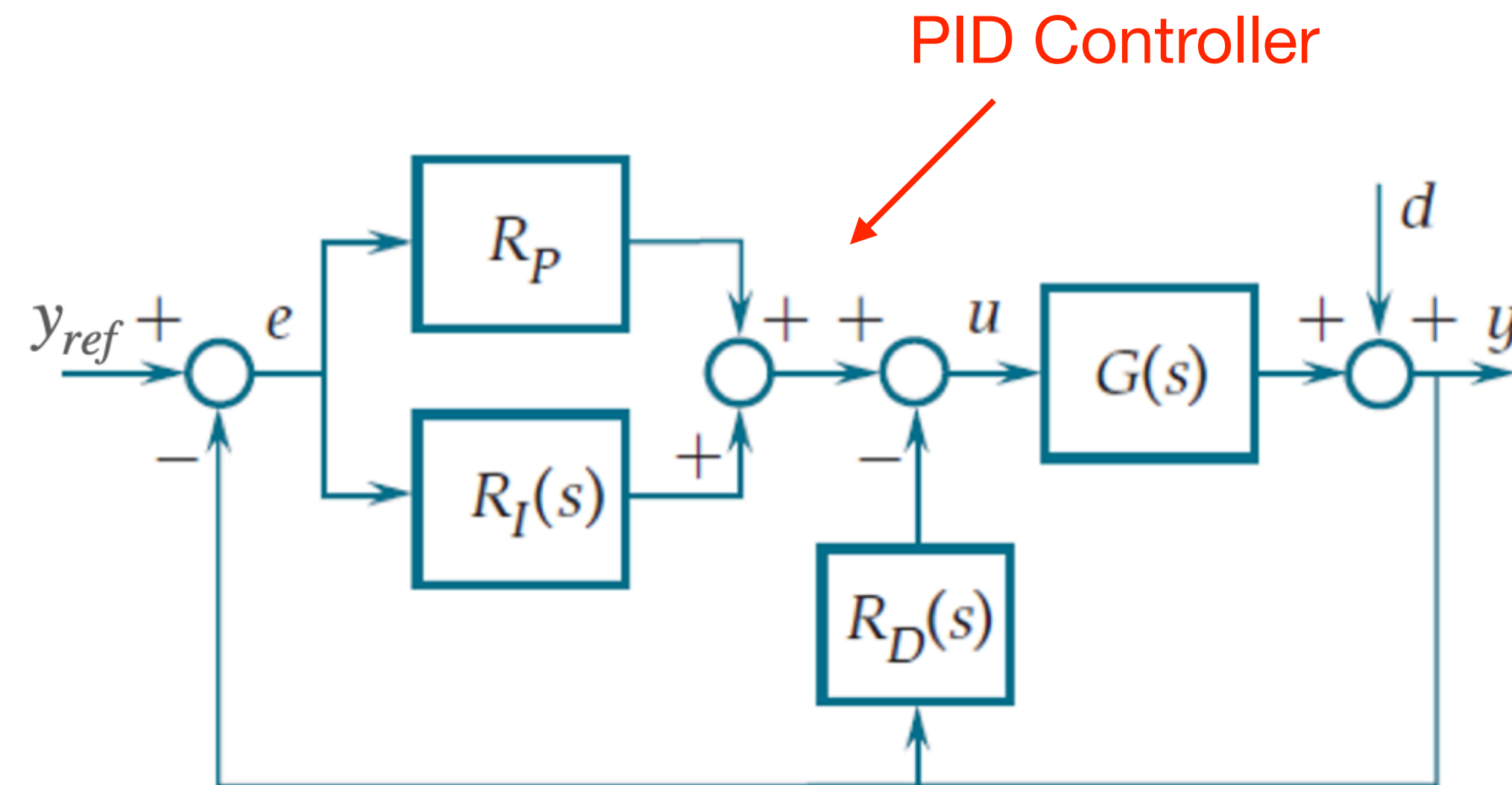
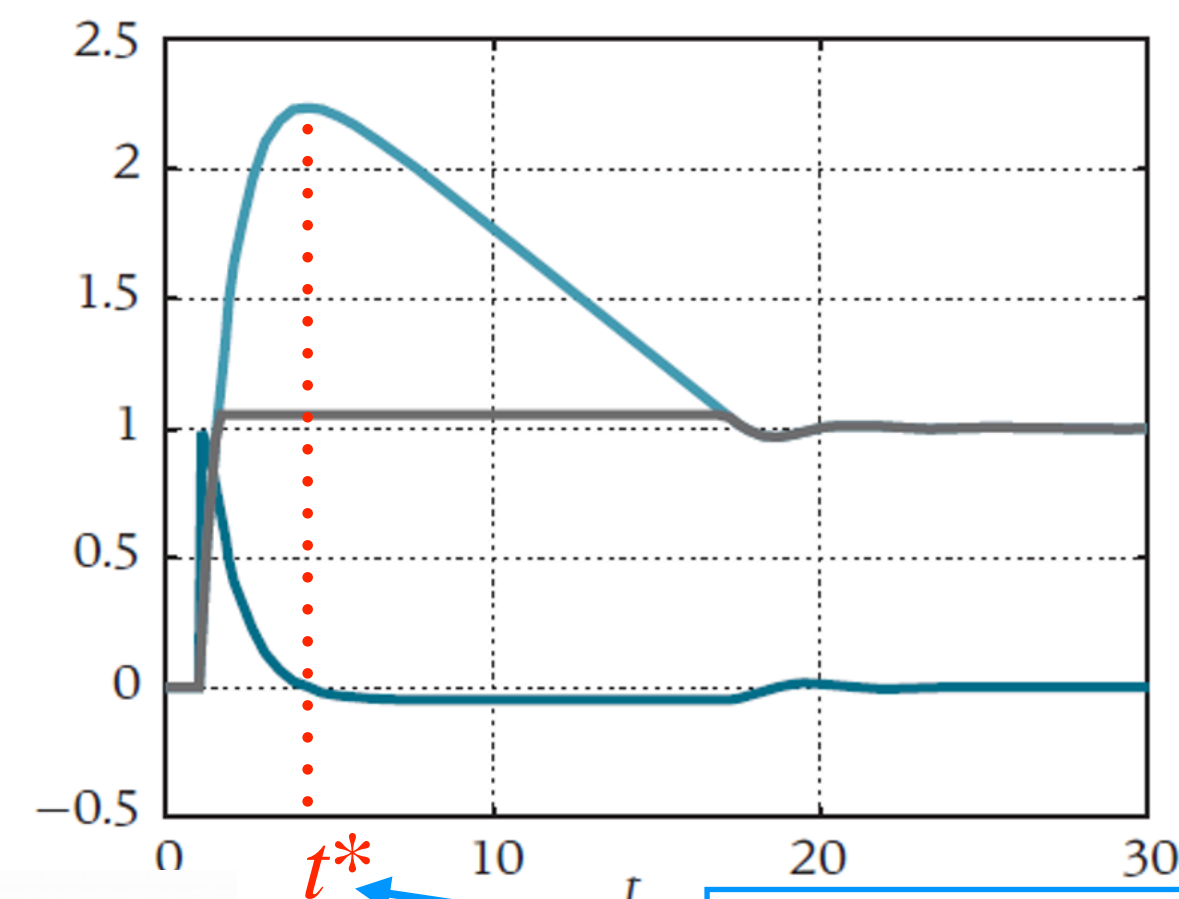
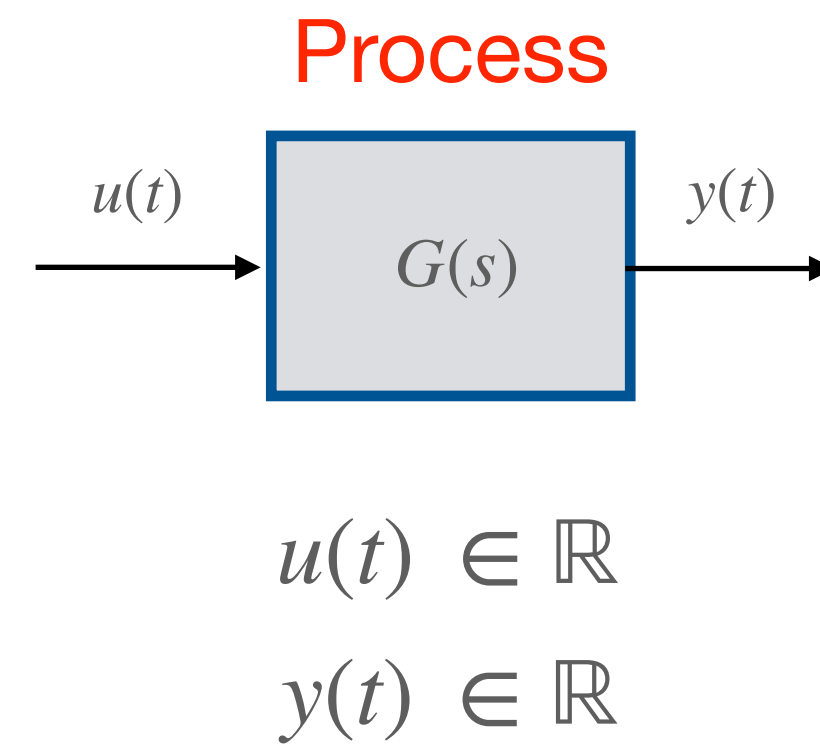


Only the integral component is considered

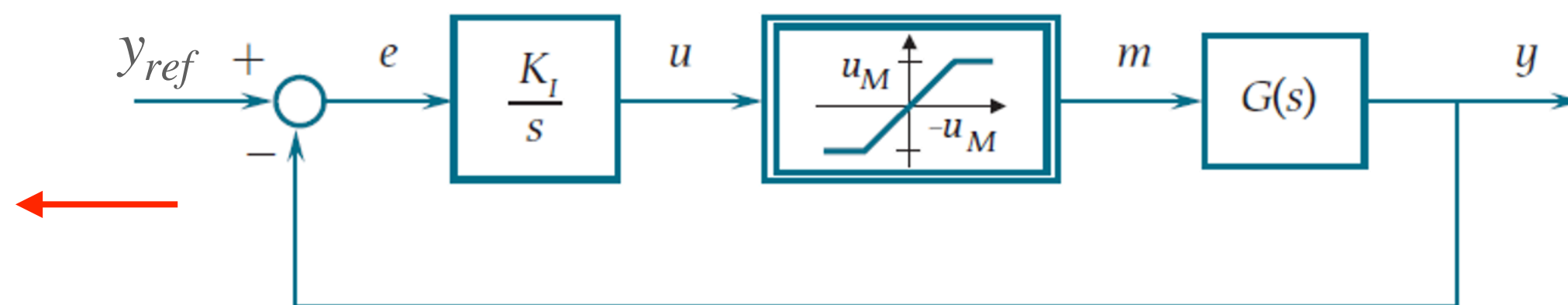
PID Controllers: **Wind-up Effect****Realistic situation**

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PID Controllers: Wind-up Effect

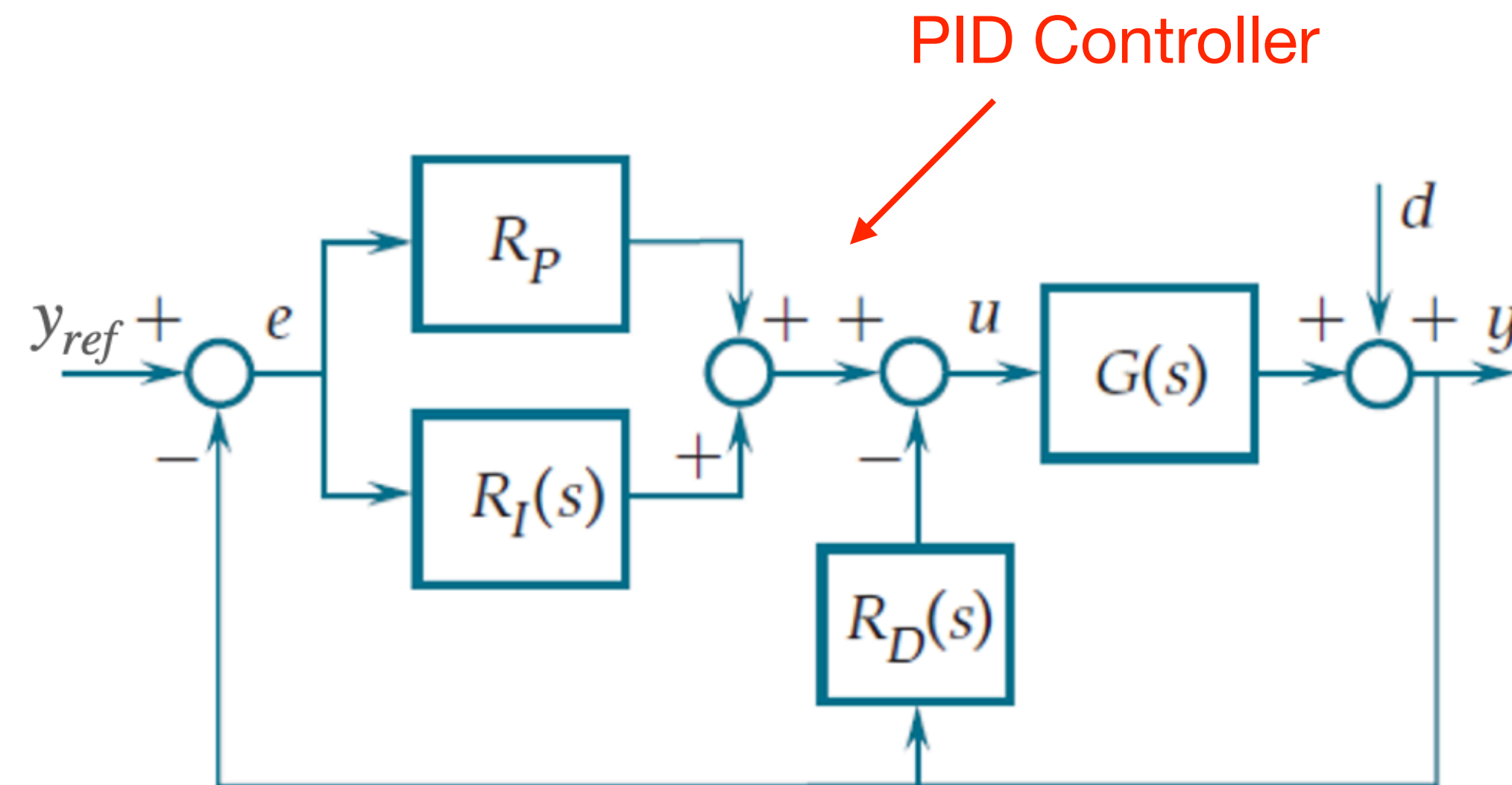
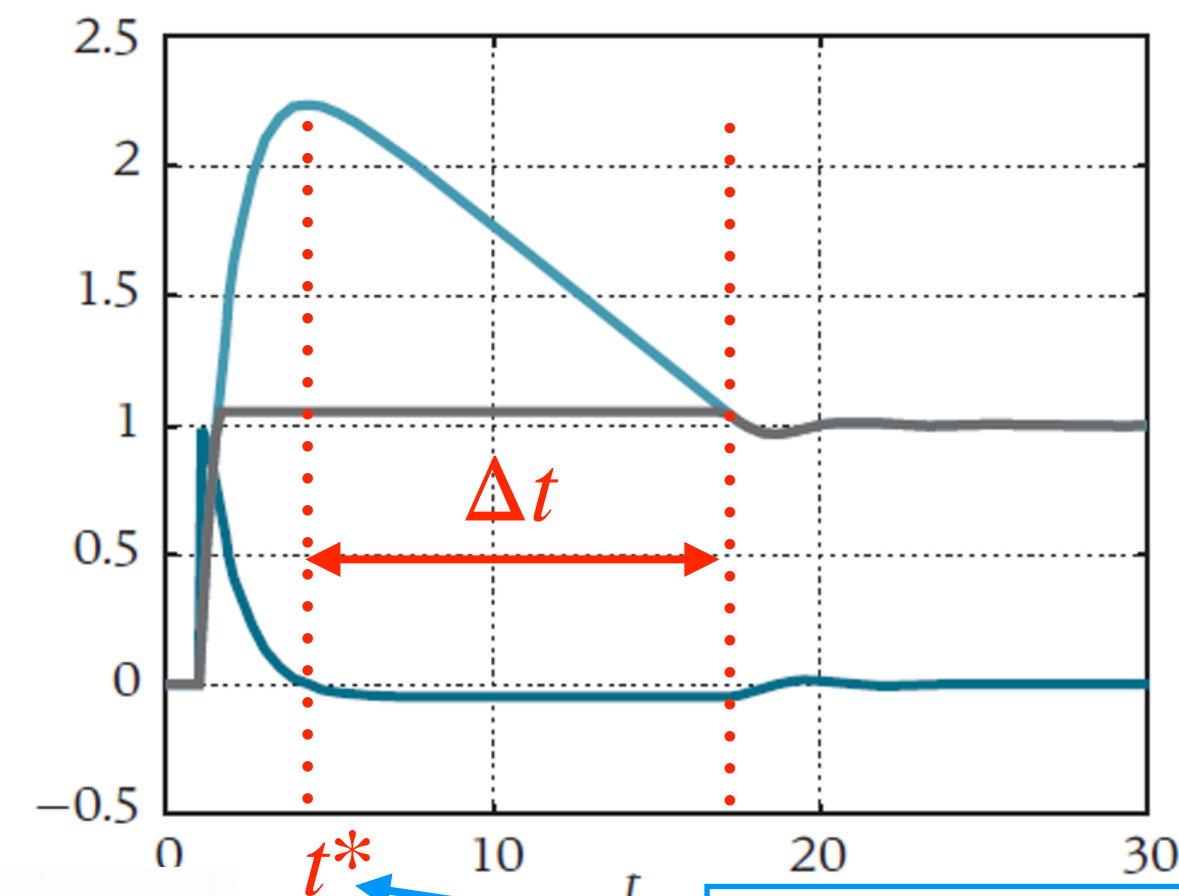
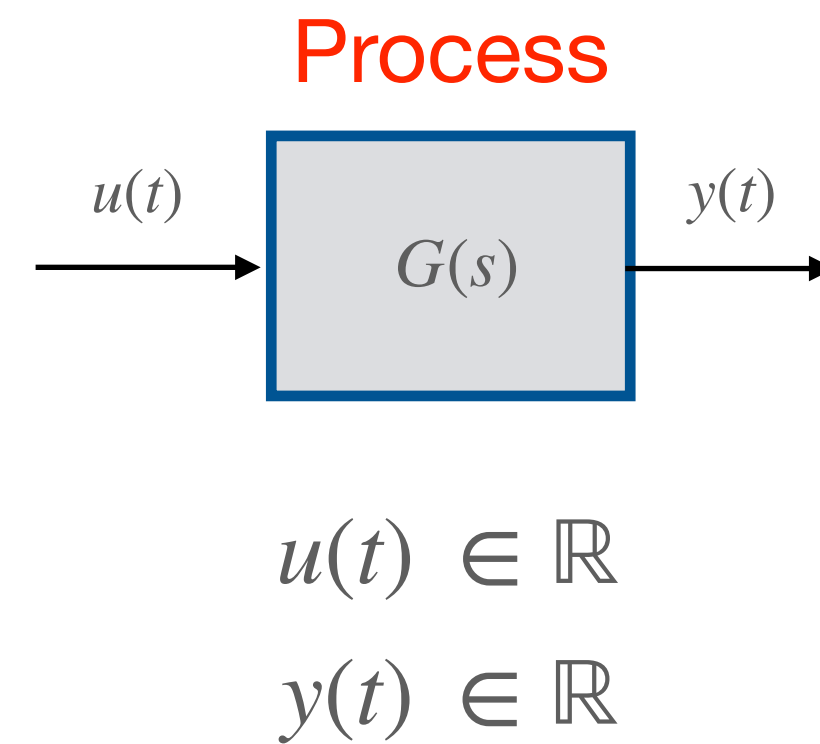


Realistic situation

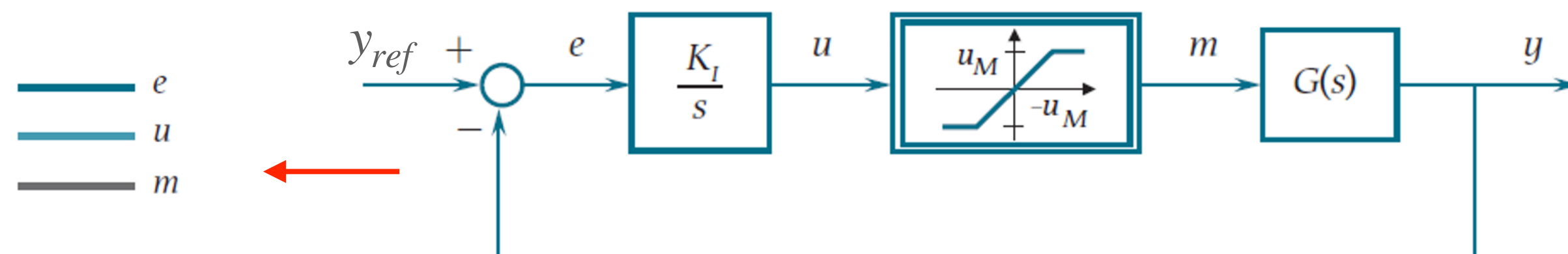


Ideal situation: the control action exits the saturation mode when e changes its sign (no useless waiting time due to wind-up)

PID Controllers: Wind-up Effect



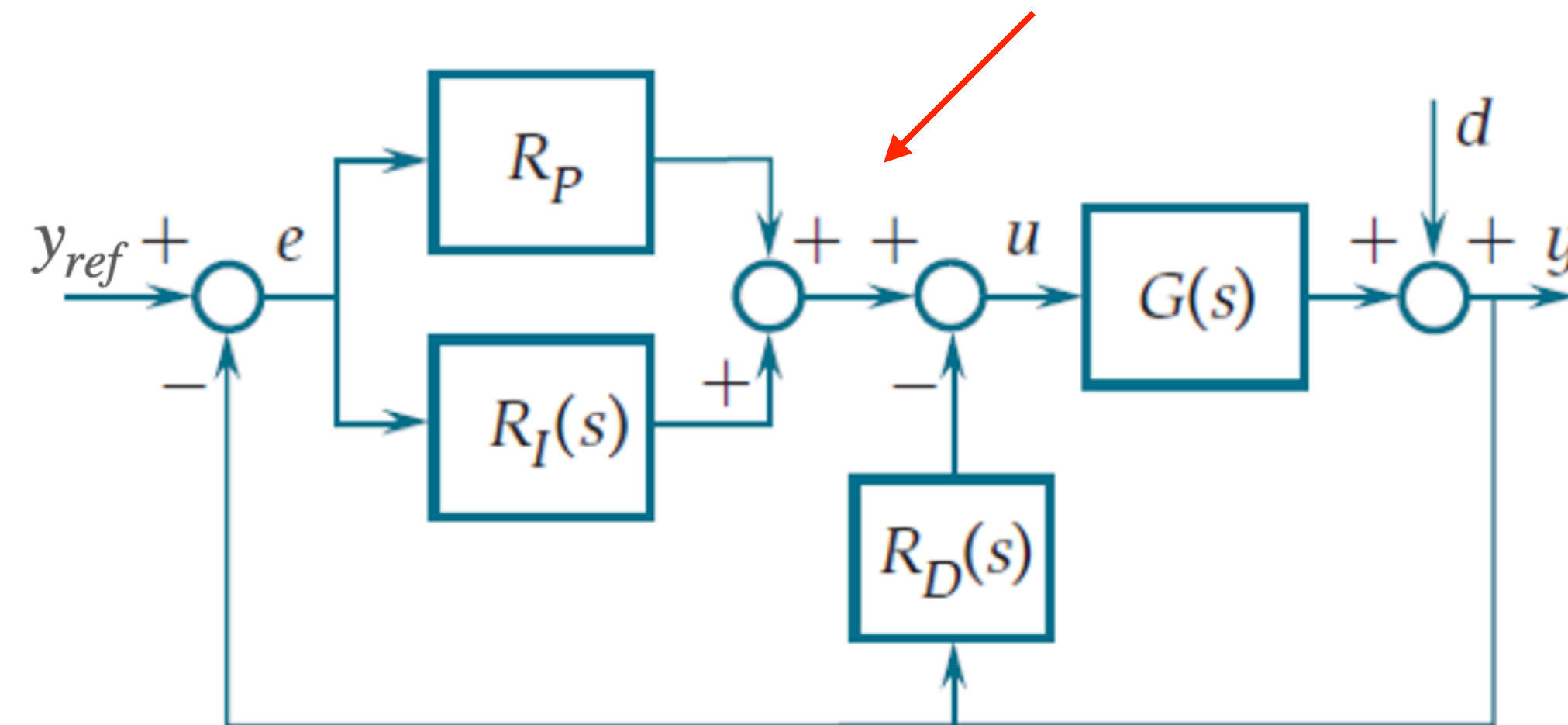
Realistic situation



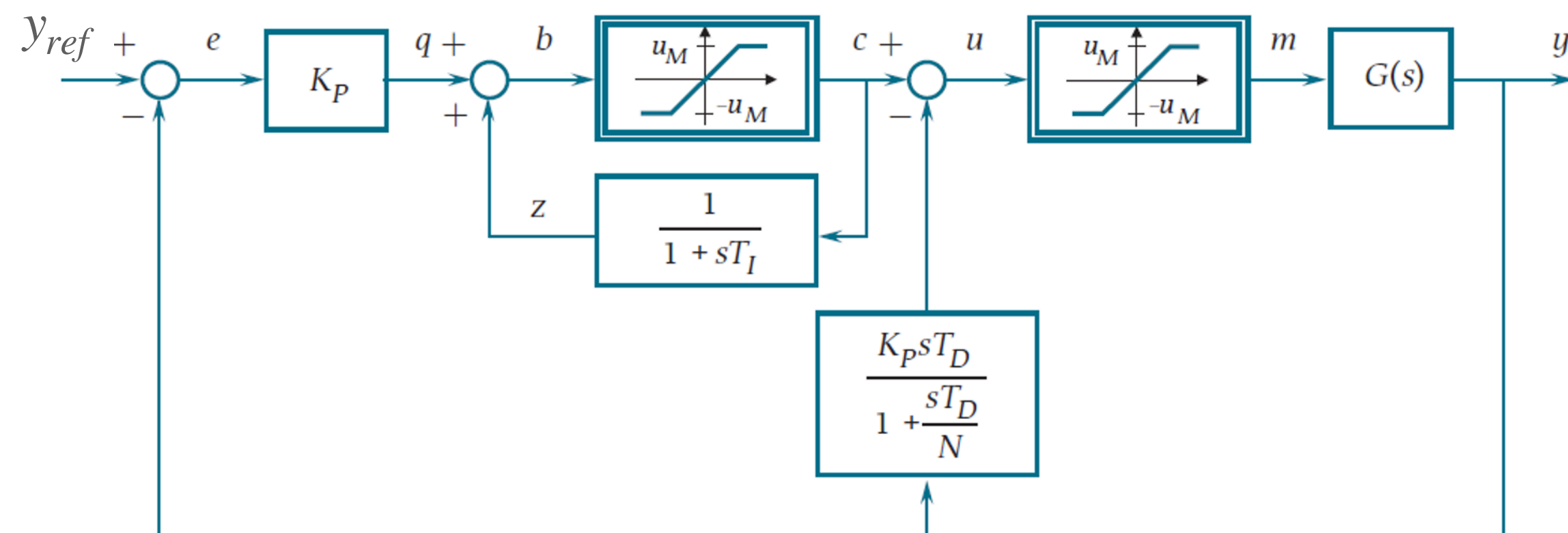
Ideal situation: the control action exits the saturation mode when e changes its sign (no useless waiting time due to wind-up)

PID Controllers: Wind-up Effect

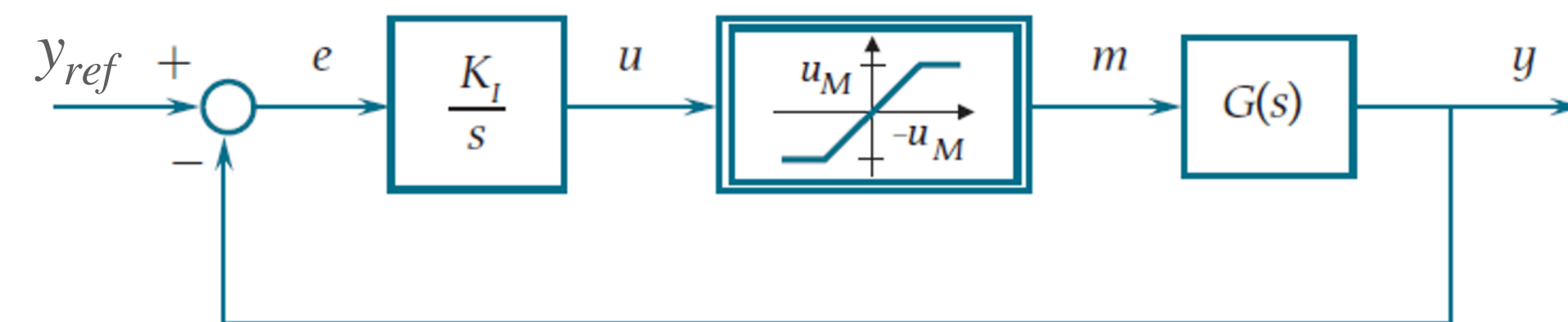
PID Controller



Solution: Anti-wind-up scheme



Realistic situation

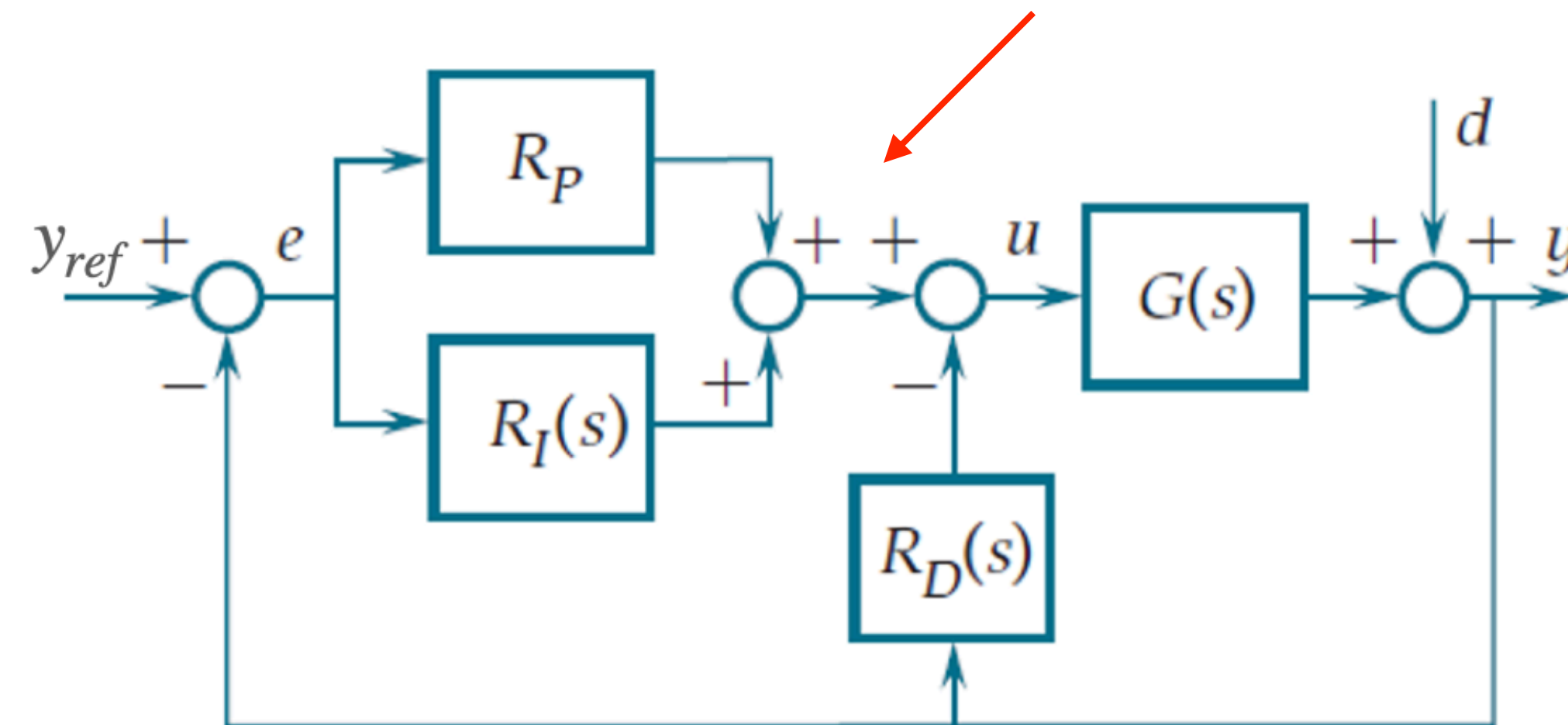


PID Controllers: Wind-up Effect

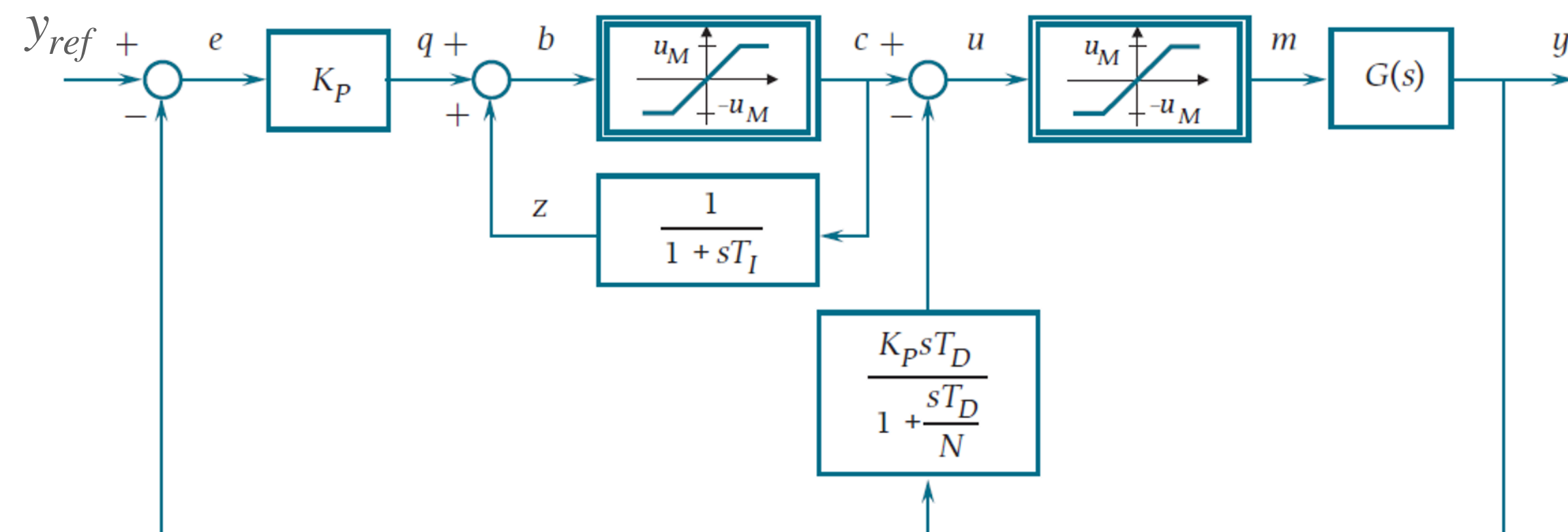
Case 1: The system operates in non-saturation mode



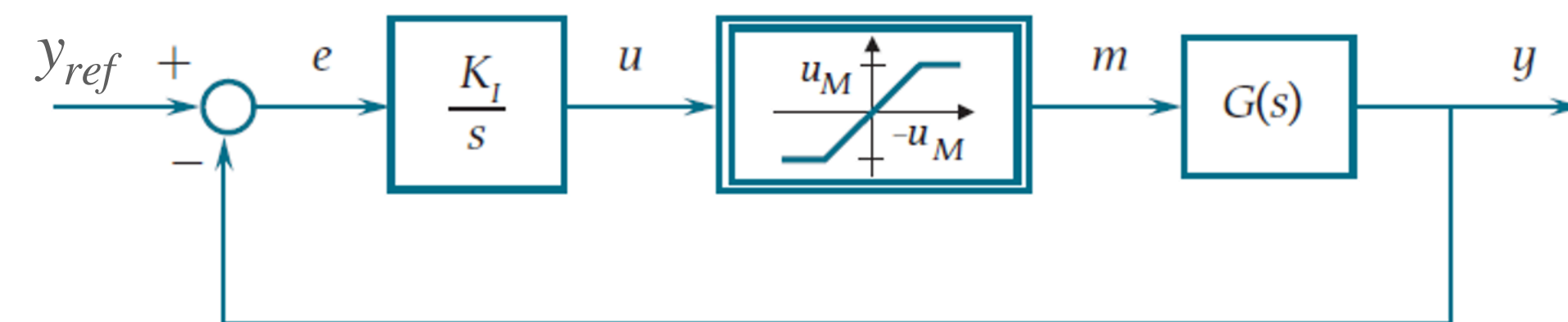
PID Controller



Solution: Anti-wind-up scheme



Realistic situation



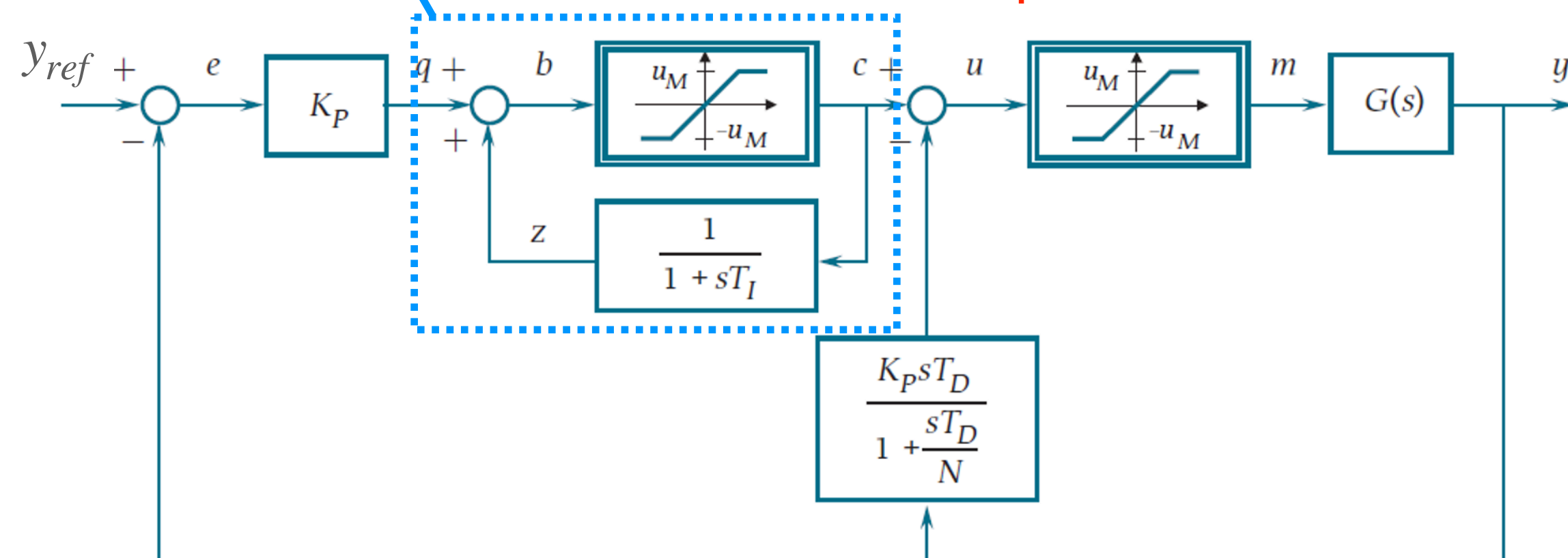
PID Controllers: Wind-up Effect

Case 1: The system operates in non-saturation mode

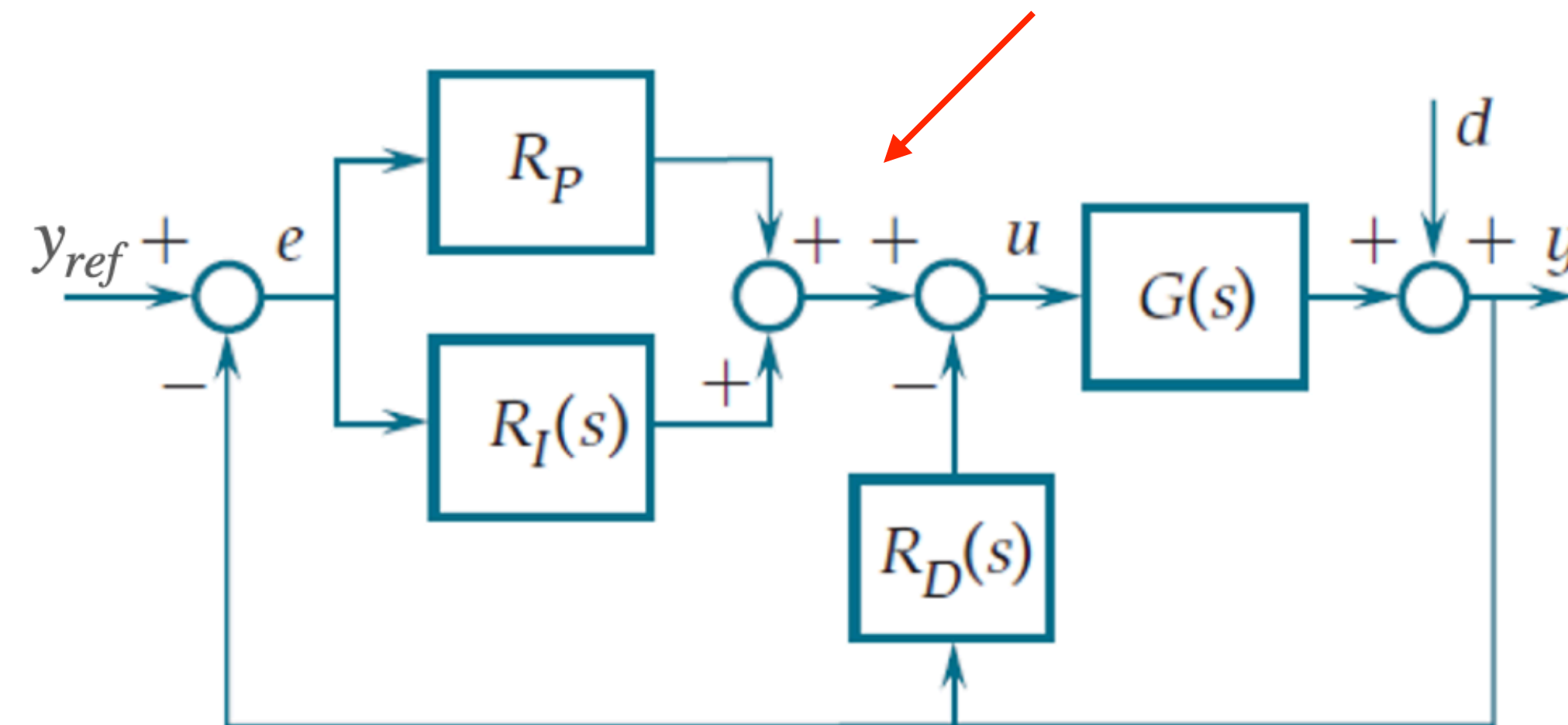


$$\frac{1}{1 - \frac{1}{1 + sT_I}} = \frac{1 + sT_I}{1 + sT_I - 1} = \frac{1 + sT_I}{sT_I} = 1 + \frac{1}{sT_I}$$

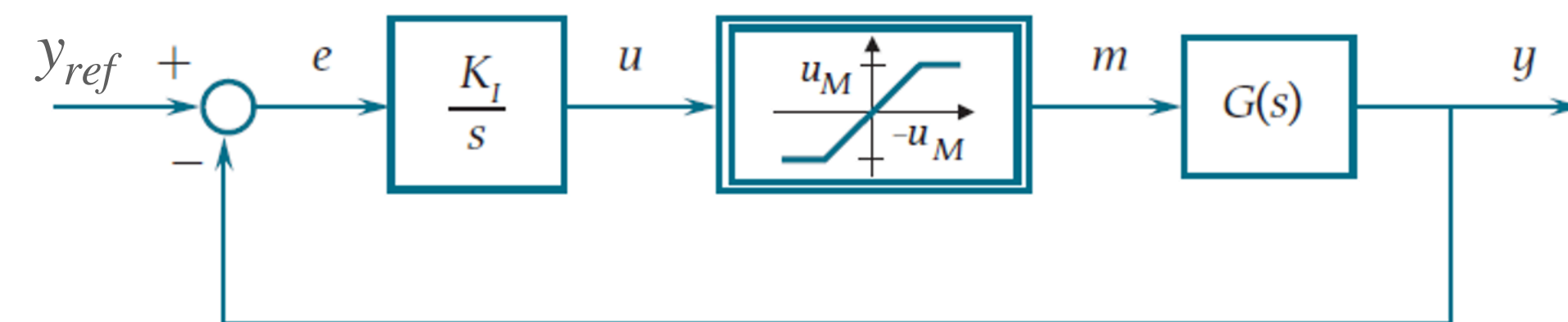
Solution: Anti-wind-up scheme



PID Controller

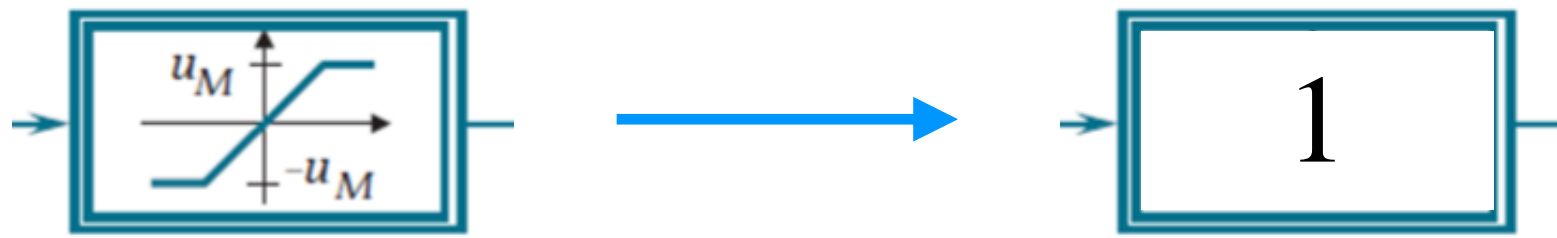


Realistic situation



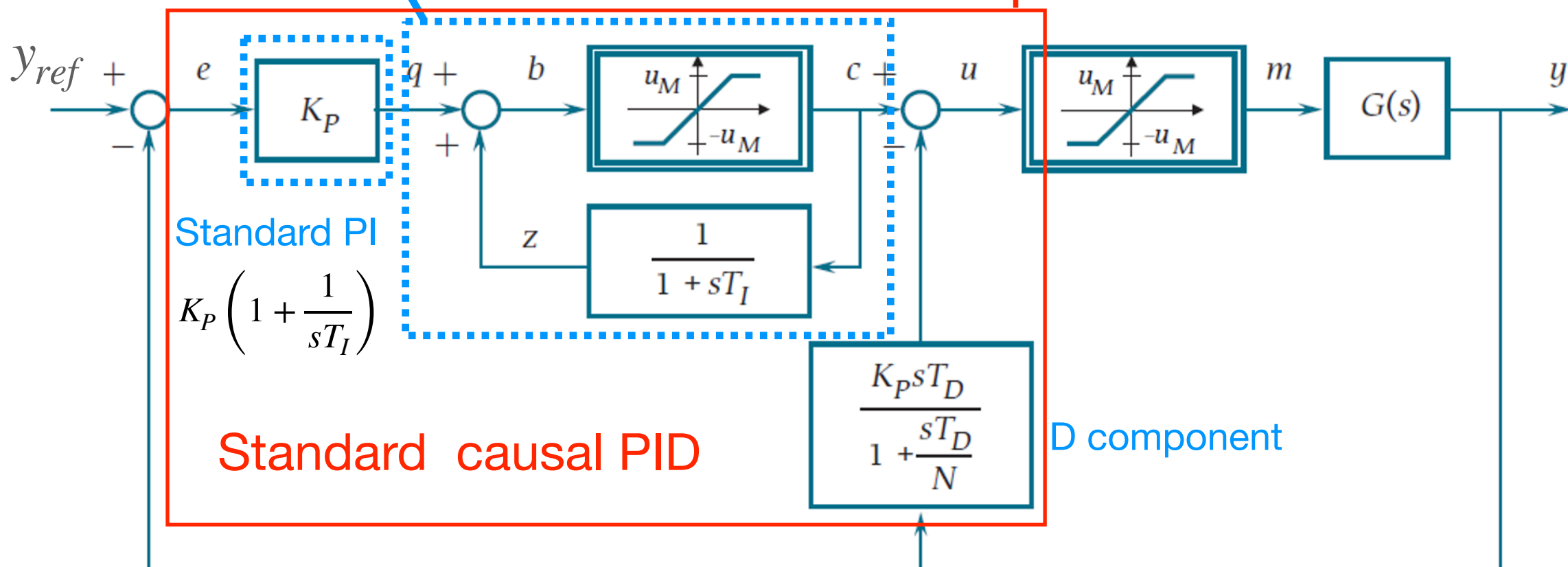
PID Controllers: Wind-up Effect

Case 1: The system operates in non-saturation mode

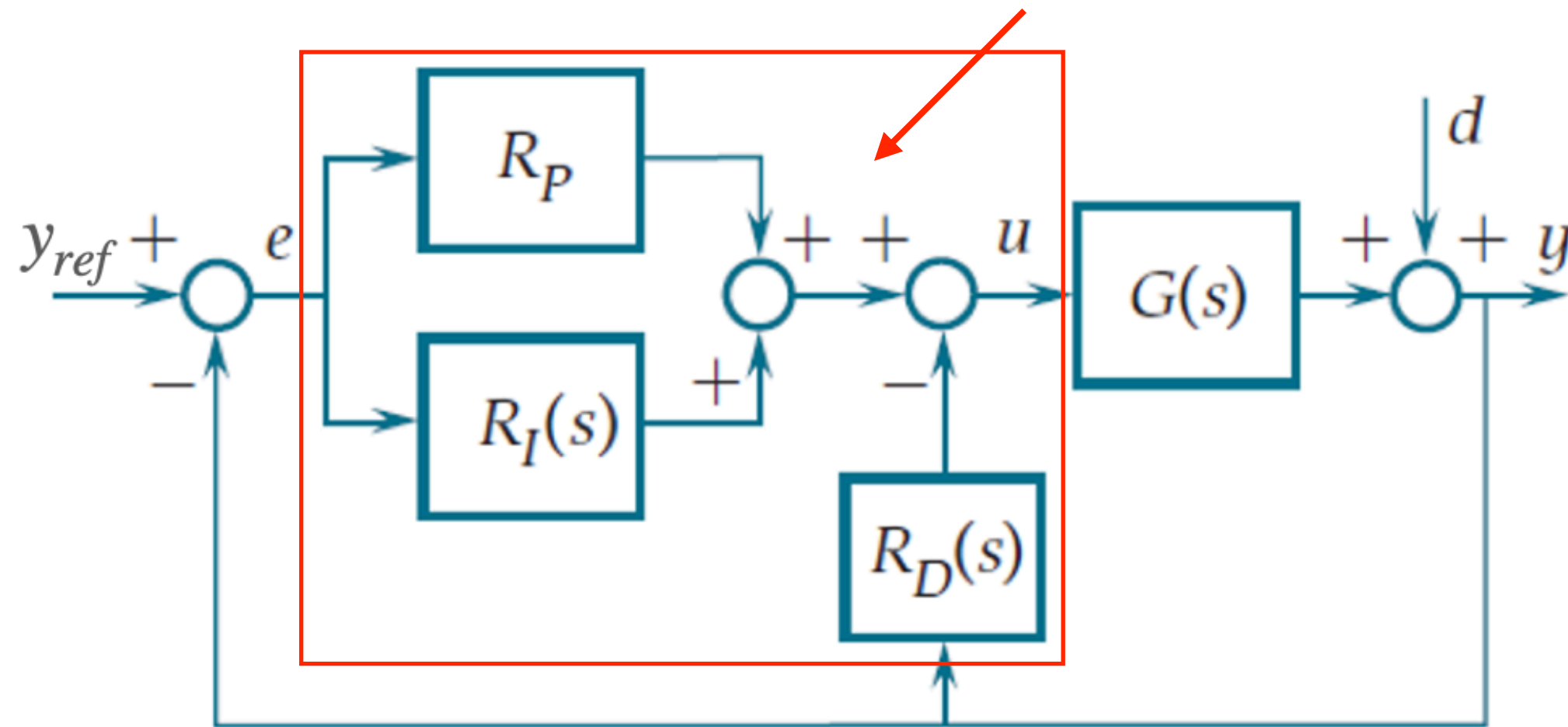


$$\frac{1}{1 - \frac{1}{1 + sT_I}} = \frac{1 + sT_I}{1 + sT_I - 1} = \frac{1 + sT_I}{sT_I} = 1 + \frac{1}{sT_I}$$

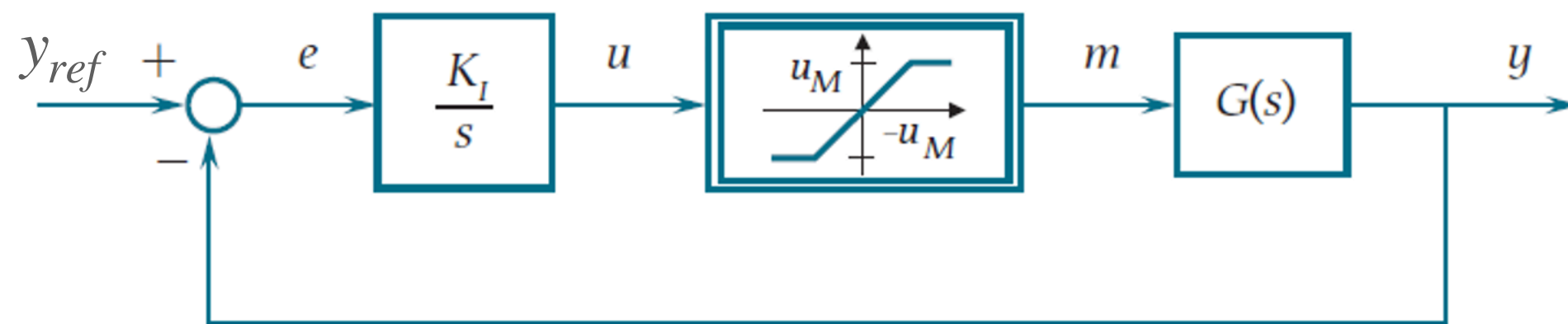
Solution: Anti-wind-up scheme



PID Controller

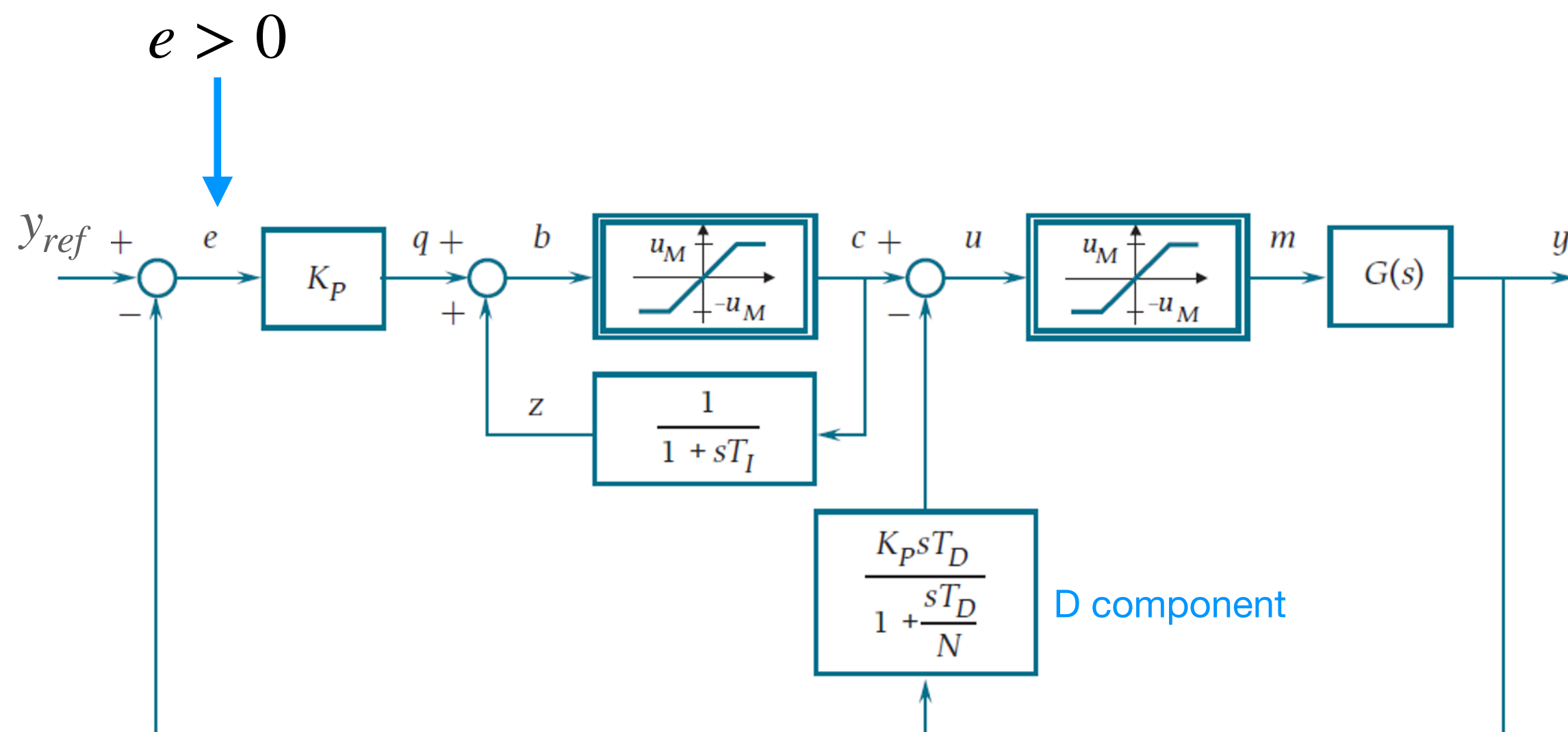
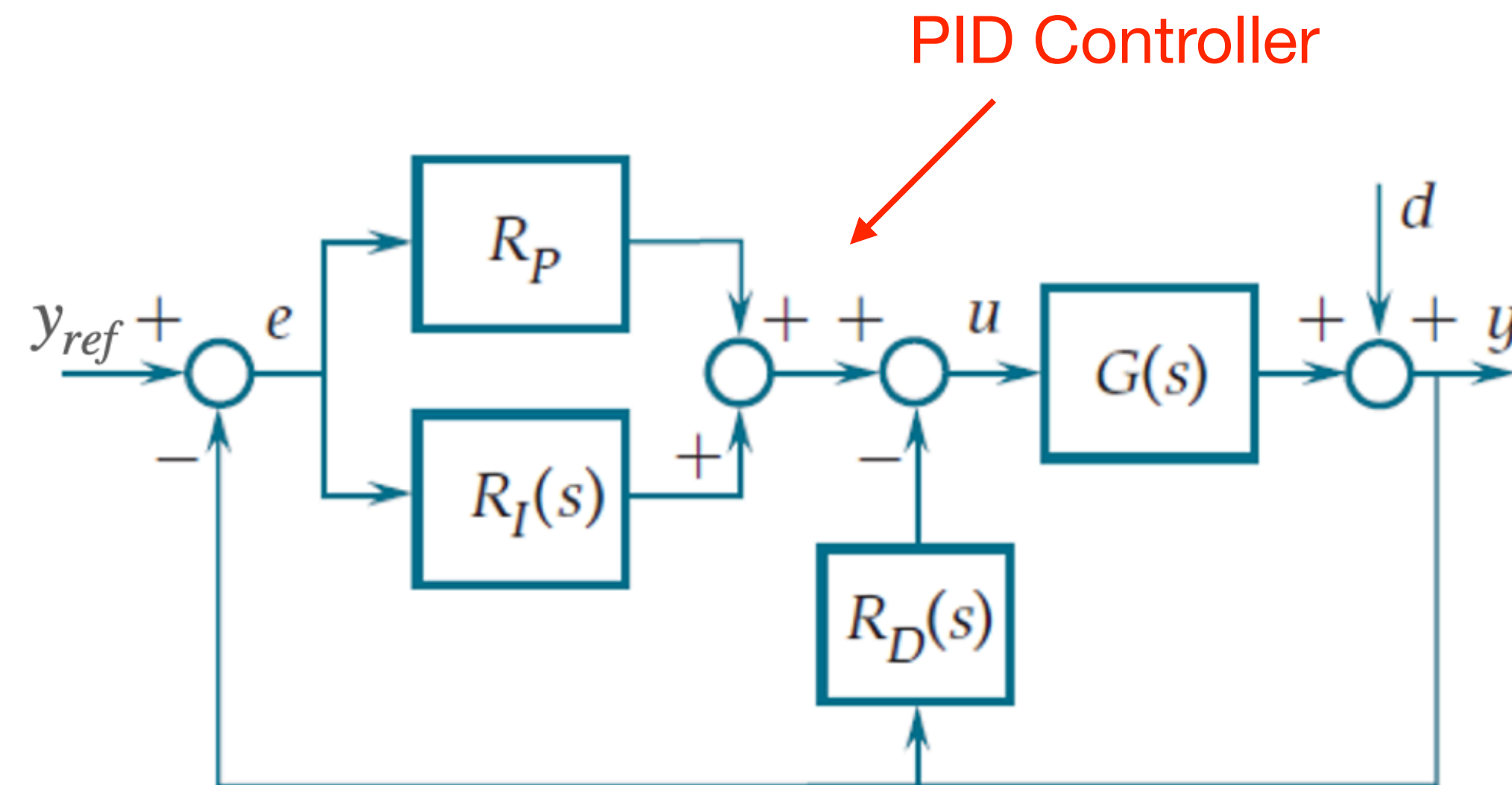
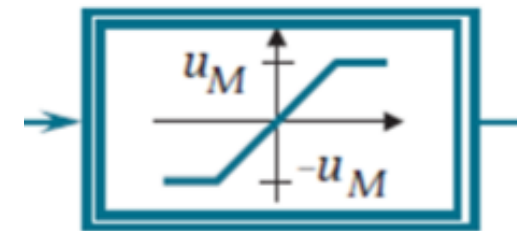


Realistic situation

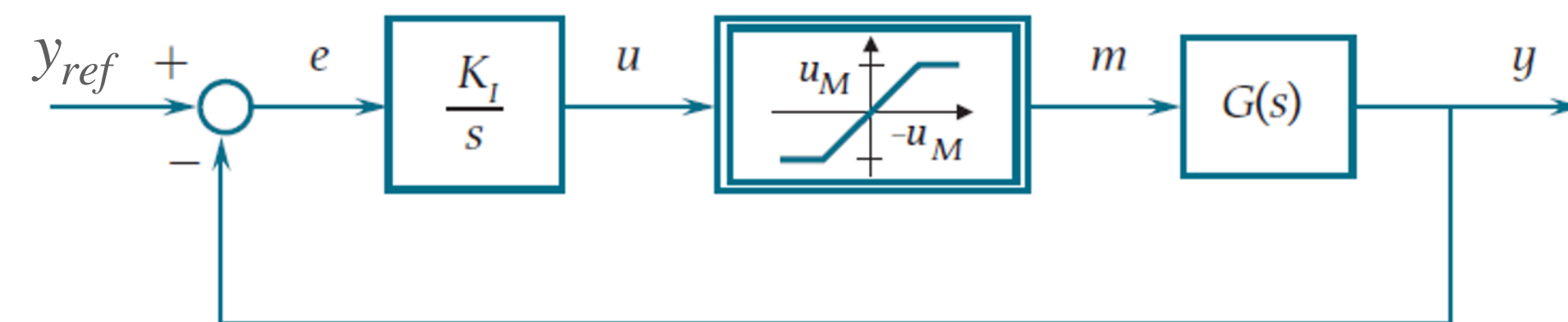


PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

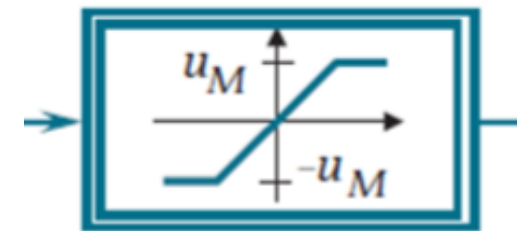


Realistic situation



PID Controllers: Wind-up Effect

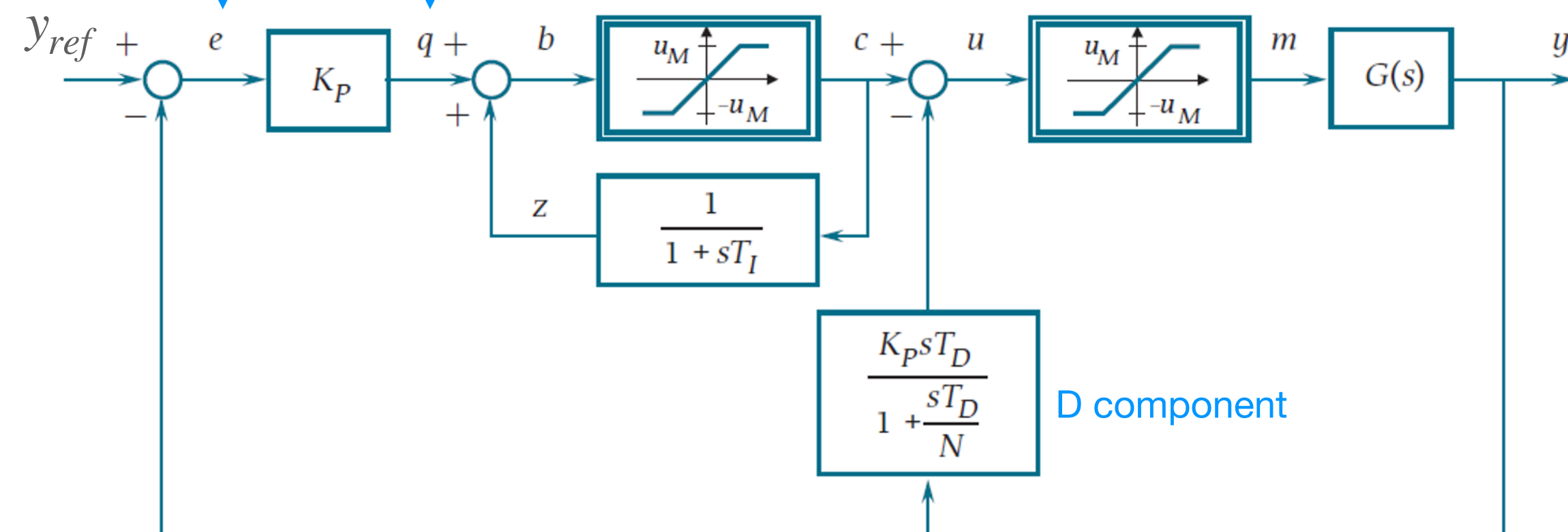
Case 2: The system operates in saturation mode



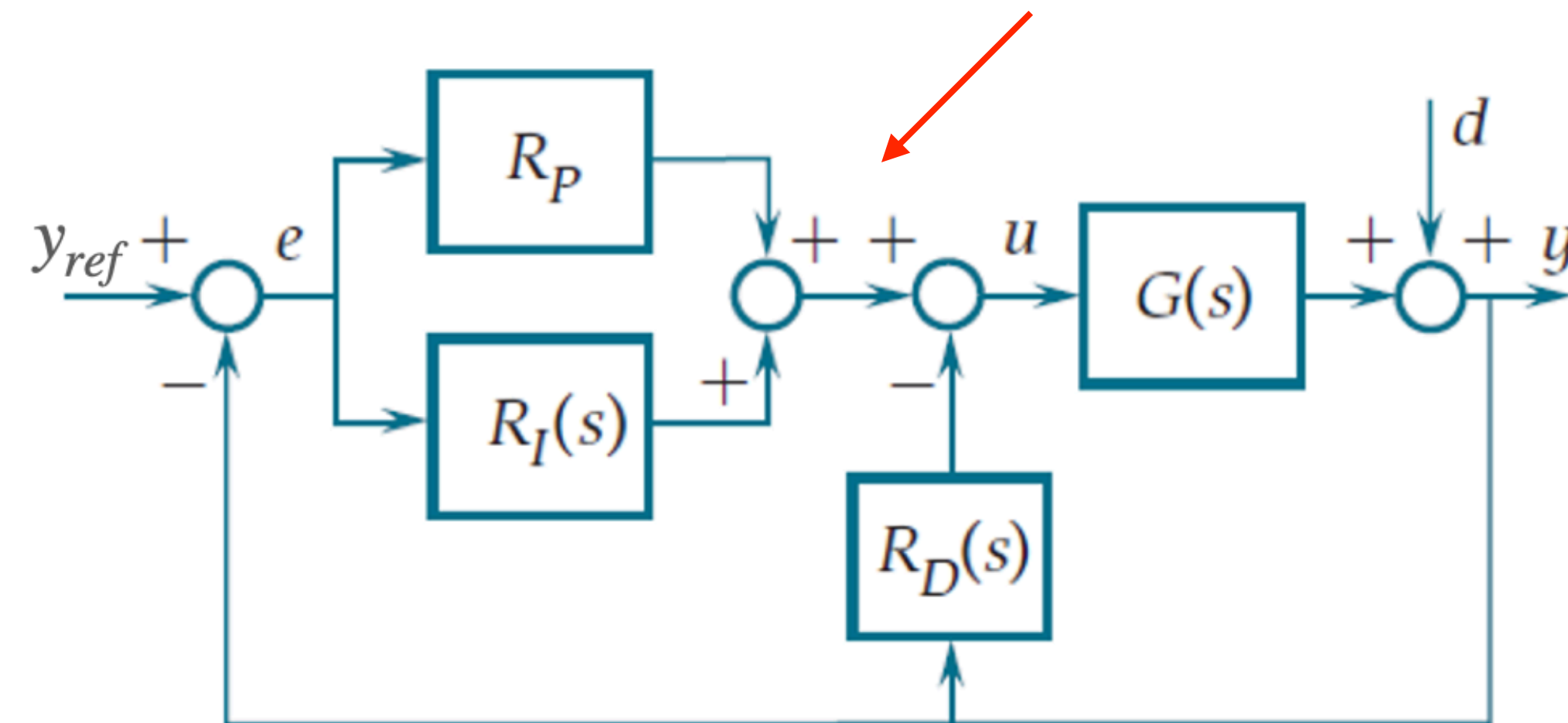
Assumption

$$q > 0, \boxed{q > U_M}$$

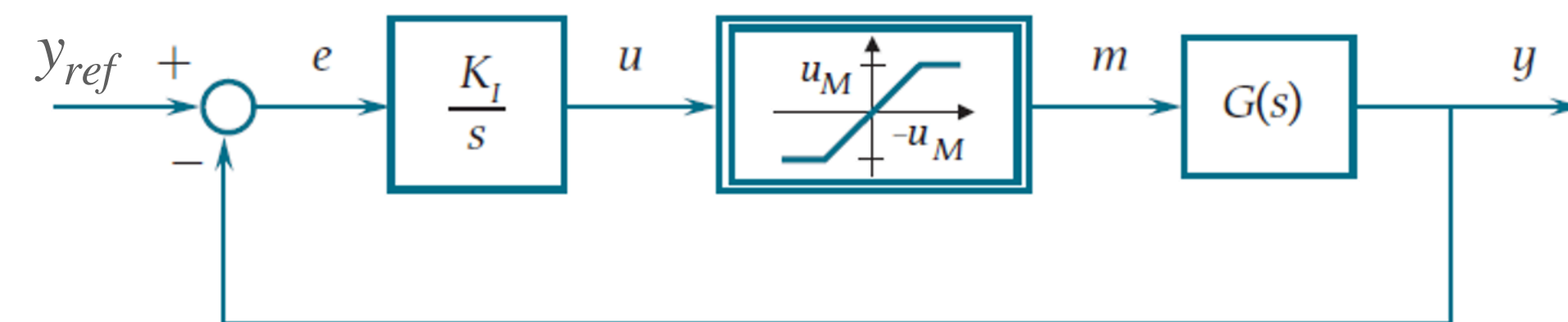
$$e > 0$$



PID Controller

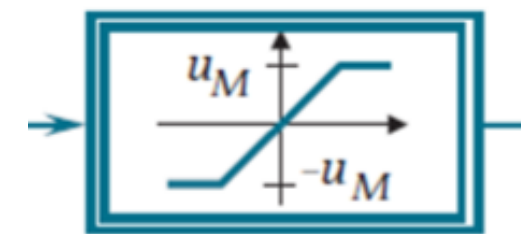


Realistic situation



PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

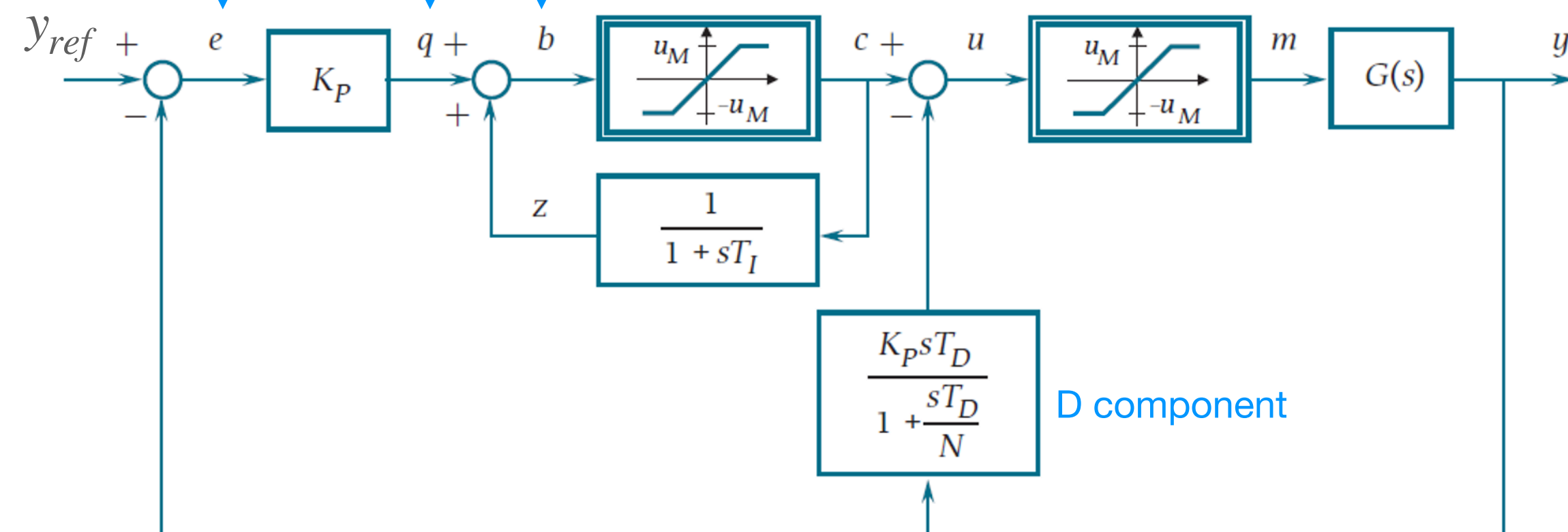


Assumption

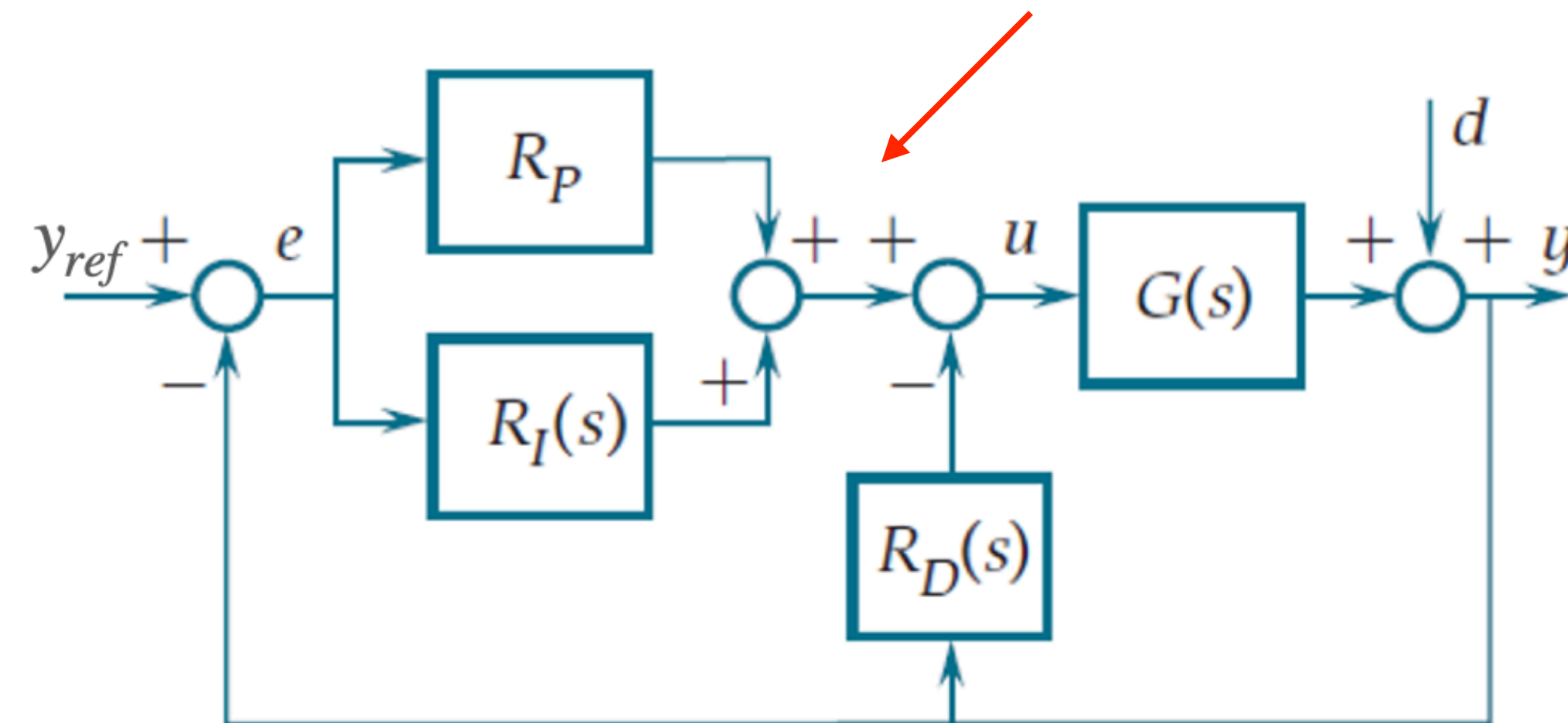
$$q > 0, q > U_M$$

$$b > 0, b > U_M$$

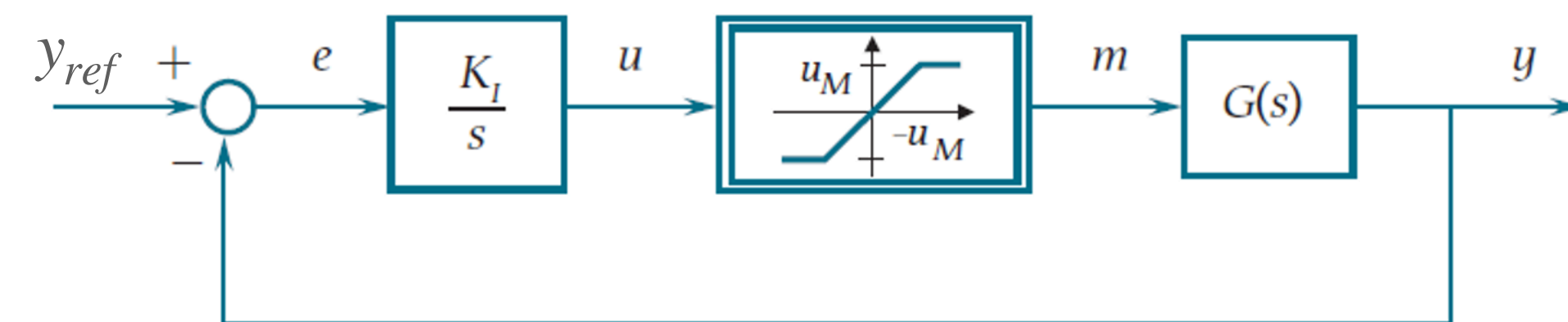
$$e > 0$$



PID Controller

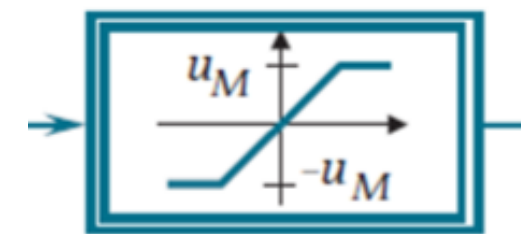


Realistic situation



PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

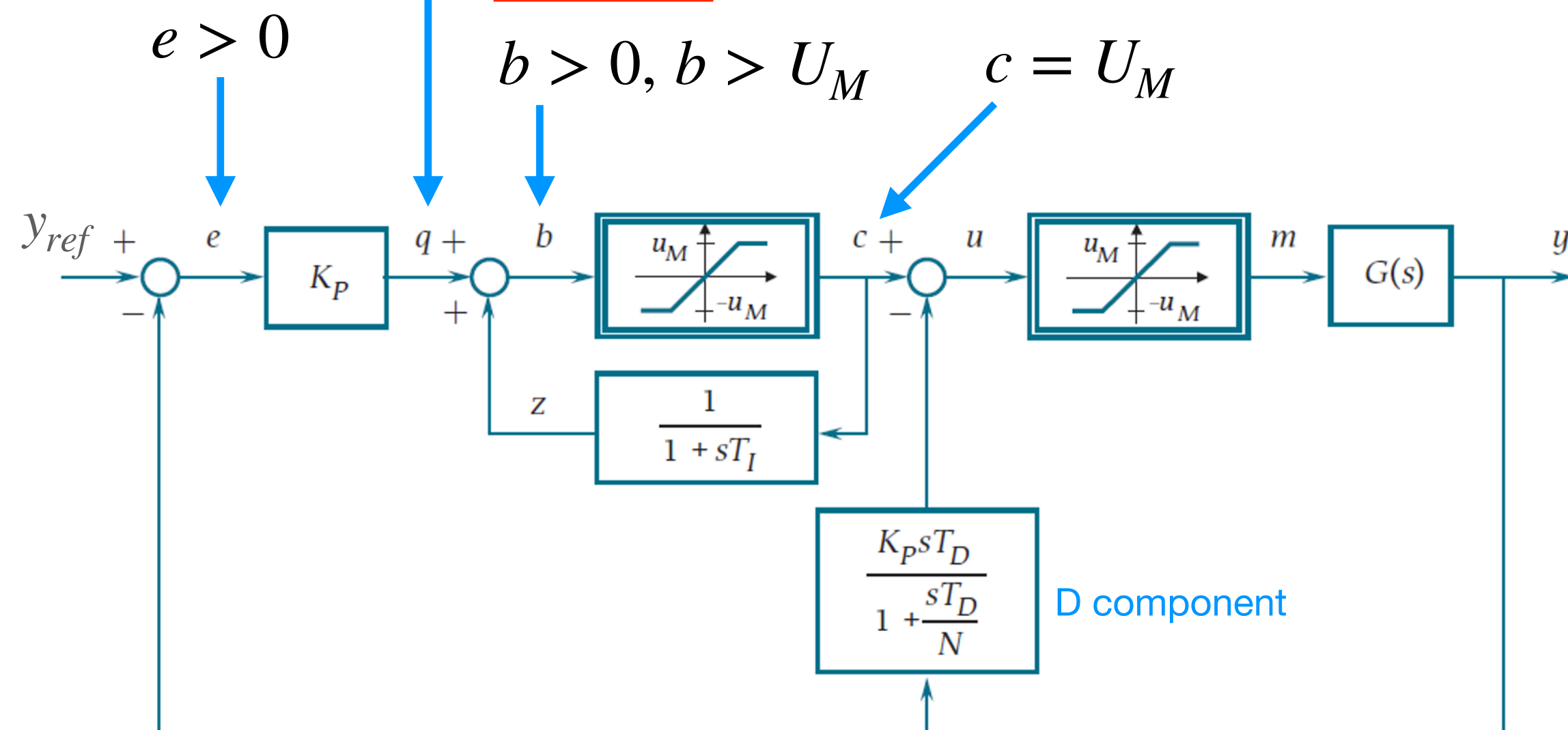


Assumption

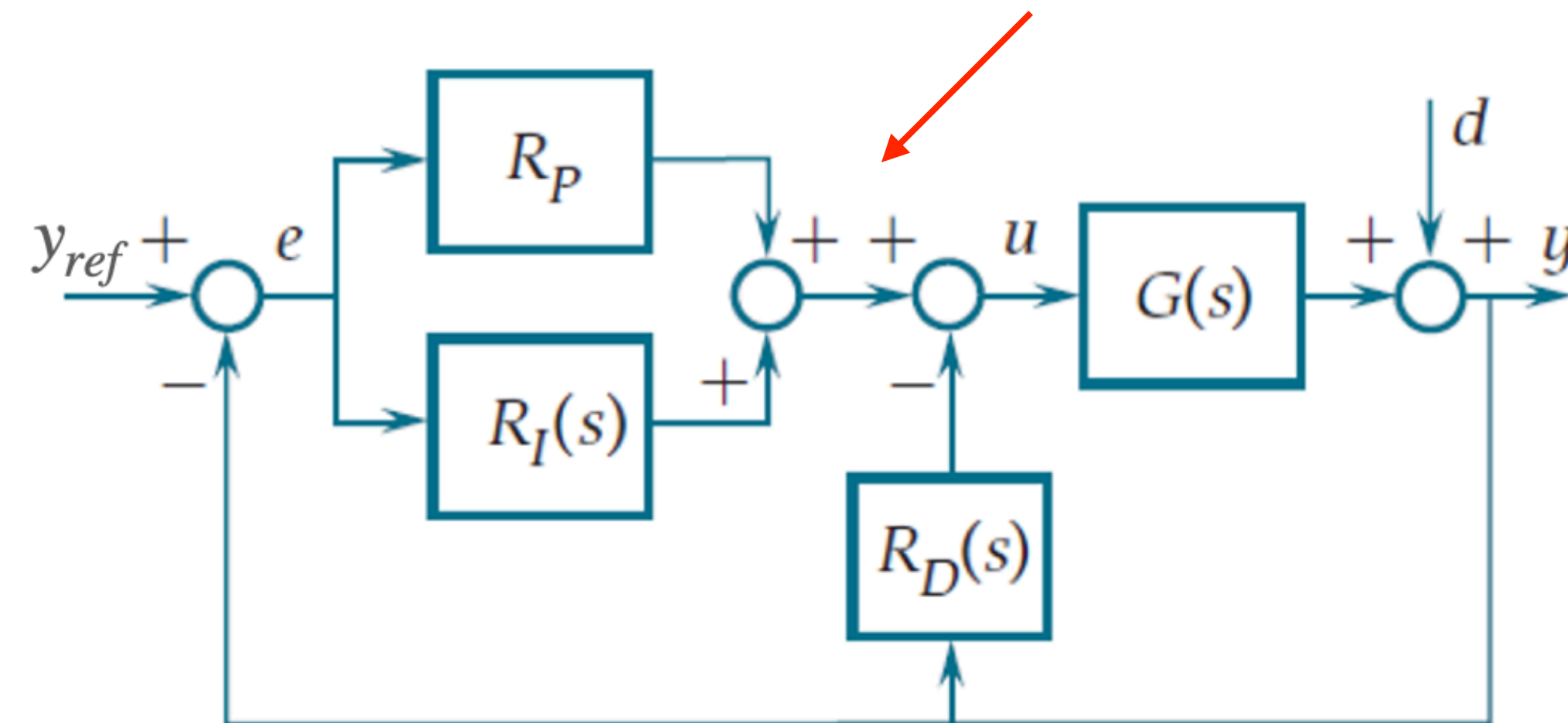
$$q > 0, q > U_M$$

$$b > 0, b > U_M$$

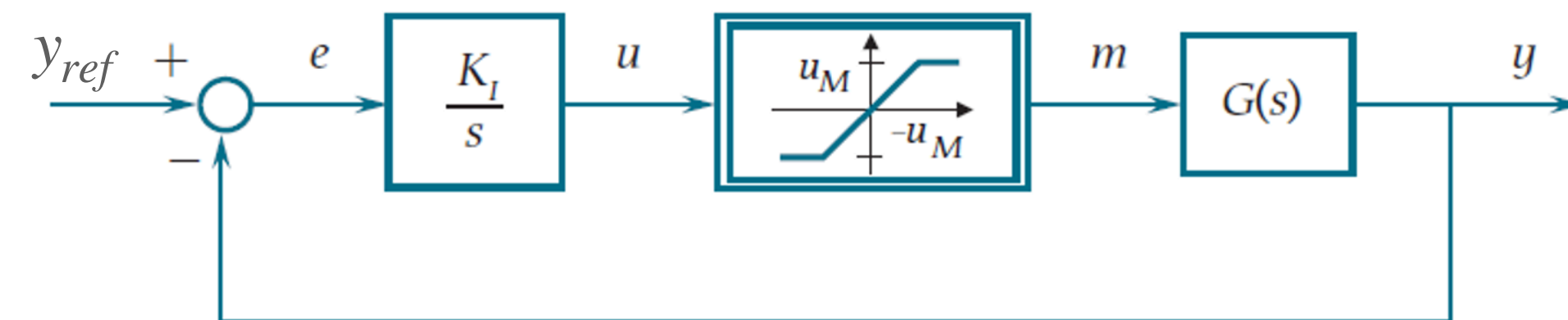
$$c = U_M$$



PID Controller

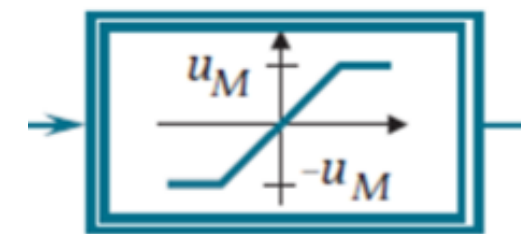


Realistic situation



PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode



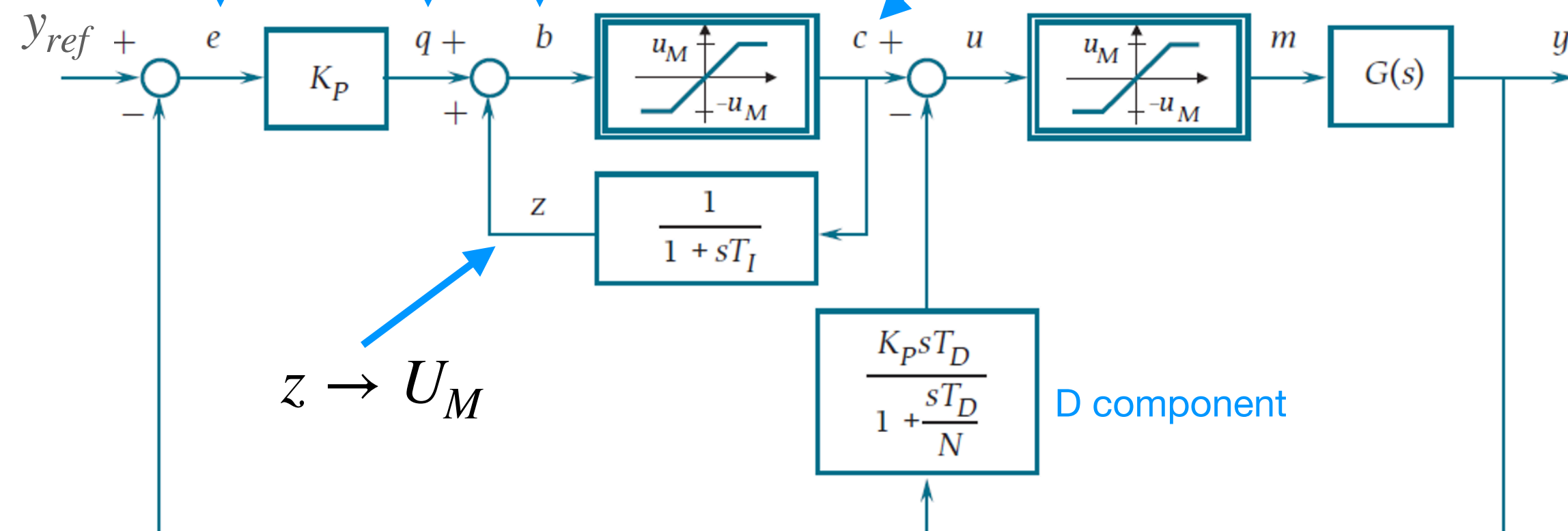
Assumption

$$q > 0, q > U_M$$

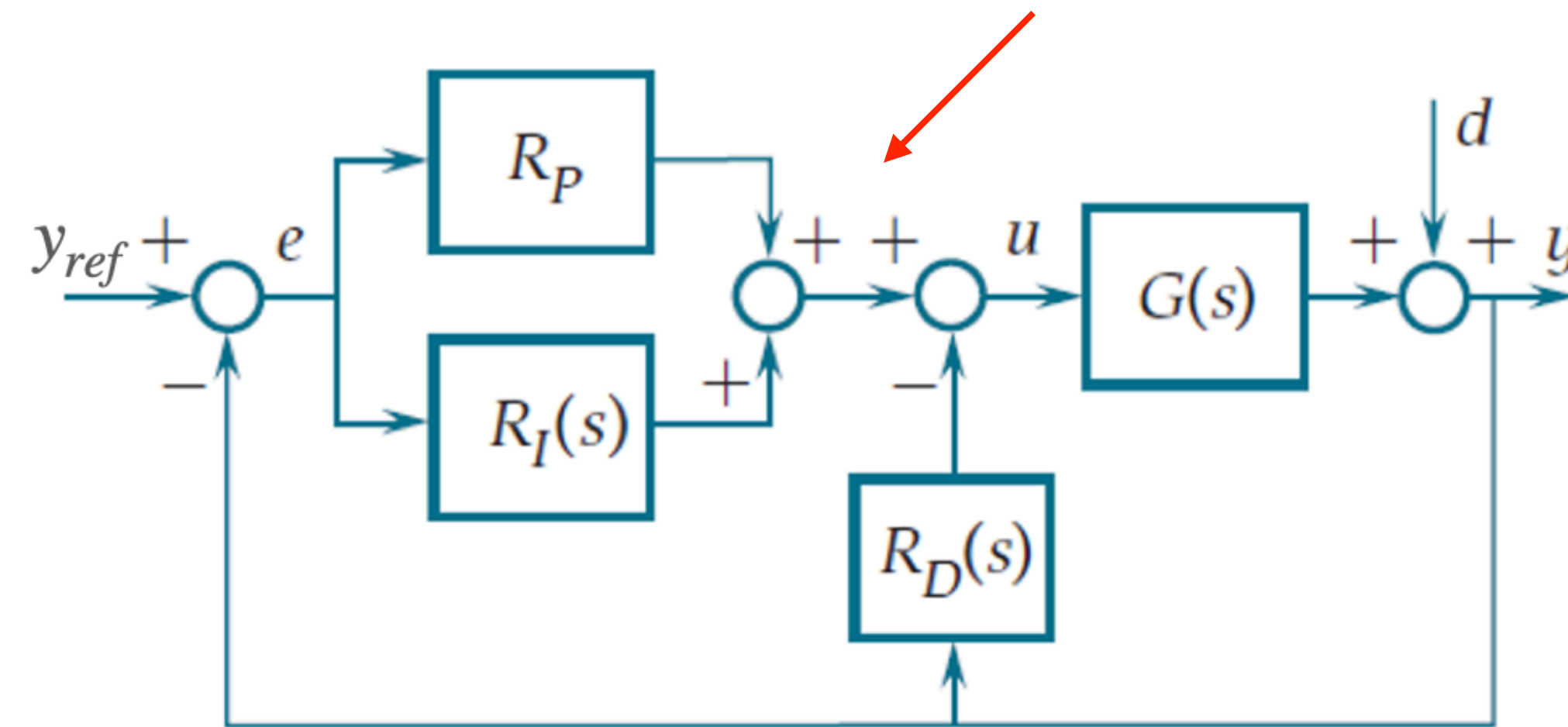
$$b > 0, b > U_M$$

$$c = U_M$$

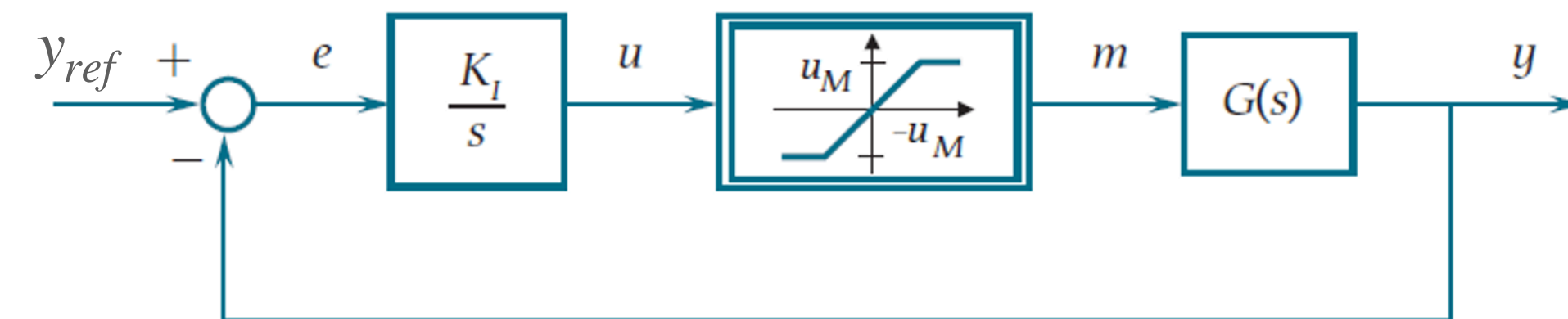
$$e > 0$$



PID Controller

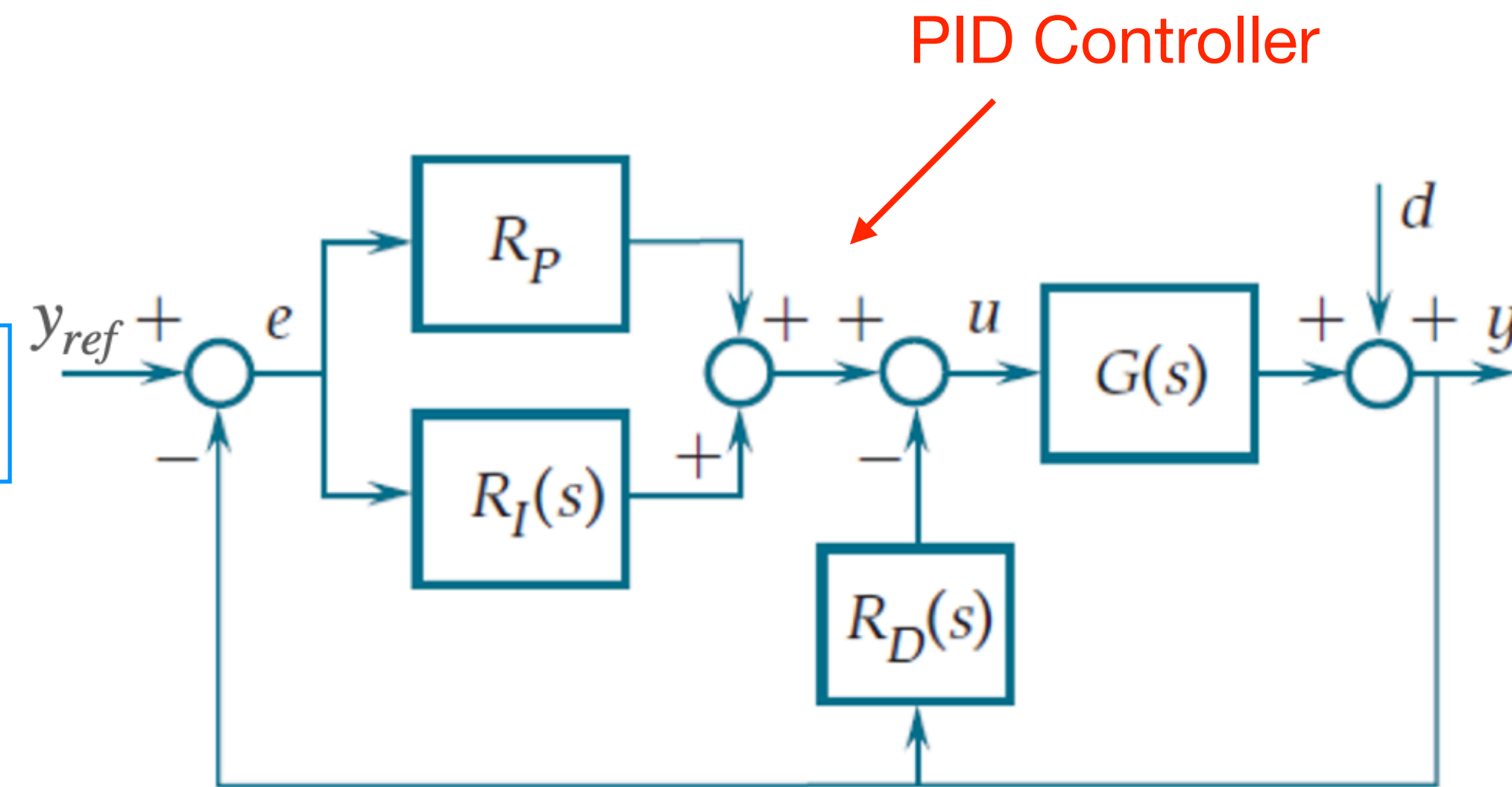
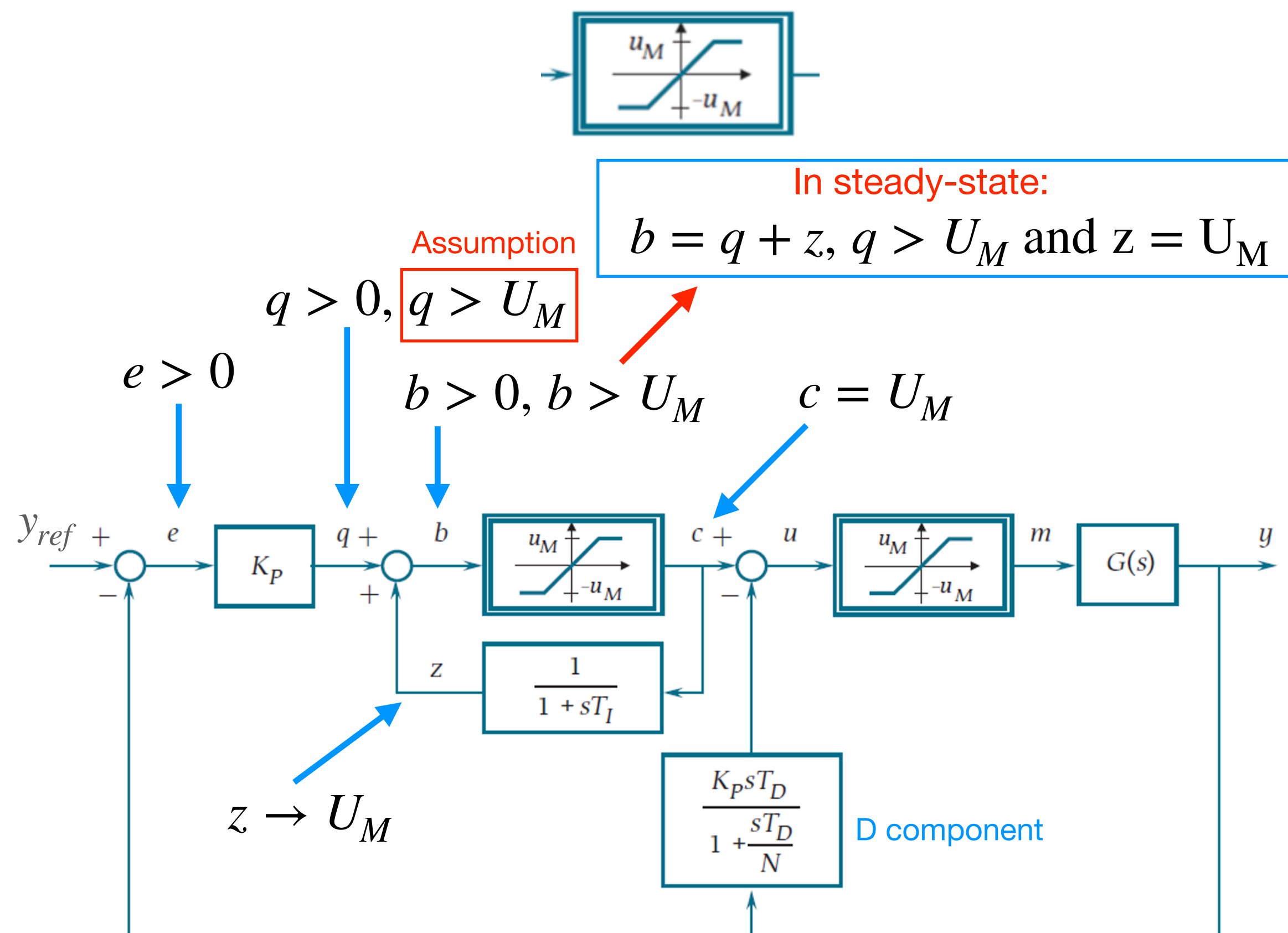


Realistic situation

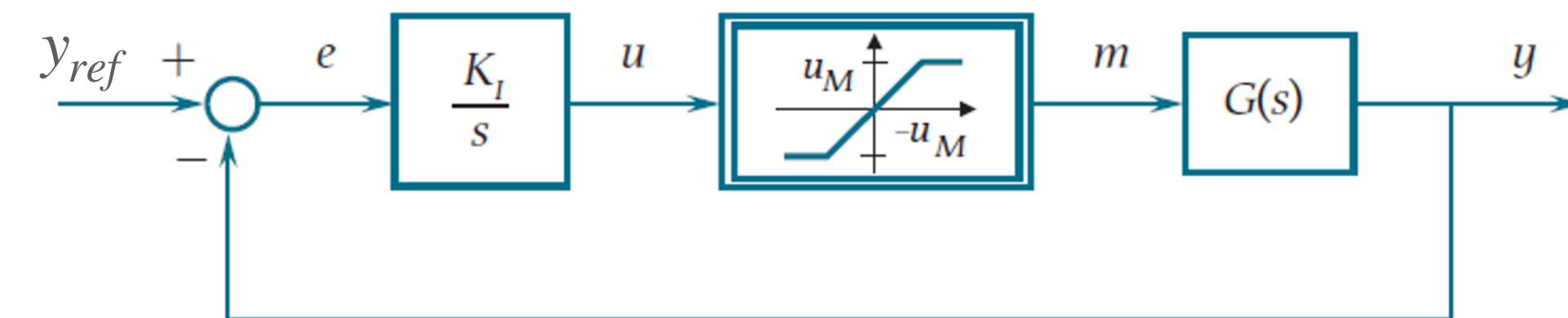


PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

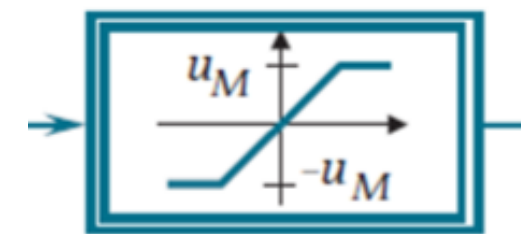


Realistic situation

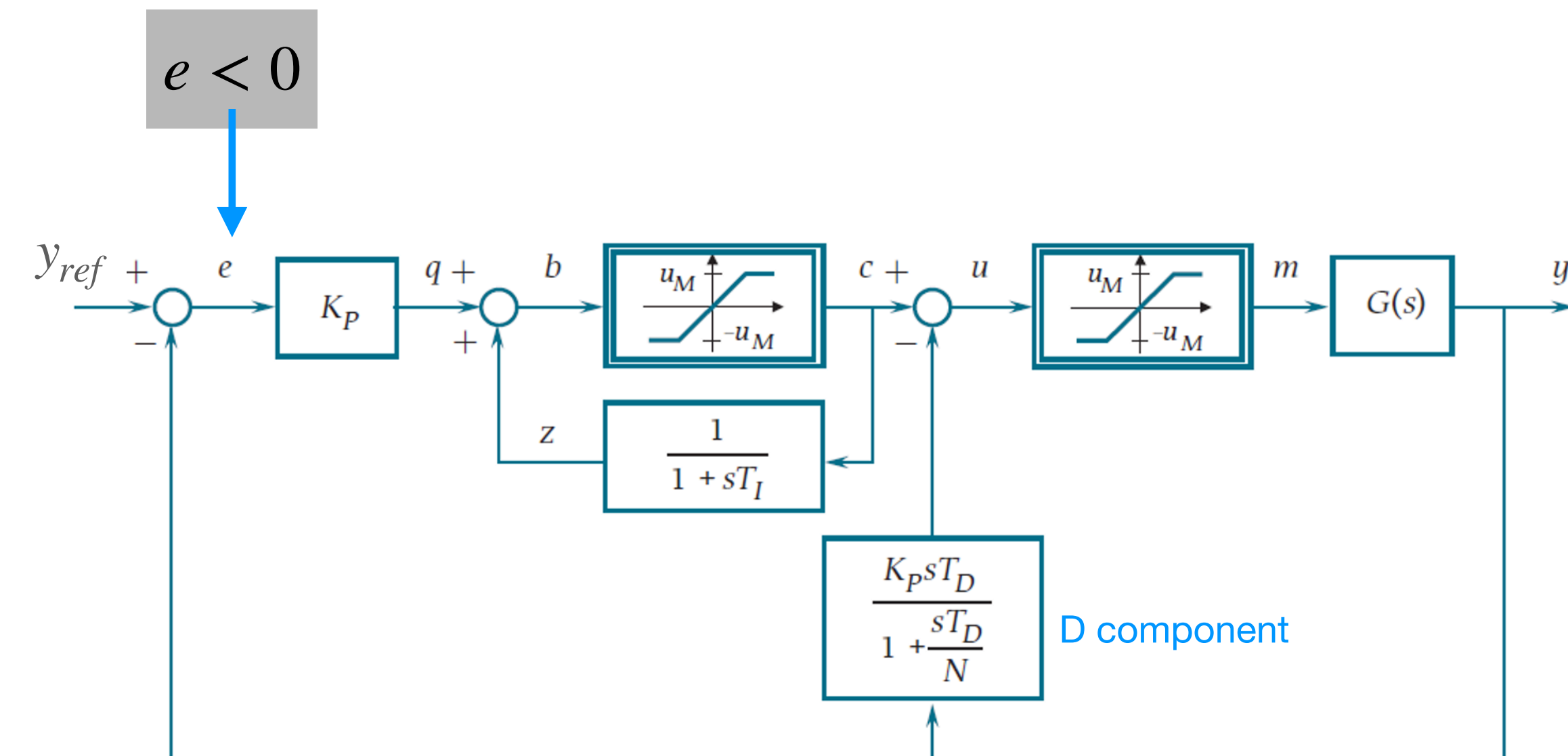
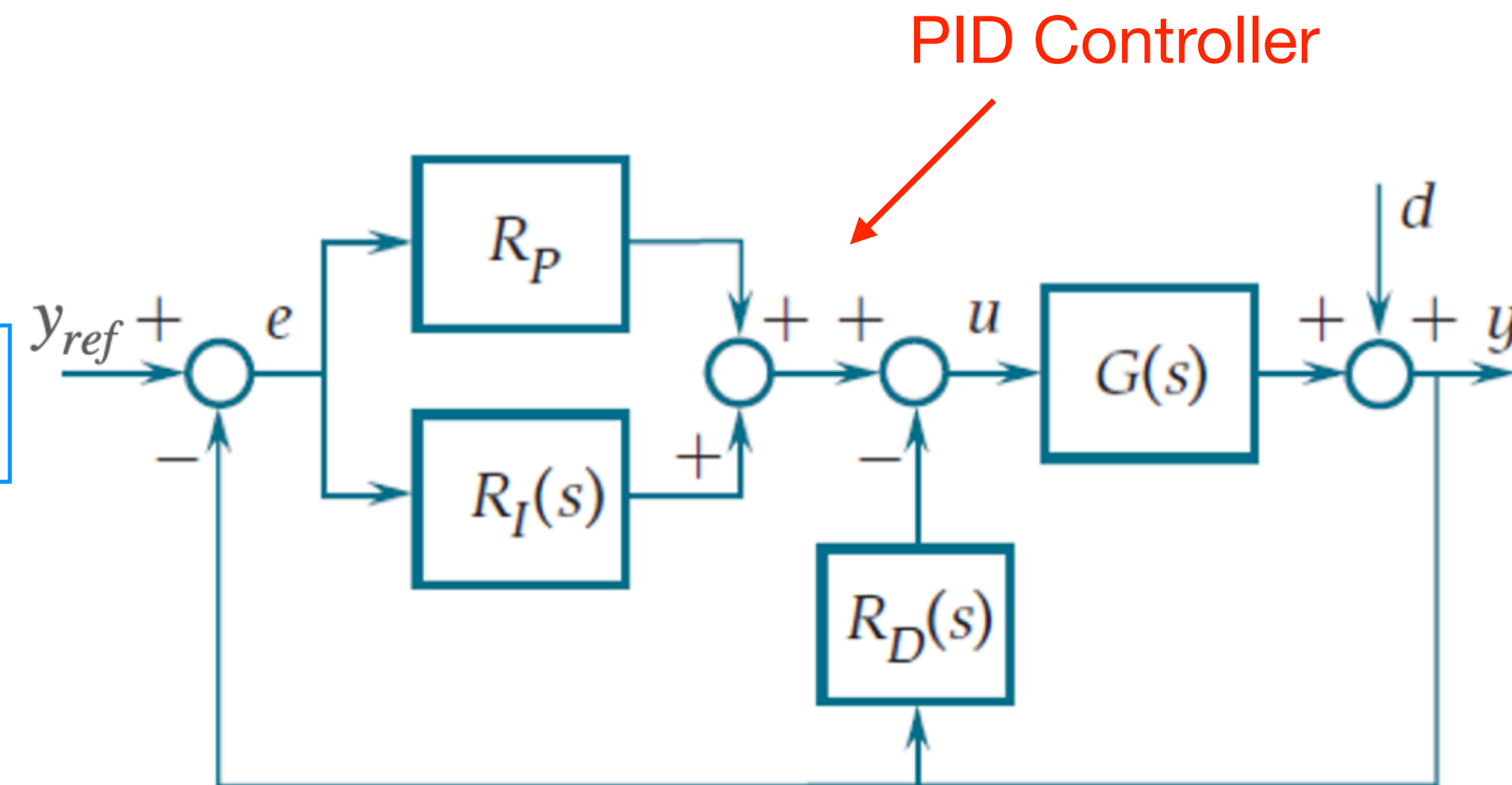


PID Controllers: Wind-up Effect

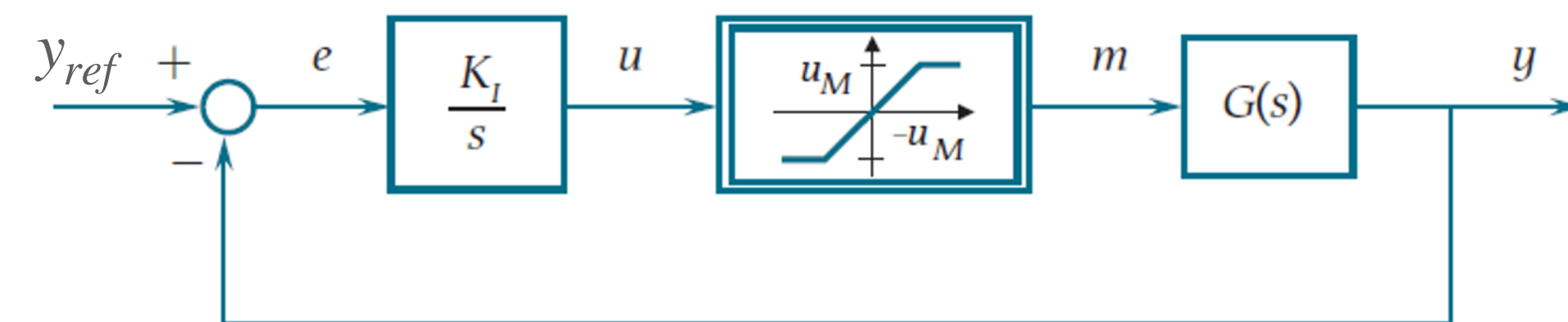
Case 2: The system operates in saturation mode



In steady-state:
 $b = q + z$, $q > U_M$ and $z = U_M$

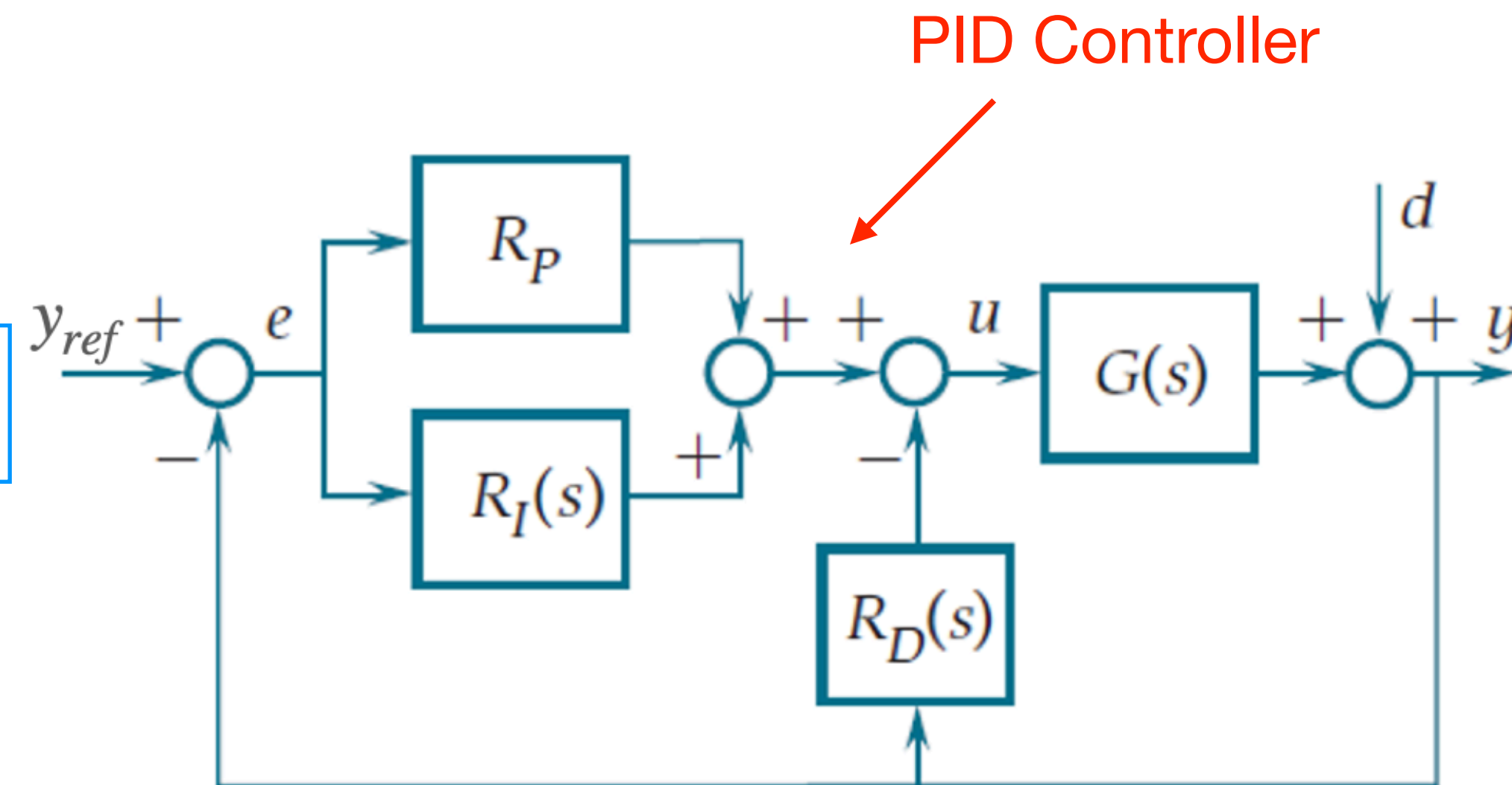
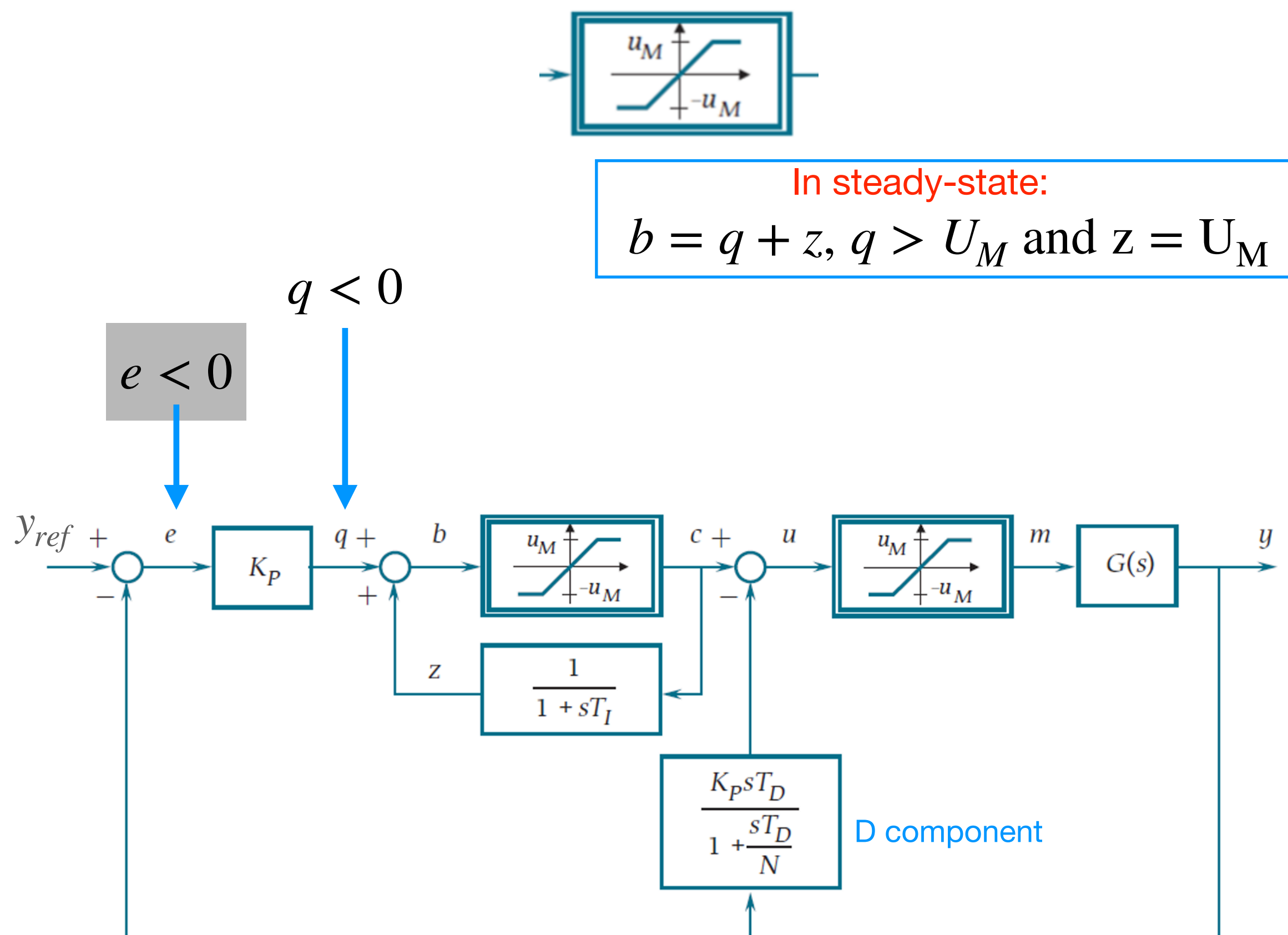


Realistic situation

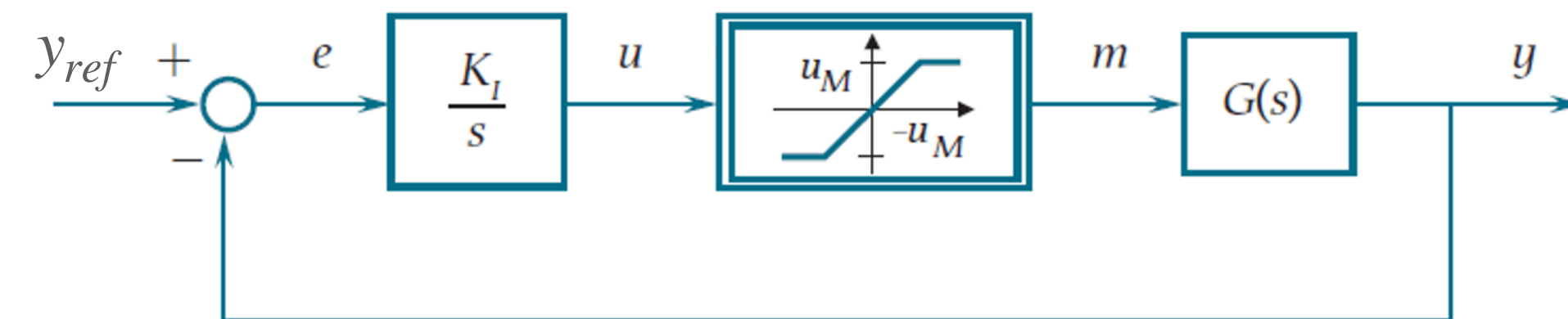


PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

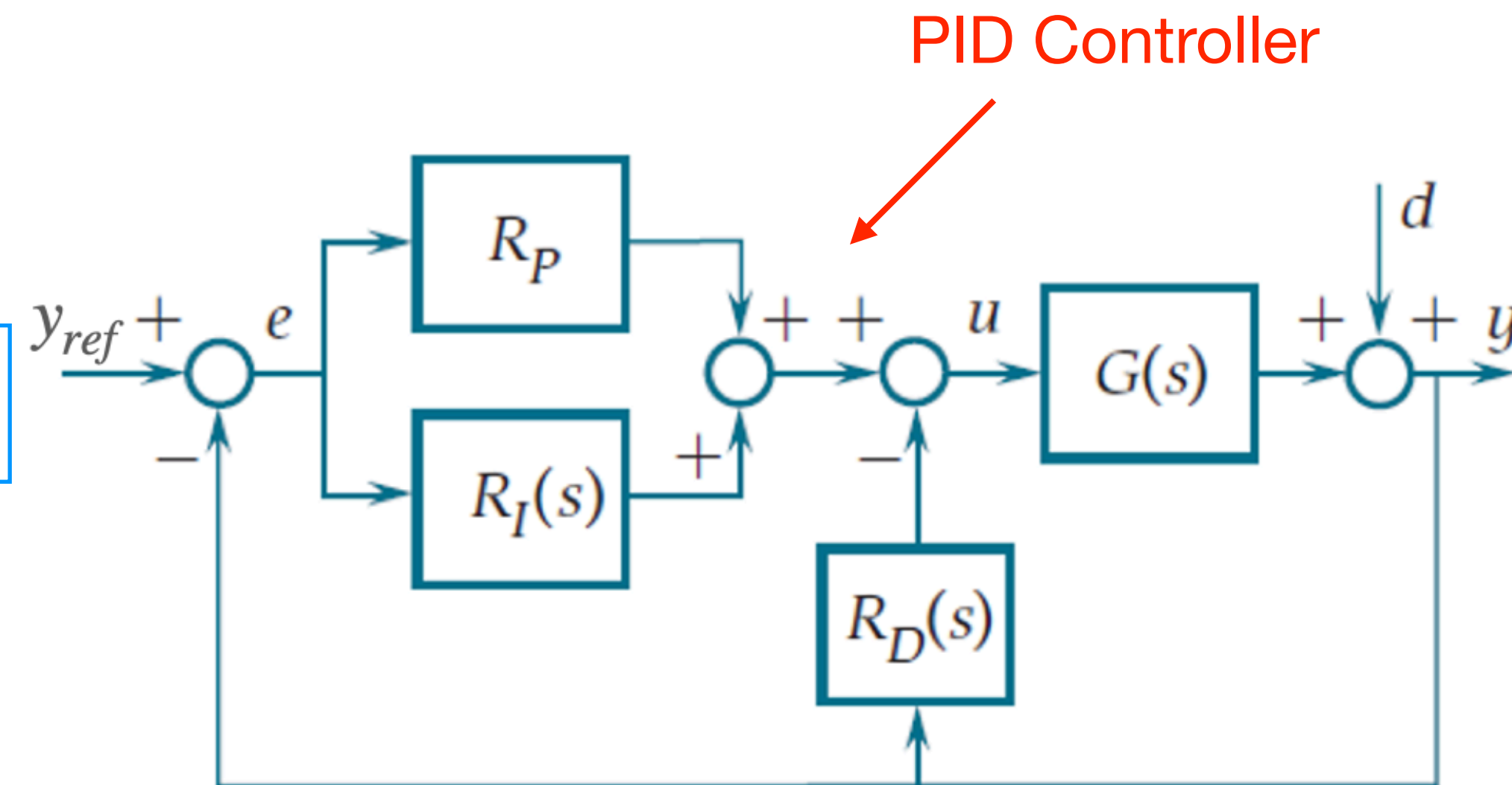
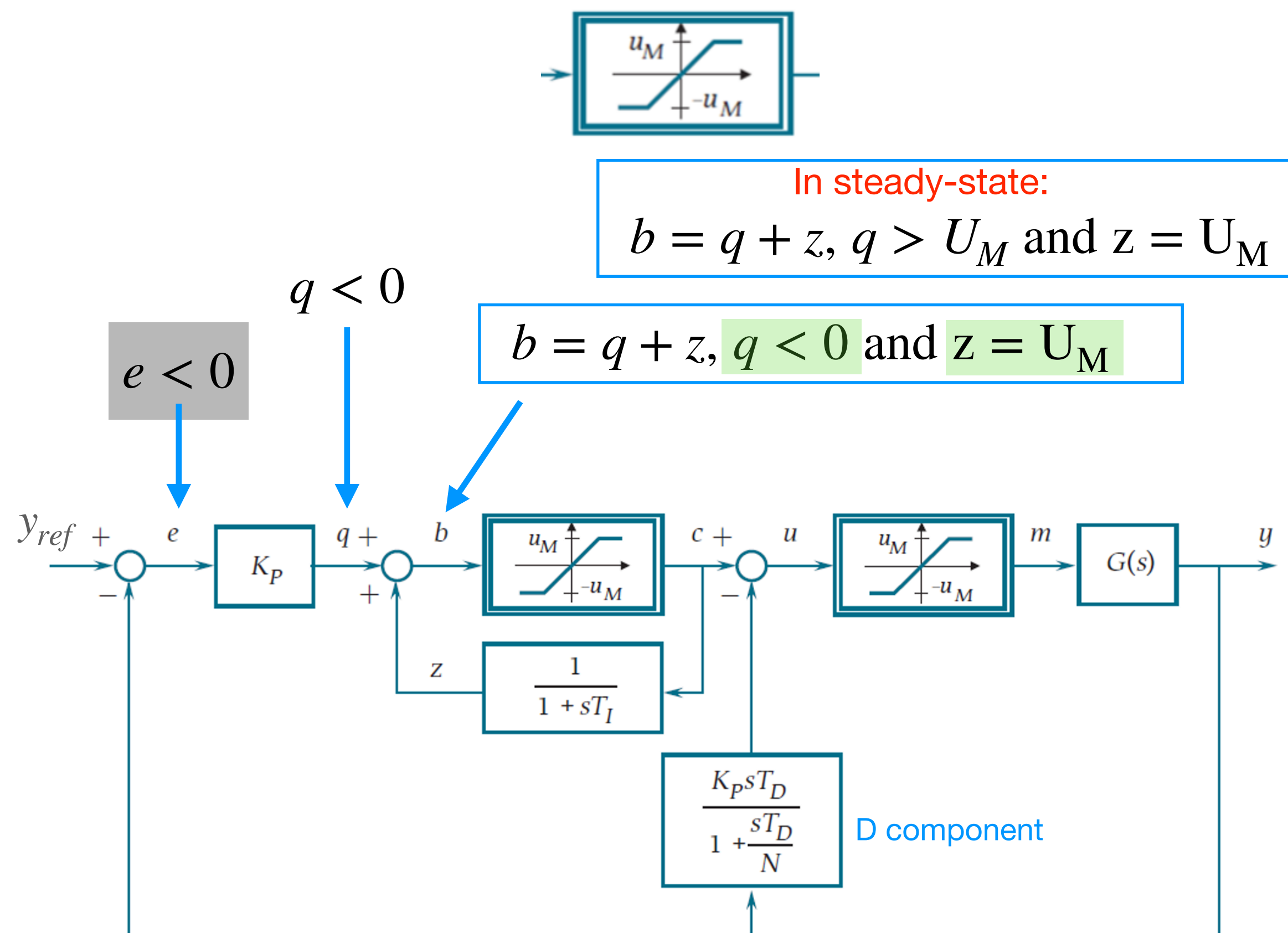


Realistic situation

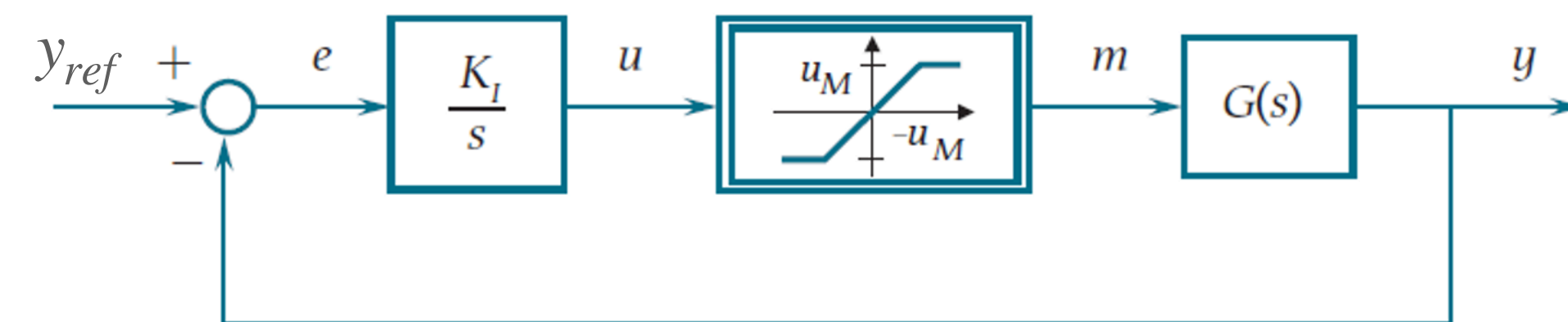


PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

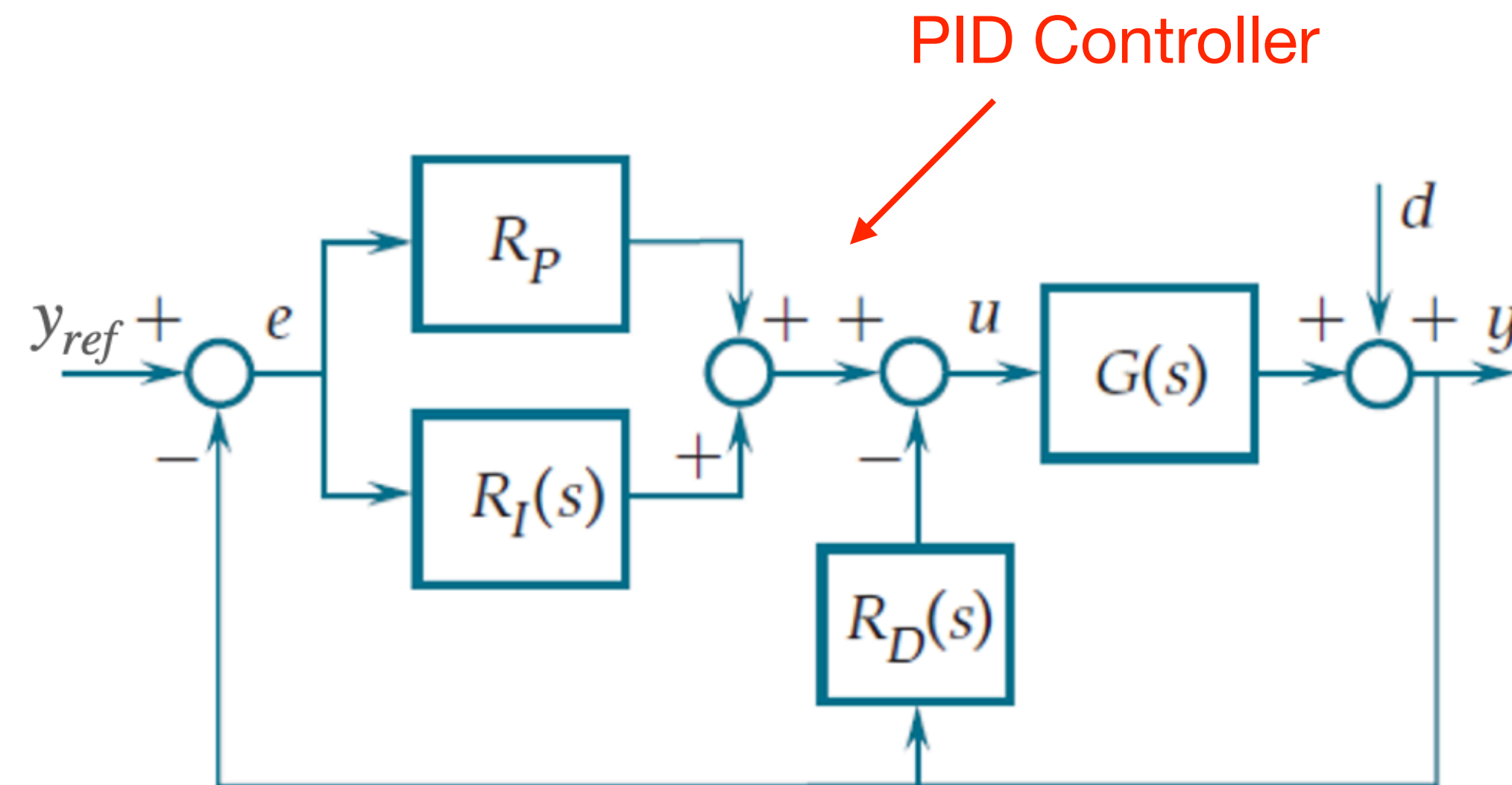
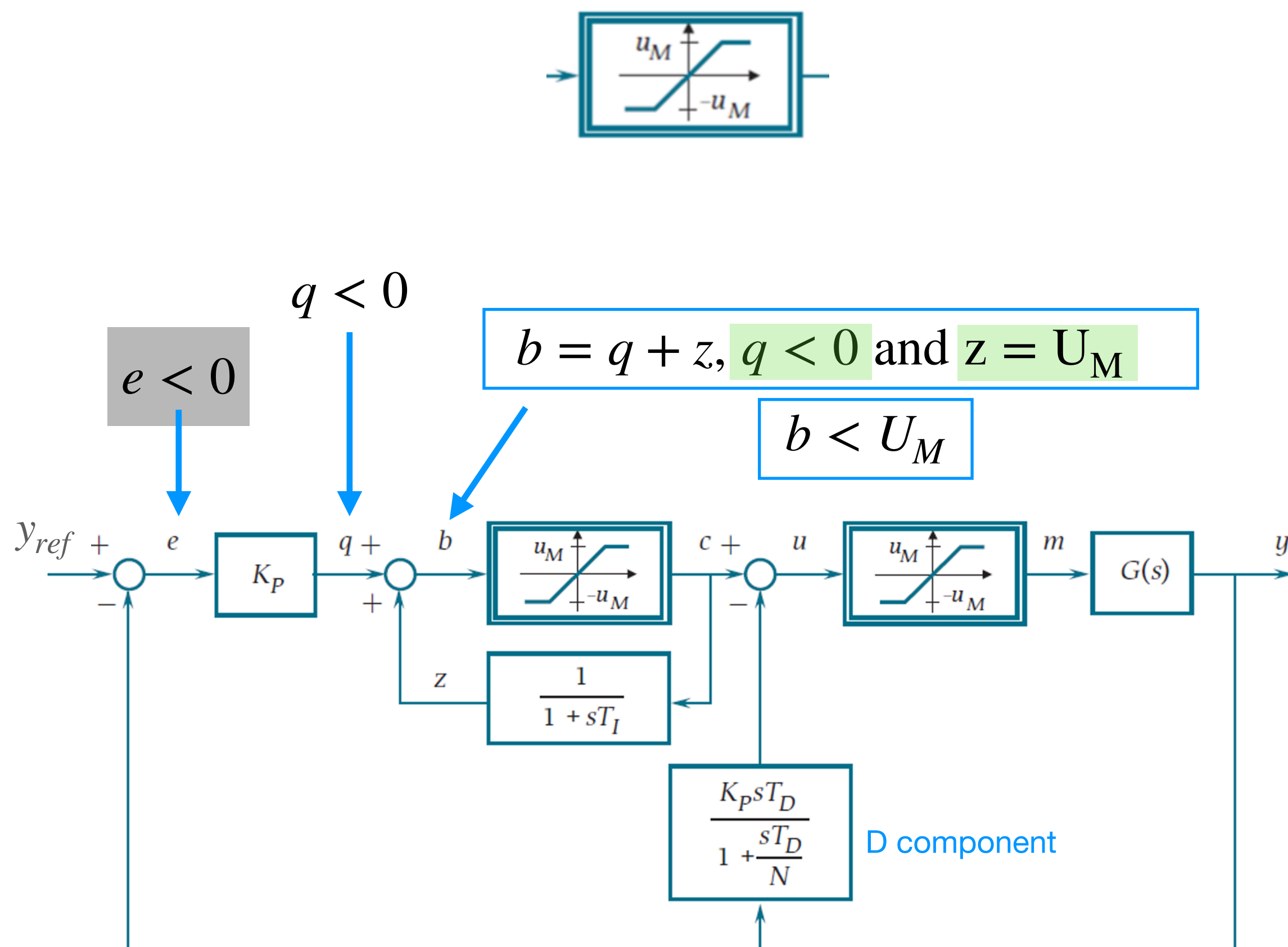


Realistic situation

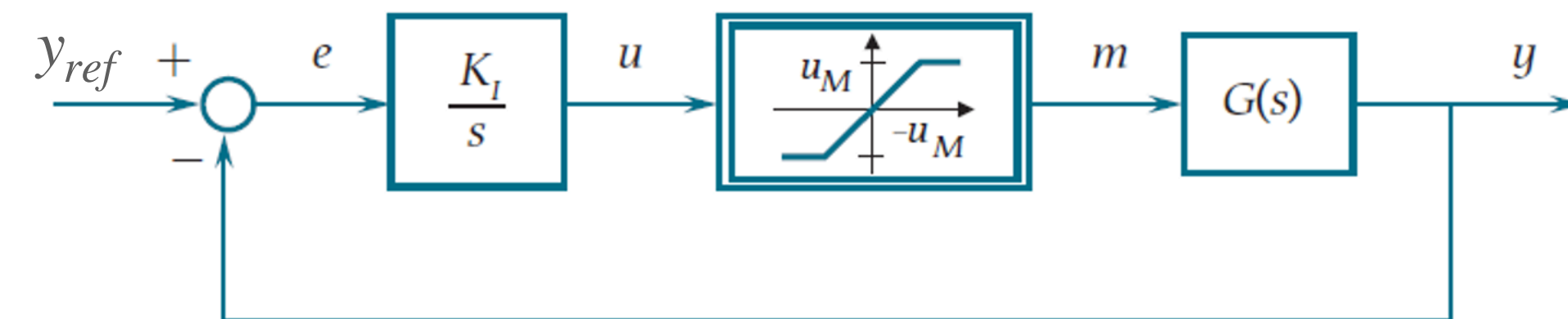


PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

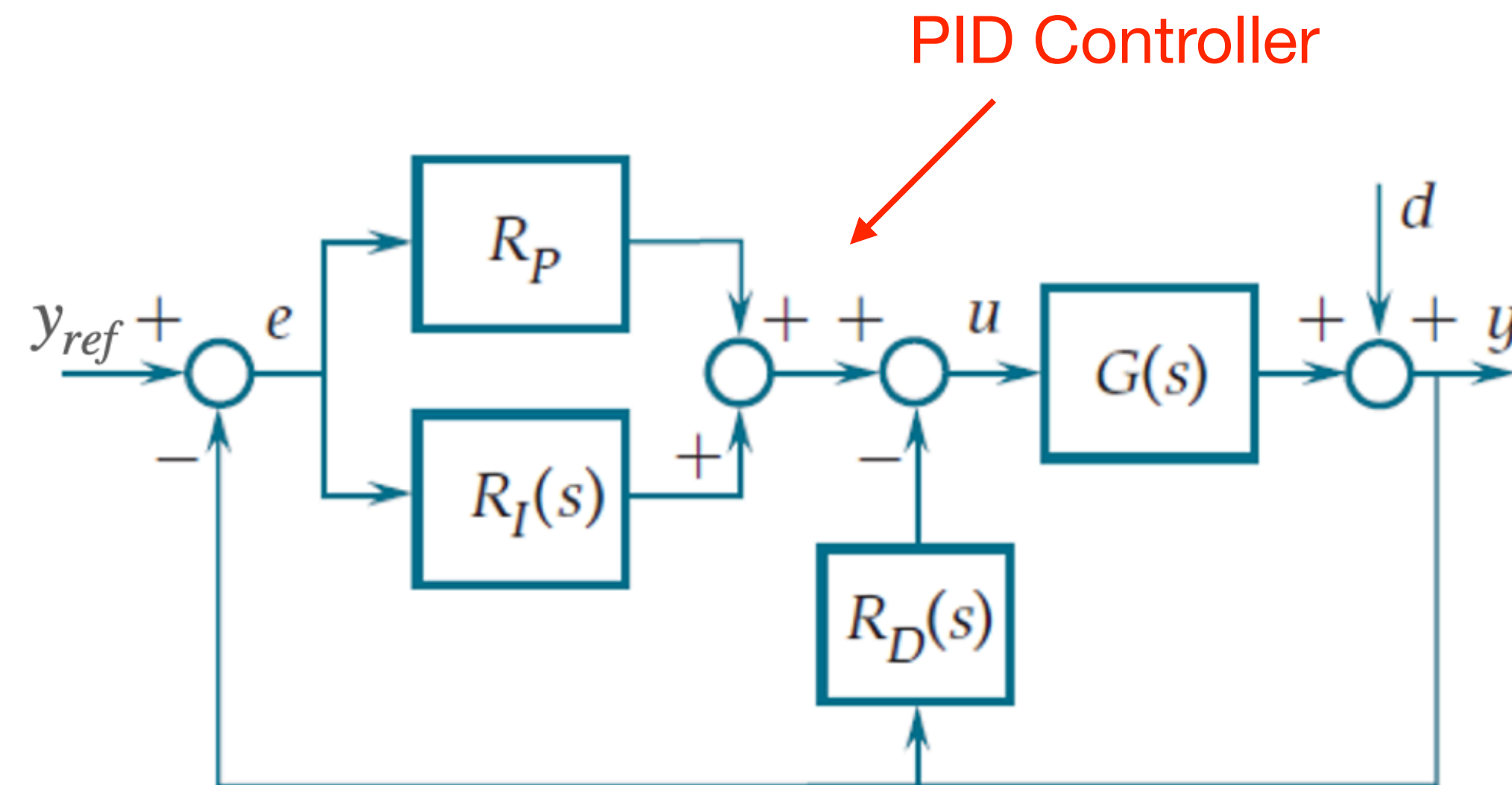
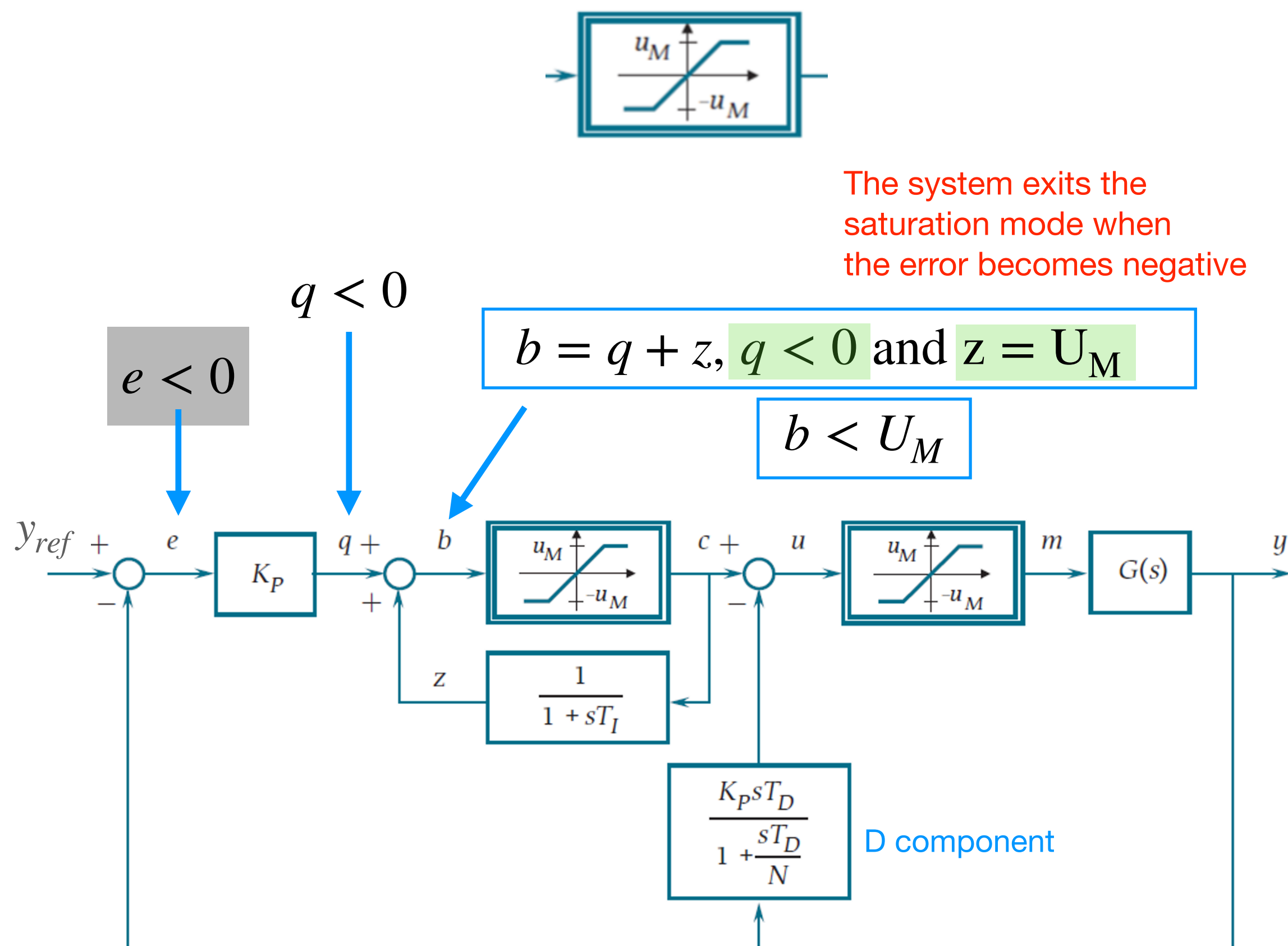


Realistic situation

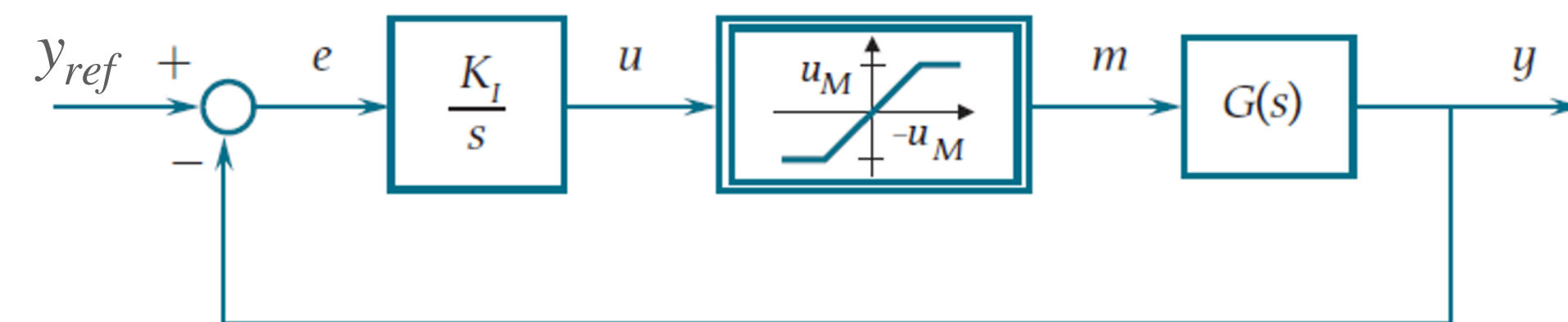


PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

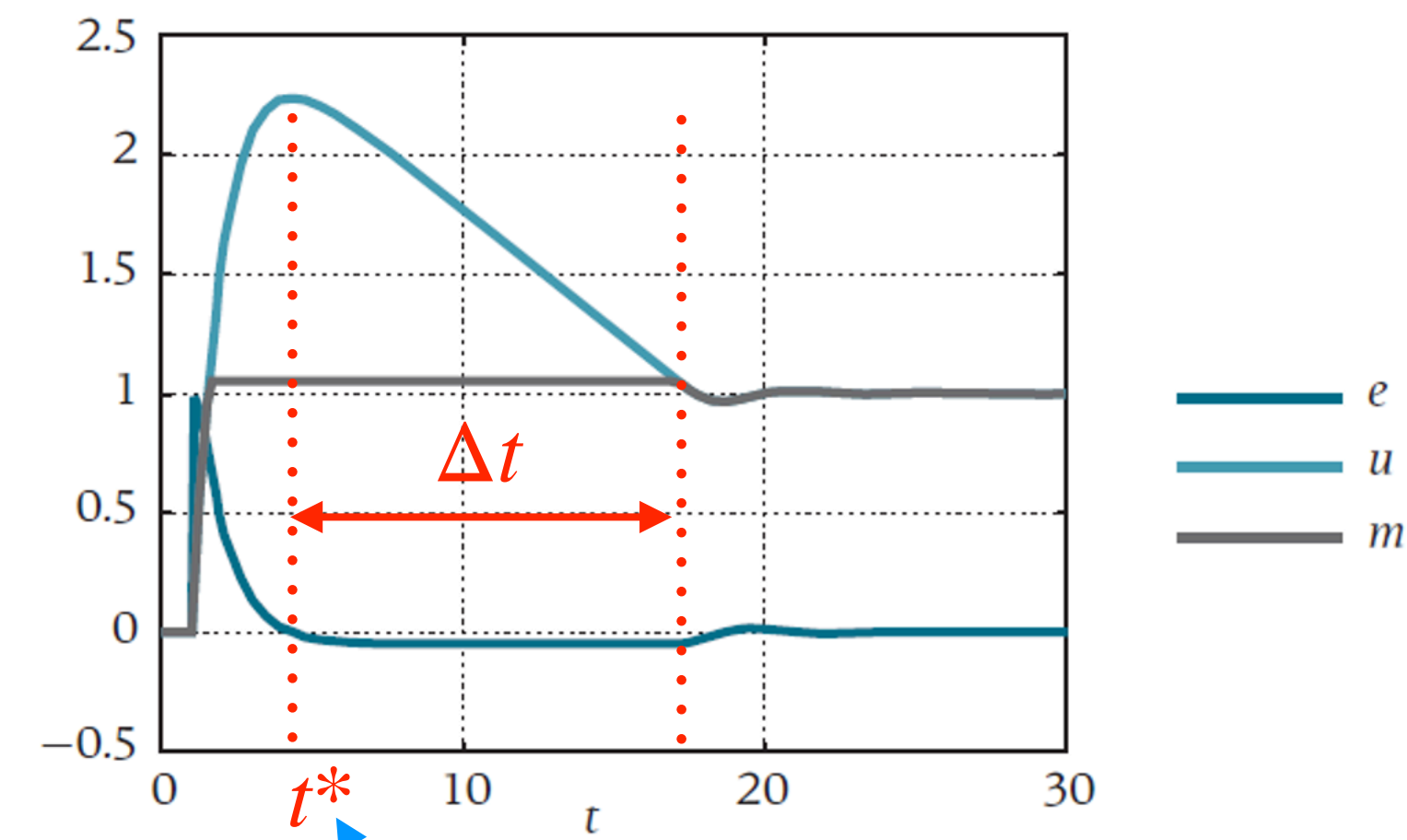
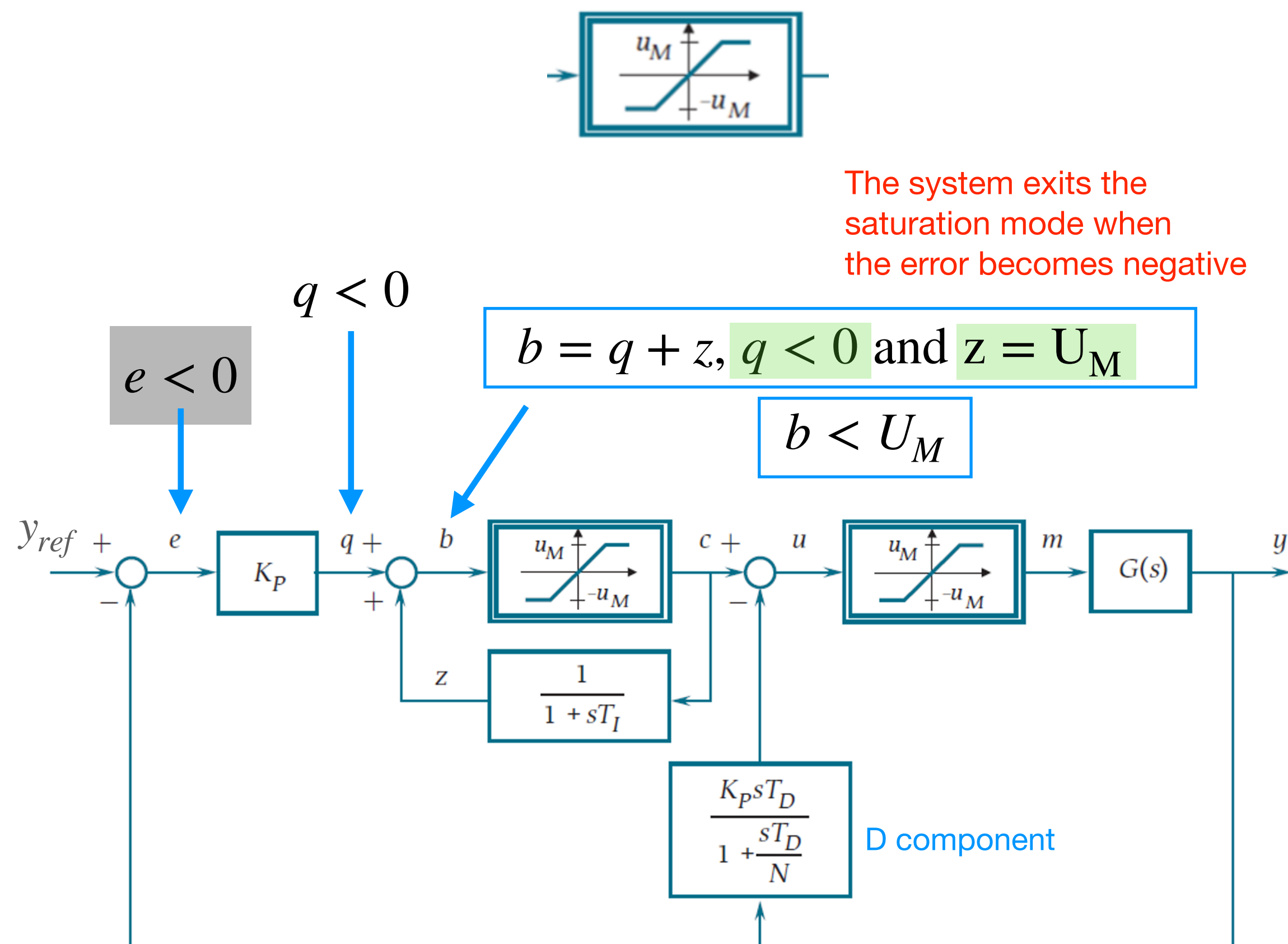


Realistic situation



PID Controllers: Wind-up Effect

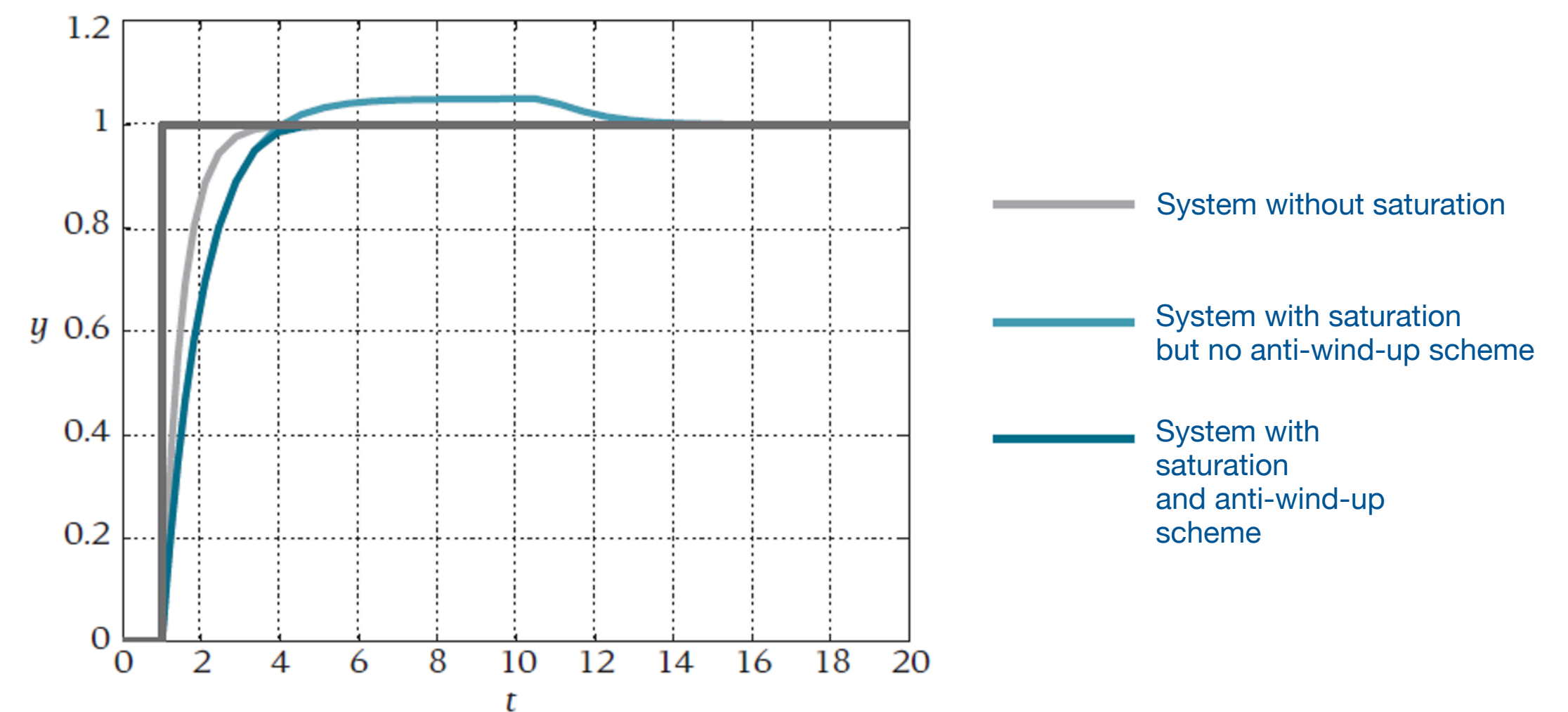
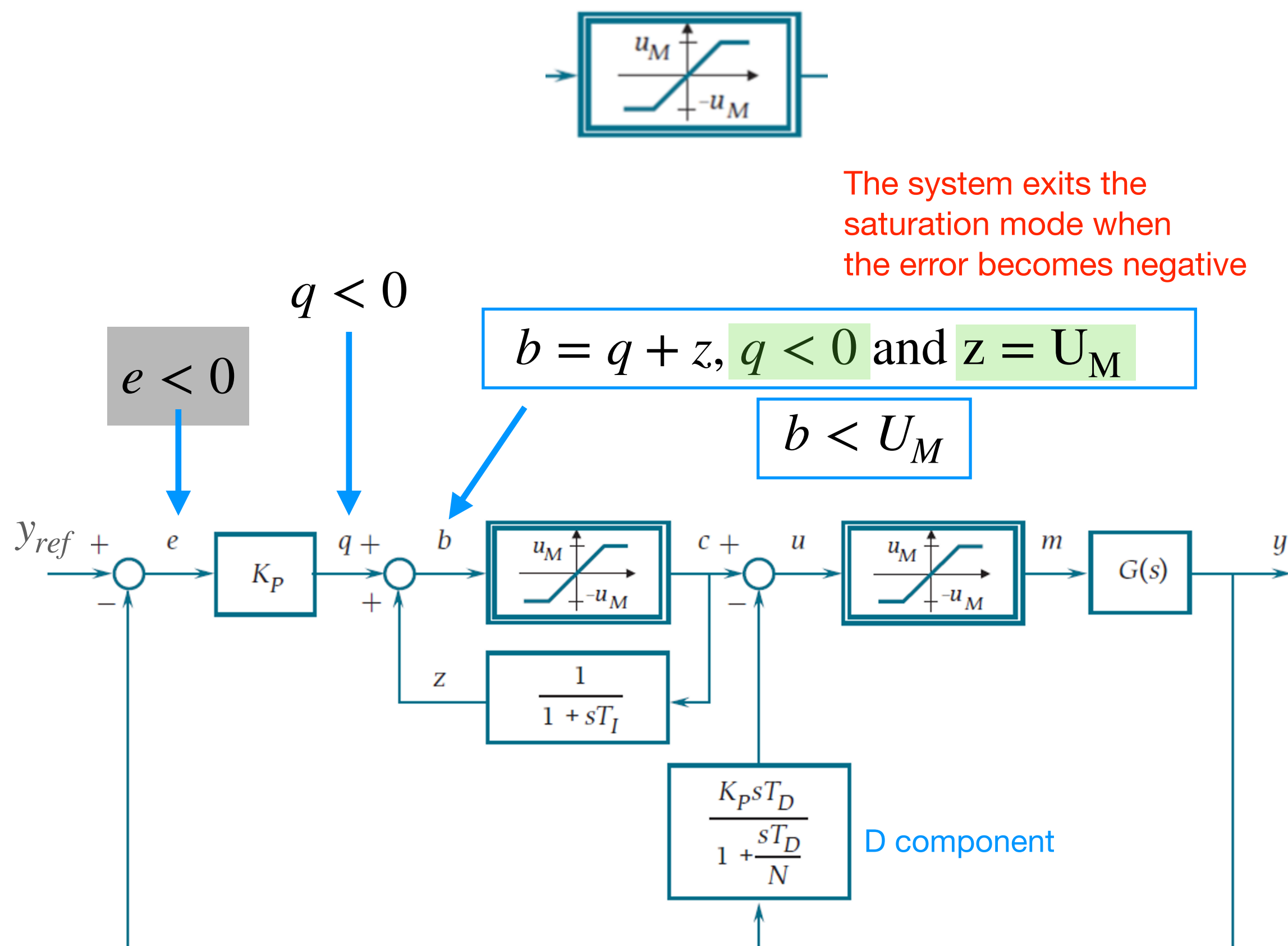
Case 2: The system operates in saturation mode



Ideal situation: the control action exits the saturation mode when e changes its sign (no useless waiting time due to wind-up)

PID Controllers: Wind-up Effect

Case 2: The system operates in saturation mode

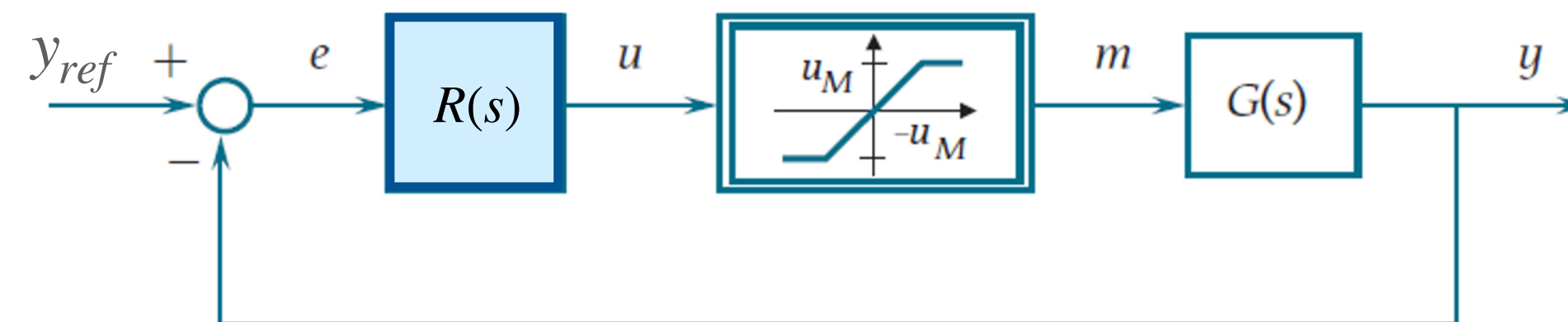


PID Controllers: Wind-up Effect

Generic integral controller

$$R(s) = \frac{N_R(s)}{D_R(s)}, \quad D_R(0) = 0$$

Realistic situation

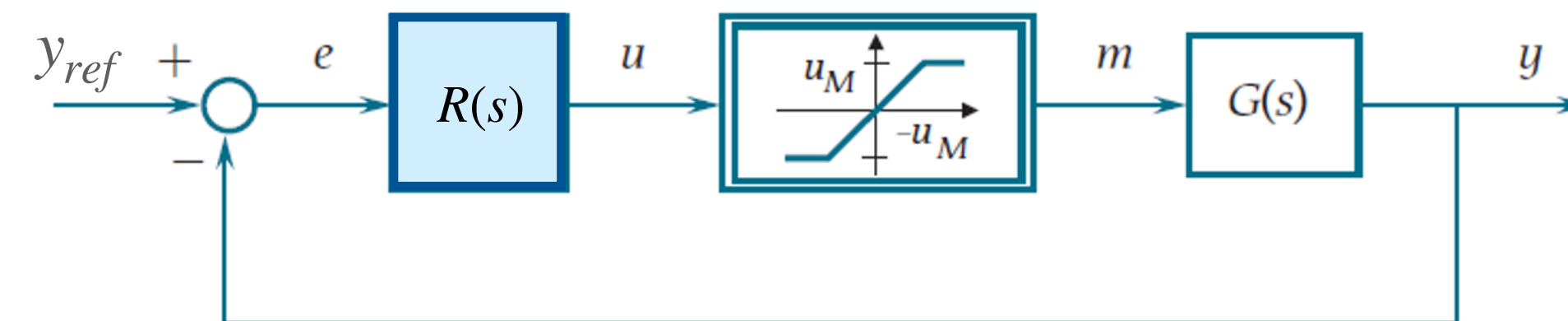


PID Controllers: Wind-up Effect

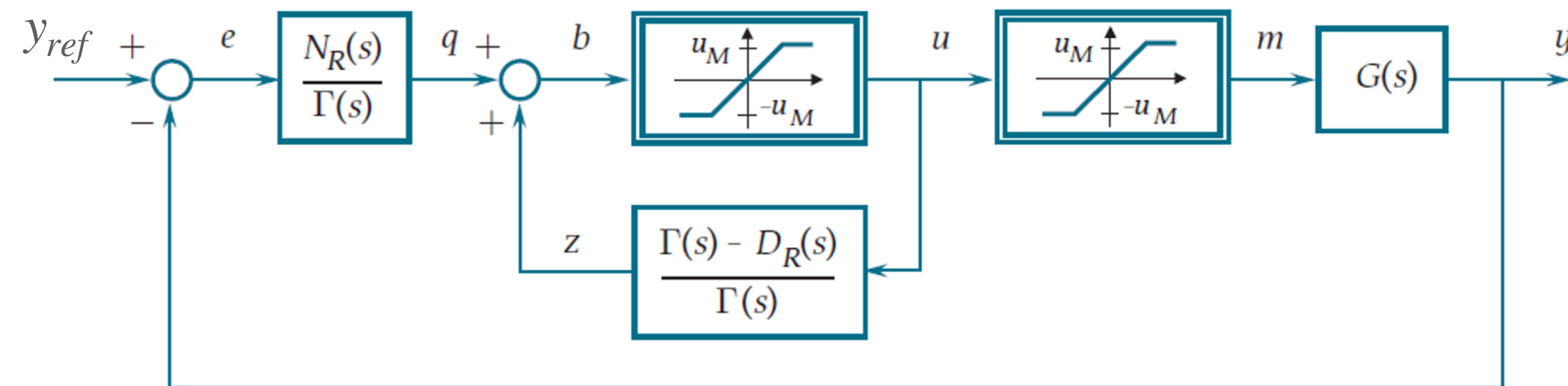
Generic integral controller

$$R(s) = \frac{N_R(s)}{D_R(s)}, \quad D_R(0) = 0$$

Realistic situation



Solution: Anti-wind-up scheme for generic integral controller

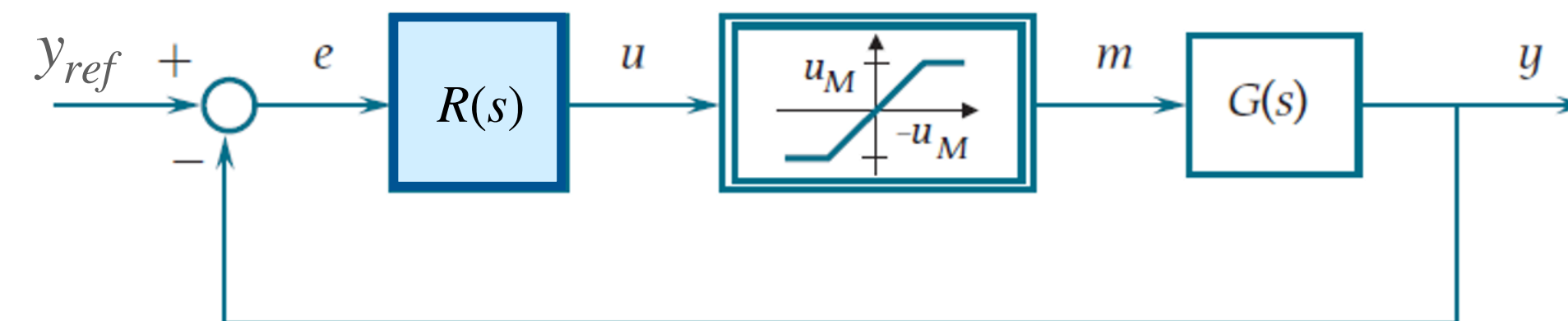


PID Controllers: Wind-up Effect

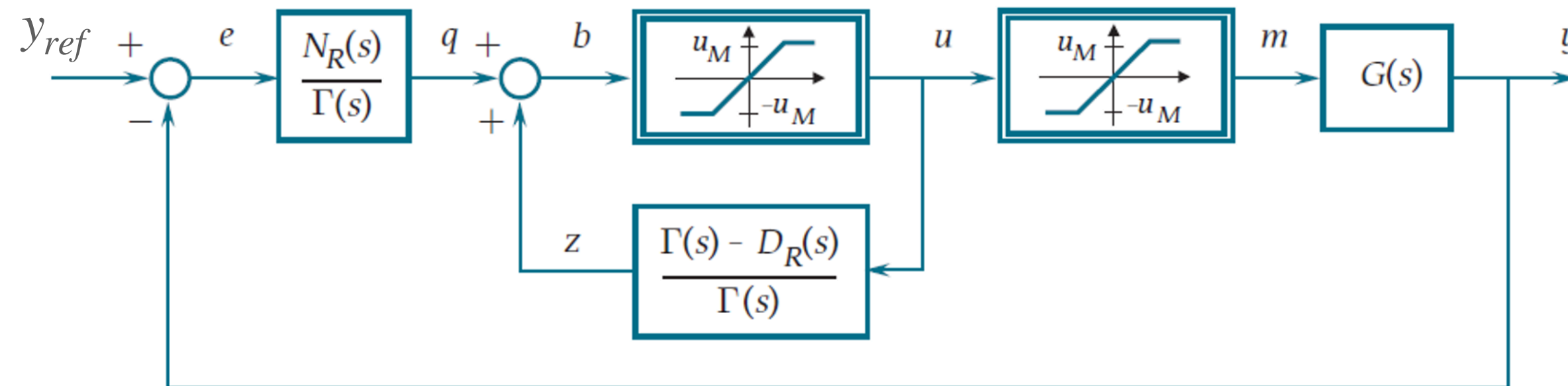
Generic integral controller

$$R(s) = \frac{N_R(s)}{D_R(s)}, \quad D_R(0) = 0$$

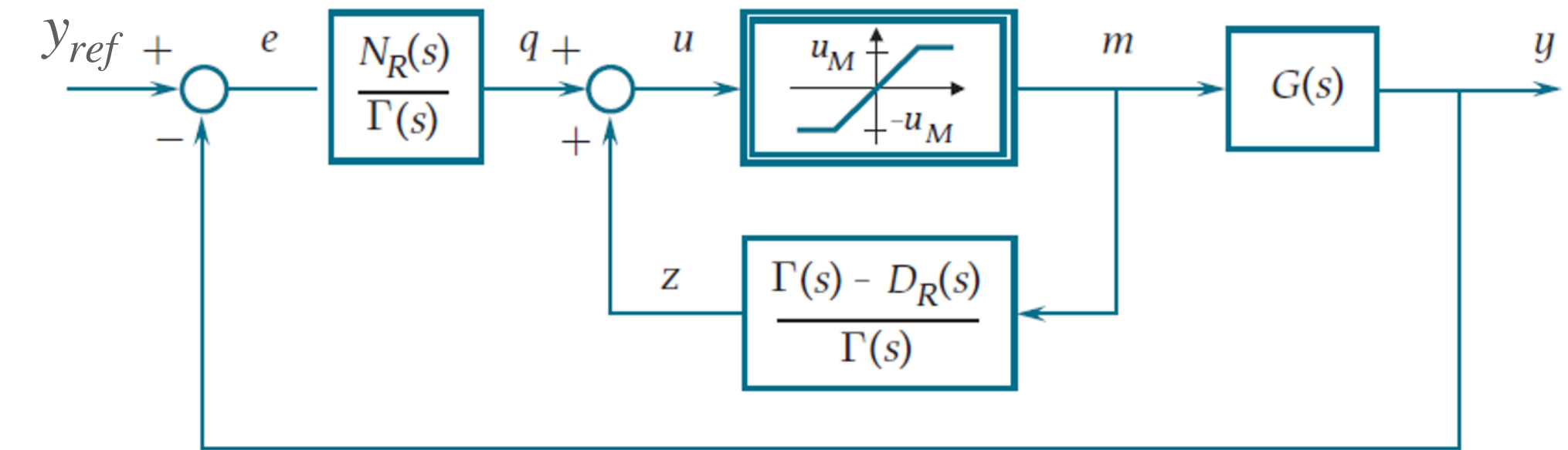
Realistic situation



Solution: Anti-wind-up scheme for generic integral controller



Alternative (accessible signal downstream of saturation)



General Information

Prof. Antonella Ferrara

<https://scholar.google.com/citations?user=r5JuMskAAAAJ&hl=en>

Course Teaching Material:

KIRO UNIPV <https://elearning.unipv.it/>

- [504462 - PROCESS CONTROL 2025-26 - PROF.SSA FERRARA ANTONELLA](#)

Lecture Time-table:

<http://www-3.unipv.it/ingserv/orario2526/1sem/insegnamenti/ProCont.html>

Exams:

<https://studentionline.unipv.it/esse3/Home.do>

<https://kirotesting.unipv.it/>

Introduction

- Program of the course:

Advanced SISO control schemes:

Pre-filters and parallel compensators, two degrees of freedom control schemes, compensation of measurable disturbances, systems with delays and Smith Predictor, Padé approximation, decoupling in the frequency domain, control of open loop unstable systems.

Advanced MIMO control schemes:

Decoupling based control schemes, decentralized control, relative gain array.

PID controllers:

Features and properties. Rules for the empirical calibration. Wind-up and anti wind-up schemes.

Digital control:

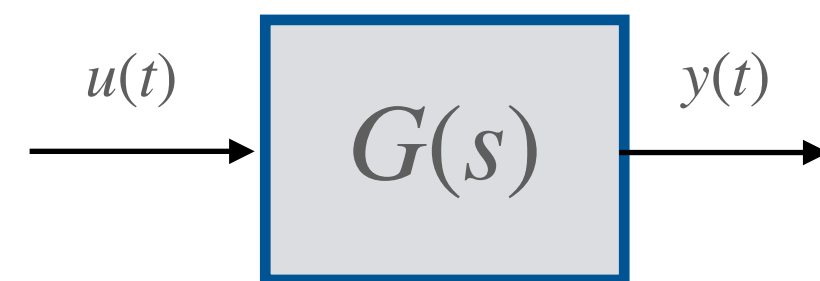
Discrete-time systems. The concept of equilibrium for discrete-time systems. Stability of linear time-invariant discrete-time systems. Jury test. Digital control schemes. Zeta transform and its properties. Transfer functions in the z domain. Sampling and aliasing. Choice of the sampling time. Zero-order-Hold. Discretization of continuous-time controllers. Bilinear transformation, Euler and Tustin methods.

Introduction

- Some of the figures in these slides, kindly provided by McGraw-Hill, are those of the Textbook:



Discrete-time Systems



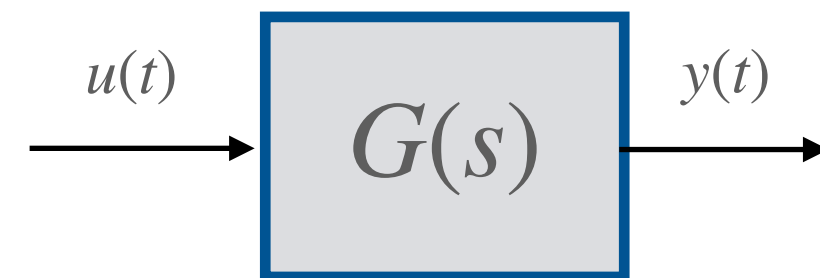
$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

Discrete-time Systems



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

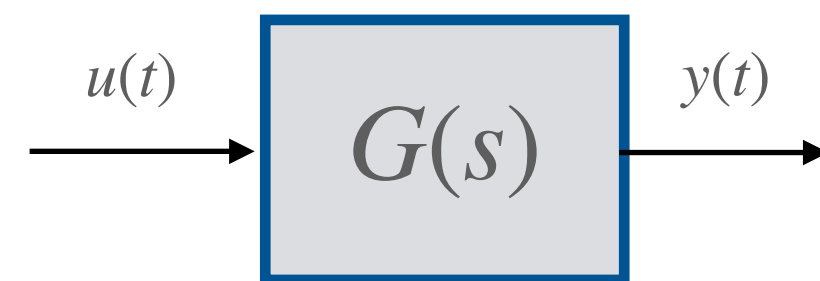


$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

Discrete-time Systems



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

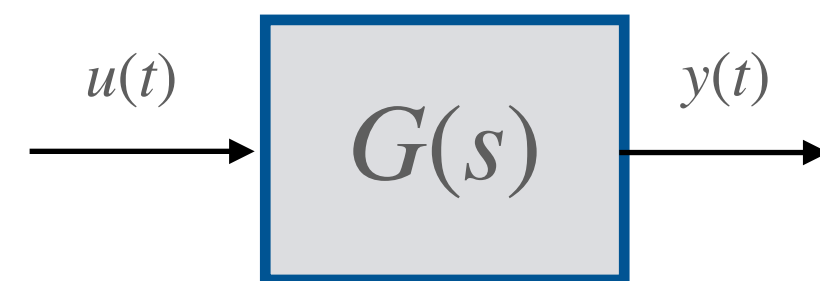
Discrete-time System

$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Discrete-time Systems

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$

$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

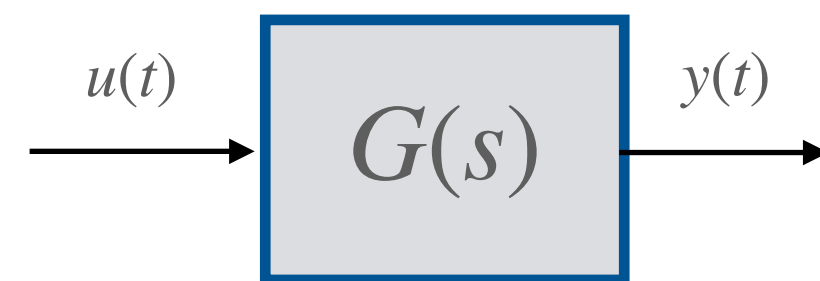
Discrete-time System

$$\begin{aligned} x(k+1) &= f(x(k), u(k), k) \\ y(k) &= g(x(k), u(k), k) \end{aligned}$$

Discrete-time Systems

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$

$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

$$x(k+1) = f(x(k), u(k), k)$$

$$y(k) = g(x(k), u(k), k)$$

$$x(1) = f(x_0, u(0)),$$

$$x(2) = f(x(1), u(1)),$$

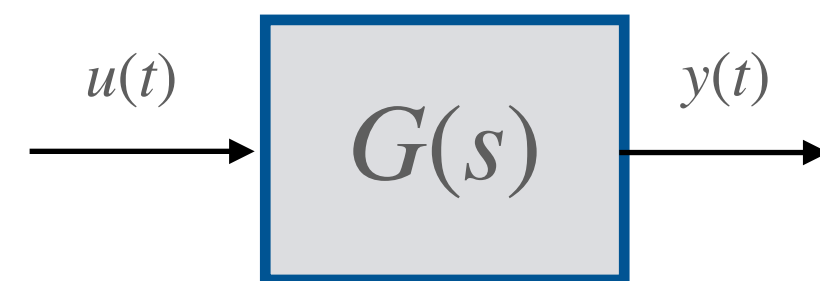
$$\vdots$$

$$x(i) = f(x(i-1), u(i-1))$$

Discrete-time Systems

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$

$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

$$x(k+1) = f(x(k), u(k), k)$$

$$y(k) = g(x(k), u(k), k)$$

$$x(1) = f(x_0, u(0), 0),$$

$$x(2) = f(x(1), u(1), 1),$$

$$\vdots$$

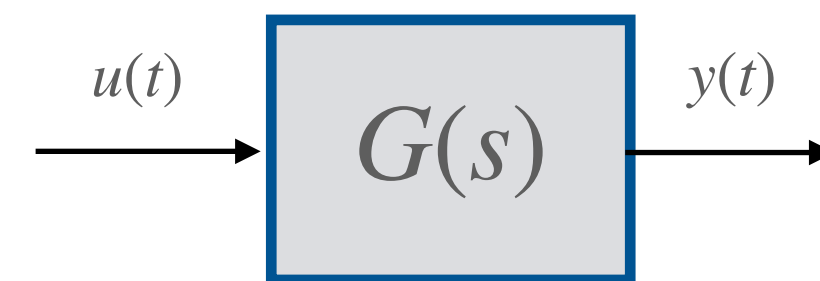
$$x(i) = f(x(i-1), u(i-1), i)$$

$$y(k) = g(x(k), u(k), k), \quad k \geq 0$$

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

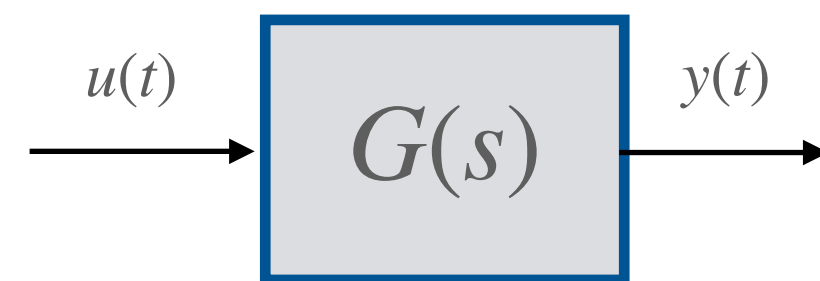
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

MIMO

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}$$
$$y(k) \in \mathbb{R}$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

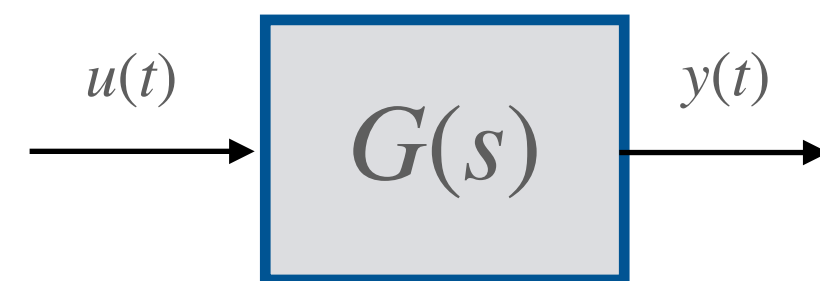
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

SISO

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

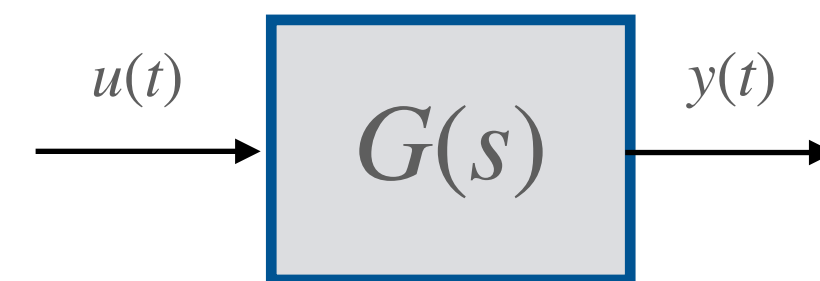
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Proper

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

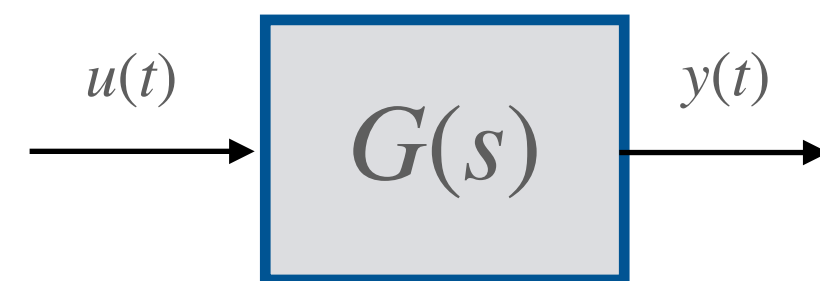
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), k)$$

Strictly proper

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

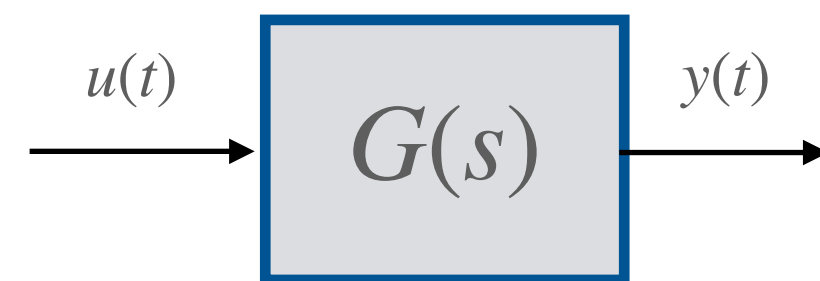
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Time-varying

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

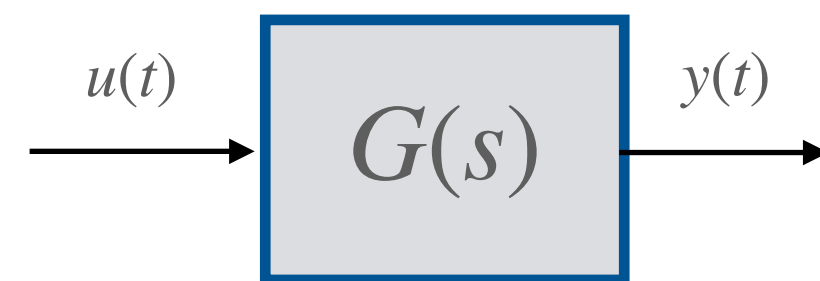
$$x(k+1) = f(x(k), u(k))$$
$$y(k) = g(x(k), u(k))$$

Time-invariant

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

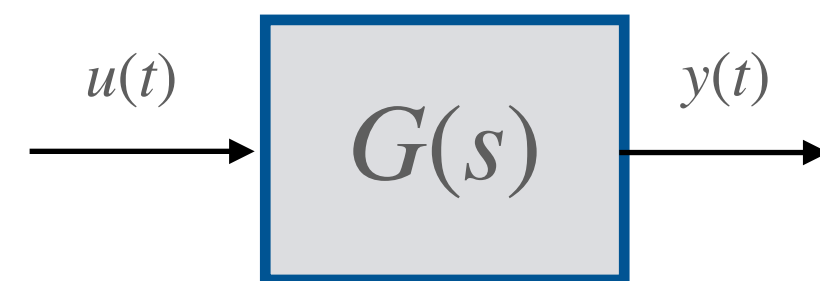
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Nonlinear

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

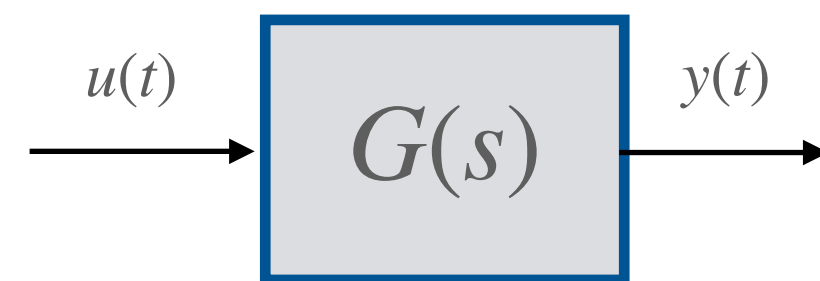
$$x(k+1) = A(k)x(k) + B(k)u(k)$$
$$y(k) = C(k)x(k) + D(k)u(k)$$

Linear Time-varying

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

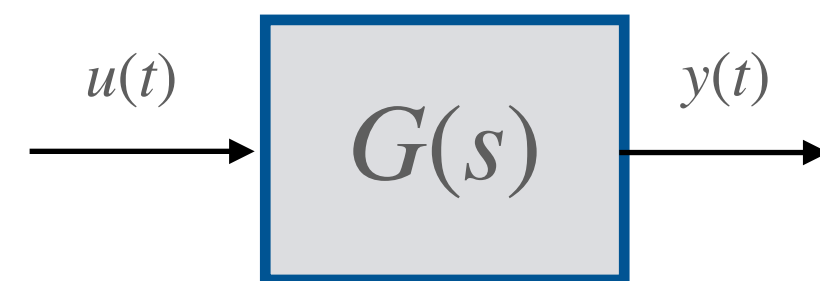
$$x(k+1) = Ax(k) + Bu(k)$$
$$y(k) = Cx(k) + Du(k)$$

Linear Time-invariant

Discrete-time Systems: **Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

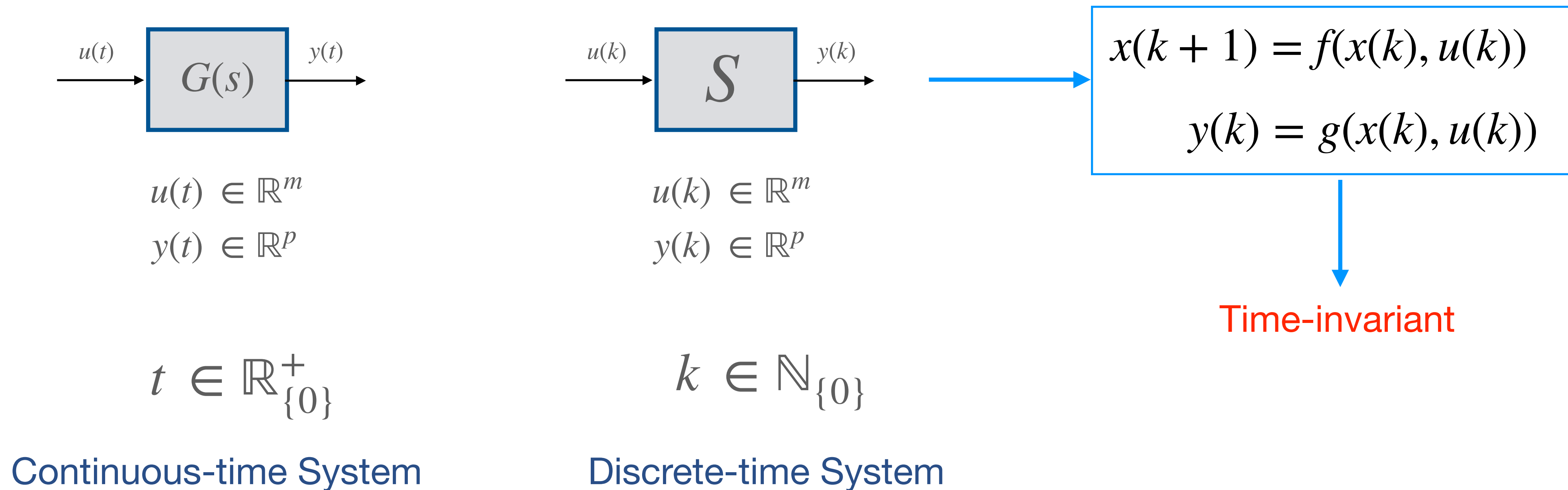
$$x(k+1) = f(x(k), u(k))$$
$$y(k) = g(x(k), u(k))$$

Time-invariant

Discrete-time Systems: **Equilibrium**

Given:

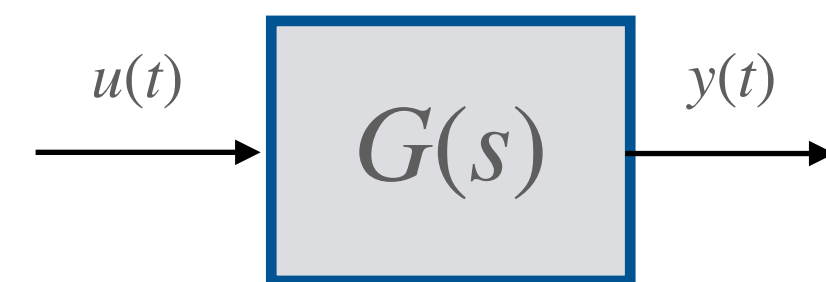
$$u(k) = \bar{u}, \quad \forall k \geq 0$$



Discrete-time Systems: **Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$



$$\begin{aligned} u(t) &\in \mathbb{R}^m \\ y(t) &\in \mathbb{R}^p \end{aligned}$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$\begin{aligned} u(k) &\in \mathbb{R}^m \\ y(k) &\in \mathbb{R}^p \end{aligned}$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$

Time-invariant

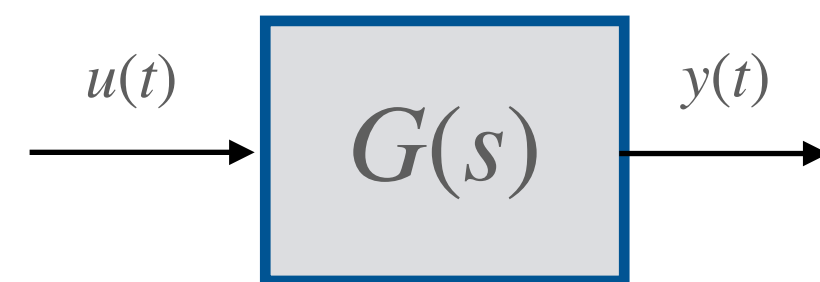
$$\dot{x}(t) = 0$$

$$x(k+1) = x(k)$$

Discrete-time Systems: **Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$



$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$

$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

$$x(k+1) = f(x(k), u(k))$$

$$y(k) = g(x(k), u(k))$$

Time-invariant

$$\dot{x}(t) = 0$$

$$x(k+1) = x(k)$$

If the solution exists:

$$x(k) = \bar{x}, \quad \forall k \geq 0$$

$$\bar{y}(k) = g(\bar{x}, \bar{u})$$

Discrete-time Systems: Stability of Equilibrium

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$

An equilibrium state is STABLE if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad \|x(0) - \bar{x}\| < \delta$$

$$\implies \|x(k) - \bar{x}\| < \varepsilon, \quad \forall k \geq 0$$

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$



Time-invariant

If the solution exists:

$$\begin{aligned} x(k) &= \bar{x}, \quad \forall k \geq 0 \\ \bar{y}(k) &= g(\bar{x}, \bar{u}) \end{aligned}$$

Discrete-time Systems: Stability of Equilibrium

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$

An equilibrium state is AS. STABLE if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad \|x(0) - \bar{x}\| < \delta$$

$$\implies \|x(k) - \bar{x}\| < \varepsilon, \quad \forall k \geq 0$$

and in addition:

$$\lim_{k \rightarrow \infty} \|x(k) - \bar{x}\| = 0$$

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$



Time-invariant

If the solution exists:

$$\begin{aligned} x(k) &= \bar{x}, \quad \forall k \geq 0 \\ \bar{y}(k) &= g(\bar{x}, \bar{u}) \end{aligned}$$

Discrete-time Systems: **Stability of Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$

An equilibrium state is UNSTABLE if:

it is not stable

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$



Time-invariant

If the solution exists:

$$\begin{aligned} x(k) &= \bar{x}, \quad \forall k \geq 0 \\ \bar{y}(k) &= g(\bar{x}, \bar{u}) \end{aligned}$$

Discrete-time Systems: Stability of a Generic Motion of the State

Given:

$$\tilde{x}(0) = \tilde{x}_0, \quad u(k) = \tilde{u}(k)$$

An motion of the state $\tilde{x}(k)$ is STABLE if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad \|x(0) - \tilde{x}(0)\| < \delta$$

$$\implies \|x(k) - \tilde{x}(k)\| < \varepsilon, \quad \forall k \geq 0$$

An motion of the state $\tilde{x}(k)$ is AS. STABLE if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad \|x(0) - \tilde{x}(0)\| < \delta$$

$$\implies \|x(k) - \tilde{x}(k)\| < \varepsilon, \quad \forall k \geq 0 \quad \text{and in addition:}$$

$$\lim_{k \rightarrow \infty} \|x(k) - \tilde{x}(k)\| = 0$$

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$



Time-invariant

Discrete-time Systems: Stability of a Generic Motion of the State

Given:

$$\tilde{x}(0) = \tilde{x}_0, \quad u(k) = \tilde{u}(k)$$

$$x(k+1) = f(x(k), u(k))$$

$$y(k) = g(x(k), u(k))$$



Time-invariant

An motion of the state $\tilde{x}(k)$ is UNSTABLE if:

it is not stable

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$



Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Given:

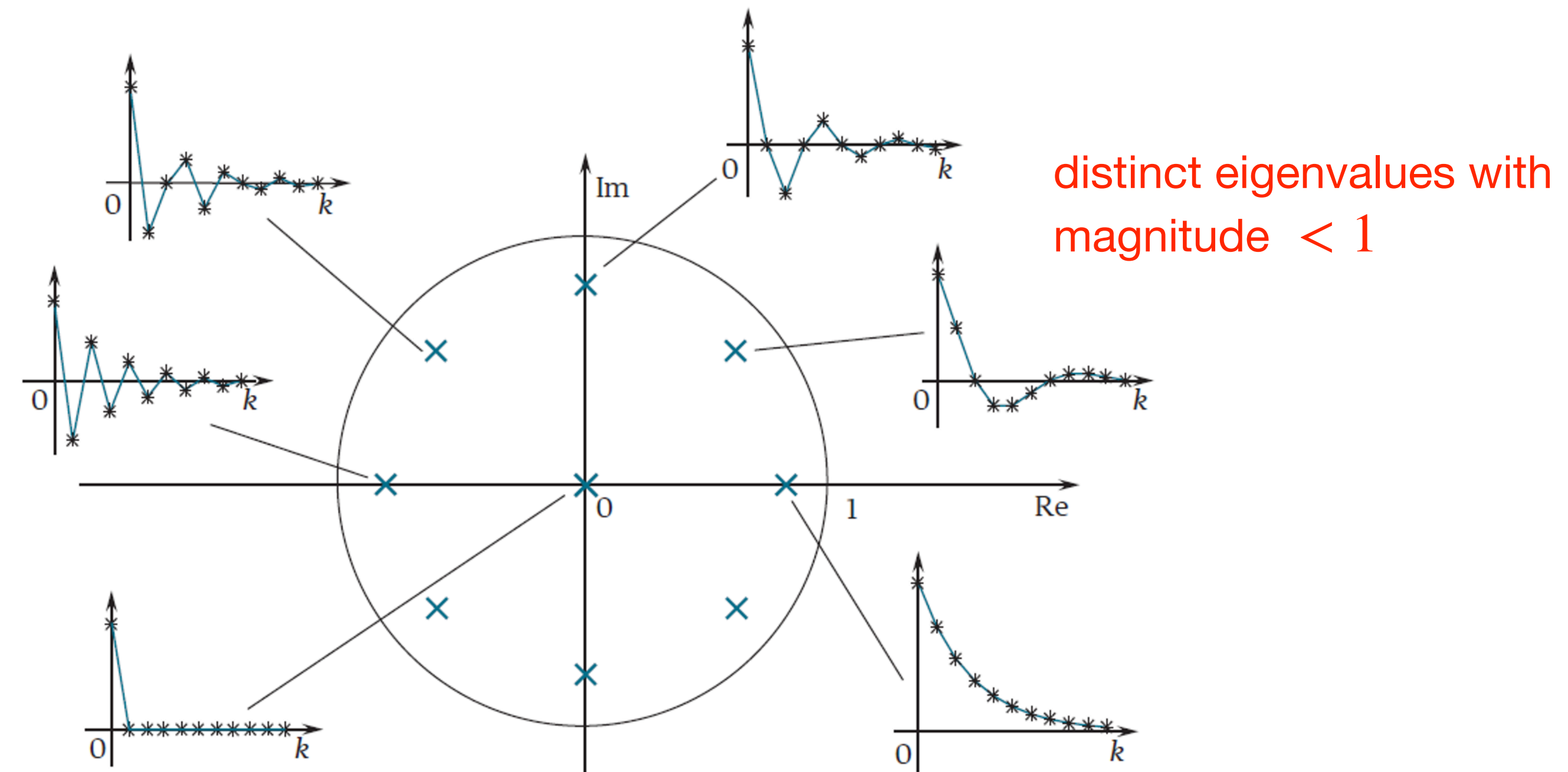
$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A



Discrete-time Systems: Stability of LTI Discrete-Time Systems

Given:

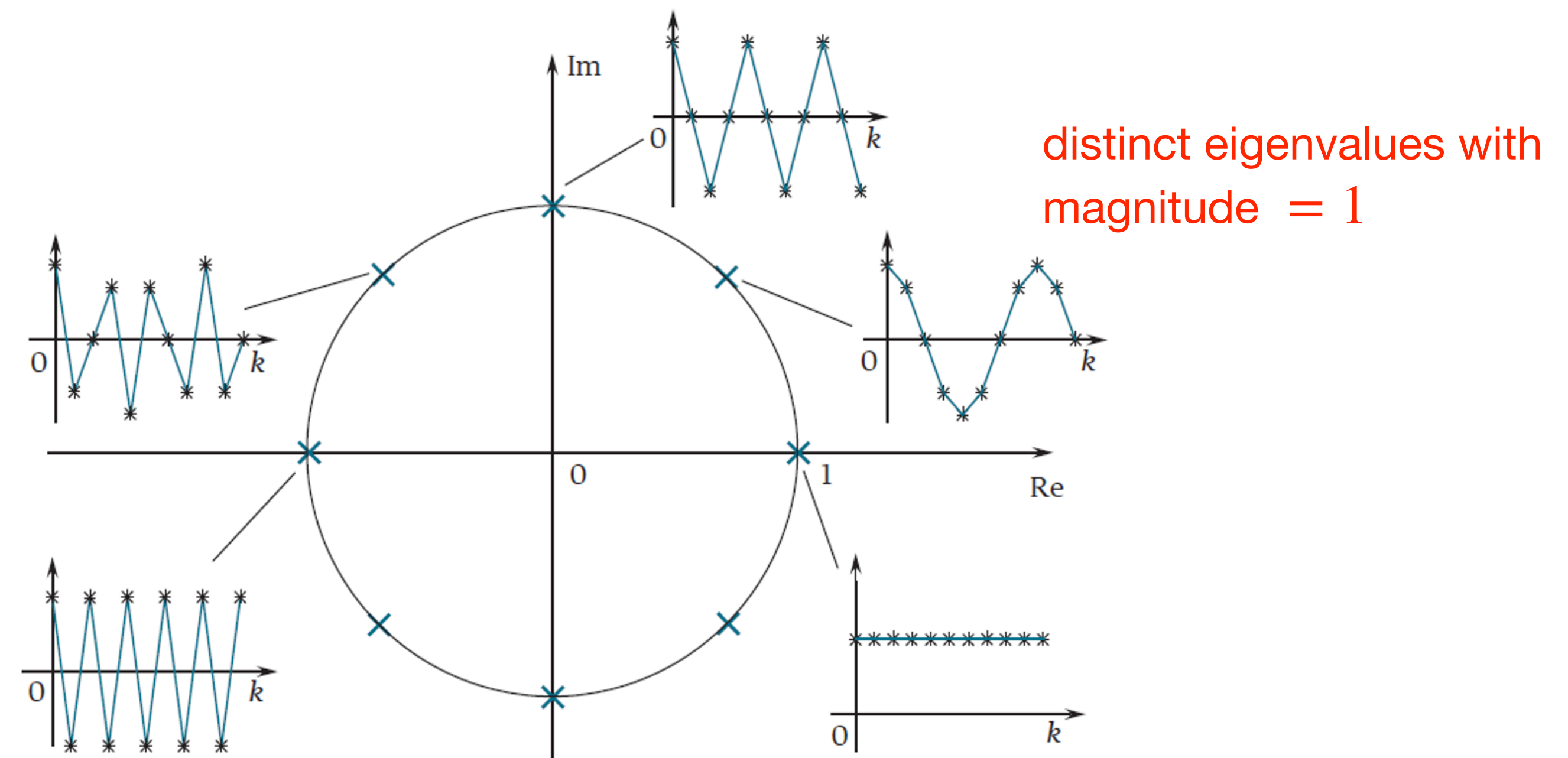
$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A



Discrete-time Systems: Stability of LTI Discrete-Time Systems

Given:

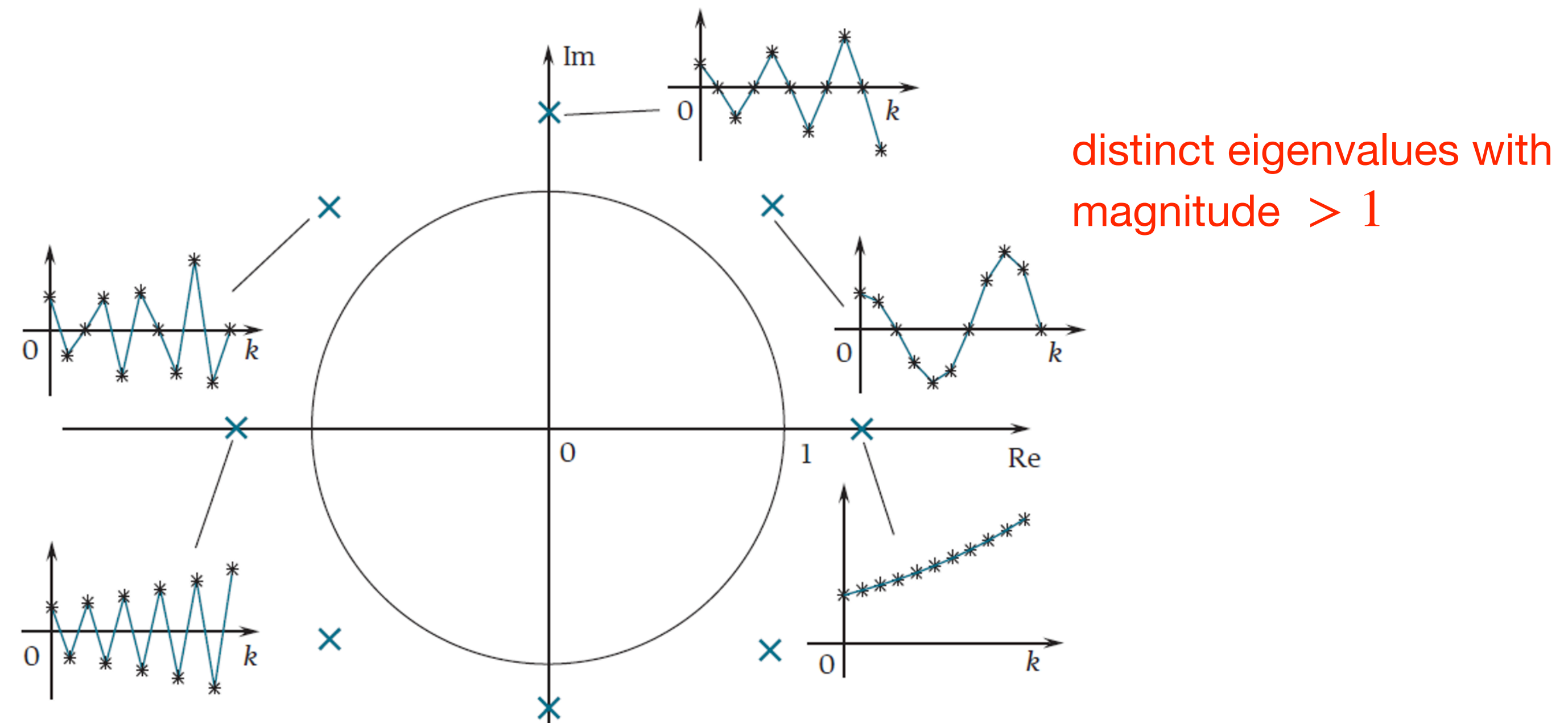
$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A



Discrete-time Systems: Stability of LTI Discrete-Time Systems

Given:

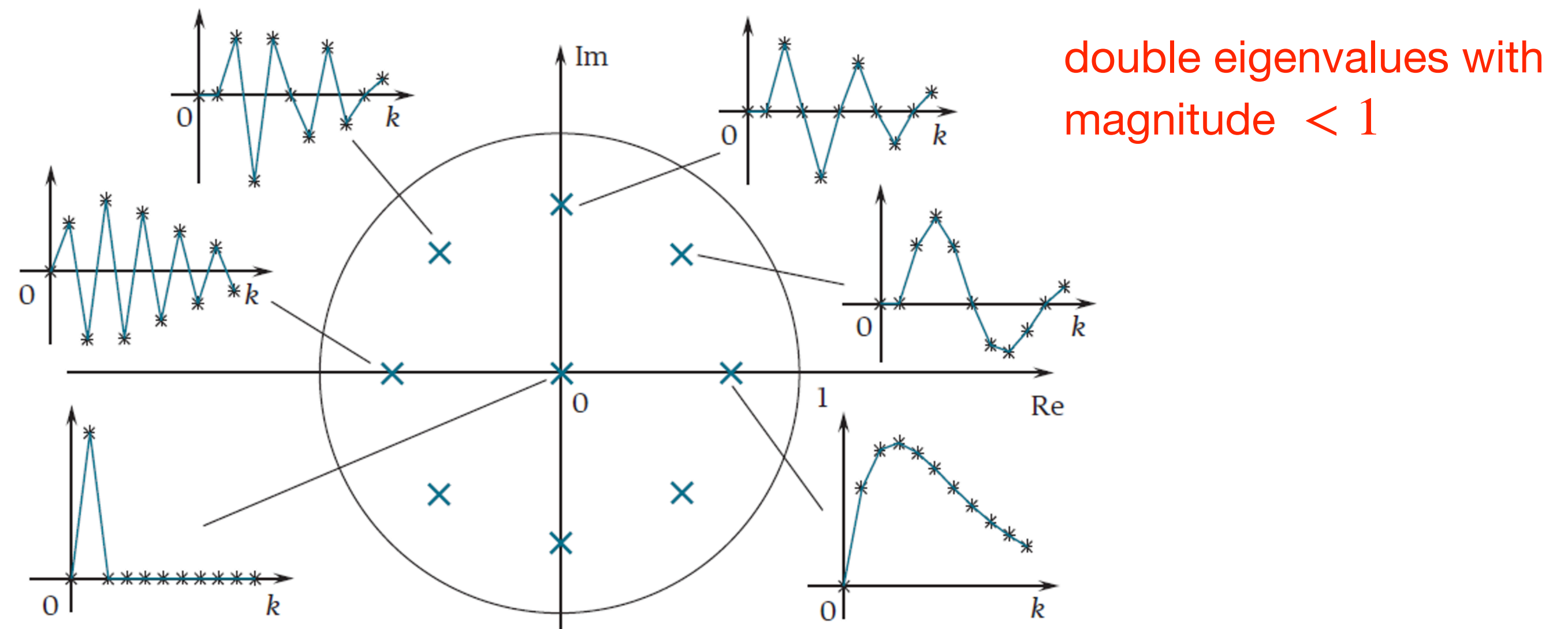
$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

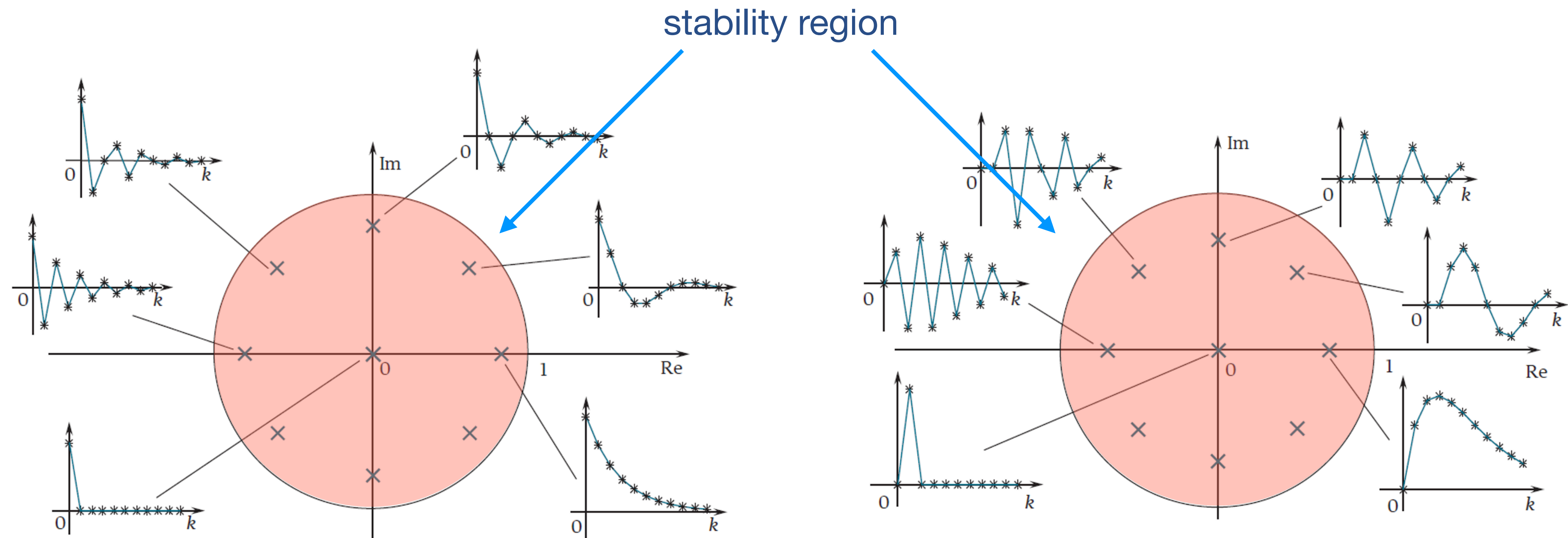
Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A



Discrete-time Systems: Stability of LTI Discrete-Time Systems



distinct eigenvalues with
magnitude < 1

double eigenvalues with
magnitude < 1

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n \longrightarrow$$

To find the eigenvalues:

$$\varphi(z) = \det(zI - A) = 0$$



not straightforward if $n > 2$

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n \longrightarrow$$

To find the eigenvalues:

$$\varphi(z) = \det(zI - A) = 0$$

Alternatives:

Necessary or sufficient conditions based on $\varphi(z)$

Stability test (Jury's Test) based on $\varphi(z)$

not straightforward if $n > 2$

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Necessary conditions based on $\varphi(z)$:

1. A necessary condition for a discrete-time dynamical system to be asymptotically stable is that

$$\left| \frac{\varphi_n}{\varphi_0} \right| < 1$$

2. A necessary condition for a discrete-time dynamical system to be asymptotically stable is that

$$\varphi_0 \varphi(1) > 0$$

3. A necessary condition for a discrete-time dynamical system to be asymptotically stable is that

$$(-1)^n \varphi_0 \varphi(-1) > 0$$

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Sufficient conditions based on $\varphi(z)$:

1. A sufficient condition for a discrete-time dynamical system to be asymptotically stable is that

$$\varphi_0 > \varphi_1 > \varphi_2 > \cdots > \varphi_{n-1} > \varphi_n > 0$$

2. A sufficient condition for a discrete-time dynamical system to be asymptotically stable is that

$$\sum_{i=1}^n |\varphi_i| < |\varphi_0|$$

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

$$\varphi_0 \quad \varphi_1 \quad \varphi_2 \quad \cdots \quad \varphi_{n-1} \quad \varphi_n$$

.....

$$h_1 \quad h_2 \quad \cdots \quad h_{\nu-1} \quad h_{\nu}$$

$$l_1 \quad l_2 \quad \cdots \quad l_{\nu-1}$$

.....



Build the Jury's Table

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

$$\varphi_0 \quad \varphi_1 \quad \varphi_2 \quad \cdots \quad \varphi_{n-1} \quad \varphi_n$$

.....

$$h_1 \quad h_2 \quad \cdots \quad h_{\nu-1} \quad h_{\nu}$$

$$l_1 \quad l_2 \quad \cdots \quad l_{\nu-1}$$

.....

Fill the generic line

$$l_i = \frac{1}{h_1} \det \begin{bmatrix} h_1 & h_{\nu-i+1} \\ h_{\nu} & h_i \end{bmatrix}$$

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

φ_0	φ_1	φ_2	$\cdots$	φ_{n-1}	φ_n	Check the first column
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	
h_1	h_2	$\dots$	$h_{\nu-1}$	h_{ν}		
l_1	l_2	$\dots$	$l_{\nu-1}$			
$\dots$	$\dots$	$\dots$	$\dots$			

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \dots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

φ_0	φ_1	φ_2	$\dots$	φ_{n-1}	φ_n	Check the first column
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	
h_1	h_2	$\dots$	$h_{\nu-1}$	h_{ν}		
l_1	l_2	$\dots$	$l_{\nu-1}$			
$\dots$	$\dots$	$\dots$	$\dots$			

Jury's Criterion:

A LTI system is asymptotically stable if and only if the Yuri Table associated with its characteristic polynomial is well defined and the elements appearing in the first column have all the same sign

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Example

Characteristic Polynomial $\varphi(z) = \det(zI - A) = z^3 + 2z^2 + z + 3$

Jury's Test:

$$\begin{array}{cccc} 1 & 2 & 1 & 3 \\ -8 & & & \end{array}$$

Fill the second line

$$l_1 = \frac{1}{1} \det \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$$

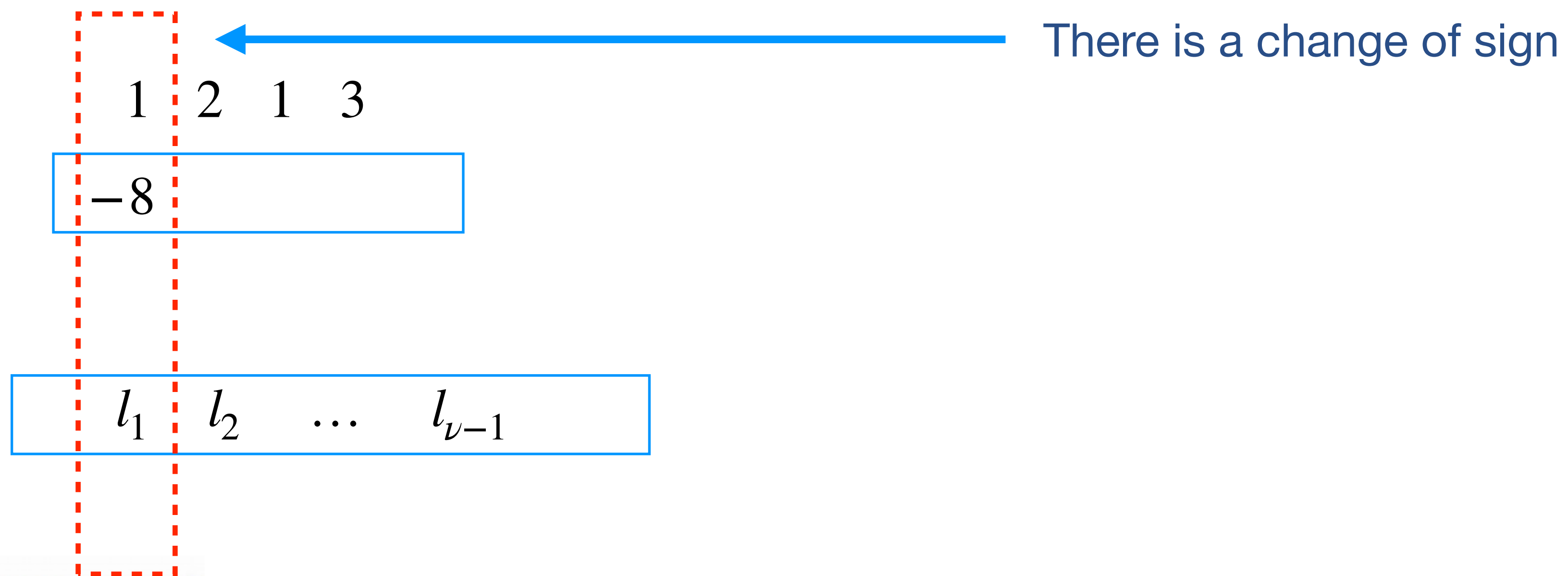
$$\begin{array}{cccc} l_1 & l_2 & \dots & l_{\nu-1} \end{array}$$

Fill the generic line

$$l_i = \frac{1}{h_1} \det \begin{bmatrix} h_1 & h_{\nu-i+1} \\ h_{\nu} & h_i \end{bmatrix}$$

Discrete-time Systems: Stability of LTI Discrete-Time Systems**Example**

Characteristic Polynomial $\varphi(z) = \det(zI - A) = z^3 + 2z^2 + z + 3$

Jury's Test:

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Example

Characteristic Polynomial $\varphi(z) = \det(zI - A) = z^3 + 2z^2 + z + 3$

Jury's Test:

1	2	1	3
-8			
l_1	l_2	$\dots$	$l_{\nu-1}$

There is a change of sign

Jury's Criterion:

A LTI system is asymptotically stable if and only if the Yuri Table associated with its characteristic polynomial is well defined and the elements appearing in the first column have all the same sign

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Example

Characteristic Polynomial $\varphi(z) = \det(zI - A) = z^3 + 2z^2 + z + 3$

Jury's Test:

1	2	1	3
-8			
l_1	l_2	$\dots$	$l_{\nu-1}$

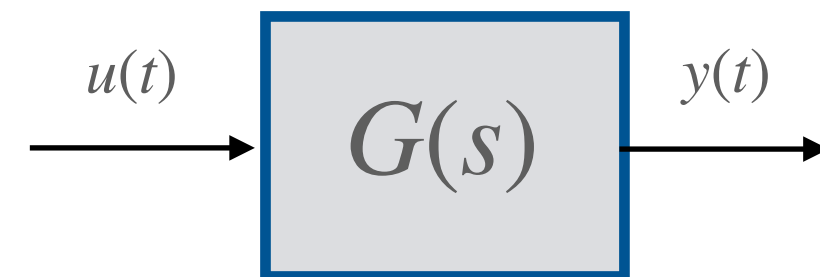
There is a change of sign

The system with characteristic polynomial $\varphi(z)$ is NOT As. Stable

Jury's Criterion:

A LTI system is asymptotically stable if and only if the Yuri Table associated with its characteristic polynomial is well defined and the elements appearing in the first column have all the same sign

Digital Control Schemes



$$u(t) \in \mathbb{R}^m$$

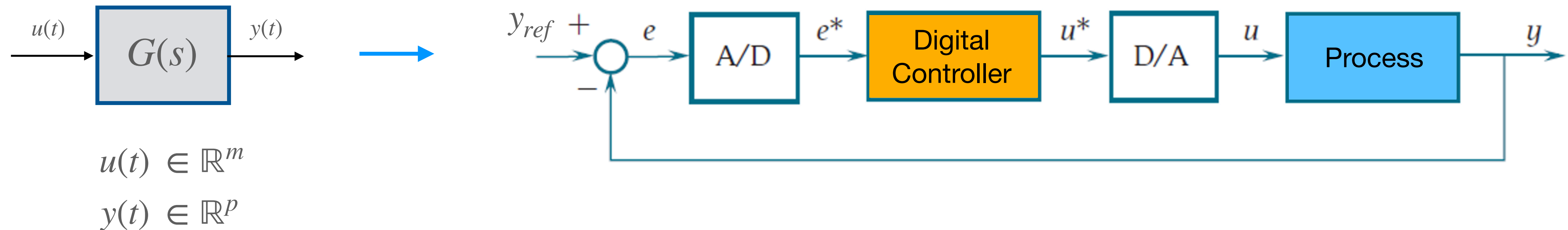
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

Digital Control Schemes

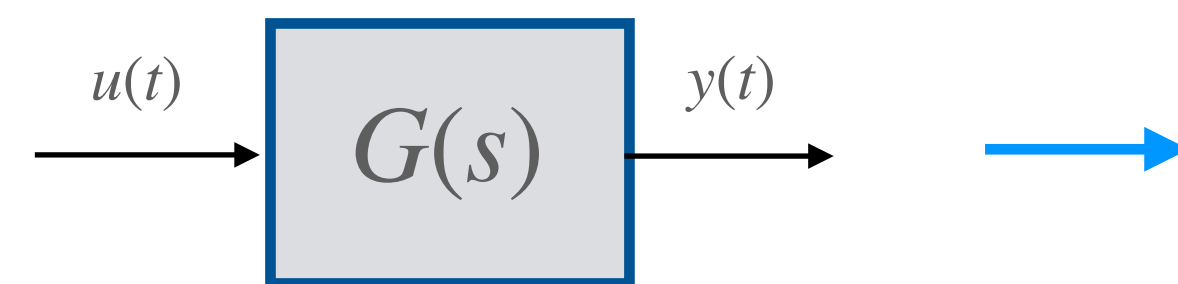
Digital control scheme with error sampling



$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

Digital Control Schemes



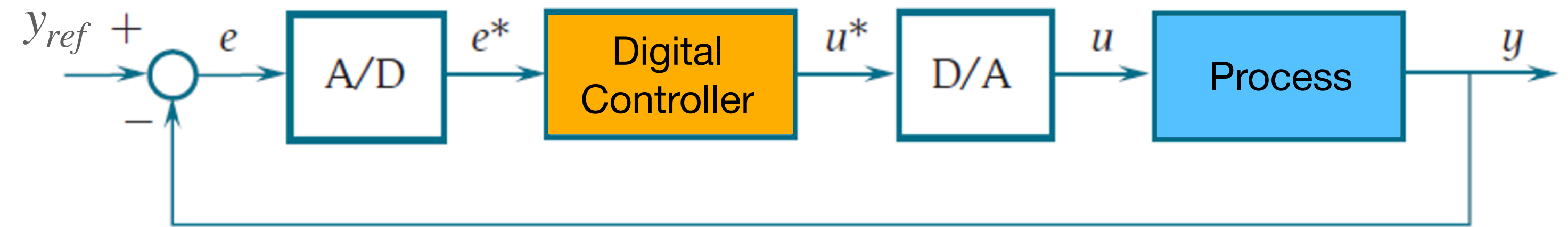
$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

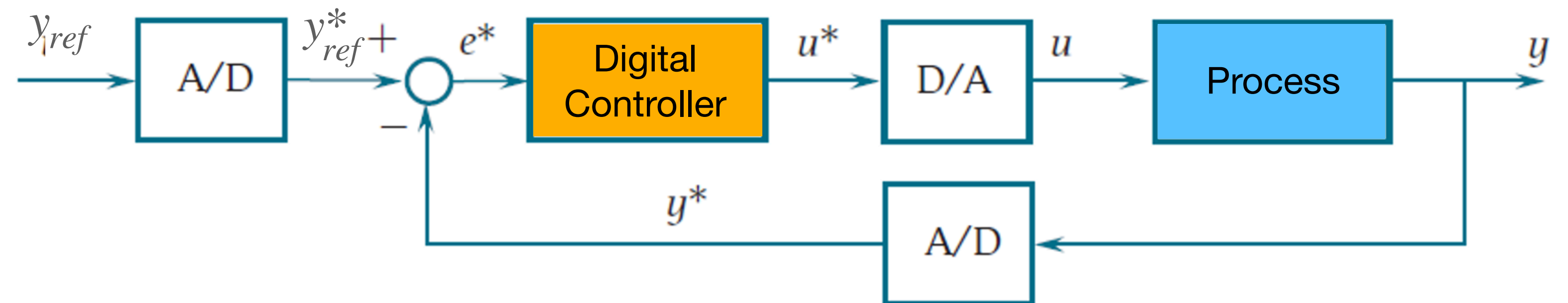
$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

Digital control scheme with error sampling



Digital control scheme with output and reference sampling



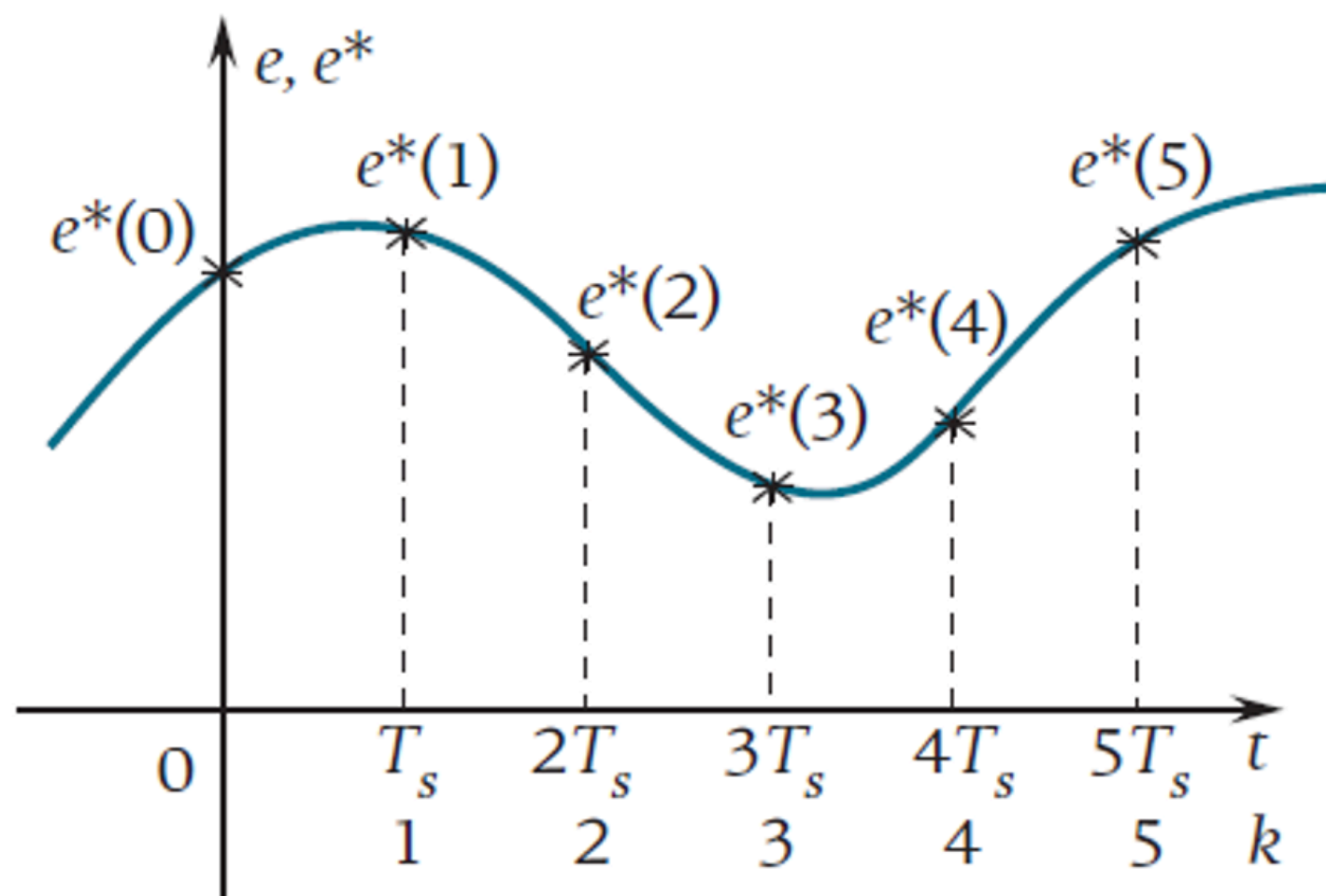
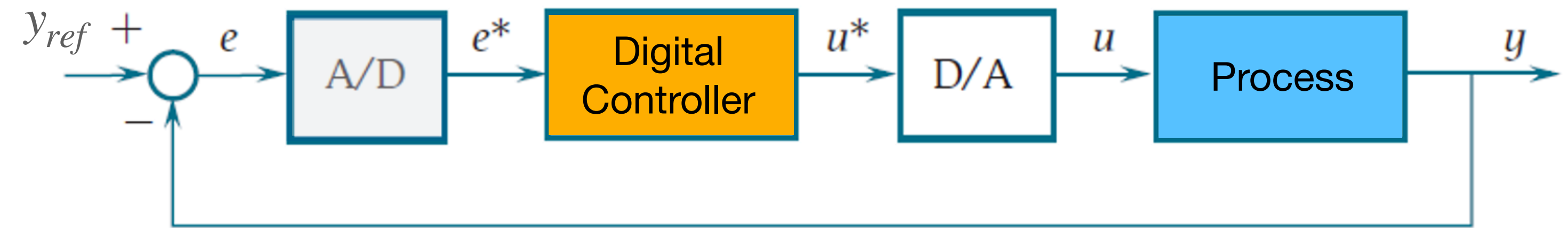
Digital Control Schemes

 $\{e^*(k)\}, \{u^*(k)\}$ sequences

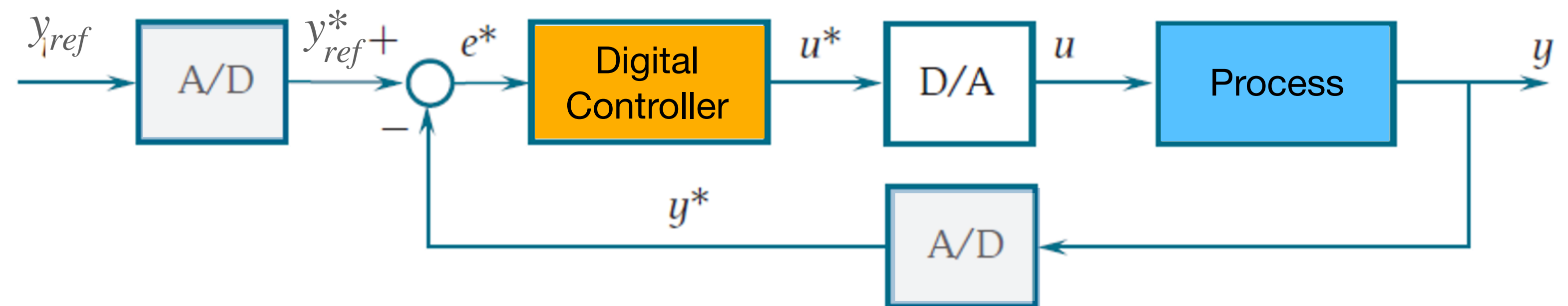
A/D converter: it is a sampler



Digital control scheme with error sampling

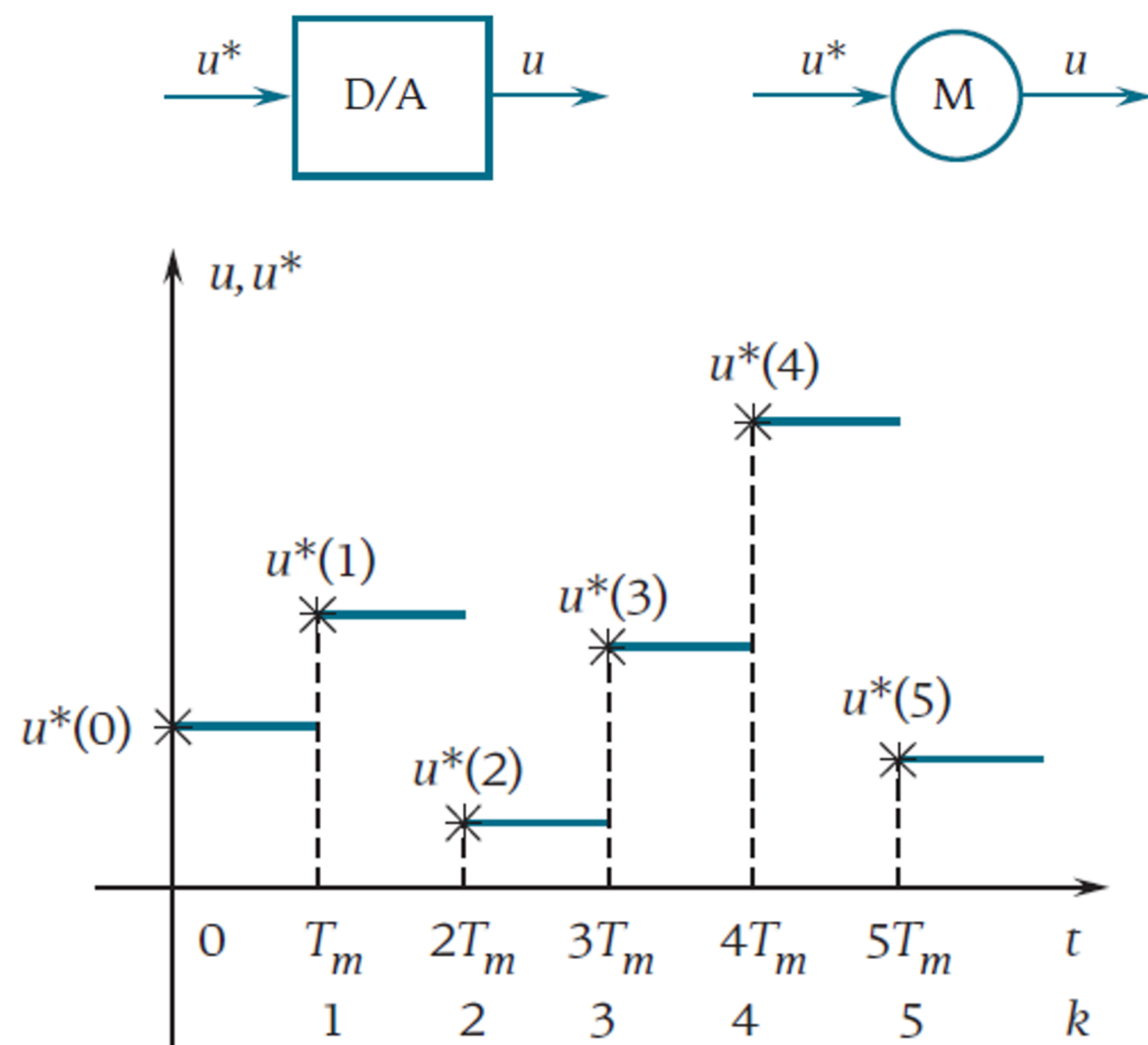


Digital control scheme with output and reference sampling

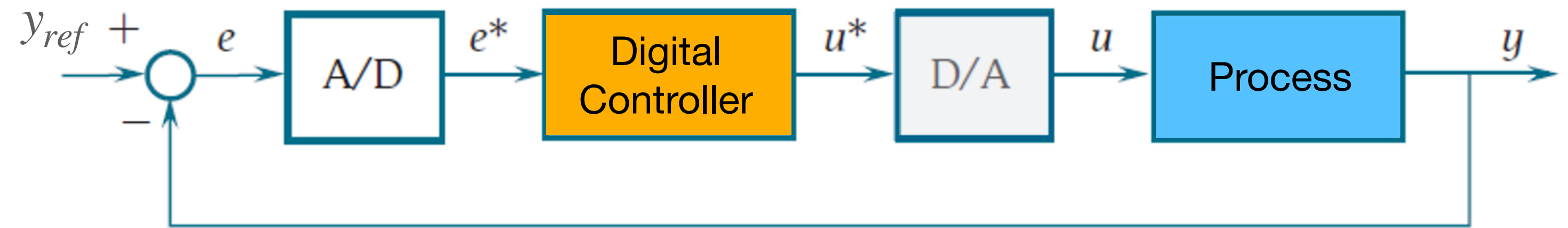

 $\{y_{ref}^*(k)\}, \{y^*(k)\}$ sequences

Digital Control Schemes

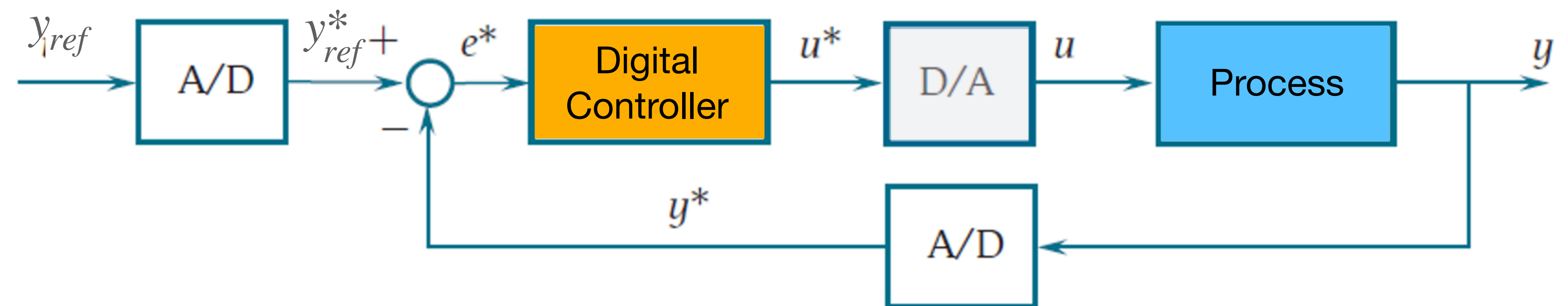
D/A converter: it is a hold



Digital control scheme with error sampling

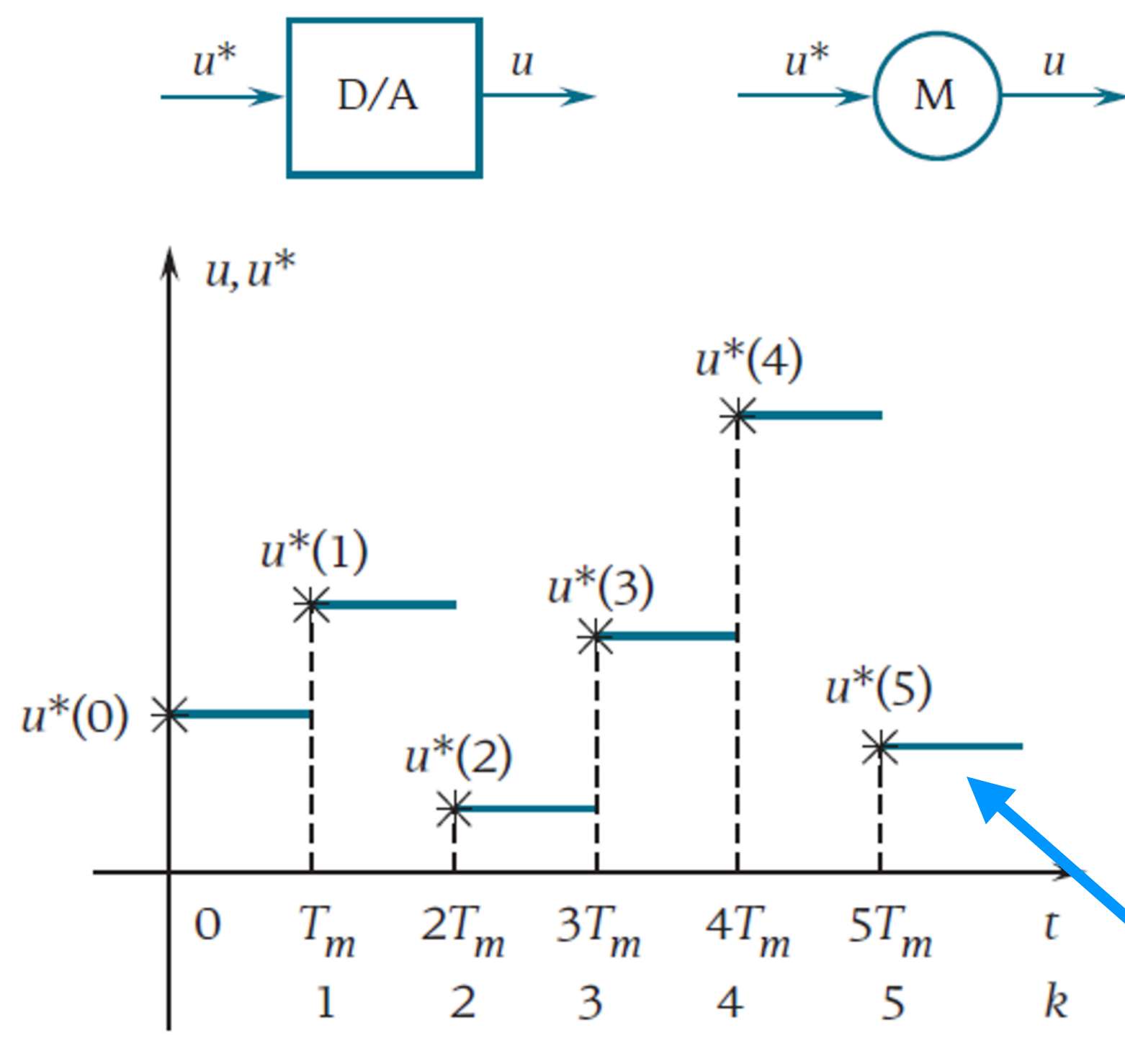


Digital control scheme with output and reference sampling

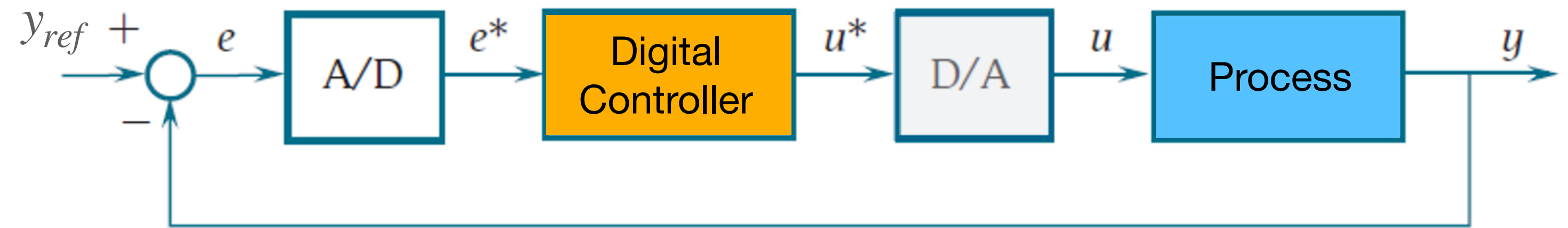


Digital Control Schemes

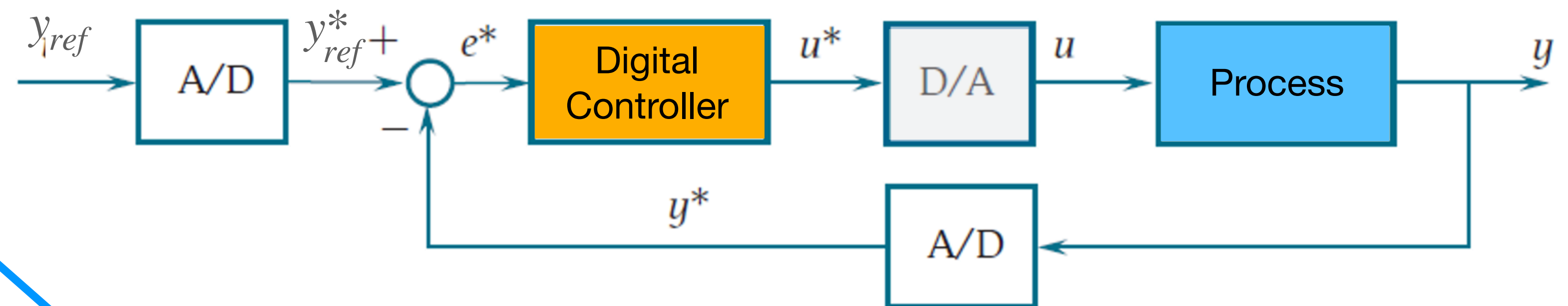
D/A converter: it is a hold



Digital control scheme with error sampling



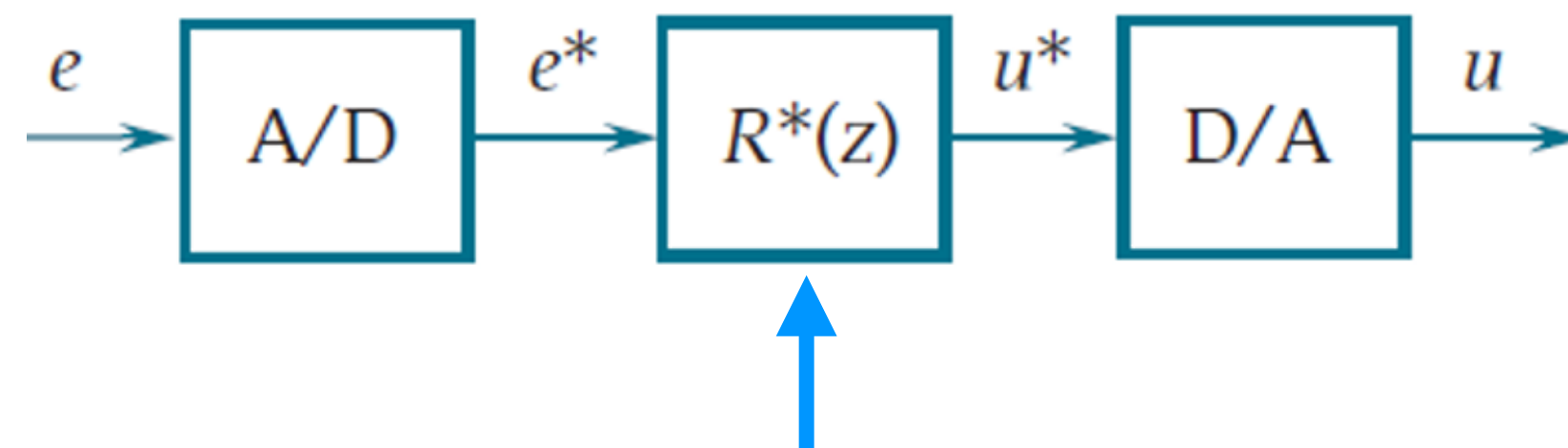
Digital control scheme with output and reference sampling



ZOH: zero-order hold

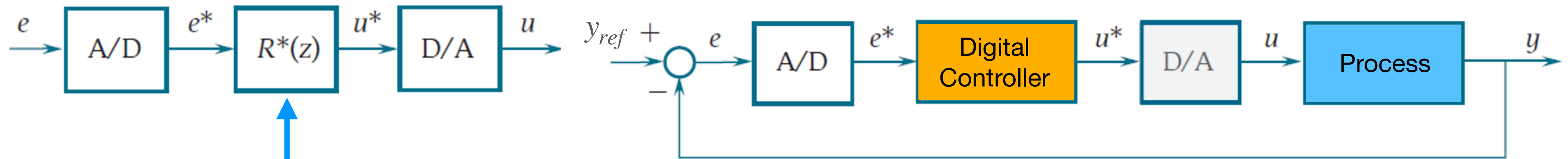
Digital Control Schemes

The control element:

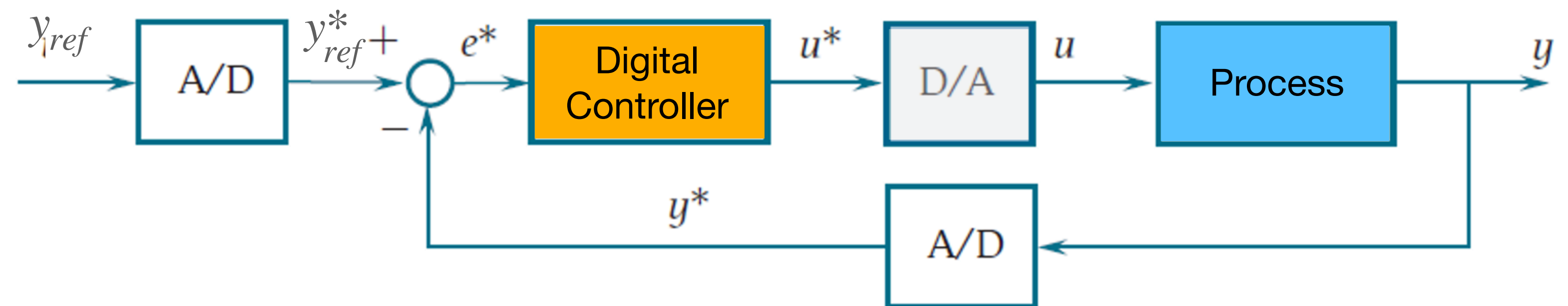


Digital Controller

Digital control scheme with error sampling

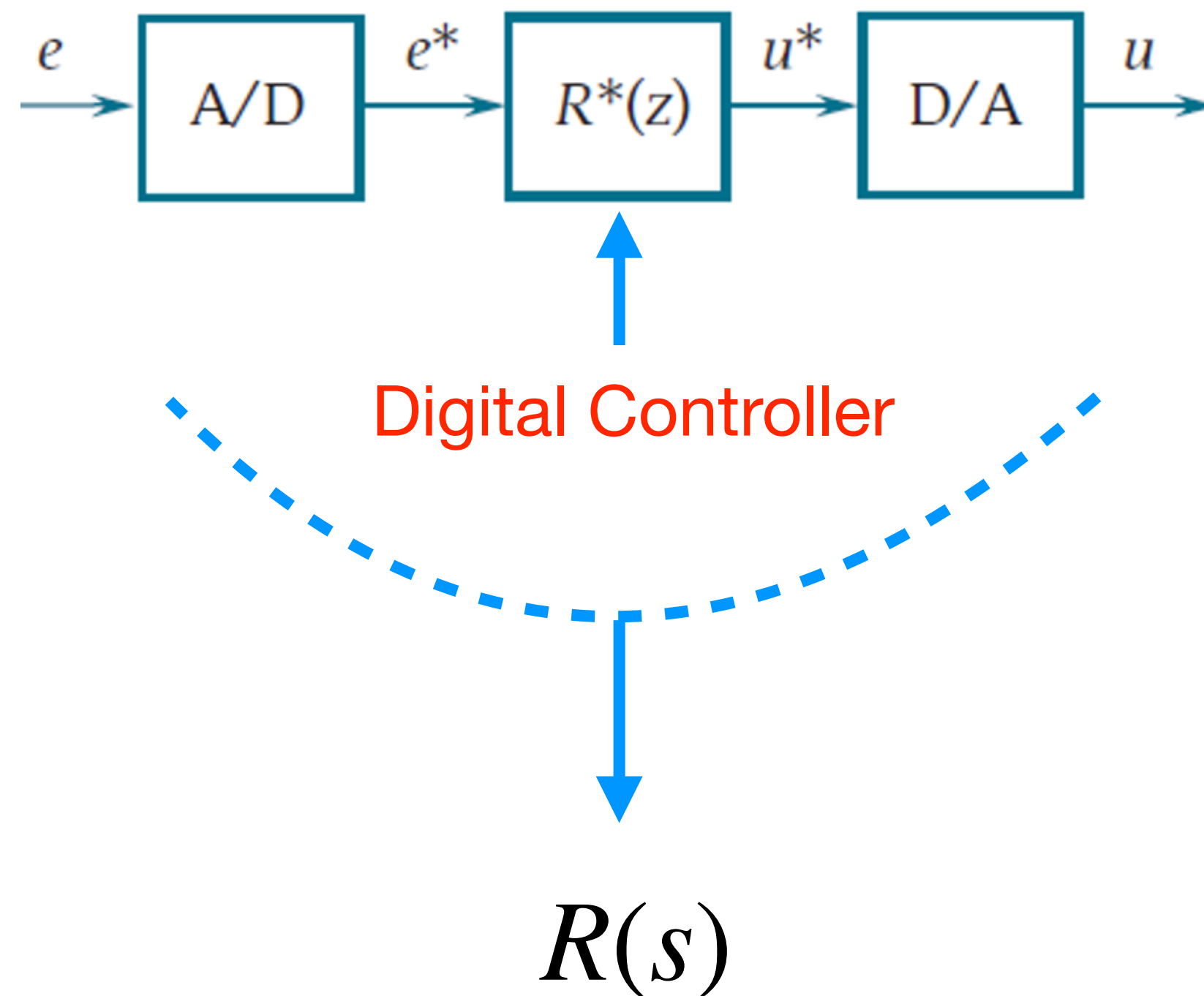


Digital control scheme with output and reference sampling

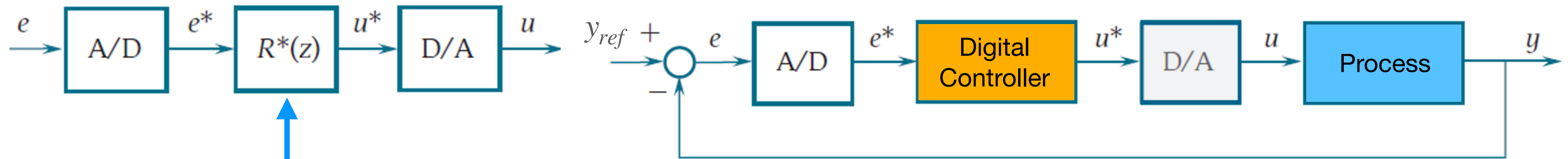


Digital Control Schemes

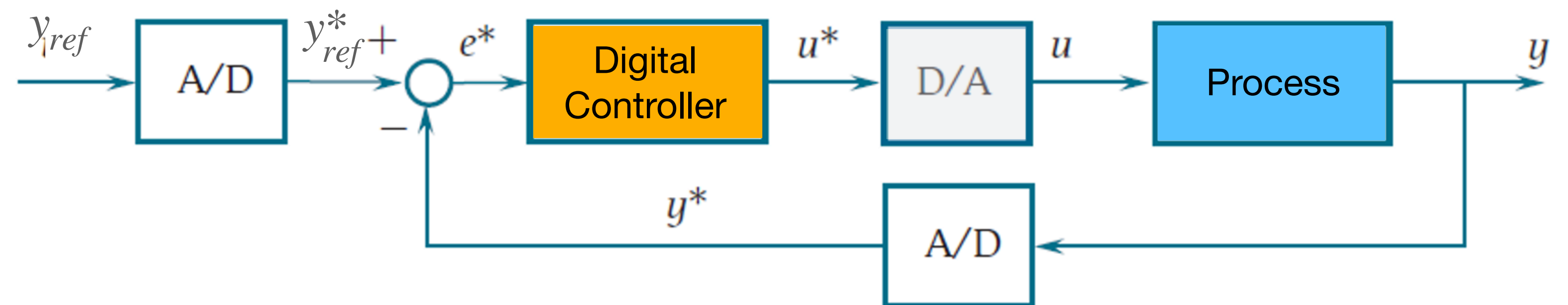
The control element:



Digital control scheme with error sampling

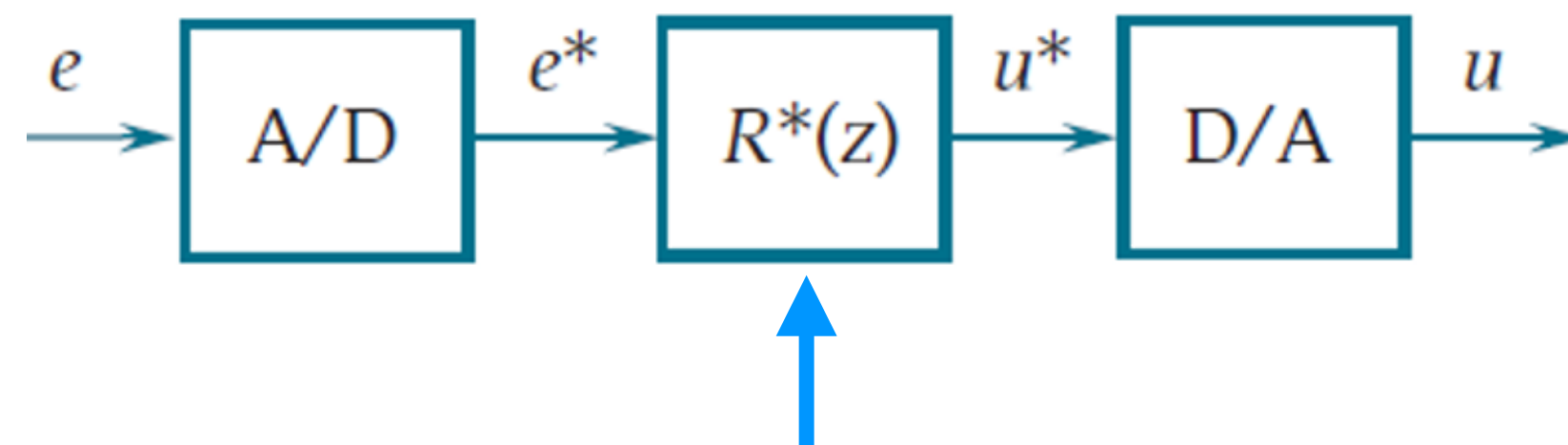


Digital control scheme with output and reference sampling



Digital Control Schemes

The control element:

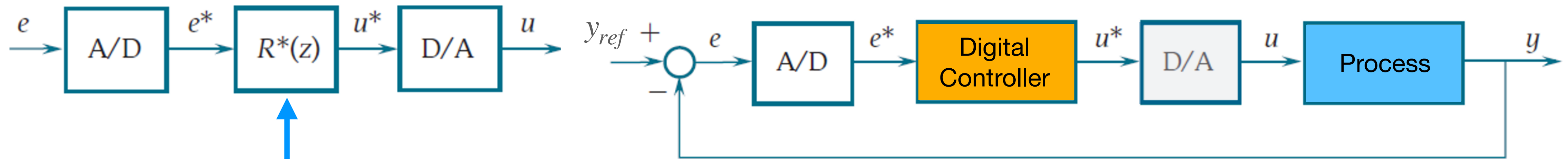


Digital Controller

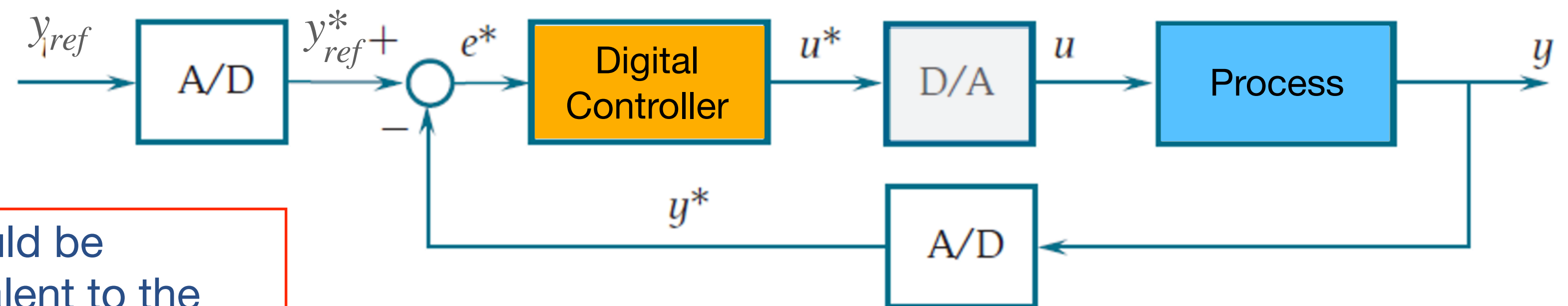
 $R(s)$

it should be
equivalent to the
original continuous-
time controller

Digital control scheme with error sampling

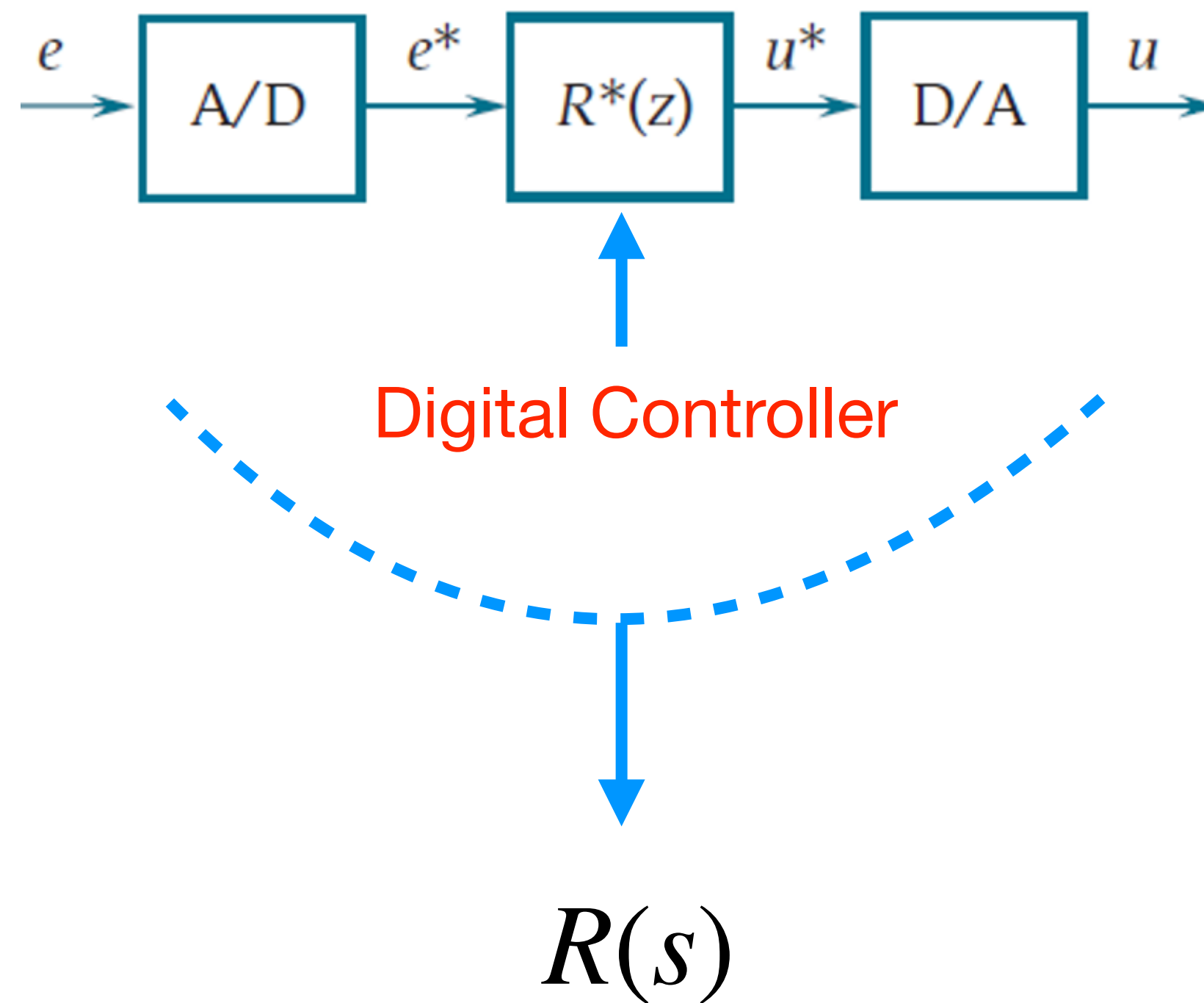


Digital control scheme with output and reference sampling



Digital Control Schemes: **Problems to solve**

The control element:



A/D converter: it is a sampler



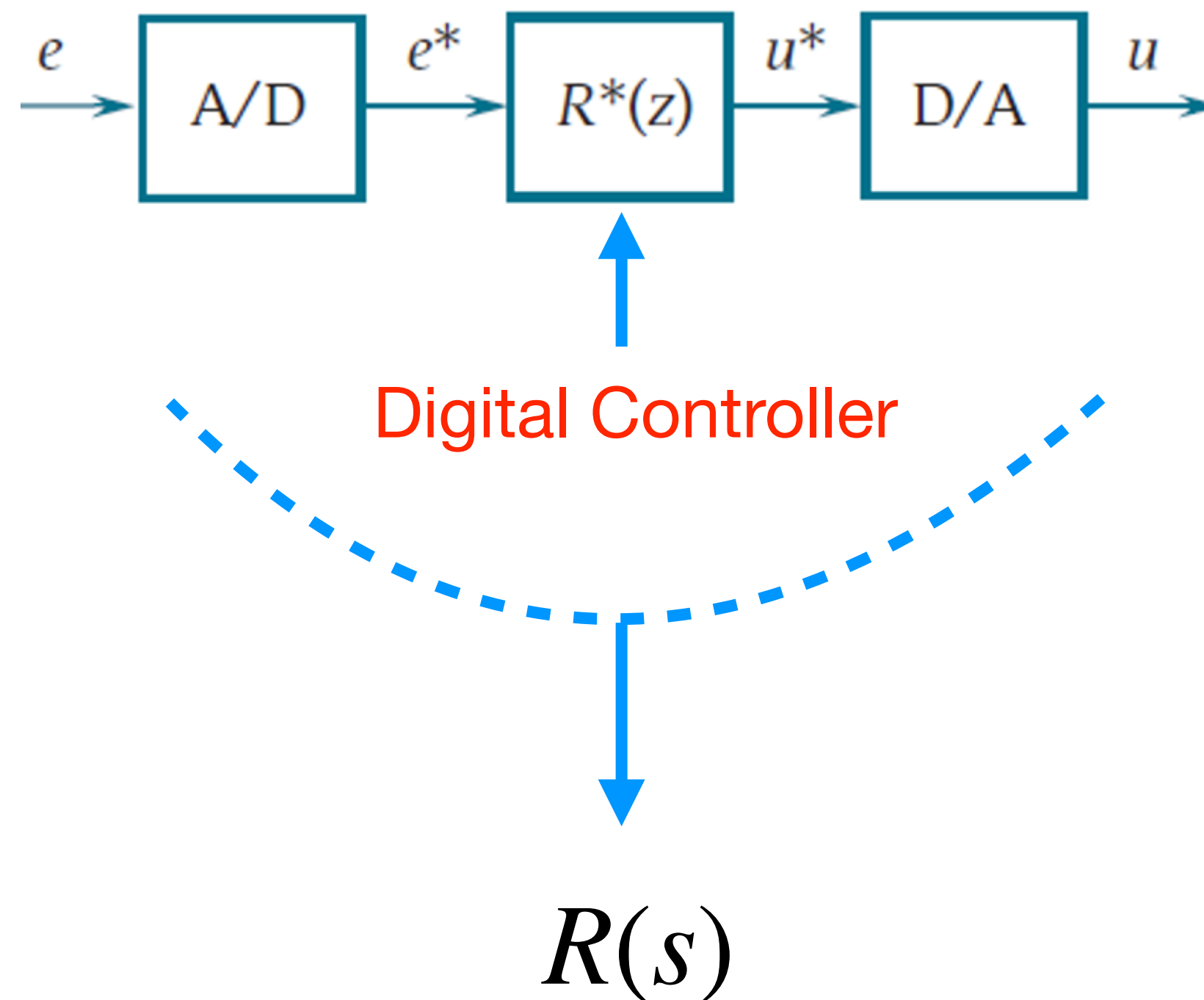
D/A converter: it is a hold



Digital Control Schemes: **Problems to solve**

1. How to determine the transfer function of a discrete-time LTI system

The control element:



A/D converter: it is a sampler



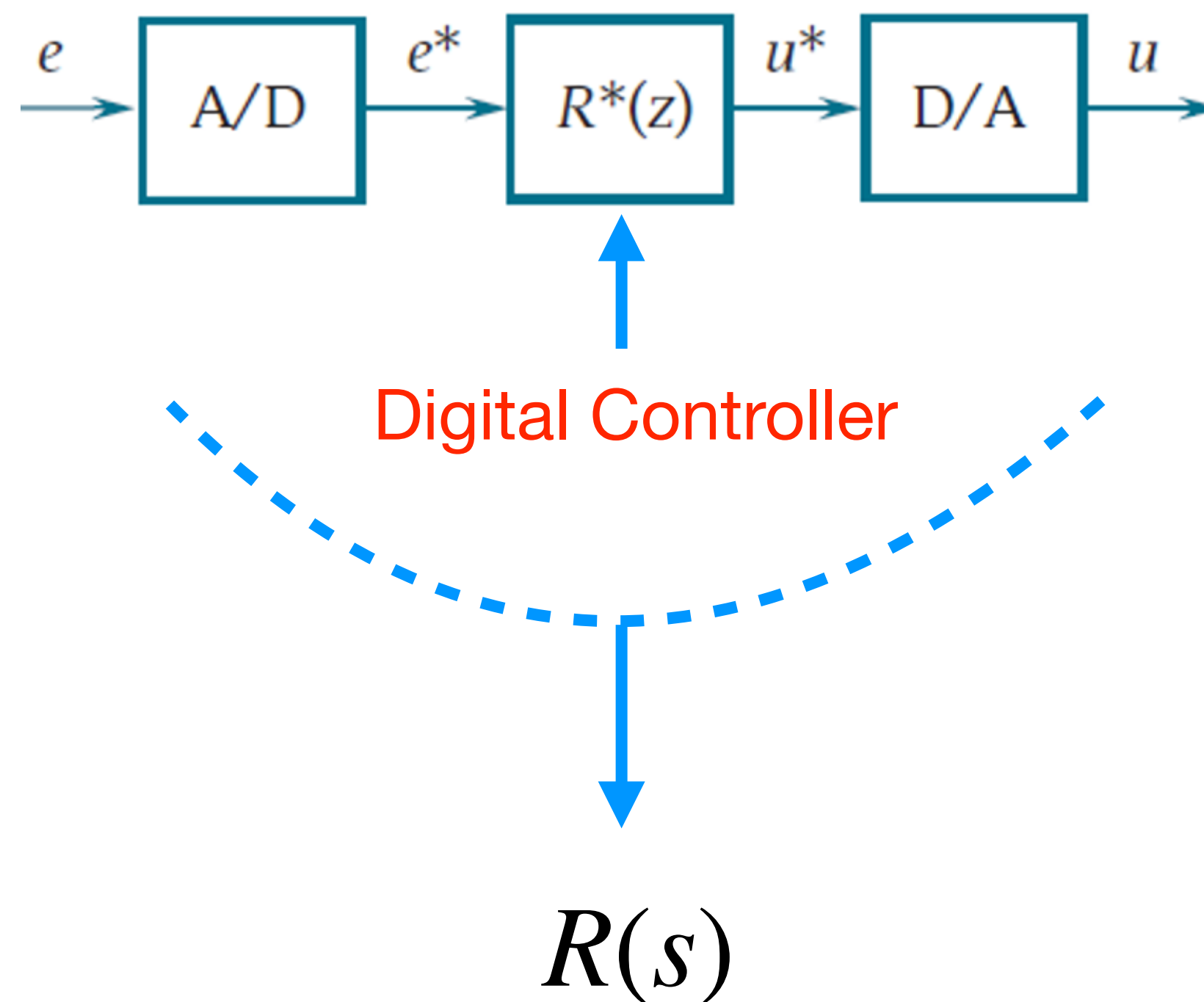
D/A converter: it is a hold



Digital Control Schemes: **Problems to solve**

1. How to determine the transfer function of a discrete-time LTI system

The control element:



A/D converter: it is a sampler



2. How to determine the sampling time T_s

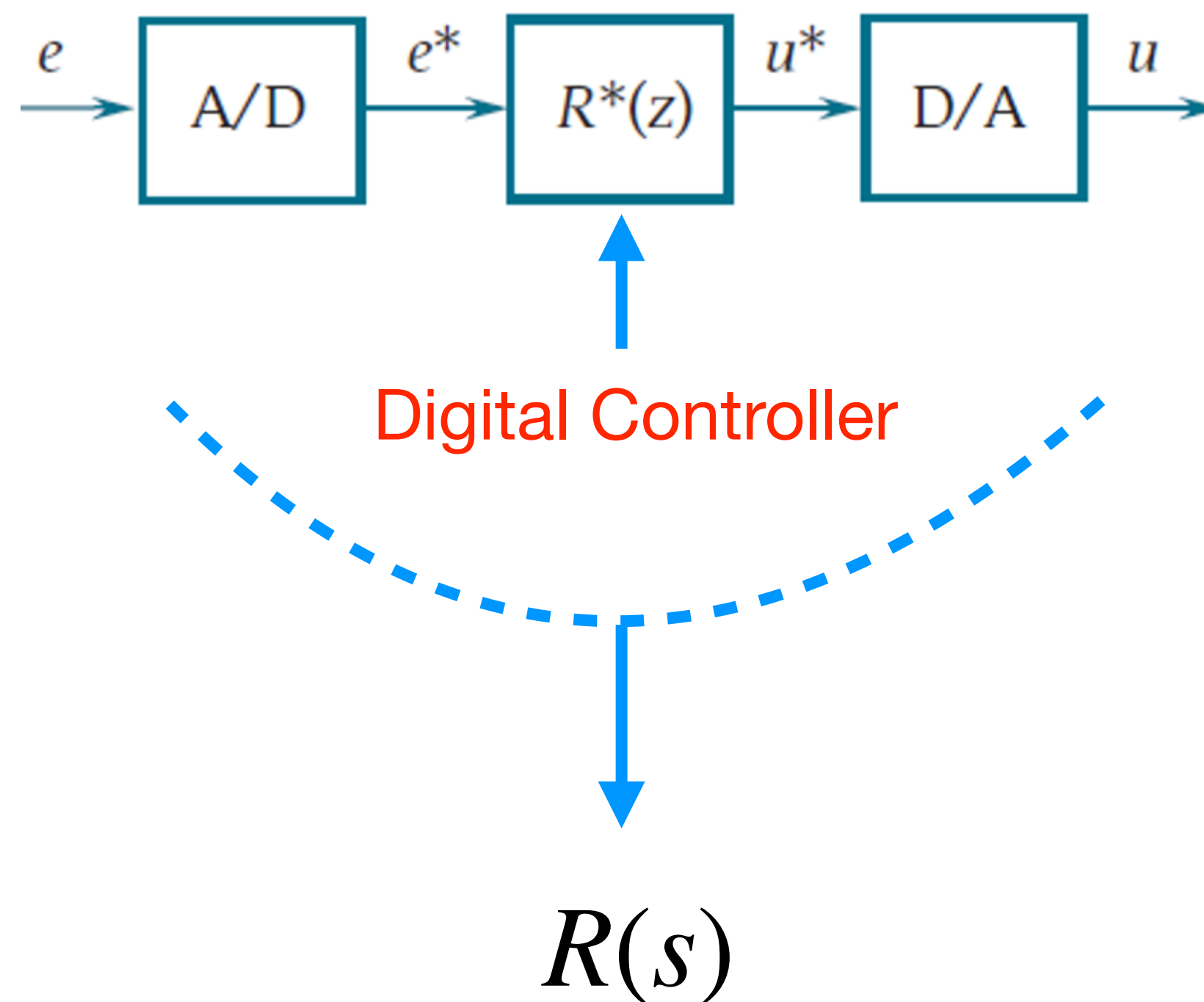
D/A converter: it is a hold



Digital Control Schemes: Problems to solve

1. How to determine the transfer function of a discrete-time LTI system

The control element:



A/D converter: it is a sampler



2. How to determine the sampling time T_s

D/A converter: it is a hold

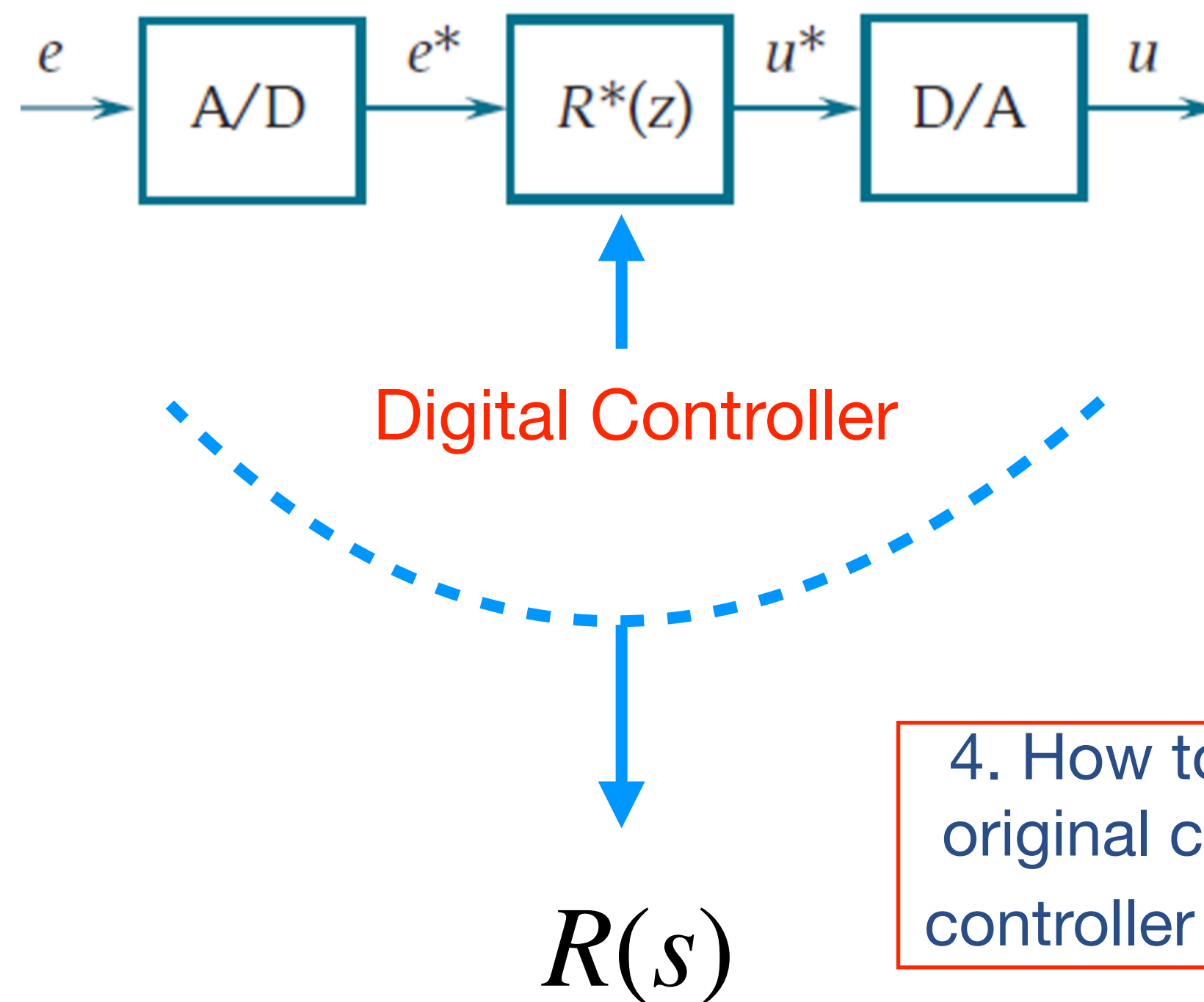


3. How to model the ZOH effect

Digital Control Schemes: Problems to solve

1. How to determine the transfer function of a discrete-time LTI system

The control element:



4. How to discretize the original continuous-time controller to obtain $R^*(z)$

A/D converter: it is a sampler



2. How to determine the sampling time T_s

D/A converter: it is a hold

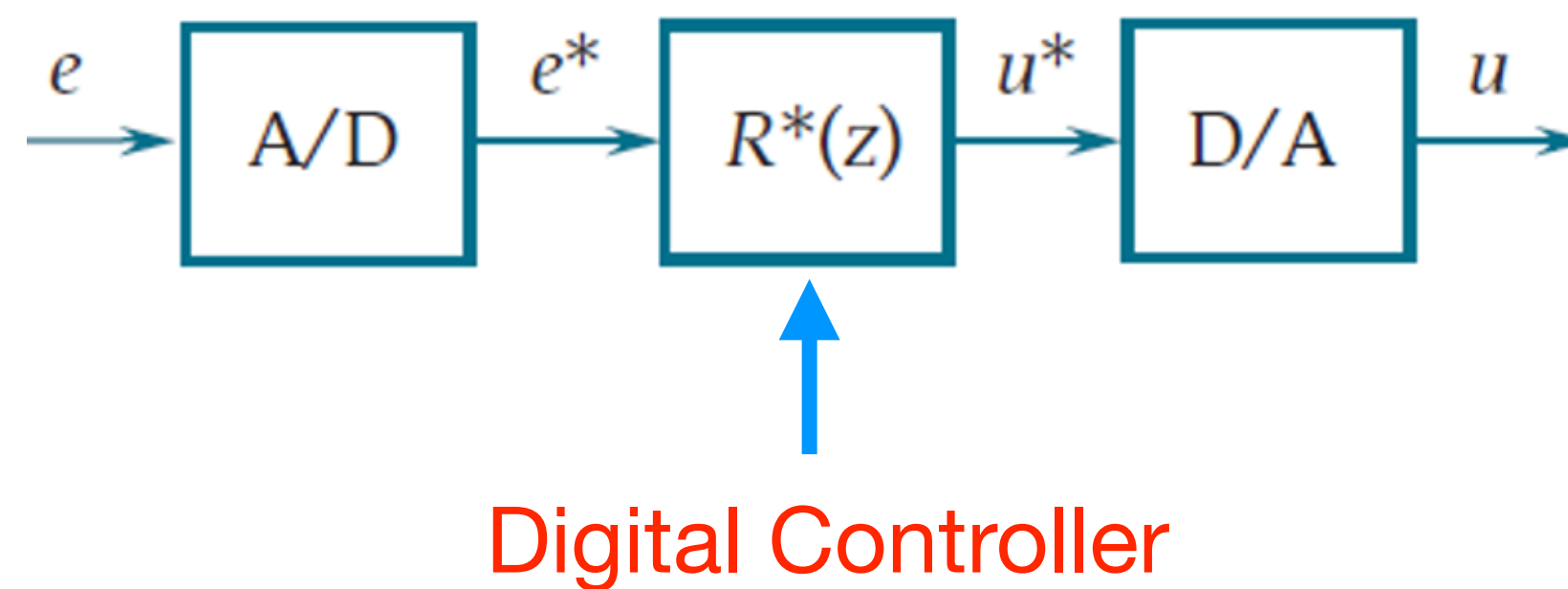


3. How to model the ZOH effect

Digital Control Schemes: Z-transform

1. How to determine the transfer function of a discrete-time LTI system

The control element:



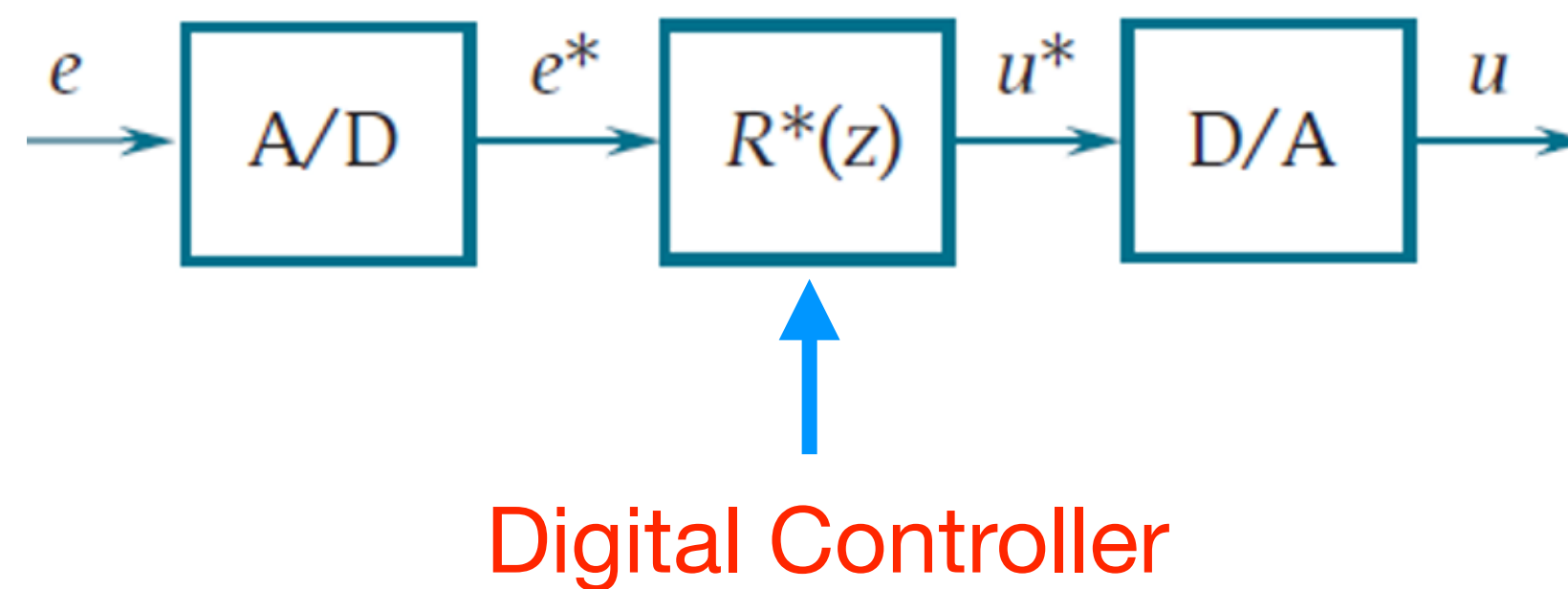
Given the discrete sequence $\{x(k)\}$, $k \geq 0$, its Z-transform is defined as:

$$X(z) = \mathcal{Z} [\{x(k)\}] = \sum_{k=0}^{\infty} x(k) z^{-k}, \text{ with } z \in \mathbb{C}$$

Digital Control Schemes: Z-transform

1. How to determine the transfer function of a discrete-time LTI system

The control element:



Given the discrete sequence $\{x(k)\}$, $k \geq 0$, its Z-transform is defined as:

$$X(z) = \mathcal{Z} [\{x(k)\}] = \sum_{k=0}^{\infty} x(k) z^{-k}, \text{ with } z \in \mathbb{C}$$

Properties:

$$1) \mathcal{Z} [a \{x(k)\} + b \{y(k)\}] = a X(z) + b Y(z)$$

$$2) \mathcal{Z} [\{x(k-n)\}] = z^{-n} X(z)$$

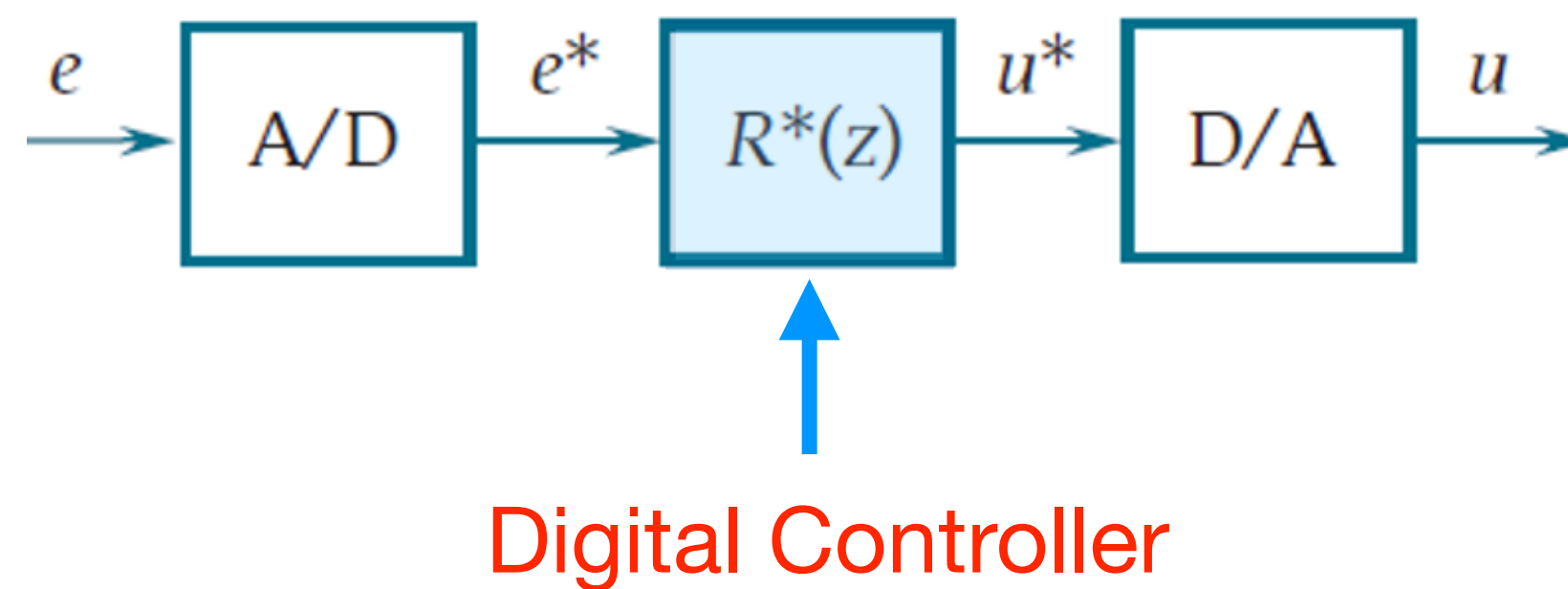
$$3) \mathcal{Z} [\{x(k+1)\}] = z X(z)$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



The control element:



Given the discrete sequence $\{x(k)\}$, $k \geq 0$, its Z-transform is defined as:

$$X(z) = \mathcal{Z} [\{x(k)\}] = \sum_{k=0}^{\infty} x(k) z^{-k}, \text{ with } z \in \mathbb{C}$$

Properties:

- 1) $\mathcal{Z} [a \{x(k)\} + b \{y(k)\}] = a X(z) + b Y(z)$
- 2) $\mathcal{Z} [\{x(k - n)\}] = z^{-n} X(z)$
- 3) $\mathcal{Z} [\{x(k + 1)\}] = z X(z)$

Transfer Function:

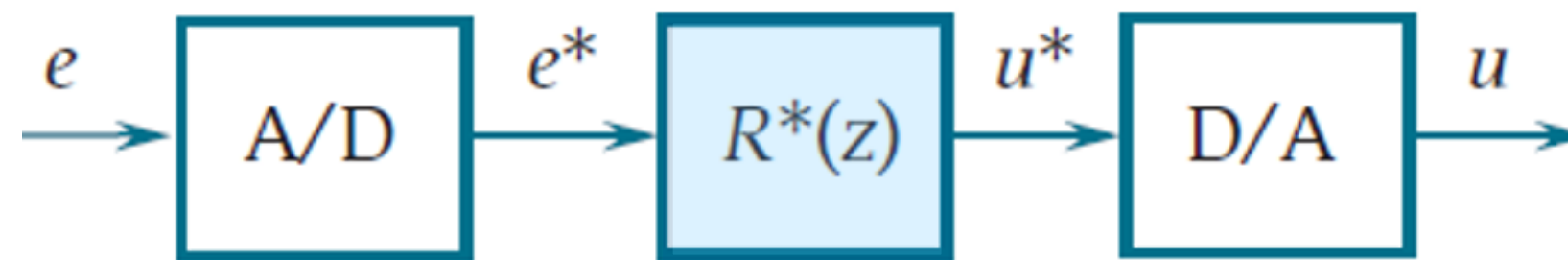
$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



The control element:



1bis. How to determine the finite difference equation to compute the control u at any step k ?

Digital Controller

Transfer Function:

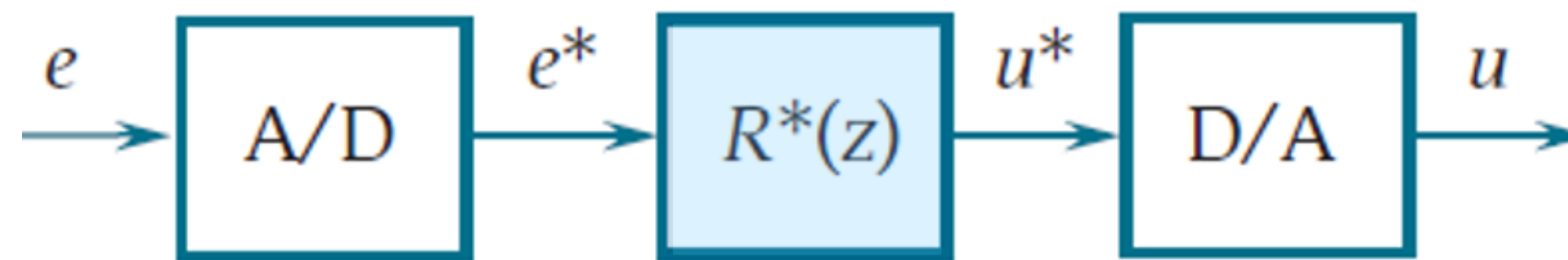
$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



The control element:



1bis. How to determine the finite difference equation to compute the control u at any step k ?

Start from the transfer function :

$$U^*(z) = R^*(z)E^*(z)$$

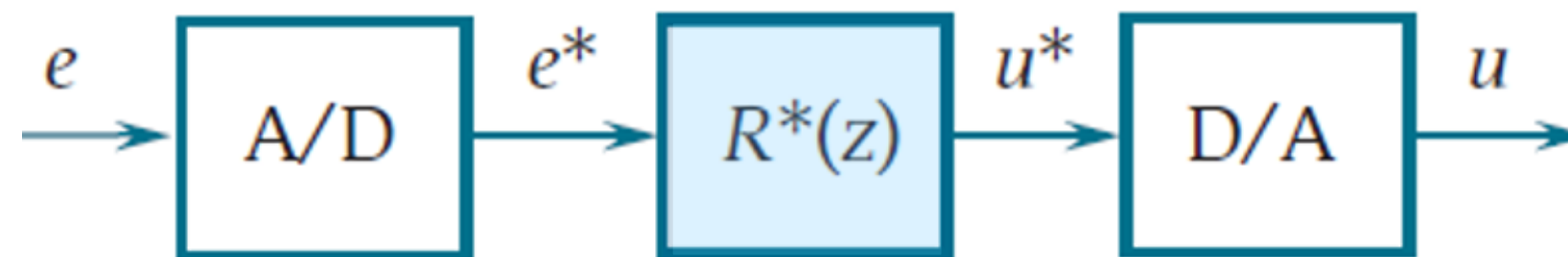
$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



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Start from the transfer function :

$$U^*(z) = R^*(z)E^*(z)$$

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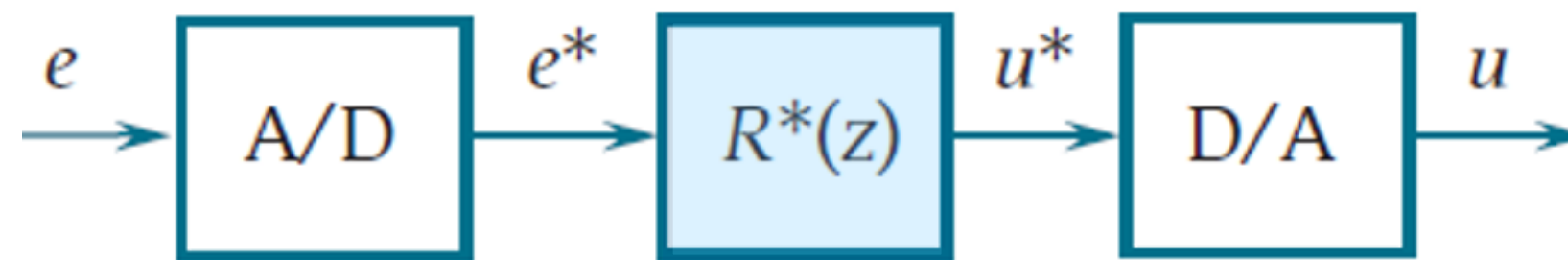
$$U^*(z) = \frac{\beta_m z^m + \beta_{m-1} z^{m-1} + \dots + \beta_1 z^1 + \beta_0}{z^m + \alpha_{m-1} z^{m-1} + \dots + \alpha_1 z^1 + \alpha_0} E^*(z)$$

Digital Control Schemes: Transfer Function

1. How to determine the transfer function of a discrete-time LTI system



The control element:



1bis. How to determine the finite difference equation to compute the control u at any step k ?

Start from the transfer function :

$$U^*(z) = R^*(z)E^*(z)$$

$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

$$U^*(z) = \frac{\beta_m z^m + \beta_{m-1} z^{m-1} + \dots + \beta_1 z^1 + \beta_0}{z^m + \alpha_{m-1} z^{m-1} + \dots + \alpha_1 z^1 + \alpha_0} E^*(z)$$

$$U^*(z)[z^m + \alpha_{m-1} z^{m-1} + \dots + \alpha_1 z^1 + \alpha_0] = [\beta_m z^m + \beta_{m-1} z^{m-1} + \dots + \beta_1 z^1 + \beta_0] E^*(z)$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



1bis. How to determine the finite difference equation to compute the control u at any step k ?

$$U^*(z)[z^m + \alpha_{m-1}z^{m-1} + \dots + \alpha_1z + \alpha_0] = [\beta_mz^m + \beta_{m-1}z^{m-1} + \dots + \beta_1z + \beta_0] E^*(z)$$

$$\begin{aligned} z^m U^*(z) = & -\alpha_{m-1}z^{m-1}U^*(z) - \dots - \alpha_1zU^*(z) - \alpha_0U^*(z) + \beta_mz^mE^*(z) + \beta_{m-1}z^{m-1}E^*(z) \\ & + \dots + \beta_1z^1E^*(z) + \beta_0E^*(z) \end{aligned}$$

$$\begin{aligned} U^*(z) = & -\alpha_{m-1}z^{-1}U^*(z) - \dots - \alpha_1z^{1-m}U^*(z) - \alpha_0z^{-m}U^*(z) + \beta_mE^*(z) + \beta_{m-1}z^{-1}E^*(z) \\ & + \dots + \beta_1z^{1-m}E^*(z) + \beta_0z^{-m}E^*(z) \end{aligned}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



1bis. How to determine the finite difference equation to compute the control u at any step k ?

$$U^*(z)[z^m + \alpha_{m-1}z^{m-1} + \dots + \alpha_1z + \alpha_0] = [\beta_mz^m + \beta_{m-1}z^{m-1} + \dots + \beta_1z + \beta_0] E^*(z)$$

$$\begin{aligned} z^m U^*(z) = & -\alpha_{m-1}z^{m-1}U^*(z) - \dots - \alpha_1zU^*(z) - \alpha_0U^*(z) + \beta_mz^mE^*(z) + \beta_{m-1}z^{m-1}E^*(z) \\ & + \dots + \beta_1zE^*(z) + \beta_0E^*(z) \end{aligned}$$

Anti Z-Transform

$\mathcal{Z}^{-1}[\cdot]$

$$\begin{aligned} U^*(z) = & -\alpha_{m-1}z^{-1}U^*(z) - \dots - \alpha_1z^{1-m}U^*(z) - \alpha_0z^{-m}U^*(z) + \beta_mE^*(z) + \beta_{m-1}z^{-1}E^*(z) \\ & + \dots + \beta_1z^{1-m}E^*(z) + \beta_0z^{-m}E^*(z) \end{aligned}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



Properties:

$$1) \mathcal{Z} [a \{x(k)\} + b \{y(k)\}] = a X(z) + b Y(z)$$

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1bis. How to determine the finite difference equation to compute the control u at any step k ?

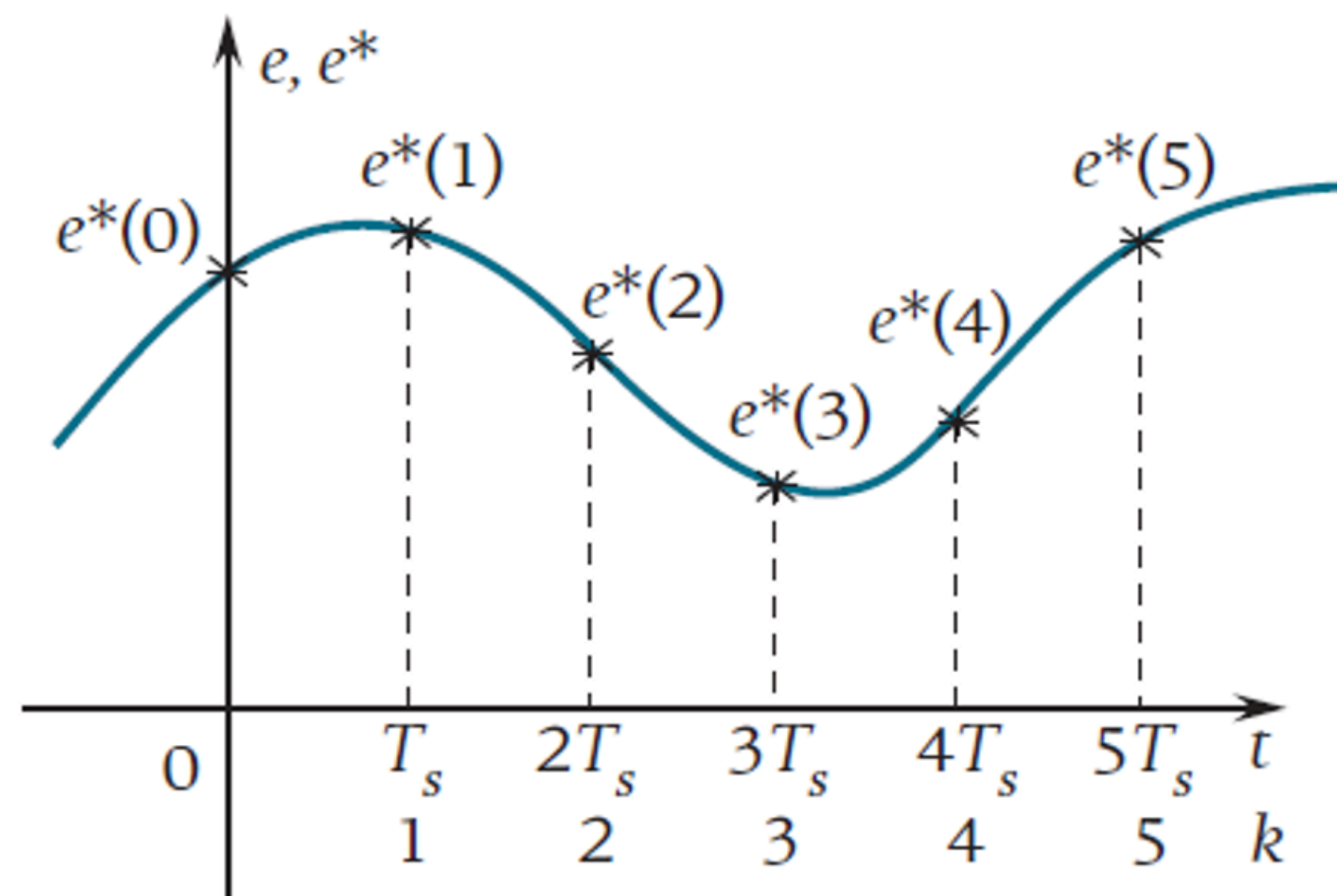


$$u^*(k) = -\alpha_{m-1}u^*(k-1) - \dots - \alpha_1u^*(k-m+1) \\ - \alpha_0u^*(k-m) + \beta_me^*(k) + \beta_{m-1}e^*(k-1) \\ + \dots + \beta_1z^{1-m}e^*(k-m+1) + \beta_0e^*(k-m)$$

$\mathcal{Z}^{-1}[\cdot]$



$$U^*(z) = -\alpha_{m-1}z^{-1}U^*(z) - \dots - \alpha_1z^{1-m} - \alpha_0z^{-m}U^*(z) + \beta_mE^*(z) + \beta_{m-1}z^{-1}E^*(z) \\ + \dots + \beta_1z^{1-m}E^*(z) + \beta_0z^{-m}E^*(z)$$

Digital Control Schemes: Problems to solve

A/D converter: it is a sampler

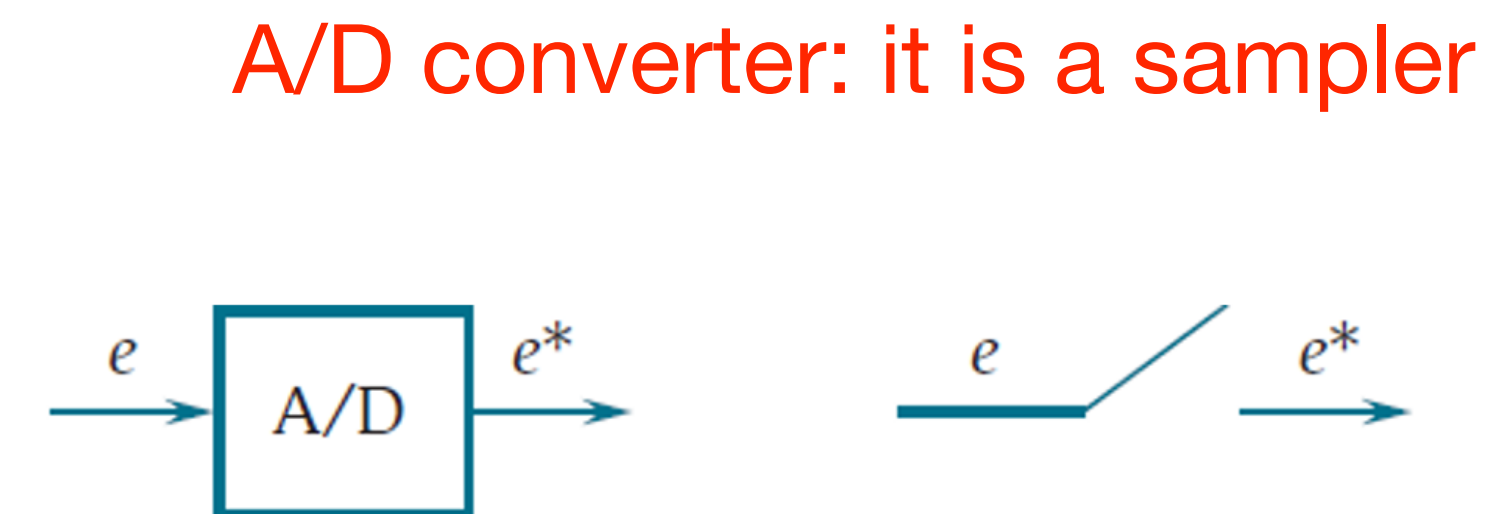


2. How to determine the sampling time T_s

Digital Control Schemes: **Fourier Transform**

Fourier Transform for continuous-time signals:

$$F(j\omega) = \mathcal{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

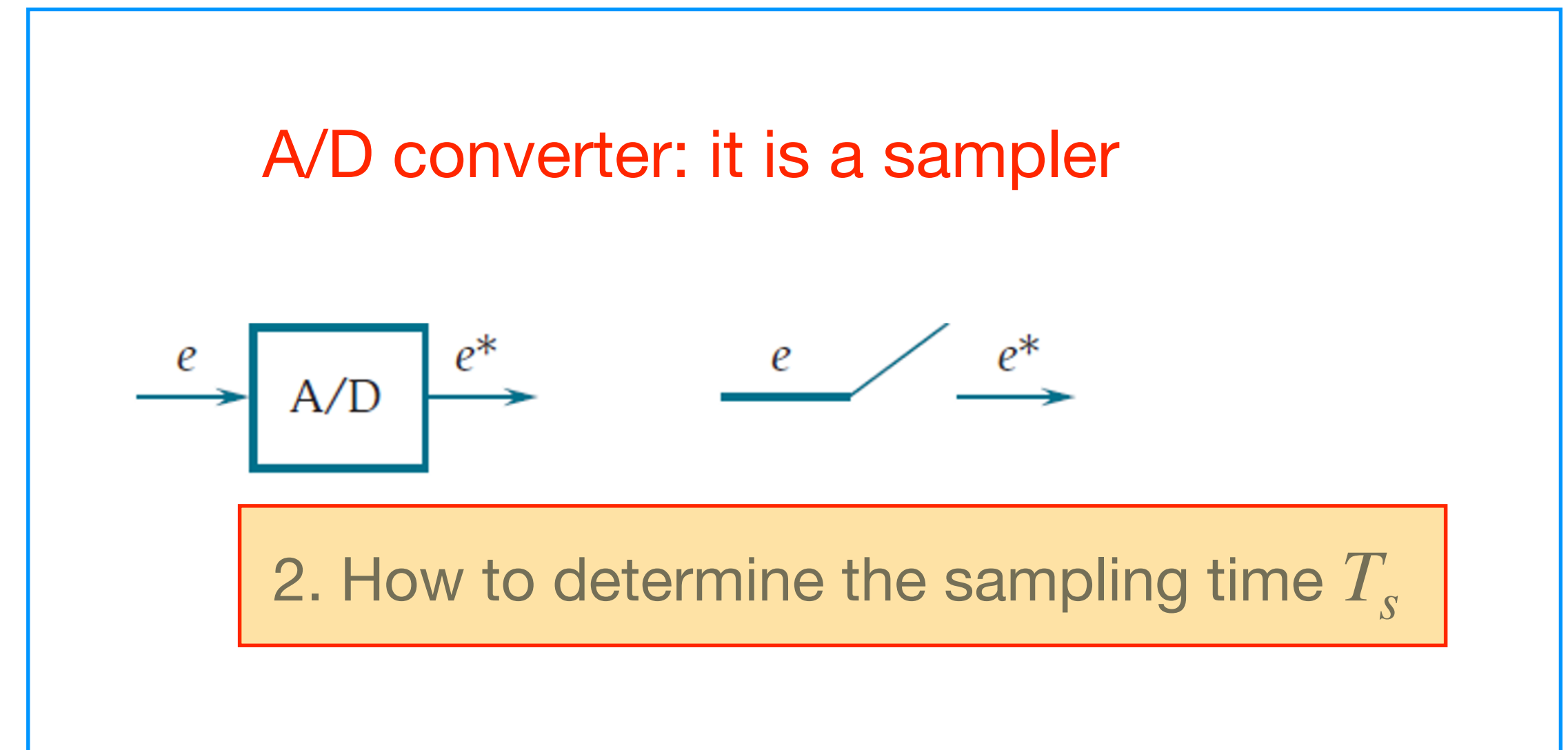


2. How to determine the sampling time T_s

Digital Control Schemes: **Fourier Transform**

Fourier Transform for continuous-time signals:

$$F(j\omega) = \mathcal{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$



Fourier Transform for discrete-time signals:

$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)]$$

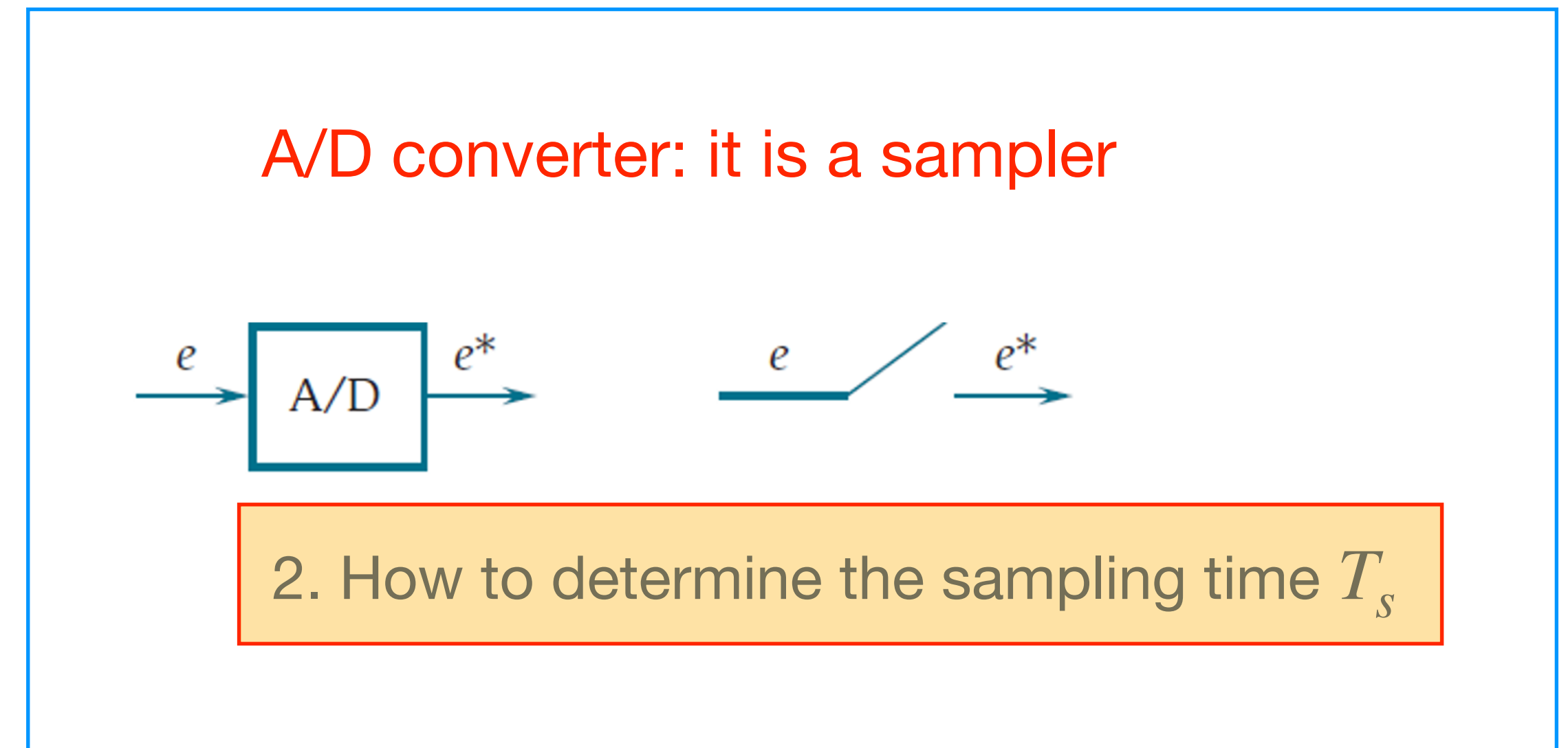
Digital Control Schemes: **Fourier Transform**

Fourier Transform for continuous-time signals:

$$F(j\omega) = \mathcal{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$



which relationship?



Fourier Transform for discrete-time signals:

$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)]$$

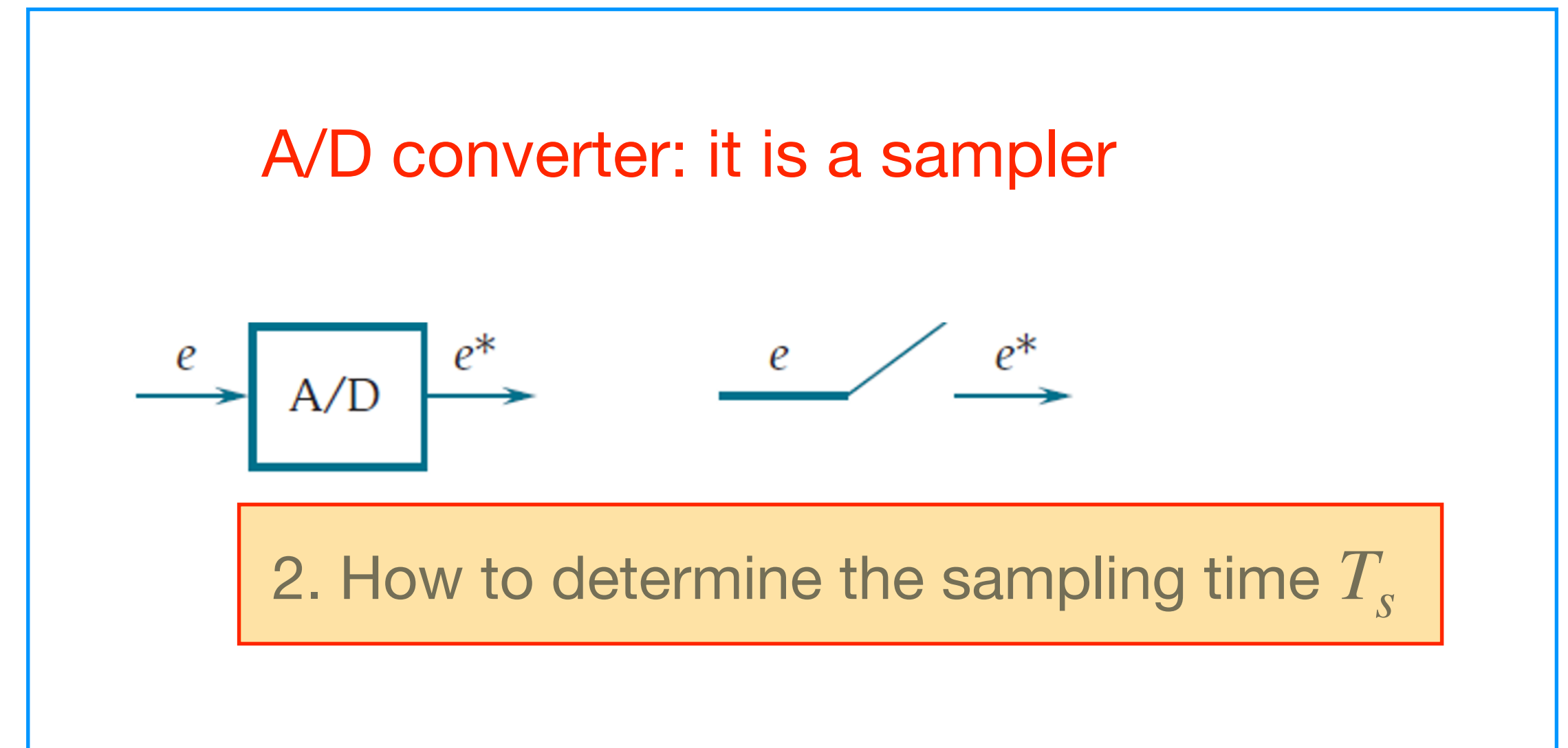
Digital Control Schemes: **Fourier Transform**

Fourier Transform for continuous-time signals:

$$F(j\omega) = \mathcal{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$



which relationship?



Fourier Transform for discrete-time signals:

$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega)$$

Digital Control Schemes: **Fourier Transform**

Fourier Transform for continuous-time signals:

$$F(j\omega) = \mathcal{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$



which relationship?

A/D converter: it is a sampler



2. How to determine the sampling time T_s

Fourier Transform for discrete-time signals:

$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{+\infty} F(j(\omega + h\omega_s))$$

Digital Control Schemes: Fourier Transform

Fourier Transform for continuous-time signals:

$$F(j\omega) = \mathcal{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$



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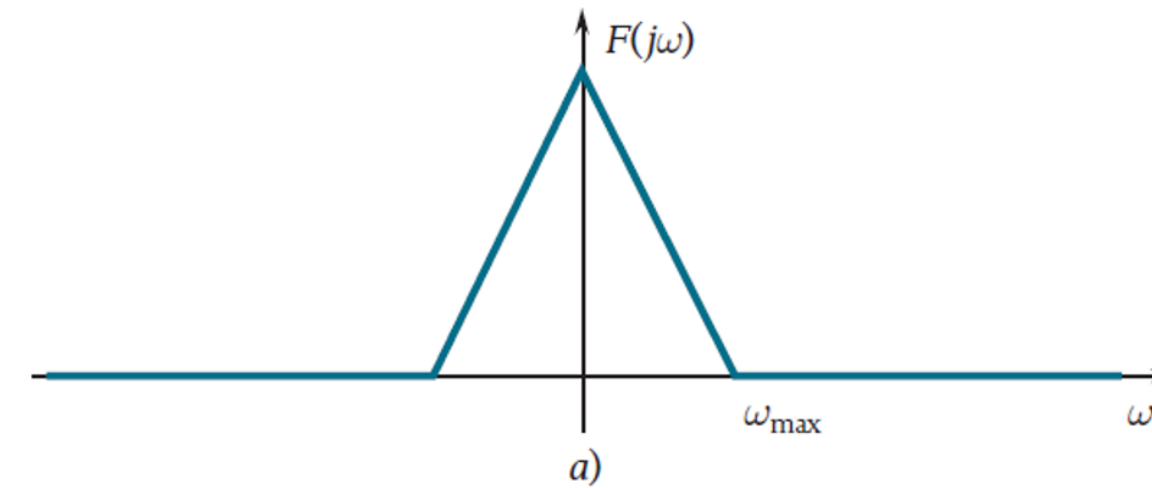
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Fourier Transform for discrete-time signals:

$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{+\infty} F(j(\omega + h\omega_s)) \quad \omega_s = \frac{2\pi}{T_s}$$

Digital Control Schemes: Fourier Transform

Example:

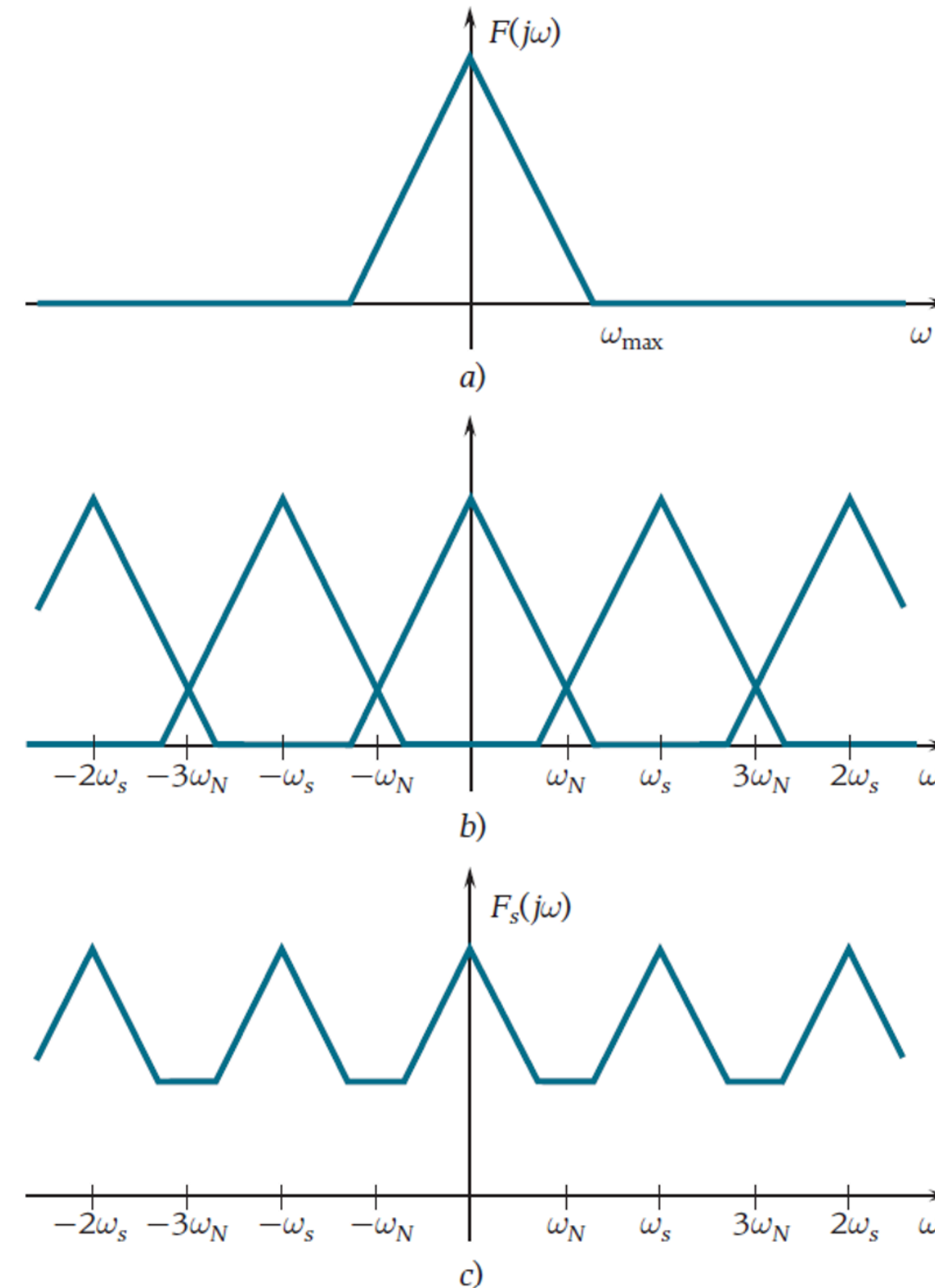


Fourier Transform for
the continuous-time
signal $f(t)$

$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{+\infty} F(j(\omega + h\omega_s)) \quad \omega_s = \frac{2\pi}{T_s}$$

Digital Control Schemes: Fourier Transform

Example:



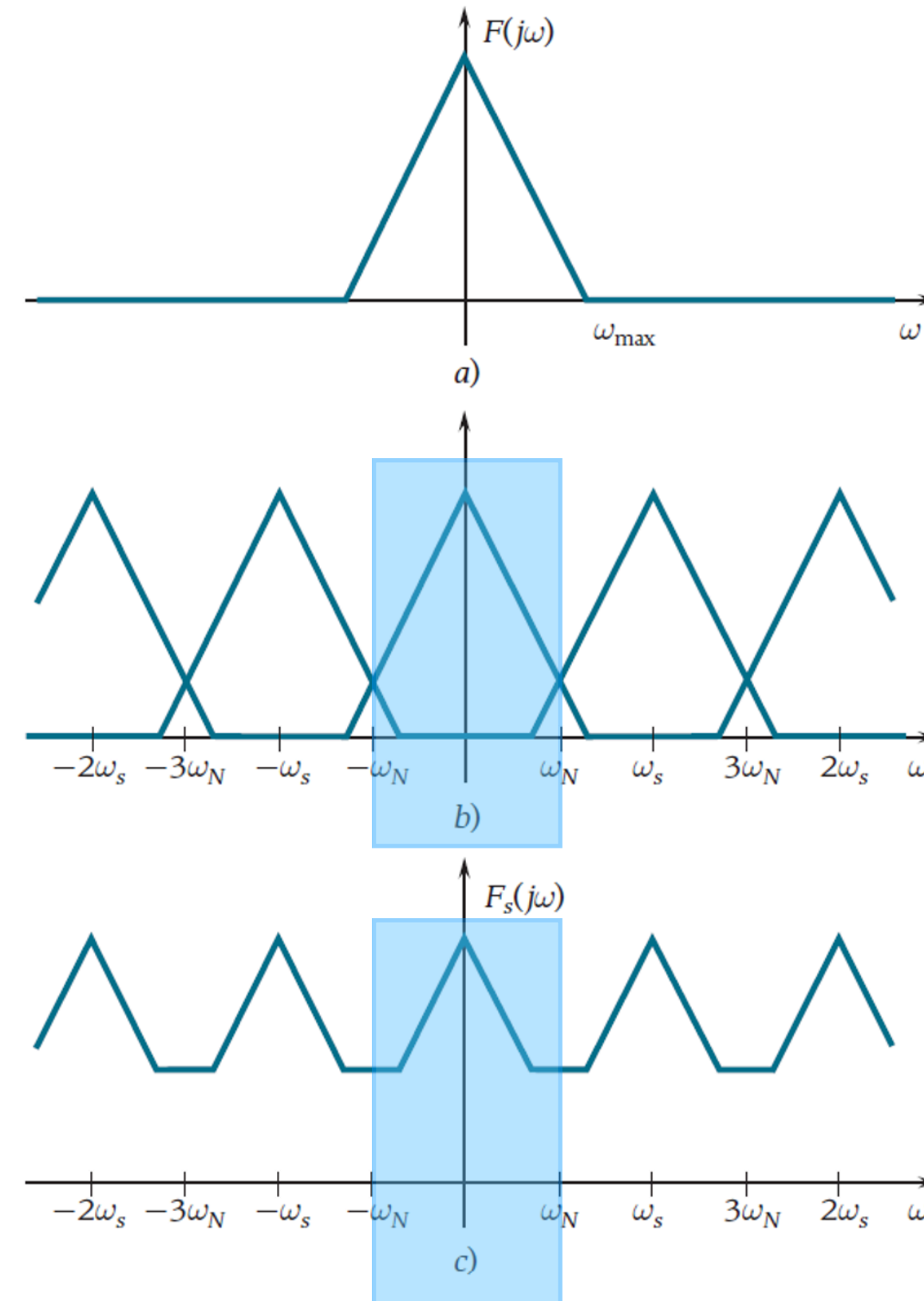
Fourier Transform for
the continuous-time
signal $f(t)$

Fourier Transform for
the discrete-time signal
 $f^*(k)$

$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{+\infty} F(j(\omega + h\omega_s)) \quad \omega_s = \frac{2\pi}{T_s}$$

Digital Control Schemes: Fourier Transform

Example:



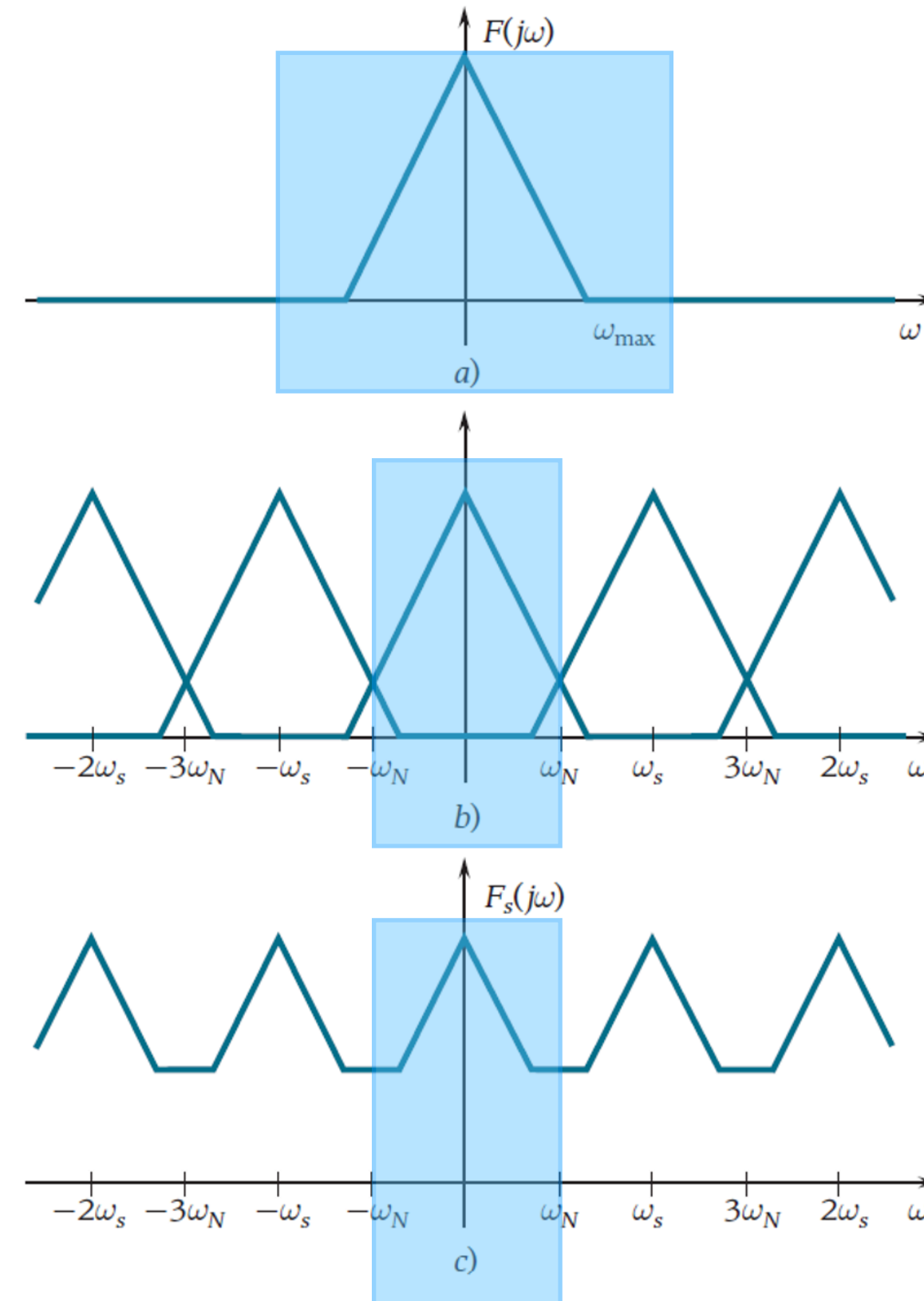
Fourier Transform for
the continuous-time
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Fourier Transform for
the discrete-time signal
 $f^*(k)$

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Digital Control Schemes: **Fourier Transform**

Example:



Fourier Transform for
the continuous-time
signal $f(t)$

Fourier Transform for
the discrete-time signal
 $f^*(k)$

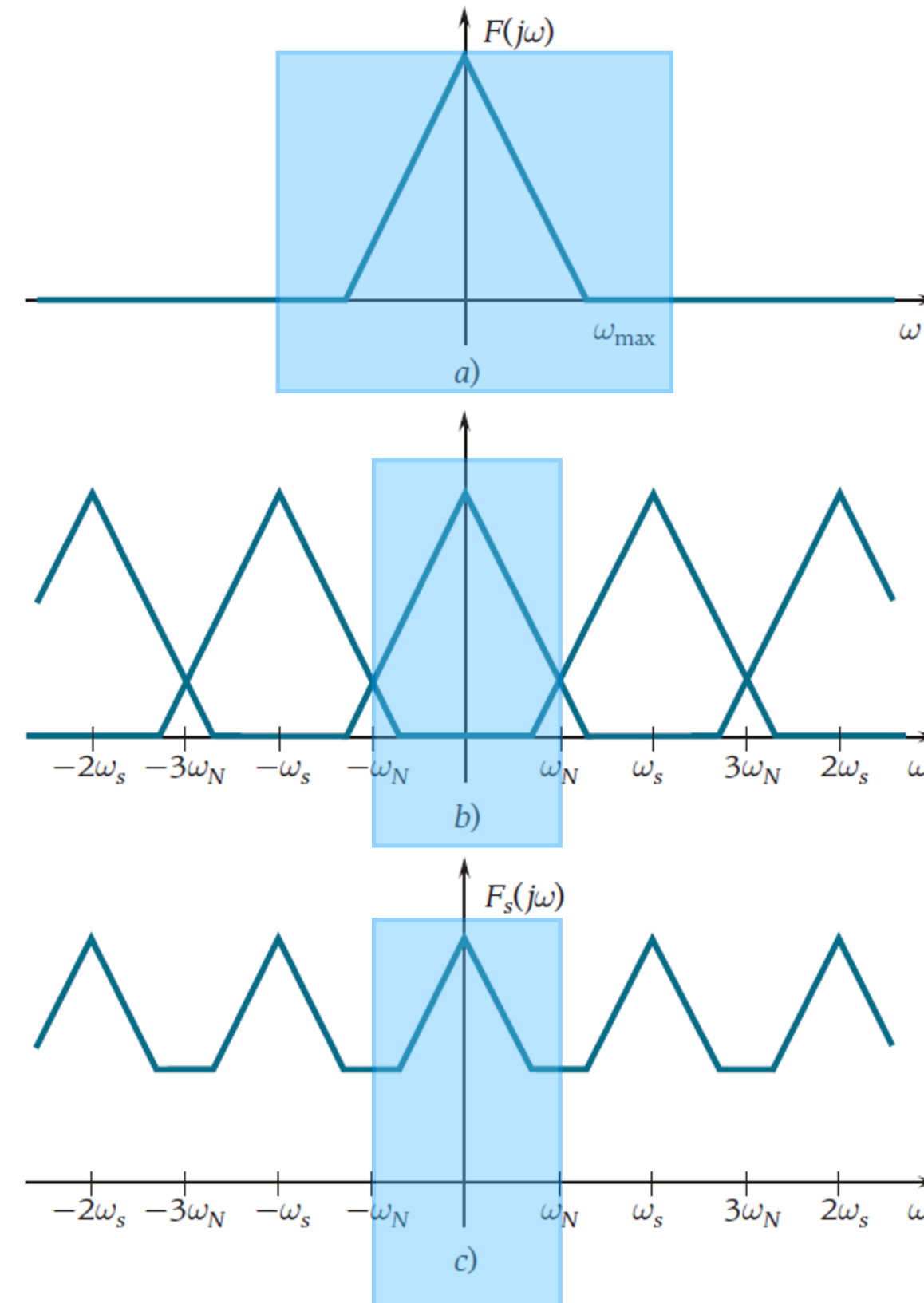
$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{+\infty} F(j(\omega + h\omega_s)) \quad \omega_s = \frac{2\pi}{T_s}$$

Digital Control Schemes: Aliasing

Example:

They are different:

Aliasing!



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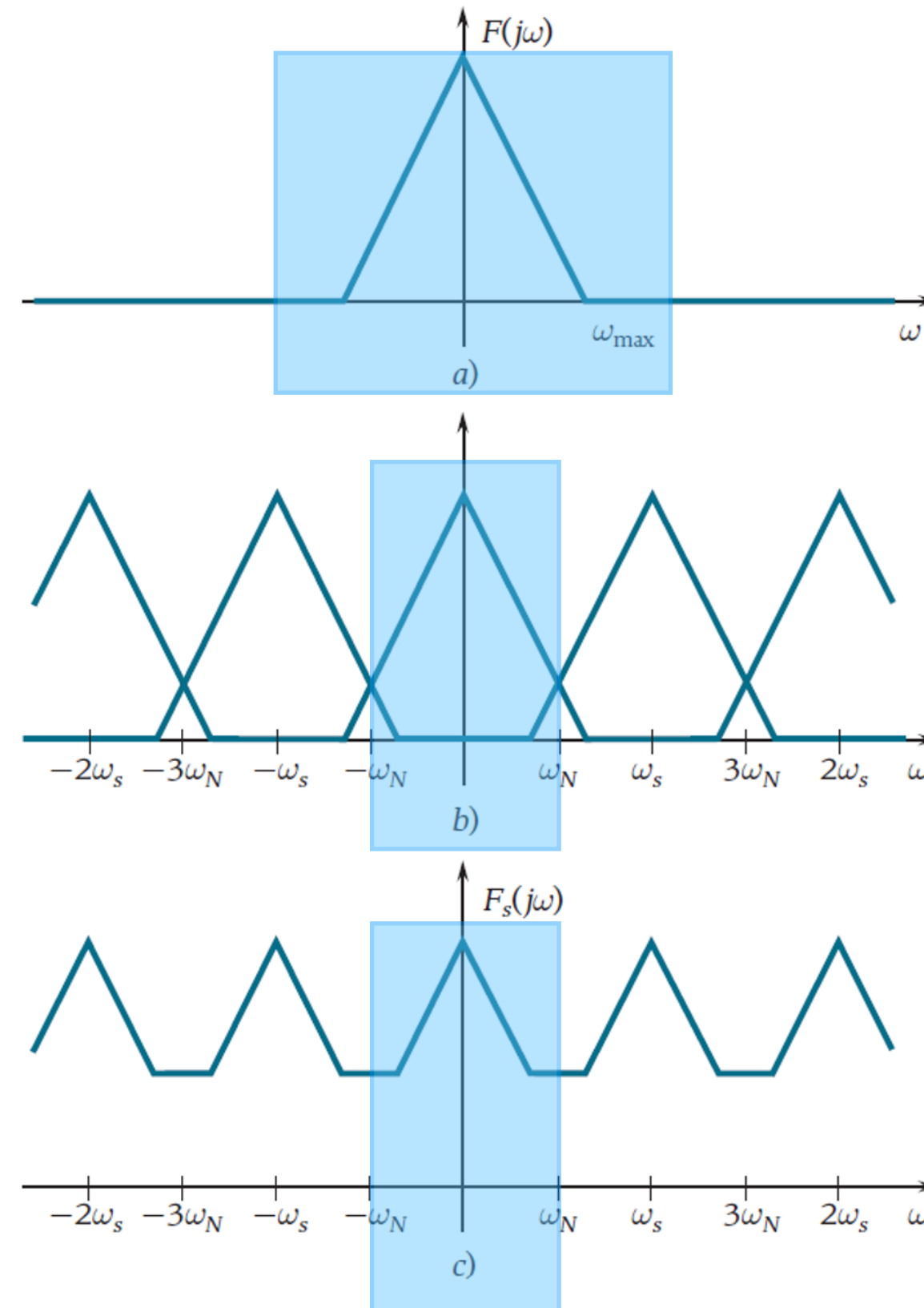
Digital Control Schemes: Aliasing

Example:

They are different:

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To solve set $\omega_N > \omega_{max}$



Fourier Transform for the continuous-time signal $f(t)$

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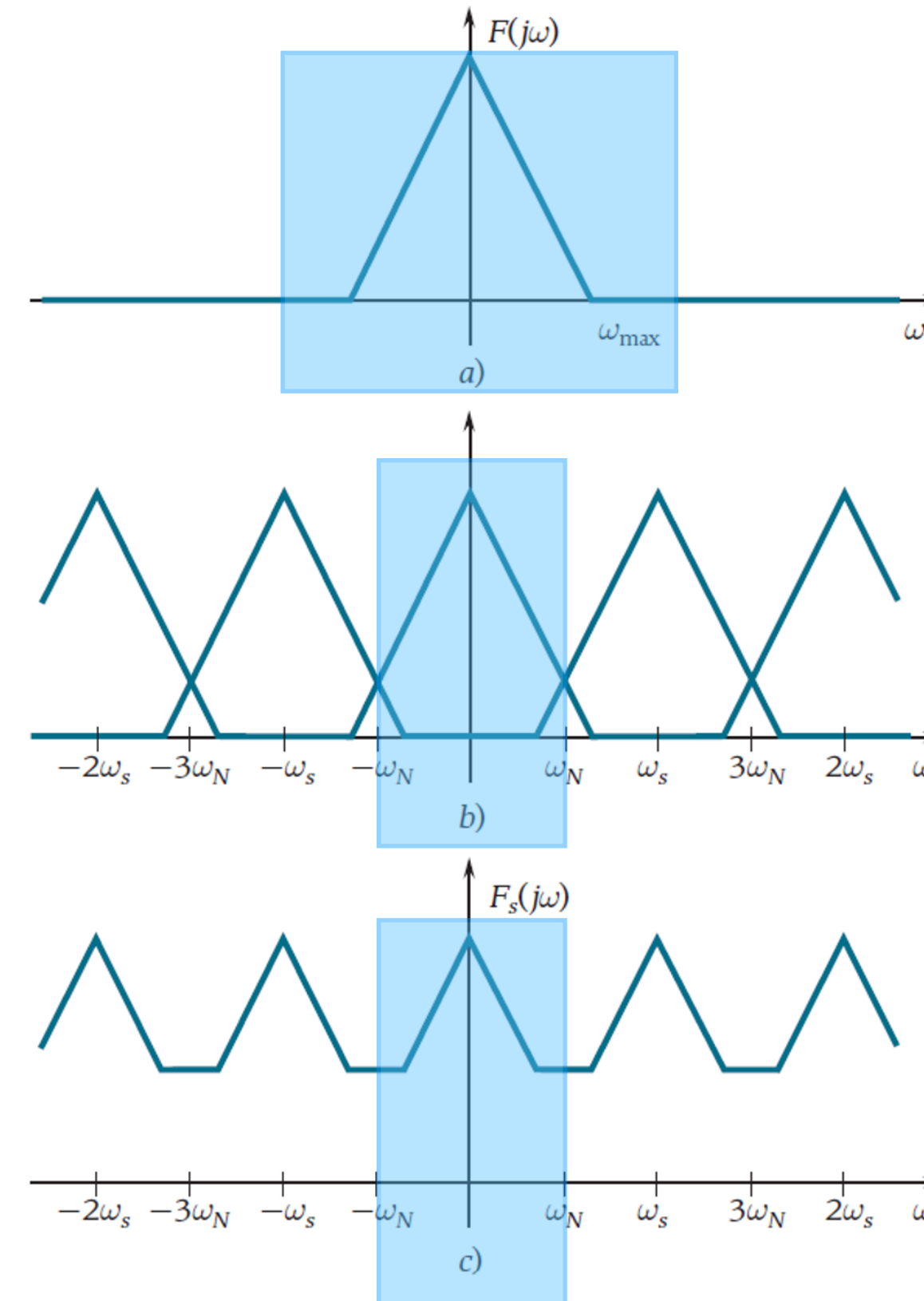
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Fourier Transform for the continuous-time signal $f(t)$

Fourier Transform for the discrete-time signal $f^*(k)$

$$\omega_N > \omega_{max} \rightarrow \frac{\omega_s}{2} > \omega_{max} \rightarrow \omega_s > 2\omega_{max}$$

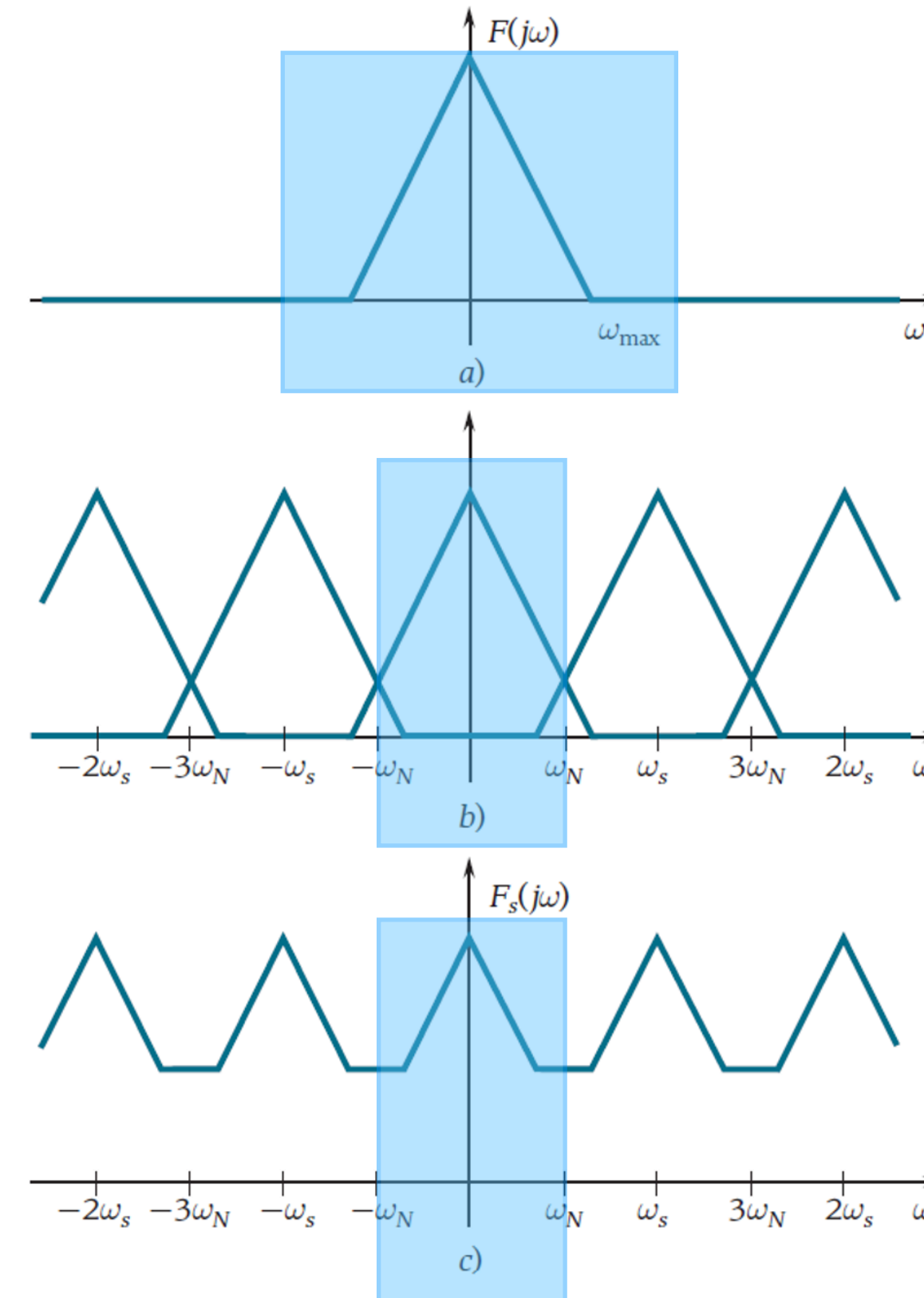
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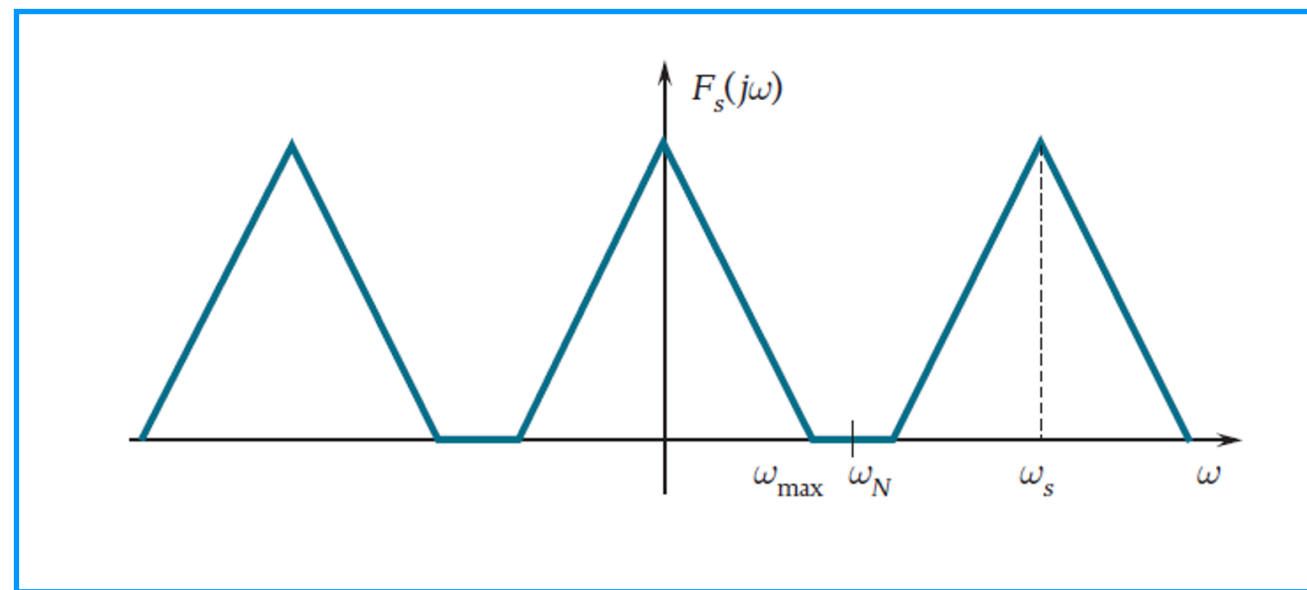
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Fourier Transform for the discrete-time signal $f^*(k)$

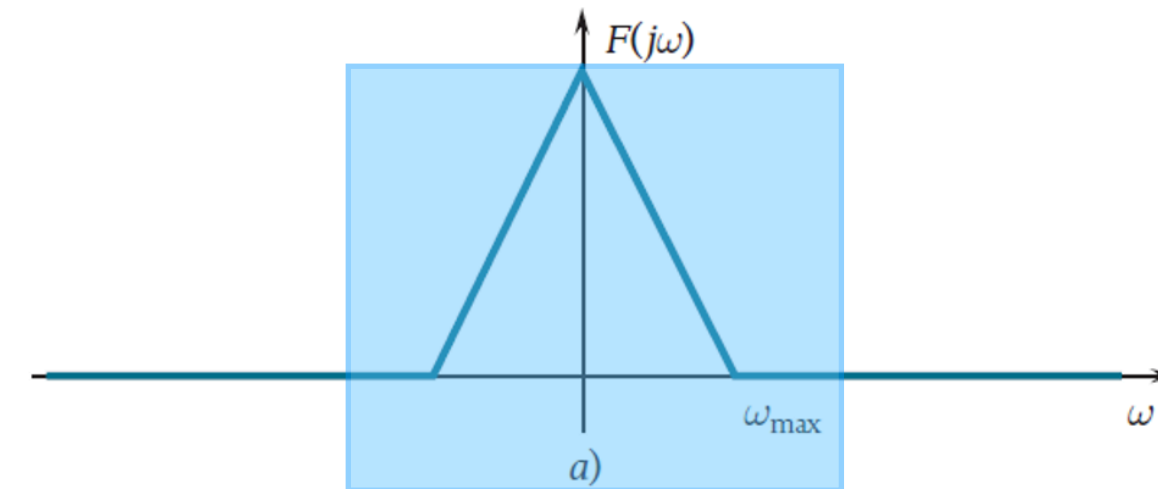
$$\omega_N > \omega_{max} \rightarrow \frac{\omega_s}{2} > \omega_{max} \rightarrow \boxed{\omega_s > 2\omega_{max}} \quad \text{Shannon Theorem}$$

Digital Control Schemes: **Aliasing**

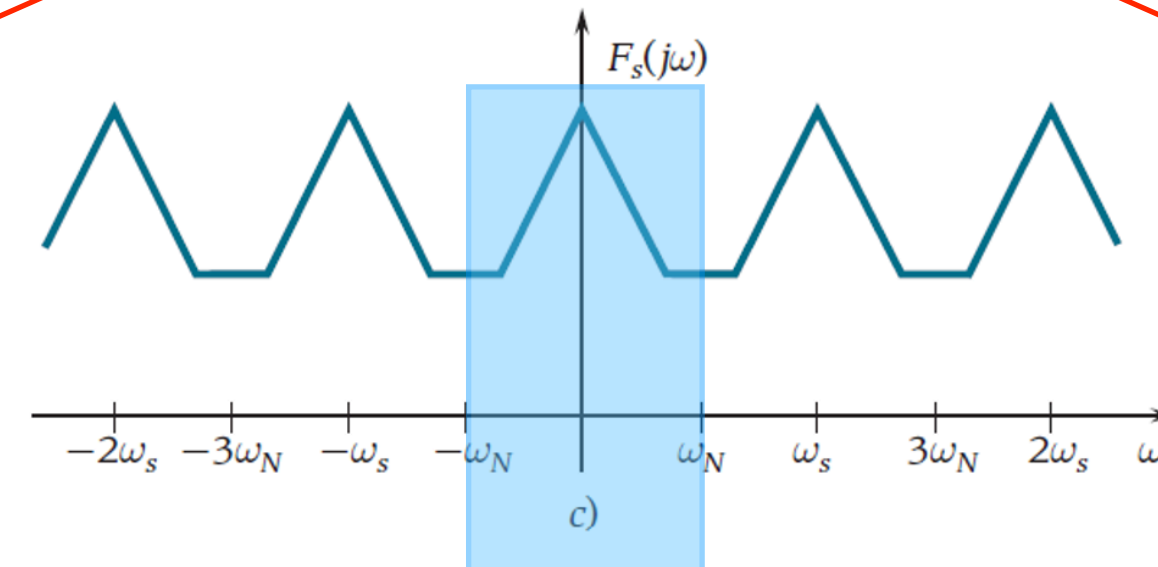
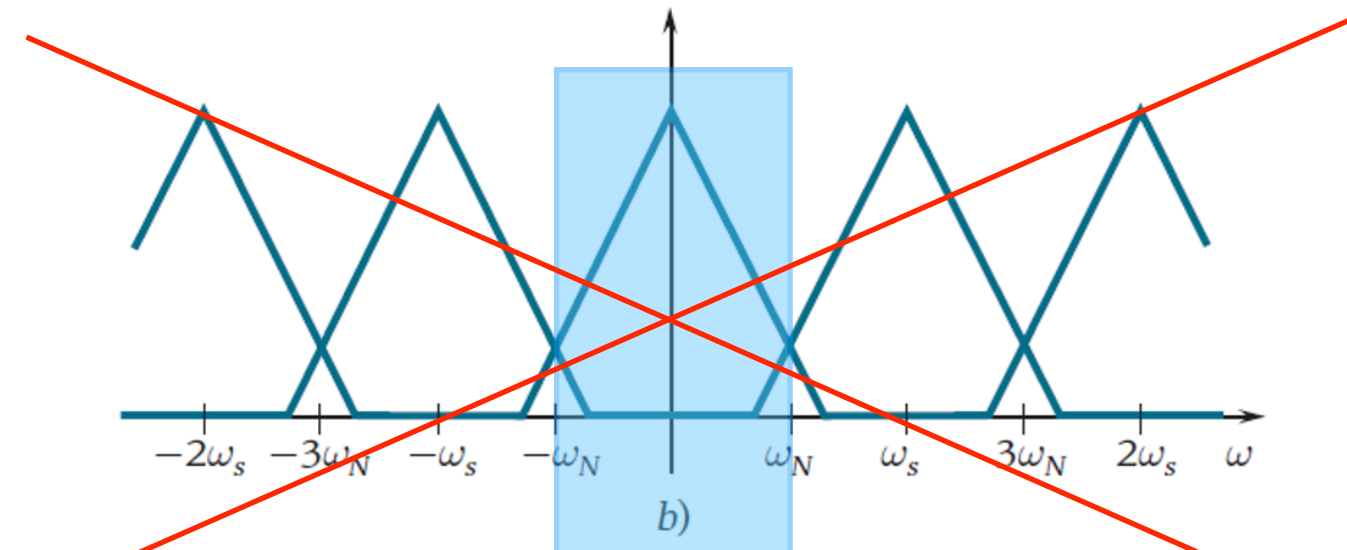
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Fourier Transform for the continuous-time signal $f(t)$



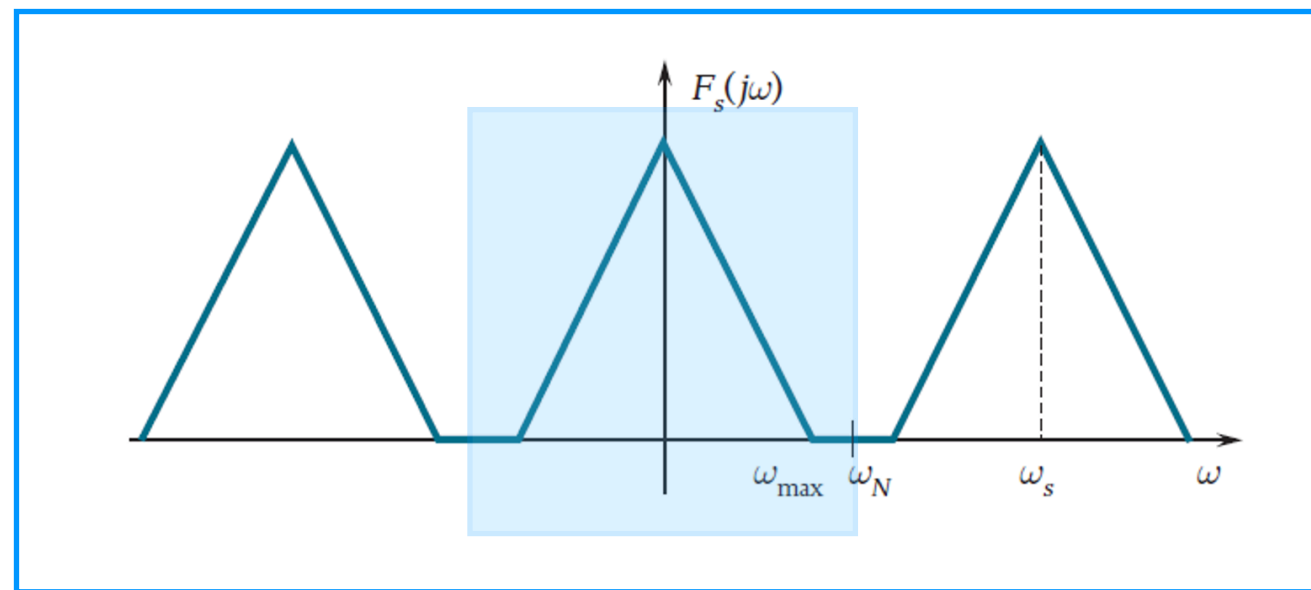
Fourier Transform for the discrete-time signal $f^*(k)$

$$\omega_N > \omega_{max} \rightarrow \frac{\omega_s}{2} > \omega_{max} \rightarrow \boxed{\omega_s > 2\omega_{max}}$$

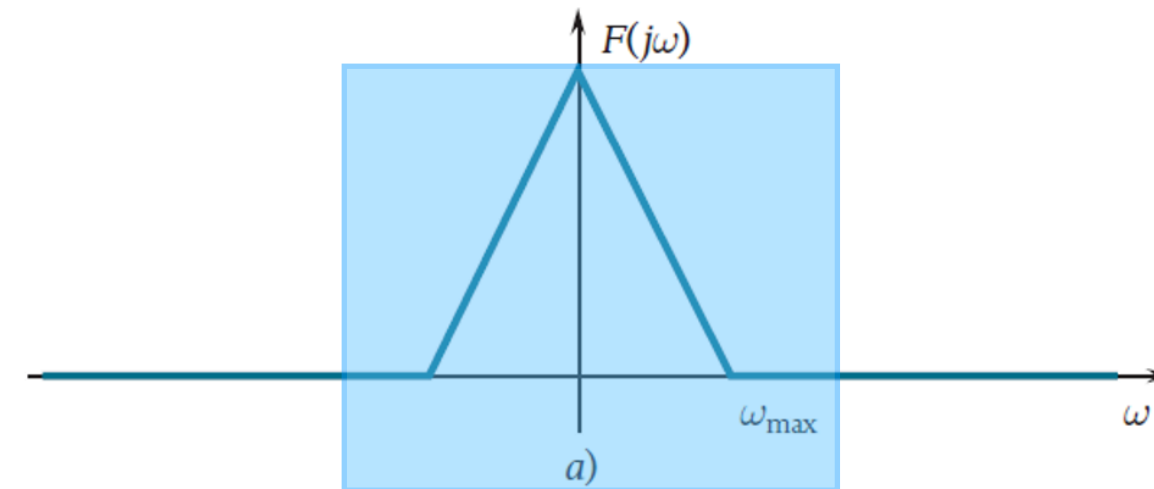
Shannon Theorem

Digital Control Schemes: Aliasing

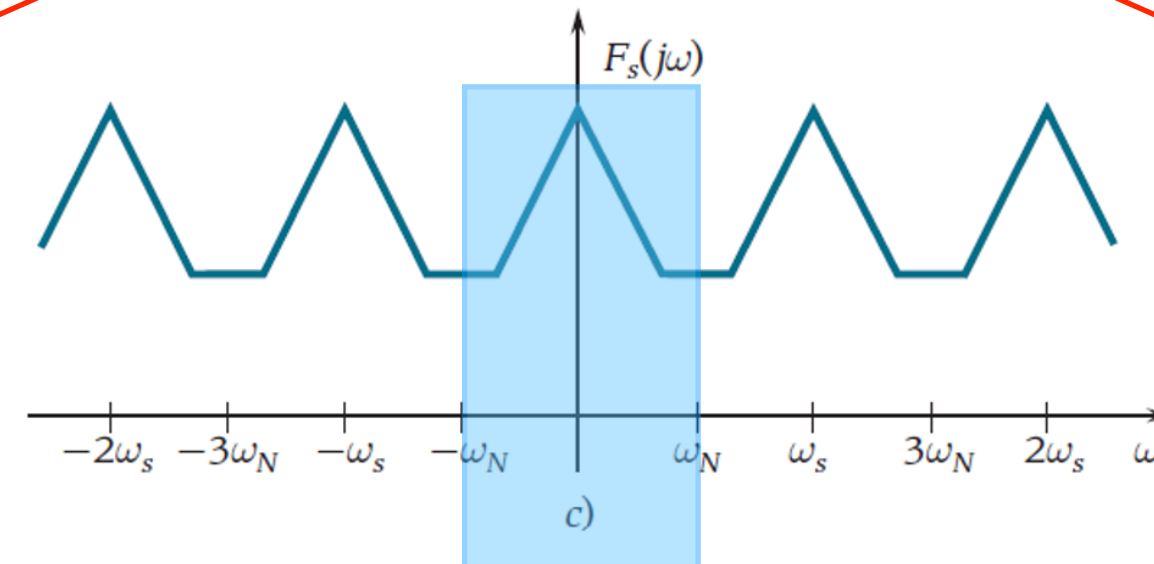
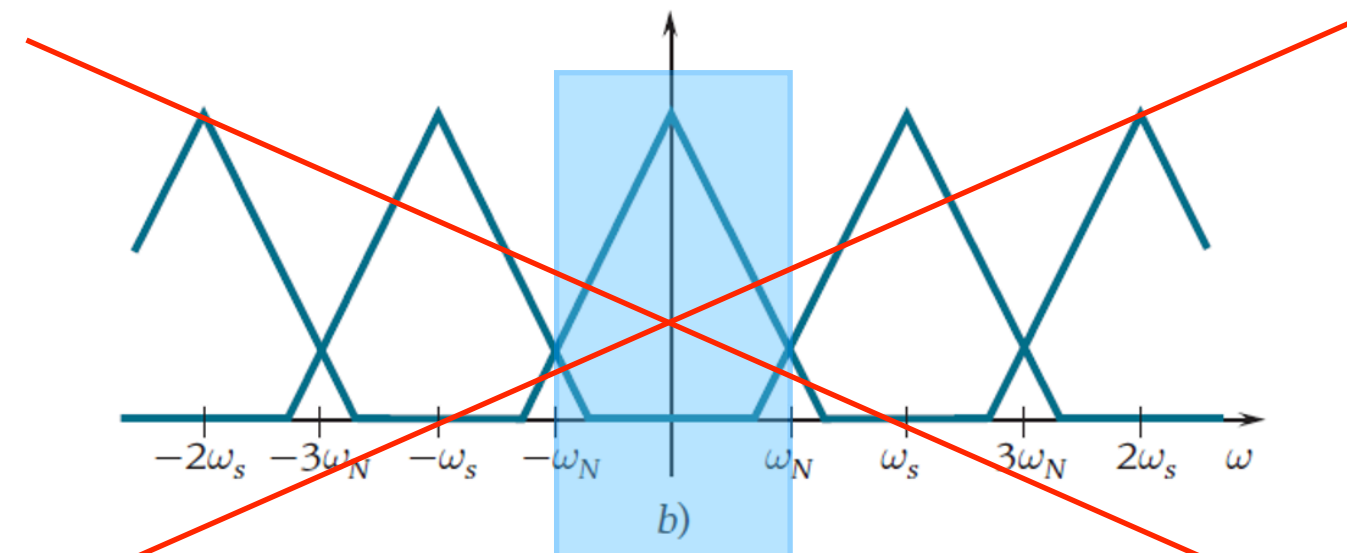
Example:



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Fourier Transform for the discrete-time signal $f^*(k)$

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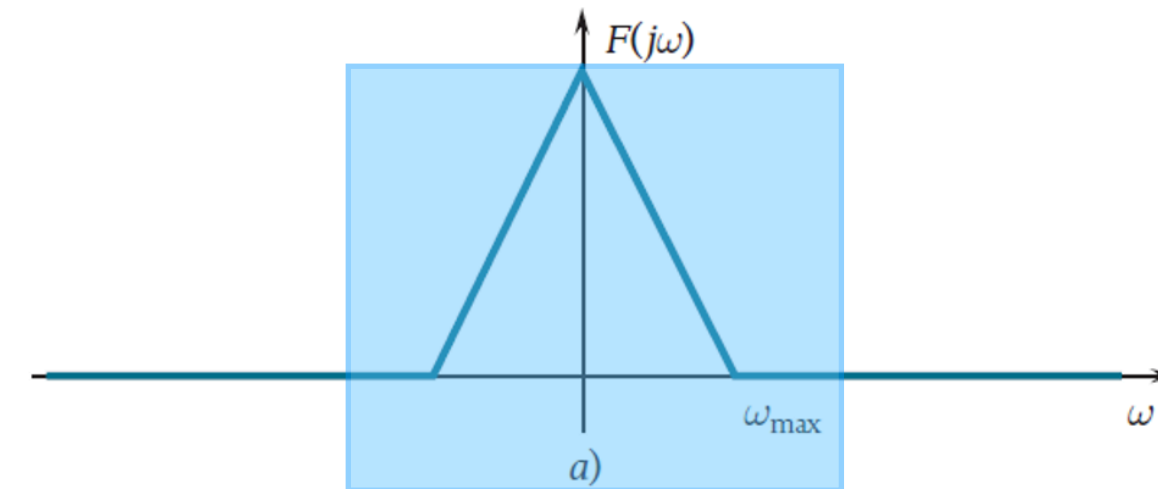
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Digital Control Schemes: Shannon Theorem

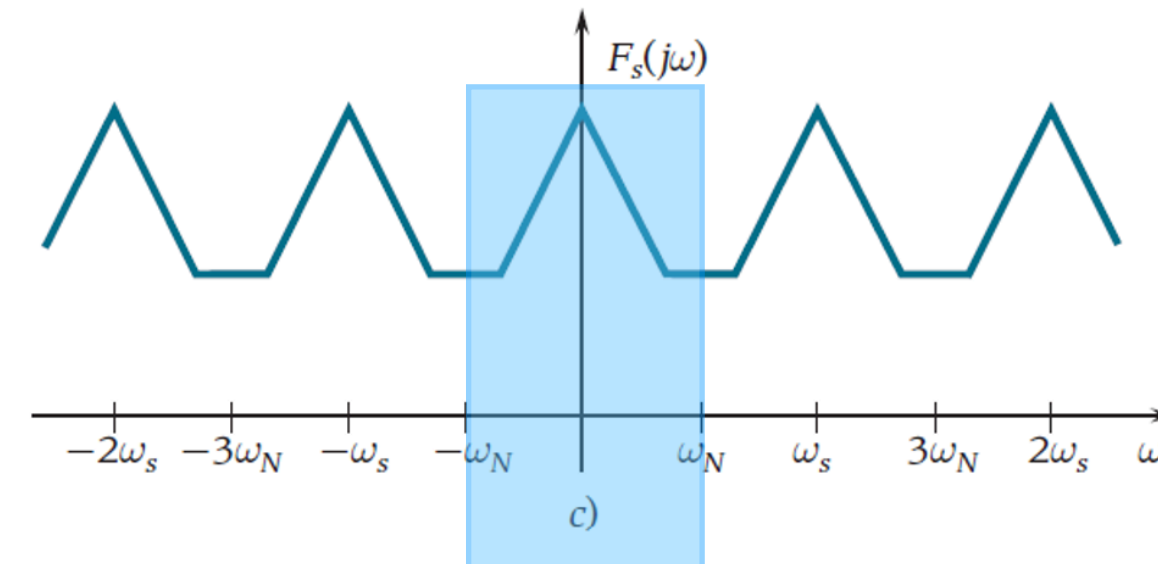
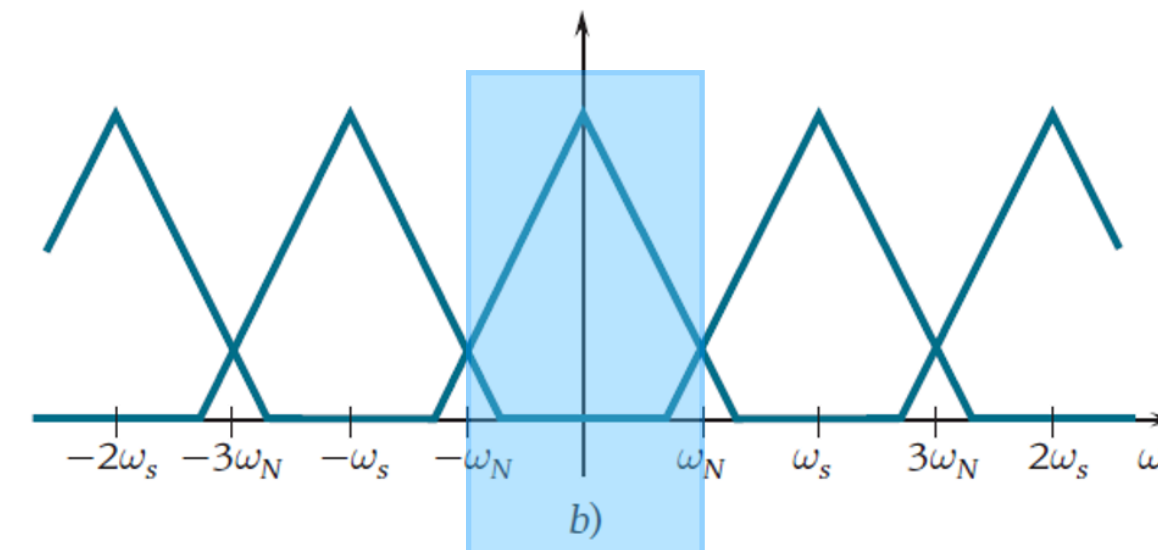
Example:

For control systems:

$$\omega_{max} = \omega_{BW_{CL}}$$



Fourier Transform for the continuous-time signal $f(t)$



Fourier Transform for the discrete-time signal $f^*(k)$

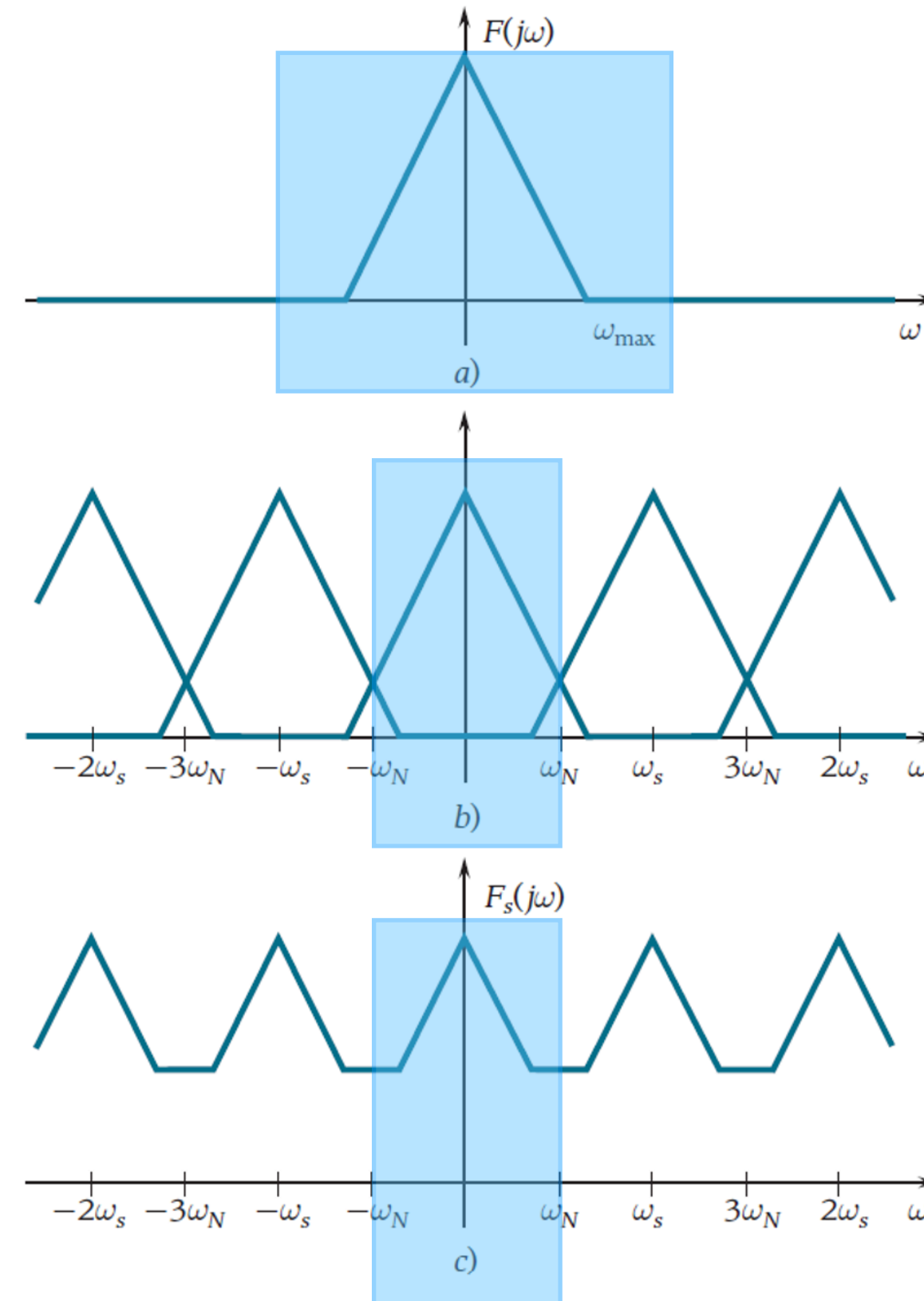
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Digital Control Schemes: Shannon Theorem

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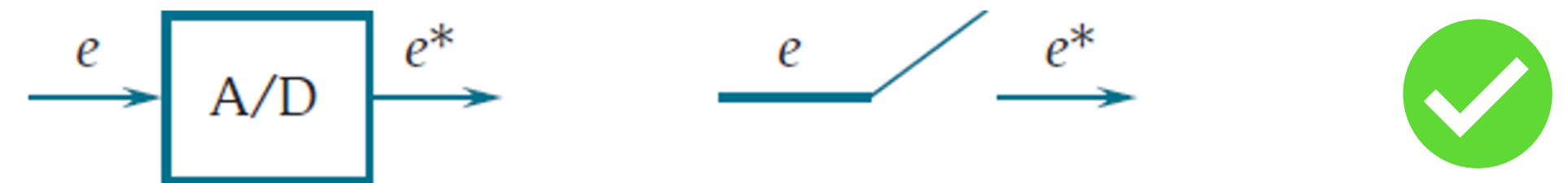
$$\omega_N > \omega_{BW_{CL}} \rightarrow \frac{\omega_s}{2} > \omega_{BW_{CL}} \rightarrow \omega_s > 2\omega_{BW_{CL}} \quad \text{Shannon Theorem}$$

Digital Control Schemes: **Choice of the Sampling Time**

For control systems:

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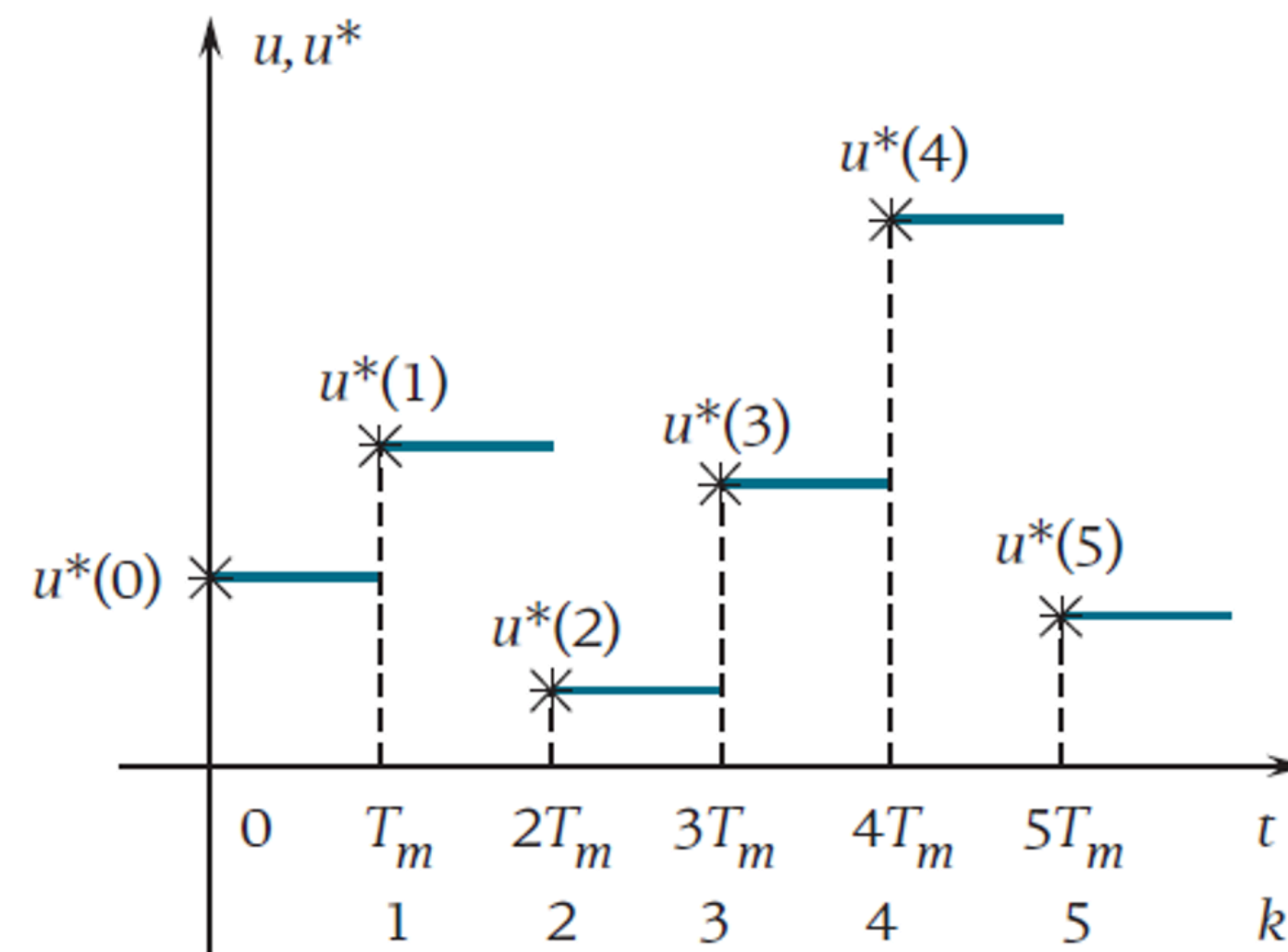
A/D converter: it is a sampler

2. How to determine the sampling time T_s

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$\omega_s = \frac{2\pi}{T_s} \rightarrow T_s = \frac{2\pi}{\omega_s}$

Digital Control Schemes: Problems to solve

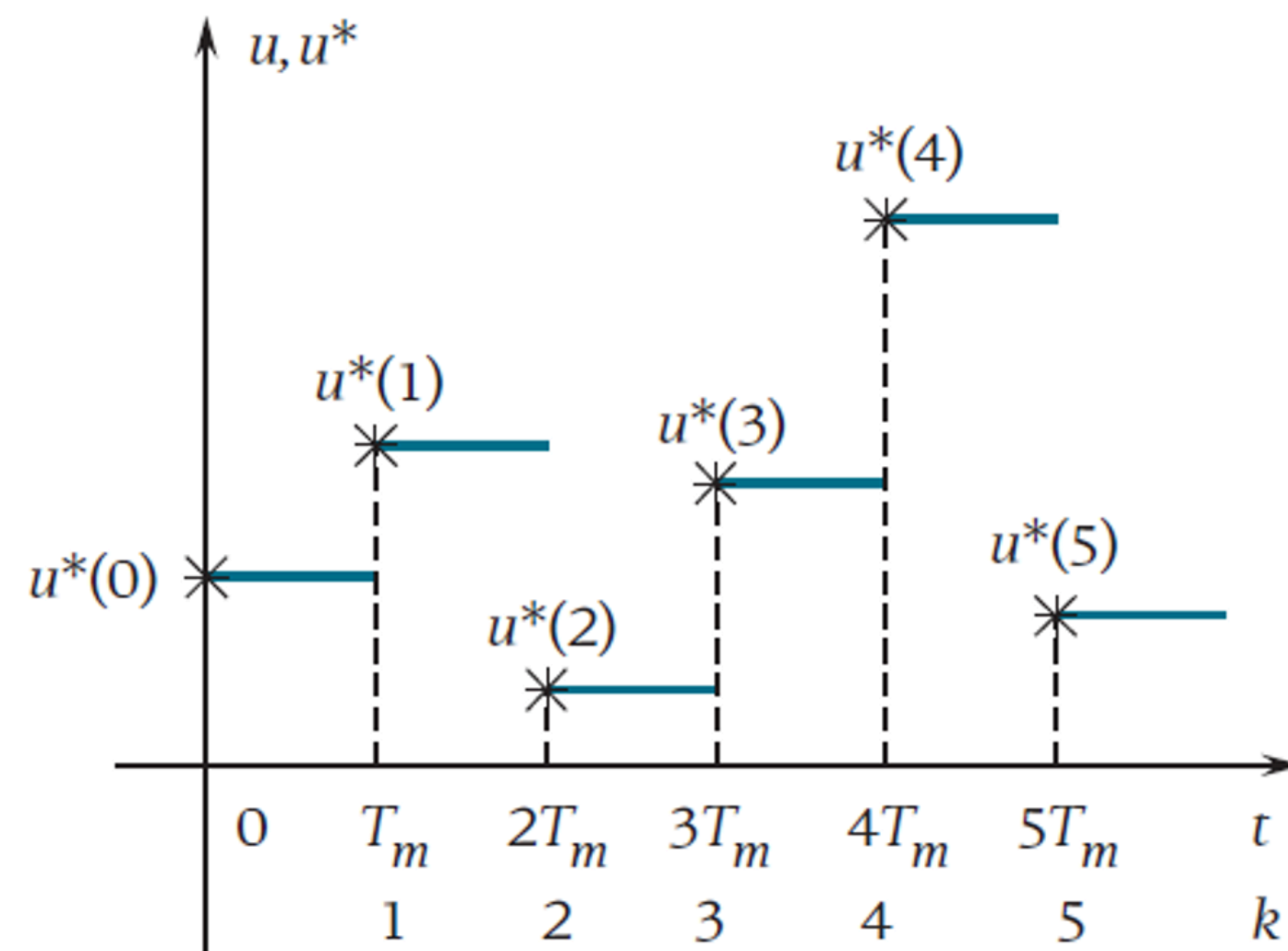


D/A converter: it is a hold



3. How to model the ZOH effect

Digital Control Schemes: Effect of the ZOH



D/A converter: it is a hold

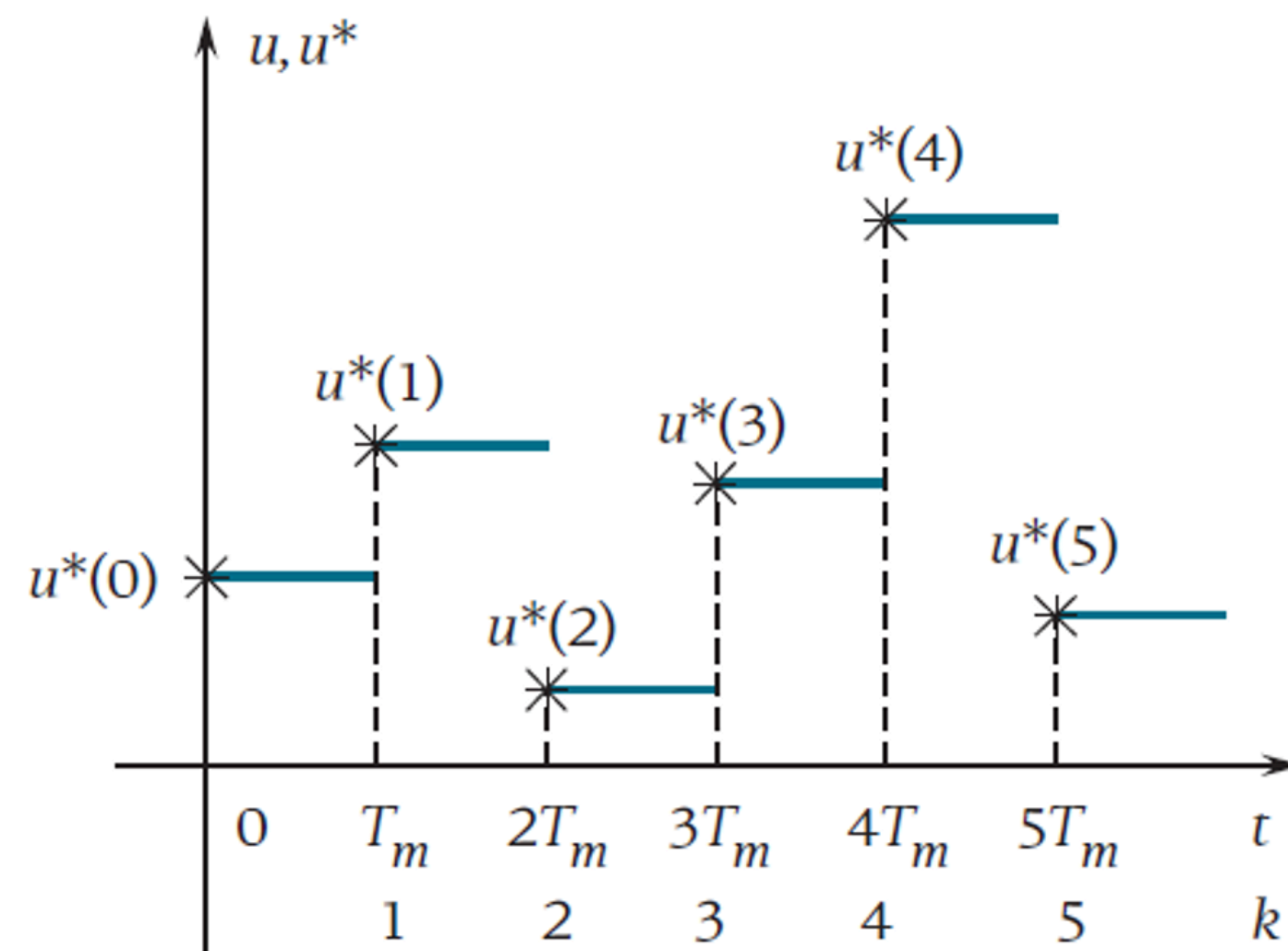


3. How to model the ZOH effect

The ZOH can be regarded as a LOW PASS FILTER with bandwidth $[0, \omega_N]$.
In that interval, the following approximation can be considered valid:

$$H_0(j\omega) \approx T_s e^{-j\omega \frac{T_s}{2}}, \quad \omega \in [0, \omega_N]$$

Digital Control Schemes: Effect of the ZOH



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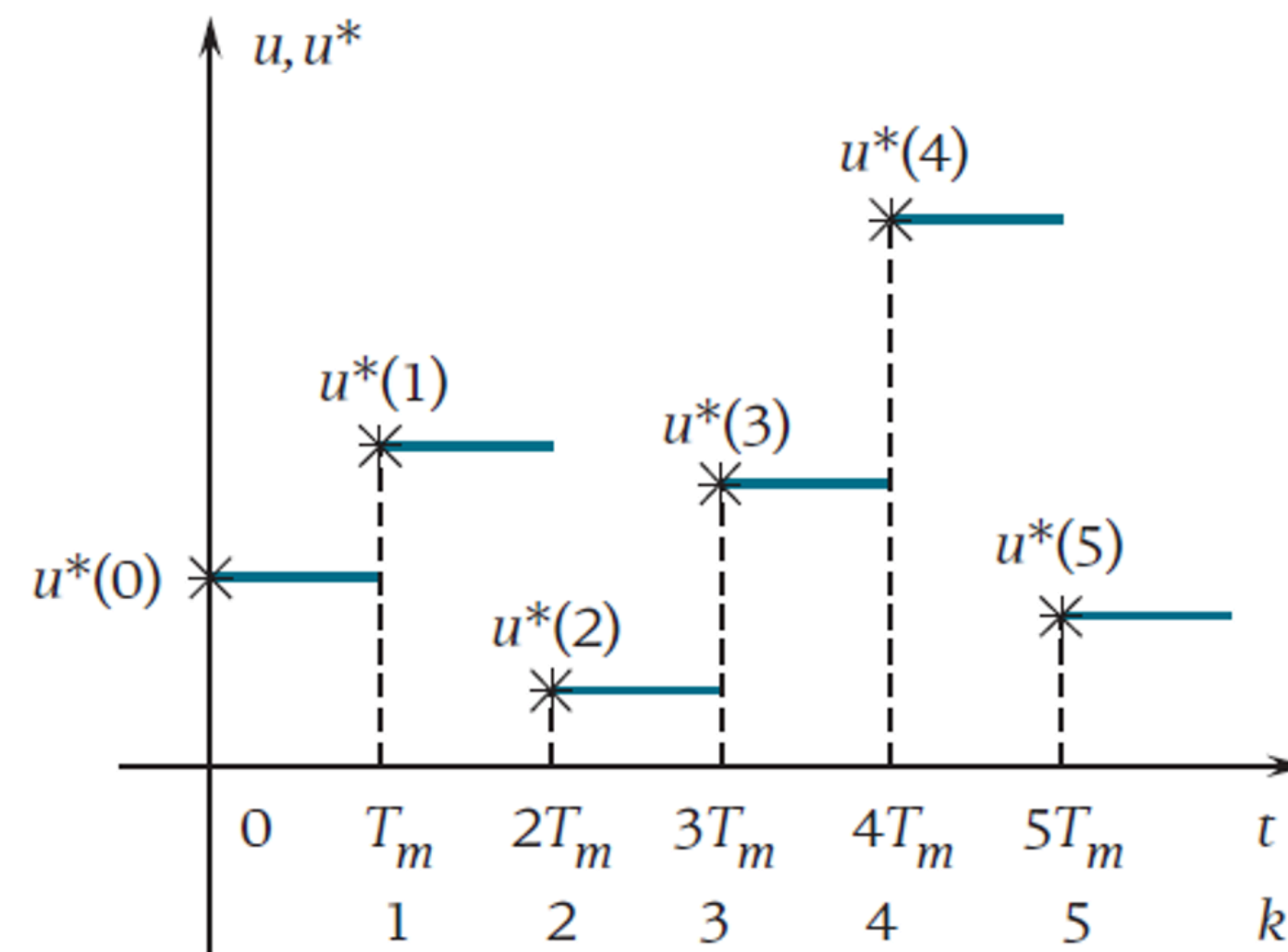
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Digital Control Schemes: Effect of the ZOH



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3. How to model the ZOH effect



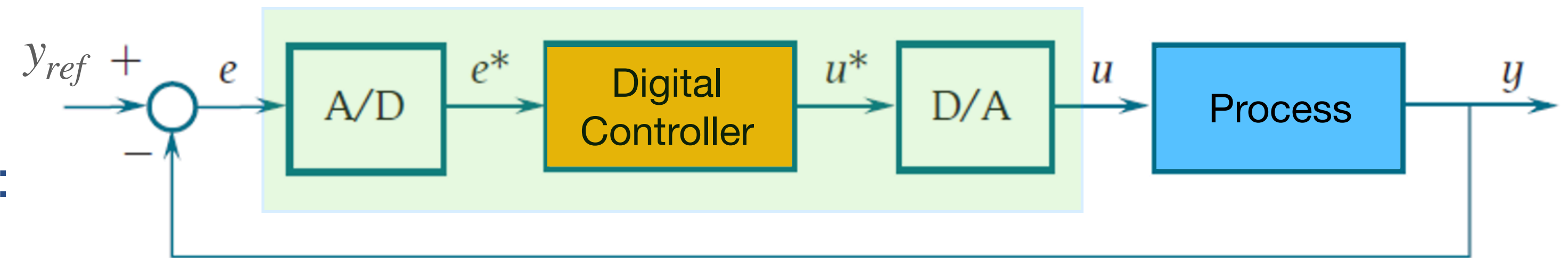
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impact on the cascade
connection of
sampler-digital controller-ZOH

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Digital Control Schemes: **Effect of the ZOH**

If the error e is a signal with limited bandwidth ω_{max} , with $\omega_{max} < \omega_N$, then, it can be shown that $\forall \omega \in [0, \omega_N]$:



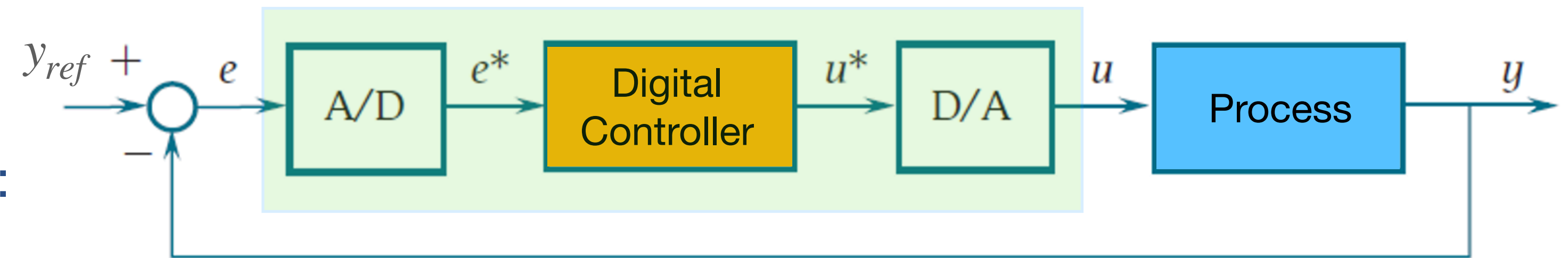
$$U(j\omega) = \frac{1}{T_s} H_0(j\omega) R^*(e^{j\omega T_s}) E(j\omega)$$

impact on the cascade
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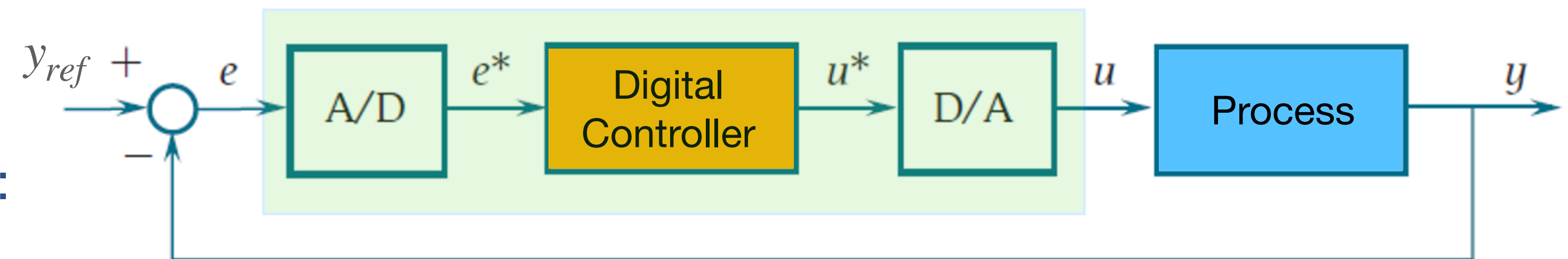
$$\frac{U(j\omega)}{E(j\omega)} = \frac{1}{T_s} H_0(j\omega) R^*(e^{j\omega T_s})$$

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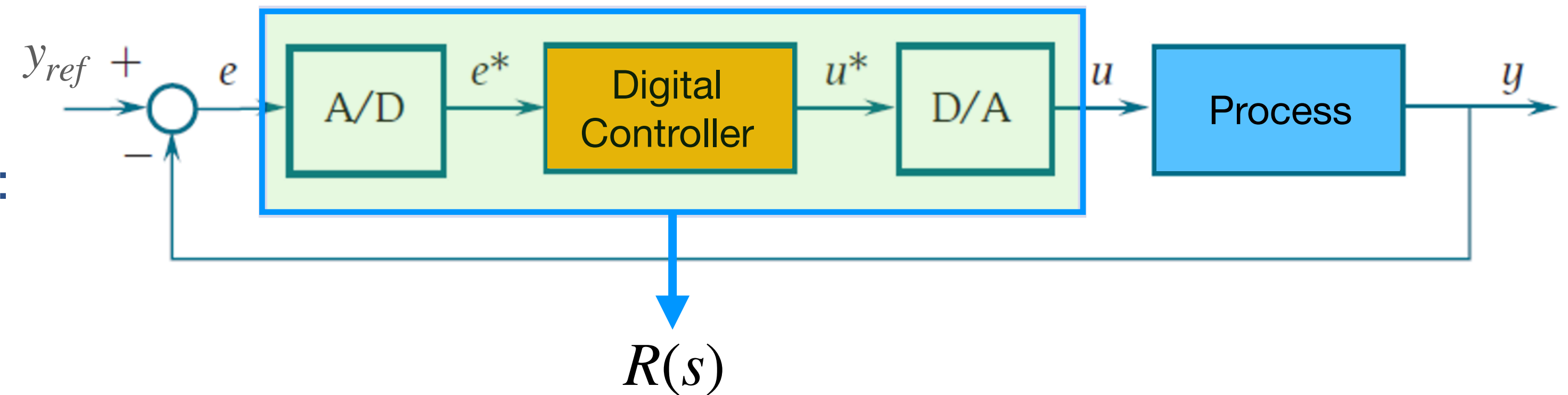
$$\frac{U(j\omega)}{E(j\omega)} = \frac{1}{T_s} H_0(j\omega) R^*(e^{j\omega T_s}) \longrightarrow R(j\omega) \approx \frac{1}{T_s} T_s e^{-j\omega \frac{T_s}{2}} R^*(e^{j\omega T_s})$$

impact on the cascade
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Digital Control Schemes: Effect of the ZOH

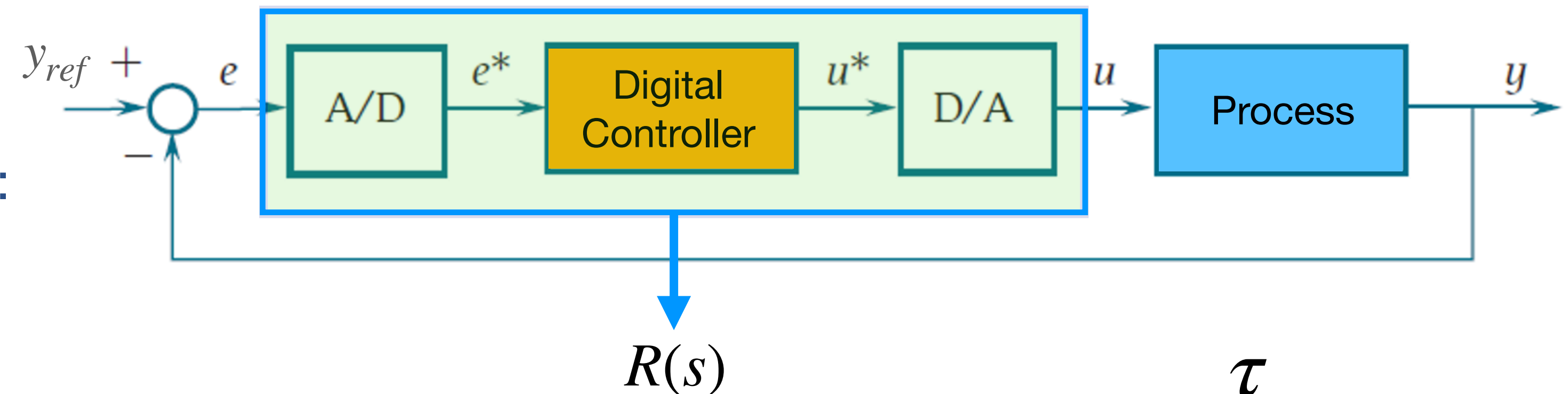
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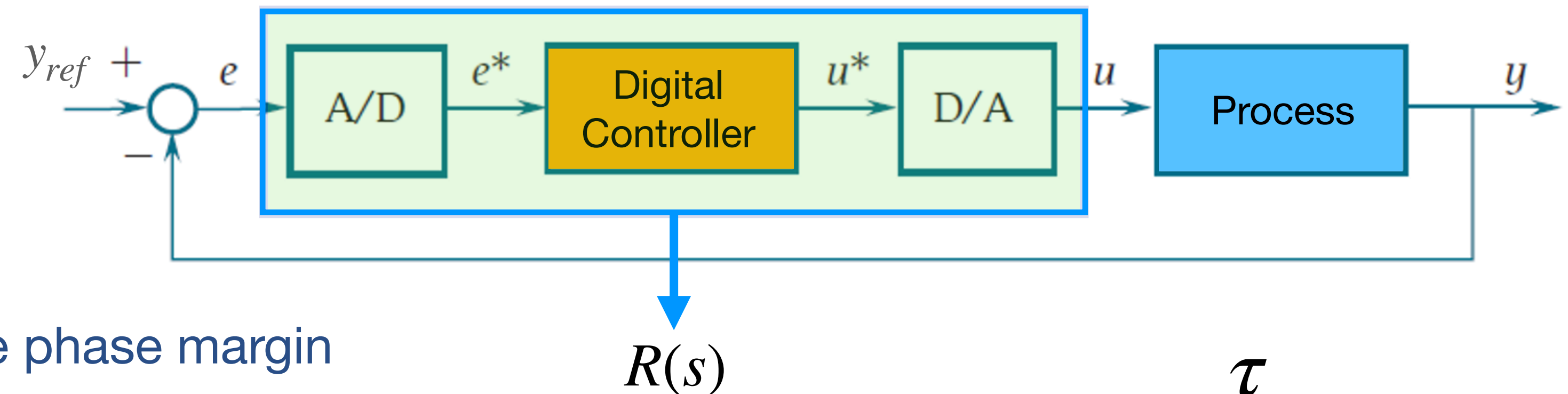
impact on the cascade
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$$H_0(j\omega) \approx T_s e^{-j\omega \frac{T_s}{2}}, \quad \omega \in [0, \omega_N]$$



τ_{ZOH}

Digital Control Schemes: Effect of the ZOH



Warning:

If the original $R_o(s)$ was such that the phase margin φ_M was acceptable, the analog controller realized via the cascade connection of sampler-digital controller-ZOH, i.e. $R(s)$, has a phase margin equal to $\varphi_M + \varphi_{ZOH}$ with:

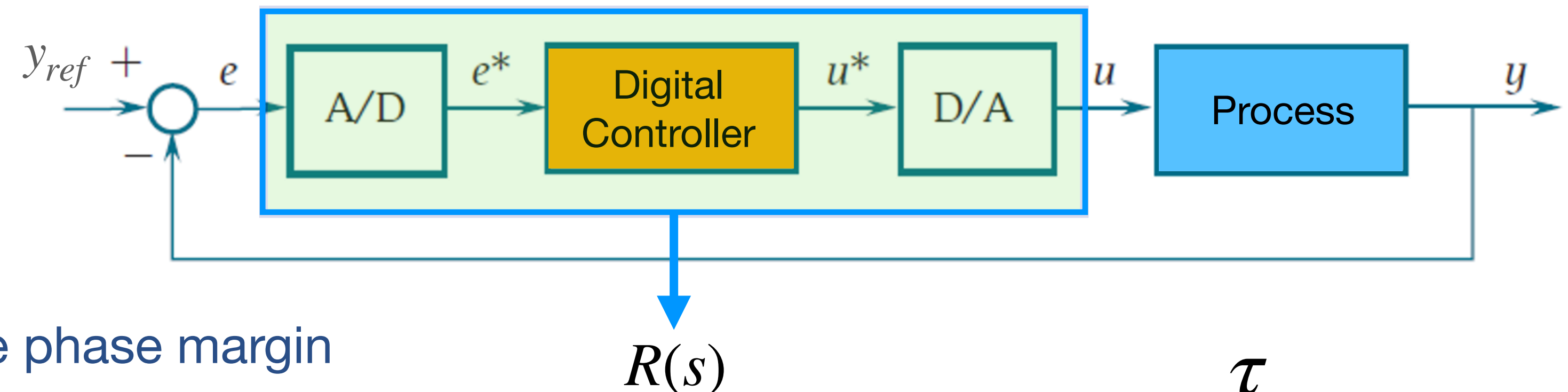
$$R(s) \approx e^{-\frac{T_s}{2}s} R^*(e^{sT_s})$$

τ_{ZOH}

$$\varphi_{ZOH} = -\frac{\omega_c T_s}{2} \frac{180^\circ}{\pi}$$

$$H_0(j\omega) \approx T_s e^{-j\omega \frac{T_s}{2}}, \quad \omega \in [0, \omega_N]$$

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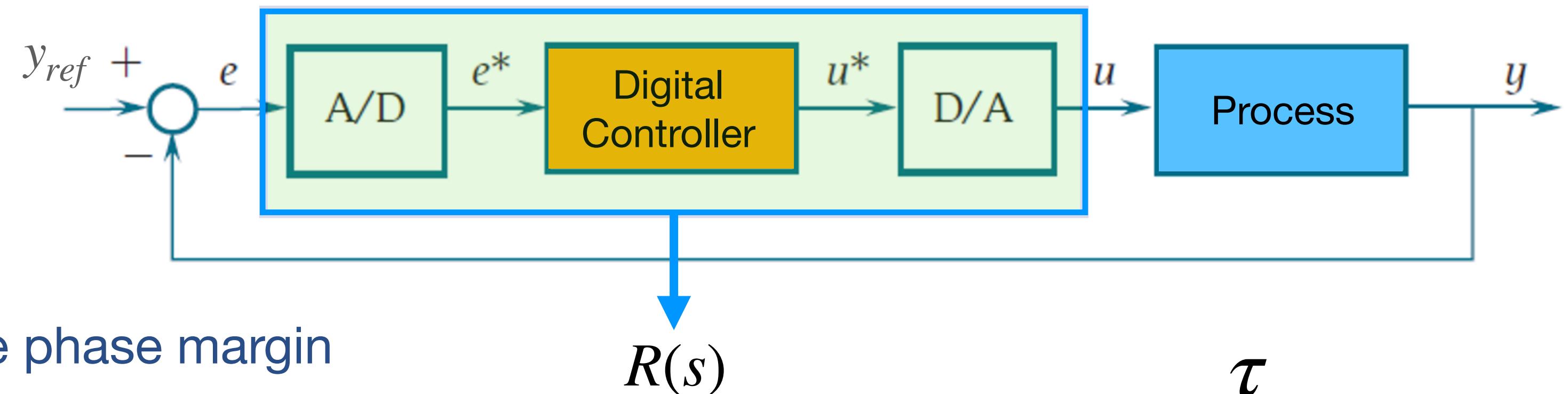
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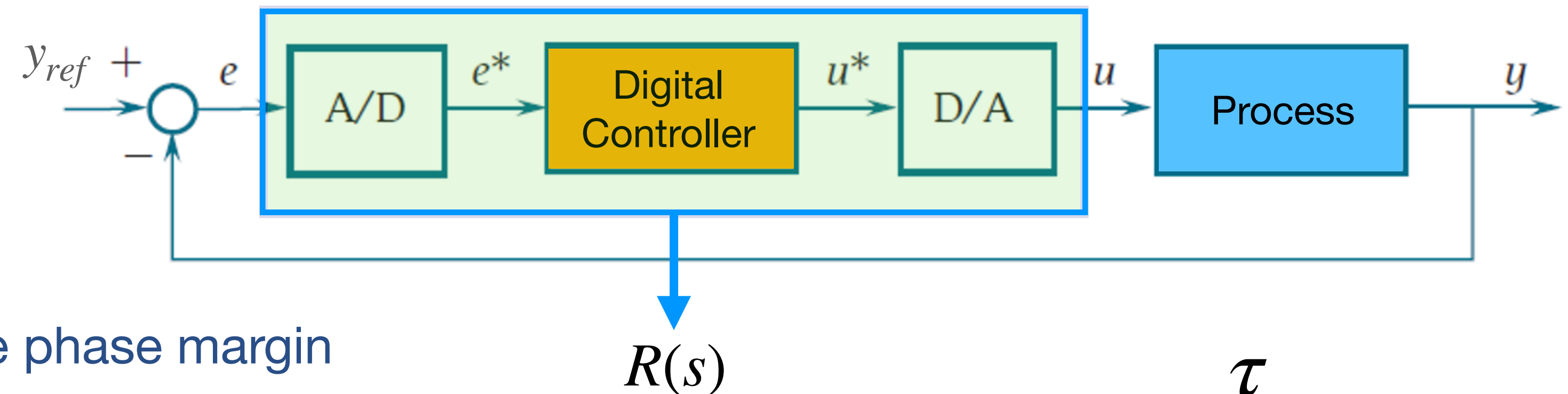
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Example:

$$\varphi_M = 40^\circ, \omega_c = 10 \text{ [rad/s]}, \omega_s = 10^2 \text{ [rad/s]}$$

Digital Control Schemes: Effect of the ZOH



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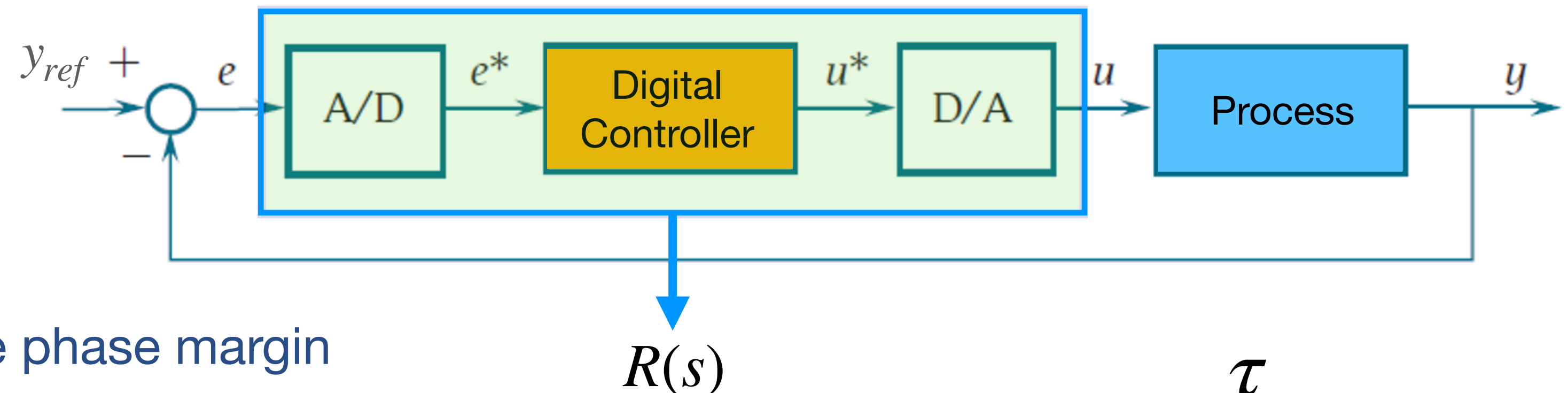
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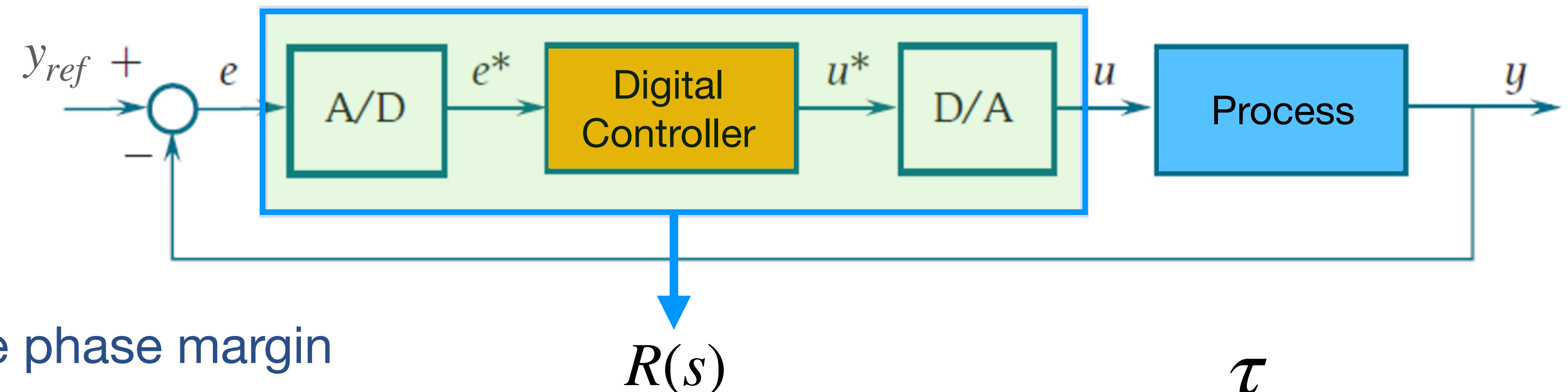
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$$\varphi_M + \varphi_{ZOH} = 40^\circ - 18^\circ = 22^\circ$$

Digital Control Schemes: Effect of the ZOH



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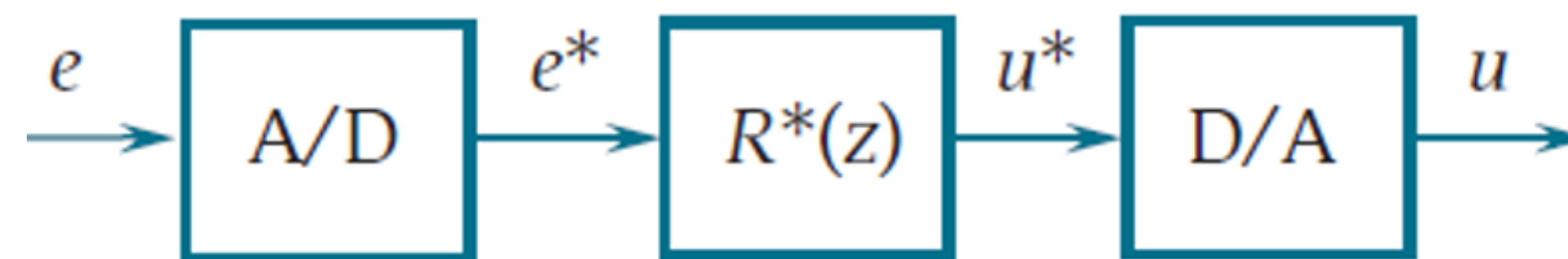
$$\varphi_M + \varphi_{ZOH} = 40^\circ - 18^\circ = 22^\circ$$

Digital Control Schemes: **Problems to solve**

1. How to determine the transfer function of a discrete-time LTI system



The control element:



Digital Controller

$R(s)$

4. How to discretize the original continuous-time controller to obtain $R^*(z)$

A/D converter: it is a sampler



2. How to determine the sampling time T_s

D/A converter: it is a hold



3. How to model the ZOH effect

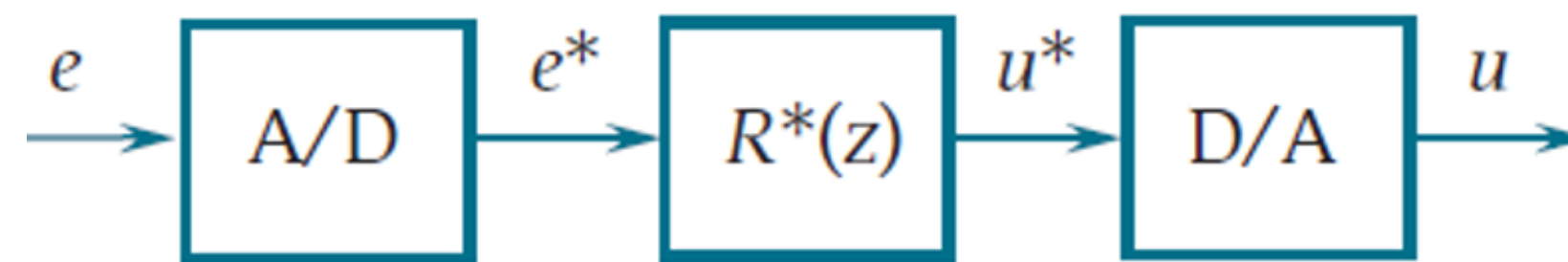


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Digital Control Schemes: Discretization of the analog controller

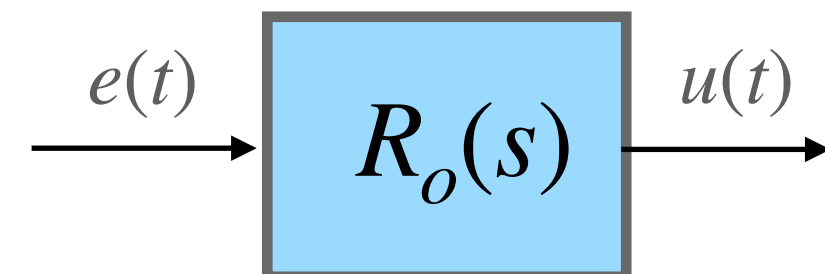
Consider the original continuous-time controller $R_o(s)$ and its realization in state-form:

$$R_o(s) = C(sI - A)^{-1}B + D$$



$$\dot{x} = Ax + Be$$

$$u = Cx + De$$



Digital Control Schemes: Discretization of the analog controller

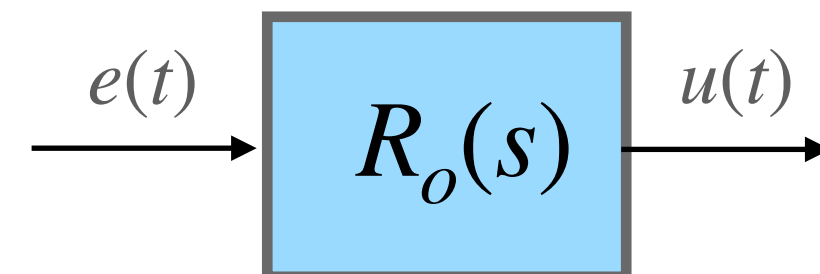
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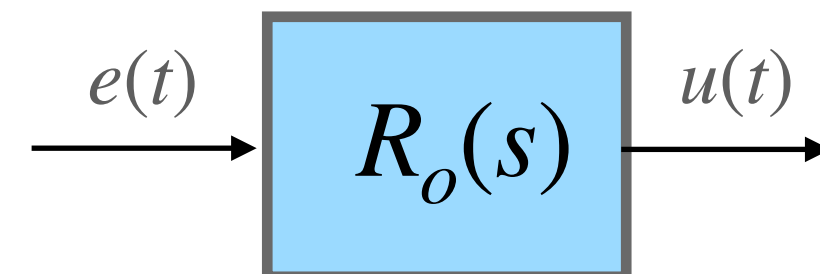
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↓

$$\begin{aligned}\dot{x} &= Ax + Be \\ u &= Cx + De\end{aligned}$$

→

$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} (Ax + Be) dt$$

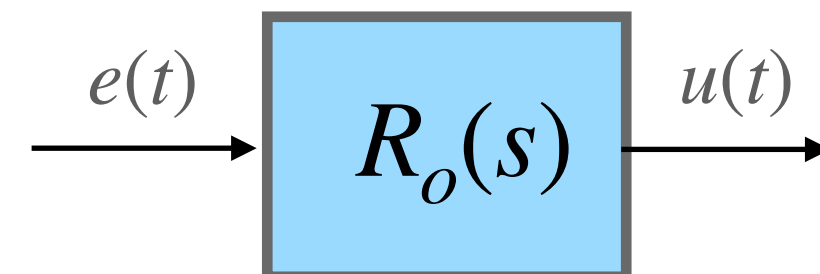


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$$R_o(s) = C(sI - A)^{-1}B + D$$

$$\begin{aligned} \dot{x} &= Ax + Be \\ u &= Cx + De \end{aligned}$$



$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} (Ax + Be) dt$$

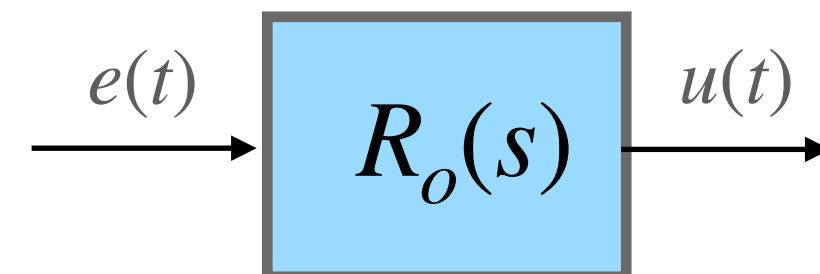
$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} A x dt + \int_{kT_s}^{(k+1)T_s} B e dt$$

Digital Control Schemes: Discretization of the analog controller

Consider the original continuous-time controller $R_o(s)$ and its realization in state-form:

$$R_o(s) = C(sI - A)^{-1}B + D$$

$$\begin{aligned} \dot{x} &= Ax + Be \\ u &= Cx + De \end{aligned}$$



$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} (Ax + Be) dt$$

$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} A x dt + \int_{kT_s}^{(k+1)T_s} B e dt$$

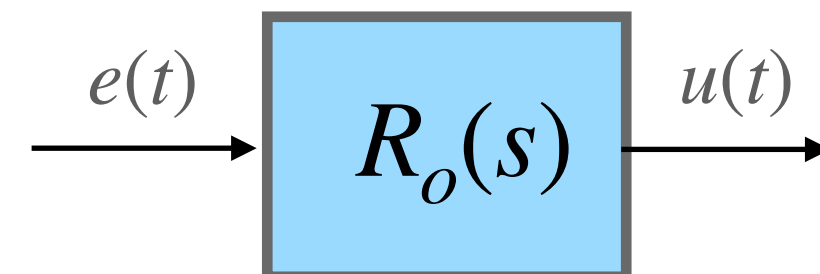
$$x((k+1)T_s) - x(kT_s) = A \int_{kT_s}^{(k+1)T_s} x dt + B \int_{kT_s}^{(k+1)T_s} e dt$$

Digital Control Schemes: Discretization of the analog controller

Consider the original continuous-time controller $R_o(s)$ and its realization in state-form:

$$R_o(s) = C(sI - A)^{-1}B + D$$

$$\begin{aligned} \dot{x} &= Ax + Be \\ u &= Cx + De \end{aligned}$$



$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} (Ax + Be) dt$$

$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} A x dt + \int_{kT_s}^{(k+1)T_s} B e dt$$

$$x((k+1)T_s) - x(kT_s) = A \int_{kT_s}^{(k+1)T_s} x dt + B \int_{kT_s}^{(k+1)T_s} e dt$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A \int_{kT_s}^{(k+1)T_s} x dt + B \int_{kT_s}^{(k+1)T_s} e dt$$

$$\int_{kT_s}^{(k+1)T_s} f(t) dt =$$

Digital Control Schemes: Discretization of the analog controller

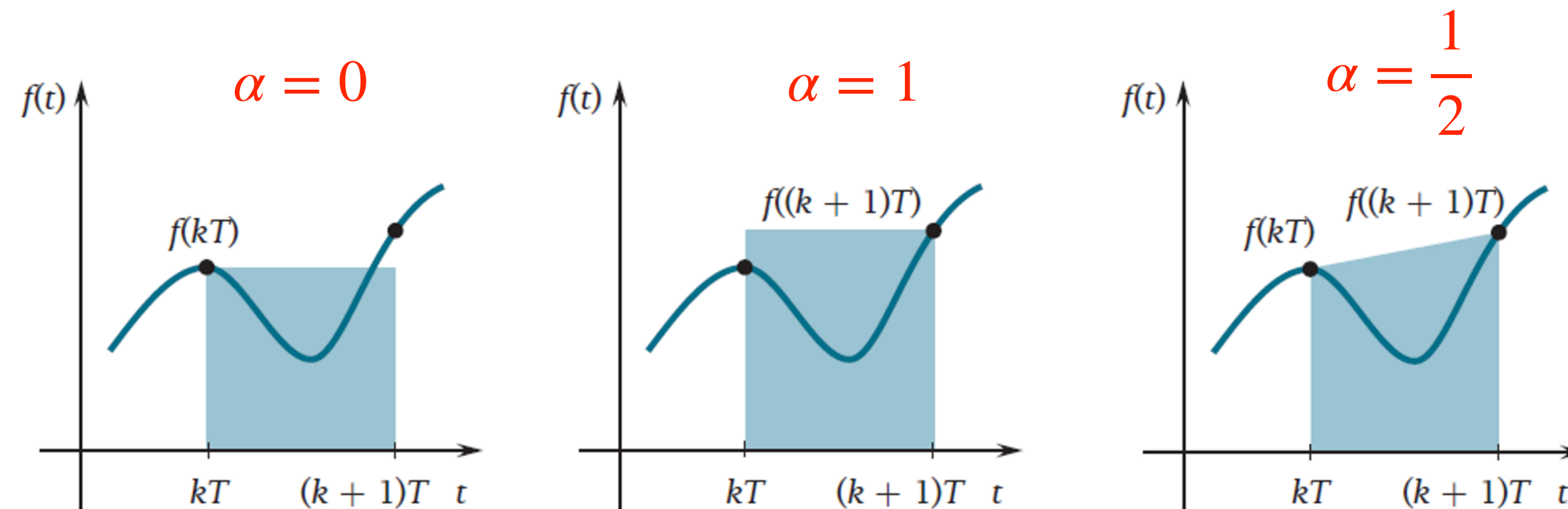
$$x((k+1)T_s) - x(kT_s) = A \int_{kT_s}^{(k+1)T_s} x dt + B \int_{kT_s}^{(k+1)T_s} e dt$$

$$\int_{kT_s}^{(k+1)T_s} f(t) dt = [(1 - \alpha)f(kT_s) + \alpha f((k+1)T_s)] T_s, \quad 0 \leq \alpha \leq 1$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A \int_{kT_s}^{(k+1)T_s} x dt + B \int_{kT_s}^{(k+1)T_s} e dt$$

$$\int_{kT_s}^{(k+1)T_s} f(t) dt = [(1 - \alpha)f(kT_s) + \alpha f((k+1)T_s)] T_s, \quad 0 \leq \alpha \leq 1$$



Forward Euler Method

Backward Euler Method

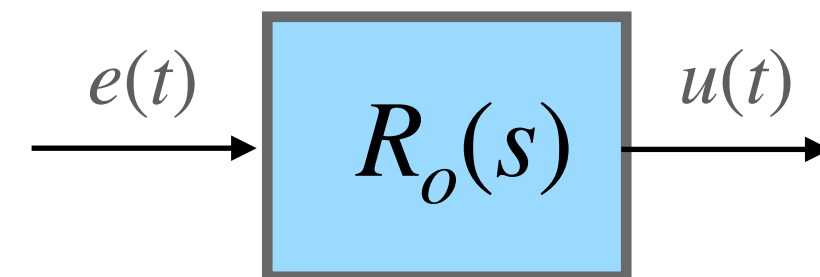
Tustin Method

Digital Control Schemes: Discretization of the analog controller

Consider the original continuous-time controller $R_o(s)$ and its realization in state-form:

$$R_o(s) = C(sI - A)^{-1}B + D$$

$$\begin{aligned} \dot{x} &= Ax + Be \\ u &= Cx + De \end{aligned}$$



$$\longrightarrow x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} (Ax + Be) dt$$

$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} Ax dt + \int_{kT_s}^{(k+1)T_s} Be dt$$

$$x((k+1)T_s) - x(kT_s) = A \int_{kT_s}^{(k+1)T_s} x dt + B \int_{kT_s}^{(k+1)T_s} e dt$$

$$x((k+1)T_s) - x(kT_s) = A [(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)] T_s + B [(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)] T_s \quad 0 \leq \alpha \leq 1$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A [(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)] T_s + B [(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)] T_s \quad 0 \leq \alpha \leq 1$$

 $\mathcal{Z}[\cdot]$ 

$$zX(z) - X(z) = A [(1-\alpha)X(z) + \alpha zX(z)] T_s + B [(1-\alpha)E(z) + \alpha zE(z)] T_s \quad 0 \leq \alpha \leq 1$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A [(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)] T_s + B [(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)] T_s \quad 0 \leq \alpha \leq 1$$

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$$zX(z) - X(z) = A [(1-\alpha)X(z) + \alpha zX(z)] T_s + B [(1-\alpha)E(z) + \alpha zE(z)] T_s \quad 0 \leq \alpha \leq 1$$

$$(z-1)X(z) = A(1-\alpha + \alpha z) T_s X(z) + B(1-\alpha + \alpha z) T_s E(z) \quad 0 \leq \alpha \leq 1$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A [(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)] T_s + B [(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)] T_s \quad 0 \leq \alpha \leq 1$$

 $\mathcal{Z}[\cdot]$ 

$$zX(z) - X(z) = A [(1-\alpha)X(z) + \alpha zX(z)] T_s + B [(1-\alpha)E(z) + \alpha zE(z)] T_s \quad 0 \leq \alpha \leq 1$$

$$(z-1)X(z) = A(1-\alpha + \alpha z)T_s X(z) + B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$(z-1)X(z) - A(1-\alpha + \alpha z)T_s X(z) = B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A [(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)] T_s + B [(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)] T_s \quad 0 \leq \alpha \leq 1$$

 $\mathcal{Z}[\cdot]$ 

$$zX(z) - X(z) = A [(1-\alpha)X(z) + \alpha zX(z)] T_s + B [(1-\alpha)E(z) + \alpha zE(z)] T_s \quad 0 \leq \alpha \leq 1$$

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$$(z-1)X(z) - A(1-\alpha + \alpha z)T_s X(z) = B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$[(z-1)I - A(1-\alpha + \alpha z)T_s]X(z) = B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A [(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)] T_s + B [(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)] T_s \quad 0 \leq \alpha \leq 1$$

 $\mathcal{Z}[\cdot]$ 

$$zX(z) - X(z) = A [(1-\alpha)X(z) + \alpha zX(z)] T_s + B [(1-\alpha)E(z) + \alpha zE(z)] T_s \quad 0 \leq \alpha \leq 1$$

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$$(z-1)X(z) - A(1-\alpha + \alpha z)T_s X(z) = B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

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Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A [(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)] T_s + B [(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)] T_s \quad 0 \leq \alpha \leq 1$$

 $\mathcal{Z}[\cdot]$


$$zX(z) - X(z) = A [(1-\alpha)X(z) + \alpha zX(z)] T_s + B [(1-\alpha)E(z) + \alpha zE(z)] T_s \quad 0 \leq \alpha \leq 1$$

$$(z-1)X(z) = A(1-\alpha + \alpha z) T_s X(z) + B(1-\alpha + \alpha z) T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$(z-1)X(z) - A(1-\alpha + \alpha z) T_s X(z) = B(1-\alpha + \alpha z) T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$[(z-1)I - A(1-\alpha + \alpha z) T_s] X(z) = B(1-\alpha + \alpha z) T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$\left[\frac{1}{T_s} \frac{z-1}{1-\alpha + \alpha z} I - A \right] X(z) = B E(z) \quad 0 \leq \alpha \leq 1$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A[(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)]T_s + B[(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)]T_s \quad 0 \leq \alpha \leq 1$$

 $\mathcal{Z}[\cdot]$


$$zX(z) - X(z) = A[(1-\alpha)X(z) + \alpha zX(z)]T_s + B[(1-\alpha)E(z) + \alpha zE(z)]T_s \quad 0 \leq \alpha \leq 1$$

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$$[(z-1)I - A(1-\alpha + \alpha z)T_s]X(z) = B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$\left[\frac{1}{T_s} \frac{z-1}{1-\alpha + \alpha z} I - A \right] X(z) = B E(z) \quad 0 \leq \alpha \leq 1 \quad \longrightarrow \quad X(z) = \left[\frac{1}{T_s} \frac{z-1}{1-\alpha + \alpha z} I - A \right]^{-1} B E(z)$$

Digital Control Schemes: Discretization of the analog controller

$$x((k+1)T_s) - x(kT_s) = A[(1-\alpha)x(kT_s) + \alpha x((k+1)T_s)]T_s + B[(1-\alpha)e(kT_s) + \alpha e((k+1)T_s)]T_s \quad 0 \leq \alpha \leq 1$$

 $\mathcal{Z}[\cdot]$


$$zX(z) - X(z) = A[(1-\alpha)X(z) + \alpha zX(z)]T_s + B[(1-\alpha)E(z) + \alpha zE(z)]T_s \quad 0 \leq \alpha \leq 1$$

$$(z-1)X(z) = A(1-\alpha + \alpha z)T_s X(z) + B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$(z-1)X(z) - A(1-\alpha + \alpha z)T_s X(z) = B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$[(z-1)I - A(1-\alpha + \alpha z)T_s]X(z) = B(1-\alpha + \alpha z)T_s E(z) \quad 0 \leq \alpha \leq 1$$

$$\left[\frac{1}{T_s} \frac{z-1}{1-\alpha + \alpha z} I - A \right] X(z) = B E(z) \quad 0 \leq \alpha \leq 1 \quad \longrightarrow$$

$$X(z) = \left[\frac{1}{T_s} \frac{z-1}{1-\alpha + \alpha z} I - A \right]^{-1} B E(z)$$

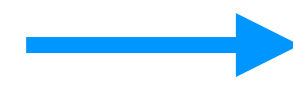
Digital Control Schemes: Discretization of the analog controller

$$\boxed{u = Cx + De} \longrightarrow u(kT_s) = Cx(kT_s) + De(kT_s)$$

$$X(z) = \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z)$$

Digital Control Schemes: Discretization of the analog controller

$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

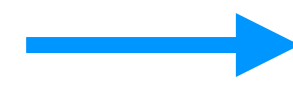
 $\mathcal{Z}[\cdot]$ 

$$U(z) = CX(z) + DE(z)$$

$$X(z) = \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z)$$

Digital Control Schemes: Discretization of the analog controller

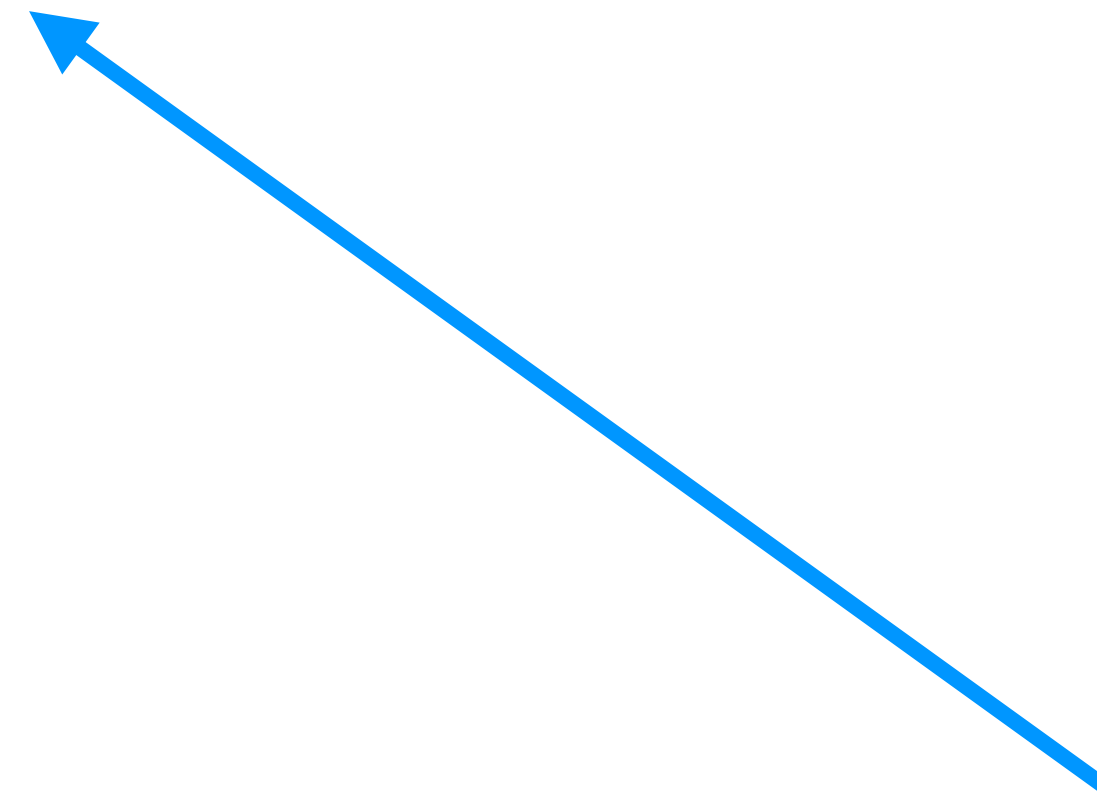
$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

 $\mathcal{Z}[\cdot]$ 

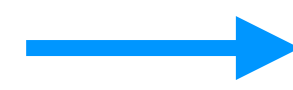
$$U(z) = CX(z) + DE(z)$$



$$X(z) = \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z)$$

Digital Control Schemes: Discretization of the analog controller

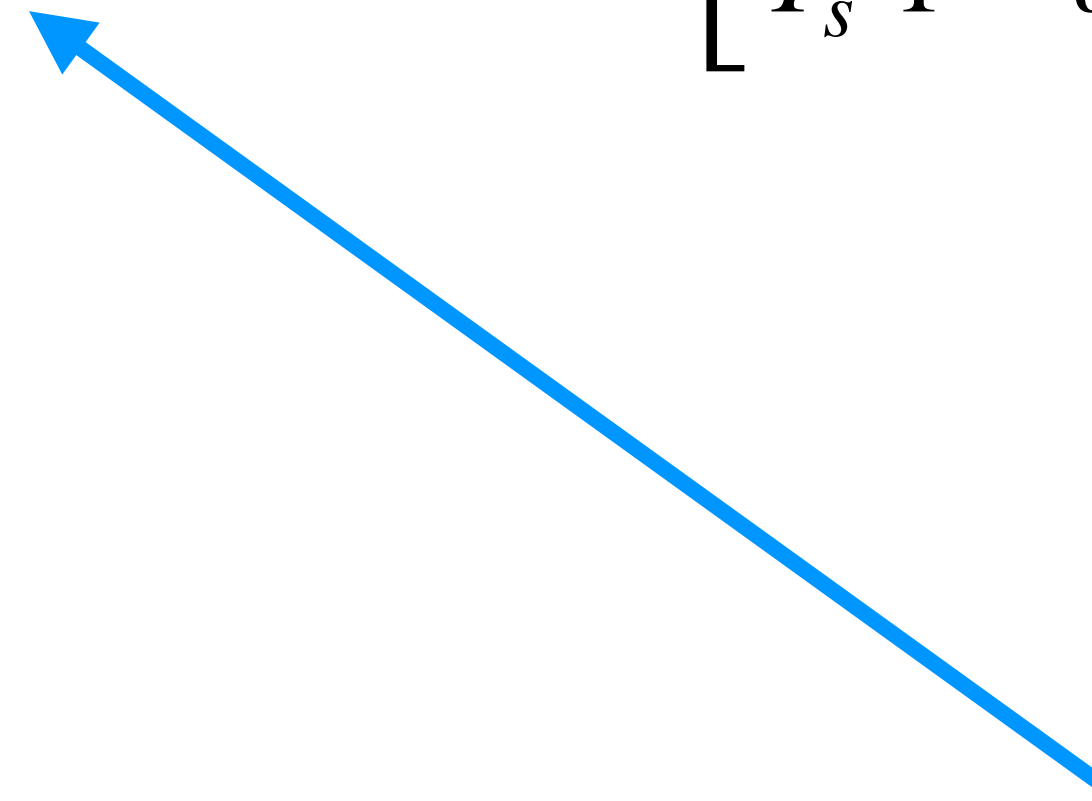
$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

 $\mathcal{Z}[\cdot]$ 

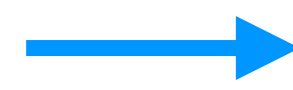
$$U(z) = CX(z) + DE(z) = C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z) + DE(z)$$



$$X(z) = \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z)$$

Digital Control Schemes: Discretization of the analog controller

$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

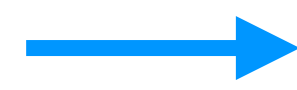
 $\mathcal{Z}[\cdot]$ 

$$U(z) = CX(z) + DE(z) = C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z) + DE(z)$$

$$U(z) = \left\{ C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B + D \right\} E(z)$$

Digital Control Schemes: Discretization of the analog controller

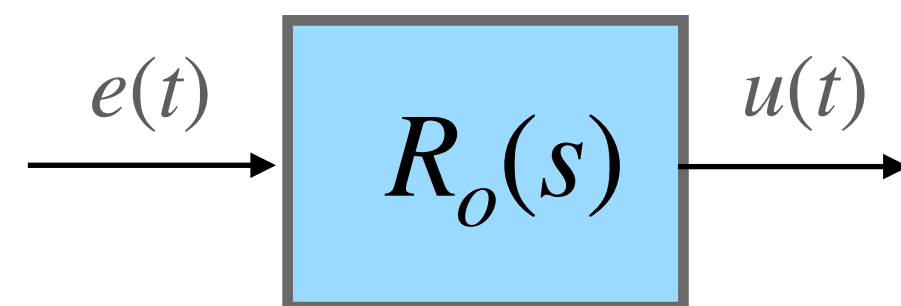
$$u = Cx + De$$



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 $\mathcal{Z}[\cdot]$ 

$$U(z) = CX(z) + DE(z) = C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z) + DE(z)$$

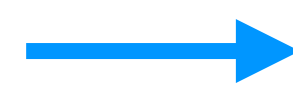


$$U(z) = \left\{ C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B + D \right\} E(z)$$


$$R_o(s) = C (sI - A)^{-1} B + D$$

Digital Control Schemes: Discretization of the analog controller

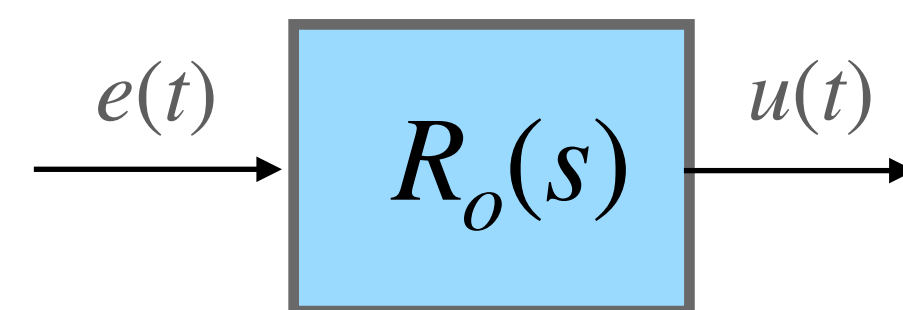
$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

 $\mathcal{Z}[\cdot]$


$$U(z) = CX(z) + DE(z) = C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z) + DE(z)$$

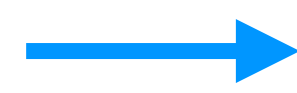


$$U(z) = \left\{ \begin{array}{c} \text{C} \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} \text{I} - \text{A} \right]^{-1} \text{B} + \text{D} \end{array} \right\} E(z)$$

$$R_o(s) = C (sI - A)^{-1} B + D$$

Digital Control Schemes: Discretization of the analog controller

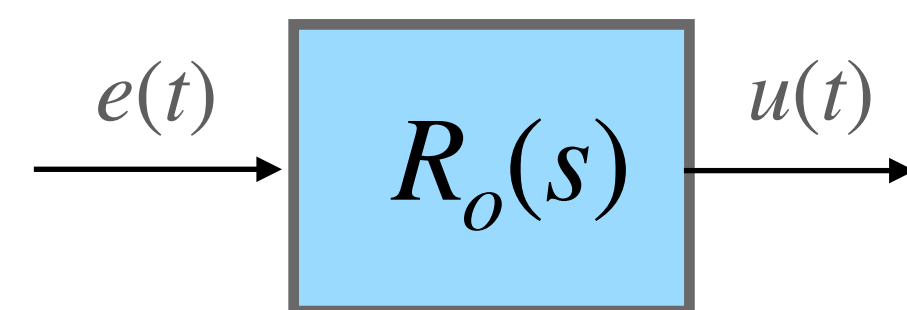
$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

 $\mathcal{Z}[\cdot]$


$$U(z) = CX(z) + DE(z) = C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z) + DE(z)$$



$$U(z) = \left\{ C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B + D \right\} E(z)$$

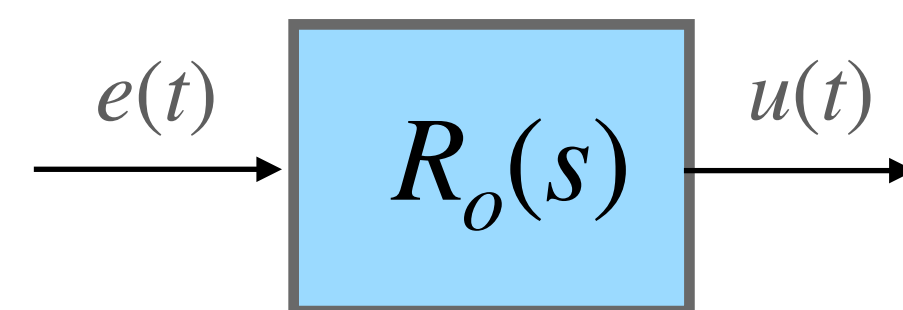
$$R_o(s) = C (sI - A)^{-1} B + D$$

Digital Control Schemes: Bilinear Transformation Method

the original continuous-time controller $R_o(s)$ is discretized using:

$$s = \frac{1}{T_s} \frac{z - 1}{1 - \alpha + \alpha z}$$

the digital controller $R(z)$ (or $R^*(z)$, depending on the notation) is obtained



$$U(z) = \left\{ C \left[\frac{1}{T_s} \frac{z - 1}{1 - \alpha + \alpha z} I - A \right]^{-1} B + D \right\} E(z)$$

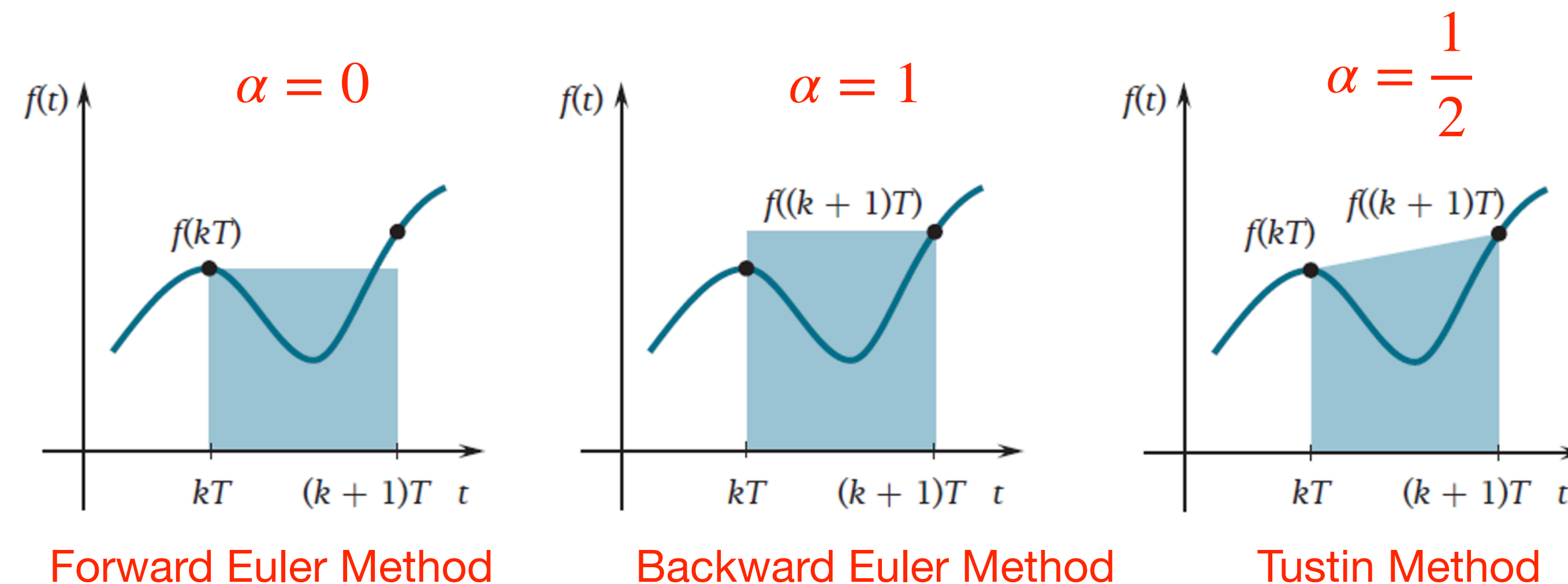
$$R_o(s) = C (sI - A)^{-1} B + D$$

Digital Control Schemes: Bilinear Transformation Method

the original continuous-time controller $R_o(s)$ is discretized using:

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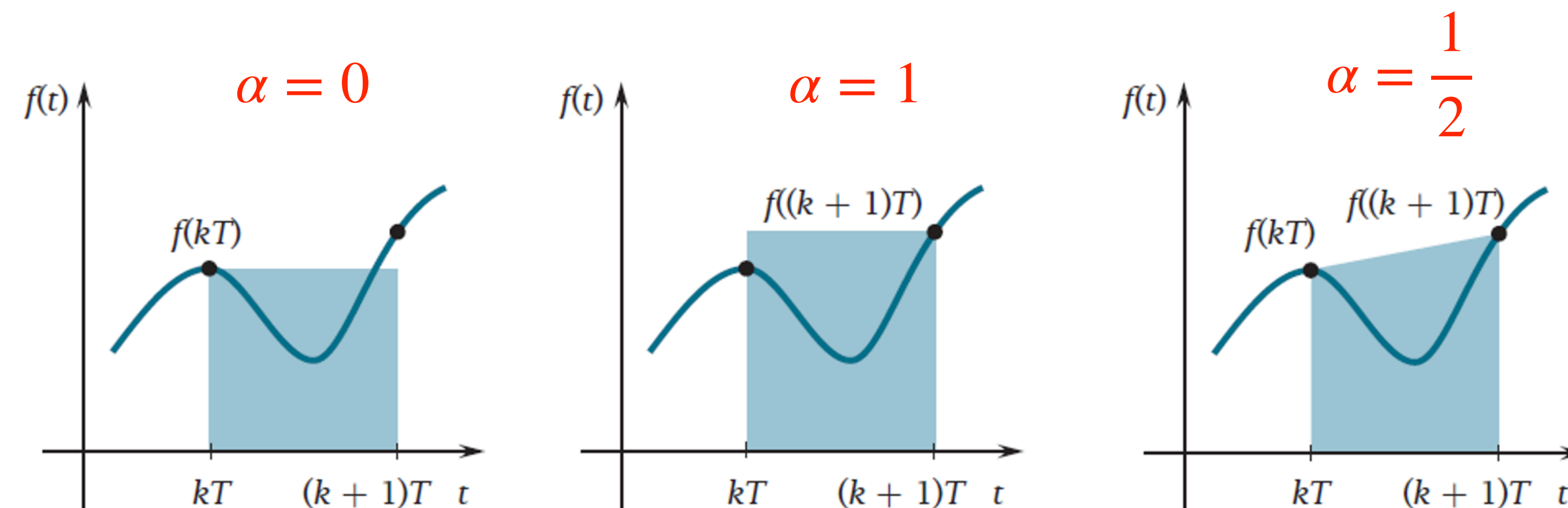
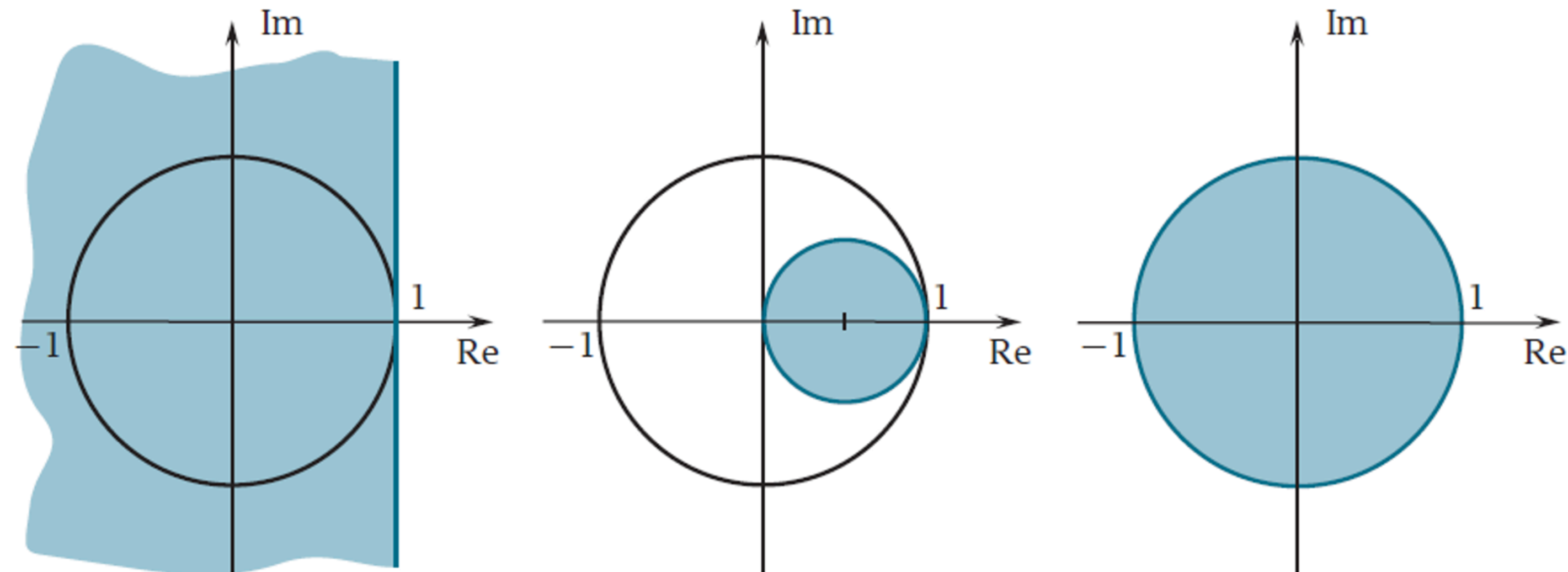
the digital controller $R(z)$ (or $R^*(z)$, depending on the notation) is obtained



Digital Control Schemes: Bilinear Transformation Method

Warning:

The stability properties of the original continuous-time controller can be lost because of the discretization!



Forward Euler Method

Backward Euler Method

Tustin Method